

PROJECT MANUAL  
  
FOR  
  
FIRE STATION NO. 5  
  
FOR THE  
  
**CITY OF EDINBURG**



2019

Page Intentionally Blank

City of Edinburg  
415 W. University Drive  
Edinburg, Texas 78539  
(956) 388-8211

Document 00001

TITLE SHEET  
PROJECT MANUAL  
FOR  
CITY OF EDINBURG  
FIRE STATION #5

FOR

EDINBURG, TEXAS

**CITY ENGINEER**

---

4/30/2019

END OF DOCUMENT

Page Intentionally Blank



Document 00003

**TABLE OF CONTENTS**

| <b><u>Document</u></b>            | <b><u>Title</u></b>                                                                 | <b><u>No. of Pages</u></b> |
|-----------------------------------|-------------------------------------------------------------------------------------|----------------------------|
| <b>INTRODUCTORY INFORMATION</b>   |                                                                                     |                            |
| 00001                             | Title Sheet.....                                                                    | 2                          |
| 00003                             | Table of Contents.....                                                              | 2                          |
| 00004                             | List of Drawings .....                                                              | 2                          |
| 00010                             | Request For Proposals .....                                                         | 2                          |
| 00020                             | Notice to Bidders.....                                                              | 4                          |
| 00100                             | Instruction to Bidders .....                                                        | 10                         |
| <b>BIDDING REQUIREMENTS</b>       |                                                                                     |                            |
| <b>BID FORMS</b>                  |                                                                                     |                            |
| 00300                             | Intent to Respond.....                                                              | 2                          |
| 00310                             | Form of Proposal.....                                                               | 4                          |
| <b>SUPPLEMENTS TO BID FORMS</b>   |                                                                                     |                            |
| 00411                             | Bid Bond .....                                                                      | 2                          |
| 00420                             | Statement of Bidders Qualifications.....                                            | 14                         |
| 00423                             | Certificate of Bidder's Experience & Qualifications.....                            | 2                          |
| 00425                             | Equipment & Material Suppliers List .....                                           | 4                          |
| 00429                             | Non-Bribery Model Form.....                                                         | 2                          |
| <b>POST-BID PROCEDURES</b>        |                                                                                     |                            |
| 00450                             | Post Bid Procedures .....                                                           | 2                          |
| 00460                             | Noncollusion Affidavit.....                                                         | 2                          |
| <b>CONTRACT FORMS</b>             |                                                                                     |                            |
| 00500                             | Agreement.....                                                                      | 54                         |
| 00510                             | Notice of Award.....                                                                | 4                          |
| 00550                             | Notice to Proceed .....                                                             | 2                          |
| 00610                             | Performance Bond .....                                                              | 4                          |
| 00620                             | Payment Bond .....                                                                  | 4                          |
| 00625                             | Affidavit of Insurance .....                                                        | 2                          |
| 00630                             | Forms of Business .....                                                             | 2                          |
| 00631                             | Resolution of Corporation .....                                                     | 2                          |
| 00632                             | Contractor's Resolution on Authorized Representative .....                          | 2                          |
| 00635                             | Contractor's Act of Assurance .....                                                 | 2                          |
| 00640                             | Certification Regarding Department Suspension and Other Responsibility Matters..... | 2                          |
| <b>CONDITIONS OF THE CONTRACT</b> |                                                                                     |                            |
| <b>GENERAL</b>                    |                                                                                     |                            |
| 00700                             | General Conditions .....                                                            | 80                         |
| <b>SUPPLEMENTARY CONDITIONS</b>   |                                                                                     |                            |
| 00800                             | Supplementary Conditions.....                                                       | 14                         |
| 00811                             | Wage Rates .....                                                                    | 2                          |
| 00830                             | Warranty.....                                                                       | 4                          |

00003-1 of 2

**ADDENDA**

|       |                            |   |
|-------|----------------------------|---|
| 00900 | Addenda (For filing) ..... | 4 |
| 00910 | Modifications .....        | 2 |

(For filing) Documents listed "for filing" are to be provided by the Bidder and are not included in this Project Manual unless indicated for example only. The Document numbers and titles hold places for actual documents to be submitted by the Contractor during the bid, post-bid, or construction phase of the Project.

\*1 For newspaper publication; not included as part of Project Manual.

**END OF DOCUMENT**

Document 00004

**LIST OF DRAWINGS**

The list of Drawings is provided on the Sheet Index page for Fire Station No. 5

|        |                                             |
|--------|---------------------------------------------|
| A0.0   | COVER SHEET                                 |
| AS1.0  | SITE PLAN                                   |
| AS1.1  | PROPOSED LANDSCAPE PLAN                     |
| C-1    | GENERAL NOTES                               |
| C-2    | EXISTING CONDITIONS LAYOUT                  |
| C-3    | NEW SITE LAYOUT                             |
| C-4    | NEW UTILITIES LAYOUT                        |
| C-5    | NEW GRADING & DRAINAGE LAYOUT               |
| C-6    | SUGGESTED EROSION & SEDIMENT CONTROL LAYOUT |
| C-7    | WATER DETAILS                               |
| C-8    | SANITARY SEWER DETAILS                      |
| C-9    | PAVING – SIDEWALK – DRAINAGE DETAILS        |
| C-10   | EROSION & SEDIMENT CONTROL DETAILS          |
| S1.1   | GENERAL NOTES                               |
| S1.2   | GENERAL NOTES                               |
| S1.3   | STRUCTURAL TYPICAL DETAILS                  |
| S1.4   | STRUCTURAL TYPICAL DETAILS                  |
| S2.1   | FOUNDATION PLAN                             |
| S3.1   | FRAMING PLAN                                |
| S4.1   | STRUCTURAL FOUNDATION DETAILS               |
| S5.1   | STRUCTURAL FRAMING DETAILS                  |
| S5.2   | STRUCTURAL FRAMING DETAILS                  |
| A1.0   | FLOOR PLAN                                  |
| A2.0   | EXTERIOR ELEVATIONS                         |
| A3.0   | ENLARGED RR PLANS & DETAILS                 |
| A3.1   | RESTROOM DETAILS                            |
| A4.0   | REFLECTED CEILING PLAN                      |
| A5.0   | BUILDING SECTION                            |
| A5.1   | WALL DETAILS                                |
| A5.2   | ROOF PLAN                                   |
| A6.0   | MILLWORK ELEVATIONS & DETAILS               |
| A6.1   | MILLWORK SECTIONS                           |
| A7.0   | DOOR SCHEDULE & DOOR DETAILS                |
| A7.1   | FINISH FLOOR PLAN & SCHEDULE                |
| A8.0   | ADA SHEET                                   |
| MEP2.1 | MEP SITE PLAN                               |
| MEP2.2 | ROOF PLAN                                   |
| M0.0   | MECHANICAL LEGEND                           |
| M2.1   | MECHANICAL FLOOR PLAN                       |
| M4.0   | MECHANICAL DETAILS                          |
| M5.0   | MECHANICAL SCHEDULES                        |
| E0.0   | ELECTRICAL SYMBOLS & ABBRV.                 |
| EL2.1  | ELECTRICAL LIGHTING PLAN                    |
| EP2.1  | ELECTRICAL POWER PLAN                       |
| EP2.2  | POWER PLAN ROOF                             |
| E4.1   | ELECTRICAL ONE LINE DIAGRAM                 |

00004-1 of 2

|      |                              |
|------|------------------------------|
| E5.1 | ELECTRICAL SCHEDULES         |
| E6.1 | ELECTRICAL DETAILS           |
| E6.2 | ELECTRICAL DETAILS           |
| P0.0 | PLUMBING SYMBOLS & ABBRV.    |
| P2.1 | PLUMBING PLAN                |
| P3.1 | ENLARGED PLUMBING PLAN       |
| P4.1 | PLUMBING SCHEDULE            |
| P4.2 | PLUMBING DETAILS             |
| P4.3 | PLUMBING DETAILS             |
| P5.1 | PLUMBING RISERS              |
| P5.2 | PLUMBING RISERS              |
| P5.3 | PLUMBING RISERS              |
| T0.0 | TECH. & SEC. NOTES & LEGENDS |
| T2.1 | TECHNOLOGY PLAN              |
| T5.1 | TECHNOLOGY DETAILS           |
| T5.2 | TECHNOLOGY DETAILS           |
| T5.3 | TECHNOLOGY DETAILS           |
| T5.4 | ACCESS CONTROL DETAILS       |

END OF DOCUMENT



## REQUEST FOR PROPOSALS

The City of Edinburg is soliciting sealed Request for Bids; hereinafter referred to as RFB, to be received by the City Secretary's Office located at 415 W. University Drive, Edinburg, Texas 78539. City of Edinburg normal business days are Monday through Friday between the hours of 8:00 a.m. to 5:00 p.m. and shall be closed on recognized holidays. A **pre-proposal conference** will be conducted by the Owner /Architect on May 8, 2019 at 2:00 p.m. The pre-proposal conference shall be conducted at the City of Edinburg City Hall Community Room: located at 415 W. University Drive Edinburg, Texas 78539. Attendance by prospective Bidders is recommended for all general contractors submitting proposals. Sub-contractors, suppliers, and equipment suppliers may attend.

RFP'S will be received until 3:00 p.m. Central Time, on Monday, May 20, 2019, shortly thereafter all submitted RFP'S will be gathered and taken to the Edinburg City Hall Community Room, 1st Floor, to be publicly opened and read aloud. Any RFP received after the closing time will not be accepted and will be returned to the submitter unopened. It is the responsibility of the submitter to see that any RFP submitted shall have sufficient time to be received by the City Secretary's Office prior to the RFP opening date and time. The receiving time in the City Secretary's Office will be the governing time for acceptability of the RFP's. RFP's will not be accepted by telephone or facsimile machine. All RFP'S must bear original signatures and figures. The RFP shall be for:

### RFB # 2019-20 FIRE STATION NO. 5

**Bidders receiving a "NOTICE TO BIDDERS" and/or "REQUEST FOR PROPOSALS" notice in the mail or reading same in the newspaper are advised that the bidding documents may be obtained from the City of Edinburg web page address: [www.cityofedinburg.com](http://www.cityofedinburg.com), or may obtain copies by contacting the office of: LORENA FUENTES, PURCHASING AGENT, LOCATED AT 415 W. UNIVERSITY DRIVE, EDINBURG, TEXAS 78539 by calling (956) 388-1895 or by emailing your request to the following address: [lfuentes@cityofedinburg.com](mailto:lfuentes@cityofedinburg.com). General and/or Prime Contractors submitting bids and/or proposals to the City of Edinburg shall be non-refundable.**

Plans, proposal forms, specifications, and contract documents may be purchased from the Engineering Department, Architect or are available for printing at [http://cityofedinburg.com/departments/finance/open\\_bid\\_notices.php](http://cityofedinburg.com/departments/finance/open_bid_notices.php). Copies of the plans and specifications may be examined without charge at the following location:

**City of Edinburg  
Engineering Department – 2nd Floor  
415 W. University Drive  
Edinburg, Texas 78539**

Hand Delivered RFB'S:

415 W. University Drive  
C/o City Secretary Department (1<sup>st</sup> Floor)

If using Land Courier (i.e. FedEx, UPS):

City of Edinburg  
C/o City Secretary  
415 W. University Drive  
Edinburg, Texas 78539

If Mailing Proposals:

City of Edinburg  
C/o City Secretary  
P.O. Box 1079  
Edinburg, Texas 78540-1079

The City of Edinburg reserves the right to refuse and reject any or all RFP's and to waive any or all formalities or technicalities and to accept the RFP deemed most advantageous to the City, and hold the RFP's for a period of **90** days without taking action.

RFP's must be submitted in an envelope sealed with tape and prominently marked on the lower left hand corner of the envelope with corresponding RFP number and title.

Please read your requirements thoroughly and be sure that the RFP offered complies with all requirements/specifications noted. Any variation from the solicitation requirements/specifications must be clearly indicated by letter, on a point by point basis, attached to and made a part of your RFP. If no exceptions are noted, and you are the successful respondent, it will be required that the service(s) be provided as specified.

**PURPOSE**

(1) The purpose of these solicitation documents is to provide a proposal for construction of:

**FIRE STATION NO. 5**

**INTENT**

(2) The services to be provided under this RFP shall be in accordance with and shall meet all specifications and/or requirements as shown in this solicitation for RFP. There is no intention to disqualify any respondent who can meet the requirements.

**SUBMITTAL OF RFP**

(3) RFPs shall be submitted in sealed envelopes as referenced on the attached solicitation. Four (4) complete sets of the response, one (1) original marked "**ORIGINAL**," and three (3) copies marked "**COPY**". RFPs submitted by facsimile (fax) or electronically shall **NOT** be accepted. Submittal of an RFP in response to this solicitation constitutes an offer by the respondent. Once submitted, RFP's become the property of the City of Edinburg and as such the City reserves the right to use any ideas contained in any RFP regardless of whether that respondent/firm is selected. Submission of a RFP in response to this solicitation, by any respondent, shall indicate that the respondent(s) has/have accepted the conditions contained in the RFP, unless clearly and specifically noted in the RFP submitted and confirmed in the contract between the City and the successful respondent otherwise. RFPs which do not comply with these requirements may be rejected at the option of the City. RFPs must be filed with the City of Edinburg before the deadline day and hour. No late RFPs will be accepted. They will be returned to respondent unopened (if properly identified). Failure to meet RFP requirements may be grounds for disqualification.

Hand Delivered RFP'S:

415 W. University Drive  
c/o City Secretary Department (1<sup>st</sup> Floor)

If using Land Courier (i.e. FedEx, UPS):

City of Edinburg

c/o City Secretary  
415 W. University Drive  
Edinburg, Texas 78541

If Mailing RFP's:

City of Edinburg  
c/o City Secretary  
P.O. Box 1079  
Edinburg, Texas 78540-1079

**RFP DOCUMENTS:** Copies of the RFP Documents, including Drawings, Contract Documents and Technical Specifications may be obtained at Milnet Architectural Services, PLLC at 608 S 12<sup>th</sup> Street, McAllen, Texas 78501 office for a ***non-refundable*** deposit payable to Milnet Architectural Services, PLLC.

**TIME ALLOWED FOR ACTION TAKEN**

(4) The City of Edinburg may hold RFP/s 90 days after deadline without taking action. Respondents are required to hold their RFP/s firm for same period of time.

**RIGHT TO REJECT/AWARD**

(5) The City of Edinburg reserves the right to reject any or all RFPs, to waive any or all formalities or technicalities, and to make such awards of contract as may be deemed to be the best and most advantageous to the City of Edinburg.

**ASSIGNMENT**

(6) Respondents are advised that the City of Edinburg shall not allow the successful respondent to sell, assign, transfer, or convey any part of any contract resulting from this RFP in whole or in part, to a third party without the written approval of the City of Edinburg.

**AWARD**

(7) Respondents are advised that the City of Edinburg is soliciting RFPs and award shall be made to the respondent that in the opinion of the City of Edinburg is the best qualified.

**NUMBER OF CONTRACTS**

(8) THE CITY reserves the right to award one or no contract in response to this RFP.

**STATUTORY REQUIREMENTS**

(9) It shall be the responsibility of the successful respondent to comply with all applicable State & Federal laws, Executive Orders and Municipal Ordinances, and the Rules and Regulations of all authorities having jurisdiction over the work to be performed hereunder and such shall apply to the contract throughout, and that they will be deemed to be included in the contract as though written out in full in the contract documents.

**ALTERATIONS/AMENDMENTS TO RFP**

(10) RFP **CANNOT** be altered or amended after opening time. Alterations made before opening time must be initialed by respondent guaranteeing authenticity. No RFP may be withdrawn after opening time without acceptable reason in writing and only after approval by the City of Edinburg.

**NO RESPONSE TO RFP**

(11) If unable to submit a RFP, respondent should return inquiry giving reasons.

**LIST OF EXCEPTIONS**

(12) The respondent shall attach to his/her RFP a list of any exceptions to the specifications/requirements.

**PAYMENT**

(13) The City of Edinburg will execute payment by mail in accordance with the State of Texas Pay Law after SERVICES have been completed, introduced to the City, and found to meet City of Edinburg specifications/requirements. No other method of payment will be considered.

**SYNONYM**

(14) Where in this solicitation package SERVICES is used, its meaning shall refer to the request for the FIRE STATION NO. 5 as specified.

**RESPONDENT'S EMPLOYEES**

(15) Neither the Respondent nor his/her employees engaged in fulfilling the terms and conditions of this Service Contract shall be considered employees of the City. The method and manner of performance of such undertakings shall be under the exclusive control of the vendor on contract. The City shall have the right of inspection of said undertakings at any time.

**INDEMNIFICATION CLAUSE**

(16) The Respondent agrees to indemnify and save harmless the City, from all suits and actions of every nature and description brought against them or any of them, for or on account of the use of patented appliances, products or processes, and he shall pay all royalties and charges which are legal and equitable. Evidence of such payment or satisfaction shall be submitted upon request of the Purchasing Agent, as a necessary requirement in connection with the final estimate for payment in which such patented appliance, products or processes are used

**INTERPRETATIONS**

(17) Any questions concerning the project and/or specifications/requirements with regards to this solicitation for statement(s) of qualifications shall be directed to the designated individuals as outlined in the RFP. Such interpretations, which may affect the eventual outcome of this request for statements of qualifications, shall be furnished in writing to all prospective Respondents via Addendum. No interpretation shall be considered binding unless provided in writing by the City of Edinburg in accordance with paragraph entitled "Addenda and Modifications".

**VERBAL THREATS AND OFFICIAL CONTACT**

(18) Any threats made to any employee of the City, be it verbal or written, to discontinue the providing of item/material/services for whatever reason and/or reasons shall be considered a breach of contract and the City will immediately sever the contract with the Respondent/Consultant on contract.

Respondents shall not offer gratuities, favors or any monetary value to any official or employee of the City for purpose of influencing the selection. Any attempt by any Respondent to influence the selection process by any means, other than disclosure of qualifications and credentials through the proper channels, shall be grounds from exclusion from the selection process. Once the project is



advertised, there shall be no contact with any city official or employee unless using the formal process through the Purchasing Department. Failure to comply will result in the firm being disqualified from the process.

Questions and answers that change or substantially clarify the Request for Proposals will be affirmed in writing and copies will be provided to all firms on record responding to RFP. Any inquiries to this RFP must be submitted Ms. Lorena Fuentes, Purchasing Agent, at (956) 388-1895 or at the following e-mail address: [lfuentes@cityofedinburg.com](mailto:lfuentes@cityofedinburg.com) no later than **May 15, 2019 by 5:00 pm**.

#### **CONFIDENTIAL INFORMATION**

(19) Any information deemed to be confidential by the respondent should be clearly noted on the pages where confidential information is contained; however, the City cannot guarantee that it will not be compelled to disclose all or part of any public record under Texas Public Information Act, since information deemed to be confidential by the respondent may not be considered confidential under Texas Law, or pursuant to a Court order.

#### **PAST PERFORMANCE**

(20) Respondent's past performance shall be taken into consideration in the evaluation of RFP submittal.

#### **JURISDICTION**

(21) Contract(s) executed as part of this solicitation shall be subject to and governed under the laws of the State of Texas. Any and all obligations and payments are due and performable and payable in Hidalgo County, Texas.

#### **RIGHT TO AUDIT**

(22) The City of Edinburg reserves the right to audit the vendor's books and records relating to the performance of this contract. The City of Edinburg, at its own expense, shall have the right at all reasonable times during normal business hours and upon at least twenty-four (24) hours' advance notice, to audit, to examine, and to make copies of or extracts from the books of account and records maintained by the vendor(s) with respect to the Supply/Service and/or Purchase Contract. If such audit shall disclose overpayment by City to vendor, written notice of such overpayment shall be provided to the vendor and the amount of overpayment shall be promptly reimbursed by vendor to the City. In the event any such overpayment is not paid within ten (10) business days after receipt of such notice, the unpaid amount of such overpayment shall bear interest at the rate of one percent (1%) per month from the date of such notice until paid.

#### **VENUE**

(23) The parties agree that venue for purposes of any and all lawsuits, cause of action, and/or any other dispute(s) shall be in Hidalgo County, Texas.

**IF YOU HAVE ANY QUESTIONS ABOUT COMPLIANCE, PLEASE CONSULT YOUR OWN LEGAL COUNSEL. COMPLIANCE IS THE INDIVIDUAL RESPONSIBILITY OF EACH PERSON OR AGENT OF A PERSON WHO IS SUBJECT TO THE FILING REQUIREMENT. AN OFFENSE UNDER CHAPTER 176 IS A CLASS "C" MISDEMEANOR.**

### **CONFLICT OF INTEREST**

#### **(24) CHAPTER 176 OF THE TEXAS LOCAL GOVERNMENT CODE**

Effective January 1, 2006, Chapter 176 of the Texas Local Government Code requires that any vendor or person considering doing business with a local government entity disclose in the Questionnaire Form CIQ, the vendor or person's affiliation or business relationship that might cause a conflict of interest with a local government entity. By law, this questionnaire must be filed with the records administrator of the City of Edinburg not later than the 7<sup>th</sup> business day after the date the person becomes aware of facts that require the statement be filed. See Section 176.006, Local Government Code. A person commits an offense if the person violates Section 176.006, Local Government Code. An offense under this section is a Class C misdemeanor. For more information or to obtain Questionnaire CIQ go to the Texas Ethics Commission web page at [www.ethics.state.tx.us/forms/CIQ.pdf](http://www.ethics.state.tx.us/forms/CIQ.pdf).

### **CERTIFICATE OF INTERESTED PARTIES (Form 1295)**

(25) In 2015, the Texas Legislature adopted [House Bill 1295](#), which added section 2252.908 of the Government Code. The law states that a governmental entity or state agency may not enter into certain contracts with a business entity unless the business entity submits a disclosure of interested parties to the governmental entity or state agency at the time the business entity submits the signed contract to the governmental entity or state agency. The law applies only to a contract of a governmental entity or state agency that either (1) requires an action or vote by the governing body of the entity or agency before the contract may be signed or (2) has a value of at least \$1 million. The disclosure requirement applies to a contract entered into on or after January 1, 2016. For more information go to the Texas Ethics Commission web page at [www.ethics.state.tx.us/forms/CIQ.pdf](http://www.ethics.state.tx.us/forms/CIQ.pdf).

### **CONFIDENTIALITY OF INFORMATION AND SECURITY**

(26) Should the successful respondent become the holder of and have access to confidential information in the process of fulfilling its responsibilities in connection with an awarded contract the successful respondent agrees that it shall keep such information confidential and will comply fully with the laws and regulations of the State of Texas, ordinances and regulations of the City, and any applicable federal laws and regulations relating to confidentiality.

### **TERMINATION OF CONTRACT**

(27) The City of Edinburg reserves the right to terminate the contract if, in the opinion of the City of Edinburg, the successful vendor's performance is not acceptable, no funds are available, or if the City wishes, without cause, to discontinue this contract. Termination will be in written form allowing a 30-day notice.

### **RESPONSE DEADLINE**

(28) Responses to the RFP must be addressed to City Secretary, City of Edinburg, 415 W. University Drive by **Monday, May 20, 2019 until 3:00 p.m.** for consideration. **An (1) original and three (3) copies** of complete sets of the response must be submitted no later than this date and time in a **sealed envelope** indicating that its contents are in response to the RFP for the **"FIRE STATION NO. 5"**. **Respondents are advised that all confidential records must be submitted in a separate sealed envelope and marked accordingly.**

Hand Delivered RFP's:

415 W. University Drive

c/o City Secretary Department (1<sup>st</sup> Floor)

If using Land Courier (i.e. FedEx, UPS):

City of Edinburg  
c/o City Secretary  
415 W. University Drive  
Edinburg, Texas 78541

If Mailing RFPs:

City of Edinburg  
c/o City Secretary  
P.O. Box 1079  
Edinburg, Texas 78540-1079

### **ADDENDA AND MODIFICATIONS**

(29) Any changes, additions, or clarifications to the RFP are made by amendments (addenda). Any respondent in doubt as to the true meaning of any part of the RFP or other documents may request an interpretation from the Purchasing Division. At the request of the respondent, or in the event the Purchasing Division deems the interpretation to be substantive, the interpretation will be made by written addendum. Said Addenda shall be mailed, e-mailed, hand delivered and/or faxed, to all prospective respondents. All Addenda issued in respect to this RFP shall be considered official changes to the original documents. Verbal statements in response to inquiries and/or requests for explanations shall not be authoritative or binding. It shall be the respondent's responsibility to ensure that they have received all Addenda in respect to this project. Furthermore, respondents are advised that they must recognize, comply with, and attach a signed copy of each Addendum which shall be made part of their RFP Submittal. Respondent(s) signature on Addenda shall be interpreted as the respondent's "recognition and compliance to" official changes as outlined by the City of Edinburg and as such are made part of the original solicitation documents. Failure of any respondent to receive any such addendum or interpretation shall not relieve such respondent from its terms and requirements. The City may issue a written addendum no later than five calendar days prior to the date bids must be received. Addendums are available online at [www.cityofedinburg.com](http://www.cityofedinburg.com).

### **RFP PREPARATION COSTS**

(30) The City of Edinburg shall not be held liable for any costs incurred by any respondent for work performed in the preparation of and production of a RFP or for any work performed prior to execution of contract.

### **EQUAL EMPLOYMENT OPPORTUNITY**

(31) Respondent agrees that they will not discriminate in hiring, promotion, treatment, or other terms and conditions of employment based on race, sex, national origin, age, disability, or in any way violate Title VII of 1964 Civil Rights Act and amendments, except as permitted by said laws.

### **AUTHORIZATION TO BIND RESPONDENT TO RFP**

(32) RFPs MUST give full firm name and address of respondent, and be manually signed. Failure to do so will disqualify your RFP. Person signing bid must show title or AUTHORITY TO BIND HIS/HER FIRM IN A CONTRACT. Firm name and authorized signature must appear on each page that calls for this information. The legal status of the Respondent whether corporation, partnership,

or individual, shall also be stated in the RFP. A corporation shall execute the RFP by its duly authorized officers in accordance with its corporate by-laws and shall also list the state in which it is incorporated. A partnership Respondent shall give full names and addresses of all partners. All partners shall execute the RFP. Partnership and Individual Respondent shall state in the proposal the names and addresses of all persons with a vested interest therein. The place of residence of each Respondent, or the office address in the case of a firm or company, with county and state and telephone number, shall be given after the signature.

#### **BRAND OR MANUFACTURER REFERENCE**

(33) Unless otherwise specified, any catalog or manufacturer's reference or brand name used in describing an item is merely descriptive, and not restrictive, and is used only to indicate type and style of product desired. Proposals on alternate brands will be considered if they meet specification requirements. If a bidder quotes on equipment other than the one(s) specified in the bid, sufficient specifications and descriptive (pictured literature) data must accompany same to permit thorough evaluation. In the absence of these qualifications, he/she will be expected to furnish the product called for.

#### **COOPERATIVE PRICING**

(34) Bidders are advised that in addition to responding to our "local" solicitation for bids/Bids with Dealer pricing, vendors/contractors are encouraged to provide pricing on the below referenced items/products/services based on BuyBoard, TX-MAS, H-GAC and/or any other State of Texas recognized and approved cooperative which has complied with the bidding requirements for the State of Texas. If bidding other than or in addition to "dealer" pricing, kindly duplicate the bid forms for each bid being provided from a cooperative contract. Any and all applicable fees must be included. All cooperative pricing must be submitted on or before bid opening date and hour.

#### **HB 89**

(35) The 85th Texas Legislature approved new legislation, effective Sept. 1, 2017, which amends Texas Local Government Code Section 1. Subtitle F, Title 10, Government Code by adding Chapter 2270 which states that a governmental entity may not enter into a contract with a company for goods or services unless the contract contains a written verification from the company that it:

- 1) does not boycott Israel; and
- 2) will not boycott Israel during the term of the contract

**Confidential Information** Respondents are advised that all confidential records must be submitted in a separate sealed envelope and marked accordingly.

## **SECTION I SCOPE OF THE PROPOSAL**

### **INTRODUCTION**

The purpose of the RFP is to solicit and obtain from interested parties (also referred to herein as "Vendor" or "Vendors") the best possible proposal for the FIRE STATION NO. 5. The City of Edinburg intends to select the most competitive proposal that meets the City's requirements and specifications listed within the proposal and then enter into negotiations with the Vendor/s for purposes of reaching a satisfactory agreement for the City for the FIRE STATION NO. 5.

### **BACKGROUND**

The City of Edinburg FIRE STATION NO. 5 project will mitigate drainage issues occurring within the project area.

### **SCOPE OF WORK**

The City is soliciting competitive proposals from experienced and qualified companies for the FIRE STATION NO. 5. The project will improve the drainage in the area.

### **ADDITIONAL INFORMATION**

The City of Edinburg is requesting that RFP's (Request for Proposal) be routed to: The CITY Secretary, at 415 West University, Edinburg, Texas 78541.

### **NON-COLLUSION**

Submitters, by submitting a signed submission, certify that the accompanying submission is not the result of, or affected by, any unlawful act of collusion with any other person or company engaged in the same line of business or commerce, or any other fraudulent act punishable under Texas or United States law.

### **NON-DISCRIMINATION**

Submitters, during the performance of this contract, will not discriminate against any employee or applicant for employment because of race, religion, sex, national origin or disability except where religion, sex, national origin or disability is a bona fide occupational qualification reasonably necessary to the normal operation of the contractor.

### **PROCESSING TIME FOR PAYMENT**

Submitters are advised that a minimum of thirty (30) days is required to process invoices for payment.

### **ELECTRONIC SUBMISSION OF BIDS**

The City of Edinburg's City Secretary Department will not accept telegraphic or electronically transmitted submissions.

### **PROOF OF FINANCIAL AND BUSINESS CAPABILITY**

Submitters must, upon request, furnish satisfactory evidence of their ability to furnish products or services in accordance with the terms and conditions of these requirements. The CITY will make the final determination as to the submitter's ability.

### **SUBMITTER DEFAULT**

The City of Edinburg reserves the right, in case of submitter default, to procure the articles or services from other sources and hold the defaulting submitter responsible for any excess costs occasioned thereby.

### **RESTRICTIVE OR AMBIGUOUS REQUIREMENTS**

It is the responsibility of the submitter to review the Request for Proposals (RFP) packet and to notify the City Engineering Department if the requirements are formulated in a manner that would unnecessarily restrict competition. Any such protest or question regarding the requirements or bidding procedures must be received in the City Secretary Department not less than seventy-two hours prior to the time set for the opening. These criteria also apply to requirements that are ambiguous.

### **RFP DELIVERY**

The City of Edinburg requires submitters, when hand-delivering proposals by **3:00 pm on May 20, 2019**, to have a City Secretary Department representative time/date stamp and initial the envelope.

### **SIGNING OF PROPOSALS**

In order to be considered, all submittals **must** be signed.

### **WAIVING OF INFORMALITIES**

THE CITY reserves the right to waive minor informalities or technicalities when it is in the best interest of THE CITY.

### **SUBCONTRACTING**

The successful submitter may not subcontract the award without the written consent of the City.

### **BIDDER RESPONSIBILITY**

It is the responsibility of each vendor before submitting a proposal:

- To examine thoroughly the contract documents and other related data identified in the proposal documents.
- To visit the site to become familiar with and satisfy vendor as to the general, local, and site conditions that may affect cost, progress, performance, etc.
- To consider federal, state, and local laws and regulations that may affect costs, progress, performance or furnishing of the work.
- To study and carefully correlate vendor's knowledge and observations with the contract documents and such other related data.
- To promptly notify THE CITY Purchasing of all conflicts, errors, ambiguities, or discrepancies which vendor has discovered in or between the contract documents and such other related documents.

**TERMINATION**

THE CITY has the authority and express right to terminate any Agreement awarded under this RFP or any Work Order resulting from the Agreement at any time for any reason, including but not limited to, instances where THE CITY finds that the Contractor's work is negligent, not satisfactory, or not in accordance with the Agreement requirements.

## **SECTION II RFP REQUIREMENTS**

### **PURPOSE**

The intent of this Request for Proposal and resulting contract is to obtain proposals to FIRE STATION NO. 5.

### **REQUEST FOR PROPOSALS**

The required contents and limitations for the preparation of the RFP are described in this section. Failure to provide the requested information or adhere to any of The CITY limitations will result in disqualification of the submitted RFP. A total of **one (1) original and three (3) copies** of the RFP shall be submitted to the address on the cover letter. Letter of Intent from Surety Company to provide Payment and Performance Bonds shall also be required from the proposer as part of RFP.

### **SUBMITTAL**

For proper comparison and evaluation, THE CITY requests that proposals address, at a minimum, the following format.

- 1) **Cover Letter** - A brief introductory letter of representation.
- 2) **Executive Summary** - A brief summary highlighting the most important points of the proposal. If used, the Summary should not exceed five pages.
- 3) **Degree of Compliance** - A statement that all products and services quoted in proposal is in full accord with the specifications or a brief listing of all those specification sections to which the Proposer takes exception. All explanations, exceptions, comments, etc., pertaining to the specific sections of the specifications shall be listed and numbered in order of the respective article of the specification.

### **CONTENTS**

The required contents for the RFP are presented below in the order they should be incorporated into the submitted document.

- 1) **UNDERSTANDING OF THE PROJECT:** This section should demonstrate the submitter's understanding of the project's needs, the work required, and any local issues or concerns. This description should be concise, candid, and limited to 2 pages in length.
- 2) **FIRM QUALIFICATIONS, PERSONNEL AND STAFFING (00420 Statement of Bidder's Qualifications):** The CITY is seeking a contract with a competent firm(s); with a minimum of 5 years' experience of installation of the FIRE STATION NO. 5.
  - a) **Qualifications:**
    - i) List Firm's qualifications and ability to perform the service requirements.
    - ii) List qualifications of key personnel to be assigned to this project, including but not limited to education, training, registrations, certifications and licenses.



**b) Experience:**

- i) Number of years of experience as a General Contractor.
- ii) Relevant experience with projects of similar size and scope performed over the past five (5) years. For each project listed, date services provided and name, titles, and telephone numbers of each client or client's representative.
- iii) Specific experience with public entity clients, especially large municipalities. If company submitting proposal for new construction has provided services to the CITY in the past, identify the name of the project and the department for which services were provided.
- iv) If company submitting proposal for this project is submitting as a team or joint venture, provide the same information for each member of the team or joint venture.
- v) Provide the following information for key personnel to be assigned to this project:
  - (1) Total years' experience.
  - (2) Primary work assignment for the projects outlined in this RFP.
  - (3) Relevant experience with projects of similar size and scope.

**c) Previous Project Performance:**

- i) Provide evidence of satisfactory performance on past projects
- ii) List past assignments over the past five (5) years
- iii) Provide copies of outstanding service letters, letters of commendation, service awards, etc.
- iv) Provide five recent references who may be contacted to verify performance of similar services. For each reference, provide a current phone number and e-mail address. References may not be present or former CITY employees.

**d) Quality of Service:**

- i) Company submitting proposal for the FIRE STATION NO. 5 – Availability: Identify any concurrent or near future commitment that would impede the firm's ability to perform this contract.
- ii) Describe company submitting proposal for the FIRE STATION NO. 5 policies, procedures and plans to ensure quality services (continuing education, on-going training, internal quality practices, etc.)
- iii) If company submitting proposal for the FIRE STATION NO. 5 has ever had a contract terminated or has been dismissed due to alleged unsatisfactory performance, state when, where and why the contract was terminated and/or Security Consultant dismissed, the client's name, and the contact person's phone number.

- 3) **Proposal Pricing/Delivery** - Pricing shall be inclusive for all items requested in this proposal. Brief notes referencing specific line items may be included, if necessary, for explanation. Proposal shall state all labor, materials and equipment necessary to complete the project as stated in the SCOPE OF WORK (Page 2).
- 4) **Contractor Background Information** - This section should include a description of the Proposer experience with other services similar to the one described herein. This information should include scope of several similar jobs including magnitude and cost, customer contacts and other information that THE CITY can use as a basis for performance evaluation. This section should also include information on your organization and staff assigned to the project.
- 5) **References** - Proposer shall submit with this proposal a list of at least three (3) references where like services or similar projects have been performed by their firm. Include name of firm, address, telephone number and name of representative.
- 6) **Schedule** – Proposer shall submit the amount of working days that will take company to complete project.

### **SECTION III SELECTION AND SCHEDULES**

#### **SELECTION PROCEDURES**

The RFP shall be submitted according to the schedule below.

#### **PROPOSAL RANKING**

A selection committee will evaluate and rank the written RFPs on a per project basis. After the RFPs have been ranked, the committee will make a recommendation to the CITY Council.

#### **RFP SUBMITTED TO**

An original and four (4) copies of RFPs should be submitted to:

**City of Edinburg  
c/o City Secretary  
415 West University  
P.O. Box 1079  
Edinburg, Texas 78541**

RFPs must be submitted by no later than 3:00 p.m. on Monday, May 20, 2019.

## **SECTION IV FIRM and RFP EVALUATION**

### **RFP – EVALUATION**

The evaluation system consists of a 100 Point system. The RFP will be ranked after evaluation. All RFP's submitted will be ranked and evaluated based on specified RFP criteria. The submittal evaluation will be based on the following criteria.

- **40 Points: Proposer's itemized and total proposed price**
  - Total estimated cost for base bid submitted\*

\*Alternates might be included based on what is most advantageous to City.
- **40 Points: Proposer's qualifications/experience and performance/references**
  - Demonstrated prior experience for similar projects (20 points)
  - Number of years in business (5 points)
  - Litigation History/Lawsuit History (5 points)
  - References (10 points)
- **10 Points: The Proposer's Team and Subcontractors.**
  - Resumes for Key Individuals (5 points)
    - Project Superintendent
    - Project Manager
  - List of Subcontractors (5 points)
- **10 Points: Schedule.**
  - Lowest total days (10 points)
  - Within 30 days of lowest (8 points)
  - Within 60 days of lowest (6 points)
  - More than 60 days from lowest (5 points)

### **Proposed Price (40 points):**

The price will be evaluated and scored based on the main proposal cost. The City reserves the right to include any and all alternate price proposals in the price evaluation process. The established budget will determine which, if any, alternates will be recommended and accepted as part of the overall price ranking evaluation. After the highest ranked firm is selected, negotiations on price and changes on the scope of work may occur with the firm that provides the best value to the City.

Points will be awarded based upon the total number of offers submitted. The lowest offeror will receive the maximum number of points and the highest offeror will receive the minimum number of points. A point spread system will be established once all the offers are tabulated. The closer the prices of the offers, the larger the point spread will be.

### SAMPLE: Utilizing the 80% Spread Formula

| Contractor     | Price          | Points |
|----------------|----------------|--------|
| Offeror No. 01 | \$1,000,000.00 | 40.0   |
| Offeror No. 02 | \$1,050,000.00 | 37.33  |
| Offeror No. 03 | \$1,100,000.00 | 34.67  |
| Offeror No. 04 | \$1,150,000.00 | 32.0   |

70% spread:  $40 \times 70\% = 28.0$  points   Results: 12 points spread  
75% spread:  $40 \times 75\% = 30.0$  points   Results: 10 points spread  
80% spread:  $40 \times 80\% = 32.0$  points   Results: 8 points spread  
85% spread:  $40 \times 85\% = 34.0$  points   Results: 6 points spread  
90% spread:  $40 \times 90\% = 36.0$  points   Results: 4 points spread  
95% spread:  $40 \times 95\% = 38.0$  points   Results: 2 points spread

If the committee decided to utilize the 90% spread formula, Offeror No. 04 is only 4 points away from Offeror No. 1. The committee may feel that a 4 point difference is too close, and is unfair to the lowest price offeror. A 70% spread, or 12 point difference, may be too far spread out and may be considered unfair to the highest price offer. Especially since the prices are not too far apart on a \$1 Million project. The point spread could be very different on a \$300,000.00 project budget versus a \$30 million project budget.

After the percentage spread is agreed upon, in this case the 80% formula, the lowest offeror gets the maximum 40 points and the highest offeror gets 32 points. Everyone else in the middle will get their points scored proportionately (extrapolated). This is the scoring system which will be utilized by the ranking committee on the price category for all construction projects. The point system will vary from project to project depending on the project budget ranges, on the number of offers submitted, and on the price spread differences between all offerors.

### RESPONDENT – EVALUATION

The evaluation system consists of a 100-point system. The firms will be ranked after evaluation. Categories under the 100-point system include response to RFP. RFP submittal evaluation will be based on the following criteria.

### STAFFING OF PROJECT TEAM

The firms should provide information on their proposed professional team members, i.e., applicable certifications/registrations and other pertinent information that demonstrates their qualifications to perform the contract. The professional team members shall have experience in performing similar contracts for counties, cities, irrigation districts, TX DOT or other clients as stated in the Request for Proposals (RFP). Similar experience gained through other clients should be substantiated by reference. A list and scope of the various projects for comparative purposes shall be included in an appendix.

### EXPERIENCE OF PROJECT TEAM/ABILITY TO COMMIT RESOURCES

The provider shall designate experienced staff to completely and efficiently perform the work. Also, in this section, outline the firm's contingency plans for servicing the project in the event that one or more key personnel are not available for any reason during the period of performance.

### **METHODOLOGY**

The RFP should provide a description of the firm's approach to the methodology and management to the scope of services for the project.

### **UNDERSTANDING OF PROJECT/SIMILAR PROJECTS**

The proposal shall include the following:

1. Address appropriate Federal/State/Local regulations and policies
2. Identify information to be gathered or obtained

The respondents should provide as much background information as to its experience in providing similar services to State, CITY, County or any other governmental agencies. Reference information should be as current as possible, especially contact persons and telephone numbers.

### **FAMILIARITY WITH APPLICABLE RULES AND REGULATIONS**

The RFP should indicate, through past experience of the proposed Team, that they possess sufficient knowledge of governmental regulations, appropriate codes, guidelines, professional standards and policies (as required).

## **SECTION V**

### **AWARD OF CONTRACT, RESERVATION OF RIGHTS**

#### **Number of Contracts**

The CITY reserves the right to award one or no contract(s) in response to this RFP.

#### **Advantageous Contract**

The Contract/s, if awarded, will be awarded to the vendor/s submitting proposal for the FIRE STATION NO. 5 whose Submittal(s) is/are deemed most advantageous to the CITY and, as determined by the selection committee, upon approval of the CITY Council.

#### **Final Selection and City Council Approval**

The CITY may accept any Submittal in whole or in part. If subsequent negotiations are conducted, they shall not constitute a rejection or alternate RFP on the part of THE CITY. However, final selection of a company submitting proposal for the FIRE STATION NO. 5 is subject to City Council approval.

#### **Remedy of Technical Errors**

The CITY reserves the right to accept one or more submittals or reject any or all submittals received in response to this RFP, and to waive informalities and irregularities in the submittals received. The CITY also reserves the right to terminate this RFP, and reissue a subsequent solicitation, and/or remedy technical errors in the RFP process.

#### **Preparation Costs**

This RFP does not commit the CITY to enter into a Contract, award any services related to this RFP, nor does it obligate the CITY to pay any costs incurred in preparation or submission of a submittal or in anticipation of a contract.

#### **Insurance and Indemnity**

If selected, vendor/s submitting proposal for FIRE STATION NO. 5 will be required to comply with the Insurance and Indemnity Requirements established herein.

#### **Independent Contractor**

The company/s submitting proposal for the FIRE STATION NO. 5 agrees and understands that, if selected, it and all persons designated by it to provide services in connection with a contract, is (are) and shall be deemed to be (an) independent contractor(s), responsible for its (their) respective acts or omissions, and that THE CITY shall in no way be responsible for company submitting proposal for the FIRE STATION NO. 5 actions, and that none of the parties hereto will have authority to bind the other or to hold out to third parties.

#### **Purchase Orders, As Needed**

Execution of a contract does not obligate the CITY to engage any delivery orders, Purchase Orders, or other commitments for services. Service delivery shall be at the CITY's discretion, as needed, and will be communicated to the company submitting proposal for the FIRE STATION NO. 5 through individual Purchase Orders.

## **ATTACHMENT I**

### **Insurance Requirements**

The Respondent awarded the contract shall furnish proof of insurance, which will also include any subcontractor that is subcontracted by the bidder in at least the following limits, to be in place prior to providing any services under this Contract and to continue in effect at all times during the term of this Contract:

- 1 Professional liability insurance policy with limits of at least One Million Dollars (\$1,000,000) per occurrence, or limited to claims made, include at least a five (5) year extended reporting period.
- 1 Automobile liability insurance policy with limits of at least Three Hundred Thousand Dollars (\$300,000) per person and \$500,000 per occurrence consistent with potential exposure to The CITY under the Texas Tort Claims Act. Coverage should include injury to or death of persons and property damage claims (with limits up to \$500,000) arising out of the services provided to The CITY hereunder.
- 1 Uninsured/Underinsured motorist coverage in an amount equal to the bodily injury limits set forth immediately above;
- 1 A Five Hundred Thousand Dollar (\$500,000) Comprehensive General Liability insurance policy providing additional coverage to all underlying liabilities of The CITY consistent with potential exposure of The CITY under the Texas Tort Claims Act;
- 1 Workers' compensation insurance in amounts established by Texas law, unless the Bidder is specifically exempted from the Texas Workers' Compensation Act, Texas Labor Code Chapter 401, et. Seq.

Certificates of insurance naming The CITY as an additional insured shall be submitted to The CITY for approval prior to any services being performed by Contractor. Each policy of insurance required hereunder shall extend for a period equivalent to, or longer than the term of the Contract, and any insurer hereunder shall be required to give at least thirty (30) days written notice to The CITY prior to the cancellation of any such coverage on the termination date, or otherwise. This Contract shall be automatically suspended upon the cancellation, or other termination, of any required policy of insurance hereunder, and such suspension shall continue until evidence that adequate replacement coverage is provided to The CITY. If replacement coverage is not provided within thirty (30) days following suspension of the Contract, the Contract shall automatically terminate.



**ATTACHMENT II**  
**Insurance Requirement Acknowledgement**

I, \_\_\_\_\_, authorized representative for  
\_\_\_\_\_,

Company/Vendor

Hereby acknowledge the receipt of The CITY's required insurance limits. Said requirements:

- ☐ Will be acquired within 10 working days after notification from the Engineering Department of proposal awarded by The CITY of Edinburg; (\*An insurance certificate for the required insurance limits shall be provided to the City Engineer in order to qualify for award of bid and to execute a contract between the Company and The CITY.)
- ☐ Will acquire additional amount needed to meet The CITY's requirements within 10 working days after notification from the Engineering Department of bid awarded by The CITY of Edinburg; currently carry the following:

Professional Liability (Errors & Omissions):  
\$ \_\_\_\_\_

Automobile Liability: \$ \_\_\_\_\_ General Liability:  
\$ \_\_\_\_\_

(\* An insurance certificate for the required insurance limits shall be provided to the City Engineer in order to qualify for award of bid and to execute a contract between the Company and The CITY.) **OR**

- ☐ Have already been met (see attached copy of insurance certificate).

\_\_\_\_\_  
Authorized Representative

\_\_\_\_\_  
Date

**Notice to Bidder:** Failure to provide Certificates of Insurance to the City Engineer will cause the bid award to be rescinded and then awarded to next lowest bidder. Certificates of Insurance will be monitored/verified on a **quarterly basis** to ensure that coverage policy is in place. It is the Company's obligation to maintain the appropriate insurance coverage throughout the term of the contract.

---

THIS FORM MUST ACCOMPANY BID PACKET

**ATTACHMENT III**

**Project Requirements  
Acknowledgement**

This is to certify that I, \_\_\_\_\_, possess all of the  
**APPLICABLE:**

1. Licenses:

\_\_\_\_\_

2. Bonds:

\_\_\_\_\_

3. Certificates:

\_\_\_\_\_

4. Permits:

\_\_\_\_\_

5. Other:

\_\_\_\_\_

necessary to carry out the required project. Furthermore, I am providing copies of the required documentation, so that if my company is awarded the bid, I may be eligible to enter a contract with the CITY and proceed to complete the project in a timely manner.

**\* Any license, bonds, certificates, permits, etc. which are required must be presented as part of the bid packet in order to expedite the bid evaluation process. Failure to provide said documentation will result in the disqualification of your bid.**

\_\_\_\_\_

\_\_\_\_\_  
Authorized Signature

\_\_\_\_\_  
Date

\_\_\_\_\_  
Company

\_\_\_\_\_  
Address

\_\_\_\_\_  
City, State, Zip

## ATTACHMENT IV

### LITIGATION DISCLOSURE FORM

**Failure to fully and truthfully disclose the information required by this Litigation Disclosure form may result in the disqualification of your submittal from consideration or termination of the contract, once awarded.**

1. Have you or any member of your Firm or Team to be assigned to this engagement ever been indicted or convicted of a felony or misdemeanor greater than a Class C in the last five (5) years?

**Circle One**                      YES                      NO

2. Have you or any member of your Firm or Team to be assigned to this engagement ever been terminated (for cause or otherwise) from any work being performed for the CITY or any other Federal, State or Local Government, or Private Entity?

**Circle One**                      YES                      NO

3. Have you or any member of your Firm or Team to be assigned to this engagement ever been involved in any claim or litigation with the CITY or any other Federal, State or Local Government, or Private Entity during the last ten (10) years?

**Circle One**                      YES                      NO

**If you have answered "Yes" to any of the above questions, please indicate the name(s) of the person(s), the nature, and the status and/or outcome of the information, indictment, conviction, termination, claim or litigation, as applicable. Any such information should be provided on a separate page, attached to this form and submitted with your submittal.**

**ATTACHMENT V**

**VENDOR/S PROVIDING PROPOSAL FOR FIRE STATION NO. 5  
QUALIFICATIONS  
GENERAL QUESTIONNAIRE**

1 Name/Name of Agency/Company: \_\_\_\_\_

(Full, correct legal name)

2 Address: \_\_\_\_\_

3. Telephone/Fax: \_\_\_\_\_

4. Does your Company anticipate any mergers, transfer of organization ownership, management reorganization, or departure of key personnel within the next twelve (12) months that may affect the organization's ability to carry out its submittal?

Yes\_\_\_\_ No\_\_\_\_

5. Is your Company authorized and/or licensed to do business in Texas?

Yes\_\_\_\_ No\_\_\_\_

6. Where is the Company's corporate headquarters located? \_\_\_\_\_

7. a. Does the Company have an office located in Edinburg, Texas?

Yes\_\_\_\_ No\_\_\_\_

b. If the answer to the previous question is "yes", how long has the Company conducted business from its Edinburg office?

\_\_\_\_ (years)      \_\_\_\_ (months)

c. State the number of full-time employees at the Edinburg office. \_\_\_\_\_

8. a. If the Company does not have an Edinburg office, does the Company have an office located in Hidalgo County, Texas?

Yes\_\_\_\_ No\_\_\_\_

b. If the answer to the previous question is yes, how long has the Company conducted business from its Hidalgo County office?

\_\_\_\_\_ (years)      \_\_\_\_\_ (months)

c. State the number of full-time employees at the Hidalgo County office. \_\_\_\_\_  
\_\_\_\_\_

9. Has the Company or any of its principals been debarred or suspended from contracting with any public entity? Yes \_\_\_\_\_ No \_\_\_\_\_

If yes, identify the public entity and the name and current phone number of a representative of the public entity familiar with the debarment or suspension, and state the reason for or circumstances surrounding the debarment or suspension, including but not limited to the period of time for such debarment or suspension. \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

10. Indicate person whom The CITY may contact concerning your submittal or setting dates for meetings.

Name: \_\_\_\_\_  
Address: \_\_\_\_\_  
Telephone: \_\_\_\_\_  
Fax: \_\_\_\_\_  
Email: \_\_\_\_\_

#### 11. Surety Information

Have you or the Company ever had a bond or surety instrument "called," canceled, or forfeited?

Yes ( ) No ( ).

If yes, state the name of the bonding company, date, amount of bond and reason for such bond being "called," or its cancellation or forfeiture. \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

#### 12. Bankruptcy Information

Have you or the Company ever been declared bankrupt or filed for protection from creditors under state or federal proceedings? Yes ( ) No ( )

If yes, state the date, court, jurisdiction, cause number, amount of liabilities

and amount of assets. \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

13. Provide any other names under which your business has operated within the last 10 years.

\_\_\_\_\_  
\_\_\_\_\_

**ATTACHMENT VI**  
**HOUSE BILL 89 VERIFICATION**

I, \_\_\_\_\_, the undersigned representative  
of \_\_\_\_\_,

\_\_\_\_\_, (Company or Business name)  
(hereafter referred to as company) **being an adult over the age of eighteen (18) years of age,**  
**verify that the company named-above, under the provisions of Subtitle F, Title 10,**  
**Government Code Chapter 2270:**

- 1. Does not boycott Israel currently; and**
- 2. Will not boycott Israel during the term of the contract.**
- 3) Is not currently listed on the State of Texas Comptroller's Companies that Boycott**  
**Israel                                      List                                      located                                      at**  
**<https://comptroller.texas.gov/purchasing/publications/divestment.php>**

*Pursuant to Section 2270.001, Texas Government Code:*

- 1. "Boycott Israel" means refusing to deal with, terminating business activities with, or otherwise taking any action that is intended to penalize, inflict economic harm on, or limit commercial relations specifically with Israel, or with a person or entity doing business in Israel or in an Israeli-controlled territory, but does not include an action made for ordinary business purposes; and*
  - 2. "Company" means a for-profit sole proprietorship, organization, association, corporation, partnership, joint venture, limited partnership, limited liability partnership, or any limited liability company, including a wholly owned subsidiary, majority-owned subsidiary, parent company or affiliate of those entities or business associations that exist to make a profit.*
- 

\_\_\_\_\_  
**SIGNATURE OF COMPANY REPRESENTATIVE:**

\_\_\_\_\_  
**TYPE/PRINT NAME AND TITLE:**

\_\_\_\_\_  
**DATE:**

\_\_\_\_\_



## ATTACHMENT VII

### SUBMITTAL CHECKLIST

**This checklist is to help the company submitting proposal for the FIRE STATION NO. 5 ensure that all required documents have been included in its submittal.**

| Document and Location in Submittal                                                                                        | Check or Initial to Indicate Document is Attached to Submittal |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Tab A – Interest Statement                                                                                                |                                                                |
| Tab B – Company submitting proposal for the FIRE STATION NO. 5 Qualification General Questionnaire (Attachment VI in RFP) |                                                                |
| Tab C – *Project Requirements Acknowledgement (Attachment V in RFP)                                                       |                                                                |
| Tab D – Litigation Disclosure (Attachment IV in RFP)                                                                      |                                                                |
| Tab E – Proof of Insurability (Letter from Insurance Provider and copy of current Insurance Certificate)                  |                                                                |
| Tab F – *Insurance Requirement Acknowledgement (Attachment II in RFP)                                                     |                                                                |
| Tab G – Letter of Intent from Surety Company to provide Payment and Performance Bonds. (Section II in RFP Requirements)   |                                                                |
| Tab H – Submittal Checklist (Attachment VI in RFP)                                                                        |                                                                |
| Tab I - *House Bill 89 Verification (Attachment VI)                                                                       |                                                                |
| Tab J- *Formal Proposal for the FIRE STATION NO. 5                                                                        |                                                                |
| 1 Original* and 2 Copies of Submittal                                                                                     |                                                                |

**\*Documents marked with an asterisk on this checklist require a signature. Be sure they are signed prior to submittal.**

Page Intentionally Blank

Document 00020

**NOTICE TO BIDDERS**

**Owner:** City of Edinburg  
415 W. University Drive  
Edinburg, Texas 78539  
Phone: (956) 388-8211  
Fax: (956) 383-7111

**Architect:** Milnet Architectural Services, PLLC  
608 S. 12<sup>th</sup> St  
McAllen, TEXAS 78501  
Phone: 956-688-5656  
Fax: (956)

**1.00 INVITATION**

- A. Bidders are invited to submit an offer for performance of a Contract to the City of Edinburg located at the above address, for the following construction Project:  
Project: Fire Station No. 5  
Located: Davis Road, Edinburg, Texas
- B. Work of the Project consists of the construction of New Fire Station No. 5. 10,386 square feet concrete slab with stucco finish CMU walls and integrated insulation standing seam metal roof.
- C. The Contract Documents are identified Fire Station No. 5 as listed in the Project Manual, issued by the City of Edinburg/Milnet Architectural Services, PLLC.
- D. The bidder shall bear all costs associated with the preparation and submission of its bid, and the Owner will in no case be responsible or liable for those costs, regardless of the conduct or outcome of the bidding process.
- E. When requested, the successful Bidder shall present satisfactory evidence that Bidder has regularly engaged in furnishing products and performing construction work as proposed, and has the capital, labor, equipment, and material to execute the Work required by Contract Documents.

**2.00 BID SUBMISSION**

- A. Bids signed by an officer of the company and dated will be received at the City Secretary's Office, at 415 W. University Drive, Edinburg, TX 78539 until 3:00 pm local time, on May 20, 2019
- B. Bids submitted after the above time will be returned to the Bidder unopened.
- C. Bids shall be submitted in United States Currency and the English language on the Bid Forms and Supplements to Bid Forms provided with this Project Manual.
- D. Oral, telephonic, facsimile, or telegraphic bids are invalid and will not receive consideration.
- E. Bids will be opened and publicly read in the City of Edinburg City Hall Community Room at 415 W. University Drive, Edinburg, TX 78539 on the same date bids are received.
- F. Bids will be irrevocable for 90 **days** from the bid date. Bidder may withdraw after 90 days without penalty if no mutual agreement can be reached.

00020-1 of 4

**3.00 MODIFICATION OR WITHDRAWAL**

- A. Bids submitted early may be modified or withdrawn by notice to the City of Edinburg at the place and prior to the time designated for receipt of Bids. Such notice shall be in writing over the signature of the Bidder and shall be so worded as not to reveal the amount of the original Bid.
- B. Oral, telephonic, facsimile, or telegraphic modification of Bids will not receive consideration.
- C. Withdrawn Bids may be resubmitted up to the time designated for receipt of Bids.

**4.00 CONTRACT TIME**

- A. The Work shall be performed within the date established in the Notice to Proceed.
- B. Contractor shall pay liquidated damages in the amounts stated in Document 00500 - Agreement for failure to complete the Work within the Contract Time.
- C. The work is to be performed only during weekdays 8:00 AM to 5:00 PM (Monday to Friday). City recognized holidays are recommended to be avoided. Work performed during weekends (Saturday-Sunday) and holidays will incur a Contractor payment of **\$50** per hour to Owner for onsite inspection.

**5.00 SECURITY DEPOSIT REQUIREMENTS**

- A. Bids shall be accompanied by a security deposit as stated in Document 00100 - Instructions to Bidders.

**6.00 EXAMINATION**

- A. Bid Documents are on display on the City of Edinburg website, may be examined at the location below or purchased from the Engineer of Record:

**City of Edinburg  
Engineering Department – 2<sup>nd</sup> Floor  
415 W. University Drive  
Edinburg, Texas 78539**

**7.00 AVAILABILITY**

- A. Bid Documents may be purchased from the Engineering Department, Engineer of Record or are available for printing at [http://cityofedinburg.com/departments/finance/open\\_bid\\_notices.php](http://cityofedinburg.com/departments/finance/open_bid_notices.php).
- B. All official notifications, addenda, and other Bidding Documents will be offered only through the designated website. Neither Owner nor Engineer will be responsible for Bidding Documents, including addenda, if any, obtained from sources other than the designated website.
- C. Bid Documents may be purchased by bidders upon receipt of a cashier's check, certified check, money order, company check, or personal check in the amount established by the City of Edinburg or Architect of Record. The cost includes the Project Manual w/ Specifications and one full sized set of Drawings. They can also be downloaded at no cost, as specified on 7(A).
- D. **The cost for the bid documents will not be refunded.**
- E. Bid Documents are made available only for the purpose of obtaining offers for this Project. Purchase of Bid Documents does not grant a license for other purposes.

- E. On receipt of Bid Documents, verify that documents are legible and complete. Compare contents of Project Manual with Table of Contents; see that all drawings listed in the List of Drawings are included. Notify City of Edinburg should the documents be incomplete as issued.

## 8.00 QUESTIONS AND INTERPRETATIONS

- A. Bidder is required to study Bid Documents, the site, and conditions affecting the Work, and submit written questions on interpretation of those documents and conditions, or other factors affecting the Work, to the City of Edinburg.
- B. Written questions may be submitted by facsimile or email, addressed to the Engineer. **No questions will be accepted after 5:00 PM, May 15, 2019.** All facsimile communications shall be confirmed by mailing the original correspondence to the City of Edinburg Purchasing Department, if applicable.
- C. Immediately notify the Architect upon finding discrepancies or omissions in the Bid Documents.

## 9.00 ACCEPTANCE/REJECTION OF BIDS

- A. The Owner reserves the right to reject or accept any bids as stated in Document 00100 - Instructions to Bidders.

## 10.00 PRE-BID CONFERENCE

- A. One (1) pre-bid conference will be conducted by the Owner on May 8th at 2:00 pm. The pre-bid conference shall be conducted at the City of Edinburg Engineering Conference Room: located at 415 W. University Drive Edinburg, Texas 78539.
- B. **Attendance by prospective Bidders is highly recommended.** Sub-contractors, suppliers, and equipment suppliers may attend.
- C. Recognizing that free and open communication will benefit all participants, the Owner does not intend to limit or curtail the exchange of information between the Engineer and the prospective Bidders. However, the pre-bid conference is conducted primarily for the benefit of prospective Bidders. As such, a specific procedure will be followed during the conference:
  - a. All attendees will sign-in, indicating their role with the project: contractor, supplier, manufacturer, etc.
  - b. Seating priority will be given to Prospective Bidders. Sub-contractors, suppliers, and manufacturer's representatives shall remain behind the contractor area.
  - c. The Owner will make introductions of his staff and consultants.
  - d. The Owner and consultants will give a brief description of the project.
  - e. Only Contracting firms (Prospective Bidders) are permitted to ask questions. Sub-contractors suppliers, and manufacturer's shall deliver their questions to the Contractor they are working with for presentation.
  - f. Questions and answers will be recorded and developed into Meeting Minutes. Meeting Minutes will be distributed to meeting attendees. The Owner reserves the right to use electronic recording, or some other method to record the meeting.

- D. The meeting will be conducted in English. Translators will not be provided.
- E. If necessary, written clarifications or instructions will be issued in the form of an Addendum. Refer to Section 00100 – Instructions to Bidders for specific information concerning Addendums.

**END OF DOCUMENT**

Page Intentionally Blank

Document 00100

**INSTRUCTIONS TO BIDDERS****1.00 SUMMARY****1.01 DOCUMENT INCLUDES**

- A. Bid Documents and Contract Documents.
- B. Site Assessment.
- C. Subcontractors/Suppliers/Others.
- D. Bid Submission.
- E. Bid Enclosure Requirements.
- F. Offer, Acceptance, Rejection.

**1.02 RELATED DOCUMENTS**

- A. Document 00020 - Notice to Bidders: Date, time and place for receipt of bids; Contract Time.
- B. Document 00310 - Form of Proposal.
- C. Document 00405 - Schedule of Unit Price Work.
- D. Document 00450 - Post Bid Procedures.
- E. Document 00500 - Agreement.
- F. Document 00700 - General Conditions.
- G. Document 00800 - Supplementary Conditions.

**2.00 BID DOCUMENTS AND CONTRACT DOCUMENTS****2.01 DEFINITIONS**

- A. Definitions set forth in Document 00700 – General Conditions and in other Contract Documents, are applicable to the Bid Documents.
- B. Addenda: Written or graphic instruments issued prior to the opening of Bids, which clarify, modify, correct, or change the Bid Documents.
- C. Alternate Bid: The total amount bid for additions to the Work, as described in the Bid Documents. Each Alternate Bid shall include the cost of effects on adjacent or related components, and the Contractor's overhead and profit.
- D. Bid Documents: The Project Manual and Drawings, including Addenda, plus Notice to Bidders, Instructions to Bidders, and Supplements to Bid Forms identified in Document 00310 - Form of Proposal.
- E. Bidder: A person or entity who submits a Bid.
- F. Low Bidder: The apparent successful Bidder who qualifies as a responsible Bidder and who submits the Bid with the lowest Total Bid Price.

00100- 1 of 10

- G. Bid, Offer, Bidding: The act of submitting a complete and properly signed offer in accordance with these Instructions to Bidders. The Bid will be in the English language.
- H. Total Bid Price: The monetary amount for performing the Work as identified by the Bidder in Document 00310 - Form of Proposal, which amount includes Cash Allowances and Alternate Bids, if any. Bid Price(s) will be in United States.
- I. Security Deposit: A certified check, cashiers check or bid bond in at least the sum of 5 percent of the Total Bid Price which includes Cash Allowances and Alternate Bids, if any.

## **2.02 QUESTIONS, INTERPRETATIONS**

- A. Bidder shall: 1) carefully study the Bid Documents and compare them with each other, 2) examine the site, conditions thereon, and local conditions, and 3) report at once to the Architect any errors, inconsistencies or ambiguities discovered.
- B. Bidder assumes sole responsibility for errors or misinterpretations resulting from the use of incomplete documents, by Bidder itself or by its prospective Subcontractors and Suppliers.
- C. Direct questions to Architect.
- D. Verbal discussions and answers are not binding. Requests from Bidders for clarifications and interpretations of content of documents must be in writing (mail or facsimile transmission only), and must be received not less than 5 business days before the date set for receipt of Bids.
- E. The reply will be by Addendum.

## **2.03 ADDENDA**

- A. Addenda issued to Bidding Requirements are applicable only during the bidding period. Addenda to the Post-Bid Procedures are applicable only through the issuance of the Notice to Proceed. Any Addenda issued to Contract Forms, Conditions of the Contract, Specifications or Drawings become a part of the Contract Documents. Include resultant costs in the Total Bid Price.
- B. Addenda will be issued by the Architect to Bidders of record by email. Addenda will also be posted on the City website.
- C. Each Bidder shall ascertain, prior to submitting a Bid that the Bidder has received all Addenda issued. The Bidder shall acknowledge their receipt in the place indicated in Document 00310 - Form of Proposal.

## **2.04 SUBSTITUTIONS OF MATERIALS/EQUIPMENT**

- A. No substitutions will be considered on this Project during the bidding period.
- B. Voluntary substitutions by the Bidder will not be considered.

## **3.00 SITE ASSESSMENT**

- A. Bidders shall examine the Project site before submitting a Bid, become familiar with local conditions under which the Work will be performed, conduct appropriate explorations, and correlate personal observations with requirements of the Bid Documents. Work will be



performed in public right-of-way and City property. The site may be examined at any time during daylight hours.

- B. Bidder shall make site investigations to the extent Bidder deems necessary to ascertain the extent of subsurface conditions and variations thereof.
- C. Failure to perform such investigations during the bid period shall not relieve Bidder from responsibility for investigations, interpretations and proper use of available information in preparation of Bidder's proposal.
- D. Publications by the United States Department of Agriculture, Soil Conservation Service and others may be helpful to the bidder in his subsurface site investigation.
- E. Geotechnical investigation reports for the proposed project site may also be helpful to the bidder in his subsurface site investigation.

#### 4.00 SUBCONTRACTORS/SUPPLIERS/OTHERS

- A. The Owner reserves the right to reject a proposed Subcontractor or Supplier for reasonable cause.

#### 5.00 BID SUBMISSION

##### 5.01 SUBMISSION PROCEDURES

- A. Bidders shall be solely responsible for the delivery of their Bids in the manner and time prescribed in Document 00020 - Notice to Bidders.
- B. Submit **one copy of the original executed offer** on the bid forms provided, properly signed, with required Security Deposit, and other Supplements to Bid Forms, in a sealed, opaque envelope. On the outside of the envelope, clearly indicate that it is a sealed bid and include the Bidder's name, Project name and Owner name. Bids submitted by mail shall be enclosed in a separate envelope addressed for mailing, and identifying the enclosure as a bid. In addition, **three copies must also be submitted.**
- C. Fill in all blanks in the Bid forms. Acknowledge receipt of Addenda. Bid all Alternate Bids required by Bid Documents.
- D. A summary of submitted Bids will be made available to Bidders following the Bid opening.
- E. All costs and expenses incurred by the Bidder that are associated with preparation of the Bid shall be paid by and be the sole responsibility of the Bidder.

##### 5.02 BID INELIGIBILITY

- A. Failure to provide required Security Deposit in the proper amount will be cause to declare the Bid invalid.
- B. Improperly completed information may be cause for declaring the Bid invalid.
- C. Bids that are unsigned, improperly signed, illegible, obscure, altered, or which contain qualifications or irregularities of any kind, may be declared invalid. Document 00310 - Form of Proposal, Supplements to the Bid Forms identified in the Form of Proposal, or enclosures which are improperly prepared, may be declared invalid.

**6.00 BID ENCLOSURE REQUIREMENTS****6.01 SUPPLEMENTS TO BID FORMS**

- A. Bid submittals shall include any other documents specified in Document 00310 - Form of Proposal.

**6.02 SECURITY DEPOSIT**

- A. Bids shall be accompanied by a Security Deposit.
- B. The Security Deposit of the Bidders will be retained until after the Contract is executed.
- C. After execution of the Contract, Security Deposits will be returned to the Bidders.
- D. If no Contract is awarded, all Security Deposits will be returned to the respective Bidders.

**6.03 CERTIFIED CHECK/CASHIER'S CHECK**

- A. Make certified check or cashier's check (security checks) payable to the Owner.
- B. The security checks are submitted on the condition that if the Bidder is named apparent Low Bidder and then fails either to timely execute the Agreement or to timely provide any required bonds, or to do both, then in that event the Owner will cash the security check.
- C. The Owner will retain an amount equal to the difference between the Bid of the Bidder providing the security check and the Bid of the Bidder who is finally awarded the Contract and who executes the Agreement and provides the required bonds.
- D. Any balance remaining will be reimbursed by the Owner to the Bidder who provided the security check.

**6.04 BID BOND**

- A. The bid bond must be a valid and enforceable bond, executed by a corporate Surety authorized by the Texas State Board of Insurance to conduct insurance business in the State of Texas and shall comply with other requirements set out by law or included in the Bid Documents.
- B. Endorse the bid bond in the name of the Owner as obligee, signed by the Contractor as principal and executed, signed and sealed by the Surety.
- C. The bid bond must be conditioned such that if the Bidder is named apparent Low Bidder and then fails either to execute the Agreement timely or to provide any required bonds timely, or to do both, then in that event the Surety will be obligated to pay to the Owner an amount equal to the difference between the Bid of the Bidder on whom the bond was written and the Bid of the Bidder who is finally awarded the Contract and who executes the Agreement and provides the required bonds, up to the penal sum of the Bond.
- D. In addition, the Owner expressly reserves the right to reject any Bid if the Bid Bond (or Bid Bond rider) conditions the Bid in a way inconsistent with the Bid Documents. Examples include but are not limited to:
  - 1. a condition prohibiting the Owner from making a Claim against the Performance Bond Surety that would be allowable under the Contract and Performance Bond form published in the Bid Documents;

2. a condition that provides that the Performance Bond Surety cannot be held liable for completing the Contract in case of default; or
  3. a condition limiting the Performance Bond Surety's liability for damages inconsistent with the Contract and Performance Bond form published in the Bid Documents.
- E. On all contracts that will equal to or exceed \$100,000, the performance bond and the payment bond must be provided by a surety that has a rating of "A" from AM BEST, MOODY'S or STANDARD & POORS.

In the event that the total bid amount is \$50,000 or less, the successful contractor has the option to enter into a single payment contract with the City of Edinburg in lieu of a Performance Bond, provided that no money shall be paid to the contractor until completion of the work by the contractor and accepted of same by the City of Edinburg. In the event that the total bid amount is \$25,000 or less, the successful contractor has the option to enter into a single payment contract with the City of Edinburg in lieu of a Payment and Performance Bond.

#### **6.05 BID FORM SIGNATURE**

- A. Document 00310 - Form of Proposal shall be signed by the Bidder as follows:
1. Sole Proprietorship: Full name, address, and signature of sole proprietor, signed in the presence of a witness who will also sign. Insert the words "Sole Proprietor" under the signature.
  2. Partnership: Name and address of the firm, signature of each partner in the presence of a witness who will also sign. The full name and address of each partner shall be given.
  3. Corporation: Signature of duly authorized officer.
  4. Joint Venture: Each party of the joint venture shall execute Document 00310 - Form of Proposal under their respective seals in a manner appropriate to such party as described above, similar to the requirements of a Partnership.

#### **7.00 DETERMINING LOWEST RESPONSIVE, RESPONSIBLE BIDDER**

##### **7.01 BIDDERS QUALIFICATIONS**

- A. Bids must contain evidence of Bidder's qualifications to do business in the state of Texas. To demonstrate that the Bidder is responsible and able to perform the Work, funding policies dictate each Bidder must submit, as a part of the Bidding Documents, all of the items listed below:

00310 Form of Proposal  
00405 Schedule of Unit Price Work  
00411 Bid Bond  
00420 Statement of Bidder's Qualifications  
00423 Certification of Bidder's Qualifications  
00425 Equipment & Material Suppliers List  
00460 Non-Collusion Affidavit  
00429 Non-Bribery Model Form

00100- 5 of 10

- B. Only the above data/information provided with the Bidding Documents may be used for evaluation and developing the Recommendation to Award by the Architect. Bidders will not be allowed to substitute any "Key Personnel" other than alternates presented in the bid or examples of previous projects submitted in the bid package. Minor clarifications of submitted materials will be permitted after bid opening. Such request for clarifications will only be initiated by the Architect in writing and only written responses will be accepted.
- C. In determining the lowest responsible, responsive Bidder, in addition to price, the following elements will be considered:
1. The quality, availability, and adaptability of the supplies, materials, equipment, or contractual services, to the particular use required;
  2. The ability, capacity and skill of the bidder to perform the contract or to provide the service required;
  3. Whether the bidder can perform the contract and provide the service promptly, or within the time required, without delay or interference;
  4. The character, responsibility, integrity, reputation, and experience of the bidder;
  5. The quality of performance of previous services, or contracts;
  6. The previous and existing compliance by the bidder with laws relating to the contract or service;
  7. Any previous or existing noncompliance by the bidder with specifications, or requirements relating to time of submission of specified data such as samples, models, drawings, certificates, or other information;
  8. The sufficiency of the financial resources and ability of the bidder to perform the contract or to provide the service; and
  9. The ability of the bidder to provide competent personnel for the job, as demonstrated by the submitted listing of the names and the skills of experienced personnel, including potential alternates, whom the bidder currently employs and who will be available for performing this work;
  10. The experience of the bidder in performing work similar in type, size and complexity to this project, as demonstrated by a listing of projects, with verifiable references (names, addresses, phone numbers, etc.), successfully completed.
  11. Bidder shall provide with the Bid an experience statement with pertinent information regarding similar projects and other evidence of qualifications for each such Subcontractor, Supplier, person, or organization.

**7.02 BIDDER MUST MEET THE FOLLOWING MINIMUM CRITERIA:**

- (A) The Bidder must demonstrate \*\*Successful Completion during the last five (5) years of at least one project comparable in nature and scope to this project. The comparable scope shall be at least 1/4 the size of the proposed project.
- (B) At least two \*Key Personnel, and their potential alternate, employed by the Bidder must have a minimum of five (5) years experience in similar construction projects.

- (C) The Bidder must have an employee, to be dedicated to this project, who is experienced in scheduling, with demonstrated ability in employing scheduling techniques similar to those to be used for this project.
- (D) Bidder may, at its discretion, include resumes of alternates for Key Personnel, and if in the process of bid evaluation, the Owner rejects any Key Personnel, the Owner will consider the alternates.

\* KEY PERSONNEL: Individuals who will be directly assigned to this project. Resumes of Key Personnel must be submitted with the Bid (include in Document 00420) and accepted by the Owner in order for Bidder to receive the Award. At the minimum, the resumes for the following personnel that are to be assigned to this Project are to be submitted.

- (a) Owner or Principals of the Bidder
- (b) The Project Manager
- (c) The Project Superintendent
- (d) The Project Scheduler
- (e) Minimum of two Foremen

\*\*SUCCESSFUL COMPLETION: Defined as completion of a project on time, no more than thirty (30) days later than the original contract time, and within budget, within 5% of the original contract price. If there is any project submitted by the Bidder as qualifying, but which does not meet these requirements, in order to be fully responsible, the Bidder is required to submit detailed information on that project demonstrating what caused the increases to cost or time. The name and telephone numbers of the Design Architect and the Client are to be provided for evaluation as to whether the project may be considered "successful". For any project where liquidated damages were assessed, the Bidder will not be considered to have been on time.

### 7.03 BIDDERS ARE REQUIRED TO SUBMIT WITH THEIR BID:

00310 Form of Proposal

00405 Schedule of Unit Price Work

00411 Bid Bond

00420 Statement of Bidder's Qualifications

00423 Certification of Bidder's Qualifications

00425 Equipment & Material Suppliers List

00429 Non-Bribery Model Form

00460 Non-Collusion Affidavit

- (A) Failure to submit these items with the bid will result in a finding that the bid is non-responsive and the bid will be disqualified.

### 7.04 The Owner will evaluate and compare only the bids determined to be responsive in accordance with the following:

- (a) Is the bid complete (all Bidding Documents submitted);
- (b) Have documents been properly signed;
- (c) Are the required bid securities part of the bid package; and

(d) Are there any computational errors present?

**7.05 The Owner reserves the right to accept or reject any variation, deviation, or alternative offer which is not submitted in accordance with the bidding documents. Variations, deviations, alternative offers, and other factors that are in excess of the requirements of the bidding documents or which otherwise result in unsolicited benefits for the Owner, shall not be taken into account in bid evaluation.**

**7.06 In evaluating the bids, the Owner will determine for each bid, the evaluated bid price by adjusting the bid price as follows:**

- A. Making any correction for errors;
- B. Excluding provisional sums and the provision, if any, for contingencies in the price schedules;
- C. Taking an appropriate adjustment for any other quantifiable acceptable non-material variations, deviations or alternative offers; and
- D. Making appropriate adjustments to reflect additional factors in the manner and to the extent indicated in the Bidding Documents.

**7.07 The Owner will award the contract to the bidder whose bid has been determined to be substantially responsive to the bidding documents and who has offered the lowest evaluated bid price provided that such bidder has been determined to be qualified to perform the contract satisfactorily in accordance with the provisions of the Bidding Documents.**

## **8.00 OFFER ACCEPTANCE, REJECTION**

### **8.01 ACCEPTANCE**

- A. The Owner will give notice of intent to award the Contract to the Low Bidder. Acceptance by the Owner is conditioned upon Bidder's submission of information for establishing satisfactory qualifications, if required; and execution of submittals required in Document 00450 - Post-Bid Procedures.
- B. The Bid shall remain open to acceptance and shall be irrevocable for the Period for Bid Acceptance stated in Document 00020 – Notice to Bidders.
- C. Additional time taken by Contractor to fulfill requirements for submittals, including review and re-submittal, shall be added to the acceptance period.

### **8.02 REJECTION**

- A. The Owner reserves the right to reject any and all Bids or to accept any Bid deemed advantageous to it.

### **8.03 BID TABULATION**

- A. The Architect will tabulate, record, and evaluate the Bids of all responsible Bidders after the Bid opening.
- B. In tabulating Bids, the amount written for a unit price governs over the total amount calculated. Therefore, the Architect may correct any mathematical errors in the extension of the total amount based on the unit price given by a Bidder and adjust their Total Bid Price.

## **9.00 APPROVAL BY THE FUNDING AGENCIES**

- A. All addenda, contracts, work directives, change orders, time extensions, and other matters specified in the Contract Documents are not valid until approved in accordance with the City of Edinburg's Purchasing Policies and Procedures Manual.

**END OF DOCUMENT**

Page Intentionally Blank



Document 00300

**CONTRACTOR NOTICE OF INTENT TO RESPOND**

Firms interested in submitting a bid on the project as outlined in the specifications, should indicate their intention by signing, dating and returning the form to the address below prior to May 15, 2019, so that they may receive any addendums to the specifications should the need arise.

**Owner:** City of Edinburg  
Attn: Finance Department  
415 W. University Drive  
Edinburg, Texas 78539

**Architect:** Milnet Architectural Services, PLLC  
Attn: Rodolfo R. Molina, AIA  
608 S. 12<sup>th</sup> St.  
McAllen, Texas 78501

Bidder: \_\_\_\_\_  
[Please print or type the full name of your proprietorship, partnership, corporation, or joint venture.\*)

Contact Name: \_\_\_\_\_  
[Please print or type name] [Title]

Address: \_\_\_\_\_  
[Mailing]  
\_\_\_\_\_  
[Street, if different]

Telephone: \_\_\_\_\_  
[Print or type telephone number]

Fax: \_\_\_\_\_  
[Print or type telephone number]

Email: \_\_\_\_\_  
[Print or type telephone number]

**END OF DOCUMENT**

Page Intentionally Blank

Document 00310

**FORM OF PROPOSAL**To: CITY OF EDINBURG

---

Project No.: 2019-20

Project: Fire Station No. 5

Bidder: \_\_\_\_\_

[Print or type full name of proprietorship, partnership, corporation, or joint venture]

**1.0 OFFER**

Having examined the place of the Work and all matters referred to in the Bid Documents, and the Contract Documents prepared by or approved by the Engineer for the named Project, we, the undersigned, hereby offer to enter into a Contract to perform the Work for the Total Bid Price of:

\_\_\_\_\_(Dollars)

[Print or type in words, Bidder's Total Bid Price]

(\$ \_\_\_\_\_)

[Print or type in figures, Bidder's Total Bid Price]

**Unit Price or Combination Stipulated Price and Unit Price Contract.** If the Bid is for a Unit Price Contract or a combination of Stipulated Price and Unit Price Contract, the Total Bid Price, including Cash Allowances, if any, is tabulated in: Document 00405 - Schedule of Unit Price Work for a Project with no Alternate Bids, or Document 00407 - Schedule of Alternates for a Project with Alternate Bids.

**Cash Allowances.** All Cash Allowances, totaled in either Document 00405 - Schedule of Unit Price Work, as applicable, and described in the Bid Documents are included in the Total Bid Price.

**Changes in Contract Price Due to Variations in Actual Quantities.** For items quoted in Document 00405 - Schedule of Unit Price Work, the Total Bid Price is based in whole or in part on the Unit Price multiplied by the quantity for each of the items listed. The Contract Price is subject to change due to variation in the actual quantities of each item in the completed Work in accordance with the Contract Documents.

**Alternate Bids.** Alternate Bid work, as described in the Bid Documents, will be performed for an amount added or deducted to the Total Bid Price for each Alternate Bid that is accepted by the Owner. The Owner may accept or reject any or all Alternate Bids.

**Security Deposit.** Included herewith is a Security Deposit in the amount of 5 percent of the greatest amount of the Total Bid Price, or Total Alternate Bid Price(s).

**Period for Bid Acceptance.** This offer shall be open to acceptance and is irrevocable for 90 days from the Bid date. That period may be extended by mutual written agreement of the Owner and the Bidder. After 90 days, the Bidder may withdraw without penalty if no mutual agreement can be reached.

**2.0 CONTRACT TIME**

If this offer is accepted, Substantial Completion of the Work will be achieved within the time stated in Document 00020 - Notice to Bidders. The Date of Commencement will be established by the Notice to Proceed.

**3.0 ADDENDA**

The following Addenda have been received. The modifications to the Bid Documents noted therein have been considered and all costs relating thereto are included in the Bid Price:

Addendum No. \_\_\_\_\_, dated \_\_\_\_\_

Addendum No. \_\_\_\_\_, dated \_\_\_\_\_

Addendum No. \_\_\_\_\_, dated \_\_\_\_\_

Addendum No. \_\_\_\_\_, dated \_\_\_\_\_

Addendum No. \_\_\_\_\_, dated \_\_\_\_\_

Addendum No. \_\_\_\_\_, dated \_\_\_\_\_

**4.0 SUPPLEMENTS TO THIS BID:**

The following Supplements are attached as an integral part of this Bid:

- ☐ Document 00405 - Schedule of Unit Price Work, if applicable
- ☐ Document 00411 – Bid Bond (*Form supplied by Bidder*)
- ☐ Document 00420 – Statement of Bidder's Qualifications
- ☐ Document 00423 – Certification to Bidder's Experience & Qualifications
- ☐ Document 00425 – Equipment & Material Suppliers List

**5.0 SIGNATURES:**

Bidder: \_\_\_\_\_  
[Please print or type the full name of your proprietorship, partnership, corporation, or joint venture.\*]

By: \_\_\_\_\_  
[Signature]\*\* [Date]

Name: \_\_\_\_\_  
[Please print or type name] [Title]

Address: \_\_\_\_\_  
[Mailing]

\_\_\_\_\_  
[Street, if different]

Telephone: \_\_\_\_\_  
[Print or type telephone number]

\* *If the Bid is a joint venture, add additional Bid form signature sheets for each member of the joint venture.*

\*\* *The undersigned, as bidder, certifies that the only person or parties interested in this proposal as principals are those named herein; that the Bidder has not, either directly or indirectly, entered into any agreement, participated in any collusion, or otherwise taken any action in restraint of free competitive bidding in connection with the Contract for the Project.*

*Note: This document constitutes a government record, as defined by § 37.01 of the Texas Penal Code. Submission of a false government record is punishable as provided in § 37.10 of the Texas Penal Code.*

**END OF DOCUMENT**

Page Intentionally Blank

Document 00411

**BID BOND****PART 1 - GENERAL****1.01 SECTION INCLUDES**

This section describes the standardized bid bond form to be submitted with the bid on the project.

**1.02 REFERENCES – Not Used****1.03 DEFINITIONS – Section 0700****1.04 BID BOND FORMS**

Bidder is to inset an original bid bond or a copy of cashiers check provided for bid bond Purposes. Original check is to be submitted along with bid.

**PART 2 - PRODUCT – Not Used****PART 3 - EXECUTION**

STANDARIZED FORMS FOLLOW

## BID BOND (PENAL SUM FORM)

|                                                                                                                                                                                                               |                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bidder</b><br>Name: <b>[Full formal name of Bidder]</b><br>Address <i>(principal place of business)</i> :<br><b>[Address of Bidder's principal place of business]</b>                                      | <b>Surety</b><br>Name: <b>[Full formal name of Surety]</b><br>Address <i>(principal place of business)</i> :<br><b>[Address of Surety's principal place of business]</b>    |
| <b>Owner</b><br>Name: <b>[Full formal name of Owner]</b><br>Address <i>(principal place of business)</i> :<br><b>[Address of Owner's principal place of business]</b>                                         | <b>Bid</b><br>Project <i>(name and location)</i> :<br><b>[Owner project/contract name, and location of the project]</b><br><br>Bid Due Date: <b>[Enter date bid is due]</b> |
| <b>Bond</b><br>Penal Sum: <b>[Amount]</b><br>Date of Bond: <b>[Date]</b>                                                                                                                                      |                                                                                                                                                                             |
| Surety and Bidder, intending to be legally bound hereby, subject to the terms set forth in this Bid Bond, do each cause this Bid Bond to be duly executed by an authorized officer, agent, or representative. |                                                                                                                                                                             |
| <b>Bidder</b>                                                                                                                                                                                                 | <b>Surety</b>                                                                                                                                                               |
| <i>(Full formal name of Bidder)</i>                                                                                                                                                                           | <i>(Full formal name of Surety) (corporate seal)</i>                                                                                                                        |
| By: _____<br><i>(Signature)</i>                                                                                                                                                                               | By: _____<br><i>(Signature) (Attach Power of Attorney)</i>                                                                                                                  |
| Name: _____<br><i>(Printed or typed)</i>                                                                                                                                                                      | Name: _____<br><i>(Printed or typed)</i>                                                                                                                                    |
| Title: _____                                                                                                                                                                                                  | Title: _____                                                                                                                                                                |
| Attest: _____<br><i>(Signature)</i>                                                                                                                                                                           | Attest: _____<br><i>(Signature)</i>                                                                                                                                         |
| Name: _____<br><i>(Printed or typed)</i>                                                                                                                                                                      | Name: _____<br><i>(Printed or typed)</i>                                                                                                                                    |
| Title: _____                                                                                                                                                                                                  | Title: _____                                                                                                                                                                |
| Notes: (1) Note: Addresses are to be used for giving any required notice. (2) Provide execution by any additional parties, such as joint venturers, if necessary.                                             |                                                                                                                                                                             |



1. Bidder and Surety, jointly and severally, bind themselves, their heirs, executors, administrators, successors, and assigns to pay to Owner upon default of Bidder the penal sum set forth on the face of this Bond. Payment of the penal sum is the extent of Bidder's and Surety's liability. Recovery of such penal sum under the terms of this Bond will be Owner's sole and exclusive remedy upon default of Bidder.
2. Default of Bidder occurs upon the failure of Bidder to deliver within the time required by the Bidding Documents (or any extension thereof agreed to in writing by Owner) the executed Agreement required by the Bidding Documents and any performance and payment bonds required by the Bidding Documents.
3. This obligation will be null and void if:
  - 3.1. Owner accepts Bidder's Bid and Bidder delivers within the time required by the Bidding Documents (or any extension thereof agreed to in writing by Owner) the executed Agreement required by the Bidding Documents and any performance and payment bonds required by the Bidding Documents, or
  - 3.2. All Bids are rejected by Owner, or
  - 3.3. Owner fails to issue a Notice of Award to Bidder within the time specified in the Bidding Documents (or any extension thereof agreed to in writing by Bidder and, if applicable, consented to by Surety when required by Paragraph 5 hereof).
4. Payment under this Bond will be due and payable upon default of Bidder and within 30 calendar days after receipt by Bidder and Surety of written notice of default from Owner, which notice will be given with reasonable promptness, identifying this Bond and the Project and including a statement of the amount due.
5. Surety waives notice of any and all defenses based on or arising out of any time extension to issue Notice of Award agreed to in writing by Owner and Bidder, provided that the total time for issuing Notice of Award including extensions does not in the aggregate exceed 120 days from the Bid due date without Surety's written consent.
6. No suit or action will be commenced under this Bond prior to 30 calendar days after the notice of default required in Paragraph 4 above is received by Bidder and Surety, and in no case later than one year after the Bid due date.
7. Any suit or action under this Bond will be commenced only in a court of competent jurisdiction located in the state in which the Project is located.
8. Notices required hereunder must be in writing and sent to Bidder and Surety at their respective addresses shown on the face of this Bond. Such notices may be sent by personal delivery, commercial courier, or by United States Postal Service registered or certified mail, return receipt requested, postage pre-paid, and will be deemed to be effective upon receipt by the party concerned.
9. Surety shall cause to be attached to this Bond a current and effective Power of Attorney evidencing the authority of the officer, agent, or representative who executed this Bond on behalf of Surety to execute, seal, and deliver such Bond and bind the Surety thereby.
10. This Bond is intended to conform to all applicable statutory requirements. Any applicable requirement of any applicable statute that has been omitted from this Bond will be deemed to be included

herein as if set forth at length. If any provision of this Bond conflicts with any applicable statute, then the provision of said statute governs and the remainder of this Bond that is not in conflict therewith continues in full force and effect.

11. The term “Bid” as used herein includes a Bid, offer, or proposal as applicable.

**END OF SECTION**

DOCUMENT 00420

**STATEMENT OF BIDDER'S QUALIFICATIONS****ARTICLE 1—GENERAL INFORMATION****1.01 Provide contact information for the Business:**

|                                       |  |                |  |
|---------------------------------------|--|----------------|--|
| Legal Name of Business:               |  |                |  |
| Corporate Office                      |  |                |  |
| Name:                                 |  | Phone number:  |  |
| Title:                                |  | Email address: |  |
| Business address of corporate office: |  |                |  |
|                                       |  |                |  |
|                                       |  |                |  |
| Local Office                          |  |                |  |
| Name:                                 |  | Phone number:  |  |
| Title:                                |  | Email address: |  |
| Business address of local office:     |  |                |  |
|                                       |  |                |  |
|                                       |  |                |  |

**1.02 Provide information on the Business's organizational structure:**

|                                                                                                                                 |                                                                                                                        |                                                                                           |  |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--|
| Form of Business:                                                                                                               | <input type="checkbox"/> Sole Proprietorship <input type="checkbox"/> Partnership <input type="checkbox"/> Corporation |                                                                                           |  |
| <input type="checkbox"/> Limited Liability Company <input type="checkbox"/> Joint Venture comprised of the following companies: |                                                                                                                        |                                                                                           |  |
| 1.                                                                                                                              |                                                                                                                        |                                                                                           |  |
| 2.                                                                                                                              |                                                                                                                        |                                                                                           |  |
| 3.                                                                                                                              |                                                                                                                        |                                                                                           |  |
| Provide a separate Qualification Statement for each Joint Venturer.                                                             |                                                                                                                        |                                                                                           |  |
| Date Business was formed:                                                                                                       |                                                                                                                        | State in which Business was formed:                                                       |  |
| Is this Business authorized to operate in the Project location?                                                                 |                                                                                                                        | <input type="checkbox"/> Yes <input type="checkbox"/> No <input type="checkbox"/> Pending |  |

- 1.03 Identify all businesses that own Business in whole or in part (25% or greater), or that are wholly or partly (25% or greater) owned by Business:

|                   |  |              |  |
|-------------------|--|--------------|--|
| Name of business: |  | Affiliation: |  |
| Address:          |  |              |  |
| Name of business: |  | Affiliation: |  |
| Address:          |  |              |  |
| Name of business: |  | Affiliation: |  |
| Address:          |  |              |  |

- 1.04 Provide information regarding the Business's officers, partners, and limits of authority.

|                                                                                        |  |                     |    |
|----------------------------------------------------------------------------------------|--|---------------------|----|
| Name:                                                                                  |  | Title:              |    |
| Authorized to sign contracts: <input type="checkbox"/> Yes <input type="checkbox"/> No |  | Limit of Authority: | \$ |
| Name:                                                                                  |  | Title:              |    |
| Authorized to sign contracts: <input type="checkbox"/> Yes <input type="checkbox"/> No |  | Limit of Authority: | \$ |
| Name:                                                                                  |  | Title:              |    |
| Authorized to sign contracts: <input type="checkbox"/> Yes <input type="checkbox"/> No |  | Limit of Authority: | \$ |
| Name:                                                                                  |  | Title:              |    |

## ARTICLE 2—LICENSING

- 2.01 Provide information regarding licensure for Business, if applicable:

|                   |  |                  |  |
|-------------------|--|------------------|--|
| Name of License:  |  |                  |  |
| Licensing Agency: |  |                  |  |
| License No:       |  | Expiration Date: |  |
| Name of License:  |  |                  |  |
| Licensing Agency: |  |                  |  |
| License No:       |  | Expiration Date: |  |

**ARTICLE 3—DIVERSE BUSINESS CERTIFICATIONS**

- 3.01 Provide information regarding Business's Diverse Business Certification, if any. Provide evidence of current certification.

| Certification                                                                   | Certifying Agency | Certification Date |
|---------------------------------------------------------------------------------|-------------------|--------------------|
| <input type="checkbox"/> Disadvantaged Business Enterprise                      |                   |                    |
| <input type="checkbox"/> Minority Business Enterprise                           |                   |                    |
| <input type="checkbox"/> Woman-Owned Business Enterprise                        |                   |                    |
| <input type="checkbox"/> Small Business Enterprise                              |                   |                    |
| <input type="checkbox"/> Disabled Business Enterprise                           |                   |                    |
| <input type="checkbox"/> Veteran-Owned Business Enterprise                      |                   |                    |
| <input type="checkbox"/> Service-Disabled Veteran-Owned Business                |                   |                    |
| <input type="checkbox"/> HUBZone Business (Historically Underutilized) Business |                   |                    |
| <input type="checkbox"/> Other                                                  |                   |                    |
| <input type="checkbox"/> None                                                   |                   |                    |

**ARTICLE 4—SAFETY**

- 4.01 Provide information regarding Business's safety organization and safety performance.

|                                    |                |            |
|------------------------------------|----------------|------------|
| Name of Business's Safety Officer: |                |            |
| Safety Certifications              |                |            |
| Certification Name                 | Issuing Agency | Expiration |
|                                    |                |            |
|                                    |                |            |

- 4.02 Provide Worker's Compensation Insurance Experience Modification Rate (EMR), Total Recordable Frequency Rate (TRFR) for incidents, and Total Number of Recorded Manhours (MH) for the last 3 years and the EMR, TRFR, and MH history for the last 3 years of any proposed Subcontractor(s) that will provide Work valued at 10% or more of the Contract Price. Provide documentation of the EMR history for Business and Subcontractor(s).

| Year    |     |      |    |     |      |    |     |      |    |
|---------|-----|------|----|-----|------|----|-----|------|----|
| Company | EMR | TRFR | MH | EMR | TRFR | MH | EMR | TRFR | MH |
|         |     |      |    |     |      |    |     |      |    |
|         |     |      |    |     |      |    |     |      |    |

**ARTICLE 5—FINANCIAL**

- 5.01 Provide information regarding the Business's financial stability. Provide the most recent audited financial statement, and if such audited financial statement is not current, also provide the most current financial statement.

|                                                                                                                             |  |                                   |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------|--|
| Financial Institution:                                                                                                      |  |                                   |  |
| Business address:                                                                                                           |  |                                   |  |
| Date of Business's most recent financial statement:                                                                         |  | <input type="checkbox"/> Attached |  |
| Date of Business's most recent audited financial statement:                                                                 |  | <input type="checkbox"/> Attached |  |
| Financial indicators from the most recent financial statement                                                               |  |                                   |  |
| Contractor's Current Ratio (Current Assets ÷ Current Liabilities)                                                           |  |                                   |  |
| Contractor's Quick Ratio ((Cash and Cash Equivalents + Accounts Receivable + Short Term Investments) ÷ Current Liabilities) |  |                                   |  |

**ARTICLE 6—SURETY INFORMATION**

- 6.01 Provide information regarding the surety company that will issue required bonds on behalf of the Business, including but not limited to performance and payment bonds.

|                                                                                                                                                                                                                                                                                                                                      |                                                          |                 |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------|--|
| Surety Name:                                                                                                                                                                                                                                                                                                                         |                                                          |                 |  |
| Surety is a corporation organized and existing under the laws of the state of:                                                                                                                                                                                                                                                       |                                                          |                 |  |
| Is surety authorized to provide surety bonds in the Project location?                                                                                                                                                                                                                                                                | <input type="checkbox"/> Yes <input type="checkbox"/> No |                 |  |
| Is surety listed in "Companies Holding Certificates of Authority as Acceptable Sureties on Federal Bonds and as Acceptable Reinsuring Companies" published in Department Circular 570 (as amended) by the Bureau of the Fiscal Service, U.S. Department of the Treasury?<br><input type="checkbox"/> Yes <input type="checkbox"/> No |                                                          |                 |  |
| Mailing Address<br>(principal place of business):                                                                                                                                                                                                                                                                                    |                                                          |                 |  |
|                                                                                                                                                                                                                                                                                                                                      |                                                          |                 |  |
|                                                                                                                                                                                                                                                                                                                                      |                                                          |                 |  |
| Physical Address<br>(principal place of business):                                                                                                                                                                                                                                                                                   |                                                          |                 |  |
|                                                                                                                                                                                                                                                                                                                                      |                                                          |                 |  |
|                                                                                                                                                                                                                                                                                                                                      |                                                          |                 |  |
| Phone (main):                                                                                                                                                                                                                                                                                                                        |                                                          | Phone (claims): |  |

**ARTICLE 7—INSURANCE**

- 7.01 Provide information regarding Business's insurance company(s), including but not limited to its Commercial General Liability carrier. Provide information for each provider.

|                                                                                 |  |                                                          |  |
|---------------------------------------------------------------------------------|--|----------------------------------------------------------|--|
| Name of insurance provider, and type of policy (CLE, auto, etc.):               |  |                                                          |  |
| Insurance Provider                                                              |  | Type of Policy (Coverage Provided)                       |  |
|                                                                                 |  |                                                          |  |
|                                                                                 |  |                                                          |  |
|                                                                                 |  |                                                          |  |
| Are providers licensed or authorized to issue policies in the Project location? |  | <input type="checkbox"/> Yes <input type="checkbox"/> No |  |
| Does provider have an A.M. Best Rating of A-VII or better?                      |  | <input type="checkbox"/> Yes <input type="checkbox"/> No |  |
| Mailing Address<br>(principal place of business):                               |  |                                                          |  |
|                                                                                 |  |                                                          |  |
|                                                                                 |  |                                                          |  |
| Physical Address<br>(principal place of business):                              |  |                                                          |  |
|                                                                                 |  |                                                          |  |
|                                                                                 |  |                                                          |  |
| Phone (main):                                                                   |  | Phone (claims):                                          |  |

**ARTICLE 8—CONSTRUCTION EXPERIENCE**

- 8.01 Provide information that will identify the overall size and capacity of the Business.

|                                                |  |
|------------------------------------------------|--|
| Average number of current full-time employees: |  |
| Estimate of revenue for the current year:      |  |
| Estimate of revenue for the previous year:     |  |

- 8.02 Provide information regarding the Business's previous contracting experience.

|                                                                                                                                                           |  |                      |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|--|
| Years of experience with projects like the proposed project:                                                                                              |  |                      |  |
| As a general contractor:                                                                                                                                  |  | As a joint venturer: |  |
| Has Business, or a predecessor in interest, or an affiliate identified in Paragraph 1.03:                                                                 |  |                      |  |
| Been disqualified as a bidder by any local, state, or federal agency within the last 5 years?<br><input type="checkbox"/> Yes <input type="checkbox"/> No |  |                      |  |
| Been barred from contracting by any local, state, or federal agency within the last 5 years?<br><input type="checkbox"/> Yes <input type="checkbox"/> No  |  |                      |  |
| Been released from a bid in the past 5 years? <input type="checkbox"/> Yes <input type="checkbox"/> No                                                    |  |                      |  |
| Defaulted on a project or failed to complete any contract awarded to it? <input type="checkbox"/> Yes <input type="checkbox"/> No                         |  |                      |  |

|                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refused to construct or refused to provide materials defined in the contract documents or in a change order? <input type="checkbox"/> Yes <input type="checkbox"/> No |
| Been a party to any currently pending litigation or arbitration? <input type="checkbox"/> Yes <input type="checkbox"/> No                                             |
| Provide full details in a separate attachment if the response to any of these questions is Yes.                                                                       |

- 8.03 List all projects currently under contract in Schedule A and provide indicated information.
- 8.04 List a minimum of three and a maximum of six projects completed in the last 5 years in Schedule B and provide indicated information to demonstrate the Business's experience with projects similar in type and cost of construction.
- 8.05 In Schedule C, provide information on key individuals whom Business intends to assign to the Project. Provide resumes for those individuals included in Schedule C. Key individuals include the Project Manager, Project Superintendent, Quality Manager, and Safety Manager. Resumes may be provided for Business's key leaders as well.

#### ARTICLE 9—REQUIRED ATTACHMENTS

- 9.01 Provide the following information with the Statement of Qualifications:
- A. If Business is a Joint Venture, separate Qualifications Statements for each Joint Venturer, as required in Paragraph 1.02.
  - B. Diverse Business Certifications if required by Paragraph 3.01.
  - C. Certification of Business's safety performance if required by Paragraph 4.02.
  - D. Financial statements as required by Paragraph 5.01.
  - E. Attachments providing additional information as required by Paragraph 8.02.
  - F. Schedule A (Current Projects) as required by Paragraph 8.03.
  - G. Schedule B (Previous Experience with Similar Projects) as required by Paragraph 8.04.
  - H. Schedule C (Key Individuals) and resumes for the key individuals listed, as required by Paragraph 8.05.
  - I. Additional items as pertinent.



This Statement of Qualifications is offered by:

Business: \_\_\_\_\_  
(typed or printed name of organization)

By: \_\_\_\_\_  
(individual's signature)

Name: \_\_\_\_\_  
(typed or printed)

Title: \_\_\_\_\_  
(typed or printed)

Date: \_\_\_\_\_  
(date signed)

*(If Business is a corporation, a partnership, or a joint venture, attach evidence of authority to sign.)*

Attest: \_\_\_\_\_  
(individual's signature)

Name: \_\_\_\_\_  
(typed or printed)

Title: \_\_\_\_\_  
(typed or printed)

Address for giving notices:

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

Designated Representative:

Name: \_\_\_\_\_  
(typed or printed)

Title: \_\_\_\_\_  
(typed or printed)

Address: \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

Phone: \_\_\_\_\_

Email: \_\_\_\_\_

### Schedule A—Current Projects

|                                                                                                                     |                 |                        |                |                         |       |
|---------------------------------------------------------------------------------------------------------------------|-----------------|------------------------|----------------|-------------------------|-------|
| Name of Organization                                                                                                |                 |                        |                |                         |       |
| Project Owner                                                                                                       |                 |                        | Project Name   |                         |       |
| General Description of Project                                                                                      |                 |                        |                |                         |       |
| Project Cost                                                                                                        |                 |                        | Date Project   |                         |       |
| Key Project Personnel                                                                                               | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |       |
| Name                                                                                                                |                 |                        |                |                         |       |
| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |                 |                        |                |                         |       |
|                                                                                                                     | Name            | Title/Position         | Organization   | Telephone               | Email |
| Owner                                                                                                               |                 |                        |                |                         |       |
| Designer                                                                                                            |                 |                        |                |                         |       |
| Construction Manager                                                                                                |                 |                        |                |                         |       |

|                                                                                                                     |                 |                        |                |                         |       |
|---------------------------------------------------------------------------------------------------------------------|-----------------|------------------------|----------------|-------------------------|-------|
| Project Owner                                                                                                       |                 |                        | Project Name   |                         |       |
| General Description of Project                                                                                      |                 |                        |                |                         |       |
| Project Cost                                                                                                        |                 |                        | Date Project   |                         |       |
| Key Project Personnel                                                                                               | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |       |
| Name                                                                                                                |                 |                        |                |                         |       |
| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |                 |                        |                |                         |       |
|                                                                                                                     | Name            | Title/Position         | Organization   | Telephone               | Email |
| Owner                                                                                                               |                 |                        |                |                         |       |
| Designer                                                                                                            |                 |                        |                |                         |       |
| Construction Manager                                                                                                |                 |                        |                |                         |       |

|                                                                                                                     |                 |                        |                |                         |       |
|---------------------------------------------------------------------------------------------------------------------|-----------------|------------------------|----------------|-------------------------|-------|
| Project Owner                                                                                                       |                 |                        | Project Name   |                         |       |
| General Description of Project                                                                                      |                 |                        |                |                         |       |
| Project Cost                                                                                                        |                 |                        | Date Project   |                         |       |
| Key Project Personnel                                                                                               | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |       |
| Name                                                                                                                |                 |                        |                |                         |       |
| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |                 |                        |                |                         |       |
|                                                                                                                     | Name            | Title/Position         | Organization   | Telephone               | Email |
| Owner                                                                                                               |                 |                        |                |                         |       |
| Designer                                                                                                            |                 |                        |                |                         |       |

|                      |  |  |  |  |  |
|----------------------|--|--|--|--|--|
| Construction Manager |  |  |  |  |  |
|----------------------|--|--|--|--|--|

### Schedule B—Previous Experience with Similar Projects

|                                                                                                                     |                 |                        |                |                         |       |
|---------------------------------------------------------------------------------------------------------------------|-----------------|------------------------|----------------|-------------------------|-------|
| Name of Organization                                                                                                |                 |                        |                |                         |       |
| Project Owner                                                                                                       |                 |                        | Project Name   |                         |       |
| General Description of Project                                                                                      |                 |                        |                |                         |       |
| Project Cost                                                                                                        |                 |                        | Date Project   |                         |       |
| Key Project Personnel                                                                                               | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |       |
| Name                                                                                                                |                 |                        |                |                         |       |
| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |                 |                        |                |                         |       |
|                                                                                                                     | Name            | Title/Position         | Organization   | Telephone               | Email |
| Owner                                                                                                               |                 |                        |                |                         |       |
| Designer                                                                                                            |                 |                        |                |                         |       |
| Construction Manager                                                                                                |                 |                        |                |                         |       |

|                                                                                                                     |                 |                        |                |                         |       |
|---------------------------------------------------------------------------------------------------------------------|-----------------|------------------------|----------------|-------------------------|-------|
| Project Owner                                                                                                       |                 |                        | Project Name   |                         |       |
| General Description of Project                                                                                      |                 |                        |                |                         |       |
| Project Cost                                                                                                        |                 |                        | Date Project   |                         |       |
| Key Project Personnel                                                                                               | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |       |
| Name                                                                                                                |                 |                        |                |                         |       |
| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |                 |                        |                |                         |       |
|                                                                                                                     | Name            | Title/Position         | Organization   | Telephone               | Email |
| Owner                                                                                                               |                 |                        |                |                         |       |
| Designer                                                                                                            |                 |                        |                |                         |       |
| Construction Manager                                                                                                |                 |                        |                |                         |       |

|                                                                                                                     |                 |                        |                |                         |       |
|---------------------------------------------------------------------------------------------------------------------|-----------------|------------------------|----------------|-------------------------|-------|
| Project Owner                                                                                                       |                 |                        | Project Name   |                         |       |
| General Description of Project                                                                                      |                 |                        |                |                         |       |
| Project Cost                                                                                                        |                 |                        | Date Project   |                         |       |
| Key Project Personnel                                                                                               | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |       |
| Name                                                                                                                |                 |                        |                |                         |       |
| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |                 |                        |                |                         |       |
|                                                                                                                     | Name            | Title/Position         | Organization   | Telephone               | Email |

|                      |  |  |  |  |  |
|----------------------|--|--|--|--|--|
| Owner                |  |  |  |  |  |
| Designer             |  |  |  |  |  |
| Construction Manager |  |  |  |  |  |

### Schedule B—Previous Experience with Similar Projects

|                                                                                                                     |                 |                        |                |                         |       |
|---------------------------------------------------------------------------------------------------------------------|-----------------|------------------------|----------------|-------------------------|-------|
| Name of Organization                                                                                                |                 |                        |                |                         |       |
| Project Owner                                                                                                       |                 |                        | Project Name   |                         |       |
| General Description of Project                                                                                      |                 |                        |                |                         |       |
| Project Cost                                                                                                        |                 |                        | Date Project   |                         |       |
| Key Project Personnel                                                                                               | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |       |
| Name                                                                                                                |                 |                        |                |                         |       |
| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |                 |                        |                |                         |       |
|                                                                                                                     | Name            | Title/Position         | Organization   | Telephone               | Email |
| Owner                                                                                                               |                 |                        |                |                         |       |
| Designer                                                                                                            |                 |                        |                |                         |       |
| Construction Manager                                                                                                |                 |                        |                |                         |       |

|                                                                                                                     |                 |                        |                |                         |       |
|---------------------------------------------------------------------------------------------------------------------|-----------------|------------------------|----------------|-------------------------|-------|
| Project Owner                                                                                                       |                 |                        | Project Name   |                         |       |
| General Description of Project                                                                                      |                 |                        |                |                         |       |
| Project Cost                                                                                                        |                 |                        | Date Project   |                         |       |
| Key Project Personnel                                                                                               | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |       |
| Name                                                                                                                |                 |                        |                |                         |       |
| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |                 |                        |                |                         |       |
|                                                                                                                     | Name            | Title/Position         | Organization   | Telephone               | Email |
| Owner                                                                                                               |                 |                        |                |                         |       |
| Designer                                                                                                            |                 |                        |                |                         |       |
| Construction Manager                                                                                                |                 |                        |                |                         |       |

|                                |                 |                        |                |                         |  |
|--------------------------------|-----------------|------------------------|----------------|-------------------------|--|
| Project Owner                  |                 |                        | Project Name   |                         |  |
| General Description of Project |                 |                        |                |                         |  |
| Project Cost                   |                 |                        | Date Project   |                         |  |
| Key Project Personnel          | Project Manager | Project Superintendent | Safety Manager | Quality Control Manager |  |
| Name                           |                 |                        |                |                         |  |

| Reference Contact Information (listing names indicates approval to contacting the names individuals as a reference) |      |                |              |           |       |
|---------------------------------------------------------------------------------------------------------------------|------|----------------|--------------|-----------|-------|
|                                                                                                                     | Name | Title/Position | Organization | Telephone | Email |
| Owner                                                                                                               |      |                |              |           |       |
| Designer                                                                                                            |      |                |              |           |       |
| Construction Manager                                                                                                |      |                |              |           |       |

## Schedule C—Key Individuals

|                                                                                                              |  |                                       |                                   |
|--------------------------------------------------------------------------------------------------------------|--|---------------------------------------|-----------------------------------|
| <b>Project Manager</b>                                                                                       |  |                                       |                                   |
| Name of individual                                                                                           |  |                                       |                                   |
| Years of experience as project manager                                                                       |  |                                       |                                   |
| Years of experience with this organization                                                                   |  |                                       |                                   |
| Number of similar projects as project manager                                                                |  |                                       |                                   |
| Number of similar projects in other positions                                                                |  |                                       |                                   |
| Current Project Assignments                                                                                  |  |                                       |                                   |
| Name of assignment                                                                                           |  | Percent of time used for this project | Estimated project completion date |
|                                                                                                              |  |                                       |                                   |
|                                                                                                              |  |                                       |                                   |
|                                                                                                              |  |                                       |                                   |
| Reference Contact Information (listing names indicates approval to contact named individuals as a reference) |  |                                       |                                   |
| Name                                                                                                         |  | Name                                  |                                   |
| Title/Position                                                                                               |  | Title/Position                        |                                   |
| Organization                                                                                                 |  | Organization                          |                                   |
| Telephone                                                                                                    |  | Telephone                             |                                   |
| Email                                                                                                        |  | Email                                 |                                   |
| Project                                                                                                      |  | Project                               |                                   |
| Candidate's role on project                                                                                  |  | Candidate's role on project           |                                   |
| <b>Project Superintendent</b>                                                                                |  |                                       |                                   |
| Name of individual                                                                                           |  |                                       |                                   |
| Years of experience as project superintendent                                                                |  |                                       |                                   |
| Years of experience with this organization                                                                   |  |                                       |                                   |
| Number of similar projects as project superintendent                                                         |  |                                       |                                   |
| Number of similar projects in other positions                                                                |  |                                       |                                   |
| Current Project Assignments                                                                                  |  |                                       |                                   |
| Name of assignment                                                                                           |  | Percent of time used for this project | Estimated project completion date |
|                                                                                                              |  |                                       |                                   |
|                                                                                                              |  |                                       |                                   |
|                                                                                                              |  |                                       |                                   |
| Reference Contact Information (listing names indicates approval to contact named individuals as a reference) |  |                                       |                                   |
| Name                                                                                                         |  | Name                                  |                                   |
| Title/Position                                                                                               |  | Title/Position                        |                                   |
| Organization                                                                                                 |  | Organization                          |                                   |
| Telephone                                                                                                    |  | Telephone                             |                                   |
| Email                                                                                                        |  | Email                                 |                                   |
| Project                                                                                                      |  | Project                               |                                   |
| Candidate's role on project                                                                                  |  | Candidate's role on project           |                                   |

|                                                                                                              |  |                                       |                                   |
|--------------------------------------------------------------------------------------------------------------|--|---------------------------------------|-----------------------------------|
| <b>Safety Manager</b>                                                                                        |  |                                       |                                   |
| Name of individual                                                                                           |  |                                       |                                   |
| Years of experience as project manager                                                                       |  |                                       |                                   |
| Years of experience with this organization                                                                   |  |                                       |                                   |
| Number of similar projects as project manager                                                                |  |                                       |                                   |
| Number of similar projects in other positions                                                                |  |                                       |                                   |
| Current Project Assignments                                                                                  |  |                                       |                                   |
| Name of assignment                                                                                           |  | Percent of time used for this project | Estimated project completion date |
|                                                                                                              |  |                                       |                                   |
|                                                                                                              |  |                                       |                                   |
|                                                                                                              |  |                                       |                                   |
| Reference Contact Information (listing names indicates approval to contact named individuals as a reference) |  |                                       |                                   |
| Name                                                                                                         |  | Name                                  |                                   |
| Title/Position                                                                                               |  | Title/Position                        |                                   |
| Organization                                                                                                 |  | Organization                          |                                   |
| Telephone                                                                                                    |  | Telephone                             |                                   |
| Email                                                                                                        |  | Email                                 |                                   |
| Project                                                                                                      |  | Project                               |                                   |
| Candidate's role on project                                                                                  |  | Candidate's role on project           |                                   |
| <b>Quality Control Manager</b>                                                                               |  |                                       |                                   |
| Name of individual                                                                                           |  |                                       |                                   |
| Years of experience as project superintendent                                                                |  |                                       |                                   |
| Years of experience with this organization                                                                   |  |                                       |                                   |
| Number of similar projects as project superintendent                                                         |  |                                       |                                   |
| Number of similar projects in other positions                                                                |  |                                       |                                   |
| Current Project Assignments                                                                                  |  |                                       |                                   |
| Name of assignment                                                                                           |  | Percent of time used for this project | Estimated project completion date |
|                                                                                                              |  |                                       |                                   |
|                                                                                                              |  |                                       |                                   |
|                                                                                                              |  |                                       |                                   |
| Reference Contact Information (listing names indicates approval to contact named individuals as a reference) |  |                                       |                                   |
| Name                                                                                                         |  | Name                                  |                                   |
| Title/Position                                                                                               |  | Title/Position                        |                                   |
| Organization                                                                                                 |  | Organization                          |                                   |
| Telephone                                                                                                    |  | Telephone                             |                                   |
| Email                                                                                                        |  | Email                                 |                                   |
| Project                                                                                                      |  | Project                               |                                   |
| Candidate's role on project                                                                                  |  | Candidate's role on project           |                                   |

**END OF DOCUMENT**



DOCUMENT 00423

**CERTIFICATE OF BIDDER'S  
EXPERIENCE & QUALIFICATIONS**

The undersigned bidder certifies that he is, at the time of bidding, and shall be, throughout the period of the contract, licensed by the State of Texas to do the type of work required under terms of the contract documents. Bidder further certifies that he is skilled and regularly engaged in the general class and type of work called for in the contract documents.

The bidder represents that he is competent, knowledgeable and has special skills on the nature, extent and inherent conditions of the work to be performed. Bidder further acknowledges that there are certain peculiar and inherent conditions existent in the construction of the particular facilities which may create, during the construction program, unusual or peculiar unsafe conditions hazardous to persons and property. Bidder expressly acknowledges that he is aware of such peculiar risks and that he has the skill and experience to foresee and to adopt protective measures to adequately and safely perform the construction work with respect to such hazards.

Signed this \_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

---

Name of Bidder

---

Signature of Bidder

---

Title of Signatory**END OF SECTION**

00423 - 1 of 2

Page Intentionally Blank

DOCUMENT 00425

**EQUIPMENT & MATERIAL SUPPLIERS LIST**

**PURPOSE:** To assist the Owner in determining the ability of each Bidder to properly fulfill the requirements of this proposed contract, the Bidder shall complete the following items. All questions must be answered and the data given must be clear and comprehensive. If necessary, questions may be answered on separate attached sheets as specified by 00420 Statement of Bidder's Qualifications. If, in the course of evaluating the bids, the Owner discovers that answers to these questions are false or misleading then the Owner reserves the right to reject the bid based on non-responsiveness. **This statement must be notarized.**

The undersigned hereby authorizes and requests any person, firm, or corporation to furnish any information requested by the Owner in verification of the recitals comprising this Statement of Bidder's Qualifications.

**A. EQUIPMENT AVAILABLE FOR THIS CONTRACT:** The Bidder shall provide below a list of equipment available for use on this contract:

| EQUIPMENT | OWN | RENT/LEASE<br>(Supplier & Phone #) |
|-----------|-----|------------------------------------|
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |
|           |     |                                    |

**B. MATERIALS AND MAJOR EQUIPMENT:** The Bidder shall provide below a list of manufacturers and suppliers of major equipment and materials proposed on this contract:

[illegible]

**BIDDER**

Executed this: \_\_\_\_\_ Day of : \_\_\_\_\_ 20. \_\_\_\_\_

By: \_\_\_\_\_

BIDDER

Title: \_\_\_\_\_

**NOTARY PUBLIC**

State of Texas

County of:

Subscribed and sworn to before me this: \_\_\_\_\_

\_\_\_\_\_  
NOTARY PUBLIC

**END OF SECTION**

Page Intentionally Blank

DOCUMENT 00429

## CITY OF EDINBURG NON-BRIBERY MODEL FORM

[ *Bidder's letterhead* ][ *Date* ][ *Name and address* ]Dear [ *Name of Owner* ] :

The undersigned party certifies that [ *Name of bidding company* ] complies with the following criteria:

1. They have not engaged and will not engage in bribery of officials related to potential or active City of Edinburg projects.
2. Respondents shall not offer gratuities, favors or any monetary value to any official or employee of the City for purpose of influencing the selection. Any attempt by any Respondent to influence the selection process by any means, other than disclosure of qualifications and credentials through the proper channels, shall be grounds from exclusion from the selection process. Once the project is advertised, there shall be no contact with any city official or employee unless using the formal process through the Purchasing Department. Failure to comply will result in the firm being disqualified from the process.
3. They have corporate policies that clearly prohibit the use of any bribery in a corporate activity.
4. They have neither been convicted of (nor found by a civil judgment to have committed) bribery of domestic officials, fraud, embezzlement, theft, forgery, destruction of records, making false statements to government officials, receiving stolen property, or any other offense indicating a lack of business integrity or business honesty, within five years of the date of this certification.

---

*Printed name*

---

*Signature*

\_\_\_\_\_  
*Position in bidding company*

\_\_\_\_\_  
*Date*

END OF SECTION



Document 00450

**POST-BID PROCEDURES****1.0 DOCUMENT INCLUDES**

- A. Notice of Intent to Award.
- B. Agreement.
- C. Requirements of Bidder.
- D. Failure of Bidder to comply with requirements.
- E. Notice to Proceed.
- F. Pre-construction Conference.
- G. Starting the Project.

**2.0 NOTICE OF INTENT TO AWARD**

- A. Owner will provide written Notice of Intent to Award (the Contract) to the selected bid or proposal, stating that upon compliance with the conditions listed herein within 14 days after receipt of the notice, and on approval by Owner, Owner will execute and deliver the Agreement.

**3.0 FORM OF AGREEMENT**

- A. The Agreement shall be Document 00500 - Agreement between the Owner and Contractor, together with Supplements enumerated in and attached thereto.

**4.0 REQUIREMENTS OF BIDDER**

- A. Within 14 days of receipt of the Notice of Intent to Award, the selected bidder or proposal shall execute and deliver to the Engineer for the Owner's approval those documents indicated by an "X" below:

- ☒ Document 00500 - Agreement Between the Owner and Contractor
- ☒ Document 00610 - Performance Bond (100% of the Contract Amount)
- ☒ Document 00620 - Payment Bond (100% of the Contract Amount)
- ☒ Document 00625 - Affidavit of Insurance (*with Certificate of Insurance attached*)

**5.0 FAILURE OF BIDDER TO COMPLY WITH REQUIREMENTS**

- A. Should the Bidder on receipt of the Notice of Intent to Award fail to comply with requirements of this Document 00450 within the stated time, the Owner may declare the award in default and require forfeiture of the Security Deposit.
- B. After Owner's written notice of default to the Bidder or proposal, Owner may award the Contract to the responsible Bidder whose offer is the next advantageous bid or proposal, and the Security Deposit of

the Bidder in default shall be forfeited to the Owner in accordance with the provisions of Document 00100 - Instructions to Bidders.

**6.0 NOTICE TO PROCEED**

- A. Upon Owner's execution of the Agreement and delivery to Contractor, the Engineer shall give the Contractor Notice to Proceed within 30 days after the Effective Date of the Agreement, which notice shall establish the Date of Commencement of the Work.

**7.0 PRE-CONSTRUCTION CONFERENCE**

- A. Not later than 10 days after the date of Notice to Proceed, but before Contractor starts work at the site, Owner will convene a Pre-construction Conference as specified in Section 01312 - Coordination and Meetings.

**8.0 STARTING THE PROJECT**

- A. Contractor shall start performance of the Work at the site on the Date of the Commencement of the Work, but no Work shall be done at the site prior to that date.
- B. As Contractor, verify that you and all Subcontractors pay the Prevailing Wage.

**END OF DOCUMENT**

## SECTION 00460

**NONCOLLUSION AFFIDAVIT****PART 1 - GENERAL****1.01 SECTION INCLUDES**

This section describes the standardized forms for use in Bidder and Contractor representations and certifications for the project.

**1.02 REFERENCES – Not Used****1.03 DEFINITIONS – Section 0700****1.04 REPRESENTATIONS AND CERTIFICATIONS**

- A. Affidavit of Non-collusion
- B. Historically Underutilized Business (HUB) Certification (Bidder to insert appropriate certification notice at the end of this Section).

**PART 2 - PRODUCT – Not Used****PART 3 - EXECUTION – Not Used**

STANDARIZED FORMS FOLLOW

## NONCOLLUSION AFFIDAVIT OF PRIME BIDDER

STATE OF TEXAS  
COUNTY OF HIDALGO

\_\_\_\_\_, being first duly sworn, deposes and says that:

- (1) He is President of \_\_\_\_\_, the Bidder that has submitted the attached Bid;  
(Name)  
(Company)
- (2) He is fully informed respecting the preparation and contents of the attached Bid and of all pertinent circumstances respecting such Bid;
- (3) Such Bid is genuine and is not a collusive or sham Bid.
- (4) Neither said Bidder nor any of its officers, partners, owners, agents, representatives, employees or parties in interest, including this affiant, has in any way colluded, conspired, connived or agreed directly or indirectly with another Bidder, firm or person to submit a collusive or sham Bid in connection with the Contract for which the attached Bid has been submitted or to refrain from bidding in connection with such contract, or has in any manner, directly or indirectly sought by agreement or collusion or communication or conference with any other Bidder, firm or person to fix the price or prices in the attached Bid or of any other Bidder, or to fix an overhead, profit or cost element of the Bid price or the Bid price of any other Bidder, or to secure through any collusion, conspiracy, connivance or unlawful agreement any advantage against the **CITY OF EDINBURG**, or any person interested in the proposed Contract; and
- (5) The price or prices quoted in the attached Bid are fair and proper and are not tainted by any collusion, conspiracy, connivance or unlawful agreement on the part of the Bidder or any of its agents, representatives, owners, employees, or parties in interest, including affiant.

Sign \_\_\_\_\_

Title \_\_\_\_\_

Subscribed and sworn to me this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

By:

Notary Public  
My commission expires

**END OF SECTION**

Document 00500

## **AGREEMENT BETWEEN OWNER AND CONTRACTOR FOR CONSTRUCTION CONTRACT**

1. A101\_2017 Standard form of Agreement owner and contractor
2. A101 ExhibitA\_2017 Insurances and Bonds
3. A201\_2017 General Conditions of the contract for Construction



# AIA<sup>®</sup> Document A101<sup>™</sup> – 2017

## Standard Form of Agreement Between Owner and Contractor where the basis of payment is a Stipulated Sum

AGREEMENT made as of the \_\_\_\_\_ day of \_\_\_\_\_ in the year \_\_\_\_\_  
(In words, indicate day, month and year.)

BETWEEN the Owner:  
(Name, legal status, address and other information)

and the Contractor:  
(Name, legal status, address and other information)

for the following Project:  
(Name, location and detailed description)

The Architect:  
(Name, legal status, address and other information)

This document has important legal consequences. Consultation with an attorney is encouraged with respect to its completion or modification.

The parties should complete A101<sup>™</sup>–2017, Exhibit A, Insurance and Bonds, contemporaneously with this Agreement.

AIA Document A201<sup>™</sup>–2017, General Conditions of the Contract for Construction, is adopted in this document by reference. Do not use with other general conditions unless this document is modified.

The Owner and Contractor agree as follows.

## TABLE OF ARTICLES

- 1 THE CONTRACT DOCUMENTS
- 2 THE WORK OF THIS CONTRACT
- 3 DATE OF COMMENCEMENT AND SUBSTANTIAL COMPLETION
- 4 CONTRACT SUM
- 5 PAYMENTS
- 6 DISPUTE RESOLUTION
- 7 TERMINATION OR SUSPENSION
- 8 MISCELLANEOUS PROVISIONS
- 9 ENUMERATION OF CONTRACT DOCUMENTS

## EXHIBIT A INSURANCE AND BONDS

### ARTICLE 1 THE CONTRACT DOCUMENTS

The Contract Documents consist of this Agreement, Conditions of the Contract (General, Supplementary, and other Conditions), Drawings, Specifications, Addenda issued prior to execution of this Agreement, other documents listed in this Agreement, and Modifications issued after execution of this Agreement, all of which form the Contract, and are as fully a part of the Contract as if attached to this Agreement or repeated herein. The Contract represents the entire and integrated agreement between the parties hereto and supersedes prior negotiations, representations, or agreements, either written or oral. An enumeration of the Contract Documents, other than a Modification, appears in Article 9.

### ARTICLE 2 THE WORK OF THIS CONTRACT

The Contractor shall fully execute the Work described in the Contract Documents, except as specifically indicated in the Contract Documents to be the responsibility of others.

### ARTICLE 3 DATE OF COMMENCEMENT AND SUBSTANTIAL COMPLETION

§ 3.1 The date of commencement of the Work shall be:

*(Check one of the following boxes.)*

- ☐ The date of this Agreement.
- ☐ A date set forth in a notice to proceed issued by the Owner.
- ☐ Established as follows:  
*(Insert a date or a means to determine the date of commencement of the Work.)*

If a date of commencement of the Work is not selected, then the date of commencement shall be the date of this Agreement.

§ 3.2 The Contract Time shall be measured from the date of commencement of the Work.

#### § 3.3 Substantial Completion

§ 3.3.1 Subject to adjustments of the Contract Time as provided in the Contract Documents, the Contractor shall achieve Substantial Completion of the entire Work:

*(Check one of the following boxes and complete the necessary information.)*

- ☐ Not later than ( ) calendar days from the date of commencement of the Work.

☐ By the following date:

§ 3.3.2 Subject to adjustments of the Contract Time as provided in the Contract Documents, if portions of the Work are to be completed prior to Substantial Completion of the entire Work, the Contractor shall achieve Substantial Completion of such portions by the following dates:

Portion of Work

Substantial Completion Date

§ 3.3.3 If the Contractor fails to achieve Substantial Completion as provided in this Section 3.3, liquidated damages, if any, shall be assessed as set forth in Section 4.5.

#### ARTICLE 4 CONTRACT SUM

§ 4.1 The Owner shall pay the Contractor the Contract Sum in current funds for the Contractor's performance of the Contract. The Contract Sum shall be ( \$ ), subject to additions and deductions as provided in the Contract Documents.

#### § 4.2 Alternates

§ 4.2.1 Alternates, if any, included in the Contract Sum:

Item

Price

§ 4.2.2 Subject to the conditions noted below, the following alternates may be accepted by the Owner following execution of this Agreement. Upon acceptance, the Owner shall issue a Modification to this Agreement.  
(Insert below each alternate and the conditions that must be met for the Owner to accept the alternate.)

Item

Price

Conditions for Acceptance

§ 4.3 Allowances, if any, included in the Contract Sum:  
(Identify each allowance.)

Item

Price

#### § 4.4 Unit prices, if any:

(Identify the item and state the unit price and quantity limitations, if any, to which the unit price will be applicable.)

Item

Units and Limitations

Price per Unit (\$0.00)

#### § 4.5 Liquidated damages, if any:

(Insert terms and conditions for liquidated damages, if any.)

#### § 4.6 Other:

(Insert provisions for bonus or other incentives, if any, that might result in a change to the Contract Sum.)



## ARTICLE 5 PAYMENTS

### § 5.1 Progress Payments

§ 5.1.1 Based upon Applications for Payment submitted to the Architect by the Contractor and Certificates for Payment issued by the Architect, the Owner shall make progress payments on account of the Contract Sum to the Contractor as provided below and elsewhere in the Contract Documents.

§ 5.1.2 The period covered by each Application for Payment shall be one calendar month ending on the last day of the month, or as follows:

§ 5.1.3 Provided that an Application for Payment is received by the Architect not later than the       day of a       month, the Owner shall make payment of the amount certified to the Contractor not later than the       day of the       month. If an Application for Payment is received by the Architect after the application date fixed above, payment of the amount certified shall be made by the Owner not later than       (       ) days after the Architect receives the Application for Payment.

*(Federal, state or local laws may require payment within a certain period of time.)*

§ 5.1.4 Each Application for Payment shall be based on the most recent schedule of values submitted by the Contractor in accordance with the Contract Documents. The schedule of values shall allocate the entire Contract Sum among the various portions of the Work. The schedule of values shall be prepared in such form, and supported by such data to substantiate its accuracy, as the Architect may require. This schedule of values shall be used as a basis for reviewing the Contractor's Applications for Payment.

§ 5.1.5 Applications for Payment shall show the percentage of completion of each portion of the Work as of the end of the period covered by the Application for Payment.

§ 5.1.6 In accordance with AIA Document A201™–2017, General Conditions of the Contract for Construction, and subject to other provisions of the Contract Documents, the amount of each progress payment shall be computed as follows:

§ 5.1.6.1 The amount of each progress payment shall first include:

- .1 That portion of the Contract Sum properly allocable to completed Work;
- .2 That portion of the Contract Sum properly allocable to materials and equipment delivered and suitably stored at the site for subsequent incorporation in the completed construction, or, if approved in advance by the Owner, suitably stored off the site at a location agreed upon in writing; and
- .3 That portion of Construction Change Directives that the Architect determines, in the Architect's professional judgment, to be reasonably justified.

§ 5.1.6.2 The amount of each progress payment shall then be reduced by:

- .1 The aggregate of any amounts previously paid by the Owner;
- .2 The amount, if any, for Work that remains uncorrected and for which the Architect has previously withheld a Certificate for Payment as provided in Article 9 of AIA Document A201–2017;
- .3 Any amount for which the Contractor does not intend to pay a Subcontractor or material supplier, unless the Work has been performed by others the Contractor intends to pay;
- .4 For Work performed or defects discovered since the last payment application, any amount for which the Architect may withhold payment, or nullify a Certificate of Payment in whole or in part, as provided in Article 9 of AIA Document A201–2017; and
- .5 Retainage withheld pursuant to Section 5.1.7.

### § 5.1.7 Retainage

§ 5.1.7.1 For each progress payment made prior to Substantial Completion of the Work, the Owner may withhold the following amount, as retainage, from the payment otherwise due:

*(Insert a percentage or amount to be withheld as retainage from each Application for Payment. The amount of retainage may be limited by governing law.)*

§ 5.1.7.1.1 The following items are not subject to retainage:  
(Insert any items not subject to the withholding of retainage, such as general conditions, insurance, etc.)

§ 5.1.7.2 Reduction or limitation of retainage, if any, shall be as follows:  
(If the retainage established in Section 5.1.7.1 is to be modified prior to Substantial Completion of the entire Work, including modifications for Substantial Completion of portions of the Work as provided in Section 3.3.2, insert provisions for such modifications.)

§ 5.1.7.3 Except as set forth in this Section 5.1.7.3, upon Substantial Completion of the Work, the Contractor may submit an Application for Payment that includes the retainage withheld from prior Applications for Payment pursuant to this Section 5.1.7. The Application for Payment submitted at Substantial Completion shall not include retainage as follows:  
(Insert any other conditions for release of retainage upon Substantial Completion.)

§ 5.1.8 If final completion of the Work is materially delayed through no fault of the Contractor, the Owner shall pay the Contractor any additional amounts in accordance with Article 9 of AIA Document A201–2017.

§ 5.1.9 Except with the Owner's prior approval, the Contractor shall not make advance payments to suppliers for materials or equipment which have not been delivered and stored at the site.

## § 5.2 Final Payment

§ 5.2.1 Final payment, constituting the entire unpaid balance of the Contract Sum, shall be made by the Owner to the Contractor when

- .1 the Contractor has fully performed the Contract except for the Contractor's responsibility to correct Work as provided in Article 12 of AIA Document A201–2017, and to satisfy other requirements, if any, which extend beyond final payment; and
- .2 a final Certificate for Payment has been issued by the Architect.

§ 5.2.2 The Owner's final payment to the Contractor shall be made no later than 30 days after the issuance of the Architect's final Certificate for Payment, or as follows:

## § 5.3 Interest

Payments due and unpaid under the Contract shall bear interest from the date payment is due at the rate stated below, or in the absence thereof, at the legal rate prevailing from time to time at the place where the Project is located.

(Insert rate of interest agreed upon, if any.)

\_\_\_\_\_ %

## ARTICLE 6 DISPUTE RESOLUTION

### § 6.1 Initial Decision Maker

The Architect will serve as the Initial Decision Maker pursuant to Article 15 of AIA Document A201–2017, unless the parties appoint below another individual, not a party to this Agreement, to serve as the Initial Decision Maker.

(If the parties mutually agree, insert the name, address and other contact information of the Initial Decision Maker, if other than the Architect.)

## § 6.2 Binding Dispute Resolution

For any Claim subject to, but not resolved by, mediation pursuant to Article 15 of AIA Document A201–2017, the method of binding dispute resolution shall be as follows:

*(Check the appropriate box.)*

- ☐ Arbitration pursuant to Section 15.4 of AIA Document A201–2017
- ☐ Litigation in a court of competent jurisdiction
- ☐ Other *(Specify)*

If the Owner and Contractor do not select a method of binding dispute resolution, or do not subsequently agree in writing to a binding dispute resolution method other than litigation, Claims will be resolved by litigation in a court of competent jurisdiction.

## ARTICLE 7 TERMINATION OR SUSPENSION

§ 7.1 The Contract may be terminated by the Owner or the Contractor as provided in Article 14 of AIA Document A201–2017.

§ 7.1.1 If the Contract is terminated for the Owner’s convenience in accordance with Article 14 of AIA Document A201–2017, then the Owner shall pay the Contractor a termination fee as follows:

*(Insert the amount of, or method for determining, the fee, if any, payable to the Contractor following a termination for the Owner’s convenience.)*

§ 7.2 The Work may be suspended by the Owner as provided in Article 14 of AIA Document A201–2017.

## ARTICLE 8 MISCELLANEOUS PROVISIONS

§ 8.1 Where reference is made in this Agreement to a provision of AIA Document A201–2017 or another Contract Document, the reference refers to that provision as amended or supplemented by other provisions of the Contract Documents.

§ 8.2 The Owner’s representative:

*(Name, address, email address, and other information)*

§ 8.3 The Contractor’s representative:

*(Name, address, email address, and other information)*

§ 8.4 Neither the Owner’s nor the Contractor’s representative shall be changed without ten days’ prior notice to the other party.

## § 8.5 Insurance and Bonds

§ 8.5.1 The Owner and the Contractor shall purchase and maintain insurance as set forth in AIA Document A101™–2017, Standard Form of Agreement Between Owner and Contractor where the basis of payment is a Stipulated Sum, Exhibit A, Insurance and Bonds, and elsewhere in the Contract Documents.

§ 8.5.2 The Contractor shall provide bonds as set forth in AIA Document A101™–2017 Exhibit A, and elsewhere in the Contract Documents.

§ 8.6 Notice in electronic format, pursuant to Article 1 of AIA Document A201–2017, may be given in accordance with AIA Document E203™–2013, Building Information Modeling and Digital Data Exhibit, if completed, or as otherwise set forth below:

*(If other than in accordance with AIA Document E203–2013, insert requirements for delivering notice in electronic format such as name, title, and email address of the recipient and whether and how the system will be required to generate a read receipt for the transmission.)*

§ 8.7 Other provisions:

## ARTICLE 9 ENUMERATION OF CONTRACT DOCUMENTS

§ 9.1 This Agreement is comprised of the following documents:

- .1 AIA Document A101™–2017, Standard Form of Agreement Between Owner and Contractor
- .2 AIA Document A101™–2017, Exhibit A, Insurance and Bonds
- .3 AIA Document A201™–2017, General Conditions of the Contract for Construction
- .4 AIA Document E203™–2013, Building Information Modeling and Digital Data Exhibit, dated as indicated below:

*(Insert the date of the E203-2013 incorporated into this Agreement.)*

- .5 Drawings

| Number | Title | Date |
|--------|-------|------|
|--------|-------|------|

- .6 Specifications

| Section | Title | Date | Pages |
|---------|-------|------|-------|
|---------|-------|------|-------|

- .7 Addenda, if any:

| Number | Date | Pages |
|--------|------|-------|
|--------|------|-------|

Portions of Addenda relating to bidding or proposal requirements are not part of the Contract Documents unless the bidding or proposal requirements are also enumerated in this Article 9.

- .8 Other Exhibits:

*(Check all boxes that apply and include appropriate information identifying the exhibit where required.)*

☐ AIA Document E204™–2017, Sustainable Projects Exhibit, dated as indicated below:  
*(Insert the date of the E204-2017 incorporated into this Agreement.)*

☐ The Sustainability Plan:

| Title | Date | Pages |
|-------|------|-------|
|-------|------|-------|

☐ Supplementary and other Conditions of the Contract:

| Document | Title | Date | Pages |
|----------|-------|------|-------|
|----------|-------|------|-------|

.9 Other documents, if any, listed below:

*(List here any additional documents that are intended to form part of the Contract Documents. AIA Document A201™–2017 provides that the advertisement or invitation to bid, Instructions to Bidders, sample forms, the Contractor's bid or proposal, portions of Addenda relating to bidding or proposal requirements, and other information furnished by the Owner in anticipation of receiving bids or proposals, are not part of the Contract Documents unless enumerated in this Agreement. Any such documents should be listed here only if intended to be part of the Contract Documents.)*

This Agreement entered into as of the day and year first written above.

\_\_\_\_\_  
OWNER (Signature)

\_\_\_\_\_  
CONTRACTOR (Signature)

\_\_\_\_\_  
(Printed name and title)

\_\_\_\_\_  
(Printed name and title)



# AIA<sup>®</sup> Document A101<sup>™</sup> – 2017 Exhibit A

## Insurance and Bonds

This Insurance and Bonds Exhibit is part of the Agreement, between the Owner and the Contractor, dated the \_\_\_\_\_ day of \_\_\_\_\_ in the year \_\_\_\_\_  
(In words, indicate day, month and year.)

for the following PROJECT:  
(Name and location or address)

THE OWNER:  
(Name, legal status and address)

THE CONTRACTOR:  
(Name, legal status and address)

This document has important legal consequences. Consultation with an attorney is encouraged with respect to its completion or modification.

This document is intended to be used in conjunction with AIA Document A201<sup>™</sup>–2017, General Conditions of the Contract for Construction. Article 11 of A201<sup>™</sup>–2017 contains additional insurance provisions.

### TABLE OF ARTICLES

- A.1 GENERAL
- A.2 OWNER'S INSURANCE
- A.3 CONTRACTOR'S INSURANCE AND BONDS
- A.4 SPECIAL TERMS AND CONDITIONS

#### ARTICLE A.1 GENERAL

The Owner and Contractor shall purchase and maintain insurance, and provide bonds, as set forth in this Exhibit. As used in this Exhibit, the term General Conditions refers to AIA Document A201<sup>™</sup>–2017, General Conditions of the Contract for Construction.

#### ARTICLE A.2 OWNER'S INSURANCE

##### § A.2.1 General

Prior to commencement of the Work, the Owner shall secure the insurance, and provide evidence of the coverage, required under this Article A.2 and, upon the Contractor's request, provide a copy of the property insurance policy or policies required by Section A.2.3. The copy of the policy or policies provided shall contain all applicable conditions, definitions, exclusions, and endorsements.

##### § A.2.2 Liability Insurance

The Owner shall be responsible for purchasing and maintaining the Owner's usual general liability insurance.

##### § A.2.3 Required Property Insurance

§ A.2.3.1 Unless this obligation is placed on the Contractor pursuant to Section A.3.3.2.1, the Owner shall purchase and maintain, from an insurance company or insurance companies lawfully authorized to issue insurance in the jurisdiction where the Project is located, property insurance written on a builder's risk "all-risks" completed value or equivalent policy form and sufficient to cover the total value of the entire Project on a replacement cost basis. The Owner's

property insurance coverage shall be no less than the amount of the initial Contract Sum, plus the value of subsequent Modifications and labor performed and materials or equipment supplied by others. The property insurance shall be maintained until Substantial Completion and thereafter as provided in Section A.2.3.1.3, unless otherwise provided in the Contract Documents or otherwise agreed in writing by the parties to this Agreement. This insurance shall include the interests of the Owner, Contractor, Subcontractors, and Sub-subcontractors in the Project as insureds. This insurance shall include the interests of mortgagees as loss payees.

**§ A.2.3.1.1 Causes of Loss.** The insurance required by this Section A.2.3.1 shall provide coverage for direct physical loss or damage, and shall not exclude the risks of fire, explosion, theft, vandalism, malicious mischief, collapse, earthquake, flood, or windstorm. The insurance shall also provide coverage for ensuing loss or resulting damage from error, omission, or deficiency in construction methods, design, specifications, workmanship, or materials. Sub-limits, if any, are as follows:

*(Indicate below the cause of loss and any applicable sub-limit.)*

| Cause of Loss | Sub-Limit |
|---------------|-----------|
|---------------|-----------|

**§ A.2.3.1.2 Specific Required Coverages.** The insurance required by this Section A.2.3.1 shall provide coverage for loss or damage to falsework and other temporary structures, and to building systems from testing and startup. The insurance shall also cover debris removal, including demolition occasioned by enforcement of any applicable legal requirements, and reasonable compensation for the Architect's and Contractor's services and expenses required as a result of such insured loss, including claim preparation expenses. Sub-limits, if any, are as follows:

*(Indicate below type of coverage and any applicable sub-limit for specific required coverages.)*

| Coverage | Sub-Limit |
|----------|-----------|
|----------|-----------|

**§ A.2.3.1.3** Unless the parties agree otherwise, upon Substantial Completion, the Owner shall continue the insurance required by Section A.2.3.1 or, if necessary, replace the insurance policy required under Section A.2.3.1 with property insurance written for the total value of the Project that shall remain in effect until expiration of the period for correction of the Work set forth in Section 12.2.2 of the General Conditions.

**§ A.2.3.1.4 Deductibles and Self-Insured Retentions.** If the insurance required by this Section A.2.3 is subject to deductibles or self-insured retentions, the Owner shall be responsible for all loss not covered because of such deductibles or retentions.

**§ A.2.3.2 Occupancy or Use Prior to Substantial Completion.** The Owner's occupancy or use of any completed or partially completed portion of the Work prior to Substantial Completion shall not commence until the insurance company or companies providing the insurance under Section A.2.3.1 have consented in writing to the continuance of coverage. The Owner and the Contractor shall take no action with respect to partial occupancy or use that would cause cancellation, lapse, or reduction of insurance, unless they agree otherwise in writing.

**§ A.2.3.3 Insurance for Existing Structures**

If the Work involves remodeling an existing structure or constructing an addition to an existing structure, the Owner shall purchase and maintain, until the expiration of the period for correction of Work as set forth in Section 12.2.2 of the General Conditions, "all-risks" property insurance, on a replacement cost basis, protecting the existing structure against direct physical loss or damage from the causes of loss identified in Section A.2.3.1, notwithstanding the undertaking of the Work. The Owner shall be responsible for all co-insurance penalties.

**§ A.2.4 Optional Extended Property Insurance.**

The Owner shall purchase and maintain the insurance selected and described below.

*(Select the types of insurance the Owner is required to purchase and maintain by placing an X in the box(es) next to the description(s) of selected insurance. For each type of insurance selected, indicate applicable limits of coverage or other conditions in the fill point below the selected item.)*



- ☐ § A.2.4.1 Loss of Use, Business Interruption, and Delay in Completion Insurance, to reimburse the Owner for loss of use of the Owner's property, or the inability to conduct normal operations due to a covered cause of loss.
- ☐ § A.2.4.2 Ordinance or Law Insurance, for the reasonable and necessary costs to satisfy the minimum requirements of the enforcement of any law or ordinance regulating the demolition, construction, repair, replacement or use of the Project.
- ☐ § A.2.4.3 Expediting Cost Insurance, for the reasonable and necessary costs for the temporary repair of damage to insured property, and to expedite the permanent repair or replacement of the damaged property.
- ☐ § A.2.4.4 Extra Expense Insurance, to provide reimbursement of the reasonable and necessary excess costs incurred during the period of restoration or repair of the damaged property that are over and above the total costs that would normally have been incurred during the same period of time had no loss or damage occurred.
- ☐ § A.2.4.5 Civil Authority Insurance, for losses or costs arising from an order of a civil authority prohibiting access to the Project, provided such order is the direct result of physical damage covered under the required property insurance.
- ☐ § A.2.4.6 Ingress/Egress Insurance, for loss due to the necessary interruption of the insured's business due to physical prevention of ingress to, or egress from, the Project as a direct result of physical damage.
- ☐ § A.2.4.7 Soft Costs Insurance, to reimburse the Owner for costs due to the delay of completion of the Work, arising out of physical loss or damage covered by the required property insurance: including construction loan fees; leasing and marketing expenses; additional fees, including those of architects, engineers, consultants, attorneys and accountants, needed for the completion of the construction, repairs, or reconstruction; and carrying costs such as property taxes, building permits, additional interest on loans, realty taxes, and insurance premiums over and above normal expenses.

#### § A.2.5 Other Optional Insurance.

The Owner shall purchase and maintain the insurance selected below.

*(Select the types of insurance the Owner is required to purchase and maintain by placing an X in the box(es) next to the description(s) of selected insurance.)*

- ☐ § A.2.5.1 Cyber Security Insurance for loss to the Owner due to data security and privacy breach, including costs of investigating a potential or actual breach of confidential or private information.  
*(Indicate applicable limits of coverage or other conditions in the fill point below.)*



- ☐ § A.2.5.2 Other Insurance  
(List below any other insurance coverage to be provided by the Owner and any applicable limits.)

Coverage

Limits

## ARTICLE A.3 CONTRACTOR'S INSURANCE AND BONDS

### § A.3.1 General

§ A.3.1.1 **Certificates of Insurance.** The Contractor shall provide certificates of insurance acceptable to the Owner evidencing compliance with the requirements in this Article A.3 at the following times: (1) prior to commencement of the Work; (2) upon renewal or replacement of each required policy of insurance; and (3) upon the Owner's written request. An additional certificate evidencing continuation of commercial liability coverage, including coverage for completed operations, shall be submitted with the final Application for Payment and thereafter upon renewal or replacement of such coverage until the expiration of the periods required by Section A.3.2.1 and Section A.3.3.1. The certificates will show the Owner as an additional insured on the Contractor's Commercial General Liability and excess or umbrella liability policy or policies.

§ A.3.1.2 **Deductibles and Self-Insured Retentions.** The Contractor shall disclose to the Owner any deductible or self-insured retentions applicable to any insurance required to be provided by the Contractor.

§ A.3.1.3 **Additional Insured Obligations.** To the fullest extent permitted by law, the Contractor shall cause the commercial general liability coverage to include (1) the Owner, the Architect, and the Architect's consultants as additional insureds for claims caused in whole or in part by the Contractor's negligent acts or omissions during the Contractor's operations; and (2) the Owner as an additional insured for claims caused in whole or in part by the Contractor's negligent acts or omissions for which loss occurs during completed operations. The additional insured coverage shall be primary and non-contributory to any of the Owner's general liability insurance policies and shall apply to both ongoing and completed operations. To the extent commercially available, the additional insured coverage shall be no less than that provided by Insurance Services Office, Inc. (ISO) forms CG 20 10 07 04, CG 20 37 07 04, and, with respect to the Architect and the Architect's consultants, CG 20 32 07 04.

### § A.3.2 Contractor's Required Insurance Coverage

§ A.3.2.1 The Contractor shall purchase and maintain the following types and limits of insurance from an insurance company or insurance companies lawfully authorized to issue insurance in the jurisdiction where the Project is located. The Contractor shall maintain the required insurance until the expiration of the period for correction of Work as set forth in Section 12.2.2 of the General Conditions, unless a different duration is stated below:  
(If the Contractor is required to maintain insurance for a duration other than the expiration of the period for correction of Work, state the duration.)

### § A.3.2.2 Commercial General Liability

§ A.3.2.2.1 Commercial General Liability insurance for the Project written on an occurrence form with policy limits of not less than \_\_\_\_\_ ( \$ ) each occurrence, \_\_\_\_\_ ( \$ ) general aggregate, and \_\_\_\_\_ ( \$ ) aggregate for products-completed operations hazard, providing coverage for claims including

- .1 damages because of bodily injury, sickness or disease, including occupational sickness or disease, and death of any person;
- .2 personal injury and advertising injury;
- .3 damages because of physical damage to, or destruction of, tangible property, including the loss of use of such property;
- .4 bodily injury or property damage arising out of completed operations; and
- .5 the Contractor's indemnity obligations under Section 3.18 of the General Conditions.

§ A.3.2.2.2 The Contractor's Commercial General Liability policy under this Section A.3.2.2 shall not contain an exclusion or restriction of coverage for the following:

- .1 Claims by one insured against another insured, if the exclusion or restriction is based solely on the fact that the claimant is an insured, and there would otherwise be coverage for the claim.
- .2 Claims for property damage to the Contractor's Work arising out of the products-completed operations hazard where the damaged Work or the Work out of which the damage arises was performed by a Subcontractor.
- .3 Claims for bodily injury other than to employees of the insured.
- .4 Claims for indemnity under Section 3.18 of the General Conditions arising out of injury to employees of the insured
- .5 Claims or loss excluded under a prior work endorsement or other similar exclusionary language.
- .6 Claims or loss due to physical damage under a prior injury endorsement or similar exclusionary language.
- .7 Claims related to residential, multi-family, or other habitational projects, if the Work is to be performed on such a project.
- .8 Claims related to roofing, if the Work involves roofing.
- .9 Claims related to exterior insulation finish systems (EIFS), synthetic stucco or similar exterior coatings or surfaces, if the Work involves such coatings or surfaces.
- .10 Claims related to earth subsidence or movement, where the work involves such hazards.
- .11 Claims related to explosion, collapse, and underground hazards, where the Work involves such hazards.

§ A.3.2.3 Automobile Liability covering vehicles owned, and non-owned vehicles used, by the Contractor, with policy limits of not less than \_\_\_\_\_ ( \$\_\_ ) per accident, for bodily injury, death of any person, and property damage arising out of the ownership, maintenance and use of those motor vehicles along with any other statutorily required automobile coverage.

§ A.3.2.4 The Contractor may achieve the required limits and coverage for Commercial General Liability and Automobile Liability through a combination of primary and excess or umbrella liability insurance, provided such primary and excess or umbrella insurance policies result in the same or greater coverage as the coverages required under Section A.3.2.2 and A.3.2.3, and in no event shall any excess or umbrella liability insurance provide narrower coverage than the primary policy. The excess policy shall not require the exhaustion of the underlying limits only through the actual payment by the underlying insurers.

§ A.3.2.5 Workers' Compensation at statutory limits.

§ A.3.2.6 Employers' Liability with policy limits not less than \_\_\_\_\_ ( \$\_\_ ) each accident, \_\_\_\_\_ ( \$\_\_ ) each employee, and \_\_\_\_\_ ( \$\_\_ ) policy limit.

§ A.3.2.7 Jones Act, and the Longshore & Harbor Workers' Compensation Act, as required, if the Work involves hazards arising from work on or near navigable waterways, including vessels and docks

§ A.3.2.8 If the Contractor is required to furnish professional services as part of the Work, the Contractor shall procure Professional Liability insurance covering performance of the professional services, with policy limits of not less than \_\_\_\_\_ ( \$\_\_ ) per claim and \_\_\_\_\_ ( \$\_\_ ) in the aggregate.

§ A.3.2.9 If the Work involves the transport, dissemination, use, or release of pollutants, the Contractor shall procure Pollution Liability insurance, with policy limits of not less than \_\_\_\_\_ ( \$\_\_ ) per claim and \_\_\_\_\_ ( \$\_\_ ) in the aggregate.

§ A.3.2.10 Coverage under Sections A.3.2.8 and A.3.2.9 may be procured through a Combined Professional Liability and Pollution Liability insurance policy, with combined policy limits of not less than \_\_\_\_\_ ( \$\_\_ ) per claim and \_\_\_\_\_ ( \$\_\_ ) in the aggregate.

§ A.3.2.11 Insurance for maritime liability risks associated with the operation of a vessel, if the Work requires such activities, with policy limits of not less than \_\_\_\_\_ ( \$\_\_ ) per claim and \_\_\_\_\_ ( \$\_\_ ) in the aggregate.

§ A.3.2.12 Insurance for the use or operation of manned or unmanned aircraft, if the Work requires such activities, with policy limits of not less than \_\_\_\_\_ ( \$\_\_ ) per claim and \_\_\_\_\_ ( \$\_\_ ) in the aggregate.

### § A.3.3 Contractor's Other Insurance Coverage

§ A.3.3.1 Insurance selected and described in this Section A.3.3 shall be purchased from an insurance company or insurance companies lawfully authorized to issue insurance in the jurisdiction where the Project is located. The Contractor shall maintain the required insurance until the expiration of the period for correction of Work as set forth in Section 12.2.2 of the General Conditions, unless a different duration is stated below:

*(If the Contractor is required to maintain any of the types of insurance selected below for a duration other than the expiration of the period for correction of Work, state the duration.)*

§ A.3.3.2 The Contractor shall purchase and maintain the following types and limits of insurance in accordance with Section A.3.3.1.

*(Select the types of insurance the Contractor is required to purchase and maintain by placing an X in the box(es) next to the description(s) of selected insurance. Where policy limits are provided, include the policy limit in the appropriate fill point.)*

- ☐ § A.3.3.2.1 Property insurance of the same type and scope satisfying the requirements identified in Section A.2.3, which, if selected in this section A.3.3.2.1, relieves the Owner of the responsibility to purchase and maintain such insurance except insurance required by Section A.2.3.1.3 and Section A.2.3.3. The Contractor shall comply with all obligations of the Owner under Section A.2.3 except to the extent provided below. The Contractor shall disclose to the Owner the amount of any deductible, and the Owner shall be responsible for losses within the deductible. Upon request, the Contractor shall provide the Owner with a copy of the property insurance policy or policies required. The Owner shall adjust and settle the loss with the insurer and be the trustee of the proceeds of the property insurance in accordance with Article 11 of the General Conditions unless otherwise set forth below.

*(Where the Contractor's obligation to provide property insurance differs from the Owner's obligations as described under Section A.2.3, indicate such differences in the space below. Additionally, if a party other than the Owner will be responsible for adjusting and settling a loss with the insurer and acting as the trustee of the proceeds of property insurance in accordance with Article 11 of the General Conditions, indicate the responsible party below.)*

- ☐ § A.3.3.2.2 Railroad Protective Liability Insurance, with policy limits of not less than \_\_\_\_\_ ( \$\_\_ ) per claim and \_\_\_\_\_ ( \$\_\_ ) in the aggregate, for Work within fifty (50) feet of railroad property.
- ☐ § A.3.3.2.3 Asbestos Abatement Liability Insurance, with policy limits of not less than \_\_\_\_\_ ( \$\_\_ ) per claim and \_\_\_\_\_ ( \$\_\_ ) in the aggregate, for liability arising from the encapsulation, removal, handling, storage, transportation, and disposal of asbestos-containing materials.
- ☐ § A.3.3.2.4 Insurance for physical damage to property while it is in storage and in transit to the construction site on an "all-risks" completed value form.
- ☐ § A.3.3.2.5 Property insurance on an "all-risks" completed value form, covering property owned by the Contractor and used on the Project, including scaffolding and other equipment.
- ☐ § A.3.3.2.6 Other Insurance  
*(List below any other insurance coverage to be provided by the Contractor and any applicable limits.)*

Coverage

Limits

#### § A.3.4 Performance Bond and Payment Bond

The Contractor shall provide surety bonds, from a company or companies lawfully authorized to issue surety bonds in the jurisdiction where the Project is located, as follows:

*(Specify type and penal sum of bonds.)*

| Type             | Penal Sum (\$0.00) |
|------------------|--------------------|
| Payment Bond     |                    |
| Performance Bond |                    |

Payment and Performance Bonds shall be AIA Document A312™, Payment Bond and Performance Bond, or contain provisions identical to AIA Document A312™, current as of the date of this Agreement.

#### ARTICLE A.4 SPECIAL TERMS AND CONDITIONS

Special terms and conditions that modify this Insurance and Bonds Exhibit, if any, are as follows:



# AIA<sup>®</sup> Document A201<sup>™</sup> – 2017

## General Conditions of the Contract for Construction

for the following PROJECT:  
(Name and location or address)

THE OWNER:  
(Name, legal status and address)

THE ARCHITECT:  
(Name, legal status and address)

This document has important legal consequences. Consultation with an attorney is encouraged with respect to its completion or modification.

For guidance in modifying this document to include supplementary conditions, see AIA Document A503<sup>™</sup>, Guide for Supplementary Conditions.

### TABLE OF ARTICLES

- |    |                                                  |
|----|--------------------------------------------------|
| 1  | GENERAL PROVISIONS                               |
| 2  | OWNER                                            |
| 3  | CONTRACTOR                                       |
| 4  | ARCHITECT                                        |
| 5  | SUBCONTRACTORS                                   |
| 6  | CONSTRUCTION BY OWNER OR BY SEPARATE CONTRACTORS |
| 7  | CHANGES IN THE WORK                              |
| 8  | TIME                                             |
| 9  | PAYMENTS AND COMPLETION                          |
| 10 | PROTECTION OF PERSONS AND PROPERTY               |
| 11 | INSURANCE AND BONDS                              |
| 12 | UNCOVERING AND CORRECTION OF WORK                |
| 13 | MISCELLANEOUS PROVISIONS                         |
| 14 | TERMINATION OR SUSPENSION OF THE CONTRACT        |
| 15 | CLAIMS AND DISPUTES                              |

## INDEX

(Topics and numbers in bold are Section headings.)

### **Acceptance of Nonconforming Work**

9.6.6, 9.9.3, **12.3**

Acceptance of Work

9.6.6, 9.8.2, 9.9.3, 9.10.1, 9.10.3, **12.3**

### **Access to Work**

**3.16**, 6.2.1, **12.1**

Accident Prevention

10

Acts and Omissions

3.2, 3.3.2, 3.12.8, 3.18, 4.2.3, 8.3.1, 9.5.1, 10.2.5,  
10.2.8, 13.3.2, 14.1, 15.1.2, 15.2

Addenda

1.1.1

Additional Costs, Claims for

3.7.4, 3.7.5, 10.3.2, 15.1.5

### **Additional Inspections and Testing**

9.4.2, 9.8.3, 12.2.1, **13.4**

### **Additional Time, Claims for**

3.2.4, 3.7.4, 3.7.5, 3.10.2, 8.3.2, **15.1.6**

### **Administration of the Contract**

3.1.3, **4.2**, 9.4, 9.5

Advertisement or Invitation to Bid

1.1.1

Aesthetic Effect

4.2.13

### **Allowances**

**3.8**

### **Applications for Payment**

4.2.5, 7.3.9, 9.2, **9.3**, 9.4, 9.5.1, 9.5.4, 9.6.3, 9.7, 9.10

Approvals

2.1.1, 2.3.1, 2.5, 3.1.3, 3.10.2, 3.12.8, 3.12.9, 3.12.10.1,  
4.2.7, 9.3.2, 13.4.1

### **Arbitration**

8.3.1, 15.3.2, **15.4**

## **ARCHITECT**

**4**

**Architect**, Definition of

### **4.1.1**

Architect, Extent of Authority

2.5, 3.12.7, 4.1.2, 4.2, 5.2, 6.3, 7.1.2, 7.3.4, 7.4, 9.2,  
9.3.1, 9.4, 9.5, 9.6.3, 9.8, 9.10.1, 9.10.3, 12.1, 12.2.1,  
13.4.1, 13.4.2, 14.2.2, 14.2.4, 15.1.4, 15.2.1

Architect, Limitations of Authority and Responsibility

2.1.1, 3.12.4, 3.12.8, 3.12.10, 4.1.2, 4.2.1, 4.2.2, 4.2.3,  
4.2.6, 4.2.7, 4.2.10, 4.2.12, 4.2.13, 5.2.1, 7.4, 9.4.2,  
9.5.4, 9.6.4, 15.1.4, 15.2

Architect's Additional Services and Expenses

2.5, 12.2.1, 13.4.2, 13.4.3, 14.2.4

Architect's Administration of the Contract

3.1.3, 3.7.4, 15.2, 9.4.1, 9.5

Architect's Approvals

2.5, 3.1.3, 3.5, 3.10.2, 4.2.7

Architect's Authority to Reject Work

3.5, 4.2.6, 12.1.2, 12.2.1

Architect's Copyright

1.1.7, 1.5

Architect's Decisions

3.7.4, 4.2.6, 4.2.7, 4.2.11, 4.2.12, 4.2.13, 4.2.14, 6.3,  
7.3.4, 7.3.9, 8.1.3, 8.3.1, 9.2, 9.4.1, 9.5, 9.8.4, 9.9.1,  
13.4.2, 15.2

Architect's Inspections

3.7.4, 4.2.2, 4.2.9, 9.4.2, 9.8.3, 9.9.2, 9.10.1, 13.4

Architect's Instructions

3.2.4, 3.3.1, 4.2.6, 4.2.7, 13.4.2

Architect's Interpretations

4.2.11, 4.2.12

Architect's Project Representative

4.2.10

Architect's Relationship with Contractor

1.1.2, 1.5, 2.3.3, 3.1.3, 3.2.2, 3.2.3, 3.2.4, 3.3.1, 3.4.2,  
3.5, 3.7.4, 3.7.5, 3.9.2, 3.9.3, 3.10, 3.11, 3.12, 3.16,  
3.18, 4.1.2, 4.2, 5.2, 6.2.2, 7, 8.3.1, 9.2, 9.3, 9.4, 9.5,  
9.7, 9.8, 9.9, 10.2.6, 10.3, 11.3, 12, 13.3.2, 13.4, 15.2

Architect's Relationship with Subcontractors

1.1.2, 4.2.3, 4.2.4, 4.2.6, 9.6.3, 9.6.4, 11.3

Architect's Representations

9.4.2, 9.5.1, 9.10.1

Architect's Site Visits

3.7.4, 4.2.2, 4.2.9, 9.4.2, 9.5.1, 9.9.2, 9.10.1, 13.4

Asbestos

10.3.1

Attorneys' Fees

3.18.1, 9.6.8, 9.10.2, 10.3.3

Award of Separate Contracts

6.1.1, 6.1.2

### **Award of Subcontracts and Other Contracts for Portions of the Work**

**5.2**

## **Basic Definitions**

**1.1**

Bidding Requirements

1.1.1

Binding Dispute Resolution

8.3.1, 9.7, 11.5, 13.1, 15.1.2, 15.1.3, 15.2.1, 15.2.5,  
15.2.6.1, 15.3.1, 15.3.2, 15.3.3, 15.4.1

Bonds, Lien

7.3.4.4, 9.6.8, 9.10.2, 9.10.3

### **Bonds, Performance, and Payment**

7.3.4.4, 9.6.7, 9.10.3, **11.1.2**, 11.1.3, **11.5**

### **Building Information Models Use and Reliance**

**1.8**

Building Permit

3.7.1

### **Capitalization**

**1.3**

Certificate of Substantial Completion

9.8.3, 9.8.4, 9.8.5

### **Certificates for Payment**

4.2.1, 4.2.5, 4.2.9, 9.3.3, **9.4**, 9.5, 9.6.1, 9.6.6, 9.7,  
9.10.1, 9.10.3, 14.1.1.3, 14.2.4, 15.1.4

Certificates of Inspection, Testing or Approval

13.4.4



Certificates of Insurance  
9.10.2

#### **Change Orders**

1.1.1, 3.4.2, 3.7.4, 3.8.2.3, 3.11, 3.12.8, 4.2.8, 5.2.3,  
7.1.2, 7.1.3, 7.2, 7.3.2, 7.3.7, 7.3.9, 7.3.10, 8.3.1,  
9.3.1.1, 9.10.3, 10.3.2, 11.2, 11.5, 12.1.2

**Change Orders**, Definition of  
7.2.1

#### **CHANGES IN THE WORK**

2.2.2, 3.11, 4.2.8, 7, 7.2.1, 7.3.1, 7.4, 8.3.1, 9.3.1.1,  
11.5

**Claims**, Definition of

#### **15.1.1**

Claims, Notice of  
1.6.2, 15.1.3

#### **CLAIMS AND DISPUTES**

3.2.4, 6.1.1, 6.3, 7.3.9, 9.3.3, 9.10.4, 10.3.3, 15, 15.4  
Claims and Timely Assertion of Claims  
15.4.1

#### **Claims for Additional Cost**

3.2.4, 3.3.1, 3.7.4, 7.3.9, 9.5.2, 10.2.5, 10.3.2, 15.1.5

#### **Claims for Additional Time**

3.2.4, 3.3.1, 3.7.4, 6.1.1, 8.3.2, 9.5.2, 10.3.2, 15.1.6

#### **Concealed or Unknown Conditions, Claims for 3.7.4**

Claims for Damages

3.2.4, 3.18, 8.3.3, 9.5.1, 9.6.7, 10.2.5, 10.3.3, 11.3,  
11.3.2, 14.2.4, 15.1.7

Claims Subject to Arbitration  
15.4.1

#### **Cleaning Up**

3.15, 6.3

Commencement of the Work, Conditions Relating to  
2.2.1, 3.2.2, 3.4.1, 3.7.1, 3.10.1, 3.12.6, 5.2.1, 5.2.3,  
6.2.2, 8.1.2, 8.2.2, 8.3.1, 11.1, 11.2, 15.1.5

**Commencement of the Work**, Definition of  
8.1.2

#### **Communications**

3.9.1, 4.2.4

Completion, Conditions Relating to  
3.4.1, 3.11, 3.15, 4.2.2, 4.2.9, 8.2, 9.4.2, 9.8, 9.9.1,  
9.10, 12.2, 14.1.2, 15.1.2

#### **COMPLETION, PAYMENTS AND 9**

Completion, Substantial

3.10.1, 4.2.9, 8.1.1, 8.1.3, 8.2.3, 9.4.2, 9.8, 9.9.1,  
9.10.3, 12.2, 15.1.2

Compliance with Laws

2.3.2, 3.2.3, 3.6, 3.7, 3.12.10, 3.13, 9.6.4, 10.2.2, 13.1,  
13.3, 13.4.1, 13.4.2, 13.5, 14.1.1, 14.2.1.3, 15.2.8,  
15.4.2, 15.4.3

Concealed or Unknown Conditions

3.7.4, 4.2.8, 8.3.1, 10.3

Conditions of the Contract

1.1.1, 6.1.1, 6.1.4

Consent, Written

3.4.2, 3.14.2, 4.1.2, 9.8.5, 9.9.1, 9.10.2, 9.10.3, 13.2,  
15.4.4.2

#### **Consolidation or Joinder**

#### **15.4.4**

#### **CONSTRUCTION BY OWNER OR BY SEPARATE CONTRACTORS**

1.1.4, 6

**Construction Change Directive**, Definition of  
7.3.1

#### **Construction Change Directives**

1.1.1, 3.4.2, 3.11, 3.12.8, 4.2.8, 7.1.1, 7.1.2, 7.1.3, 7.3,  
9.3.1.1

Construction Schedules, Contractor's

3.10, 3.11, 3.12.1, 3.12.2, 6.1.3, 15.1.6.2

#### **Contingent Assignment of Subcontracts**

5.4, 14.2.2.2

#### **Continuing Contract Performance**

#### **15.1.4**

**Contract**, Definition of  
1.1.2

#### **CONTRACT, TERMINATION OR SUSPENSION OF THE**

5.4.1.1, 5.4.2, 11.5, 14

Contract Administration

3.1.3, 4, 9.4, 9.5

Contract Award and Execution, Conditions Relating to  
3.7.1, 3.10, 5.2, 6.1

Contract Documents, Copies Furnished and Use of  
1.5.2, 2.3.6, 5.3

**Contract Documents**, Definition of  
1.1.1

#### **Contract Sum**

2.2.2, 2.2.4, 3.7.4, 3.7.5, 3.8, 3.10.2, 5.2.3, 7.3, 7.4, 9.1,  
9.2, 9.4.2, 9.5.1.4, 9.6.7, 9.7, 10.3.2, 11.5, 12.1.2, 12.3,  
14.2.4, 14.3.2, 15.1.4.2, 15.1.5, 15.2.5

**Contract Sum**, Definition of  
9.1

Contract Time

1.1.4, 2.2.1, 2.2.2, 3.7.4, 3.7.5, 3.10.2, 5.2.3, 6.1.5,  
7.2.1.3, 7.3.1, 7.3.5, 7.3.6, 7, 7, 7.3.10, 7.4, 8.1.1, 8.2.1,  
8.2.3, 8.3.1, 9.5.1, 9.7, 10.3.2, 12.1.1, 12.1.2, 14.3.2,  
15.1.4.2, 15.1.6.1, 15.2.5

**Contract Time**, Definition of  
8.1.1

#### **CONTRACTOR**

#### **3**

Contractor, Definition of

3.1, 6.1.2

#### **Contractor's Construction and Submittal Schedules**

3.10, 3.12.1, 3.12.2, 4.2.3, 6.1.3, 15.1.6.2

Contractor's Employees

2.2.4, 3.3.2, 3.4.3, 3.8.1, 3.9, 3.18.2, 4.2.3, 4.2.6, 10.2,  
10.3, 11.3, 14.1, 14.2.1.1

#### **Contractor's Liability Insurance 11.1**

Contractor's Relationship with Separate Contractors  
and Owner's Forces

3.12.5, 3.14.2, 4.2.4, 6, 11.3, 12.2.4

Contractor's Relationship with Subcontractors  
1.2.2, 2.2.4, 3.3.2, 3.18.1, 3.18.2, 4.2.4, 5, 9.6.2, 9.6.7, 9.10.2, 11.2, 11.3, 11.4

Contractor's Relationship with the Architect  
1.1.2, 1.5, 2.3.3, 3.1.3, 3.2.2, 3.2.3, 3.2.4, 3.3.1, 3.4.2, 3.5.1, 3.7.4, 3.10, 3.11, 3.12, 3.16, 3.18, 4.2, 5.2, 6.2.2, 7, 8.3.1, 9.2, 9.3, 9.4, 9.5, 9.7, 9.8, 9.9, 10.2.6, 10.3, 11.3, 12, 13.4, 15.1.3, 15.2.1

Contractor's Representations  
3.2.1, 3.2.2, 3.5, 3.12.6, 6.2.2, 8.2.1, 9.3.3, 9.8.2

Contractor's Responsibility for Those Performing the Work  
3.3.2, 3.18, 5.3, 6.1.3, 6.2, 9.5.1, 10.2.8

Contractor's Review of Contract Documents  
3.2

Contractor's Right to Stop the Work  
2.2.2, 9.7

Contractor's Right to Terminate the Contract  
14.1

Contractor's Submittals  
3.10, 3.11, 3.12, 4.2.7, 5.2.1, 5.2.3, 9.2, 9.3, 9.8.2, 9.8.3, 9.9.1, 9.10.2, 9.10.3

Contractor's Superintendent  
3.9, 10.2.6

Contractor's Supervision and Construction Procedures  
1.2.2, 3.3, 3.4, 3.12.10, 4.2.2, 4.2.7, 6.1.3, 6.2.4, 7.1.3, 7.3.4, 7.3.6, 8.2, 10, 12, 14, 15.1.4

Coordination and Correlation  
1.2, 3.2.1, 3.3.1, 3.10, 3.12.6, 6.1.3, 6.2.1

Copies Furnished of Drawings and Specifications  
1.5, 2.3.6, 3.11

Copyrights  
1.5, **3.17**

Correction of Work  
2.5, 3.7.3, 9.4.2, 9.8.2, 9.8.3, 9.9.1, 12.1.2, **12.2**, 12.3, 15.1.3.1, 15.1.3.2, 15.2.1

**Correlation and Intent of the Contract Documents**  
**1.2**

**Cost**, Definition of  
**7.3.4**

Costs  
2.5, 3.2.4, 3.7.3, 3.8.2, 3.15.2, 5.4.2, 6.1.1, 6.2.3, 7.3.3.3, 7.3.4, 7.3.8, 7.3.9, 9.10.2, 10.3.2, 10.3.6, 11.2, 12.1.2, 12.2.1, 12.2.4, 13.4, 14

**Cutting and Patching**  
**3.14**, 6.2.5

Damage to Construction of Owner or Separate Contractors  
3.14.2, 6.2.4, 10.2.1.2, 10.2.5, 10.4, 12.2.4

Damage to the Work  
3.14.2, 9.9.1, 10.2.1.2, 10.2.5, 10.4, 12.2.4

Damages, Claims for  
3.2.4, 3.18, 6.1.1, 8.3.3, 9.5.1, 9.6.7, 10.3.3, 11.3.2, 11.3, 14.2.4, 15.1.7

Damages for Delay  
6.2.3, 8.3.3, 9.5.1.6, 9.7, 10.3.2, 14.3.2

**Date of Commencement of the Work**, Definition of  
**8.1.2**

**Date of Substantial Completion**, Definition of  
**8.1.3**

**Day**, Definition of  
**8.1.4**

Decisions of the Architect  
3.7.4, 4.2.6, 4.2.7, 4.2.11, 4.2.12, 4.2.13, 6.3, 7.3.4, 7.3.9, 8.1.3, 8.3.1, 9.2, 9.4, 9.5.1, 9.8.4, 9.9.1, 13.4.2, 14.2.2, 14.2.4, 15.1, 15.2

**Decisions to Withhold Certification**  
9.4.1, **9.5**, 9.7, 14.1.1.3

Defective or Nonconforming Work, Acceptance, Rejection and Correction of  
2.5, 3.5, 4.2.6, 6.2.3, 9.5.1, 9.5.3, 9.6.6, 9.8.2, 9.9.3, 9.10.4, 12.2.1

Definitions  
1.1, 2.1.1, 3.1.1, 3.5, 3.12.1, 3.12.2, 3.12.3, 4.1.1, 5.1, 6.1.2, 7.2.1, 7.3.1, 8.1, 9.1, 9.8.1, 15.1.1

**Delays and Extensions of Time**  
**3.2**, **3.7.4**, 5.2.3, 7.2.1, 7.3.1, **7.4**, **8.3**, 9.5.1, **9.7**, 10.3.2, **10.4**, 14.3.2, **15.1.6**, 15.2.5

**Digital Data Use and Transmission**  
**1.7**

Disputes  
6.3, 7.3.9, 15.1, 15.2

**Documents and Samples at the Site**  
**3.11**

**Drawings**, Definition of  
**1.1.5**

Drawings and Specifications, Use and Ownership of  
3.11

Effective Date of Insurance  
8.2.2

**Emergencies**  
**10.4**, 14.1.1.2, **15.1.5**

Employees, Contractor's  
3.3.2, 3.4.3, 3.8.1, 3.9, 3.18.2, 4.2.3, 4.2.6, 10.2, 10.3.3, 11.3, 14.1, 14.2.1.1

Equipment, Labor, or Materials  
1.1.3, 1.1.6, 3.4, 3.5, 3.8.2, 3.8.3, 3.12, 3.13, 3.15.1, 4.2.6, 4.2.7, 5.2.1, 6.2.1, 7.3.4, 9.3.2, 9.3.3, 9.5.1.3, 9.10.2, 10.2.1, 10.2.4, 14.2.1.1, 14.2.1.2

Execution and Progress of the Work  
1.1.3, 1.2.1, 1.2.2, 2.3.4, 2.3.6, 3.1, 3.3.1, 3.4.1, 3.7.1, 3.10.1, 3.12, 3.14, 4.2, 6.2.2, 7.1.3, 7.3.6, 8.2, 9.5.1, 9.9.1, 10.2, 10.3, 12.1, 12.2, 14.2, 14.3.1, 15.1.4

Extensions of Time  
3.2.4, 3.7.4, 5.2.3, 7.2.1, 7.3, 7.4, 9.5.1, 9.7, 10.3.2, 10.4, 14.3, 15.1.6, **15.2.5**

**Failure of Payment**  
9.5.1.3, **9.7**, 9.10.2, 13.5, 14.1.1.3, 14.2.1.2

Faulty Work  
(See Defective or Nonconforming Work)

**Final Completion and Final Payment**  
4.2.1, 4.2.9, 9.8.2, **9.10**, 12.3, 14.2.4, 14.4.3

Financial Arrangements, Owner's  
2.2.1, 13.2.2, 14.1.1.4

**GENERAL PROVISIONS**  
**1**



## **Governing Law**

### **13.1**

Guarantees (See Warranty)

## **Hazardous Materials and Substances**

### **10.2.4, 10.3**

Identification of Subcontractors and Suppliers

### **5.2.1**

## **Indemnification**

3.17, **3.18**, 9.6.8, 9.10.2, 10.3.3, 11.3

## **Information and Services Required of the Owner**

2.1.2, **2.2**, 2.3, 3.2.2, 3.12.10.1, 6.1.3, 6.1.4, 6.2.5,

9.6.1, 9.9.2, 9.10.3, 10.3.3, 11.2, 13.4.1, 13.4.2,

14.1.1.4, 14.1.4, 15.1.4

## **Initial Decision**

### **15.2**

## **Initial Decision Maker, Definition of**

### **1.1.8**

Initial Decision Maker, Decisions

14.2.4, 15.1.4.2, 15.2.1, 15.2.2, 15.2.3, 15.2.4, 15.2.5

Initial Decision Maker, Extent of Authority

14.2.4, 15.1.4.2, 15.2.1, 15.2.2, 15.2.3, 15.2.4, 15.2.5

## **Injury or Damage to Person or Property**

### **10.2.8, 10.4**

Inspections

3.1.3, 3.3.3, 3.7.1, 4.2.2, 4.2.6, 4.2.9, 9.4.2, 9.8.3, 9.9.2, 9.10.1, 12.2.1, 13.4

Instructions to Bidders

### **1.1.1**

Instructions to the Contractor

3.2.4, 3.3.1, 3.8.1, 5.2.1, 7, 8.2.2, 12, 13.4.2

## **Instruments of Service, Definition of**

### **1.1.7**

Insurance

6.1.1, 7.3.4, 8.2.2, 9.3.2, 9.8.4, 9.9.1, 9.10.2, 10.2.5, **11**

Insurance, Notice of Cancellation or Expiration

11.1.4, 11.2.3

## **Insurance, Contractor's Liability**

### **11.1**

Insurance, Effective Date of  
8.2.2, 14.4.2

## **Insurance, Owner's Liability**

### **11.2**

## **Insurance, Property**

**10.2.5**, 11.2, 11.4, 11.5

Insurance, Stored Materials

### **9.3.2**

## **INSURANCE AND BONDS**

### **11**

Insurance Companies, Consent to Partial Occupancy

9.9.1

Insured loss, Adjustment and Settlement of

11.5

Intent of the Contract Documents

1.2.1, 4.2.7, 4.2.12, 4.2.13

## **Interest**

### **13.5**

## **Interpretation**

1.1.8, 1.2.3, **1.4**, 4.1.1, 5.1, 6.1.2, 15.1.1

Interpretations, Written

4.2.11, 4.2.12

Judgment on Final Award

15.4.2

## **Labor and Materials, Equipment**

1.1.3, 1.1.6, **3.4**, 3.5, 3.8.2, 3.8.3, 3.12, 3.13, 3.15.1,

5.2.1, 6.2.1, 7.3.4, 9.3.2, 9.3.3, 9.5.1.3, 9.10.2, 10.2.1,

10.2.4, 14.2.1.1, 14.2.1.2

Labor Disputes

### **8.3.1**

Laws and Regulations

1.5, 2.3.2, 3.2.3, 3.2.4, 3.6, 3.7, 3.12.10, 3.13, 9.6.4, 9.9.1, 10.2.2, 13.1, 13.3.1, 13.4.2, 13.5, 14, 15.2.8, 15.4

Liens

2.1.2, 9.3.1, 9.3.3, 9.6.8, 9.10.2, 9.10.4, 15.2.8

Limitations, Statutes of

12.2.5, 15.1.2, 15.4.1.1

Limitations of Liability

3.2.2, 3.5, 3.12.10, 3.12.10.1, 3.17, 3.18.1, 4.2.6, 4.2.7,

6.2.2, 9.4.2, 9.6.4, 9.6.7, 9.6.8, 10.2.5, 10.3.3, 11.3,

12.2.5, 13.3.1

Limitations of Time

2.1.2, 2.2, 2.5, 3.2.2, 3.10, 3.11, 3.12.5, 3.15.1, 4.2.7,

5.2, 5.3, 5.4.1, 6.2.4, 7.3, 7.4, 8.2, 9.2, 9.3.1, 9.3.3,

9.4.1, 9.5, 9.6, 9.7, 9.8, 9.9, 9.10, 12.2, 13.4, 14, 15,

15.1.2, 15.1.3, 15.1.5

## **Materials, Hazardous**

### **10.2.4, 10.3**

Materials, Labor, Equipment and

1.1.3, 1.1.6, 3.4.1, 3.5, 3.8.2, 3.8.3, 3.12, 3.13, 3.15.1,

5.2.1, 6.2.1, 7.3.4, 9.3.2, 9.3.3, 9.5.1.3, 9.10.2, 10.2.1.2,

10.2.4, 14.2.1.1, 14.2.1.2

Means, Methods, Techniques, Sequences and

Procedures of Construction

3.3.1, 3.12.10, 4.2.2, 4.2.7, 9.4.2

Mechanic's Lien

2.1.2, 9.3.1, 9.3.3, 9.6.8, 9.10.2, 9.10.4, 15.2.8

## **Mediation**

8.3.1, 15.1.3.2, 15.2.1, 15.2.5, 15.2.6, **15.3**, 15.4.1, 15.4.1.1

## **Minor Changes in the Work**

1.1.1, 3.4.2, 3.12.8, 4.2.8, 7.1, **7.4**

## **MISCELLANEOUS PROVISIONS**

### **13**

## **Modifications, Definition of**

### **1.1.1**

Modifications to the Contract

1.1.1, 1.1.2, 2.5, 3.11, 4.1.2, 4.2.1, 5.2.3, 7, 8.3.1, 9.7, 10.3.2

## **Mutual Responsibility**

### **6.2**

## **Nonconforming Work, Acceptance of**

9.6.6, 9.9.3, **12.3**

Nonconforming Work, Rejection and Correction of

2.4, 2.5, 3.5, 4.2.6, 6.2.4, 9.5.1, 9.8.2, 9.9.3, 9.10.4,

12.2

## **Notice**

1.6, 1.6.1, 1.6.2, 2.1.2, 2.2.2., 2.2.3, 2.2.4, 2.5, 3.2.4, 3.3.1, 3.7.4, 3.7.5, 3.9.2, 3.12.9, 3.12.10, 5.2.1, 7.4, 8.2.2, 9.6.8, 9.7, 9.10.1, 10.2.8, 10.3.2, 11.5, 12.2.2.1, 13.4.1, 13.4.2, 14.1, 14.2.2, 14.4.2, 15.1.3, 15.1.5, 15.1.6, 15.4.1

Notice of Cancellation or Expiration of Insurance  
11.1.4, 11.2.3

## **Notice of Claims**

1.6.2, 2.1.2, 3.7.4, 9.6.8, 10.2.8, 15.1.3, 15.1.5, 15.1.6, 15.2.8, 15.3.2, 15.4.1

Notice of Testing and Inspections  
13.4.1, 13.4.2

Observations, Contractor's  
3.2, 3.7.4

Occupancy  
2.3.1, 9.6.6, 9.8

Orders, Written  
1.1.1, 2.4, 3.9.2, 7, 8.2.2, 11.5, 12.1, 12.2.2.1, 13.4.2, 14.3.1

## **OWNER**

### **2**

**Owner**, Definition of  
2.1.1

**Owner**, Evidence of Financial Arrangements  
2.2, 13.2.2, 14.1.1.4

**Owner**, Information and Services Required of the  
2.1.2, 2.2, 2.3, 3.2.2, 3.12.10, 6.1.3, 6.1.4, 6.2.5, 9.3.2, 9.6.1, 9.6.4, 9.9.2, 9.10.3, 10.3.3, 11.2, 13.4.1, 13.4.2, 14.1.1.4, 14.1.4, 15.1.4

**Owner's Authority**  
1.5, 2.1.1, 2.3.32.4, 2.5, 3.4.2, 3.8.1, 3.12.10, 3.14.2, 4.1.2, 4.2.4, 4.2.9, 5.2.1, 5.2.4, 5.4.1, 6.1, 6.3, 7.2.1, 7.3.1, 8.2.2, 8.3.1, 9.3.2, 9.5.1, 9.6.4, 9.9.1, 9.10.2, 10.3.2, 11.4, 11.5, 12.2.2, 12.3, 13.2.2, 14.3, 14.4, 15.2.7

**Owner's Insurance**  
11.2

**Owner's Relationship with Subcontractors**  
1.1.2, 5.2, 5.3, 5.4, 9.6.4, 9.10.2, 14.2.2

**Owner's Right to Carry Out the Work**  
2.5, 14.2.2

**Owner's Right to Clean Up**  
6.3

**Owner's Right to Perform Construction and to Award Separate Contracts**

### **6.1**

**Owner's Right to Stop the Work**  
2.4

**Owner's Right to Suspend the Work**  
14.3

**Owner's Right to Terminate the Contract**  
14.2, 14.4

**Ownership and Use of Drawings, Specifications and Other Instruments of Service**

1.1.1, 1.1.6, 1.1.7, 1.5, 2.3.6, 3.2.2, 3.11, 3.17, 4.2.12, 5.3

## **Partial Occupancy or Use**

9.6.6, 9.9

**Patching, Cutting and**  
3.14, 6.2.5

Patents  
3.17

## **Payment, Applications for**

4.2.5, 7.3.9, 9.2, 9.3, 9.4, 9.5, 9.6.3, 9.7, 9.8.5, 9.10.1, 14.2.3, 14.2.4, 14.4.3

## **Payment, Certificates for**

4.2.5, 4.2.9, 9.3.3, 9.4, 9.5, 9.6.1, 9.6.6, 9.7, 9.10.1, 9.10.3, 14.1.1.3, 14.2.4

## **Payment, Failure of**

9.5.1.3, 9.7, 9.10.2, 13.5, 14.1.1.3, 14.2.1.2  
Payment, Final

4.2.1, 4.2.9, 9.10, 12.3, 14.2.4, 14.4.3

**Payment Bond, Performance Bond and**  
7.3.4.4, 9.6.7, 9.10.3, 11.1.2

## **Payments, Progress**

9.3, 9.6, 9.8.5, 9.10.3, 14.2.3, 15.1.4

## **PAYMENTS AND COMPLETION**

### **9**

Payments to Subcontractors  
5.4.2, 9.5.1.3, 9.6.2, 9.6.3, 9.6.4, 9.6.7, 14.2.1.2

PCB

10.3.1

## **Performance Bond and Payment Bond**

7.3.4.4, 9.6.7, 9.10.3, 11.1.2

**Permits, Fees, Notices and Compliance with Laws**  
2.3.1, 3.7, 3.13, 7.3.4.4, 10.2.2

## **PERSONS AND PROPERTY, PROTECTION OF**

### **10**

Polychlorinated Biphenyl  
10.3.1

**Product Data**, Definition of  
3.12.2

**Product Data and Samples, Shop Drawings**  
3.11, 3.12, 4.2.7

## **Progress and Completion**

4.2.2, 8.2, 9.8, 9.9.1, 14.1.4, 15.1.4

## **Progress Payments**

9.3, 9.6, 9.8.5, 9.10.3, 14.2.3, 15.1.4

**Project**, Definition of  
1.1.4

Project Representatives  
4.2.10

## **Property Insurance**

10.2.5, 11.2

Proposal Requirements

1.1.1

## **PROTECTION OF PERSONS AND PROPERTY**

### **10**

Regulations and Laws

1.5, 2.3.2, 3.2.3, 3.6, 3.7, 3.12.10, 3.13, 9.6.4, 9.9.1, 10.2.2, 13.1, 13.3, 13.4.1, 13.4.2, 13.5, 14, 15.2.8, 15.4

Rejection of Work

4.2.6, 12.2.1

Releases and Waivers of Liens  
9.3.1, 9.10.2

Representations  
3.2.1, 3.5, 3.12.6, 8.2.1, 9.3.3, 9.4.2, 9.5.1, 9.10.1

Representatives  
2.1.1, 3.1.1, 3.9, 4.1.1, 4.2.10, 13.2.1

Responsibility for Those Performing the Work  
3.3.2, 3.18, 4.2.2, 4.2.3, 5.3, 6.1.3, 6.2, 6.3, 9.5.1, 10

Retainage  
9.3.1, 9.6.2, 9.8.5, 9.9.1, 9.10.2, 9.10.3

**Review of Contract Documents and Field Conditions by Contractor**  
3.2, 3.12.7, 6.1.3

Review of Contractor's Submittals by Owner and Architect  
3.10.1, 3.10.2, 3.11, 3.12, 4.2, 5.2, 6.1.3, 9.2, 9.8.2

Review of Shop Drawings, Product Data and Samples by Contractor  
3.12

**Rights and Remedies**  
1.1.2, 2.4, 2.5, 3.5, 3.7.4, 3.15.2, 4.2.6, 5.3, 5.4, 6.1, 6.3, 7.3.1, 8.3, 9.5.1, 9.7, 10.2.5, 10.3, 12.2.1, 12.2.2, 12.2.4, **13.3**, 14, 15.4

**Royalties, Patents and Copyrights**  
3.17

Rules and Notices for Arbitration  
15.4.1

**Safety of Persons and Property**  
10.2, 10.4

**Safety Precautions and Programs**  
3.3.1, 4.2.2, 4.2.7, 5.3, **10.1**, 10.2, 10.4

**Samples, Definition of**  
3.12.3

**Samples, Shop Drawings, Product Data and**  
3.11, 3.12, 4.2.7

**Samples at the Site, Documents and**  
3.11

**Schedule of Values**  
9.2, 9.3.1

Schedules, Construction  
3.10, 3.12.1, 3.12.2, 6.1.3, 15.1.6.2

Separate Contracts and Contractors  
1.1.4, 3.12.5, 3.14.2, 4.2.4, 4.2.7, 6, 8.3.1, 12.1.2

**Separate Contractors, Definition of**  
6.1.1

**Shop Drawings, Definition of**  
3.12.1

**Shop Drawings, Product Data and Samples**  
3.11, 3.12, 4.2.7

**Site, Use of**  
3.13, 6.1.1, 6.2.1

Site Inspections  
3.2.2, 3.3.3, 3.7.1, 3.7.4, 4.2, 9.9.2, 9.4.2, 9.10.1, 13.4

Site Visits, Architect's  
3.7.4, 4.2.2, 4.2.9, 9.4.2, 9.5.1, 9.9.2, 9.10.1, 13.4

Special Inspections and Testing  
4.2.6, 12.2.1, 13.4

**Specifications, Definition of**  
1.1.6

**Specifications**  
1.1.1, **1.1.6**, 1.2.2, 1.5, 3.12.10, 3.17, 4.2.14

Statute of Limitations  
15.1.2, 15.4.1.1

Stopping the Work  
2.2.2, 2.4, 9.7, 10.3, 14.1

Stored Materials  
6.2.1, 9.3.2, 10.2.1.2, 10.2.4

**Subcontractor, Definition of**  
5.1.1

**SUBCONTRACTORS**  
5

Subcontractors, Work by  
1.2.2, 3.3.2, 3.12.1, 3.18, 4.2.3, 5.2.3, 5.3, 5.4, 9.3.1.2, 9.6.7

**Subcontractual Relations**  
5.3, 5.4, 9.3.1.2, 9.6, 9.10, 10.2.1, 14.1, 14.2.1

Submittals  
3.10, 3.11, 3.12, 4.2.7, 5.2.1, 5.2.3, 7.3.4, 9.2, 9.3, 9.8, 9.9.1, 9.10.2, 9.10.3

Submittal Schedule  
3.10.2, 3.12.5, 4.2.7

**Subrogation, Waivers of**  
6.1.1, **11.3**

**Substances, Hazardous**  
10.3

**Substantial Completion**  
4.2.9, 8.1.1, 8.1.3, 8.2.3, 9.4.2, **9.8**, 9.9.1, 9.10.3, 12.2, 15.1.2

**Substantial Completion, Definition of**  
9.8.1

Substitution of Subcontractors  
5.2.3, 5.2.4

Substitution of Architect  
2.3.3

Substitutions of Materials  
3.4.2, 3.5, 7.3.8

**Sub-subcontractor, Definition of**  
5.1.2

Subsurface Conditions  
3.7.4

**Successors and Assigns**  
13.2

**Superintendent**  
3.9, 10.2.6

**Supervision and Construction Procedures**  
1.2.2, **3.3**, 3.4, 3.12.10, 4.2.2, 4.2.7, 6.1.3, 6.2.4, 7.1.3, 7.3.4, 8.2, 8.3.1, 9.4.2, 10, 12, 14, 15.1.4

Suppliers  
1.5, 3.12.1, 4.2.4, 4.2.6, 5.2.1, 9.3, 9.4.2, 9.5.4, 9.6, 9.10.5, 14.2.1

Surety  
5.4.1.2, 9.6.8, 9.8.5, 9.10.2, 9.10.3, 11.1.2, 14.2.2, 15.2.7

Surety, Consent of  
9.8.5, 9.10.2, 9.10.3

Surveys  
1.1.7, 2.3.4  
**Suspension by the Owner for Convenience**  
**14.3**

Suspension of the Work  
3.7.5, 5.4.2, 14.3  
Suspension or Termination of the Contract  
5.4.1.1, 14

**Taxes**

3.6, 3.8.2.1, 7.3.4.4

**Termination by the Contractor**

**14.1**, 15.1.7

**Termination by the Owner for Cause**

5.4.1.1, **14.2**, 15.1.7

**Termination by the Owner for Convenience**

**14.4**

Termination of the Architect

2.3.3

Termination of the Contractor Employment

14.2.2

**TERMINATION OR SUSPENSION OF THE CONTRACT**

**14**

**Tests and Inspections**

3.1.3, 3.3.3, 3.7.1, 4.2.2, 4.2.6, 4.2.9, 9.4.2, 9.8.3, 9.9.2, 9.10.1, 10.3.2, 12.2.1, **13.4**

**TIME**

**8**

**Time, Delays and Extensions of**

3.2.4, 3.7.4, 5.2.3, 7.2.1, 7.3.1, 7.4, **8.3**, 9.5.1, 9.7, 10.3.2, 10.4, 14.3.2, 15.1.6, 15.2.5

**Time Limits**

2.1.2, 2.2, 2.5, 3.2.2, 3.10, 3.11, 3.12.5, 3.15.1, 4.2, 5.2, 5.3, 5.4, 6.2.4, 7.3, 7.4, 8.2, 9.2, 9.3.1, 9.3.3, 9.4.1, 9.5, 9.6, 9.7, 9.8, 9.9, 9.10, 12.2, 13.4, 14, 15.1.2, 15.1.3, 15.4

**Time Limits on Claims**

3.7.4, 10.2.8, 15.1.2, 15.1.3

Title to Work

9.3.2, 9.3.3

**UNCOVERING AND CORRECTION OF WORK**  
**12**

**Uncovering of Work**

**12.1**

Unforeseen Conditions, Concealed or Unknown

3.7.4, 8.3.1, 10.3

Unit Prices

7.3.3.2, 9.1.2

Use of Documents

1.1.1, 1.5, 2.3.6, 3.12.6, 5.3

**Use of Site**

**3.13**, 6.1.1, 6.2.1

**Values, Schedule of**

**9.2**, 9.3.1

Waiver of Claims by the Architect

13.3.2

Waiver of Claims by the Contractor

9.10.5, 13.3.2, **15.1.7**

Waiver of Claims by the Owner

9.9.3, 9.10.3, 9.10.4, 12.2.2.1, 13.3.2, 14.2.4, **15.1.7**

Waiver of Consequential Damages

14.2.4, 15.1.7

Waiver of Liens

9.3, 9.10.2, 9.10.4

**Waivers of Subrogation**

6.1.1, **11.3**

**Warranty**

**3.5**, 4.2.9, 9.3.3, 9.8.4, 9.9.1, 9.10.2, 9.10.4, 12.2.2, 15.1.2

Weather Delays

8.3, 15.1.6.2

**Work, Definition of**

**1.1.3**

Written Consent

1.5.2, 3.4.2, 3.7.4, 3.12.8, 3.14.2, 4.1.2, 9.3.2, 9.10.3, 13.2, 13.3.2, 15.4.4.2

Written Interpretations

4.2.11, 4.2.12

Written Orders

1.1.1, 2.4, 3.9, 7, 8.2.2, 12.1, 12.2, 13.4.2, 14.3.1

## ARTICLE 1 GENERAL PROVISIONS

### § 1.1 Basic Definitions

#### § 1.1.1 The Contract Documents

The Contract Documents are enumerated in the Agreement between the Owner and Contractor (hereinafter the Agreement) and consist of the Agreement, Conditions of the Contract (General, Supplementary and other Conditions), Drawings, Specifications, Addenda issued prior to execution of the Contract, other documents listed in the Agreement, and Modifications issued after execution of the Contract. A Modification is (1) a written amendment to the Contract signed by both parties, (2) a Change Order, (3) a Construction Change Directive, or (4) a written order for a minor change in the Work issued by the Architect. Unless specifically enumerated in the Agreement, the Contract Documents do not include the advertisement or invitation to bid, Instructions to Bidders, sample forms, other information furnished by the Owner in anticipation of receiving bids or proposals, the Contractor's bid or proposal, or portions of Addenda relating to bidding or proposal requirements.

#### § 1.1.2 The Contract

The Contract Documents form the Contract for Construction. The Contract represents the entire and integrated agreement between the parties hereto and supersedes prior negotiations, representations, or agreements, either written or oral. The Contract may be amended or modified only by a Modification. The Contract Documents shall not be construed to create a contractual relationship of any kind (1) between the Contractor and the Architect or the Architect's consultants, (2) between the Owner and a Subcontractor or a Sub-subcontractor, (3) between the Owner and the Architect or the Architect's consultants, or (4) between any persons or entities other than the Owner and the Contractor. The Architect shall, however, be entitled to performance and enforcement of obligations under the Contract intended to facilitate performance of the Architect's duties.

#### § 1.1.3 The Work

The term "Work" means the construction and services required by the Contract Documents, whether completed or partially completed, and includes all other labor, materials, equipment, and services provided or to be provided by the Contractor to fulfill the Contractor's obligations. The Work may constitute the whole or a part of the Project.

#### § 1.1.4 The Project

The Project is the total construction of which the Work performed under the Contract Documents may be the whole or a part and which may include construction by the Owner and by Separate Contractors.

#### § 1.1.5 The Drawings

The Drawings are the graphic and pictorial portions of the Contract Documents showing the design, location and dimensions of the Work, generally including plans, elevations, sections, details, schedules, and diagrams.

#### § 1.1.6 The Specifications

The Specifications are that portion of the Contract Documents consisting of the written requirements for materials, equipment, systems, standards and workmanship for the Work, and performance of related services.

#### § 1.1.7 Instruments of Service

Instruments of Service are representations, in any medium of expression now known or later developed, of the tangible and intangible creative work performed by the Architect and the Architect's consultants under their respective professional services agreements. Instruments of Service may include, without limitation, studies, surveys, models, sketches, drawings, specifications, and other similar materials.

#### § 1.1.8 Initial Decision Maker

The Initial Decision Maker is the person identified in the Agreement to render initial decisions on Claims in accordance with Section 15.2. The Initial Decision Maker shall not show partiality to the Owner or Contractor and shall not be liable for results of interpretations or decisions rendered in good faith.

### § 1.2 Correlation and Intent of the Contract Documents

§ 1.2.1 The intent of the Contract Documents is to include all items necessary for the proper execution and completion of the Work by the Contractor. The Contract Documents are complementary, and what is required by one shall be as binding as if required by all; performance by the Contractor shall be required only to the extent consistent with the Contract Documents and reasonably inferable from them as being necessary to produce the indicated results.

§ 1.2.1.1 The invalidity of any provision of the Contract Documents shall not invalidate the Contract or its remaining

provisions. If it is determined that any provision of the Contract Documents violates any law, or is otherwise invalid or unenforceable, then that provision shall be revised to the extent necessary to make that provision legal and enforceable. In such case the Contract Documents shall be construed, to the fullest extent permitted by law, to give effect to the parties' intentions and purposes in executing the Contract.

§ 1.2.2 Organization of the Specifications into divisions, sections and articles, and arrangement of Drawings shall not control the Contractor in dividing the Work among Subcontractors or in establishing the extent of Work to be performed by any trade.

§ 1.2.3 Unless otherwise stated in the Contract Documents, words that have well-known technical or construction industry meanings are used in the Contract Documents in accordance with such recognized meanings.

### § 1.3 Capitalization

Terms capitalized in these General Conditions include those that are (1) specifically defined, (2) the titles of numbered articles, or (3) the titles of other documents published by the American Institute of Architects.

### § 1.4 Interpretation

In the interest of brevity the Contract Documents frequently omit modifying words such as "all" and "any" and articles such as "the" and "an," but the fact that a modifier or an article is absent from one statement and appears in another is not intended to affect the interpretation of either statement.

### § 1.5 Ownership and Use of Drawings, Specifications, and Other Instruments of Service

§ 1.5.1 The Architect and the Architect's consultants shall be deemed the authors and owners of their respective Instruments of Service, including the Drawings and Specifications, and retain all common law, statutory, and other reserved rights in their Instruments of Service, including copyrights. The Contractor, Subcontractors, Sub-subcontractors, and suppliers shall not own or claim a copyright in the Instruments of Service. Submittal or distribution to meet official regulatory requirements or for other purposes in connection with the Project is not to be construed as publication in derogation of the Architect's or Architect's consultants' reserved rights.

§ 1.5.2 The Contractor, Subcontractors, Sub-subcontractors, and suppliers are authorized to use and reproduce the Instruments of Service provided to them, subject to any protocols established pursuant to Sections 1.7 and 1.8, solely and exclusively for execution of the Work. All copies made under this authorization shall bear the copyright notice, if any, shown on the Instruments of Service. The Contractor, Subcontractors, Sub-subcontractors, and suppliers may not use the Instruments of Service on other projects or for additions to the Project outside the scope of the Work without the specific written consent of the Owner, Architect, and the Architect's consultants.

### § 1.6 Notice

§ 1.6.1 Except as otherwise provided in Section 1.6.2, where the Contract Documents require one party to notify or give notice to the other party, such notice shall be provided in writing to the designated representative of the party to whom the notice is addressed and shall be deemed to have been duly served if delivered in person, by mail, by courier, or by electronic transmission if a method for electronic transmission is set forth in the Agreement.

§ 1.6.2 Notice of Claims as provided in Section 15.1.3 shall be provided in writing and shall be deemed to have been duly served only if delivered to the designated representative of the party to whom the notice is addressed by certified or registered mail, or by courier providing proof of delivery.

### § 1.7 Digital Data Use and Transmission

The parties shall agree upon protocols governing the transmission and use of Instruments of Service or any other information or documentation in digital form. The parties will use AIA Document E203™–2013, Building Information Modeling and Digital Data Exhibit, to establish the protocols for the development, use, transmission, and exchange of digital data.

### § 1.8 Building Information Models Use and Reliance

Any use of, or reliance on, all or a portion of a building information model without agreement to protocols governing the use of, and reliance on, the information contained in the model and without having those protocols set forth in AIA Document E203™–2013, Building Information Modeling and Digital Data Exhibit, and the requisite AIA Document G202™–2013, Project Building Information Modeling Protocol Form, shall be at the using or relying party's sole risk and without liability to the other party and its contractors or consultants, the authors of, or contributors to, the building



information model, and each of their agents and employees.

## ARTICLE 2 OWNER

### § 2.1 General

§ 2.1.1 The Owner is the person or entity identified as such in the Agreement and is referred to throughout the Contract Documents as if singular in number. The Owner shall designate in writing a representative who shall have express authority to bind the Owner with respect to all matters requiring the Owner's approval or authorization. Except as otherwise provided in Section 4.2.1, the Architect does not have such authority. The term "Owner" means the Owner or the Owner's authorized representative.

§ 2.1.2 The Owner shall furnish to the Contractor, within fifteen days after receipt of a written request, information necessary and relevant for the Contractor to evaluate, give notice of, or enforce mechanic's lien rights. Such information shall include a correct statement of the record legal title to the property on which the Project is located, usually referred to as the site, and the Owner's interest therein.

### § 2.2 Evidence of the Owner's Financial Arrangements

§ 2.2.1 Prior to commencement of the Work and upon written request by the Contractor, the Owner shall furnish to the Contractor reasonable evidence that the Owner has made financial arrangements to fulfill the Owner's obligations under the Contract. The Contractor shall have no obligation to commence the Work until the Owner provides such evidence. If commencement of the Work is delayed under this Section 2.2.1, the Contract Time shall be extended appropriately.

§ 2.2.2 Following commencement of the Work and upon written request by the Contractor, the Owner shall furnish to the Contractor reasonable evidence that the Owner has made financial arrangements to fulfill the Owner's obligations under the Contract only if (1) the Owner fails to make payments to the Contractor as the Contract Documents require; (2) the Contractor identifies in writing a reasonable concern regarding the Owner's ability to make payment when due; or (3) a change in the Work materially changes the Contract Sum. If the Owner fails to provide such evidence, as required, within fourteen days of the Contractor's request, the Contractor may immediately stop the Work and, in that event, shall notify the Owner that the Work has stopped. However, if the request is made because a change in the Work materially changes the Contract Sum under (3) above, the Contractor may immediately stop only that portion of the Work affected by the change until reasonable evidence is provided. If the Work is stopped under this Section 2.2.2, the Contract Time shall be extended appropriately and the Contract Sum shall be increased by the amount of the Contractor's reasonable costs of shutdown, delay and start-up, plus interest as provided in the Contract Documents.

§ 2.2.3 After the Owner furnishes evidence of financial arrangements under this Section 2.2, the Owner shall not materially vary such financial arrangements without prior notice to the Contractor.

§ 2.2.4 Where the Owner has designated information furnished under this Section 2.2 as "confidential," the Contractor shall keep the information confidential and shall not disclose it to any other person. However, the Contractor may disclose "confidential" information, after seven (7) days' notice to the Owner, where disclosure is required by law, including a subpoena or other form of compulsory legal process issued by a court or governmental entity, or by court or arbitrator(s) order. The Contractor may also disclose "confidential" information to its employees, consultants, sureties, Subcontractors and their employees, Sub-subcontractors, and others who need to know the content of such information solely and exclusively for the Project and who agree to maintain the confidentiality of such information.

### § 2.3 Information and Services Required of the Owner

§ 2.3.1 Except for permits and fees that are the responsibility of the Contractor under the Contract Documents, including those required under Section 3.7.1, the Owner shall secure and pay for necessary approvals, easements, assessments and charges required for construction, use or occupancy of permanent structures or for permanent changes in existing facilities.

§ 2.3.2 The Owner shall retain an architect lawfully licensed to practice architecture, or an entity lawfully practicing architecture, in the jurisdiction where the Project is located. That person or entity is identified as the Architect in the Agreement and is referred to throughout the Contract Documents as if singular in number.

§ 2.3.3 If the employment of the Architect terminates, the Owner shall employ a successor to whom the Contractor has no reasonable objection and whose status under the Contract Documents shall be that of the Architect.

§ 2.3.4 The Owner shall furnish surveys describing physical characteristics, legal limitations and utility locations for the

site of the Project, and a legal description of the site. The Contractor shall be entitled to rely on the accuracy of information furnished by the Owner but shall exercise proper precautions relating to the safe performance of the Work.

§ 2.3.5 The Owner shall furnish information or services required of the Owner by the Contract Documents with reasonable promptness. The Owner shall also furnish any other information or services under the Owner's control and relevant to the Contractor's performance of the Work with reasonable promptness after receiving the Contractor's written request for such information or services.

§ 2.3.6 Unless otherwise provided in the Contract Documents, the Owner shall furnish to the Contractor one copy of the Contract Documents for purposes of making reproductions pursuant to Section 1.5.2.

#### § 2.4 Owner's Right to Stop the Work

If the Contractor fails to correct Work that is not in accordance with the requirements of the Contract Documents as required by Section 12.2 or repeatedly fails to carry out Work in accordance with the Contract Documents, the Owner may issue a written order to the Contractor to stop the Work, or any portion thereof, until the cause for such order has been eliminated; however, the right of the Owner to stop the Work shall not give rise to a duty on the part of the Owner to exercise this right for the benefit of the Contractor or any other person or entity, except to the extent required by Section 6.1.3.

#### § 2.5 Owner's Right to Carry Out the Work

If the Contractor defaults or neglects to carry out the Work in accordance with the Contract Documents and fails within a ten-day period after receipt of notice from the Owner to commence and continue correction of such default or neglect with diligence and promptness, the Owner may, without prejudice to other remedies the Owner may have, correct such default or neglect. Such action by the Owner and amounts charged to the Contractor are both subject to prior approval of the Architect and the Architect may, pursuant to Section 9.5.1, withhold or nullify a Certificate for Payment in whole or in part, to the extent reasonably necessary to reimburse the Owner for the reasonable cost of correcting such deficiencies, including Owner's expenses and compensation for the Architect's additional services made necessary by such default, neglect, or failure. If current and future payments are not sufficient to cover such amounts, the Contractor shall pay the difference to the Owner. If the Contractor disagrees with the actions of the Owner or the Architect, or the amounts claimed as costs to the Owner, the Contractor may file a Claim pursuant to Article 15.

### ARTICLE 3 CONTRACTOR

#### § 3.1 General

§ 3.1.1 The Contractor is the person or entity identified as such in the Agreement and is referred to throughout the Contract Documents as if singular in number. The Contractor shall be lawfully licensed, if required in the jurisdiction where the Project is located. The Contractor shall designate in writing a representative who shall have express authority to bind the Contractor with respect to all matters under this Contract. The term "Contractor" means the Contractor or the Contractor's authorized representative.

§ 3.1.2 The Contractor shall perform the Work in accordance with the Contract Documents.

§ 3.1.3 The Contractor shall not be relieved of its obligations to perform the Work in accordance with the Contract Documents either by activities or duties of the Architect in the Architect's administration of the Contract, or by tests, inspections or approvals required or performed by persons or entities other than the Contractor.

#### § 3.2 Review of Contract Documents and Field Conditions by Contractor

§ 3.2.1 Execution of the Contract by the Contractor is a representation that the Contractor has visited the site, become generally familiar with local conditions under which the Work is to be performed, and correlated personal observations with requirements of the Contract Documents.

§ 3.2.2 Because the Contract Documents are complementary, the Contractor shall, before starting each portion of the Work, carefully study and compare the various Contract Documents relative to that portion of the Work, as well as the information furnished by the Owner pursuant to Section 2.3.4, shall take field measurements of any existing conditions related to that portion of the Work, and shall observe any conditions at the site affecting it. These obligations are for the purpose of facilitating coordination and construction by the Contractor and are not for the purpose of discovering errors, omissions, or inconsistencies in the Contract Documents; however, the Contractor shall promptly report to the Architect any errors, inconsistencies or omissions discovered by or made known to the Contractor as a request for information in such form as the Architect may require. It is recognized that the Contractor's review is made in the Contractor's



capacity as a contractor and not as a licensed design professional, unless otherwise specifically provided in the Contract Documents.

§ 3.2.3 The Contractor is not required to ascertain that the Contract Documents are in accordance with applicable laws, statutes, ordinances, codes, rules and regulations, or lawful orders of public authorities, but the Contractor shall promptly report to the Architect any nonconformity discovered by or made known to the Contractor as a request for information in such form as the Architect may require.

§ 3.2.4 If the Contractor believes that additional cost or time is involved because of clarifications or instructions the Architect issues in response to the Contractor's notices or requests for information pursuant to Sections 3.2.2 or 3.2.3, the Contractor shall submit Claims as provided in Article 15. If the Contractor fails to perform the obligations of Sections 3.2.2 or 3.2.3, the Contractor shall pay such costs and damages to the Owner, subject to Section 15.1.7, as would have been avoided if the Contractor had performed such obligations. If the Contractor performs those obligations, the Contractor shall not be liable to the Owner or Architect for damages resulting from errors, inconsistencies or omissions in the Contract Documents, for differences between field measurements or conditions and the Contract Documents, or for nonconformities of the Contract Documents to applicable laws, statutes, ordinances, codes, rules and regulations, and lawful orders of public authorities.

### § 3.3 Supervision and Construction Procedures

§ 3.3.1 The Contractor shall supervise and direct the Work, using the Contractor's best skill and attention. The Contractor shall be solely responsible for, and have control over, construction means, methods, techniques, sequences, and procedures, and for coordinating all portions of the Work under the Contract. If the Contract Documents give specific instructions concerning construction means, methods, techniques, sequences, or procedures, the Contractor shall evaluate the jobsite safety thereof and shall be solely responsible for the jobsite safety of such means, methods, techniques, sequences, or procedures. If the Contractor determines that such means, methods, techniques, sequences or procedures may not be safe, the Contractor shall give timely notice to the Owner and Architect, and shall propose alternative means, methods, techniques, sequences, or procedures. The Architect shall evaluate the proposed alternative solely for conformance with the design intent for the completed construction. Unless the Architect objects to the Contractor's proposed alternative, the Contractor shall perform the Work using its alternative means, methods, techniques, sequences, or procedures.

§ 3.3.2 The Contractor shall be responsible to the Owner for acts and omissions of the Contractor's employees, Subcontractors and their agents and employees, and other persons or entities performing portions of the Work for, or on behalf of, the Contractor or any of its Subcontractors.

§ 3.3.3 The Contractor shall be responsible for inspection of portions of Work already performed to determine that such portions are in proper condition to receive subsequent Work.

### § 3.4 Labor and Materials

§ 3.4.1 Unless otherwise provided in the Contract Documents, the Contractor shall provide and pay for labor, materials, equipment, tools, construction equipment and machinery, water, heat, utilities, transportation, and other facilities and services necessary for proper execution and completion of the Work, whether temporary or permanent and whether or not incorporated or to be incorporated in the Work.

§ 3.4.2 Except in the case of minor changes in the Work approved by the Architect in accordance with Section 3.12.8 or ordered by the Architect in accordance with Section 7.4, the Contractor may make substitutions only with the consent of the Owner, after evaluation by the Architect and in accordance with a Change Order or Construction Change Directive.

§ 3.4.3 The Contractor shall enforce strict discipline and good order among the Contractor's employees and other persons carrying out the Work. The Contractor shall not permit employment of unfit persons or persons not properly skilled in tasks assigned to them.

### § 3.5 Warranty

§ 3.5.1 The Contractor warrants to the Owner and Architect that materials and equipment furnished under the Contract will be of good quality and new unless the Contract Documents require or permit otherwise. The Contractor further warrants that the Work will conform to the requirements of the Contract Documents and will be free from defects, except for those inherent in the quality of the Work the Contract Documents require or permit. Work, materials, or equipment not conforming to these requirements may be considered defective. The Contractor's warranty excludes

remedy for damage or defect caused by abuse, alterations to the Work not executed by the Contractor, improper or insufficient maintenance, improper operation, or normal wear and tear and normal usage. If required by the Architect, the Contractor shall furnish satisfactory evidence as to the kind and quality of materials and equipment.

§ 3.5.2 All material, equipment, or other special warranties required by the Contract Documents shall be issued in the name of the Owner, or shall be transferable to the Owner, and shall commence in accordance with Section 9.8.4.

### § 3.6 Taxes

The Contractor shall pay sales, consumer, use and similar taxes for the Work provided by the Contractor that are legally enacted when bids are received or negotiations concluded, whether or not yet effective or merely scheduled to go into effect.

### § 3.7 Permits, Fees, Notices and Compliance with Laws

§ 3.7.1 Unless otherwise provided in the Contract Documents, the Contractor shall secure and pay for the building permit as well as for other permits, fees, licenses, and inspections by government agencies necessary for proper execution and completion of the Work that are customarily secured after execution of the Contract and legally required at the time bids are received or negotiations concluded.

§ 3.7.2 The Contractor shall comply with and give notices required by applicable laws, statutes, ordinances, codes, rules and regulations, and lawful orders of public authorities applicable to performance of the Work.

§ 3.7.3 If the Contractor performs Work knowing it to be contrary to applicable laws, statutes, ordinances, codes, rules and regulations, or lawful orders of public authorities, the Contractor shall assume appropriate responsibility for such Work and shall bear the costs attributable to correction.

### § 3.7.4 Concealed or Unknown Conditions

If the Contractor encounters conditions at the site that are (1) subsurface or otherwise concealed physical conditions that differ materially from those indicated in the Contract Documents or (2) unknown physical conditions of an unusual nature that differ materially from those ordinarily found to exist and generally recognized as inherent in construction activities of the character provided for in the Contract Documents, the Contractor shall promptly provide notice to the Owner and the Architect before conditions are disturbed and in no event later than 14 days after first observance of the conditions. The Architect will promptly investigate such conditions and, if the Architect determines that they differ materially and cause an increase or decrease in the Contractor's cost of, or time required for, performance of any part of the Work, will recommend that an equitable adjustment be made in the Contract Sum or Contract Time, or both. If the Architect determines that the conditions at the site are not materially different from those indicated in the Contract Documents and that no change in the terms of the Contract is justified, the Architect shall promptly notify the Owner and Contractor, stating the reasons. If either party disputes the Architect's determination or recommendation, that party may submit a Claim as provided in Article 15.

§ 3.7.5 If, in the course of the Work, the Contractor encounters human remains or recognizes the existence of burial markers, archaeological sites or wetlands not indicated in the Contract Documents, the Contractor shall immediately suspend any operations that would affect them and shall notify the Owner and Architect. Upon receipt of such notice, the Owner shall promptly take any action necessary to obtain governmental authorization required to resume the operations. The Contractor shall continue to suspend such operations until otherwise instructed by the Owner but shall continue with all other operations that do not affect those remains or features. Requests for adjustments in the Contract Sum and Contract Time arising from the existence of such remains or features may be made as provided in Article 15.

### § 3.8 Allowances

§ 3.8.1 The Contractor shall include in the Contract Sum all allowances stated in the Contract Documents. Items covered by allowances shall be supplied for such amounts and by such persons or entities as the Owner may direct, but the Contractor shall not be required to employ persons or entities to whom the Contractor has reasonable objection.

§ 3.8.2 Unless otherwise provided in the Contract Documents,

- .1 allowances shall cover the cost to the Contractor of materials and equipment delivered at the site and all required taxes, less applicable trade discounts;
- .2 Contractor's costs for unloading and handling at the site, labor, installation costs, overhead, profit, and other expenses contemplated for stated allowance amounts shall be included in the Contract Sum but not in the allowances; and

- .3 whenever costs are more than or less than allowances, the Contract Sum shall be adjusted accordingly by Change Order. The amount of the Change Order shall reflect (1) the difference between actual costs and the allowances under Section 3.8.2.1 and (2) changes in Contractor's costs under Section 3.8.2.2.

§ 3.8.3 Materials and equipment under an allowance shall be selected by the Owner with reasonable promptness.

### § 3.9 Superintendent

§ 3.9.1 The Contractor shall employ a competent superintendent and necessary assistants who shall be in attendance at the Project site during performance of the Work. The superintendent shall represent the Contractor, and communications given to the superintendent shall be as binding as if given to the Contractor.

§ 3.9.2 The Contractor, as soon as practicable after award of the Contract, shall notify the Owner and Architect of the name and qualifications of a proposed superintendent. Within 14 days of receipt of the information, the Architect may notify the Contractor, stating whether the Owner or the Architect (1) has reasonable objection to the proposed superintendent or (2) requires additional time for review. Failure of the Architect to provide notice within the 14-day period shall constitute notice of no reasonable objection.

§ 3.9.3 The Contractor shall not employ a proposed superintendent to whom the Owner or Architect has made reasonable and timely objection. The Contractor shall not change the superintendent without the Owner's consent, which shall not unreasonably be withheld or delayed.

### § 3.10 Contractor's Construction and Submittal Schedules

§ 3.10.1 The Contractor, promptly after being awarded the Contract, shall submit for the Owner's and Architect's information a Contractor's construction schedule for the Work. The schedule shall contain detail appropriate for the Project, including (1) the date of commencement of the Work, interim schedule milestone dates, and the date of Substantial Completion; (2) an apportionment of the Work by construction activity; and (3) the time required for completion of each portion of the Work. The schedule shall provide for the orderly progression of the Work to completion and shall not exceed time limits current under the Contract Documents. The schedule shall be revised at appropriate intervals as required by the conditions of the Work and Project.

§ 3.10.2 The Contractor, promptly after being awarded the Contract and thereafter as necessary to maintain a current submittal schedule, shall submit a submittal schedule for the Architect's approval. The Architect's approval shall not be unreasonably delayed or withheld. The submittal schedule shall (1) be coordinated with the Contractor's construction schedule, and (2) allow the Architect reasonable time to review submittals. If the Contractor fails to submit a submittal schedule, or fails to provide submittals in accordance with the approved submittal schedule, the Contractor shall not be entitled to any increase in Contract Sum or extension of Contract Time based on the time required for review of submittals.

§ 3.10.3 The Contractor shall perform the Work in general accordance with the most recent schedules submitted to the Owner and Architect.

### § 3.11 Documents and Samples at the Site

The Contractor shall make available, at the Project site, the Contract Documents, including Change Orders, Construction Change Directives, and other Modifications, in good order and marked currently to indicate field changes and selections made during construction, and the approved Shop Drawings, Product Data, Samples, and similar required submittals. These shall be in electronic form or paper copy, available to the Architect and Owner, and delivered to the Architect for submittal to the Owner upon completion of the Work as a record of the Work as constructed.

### § 3.12 Shop Drawings, Product Data and Samples

§ 3.12.1 Shop Drawings are drawings, diagrams, schedules, and other data specially prepared for the Work by the Contractor or a Subcontractor, Sub-subcontractor, manufacturer, supplier, or distributor to illustrate some portion of the Work.

§ 3.12.2 Product Data are illustrations, standard schedules, performance charts, instructions, brochures, diagrams, and other information furnished by the Contractor to illustrate materials or equipment for some portion of the Work.

§ 3.12.3 Samples are physical examples that illustrate materials, equipment, or workmanship, and establish standards by which the Work will be judged.

§ 3.12.4 Shop Drawings, Product Data, Samples, and similar submittals are not Contract Documents. Their purpose is to demonstrate how the Contractor proposes to conform to the information given and the design concept expressed in the Contract Documents for those portions of the Work for which the Contract Documents require submittals. Review by the Architect is subject to the limitations of Section 4.2.7. Informational submittals upon which the Architect is not expected to take responsive action may be so identified in the Contract Documents. Submittals that are not required by the Contract Documents may be returned by the Architect without action.

§ 3.12.5 The Contractor shall review for compliance with the Contract Documents, approve, and submit to the Architect, Shop Drawings, Product Data, Samples, and similar submittals required by the Contract Documents, in accordance with the submittal schedule approved by the Architect or, in the absence of an approved submittal schedule, with reasonable promptness and in such sequence as to cause no delay in the Work or in the activities of the Owner or of Separate Contractors.

§ 3.12.6 By submitting Shop Drawings, Product Data, Samples, and similar submittals, the Contractor represents to the Owner and Architect that the Contractor has (1) reviewed and approved them, (2) determined and verified materials, field measurements and field construction criteria related thereto, or will do so, and (3) checked and coordinated the information contained within such submittals with the requirements of the Work and of the Contract Documents.

§ 3.12.7 The Contractor shall perform no portion of the Work for which the Contract Documents require submittal and review of Shop Drawings, Product Data, Samples, or similar submittals, until the respective submittal has been approved by the Architect.

§ 3.12.8 The Work shall be in accordance with approved submittals except that the Contractor shall not be relieved of responsibility for deviations from the requirements of the Contract Documents by the Architect's approval of Shop Drawings, Product Data, Samples, or similar submittals, unless the Contractor has specifically notified the Architect of such deviation at the time of submittal and (1) the Architect has given written approval to the specific deviation as a minor change in the Work, or (2) a Change Order or Construction Change Directive has been issued authorizing the deviation. The Contractor shall not be relieved of responsibility for errors or omissions in Shop Drawings, Product Data, Samples, or similar submittals, by the Architect's approval thereof.

§ 3.12.9 The Contractor shall direct specific attention, in writing or on resubmitted Shop Drawings, Product Data, Samples, or similar submittals, to revisions other than those requested by the Architect on previous submittals. In the absence of such notice, the Architect's approval of a resubmission shall not apply to such revisions.

§ 3.12.10 The Contractor shall not be required to provide professional services that constitute the practice of architecture or engineering unless such services are specifically required by the Contract Documents for a portion of the Work or unless the Contractor needs to provide such services in order to carry out the Contractor's responsibilities for construction means, methods, techniques, sequences, and procedures. The Contractor shall not be required to provide professional services in violation of applicable law.

§ 3.12.10.1 If professional design services or certifications by a design professional related to systems, materials, or equipment are specifically required of the Contractor by the Contract Documents, the Owner and the Architect will specify all performance and design criteria that such services must satisfy. The Contractor shall be entitled to rely upon the adequacy and accuracy of the performance and design criteria provided in the Contract Documents. The Contractor shall cause such services or certifications to be provided by an appropriately licensed design professional, whose signature and seal shall appear on all drawings, calculations, specifications, certifications, Shop Drawings, and other submittals prepared by such professional. Shop Drawings, and other submittals related to the Work, designed or certified by such professional, if prepared by others, shall bear such professional's written approval when submitted to the Architect. The Owner and the Architect shall be entitled to rely upon the adequacy and accuracy of the services, certifications, and approvals performed or provided by such design professionals, provided the Owner and Architect have specified to the Contractor the performance and design criteria that such services must satisfy. Pursuant to this Section 3.12.10, the Architect will review and approve or take other appropriate action on submittals only for the limited purpose of checking for conformance with information given and the design concept expressed in the Contract Documents.

§ 3.12.10.2 If the Contract Documents require the Contractor's design professional to certify that the Work has been performed in accordance with the design criteria, the Contractor shall furnish such certifications to the Architect at the

time and in the form specified by the Architect.

### § 3.13 Use of Site

The Contractor shall confine operations at the site to areas permitted by applicable laws, statutes, ordinances, codes, rules and regulations, lawful orders of public authorities, and the Contract Documents and shall not unreasonably encumber the site with materials or equipment.

### § 3.14 Cutting and Patching

§ 3.14.1 The Contractor shall be responsible for cutting, fitting, or patching required to complete the Work or to make its parts fit together properly. All areas requiring cutting, fitting, or patching shall be restored to the condition existing prior to the cutting, fitting, or patching, unless otherwise required by the Contract Documents.

§ 3.14.2 The Contractor shall not damage or endanger a portion of the Work or fully or partially completed construction of the Owner or Separate Contractors by cutting, patching, or otherwise altering such construction, or by excavation. The Contractor shall not cut or otherwise alter construction by the Owner or a Separate Contractor except with written consent of the Owner and of the Separate Contractor. Consent shall not be unreasonably withheld. The Contractor shall not unreasonably withhold, from the Owner or a Separate Contractor, its consent to cutting or otherwise altering the Work.

### § 3.15 Cleaning Up

§ 3.15.1 The Contractor shall keep the premises and surrounding area free from accumulation of waste materials and rubbish caused by operations under the Contract. At completion of the Work, the Contractor shall remove waste materials, rubbish, the Contractor's tools, construction equipment, machinery, and surplus materials from and about the Project.

§ 3.15.2 If the Contractor fails to clean up as provided in the Contract Documents, the Owner may do so and the Owner shall be entitled to reimbursement from the Contractor.

### § 3.16 Access to Work

The Contractor shall provide the Owner and Architect with access to the Work in preparation and progress wherever located.

### § 3.17 Royalties, Patents and Copyrights

The Contractor shall pay all royalties and license fees. The Contractor shall defend suits or claims for infringement of copyrights and patent rights and shall hold the Owner and Architect harmless from loss on account thereof, but shall not be responsible for defense or loss when a particular design, process, or product of a particular manufacturer or manufacturers is required by the Contract Documents, or where the copyright violations are contained in Drawings, Specifications, or other documents prepared by the Owner or Architect. However, if an infringement of a copyright or patent is discovered by, or made known to, the Contractor, the Contractor shall be responsible for the loss unless the information is promptly furnished to the Architect.

### § 3.18 Indemnification

§ 3.18.1 To the fullest extent permitted by law, the Contractor shall indemnify and hold harmless the Owner, Architect, Architect's consultants, and agents and employees of any of them from and against claims, damages, losses, and expenses, including but not limited to attorneys' fees, arising out of or resulting from performance of the Work, provided that such claim, damage, loss, or expense is attributable to bodily injury, sickness, disease or death, or to injury to or destruction of tangible property (other than the Work itself), but only to the extent caused by the negligent acts or omissions of the Contractor, a Subcontractor, anyone directly or indirectly employed by them, or anyone for whose acts they may be liable, regardless of whether or not such claim, damage, loss, or expense is caused in part by a party indemnified hereunder. Such obligation shall not be construed to negate, abridge, or reduce other rights or obligations of indemnity that would otherwise exist as to a party or person described in this Section 3.18.

§ 3.18.2 In claims against any person or entity indemnified under this Section 3.18 by an employee of the Contractor, a Subcontractor, anyone directly or indirectly employed by them, or anyone for whose acts they may be liable, the indemnification obligation under Section 3.18.1 shall not be limited by a limitation on amount or type of damages, compensation, or benefits payable by or for the Contractor or a Subcontractor under workers' compensation acts, disability benefit acts, or other employee benefit acts.



## ARTICLE 4 ARCHITECT

### § 4.1 General

§ 4.1.1 The Architect is the person or entity retained by the Owner pursuant to Section 2.3.2 and identified as such in the Agreement.

§ 4.1.2 Duties, responsibilities, and limitations of authority of the Architect as set forth in the Contract Documents shall not be restricted, modified, or extended without written consent of the Owner, Contractor, and Architect. Consent shall not be unreasonably withheld.

### § 4.2 Administration of the Contract

§ 4.2.1 The Architect will provide administration of the Contract as described in the Contract Documents and will be an Owner's representative during construction until the date the Architect issues the final Certificate for Payment. The Architect will have authority to act on behalf of the Owner only to the extent provided in the Contract Documents.

§ 4.2.2 The Architect will visit the site at intervals appropriate to the stage of construction, or as otherwise agreed with the Owner, to become generally familiar with the progress and quality of the portion of the Work completed, and to determine in general if the Work observed is being performed in a manner indicating that the Work, when fully completed, will be in accordance with the Contract Documents. However, the Architect will not be required to make exhaustive or continuous on-site inspections to check the quality or quantity of the Work. The Architect will not have control over, charge of, or responsibility for the construction means, methods, techniques, sequences or procedures, or for the safety precautions and programs in connection with the Work, since these are solely the Contractor's rights and responsibilities under the Contract Documents.

§ 4.2.3 On the basis of the site visits, the Architect will keep the Owner reasonably informed about the progress and quality of the portion of the Work completed, and promptly report to the Owner (1) known deviations from the Contract Documents, (2) known deviations from the most recent construction schedule submitted by the Contractor, and (3) defects and deficiencies observed in the Work. The Architect will not be responsible for the Contractor's failure to perform the Work in accordance with the requirements of the Contract Documents. The Architect will not have control over or charge of, and will not be responsible for acts or omissions of, the Contractor, Subcontractors, or their agents or employees, or any other persons or entities performing portions of the Work.

### § 4.2.4 Communications

The Owner and Contractor shall include the Architect in all communications that relate to or affect the Architect's services or professional responsibilities. The Owner shall promptly notify the Architect of the substance of any direct communications between the Owner and the Contractor otherwise relating to the Project. Communications by and with the Architect's consultants shall be through the Architect. Communications by and with Subcontractors and suppliers shall be through the Contractor. Communications by and with Separate Contractors shall be through the Owner. The Contract Documents may specify other communication protocols.

§ 4.2.5 Based on the Architect's evaluations of the Contractor's Applications for Payment, the Architect will review and certify the amounts due the Contractor and will issue Certificates for Payment in such amounts.

§ 4.2.6 The Architect has authority to reject Work that does not conform to the Contract Documents. Whenever the Architect considers it necessary or advisable, the Architect will have authority to require inspection or testing of the Work in accordance with Sections 13.4.2 and 13.4.3, whether or not the Work is fabricated, installed or completed. However, neither this authority of the Architect nor a decision made in good faith either to exercise or not to exercise such authority shall give rise to a duty or responsibility of the Architect to the Contractor, Subcontractors, suppliers, their agents or employees, or other persons or entities performing portions of the Work.

§ 4.2.7 The Architect will review and approve, or take other appropriate action upon, the Contractor's submittals such as Shop Drawings, Product Data, and Samples, but only for the limited purpose of checking for conformance with information given and the design concept expressed in the Contract Documents. The Architect's action will be taken in accordance with the submittal schedule approved by the Architect or, in the absence of an approved submittal schedule, with reasonable promptness while allowing sufficient time in the Architect's professional judgment to permit adequate review. Review of such submittals is not conducted for the purpose of determining the accuracy and completeness of other details such as dimensions and quantities, or for substantiating instructions for installation or performance of equipment or systems, all of which remain the responsibility of the Contractor as required by the Contract Documents. The Architect's review of the Contractor's submittals shall not relieve the Contractor of the obligations under

Sections 3.3, 3.5, and 3.12. The Architect's review shall not constitute approval of safety precautions or of any construction means, methods, techniques, sequences, or procedures. The Architect's approval of a specific item shall not indicate approval of an assembly of which the item is a component.

§ 4.2.8 The Architect will prepare Change Orders and Construction Change Directives, and may order minor changes in the Work as provided in Section 7.4. The Architect will investigate and make determinations and recommendations regarding concealed and unknown conditions as provided in Section 3.7.4.

§ 4.2.9 The Architect will conduct inspections to determine the date or dates of Substantial Completion and the date of final completion; issue Certificates of Substantial Completion pursuant to Section 9.8; receive and forward to the Owner, for the Owner's review and records, written warranties and related documents required by the Contract and assembled by the Contractor pursuant to Section 9.10; and issue a final Certificate for Payment pursuant to Section 9.10.

§ 4.2.10 If the Owner and Architect agree, the Architect will provide one or more Project representatives to assist in carrying out the Architect's responsibilities at the site. The Owner shall notify the Contractor of any change in the duties, responsibilities and limitations of authority of the Project representatives.

§ 4.2.11 The Architect will interpret and decide matters concerning performance under, and requirements of, the Contract Documents on written request of either the Owner or Contractor. The Architect's response to such requests will be made in writing within any time limits agreed upon or otherwise with reasonable promptness.

§ 4.2.12 Interpretations and decisions of the Architect will be consistent with the intent of, and reasonably inferable from, the Contract Documents and will be in writing or in the form of drawings. When making such interpretations and decisions, the Architect will endeavor to secure faithful performance by both Owner and Contractor, will not show partiality to either, and will not be liable for results of interpretations or decisions rendered in good faith.

§ 4.2.13 The Architect's decisions on matters relating to aesthetic effect will be final if consistent with the intent expressed in the Contract Documents.

§ 4.2.14 The Architect will review and respond to requests for information about the Contract Documents. The Architect's response to such requests will be made in writing within any time limits agreed upon or otherwise with reasonable promptness. If appropriate, the Architect will prepare and issue supplemental Drawings and Specifications in response to the requests for information.

## ARTICLE 5 SUBCONTRACTORS

### § 5.1 Definitions

§ 5.1.1 A Subcontractor is a person or entity who has a direct contract with the Contractor to perform a portion of the Work at the site. The term "Subcontractor" is referred to throughout the Contract Documents as if singular in number and means a Subcontractor or an authorized representative of the Subcontractor. The term "Subcontractor" does not include a Separate Contractor or the subcontractors of a Separate Contractor.

§ 5.1.2 A Sub-subcontractor is a person or entity who has a direct or indirect contract with a Subcontractor to perform a portion of the Work at the site. The term "Sub-subcontractor" is referred to throughout the Contract Documents as if singular in number and means a Sub-subcontractor or an authorized representative of the Sub-subcontractor.

### § 5.2 Award of Subcontracts and Other Contracts for Portions of the Work

§ 5.2.1 Unless otherwise stated in the Contract Documents, the Contractor, as soon as practicable after award of the Contract, shall notify the Owner and Architect of the persons or entities proposed for each principal portion of the Work, including those who are to furnish materials or equipment fabricated to a special design. Within 14 days of receipt of the information, the Architect may notify the Contractor whether the Owner or the Architect (1) has reasonable objection to any such proposed person or entity or (2) requires additional time for review. Failure of the Architect to provide notice within the 14-day period shall constitute notice of no reasonable objection.

§ 5.2.2 The Contractor shall not contract with a proposed person or entity to whom the Owner or Architect has made reasonable and timely objection. The Contractor shall not be required to contract with anyone to whom the Contractor has made reasonable objection.

§ 5.2.3 If the Owner or Architect has reasonable objection to a person or entity proposed by the Contractor, the

Contractor shall propose another to whom the Owner or Architect has no reasonable objection. If the proposed but rejected Subcontractor was reasonably capable of performing the Work, the Contract Sum and Contract Time shall be increased or decreased by the difference, if any, occasioned by such change, and an appropriate Change Order shall be issued before commencement of the substitute Subcontractor's Work. However, no increase in the Contract Sum or Contract Time shall be allowed for such change unless the Contractor has acted promptly and responsively in submitting names as required.

§ 5.2.4 The Contractor shall not substitute a Subcontractor, person, or entity for one previously selected if the Owner or Architect makes reasonable objection to such substitution.

### § 5.3 Subcontractual Relations

By appropriate written agreement, the Contractor shall require each Subcontractor, to the extent of the Work to be performed by the Subcontractor, to be bound to the Contractor by terms of the Contract Documents, and to assume toward the Contractor all the obligations and responsibilities, including the responsibility for safety of the Subcontractor's Work that the Contractor, by these Contract Documents, assumes toward the Owner and Architect. Each subcontract agreement shall preserve and protect the rights of the Owner and Architect under the Contract Documents with respect to the Work to be performed by the Subcontractor so that subcontracting thereof will not prejudice such rights, and shall allow to the Subcontractor, unless specifically provided otherwise in the subcontract agreement, the benefit of all rights, remedies, and redress against the Contractor that the Contractor, by the Contract Documents, has against the Owner. Where appropriate, the Contractor shall require each Subcontractor to enter into similar agreements with Sub-subcontractors. The Contractor shall make available to each proposed Subcontractor, prior to the execution of the subcontract agreement, copies of the Contract Documents to which the Subcontractor will be bound, and, upon written request of the Subcontractor, identify to the Subcontractor terms and conditions of the proposed subcontract agreement that may be at variance with the Contract Documents. Subcontractors will similarly make copies of applicable portions of such documents available to their respective proposed Sub-subcontractors.

### § 5.4 Contingent Assignment of Subcontracts

§ 5.4.1 Each subcontract agreement for a portion of the Work is assigned by the Contractor to the Owner, provided that

- .1 assignment is effective only after termination of the Contract by the Owner for cause pursuant to Section 14.2 and only for those subcontract agreements that the Owner accepts by notifying the Subcontractor and Contractor; and
- .2 assignment is subject to the prior rights of the surety, if any, obligated under bond relating to the Contract.

When the Owner accepts the assignment of a subcontract agreement, the Owner assumes the Contractor's rights and obligations under the subcontract.

§ 5.4.2 Upon such assignment, if the Work has been suspended for more than 30 days, the Subcontractor's compensation shall be equitably adjusted for increases in cost resulting from the suspension.

§ 5.4.3 Upon assignment to the Owner under this Section 5.4, the Owner may further assign the subcontract to a successor contractor or other entity. If the Owner assigns the subcontract to a successor contractor or other entity, the Owner shall nevertheless remain legally responsible for all of the successor contractor's obligations under the subcontract.

## ARTICLE 6 CONSTRUCTION BY OWNER OR BY SEPARATE CONTRACTORS

### § 6.1 Owner's Right to Perform Construction and to Award Separate Contracts

§ 6.1.1 The term "Separate Contractor(s)" shall mean other contractors retained by the Owner under separate agreements. The Owner reserves the right to perform construction or operations related to the Project with the Owner's own forces, and with Separate Contractors retained under Conditions of the Contract substantially similar to those of this Contract, including those provisions of the Conditions of the Contract related to insurance and waiver of subrogation.

§ 6.1.2 When separate contracts are awarded for different portions of the Project or other construction or operations on the site, the term "Contractor" in the Contract Documents in each case shall mean the Contractor who executes each separate Owner-Contractor Agreement.

§ 6.1.3 The Owner shall provide for coordination of the activities of the Owner's own forces and of each Separate



Contractor with the Work of the Contractor, who shall cooperate with them. The Contractor shall participate with any Separate Contractors and the Owner in reviewing their construction schedules. The Contractor shall make any revisions to its construction schedule deemed necessary after a joint review and mutual agreement. The construction schedules shall then constitute the schedules to be used by the Contractor, Separate Contractors, and the Owner until subsequently revised.

§ 6.1.4 Unless otherwise provided in the Contract Documents, when the Owner performs construction or operations related to the Project with the Owner's own forces or with Separate Contractors, the Owner or its Separate Contractors shall have the same obligations and rights that the Contractor has under the Conditions of the Contract, including, without excluding others, those stated in Article 3, this Article 6, and Articles 10, 11, and 12.

## § 6.2 Mutual Responsibility

§ 6.2.1 The Contractor shall afford the Owner and Separate Contractors reasonable opportunity for introduction and storage of their materials and equipment and performance of their activities, and shall connect and coordinate the Contractor's construction and operations with theirs as required by the Contract Documents.

§ 6.2.2 If part of the Contractor's Work depends for proper execution or results upon construction or operations by the Owner or a Separate Contractor, the Contractor shall, prior to proceeding with that portion of the Work, promptly notify the Architect of apparent discrepancies or defects in the construction or operations by the Owner or Separate Contractor that would render it unsuitable for proper execution and results of the Contractor's Work. Failure of the Contractor to notify the Architect of apparent discrepancies or defects prior to proceeding with the Work shall constitute an acknowledgment that the Owner's or Separate Contractor's completed or partially completed construction is fit and proper to receive the Contractor's Work. The Contractor shall not be responsible for discrepancies or defects in the construction or operations by the Owner or Separate Contractor that are not apparent.

§ 6.2.3 The Contractor shall reimburse the Owner for costs the Owner incurs that are payable to a Separate Contractor because of the Contractor's delays, improperly timed activities or defective construction. The Owner shall be responsible to the Contractor for costs the Contractor incurs because of a Separate Contractor's delays, improperly timed activities, damage to the Work or defective construction.

§ 6.2.4 The Contractor shall promptly remedy damage that the Contractor wrongfully causes to completed or partially completed construction or to property of the Owner or Separate Contractor as provided in Section 10.2.5.

§ 6.2.5 The Owner and each Separate Contractor shall have the same responsibilities for cutting and patching as are described for the Contractor in Section 3.14.

## § 6.3 Owner's Right to Clean Up

If a dispute arises among the Contractor, Separate Contractors, and the Owner as to the responsibility under their respective contracts for maintaining the premises and surrounding area free from waste materials and rubbish, the Owner may clean up and the Architect will allocate the cost among those responsible.

## ARTICLE 7 CHANGES IN THE WORK

### § 7.1 General

§ 7.1.1 Changes in the Work may be accomplished after execution of the Contract, and without invalidating the Contract, by Change Order, Construction Change Directive or order for a minor change in the Work, subject to the limitations stated in this Article 7 and elsewhere in the Contract Documents.

§ 7.1.2 A Change Order shall be based upon agreement among the Owner, Contractor, and Architect. A Construction Change Directive requires agreement by the Owner and Architect and may or may not be agreed to by the Contractor. An order for a minor change in the Work may be issued by the Architect alone.

§ 7.1.3 Changes in the Work shall be performed under applicable provisions of the Contract Documents. The Contractor shall proceed promptly with changes in the Work, unless otherwise provided in the Change Order, Construction Change Directive, or order for a minor change in the Work.

### § 7.2 Change Orders

§ 7.2.1 A Change Order is a written instrument prepared by the Architect and signed by the Owner, Contractor, and Architect stating their agreement upon all of the following:

- .1 The change in the Work;
- .2 The amount of the adjustment, if any, in the Contract Sum; and
- .3 The extent of the adjustment, if any, in the Contract Time.

### § 7.3 Construction Change Directives

§ 7.3.1 A Construction Change Directive is a written order prepared by the Architect and signed by the Owner and Architect, directing a change in the Work prior to agreement on adjustment, if any, in the Contract Sum or Contract Time, or both. The Owner may by Construction Change Directive, without invalidating the Contract, order changes in the Work within the general scope of the Contract consisting of additions, deletions, or other revisions, the Contract Sum and Contract Time being adjusted accordingly.

§ 7.3.2 A Construction Change Directive shall be used in the absence of total agreement on the terms of a Change Order.

§ 7.3.3 If the Construction Change Directive provides for an adjustment to the Contract Sum, the adjustment shall be based on one of the following methods:

- .1 Mutual acceptance of a lump sum properly itemized and supported by sufficient substantiating data to permit evaluation;
- .2 Unit prices stated in the Contract Documents or subsequently agreed upon;
- .3 Cost to be determined in a manner agreed upon by the parties and a mutually acceptable fixed or percentage fee; or
- .4 As provided in Section 7.3.4.

§ 7.3.4 If the Contractor does not respond promptly or disagrees with the method for adjustment in the Contract Sum, the Architect shall determine the adjustment on the basis of reasonable expenditures and savings of those performing the Work attributable to the change, including, in case of an increase in the Contract Sum, an amount for overhead and profit as set forth in the Agreement, or if no such amount is set forth in the Agreement, a reasonable amount. In such case, and also under Section 7.3.3.3, the Contractor shall keep and present, in such form as the Architect may prescribe, an itemized accounting together with appropriate supporting data. Unless otherwise provided in the Contract Documents, costs for the purposes of this Section 7.3.4 shall be limited to the following:

- .1 Costs of labor, including applicable payroll taxes, fringe benefits required by agreement or custom, workers' compensation insurance, and other employee costs approved by the Architect;
- .2 Costs of materials, supplies, and equipment, including cost of transportation, whether incorporated or consumed;
- .3 Rental costs of machinery and equipment, exclusive of hand tools, whether rented from the Contractor or others;
- .4 Costs of premiums for all bonds and insurance, permit fees, and sales, use, or similar taxes, directly related to the change; and
- .5 Costs of supervision and field office personnel directly attributable to the change.

§ 7.3.5 If the Contractor disagrees with the adjustment in the Contract Time, the Contractor may make a Claim in accordance with applicable provisions of Article 15.

§ 7.3.6 Upon receipt of a Construction Change Directive, the Contractor shall promptly proceed with the change in the Work involved and advise the Architect of the Contractor's agreement or disagreement with the method, if any, provided in the Construction Change Directive for determining the proposed adjustment in the Contract Sum or Contract Time.

§ 7.3.7 A Construction Change Directive signed by the Contractor indicates the Contractor's agreement therewith, including adjustment in Contract Sum and Contract Time or the method for determining them. Such agreement shall be effective immediately and shall be recorded as a Change Order.

§ 7.3.8 The amount of credit to be allowed by the Contractor to the Owner for a deletion or change that results in a net decrease in the Contract Sum shall be actual net cost as confirmed by the Architect. When both additions and credits covering related Work or substitutions are involved in a change, the allowance for overhead and profit shall be figured on the basis of net increase, if any, with respect to that change.

§ 7.3.9 Pending final determination of the total cost of a Construction Change Directive to the Owner, the Contractor may request payment for Work completed under the Construction Change Directive in Applications for Payment. The

Architect will make an interim determination for purposes of monthly certification for payment for those costs and certify for payment the amount that the Architect determines, in the Architect's professional judgment, to be reasonably justified. The Architect's interim determination of cost shall adjust the Contract Sum on the same basis as a Change Order, subject to the right of either party to disagree and assert a Claim in accordance with Article 15.

§ 7.3.10 When the Owner and Contractor agree with a determination made by the Architect concerning the adjustments in the Contract Sum and Contract Time, or otherwise reach agreement upon the adjustments, such agreement shall be effective immediately and the Architect will prepare a Change Order. Change Orders may be issued for all or any part of a Construction Change Directive.

#### § 7.4 Minor Changes in the Work

The Architect may order minor changes in the Work that are consistent with the intent of the Contract Documents and do not involve an adjustment in the Contract Sum or an extension of the Contract Time. The Architect's order for minor changes shall be in writing. If the Contractor believes that the proposed minor change in the Work will affect the Contract Sum or Contract Time, the Contractor shall notify the Architect and shall not proceed to implement the change in the Work. If the Contractor performs the Work set forth in the Architect's order for a minor change without prior notice to the Architect that such change will affect the Contract Sum or Contract Time, the Contractor waives any adjustment to the Contract Sum or extension of the Contract Time.

### ARTICLE 8 TIME

#### § 8.1 Definitions

§ 8.1.1 Unless otherwise provided, Contract Time is the period of time, including authorized adjustments, allotted in the Contract Documents for Substantial Completion of the Work.

§ 8.1.2 The date of commencement of the Work is the date established in the Agreement.

§ 8.1.3 The date of Substantial Completion is the date certified by the Architect in accordance with Section 9.8.

§ 8.1.4 The term "day" as used in the Contract Documents shall mean calendar day unless otherwise specifically defined.

#### § 8.2 Progress and Completion

§ 8.2.1 Time limits stated in the Contract Documents are of the essence of the Contract. By executing the Agreement, the Contractor confirms that the Contract Time is a reasonable period for performing the Work.

§ 8.2.2 The Contractor shall not knowingly, except by agreement or instruction of the Owner in writing, commence the Work prior to the effective date of insurance required to be furnished by the Contractor and Owner.

§ 8.2.3 The Contractor shall proceed expeditiously with adequate forces and shall achieve Substantial Completion within the Contract Time.

#### § 8.3 Delays and Extensions of Time

§ 8.3.1 If the Contractor is delayed at any time in the commencement or progress of the Work by (1) an act or neglect of the Owner or Architect, of an employee of either, or of a Separate Contractor; (2) by changes ordered in the Work; (3) by labor disputes, fire, unusual delay in deliveries, unavoidable casualties, adverse weather conditions documented in accordance with Section 15.1.6.2, or other causes beyond the Contractor's control; (4) by delay authorized by the Owner pending mediation and binding dispute resolution; or (5) by other causes that the Contractor asserts, and the Architect determines, justify delay, then the Contract Time shall be extended for such reasonable time as the Architect may determine.

§ 8.3.2 Claims relating to time shall be made in accordance with applicable provisions of Article 15.

§ 8.3.3 This Section 8.3 does not preclude recovery of damages for delay by either party under other provisions of the Contract Documents.

### ARTICLE 9 PAYMENTS AND COMPLETION

#### § 9.1 Contract Sum

§ 9.1.1 The Contract Sum is stated in the Agreement and, including authorized adjustments, is the total amount payable

by the Owner to the Contractor for performance of the Work under the Contract Documents.

§ 9.1.2 If unit prices are stated in the Contract Documents or subsequently agreed upon, and if quantities originally contemplated are materially changed so that application of such unit prices to the actual quantities causes substantial inequity to the Owner or Contractor, the applicable unit prices shall be equitably adjusted.

## § 9.2 Schedule of Values

Where the Contract is based on a stipulated sum or Guaranteed Maximum Price, the Contractor shall submit a schedule of values to the Architect before the first Application for Payment, allocating the entire Contract Sum to the various portions of the Work. The schedule of values shall be prepared in the form, and supported by the data to substantiate its accuracy, required by the Architect. This schedule, unless objected to by the Architect, shall be used as a basis for reviewing the Contractor's Applications for Payment. Any changes to the schedule of values shall be submitted to the Architect and supported by such data to substantiate its accuracy as the Architect may require, and unless objected to by the Architect, shall be used as a basis for reviewing the Contractor's subsequent Applications for Payment.

## § 9.3 Applications for Payment

§ 9.3.1 At least ten days before the date established for each progress payment, the Contractor shall submit to the Architect an itemized Application for Payment prepared in accordance with the schedule of values, if required under Section 9.2, for completed portions of the Work. The application shall be notarized, if required, and supported by all data substantiating the Contractor's right to payment that the Owner or Architect require, such as copies of requisitions, and releases and waivers of liens from Subcontractors and suppliers, and shall reflect retainage if provided for in the Contract Documents.

§ 9.3.1.1 As provided in Section 7.3.9, such applications may include requests for payment on account of changes in the Work that have been properly authorized by Construction Change Directives, or by interim determinations of the Architect, but not yet included in Change Orders.

§ 9.3.1.2 Applications for Payment shall not include requests for payment for portions of the Work for which the Contractor does not intend to pay a Subcontractor or supplier, unless such Work has been performed by others whom the Contractor intends to pay.

§ 9.3.2 Unless otherwise provided in the Contract Documents, payments shall be made on account of materials and equipment delivered and suitably stored at the site for subsequent incorporation in the Work. If approved in advance by the Owner, payment may similarly be made for materials and equipment suitably stored off the site at a location agreed upon in writing. Payment for materials and equipment stored on or off the site shall be conditioned upon compliance by the Contractor with procedures satisfactory to the Owner to establish the Owner's title to such materials and equipment or otherwise protect the Owner's interest, and shall include the costs of applicable insurance, storage, and transportation to the site, for such materials and equipment stored off the site.

§ 9.3.3 The Contractor warrants that title to all Work covered by an Application for Payment will pass to the Owner no later than the time of payment. The Contractor further warrants that upon submittal of an Application for Payment all Work for which Certificates for Payment have been previously issued and payments received from the Owner shall, to the best of the Contractor's knowledge, information, and belief, be free and clear of liens, claims, security interests, or encumbrances, in favor of the Contractor, Subcontractors, suppliers, or other persons or entities that provided labor, materials, and equipment relating to the Work.

## § 9.4 Certificates for Payment

§ 9.4.1 The Architect will, within seven days after receipt of the Contractor's Application for Payment, either (1) issue to the Owner a Certificate for Payment in the full amount of the Application for Payment, with a copy to the Contractor; or (2) issue to the Owner a Certificate for Payment for such amount as the Architect determines is properly due, and notify the Contractor and Owner of the Architect's reasons for withholding certification in part as provided in Section 9.5.1; or (3) withhold certification of the entire Application for Payment, and notify the Contractor and Owner of the Architect's reason for withholding certification in whole as provided in Section 9.5.1.

§ 9.4.2 The issuance of a Certificate for Payment will constitute a representation by the Architect to the Owner, based on the Architect's evaluation of the Work and the data in the Application for Payment, that, to the best of the Architect's knowledge, information, and belief, the Work has progressed to the point indicated, the quality of the Work is in accordance with the Contract Documents, and that the Contractor is entitled to payment in the amount certified. The

foregoing representations are subject to an evaluation of the Work for conformance with the Contract Documents upon Substantial Completion, to results of subsequent tests and inspections, to correction of minor deviations from the Contract Documents prior to completion, and to specific qualifications expressed by the Architect. However, the issuance of a Certificate for Payment will not be a representation that the Architect has (1) made exhaustive or continuous on-site inspections to check the quality or quantity of the Work; (2) reviewed construction means, methods, techniques, sequences, or procedures; (3) reviewed copies of requisitions received from Subcontractors and suppliers and other data requested by the Owner to substantiate the Contractor's right to payment; or (4) made examination to ascertain how or for what purpose the Contractor has used money previously paid on account of the Contract Sum.

#### § 9.5 Decisions to Withhold Certification

§ 9.5.1 The Architect may withhold a Certificate for Payment in whole or in part, to the extent reasonably necessary to protect the Owner, if in the Architect's opinion the representations to the Owner required by Section 9.4.2 cannot be made. If the Architect is unable to certify payment in the amount of the Application, the Architect will notify the Contractor and Owner as provided in Section 9.4.1. If the Contractor and Architect cannot agree on a revised amount, the Architect will promptly issue a Certificate for Payment for the amount for which the Architect is able to make such representations to the Owner. The Architect may also withhold a Certificate for Payment or, because of subsequently discovered evidence, may nullify the whole or a part of a Certificate for Payment previously issued, to such extent as may be necessary in the Architect's opinion to protect the Owner from loss for which the Contractor is responsible, including loss resulting from acts and omissions described in Section 3.3.2, because of

- .1 defective Work not remedied;
- .2 third party claims filed or reasonable evidence indicating probable filing of such claims, unless security acceptable to the Owner is provided by the Contractor;
- .3 failure of the Contractor to make payments properly to Subcontractors or suppliers for labor, materials or equipment;
- .4 reasonable evidence that the Work cannot be completed for the unpaid balance of the Contract Sum;
- .5 damage to the Owner or a Separate Contractor;
- .6 reasonable evidence that the Work will not be completed within the Contract Time, and that the unpaid balance would not be adequate to cover actual or liquidated damages for the anticipated delay; or
- .7 repeated failure to carry out the Work in accordance with the Contract Documents.

§ 9.5.2 When either party disputes the Architect's decision regarding a Certificate for Payment under Section 9.5.1, in whole or in part, that party may submit a Claim in accordance with Article 15.

§ 9.5.3 When the reasons for withholding certification are removed, certification will be made for amounts previously withheld.

§ 9.5.4 If the Architect withholds certification for payment under Section 9.5.1.3, the Owner may, at its sole option, issue joint checks to the Contractor and to any Subcontractor or supplier to whom the Contractor failed to make payment for Work properly performed or material or equipment suitably delivered. If the Owner makes payments by joint check, the Owner shall notify the Architect and the Contractor shall reflect such payment on its next Application for Payment.

#### § 9.6 Progress Payments

§ 9.6.1 After the Architect has issued a Certificate for Payment, the Owner shall make payment in the manner and within the time provided in the Contract Documents, and shall so notify the Architect.

§ 9.6.2 The Contractor shall pay each Subcontractor, no later than seven days after receipt of payment from the Owner, the amount to which the Subcontractor is entitled, reflecting percentages actually retained from payments to the Contractor on account of the Subcontractor's portion of the Work. The Contractor shall, by appropriate agreement with each Subcontractor, require each Subcontractor to make payments to Sub-subcontractors in a similar manner.

§ 9.6.3 The Architect will, on request, furnish to a Subcontractor, if practicable, information regarding percentages of completion or amounts applied for by the Contractor and action taken thereon by the Architect and Owner on account of portions of the Work done by such Subcontractor.

§ 9.6.4 The Owner has the right to request written evidence from the Contractor that the Contractor has properly paid Subcontractors and suppliers amounts paid by the Owner to the Contractor for subcontracted Work. If the Contractor fails to furnish such evidence within seven days, the Owner shall have the right to contact Subcontractors and suppliers



to ascertain whether they have been properly paid. Neither the Owner nor Architect shall have an obligation to pay, or to see to the payment of money to, a Subcontractor or supplier, except as may otherwise be required by law.

§ 9.6.5 The Contractor's payments to suppliers shall be treated in a manner similar to that provided in Sections 9.6.2, 9.6.3 and 9.6.4.

§ 9.6.6 A Certificate for Payment, a progress payment, or partial or entire use or occupancy of the Project by the Owner shall not constitute acceptance of Work not in accordance with the Contract Documents.

§ 9.6.7 Unless the Contractor provides the Owner with a payment bond in the full penal sum of the Contract Sum, payments received by the Contractor for Work properly performed by Subcontractors or provided by suppliers shall be held by the Contractor for those Subcontractors or suppliers who performed Work or furnished materials, or both, under contract with the Contractor for which payment was made by the Owner. Nothing contained herein shall require money to be placed in a separate account and not commingled with money of the Contractor, create any fiduciary liability or tort liability on the part of the Contractor for breach of trust, or entitle any person or entity to an award of punitive damages against the Contractor for breach of the requirements of this provision.

§ 9.6.8 Provided the Owner has fulfilled its payment obligations under the Contract Documents, the Contractor shall defend and indemnify the Owner from all loss, liability, damage or expense, including reasonable attorney's fees and litigation expenses, arising out of any lien claim or other claim for payment by any Subcontractor or supplier of any tier. Upon receipt of notice of a lien claim or other claim for payment, the Owner shall notify the Contractor. If approved by the applicable court, when required, the Contractor may substitute a surety bond for the property against which the lien or other claim for payment has been asserted.

#### § 9.7 Failure of Payment

If the Architect does not issue a Certificate for Payment, through no fault of the Contractor, within seven days after receipt of the Contractor's Application for Payment, or if the Owner does not pay the Contractor within seven days after the date established in the Contract Documents, the amount certified by the Architect or awarded by binding dispute resolution, then the Contractor may, upon seven additional days' notice to the Owner and Architect, stop the Work until payment of the amount owing has been received. The Contract Time shall be extended appropriately and the Contract Sum shall be increased by the amount of the Contractor's reasonable costs of shutdown, delay and start-up, plus interest as provided for in the Contract Documents.

#### § 9.8 Substantial Completion

§ 9.8.1 Substantial Completion is the stage in the progress of the Work when the Work or designated portion thereof is sufficiently complete in accordance with the Contract Documents so that the Owner can occupy or utilize the Work for its intended use.

§ 9.8.2 When the Contractor considers that the Work, or a portion thereof which the Owner agrees to accept separately, is substantially complete, the Contractor shall prepare and submit to the Architect a comprehensive list of items to be completed or corrected prior to final payment. Failure to include an item on such list does not alter the responsibility of the Contractor to complete all Work in accordance with the Contract Documents.

§ 9.8.3 Upon receipt of the Contractor's list, the Architect will make an inspection to determine whether the Work or designated portion thereof is substantially complete. If the Architect's inspection discloses any item, whether or not included on the Contractor's list, which is not sufficiently complete in accordance with the Contract Documents so that the Owner can occupy or utilize the Work or designated portion thereof for its intended use, the Contractor shall, before issuance of the Certificate of Substantial Completion, complete or correct such item upon notification by the Architect. In such case, the Contractor shall then submit a request for another inspection by the Architect to determine Substantial Completion.

§ 9.8.4 When the Work or designated portion thereof is substantially complete, the Architect will prepare a Certificate of Substantial Completion that shall establish the date of Substantial Completion; establish responsibilities of the Owner and Contractor for security, maintenance, heat, utilities, damage to the Work and insurance; and fix the time within which the Contractor shall finish all items on the list accompanying the Certificate. Warranties required by the Contract Documents shall commence on the date of Substantial Completion of the Work or designated portion thereof unless otherwise provided in the Certificate of Substantial Completion.

§ 9.8.5 The Certificate of Substantial Completion shall be submitted to the Owner and Contractor for their written acceptance of responsibilities assigned to them in the Certificate. Upon such acceptance, and consent of surety if any, the Owner shall make payment of retainage applying to the Work or designated portion thereof. Such payment shall be adjusted for Work that is incomplete or not in accordance with the requirements of the Contract Documents.

#### § 9.9 Partial Occupancy or Use

§ 9.9.1 The Owner may occupy or use any completed or partially completed portion of the Work at any stage when such portion is designated by separate agreement with the Contractor, provided such occupancy or use is consented to by the insurer and authorized by public authorities having jurisdiction over the Project. Such partial occupancy or use may commence whether or not the portion is substantially complete, provided the Owner and Contractor have accepted in writing the responsibilities assigned to each of them for payments, retainage, if any, security, maintenance, heat, utilities, damage to the Work and insurance, and have agreed in writing concerning the period for correction of the Work and commencement of warranties required by the Contract Documents. When the Contractor considers a portion substantially complete, the Contractor shall prepare and submit a list to the Architect as provided under Section 9.8.2. Consent of the Contractor to partial occupancy or use shall not be unreasonably withheld. The stage of the progress of the Work shall be determined by written agreement between the Owner and Contractor or, if no agreement is reached, by decision of the Architect.

§ 9.9.2 Immediately prior to such partial occupancy or use, the Owner, Contractor, and Architect shall jointly inspect the area to be occupied or portion of the Work to be used in order to determine and record the condition of the Work.

§ 9.9.3 Unless otherwise agreed upon, partial occupancy or use of a portion or portions of the Work shall not constitute acceptance of Work not complying with the requirements of the Contract Documents.

#### § 9.10 Final Completion and Final Payment

§ 9.10.1 Upon receipt of the Contractor's notice that the Work is ready for final inspection and acceptance and upon receipt of a final Application for Payment, the Architect will promptly make such inspection. When the Architect finds the Work acceptable under the Contract Documents and the Contract fully performed, the Architect will promptly issue a final Certificate for Payment stating that to the best of the Architect's knowledge, information and belief, and on the basis of the Architect's on-site visits and inspections, the Work has been completed in accordance with the Contract Documents and that the entire balance found to be due the Contractor and noted in the final Certificate is due and payable. The Architect's final Certificate for Payment will constitute a further representation that conditions listed in Section 9.10.2 as precedent to the Contractor's being entitled to final payment have been fulfilled.

§ 9.10.2 Neither final payment nor any remaining retained percentage shall become due until the Contractor submits to the Architect (1) an affidavit that payrolls, bills for materials and equipment, and other indebtedness connected with the Work for which the Owner or the Owner's property might be responsible or encumbered (less amounts withheld by Owner) have been paid or otherwise satisfied, (2) a certificate evidencing that insurance required by the Contract Documents to remain in force after final payment is currently in effect, (3) a written statement that the Contractor knows of no reason that the insurance will not be renewable to cover the period required by the Contract Documents, (4) consent of surety, if any, to final payment, (5) documentation of any special warranties, such as manufacturers' warranties or specific Subcontractor warranties, and (6) if required by the Owner, other data establishing payment or satisfaction of obligations, such as receipts and releases and waivers of liens, claims, security interests, or encumbrances arising out of the Contract, to the extent and in such form as may be designated by the Owner. If a Subcontractor refuses to furnish a release or waiver required by the Owner, the Contractor may furnish a bond satisfactory to the Owner to indemnify the Owner against such lien, claim, security interest, or encumbrance. If a lien, claim, security interest, or encumbrance remains unsatisfied after payments are made, the Contractor shall refund to the Owner all money that the Owner may be compelled to pay in discharging the lien, claim, security interest, or encumbrance, including all costs and reasonable attorneys' fees.

§ 9.10.3 If, after Substantial Completion of the Work, final completion thereof is materially delayed through no fault of the Contractor or by issuance of Change Orders affecting final completion, and the Architect so confirms, the Owner shall, upon application by the Contractor and certification by the Architect, and without terminating the Contract, make payment of the balance due for that portion of the Work fully completed, corrected, and accepted. If the remaining balance for Work not fully completed or corrected is less than retainage stipulated in the Contract Documents, and if bonds have been furnished, the written consent of the surety to payment of the balance due for that portion of the Work fully completed and accepted shall be submitted by the Contractor to the Architect prior to certification of such payment. Such payment shall be made under terms and conditions governing final payment, except that it shall not

constitute a waiver of Claims.

§ 9.10.4 The making of final payment shall constitute a waiver of Claims by the Owner except those arising from

- .1 liens, Claims, security interests, or encumbrances arising out of the Contract and unsettled;
- .2 failure of the Work to comply with the requirements of the Contract Documents;
- .3 terms of special warranties required by the Contract Documents; or
- .4 audits performed by the Owner, if permitted by the Contract Documents, after final payment.

§ 9.10.5 Acceptance of final payment by the Contractor, a Subcontractor, or a supplier, shall constitute a waiver of claims by that payee except those previously made in writing and identified by that payee as unsettled at the time of final Application for Payment.

## ARTICLE 10 PROTECTION OF PERSONS AND PROPERTY

### § 10.1 Safety Precautions and Programs

The Contractor shall be responsible for initiating, maintaining, and supervising all safety precautions and programs in connection with the performance of the Contract.

### § 10.2 Safety of Persons and Property

§ 10.2.1 The Contractor shall take reasonable precautions for safety of, and shall provide reasonable protection to prevent damage, injury, or loss to

- .1 employees on the Work and other persons who may be affected thereby;
- .2 the Work and materials and equipment to be incorporated therein, whether in storage on or off the site, under care, custody, or control of the Contractor, a Subcontractor, or a Sub-subcontractor; and
- .3 other property at the site or adjacent thereto, such as trees, shrubs, lawns, walks, pavements, roadways, structures, and utilities not designated for removal, relocation, or replacement in the course of construction.

§ 10.2.2 The Contractor shall comply with, and give notices required by applicable laws, statutes, ordinances, codes, rules and regulations, and lawful orders of public authorities, bearing on safety of persons or property or their protection from damage, injury, or loss.

§ 10.2.3 The Contractor shall implement, erect, and maintain, as required by existing conditions and performance of the Contract, reasonable safeguards for safety and protection, including posting danger signs and other warnings against hazards; promulgating safety regulations; and notifying the owners and users of adjacent sites and utilities of the safeguards.

§ 10.2.4 When use or storage of explosives or other hazardous materials or equipment, or unusual methods are necessary for execution of the Work, the Contractor shall exercise utmost care and carry on such activities under supervision of properly qualified personnel.

§ 10.2.5 The Contractor shall promptly remedy damage and loss (other than damage or loss insured under property insurance required by the Contract Documents) to property referred to in Sections 10.2.1.2 and 10.2.1.3 caused in whole or in part by the Contractor, a Subcontractor, a Sub-subcontractor, or anyone directly or indirectly employed by any of them, or by anyone for whose acts they may be liable and for which the Contractor is responsible under Sections 10.2.1.2 and 10.2.1.3. The Contractor may make a Claim for the cost to remedy the damage or loss to the extent damage or loss is attributable to acts or omissions of the Owner or Architect or anyone directly or indirectly employed by either of them, or by anyone for whose acts either of them may be liable, and not attributable to the fault or negligence of the Contractor. The foregoing obligations of the Contractor are in addition to the Contractor's obligations under Section 3.18.

§ 10.2.6 The Contractor shall designate a responsible member of the Contractor's organization at the site whose duty shall be the prevention of accidents. This person shall be the Contractor's superintendent unless otherwise designated by the Contractor in writing to the Owner and Architect.

§ 10.2.7 The Contractor shall not permit any part of the construction or site to be loaded so as to cause damage or create an unsafe condition.



### § 10.2.8 Injury or Damage to Person or Property

If either party suffers injury or damage to person or property because of an act or omission of the other party, or of others for whose acts such party is legally responsible, notice of the injury or damage, whether or not insured, shall be given to the other party within a reasonable time not exceeding 21 days after discovery. The notice shall provide sufficient detail to enable the other party to investigate the matter.

### § 10.3 Hazardous Materials and Substances

§ 10.3.1 The Contractor is responsible for compliance with any requirements included in the Contract Documents regarding hazardous materials or substances. If the Contractor encounters a hazardous material or substance not addressed in the Contract Documents and if reasonable precautions will be inadequate to prevent foreseeable bodily injury or death to persons resulting from a material or substance, including but not limited to asbestos or polychlorinated biphenyl (PCB), encountered on the site by the Contractor, the Contractor shall, upon recognizing the condition, immediately stop Work in the affected area and notify the Owner and Architect of the condition.

§ 10.3.2 Upon receipt of the Contractor's notice, the Owner shall obtain the services of a licensed laboratory to verify the presence or absence of the material or substance reported by the Contractor and, in the event such material or substance is found to be present, to cause it to be rendered harmless. Unless otherwise required by the Contract Documents, the Owner shall furnish in writing to the Contractor and Architect the names and qualifications of persons or entities who are to perform tests verifying the presence or absence of the material or substance or who are to perform the task of removal or safe containment of the material or substance. The Contractor and the Architect will promptly reply to the Owner in writing stating whether or not either has reasonable objection to the persons or entities proposed by the Owner. If either the Contractor or Architect has an objection to a person or entity proposed by the Owner, the Owner shall propose another to whom the Contractor and the Architect have no reasonable objection. When the material or substance has been rendered harmless, Work in the affected area shall resume upon written agreement of the Owner and Contractor. By Change Order, the Contract Time shall be extended appropriately and the Contract Sum shall be increased by the amount of the Contractor's reasonable additional costs of shutdown, delay, and start-up.

§ 10.3.3 To the fullest extent permitted by law, the Owner shall indemnify and hold harmless the Contractor, Subcontractors, Architect, Architect's consultants, and agents and employees of any of them from and against claims, damages, losses, and expenses, including but not limited to attorneys' fees, arising out of or resulting from performance of the Work in the affected area if in fact the material or substance presents the risk of bodily injury or death as described in Section 10.3.1 and has not been rendered harmless, provided that such claim, damage, loss, or expense is attributable to bodily injury, sickness, disease or death, or to injury to or destruction of tangible property (other than the Work itself), except to the extent that such damage, loss, or expense is due to the fault or negligence of the party seeking indemnity.

§ 10.3.4 The Owner shall not be responsible under this Section 10.3 for hazardous materials or substances the Contractor brings to the site unless such materials or substances are required by the Contract Documents. The Owner shall be responsible for hazardous materials or substances required by the Contract Documents, except to the extent of the Contractor's fault or negligence in the use and handling of such materials or substances.

§ 10.3.5 The Contractor shall reimburse the Owner for the cost and expense the Owner incurs (1) for remediation of hazardous materials or substances the Contractor brings to the site and negligently handles, or (2) where the Contractor fails to perform its obligations under Section 10.3.1, except to the extent that the cost and expense are due to the Owner's fault or negligence.

§ 10.3.6 If, without negligence on the part of the Contractor, the Contractor is held liable by a government agency for the cost of remediation of a hazardous material or substance solely by reason of performing Work as required by the Contract Documents, the Owner shall reimburse the Contractor for all cost and expense thereby incurred.

### § 10.4 Emergencies

In an emergency affecting safety of persons or property, the Contractor shall act, at the Contractor's discretion, to prevent threatened damage, injury, or loss. Additional compensation or extension of time claimed by the Contractor on account of an emergency shall be determined as provided in Article 15 and Article 7.

## ARTICLE 11 INSURANCE AND BONDS

### § 11.1 Contractor's Insurance and Bonds

§ 11.1.1 The Contractor shall purchase and maintain insurance of the types and limits of liability, containing the

endorsements, and subject to the terms and conditions, as described in the Agreement or elsewhere in the Contract Documents. The Contractor shall purchase and maintain the required insurance from an insurance company or insurance companies lawfully authorized to issue insurance in the jurisdiction where the Project is located. The Owner, Architect, and Architect's consultants shall be named as additional insureds under the Contractor's commercial general liability policy or as otherwise described in the Contract Documents.

§ 11.1.2 The Contractor shall provide surety bonds of the types, for such penal sums, and subject to such terms and conditions as required by the Contract Documents. The Contractor shall purchase and maintain the required bonds from a company or companies lawfully authorized to issue surety bonds in the jurisdiction where the Project is located.

§ 11.1.3 Upon the request of any person or entity appearing to be a potential beneficiary of bonds covering payment of obligations arising under the Contract, the Contractor shall promptly furnish a copy of the bonds or shall authorize a copy to be furnished.

§ 11.1.4 Notice of Cancellation or Expiration of Contractor's Required Insurance. Within three (3) business days of the date the Contractor becomes aware of an impending or actual cancellation or expiration of any insurance required by the Contract Documents, the Contractor shall provide notice to the Owner of such impending or actual cancellation or expiration. Upon receipt of notice from the Contractor, the Owner shall, unless the lapse in coverage arises from an act or omission of the Owner, have the right to stop the Work until the lapse in coverage has been cured by the procurement of replacement coverage by the Contractor. The furnishing of notice by the Contractor shall not relieve the Contractor of any contractual obligation to provide any required coverage.

## § 11.2 Owner's Insurance

§ 11.2.1 The Owner shall purchase and maintain insurance of the types and limits of liability, containing the endorsements, and subject to the terms and conditions, as described in the Agreement or elsewhere in the Contract Documents. The Owner shall purchase and maintain the required insurance from an insurance company or insurance companies lawfully authorized to issue insurance in the jurisdiction where the Project is located.

§ 11.2.2 Failure to Purchase Required Property Insurance. If the Owner fails to purchase and maintain the required property insurance, with all of the coverages and in the amounts described in the Agreement or elsewhere in the Contract Documents, the Owner shall inform the Contractor in writing prior to commencement of the Work. Upon receipt of notice from the Owner, the Contractor may delay commencement of the Work and may obtain insurance that will protect the interests of the Contractor, Subcontractors, and Sub-Subcontractors in the Work. When the failure to provide coverage has been cured or resolved, the Contract Sum and Contract Time shall be equitably adjusted. In the event the Owner fails to procure coverage, the Owner waives all rights against the Contractor, Subcontractors, and Sub-subcontractors to the extent the loss to the Owner would have been covered by the insurance to have been procured by the Owner. The cost of the insurance shall be charged to the Owner by a Change Order. If the Owner does not provide written notice, and the Contractor is damaged by the failure or neglect of the Owner to purchase or maintain the required insurance, the Owner shall reimburse the Contractor for all reasonable costs and damages attributable thereto.

§ 11.2.3 Notice of Cancellation or Expiration of Owner's Required Property Insurance. Within three (3) business days of the date the Owner becomes aware of an impending or actual cancellation or expiration of any property insurance required by the Contract Documents, the Owner shall provide notice to the Contractor of such impending or actual cancellation or expiration. Unless the lapse in coverage arises from an act or omission of the Contractor: (1) the Contractor, upon receipt of notice from the Owner, shall have the right to stop the Work until the lapse in coverage has been cured by the procurement of replacement coverage by either the Owner or the Contractor; (2) the Contract Time and Contract Sum shall be equitably adjusted; and (3) the Owner waives all rights against the Contractor, Subcontractors, and Sub-subcontractors to the extent any loss to the Owner would have been covered by the insurance had it not expired or been cancelled. If the Contractor purchases replacement coverage, the cost of the insurance shall be charged to the Owner by an appropriate Change Order. The furnishing of notice by the Owner shall not relieve the Owner of any contractual obligation to provide required insurance.

## § 11.3 Waivers of Subrogation

§ 11.3.1 The Owner and Contractor waive all rights against (1) each other and any of their subcontractors, sub-subcontractors, agents, and employees, each of the other; (2) the Architect and Architect's consultants; and (3) Separate Contractors, if any, and any of their subcontractors, sub-subcontractors, agents, and employees, for damages caused by fire, or other causes of loss, to the extent those losses are covered by property insurance required by the Agreement or other property insurance applicable to the Project, except such rights as they have to proceeds of such insurance. The

Owner or Contractor, as appropriate, shall require similar written waivers in favor of the individuals and entities identified above from the Architect, Architect's consultants, Separate Contractors, subcontractors, and sub-subcontractors. The policies of insurance purchased and maintained by each person or entity agreeing to waive claims pursuant to this section 11.3.1 shall not prohibit this waiver of subrogation. This waiver of subrogation shall be effective as to a person or entity (1) even though that person or entity would otherwise have a duty of indemnification, contractual or otherwise, (2) even though that person or entity did not pay the insurance premium directly or indirectly, or (3) whether or not the person or entity had an insurable interest in the damaged property.

§ 11.3.2 If during the Project construction period the Owner insures properties, real or personal or both, at or adjacent to the site by property insurance under policies separate from those insuring the Project, or if after final payment property insurance is to be provided on the completed Project through a policy or policies other than those insuring the Project during the construction period, to the extent permissible by such policies, the Owner waives all rights in accordance with the terms of Section 11.3.1 for damages caused by fire or other causes of loss covered by this separate property insurance.

#### § 11.4 Loss of Use, Business Interruption, and Delay in Completion Insurance

The Owner, at the Owner's option, may purchase and maintain insurance that will protect the Owner against loss of use of the Owner's property, or the inability to conduct normal operations, due to fire or other causes of loss. The Owner waives all rights of action against the Contractor and Architect for loss of use of the Owner's property, due to fire or other hazards however caused.

#### § 11.5 Adjustment and Settlement of Insured Loss

§ 11.5.1 A loss insured under the property insurance required by the Agreement shall be adjusted by the Owner as fiduciary and made payable to the Owner as fiduciary for the insureds, as their interests may appear, subject to requirements of any applicable mortgagee clause and of Section 11.5.2. The Owner shall pay the Architect and Contractor their just shares of insurance proceeds received by the Owner, and by appropriate agreements the Architect and Contractor shall make payments to their consultants and Subcontractors in similar manner.

§ 11.5.2 Prior to settlement of an insured loss, the Owner shall notify the Contractor of the terms of the proposed settlement as well as the proposed allocation of the insurance proceeds. The Contractor shall have 14 days from receipt of notice to object to the proposed settlement or allocation of the proceeds. If the Contractor does not object, the Owner shall settle the loss and the Contractor shall be bound by the settlement and allocation. Upon receipt, the Owner shall deposit the insurance proceeds in a separate account and make the appropriate distributions. Thereafter, if no other agreement is made or the Owner does not terminate the Contract for convenience, the Owner and Contractor shall execute a Change Order for reconstruction of the damaged or destroyed Work in the amount allocated for that purpose. If the Contractor timely objects to either the terms of the proposed settlement or the allocation of the proceeds, the Owner may proceed to settle the insured loss, and any dispute between the Owner and Contractor arising out of the settlement or allocation of the proceeds shall be resolved pursuant to Article 15. Pending resolution of any dispute, the Owner may issue a Construction Change Directive for the reconstruction of the damaged or destroyed Work.

### ARTICLE 12 UNCOVERING AND CORRECTION OF WORK

#### § 12.1 Uncovering of Work

§ 12.1.1 If a portion of the Work is covered contrary to the Architect's request or to requirements specifically expressed in the Contract Documents, it must, if requested in writing by the Architect, be uncovered for the Architect's examination and be replaced at the Contractor's expense without change in the Contract Time.

§ 12.1.2 If a portion of the Work has been covered that the Architect has not specifically requested to examine prior to its being covered, the Architect may request to see such Work and it shall be uncovered by the Contractor. If such Work is in accordance with the Contract Documents, the Contractor shall be entitled to an equitable adjustment to the Contract Sum and Contract Time as may be appropriate. If such Work is not in accordance with the Contract Documents, the costs of uncovering the Work, and the cost of correction, shall be at the Contractor's expense.

#### § 12.2 Correction of Work

##### § 12.2.1 Before Substantial Completion

The Contractor shall promptly correct Work rejected by the Architect or failing to conform to the requirements of the Contract Documents, discovered before Substantial Completion and whether or not fabricated, installed or completed. Costs of correcting such rejected Work, including additional testing and inspections, the cost of uncovering and replacement, and compensation for the Architect's services and expenses made necessary thereby, shall be at the

Contractor's expense.

#### § 12.2.2 After Substantial Completion

§ 12.2.2.1 In addition to the Contractor's obligations under Section 3.5, if, within one year after the date of Substantial Completion of the Work or designated portion thereof or after the date for commencement of warranties established under Section 9.9.1, or by terms of any applicable special warranty required by the Contract Documents, any of the Work is found to be not in accordance with the requirements of the Contract Documents, the Contractor shall correct it promptly after receipt of notice from the Owner to do so, unless the Owner has previously given the Contractor a written acceptance of such condition. The Owner shall give such notice promptly after discovery of the condition. During the one-year period for correction of Work, if the Owner fails to notify the Contractor and give the Contractor an opportunity to make the correction, the Owner waives the rights to require correction by the Contractor and to make a claim for breach of warranty. If the Contractor fails to correct nonconforming Work within a reasonable time during that period after receipt of notice from the Owner or Architect, the Owner may correct it in accordance with Section 2.5.

§ 12.2.2.2 The one-year period for correction of Work shall be extended with respect to portions of Work first performed after Substantial Completion by the period of time between Substantial Completion and the actual completion of that portion of the Work.

§ 12.2.2.3 The one-year period for correction of Work shall not be extended by corrective Work performed by the Contractor pursuant to this Section 12.2.

§ 12.2.3 The Contractor shall remove from the site portions of the Work that are not in accordance with the requirements of the Contract Documents and are neither corrected by the Contractor nor accepted by the Owner.

§ 12.2.4 The Contractor shall bear the cost of correcting destroyed or damaged construction of the Owner or Separate Contractors, whether completed or partially completed, caused by the Contractor's correction or removal of Work that is not in accordance with the requirements of the Contract Documents.

§ 12.2.5 Nothing contained in this Section 12.2 shall be construed to establish a period of limitation with respect to other obligations the Contractor has under the Contract Documents. Establishment of the one-year period for correction of Work as described in Section 12.2.2 relates only to the specific obligation of the Contractor to correct the Work, and has no relationship to the time within which the obligation to comply with the Contract Documents may be sought to be enforced, nor to the time within which proceedings may be commenced to establish the Contractor's liability with respect to the Contractor's obligations other than specifically to correct the Work.

#### § 12.3 Acceptance of Nonconforming Work

If the Owner prefers to accept Work that is not in accordance with the requirements of the Contract Documents, the Owner may do so instead of requiring its removal and correction, in which case the Contract Sum will be reduced as appropriate and equitable. Such adjustment shall be effected whether or not final payment has been made.

### ARTICLE 13 MISCELLANEOUS PROVISIONS

#### § 13.1 Governing Law

The Contract shall be governed by the law of the place where the Project is located, excluding that jurisdiction's choice of law rules. If the parties have selected arbitration as the method of binding dispute resolution, the Federal Arbitration Act shall govern Section 15.4.

#### § 13.2 Successors and Assigns

§ 13.2.1 The Owner and Contractor respectively bind themselves, their partners, successors, assigns, and legal representatives to covenants, agreements, and obligations contained in the Contract Documents. Except as provided in Section 13.2.2, neither party to the Contract shall assign the Contract as a whole without written consent of the other. If either party attempts to make an assignment without such consent, that party shall nevertheless remain legally responsible for all obligations under the Contract.

§ 13.2.2 The Owner may, without consent of the Contractor, assign the Contract to a lender providing construction financing for the Project, if the lender assumes the Owner's rights and obligations under the Contract Documents. The Contractor shall execute all consents reasonably required to facilitate the assignment.



### § 13.3 Rights and Remedies

§ 13.3.1 Duties and obligations imposed by the Contract Documents and rights and remedies available thereunder shall be in addition to and not a limitation of duties, obligations, rights, and remedies otherwise imposed or available by law.

§ 13.3.2 No action or failure to act by the Owner, Architect, or Contractor shall constitute a waiver of a right or duty afforded them under the Contract, nor shall such action or failure to act constitute approval of or acquiescence in a breach thereunder, except as may be specifically agreed upon in writing.

### § 13.4 Tests and Inspections

§ 13.4.1 Tests, inspections, and approvals of portions of the Work shall be made as required by the Contract Documents and by applicable laws, statutes, ordinances, codes, rules, and regulations or lawful orders of public authorities. Unless otherwise provided, the Contractor shall make arrangements for such tests, inspections, and approvals with an independent testing laboratory or entity acceptable to the Owner, or with the appropriate public authority, and shall bear all related costs of tests, inspections, and approvals. The Contractor shall give the Architect timely notice of when and where tests and inspections are to be made so that the Architect may be present for such procedures. The Owner shall bear costs of tests, inspections, or approvals that do not become requirements until after bids are received or negotiations concluded. The Owner shall directly arrange and pay for tests, inspections, or approvals where building codes or applicable laws or regulations so require.

§ 13.4.2 If the Architect, Owner, or public authorities having jurisdiction determine that portions of the Work require additional testing, inspection, or approval not included under Section 13.4.1, the Architect will, upon written authorization from the Owner, instruct the Contractor to make arrangements for such additional testing, inspection, or approval, by an entity acceptable to the Owner, and the Contractor shall give timely notice to the Architect of when and where tests and inspections are to be made so that the Architect may be present for such procedures. Such costs, except as provided in Section 13.4.3, shall be at the Owner's expense.

§ 13.4.3 If procedures for testing, inspection, or approval under Sections 13.4.1 and 13.4.2 reveal failure of the portions of the Work to comply with requirements established by the Contract Documents, all costs made necessary by such failure, including those of repeated procedures and compensation for the Architect's services and expenses, shall be at the Contractor's expense.

§ 13.4.4 Required certificates of testing, inspection, or approval shall, unless otherwise required by the Contract Documents, be secured by the Contractor and promptly delivered to the Architect.

§ 13.4.5 If the Architect is to observe tests, inspections, or approvals required by the Contract Documents, the Architect will do so promptly and, where practicable, at the normal place of testing.

§ 13.4.6 Tests or inspections conducted pursuant to the Contract Documents shall be made promptly to avoid unreasonable delay in the Work.

### § 13.5 Interest

Payments due and unpaid under the Contract Documents shall bear interest from the date payment is due at the rate the parties agree upon in writing or, in the absence thereof, at the legal rate prevailing from time to time at the place where the Project is located.

## ARTICLE 14 TERMINATION OR SUSPENSION OF THE CONTRACT

### § 14.1 Termination by the Contractor

§ 14.1.1 The Contractor may terminate the Contract if the Work is stopped for a period of 30 consecutive days through no act or fault of the Contractor, a Subcontractor, a Sub-subcontractor, their agents or employees, or any other persons or entities performing portions of the Work, for any of the following reasons:

- .1 Issuance of an order of a court or other public authority having jurisdiction that requires all Work to be stopped;
- .2 An act of government, such as a declaration of national emergency, that requires all Work to be stopped;
- .3 Because the Architect has not issued a Certificate for Payment and has not notified the Contractor of the reason for withholding certification as provided in Section 9.4.1, or because the Owner has not made payment on a Certificate for Payment within the time stated in the Contract Documents; or
- .4 The Owner has failed to furnish to the Contractor reasonable evidence as required by Section 2.2.

§ 14.1.2 The Contractor may terminate the Contract if, through no act or fault of the Contractor, a Subcontractor, a Sub-subcontractor, their agents or employees, or any other persons or entities performing portions of the Work, repeated suspensions, delays, or interruptions of the entire Work by the Owner as described in Section 14.3, constitute in the aggregate more than 100 percent of the total number of days scheduled for completion, or 120 days in any 365-day period, whichever is less.

§ 14.1.3 If one of the reasons described in Section 14.1.1 or 14.1.2 exists, the Contractor may, upon seven days' notice to the Owner and Architect, terminate the Contract and recover from the Owner payment for Work executed, as well as reasonable overhead and profit on Work not executed, and costs incurred by reason of such termination.

§ 14.1.4 If the Work is stopped for a period of 60 consecutive days through no act or fault of the Contractor, a Subcontractor, a Sub-subcontractor, or their agents or employees or any other persons or entities performing portions of the Work because the Owner has repeatedly failed to fulfill the Owner's obligations under the Contract Documents with respect to matters important to the progress of the Work, the Contractor may, upon seven additional days' notice to the Owner and the Architect, terminate the Contract and recover from the Owner as provided in Section 14.1.3.

## § 14.2 Termination by the Owner for Cause

§ 14.2.1 The Owner may terminate the Contract if the Contractor

- .1 repeatedly refuses or fails to supply enough properly skilled workers or proper materials;
- .2 fails to make payment to Subcontractors or suppliers in accordance with the respective agreements between the Contractor and the Subcontractors or Suppliers;
- .3 repeatedly disregards applicable laws, statutes, ordinances, codes, rules and regulations, or lawful orders of a public authority; or
- .4 otherwise is guilty of substantial breach of a provision of the Contract Documents.

§ 14.2.2 When any of the reasons described in Section 14.2.1 exist, and upon certification by the Architect that sufficient cause exists to justify such action, the Owner may, without prejudice to any other rights or remedies of the Owner and after giving the Contractor and the Contractor's surety, if any, seven days' notice, terminate employment of the Contractor and may, subject to any prior rights of the surety:

- .1 Exclude the Contractor from the site and take possession of all materials, equipment, tools, and construction equipment and machinery thereon owned by the Contractor;
- .2 Accept assignment of subcontracts pursuant to Section 5.4; and
- .3 Finish the Work by whatever reasonable method the Owner may deem expedient. Upon written request of the Contractor, the Owner shall furnish to the Contractor a detailed accounting of the costs incurred by the Owner in finishing the Work.

§ 14.2.3 When the Owner terminates the Contract for one of the reasons stated in Section 14.2.1, the Contractor shall not be entitled to receive further payment until the Work is finished.

§ 14.2.4 If the unpaid balance of the Contract Sum exceeds costs of finishing the Work, including compensation for the Architect's services and expenses made necessary thereby, and other damages incurred by the Owner and not expressly waived, such excess shall be paid to the Contractor. If such costs and damages exceed the unpaid balance, the Contractor shall pay the difference to the Owner. The amount to be paid to the Contractor or Owner, as the case may be, shall be certified by the Initial Decision Maker, upon application, and this obligation for payment shall survive termination of the Contract.

## § 14.3 Suspension by the Owner for Convenience

§ 14.3.1 The Owner may, without cause, order the Contractor in writing to suspend, delay or interrupt the Work, in whole or in part for such period of time as the Owner may determine.

§ 14.3.2 The Contract Sum and Contract Time shall be adjusted for increases in the cost and time caused by suspension, delay, or interruption under Section 14.3.1. Adjustment of the Contract Sum shall include profit. No adjustment shall be made to the extent

- .1 that performance is, was, or would have been, so suspended, delayed, or interrupted, by another cause for which the Contractor is responsible; or
- .2 that an equitable adjustment is made or denied under another provision of the Contract.

#### § 14.4 Termination by the Owner for Convenience

§ 14.4.1 The Owner may, at any time, terminate the Contract for the Owner's convenience and without cause.

§ 14.4.2 Upon receipt of notice from the Owner of such termination for the Owner's convenience, the Contractor shall

- .1 cease operations as directed by the Owner in the notice;
- .2 take actions necessary, or that the Owner may direct, for the protection and preservation of the Work; and
- .3 except for Work directed to be performed prior to the effective date of termination stated in the notice, terminate all existing subcontracts and purchase orders and enter into no further subcontracts and purchase orders.

§ 14.4.3 In case of such termination for the Owner's convenience, the Owner shall pay the Contractor for Work properly executed; costs incurred by reason of the termination, including costs attributable to termination of Subcontracts; and the termination fee, if any, set forth in the Agreement.

### ARTICLE 15 CLAIMS AND DISPUTES

#### § 15.1 Claims

##### § 15.1.1 Definition

A Claim is a demand or assertion by one of the parties seeking, as a matter of right, payment of money, a change in the Contract Time, or other relief with respect to the terms of the Contract. The term "Claim" also includes other disputes and matters in question between the Owner and Contractor arising out of or relating to the Contract. The responsibility to substantiate Claims shall rest with the party making the Claim. This Section 15.1.1 does not require the Owner to file a Claim in order to impose liquidated damages in accordance with the Contract Documents.

##### § 15.1.2 Time Limits on Claims

The Owner and Contractor shall commence all Claims and causes of action against the other and arising out of or related to the Contract, whether in contract, tort, breach of warranty or otherwise, in accordance with the requirements of the binding dispute resolution method selected in the Agreement and within the period specified by applicable law, but in any case not more than 10 years after the date of Substantial Completion of the Work. The Owner and Contractor waive all Claims and causes of action not commenced in accordance with this Section 15.1.2.

##### § 15.1.3 Notice of Claims

§ 15.1.3.1 Claims by either the Owner or Contractor, where the condition giving rise to the Claim is first discovered prior to expiration of the period for correction of the Work set forth in Section 12.2.2, shall be initiated by notice to the other party and to the Initial Decision Maker with a copy sent to the Architect, if the Architect is not serving as the Initial Decision Maker. Claims by either party under this Section 15.1.3.1 shall be initiated within 21 days after occurrence of the event giving rise to such Claim or within 21 days after the claimant first recognizes the condition giving rise to the Claim, whichever is later.

§ 15.1.3.2 Claims by either the Owner or Contractor, where the condition giving rise to the Claim is first discovered after expiration of the period for correction of the Work set forth in Section 12.2.2, shall be initiated by notice to the other party. In such event, no decision by the Initial Decision Maker is required.

##### § 15.1.4 Continuing Contract Performance

§ 15.1.4.1 Pending final resolution of a Claim, except as otherwise agreed in writing or as provided in Section 9.7 and Article 14, the Contractor shall proceed diligently with performance of the Contract and the Owner shall continue to make payments in accordance with the Contract Documents.

§ 15.1.4.2 The Contract Sum and Contract Time shall be adjusted in accordance with the Initial Decision Maker's decision, subject to the right of either party to proceed in accordance with this Article 15. The Architect will issue Certificates for Payment in accordance with the decision of the Initial Decision Maker.

##### § 15.1.5 Claims for Additional Cost

If the Contractor wishes to make a Claim for an increase in the Contract Sum, notice as provided in Section 15.1.3 shall be given before proceeding to execute the portion of the Work that is the subject of the Claim. Prior notice is not required for Claims relating to an emergency endangering life or property arising under Section 10.4.

##### § 15.1.6 Claims for Additional Time

§ 15.1.6.1 If the Contractor wishes to make a Claim for an increase in the Contract Time, notice as provided in Section

15.1.3 shall be given. The Contractor's Claim shall include an estimate of cost and of probable effect of delay on progress of the Work. In the case of a continuing delay, only one Claim is necessary.

§ 15.1.6.2 If adverse weather conditions are the basis for a Claim for additional time, such Claim shall be documented by data substantiating that weather conditions were abnormal for the period of time, could not have been reasonably anticipated, and had an adverse effect on the scheduled construction.

#### § 15.1.7 Waiver of Claims for Consequential Damages

The Contractor and Owner waive Claims against each other for consequential damages arising out of or relating to this Contract. This mutual waiver includes

- .1 damages incurred by the Owner for rental expenses, for losses of use, income, profit, financing, business and reputation, and for loss of management or employee productivity or of the services of such persons; and
- .2 damages incurred by the Contractor for principal office expenses including the compensation of personnel stationed there, for losses of financing, business and reputation, and for loss of profit, except anticipated profit arising directly from the Work.

This mutual waiver is applicable, without limitation, to all consequential damages due to either party's termination in accordance with Article 14. Nothing contained in this Section 15.1.7 shall be deemed to preclude assessment of liquidated damages, when applicable, in accordance with the requirements of the Contract Documents.

#### § 15.2 Initial Decision

§ 15.2.1 Claims, excluding those where the condition giving rise to the Claim is first discovered after expiration of the period for correction of the Work set forth in Section 12.2.2 or arising under Sections 10.3, 10.4, and 11.5, shall be referred to the Initial Decision Maker for initial decision. The Architect will serve as the Initial Decision Maker, unless otherwise indicated in the Agreement. Except for those Claims excluded by this Section 15.2.1, an initial decision shall be required as a condition precedent to mediation of any Claim. If an initial decision has not been rendered within 30 days after the Claim has been referred to the Initial Decision Maker, the party asserting the Claim may demand mediation and binding dispute resolution without a decision having been rendered. Unless the Initial Decision Maker and all affected parties agree, the Initial Decision Maker will not decide disputes between the Contractor and persons or entities other than the Owner.

§ 15.2.2 The Initial Decision Maker will review Claims and within ten days of the receipt of a Claim take one or more of the following actions: (1) request additional supporting data from the claimant or a response with supporting data from the other party, (2) reject the Claim in whole or in part, (3) approve the Claim, (4) suggest a compromise, or (5) advise the parties that the Initial Decision Maker is unable to resolve the Claim if the Initial Decision Maker lacks sufficient information to evaluate the merits of the Claim or if the Initial Decision Maker concludes that, in the Initial Decision Maker's sole discretion, it would be inappropriate for the Initial Decision Maker to resolve the Claim.

§ 15.2.3 In evaluating Claims, the Initial Decision Maker may, but shall not be obligated to, consult with or seek information from either party or from persons with special knowledge or expertise who may assist the Initial Decision Maker in rendering a decision. The Initial Decision Maker may request the Owner to authorize retention of such persons at the Owner's expense.

§ 15.2.4 If the Initial Decision Maker requests a party to provide a response to a Claim or to furnish additional supporting data, such party shall respond, within ten days after receipt of the request, and shall either (1) provide a response on the requested supporting data, (2) advise the Initial Decision Maker when the response or supporting data will be furnished, or (3) advise the Initial Decision Maker that no supporting data will be furnished. Upon receipt of the response or supporting data, if any, the Initial Decision Maker will either reject or approve the Claim in whole or in part.

§ 15.2.5 The Initial Decision Maker will render an initial decision approving or rejecting the Claim, or indicating that the Initial Decision Maker is unable to resolve the Claim. This initial decision shall (1) be in writing; (2) state the reasons therefor; and (3) notify the parties and the Architect, if the Architect is not serving as the Initial Decision Maker, of any change in the Contract Sum or Contract Time or both. The initial decision shall be final and binding on the parties but subject to mediation and, if the parties fail to resolve their dispute through mediation, to binding dispute resolution.

§ 15.2.6 Either party may file for mediation of an initial decision at any time, subject to the terms of Section 15.2.6.1.



§ 15.2.6.1 Either party may, within 30 days from the date of receipt of an initial decision, demand in writing that the other party file for mediation. If such a demand is made and the party receiving the demand fails to file for mediation within 30 days after receipt thereof, then both parties waive their rights to mediate or pursue binding dispute resolution proceedings with respect to the initial decision.

§ 15.2.7 In the event of a Claim against the Contractor, the Owner may, but is not obligated to, notify the surety, if any, of the nature and amount of the Claim. If the Claim relates to a possibility of a Contractor's default, the Owner may, but is not obligated to, notify the surety and request the surety's assistance in resolving the controversy.

§ 15.2.8 If a Claim relates to or is the subject of a mechanic's lien, the party asserting such Claim may proceed in accordance with applicable law to comply with the lien notice or filing deadlines.

### § 15.3 Mediation

§ 15.3.1 Claims, disputes, or other matters in controversy arising out of or related to the Contract, except those waived as provided for in Sections 9.10.4, 9.10.5, and 15.1.7, shall be subject to mediation as a condition precedent to binding dispute resolution.

§ 15.3.2 The parties shall endeavor to resolve their Claims by mediation which, unless the parties mutually agree otherwise, shall be administered by the American Arbitration Association in accordance with its Construction Industry Mediation Procedures in effect on the date of the Agreement. A request for mediation shall be made in writing, delivered to the other party to the Contract, and filed with the person or entity administering the mediation. The request may be made concurrently with the filing of binding dispute resolution proceedings but, in such event, mediation shall proceed in advance of binding dispute resolution proceedings, which shall be stayed pending mediation for a period of 60 days from the date of filing, unless stayed for a longer period by agreement of the parties or court order. If an arbitration is stayed pursuant to this Section 15.3.2, the parties may nonetheless proceed to the selection of the arbitrator(s) and agree upon a schedule for later proceedings.

§ 15.3.3 Either party may, within 30 days from the date that mediation has been concluded without resolution of the dispute or 60 days after mediation has been demanded without resolution of the dispute, demand in writing that the other party file for binding dispute resolution. If such a demand is made and the party receiving the demand fails to file for binding dispute resolution within 60 days after receipt thereof, then both parties waive their rights to binding dispute resolution proceedings with respect to the initial decision.

§ 15.3.4 The parties shall share the mediator's fee and any filing fees equally. The mediation shall be held in the place where the Project is located, unless another location is mutually agreed upon. Agreements reached in mediation shall be enforceable as settlement agreements in any court having jurisdiction thereof.

### § 15.4 Arbitration

§ 15.4.1 If the parties have selected arbitration as the method for binding dispute resolution in the Agreement, any Claim subject to, but not resolved by, mediation shall be subject to arbitration which, unless the parties mutually agree otherwise, shall be administered by the American Arbitration Association in accordance with its Construction Industry Arbitration Rules in effect on the date of the Agreement. The Arbitration shall be conducted in the place where the Project is located, unless another location is mutually agreed upon. A demand for arbitration shall be made in writing, delivered to the other party to the Contract, and filed with the person or entity administering the arbitration. The party filing a notice of demand for arbitration must assert in the demand all Claims then known to that party on which arbitration is permitted to be demanded.

§ 15.4.1.1 A demand for arbitration shall be made no earlier than concurrently with the filing of a request for mediation, but in no event shall it be made after the date when the institution of legal or equitable proceedings based on the Claim would be barred by the applicable statute of limitations. For statute of limitations purposes, receipt of a written demand for arbitration by the person or entity administering the arbitration shall constitute the institution of legal or equitable proceedings based on the Claim.

§ 15.4.2 The award rendered by the arbitrator or arbitrators shall be final, and judgment may be entered upon it in accordance with applicable law in any court having jurisdiction thereof.

§ 15.4.3 The foregoing agreement to arbitrate and other agreements to arbitrate with an additional person or entity duly

consented to by parties to the Agreement, shall be specifically enforceable under applicable law in any court having jurisdiction thereof.

#### § 15.4.4 Consolidation or Joinder

§ 15.4.4.1 Subject to the rules of the American Arbitration Association or other applicable arbitration rules, either party may consolidate an arbitration conducted under this Agreement with any other arbitration to which it is a party provided that (1) the arbitration agreement governing the other arbitration permits consolidation, (2) the arbitrations to be consolidated substantially involve common questions of law or fact, and (3) the arbitrations employ materially similar procedural rules and methods for selecting arbitrator(s).

§ 15.4.4.2 Subject to the rules of the American Arbitration Association or other applicable arbitration rules, either party may include by joinder persons or entities substantially involved in a common question of law or fact whose presence is required if complete relief is to be accorded in arbitration, provided that the party sought to be joined consents in writing to such joinder. Consent to arbitration involving an additional person or entity shall not constitute consent to arbitration of any claim, dispute or other matter in question not described in the written consent.

§ 15.4.4.3 The Owner and Contractor grant to any person or entity made a party to an arbitration conducted under this Section 15.4, whether by joinder or consolidation, the same rights of joinder and consolidation as those of the Owner and Contractor under this Agreement.

SECTION 00510

**NOTICE OF AWARD**

**PART 1 - GENERAL**

**1.01 SECTION INCLUDES**

This section describes the standardized Notice of Award form for use in the project.

**1.02 REFERENCES – Not Used**

**1.03 DEFINITIONS – Section 0700**

**PART 2 - PRODUCT – Not Used**

**PART 3 - EXECUTION (FORMS ON FOLLOWING PAGES)**

STANDARDIZED FORM FOLLOWS

**NOTICE OF AWARD**

Date

Owner: City of Edinburg

Owner's Project No.: 2019-20

Architect: Rodolfo R. Molina, AIA

Project: Fire Station No.5

Bidder: *Name*

Bidder's Address: *Address*

You are notified that Owner has accepted your Proposal dated **(Date)** for the above Contract, and that you are the Successful Proposal and are awarded a Contract for:

**Base Proposal**

The Contract Price of the awarded Contract is **(Contract Amount)**. Contract Price is subject to adjustment based on the provisions of the Contract, including but not limited to those governing changes, Unit Price Work, and Work performed on a cost-plus-fee basis, as applicable.

**Four (4)** unexecuted counterparts of the Agreement accompany this Notice of Award, and one copy of the Contract Documents accompanies this Notice of Award, or has been transmitted or made available to Bidder electronically.

☒ Drawings will be delivered separately from the other Contract Documents by Architect of Record.

You must comply with the following conditions precedent within 15 days of the date of receipt of this Notice of Award:

1. Deliver to Owner **four (4)** counterparts of the Agreement, signed by Bidder (as Contractor).
2. Deliver with the signed Agreement(s) the Contract security (such as required performance and payment bonds) and insurance documentation, as specified in the Instructions to Bidders and in the General Conditions, Articles 2 and 6.

Failure to comply with these conditions within the time specified will entitle Owner to consider you in default, annul this Notice of Award, and declare your Bid security forfeited.

Within 10 days after you comply with the above conditions, Owner will return to you one fully signed counterpart of the Agreement, together with any additional copies of the Contract Documents as indicated in Paragraph 2.02 of the General Conditions.

Owner: **City of Edinburg**

By (signature):

Name (printed): Mardoqueo Hinojosa, P.E., CFM, CPM

Title: City Engineer

Cc: Milnet Architectural Services, PLLC.

**END OF SECTION**

Page Intentionally Blank

## SECTION 00550

**NOTICE TO PROCEED****PART 1 - GENERAL****1.01 SECTION INCLUDES**

This section describes the standardized Notice to Proceed form for use in the project.

**1.02 REFERENCES – Not Used****1.03 DEFINITIONS – Section 0700****PART 2 - PRODUCT – Not Used****PART 3 - EXECUTION**

**TO BE ISSUED BY ARCHITECT**

**NOTICE TO PROCEED****Date:****To:****Project No.:** 2019-20**Project:** Fire Station No. 5

You are notified that the Contract Time under the above contract will commence to run on \_\_\_\_\_. By this date you are to start performing your obligations under the Contract Documents.

In accordance with the Agreement the date of Substantial Completion is \_\_\_\_\_ and Final Completion is \_\_\_\_\_, respectively.

Before you may start any Work at the site, the General Conditions and Contract Documents provides that you and Owner must each deliver to the other (with copies to Architect) certificates of insurance which each is required to purchase and maintain in accordance with the Contract Documents.

Also before you may start any work at the site you must

1. Notify the City 48 hours prior to beginning construction.
2. Setup construction barricades.
3. Setup erosion control measures.
4. \_\_\_\_\_
5. \_\_\_\_\_

Copy to Architect:

ARCHITECT: Milnet Architectural Services, PLLC.

By \_\_\_\_\_  
Rodolfo R. Molina, AIA

\_\_\_\_\_  
President  
Title

OWNER: City of Edinburg

By \_\_\_\_\_  
Juan G. Guerra

\_\_\_\_\_  
City Manager  
Title

**ACCEPTANCE OF NOTICE BY BIDDER**

Receipt of the above NOTICE TO PROCEED is hereby acknowledged by \_\_\_\_\_,  
this the \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_. (Contractor)

BY: \_\_\_\_\_

TITLE: \_\_\_\_\_

**END OF SECTION**



**PERFORMANCE BOND**

|                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contractor</b><br>Name: <b>[Full formal name of Contractor]</b><br>Address <i>(principal place of business)</i> :<br><b>[Address of Contractor's principal place of business]</b>                                                                         | <b>Surety</b><br>Name: <b>[Full formal name of Surety]</b><br>Address <i>(principal place of business)</i> :<br><b>[Address of Surety's principal place of business]</b>                                                                               |
| <b>Owner</b><br>Name: <b>City of Edinburg</b><br>Mailing address <i>(principal place of business)</i> :<br><b>415 W. University Drive</b><br><b>Edinburg, TX. 78539</b>                                                                                      | <b>Contract</b><br>Description <i>(name and location)</i> :<br><b>2019-20 / Fire Station No. 5</b><br><b>Davis Road, Edinburg, TX.</b><br><br>Contract Price: <b>[Amount from Contract]</b><br>Effective Date of Contract: <b>[Date from Contract]</b> |
| <b>Bond</b><br>Bond <b>[Amount]</b><br>Date of Bond: <b>[Date]</b><br><i>(Date of Bond cannot be earlier than Effective Date of Contract)</i><br>Modifications to this Bond form:<br><input type="checkbox"/> None <input type="checkbox"/> See Paragraph 16 |                                                                                                                                                                                                                                                        |
| Surety and Contractor, intending to be legally bound hereby, subject to the terms set forth in this Performance Bond, do each cause this Performance Bond to be duly executed by an authorized officer, agent, or representative.                            |                                                                                                                                                                                                                                                        |
| Contractor as Principal                                                                                                                                                                                                                                      | Surety                                                                                                                                                                                                                                                 |
| <i>(Full formal name of Contractor)</i>                                                                                                                                                                                                                      | <i>(Full formal name of Surety) (corporate seal)</i>                                                                                                                                                                                                   |
| By: _____<br><i>(Signature)</i>                                                                                                                                                                                                                              | By: _____<br><i>(Signature)(Attach Power of Attorney)</i>                                                                                                                                                                                              |
| Name: _____<br><i>(Printed or typed)</i>                                                                                                                                                                                                                     | Name: _____<br><i>(Printed or typed)</i>                                                                                                                                                                                                               |
| Title: _____                                                                                                                                                                                                                                                 | Title: _____                                                                                                                                                                                                                                           |
| Attest: _____<br><i>(Signature)</i>                                                                                                                                                                                                                          | Attest: _____<br><i>(Signature)</i>                                                                                                                                                                                                                    |
| Name: _____<br><i>(Printed or typed)</i>                                                                                                                                                                                                                     | Name: _____<br><i>(Printed or typed)</i>                                                                                                                                                                                                               |
| Title: _____                                                                                                                                                                                                                                                 | Title: _____                                                                                                                                                                                                                                           |
| <i>Notes: (1) Provide supplemental execution by any additional parties, such as joint venturers. (2) Any singular reference to Contractor, Surety, Owner, or other party is considered plural where applicable.</i>                                          |                                                                                                                                                                                                                                                        |

1. The Contractor and Surety, jointly and severally, bind themselves, their heirs, executors, administrators, successors, and assigns to the Owner for the performance of the Construction Contract, which is incorporated herein by reference.
2. If the Contractor performs the Construction Contract, the Surety and the Contractor shall have no obligation under this Bond, except when applicable to participate in a conference as provided in Paragraph 3.
3. If there is no Owner Default under the Construction Contract, the Surety's obligation under this Bond will arise after:
  - 3.1. The Owner first provides notice to the Contractor and the Surety that the Owner is considering declaring a Contractor Default. Such notice may indicate whether the Owner is requesting a conference among the Owner, Contractor, and Surety to discuss the Contractor's performance. If the Owner does not request a conference, the Surety may, within five (5) business days after receipt of the Owner's notice, request such a conference. If the Surety timely requests a conference, the Owner shall attend. Unless the Owner agrees otherwise, any conference requested under this Paragraph 3.1 will be held within ten (10) business days of the Surety's receipt of the Owner's notice. If the Owner, the Contractor, and the Surety agree, the Contractor shall be allowed a reasonable time to perform the Construction Contract, but such an agreement does not waive the Owner's right, if any, subsequently to declare a Contractor Default;
  - 3.2. The Owner declares a Contractor Default, terminates the Construction Contract and notifies the Surety; and
  - 3.3. The Owner has agreed to pay the Balance of the Contract Price in accordance with the terms of the Construction Contract to the Surety or to a contractor selected to perform the Construction Contract.
4. Failure on the part of the Owner to comply with the notice requirement in Paragraph 3.1 does not constitute a failure to comply with a condition precedent to the Surety's obligations, or release the Surety from its obligations, except to the extent the Surety demonstrates actual prejudice.
5. When the Owner has satisfied the conditions of Paragraph 3, the Surety shall promptly and at the Surety's expense take one of the following actions:
  - 5.1. Arrange for the Contractor, with the consent of the Owner, to perform and complete the Construction Contract;
  - 5.2. Undertake to perform and complete the Construction Contract itself, through its agents or independent contractors;
  - 5.3. Obtain bids or negotiated proposals from qualified contractors acceptable to the Owner for a contract for performance and completion of the Construction Contract, arrange for a contract to be prepared for execution by the Owner and a contractor selected with the Owners concurrence, to be secured with performance and payment bonds executed by a qualified surety equivalent to the bonds issued on the Construction Contract, and pay to the Owner the amount of damages as described in Paragraph 7 in excess of the Balance of the Contract Price incurred by the Owner as a result of the Contractor Default; or

- 5.4. Waive its right to perform and complete, arrange for completion, or obtain a new contractor, and with reasonable promptness under the circumstances:
  - 5.4.1 After investigation, determine the amount for which it may be liable to the Owner and, as soon as practicable after the amount is determined, make payment to the Owner; or
  - 5.4.2 Deny liability in whole or in part and notify the Owner, citing the reasons for denial.
6. If the Surety does not proceed as provided in Paragraph 5 with reasonable promptness, the Surety shall be deemed to be in default on this Bond seven days after receipt of an additional written notice from the Owner to the Surety demanding that the Surety perform its obligations under this Bond, and the Owner shall be entitled to enforce any remedy available to the Owner. If the Surety proceeds as provided in Paragraph 5.4, and the Owner refuses the payment, or the Surety has denied liability, in whole or in part, without further notice, the Owner shall be entitled to enforce any remedy available to the Owner.
7. If the Surety elects to act under Paragraph 5.1, 5.2, or 5.3, then the responsibilities of the Surety to the Owner will not be greater than those of the Contractor under the Construction Contract, and the responsibilities of the Owner to the Surety will not be greater than those of the Owner under the Construction Contract. Subject to the commitment by the Owner to pay the Balance of the Contract Price, the Surety is obligated, without duplication for:
  - 7.1. the responsibilities of the Contractor for correction of defective work and completion of the Construction Contract;
  - 7.2. additional legal, design professional, and delay costs resulting from the Contractor's Default, and resulting from the actions or failure to act of the Surety under Paragraph 5; and
  - 7.3. liquidated damages, or if no liquidated damages are specified in the Construction Contract, actual damages caused by delayed performance or non-performance of the Contractor.
8. If the Surety elects to act under Paragraph 5.1, 5.3, or 5.4, the Surety's liability is limited to the amount of this Bond.
9. The Surety shall not be liable to the Owner or others for obligations of the Contractor that are unrelated to the Construction Contract, and the Balance of the Contract Price will not be reduced or set off on account of any such unrelated obligations. No right of action will accrue on this Bond to any person or entity other than the Owner or its heirs, executors, administrators, successors, and assigns.
10. The Surety hereby waives notice of any change, including changes of time, to the Construction Contract or to related subcontracts, purchase orders, and other obligations.
11. Any proceeding, legal or equitable, under this Bond must be instituted in any court of competent jurisdiction in the location in which the work or part of the work is located and must be instituted within two years after a declaration of Contractor Default or within two years after the Contractor ceased working or within two years after the Surety refuses or fails to perform its obligations under this Bond, whichever occurs first. If the provisions of this

paragraph are void or prohibited by law, the minimum periods of limitations available to sureties as a defense in the jurisdiction of the suit will be applicable.

12. Notice to the Surety, the Owner, or the Contractor must be mailed or delivered to the address shown on the page on which their signature appears.

13. When this Bond has been furnished to comply with a statutory or other legal requirement in the location where the construction was to be performed, any provision in this Bond conflicting with said statutory or legal requirement will be deemed deleted therefrom and provisions conforming to such statutory or other legal requirement will be deemed incorporated herein. When so furnished, the intent is that this Bond will be construed as a statutory bond and not as a common law bond.

14. Definitions

14.1. *Balance of the Contract Price*—The total amount payable by the Owner to the Contractor under the Construction Contract after all proper adjustments have been made including allowance for the Contractor for any amounts received or to be received by the Owner in settlement of insurance or other claims for damages to which the Contractor is entitled, reduced by all valid and proper payments made to or on behalf of the Contractor under the Construction Contract.

14.2. *Construction Contract*—The agreement between the Owner and Contractor identified on the cover page, including all Contract Documents and changes made to the agreement and the Contract Documents.

14.3. *Contractor Default*—Failure of the Contractor, which has not been remedied or waived, to perform or otherwise to comply with a material term of the Construction Contract.

14.4. *Owner Default*—Failure of the Owner, which has not been remedied or waived, to pay the Contractor as required under the Construction Contract or to perform and complete or comply with the other material terms of the Construction Contract.

14.5. *Contract Documents*—All the documents that comprise the agreement between the Owner and Contractor.

15. If this Bond is issued for an agreement between a contractor and subcontractor, the term Contractor in this Bond will be deemed to be Subcontractor and the term Owner will be deemed to be Contractor.

16. Modifications to this Bond are as follows: **[None]**

**END OF DOCUMENT**

**PAYMENT BOND**

|                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contractor</b><br>Name: <b>[Full formal name of Contractor]</b><br>Address <i>(principal place of business)</i> :<br><b>[Address of Contractor's principal place of business]</b>                                                                         | <b>Surety</b><br>Name: <b>[Full formal name of Surety]</b><br>Address <i>(principal place of business)</i> :<br><b>[Address of Surety's principal place of business]</b>                                                                          |
| <b>Owner</b><br>Name: <b>City of Edinburg</b><br>Mailing address <i>(principal place of business)</i> :<br><b>415 W University Drive</b><br><b>Edinburg, TX. 78539</b>                                                                                       | <b>Contract</b><br>Description <i>(name and location)</i> :<br><b>2019-20/Fire Station No. 5</b><br><b>Davis Road, Edinburg</b><br><br>Contract Price: <b>[Amount, from Contract]</b><br>Effective Date of Contract: <b>[Date, from Contract]</b> |
| <b>Bond</b><br>Bond <b>[Amount]</b><br>Date of Bond: <b>[Date]</b><br><i>(Date of Bond cannot be earlier than Effective Date of Contract)</i><br>Modifications to this Bond form:<br><input type="checkbox"/> None <input type="checkbox"/> See Paragraph 18 |                                                                                                                                                                                                                                                   |
| Surety and Contractor, intending to be legally bound hereby, subject to the terms set forth in this Payment Bond, do each cause this Payment Bond to be duly executed by an authorized officer, agent, or representative.                                    |                                                                                                                                                                                                                                                   |
| Contractor as Principal                                                                                                                                                                                                                                      | Surety                                                                                                                                                                                                                                            |
| <i>(Full formal name of Contractor)</i><br><hr/>                                                                                                                                                                                                             | <i>(Full formal name of Surety) (corporate seal)</i><br><hr/>                                                                                                                                                                                     |
| By: <hr/> <div style="text-align: center;"><i>(Signature)</i></div>                                                                                                                                                                                          | By: <hr/> <div style="text-align: center;"><i>(Signature)(Attach Power of Attorney)</i></div>                                                                                                                                                     |
| Name: <hr/> <div style="text-align: center;"><i>(Printed or typed)</i></div>                                                                                                                                                                                 | Name: <hr/> <div style="text-align: center;"><i>(Printed or typed)</i></div>                                                                                                                                                                      |
| Title: <hr/>                                                                                                                                                                                                                                                 | Title: <hr/>                                                                                                                                                                                                                                      |
| Attest: <hr/> <div style="text-align: center;"><i>(Signature)</i></div>                                                                                                                                                                                      | Attest: <hr/> <div style="text-align: center;"><i>(Signature)</i></div>                                                                                                                                                                           |
| Name: <hr/> <div style="text-align: center;"><i>(Printed or typed)</i></div>                                                                                                                                                                                 | Name: <hr/> <div style="text-align: center;"><i>(Printed or typed)</i></div>                                                                                                                                                                      |
| Title: <hr/>                                                                                                                                                                                                                                                 | Title: <hr/>                                                                                                                                                                                                                                      |
| <i>Notes: (1) Provide supplemental execution by any additional parties, such as joint venturers. (2) Any singular reference to Contractor, Surety, Owner, or other party is considered plural where applicable.</i>                                          |                                                                                                                                                                                                                                                   |

1. The Contractor and Surety, jointly and severally, bind themselves, their heirs, executors, administrators, successors, and assigns to the Owner to pay for labor, materials, and equipment furnished for use in the performance of the Construction Contract, which is incorporated herein by reference, subject to the following terms.
2. If the Contractor promptly makes payment of all sums due to Claimants, and defends, indemnifies, and holds harmless the Owner from claims, demands, liens, or suits by any person or entity seeking payment for labor, materials, or equipment furnished for use in the performance of the Construction Contract, then the Surety and the Contractor shall have no obligation under this Bond.
3. If there is no Owner Default under the Construction Contract, the Surety's obligation to the Owner under this Bond will arise after the Owner has promptly notified the Contractor and the Surety (at the address described in Paragraph 13) of claims, demands, liens, or suits against the Owner or the Owner's property by any person or entity seeking payment for labor, materials, or equipment furnished for use in the performance of the Construction Contract, and tendered defense of such claims, demands, liens, or suits to the Contractor and the Surety.
4. When the Owner has satisfied the conditions in Paragraph 3, the Surety shall promptly and at the Surety's expense defend, indemnify, and hold harmless the Owner against a duly tendered claim, demand, lien, or suit.
5. The Surety's obligations to a Claimant under this Bond will arise after the following:
  - 5.1. Claimants who do not have a direct contract with the Contractor
    - 5.1.1. have furnished a written notice of non-payment to the Contractor, stating with substantial accuracy the amount claimed and the name of the party to whom the materials were, or equipment was, furnished or supplied or for whom the labor was done or performed, within ninety (90) days after having last performed labor or last furnished materials or equipment included in the Claim; and
    - 5.1.2. have sent a Claim to the Surety (at the address described in Paragraph 13).
  - 5.2. Claimants who are employed by or have a direct contract with the Contractor have sent a Claim to the Surety (at the address described in Paragraph 13).
6. If a notice of non-payment required by Paragraph 5.1.1 is given by the Owner to the Contractor, that is sufficient to satisfy a Claimant's obligation to furnish a written notice of non-payment under Paragraph 5.1.1.
7. When a Claimant has satisfied the conditions of Paragraph 5.1 or 5.2, whichever is applicable, the Surety shall promptly and at the Surety's expense take the following actions:
  - 7.1. Send an answer to the Claimant, with a copy to the Owner, within sixty (60) days after receipt of the Claim, stating the amounts that are undisputed and the basis for challenging any amounts that are disputed; and
  - 7.2. Pay or arrange for payment of any undisputed amounts.
  - 7.3. The Surety's failure to discharge its obligations under Paragraph 7.1 or 7.2 will not be deemed to constitute a waiver of defenses the Surety or Contractor may have or acquire as to a Claim, except as to undisputed amounts for which the Surety and Claimant have

reached agreement. If, however, the Surety fails to discharge its obligations under Paragraph 7.1 or 7.2, the Surety shall indemnify the Claimant for the reasonable attorney's fees the Claimant incurs thereafter to recover any sums found to be due and owing to the Claimant.

8. The Surety's total obligation will not exceed the amount of this Bond, plus the amount of reasonable attorney's fees provided under Paragraph 7.3, and the amount of this Bond will be credited for any payments made in good faith by the Surety.
9. Amounts owed by the Owner to the Contractor under the Construction Contract will be used for the performance of the Construction Contract and to satisfy claims, if any, under any construction performance bond. By the Contractor furnishing and the Owner accepting this Bond, they agree that all funds earned by the Contractor in the performance of the Construction Contract are dedicated to satisfying obligations of the Contractor and Surety under this Bond, subject to the Owner's priority to use the funds for the completion of the work.
10. The Surety shall not be liable to the Owner, Claimants, or others for obligations of the Contractor that are unrelated to the Construction Contract. The Owner shall not be liable for the payment of any costs or expenses of any Claimant under this Bond, and shall have under this Bond no obligation to make payments to or give notice on behalf of Claimants, or otherwise have any obligations to Claimants under this Bond.
11. The Surety hereby waives notice of any change, including changes of time, to the Construction Contract or to related subcontracts, purchase orders, and other obligations.
12. No suit or action will be commenced by a Claimant under this Bond other than in a court of competent jurisdiction in the state in which the project that is the subject of the Construction Contract is located or after the expiration of one year from the date (1) on which the Claimant sent a Claim to the Surety pursuant to Paragraph 5.1.2 or 5.2, or (2) on which the last labor or service was performed by anyone or the last materials or equipment were furnished by anyone under the Construction Contract, whichever of (1) or (2) first occurs. If the provisions of this paragraph are void or prohibited by law, the minimum period of limitation available to sureties as a defense in the jurisdiction of the suit will be applicable.
13. Notice and Claims to the Surety, the Owner, or the Contractor must be mailed or delivered to the address shown on the page on which their signature appears. Actual receipt of notice or Claims, however accomplished, will be sufficient compliance as of the date received.
14. When this Bond has been furnished to comply with a statutory or other legal requirement in the location where the construction was to be performed, any provision in this Bond conflicting with said statutory or legal requirement will be deemed deleted here from and provisions conforming to such statutory or other legal requirement will be deemed incorporated herein. When so furnished, the intent is that this Bond will be construed as a statutory bond and not as a common law bond.
15. Upon requests by any person or entity appearing to be a potential beneficiary of this Bond, the Contractor and Owner shall promptly furnish a copy of this Bond or shall permit a copy to be made.

16. Definitions

16.1. *Claim*—A written statement by the Claimant including at a minimum:



- 16.1.1. The name of the Claimant;
  - 16.1.2. The name of the person for whom the labor was done, or materials or equipment furnished;
  - 16.1.3. A copy of the agreement or purchase order pursuant to which labor, materials, or equipment was furnished for use in the performance of the Construction Contract;
  - 16.1.4. A brief description of the labor, materials, or equipment furnished;
  - 16.1.5. The date on which the Claimant last performed labor or last furnished materials or equipment for use in the performance of the Construction Contract;
  - 16.1.6. The total amount earned by the Claimant for labor, materials, or equipment furnished as of the date of the Claim;
  - 16.1.7. The total amount of previous payments received by the Claimant; and
  - 16.1.8. The total amount due and unpaid to the Claimant for labor, materials, or equipment furnished as of the date of the Claim.
- 16.2. *Claimant*—An individual or entity having a direct contract with the Contractor or with a subcontractor of the Contractor to furnish labor, materials, or equipment for use in the performance of the Construction Contract. The term Claimant also includes any individual or entity that has rightfully asserted a claim under an applicable mechanic's lien or similar statute against the real property upon which the Project is located. The intent of this Bond is to include without limitation in the terms of "labor, materials, or equipment" that part of the water, gas, power, light, heat, oil, gasoline, telephone service, or rental equipment used in the Construction Contract, architectural and engineering services required for performance of the work of the Contractor and the Contractor's subcontractors, and all other items for which a mechanic's lien may be asserted in the jurisdiction where the labor, materials, or equipment were furnished.
- 16.3. *Construction Contract*—The agreement between the Owner and Contractor identified on the cover page, including all Contract Documents and all changes made to the agreement and the Contract Documents.
- 16.4. *Owner Default*—Failure of the Owner, which has not been remedied or waived, to pay the Contractor as required under the Construction Contract or to perform and complete or comply with the other material terms of the Construction Contract.
- 16.5. *Contract Documents*—All the documents that comprise the agreement between the Owner and Contractor.
17. If this Bond is issued for an agreement between a contractor and subcontractor, the term Contractor in this Bond will be deemed to be Subcontractor and the term Owner will be deemed to be Contractor.

Modifications to this Bond are as follows: **[None]**



Document 00625

**AFFIDAVIT OF INSURANCE**

THE STATE OF TEXAS

§  
§ KNOW ALL MEN BY THESE PRESENTS:

THE COUNTY OF \_\_\_\_\_

**BEFORE ME**, the undersigned authority, on this day personally appeared\_\_\_\_\_, who  
[Affiant]being by me duly sworn on his oath stated that he is \_\_\_\_\_, of  
[Title]\_\_\_\_\_  
[Contractor's Company Name]

the Contractor named and referred to within the Contract Documents; that he is fully competent and authorized to give this affidavit and that the attached original insurance certificate truly and accurately reflects the insurance coverage that is now available and will be available during the term of the Agreement.

\_\_\_\_\_  
[Affiant's Signature]SWORN AND SUBSCRIBED before me on \_\_\_\_\_.  
[Date]\_\_\_\_\_  
Notary Public in and for the State of TEXAS\_\_\_\_\_  
[Print or type Notary Public name]

[Notary Seal]

My Commission Expires: \_\_\_\_\_  
[Expiration Date]**END OF DOCUMENT**

Page Intentionally Blank

Document 00630

**FORM OF BUSINESS**

Please, fill in the appropriate area describing your firm's form of business and include the relevant attachments.

**Corporation:**

Corporate Name: \_\_\_\_\_

State of Incorporation: \_\_\_\_\_

Mailing Address: \_\_\_\_\_

- Certificate of Assumed Name, if operating under a name different than that on the corporate charter (the Certificate must have been issued within the past ten years to be valid)
- Certificate of Good Standing\*
- Certificate of Existence (if non-Texas corporation, Certificate of Authority) \*

**Partnership/Joint Venture:**

Partnership/Joint Venture Name: \_\_\_\_\_

Mailing Address: \_\_\_\_\_

- Copy of the Partnership or Joint Venture Agreement, or Affidavit with the name of the partnership or joint venture, the names of the individual partners or participants in the joint venture, and a statement that the partnership or joint venture is in existence
- Certificate of Assumed Name, (the Certificate must have been issued within the past ten years to be valid)
- If firm is a limited partnership, the Certificate of Limited Partnership
- If any partner or joint venturer is a corporation, the above information relating to corporation must be included as to each sum partner or joint venturer.

**Sole Proprietorship**

Name: \_\_\_\_\_

Mailing Address: \_\_\_\_\_

- Certificate of Assumed Name, if operating under a name different than that of the sole proprietor (the Certificate must have been issued within the past ten years to be valid)

\* *Must be furnished upon request of the Owner and must be less than 90 days old.*

\_\_\_\_\_  
[Typed Name and Title of Authorized Representative]\_\_\_\_\_  
[Signature of Authorized Representative]\_\_\_\_\_  
[Typed Date]**END OF DOCUMENT**

00630- 1 of 2

Page Intentionally Blank

Document 00631

**RESOLUTION OF CORPORATION**

I hereby certify that it was RESOLVED by a quorum of the directors of

\_\_\_\_\_  
[Name of Corporation / Contractor]

meeting on this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_, that \_\_\_\_\_,  
[Corporate Representative]

be, and hereby is, authorized to act on behalf of the Corporation, as its representative, in all business transactions conducted in the State of Texas, and that the above resolution was unanimously ratified by the Board of Directors at said meeting and that the resolution has not been rescinded or amended and is now in full force and effect; and in authentication of the adoption of this resolution, I subscribe my name and affix the seal of the Corporation on this

\_\_\_\_\_ day of, \_\_\_\_\_, 20\_\_\_\_\_.

\_\_\_\_\_  
Secretary/Assistant Secretary

\_\_\_\_\_  
[Seal]

**END OF DOCUMENT**

00631- 1 of 2

Page Intentionally Blank

**CONTRACTOR'S RESOLUTION  
ON  
AUTHORIZED REPRESENTATIVE (ED-104)**

---

Name or Names

I hereby certify that it was RESOLVED by a quorum of the directors of the

\_\_\_\_\_, meeting  
name of corporation

on the day of \_\_\_\_\_, 20\_\_\_\_, that \_\_\_\_\_,  
\_\_\_\_\_, and \_\_\_\_\_, be, and hereby is,  
authorized to act on behalf of \_\_\_\_\_, as its  
name of corporation

representative, in all business transactions conducted in the State of Texas, and;

That all above resolution was unanimously ratified by the Board of Directors at said  
meeting and that the resolution has not been rescinded or amended and is now in full forces  
and effect; and;

In authentication of the adoption of this resolution, I subscribe my name and  
affix the seal of the corporation this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

---

Secretary

(seal)

Page Intentionally Blank



Document 00635

**CONTRACTOR'S ACT OF ASSURANCE**

THE STATE OF TEXAS

KNOW ALL MEN BY THESE PRESENTS:

THE COUNTY OF \_\_\_\_\_

BEFORE ME, the undersigned authority, a Notary Public in and for the State of Texas,

on this day personally appeared \_\_\_\_\_, Affiant,  
[Affiant]who being by me duly sworn on his oath stated that he is \_\_\_\_\_, of  
[Title]the \_\_\_\_\_, Contractor, that he is authorized to represent Contractor  
[Contractor]pursuant to provisions of a resolution adopted on this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_. A duly  
certified copy of such resolution is attached to and is hereby made a part of this document.Affiant, in such capacity declares and assures the City of Edinburg that Contractor will construct the Project  
in accordance with sound construction practice and all laws of the State of Texas.\_\_\_\_\_  
[Affiant]

SWORN AND SUBSCRIBED before me on this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

\_\_\_\_\_  
Notary Public in and for the State of TEXAS\_\_\_\_\_  
[Print or Type Notary Public Name]My Commission Expires: \_\_\_\_\_  
[Expiration Date]\_\_\_\_\_  
[Seal]**END OF DOCUMENT**

Page Intentionally Blank

Document 00640

**CERTIFICATION REGARDING DEPARTMENT,  
SUSPENSION AND OTHER RESPONSIBILITY MATTERS**

The prospective participant certifies to the best of its knowledge and belief that it and its principals:

- (a) Are not presently debarred, suspended, proposed for debarment, declared ineligible, or voluntarily excluded from covered transactions by any Federal, State, or local department or agency;
- (b) Have not within a three-year period preceding this proposal been convicted of or had a civil judgment rendered against them for commission of fraud or a criminal offense in connection with obtaining, attempting to obtain, or performing a public (Federal, State, or local) transaction or contract under a public transaction: violation of Federal or State antitrust statutes or commission of embezzlement, theft, forgery, bribery, falsification or destruction of records, making false statements, or receiving stolen property;
- (c) Are not presently indicted for or otherwise criminally or civilly charged by a government entity (Federal, State, or local) with commission of any of the offenses enumerated in paragraph (b) of this certification; and
- (d) Have not within a three-year period preceding this application/proposal had one or more public transactions (Federal, State, or local) terminated for cause or default.

I understand that a false statement on this certification may be grounds for rejection of this proposal or termination of the award. In addition, under 18 USC Section 1001, a false statement may result in a fine of up to \$10,000 or imprisonment for up to 5 years, or both.

---

[Typed Name of Company:]

---

[Typed Name & Title of Authorized Representative]

---

[Signature of Authorized Representative]

---

[Date]

If unable certify the above statements, explanation is attached.

**END OF DOCUMENT**

Page Intentionally Blank

This document has important legal consequences; consultation with an attorney is encouraged with respect to its use or modification. This document should be adapted to the particular circumstances of the contemplated Project and the controlling Laws and Regulations.

## STANDARD GENERAL CONDITIONS OF THE CONSTRUCTION CONTRACT

Prepared By



Endorsed By



Copyright© 2018

National Society of Professional Engineers  
1420 King Street, Alexandria, VA 22314-2794  
(703) 684-2882  
[www.nspe.org](http://www.nspe.org)

American Council of Engineering Companies  
1015 15th Street N.W., Washington, DC 20005  
(202) 347-7474  
[www.acec.org](http://www.acec.org)

American Society of Civil Engineers  
1801 Alexander Bell Drive, Reston, VA 20191-4400  
(800) 548-2723  
[www.asce.org](http://www.asce.org)

The copyright for this EJCDC document is owned jointly by the three sponsoring organizations listed above. The National Society of Professional Engineers is the Copyright Administrator for the EJCDC documents; please direct all inquiries regarding EJCDC copyrights to NSPE.

NOTE: EJCDC publications may be purchased at [www.ejcdc.org](http://www.ejcdc.org), or from any of the sponsoring organizations above.

# STANDARD GENERAL CONDITIONS OF THE CONSTRUCTION CONTRACT

## TABLE OF CONTENTS

|                                                                                              | Page |
|----------------------------------------------------------------------------------------------|------|
| Article 1—Definitions and Terminology.....                                                   | 1    |
| 1.01 Defined Terms.....                                                                      | 1    |
| 1.02 Terminology .....                                                                       | 6    |
| Article 2—Preliminary Matters .....                                                          | 7    |
| 2.01 Delivery of Performance and Payment Bonds; Evidence of Insurance.....                   | 7    |
| 2.02 Copies of Documents .....                                                               | 7    |
| 2.03 Before Starting Construction .....                                                      | 8    |
| 2.04 Preconstruction Conference; Designation of Authorized Representatives .....             | 8    |
| 2.05 Acceptance of Schedules .....                                                           | 8    |
| 2.06 Electronic Transmittals .....                                                           | 9    |
| Article 3—Contract Documents: Intent, Requirements, Reuse .....                              | 9    |
| 3.01 Intent.....                                                                             | 9    |
| 3.02 Reference Standards .....                                                               | 10   |
| 3.03 Reporting and Resolving Discrepancies .....                                             | 10   |
| 3.04 Requirements of the Contract Documents .....                                            | 11   |
| 3.05 Reuse of Documents .....                                                                | 11   |
| Article 4—Commencement and Progress of the Work .....                                        | 12   |
| 4.01 Commencement of Contract Times; Notice to Proceed .....                                 | 12   |
| 4.02 Starting the Work.....                                                                  | 12   |
| 4.03 Reference Points .....                                                                  | 12   |
| 4.04 Progress Schedule .....                                                                 | 12   |
| 4.05 Delays in Contractor’s Progress .....                                                   | 12   |
| Article 5—Site; Subsurface and Physical Conditions; Hazardous Environmental Conditions ..... | 14   |
| 5.01 Availability of Lands .....                                                             | 14   |
| 5.02 Use of Site and Other Areas.....                                                        | 14   |
| 5.03 Subsurface and Physical Conditions.....                                                 | 15   |
| 5.04 Differing Subsurface or Physical Conditions .....                                       | 16   |

|                                               |                                                              |    |
|-----------------------------------------------|--------------------------------------------------------------|----|
| 5.05                                          | Underground Facilities .....                                 | 18 |
| 5.06                                          | Hazardous Environmental Conditions at Site .....             | 20 |
| Article 6—Bonds and Insurance.....            |                                                              | 22 |
| 6.01                                          | Performance, Payment, and Other Bonds .....                  | 22 |
| 6.02                                          | Insurance—General Provisions .....                           | 23 |
| 6.03                                          | Contractor’s Insurance .....                                 | 24 |
| 6.04                                          | Builder’s Risk and Other Property Insurance .....            | 25 |
| 6.05                                          | Property Losses; Subrogation .....                           | 26 |
| 6.06                                          | Receipt and Application of Property Insurance Proceeds ..... | 27 |
| Article 7—Contractor’s Responsibilities ..... |                                                              | 28 |
| 7.01                                          | Contractor’s Means and Methods of Construction .....         | 28 |
| 7.02                                          | Supervision and Superintendence .....                        | 28 |
| 7.03                                          | Labor; Working Hours .....                                   | 28 |
| 7.04                                          | Services, Materials, and Equipment .....                     | 28 |
| 7.05                                          | “Or Equals” .....                                            | 29 |
| 7.06                                          | Substitutes .....                                            | 30 |
| 7.07                                          | Concerning Subcontractors and Suppliers .....                | 31 |
| 7.08                                          | Patent Fees and Royalties .....                              | 33 |
| 7.09                                          | Permits .....                                                | 33 |
| 7.10                                          | Taxes .....                                                  | 34 |
| 7.11                                          | Laws and Regulations.....                                    | 34 |
| 7.12                                          | Record Documents.....                                        | 34 |
| 7.13                                          | Safety and Protection.....                                   | 34 |
| 7.14                                          | Hazard Communication Programs .....                          | 36 |
| 7.15                                          | Emergencies .....                                            | 36 |
| 7.16                                          | Submittals .....                                             | 36 |
| 7.17                                          | Contractor’s General Warranty and Guarantee .....            | 39 |
| 7.18                                          | Indemnification .....                                        | 40 |
| 7.19                                          | Delegation of Professional Design Services .....             | 40 |
| Article 8—Other Work at the Site .....        |                                                              | 41 |
| 8.01                                          | Other Work .....                                             | 41 |
| 8.02                                          | Coordination .....                                           | 42 |
| 8.03                                          | Legal Relationships.....                                     | 42 |



|                                                                                       |    |
|---------------------------------------------------------------------------------------|----|
| Article 9—Owner’s Responsibilities .....                                              | 43 |
| 9.01 Communications to Contractor .....                                               | 43 |
| 9.02 Replacement of Engineer/Architect .....                                          | 43 |
| 9.03 Furnish Data .....                                                               | 44 |
| 9.04 Pay When Due.....                                                                | 44 |
| 9.05 Lands and Easements; Reports, Tests, and Drawings .....                          | 44 |
| 9.06 Insurance.....                                                                   | 44 |
| 9.07 Change Orders .....                                                              | 44 |
| 9.08 Inspections, Tests, and Approvals.....                                           | 44 |
| 9.09 Limitations on Owner’s Responsibilities .....                                    | 44 |
| 9.10 Undisclosed Hazardous Environmental Condition.....                               | 44 |
| 9.11 Evidence of Financial Arrangements.....                                          | 44 |
| 9.12 Safety Programs .....                                                            | 44 |
| Article 10—Engineer/Architect’s Status During Construction .....                      | 45 |
| 10.01 Owner’s Representative.....                                                     | 45 |
| 10.02 Visits to Site.....                                                             | 45 |
| 10.03 Resident Project Representative.....                                            | 45 |
| 10.04 Engineer/Architect’s Authority .....                                            | 45 |
| 10.05 Determinations for Unit Price Work .....                                        | 46 |
| 10.06 Decisions on Requirements of Contract Documents and Acceptability of Work ..... | 46 |
| 10.07 Limitations on Engineer/Architect’s Authority and Responsibilities .....        | 46 |
| 10.08 Compliance with Safety Program.....                                             | 47 |
| Article 11—Changes to the Contract .....                                              | 47 |
| 11.01 Amending and Supplementing the Contract .....                                   | 47 |
| 11.02 Change Orders .....                                                             | 47 |
| 11.03 Work Change Directives.....                                                     | 47 |
| 11.04 Field Orders.....                                                               | 48 |
| 11.05 Owner-Authorized Changes in the Work.....                                       | 48 |
| 11.06 Unauthorized Changes in the Work.....                                           | 48 |
| 11.07 Change of Contract Price .....                                                  | 49 |
| 11.08 Change of Contract Times.....                                                   | 50 |
| 11.09 Change Proposals.....                                                           | 50 |
| 11.10 Notification to Surety.....                                                     | 51 |

|                                                                                              |    |
|----------------------------------------------------------------------------------------------|----|
| Article 12—Claims.....                                                                       | 51 |
| 12.01    Claims.....                                                                         | 51 |
| Article 13—Cost of the Work; Allowances; Unit Price Work .....                               | 53 |
| 13.01    Cost of the Work .....                                                              | 53 |
| 13.02    Allowances .....                                                                    | 56 |
| 13.03    Unit Price Work.....                                                                | 57 |
| Article 14—Tests and Inspections; Correction, Removal, or Acceptance of Defective Work ..... | 57 |
| 14.01    Access to Work.....                                                                 | 57 |
| 14.02    Tests, Inspections, and Approvals.....                                              | 58 |
| 14.03    Defective Work .....                                                                | 59 |
| 14.04    Acceptance of Defective Work.....                                                   | 59 |
| 14.05    Uncovering Work .....                                                               | 59 |
| 14.06    Owner May Stop the Work .....                                                       | 60 |
| 14.07    Owner May Correct Defective Work.....                                               | 60 |
| Article 15—Payments to Contractor; Set-Offs; Completion; Correction Period .....             | 61 |
| 15.01    Progress Payments.....                                                              | 61 |
| 15.02    Contractor’s Warranty of Title .....                                                | 64 |
| 15.03    Substantial Completion.....                                                         | 64 |
| 15.04    Partial Use or Occupancy .....                                                      | 65 |
| 15.05    Final Inspection .....                                                              | 66 |
| 15.06    Final Payment.....                                                                  | 66 |
| 15.07    Waiver of Claims .....                                                              | 67 |
| 15.08    Correction Period .....                                                             | 67 |
| Article 16—Suspension of Work and Termination .....                                          | 68 |
| 16.01    Owner May Suspend Work .....                                                        | 68 |
| 16.02    Owner May Terminate for Cause.....                                                  | 69 |
| 16.03    Owner May Terminate for Convenience.....                                            | 70 |
| 16.04    Contractor May Stop Work or Terminate .....                                         | 70 |
| Article 17—Final Resolution of Disputes .....                                                | 71 |
| 17.01    Methods and Procedures.....                                                         | 71 |
| Article 18—Miscellaneous .....                                                               | 71 |
| 18.01    Giving Notice .....                                                                 | 71 |
| 18.02    Computation of Times .....                                                          | 71 |

|       |                               |    |
|-------|-------------------------------|----|
| 18.03 | Cumulative Remedies .....     | 71 |
| 18.04 | Limitation of Damages .....   | 72 |
| 18.05 | No Waiver .....               | 72 |
| 18.06 | Survival of Obligations ..... | 72 |
| 18.07 | Controlling Law .....         | 72 |
| 18.08 | Assignment of Contract .....  | 72 |
| 18.09 | Successors and Assigns .....  | 72 |
| 18.10 | Headings.....                 | 72 |

# STANDARD GENERAL CONDITIONS OF THE CONSTRUCTION CONTRACT

## ARTICLE 1—DEFINITIONS AND TERMINOLOGY

### 1.01 *Defined Terms*

- A. Wherever used in the Bidding Requirements or Contract Documents, a term printed with initial capital letters, including the term's singular and plural forms, will have the meaning indicated in the definitions below. In addition to terms specifically defined, terms with initial capital letters in the Contract Documents include references to identified articles and paragraphs, and the titles of other documents or forms.
1. *Addenda*—Written or graphic instruments issued prior to the opening of Bids which clarify, correct, or change the Bidding Requirements or the proposed Contract Documents.
  2. *Agreement*—The written instrument, executed by Owner and Contractor, that sets forth the Contract Price and Contract Times, identifies the parties and the Engineer/Architect, and designates the specific items that are Contract Documents.
  3. *Application for Payment*—The document prepared by Contractor, in a form acceptable to Engineer/Architect, to request progress or final payments, and which is to be accompanied by such supporting documentation as is required by the Contract Documents.
  4. *Bid*—The offer of a Bidder submitted on the prescribed form setting forth the prices for the Work to be performed.
  5. *Bidder*—An individual or entity that submits a Bid to Owner.
  6. *Bidding Documents*—The Bidding Requirements, the proposed Contract Documents, and all Addenda.
  7. *Bidding Requirements*—The Advertisement or invitation to bid, Instructions to Bidders, Bid Bond or other Bid security, if any, the Bid Form, and the Bid with any attachments.
  8. *Change Order*—A document which is signed by Contractor and Owner and authorizes an addition, deletion, or revision in the Work or an adjustment in the Contract Price or the Contract Times, or other revision to the Contract, issued on or after the Effective Date of the Contract.
  9. *Change Proposal*—A written request by Contractor, duly submitted in compliance with the procedural requirements set forth herein, seeking an adjustment in Contract Price or Contract Times; contesting an initial decision by Engineer/Architect concerning the requirements of the Contract Documents or the acceptability of Work under the Contract Documents; challenging a set-off against payments due; or seeking other relief with respect to the terms of the Contract.
  10. *Claim*
    - a. A demand or assertion by Owner directly to Contractor, duly submitted in compliance with the procedural requirements set forth herein, seeking an

adjustment of Contract Price or Contract Times; contesting an initial decision by Engineer/Architect concerning the requirements of the Contract Documents or the acceptability of Work under the Contract Documents; contesting Engineer/Architect's decision regarding a Change Proposal; seeking resolution of a contractual issue that Engineer/Architect has declined to address; or seeking other relief with respect to the terms of the Contract.

- b. A demand or assertion by Contractor directly to Owner, duly submitted in compliance with the procedural requirements set forth herein, contesting Engineer/Architect's decision regarding a Change Proposal, or seeking resolution of a contractual issue that Engineer/Architect has declined to address.
  - c. A demand or assertion by Owner or Contractor, duly submitted in compliance with the procedural requirements set forth herein, made pursuant to Paragraph 12.01.A.4, concerning disputes arising after Engineer/Architect has issued a recommendation of final payment.
  - d. A demand for money or services by a third party is not a Claim.
- 11. *Constituent of Concern*—Asbestos, petroleum, radioactive materials, polychlorinated biphenyls (PCBs), lead-based paint (as defined by the HUD/EPA standard), hazardous waste, and any substance, product, waste, or other material of any nature whatsoever that is or becomes listed, regulated, or addressed pursuant to Laws and Regulations regulating, relating to, or imposing liability or standards of conduct concerning, any hazardous, toxic, or dangerous waste, substance, or material.
  - 12. *Contract*—The entire and integrated written contract between Owner and Contractor concerning the Work.
  - 13. *Contract Documents*—Those items so designated in the Agreement, and which together comprise the Contract.
  - 14. *Contract Price*—The money that Owner has agreed to pay Contractor for completion of the Work in accordance with the Contract Documents.
  - 15. *Contract Times*—The number of days or the dates by which Contractor shall: (a) achieve Milestones, if any; (b) achieve Substantial Completion; and (c) complete the Work.
  - 16. *Contractor*—The individual or entity with which Owner has contracted for performance of the Work.
  - 17. *Cost of the Work*—See Paragraph 13.01 for definition.
  - 18. *Drawings*—The part of the Contract that graphically shows the scope, extent, and character of the Work to be performed by Contractor.
  - 19. *Effective Date of the Contract*—The date, indicated in the Agreement, on which the Contract becomes effective.
  - 20. *Electronic Document*—Any Project-related correspondence, attachments to correspondence, data, documents, drawings, information, or graphics, including but not limited to Shop Drawings and other Submittals, that are in an electronic or digital format.

21. *Electronic Means*—Electronic mail (email), upload/download from a secure Project website, or other communications methods that allow: (a) the transmission or communication of Electronic Documents; (b) the documentation of transmissions, including sending and receipt; (c) printing of the transmitted Electronic Document by the recipient; (d) the storage and archiving of the Electronic Document by sender and recipient; and (e) the use by recipient of the Electronic Document for purposes permitted by this Contract. Electronic Means does not include the use of text messaging, or of Facebook, Twitter, Instagram, or similar social media services for transmission of Electronic Documents.
22. *Engineer/Architect*—The individual or entity named as such in the Agreement.
23. *Field Order*—A written order issued by Engineer/Architect which requires minor changes in the Work but does not change the Contract Price or the Contract Times.
24. *Hazardous Environmental Condition*—The presence at the Site of Constituents of Concern in such quantities or circumstances that may present a danger to persons or property exposed thereto.
- a. The presence at the Site of materials that are necessary for the execution of the Work, or that are to be incorporated into the Work, and that are controlled and contained pursuant to industry practices, Laws and Regulations, and the requirements of the Contract, is not a Hazardous Environmental Condition.
  - b. The presence of Constituents of Concern that are to be removed or remediated as part of the Work is not a Hazardous Environmental Condition.
  - c. The presence of Constituents of Concern as part of the routine, anticipated, and obvious working conditions at the Site, is not a Hazardous Environmental Condition.
25. *Laws and Regulations; Laws or Regulations*—Any and all applicable laws, statutes, rules, regulations, ordinances, codes, and binding decrees, resolutions, and orders of any and all governmental bodies, agencies, authorities, and courts having jurisdiction.
26. *Liens*—Charges, security interests, or encumbrances upon Contract-related funds, real property, or personal property.
27. *Milestone*—A principal event in the performance of the Work that the Contract requires Contractor to achieve by an intermediate completion date, or by a time prior to Substantial Completion of all the Work.
28. *Notice of Award*—The written notice by Owner to a Bidder of Owner's acceptance of the Bid.
29. *Notice to Proceed*—A written notice by Owner to Contractor fixing the date on which the Contract Times will commence to run and on which Contractor shall start to perform the Work.
30. *Owner*—The individual or entity with which Contractor has contracted regarding the Work, and which has agreed to pay Contractor for the performance of the Work, pursuant to the terms of the Contract.
31. *Progress Schedule*—A schedule, prepared and maintained by Contractor, describing the sequence and duration of the activities comprising Contractor's plan to accomplish the Work within the Contract Times.

32. *Project*—The total undertaking to be accomplished for Owner by Engineer/Architects, contractors, and others, including planning, study, design, construction, testing, commissioning, and start-up, and of which the Work to be performed under the Contract Documents is a part.
33. *Resident Project Representative*—The authorized representative of Engineer/Architect assigned to assist Engineer/Architect at the Site. As used herein, the term Resident Project Representative (RPR) includes any assistants or field staff of Resident Project Representative.
34. *Samples*—Physical examples of materials, equipment, or workmanship that are representative of some portion of the Work and that establish the standards by which such portion of the Work will be judged.
35. *Schedule of Submittals*—A schedule, prepared and maintained by Contractor, of required submittals and the time requirements for Engineer/Architect's review of the submittals.
36. *Schedule of Values*—A schedule, prepared and maintained by Contractor, allocating portions of the Contract Price to various portions of the Work and used as the basis for reviewing Contractor's Applications for Payment.
37. *Shop Drawings*—All drawings, diagrams, illustrations, schedules, and other data or information that are specifically prepared or assembled by or for Contractor and submitted by Contractor to illustrate some portion of the Work. Shop Drawings, whether approved or not, are not Drawings and are not Contract Documents.
38. *Site*—Lands or areas indicated in the Contract Documents as being furnished by Owner upon which the Work is to be performed, including rights-of-way and easements, and such other lands or areas furnished by Owner which are designated for the use of Contractor.
39. *Specifications*—The part of the Contract that consists of written requirements for materials, equipment, systems, standards, and workmanship as applied to the Work, and certain administrative requirements and procedural matters applicable to the Work.
40. *Subcontractor*—An individual or entity having a direct contract with Contractor or with any other Subcontractor for the performance of a part of the Work.
41. *Submittal*—A written or graphic document, prepared by or for Contractor, which the Contract Documents require Contractor to submit to Engineer/Architect, or that is indicated as a Submittal in the Schedule of Submittals accepted by Engineer/Architect. Submittals may include Shop Drawings and Samples; schedules; product data; Owner-delegated designs; sustainable design information; information on special procedures; testing plans; results of tests and evaluations, source quality-control testing and inspections, and field or Site quality-control testing and inspections; warranties and certifications; Suppliers' instructions and reports; records of delivery of spare parts and tools; operations and maintenance data; Project photographic documentation; record documents; and other such documents required by the Contract Documents. Submittals, whether or not approved or accepted by Engineer/Architect, are not Contract Documents. Change Proposals, Change Orders, Claims, notices, Applications for Payment, and requests for interpretation or clarification are not Submittals.

42. *Substantial Completion*—The time at which the Work (or a specified part thereof) has progressed to the point where, in the opinion of Engineer/Architect, the Work (or a specified part thereof) is sufficiently complete, in accordance with the Contract Documents, so that the Work (or a specified part thereof) can be utilized for the purposes for which it is intended. The terms “substantially complete” and “substantially completed” as applied to all or part of the Work refer to Substantial Completion of such Work.
43. *Successful Bidder*—The Bidder to which the Owner makes an award of contract.
44. *Supplementary Conditions*—The part of the Contract that amends or supplements these General Conditions.
45. *Supplier*—A manufacturer, fabricator, supplier, distributor, or vendor having a direct contract with Contractor or with any Subcontractor to furnish materials or equipment to be incorporated in the Work by Contractor or a Subcontractor.
46. *Technical Data*
- a. Those items expressly identified as Technical Data in the Supplementary Conditions, with respect to either (1) existing subsurface conditions at or adjacent to the Site, or existing physical conditions at or adjacent to the Site including existing surface or subsurface structures (except Underground Facilities) or (2) Hazardous Environmental Conditions at the Site.
  - b. If no such express identifications of Technical Data have been made with respect to conditions at the Site, then Technical Data is defined, with respect to conditions at the Site under Paragraphs 5.03, 5.04, and 5.06, as the data contained in boring logs, recorded measurements of subsurface water levels, assessments of the condition of subsurface facilities, laboratory test results, and other factual, objective information regarding conditions at the Site that are set forth in any geotechnical, environmental, or other Site or facilities conditions report prepared for the Project and made available to Contractor.
  - c. Information and data regarding the presence or location of Underground Facilities are not intended to be categorized, identified, or defined as Technical Data, and instead Underground Facilities are shown or indicated on the Drawings.
47. *Underground Facilities*—All active or not-in-service underground lines, pipelines, conduits, ducts, encasements, cables, wires, manholes, vaults, tanks, tunnels, or other such facilities or systems at the Site, including but not limited to those facilities or systems that produce, transmit, distribute, or convey telephone or other communications, cable television, fiber optic transmissions, power, electricity, light, heat, gases, oil, crude oil products, liquid petroleum products, water, steam, waste, wastewater, storm water, other liquids or chemicals, or traffic or other control systems. An abandoned facility or system is not an Underground Facility.
48. *Unit Price Work*—Work to be paid for on the basis of unit prices.
49. *Work*—The entire construction or the various separately identifiable parts thereof required to be provided under the Contract Documents. Work includes and is the result of performing or providing all labor, services, and documentation necessary to produce such construction; furnishing, installing, and incorporating all materials and equipment



into such construction; and may include related services such as testing, start-up, and commissioning, all as required by the Contract Documents.

50. *Work Change Directive*—A written directive to Contractor issued on or after the Effective Date of the Contract, signed by Owner and recommended by Engineer/Architect, ordering an addition, deletion, or revision in the Work.

## 1.02 Terminology

- A. The words and terms discussed in Paragraphs 1.02.B, C, D, and E are not defined terms that require initial capital letters, but, when used in the Bidding Requirements or Contract Documents, have the indicated meaning.
- B. *Intent of Certain Terms or Adjectives*: The Contract Documents include the terms “as allowed,” “as approved,” “as ordered,” “as directed” or terms of like effect or import to authorize an exercise of professional judgment by Engineer/Architect. In addition, the adjectives “reasonable,” “suitable,” “acceptable,” “proper,” “satisfactory,” or adjectives of like effect or import are used to describe an action or determination of Engineer/Architect as to the Work. It is intended that such exercise of professional judgment, action, or determination will be solely to evaluate, in general, the Work for compliance with the information in the Contract Documents and with the design concept of the Project as a functioning whole as shown or indicated in the Contract Documents (unless there is a specific statement indicating otherwise). The use of any such term or adjective is not intended to and shall not be effective to assign to Engineer/Architect any duty or authority to supervise or direct the performance of the Work, or any duty or authority to undertake responsibility contrary to the provisions of Article 10 or any other provision of the Contract Documents.
- C. *Day*: The word “day” means a calendar day of 24 hours measured from midnight to the next midnight.
- D. *Defective*: The word “defective,” when modifying the word “Work,” refers to Work that is unsatisfactory, faulty, or deficient in that it:
  - 1. does not conform to the Contract Documents;
  - 2. does not meet the requirements of any applicable inspection, reference standard, test, or approval referred to in the Contract Documents; or
  - 3. has been damaged prior to Engineer/Architect’s recommendation of final payment (unless responsibility for the protection thereof has been assumed by Owner at Substantial Completion in accordance with Paragraph 15.03 or Paragraph 15.04).
- E. *Furnish, Install, Perform, Provide*
  - 1. The word “furnish,” when used in connection with services, materials, or equipment, means to supply and deliver said services, materials, or equipment to the Site (or some other specified location) ready for use or installation and in usable or operable condition.
  - 2. The word “install,” when used in connection with services, materials, or equipment, means to put into use or place in final position said services, materials, or equipment complete and ready for intended use.

3. The words “perform” or “provide,” when used in connection with services, materials, or equipment, means to furnish and install said services, materials, or equipment complete and ready for intended use.
  4. If the Contract Documents establish an obligation of Contractor with respect to specific services, materials, or equipment, but do not expressly use any of the four words “furnish,” “install,” “perform,” or “provide,” then Contractor shall furnish and install said services, materials, or equipment complete and ready for intended use.
- F. *Contract Price or Contract Times*: References to a change in “Contract Price or Contract Times” or “Contract Times or Contract Price” or similar, indicate that such change applies to (1) Contract Price, (2) Contract Times, or (3) both Contract Price and Contract Times, as warranted, even if the term “or both” is not expressed.
- G. Unless stated otherwise in the Contract Documents, words or phrases that have a well-known technical or construction industry or trade meaning are used in the Contract Documents in accordance with such recognized meaning.

## **ARTICLE 2—PRELIMINARY MATTERS**

### **2.01 *Delivery of Performance and Payment Bonds; Evidence of Insurance***

- A. *Performance and Payment Bonds*: When Contractor delivers the signed counterparts of the Agreement to Owner, Contractor shall also deliver to Owner the performance bond and payment bond (if the Contract requires Contractor to furnish such bonds).
- B. *Evidence of Contractor’s Insurance*: When Contractor delivers the signed counterparts of the Agreement to Owner, Contractor shall also deliver to Owner, with copies to each additional insured (as identified in the Contract), the certificates, endorsements, and other evidence of insurance required to be provided by Contractor in accordance with Article 6, except to the extent the Supplementary Conditions expressly establish other dates for delivery of specific insurance policies.
- C. *Evidence of Owner’s Insurance*: After receipt of the signed counterparts of the Agreement and all required bonds and insurance documentation, Owner shall promptly deliver to Contractor, with copies to each additional insured (as identified in the Contract), the certificates and other evidence of insurance required to be provided by Owner under Article 6.

### **2.02 *Copies of Documents***

- A. Owner shall furnish to Contractor four printed copies of the Contract (including one fully signed counterpart of the Agreement), and one copy in electronic portable document format (PDF). Additional printed copies will be furnished upon request at the cost of reproduction.
- B. Owner shall maintain and safeguard at least one original printed record version of the Contract, including Drawings and Specifications signed and sealed by Engineer/Architect and other design professionals. Owner shall make such original printed record version of the Contract available to Contractor for review. Owner may delegate the responsibilities under this provision to Engineer/Architect.

### 2.03 *Before Starting Construction*

- A. *Preliminary Schedules:* Within 10 days after the Effective Date of the Contract (or as otherwise required by the Contract Documents), Contractor shall submit to Engineer/Architect for timely review:
1. a preliminary Progress Schedule indicating the times (numbers of days or dates) for starting and completing the various stages of the Work, including any Milestones specified in the Contract;
  2. a preliminary Schedule of Submittals; and
  3. a preliminary Schedule of Values for all of the Work which includes quantities and prices of items which when added together equal the Contract Price and subdivides the Work into component parts in sufficient detail to serve as the basis for progress payments during performance of the Work. Such prices will include an appropriate amount of overhead and profit applicable to each item of Work.

### 2.04 *Preconstruction Conference; Designation of Authorized Representatives*

- A. Before any Work at the Site is started, a conference attended by Owner, Contractor, Engineer/Architect, and others as appropriate will be held to establish a working understanding among the parties as to the Work, and to discuss the schedules referred to in Paragraph 2.03.A, procedures for handling Shop Drawings, Samples, and other Submittals, processing Applications for Payment, electronic or digital transmittals, and maintaining required records.
- B. At this conference Owner and Contractor each shall designate, in writing, a specific individual to act as its authorized representative with respect to the services and responsibilities under the Contract. Such individuals shall have the authority to transmit and receive information, render decisions relative to the Contract, and otherwise act on behalf of each respective party.

### 2.05 *Acceptance of Schedules*

- A. At least 10 days before submission of the first Application for Payment a conference, attended by Contractor, Engineer/Architect, and others as appropriate, will be held to review the schedules submitted in accordance with Paragraph 2.03.A. No progress payment will be made to Contractor until acceptable schedules are submitted to Engineer/Architect.
1. The Progress Schedule will be acceptable to Engineer/Architect if it provides an orderly progression of the Work to completion within the Contract Times. Such acceptance will not impose on Engineer/Architect responsibility for the Progress Schedule, for sequencing, scheduling, or progress of the Work, nor interfere with or relieve Contractor from Contractor's full responsibility therefor.
  2. Contractor's Schedule of Submittals will be acceptable to Engineer/Architect if it provides a workable arrangement for reviewing and processing the required submittals.
  3. Contractor's Schedule of Values will be acceptable to Engineer/Architect as to form and substance if it provides a reasonable allocation of the Contract Price to the component parts of the Work.
  4. If a schedule is not acceptable, Contractor will have an additional 10 days to revise and resubmit the schedule.

## 2.06 *Electronic Transmittals*

- A. Except as otherwise stated elsewhere in the Contract, the Owner, Engineer/Architect, and Contractor may send, and shall accept, Electronic Documents transmitted by Electronic Means.
- B. If the Contract does not establish protocols for Electronic Means, then Owner, Engineer/Architect, and Contractor shall jointly develop such protocols.
- C. Subject to any governing protocols for Electronic Means, when transmitting Electronic Documents by Electronic Means, the transmitting party makes no representations as to long-term compatibility, usability, or readability of the Electronic Documents resulting from the recipient's use of software application packages, operating systems, or computer hardware differing from those used in the drafting or transmittal of the Electronic Documents.

## **ARTICLE 3—CONTRACT DOCUMENTS: INTENT, REQUIREMENTS, REUSE**

### 3.01 *Intent*

- A. The Contract Documents are complementary; what is required by one Contract Document is as binding as if required by all.
- B. It is the intent of the Contract Documents to describe a functionally complete Project (or part thereof) to be constructed in accordance with the Contract Documents.
- C. Unless otherwise stated in the Contract Documents, if there is a discrepancy between the electronic versions of the Contract Documents (including any printed copies derived from such electronic versions) and the printed record version, the printed record version will govern.
- D. The Contract supersedes prior negotiations, representations, and agreements, whether written or oral.
- E. Engineer/Architect will issue clarifications and interpretations of the Contract Documents as provided herein.
- F. Any provision or part of the Contract Documents held to be void or unenforceable under any Law or Regulation will be deemed stricken, and all remaining provisions will continue to be valid and binding upon Owner and Contractor, which agree that the Contract Documents will be reformed to replace such stricken provision or part thereof with a valid and enforceable provision that comes as close as possible to expressing the intention of the stricken provision.
- G. Nothing in the Contract Documents creates:
  - 1. any contractual relationship between Owner or Engineer/Architect and any Subcontractor, Supplier, or other individual or entity performing or furnishing any of the Work, for the benefit of such Subcontractor, Supplier, or other individual or entity; or
  - 2. any obligation on the part of Owner or Engineer/Architect to pay or to see to the payment of any money due any such Subcontractor, Supplier, or other individual or entity, except as may otherwise be required by Laws and Regulations.

### 3.02 *Reference Standards*

#### A. *Standards Specifications, Codes, Laws and Regulations*

1. Reference in the Contract Documents to standard specifications, manuals, reference standards, or codes of any technical society, organization, or association, or to Laws or Regulations, whether such reference be specific or by implication, means the standard specification, manual, reference standard, code, or Laws or Regulations in effect at the time of opening of Bids (or on the Effective Date of the Contract if there were no Bids), except as may be otherwise specifically stated in the Contract Documents.
2. No provision of any such standard specification, manual, reference standard, or code, and no instruction of a Supplier, will be effective to change the duties or responsibilities of Owner, Contractor, or Engineer/Architect from those set forth in the part of the Contract Documents prepared by or for Engineer/Architect. No such provision or instruction shall be effective to assign to Owner or Engineer/Architect any duty or authority to supervise or direct the performance of the Work, or any duty or authority to undertake responsibility inconsistent with the provisions of the part of the Contract Documents prepared by or for Engineer/Architect.

### 3.03 *Reporting and Resolving Discrepancies*

#### A. *Reporting Discrepancies*

1. *Contractor's Verification of Figures and Field Measurements:* Before undertaking each part of the Work, Contractor shall carefully study the Contract Documents, and check and verify pertinent figures and dimensions therein, particularly with respect to applicable field measurements. Contractor shall promptly report in writing to Engineer/Architect any conflict, error, ambiguity, or discrepancy that Contractor discovers, or has actual knowledge of, and shall not proceed with any Work affected thereby until the conflict, error, ambiguity, or discrepancy is resolved by a clarification or interpretation by Engineer/Architect, or by an amendment or supplement to the Contract issued pursuant to Paragraph 11.01.
2. *Contractor's Review of Contract Documents:* If, before or during the performance of the Work, Contractor discovers any conflict, error, ambiguity, or discrepancy within the Contract Documents, or between the Contract Documents and (a) any applicable Law or Regulation, (b) actual field conditions, (c) any standard specification, manual, reference standard, or code, or (d) any instruction of any Supplier, then Contractor shall promptly report it to Engineer/Architect in writing. Contractor shall not proceed with the Work affected thereby (except in an emergency as required by Paragraph 7.15) until the conflict, error, ambiguity, or discrepancy is resolved, by a clarification or interpretation by Engineer/Architect, or by an amendment or supplement to the Contract issued pursuant to Paragraph 11.01.
3. Contractor shall not be liable to Owner or Engineer/Architect for failure to report any conflict, error, ambiguity, or discrepancy in the Contract Documents unless Contractor had actual knowledge thereof.

#### B. *Resolving Discrepancies*

1. Except as may be otherwise specifically stated in the Contract Documents, the provisions of the part of the Contract Documents prepared by or for Engineer/Architect

take precedence in resolving any conflict, error, ambiguity, or discrepancy between such provisions of the Contract Documents and:

- a. the provisions of any standard specification, manual, reference standard, or code, or the instruction of any Supplier (whether or not specifically incorporated by reference as a Contract Document); or
- b. the provisions of any Laws or Regulations applicable to the performance of the Work (unless such an interpretation of the provisions of the Contract Documents would result in violation of such Law or Regulation).

### 3.04 *Requirements of the Contract Documents*

- A. During the performance of the Work and until final payment, Contractor and Owner shall submit to the Engineer/Architect in writing all matters in question concerning the requirements of the Contract Documents (sometimes referred to as requests for information or interpretation—RFIs), or relating to the acceptability of the Work under the Contract Documents, as soon as possible after such matters arise. Engineer/Architect will be the initial interpreter of the requirements of the Contract Documents, and judge of the acceptability of the Work.
- B. Engineer/Architect will, with reasonable promptness, render a written clarification, interpretation, or decision on the issue submitted, or initiate an amendment or supplement to the Contract Documents. Engineer/Architect's written clarification, interpretation, or decision will be final and binding on Contractor, unless it appeals by submitting a Change Proposal, and on Owner, unless it appeals by filing a Claim.
- C. If a submitted matter in question concerns terms and conditions of the Contract Documents that do not involve (1) the performance or acceptability of the Work under the Contract Documents, (2) the design (as set forth in the Drawings, Specifications, or otherwise), or (3) other engineering/architectural or technical matters, then Engineer/Architect will promptly notify Owner and Contractor in writing that Engineer/Architect is unable to provide a decision or interpretation. If Owner and Contractor are unable to agree on resolution of such a matter in question, either party may pursue resolution as provided in Article 12.

### 3.05 *Reuse of Documents*

- A. Contractor and its Subcontractors and Suppliers shall not:
  1. have or acquire any title to or ownership rights in any of the Drawings, Specifications, or other documents (or copies of any thereof) prepared by or bearing the seal of Engineer/Architect or its consultants, including electronic media versions, or reuse any such Drawings, Specifications, other documents, or copies thereof on extensions of the Project or any other project without written consent of Owner and Engineer/Architect and specific written verification or adaptation by Engineer/Architect; or
  2. have or acquire any title or ownership rights in any other Contract Documents, reuse any such Contract Documents for any purpose without Owner's express written consent, or violate any copyrights pertaining to such Contract Documents.
- B. The prohibitions of this Paragraph 3.05 will survive final payment, or termination of the Contract. Nothing herein precludes Contractor from retaining copies of the Contract Documents for record purposes.

## ARTICLE 4—COMMENCEMENT AND PROGRESS OF THE WORK

### 4.01 *Commencement of Contract Times; Notice to Proceed*

- A. The Contract Times will commence to run on the 30th day after the Effective Date of the Contract or, if a Notice to Proceed is given, on the day indicated in the Notice to Proceed. A Notice to Proceed may be given at any time within 30 days after the Effective Date of the Contract. In no event will the Contract Times commence to run later than the 60th day after the day of Bid opening or the 30th day after the Effective Date of the Contract, whichever date is earlier.

### 4.02 *Starting the Work*

- A. Contractor shall start to perform the Work on the date when the Contract Times commence to run. No Work may be done at the Site prior to such date.

### 4.03 *Reference Points*

- A. Owner shall provide engineering surveys to establish reference points for construction which in Engineer/Architect's judgment are necessary to enable Contractor to proceed with the Work. Contractor shall be responsible for laying out the Work, shall protect and preserve the established reference points and property monuments, and shall make no changes or relocations without the prior written approval of Owner. Contractor shall report to Engineer/Architect whenever any reference point or property monument is lost or destroyed or requires relocation because of necessary changes in grades or locations, and shall be responsible for the accurate replacement or relocation of such reference points or property monuments by professionally qualified personnel.

### 4.04 *Progress Schedule*

- A. Contractor shall adhere to the Progress Schedule established in accordance with Paragraph 2.05 as it may be adjusted from time to time as provided below.
  - 1. Contractor shall submit to Engineer/Architect for acceptance (to the extent indicated in Paragraph 2.05) proposed adjustments in the Progress Schedule that will not result in changing the Contract Times.
  - 2. Proposed adjustments in the Progress Schedule that will change the Contract Times must be submitted in accordance with the requirements of Article 11.
- B. Contractor shall carry on the Work and adhere to the Progress Schedule during all disputes or disagreements with Owner. No Work will be delayed or postponed pending resolution of any disputes or disagreements, or during any appeal process, except as permitted by Paragraph 16.04, or as Owner and Contractor may otherwise agree in writing.

### 4.05 *Delays in Contractor's Progress*

- A. If Owner, Engineer/Architect, or anyone for whom Owner is responsible, delays, disrupts, or interferes with the performance or progress of the Work, then Contractor shall be entitled to an equitable adjustment in Contract Price or Contract Times.
- B. Contractor shall not be entitled to an adjustment in Contract Price or Contract Times for delay, disruption, or interference caused by or within the control of Contractor. Delay, disruption, and interference attributable to and within the control of a Subcontractor or Supplier shall be deemed to be within the control of Contractor.

- C. If Contractor's performance or progress is delayed, disrupted, or interfered with by unanticipated causes not the fault of and beyond the control of Owner, Contractor, and those for which they are responsible, then Contractor shall be entitled to an equitable adjustment in Contract Times. Such an adjustment will be Contractor's sole and exclusive remedy for the delays, disruption, and interference described in this paragraph. Causes of delay, disruption, or interference that may give rise to an adjustment in Contract Times under this paragraph include but are not limited to the following:
1. Severe and unavoidable natural catastrophes such as fires, floods, epidemics, and earthquakes;
  2. Abnormal weather conditions;
  3. Acts or failures to act of third-party utility owners or other third-party entities (other than those third-party utility owners or other third-party entities performing other work at or adjacent to the Site as arranged by or under contract with Owner, as contemplated in Article 8); and
  4. Acts of war or terrorism.
- D. Contractor's entitlement to an adjustment of Contract Times or Contract Price is limited as follows:
1. Contractor's entitlement to an adjustment of the Contract Times is conditioned on the delay, disruption, or interference adversely affecting an activity on the critical path to completion of the Work, as of the time of the delay, disruption, or interference.
  2. Contractor shall not be entitled to an adjustment in Contract Price for any delay, disruption, or interference if such delay is concurrent with a delay, disruption, or interference caused by or within the control of Contractor. Such a concurrent delay by Contractor shall not preclude an adjustment of Contract Times to which Contractor is otherwise entitled.
  3. Adjustments of Contract Times or Contract Price are subject to the provisions of Article 11.
- E. Each Contractor request or Change Proposal seeking an increase in Contract Times or Contract Price must be supplemented by supporting data that sets forth in detail the following:
1. The circumstances that form the basis for the requested adjustment;
  2. The date upon which each cause of delay, disruption, or interference began to affect the progress of the Work;
  3. The date upon which each cause of delay, disruption, or interference ceased to affect the progress of the Work;
  4. The number of days' increase in Contract Times claimed as a consequence of each such cause of delay, disruption, or interference; and
  5. The impact on Contract Price, in accordance with the provisions of Paragraph 11.07.

Contractor shall also furnish such additional supporting documentation as Owner or Engineer/Architect may require including, where appropriate, a revised progress schedule indicating all the activities affected by the delay, disruption, or interference, and an



explanation of the effect of the delay, disruption, or interference on the critical path to completion of the Work.

- F. Delays, disruption, and interference to the performance or progress of the Work resulting from the existence of a differing subsurface or physical condition, an Underground Facility that was not shown or indicated by the Contract Documents, or not shown or indicated with reasonable accuracy, and those resulting from Hazardous Environmental Conditions, are governed by Article 5, together with the provisions of Paragraphs 4.05.D and 4.05.E.
- G. Paragraph 8.03 addresses delays, disruption, and interference to the performance or progress of the Work resulting from the performance of certain other work at or adjacent to the Site.

## **ARTICLE 5—SITE; SUBSURFACE AND PHYSICAL CONDITIONS; HAZARDOUS ENVIRONMENTAL CONDITIONS**

### **5.01 *Availability of Lands***

- A. Owner shall furnish the Site. Owner shall notify Contractor in writing of any encumbrances or restrictions not of general application but specifically related to use of the Site with which Contractor must comply in performing the Work.
- B. Upon reasonable written request, Owner shall furnish Contractor with a current statement of record legal title and legal description of the lands upon which permanent improvements are to be made and Owner's interest therein as necessary for giving notice of or filing a mechanic's or construction lien against such lands in accordance with applicable Laws and Regulations.
- C. Contractor shall provide for all additional lands and access thereto that may be required for temporary construction facilities or storage of materials and equipment.

### **5.02 *Use of Site and Other Areas***

#### **A. *Limitation on Use of Site and Other Areas***

- 1. Contractor shall confine construction equipment, temporary construction facilities, the storage of materials and equipment, and the operations of workers to the Site, adjacent areas that Contractor has arranged to use through construction easements or otherwise, and other adjacent areas permitted by Laws and Regulations, and shall not unreasonably encumber the Site and such other adjacent areas with construction equipment or other materials or equipment. Contractor shall assume full responsibility for (a) damage to the Site; (b) damage to any such other adjacent areas used for Contractor's operations; (c) damage to any other adjacent land or areas, or to improvements, structures, utilities, or similar facilities located at such adjacent lands or areas; and (d) for injuries and losses sustained by the owners or occupants of any such land or areas; provided that such damage or injuries result from the performance of the Work or from other actions or conduct of the Contractor or those for which Contractor is responsible.
- 2. If a damage or injury claim is made by the owner or occupant of any such land or area because of the performance of the Work, or because of other actions or conduct of the Contractor or those for which Contractor is responsible, Contractor shall (a) take immediate corrective or remedial action as required by Paragraph 7.13, or otherwise;

(b) promptly attempt to settle the claim as to all parties through negotiations with such owner or occupant, or otherwise resolve the claim by arbitration or other dispute resolution proceeding, or in a court of competent jurisdiction; and (c) to the fullest extent permitted by Laws and Regulations, indemnify and hold harmless Owner and Engineer/Architect, and the officers, directors, members, partners, employees, agents, consultants and subcontractors of each and any of them, from and against any such claim, and against all costs, losses, and damages (including but not limited to all fees and charges of engineers, architects, attorneys, and other professionals and all court or arbitration or other dispute resolution costs) arising out of or relating to any claim or action, legal or equitable, brought by any such owner or occupant against Owner, Engineer/Architect, or any other party indemnified hereunder to the extent caused directly or indirectly, in whole or in part by, or based upon, Contractor's performance of the Work, or because of other actions or conduct of the Contractor or those for which Contractor is responsible.

- B. *Removal of Debris During Performance of the Work:* During the progress of the Work the Contractor shall keep the Site and other adjacent areas free from accumulations of waste materials, rubbish, and other debris. Removal and disposal of such waste materials, rubbish, and other debris will conform to applicable Laws and Regulations.
- C. *Cleaning:* Prior to Substantial Completion of the Work Contractor shall clean the Site and the Work and make it ready for utilization by Owner. At the completion of the Work Contractor shall remove from the Site and adjacent areas all tools, appliances, construction equipment and machinery, and surplus materials and shall restore to original condition all property not designated for alteration by the Contract Documents.
- D. *Loading of Structures:* Contractor shall not load nor permit any part of any structure to be loaded in any manner that will endanger the structure, nor shall Contractor subject any part of the Work or adjacent structures or land to stresses or pressures that will endanger them.

#### 5.03 *Subsurface and Physical Conditions*

- A. *Reports and Drawings:* The Supplementary Conditions identify:
  - 1. Those reports of explorations and tests of subsurface conditions at or adjacent to the Site that contain Technical Data;
  - 2. Those drawings of existing physical conditions at or adjacent to the Site, including those drawings depicting existing surface or subsurface structures at or adjacent to the Site (except Underground Facilities), that contain Technical Data; and
  - 3. Technical Data contained in such reports and drawings.
- B. *Underground Facilities:* Underground Facilities are shown or indicated on the Drawings, pursuant to Paragraph 5.05, and not in the drawings referred to in Paragraph 5.03.A. Information and data regarding the presence or location of Underground Facilities are not intended to be categorized, identified, or defined as Technical Data.
- C. *Reliance by Contractor on Technical Data:* Contractor may rely upon the accuracy of the Technical Data expressly identified in the Supplementary Conditions with respect to such reports and drawings, but such reports and drawings are not Contract Documents. If no such express identification has been made, then Contractor may rely upon the accuracy of the Technical Data as defined in Paragraph 1.01.A.46.b.

- D. *Limitations of Other Data and Documents:* Except for such reliance on Technical Data, Contractor may not rely upon or make any claim against Owner or Engineer/Architect, or any of their officers, directors, members, partners, employees, agents, consultants, or subcontractors, with respect to:
1. the completeness of such reports and drawings for Contractor's purposes, including, but not limited to, any aspects of the means, methods, techniques, sequences, and procedures of construction to be employed by Contractor, and safety precautions and programs incident thereto;
  2. other data, interpretations, opinions, and information contained in such reports or shown or indicated in such drawings;
  3. the contents of other Site-related documents made available to Contractor, such as record drawings from other projects at or adjacent to the Site, or Owner's archival documents concerning the Site; or
  4. any Contractor interpretation of or conclusion drawn from any Technical Data or any such other data, interpretations, opinions, or information.

5.04 *Differing Subsurface or Physical Conditions*

- A. *Notice by Contractor:* If Contractor believes that any subsurface or physical condition that is uncovered or revealed at the Site:
1. is of such a nature as to establish that any Technical Data on which Contractor is entitled to rely as provided in Paragraph 5.03 is materially inaccurate;
  2. is of such a nature as to require a change in the Drawings or Specifications;
  3. differs materially from that shown or indicated in the Contract Documents; or
  4. is of an unusual nature, and differs materially from conditions ordinarily encountered and generally recognized as inherent in work of the character provided for in the Contract Documents;

then Contractor shall, promptly after becoming aware thereof and before further disturbing the subsurface or physical conditions or performing any Work in connection therewith (except in an emergency as required by Paragraph 7.15), notify Owner and Engineer/Architect in writing about such condition. Contractor shall not further disturb such condition or perform any Work in connection therewith (except with respect to an emergency) until receipt of a written statement permitting Contractor to do so.

- B. *Engineer/Architect's Review:* After receipt of written notice as required by the preceding paragraph, Engineer/Architect will promptly review the subsurface or physical condition in question; determine whether it is necessary for Owner to obtain additional exploration or tests with respect to the condition; conclude whether the condition falls within any one or more of the differing site condition categories in Paragraph 5.04.A; obtain any pertinent cost or schedule information from Contractor; prepare recommendations to Owner regarding the Contractor's resumption of Work in connection with the subsurface or physical condition in question and the need for any change in the Drawings or Specifications; and advise Owner in writing of Engineer/Architect's findings, conclusions, and recommendations.

- C. *Owner's Statement to Contractor Regarding Site Condition:* After receipt of Engineer/Architect's written findings, conclusions, and recommendations, Owner shall issue a written statement to Contractor (with a copy to Engineer/Architect) regarding the subsurface or physical condition in question, addressing the resumption of Work in connection with such condition, indicating whether any change in the Drawings or Specifications will be made, and adopting or rejecting Engineer/Architect's written findings, conclusions, and recommendations, in whole or in part.
- D. *Early Resumption of Work:* If at any time Engineer/Architect determines that Work in connection with the subsurface or physical condition in question may resume prior to completion of Engineer/Architect's review or Owner's issuance of its statement to Contractor, because the condition in question has been adequately documented, and analyzed on a preliminary basis, then the Engineer/Architect may at its discretion instruct Contractor to resume such Work.
- E. *Possible Price and Times Adjustments*
1. Contractor shall be entitled to an equitable adjustment in Contract Price or Contract Times, to the extent that the existence of a differing subsurface or physical condition, or any related delay, disruption, or interference, causes an increase or decrease in Contractor's cost of, or time required for, performance of the Work; subject, however, to the following:
    - a. Such condition must fall within any one or more of the categories described in Paragraph 5.04.A;
    - b. With respect to Work that is paid for on a unit price basis, any adjustment in Contract Price will be subject to the provisions of Paragraph 13.03; and,
    - c. Contractor's entitlement to an adjustment of the Contract Times is subject to the provisions of Paragraphs 4.05.D and 4.05.E.
  2. Contractor shall not be entitled to any adjustment in the Contract Price or Contract Times with respect to a subsurface or physical condition if:
    - a. Contractor knew of the existence of such condition at the time Contractor made a commitment to Owner with respect to Contract Price and Contract Times by the submission of a Bid or becoming bound under a negotiated contract, or otherwise;
    - b. The existence of such condition reasonably could have been discovered or revealed as a result of any examination, investigation, exploration, test, or study of the Site and contiguous areas expressly required by the Bidding Requirements or Contract Documents to be conducted by or for Contractor prior to Contractor's making such commitment; or
    - c. Contractor failed to give the written notice required by Paragraph 5.04.A.
  3. If Owner and Contractor agree regarding Contractor's entitlement to and the amount or extent of any adjustment in the Contract Price or Contract Times, then any such adjustment will be set forth in a Change Order.
  4. Contractor may submit a Change Proposal regarding its entitlement to or the amount or extent of any adjustment in the Contract Price or Contract Times, no later than 30 days

after Owner's issuance of the Owner's written statement to Contractor regarding the subsurface or physical condition in question.

- F. *Underground Facilities; Hazardous Environmental Conditions*: Paragraph 5.05 governs rights and responsibilities regarding the presence or location of Underground Facilities. Paragraph 5.06 governs rights and responsibilities regarding Hazardous Environmental Conditions. The provisions of Paragraphs 5.03 and 5.04 are not applicable to the presence or location of Underground Facilities, or to Hazardous Environmental Conditions.

5.05 *Underground Facilities*

- A. *Contractor's Responsibilities*: Unless it is otherwise expressly provided in the Supplementary Conditions, the cost of all of the following are included in the Contract Price, and Contractor shall have full responsibility for:
1. reviewing and checking all information and data regarding existing Underground Facilities at the Site;
  2. complying with applicable state and local utility damage prevention Laws and Regulations;
  3. verifying the actual location of those Underground Facilities shown or indicated in the Contract Documents as being within the area affected by the Work, by exposing such Underground Facilities during the course of construction;
  4. coordination of the Work with the owners (including Owner) of such Underground Facilities, during construction; and
  5. the safety and protection of all existing Underground Facilities at the Site, and repairing any damage thereto resulting from the Work.
- B. *Notice by Contractor*: If Contractor believes that an Underground Facility that is uncovered or revealed at the Site was not shown or indicated on the Drawings, or was not shown or indicated on the Drawings with reasonable accuracy, then Contractor shall, promptly after becoming aware thereof and before further disturbing conditions affected thereby or performing any Work in connection therewith (except in an emergency as required by Paragraph 7.15), notify Owner and Engineer/Architect in writing regarding such Underground Facility.
- C. *Engineer/Architect's Review*: Engineer/Architect will:
1. promptly review the Underground Facility and conclude whether such Underground Facility was not shown or indicated on the Drawings, or was not shown or indicated with reasonable accuracy;
  2. identify and communicate with the owner of the Underground Facility; prepare recommendations to Owner (and if necessary issue any preliminary instructions to Contractor) regarding the Contractor's resumption of Work in connection with the Underground Facility in question;
  3. obtain any pertinent cost or schedule information from Contractor; determine the extent, if any, to which a change is required in the Drawings or Specifications to reflect and document the consequences of the existence or location of the Underground Facility; and

4. advise Owner in writing of Engineer/Architect's findings, conclusions, and recommendations.

During such time, Contractor shall be responsible for the safety and protection of such Underground Facility.

- D. *Owner's Statement to Contractor Regarding Underground Facility:* After receipt of Engineer/Architect's written findings, conclusions, and recommendations, Owner shall issue a written statement to Contractor (with a copy to Engineer/Architect) regarding the Underground Facility in question addressing the resumption of Work in connection with such Underground Facility, indicating whether any change in the Drawings or Specifications will be made, and adopting or rejecting Engineer/Architect's written findings, conclusions, and recommendations in whole or in part.
- E. *Early Resumption of Work:* If at any time Engineer/Architect determines that Work in connection with the Underground Facility may resume prior to completion of Engineer/Architect's review or Owner's issuance of its statement to Contractor, because the Underground Facility in question and conditions affected by its presence have been adequately documented, and analyzed on a preliminary basis, then the Engineer/Architect may at its discretion instruct Contractor to resume such Work.
- F. *Possible Price and Times Adjustments*
  1. Contractor shall be entitled to an equitable adjustment in the Contract Price or Contract Times, to the extent that any existing Underground Facility at the Site that was not shown or indicated on the Drawings, or was not shown or indicated with reasonable accuracy, or any related delay, disruption, or interference, causes an increase or decrease in Contractor's cost of, or time required for, performance of the Work; subject, however, to the following:
    - a. With respect to Work that is paid for on a unit price basis, any adjustment in Contract Price will be subject to the provisions of Paragraph 13.03;
    - b. Contractor's entitlement to an adjustment of the Contract Times is subject to the provisions of Paragraphs 4.05.D and 4.05.E; and
    - c. Contractor gave the notice required in Paragraph 5.05.B.
  2. If Owner and Contractor agree regarding Contractor's entitlement to and the amount or extent of any adjustment in the Contract Price or Contract Times, then any such adjustment will be set forth in a Change Order.
  3. Contractor may submit a Change Proposal regarding its entitlement to or the amount or extent of any adjustment in the Contract Price or Contract Times, no later than 30 days after Owner's issuance of the Owner's written statement to Contractor regarding the Underground Facility in question.
  4. The information and data shown or indicated on the Drawings with respect to existing Underground Facilities at the Site is based on information and data (a) furnished by the owners of such Underground Facilities, or by others, (b) obtained from available records, or (c) gathered in an investigation conducted in accordance with the current edition of ASCE 38, Standard Guideline for the Collection and Depiction of Existing Subsurface Utility Data, by the American Society of Civil Engineer/Architects. If such

information or data is incorrect or incomplete, Contractor's remedies are limited to those set forth in this Paragraph 5.05.F.

5.06 *Hazardous Environmental Conditions at Site*

A. *Reports and Drawings*: The Supplementary Conditions identify:

1. those reports known to Owner relating to Hazardous Environmental Conditions that have been identified at or adjacent to the Site;
2. drawings known to Owner relating to Hazardous Environmental Conditions that have been identified at or adjacent to the Site; and
3. Technical Data contained in such reports and drawings.

B. *Reliance by Contractor on Technical Data Authorized*: Contractor may rely upon the accuracy of the Technical Data expressly identified in the Supplementary Conditions with respect to such reports and drawings, but such reports and drawings are not Contract Documents. If no such express identification has been made, then Contractor may rely on the accuracy of the Technical Data as defined in Paragraph 1.01.A.46.b. Except for such reliance on Technical Data, Contractor may not rely upon or make any claim against Owner or Engineer/Architect, or any of their officers, directors, members, partners, employees, agents, consultants, or subcontractors, with respect to:

1. the completeness of such reports and drawings for Contractor's purposes, including, but not limited to, any aspects of the means, methods, techniques, sequences and procedures of construction to be employed by Contractor, and safety precautions and programs incident thereto;
2. other data, interpretations, opinions, and information contained in such reports or shown or indicated in such drawings; or
3. any Contractor interpretation of or conclusion drawn from any Technical Data or any such other data, interpretations, opinions or information.

C. Contractor shall not be responsible for removing or remediating any Hazardous Environmental Condition encountered, uncovered, or revealed at the Site unless such removal or remediation is expressly identified in the Contract Documents to be within the scope of the Work.

D. Contractor shall be responsible for controlling, containing, and duly removing all Constituents of Concern brought to the Site by Contractor, Subcontractors, Suppliers, or anyone else for whom Contractor is responsible, and for any associated costs; and for the costs of removing and remediating any Hazardous Environmental Condition created by the presence of any such Constituents of Concern.

E. If Contractor encounters, uncovers, or reveals a Hazardous Environmental Condition whose removal or remediation is not expressly identified in the Contract Documents as being within the scope of the Work, or if Contractor or anyone for whom Contractor is responsible creates a Hazardous Environmental Condition, then Contractor shall immediately: (1) secure or otherwise isolate such condition; (2) stop all Work in connection with such condition and in any area affected thereby (except in an emergency as required by Paragraph 7.15); and (3) notify Owner and Engineer/Architect (and promptly thereafter confirm such notice in writing). Owner shall promptly consult with Engineer/Architect concerning the necessity for

Owner to retain a qualified expert to evaluate such condition or take corrective action, if any. Promptly after consulting with Engineer/Architect, Owner shall take such actions as are necessary to permit Owner to timely obtain required permits and provide Contractor the written notice required by Paragraph 5.06.F. If Contractor or anyone for whom Contractor is responsible created the Hazardous Environmental Condition in question, then Owner may remove and remediate the Hazardous Environmental Condition, and impose a set-off against payments to account for the associated costs.

- F. Contractor shall not resume Work in connection with such Hazardous Environmental Condition or in any affected area until after Owner has obtained any required permits related thereto, and delivered written notice to Contractor either (1) specifying that such condition and any affected area is or has been rendered safe for the resumption of Work, or (2) specifying any special conditions under which such Work may be resumed safely.
- G. If Owner and Contractor cannot agree as to entitlement to or on the amount or extent, if any, of any adjustment in Contract Price or Contract Times, as a result of such Work stoppage, such special conditions under which Work is agreed to be resumed by Contractor, or any costs or expenses incurred in response to the Hazardous Environmental Condition, then within 30 days of Owner's written notice regarding the resumption of Work, Contractor may submit a Change Proposal, or Owner may impose a set-off. Entitlement to any such adjustment is subject to the provisions of Paragraphs 4.05.D, 4.05.E, 11.07, and 11.08.
- H. If, after receipt of such written notice, Contractor does not agree to resume such Work based on a reasonable belief it is unsafe, or does not agree to resume such Work under such special conditions, then Owner may order the portion of the Work that is in the area affected by such condition to be deleted from the Work, following the contractual change procedures in Article 11. Owner may have such deleted portion of the Work performed by Owner's own forces or others in accordance with Article 8.
- I. To the fullest extent permitted by Laws and Regulations, Owner shall indemnify and hold harmless Contractor, Subcontractors, and Engineer/Architect, and the officers, directors, members, partners, employees, agents, consultants, and subcontractors of each and any of them, from and against all claims, costs, losses, and damages (including but not limited to all fees and charges of engineers, architects, attorneys, and other professionals, and all court, arbitration, or other dispute resolution costs) arising out of or relating to a Hazardous Environmental Condition, provided that such Hazardous Environmental Condition (1) was not shown or indicated in the Drawings, Specifications, or other Contract Documents, identified as Technical Data entitled to limited reliance pursuant to Paragraph 5.06.B, or identified in the Contract Documents to be included within the scope of the Work, and (2) was not created by Contractor or by anyone for whom Contractor is responsible. Nothing in this Paragraph 5.06.I obligates Owner to indemnify any individual or entity from and against the consequences of that individual's or entity's own negligence.
- J. To the fullest extent permitted by Laws and Regulations, Contractor shall indemnify and hold harmless Owner and Engineer/Architect, and the officers, directors, members, partners, employees, agents, consultants, and subcontractors of each and any of them, from and against all claims, costs, losses, and damages (including but not limited to all fees and charges of engineers, architects, attorneys, and other professionals and all court or arbitration or other dispute resolution costs) arising out of or relating to the failure to control, contain, or remove a Constituent of Concern brought to the Site by Contractor or by



anyone for whom Contractor is responsible, or to a Hazardous Environmental Condition created by Contractor or by anyone for whom Contractor is responsible. Nothing in this Paragraph 5.06.J obligates Contractor to indemnify any individual or entity from and against the consequences of that individual's or entity's own negligence.

- K. The provisions of Paragraphs 5.03, 5.04, and 5.05 do not apply to the presence of Constituents of Concern or to a Hazardous Environmental Condition uncovered or revealed at the Site.

## **ARTICLE 6—BONDS AND INSURANCE**

### **6.01 *Performance, Payment, and Other Bonds***

- A. Contractor shall furnish a performance bond and a payment bond, each in an amount at least equal to the Contract Price, as security for the faithful performance and payment of Contractor's obligations under the Contract. These bonds must remain in effect until one year after the date when final payment becomes due or until completion of the correction period specified in Paragraph 15.08, whichever is later, except as provided otherwise by Laws or Regulations, the terms of a prescribed bond form, the Supplementary Conditions, or other provisions of the Contract.
- B. Contractor shall also furnish such other bonds (if any) as are required by the Supplementary Conditions or other provisions of the Contract.
- C. All bonds must be in the form included in the Bidding Documents or otherwise specified by Owner prior to execution of the Contract, except as provided otherwise by Laws or Regulations, and must be issued and signed by a surety named in "Companies Holding Certificates of Authority as Acceptable Sureties on Federal Bonds and as Acceptable Reinsuring Companies" as published in Department Circular 570 (as amended and supplemented) by the Bureau of the Fiscal Service, U.S. Department of the Treasury. A bond signed by an agent or attorney-in-fact must be accompanied by a certified copy of that individual's authority to bind the surety. The evidence of authority must show that it is effective on the date the agent or attorney-in-fact signed the accompanying bond.
- D. Contractor shall obtain the required bonds from surety companies that are duly licensed or authorized, in the state or jurisdiction in which the Project is located, to issue bonds in the required amounts.
- E. If the surety on a bond furnished by Contractor is declared bankrupt or becomes insolvent, or the surety ceases to meet the requirements above, then Contractor shall promptly notify Owner and Engineer/Architect in writing and shall, within 20 days after the event giving rise to such notification, provide another bond and surety, both of which must comply with the bond and surety requirements above.
- F. If Contractor has failed to obtain a required bond, Owner may exclude the Contractor from the Site and exercise Owner's termination rights under Article 16.
- G. Upon request to Owner from any Subcontractor, Supplier, or other person or entity claiming to have furnished labor, services, materials, or equipment used in the performance of the Work, Owner shall provide a copy of the payment bond to such person or entity.

- H. Upon request to Contractor from any Subcontractor, Supplier, or other person or entity claiming to have furnished labor, services, materials, or equipment used in the performance of the Work, Contractor shall provide a copy of the payment bond to such person or entity.

#### 6.02 *Insurance—General Provisions*

- A. Owner and Contractor shall obtain and maintain insurance as required in this article and in the Supplementary Conditions.
- B. All insurance required by the Contract to be purchased and maintained by Owner or Contractor shall be obtained from insurance companies that are duly licensed or authorized in the state or jurisdiction in which the Project is located to issue insurance policies for the required limits and coverages. Unless a different standard is indicated in the Supplementary Conditions, all companies that provide insurance policies required under this Contract shall have an A.M. Best rating of A-VII or better.
- C. Alternative forms of insurance coverage, including but not limited to self-insurance and “Occupational Accident and Excess Employer’s Indemnity Policies,” are not sufficient to meet the insurance requirements of this Contract, unless expressly allowed in the Supplementary Conditions.
- D. Contractor shall deliver to Owner, with copies to each additional insured identified in the Contract, certificates of insurance and endorsements establishing that Contractor has obtained and is maintaining the policies and coverages required by the Contract. Upon request by Owner or any other insured, Contractor shall also furnish other evidence of such required insurance, including but not limited to copies of policies, documentation of applicable self-insured retentions (if allowed) and deductibles, full disclosure of all relevant exclusions, and evidence of insurance required to be purchased and maintained by Subcontractors or Suppliers. In any documentation furnished under this provision, Contractor, Subcontractors, and Suppliers may block out (redact) (1) any confidential premium or pricing information and (2) any wording specific to a project or jurisdiction other than those applicable to this Contract.
- E. Owner shall deliver to Contractor, with copies to each additional insured identified in the Contract, certificates of insurance and endorsements establishing that Owner has obtained and is maintaining the policies and coverages required of Owner by the Contract (if any). Upon request by Contractor or any other insured, Owner shall also provide other evidence of such required insurance (if any), including but not limited to copies of policies, documentation of applicable self-insured retentions (if allowed) and deductibles, and full disclosure of all relevant exclusions. In any documentation furnished under this provision, Owner may block out (redact) (1) any confidential premium or pricing information and (2) any wording specific to a project or jurisdiction other than those relevant to this Contract.
- F. Failure of Owner or Contractor to demand such certificates or other evidence of the other party’s full compliance with these insurance requirements, or failure of Owner or Contractor to identify a deficiency in compliance from the evidence provided, will not be construed as a waiver of the other party’s obligation to obtain and maintain such insurance.
- G. In addition to the liability insurance required to be provided by Contractor, the Owner, at Owner’s option, may purchase and maintain Owner’s own liability insurance. Owner’s liability policies, if any, operate separately and independently from policies required to be

provided by Contractor, and Contractor cannot rely upon Owner's liability policies for any of Contractor's obligations to the Owner, Engineer/Architect, or third parties.

H. Contractor shall require:

1. Subcontractors to purchase and maintain worker's compensation, commercial general liability, and other insurance that is appropriate for their participation in the Project, and to name as additional insureds Owner and Engineer/Architect (and any other individuals or entities identified in the Supplementary Conditions as additional insureds on Contractor's liability policies) on each Subcontractor's commercial general liability insurance policy; and
  2. Suppliers to purchase and maintain insurance that is appropriate for their participation in the Project.
- I. If either party does not purchase or maintain the insurance required of such party by the Contract, such party shall notify the other party in writing of such failure to purchase prior to the start of the Work, or of such failure to maintain prior to any change in the required coverage.
- J. If Contractor has failed to obtain and maintain required insurance, Contractor's entitlement to enter or remain at the Site will end immediately, and Owner may impose an appropriate set-off against payment for any associated costs (including but not limited to the cost of purchasing necessary insurance coverage), and exercise Owner's termination rights under Article 16.
- K. Without prejudice to any other right or remedy, if a party has failed to obtain required insurance, the other party may elect (but is in no way obligated) to obtain equivalent insurance to protect such other party's interests at the expense of the party who was required to provide such coverage, and the Contract Price will be adjusted accordingly.
- L. Owner does not represent that insurance coverage and limits established in this Contract necessarily will be adequate to protect Contractor or Contractor's interests. Contractor is responsible for determining whether such coverage and limits are adequate to protect its interests, and for obtaining and maintaining any additional insurance that Contractor deems necessary.
- M. The insurance and insurance limits required herein will not be deemed as a limitation on Contractor's liability, or that of its Subcontractors or Suppliers, under the indemnities granted to Owner and other individuals and entities in the Contract or otherwise.
- N. All the policies of insurance required to be purchased and maintained under this Contract will contain a provision or endorsement that the coverage afforded will not be canceled, or renewal refused, until at least 10 days prior written notice has been given to the purchasing policyholder. Within three days of receipt of any such written notice, the purchasing policyholder shall provide a copy of the notice to each other insured and Engineer/Architect.

6.03 *Contractor's Insurance*

- A. *Required Insurance:* Contractor shall purchase and maintain Worker's Compensation, Commercial General Liability, and other insurance pursuant to the specific requirements of the Supplementary Conditions.

- B. *General Provisions:* The policies of insurance required by this Paragraph 6.03 as supplemented must:
1. include at least the specific coverages required;
  2. be written for not less than the limits provided, or those required by Laws or Regulations, whichever is greater;
  3. remain in effect at least until the Work is complete (as set forth in Paragraph 15.06.D), and longer if expressly required elsewhere in this Contract, and at all times thereafter when Contractor may be correcting, removing, or replacing defective Work as a warranty or correction obligation, or otherwise, or returning to the Site to conduct other tasks arising from the Contract;
  4. apply with respect to the performance of the Work, whether such performance is by Contractor, any Subcontractor or Supplier, or by anyone directly or indirectly employed by any of them to perform any of the Work, or by anyone for whose acts any of them may be liable; and
  5. include all necessary endorsements to support the stated requirements.
- C. *Additional Insureds:* The Contractor's commercial general liability, automobile liability, employer's liability, umbrella or excess, pollution liability, and unmanned aerial vehicle liability policies, if required by this Contract, must:
1. include and list as additional insureds Owner and Engineer/Architect, and any individuals or entities identified as additional insureds in the Supplementary Conditions;
  2. include coverage for the respective officers, directors, members, partners, employees, and consultants of all such additional insureds;
  3. afford primary coverage to these additional insureds for all claims covered thereby (including as applicable those arising from both ongoing and completed operations);
  4. not seek contribution from insurance maintained by the additional insured; and
  5. as to commercial general liability insurance, apply to additional insureds with respect to liability caused in whole or in part by Contractor's acts or omissions, or the acts and omissions of those working on Contractor's behalf, in the performance of Contractor's operations.

#### 6.04 *Builder's Risk and Other Property Insurance*

- A. *Builder's Risk:* Unless otherwise provided in the Supplementary Conditions, Contractor shall purchase and maintain builder's risk insurance upon the Work on a completed value basis, in the amount of the Work's full insurable replacement cost (subject to such deductible amounts as may be provided in the Supplementary Conditions or required by Laws and Regulations). The specific requirements applicable to the builder's risk insurance are set forth in the Supplementary Conditions.
- B. *Property Insurance for Facilities of Owner Where Work Will Occur:* Owner is responsible for obtaining and maintaining property insurance covering each existing structure, building, or facility in which any part of the Work will occur, or to which any part of the Work will attach or be adjoined. Such property insurance will be written on a special perils (all-risk) form, on a replacement cost basis, providing coverage consistent with that required for the builder's

risk insurance, and will be maintained until the Work is complete, as set forth in Paragraph 15.06.D.

- C. *Property Insurance for Substantially Complete Facilities:* Promptly after Substantial Completion, and before actual occupancy or use of the substantially completed Work, Owner will obtain property insurance for such substantially completed Work, and maintain such property insurance at least until the Work is complete, as set forth in Paragraph 15.06.D. Such property insurance will be written on a special perils (all-risk) form, on a replacement cost basis, and provide coverage consistent with that required for the builder's risk insurance. The builder's risk insurance may terminate upon written confirmation of Owner's procurement of such property insurance.
- D. *Partial Occupancy or Use by Owner:* If Owner will occupy or use a portion or portions of the Work prior to Substantial Completion of all the Work, as provided in Paragraph 15.04, then Owner (directly, if it is the purchaser of the builder's risk policy, or through Contractor) will provide advance notice of such occupancy or use to the builder's risk insurer, and obtain an endorsement consenting to the continuation of coverage prior to commencing such partial occupancy or use.
- E. *Insurance of Other Property; Additional Insurance:* If the express insurance provisions of the Contract do not require or address the insurance of a property item or interest, then the entity or individual owning such property item will be responsible for insuring it. If Contractor elects to obtain other special insurance to be included in or supplement the builder's risk or property insurance policies provided under this Paragraph 6.04, it may do so at Contractor's expense.

6.05 *Property Losses; Subrogation*

- A. The builder's risk insurance policy purchased and maintained in accordance with Paragraph 6.04 (or an installation floater policy if authorized by the Supplementary Conditions), will contain provisions to the effect that in the event of payment of any loss or damage the insurer will have no rights of recovery against any insureds thereunder, or against Engineer/Architect or its consultants, or their officers, directors, members, partners, employees, agents, consultants, or subcontractors.
  - 1. Owner and Contractor waive all rights against each other and the respective officers, directors, members, partners, employees, agents, consultants, and subcontractors of each and any of them, for all losses and damages caused by, arising out of, or resulting from any of the perils, risks, or causes of loss covered by such policies and any other property insurance applicable to the Work; and, in addition, waive all such rights against Engineer/Architect, its consultants, all individuals or entities identified in the Supplementary Conditions as builder's risk or installation floater insureds, and the officers, directors, members, partners, employees, agents, consultants, and subcontractors of each and any of them, under such policies for losses and damages so caused.
  - 2. None of the above waivers extends to the rights that any party making such waiver may have to the proceeds of insurance held by Owner or Contractor as trustee or fiduciary, or otherwise payable under any policy so issued.
- B. Any property insurance policy maintained by Owner covering any loss, damage, or consequential loss to Owner's existing structures, buildings, or facilities in which any part of

the Work will occur, or to which any part of the Work will attach or adjoin; to adjacent structures, buildings, or facilities of Owner; or to part or all of the completed or substantially completed Work, during partial occupancy or use pursuant to Paragraph 15.04, after Substantial Completion pursuant to Paragraph 15.03, or after final payment pursuant to Paragraph 15.06, will contain provisions to the effect that in the event of payment of any loss or damage the insurer will have no rights of recovery against any insureds thereunder, or against Contractor, Subcontractors, or Engineer/Architect, or the officers, directors, members, partners, employees, agents, consultants, or subcontractors of each and any of them, and that the insured is allowed to waive the insurer's rights of subrogation in a written contract executed prior to the loss, damage, or consequential loss.

1. Owner waives all rights against Contractor, Subcontractors, and Engineer/Architect, and the officers, directors, members, partners, employees, agents, consultants and subcontractors of each and any of them, for all losses and damages caused by, arising out of, or resulting from fire or any of the perils, risks, or causes of loss covered by such policies.
- C. The waivers in this Paragraph 6.05 include the waiver of rights due to business interruption, loss of use, or other consequential loss extending beyond direct physical loss or damage to Owner's property or the Work caused by, arising out of, or resulting from fire or other insured peril, risk, or cause of loss.
- D. Contractor shall be responsible for assuring that each Subcontract contains provisions whereby the Subcontractor waives all rights against Owner, Contractor, all individuals or entities identified in the Supplementary Conditions as insureds, the Engineer/Architect and its consultants, and the officers, directors, members, partners, employees, agents, consultants, and subcontractors of each and any of them, for all losses and damages caused by, arising out of, relating to, or resulting from fire or other peril, risk, or cause of loss covered by builder's risk insurance, installation floater, and any other property insurance applicable to the Work.

#### 6.06 *Receipt and Application of Property Insurance Proceeds*

- A. Any insured loss under the builder's risk and other policies of property insurance required by Paragraph 6.04 will be adjusted and settled with the named insured that purchased the policy. Such named insured shall act as fiduciary for the other insureds, and give notice to such other insureds that adjustment and settlement of a claim is in progress. Any other insured may state its position regarding a claim for insured loss in writing within 15 days after notice of such claim.
- B. Proceeds for such insured losses may be made payable by the insurer either jointly to multiple insureds, or to the named insured that purchased the policy in its own right and as fiduciary for other insureds, subject to the requirements of any applicable mortgage clause. A named insured receiving insurance proceeds under the builder's risk and other policies of insurance required by Paragraph 6.04 shall maintain such proceeds in a segregated account, and distribute such proceeds in accordance with such agreement as the parties in interest may reach, or as otherwise required under the dispute resolution provisions of this Contract or applicable Laws and Regulations.
- C. If no other special agreement is reached, Contractor shall repair or replace the damaged Work, using allocated insurance proceeds.

## ARTICLE 7—CONTRACTOR'S RESPONSIBILITIES

### 7.01 *Contractor's Means and Methods of Construction*

- A. Contractor shall be solely responsible for the means, methods, techniques, sequences, and procedures of construction.
- B. If the Contract Documents note, or Contractor determines, that professional engineering/architectural or other design services are needed to carry out Contractor's responsibilities for construction means, methods, techniques, sequences, and procedures, or for Site safety, then Contractor shall cause such services to be provided by a properly licensed design professional, at Contractor's expense. Such services are not Owner-delegated professional design services under this Contract, and neither Owner nor Engineer/Architect has any responsibility with respect to (1) Contractor's determination of the need for such services, (2) the qualifications or licensing of the design professionals retained or employed by Contractor, (3) the performance of such services, or (4) any errors, omissions, or defects in such services.

### 7.02 *Supervision and Superintendence*

- A. Contractor shall supervise, inspect, and direct the Work competently and efficiently, devoting such attention thereto and applying such skills and expertise as may be necessary to perform the Work in accordance with the Contract Documents.
- B. At all times during the progress of the Work, Contractor shall assign a competent resident superintendent who will not be replaced without written notice to Owner and Engineer/Architect except under extraordinary circumstances.

### 7.03 *Labor; Working Hours*

- A. Contractor shall provide competent, suitably qualified personnel to survey and lay out the Work and perform construction as required by the Contract Documents. Contractor shall maintain good discipline and order at the Site.
- B. Contractor shall be fully responsible to Owner and Engineer/Architect for all acts and omissions of Contractor's employees; of Suppliers and Subcontractors, and their employees; and of any other individuals or entities performing or furnishing any of the Work, just as Contractor is responsible for Contractor's own acts and omissions.
- C. Except as otherwise required for the safety or protection of persons or the Work or property at the Site or adjacent thereto, and except as otherwise stated in the Contract Documents, all Work at the Site will be performed during regular working hours, Monday through Friday. Contractor will not perform Work on a Saturday, Sunday, or any legal holiday. Contractor may perform Work outside regular working hours or on Saturdays, Sundays, or legal holidays only with Owner's written consent, which will not be unreasonably withheld.

### 7.04 *Services, Materials, and Equipment*

- A. Unless otherwise specified in the Contract Documents, Contractor shall provide and assume full responsibility for all services, materials, equipment, labor, transportation, construction equipment and machinery, tools, appliances, fuel, power, light, heat, telephone, water, sanitary facilities, temporary facilities, and all other facilities and incidentals necessary for the performance, testing, start up, and completion of the Work, whether or not such items are specifically called for in the Contract Documents.

- B. All materials and equipment incorporated into the Work must be new and of good quality, except as otherwise provided in the Contract Documents. All special warranties and guarantees required by the Specifications will expressly run to the benefit of Owner. If required by Engineer/Architect, Contractor shall furnish satisfactory evidence (including reports of required tests) as to the source, kind, and quality of materials and equipment.
- C. All materials and equipment must be stored, applied, installed, connected, erected, protected, used, cleaned, and conditioned in accordance with instructions of the applicable Supplier, except as otherwise may be provided in the Contract Documents.

7.05 *“Or Equals”*

- A. *Contractor’s Request; Governing Criteria:* Whenever an item of equipment or material is specified or described in the Contract Documents by using the names of one or more proprietary items or specific Suppliers, the Contract Price has been based upon Contractor furnishing such item as specified. The specification or description of such an item is intended to establish the type, function, appearance, and quality required. Unless the specification or description contains or is followed by words reading that no like, equivalent, or “or equal” item is permitted, Contractor may request that Engineer/Architect authorize the use of other items of equipment or material, or items from other proposed Suppliers, under the circumstances described below.
  - 1. If Engineer/Architect in its sole discretion determines that an item of equipment or material proposed by Contractor is functionally equal to that named and sufficiently similar so that no change in related Work will be required, Engineer/Architect will deem it an “or equal” item. For the purposes of this paragraph, a proposed item of equipment or material will be considered functionally equal to an item so named if:
    - a. in the exercise of reasonable judgment Engineer/Architect determines that the proposed item:
      - 1) is at least equal in materials of construction, quality, durability, appearance, strength, and design characteristics;
      - 2) will reliably perform at least equally well the function and achieve the results imposed by the design concept of the completed Project as a functioning whole;
      - 3) has a proven record of performance and availability of responsive service; and
      - 4) is not objectionable to Owner.
    - b. Contractor certifies that, if the proposed item is approved and incorporated into the Work:
      - 1) there will be no increase in cost to the Owner or increase in Contract Times; and
      - 2) the item will conform substantially to the detailed requirements of the item named in the Contract Documents.
- B. *Contractor’s Expense:* Contractor shall provide all data in support of any proposed “or equal” item at Contractor’s expense.
- C. *Engineer/Architect’s Evaluation and Determination:* Engineer/Architect will be allowed a reasonable time to evaluate each “or-equal” request. Engineer/Architect may require Contractor to furnish additional data about the proposed “or-equal” item.



Engineer/Architect will be the sole judge of acceptability. No “or-equal” item will be ordered, furnished, installed, or utilized until Engineer/Architect’s review is complete and Engineer/Architect determines that the proposed item is an “or-equal,” which will be evidenced by an approved Shop Drawing or other written communication. Engineer/Architect will advise Contractor in writing of any negative determination.

- D. *Effect of Engineer/Architect’s Determination:* Neither approval nor denial of an “or-equal” request will result in any change in Contract Price. The Engineer/Architect’s denial of an “or-equal” request will be final and binding, and may not be reversed through an appeal under any provision of the Contract.
- E. *Treatment as a Substitution Request:* If Engineer/Architect determines that an item of equipment or material proposed by Contractor does not qualify as an “or-equal” item, Contractor may request that Engineer/Architect consider the item a proposed substitute pursuant to Paragraph 7.06.

#### 7.06 Substitutes

- A. *Contractor’s Request; Governing Criteria:* Unless the specification or description of an item of equipment or material required to be furnished under the Contract Documents contains or is followed by words reading that no substitution is permitted, Contractor may request that Engineer/Architect authorize the use of other items of equipment or material under the circumstances described below. To the extent possible such requests must be made before commencement of related construction at the Site.
  - 1. Contractor shall submit sufficient information as provided below to allow Engineer/Architect to determine if the item of material or equipment proposed is functionally equivalent to that named and an acceptable substitute therefor. Engineer/Architect will not accept requests for review of proposed substitute items of equipment or material from anyone other than Contractor.
  - 2. The requirements for review by Engineer/Architect will be as set forth in Paragraph 7.06.B, as supplemented by the Specifications, and as Engineer/Architect may decide is appropriate under the circumstances.
  - 3. Contractor shall make written application to Engineer/Architect for review of a proposed substitute item of equipment or material that Contractor seeks to furnish or use. The application:
    - a. will certify that the proposed substitute item will:
      - 1) perform adequately the functions and achieve the results called for by the general design;
      - 2) be similar in substance to the item specified; and
      - 3) be suited to the same use as the item specified.
    - b. will state:
      - 1) the extent, if any, to which the use of the proposed substitute item will necessitate a change in Contract Times;
      - 2) whether use of the proposed substitute item in the Work will require a change in any of the Contract Documents (or in the provisions of any other direct

contract with Owner for other work on the Project) to adapt the design to the proposed substitute item; and

- 3) whether incorporation or use of the proposed substitute item in connection with the Work is subject to payment of any license fee or royalty.

c. will identify:

- 1) all variations of the proposed substitute item from the item specified; and
- 2) available engineering, sales, maintenance, repair, and replacement services.

d. will contain an itemized estimate of all costs or credits that will result directly or indirectly from use of such substitute item, including but not limited to changes in Contract Price, shared savings, costs of redesign, and claims of other contractors affected by any resulting change.

- B. *Engineer/Architect's Evaluation and Determination:* Engineer/Architect will be allowed a reasonable time to evaluate each substitute request, and to obtain comments and direction from Owner. Engineer/Architect may require Contractor to furnish additional data about the proposed substitute item. Engineer/Architect will be the sole judge of acceptability. No substitute will be ordered, furnished, installed, or utilized until Engineer/Architect's review is complete and Engineer/Architect determines that the proposed item is an acceptable substitute. Engineer/Architect's determination will be evidenced by a Field Order or a proposed Change Order accounting for the substitution itself and all related impacts, including changes in Contract Price or Contract Times. Engineer/Architect will advise Contractor in writing of any negative determination.
- C. *Special Guarantee:* Owner may require Contractor to furnish at Contractor's expense a special performance guarantee or other surety with respect to any substitute.
- D. *Reimbursement of Engineer/Architect's Cost:* Engineer/Architect will record Engineer/Architect's costs in evaluating a substitute proposed or submitted by Contractor. Whether or not Engineer/Architect approves a substitute so proposed or submitted by Contractor, Contractor shall reimburse Owner for the reasonable charges of Engineer/Architect for evaluating each such proposed substitute. Contractor shall also reimburse Owner for the reasonable charges of Engineer/Architect for making changes in the Contract Documents (or in the provisions of any other direct contract with Owner) resulting from the acceptance of each proposed substitute.
- E. *Contractor's Expense:* Contractor shall provide all data in support of any proposed substitute at Contractor's expense.
- F. *Effect of Engineer/Architect's Determination:* If Engineer/Architect approves the substitution request, Contractor shall execute the proposed Change Order and proceed with the substitution. The Engineer/Architect's denial of a substitution request will be final and binding, and may not be reversed through an appeal under any provision of the Contract. Contractor may challenge the scope of reimbursement costs imposed under Paragraph 7.06.D, by timely submittal of a Change Proposal.

#### 7.07 Concerning Subcontractors and Suppliers

- A. Contractor may retain Subcontractors and Suppliers for the performance of parts of the Work. Such Subcontractors and Suppliers must be acceptable to Owner. The Contractor's

retention of a Subcontractor or Supplier for the performance of parts of the Work will not relieve Contractor's obligation to Owner to perform and complete the Work in accordance with the Contract Documents.

- B. Contractor shall retain specific Subcontractors and Suppliers for the performance of designated parts of the Work if required by the Contract to do so.
- C. Subsequent to the submittal of Contractor's Bid or final negotiation of the terms of the Contract, Owner may not require Contractor to retain any Subcontractor or Supplier to furnish or perform any of the Work against which Contractor has reasonable objection.
- D. Prior to entry into any binding subcontract or purchase order, Contractor shall submit to Owner the identity of the proposed Subcontractor or Supplier (unless Owner has already deemed such proposed Subcontractor or Supplier acceptable during the bidding process or otherwise). Such proposed Subcontractor or Supplier shall be deemed acceptable to Owner unless Owner raises a substantive, reasonable objection within 5 days.
- E. Owner may require the replacement of any Subcontractor or Supplier. Owner also may require Contractor to retain specific replacements; provided, however, that Owner may not require a replacement to which Contractor has a reasonable objection. If Contractor has submitted the identity of certain Subcontractors or Suppliers for acceptance by Owner, and Owner has accepted it (either in writing or by failing to make written objection thereto), then Owner may subsequently revoke the acceptance of any such Subcontractor or Supplier so identified solely on the basis of substantive, reasonable objection after due investigation. Contractor shall submit an acceptable replacement for the rejected Subcontractor or Supplier.
- F. If Owner requires the replacement of any Subcontractor or Supplier retained by Contractor to perform any part of the Work, then Contractor shall be entitled to an adjustment in Contract Price or Contract Times, with respect to the replacement; and Contractor shall initiate a Change Proposal for such adjustment within 30 days of Owner's requirement of replacement.
- G. No acceptance by Owner of any such Subcontractor or Supplier, whether initially or as a replacement, will constitute a waiver of the right of Owner to the completion of the Work in accordance with the Contract Documents.
- H. On a monthly basis, Contractor shall submit to Engineer/Architect a complete list of all Subcontractors and Suppliers having a direct contract with Contractor, and of all other Subcontractors and Suppliers known to Contractor at the time of submittal.
- I. Contractor shall be solely responsible for scheduling and coordinating the work of Subcontractors and Suppliers.
- J. The divisions and sections of the Specifications and the identifications of any Drawings do not control Contractor in dividing the Work among Subcontractors or Suppliers, or in delineating the Work to be performed by any specific trade.
- K. All Work performed for Contractor by a Subcontractor or Supplier must be pursuant to an appropriate contractual agreement that specifically binds the Subcontractor or Supplier to the applicable terms and conditions of the Contract for the benefit of Owner and Engineer/Architect.

- L. Owner may furnish to any Subcontractor or Supplier, to the extent practicable, information about amounts paid to Contractor for Work performed for Contractor by the Subcontractor or Supplier.
- M. Contractor shall restrict all Subcontractors and Suppliers from communicating with Engineer/Architect or Owner, except through Contractor or in case of an emergency, or as otherwise expressly allowed in this Contract.

#### 7.08 *Patent Fees and Royalties*

- A. Contractor shall pay all license fees and royalties and assume all costs incident to the use in the performance of the Work or the incorporation in the Work of any invention, design, process, product, or device which is the subject of patent rights or copyrights held by others. If an invention, design, process, product, or device is specified in the Contract Documents for use in the performance of the Work and if, to the actual knowledge of Owner or Engineer/Architect, its use is subject to patent rights or copyrights calling for the payment of any license fee or royalty to others, the existence of such rights will be disclosed in the Contract Documents.
- B. To the fullest extent permitted by Laws and Regulations, Owner shall indemnify and hold harmless Contractor, and its officers, directors, members, partners, employees, agents, consultants, and subcontractors, from and against all claims, costs, losses, and damages (including but not limited to all fees and charges of Engineer/Architects, architects, attorneys, and other professionals, and all court or arbitration or other dispute resolution costs) arising out of or relating to any infringement of patent rights or copyrights incident to the use in the performance of the Work or resulting from the incorporation in the Work of any invention, design, process, product, or device specified in the Contract Documents, but not identified as being subject to payment of any license fee or royalty to others required by patent rights or copyrights.
- C. To the fullest extent permitted by Laws and Regulations, Contractor shall indemnify and hold harmless Owner and Engineer/Architect, and the officers, directors, members, partners, employees, agents, consultants and subcontractors of each and any of them, from and against all claims, costs, losses, and damages (including but not limited to all fees and charges of engineers, architects, attorneys, and other professionals and all court or arbitration or other dispute resolution costs) arising out of or relating to any infringement of patent rights or copyrights incident to the use in the performance of the Work or resulting from the incorporation in the Work of any invention, design, process, product, or device not specified in the Contract Documents.

#### 7.09 *Permits*

- A. Unless otherwise provided in the Contract Documents, Contractor shall obtain and pay for all construction permits, licenses, and certificates of occupancy. Owner shall assist Contractor, when necessary, in obtaining such permits and licenses. Contractor shall pay all governmental charges and inspection fees necessary for the prosecution of the Work which are applicable at the time of the submission of Contractor's Bid (or when Contractor became bound under a negotiated contract). Owner shall pay all charges of utility owners for connections for providing permanent service to the Work.

#### 7.10 *Taxes*

- A. Contractor shall pay all sales, consumer, use, and other similar taxes required to be paid by Contractor in accordance with the Laws and Regulations of the place of the Project which are applicable during the performance of the Work.

#### 7.11 *Laws and Regulations*

- A. Contractor shall give all notices required by and shall comply with all Laws and Regulations applicable to the performance of the Work. Neither Owner nor Engineer/Architect shall be responsible for monitoring Contractor's compliance with any Laws or Regulations.
- B. If Contractor performs any Work or takes any other action knowing or having reason to know that it is contrary to Laws or Regulations, Contractor shall bear all resulting costs and losses, and shall indemnify and hold harmless Owner and Engineer/Architect, and the officers, directors, members, partners, employees, agents, consultants, and subcontractors of each and any of them, from and against all claims, costs, losses, and damages (including but not limited to all fees and charges of engineers, architects, attorneys, and other professionals and all court or arbitration or other dispute resolution costs) arising out of or relating to such Work or other action. It is not Contractor's responsibility to make certain that the Work described in the Contract Documents is in accordance with Laws and Regulations, but this does not relieve Contractor of its obligations under Paragraph 3.03.
- C. Owner or Contractor may give written notice to the other party of any changes after the submission of Contractor's Bid (or after the date when Contractor became bound under a negotiated contract) in Laws or Regulations having an effect on the cost or time of performance of the Work, including but not limited to changes in Laws or Regulations having an effect on procuring permits and on sales, use, value-added, consumption, and other similar taxes. If Owner and Contractor are unable to agree on entitlement to or on the amount or extent, if any, of any adjustment in Contract Price or Contract Times resulting from such changes, then within 30 days of such written notice Contractor may submit a Change Proposal, or Owner may initiate a Claim.

#### 7.12 *Record Documents*

- A. Contractor shall maintain in a safe place at the Site one printed record copy of all Drawings, Specifications, Addenda, Change Orders, Work Change Directives, Field Orders, written interpretations and clarifications, and approved Shop Drawings. Contractor shall keep such record documents in good order and annotate them to show changes made during construction. These record documents, together with all approved Samples, will be available to Engineer/Architect for reference. Upon completion of the Work, Contractor shall deliver these record documents to Engineer/Architect.

#### 7.13 *Safety and Protection*

- A. Contractor shall be solely responsible for initiating, maintaining, and supervising all safety precautions and programs in connection with the Work. Such responsibility does not relieve Subcontractors of their responsibility for the safety of persons or property in the performance of their work, nor for compliance with applicable safety Laws and Regulations.
- B. Contractor shall designate a qualified and experienced safety representative whose duties and responsibilities are the prevention of Work-related accidents and the maintenance and supervision of safety precautions and programs.

- C. Contractor shall take all necessary precautions for the safety of, and shall provide the necessary protection to prevent damage, injury, or loss to:
  - 1. all persons on the Site or who may be affected by the Work;
  - 2. all the Work and materials and equipment to be incorporated therein, whether in storage on or off the Site; and
  - 3. other property at the Site or adjacent thereto, including trees, shrubs, lawns, walks, pavements, roadways, structures, other work in progress, utilities, and Underground Facilities not designated for removal, relocation, or replacement in the course of construction.
- D. All damage, injury, or loss to any property referred to in Paragraph 7.13.C.2 or 7.13.C.3 caused, directly or indirectly, in whole or in part, by Contractor, any Subcontractor, Supplier, or any other individual or entity directly or indirectly employed by any of them to perform any of the Work, or anyone for whose acts any of them may be liable, shall be remedied by Contractor at its expense (except damage or loss attributable to the fault of Drawings or Specifications or to the acts or omissions of Owner or Engineer/Architect or anyone employed by any of them, or anyone for whose acts any of them may be liable, and not attributable, directly or indirectly, in whole or in part, to the fault or negligence of Contractor or any Subcontractor, Supplier, or other individual or entity directly or indirectly employed by any of them).
- E. Contractor shall comply with all applicable Laws and Regulations relating to the safety of persons or property, or to the protection of persons or property from damage, injury, or loss; and shall erect and maintain all necessary safeguards for such safety and protection.
- F. Contractor shall notify Owner; the owners of adjacent property; the owners of Underground Facilities and other utilities (if the identity of such owners is known to Contractor); and other contractors and utility owners performing work at or adjacent to the Site, in writing, when Contractor knows that prosecution of the Work may affect them, and shall cooperate with them in the protection, removal, relocation, and replacement of their property or work in progress.
- G. Contractor shall comply with the applicable requirements of Owner's safety programs, if any. Any Owner's safety programs that are applicable to the Work are identified or included in the Supplementary Conditions or Specifications.
- H. Contractor shall inform Owner and Engineer/Architect of the specific requirements of Contractor's safety program with which Owner's and Engineer/Architect's employees and representatives must comply while at the Site.
- I. Contractor's duties and responsibilities for safety and protection will continue until all the Work is completed, Engineer/Architect has issued a written notice to Owner and Contractor in accordance with Paragraph 15.06.C that the Work is acceptable, and Contractor has left the Site (except as otherwise expressly provided in connection with Substantial Completion).
- J. Contractor's duties and responsibilities for safety and protection will resume whenever Contractor or any Subcontractor or Supplier returns to the Site to fulfill warranty or correction obligations, or to conduct other tasks arising from the Contract Documents.

#### 7.14 *Hazard Communication Programs*

- A. Contractor shall be responsible for coordinating any exchange of safety data sheets (formerly known as material safety data sheets) or other hazard communication information required to be made available to or exchanged between or among employers at the Site in accordance with Laws or Regulations.

#### 7.15 *Emergencies*

- A. In emergencies affecting the safety or protection of persons or the Work or property at the Site or adjacent thereto, Contractor is obligated to act to prevent damage, injury, or loss. Contractor shall give Engineer/Architect prompt written notice if Contractor believes that any significant changes in the Work or variations from the Contract Documents have been caused by an emergency, or are required as a result of Contractor's response to an emergency. If Engineer/Architect determines that a change in the Contract Documents is required because of an emergency or Contractor's response, a Work Change Directive or Change Order will be issued.

#### 7.16 *Submittals*

##### A. *Shop Drawing and Sample Requirements*

1. Before submitting a Shop Drawing or Sample, Contractor shall:
  - a. review and coordinate the Shop Drawing or Sample with other Shop Drawings and Samples and with the requirements of the Work and the Contract Documents;
  - b. determine and verify:
    - 1) all field measurements, quantities, dimensions, specified performance and design criteria, installation requirements, materials, catalog numbers, and similar information with respect to the Submittal;
    - 2) the suitability of all materials and equipment offered with respect to the indicated application, fabrication, shipping, handling, storage, assembly, and installation pertaining to the performance of the Work; and
    - 3) all information relative to Contractor's responsibilities for means, methods, techniques, sequences, and procedures of construction, and safety precautions and programs incident thereto;
  - c. confirm that the Submittal is complete with respect to all related data included in the Submittal.
2. Each Shop Drawing or Sample must bear a stamp or specific written certification that Contractor has satisfied Contractor's obligations under the Contract Documents with respect to Contractor's review of that Submittal, and that Contractor approves the Submittal.
3. With each Shop Drawing or Sample, Contractor shall give Engineer/Architect specific written notice of any variations that the Submittal may have from the requirements of the Contract Documents. This notice must be set forth in a written communication separate from the Submittal; and, in addition, in the case of a Shop Drawing by a specific notation made on the Shop Drawing itself.

- B. *Submittal Procedures for Shop Drawings and Samples:* Contractor shall label and submit Shop Drawings and Samples to Engineer/Architect for review and approval in accordance with the accepted Schedule of Submittals.

1. *Shop Drawings*

- a. Contractor shall submit the number of copies required in the Specifications.
- b. Data shown on the Shop Drawings must be complete with respect to quantities, dimensions, specified performance and design criteria, materials, and similar data to show Engineer/Architect the services, materials, and equipment Contractor proposes to provide, and to enable Engineer/Architect to review the information for the limited purposes required by Paragraph 7.16.C.

2. *Samples*

- a. Contractor shall submit the number of Samples required in the Specifications.
  - b. Contractor shall clearly identify each Sample as to material, Supplier, pertinent data such as catalog numbers, the use for which intended and other data as Engineer/Architect may require to enable Engineer/Architect to review the Submittal for the limited purposes required by Paragraph 7.16.C.
3. Where a Shop Drawing or Sample is required by the Contract Documents or the Schedule of Submittals, any related Work performed prior to Engineer/Architect's review and approval of the pertinent submittal will be at the sole expense and responsibility of Contractor.

C. *Engineer/Architect's Review of Shop Drawings and Samples*

1. Engineer/Architect will provide timely review of Shop Drawings and Samples in accordance with the accepted Schedule of Submittals. Engineer/Architect's review and approval will be only to determine if the items covered by the Submittals will, after installation or incorporation in the Work, comply with the requirements of the Contract Documents, and be compatible with the design concept of the completed Project as a functioning whole as indicated by the Contract Documents.
2. Engineer/Architect's review and approval will not extend to means, methods, techniques, sequences, or procedures of construction, or to safety precautions or programs incident thereto.
3. Engineer/Architect's review and approval of a separate item as such will not indicate approval of the assembly in which the item functions.
4. Engineer/Architect's review and approval of a Shop Drawing or Sample will not relieve Contractor from responsibility for any variation from the requirements of the Contract Documents unless Contractor has complied with the requirements of Paragraph 7.16.A.3 and Engineer/Architect has given written approval of each such variation by specific written notation thereof incorporated in or accompanying the Shop Drawing or Sample. Engineer/Architect will document any such approved variation from the requirements of the Contract Documents in a Field Order or other appropriate Contract modification.



5. Engineer/Architect's review and approval of a Shop Drawing or Sample will not relieve Contractor from responsibility for complying with the requirements of Paragraphs 7.16.A and B.
6. Engineer/Architect's review and approval of a Shop Drawing or Sample, or of a variation from the requirements of the Contract Documents, will not, under any circumstances, change the Contract Times or Contract Price, unless such changes are included in a Change Order.
7. Neither Engineer/Architect's receipt, review, acceptance, or approval of a Shop Drawing or Sample will result in such item becoming a Contract Document.
8. Contractor shall perform the Work in compliance with the requirements and commitments set forth in approved Shop Drawings and Samples, subject to the provisions of Paragraph 7.16.C.4.

*D. Resubmittal Procedures for Shop Drawings and Samples*

1. Contractor shall make corrections required by Engineer/Architect and shall return the required number of corrected copies of Shop Drawings and submit, as required, new Samples for review and approval. Contractor shall direct specific attention in writing to revisions other than the corrections called for by Engineer/Architect on previous Submittals.
2. Contractor shall furnish required Shop Drawing and Sample submittals with sufficient information and accuracy to obtain required approval of an item with no more than two resubmittals. Engineer/Architect will record Engineer/Architect's time for reviewing a third or subsequent resubmittal of a Shop Drawing or Sample, and Contractor shall be responsible for Engineer/Architect's charges to Owner for such time. Owner may impose a set-off against payments due Contractor to secure reimbursement for such charges.
3. If Contractor requests a change of a previously approved Shop Drawing or Sample, Contractor shall be responsible for Engineer/Architect's charges to Owner for its review time, and Owner may impose a set-off against payments due Contractor to secure reimbursement for such charges, unless the need for such change is beyond the control of Contractor.

*E. Submittals Other than Shop Drawings, Samples, and Owner-Delegated Designs*

1. The following provisions apply to all Submittals other than Shop Drawings, Samples, and Owner-delegated designs:
  - a. Contractor shall submit all such Submittals to the Engineer/Architect in accordance with the Schedule of Submittals and pursuant to the applicable terms of the Contract Documents.
  - b. Engineer/Architect will provide timely review of all such Submittals in accordance with the Schedule of Submittals and return such Submittals with a notation of either Accepted or Not Accepted. Any such Submittal that is not returned within the time established in the Schedule of Submittals will be deemed accepted.

- c. Engineer/Architect's review will be only to determine if the Submittal is acceptable under the requirements of the Contract Documents as to general form and content of the Submittal.
  - d. If any such Submittal is not accepted, Contractor shall confer with Engineer/Architect regarding the reason for the non-acceptance, and resubmit an acceptable document.
2. Procedures for the submittal and acceptance of the Progress Schedule, the Schedule of Submittals, and the Schedule of Values are set forth in Paragraphs 2.03, 2.04, and 2.05.
- F. Owner-delegated Designs: Submittals pursuant to Owner-delegated designs are governed by the provisions of Paragraph 7.19.

**7.17 Contractor's General Warranty and Guarantee**

- A. Contractor warrants and guarantees to Owner that all Work will be in accordance with the Contract Documents and will not be defective. Engineer/Architect is entitled to rely on Contractor's warranty and guarantee.
- B. Owner's rights under this warranty and guarantee are in addition to, and are not limited by, Owner's rights under the correction period provisions of Paragraph 15.08. The time in which Owner may enforce its warranty and guarantee rights under this Paragraph 7.17 is limited only by applicable Laws and Regulations restricting actions to enforce such rights; provided, however, that after the end of the correction period under Paragraph 15.08:
  - 1. Owner shall give Contractor written notice of any defective Work within 60 days of the discovery that such Work is defective; and
  - 2. Such notice will be deemed the start of an event giving rise to a Claim under Paragraph 12.01.B, such that any related Claim must be brought within 30 days of the notice.
- C. Contractor's warranty and guarantee hereunder excludes defects or damage caused by:
  - 1. abuse, or improper modification, maintenance, or operation, by persons other than Contractor, Subcontractors, Suppliers, or any other individual or entity for whom Contractor is responsible; or
  - 2. normal wear and tear under normal usage.
- D. Contractor's obligation to perform and complete the Work in accordance with the Contract Documents is absolute. None of the following will constitute an acceptance of Work that is not in accordance with the Contract Documents, a release of Contractor's obligation to perform the Work in accordance with the Contract Documents, or a release of Owner's warranty and guarantee rights under this Paragraph 7.17:
  - 1. Observations by Engineer/Architect;
  - 2. Recommendation by Engineer/Architect or payment by Owner of any progress or final payment;
  - 3. The issuance of a certificate of Substantial Completion by Engineer/Architect or any payment related thereto by Owner;
  - 4. Use or occupancy of the Work or any part thereof by Owner;

5. Any review and approval of a Shop Drawing or Sample submittal;
  6. The issuance of a notice of acceptability by Engineer/Architect;
  7. The end of the correction period established in Paragraph 15.08;
  8. Any inspection, test, or approval by others; or
  9. Any correction of defective Work by Owner.
- E. If the Contract requires the Contractor to accept the assignment of a contract entered into by Owner, then the specific warranties, guarantees, and correction obligations contained in the assigned contract will govern with respect to Contractor's performance obligations to Owner for the Work described in the assigned contract.

#### 7.18 *Indemnification*

- A. To the fullest extent permitted by Laws and Regulations, and in addition to any other obligations of Contractor under the Contract or otherwise, Contractor shall indemnify and hold harmless Owner and Engineer/Architect, and the officers, directors, members, partners, employees, agents, consultants and subcontractors of each and any of them, from losses, damages, costs, and judgments (including but not limited to all fees and charges of engineers, architects, attorneys, and other professionals, and all court or arbitration or other dispute resolution costs) arising from third-party claims or actions relating to or resulting from the performance or furnishing of the Work, provided that any such claim, action, loss, cost, judgment or damage is attributable to bodily injury, sickness, disease, or death, or to damage to or destruction of tangible property (other than the Work itself), including the loss of use resulting therefrom, but only to the extent caused by any negligent act or omission of Contractor, any Subcontractor, any Supplier, or any individual or entity directly or indirectly employed by any of them to perform any of the Work, or anyone for whose acts any of them may be liable.
- B. In any and all claims against Owner or Engineer/Architect, or any of their officers, directors, members, partners, employees, agents, consultants, or subcontractors, by any employee (or the survivor or personal representative of such employee) of Contractor, any Subcontractor, any Supplier, or any individual or entity directly or indirectly employed by any of them to perform any of the Work, or anyone for whose acts any of them may be liable, the indemnification obligation under Paragraph 7.18.A will not be limited in any way by any limitation on the amount or type of damages, compensation, or benefits payable by or for Contractor or any such Subcontractor, Supplier, or other individual or entity under workers' compensation acts, disability benefit acts, or other employee benefit acts.

#### 7.19 *Delegation of Professional Design Services*

- A. Owner may require Contractor to provide professional design services for a portion of the Work by express delegation in the Contract Documents. Such delegation will specify the performance and design criteria that such services must satisfy, and the Submittals that Contractor must furnish to Engineer/Architect with respect to the Owner-delegated design.
- B. Contractor shall cause such Owner-delegated professional design services to be provided pursuant to the professional standard of care by a properly licensed design professional, whose signature and seal must appear on all drawings, calculations, specifications, certifications, and Submittals prepared by such design professional. Such design professional must issue all certifications of design required by Laws and Regulations.

- C. If a Shop Drawing or other Submittal related to the Owner-delegated design is prepared by Contractor, a Subcontractor, or others for submittal to Engineer/Architect, then such Shop Drawing or other Submittal must bear the written approval of Contractor's design professional when submitted by Contractor to Engineer/Architect.
- D. Owner and Engineer/Architect shall be entitled to rely upon the adequacy, accuracy, and completeness of the services, certifications, and approvals performed or provided by the design professionals retained or employed by Contractor under an Owner-delegated design, subject to the professional standard of care and the performance and design criteria stated in the Contract Documents.
- E. Pursuant to this Paragraph 7.19, Engineer/Architect's review, approval, and other determinations regarding design drawings, calculations, specifications, certifications, and other Submittals furnished by Contractor pursuant to an Owner-delegated design will be only for the following limited purposes:
  - 1. Checking for conformance with the requirements of this Paragraph 7.19;
  - 2. Confirming that Contractor (through its design professionals) has used the performance and design criteria specified in the Contract Documents; and
  - 3. Establishing that the design furnished by Contractor is consistent with the design concept expressed in the Contract Documents.
- F. Contractor shall not be responsible for the adequacy of performance or design criteria specified by Owner or Engineer/Architect.
- G. Contractor is not required to provide professional services in violation of applicable Laws and Regulations.

## **ARTICLE 8—OTHER WORK AT THE SITE**

### **8.01 *Other Work***

- A. In addition to and apart from the Work under the Contract Documents, the Owner may perform other work at or adjacent to the Site. Such other work may be performed by Owner's employees, or through contracts between the Owner and third parties. Owner may also arrange to have third-party utility owners perform work on their utilities and facilities at or adjacent to the Site.
- B. If Owner performs other work at or adjacent to the Site with Owner's employees, or through contracts for such other work, then Owner shall give Contractor written notice thereof prior to starting any such other work. If Owner has advance information regarding the start of any third-party utility work that Owner has arranged to take place at or adjacent to the Site, Owner shall provide such information to Contractor.
- C. Contractor shall afford proper and safe access to the Site to each contractor that performs such other work, each utility owner performing other work, and Owner, if Owner is performing other work with Owner's employees, and provide a reasonable opportunity for the introduction and storage of materials and equipment and the execution of such other work.
- D. Contractor shall do all cutting, fitting, and patching of the Work that may be required to properly connect or otherwise make its several parts come together and properly integrate

with such other work. Contractor shall not endanger any work of others by cutting, excavating, or otherwise altering such work; provided, however, that Contractor may cut or alter others' work with the written consent of Engineer/Architect and the others whose work will be affected.

- E. If the proper execution or results of any part of Contractor's Work depends upon work performed by others, Contractor shall inspect such other work and promptly report to Engineer/Architect in writing any delays, defects, or deficiencies in such other work that render it unavailable or unsuitable for the proper execution and results of Contractor's Work. Contractor's failure to so report will constitute an acceptance of such other work as fit and proper for integration with Contractor's Work except for latent defects and deficiencies in such other work.
- F. The provisions of this article are not applicable to work that is performed by third-party utilities or other third-party entities without a contract with Owner, or that is performed without having been arranged by Owner. If such work occurs, then any related delay, disruption, or interference incurred by Contractor is governed by the provisions of Paragraph 4.05.C.3.

#### 8.02 *Coordination*

- A. If Owner intends to contract with others for the performance of other work at or adjacent to the Site, to perform other work at or adjacent to the Site with Owner's employees, or to arrange to have utility owners perform work at or adjacent to the Site, the following will be set forth in the Supplementary Conditions or provided to Contractor prior to the start of any such other work:
  - 1. The identity of the individual or entity that will have authority and responsibility for coordination of the activities among the various contractors;
  - 2. An itemization of the specific matters to be covered by such authority and responsibility; and
  - 3. The extent of such authority and responsibilities.
- B. Unless otherwise provided in the Supplementary Conditions, Owner shall have sole authority and responsibility for such coordination.

#### 8.03 *Legal Relationships*

- A. If, in the course of performing other work for Owner at or adjacent to the Site, the Owner's employees, any other contractor working for Owner, or any utility owner that Owner has arranged to perform work, causes damage to the Work or to the property of Contractor or its Subcontractors, or delays, disrupts, interferes with, or increases the scope or cost of the performance of the Work, through actions or inaction, then Contractor shall be entitled to an equitable adjustment in the Contract Price or the Contract Times. Contractor must submit any Change Proposal seeking an equitable adjustment in the Contract Price or the Contract Times under this paragraph within 30 days of the damaging, delaying, disrupting, or interfering event. The entitlement to, and extent of, any such equitable adjustment will take into account information (if any) regarding such other work that was provided to Contractor in the Contract Documents prior to the submittal of the Bid or the final negotiation of the terms of the Contract, and any remedies available to Contractor under Laws or Regulations concerning utility action or inaction. When applicable, any such

equitable adjustment in Contract Price will be conditioned on Contractor assigning to Owner all Contractor's rights against such other contractor or utility owner with respect to the damage, delay, disruption, or interference that is the subject of the adjustment. Contractor's entitlement to an adjustment of the Contract Times or Contract Price is subject to the provisions of Paragraphs 4.05.D and 4.05.E.

- B. Contractor shall take reasonable and customary measures to avoid damaging, delaying, disrupting, or interfering with the work of Owner, any other contractor, or any utility owner performing other work at or adjacent to the Site.
  - 1. If Contractor fails to take such measures and as a result damages, delays, disrupts, or interferes with the work of any such other contractor or utility owner, then Owner may impose a set-off against payments due Contractor, and assign to such other contractor or utility owner the Owner's contractual rights against Contractor with respect to the breach of the obligations set forth in this Paragraph 8.03.B.
  - 2. When Owner is performing other work at or adjacent to the Site with Owner's employees, Contractor shall be liable to Owner for damage to such other work, and for the reasonable direct delay, disruption, and interference costs incurred by Owner as a result of Contractor's failure to take reasonable and customary measures with respect to Owner's other work. In response to such damage, delay, disruption, or interference, Owner may impose a set-off against payments due Contractor.
- C. If Contractor damages, delays, disrupts, or interferes with the work of any other contractor, or any utility owner performing other work at or adjacent to the Site, through Contractor's failure to take reasonable and customary measures to avoid such impacts, or if any claim arising out of Contractor's actions, inactions, or negligence in performance of the Work at or adjacent to the Site is made by any such other contractor or utility owner against Contractor, Owner, or Engineer/Architect, then Contractor shall (1) promptly attempt to settle the claim as to all parties through negotiations with such other contractor or utility owner, or otherwise resolve the claim by arbitration or other dispute resolution proceeding or at law, and (2) indemnify and hold harmless Owner and Engineer/Architect, and the officers, directors, members, partners, employees, agents, consultants and subcontractors of each and any of them from and against any such claims, and against all costs, losses, and damages (including but not limited to all fees and charges of engineers, architects, attorneys, and other professionals and all court or arbitration or other dispute resolution costs) arising out of or relating to such damage, delay, disruption, or interference.

## **ARTICLE 9—OWNER'S RESPONSIBILITIES**

### **9.01   *Communications to Contractor***

- A. Except as otherwise provided in these General Conditions, Owner shall issue all communications to Contractor through Engineer/Architect.

### **9.02   *Replacement of Engineer/Architect***

- A. Owner may at its discretion appoint an Engineer/Architect to replace Engineer/Architect, provided Contractor makes no reasonable objection to the replacement Engineer/Architect. The replacement Engineer/Architect's status under the Contract Documents will be that of the former Engineer/Architect.

9.03 *Furnish Data*

A. Owner shall promptly furnish the data required of Owner under the Contract Documents.

9.04 *Pay When Due*

A. Owner shall make payments to Contractor when they are due as provided in the Agreement.

9.05 *Lands and Easements; Reports, Tests, and Drawings*

A. Owner's duties with respect to providing lands and easements are set forth in Paragraph 5.01.

B. Owner's duties with respect to providing engineering surveys to establish reference points are set forth in Paragraph 4.03.

C. Article 5 refers to Owner's identifying and making available to Contractor copies of reports of explorations and tests of conditions at the Site, and drawings of physical conditions relating to existing surface or subsurface structures at the Site.

9.06 *Insurance*

A. Owner's responsibilities, if any, with respect to purchasing and maintaining liability and property insurance are set forth in Article 6.

9.07 *Change Orders*

A. Owner's responsibilities with respect to Change Orders are set forth in Article 11.

9.08 *Inspections, Tests, and Approvals*

A. Owner's responsibility with respect to certain inspections, tests, and approvals is set forth in Paragraph 14.02.B.

9.09 *Limitations on Owner's Responsibilities*

A. The Owner shall not supervise, direct, or have control or authority over, nor be responsible for, Contractor's means, methods, techniques, sequences, or procedures of construction, or the safety precautions and programs incident thereto, or for any failure of Contractor to comply with Laws and Regulations applicable to the performance of the Work. Owner will not be responsible for Contractor's failure to perform the Work in accordance with the Contract Documents.

9.10 *Undisclosed Hazardous Environmental Condition*

A. Owner's responsibility in respect to an undisclosed Hazardous Environmental Condition is set forth in Paragraph 5.06.

9.11 *Evidence of Financial Arrangements*

A. Upon request of Contractor, Owner shall furnish Contractor reasonable evidence that financial arrangements have been made to satisfy Owner's obligations under the Contract (including obligations under proposed changes in the Work).

9.12 *Safety Programs*

A. While at the Site, Owner's employees and representatives shall comply with the specific applicable requirements of Contractor's safety programs of which Owner has been informed.

- B. Owner shall furnish copies of any applicable Owner safety programs to Contractor.

## **ARTICLE 10—ENGINEER/ARCHITECT’S STATUS DURING CONSTRUCTION**

### **10.01 *Owner’s Representative***

- A. Engineer/Architect will be Owner’s representative during the construction period. The duties and responsibilities and the limitations of authority of Engineer/Architect as Owner’s representative during construction are set forth in the Contract.

### **10.02 *Visits to Site***

- A. Engineer/Architect will make visits to the Site at intervals appropriate to the various stages of construction as Engineer/Architect deems necessary in order to observe, as an experienced and qualified design professional, the progress that has been made and the quality of the various aspects of Contractor’s executed Work. Based on information obtained during such visits and observations, Engineer/Architect, for the benefit of Owner, will determine, in general, if the Work is proceeding in accordance with the Contract Documents. Engineer/Architect will not be required to make exhaustive or continuous inspections on the Site to check the quality or quantity of the Work. Engineer/Architect’s efforts will be directed toward providing for Owner a greater degree of confidence that the completed Work will conform generally to the Contract Documents. On the basis of such visits and observations, Engineer/Architect will keep Owner informed of the progress of the Work and will endeavor to guard Owner against defective Work.
- B. Engineer/Architect’s visits and observations are subject to all the limitations on Engineer/Architect’s authority and responsibility set forth in Paragraph 10.07. Particularly, but without limitation, during or as a result of Engineer/Architect’s visits or observations of Contractor’s Work, Engineer/Architect will not supervise, direct, control, or have authority over or be responsible for Contractor’s means, methods, techniques, sequences, or procedures of construction, or the safety precautions and programs incident thereto, or for any failure of Contractor to comply with Laws and Regulations applicable to the performance of the Work.

### **10.03 *Resident Project Representative***

- A. If Owner and Engineer/Architect have agreed that Engineer/Architect will furnish a Resident Project Representative to represent Engineer/Architect at the Site and assist Engineer/Architect in observing the progress and quality of the Work, then the authority and responsibilities of any such Resident Project Representative will be as provided in the Supplementary Conditions, and limitations on the responsibilities thereof will be as provided in the Supplementary Conditions and in Paragraph 10.07.
- B. If Owner designates an individual or entity who is not Engineer/Architect’s consultant, agent, or employee to represent Owner at the Site, then the responsibilities and authority of such individual or entity will be as provided in the Supplementary Conditions.

### **10.04 *Engineer/Architect’s Authority***

- A. Engineer/Architect has the authority to reject Work in accordance with Article 14.
- B. Engineer/Architect’s authority as to Submittals is set forth in Paragraph 7.16.



- C. Engineer/Architect's authority as to design drawings, calculations, specifications, certifications and other Submittals from Contractor in response to Owner's delegation (if any) to Contractor of professional design services, is set forth in Paragraph 7.19.
- D. Engineer/Architect's authority as to changes in the Work is set forth in Article 11.
- E. Engineer/Architect's authority as to Applications for Payment is set forth in Article 15.

10.05 *Determinations for Unit Price Work*

- A. Engineer/Architect will determine the actual quantities and classifications of Unit Price Work performed by Contractor as set forth in Paragraph 13.03.

10.06 *Decisions on Requirements of Contract Documents and Acceptability of Work*

- A. Engineer/Architect will render decisions regarding the requirements of the Contract Documents, and judge the acceptability of the Work, pursuant to the specific procedures set forth herein for initial interpretations, Change Proposals, and acceptance of the Work. In rendering such decisions and judgments, Engineer/Architect will not show partiality to Owner or Contractor, and will not be liable to Owner, Contractor, or others in connection with any proceedings, interpretations, decisions, or judgments conducted or rendered in good faith.

10.07 *Limitations on Engineer/Architect's Authority and Responsibilities*

- A. Neither Engineer/Architect's authority or responsibility under this Article 10 or under any other provision of the Contract, nor any decision made by Engineer/Architect in good faith either to exercise or not exercise such authority or responsibility or the undertaking, exercise, or performance of any authority or responsibility by Engineer/Architect, will create, impose, or give rise to any duty in contract, tort, or otherwise owed by Engineer/Architect to Contractor, any Subcontractor, any Supplier, any other individual or entity, or to any surety for or employee or agent of any of them.
- B. Engineer/Architect will not supervise, direct, control, or have authority over or be responsible for Contractor's means, methods, techniques, sequences, or procedures of construction, or the safety precautions and programs incident thereto, or for any failure of Contractor to comply with Laws and Regulations applicable to the performance of the Work. Engineer/Architect will not be responsible for Contractor's failure to perform the Work in accordance with the Contract Documents.
- C. Engineer/Architect will not be responsible for the acts or omissions of Contractor or of any Subcontractor, any Supplier, or of any other individual or entity performing any of the Work.
- D. Engineer/Architect's review of the final Application for Payment and accompanying documentation, and all maintenance and operating instructions, schedules, guarantees, bonds, certificates of inspection, tests and approvals, and other documentation required to be delivered by Contractor under Paragraph 15.06.A, will only be to determine generally that their content complies with the requirements of, and in the case of certificates of inspections, tests, and approvals, that the results certified indicate compliance with the Contract Documents.
- E. The limitations upon authority and responsibility set forth in this Paragraph 10.07 also apply to the Resident Project Representative, if any.

#### 10.08 *Compliance with Safety Program*

- A. While at the Site, Engineer/Architect's employees and representatives will comply with the specific applicable requirements of Owner's and Contractor's safety programs of which Engineer/Architect has been informed.

### **ARTICLE 11—CHANGES TO THE CONTRACT**

#### 11.01 *Amending and Supplementing the Contract*

- A. The Contract may be amended or supplemented by a Change Order, a Work Change Directive, or a Field Order.
- B. If an amendment or supplement to the Contract includes a change in the Contract Price or the Contract Times, such amendment or supplement must be set forth in a Change Order.
- C. All changes to the Contract that involve (1) the performance or acceptability of the Work, (2) the design (as set forth in the Drawings, Specifications, or otherwise), or (3) other engineering or technical matters, must be supported by Engineer/Architect's recommendation. Owner and Contractor may amend other terms and conditions of the Contract without the recommendation of the Engineer/Architect.

#### 11.02 *Change Orders*

- A. Owner and Contractor shall execute appropriate Change Orders covering:
  - 1. Changes in Contract Price or Contract Times which are agreed to by the parties, including any undisputed sum or amount of time for Work actually performed in accordance with a Work Change Directive;
  - 2. Changes in Contract Price resulting from an Owner set-off, unless Contractor has duly contested such set-off;
  - 3. Changes in the Work which are: (a) ordered by Owner pursuant to Paragraph 11.05, (b) required because of Owner's acceptance of defective Work under Paragraph 14.04 or Owner's correction of defective Work under Paragraph 14.07, or (c) agreed to by the parties, subject to the need for Engineer/Architect's recommendation if the change in the Work involves the design (as set forth in the Drawings, Specifications, or otherwise) or other engineering or technical matters; and
  - 4. Changes that embody the substance of any final and binding results under: Paragraph 11.03.B, resolving the impact of a Work Change Directive; Paragraph 11.09, concerning Change Proposals; Article 12, Claims; Paragraph 13.02.D, final adjustments resulting from allowances; Paragraph 13.03.D, final adjustments relating to determination of quantities for Unit Price Work; and similar provisions.
- B. If Owner or Contractor refuses to execute a Change Order that is required to be executed under the terms of Paragraph 11.02.A, it will be deemed to be of full force and effect, as if fully executed.

#### 11.03 *Work Change Directives*

- A. A Work Change Directive will not change the Contract Price or the Contract Times but is evidence that the parties expect that the modification ordered or documented by a Work Change Directive will be incorporated in a subsequently issued Change Order, following

negotiations by the parties as to the Work Change Directive's effect, if any, on the Contract Price and Contract Times; or, if negotiations are unsuccessful, by a determination under the terms of the Contract Documents governing adjustments, expressly including Paragraph 11.07 regarding change of Contract Price.

- B. If Owner has issued a Work Change Directive and:
  - 1. Contractor believes that an adjustment in Contract Times or Contract Price is necessary, then Contractor shall submit any Change Proposal seeking such an adjustment no later than 30 days after the completion of the Work set out in the Work Change Directive.
  - 2. Owner believes that an adjustment in Contract Times or Contract Price is necessary, then Owner shall submit any Claim seeking such an adjustment no later than 60 days after issuance of the Work Change Directive.

#### 11.04 *Field Orders*

- A. Engineer/Architect may authorize minor changes in the Work if the changes do not involve an adjustment in the Contract Price or the Contract Times and are compatible with the design concept of the completed Project as a functioning whole as indicated by the Contract Documents. Such changes will be accomplished by a Field Order and will be binding on Owner and also on Contractor, which shall perform the Work involved promptly.
- B. If Contractor believes that a Field Order justifies an adjustment in the Contract Price or Contract Times, then before proceeding with the Work at issue, Contractor shall submit a Change Proposal as provided herein.

#### 11.05 *Owner-Authorized Changes in the Work*

- A. Without invalidating the Contract and without notice to any surety, Owner may, at any time or from time to time, order additions, deletions, or revisions in the Work. Changes involving the design (as set forth in the Drawings, Specifications, or otherwise) or other engineering or technical matters will be supported by Engineer/Architect's recommendation.
- B. Such changes in the Work may be accomplished by a Change Order, if Owner and Contractor have agreed as to the effect, if any, of the changes on Contract Times or Contract Price; or by a Work Change Directive. Upon receipt of any such document, Contractor shall promptly proceed with the Work involved; or, in the case of a deletion in the Work, promptly cease construction activities with respect to such deleted Work. Added or revised Work must be performed under the applicable conditions of the Contract Documents.
- C. Nothing in this Paragraph 11.05 obligates Contractor to undertake work that Contractor reasonably concludes cannot be performed in a manner consistent with Contractor's safety obligations under the Contract Documents or Laws and Regulations.

#### 11.06 *Unauthorized Changes in the Work*

- A. Contractor shall not be entitled to an increase in the Contract Price or an extension of the Contract Times with respect to any work performed that is not required by the Contract Documents, as amended, modified, or supplemented, except in the case of an emergency as provided in Paragraph 7.15 or in the case of uncovering Work as provided in Paragraph 14.05.C.2.

#### 11.07 *Change of Contract Price*

- A. The Contract Price may only be changed by a Change Order. Any Change Proposal for an adjustment in the Contract Price must comply with the provisions of Paragraph 11.09. Any Claim for an adjustment of Contract Price must comply with the provisions of Article 12.
- B. An adjustment in the Contract Price will be determined as follows:
  - 1. Where the Work involved is covered by unit prices contained in the Contract Documents, then by application of such unit prices to the quantities of the items involved (subject to the provisions of Paragraph 13.03);
  - 2. Where the Work involved is not covered by unit prices contained in the Contract Documents, then by a mutually agreed lump sum (which may include an allowance for overhead and profit not necessarily in accordance with Paragraph 11.07.C.2); or
  - 3. Where the Work involved is not covered by unit prices contained in the Contract Documents and the parties do not reach mutual agreement to a lump sum, then on the basis of the Cost of the Work (determined as provided in Paragraph 13.01) plus a Contractor's fee for overhead and profit (determined as provided in Paragraph 11.07.C).
- C. *Contractor's Fee*: When applicable, the Contractor's fee for overhead and profit will be determined as follows:
  - 1. A mutually acceptable fixed fee; or
  - 2. If a fixed fee is not agreed upon, then a fee based on the following percentages of the various portions of the Cost of the Work:
    - a. For costs incurred under Paragraphs 13.01.B.1 and 13.01.B.2, the Contractor's fee will be 15 percent;
    - b. For costs incurred under Paragraph 13.01.B.3, the Contractor's fee will be 5 percent;
    - c. Where one or more tiers of subcontracts are on the basis of Cost of the Work plus a fee and no fixed fee is agreed upon, the intent of Paragraphs 11.07.C.2.a and 11.07.C.2.b is that the Contractor's fee will be based on: (1) a fee of 15 percent of the costs incurred under Paragraphs 13.01.B.1 and 13.01.B.2 by the Subcontractor that actually performs the Work, at whatever tier, and (2) with respect to Contractor itself and to any Subcontractors of a tier higher than that of the Subcontractor that actually performs the Work, a fee of 5 percent of the amount (fee plus underlying costs incurred) attributable to the next lower tier Subcontractor; provided, however, that for any such subcontracted Work the maximum total fee to be paid by Owner will be no greater than 27 percent of the costs incurred by the Subcontractor that actually performs the Work;
    - d. No fee will be payable on the basis of costs itemized under Paragraphs 13.01.B.4, 13.01.B.5, and 13.01.C;
    - e. The amount of credit to be allowed by Contractor to Owner for any change which results in a net decrease in Cost of the Work will be the amount of the actual net decrease in Cost of the Work and a deduction of an additional amount equal to 5 percent of such actual net decrease in Cost of the Work; and

- f. When both additions and credits are involved in any one change or Change Proposal, the adjustment in Contractor's fee will be computed by determining the sum of the costs in each of the cost categories in Paragraph 13.01.B (specifically, payroll costs, Paragraph 13.01.B.1; incorporated materials and equipment costs, Paragraph 13.01.B.2; Subcontract costs, Paragraph 13.01.B.3; special consultants costs, Paragraph 13.01.B.4; and other costs, Paragraph 13.01.B.5) and applying to each such cost category sum the appropriate fee from Paragraphs 11.07.C.2.a through 11.07.C.2.e, inclusive.

#### 11.08 *Change of Contract Times*

- A. The Contract Times may only be changed by a Change Order. Any Change Proposal for an adjustment in the Contract Times must comply with the provisions of Paragraph 11.09. Any Claim for an adjustment in the Contract Times must comply with the provisions of Article 12.
- B. Delay, disruption, and interference in the Work, and any related changes in Contract Times, are addressed in and governed by Paragraph 4.05.

#### 11.09 *Change Proposals*

- A. *Purpose and Content:* Contractor shall submit a Change Proposal to Engineer/Architect to request an adjustment in the Contract Times or Contract Price; contest an initial decision by Engineer/Architect concerning the requirements of the Contract Documents or relating to the acceptability of the Work under the Contract Documents; challenge a set-off against payment due; or seek other relief under the Contract. The Change Proposal will specify any proposed change in Contract Times or Contract Price, or other proposed relief, and explain the reason for the proposed change, with citations to any governing or applicable provisions of the Contract Documents. Each Change Proposal will address only one issue, or a set of closely related issues.
- B. *Change Proposal Procedures*
  - 1. *Submittal:* Contractor shall submit each Change Proposal to Engineer/Architect within 30 days after the start of the event giving rise thereto, or after such initial decision.
  - 2. *Supporting Data:* The Contractor shall submit supporting data, including the proposed change in Contract Price or Contract Time (if any), to the Engineer/Architect and Owner within 15 days after the submittal of the Change Proposal.
    - a. Change Proposals based on or related to delay, interruption, or interference must comply with the provisions of Paragraphs 4.05.D and 4.05.E.
    - b. Change proposals related to a change of Contract Price must include full and detailed accounts of materials incorporated into the Work and labor and equipment used for the subject Work.

The supporting data must be accompanied by a written statement that the supporting data are accurate and complete, and that any requested time or price adjustment is the entire adjustment to which Contractor believes it is entitled as a result of said event.

- 3. *Engineer/Architect's Initial Review:* Engineer/Architect will advise Owner regarding the Change Proposal, and consider any comments or response from Owner regarding the Change Proposal. If in its discretion Engineer/Architect concludes that additional supporting data is needed before conducting a full review and making a decision

regarding the Change Proposal, then Engineer/Architect may request that Contractor submit such additional supporting data by a date specified by Engineer/Architect, prior to Engineer/Architect beginning its full review of the Change Proposal.

4. *Engineer/Architect's Full Review and Action on the Change Proposal:* Upon receipt of Contractor's supporting data (including any additional data requested by Engineer/Architect), Engineer/Architect will conduct a full review of each Change Proposal and, within 30 days after such receipt of the Contractor's supporting data, either approve the Change Proposal in whole, deny it in whole, or approve it in part and deny it in part. Such actions must be in writing, with a copy provided to Owner and Contractor. If Engineer/Architect does not take action on the Change Proposal within 30 days, then either Owner or Contractor may at any time thereafter submit a letter to the other party indicating that as a result of Engineer/Architect's inaction the Change Proposal is deemed denied, thereby commencing the time for appeal of the denial under Article 12.
5. *Binding Decision:* Engineer/Architect's decision is final and binding upon Owner and Contractor, unless Owner or Contractor appeals the decision by filing a Claim under Article 12.
- C. *Resolution of Certain Change Proposals:* If the Change Proposal does not involve the design (as set forth in the Drawings, Specifications, or otherwise), the acceptability of the Work, or other engineering or technical matters, then Engineer/Architect will notify the parties in writing that the Engineer/Architect is unable to resolve the Change Proposal. For purposes of further resolution of such a Change Proposal, such notice will be deemed a denial, and Contractor may choose to seek resolution under the terms of Article 12.
- D. *Post-Completion:* Contractor shall not submit any Change Proposals after Engineer/Architect issues a written recommendation of final payment pursuant to Paragraph 15.06.B.

#### 11.10 *Notification to Surety*

- A. If the provisions of any bond require notice to be given to a surety of any change affecting the general scope of the Work or the provisions of the Contract Documents (including, but not limited to, Contract Price or Contract Times), the giving of any such notice will be Contractor's responsibility. The amount of each applicable bond will be adjusted to reflect the effect of any such change.

### **ARTICLE 12—CLAIMS**

#### 12.01 *Claims*

- A. *Claims Process:* The following disputes between Owner and Contractor are subject to the Claims process set forth in this article:
  1. Appeals by Owner or Contractor of Engineer/Architect's decisions regarding Change Proposals;
  2. Owner demands for adjustments in the Contract Price or Contract Times, or other relief under the Contract Documents;

3. Disputes that Engineer/Architect has been unable to address because they do not involve the design (as set forth in the Drawings, Specifications, or otherwise), the acceptability of the Work, or other engineering or technical matters; and
  4. Subject to the waiver provisions of Paragraph 15.07, any dispute arising after Engineer/Architect has issued a written recommendation of final payment pursuant to Paragraph 15.06.B.
- B. *Submittal of Claim:* The party submitting a Claim shall deliver it directly to the other party to the Contract promptly (but in no event later than 30 days) after the start of the event giving rise thereto; in the case of appeals regarding Change Proposals within 30 days of the decision under appeal. The party submitting the Claim shall also furnish a copy to the Engineer/Architect, for its information only. The responsibility to substantiate a Claim rests with the party making the Claim. In the case of a Claim by Contractor seeking an increase in the Contract Times or Contract Price, Contractor shall certify that the Claim is made in good faith, that the supporting data are accurate and complete, and that to the best of Contractor's knowledge and belief the amount of time or money requested accurately reflects the full amount to which Contractor is entitled.
- C. *Review and Resolution:* The party receiving a Claim shall review it thoroughly, giving full consideration to its merits. The two parties shall seek to resolve the Claim through the exchange of information and direct negotiations. The parties may extend the time for resolving the Claim by mutual agreement. All actions taken on a Claim will be stated in writing and submitted to the other party, with a copy to Engineer/Architect.
- D. *Mediation*
1. At any time after initiation of a Claim, Owner and Contractor may mutually agree to mediation of the underlying dispute. The agreement to mediate will stay the Claim submittal and response process.
  2. If Owner and Contractor agree to mediation, then after 60 days from such agreement, either Owner or Contractor may unilaterally terminate the mediation process, and the Claim submittal and decision process will resume as of the date of the termination. If the mediation proceeds but is unsuccessful in resolving the dispute, the Claim submittal and decision process will resume as of the date of the conclusion of the mediation, as determined by the mediator.
  3. Owner and Contractor shall each pay one-half of the mediator's fees and costs.
- E. *Partial Approval:* If the party receiving a Claim approves the Claim in part and denies it in part, such action will be final and binding unless within 30 days of such action the other party invokes the procedure set forth in Article 17 for final resolution of disputes.
- F. *Denial of Claim:* If efforts to resolve a Claim are not successful, the party receiving the Claim may deny it by giving written notice of denial to the other party. If the receiving party does not take action on the Claim within 90 days, then either Owner or Contractor may at any time thereafter submit a letter to the other party indicating that as a result of the inaction, the Claim is deemed denied, thereby commencing the time for appeal of the denial. A denial of the Claim will be final and binding unless within 30 days of the denial the other party invokes the procedure set forth in Article 17 for the final resolution of disputes.

- G. *Final and Binding Results*: If the parties reach a mutual agreement regarding a Claim, whether through approval of the Claim, direct negotiations, mediation, or otherwise; or if a Claim is approved in part and denied in part, or denied in full, and such actions become final and binding; then the results of the agreement or action on the Claim will be incorporated in a Change Order or other written document to the extent they affect the Contract, including the Work, the Contract Times, or the Contract Price.

## **ARTICLE 13—COST OF THE WORK; ALLOWANCES; UNIT PRICE WORK**

### **13.01 *Cost of the Work***

- A. *Purposes for Determination of Cost of the Work*: The term Cost of the Work means the sum of all costs necessary for the proper performance of the Work at issue, as further defined below. The provisions of this Paragraph 13.01 are used for two distinct purposes:
1. To determine Cost of the Work when Cost of the Work is a component of the Contract Price, under cost-plus-fee, time-and-materials, or other cost-based terms; or
  2. When needed to determine the value of a Change Order, Change Proposal, Claim, set-off, or other adjustment in Contract Price. When the value of any such adjustment is determined on the basis of Cost of the Work, Contractor is entitled only to those additional or incremental costs required because of the change in the Work or because of the event giving rise to the adjustment.
- B. *Costs Included*: Except as otherwise may be agreed to in writing by Owner, costs included in the Cost of the Work will be in amounts no higher than those commonly incurred in the locality of the Project, will not include any of the costs itemized in Paragraph 13.01.C, and will include only the following items:
1. Payroll costs for employees in the direct employ of Contractor in the performance of the Work under schedules of job classifications agreed upon by Owner and Contractor in advance of the subject Work. Such employees include, without limitation, superintendents, foremen, safety managers, safety representatives, and other personnel employed full time on the Work. Payroll costs for employees not employed full time on the Work will be apportioned on the basis of their time spent on the Work. Payroll costs include, but are not limited to, salaries and wages plus the cost of fringe benefits, which include social security contributions, unemployment, excise, and payroll taxes, workers' compensation, health and retirement benefits, sick leave, and vacation and holiday pay applicable thereto. The expenses of performing Work outside of regular working hours, on Saturday, Sunday, or legal holidays, will be included in the above to the extent authorized by Owner.
  2. Cost of all materials and equipment furnished and incorporated in the Work, including costs of transportation and storage thereof, and Suppliers' field services required in connection therewith. All cash discounts accrue to Contractor unless Owner deposits funds with Contractor with which to make payments, in which case the cash discounts will accrue to Owner. All trade discounts, rebates, and refunds and returns from sale of surplus materials and equipment will accrue to Owner, and Contractor shall make provisions so that they may be obtained.
  3. Payments made by Contractor to Subcontractors for Work performed by Subcontractors. If required by Owner, Contractor shall obtain competitive bids from



subcontractors acceptable to Owner and Contractor and shall deliver such bids to Owner, which will then determine, with the advice of Engineer/Architect, which bids, if any, will be acceptable. If any subcontract provides that the Subcontractor is to be paid on the basis of Cost of the Work plus a fee, the Subcontractor's Cost of the Work and fee will be determined in the same manner as Contractor's Cost of the Work and fee as provided in this Paragraph 13.01.

4. Costs of special consultants (including but not limited to engineers, architects, testing laboratories, surveyors, attorneys, and accountants) employed or retained for services specifically related to the Work.
5. Other costs consisting of the following:
  - a. The proportion of necessary transportation, travel, and subsistence expenses of Contractor's employees incurred in discharge of duties connected with the Work.
  - b. Cost, including transportation and maintenance, of all materials, supplies, equipment, machinery, appliances, office, and temporary facilities at the Site, which are consumed in the performance of the Work, and cost, less market value, of such items used but not consumed which remain the property of Contractor.
    - 1) In establishing included costs for materials such as scaffolding, plating, or sheeting, consideration will be given to the actual or the estimated life of the material for use on other projects; or rental rates may be established on the basis of purchase or salvage value of such items, whichever is less. Contractor will not be eligible for compensation for such items in an amount that exceeds the purchase cost of such item.
  - c. *Construction Equipment Rental*
    - 1) Rentals of all construction equipment and machinery, and the parts thereof, in accordance with rental agreements approved by Owner as to price (including any surcharge or special rates applicable to overtime use of the construction equipment or machinery), and the costs of transportation, loading, unloading, assembly, dismantling, and removal thereof. All such costs will be in accordance with the terms of said rental agreements. The rental of any such equipment, machinery, or parts must cease when the use thereof is no longer necessary for the Work.
    - 2) Costs for equipment and machinery owned by Contractor or a Contractor-related entity will be paid at a rate shown for such equipment in the equipment rental rate book specified in the Supplementary Conditions. An hourly rate will be computed by dividing the monthly rates by 176. These computed rates will include all operating costs.
    - 3) With respect to Work that is the result of a Change Order, Change Proposal, Claim, set-off, or other adjustment in Contract Price ("changed Work"), included costs will be based on the time the equipment or machinery is in use on the changed Work and the costs of transportation, loading, unloading, assembly, dismantling, and removal when directly attributable to the changed Work. The cost of any such equipment or machinery, or parts thereof, must cease to accrue when the use thereof is no longer necessary for the changed Work.

- d. Sales, consumer, use, and other similar taxes related to the Work, and for which Contractor is liable, as imposed by Laws and Regulations.
  - e. Deposits lost for causes other than negligence of Contractor, any Subcontractor, or anyone directly or indirectly employed by any of them or for whose acts any of them may be liable, and royalty payments and fees for permits and licenses.
  - f. Losses and damages (and related expenses) caused by damage to the Work, not compensated by insurance or otherwise, sustained by Contractor in connection with the performance of the Work (except losses and damages within the deductible amounts of builder's risk or other property insurance established in accordance with Paragraph 6.04), provided such losses and damages have resulted from causes other than the negligence of Contractor, any Subcontractor, or anyone directly or indirectly employed by any of them or for whose acts any of them may be liable. Such losses include settlements made with the written consent and approval of Owner. No such losses, damages, and expenses will be included in the Cost of the Work for the purpose of determining Contractor's fee.
  - g. The cost of utilities, fuel, and sanitary facilities at the Site.
  - h. Minor expenses such as communication service at the Site, express and courier services, and similar petty cash items in connection with the Work.
  - i. The costs of premiums for all bonds and insurance that Contractor is required by the Contract Documents to purchase and maintain.
- C. *Costs Excluded:* The term Cost of the Work does not include any of the following items:
- 1. Payroll costs and other compensation of Contractor's officers, executives, principals, general managers, engineers, architects, estimators, attorneys, auditors, accountants, purchasing and contracting agents, expeditors, timekeepers, clerks, and other personnel employed by Contractor, whether at the Site or in Contractor's principal or branch office for general administration of the Work and not specifically included in the agreed upon schedule of job classifications referred to in Paragraph 13.01.B.1 or specifically covered by Paragraph 13.01.B.4. The payroll costs and other compensation excluded here are to be considered administrative costs covered by the Contractor's fee.
  - 2. The cost of purchasing, renting, or furnishing small tools and hand tools.
  - 3. Expenses of Contractor's principal and branch offices other than Contractor's office at the Site.
  - 4. Any part of Contractor's capital expenses, including interest on Contractor's capital employed for the Work and charges against Contractor for delinquent payments.
  - 5. Costs due to the negligence of Contractor, any Subcontractor, or anyone directly or indirectly employed by any of them or for whose acts any of them may be liable, including but not limited to, the correction of defective Work, disposal of materials or equipment wrongly supplied, and making good any damage to property.
  - 6. Expenses incurred in preparing and advancing Claims.
  - 7. Other overhead or general expense costs of any kind and the costs of any item not specifically and expressly included in Paragraph 13.01.B.

D. *Contractor's Fee*

1. When the Work as a whole is performed on the basis of cost-plus-a-fee, then:
    - a. Contractor's fee for the Work set forth in the Contract Documents as of the Effective Date of the Contract will be determined as set forth in the Agreement.
    - b. for any Work covered by a Change Order, Change Proposal, Claim, set-off, or other adjustment in Contract Price on the basis of Cost of the Work, Contractor's fee will be determined as follows:
      - 1) When the fee for the Work as a whole is a percentage of the Cost of the Work, the fee will automatically adjust as the Cost of the Work changes.
      - 2) When the fee for the Work as a whole is a fixed fee, the fee for any additions or deletions will be determined in accordance with Paragraph 11.07.C.2.
  2. When the Work as a whole is performed on the basis of a stipulated sum, or any other basis other than cost-plus-a-fee, then Contractor's fee for any Work covered by a Change Order, Change Proposal, Claim, set-off, or other adjustment in Contract Price on the basis of Cost of the Work will be determined in accordance with Paragraph 11.07.C.2.
- E. *Documentation and Audit*: Whenever the Cost of the Work for any purpose is to be determined pursuant to this Article 13, Contractor and pertinent Subcontractors will establish and maintain records of the costs in accordance with generally accepted accounting practices. Subject to prior written notice, Owner will be afforded reasonable access, during normal business hours, to all Contractor's accounts, records, books, correspondence, instructions, drawings, receipts, vouchers, memoranda, and similar data relating to the Cost of the Work and Contractor's fee. Contractor shall preserve all such documents for a period of three years after the final payment by Owner. Pertinent Subcontractors will afford such access to Owner, and preserve such documents, to the same extent required of Contractor.

13.02 *Allowances*

- A. It is understood that Contractor has included in the Contract Price all allowances so named in the Contract Documents and shall cause the Work so covered to be performed for such sums and by such persons or entities as may be acceptable to Owner and Engineer/Architect.
- B. *Cash Allowances*: Contractor agrees that:
1. the cash allowances include the cost to Contractor (less any applicable trade discounts) of materials and equipment required by the allowances to be delivered at the Site, and all applicable taxes; and
  2. Contractor's costs for unloading and handling on the Site, labor, installation, overhead, profit, and other expenses contemplated for the cash allowances have been included in the Contract Price and not in the allowances, and no demand for additional payment for any of the foregoing will be valid.
- C. *Owner's Contingency Allowance*: Contractor agrees that an Owner's contingency allowance, if any, is for the sole use of Owner to cover unanticipated costs.

- D. Prior to final payment, an appropriate Change Order will be issued as recommended by Engineer/Architect to reflect actual amounts due Contractor for Work covered by allowances, and the Contract Price will be correspondingly adjusted.

#### 13.03 *Unit Price Work*

- A. Where the Contract Documents provide that all or part of the Work is to be Unit Price Work, initially the Contract Price will be deemed to include for all Unit Price Work an amount equal to the sum of the unit price for each separately identified item of Unit Price Work times the estimated quantity of each item as indicated in the Agreement.
- B. The estimated quantities of items of Unit Price Work are not guaranteed and are solely for the purpose of comparison of Bids and determining an initial Contract Price. Payments to Contractor for Unit Price Work will be based on actual quantities.
- C. Each unit price will be deemed to include an amount considered by Contractor to be adequate to cover Contractor's overhead and profit for each separately identified item.
- D. Engineer/Architect will determine the actual quantities and classifications of Unit Price Work performed by Contractor. Engineer/Architect will review with Contractor the Engineer/Architect's preliminary determinations on such matters before rendering a written decision thereon (by recommendation of an Application for Payment or otherwise). Engineer/Architect's written decision thereon will be final and binding (except as modified by Engineer/Architect to reflect changed factual conditions or more accurate data) upon Owner and Contractor, and the final adjustment of Contract Price will be set forth in a Change Order, subject to the provisions of the following paragraph.
- E. *Adjustments in Unit Price*
  - 1. Contractor or Owner shall be entitled to an adjustment in the unit price with respect to an item of Unit Price Work if:
    - a. the quantity of the item of Unit Price Work performed by Contractor differs materially and significantly from the estimated quantity of such item indicated in the Agreement; and
    - b. Contractor's unit costs to perform the item of Unit Price Work have changed materially and significantly as a result of the quantity change.
  - 2. The adjustment in unit price will account for and be coordinated with any related changes in quantities of other items of Work, and in Contractor's costs to perform such other Work, such that the resulting overall change in Contract Price is equitable to Owner and Contractor.
  - 3. Adjusted unit prices will apply to all units of that item.

### **ARTICLE 14—TESTS AND INSPECTIONS; CORRECTION, REMOVAL, OR ACCEPTANCE OF DEFECTIVE WORK**

#### 14.01 *Access to Work*

- A. Owner, Engineer/Architect, their consultants and other representatives and personnel of Owner, independent testing laboratories, and authorities having jurisdiction have access to the Site and the Work at reasonable times for their observation, inspection, and testing.

Contractor shall provide them proper and safe conditions for such access and advise them of Contractor's safety procedures and programs so that they may comply with such procedures and programs as applicable.

14.02 *Tests, Inspections, and Approvals*

- A. Contractor shall give Engineer/Architect timely notice of readiness of the Work (or specific parts thereof) for all required inspections and tests, and shall cooperate with inspection and testing personnel to facilitate required inspections and tests.
- B. Owner shall retain and pay for the services of an independent inspector, testing laboratory, or other qualified individual or entity to perform all inspections and tests expressly required by the Contract Documents to be furnished and paid for by Owner, except that costs incurred in connection with tests or inspections of covered Work will be governed by the provisions of Paragraph 14.05.
- C. If Laws or Regulations of any public body having jurisdiction require any Work (or part thereof) specifically to be inspected, tested, or approved by an employee or other representative of such public body, Contractor shall assume full responsibility for arranging and obtaining such inspections, tests, or approvals, pay all costs in connection therewith, and furnish Engineer/Architect the required certificates of inspection or approval.
- D. Contractor shall be responsible for arranging, obtaining, and paying for all inspections and tests required:
  - 1. by the Contract Documents, unless the Contract Documents expressly allocate responsibility for a specific inspection or test to Owner;
  - 2. to attain Owner's and Engineer/Architect's acceptance of materials or equipment to be incorporated in the Work;
  - 3. by manufacturers of equipment furnished under the Contract Documents;
  - 4. for testing, adjusting, and balancing of mechanical, electrical, and other equipment to be incorporated into the Work; and
  - 5. for acceptance of materials, mix designs, or equipment submitted for approval prior to Contractor's purchase thereof for incorporation in the Work.

Such inspections and tests will be performed by independent inspectors, testing laboratories, or other qualified individuals or entities acceptable to Owner and Engineer/Architect.

- E. If the Contract Documents require the Work (or part thereof) to be approved by Owner, Engineer/Architect, or another designated individual or entity, then Contractor shall assume full responsibility for arranging and obtaining such approvals.
- F. If any Work (or the work of others) that is to be inspected, tested, or approved is covered by Contractor without written concurrence of Engineer/Architect, Contractor shall, if requested by Engineer/Architect, uncover such Work for observation. Such uncovering will be at Contractor's expense unless Contractor had given Engineer/Architect timely notice of Contractor's intention to cover the same and Engineer/Architect had not acted with reasonable promptness in response to such notice.

#### 14.03 *Defective Work*

- A. *Contractor's Obligation:* It is Contractor's obligation to assure that the Work is not defective.
- B. *Engineer/Architect's Authority:* Engineer/Architect has the authority to determine whether Work is defective, and to reject defective Work.
- C. *Notice of Defects:* Prompt written notice of all defective Work of which Owner or Engineer/Architect has actual knowledge will be given to Contractor.
- D. *Correction, or Removal and Replacement:* Promptly after receipt of written notice of defective Work, Contractor shall correct all such defective Work, whether or not fabricated, installed, or completed, or, if Engineer/Architect has rejected the defective Work, remove it from the Project and replace it with Work that is not defective.
- E. *Preservation of Warranties:* When correcting defective Work, Contractor shall take no action that would void or otherwise impair Owner's special warranty and guarantee, if any, on said Work.
- F. *Costs and Damages:* In addition to its correction, removal, and replacement obligations with respect to defective Work, Contractor shall pay all claims, costs, losses, and damages arising out of or relating to defective Work, including but not limited to the cost of the inspection, testing, correction, removal, replacement, or reconstruction of such defective Work, fines levied against Owner by governmental authorities because the Work is defective, and the costs of repair or replacement of work of others resulting from defective Work. Prior to final payment, if Owner and Contractor are unable to agree as to the measure of such claims, costs, losses, and damages resulting from defective Work, then Owner may impose a reasonable set-off against payments due under Article 15.

#### 14.04 *Acceptance of Defective Work*

- A. If, instead of requiring correction or removal and replacement of defective Work, Owner prefers to accept it, Owner may do so (subject, if such acceptance occurs prior to final payment, to Engineer/Architect's confirmation that such acceptance is in general accord with the design intent and applicable engineering principles, and will not endanger public safety). Contractor shall pay all claims, costs, losses, and damages attributable to Owner's evaluation of and determination to accept such defective Work (such costs to be approved by Engineer/Architect as to reasonableness), and for the diminished value of the Work to the extent not otherwise paid by Contractor. If any such acceptance occurs prior to final payment, the necessary revisions in the Contract Documents with respect to the Work will be incorporated in a Change Order. If the parties are unable to agree as to the decrease in the Contract Price, reflecting the diminished value of Work so accepted, then Owner may impose a reasonable set-off against payments due under Article 15. If the acceptance of defective Work occurs after final payment, Contractor shall pay an appropriate amount to Owner.

#### 14.05 *Uncovering Work*

- A. Engineer/Architect has the authority to require additional inspection or testing of the Work, whether or not the Work is fabricated, installed, or completed.
- B. If any Work is covered contrary to the written request of Engineer/Architect, then Contractor shall, if requested by Engineer/Architect, uncover such Work for Engineer/Architect's observation, and then replace the covering, all at Contractor's expense.

- C. If Engineer/Architect considers it necessary or advisable that covered Work be observed by Engineer/Architect or inspected or tested by others, then Contractor, at Engineer/Architect's request, shall uncover, expose, or otherwise make available for observation, inspection, or testing as Engineer/Architect may require, that portion of the Work in question, and provide all necessary labor, material, and equipment.
  - 1. If it is found that the uncovered Work is defective, Contractor shall be responsible for all claims, costs, losses, and damages arising out of or relating to such uncovering, exposure, observation, inspection, and testing, and of satisfactory replacement or reconstruction (including but not limited to all costs of repair or replacement of work of others); and pending Contractor's full discharge of this responsibility the Owner shall be entitled to impose a reasonable set-off against payments due under Article 15.
  - 2. If the uncovered Work is not found to be defective, Contractor shall be allowed an increase in the Contract Price or an extension of the Contract Times, directly attributable to such uncovering, exposure, observation, inspection, testing, replacement, and reconstruction. If the parties are unable to agree as to the amount or extent thereof, then Contractor may submit a Change Proposal within 30 days of the determination that the Work is not defective.

#### 14.06 *Owner May Stop the Work*

- A. If the Work is defective, or Contractor fails to supply sufficient skilled workers or suitable materials or equipment, or fails to perform the Work in such a way that the completed Work will conform to the Contract Documents, then Owner may order Contractor to stop the Work, or any portion thereof, until the cause for such order has been eliminated; however, this right of Owner to stop the Work will not give rise to any duty on the part of Owner to exercise this right for the benefit of Contractor, any Subcontractor, any Supplier, any other individual or entity, or any surety for, or employee or agent of any of them.

#### 14.07 *Owner May Correct Defective Work*

- A. If Contractor fails within a reasonable time after written notice from Engineer/Architect to correct defective Work, or to remove and replace defective Work as required by Engineer/Architect, then Owner may, after 7 days' written notice to Contractor, correct or remedy any such deficiency.
- B. In exercising the rights and remedies under this Paragraph 14.07, Owner shall proceed expeditiously. In connection with such corrective or remedial action, Owner may exclude Contractor from all or part of the Site, take possession of all or part of the Work and suspend Contractor's services related thereto, and incorporate in the Work all materials and equipment stored at the Site or for which Owner has paid Contractor but which are stored elsewhere. Contractor shall allow Owner, Owner's representatives, agents and employees, Owner's other contractors, and Engineer/Architect and Engineer/Architect's consultants access to the Site to enable Owner to exercise the rights and remedies under this paragraph.
- C. All claims, costs, losses, and damages incurred or sustained by Owner in exercising the rights and remedies under this Paragraph 14.07 will be charged against Contractor as set-offs against payments due under Article 15. Such claims, costs, losses and damages will include but not be limited to all costs of repair, or replacement of work of others destroyed or damaged by correction, removal, or replacement of Contractor's defective Work.

- D. Contractor shall not be allowed an extension of the Contract Times because of any delay in the performance of the Work attributable to the exercise by Owner of Owner's rights and remedies under this Paragraph 14.07.

## **ARTICLE 15—PAYMENTS TO CONTRACTOR; SET-OFFS; COMPLETION; CORRECTION PERIOD**

### **15.01 *Progress Payments***

- A. *Basis for Progress Payments:* The Schedule of Values established as provided in Article 2 will serve as the basis for progress payments and will be incorporated into a form of Application for Payment acceptable to Engineer/Architect. Progress payments for Unit Price Work will be based on the number of units completed during the pay period, as determined under the provisions of Paragraph 13.03. Progress payments for cost-based Work will be based on Cost of the Work completed by Contractor during the pay period.

B. *Applications for Payments*

1. At least 20 days before the date established in the Agreement for each progress payment (but not more often than once a month), Contractor shall submit to Engineer/Architect for review an Application for Payment filled out and signed by Contractor covering the Work completed as of the date of the Application and accompanied by such supporting documentation as is required by the Contract Documents.
2. If payment is requested on the basis of materials and equipment not incorporated in the Work but delivered and suitably stored at the Site or at another location agreed to in writing, the Application for Payment must also be accompanied by: (a) a bill of sale, invoice, copies of subcontract or purchase order payments, or other documentation establishing full payment by Contractor for the materials and equipment; (b) at Owner's request, documentation warranting that Owner has received the materials and equipment free and clear of all Liens; and (c) evidence that the materials and equipment are covered by appropriate property insurance, a warehouse bond, or other arrangements to protect Owner's interest therein, all of which must be satisfactory to Owner.
3. Beginning with the second Application for Payment, each Application must include an affidavit of Contractor stating that all previous progress payments received by Contractor have been applied to discharge Contractor's legitimate obligations associated with prior Applications for Payment.
4. The amount of retainage with respect to progress payments will be as stipulated in the Agreement.

C. *Review of Applications*

1. Engineer/Architect will, within 10 days after receipt of each Application for Payment, including each resubmittal, either indicate in writing a recommendation of payment and present the Application to Owner, or return the Application to Contractor indicating in writing Engineer/Architect's reasons for refusing to recommend payment. In the latter case, Contractor may make the necessary corrections and resubmit the Application.
2. Engineer/Architect's recommendation of any payment requested in an Application for Payment will constitute a representation by Engineer/Architect to Owner, based on



Engineer/Architect's observations of the executed Work as an experienced and qualified design professional, and on Engineer/Architect's review of the Application for Payment and the accompanying data and schedules, that to the best of Engineer/Architect's knowledge, information and belief:

- a. the Work has progressed to the point indicated;
  - b. the quality of the Work is generally in accordance with the Contract Documents (subject to an evaluation of the Work as a functioning whole prior to or upon Substantial Completion, the results of any subsequent tests called for in the Contract Documents, a final determination of quantities and classifications for Unit Price Work under Paragraph 13.03, and any other qualifications stated in the recommendation); and
  - c. the conditions precedent to Contractor's being entitled to such payment appear to have been fulfilled in so far as it is Engineer/Architect's responsibility to observe the Work.
3. By recommending any such payment Engineer/Architect will not thereby be deemed to have represented that:
- a. inspections made to check the quality or the quantity of the Work as it has been performed have been exhaustive, extended to every aspect of the Work in progress, or involved detailed inspections of the Work beyond the responsibilities specifically assigned to Engineer/Architect in the Contract; or
  - b. there may not be other matters or issues between the parties that might entitle Contractor to be paid additionally by Owner or entitle Owner to withhold payment to Contractor.
4. Neither Engineer/Architect's review of Contractor's Work for the purposes of recommending payments nor Engineer/Architect's recommendation of any payment, including final payment, will impose responsibility on Engineer/Architect:
- a. to supervise, direct, or control the Work;
  - b. for the means, methods, techniques, sequences, or procedures of construction, or the safety precautions and programs incident thereto;
  - c. for Contractor's failure to comply with Laws and Regulations applicable to Contractor's performance of the Work;
  - d. to make any examination to ascertain how or for what purposes Contractor has used the money paid by Owner; or
  - e. to determine that title to any of the Work, materials, or equipment has passed to Owner free and clear of any Liens.
5. Engineer/Architect may refuse to recommend the whole or any part of any payment if, in Engineer/Architect's opinion, it would be incorrect to make the representations to Owner stated in Paragraph 15.01.C.2.
6. Engineer/Architect will recommend reductions in payment (set-offs) necessary in Engineer/Architect's opinion to protect Owner from loss because:
- a. the Work is defective, requiring correction or replacement;

- b. the Contract Price has been reduced by Change Orders;
- c. Owner has been required to correct defective Work in accordance with Paragraph 14.07, or has accepted defective Work pursuant to Paragraph 14.04;
- d. Owner has been required to remove or remediate a Hazardous Environmental Condition for which Contractor is responsible; or
- e. Engineer/Architect has actual knowledge of the occurrence of any of the events that would constitute a default by Contractor and therefore justify termination for cause under the Contract Documents.

**D. *Payment Becomes Due***

- 1. Ten days after presentation of the Application for Payment to Owner with Engineer/Architect's recommendation, the amount recommended (subject to any Owner set-offs) will become due, and when due will be paid by Owner to Contractor.

**E. *Reductions in Payment by Owner***

- 1. In addition to any reductions in payment (set-offs) recommended by Engineer/Architect, Owner is entitled to impose a set-off against payment based on any of the following:
  - a. Claims have been made against Owner based on Contractor's conduct in the performance or furnishing of the Work, or Owner has incurred costs, losses, or damages resulting from Contractor's conduct in the performance or furnishing of the Work, including but not limited to claims, costs, losses, or damages from workplace injuries, adjacent property damage, non-compliance with Laws and Regulations, and patent infringement;
  - b. Contractor has failed to take reasonable and customary measures to avoid damage, delay, disruption, and interference with other work at or adjacent to the Site;
  - c. Contractor has failed to provide and maintain required bonds or insurance;
  - d. Owner has been required to remove or remediate a Hazardous Environmental Condition for which Contractor is responsible;
  - e. Owner has incurred extra charges or engineering/architectural costs related to submittal reviews, evaluations of proposed substitutes, tests and inspections, or return visits to manufacturing or assembly facilities;
  - f. The Work is defective, requiring correction or replacement;
  - g. Owner has been required to correct defective Work in accordance with Paragraph 14.07, or has accepted defective Work pursuant to Paragraph 14.04;
  - h. The Contract Price has been reduced by Change Orders;
  - i. An event has occurred that would constitute a default by Contractor and therefore justify a termination for cause;
  - j. Liquidated or other damages have accrued as a result of Contractor's failure to achieve Milestones, Substantial Completion, or final completion of the Work;
  - k. Liens have been filed in connection with the Work, except where Contractor has delivered a specific bond satisfactory to Owner to secure the satisfaction and discharge of such Liens; or

- I. Other items entitle Owner to a set-off against the amount recommended.
2. If Owner imposes any set-off against payment, whether based on its own knowledge or on the written recommendations of Engineer/Architect, Owner will give Contractor immediate written notice (with a copy to Engineer/Architect) stating the reasons for such action and the specific amount of the reduction, and promptly pay Contractor any amount remaining after deduction of the amount so withheld. Owner shall promptly pay Contractor the amount so withheld, or any adjustment thereto agreed to by Owner and Contractor, if Contractor remedies the reasons for such action. The reduction imposed will be binding on Contractor unless it duly submits a Change Proposal contesting the reduction.
3. Upon a subsequent determination that Owner's refusal of payment was not justified, the amount wrongfully withheld will be treated as an amount due as determined by Paragraph 15.01.D.1 and subject to interest as provided in the Agreement.

#### 15.02 *Contractor's Warranty of Title*

- A. Contractor warrants and guarantees that title to all Work, materials, and equipment furnished under the Contract will pass to Owner free and clear of (1) all Liens and other title defects, and (2) all patent, licensing, copyright, or royalty obligations, no later than 7 days after the time of payment by Owner.

#### 15.03 *Substantial Completion*

- A. When Contractor considers the entire Work ready for its intended use Contractor shall notify Owner and Engineer/Architect in writing that the entire Work is substantially complete and request that Engineer/Architect issue a certificate of Substantial Completion. Contractor shall at the same time submit to Owner and Engineer/Architect an initial draft of punch list items to be completed or corrected before final payment.
- B. Promptly after Contractor's notification, Owner, Contractor, and Engineer/Architect shall make an inspection of the Work to determine the status of completion. If Engineer/Architect does not consider the Work substantially complete, Engineer/Architect will notify Contractor in writing giving the reasons therefor.
- C. If Engineer/Architect considers the Work substantially complete, Engineer/Architect will deliver to Owner a preliminary certificate of Substantial Completion which will fix the date of Substantial Completion. Engineer/Architect shall attach to the certificate a punch list of items to be completed or corrected before final payment. Owner shall have 7 days after receipt of the preliminary certificate during which to make written objection to Engineer/Architect as to any provisions of the certificate or attached punch list. If, after considering the objections to the provisions of the preliminary certificate, Engineer/Architect concludes that the Work is not substantially complete, Engineer/Architect will, within 14 days after submission of the preliminary certificate to Owner, notify Contractor in writing that the Work is not substantially complete, stating the reasons therefor. If Owner does not object to the provisions of the certificate, or if despite consideration of Owner's objections Engineer/Architect concludes that the Work is substantially complete, then Engineer/Architect will, within said 14 days, execute and deliver to Owner and Contractor a final certificate of Substantial Completion (with a revised punch list of items to be completed or corrected) reflecting such changes from the

preliminary certificate as Engineer/Architect believes justified after consideration of any objections from Owner.

- D. At the time of receipt of the preliminary certificate of Substantial Completion, Owner and Contractor will confer regarding Owner's use or occupancy of the Work following Substantial Completion, review the builder's risk insurance policy with respect to the end of the builder's risk coverage, and confirm the transition to coverage of the Work under a permanent property insurance policy held by Owner. Unless Owner and Contractor agree otherwise in writing, Owner shall bear responsibility for security, operation, protection of the Work, property insurance, maintenance, heat, and utilities upon Owner's use or occupancy of the Work.
- E. After Substantial Completion the Contractor shall promptly begin work on the punch list of items to be completed or corrected prior to final payment. In appropriate cases Contractor may submit monthly Applications for Payment for completed punch list items, following the progress payment procedures set forth above.
- F. Owner shall have the right to exclude Contractor from the Site after the date of Substantial Completion subject to allowing Contractor reasonable access to remove its property and complete or correct items on the punch list.

#### 15.04 *Partial Use or Occupancy*

- A. Prior to Substantial Completion of all the Work, Owner may use or occupy any substantially completed part of the Work which has specifically been identified in the Contract Documents, or which Owner, Engineer/Architect, and Contractor agree constitutes a separately functioning and usable part of the Work that can be used by Owner for its intended purpose without significant interference with Contractor's performance of the remainder of the Work, subject to the following conditions:
  - 1. At any time, Owner may request in writing that Contractor permit Owner to use or occupy any such part of the Work that Owner believes to be substantially complete. If and when Contractor agrees that such part of the Work is substantially complete, Contractor, Owner, and Engineer/Architect will follow the procedures of Paragraph 15.03.A through 15.03.E for that part of the Work.
  - 2. At any time, Contractor may notify Owner and Engineer/Architect in writing that Contractor considers any such part of the Work substantially complete and request Engineer/Architect to issue a certificate of Substantial Completion for that part of the Work.
  - 3. Within a reasonable time after either such request, Owner, Contractor, and Engineer/Architect shall make an inspection of that part of the Work to determine its status of completion. If Engineer/Architect does not consider that part of the Work to be substantially complete, Engineer/Architect will notify Owner and Contractor in writing giving the reasons therefor. If Engineer/Architect considers that part of the Work to be substantially complete, the provisions of Paragraph 15.03 will apply with respect to certification of Substantial Completion of that part of the Work and the division of responsibility in respect thereof and access thereto.
  - 4. No use or occupancy or separate operation of part of the Work may occur prior to compliance with the requirements of Paragraph 6.04 regarding builder's risk or other property insurance.

#### 15.05 *Final Inspection*

- A. Upon written notice from Contractor that the entire Work or an agreed portion thereof is complete, Engineer/Architect will promptly make a final inspection with Owner and Contractor and will notify Contractor in writing of all particulars in which this inspection reveals that the Work, or agreed portion thereof, is incomplete or defective. Contractor shall immediately take such measures as are necessary to complete such Work or remedy such deficiencies.

#### 15.06 *Final Payment*

##### A. *Application for Payment*

1. After Contractor has, in the opinion of Engineer/Architect, satisfactorily completed all corrections identified during the final inspection and has delivered, in accordance with the Contract Documents, all maintenance and operating instructions, schedules, guarantees, bonds, certificates or other evidence of insurance, certificates of inspection, annotated record documents (as provided in Paragraph 7.12), and other documents, Contractor may make application for final payment.
2. The final Application for Payment must be accompanied (except as previously delivered) by:
  - a. all documentation called for in the Contract Documents;
  - b. consent of the surety, if any, to final payment;
  - c. satisfactory evidence that all title issues have been resolved such that title to all Work, materials, and equipment has passed to Owner free and clear of any Liens or other title defects, or will so pass upon final payment.
  - d. a list of all duly pending Change Proposals and Claims; and
  - e. complete and legally effective releases or waivers (satisfactory to Owner) of all Lien rights arising out of the Work, and of Liens filed in connection with the Work.
3. In lieu of the releases or waivers of Liens specified in Paragraph 15.06.A.2 and as approved by Owner, Contractor may furnish receipts or releases in full and an affidavit of Contractor that: (a) the releases and receipts include all labor, services, material, and equipment for which a Lien could be filed; and (b) all payrolls, material and equipment bills, and other indebtedness connected with the Work for which Owner might in any way be responsible, or which might in any way result in liens or other burdens on Owner's property, have been paid or otherwise satisfied. If any Subcontractor or Supplier fails to furnish such a release or receipt in full, Contractor may furnish a bond or other collateral satisfactory to Owner to indemnify Owner against any Lien, or Owner at its option may issue joint checks payable to Contractor and specified Subcontractors and Suppliers.

- B. *Engineer/Architect's Review of Final Application and Recommendation of Payment:* If, on the basis of Engineer/Architect's observation of the Work during construction and final inspection, and Engineer/Architect's review of the final Application for Payment and accompanying documentation as required by the Contract Documents, Engineer/Architect is satisfied that the Work has been completed and Contractor's other obligations under the Contract have been fulfilled, Engineer/Architect will, within 10 days after receipt of the final

Application for Payment, indicate in writing Engineer/Architect's recommendation of final payment and present the final Application for Payment to Owner for payment. Such recommendation will account for any set-offs against payment that are necessary in Engineer/Architect's opinion to protect Owner from loss for the reasons stated above with respect to progress payments. Otherwise, Engineer/Architect will return the Application for Payment to Contractor, indicating in writing the reasons for refusing to recommend final payment, in which case Contractor shall make the necessary corrections and resubmit the Application for Payment.

- C. *Notice of Acceptability:* In support of its recommendation of payment of the final Application for Payment, Engineer/Architect will also give written notice to Owner and Contractor that the Work is acceptable, subject to stated limitations in the notice and to the provisions of Paragraph 15.07.
- D. *Completion of Work:* The Work is complete (subject to surviving obligations) when it is ready for final payment as established by the Engineer/Architect's written recommendation of final payment and issuance of notice of the acceptability of the Work.
- E. *Final Payment Becomes Due:* Upon receipt from Engineer/Architect of the final Application for Payment and accompanying documentation, Owner shall set off against the amount recommended by Engineer/Architect for final payment any further sum to which Owner is entitled, including but not limited to set-offs for liquidated damages and set-offs allowed under the provisions of this Contract with respect to progress payments. Owner shall pay the resulting balance due to Contractor within 30 days of Owner's receipt of the final Application for Payment from Engineer/Architect.

#### 15.07 *Waiver of Claims*

- A. By making final payment, Owner waives its claim or right to liquidated damages or other damages for late completion by Contractor, except as set forth in an outstanding Claim, appeal under the provisions of Article 17, set-off, or express reservation of rights by Owner. Owner reserves all other claims or rights after final payment.
- B. The acceptance of final payment by Contractor will constitute a waiver by Contractor of all claims and rights against Owner other than those pending matters that have been duly submitted as a Claim, or appealed under the provisions of Article 17.

#### 15.08 *Correction Period*

- A. If within one year after the date of Substantial Completion (or such longer period of time as may be prescribed by the Supplementary Conditions or the terms of any applicable special guarantee required by the Contract Documents), Owner gives Contractor written notice that any Work has been found to be defective, or that Contractor's repair of any damages to the Site or adjacent areas has been found to be defective, then after receipt of such notice of defect Contractor shall promptly, without cost to Owner and in accordance with Owner's written instructions:
  - 1. correct the defective repairs to the Site or such adjacent areas;
  - 2. correct such defective Work;
  - 3. remove the defective Work from the Project and replace it with Work that is not defective, if the defective Work has been rejected by Owner, and

4. satisfactorily correct or repair or remove and replace any damage to other Work, to the work of others, or to other land or areas resulting from the corrective measures.
- B. Owner shall give any such notice of defect within 60 days of the discovery that such Work or repairs is defective. If such notice is given within such 60 days but after the end of the correction period, the notice will be deemed a notice of defective Work under Paragraph 7.17.B.
- C. If, after receipt of a notice of defect within 60 days and within the correction period, Contractor does not promptly comply with the terms of Owner's written instructions, or in an emergency where delay would cause serious risk of loss or damage, Owner may have the defective Work corrected or repaired or may have the rejected Work removed and replaced. Contractor shall pay all costs, losses, and damages (including but not limited to all fees and charges of engineers, architects, attorneys, and other professionals and all court or arbitration or other dispute resolution costs) arising out of or relating to such correction or repair or such removal and replacement (including but not limited to all costs of repair or replacement of work of others). Contractor's failure to pay such costs, losses, and damages within 10 days of invoice from Owner will be deemed the start of an event giving rise to a Claim under Paragraph 12.01.B, such that any related Claim must be brought within 30 days of the failure to pay.
- D. In special circumstances where a particular item of equipment is placed in continuous service before Substantial Completion of all the Work, the correction period for that item may start to run from an earlier date if so provided in the Specifications.
- E. Where defective Work (and damage to other Work resulting therefrom) has been corrected or removed and replaced under this paragraph, the correction period hereunder with respect to such Work will be extended for an additional period of one year after such correction or removal and replacement has been satisfactorily completed.
- F. Contractor's obligations under this paragraph are in addition to all other obligations and warranties. The provisions of this paragraph are not to be construed as a substitute for, or a waiver of, the provisions of any applicable statute of limitation or repose.

## **ARTICLE 16—SUSPENSION OF WORK AND TERMINATION**

### **16.01 *Owner May Suspend Work***

- A. At any time and without cause, Owner may suspend the Work or any portion thereof for a period of not more than 90 consecutive days by written notice to Contractor and Engineer/Architect. Such notice will fix the date on which Work will be resumed. Contractor shall resume the Work on the date so fixed. Contractor shall be entitled to an adjustment in the Contract Price or an extension of the Contract Times directly attributable to any such suspension. Any Change Proposal seeking such adjustments must be submitted no later than 30 days after the date fixed for resumption of Work.

#### 16.02 *Owner May Terminate for Cause*

- A. The occurrence of any one or more of the following events will constitute a default by Contractor and justify termination for cause:
  - 1. Contractor's persistent failure to perform the Work in accordance with the Contract Documents (including, but not limited to, failure to supply sufficient skilled workers or suitable materials or equipment, or failure to adhere to the Progress Schedule);
  - 2. Failure of Contractor to perform or otherwise to comply with a material term of the Contract Documents;
  - 3. Contractor's disregard of Laws or Regulations of any public body having jurisdiction; or
  - 4. Contractor's repeated disregard of the authority of Owner or Engineer/Architect.
- B. If one or more of the events identified in Paragraph 16.02.A occurs, then after giving Contractor (and any surety) 10 days' written notice that Owner is considering a declaration that Contractor is in default and termination of the Contract, Owner may proceed to:
  - 1. declare Contractor to be in default, and give Contractor (and any surety) written notice that the Contract is terminated; and
  - 2. enforce the rights available to Owner under any applicable performance bond.
- C. Subject to the terms and operation of any applicable performance bond, if Owner has terminated the Contract for cause, Owner may exclude Contractor from the Site, take possession of the Work, incorporate in the Work all materials and equipment stored at the Site or for which Owner has paid Contractor but which are stored elsewhere, and complete the Work as Owner may deem expedient.
- D. Owner may not proceed with termination of the Contract under Paragraph 16.02.B if Contractor within 7 days of receipt of notice of intent to terminate begins to correct its failure to perform and proceeds diligently to cure such failure.
- E. If Owner proceeds as provided in Paragraph 16.02.B, Contractor shall not be entitled to receive any further payment until the Work is completed. If the unpaid balance of the Contract Price exceeds the cost to complete the Work, including all related claims, costs, losses, and damages (including but not limited to all fees and charges of Engineer/Architects, architects, attorneys, and other professionals) sustained by Owner, such excess will be paid to Contractor. If the cost to complete the Work including such related claims, costs, losses, and damages exceeds such unpaid balance, Contractor shall pay the difference to Owner. Such claims, costs, losses, and damages incurred by Owner will be reviewed by Engineer/Architect as to their reasonableness and, when so approved by Engineer/Architect, incorporated in a Change Order. When exercising any rights or remedies under this paragraph, Owner shall not be required to obtain the lowest price for the Work performed.
- F. Where Contractor's services have been so terminated by Owner, the termination will not affect any rights or remedies of Owner against Contractor then existing or which may thereafter accrue, or any rights or remedies of Owner against Contractor or any surety under any payment bond or performance bond. Any retention or payment of money due Contractor by Owner will not release Contractor from liability.



- G. If and to the extent that Contractor has provided a performance bond under the provisions of Paragraph 6.01.A, the provisions of that bond will govern over any inconsistent provisions of Paragraphs 16.02.B and 16.02.D.

**16.03 *Owner May Terminate for Convenience***

- A. Upon 7 days' written notice to Contractor and Engineer/Architect, Owner may, without cause and without prejudice to any other right or remedy of Owner, terminate the Contract. In such case, Contractor shall be paid for (without duplication of any items):
  - 1. completed and acceptable Work executed in accordance with the Contract Documents prior to the effective date of termination, including fair and reasonable sums for overhead and profit on such Work;
  - 2. expenses sustained prior to the effective date of termination in performing services and furnishing labor, materials, or equipment as required by the Contract Documents in connection with uncompleted Work, plus fair and reasonable sums for overhead and profit on such expenses; and
  - 3. other reasonable expenses directly attributable to termination, including costs incurred to prepare a termination for convenience cost proposal.
- B. Contractor shall not be paid for any loss of anticipated profits or revenue, post-termination overhead costs, or other economic loss arising out of or resulting from such termination.

**16.04 *Contractor May Stop Work or Terminate***

- A. If, through no act or fault of Contractor, (1) the Work is suspended for more than 90 consecutive days by Owner or under an order of court or other public authority, or (2) Engineer/Architect fails to act on any Application for Payment within 30 days after it is submitted, or (3) Owner fails for 30 days to pay Contractor any sum finally determined to be due, then Contractor may, upon 7 days' written notice to Owner and Engineer/Architect, and provided Owner or Engineer/Architect do not remedy such suspension or failure within that time, terminate the contract and recover from Owner payment on the same terms as provided in Paragraph 16.03.
- B. In lieu of terminating the Contract and without prejudice to any other right or remedy, if Engineer/Architect has failed to act on an Application for Payment within 30 days after it is submitted, or Owner has failed for 30 days to pay Contractor any sum finally determined to be due, Contractor may, 7 days after written notice to Owner and Engineer/Architect, stop the Work until payment is made of all such amounts due Contractor, including interest thereon. The provisions of this paragraph are not intended to preclude Contractor from submitting a Change Proposal for an adjustment in Contract Price or Contract Times or otherwise for expenses or damage directly attributable to Contractor's stopping the Work as permitted by this paragraph.

## ARTICLE 17—FINAL RESOLUTION OF DISPUTES

### 17.01 *Methods and Procedures*

- A. *Disputes Subject to Final Resolution:* The following disputed matters are subject to final resolution under the provisions of this article:
  - 1. A timely appeal of an approval in part and denial in part of a Claim, or of a denial in full, pursuant to Article 12; and
  - 2. Disputes between Owner and Contractor concerning the Work, or obligations under the Contract Documents, that arise after final payment has been made.
- B. *Final Resolution of Disputes:* For any dispute subject to resolution under this article, Owner or Contractor may:
  - 1. elect in writing to invoke the dispute resolution process provided for in the Supplementary Conditions;
  - 2. agree with the other party to submit the dispute to another dispute resolution process; or
  - 3. if no dispute resolution process is provided for in the Supplementary Conditions or mutually agreed to, give written notice to the other party of the intent to submit the dispute to a court of competent jurisdiction.

## ARTICLE 18—MISCELLANEOUS

### 18.01 *Giving Notice*

- A. Whenever any provision of the Contract requires the giving of written notice to Owner, Engineer/Architect, or Contractor, it will be deemed to have been validly given only if delivered:
  - 1. in person, by a commercial courier service or otherwise, to the recipient's place of business;
  - 2. by registered or certified mail, postage prepaid, to the recipient's place of business; or
  - 3. by e-mail to the recipient, with the words "Formal Notice" or similar in the e-mail's subject line.

### 18.02 *Computation of Times*

- A. When any period of time is referred to in the Contract by days, it will be computed to exclude the first and include the last day of such period. If the last day of any such period falls on a Saturday or Sunday or on a day made a legal holiday by the law of the applicable jurisdiction, such day will be omitted from the computation.

### 18.03 *Cumulative Remedies*

- A. The duties and obligations imposed by these General Conditions and the rights and remedies available hereunder to the parties hereto are in addition to, and are not to be construed in any way as a limitation of, any rights and remedies available to any or all of them which are otherwise imposed or available by Laws or Regulations, by special warranty or guarantee, or by other provisions of the Contract. The provisions of this paragraph will be

as effective as if repeated specifically in the Contract Documents in connection with each particular duty, obligation, right, and remedy to which they apply.

18.04 *Limitation of Damages*

- A. With respect to any and all Change Proposals, Claims, disputes subject to final resolution, and other matters at issue, neither Owner nor Engineer/Architect, nor any of their officers, directors, members, partners, employees, agents, consultants, or subcontractors, shall be liable to Contractor for any claims, costs, losses, or damages sustained by Contractor on or in connection with any other project or anticipated project.

18.05 *No Waiver*

- A. A party's non-enforcement of any provision will not constitute a waiver of that provision, nor will it affect the enforceability of that provision or of the remainder of this Contract.

18.06 *Survival of Obligations*

- A. All representations, indemnifications, warranties, and guarantees made in, required by, or given in accordance with the Contract, as well as all continuing obligations indicated in the Contract, will survive final payment, completion, and acceptance of the Work or termination of the Contract or of the services of Contractor.

18.07 *Controlling Law*

- A. This Contract is to be governed by the law of the state in which the Project is located.

18.08 *Assignment of Contract*

- A. Unless expressly agreed to elsewhere in the Contract, no assignment by a party to this Contract of any rights under or interests in the Contract will be binding on the other party without the written consent of the party sought to be bound; and, specifically but without limitation, money that may become due and money that is due may not be assigned without such consent (except to the extent that the effect of this restriction may be limited by law), and unless specifically stated to the contrary in any written consent to an assignment, no assignment will release or discharge the assignor from any duty or responsibility under the Contract.

18.09 *Successors and Assigns*

- A. Owner and Contractor each binds itself, its successors, assigns, and legal representatives to the other party hereto, its successors, assigns, and legal representatives in respect to all covenants, agreements, and obligations contained in the Contract Documents.

18.10 *Headings*

- A. Article and paragraph headings are inserted for convenience only and do not constitute parts of these General Conditions.

Page Intentionally Blank

This document has important legal consequences; consultation with an attorney is encouraged with respect to its use or modification. This document should be adapted to the particular circumstances of the contemplated Project and the controlling Laws and Regulations.

## **SUPPLEMENTARY CONDITIONS OF THE CONSTRUCTION CONTRACT**

**Prepared By**



**Endorsed By**



Copyright© 2018

National Society of Professional Engineers  
1420 King Street, Alexandria, VA 22314-2794  
(703) 684-2882  
[www.nspe.org](http://www.nspe.org)

American Council of Engineering Companies  
1015 15th Street N.W., Washington, DC 20005  
(202) 347-7474  
[www.acec.org](http://www.acec.org)

American Society of Civil Engineers  
1801 Alexander Bell Drive, Reston, VA 20191-4400  
(800) 548-2723  
[www.asce.org](http://www.asce.org)

The copyright for this EJCDC document is owned jointly by the three sponsoring organizations listed above. The National Society of Professional Engineers is the Copyright Administrator for the EJCDC documents; please direct all inquiries regarding EJCDC copyrights to NSPE.

NOTE: EJCDC publications may be purchased at [www.ejcdc.org](http://www.ejcdc.org), or from any of the sponsoring organizations above.

# SUPPLEMENTARY CONDITIONS OF THE CONSTRUCTION CONTRACT

## TABLE OF CONTENTS

|                                                                                               | Page                                |
|-----------------------------------------------------------------------------------------------|-------------------------------------|
| Article 1— Definitions and Terminology.....                                                   | 1                                   |
| Article 2— Preliminary Matters .....                                                          | 1                                   |
| Article 3— Contract Documents: Intent, Requirements, Reuse .....                              | 1                                   |
| Article 4— Commencement and Progress of the Work .....                                        | 1                                   |
| Article 5— Site, Subsurface and Physical Conditions, Hazardous Environmental Conditions ..... | 2                                   |
| Article 6— Bonds and Insurance .....                                                          | 2                                   |
| Article 7— Contractor’s Responsibilities .....                                                | 6                                   |
| Article 8— Other Work at the Site .....                                                       | 7                                   |
| Article 9— Owner’s Responsibilities .....                                                     | 7                                   |
| Article 10— Engineer/Architect’s Status During Construction .....                             | 7                                   |
| Article 11— Changes to the Contract .....                                                     | 8                                   |
| Article 12— Claims .....                                                                      | 8                                   |
| Article 13— Cost of Work; Allowances, Unit Price Work.....                                    | 9                                   |
| Article 14— Tests and Inspections; Correction, Removal, or Acceptance of Defective Work.....  | 9                                   |
| Article 15— Payments to Contractor, Set Offs; Completions; Correction Period .....            | 9                                   |
| Article 16— Suspension of Work and Termination .....                                          | 10                                  |
| Article 17— Final Resolutions of Disputes .....                                               | 10                                  |
| Article 18— Miscellaneous .....                                                               | 10                                  |
| Exhibit A— Software Requirements for Electronic Document Exchange .....                       | <b>Error! Bookmark not defined.</b> |
| Exhibit B— Foreseeable Bad Weather Days .....                                                 | <b>Error! Bookmark not defined.</b> |
| Exhibit C— Geotechnical Baseline Report Supplement to the Supplementary Conditions            | <b>Error! Bookmark not defined.</b> |

# SUPPLEMENTARY CONDITIONS OF THE CONSTRUCTION CONTRACT

These Supplementary Conditions amend or supplement EJCDC® C-700, Standard General Conditions of the Construction Contract (2018). The General Conditions remain in full force and effect except as amended.

The terms used in these Supplementary Conditions have the meanings stated in the General Conditions. Additional terms used in these Supplementary Conditions have the meanings stated below, which are applicable to both the singular and plural thereof.

The address system used in these Supplementary Conditions is the same as the address system used in the General Conditions, with the prefix "SC" added—for example, "Paragraph SC-4.05."

## ARTICLE 1—DEFINITIONS AND TERMINOLOGY

No suggested Supplementary Conditions in this Article.

## ARTICLE 2—PRELIMINARY MATTERS

### 2.01 *Delivery of Bonds and Evidence of Insurance*

SC-2.01 Delete Paragraphs 2.01.B. and C. in their entirety and insert the following in their place:

- B. *Evidence of Contractor's Insurance:* When Contractor delivers the signed counterparts of the Agreement to Owner, Contractor shall also deliver to Owner copies of the policies (including all endorsements, and identification of applicable self-insured retentions and deductibles) of insurance required to be provided by Contractor in this Contract. Contractor may block out (redact) any confidential premium or pricing information contained in any policy or endorsement furnished under this provision.
- C. *Evidence of Owner's Insurance:* After receipt from Contractor of the signed counterparts of the Agreement and all required bonds and insurance documentation, Owner shall promptly deliver to Contractor copies of the policies of insurance to be provided by Owner in this Contract (if any). Owner may block out (redact) any confidential premium or pricing information contained in any policy or endorsement furnished under this provision.

## ARTICLE 3—CONTRACT DOCUMENTS: INTENT, REQUIREMENTS, REUSE

## ARTICLE 4—COMMENCEMENT AND PROGRESS OF THE WORK

### 4.05 *Delays in Contractor's Progress*

SC-4.05 Amend Paragraph 4.05.C by adding the following subparagraphs:

- 5. *Weather-Related Delays*
  - a. If "abnormal weather conditions" as set forth in Paragraph 4.05.C.2 of the General Conditions are the basis for a request for an equitable adjustment in the Contract Times, such request must be documented by data substantiating each of the



following: 1) that weather conditions were abnormal for the period of time in which the delay occurred, 2) that such weather conditions could not have been reasonably anticipated, and 3) that such weather conditions had an adverse effect on the Work as scheduled.

- b. The existence of abnormal weather conditions will be determined on a month-by-month basis in accordance with the following:
  - 1) Every workday on which one or more of the following conditions exist will be considered a “bad weather day”:
    - i) Total precipitation (as rain equivalent) occurring between 7:00 p.m. on the preceding day (regardless of whether such preceding day is a workday) through 7:00 p.m. on the workday in question equals or exceeds average precipitation as rain equivalent for previous 5 years, as determined by National Weather Service.
  - 2) Determination of actual bad weather days during performance of the Work will be based on the weather records measured and recorded by **National Weather Service** weather monitoring station at **Brownsville, TX**.
  - 3) Contractor shall anticipate the number of foreseeable bad weather days per month.
  - 4) In each month, every bad weather day exceeding the number of foreseeable bad weather days will be considered as “abnormal weather conditions.” The existence of abnormal weather conditions will not relieve Contractor of the obligation to demonstrate and document that delays caused by abnormal weather are specific to the planned work activities or that such activities thus delayed were on Contractor’s then-current Progress Schedule’s critical path for the Project.

## **ARTICLE 5—SITE, SUBSURFACE AND PHYSICAL CONDITIONS, HAZARDOUS ENVIRONMENTAL CONDITIONS**

SC 5.02 Remove wording from Paragraph 5.02 A.2 “or otherwise resolve the claim by arbitration or other dispute resolution proceeding.”

5.03 *Subsurface and Physical Conditions*

## **ARTICLE 6—BONDS AND INSURANCE**

6.01 *Performance, Payment, and Other Bonds*

SC-6.01 Add the following paragraphs immediately after Paragraph 6.01.A:

- 1. *Required Performance Bond Form:* The performance bond that Contractor furnishes will be in the form of EJCDC® C-610, Performance Bond (2010, 2013, or 2018 edition).
- 2. *Required Payment Bond Form:* The payment bond that Contractor furnishes will be in the form of EJCDC® C-615, Payment Bond (2010, 2013, or 2018 edition).

SC-6.01 Add the following paragraphs immediately after Paragraph 6.01.B:

1. After Substantial Completion, Contractor shall furnish a warranty bond issued in the form of EJCDC® C-612, Warranty Bond (2018). The warranty bond must be in a bond amount of **15** percent of the final Contract Price.
2. The warranty bond must be issued by the same surety that issues the performance bond required under Paragraph 6.01.A of the General Conditions.

6.03 *Contractor's Insurance*

SC-6.03 Supplement Paragraph 6.03 with the following provisions after Paragraph 6.03.C:

- E. *Workers' Compensation and Employer's Liability:* Contractor shall purchase and maintain workers' compensation and employer's liability insurance, including, as applicable, United States Longshoreman and Harbor Workers' Compensation Act, Jones Act, stop-gap employer's liability coverage for monopolistic states, and foreign voluntary workers' compensation (from available sources, notwithstanding the jurisdictional requirement of Paragraph 6.02.B of the General Conditions).

| <b>Workers' Compensation and Related Policies</b>                                                                                                                                           | <b>Policy limits of not less than:</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Workers' Compensation</b>                                                                                                                                                                |                                        |
| State                                                                                                                                                                                       | Statutory                              |
| Applicable Federal (e.g., Longshoreman's)                                                                                                                                                   | Statutory                              |
| Foreign voluntary workers' compensation (employer's responsibility coverage), if applicable                                                                                                 | Statutory                              |
| <b>Jones Act (if applicable)</b>                                                                                                                                                            |                                        |
| Bodily injury by accident—each accident                                                                                                                                                     | \$100,000.00                           |
| Bodily injury by disease—aggregate                                                                                                                                                          | \$500,000.00                           |
| <b>Employer's Liability</b>                                                                                                                                                                 |                                        |
| Each accident                                                                                                                                                                               | \$100,000.00                           |
| Each employee                                                                                                                                                                               | \$100,000.00                           |
| Policy limit                                                                                                                                                                                | \$500,000.00                           |
| <b>Stop-gap Liability Coverage</b>                                                                                                                                                          |                                        |
| For work performed in monopolistic states, stop-gap liability coverage must be endorsed to either the worker's compensation or commercial general liability policy with a minimum limit of: | \$500,000.00                           |

- F. *Commercial General Liability—Claims Covered:* Contractor shall purchase and maintain commercial general liability insurance, covering all operations by or on behalf of Contractor, on an occurrence basis, against claims for:
1. damages because of bodily injury, sickness or disease, or death of any person other than Contractor's employees,
  2. damages insured by reasonably available personal injury liability coverage, and
  3. damages because of injury to or destruction of tangible property wherever located, including loss of use resulting therefrom.

- G. *Commercial General Liability—Form and Content:* Contractor’s commercial liability policy must be written on a 1996 (or later) Insurance Services Organization, Inc. (ISO) commercial general liability form (occurrence form) and include the following coverages and endorsements:
1. Products and completed operations coverage.
    - a. Such insurance must be maintained for three years after final payment.
    - b. Contractor shall furnish Owner and each other additional insured (as identified in the Supplementary Conditions or elsewhere in the Contract) evidence of continuation of such insurance at final payment and three years thereafter.
  2. Blanket contractual liability coverage, including but not limited to coverage of Contractor’s contractual indemnity obligations in Paragraph 7.18.
  3. Severability of interests and no insured-versus-insured or cross-liability exclusions.
  4. Underground, explosion, and collapse coverage.
  5. Personal injury coverage.
  6. Additional insured endorsements that include both ongoing operations and products and completed operations coverage through ISO Endorsements CG 20 10 10 01 and CG 20 37 10 01 (together). If Contractor demonstrates to Owner that the specified ISO endorsements are not commercially available, then Contractor may satisfy this requirement by providing equivalent endorsements.
  7. For design professional additional insureds, ISO Endorsement CG 20 32 07 04 “Additional Insured—Engineers, Architects or Surveyors Not Engaged by the Named Insured” or its equivalent.
- H. *Commercial General Liability—Excluded Content:* The commercial general liability insurance policy, including its coverages, endorsements, and incorporated provisions, must not include any of the following:
1. Any modification of the standard definition of “insured contract” (except to delete the railroad protective liability exclusion if Contractor is required to indemnify a railroad or others with respect to Work within 50 feet of railroad property).
  2. Any exclusion for water intrusion or water damage.
  3. Any provisions resulting in the erosion of insurance limits by defense costs other than those already incorporated in ISO form CG 00 01.
  4. Any exclusion of coverage relating to earth subsidence or movement.
  5. Any exclusion for the insured’s vicarious liability, strict liability, or statutory liability (other than worker’s compensation).
  6. Any limitation or exclusion based on the nature of Contractor’s work.
  7. Any professional liability exclusion broader in effect than the most recent edition of ISO form CG 22 79.

I. *Commercial General Liability—Minimum Policy Limits*

| <b>Commercial General Liability</b>               | <b>Policy limits of not less than:</b> |
|---------------------------------------------------|----------------------------------------|
| General Aggregate                                 | \$500,000.00                           |
| Products—Completed Operations Aggregate           | \$100,000.00                           |
| Personal and Advertising Injury                   | \$250,000.00                           |
| Bodily Injury and Property Damage—Each Occurrence | \$500,000.00                           |

- J. *Automobile Liability:* Contractor shall purchase and maintain automobile liability insurance for damages because of bodily injury or death of any person or property damage arising out of the ownership, maintenance, or use of any motor vehicle. The automobile liability policy must be written on an occurrence basis.

| <b>Automobile Liability</b>                               | <b>Policy limits of not less than:</b> |
|-----------------------------------------------------------|----------------------------------------|
| <b>Bodily Injury</b>                                      |                                        |
| Each Person                                               | \$100,000.00                           |
| Each Accident                                             | \$500,000.00                           |
| <b>Property Damage</b>                                    |                                        |
| Each Accident                                             | \$100,000.00                           |
| <b>[or]</b>                                               |                                        |
| <b>Combined Single Limit</b>                              |                                        |
| Combined Single Limit (Bodily Injury and Property Damage) | \$500,000.00                           |

- K. *Umbrella or Excess Liability:* Contractor shall purchase and maintain umbrella or excess liability insurance written over the underlying employer's liability, commercial general liability, and automobile liability insurance described in the Paragraphs above. The coverage afforded must be at least as broad as that of each and every one of the underlying policies.

| <b>Excess or Umbrella Liability</b> | <b>Policy limits of not less than:</b> |
|-------------------------------------|----------------------------------------|
| Each Occurrence                     | \$250,000.00                           |
| General Aggregate                   | \$500,000.00                           |

- L. *Using Umbrella or Excess Liability Insurance to Meet CGL and Other Policy Limit Requirements:* Contractor may meet the policy limits specified for employer's liability, commercial general liability, and automobile liability through the primary policies alone, or through combinations of the primary insurance policy's policy limits and partial attribution of the policy limits of an umbrella or excess liability policy that is at least as broad in coverage as that of the underlying policy, as specified herein. If such umbrella or excess liability policy was required under this Contract, at a specified minimum policy limit, such umbrella or excess policy must retain a minimum limit of **\$500,000.00** after accounting for partial attribution of its limits to underlying policies, as allowed above.
- M. *Contractor's Pollution Liability Insurance:* Contractor shall purchase and maintain a policy covering third-party injury and property damage, including cleanup costs, as a result of

pollution conditions arising from Contractor's operations and completed operations. This insurance must be maintained for no less than three years after final completion.

| <b>Contractor's Pollution Liability</b> | <b>Policy limits of not less than:</b> |
|-----------------------------------------|----------------------------------------|
| Each Occurrence/Claim                   | \$100,000.00                           |
| General Aggregate                       | \$100,000.00                           |

- N. *Contractor's Professional Liability Insurance:* If Contractor will provide or furnish professional services under this *Contract*, through a delegation of professional design services or otherwise, then Contractor shall be responsible for purchasing and maintaining applicable professional liability insurance. This insurance must cover negligent acts, errors, or omissions in the performance of professional design or related services by the insured or others for whom the insured is legally liable. The insurance must be maintained throughout the duration of the Contract and for a minimum of two years after Substantial Completion. The retroactive date on the policy must pre-date the commencement of furnishing services on the Project.

| <b>Contractor's Professional Liability</b> | <b>Policy limits of not less than:</b> |
|--------------------------------------------|----------------------------------------|
| Each Claim                                 | \$500,000.00                           |
| Annual Aggregate                           | \$1,000,000.00                         |

## **ARTICLE 7—CONTRACTOR'S RESPONSIBILITIES**

### **7.03 Labor; Working Hours**

SC-7.03 Add the following new subparagraphs immediately after Paragraph 7.03.C:

1. Regular working hours will be **Monday-Friday 8:00 am to 5:00 pm.**
2. Owner's legal holidays are **Federal recognized holidays.**

SC-7.03 Add the following new paragraph immediately after Paragraph 7.03.C:

- D. **Contractor** shall be responsible for the cost of any overtime pay or other expense incurred by the Owner for Engineer/Architect 's services (including those of the Resident Project Representative, if any), Owner's representative, and construction observation services, occasioned by the performance of Work on Saturday, Sunday, any legal holiday, or as overtime on any regular work day. If Contractor is responsible but does not pay, or if the parties are unable to agree as to the amount owed, then Owner may impose a reasonable set-off against payments due under Article 15.

### **7.13 Safety and Protection**

SC-7.13 Insert the following at the end of the last sentence of Paragraph 7.13.F:

"with at least a 48-hour notice."

SC-7.18 Delete the following from Paragraph 7.18.A:

"or arbitration"

## **ARTICLE 8—OTHER WORK AT THE SITE**

### **8.03    *Legal Relationships***

SC-8.03      Delete the following from Paragraph 8.03.C:

“or otherwise resolve the claim by arbitration or other dispute resolution proceeding”

## **ARTICLE 9—OWNER’S RESPONSIBILITIES**

## **ARTICLE 10—ENGINEER/ARCHITECT ’S STATUS DURING CONSTRUCTION**

### **10.03   *Resident Project Representative***

SC-10.03    Add the following new paragraphs immediately after Paragraph 10.03.B:

- C. The Resident Project Representative (RPR) will be Engineer/Architect 's representative at the Site. RPR's dealings in matters pertaining to the Work in general will be with Engineer/Architect and Contractor. RPR's dealings with Subcontractors will only be through or with the full knowledge or approval of Contractor. The RPR will:
  - 1. *Conferences and Meetings:* Attend meetings with Contractor, such as preconstruction conferences, progress meetings, job conferences, and other Project-related meetings (but not including Contractor's safety meetings), and as appropriate prepare and circulate copies of minutes thereof.
  - 2. *Safety Compliance:* Comply with Site safety programs, as they apply to RPR, and if required to do so by such safety programs, receive safety training specifically related to RPR's own personal safety while at the Site.
  - 3. *Liaison*
    - a. Serve as Engineer/Architect 's liaison with Contractor. Working principally through Contractor's authorized representative or designee, assist in providing information regarding the provisions and intent of the Contract Documents.
    - b. Assist Engineer/Architect in serving as Owner's liaison with Contractor when Contractor's operations affect Owner's on-Site operations.
    - c. Assist in obtaining from Owner additional details or information, when required for Contractor's proper execution of the Work.
  - 4. *Review of Work; Defective Work*
    - a. Conduct on-Site observations of the Work to assist Engineer/Architect in determining, to the extent set forth in Paragraph 10.02, if the Work is in general proceeding in accordance with the Contract Documents.
    - b. Observe whether any Work in place appears to be defective.
    - c. Observe whether any Work in place should be uncovered for observation, or requires special testing, inspection or approval.

5. *Inspections and Tests*
    - a. Observe Contractor-arranged inspections required by Laws and Regulations, including but not limited to those performed by public or other agencies having jurisdiction over the Work.
    - b. Accompany visiting inspectors representing public or other agencies having jurisdiction over the Work.
  6. *Payment Requests: Review Applications for Payment with Contractor.*
  7. *Completion*
    - a. Participate in Engineer/Architect 's visits regarding Substantial Completion.
    - b. Assist in the preparation of a punch list of items to be completed or corrected.
    - c. Participate in Engineer/Architect 's visit to the Site in the company of Owner and Contractor regarding completion of the Work, and prepare a final punch list of items to be completed or corrected by Contractor.
    - d. Observe whether items on the final punch list have been completed or corrected.
- D. The RPR will not:
1. Authorize any deviation from the Contract Documents or substitution of materials or equipment (including "or-equal" items).
  2. Exceed limitations of Engineer/Architect 's authority as set forth in the Contract Documents.
  3. Undertake any of the responsibilities of Contractor, Subcontractors, or Suppliers.
  4. Advise on, issue directions relative to, or assume control over any aspect of the means, methods, techniques, sequences or procedures of construction.
  5. Advise on, issue directions regarding, or assume control over security or safety practices, precautions, and programs in connection with the activities or operations of Owner or Contractor.
  6. Participate in specialized field or laboratory tests or inspections conducted off-site by others except as specifically authorized by Engineer/Architect .
  7. Authorize Owner to occupy the Project in whole or in part.

## **ARTICLE 11—CHANGES TO THE CONTRACT**

No suggested Supplementary Conditions in this Article.

## **ARTICLE 12—CLAIMS**

No suggested Supplementary Conditions in this Article.

## ARTICLE 13—COST OF WORK; ALLOWANCES, UNIT PRICE WORK

### 13.01 *Cost of the Work*

SC-13.01 Supplement Paragraph 13.01.B.5.c.(2) by adding the following sentence:

The equipment rental rate book that governs the included costs for the rental of machinery and equipment owned by Contractor (or a related entity) under the Cost of the Work provisions of this Contract is the most current edition of **Rental Rate Blue Book for Construction Equipment**.

### 13.03 *Unit Price Work*

SC-13.03 Delete Paragraph 13.03.E in its entirety and insert the following in its place:

#### E. *Adjustments in Unit Price*

1. Contractor or Owner shall be entitled to an adjustment in the unit price with respect to an item of Unit Price Work if:
  - a. the extended price of a particular item of Unit Price Work amounts to **5** percent or more of the Contract Price (based on estimated quantities at the time of Contract formation) and the variation in the quantity of that particular item of Unit Price Work actually furnished or performed by Contractor differs by more than **15** percent from the estimated quantity of such item indicated in the Agreement; and
  - b. Contractor's unit costs to perform the item of Unit Price Work have changed materially and significantly as a result of the quantity change.
2. The adjustment in unit price will account for and be coordinated with any related changes in quantities of other items of Work, and in Contractor's costs to perform such other Work, such that the resulting overall change in Contract Price is equitable to Owner and Contractor.
3. Adjusted unit prices will apply to all units of that item.

## ARTICLE 14—TESTS AND INSPECTIONS; CORRECTION, REMOVAL, OR ACCEPTANCE OF DEFECTIVE WORK

No suggested Supplementary Conditions in this Article.

## ARTICLE 15—PAYMENTS TO CONTRACTOR, SET OFFS; COMPLETIONS; CORRECTION PERIOD

### 15.03 *Substantial Completion*

SC-15.03 Add the following new subparagraph to Paragraph 15.03.B:

1. If some or all of the Work has been determined not to be at a point of Substantial Completion and will require re-inspection or re-testing by Engineer/Architect, the cost of such re-inspection or re-testing, including the cost of time, travel and living expenses, will be paid by Contractor to Owner. If Contractor does not pay, or the parties are unable to agree as to the amount owed, then Owner may impose a reasonable set-off against payments due under this Article 15.



## **ARTICLE 16—SUSPENSION OF WORK AND TERMINATION**

No suggested Supplementary Conditions in this Article.

## **ARTICLE 17—FINAL RESOLUTIONS OF DISPUTES**

SC-17.02 Add the following new paragraph immediately after Paragraph 17.01.

### **17.02 *Arbitration***

- A. Arbitration as a dispute resolution is deleted in its entirety in the contract, general and supplementary conditions.

## **ARTICLE 18—MISCELLANEOUS**

Page Intentionally Blank

Document 00811

**FEDERAL WAGE RATE DECISION**

1.01 In accordance with the Davis-Bacon Act (Public Law No. 403, 7th Congress), the public body awarding this Contract does hereby specify the following to be assigned minimum wage rates which will be paid by the Contractor and all Subcontractors for this Project.

1.02 This prevailing wage rate does not prohibit the payment of more than the rates stated.

1.03 The wage scale for (Specify type of work).

**Note to Specifier: Davis-Bacon Wage Rates can be found at <http://www.wdol.gov/dba.aspx>. Use the Wage Rates for Hidalgo County. Select the WD Determination that is appropriate for the Project Copy the Wage Rate directly from the DOL website to this Section.**





## SECTION 00830

**WARRANTY****PART 1 - GENERAL****1.01 SECTION INCLUDES**

This section describes the warranty. The conditions contained in this Section are specific administrative and policy requirements in addition to the general conditions and other requirements listed in the contract documents.

**1.02 REFERENCES – Not Used****1.03 DEFINITIONS – Section 0700****1.04 CONTRACTOR'S WARRANTY OF TITLE**

CONTRACTOR warrants and guarantees that all Work, materials, and equipment covered by any Application for Payment, whether incorporated in the Project or not, will pass to OWNER no later than the time of payment free and clear of all Liens.

**1.05 SUBSTANTIAL COMPLETION**

- A. When CONTRACTOR considers the entire Work ready for its intended use CONTRACTOR shall notify OWNER and ARCHITECT in writing that the entire Work is substantially complete (except for items specifically listed by CONTRACTOR as incomplete) and request that ARCHITECT issue a certificate of Substantial Completion. Promptly thereafter, OWNER, CONTRACTOR, and ARCHITECT shall make an inspection of the Work to determine the status of completion. If ARCHITECT does not consider the Work substantially complete, ARCHITECT will notify CONTRACTOR in writing giving the reasons therefore. If ARCHITECT considers the Work substantially complete, ARCHITECT will prepare and deliver to OWNER a tentative certificate of Substantial Completion which shall fix the date of Substantial Completion. There shall be attached to the certificate a tentative list of items to be completed or corrected before final payment. OWNER shall have seven days after receipt of the tentative certificate during which to make written objection to ARCHITECT as to any provisions of the certificate or attached list. If, after considering such objections, ARCHITECT concludes that the Work is not substantially complete, ARCHITECT will within 14 days after submission of the tentative certificate to OWNER notify CONTRACTOR in writing, stating the reasons therefore. If, after consideration of OWNER's objections, ARCHITECT considers the Work substantially complete, ARCHITECT will within said 14 days execute and deliver to OWNER and CONTRACTOR a definitive certificate of Substantial Completion (with a revised tentative list of items to be completed or corrected) reflecting such changes from the tentative certificate as ARCHITECT believes justified after consideration of any objections from OWNER. At the time of delivery of the tentative certificate of Substantial Completion ARCHITECT will deliver to OWNER and CONTRACTOR a written recommendation as to division of responsibilities pending final payment between OWNER and CONTRACTOR with respect to security, operation, safety, and protection of the Work, maintenance, heat, utilities, insurance, and warranties and guarantees. Unless OWNER and CONTRACTOR agree otherwise in writing and so inform ARCHITECT in writing prior to ARCHITECT's issuing the definitive certificate of Substantial Completion, ARCHITECT's aforesaid recommendation will be binding on OWNER and CONTRACTOR until final payment.
- B. OWNER shall have the right to exclude CONTRACTOR from the Site after the date of Substantial Completion, but OWNER shall allow CONTRACTOR reasonable access to complete or correct items on the tentative list.

**1.06 PARTIAL UTILIZATION**

- A. Use by OWNER at OWNER's option of any substantially completed part of the Work which has specifically been identified in the Contract Documents, or which OWNER, ARCHITECT, and CONTRACTOR agree constitutes a separately functioning and usable part of the Work that can be used by OWNER for its intended purpose without significant interference with CONTRACTOR's performance of the remainder of the Work, may be accomplished prior to Substantial Completion of all the Work subject to the following conditions.
- B. OWNER at any time may request CONTRACTOR in writing to permit OWNER to use any such part of the Work which OWNER believes to be ready for its intended use and substantially complete. If CONTRACTOR agrees that such part of the Work is substantially complete, CONTRACTOR will certify to OWNER and ARCHITECT that such part of the Work is substantially complete and request ARCHITECT to issue a certificate of Substantial Completion for that part of the Work. CONTRACTOR at any time may notify OWNER and ARCHITECT in writing that CONTRACTOR considers any such part of the Work ready for its intended use and substantially complete and request ARCHITECT to issue a certificate of Substantial Completion for that part of the Work. Within a reasonable time after either such request, OWNER, CONTRACTOR, and ARCHITECT shall make an inspection of that part of the Work to determine its status of completion. If ARCHITECT does not consider that part of the Work to be substantially complete, ARCHITECT will notify OWNER and CONTRACTOR in writing giving the reasons therefore. If ARCHITECT considers that part of the Work to be substantially complete, the above provisions will apply with respect to certification of Substantial Completion of that part of the Work and the division of responsibility in respect thereof and access thereto. No occupancy or separate operation of part of the Work may occur prior to compliance with the requirement of regarding property insurance.

**1.07 FINAL INSPECTION**

- A. Upon written notice from CONTRACTOR that the entire Work or an agreed portion thereof is complete, ARCHITECT will promptly make a final inspection with OWNER and CONTRACTOR and will notify CONTRACTOR in writing of all particulars in which this inspection reveals that the Work is incomplete or defective. CONTRACTOR shall immediately take such measures as are necessary to complete such Work or remedy such deficiencies.

**1.08 FINAL PAYMENT**

- A. Application for Payment
1. After CONTRACTOR has, in the opinion of ARCHITECT, satisfactorily completed all corrections identified during the final inspection and has delivered, in accordance with the Contract Documents, all maintenance and operating instructions, schedules, guarantees, Bonds, certificates or other evidence of insurance certificates of inspection, marked-up record documents (as provided in paragraph 6.12), and other documents, CONTRACTOR may make application for final payment following the procedure for progress payments.
  2. The final Application for Payment shall be accompanied (except as previously delivered) by: (i) all documentation called for in the Contract Documents, including but not limited to the evidence of insurance required; (ii) consent of the surety, if any, to final payment; and (iii) complete and legally effective releases or waivers (satisfactory to OWNER) of all Lien rights arising out of or Liens filed in connection with the Work.
  3. In lieu of the releases or waivers of Liens specified above and as approved by OWNER,

CONTRACTOR may furnish receipts or releases in full and an affidavit of CONTRACTOR that: (i) the releases and receipts include all labor, services, material, and equipment for which a Lien could be filed; and (ii) all payrolls, material and equipment bills, and other indebtedness connected with the Work for which OWNER or OWNER's property might in any way be responsible have been paid or otherwise satisfied. If any Subcontractor or Supplier fails to furnish such a release or receipt in full, CONTRACTOR may furnish a Bond or other collateral satisfactory to OWNER to indemnify OWNER against any Lien.

**B. Review of Application and Acceptance**

If, on the basis of ARCHITECT's observation of the Work during construction and final inspection, and ARCHITECT's review of the final Application for Payment and accompanying documentation as required by the Contract Documents, ARCHITECT is satisfied that the Work has been completed and CONTRACTOR's other obligations under the Contract Documents have been fulfilled, ARCHITECT will, within ten days after receipt of the final Application for Payment, indicate in writing ARCHITECT's recommendation of payment and present the Application for Payment to OWNER for payment. At the same time ARCHITECT will also give written notice to OWNER and CONTRACTOR that the Work is acceptable subject to the above provisions. Otherwise, ARCHITECT will return the Application for Payment to CONTRACTOR, indicating in writing the reasons for refusing to recommend final payment, in which case CONTRACTOR shall make the necessary corrections and resubmit the Application for Payment.

**C. Payment Becomes Due**

Thirty days after the presentation to OWNER of the Application for Payment and accompanying documentation, the amount recommended by ARCHITECT will become due and, when due, will be paid by OWNER to CONTRACTOR.

**D. Final Completion Delayed**

If, through no fault of CONTRACTOR, final completion of the Work is significantly delayed, and if ARCHITECT so confirms, OWNER shall, upon receipt of CONTRACTOR's final Application for Payment and recommendation of ARCHITECT, and without terminating the Agreement, make payment of the balance due for that portion of the Work fully completed and accepted. If the remaining balance to be held by OWNER for Work not fully completed or corrected is less than the retainage stipulated in the Agreement, and if Bonds have been furnished as required above, the written consent of the surety to the payment of the balance due for that portion of the Work fully completed and accepted shall be submitted by CONTRACTOR to ARCHITECT with the Application for such payment. Such payment shall be made under the terms and conditions governing final payment, except that it shall not constitute a waiver of Claims.

**1.09 WAIVER OF CLAIMS**

**A. The making and acceptance of final payment will constitute:**

1. a waiver of all Claims by OWNER against CONTRACTOR, except Claims arising from unsettled Liens, from defective Work appearing after final inspection pursuant to the above, from failure to comply with the Contract Documents or the terms of any special guarantees specified therein, or from CONTRACTOR's continuing obligations under the Contract Documents; and
2. a waiver of all Claims by CONTRACTOR against OWNER other than those previously made in writing which are still unsettled.

END OF SECTION



Document 00900

**ADDENDUM NO. \_\_\_\_\_**  
**(Sample Form)**

Date of Addendum: \_\_\_\_\_  
[Enter date]

PROJECT NAME: Fire Station #5

PROJECT NO: Project 2019-20

BID DATE: May 20, 2019 (There is no change to the Bid Date.)

FROM: City of Edinburg  
Att: Lorena Fuentes, Purchasing Agent  
415 W. University Drive  
Edinburg, Texas 78539  
Phone: (956) 388-1895

**TO: Prospective Bidders**

This Addendum forms a part of the Bidding Documents and will be incorporated into Contract Documents, as applicable. Insofar as the original Project Manual and Drawings are inconsistent, this Addendum governs. Acknowledge receipt of the Addendum by inserting its number in Document 00310 - Form of Proposal. **FAILURE TO DO SO MAY SUBJECT BIDDER TO DISQUALIFICATION.**

\*\*\*\*\*

Use the following heading and select the appropriate wording for postponement of the Bid Date. Delete the statement beside Bid Date above which indicates that the Bid Date is unchanged. If change in Bid Date, issue as separate addendum. Delete this section entirely if there is no change in Bid Date.

\*\*\*\*\*

### CHANGE IN BID DATE

The bid date for this Project has been changed from \_\_\_\_\_ to \_\_\_\_\_  
[Date] [Date]

[Time of day and place for submittal of bid remains the same]. [Time of submittal has been changed from \_\_\_\_\_ to \_\_\_\_\_. The place for submittal remains the same].

[Time]                      [Time]

\*\*\*[OR]\*\*\*

The bid date for this project has been indefinitely postponed. Another Addendum will be issued to reset the bid date or to cancel bidding on this Project.

\*\*\*\*\*

Delete the following paragraph if the sole purpose of the Addendum is to postpone the Bid Date.  
\*\*\*\*\*

This Addendum uses the change page method: remove and replace or add pages, or Drawing sheets, as directed in the change instructions below. Change bars ( | ) are provided in the right margins of pages from the Project Manual to indicate where changes have been made; no change bars are provided in added Sections. Reissued Drawing Sheets show the Addendum number above the title block and changes in the Drawing are noted by a revision mark.

\*\*\*\*\*

Number each item of the Addendum beginning with 1 through the total number of change items in the Addendum. Sample entries are provided in brackets.

\*\*\*\*\*

### CHANGES TO PREVIOUS ADDENDA

\*\*\*\*\*

Reference Addendum Number and item number to correct clarifications or make minor corrections of changes issued by previous Addenda.

\*\*\*\*\*

**ADDENDUM NO.** \_\_\_\_\_

[1. Add item]

### CHANGES TO PROJECT MANUAL

\*\*\*\*\*

Follow this format to sequence changes to the Project Manual.

\*\*\*\*\*

### BIDDING REQUIREMENTS

\*\*\*\*\*

Give the individual change instructions for each item of change by Document number and title. List changes in order of Document number.

\*\*\*\*\*

[2. Add Item]

### CONTRACT FORMS

[3. Add Item]

### CONDITIONS OF THE CONTRACT

[4. Add Item]

### SPECIFICATIONS

[5. Add Item]

### CHANGES TO DRAWINGS

[6. Add Item]

**CLARIFICATIONS**

[7. Add Item]

**MINUTES OF PRE-BID CONFERENCE**

Minutes of the Pre-Bid Conference held on \_\_\_\_\_, \_\_\_\_\_, 20\_\_\_\_, are  
[Day] [Date]  
attached as a record and for the Bidders information.

END OF ADDENDUM NO. \_\_\_\_\_

\_\_\_\_\_  
Name, P.E. DATED: \_\_\_\_\_

**END OF DOCUMENT**

Page Intentionally Blank

## DOCUMENT 00910

**MODIFICATIONS****PART 1 - GENERAL****1.01 SECTION INCLUDES**

This section contains information pertaining to modifications and changes for the Contract Documents for the Project.

**1.02 REFERENCES – Not Used****1.03 DEFINITIONS - Section 0700****1.04 MODIFICATIONS OF CONTRACT DOCUMENTS**

- A. The Contract Documents may be amended to provide for additions, deletions, and revisions in the Work or to modify the terms and conditions thereof in one or more of the following ways: (i) a Written Amendment; (ii) a Change Order; or (iii) a Work Change Directive.
- B. The requirements of the Contract Documents may be supplemented and minor variations and deviations in the Work may be authorized, by one or more of the following ways: (i) a Field Order; (ii) Architect's approval of a Shop Drawing or Sample; or (iii) Architect's written interpretation or clarification.
- C. Contractor and any Subcontractor or Supplier or other individual or entity performing or furnishing any of the Work under a direct or indirect contract with Owner: (i) shall not have or acquire any title to or ownership rights in any of the Drawings, Specifications, or other documents (or copies of any thereof) prepared by or bearing the seal of Architect or Architect's Consultant, including electronic media editions; and (ii) shall not reuse any of such Drawings, Specifications, other documents, or copies thereof on extensions of the Project or any other project without written consent of Owner and Architect and specific written verification or adoption by Architect. This prohibition will survive final payment, completion, and acceptance of the Work, or termination or completion of the Contract. Nothing herein shall preclude Contractor from retaining copies of the Contract Documents for record purposes.

**PART 2 - PRODUCT – Not Used****PART 3 - EXECUTION – Not Used**

**END OF SECTION**

Page Intentionally Blank

# PROJECT MANUAL

**Plans and Specifications – Project No. 219003  
FOR  
CITY OF EDINBURG  
NEW FIRE STATION #5  
EDINBURG, TX. 78540**



**TEXAS BOARD OF ARCHITECTURAL EXAMINERS  
333 Guadalupe, Suite 2-350, AUSTIN, TX 78701-3942  
(Tel: 512/305-9000)  
HAS JURISDICTION OVER INDIVIDUALS LICENSED UNDER  
THE ARCHITECT'S REGISTRATION LAW  
ARTICLE 249a, VERNON'S CIVIL STATUTES".**

**MILNET ARCHITECTURAL SERVICES  
608 S. 12 St.  
McALLEN, TEXAS 78501  
(956) 688- 5656 – FAX (956) 687-9289**

# **HINOJOSA ENGINEERING, INC.**

Civil Engineering • Structural Engineering  
108 West 18<sup>th</sup> Street Mission, Texas 78572  
(956) 581-0143 Fax (956) 581-2074  
Registration Number F-908 Expiration Date 09/30/2019

## **TECHNICAL SPECIFICATIONS "CIVIL"**

### **EDINBURG FIRE STATION #5**

#### **SECTION**

#### **DESCRIPTION**

|       |                                                      |
|-------|------------------------------------------------------|
| 01570 | EROSION AND SEDIMENT CONTROL                         |
| 01731 | CUTTING AND PATCHING                                 |
| 02100 | SITE PREPARATION                                     |
| 02231 | CLEARING AND GRUBBING                                |
| 02260 | TRENCH SAFETY SYSTEM                                 |
| 02300 | EARTHWORK                                            |
| 02315 | EXCAVATION, BACKFILLING AND COMPACTING FOR UTILITIES |
| 02317 | EXCAVATION, BACKFILLING AND COMPACTING FOR PAVEMENT  |
| 02341 | SOIL LIME STABILIZATION                              |
| 02505 | PAVING BASE COURSE                                   |
| 02511 | ASPHALTIC CONCRETE PAVING                            |
| 02520 | PORTLAND CEMENT CONCRETE PAVING                      |
| 02525 | CURB & SIDEWALKS                                     |
| 02584 | PAVEMENT MARKINGS                                    |
| 02605 | SEWER STRUCTURES                                     |
| 02660 | WATER DISTRIBUTION SYSTEMS                           |
| 02720 | STORM SEWER SYSTEMS                                  |
| 02730 | SANITARY SEWER SYSTEMS                               |
| 02846 | SIGNAGE                                              |

2-28-19





**HINOJOSA ENGINEERING, INC.**  
Civil Engineering • Structural Engineering  
108 West 18<sup>th</sup> Street Mission, Texas 78572  
(956) 581-0143 Fax (956) 581-2074  
Registration Number F-908 Expiration Date 09/30/2019

**TECHNICAL SPECIFICATIONS "STRUCTURAL"**

**EDINBURG FIRE STATION #5**

**CITY OF EDINBURG**

**DIVISION 2 - EARTHWORK**

|               |                            |
|---------------|----------------------------|
| SECTION 02110 | SITE CLEARING AND GRUBBING |
| SECTION 02200 | EARTHWORK                  |
| SECTION 02282 | TERMITE CONTROL            |

**DIVISION 3 - CONCRETE**

|               |                        |
|---------------|------------------------|
| SECTION 03200 | CONCRETE REINFORCEMENT |
| SECTION 03310 | CONCRETE WORK          |

**DIVISION 4 - MASONRY**

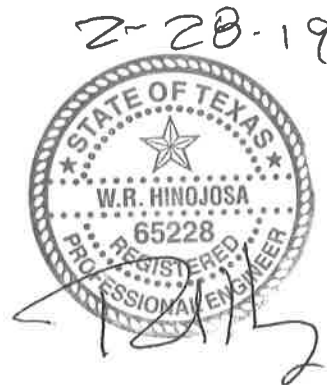
|               |                              |
|---------------|------------------------------|
| SECTION 04065 | MASONRY MORTARING & GROUTING |
| SECTION 04200 | UNIT MASONRY (CMU BLOCK)     |
| SECTION 04230 | REINFORCED UNIT MASONRY      |

**DIVISION 5 - METALS**

|               |                   |
|---------------|-------------------|
| SECTION 05100 | STRUCTURAL STEEL  |
| SECTION 05500 | METAL FABRICATION |

**DIVISION 13 - SPECIAL CONSTRUCTION**

|               |                        |
|---------------|------------------------|
| SECTION 13120 | METAL BUILDING SYSTEMS |
|---------------|------------------------|



**CITY OF EDINBURG**  
**FIRE STATION #5**  
Edinburg, Texas  
MAS Project No. 219003

| Division                                                | Section Title                                         | Pages |
|---------------------------------------------------------|-------------------------------------------------------|-------|
| <b>SERIES 0 BIDDING REQUIREMENTS AND CONTRACT FORMS</b> |                                                       |       |
| 00 61 00                                                | PERFORMANCE BOND AND LABOR AND MATERIAL PAYMENT BOND  | 1     |
| 00 62 76                                                | TAX EXEMPT ORGANIZATION CERTIFICATE                   | 1     |
| 00 73 00                                                | SUPPLEMENTARY CONDITIONS                              | 4     |
|                                                         | GEOTECHNICAL INVESTIGATION                            | 67    |
| <b>DIVISION 1 - GENERAL REQUIREMENTS</b>                |                                                       |       |
| 01 11 00                                                | SUMMARY                                               | 1     |
| 01 20 00                                                | PRICE AND PAYMENT PROCEDURES                          | 2     |
| 01 21 00                                                | ALLOWANCES                                            | 1     |
| 01 25 00                                                | SUBSTITUTION PROCEDURES                               | 3     |
| 01 30 00                                                | ADMINISTRATIVE REQUIREMENTS                           | 2     |
| 01 33 00                                                | SUBMITTALS                                            | 4     |
| 01 40 00                                                | QUALITY REQUIREMENTS                                  | 2     |
| 01 50 00                                                | TEMPORARY FACILITIES AND CONTROLS                     | 5     |
| <b>DIVISION 2 - SITE CONSTRUCTION</b>                   |                                                       |       |
| <b>DIVISION 3 - CONCRETE</b>                            |                                                       |       |
| <b>DIVISION 4 - MASONRY</b>                             |                                                       |       |
| 04 05 13                                                | MORTAR                                                | 3     |
| 04 21 13                                                | BRICK MASONRY                                         | 5     |
| <b>DIVISION 5 - METALS</b>                              |                                                       |       |
| 05 41 00                                                | LIGHT GAGE METAL FRAMING SYSTEMS AND GYPSUM SHEATHING | 4     |
| <b>DIVISION 6 - WOOD AND PLASTICS</b>                   |                                                       |       |
| 06 10 00                                                | ROUGH CARPENTRY                                       | 3     |
| 06 40 23                                                | INTERIOR ARCHITECTURAL WOODWORK                       | 12    |
| 06 61 16                                                | SOLID SURFACING                                       | 5     |
| <b>DIVISION 7 - THERMAL AND MOISTURE PROTECTION</b>     |                                                       |       |
| 07 10 00                                                | DAMPPROOFING AND WATERPROOFING                        | 4     |
| 07 21 00                                                | BUILDING INSULATION                                   | 5     |
| 07 21 19                                                | FOAMED-IN-PLACE MASONRY WALL INSULATION               | 3     |
| 07 22 00                                                | ROOF AND DECK INSULATION                              | 3     |
| 07 26 16                                                | UNDER SLAB VAPOR BARRIER                              | 3     |
| 07 41 13                                                | INSULATED METAL ROOF PANEL                            | 8     |
| 07 60 00                                                | SHEET METAL AND MISCELLANEOUS ACCESSORIES             | 7     |
| 07 92 00                                                | JOINT SEALANTS                                        | 2     |

**DIVISION 8 - DOORS AND WINDOWS**

|          |                                                  |    |
|----------|--------------------------------------------------|----|
| 08 11 13 | HOLLOW METAL DOORS AND FRAMES                    | 11 |
| 08 14 16 | FLUSH WOOD DOORS                                 | 7  |
| 08 41 13 | STOREFRONT                                       | 8  |
| 08 71 00 | DOOR HARDWARE                                    | 21 |
| 08 81 00 | GLASS AND GLAZING                                | 4  |
| 08 88 56 | BULLET RESISTANT HOLE AND BACKER WINDOW ASSEMBLY | 3  |

**DIVISION 9 - FINISHES**

|          |                             |   |
|----------|-----------------------------|---|
| 09 20 00 | LATH AND PLASTER            | 5 |
| 09 21 16 | INTERIOR DRYWALL SYSTEMS    | 8 |
| 09 22 00 | PLASTER ACCESSORIES         | 7 |
| 09 30 00 | WALL AND FLOOR TILE         | 5 |
| 09 51 00 | ACOUSTICAL TILE CEILINGS    | 4 |
| 09 65 00 | RESILIENT FLOORING AND BASE | 3 |
| 09 65 19 | SPECIALTY FLOORING          | 5 |
| 09 91 00 | PAINTING AND FINISHING      | 6 |
| 09 96 53 | ELASTOMERIC COATING         | 2 |

**DIVISION 10 - SPECIALTIES**

|          |                           |   |
|----------|---------------------------|---|
| 10 14 00 | GRAPHICS AND SIGNAGE      | 5 |
| 10 21 13 | TOILET COMPARTMENTS       | 5 |
| 10 26 41 | BULLETPROOF PANELS        | 2 |
| 10 28 13 | TOILET & BATH ACCESSORIES | 2 |
| 10 44 00 | FIRE EXTINGUISHERS        | 3 |
| 10 51 53 | LOCKER BENCHES            | 3 |
| 10 75 00 | FLAGPOLES                 | 3 |

**DIVISION 11 – EQUIPMENT****DIVISION 12 – FURNISHINGS**

|          |                          |   |
|----------|--------------------------|---|
| 12 21 13 | HORIZONTAL LOUVER BLINDS | 3 |
|----------|--------------------------|---|

**DIVISION 13 - SPECIAL CONSTRUCTION**

|          |                               |   |
|----------|-------------------------------|---|
| 13 34 19 | PRE-ENGINEERED METAL BUILDING | 4 |
|----------|-------------------------------|---|

**DIVISION 14 - CONVEYING SYSTEMS**

**TABLE OF CONTENTS**  
**Edinburg Fire Station No. 5**  
**198001.000**

DBR Engineering Consultants, Inc.  
**February 28, 2019**



| <u>DIVISION 21 – FIRE</u>       |                                                                | <u>NO. of PAGES</u> |
|---------------------------------|----------------------------------------------------------------|---------------------|
| 21 00 00                        | Fire Protection                                                | 1-3                 |
| 21 02 01                        | Coordination Drawings                                          | 1-2                 |
| 21 05 29                        | Hangers and Supports for Fire Suppression Piping and Equipment | 1-5                 |
| 21 13 13                        | Wet Pipe Sprinkler System                                      | 1-5                 |
| <u>DIVISION 22 – PLUMBING</u>   |                                                                | <u>NO. of PAGES</u> |
| 22 02 00                        | Basic Materials and Methods                                    | 1-19                |
| 22 02 01                        | Coordination Drawings                                          | 1-2                 |
| 22 05 16                        | Expansion Fittings and Loops for Plumbing Piping               | 1-4                 |
| 22 05 29                        | Hangers and Support for Plumbing Piping and Equipment          | 1-6                 |
| 22 05 53                        | Identification for Plumbing Piping and Equipment               | 1-2                 |
| 22 07 19                        | Plumbing Piping Insulation                                     | 1-4                 |
| 22 10 00                        | Plumbing Piping                                                | 1-21                |
| 22 11 19                        | Plumbing Specialties                                           | 1-8                 |
| 22 11 21                        | Natural Gas Piping Systems                                     | 1-3                 |
| 22 30 00                        | Plumbing Equipment                                             | 1-4                 |
| 22 40 00                        | Plumbing Fixtures                                              | 1-7                 |
| 22 61 00                        | Compressed Air Systems                                         | 1-5                 |
| <u>DIVISION 23 – MECHANICAL</u> |                                                                | <u>NO. of PAGES</u> |
| 23 02 00                        | Basic Materials and Methods                                    | 1-20                |
| 23 02 01                        | Coordination Drawings                                          | 1-2                 |
| 23 05 29                        | Hangers and Support for Piping and Equipment HVAC              | 1-6                 |
| 23 05 48                        | Vibration and Seismic Controls for HVAC Piping and Equipment   | 1-3                 |
| 23 05 53                        | Identification for HVAC Piping and Equipment                   | 1-3                 |
| 23 05 93                        | Testing, Adjusting, and Balancing                              | 1-6                 |
| 23 07 13                        | Duct Insulation                                                | 1-6                 |
| 23 07 19                        | HVAC Piping Insulation                                         | 1-5                 |
| 23 09 63                        | Energy Management and Control System                           | 1-30                |
| 23 23 00                        | Refrigerant Piping                                             | 1-2                 |
| 23 31 13                        | Metal Ductwork                                                 | 1-9                 |
| 23 33 00                        | Ductwork Accessories                                           | 1-6                 |
| 23 34 00                        | HVAC Fans                                                      | 1-5                 |
| 23 37 13                        | Air Distribution Devices                                       | 1-4                 |
| 23 41 00                        | Air Filters                                                    | 1-2                 |
| 23 62 13                        | Air Cooled Condensing Units                                    | 1-3                 |
| 23 82 19                        | Fan Coil Unit                                                  | 1-2                 |
| 23 82 43                        | Electric Duct Heaters                                          | 1-2                 |
| <u>DIVISION 26 – ELECTRICAL</u> |                                                                | <u>NO. of PAGES</u> |
| 26 02 00                        | Basic Materials and Methods                                    | 1-19                |
| 26 02 01                        | Coordination Drawings                                          | 1-2                 |
| 26 05 19                        | Wire, Cable, and Related Materials                             | 1-5                 |

|          |                                              |      |
|----------|----------------------------------------------|------|
| 26 05 26 | Grounding                                    | 1-3  |
| 26 05 33 | Raceways                                     | 1-9  |
| 26 08 00 | Commissioning of Electrical Systems          | 1-7  |
| 26 21 13 | Electrical Service Entrance                  | 1-2  |
| 26 24 16 | Panelboards                                  | 1-5  |
| 26 27 26 | Wiring Devices                               | 1-8  |
| 26 28 16 | Safety and Disconnect Switches               | 1-3  |
| 26 29 26 | Miscellaneous Electrical Controls and Wiring | 1-4  |
| 26 32 00 | (1000 kW) Diesel Emergency Generator         | 1-12 |
| 26 51 00 | Lighting Fixtures                            | 1-7  |

---

|                              |              |
|------------------------------|--------------|
| DIVISION 27 – COMMUNICATIONS | NO. of PAGES |
|------------------------------|--------------|

|          |                                                  |      |
|----------|--------------------------------------------------|------|
| 27 00 00 | Communications                                   | 1-18 |
| 27 05 26 | Grounding and Bonding for Communications Systems | 1-7  |
| 27 05 28 | Pathways for Communications Systems              | 1-8  |
| 27 11 00 | Communications Room Fittings                     | 1-5  |
| 27 15 00 | Communications Horizontal Cabling                | 1-11 |

---

|                                              |              |
|----------------------------------------------|--------------|
| DIVISION 28 – ELECTRONIC SAFETY AND SECURITY | NO. of PAGES |
|----------------------------------------------|--------------|

|             |                                                              |      |
|-------------|--------------------------------------------------------------|------|
| 28 00 00    | Electronic Security                                          | 1-16 |
| 28 10 00    | Access Control and Intrusion Detection                       | 1-8  |
| 28 23 00    | Video Surveillance                                           | 1-8  |
| 28 31 00 13 | Fire Alarm and Smoke Detection System (Medium Analog System) | 1-14 |

## **SECTION 01570**

### **EROSION AND SEDIMENT CONTROL**

#### **Part 1 GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Implementation of temporary and permanent erosion control systems.

##### **1.02 RELATED SECTIONS**

- A. Section 02100 – Site Preparation
- B. Erosion and Sediment Control Plan
- C. Construction Drawings
- D. Storm Water Pollution Prevention Plan (SWP3) – where required by TCEQ

##### **1.03 ENVIRONMENTAL REQUIREMENTS**

- A. The contractor shall protect adjacent properties and water resources from erosion and sediment throughout the life of the construction contract as required by governing entities.

##### **1.04 SYSTEM DESCRIPTION**

- A. Filter Fabric Barrier Type 1 and Type 2: Install to allow surface or channel runoff percolation through fabric in sheet-flow manner and to retain and accumulate sediment. Maintain Filter Fabric Barriers in proper position and configuration at all times.
- B. Hay Bale Fence: Install to allow surface runoff percolation through hay in sheet-flow manner and to retain and accumulate sediment. Maintain Hay Bale Fence in proper position and configuration at all times.
- C. Interceptor Dikes and Swales: Construct to direct surface or channel runoff around the project area or runoff from project area into sediment traps.
- D. Drop Inlet Baskets: Install to allow runoff percolation through the basket and to retain and accumulate sediment. Clean accumulation of sediment to prevent clogging and backups.
- E. Sediment Traps: Construct to pool surface runoff from construction area to allow sediment to settle onto the bottom of trap.
- F. Sand Bags: May be used during construction activities in unstabilized swales, ditches, or streambeds when the contributing drainage area is 2 acres or less. Also used as sediment barrier for inlets.
- G. Bagged Gravel Barrier: May be used during construction activities in unstabilized swales, ditches, or streambeds when the contributing drainage area is 2 acres or less. Also used as sediment barrier for inlets.

- H. Drop Inlet Insert Basket: A temporary barrier placed within a storm drain inlet consisting of a filter fabric supported by a metal frame work to prevent sediment and other pollutants from entering conveying system.
- I. Bush Berm: Is constructed at the perimeter of a distribute site within the developing area.

## **1.05 SUBMITTALS**

- A. Submit manufacturers literature for product specifications and installation instructions.
- B. Submit manufacturers catalog sheets and other product data on geotextile or filter fabrics, outlet pipes, perforated riser and connectors.

## **PART 2 PRODUCTS**

### **2.01 MATERIALS**

- A. Aggregate Materials: Use poorly graded cobbles with diameter greater than 3 inches and less than 5 inches. Provide clean cobbles and gravel consisting of crushed concrete or stone free from adherent coatings, salt, alkali, dirt, clay, loam, shale, soft or flaky material, or organic matter.
- B. Geotextile filter fabrics: Woven or nonwoven geotextile filter fabric made of either polypropylene, polyethylene, ethylene, or polyamide material, in continuous rolls of longest practical length. Grab strength 100psi in any direction and Mullen burst strength to be greater than 200psi, and an equivalent opening size between 50 and 140. Mirafi, Inc. Synthetic Industries or equivalent.
- C. Barrier: Wire barrier to be woven galvanized steel wire, 14 guage by 6 inch square mesh spacing, minimum 24 inch roll or sheet width of longest practical length. Barrier stakes to be nominal 2 by 2 inch moisture resistant treated wood or steel posts with safety caps. Length as required for minimum 8 inch bury and full height of filter fabric.
- D. Sandbags: to be woven material made of polypropylene, polyethylene, or polyamide material. Minimum unit weight of four ounces per square yard, grab strength of 100 lbs in any principal direction, ultra violet stability exceeding 70 percent after 500 hours of exposure. Length to be 18 to 24 inches, width of 12 to 18 inches, thickness 6 to 8 inches, and weight of approximately 45 pounds (not to exceed 75 pounds).
- E. Bagged gravel barriers: Minimum unit weight of four ounces per square yard, grab strength of 100 lbs in any principal direction, ultra violet stability exceeding 70 percent after 500 hours of exposure. Length to be 18 to 24 inches, width of 12 to 18 inches, thickness 6 to 8 inches, and weight of approximately 45 pounds (not to exceed 75 pounds).
- F. Hay bales: Standard baled agricultural hay bound by wire, nylon or polypropylene rope. Do not use jute or cotton binding. Hay bale stakes (where hay bale used on soil) to be No.3 reinforcing bars, deformed or smooth for a minimum length as required for a minimum 18 inch bury and full height of bales.

## **PART 3 EXECUTION**

### **3.01 PREPARATION and MAINTENANCE**

- A. Review site erosion control plan. Provide erosion and sediment control structures at locations shown on the drawings.
- B. Do not clear, grub or cut until erosion and sediment control systems are in place unless approved by Project Manager.
- C. Maintain erosion and sediment control systems located within the project site until acceptance of project, stabilization of project area, or until directed by Project Manager to remove and discard.
- D. Regularly inspect and repair or replace any damaged components of erosion and sediment control structures. Immediately repair any damage caused by construction traffic to erosion and sediment control structures.
- E. Deficiencies or changes in the erosion control plan as it is applied to current conditions will be brought to the attention of the Owner and the Engineer for remedial action.

### **3.02 EROSION CONTROL AND SLOPE PROTECTION IMPLEMENTATION**

- A. Place erosion control systems in accordance with the erosion control plan.
- B. Control surface water, runoff, and water from excavations and structures to prevent damage to the site or adjoining properties.
- C. Provide, operate and maintain facilities of adequate size to control surface water.
- D. Prevent water runoff from passing through waste collection areas, and prevent water runoff from waste collection areas to migrate outside collection area.
- E. Confine maintenance and repair of construction machinery and equipment to areas specifically designed for that purpose to prevent washing fuels, lubricants, solvents, or other potential pollutants into storm water conveyance systems. Where designated areas are not feasible, take precautions during each individual repair or maintenance operation to prevent pollution. Clean and inspect maintenance areas daily.
- F. Slopes that easily erode shall be temporarily seeded as the work progresses (see section 02922).
- G. Incorporate all permanent erosion control features into the project at the earliest practical time to minimize the need for temporary controls. Cut slopes shall be permanently seeded and mulched as the excavation proceeds to the extent considered desirable and practical.
- H. Dispose of sediments offsite, not in or adjacent to waterways or floodplains. Do not allow sediments to flush into streams nor drainage ways.
- I. Retain existing drainage patterns external to the site by constructing temporary earth berms, sedimentation basins, retaining areas, and temporary ground cover as required to control conditions.

### **3.03 REMOVAL OF CONTROLS**

- A. Remove erosion and sediment controls when the site is fully stabilized or as directed by Project Manager.



- B. Properly dispose of sediments and waste products per the requirements of the governing entities.

**END OF SECTION 01570**

## **SECTION 01731**

### **CUTTING AND PATCHING**

#### **Part 1 GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Cutting, patching and fitting of the work or work under construction.
- B. Coordinating installation or connection of the work to existing facilities, uncovering work for access, inspection or testing and related items.

##### **1.02 CUTTING AND PATCHING**

- A. Perform activities to avoid interference with facility operations and other work.
- B. Execute cutting and patching, including excavation, backfill and fitting to:
  - 1. Remove and replace defective work or work not conforming to Drawings and Specifications.
  - 2. Take samples of installed work as required for testing.
  - 3. Remove constructions required to provide for specified alterations or additions to existing work.
  - 4. Uncover work to allow inspection or re-inspection by Project Manager or regulatory agencies having jurisdiction.
  - 5. Connect uninstalled work to completed work in proper sequence.
  - 6. Remove or relocate existing utilities and pipes that obstruct work.
  - 7. Make connections or alterations to existing or new facilities.
  - 8. Provide openings, channels, chases and flues and cut, patch and finish
  - 9. Provide protection for other portions of the work
- C. Restore existing work to a condition equal to or better than that which existed prior to cutting and patching, and to standards required by contract.
- D. Support, anchor, attach, match, trim, and seal materials to work of others unless otherwise specified. Furnish and install sleeves, inserts and hangers required for execution of the work.
- E. Provide shoring, bracing and support necessary to maintain structural integrity and to protect adjacent work from damage during cutting and patching. Request written approval from Project Manager before cutting structural members such as beams, anchors, or other supports.
- F. Match new materials to existing materials by bonding, lapping, mechanically tying, anchoring or other effective means in order to prevent cracks and to minimize evidence of patching. Conceal effects of demolition and patching, and avoid breaks, joints or changes of surface appearance unless shown on drawings or authorized by Project Manager.

##### **1.03 SUBMITTALS**

- A. Submit written request to Project Manager for consent to proceed before conducting operations that may affect design function, regular operations, or other work.
- B. Notify Project Manager in writing when work will be uncovered for observation. Do not begin patching or cutting operations until notification is sent to Project Manager.

#### **1.04 CONNECTIONS TO EXISTING FACILITIES**

- A. Perform construction operations necessary to complete connections and tie-ins to existing facilities. Keep existing facilities in continuous operation unless otherwise permitted or approved in writing by Project Manager.
- B. Coordinate interruption of services requiring connection to existing facilities with Project Manager. Provide temporary pumping facilities to handle wastewater if necessary.
- C. Submit 48 hours in advance a detailed schedule of proposed connections that will include any shutdowns or bypasses. Include time and date as well as anticipated duration of work.

**END OF SECTION 01731**

## **SECTION 02100**

### **SITE PREPARATION**

#### **Part 1 GENERAL**

##### **1.01 SECTION INCLUDES:**

- A. Cleaning site of debris, grass, trees and other plant life in preparation for site excavation work.
- B. Protection of existing structures, trees or vegetation indicated on the contract documents to remain.
- C. Stripping topsoil from areas that are to be incorporated into the limits of the project and where so indicated on the construction drawings.

##### **1.02 RELATED SECTIONS**

- A. Section 01570 – Erosion and Sediment Control
- B. Section 02220 - Demolition
- C. Section 02221 - Demolition of Structures
- D. Section 02225 – Site work Demolition and Removal
- E. Section 02231 – Clearing and Grubbing
- F. Section 02300 – Earthwork
- G. Construction Drawings

##### **1.03 ENVIRONMENTAL REQUIREMENTS**

- A. Construct temporary erosion control systems as shown on the plans or as directed by the engineer to protect adjacent properties and water resources from erosion and sedimentation.
- B. In the event that site work on this project will disturb five (5) or more acres, the contractor shall NOT begin construction without a “National Pollution Discharges Elimination System” (NPDES) permit governing the discharge of storm water from the construction site for the entire construction period. The permit requires a “Storm Water Pollution Prevention Plan” (SWP) to be in place during construction which includes monitoring of storm water flows during construction.
- C. The contractor shall be responsible for conducting the storm water management practices in accordance with the NPDES permit and for any enforcement action taken or imposed by Federal or State agencies, including the cost of fines, construction delays and remedial actions resulting from the contractor's failure to comply with all provisions of the NPDES permit.

##### **1.04 JOB CONDITIONS**

- A. Conditions existing at times of inspection for bidding purposes will be maintained by owner in so far as practical.
- B. Variations to conditions or discrepancy in actual conditions as they apply to site preparation operations are to be brought to the attention of the owner prior to the commencement of any site work.

## **PART 2 PRODUCTS – Not Used**

## **PART 3 EXECUTION**

### **3.01 PREPARATION**

- A. Verify that existing plant life and clearing limits are clearly tagged, identified and marked in such manner to insure their safety throughout construction operations.

### **3.02 PROTECTION**

- A. Locate and identify existing utilities that are to remain and protect them from damage.
- B. Protect trees, plant growth and features designated to remain as final landscape.
- C. Conduct operations with minimum interference to public or private accesses and facilities. Maintain access and egress at all times and clean or sweep any roadways daily or as required by the governing authority or at such times as deemed necessary by the Project Manager. Dust control shall be provided with sprinkling systems or equipment provided by the contractor.
- D. Protect bench marks, property corners and all other survey monuments from damage or displacement. If a marker needs to be removed, it shall be referenced by a licensed land surveyor and replaced, as necessary, the same at no change in contract price.
- E. Provide traffic control as required by the contract specifications, and in accordance with the U.S. Department of Transportation "Manual of Uniform Traffic Control Devices" and the state highway department requirements.

### **3.03 CLEARING**

- A. Clear areas required for access to site and execution of work following local governing authority requirements. Contractor shall be responsible for complying with any applicable ordinances and requirements.
- B. Unless otherwise indicated on the drawings, remove trees, shrubs, grass, other vegetation, improvements, or obstructions interfering with installation of new construction. Removal includes digging out stumps and roots. Depressions caused by clearing and grubbing operations are to be filled to subgrade elevation to avoid water ponding. Satisfactory fill material shall be placed in horizontal layers not exceeding 8" loose depth, and thoroughly compacted per fill requirements of this section and other applicable sections.

- C. Remove any grass, trees, plant life, stumps or other construction debris from the site to a dump site that is suitable for handling such material according to state laws and regulations.

#### **3.04 TOPSOIL EXCAVATION**

- A. Strip topsoil from areas that are to be filled, excavated, landscaped or re-graded to such a depth that it prevents intermingling with underlying subsoil or questionable material.
- B. Cut heavy growths of grass from areas before stripping and remove with the rest of the cleared vegetative material.
- C. Topsoil shall consist of organic surficial soil found in depth of not less than 6". Satisfactory topsoil is reasonably free of subsoil, clay lumps, stones and other objects over 2" in diameter, weeds, roots, and other objectionable material.
- D. Stockpile topsoil in storage piles in areas shown or where directed. Construct storage piles to freely drain surface water. Cover storage piles as required to prevent windblown dust. Dispose of unsuitable topsoil as specified for waste material, unless otherwise specified by Project Manager. Excess topsoil shall be removed from the site by the Contractor unless specifically noted otherwise on the Drawings.

**END OF SECTION 02100**

## **SECTION 02231**

### **CLEARING AND GRUBBING**

#### **PART 1 GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Removing surface debris and rubbish
- B. Clearing the site of plant life and other vegetation
- C. Removing trees and shrubs
- D. Fence removal

##### **1.02 RELATED SECTIONS**

- A. Section 02100 – Site preparation
- B. Section 02220 – Demolition
- C. Section 02225 – Site work Removal and Demolition
- D. Construction Drawings
- E. Geotechnical Investigation Report

##### **1.03 GENERAL REQUIREMENTS**

- A. Conduct site clearing operations to insure minimum interference with roads, streets, walks and other adjacent occupied or used facilities. Do not close or obstruct streets, walks or other occupied or used facilities without permission from the district.
- B. Protection of existing improvements:
  - a. Provide protection as necessary to prevent damage to existing improvements indicated to remain in place.
  - b. Protect improvements on adjoining properties and owner's property.
  - c. Restripe, repair, replace and/or correct damaged improvements/structures to their original condition as acceptable to the Project Manager.

#### **Part 2. Materials – Not Used**

#### **Part 3 – Execution**

##### **3.01 PREPARATION AND CLEARING**

- A. Remove trees, shrubs, grass and other vegetation, improvements or obstructions interfering with installation of new construction. Remove root systems under pavements to a minimum depth of 24 inches below finished subgrade elevation in pavement areas to a minimum area of 2 feet away from back of curb or edge of pavement. Remove root systems within structure area a minimum of 24 inches below the elevation and a minimum of 2 feet away from surfaces and other areas.
- B. Carefully and cleanly cut roots and branches of trees indicated to be left standing where such roots and branches obstruct new construction. Ensure that work and contact with and around tree will not cause tree to perish.
- C. Trees that are indicated to be removed and transplanted shall be balled and replanted on site in such a way that the tree will survive. Otherwise it will be replaced with an equal tree by Contractor at Contractor's expense.
- D. All holes remaining, except in areas to be immediately excavated, after clearing and grubbing shall be backfilled and compacted to the density equal to that of the adjacent ground. The entire area shall be bladed to prevent ponding of water.
- E. Remove debris, rubbish and extracted plant material from the site in accordance with requirements for waste disposal.
- F. Topsoil is defined as friable clay loam surface soil found in a depth of not less than 4 inches. Satisfactory topsoil is reasonably free of subsoil, clay lumps, stones and other objects over 2 inches in diameter and without weeds, roots and other objectionable material.
  - a. Strip topsoil to whatever depths encountered in a manner to prevent intermingling with underlying layer.
  - b. Remove heavy growths of grass from areas before stripping.
  - c. Stockpile topsoil in storage piles in areas directed. Construct storage pile to freely drain surface water. Cover storage piles, if required to prevent wind-blow dust. Do not stockpile more than five (5) feet in height without expressed authority of Project Manager.
  - d. Disposal of unsuitable topsoil shall be considered to be the same as for waste materials.
- G. Remove existing above-grade and below-grade improvements necessary to permit construction, and other work as required. Any items interfering with construction not indicated to be removed shall be clarified with Project Manager.
- H. Removal of concrete and asphalt and any part of existing sidewalks or paving parking area shown on the plans to be removed shall be done so in leaving a neat and straight cut. Material shall be disposed of as per for waste material.

**END OF SECTION 02231**



## **SECTION 02260**

### **TRENCH SAFETY SYSTEMS**

#### **PART 1 GENERAL**

##### **A.01 SECTION INCLUDES**

- A. Trench Safety Systems for trench excavations

##### **A.02 ADDITIONAL REFERENCES**

- A. Construction Specification Manual
- B. Construction Drawings
- C. Geotechnical Investigation Report

##### **1.03 GENERAL INFORMATION**

- A. A trench shall be defined as a narrow excavation made below the surface of the ground. In general, depth is greater than width, but the width of a trench measured at the bottom is not greater than 15 feet.
- B. Trench Safety System requirements will apply to larger open excavations if the erection of structures or other installations limit the space between the excavation slope and the installation to dimensions equivalent of a trench as defined.
- C. Trench Safety Systems include but are not limited to sloping, sheeting, trench boxes, trench shields, sheet piling, cribbing, bracing, shoring, dewatering, and diversion of water for drainage.
- D. Trench Safety program shall be designed specifically for the construction of the trench excavation and be in accordance with OSHA standards governing the presence of persons and activities in and around trench excavations.
- E. Special designs shall be sealed by a licensed Engineer retained and paid by Contractor. Review by the Project Engineer will not constitute approval by the Project Engineer, nor relieve the Contractor of obligations under State and Federal safety laws.
- F. Contractor is solely responsible for complying with all applicable local, state and/or federal rules and guidelines, for designing, inspecting, implementing and constructing/providing adequate trench safety systems.

#### **Part 2. Materials – Not Used**

#### **Part 3 – Execution**

##### **3.01 INSTALLATION**

- A. Install specially designed trench safety systems in accordance with the Contractor's trench excavation safety program for the field locations and conditions identified in the project.

- B. Install and maintain trench safety systems in accordance with OSHA provisions.

### **3.02 INSPECTION**

- A. Contractor is responsible for keeping all records of inspections, making all inspections of, and for making continual inspections throughout the use of the trench safety systems, and ensuring installed systems and operations meet OSHA and other safety requirements.

### **3.03 QUALITY CONTROL**

- A. Contractor shall verify specific applicability of the selected or specially designed trench safety systems to each field condition encountered on the project.

**END OF SECTION 02260**

## **SECTION 02300**

### **EARTHWORK**

#### **PART 1 GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Protection, modification and/or installation of utilities as site work progresses paying particular attention to grade changes and any necessary staging of work.
- B. Cutting, filling and grading to required lines, dimensions, contours and proposed elevations for proposed improvements.
- C. Scarifying, compaction, drying and removal of unsuitable material to ensure proper preparation of areas for fills or proposed improvements.

##### **1.02 RELATED SECTIONS**

- A. Section 02100 – Site Preparation
- B. Section 02220 - Demolition
- C. Section 02315 - Excavation, Backfill and Compaction for Utilities
- D. Section 02316 - Excavation, Backfill and Compaction for Structures
- E. Section 02317 - Excavation Backfill and Compaction for Pavement
- F. Geotechnical Report for findings of surface materials and conditions, and recommendations
- G. Construction Drawings
- H. All applicable Project Specifications as they relate specifically to the earthwork beneath the buildings, where the requirements are more stringent than the civil requirements.

##### **1.03 REFERENCE STANDARDS**

- A. American Society for Testing and Materials (ASTM) latest edition.
  - D 422 Method for Particle Size Analysis of Soils
  - D 698 Test for Moisture-Density Relations of Soils Using 5.5 lb. (2.5 kg) Rammer and 12 inch-inch (304.8 mm) Drop (Standard Proctor)
  - D 1556 Test for Density of soil in Place by the Sand Cone Method
  - D 1557 Test for Moisture-Density Relations of Soils Using 10-lb (4.5 kg) Rammer and 18-inch (457 mm) Drop (Modified Proctor)
  - D 1559 Test Method for Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus
  - D 2167 Test for Density of Soil in Place by the Rubber Balloon Method
  - D 2216 Laboratory Determination of Moisture content of Soil
  - D 2487 Classification of Soils for Engineering Purposes
  - D 2922 Tests for Density of Soil and Soil - Aggregate in Place by Nuclear Methods (Shallow Depth)
  - D 3017 Test for Moisture Content of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)
  - D 4318 Test for Plastic Limit, Liquid Limit, and Plasticity Index of Soils

- C 25 Chemical Analysis of Limestone, Quicklime and Hydrate Lime
- C 110 Physical Testing for Quicklime and Hydrated Lime, Wet Sieve Method
- C 618 Specifications for Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Portland Cement Concrete
- C 977 Quicklime and Hydrated Lime for Soil Stabilization

- B. American Association of State Highway and Transportation Officials (AASHTO) latest edition  
T 88 Mechanical Analysis of Soils

#### 1.04 QUALITY ASSURANCE

- A. Independent Testing Laboratory paid by Contractor, unless otherwise stated in the contract, shall be retained to perform construction testing on site based on the following:
  - 1. Building Subgrade Areas, including 10'-0" Outside Exterior Building Lines: In cut areas, not less than one compaction test for every 2,500 square feet. In fill areas, same rate of testing for each 8" lift (measured loose)
  - 2. Areas of Construction exclusive of building subgrade: In cut areas, not less than one compaction test for every 10,000 square feet. In fill areas, same rate of testing for each 8" lift (measured loosed).
- B. If compaction requirements are not complied with at any time during construction process, remove and recompact deficient areas until proper compaction is obtained at no additional expense to owner. All retesting due to non-compliance shall be paid for by the contractor at no additional cost to the owner.
- C. In all areas to receive pavement, a CBR (or LBR) test shall be performed for each type of material imported from off-site.
- D. The following tests shall be performed on each type of on-site or imported soil material used as compacted fill as part of construction testing requirements.
  - 1. Moisture and Density Relationship: ASTM D 698 or ASTM D1557.
  - 2. Mechanical Analysis: AASHTO T-88
  - 3. Plasticity Index: ASTM D 4318
- E. Field density tests for in-place materials shall be performed according to one of the following standards as part of construction testing requirements.
  - 1. Sand-Cone Method: ASTM D 1556
  - 2. Ballon Method: ASTM D 2167
  - 3. Nuclear Method: ASTM D 2922 (Method B-Direct Transmission)
- F. Independent Testing Laboratory shall prepare test reports that indicate test location, elevation data, and test results. Owner, Engineer, and contractor shall be provided with copies of reports within 96 hours of time test was performed. In event that any test performed fails to meet these specifications, owner and contractor shall be notified immediately by the independent testing laboratory.
- G. All costs related to retesting due to failures shall be paid for by the contractor at no additional expense to owner. Owner reserves the right to employ an Independent

Testing Laboratory and to direct any retesting that is deemed necessary. Contractor shall provide free access to site for testing activities.

#### **1.05 SUBMITTALS**

- A. Submit a sample of each type of off-site fill materials that is to be used at the site in an air tight, 10 lb container for the testing laboratory.
- B. Submit the name of each material supplier and specific type and source of each material. Any change in source throughout the job requires approval of the engineer.
- C. For use of fabrics or geogrids, a design shall be submitted for approval by the Engineer.

### **PART 2 PRODUCTS**

#### **2.01 MATERIALS**

- 1. Excavated and re-used material for subsoil fill as specified herein and deemed suitable for intended use by Geotechnical Engineer.
- 2. Imported subsoil material approved by the Engineer and specified herein and deemed suitable for intended use by Geotechnical Engineer.
- 3. Filter/Drainage Fabrics
  - a) Mirafi 140NS
  - b) Phillips 66 Supac 4NP
  - c) Dupont Tyvar 3341
  - d) Others as approved by Engineer

### **PART 3 EXECUTION**

#### **3.01 PREPARATION**

- A. Identify required lines, levels, contours and datum. Locate and identify existing utilities that are to remain and protect them from damage. Coordinate with Section 01725 Field Surveying.
- B. Notify utility companies to remove and/or relocate any utilities that are in conflict with the proposed improvements.
- C. Protect plant life, lawns, fences, existing structures, sidewalks, paving and curbs from excavating equipment and vehicular traffic.
- D. Protect benchmarks, property corners and all other survey monuments from damage or displacement. If a marker needs to be removed it shall be referenced by a licensed land surveyor and replaced, as necessary, by the same.
- E. Remove from site material encountered in grading operations that, in opinion of Engineer, Owner or owners representative, is unsuitable or undesirable for backfilling, subgrade or foundation purposes. Dispose of in a manner satisfactory to owner. Backfill areas with layers of suitable material and compact as specified.

- F. Prior to placing fill in low areas, such as previously existing creeks, ponds, or lakes, perform following procedures:
  - 1. Drain water out by gravity with ditch having flow line lower than lowest elevation in low area. If drainage cannot be performed by gravity ditch, use adequate pump to obtain same results.
  - 2. After drainage of low area is complete, remove mulch, mud, debris, and other unsuitable material by using acceptable equipment and methods that will keep natural soils underlying low areas dry and undisturbed.
  - 3. If proposed for fill, all muck, mud, and other materials removed from above low areas shall be dried on-site by spreading in thin layers for observation by owner or owners representative. Material shall be inspected and, if found to be suitable for use as fill material, shall be incorporated into lowest elevation of site filling operation, but not under the building area or within 10'-0" of perimeter of building pad or paving subgrade. If, after observation by owner or owners representative, material is found to be unsuitable, all unsuitable material shall be removed from site.

### **3.02 EXCAVATING FOR FILLING AND GRADING**

- A. Classification of Excavation: Contractor by submitting bid/price proposal acknowledges that he has investigated the site to determine type, quantity, quality, and character of excavation work to be performed. Excavation shall be considered unclassified excavation.
- B. Perform excavation using capable, well maintained equipment and methods acceptable to governing agencies.
- C. When performing grading operations during periods of wet weather, provide adequate drainage and ground water management to control moisture of soils.
- D. Shore, brace, and drain excavations as necessary to maintain safe, secure, and free of water at all times.
- E. Excavated material containing rock or stone greater than 6" in largest dimension is unacceptable as fill to within the proposed building and paving area.
- F. Rock or stone less than 6" in largest dimension is acceptable as fill to within 24" of surface of proposed subgrade when mixed with suitable material.
- G. Rock or stone less than 2" in largest dimension and mixed with suitable material is acceptable as fill within the upper 24" of proposed subgrade.

### **3.03 FILLING AND SUBGRADE PREPARATION**

- A. Fill areas to contours and elevations shown with unfrozen materials.
- B. Place fill in continuous lifts specified herein.
- C. Refer to Section 02316 for requirements for structures.
- D. Refer to Section 02317 for requirements for pavements.

- E. Areas exposed by excavation or stripping and on which subgrade preparations are to be performed shall be scarified to minimum depth of 8" and compacted to minimum of 95% of optimum density, in accordance with ASTM D698 (or 92% of optimum density, in accordance with ASTM D1557), at a moisture content of not less than 1% below and not more than 3% above the optimum moisture content. These areas shall then be proof-rolled to detect any areas of insufficient compaction. Proof-rolling shall be accomplished by making a minimum of two (2) complete passes with a fully-loaded tandem-axle dump truck, or approved equivalent, in each of the two perpendicular directions under the supervision and direction of a field geotechnical engineer. Areas of failure shall be excavated and recompacted as stated above.
- F. Fill materials used in preparation of subgrade shall be placed in lifts or layers not to exceed 8" loose measure and compacted to a minimum density of 95% of optimum density, in accordance with ASTM D 698, (or 92% of the optimum density, in accordance with ASTM D 1557) at a moisture content of not less than 1% below and not more than 3% above the optimum moisture content.
- G. Material imported from off-site shall have a CBR (California Bearing Ratio) or LBR (Limerock Bearing Ratio) value equal to or above the pavement design subgrade CBR or LBR value indicated on the Drawings.

### **3.04 MAINTENANCE OF SUBGRADE**

- A. Finished subgrades shall be verified to ensure proper elevation and conditions for construction above subgrade.
- B. Protect subgrade from excessive wheel loading during construction, including concrete trucks and dump trucks.
- D. Remove areas of finished subgrade found to have insufficient compaction density to depth necessary and replace in manner that will comply with compaction requirements by use of material equal to or better than best subgrade material on site. Surface of subgrade after compaction shall be hard, uniform, smooth, stable, and true and cross-section.

### **3.05 RIP RAP (If Specified on Construction Drawings)**

- A. Place rip-rap in all areas where indicated on the Drawings. The stone for rip-rap shall consist of field stone or rough unhewn quarry stone as nearly uniform, in section as is practical. The stones shall be dense, resistant to the action of air and water, and suitable in all aspects for the purpose intended. Unless otherwise specified, all stones used as rip-rap shall weigh between 50 and 150 pounds each, and at least 60 percent of the stones shall weigh more than 100 pounds each.
- B. Slopes and other areas to be protected shall be dressed to the line and grade shown on the plans prior to the placing of rip-rap. Contractor shall undercut the areas to receive rip-rap to an elevation equal to the final elevation less the average diameter of the stones before placing the rip-rap.
- C. Filter fabric and bedding stone shall be installed prior to the placement of the stone if so indicated on the drawings. The bedding stone shall be quarried and crushed angular limestone in accordance with Section 02704 and shall be 6" in depth. Filter fabric shall be as specified herein and as detailed on the plans.
- D. Stones shall be placed so that the greater portion of their weight is carried by the earth and not by the adjacent stones. The stones shall be placed in a single layer with close joints. The upright areas of the stone shall make an angle of approximately 90 degree with the

embankment upward, the larger stones being placed in the lower courses. Open joints shall be filled with spalls. Stones shall be embedded in the embankment as necessary to present a uniform top surface such that the variation between tops of adjacent stones shall not exceed three inches.

### **3.06 FINISH GRADING**

- A. Grade all areas where finish grade elevations or contours are indicated on Drawings, other than paved areas and buildings, including excavated areas, filled and transition areas, and landscape areas. Graded areas shall be uniform and smooth, free from rock, debris, or irregular surface changes. Finished subgrade surface shall not be more than 0.10 feet above or below established finished subgrade elevation, and all ground surfaces shall vary uniformly between indicated elevations. Finished ditches shall be graded to allow for proper drainage without ponding and in a manner that will minimize erosion potential. For topsoil application, refer to Landscaping Sections and Specifications.
- B. Correct all settlement and eroded areas within one year after date of completion at no additional expense to owner. Bring grades to proper elevation. Replant or replace any grass, shrubs, bushes, or other vegetation that appears dead, dying or disturbed by construction activities. Refer to Section 01570 for slope protection and erosion control.
- C. Refer to Section 02341 for soil stabilization using lime, cement, fly ash and geotextile fabric methods for subbase materials.

**END OF SECTION 02300**



## **SECTION 02315**

### **EXCAVATION AND BACKFILLING FOR UTILITIES**

#### **PART 1 GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Excavation, trenching, foundation, embedment, and backfill for installation of utilities, including manholes and other pipeline structures.
- B. Borings and casings under roads

##### **1.02 RELATED SECTIONS**

- A. Section 02300 Earthwork
- B. Section 02510 Water Distribution Systems
- C. Section 02531 Sanitary Sewer Systems
- D. Section 02630 Storm Drainage Systems
- E. Geotechnical report for findings of subsurface materials and conditions, and recommendations
- F. Construction Drawings
- G. Manufacturer Installation Requirements

##### **1.03 REFERENCE STANDARDS**

- A.
  - American Society for Testing and Materials (ASTM) Latest Edition
  - D 422 Method for Particle Size Analysis
  - D 698 Test for Moisture-Density Relations of Soils Using 5.5 lb (2.5 kg) Rammer and 12 inch (304.8mm) Drop (Standard Proctor)
  - D 1556 Test for Density of Soil in Place by the Sand Cone Method
  - D 1557 Test for Moisture-Density Relations of Soils Using 10-lb (4.5 kg) Rammer and 18 inch (457 mm) Drop (Modified Proctor)
  - D 1559 Test Method for Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus
  - D 2216 Laboratory Determination of Moisture Content of Soil
  - D 2487 Classification of Soils for Engineering Purposes
  - D 2922 Tests for Density of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow depth)
  - D 3017 Test for Moisture Content of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)
  - D 4318 Test for Plastic Limit, Liquid Limit, & Plasticity Index of Soils
  - C 25 Chemical Analysis of Limestone, Quicklime and Hydrated Lime.
  - C 110 Physical Testing for Quicklime and Hydrated Lime, Wet Sieve Method
  - C 618 Specification for Fly Ash and Raw or Calcined Natural Pozzolan for Use as Mineral Admixture in Portland Cement Concrete
  - C 977 Quicklime and Hydrated Lime for Soil Stabilization
- B. American Association of State Highway and Transportation Officials (AASHTO) latest edition

T 88 Mechanical Analysis of Soils

**1.04 QUALITY ASSURANCE**

- A. Independent testing laboratory selected and paid by owner, or as specified within project contract, shall be retained to perform construction testing on backfilling operations as specified in Section 02300 and stated herein.
- B. Trenches shall not be backfilled until required tests are performed and the utility systems comply with and are accepted by applicable governing authorities.

**1.05 SUBMITTALS**

- A. Submit shop drawings and details pertaining to site utilities as required by regulatory authorities and indicate where any proposed use of materials, methods, equipment, or procedures are contrary to the Construction Drawings or Specifications. Do not perform work until required drawings and/or details have been accepted by Engineer.
- B. The Contractor shall contact all utility companies and determine if additional permits, including easements will be required to complete the project. Contractor shall provide written confirmation of the status of all permits and easements to the Owner's Construction Manager at the time of the preconstruction conference or no later than 90 days prior to the project possession date.
- C. Submit a sample of each type of offsite fill material that is to be used in backfilling in an airtight, 10 lb container for the testing laboratory or submit a gradation and certification of the aggregate material that is to be used to the testing laboratory for review.

**1.06 PROJECT RECORD DOCUMENTS**

- A. Accurately record actual locations of all subsurface utilities, structures and obstructions encountered. As-built records are to be maintained and updated daily as work progresses. Records shall contain horizontal dimensions and elevations referenced to survey control points.

**PART 2 PRODUCTS**

**2.01 MATERIALS**

- A. Bedding Material: Processed sand and gravel free from clay lumps, organic, or other deleterious material, and complying with following gradation requirements:

| <u>B. U.S. Sieve Size</u> | <u>Percent Passing (by weight)</u> |
|---------------------------|------------------------------------|
| C. 1 Inch                 | 100                                |
| D. ¾ Inch                 | 90-100                             |
| E. 3/8 Inch               | 20-55                              |
| F. No. 4                  | 0-10                               |
| G. No. 8                  | 0-5                                |

- B. Backfill material from site as specified in Section 02300 and approved by the Geotechnical Engineer or owner's representative.
- C. Backfill material from offsite as specified in Section 02300 and approved by the Geotechnical Engineer or owner's representative.

- D. Steel Casing Pipe: Comply with AWWA C-201 or C202, minimum grade B, size and wall thickness as indicated on Drawings.
- E. Materials as recommended and specified by utility line manufacturer for proper stability, strength, life, and performance of utility lines.
- F. Acceptable Stabilization Fabrics and Geogrids
  - 1. Mirafi 500X or 600X
  - 2. Phillips 66 Supac 6Ws
  - 3. Dupont Typar 3401 and 3601
  - 4. Trevira S1114 and S1120
  - 5. Tensar SS-1 and SS-2
  - 6. Exxon GTF-200 pr 350
- G. Filter/Drainage Fabrics
  - 1. Mirafi 140 NS
  - 2. Phillips 66 Supac 4Np
  - 3. Dupont Typar 3341

## **PART 3EXECUTION**

### **3.01 PREPARATION**

- A. Set all lines, elevations, and grades for utility and drainage system work and control system for duration of work, including careful maintenance of bench marks, property corners, monuments, or other reference points. Ensure compliance with Sections 01550 Traffic Control and 01725 Field Surveying.
- B. Maintain in operating condition all existing utilities, active utilities and drainage systems encountered in utility installation. Repair any surface or subsurface improvements shown on Drawings to meet specification requirements.
- C. Verify location, size, elevation, and other pertinent data required to make connections to existing utilities and drainage systems as indicated on Drawings. Contractor shall comply with local codes and regulations.
- D. Over excavate and properly prepare areas of subgrade that are not capable of supporting the proposed systems. These areas shall be stabilized by using acceptable filter fabrics and/or additional bedding material placed and compacted as specified.
- E. Install dewatering systems that will be required to construct the proposed utilities in a manner that meets project specifications and governing regulations.

### **3.02 EXCAVATION**

- A. Notify involved utility companies of the date and time excavation will occur and request their respective lines be marked in the field before excavation shall begin. Horizontal and vertical locations of various underground lines shown on drawings, including but not limited to water lines, gas lines, storm sewers, sanitary sewers, telecommunication lines, electric lines, pipelines, concrete and debris, are based on the best available information but are only approximate locations.
- B. Dig trench at proper width and depth for laying pipe, conduit, or cable. Cut trench banks vertical if possible and remove stones from bottom of trench as necessary to avoid point-bearing. Over excavate wet or unstable soil, if encountered, from trench bottom as necessary to provide suitable base for continuous and uniform bedding.
- C. All trench excavation side walls greater than 5 feet in depth shall be sloped, shored, sheeted, braced or otherwise supported by means of sufficient strength to protect the workmen within them in accordance with the applicable rules and regulations established for construction by the Department of Labor, Occupational Safety and Health Administration (OSHA), and by local ordinances. Lateral travel distance to an exit ladder or steps shall not be greater than 25 feet in trenches 4 feet or deeper. Conform to Section 02260 Trench Safety Systems.
- D. Perform excavation as indicated for specified depths. During excavation, stockpile materials suitable for backfilling in orderly manner far enough from bank of trench to avoid overloading, slides, or cave-ins.
- E. Remove excavated materials not required or not suitable for backfill or embankments and waste as specified in project specifications. Any structures discovered during excavation(s) shall be disposed of as specified in project specifications.
- F. Prevent surface water from flowing into trenches or other excavations by temporary grading or other methods, as required. Remove accumulated water in trenches or other excavations by pumping or other acceptable methods.
- G. Open cut excavation with trenching machine or backhoe. Where machines other than ladder or where-type trenching machines are used, do not use clods for backfill. Dispose of unsuitable material and provide other suitable material at no additional cost to Owner.
- H. Accurately grade trench bottom to provide uniform bearing and support for each section of pipe on bedding material at every point along entire length, except where necessary to excavate for bell holes, proper sealing of pipe joints, or other required connections. Dig bell holes and depressions for joints after trench bottom has been graded. Dig no deeper, longer, or wider than needed to make joint connection properly.
- I. Trench width requirements below the top of the pipe shall not be less than 12" or more than 18" wider than outside surface of any pipe or conduit that is to be installed to designated elevations and grades. All other trench width requirements for pipe, conduit, or cable shall be the least practical width that will allow for proper compaction of trench backfill.
- J. Trench depth requirements measured from finished grade or paved surface shall meet the following requirements or applicable codes and ordinances:
  - i. Water Mains: 30" to top of pipe barrel or 6" below the frost line (established by the local building official), whichever is deeper.
  - ii. Sanitary Sewer: Elevations, and grades as indicated on Drawings.
  - iii. Storm Sewer: Depths, elevations, and grades as shown on Drawings.
  - iv. Electrical Conduits: 24" minimum to top of conduit or as required by NEC 300-5, NEC 710-36, or the local utility company requirements, whichever is deeper.

- v. TV Conduits: 18" minimum to top of conduit or as required by the local utility company, whichever is deeper.
  - vi. Telephone Conduits: 18" minimum to top of conduit, or as required by the local company, whichever is deeper.
  - vii. Gas Mains and Service: 30" minimum to top of pipe, or as required by the local utility company, whichever is deeper.
- K. Provide sheeting and bracing, when necessary, in trenches and other excavations where protection of workmen required. Sheeting may be removed after sufficient backfilling to protect against damaging or injurious caving. Comply with 02260 Trench Safety Systems.

### **3.03 PIPE BEDDING**

- A. Accurately cut trenches for pipe or conduit that is to be installed to designated elevations 4" below bottom of pipe and to width as specified. Place 4" of bedding material, compact in bottom of trench, and accurately shape to conform to lower portion of pipe barrel.
- B. Place fabric if specified or directed by Geotechnical Engineer, Geotechnical Report or as shown on the plans and specifications.

### **3.04 PIPE EMBEDMENT, PLACEMENT, AND COMPACTION**

- A. Remove loose, sloughing, caving, or otherwise unsuitable soil from bottoms and sidewalls of trenches immediately prior to placement of embedment materials.
- B. Place embedment including bedding, haunching, and initial backfill as shown on the Construction Drawings.
- C. For pipe installation, manually spread embedment materials around pipe to provide uniform bearing and side support when compacted. Protect flexible pipe from damage during placing of pipe zone bedding material. Perform placement and compaction directly against undisturbed soils in trench sidewalls, or against sheeting which is to remain.
- D. Do not place trench shields or shoring within height of embedment zone unless means to maintain density of compacted material are used. If moveable supports are used in embedment zone, lift supports incrementally to allow placement and compaction of material against undisturbed soil.
- E. Place geotextile as specified to prevent particle migration from in-situ soil into open graded (Class I) embedment materials or drainage layers.
- F. Do not damage coatings or wrappings of pipes during backfilling and compacting operations. When embedding coated or wrapped pipes, do not use crushed stone or other sharp, angular aggregates.
- G. Place haunching material manually around pipe and compact it to provide uniform bearing and side support. If necessary hold small diameter or lightweight pipe in place during compaction of haunch areas and placement beside pipe with sand bags or other suitable means.
- H. For waterline construction embedment use specified sand, gravel or crushed limestone, and adhere to the following:

- a. Class I, II, and III embedment materials: Maximum 6 inch compacted lift thickness. Compact to achieve a minimum of 95 percent of maximum dry density as determined according to ASTM D 698. Moisture content to be within -3 percent to +5 percent of optimum as determined according to ASTM D 698, unless otherwise approved by Engineer.
  - b. Cement Stabilized Sand, where required for special installations: Maximum 6 inches compacted lift thickness. Compact to achieve a minimum of 95 percent of maximum dry density as determined according to ASTM D 698. Moisture content to be on dry side of optimum as determined according to ASTM D 698, but sufficient for hydration.
- I. For sanitary and storm sewers adhere to the following subparagraphs a and b. This provision does not apply to HDPE pipe installed under pavement.
- a. Class I Embedment Materials: Maximum 6 inches compacted lift thickness. Systematic compaction by at least two passes of vibrating equipment. Increase compaction effort as necessary to effectively embed pipe to meet test criteria. Moisture content to be as determined by Contractor for effective compaction without softening soil of trench bottom, foundation, or trench walls.
  - b. Class II Embedment and Cement Stabilized Sand: Maximum 6 inch compacted lift thickness. Compact by methods determined by contractor to achieve a minimum of 95 percent of maximum dry density as determined according to ASTM D 698 and according to ASTM 558 for cement stabilized materials. Moisture content to be within 3 percent of optimum as determined according to ASTM D 698. Moisture content of cement stabilized sands on dry side of optimum content as determined according to ASTM 558 but sufficient for hydration.
- J. For storm sewers constructed of HDPE pipe and installed under pavement provide flowable fill pipe embedment.
- K. Place trench dams in Class I embedment in line segments longer than 100 feet between manholes, and not less than one in every 500 feet of pipe placed. Install additional dams as needed to achieve workable conditions. Do not place dams closer than 5 feet from manholes.
- L. After pipe installation, place backfill as specified below and compact in maximum 8" layers measured loose to the top of the trench. An independent testing laboratory shall perform testing at intervals not exceeding 200' of trench for the first and every other lift of 8" compacted trench backfill and furnish copies of test results as specified.

### **3.05 TRENCH BACKFILL PLACEMENT**

- A. After pipe or conduit has been installed, bedded, and tested as specified, backfill trench or structure excavation to the contours and elevations shown on the plans with approved and unfrozen materials. Systematically backfill to allow maximum time for natural settlement. Do not backfill over porous, wet, frozen or spongy subgrade surfaces.
- B. Exercise proper caution when compacting immediately over top of pipes or conduits. **WATER JETTING OR FLOODING IS NOT PERMITTED AS A METHOD OF COMPACTION.**
- C. Place backfill for pipe or conduits and restore surface as soon as practicable. Leave only minimum length of trench open as necessary for construction.
- D. For water lines under pavement and to within one foot of back of curb, use backfill materials as described below:

- a. For lines 20 inches in diameter and smaller, use bank run sand or select backfill materials up to pavement base or subgrade.
  - b. For lines 24 inches in diameter and larger, backfill with suitable on-site material (random backfill) up to 12 inches below pavement base or subgrade. Place a minimum of 12 inches of select backfill below pavement base or subgrade.
- E. For sewer pipes (storm and sanitary) use backfill materials described by trench limits. For trench zone backfill under pavement and to within one foot of back of curb, use cement stabilized sand for pipes of nominal sizes 36 inches in diameter and smaller to a level 12 inches below the pavement. For sewer pipes 42 inches in diameter and larger, and under pavement or natural ground in satisfactory soil conditions, backfill from 12 inches above top of pipe to 12 inches below pavement with suitable on-site material or select backfill. Use select backfill for rigid pavements or flexible base material for asphalt pavements for 12 inch backfill directly under pavement. For backfill material reference specification sections on materials and geotechnical report. This provision does not apply where storm sewer is constructed of HDPE pipe.
- F. For storm sewers constructed of HDPE pipe and installed under pavement provide flowable fill as specified in flowable fill specification sections or as directed by engineer. For installations not under pavement provide cement stabilized sand.
- G. Unless otherwise specified in the drawings, use one of the following trench zone backfills under pavement and within one foot of edge of pavement. Place trench zone backfill in lifts and compact. Fully compact each lift before placement of the next lift.
  - a. Class I, II, III, or combination thereof: Place in maximum 8 inch compacted lifts. Compact to a minimum of 95 percent of maximum dry density determined according to ASTM D 698. Moisture content within zero to 5 percent above optimum determined according to ASTM D 698, unless otherwise approved by Engineer.
  - b. Cement Stabilized: Maximum lift thickness determined by Contractor to achieve uniform placement and required compaction, but not to exceed 12 inches. Compact to 95 percent of maximum dry density as determined according to ASTM D 558. Moisture content to be on the dry side of optimum determined according to ASTM D 558 but sufficient for cement hydration.
  - c. Class IVA and IVB (clay soils): Place in maximum 8 inch lifts. Compaction to a minimum of 95 percent of maximum dry density, and moisture content within 0 to 5 percent above optimum, unless otherwise approved by Engineer, as determined according to ASTM D 698.
- H. Unless otherwise shown, trench excavations not under pavement, nor within 12 inches of back of curb, may be backfilled with suitable random backfill material. This provision does not apply to HDPE storm sewers.
  - a. Maximum 8 inch compacted lifts for clayey soils and 12 inches for granular soils. Compact to a minimum of 90 percent of maximum dry density determined according to ASTM D 698. Moisture content as necessary to achieve density. Water jetting nor flooding are permitted as forms of compaction unless approved by Engineer in writing.

### **3.06 BORINGS AND CASINGS UNDER ROADS, HIGHWAYS AND RAILROAD CROSSINGS**

- A. When indicated by Drawings and specifications, all street, road, highway, or railroad crossings for utility mains installed by the jacking and boring method shall be in accordance with area specifications and governing authorities.

- B. Excavation of approach pits and trenches within right-of-way of street, road, highway, or railroad shall be of sufficient distance from paving or railroad tracks to permit traffic to pass without interference. Tamp backfill for approach pits and trenches within right-of-way in layers not greater than 6" thick for entire length and depth of trench or pit. Compact backfill to 95% of maximum density obtained at optimum moisture as determined by AASHTO 180-57, Method A. Mechanical tampers may be used after cover of 6" has been obtained over top of barrel of pipe.
- C. Accomplish boring operation using commercial type boring rig and hole shall be bored to proper alignment and grade and within 2" of same diameter as largest outside joint diameter of pipe installed. Install pipe in hole immediately after bore has been made, and in no instance shall hole be left open while unattended.
- D. In event subsurface operations result in failure or damage to pavement within one year of construction, Contractor shall make necessary repair to pavement at no additional cost to Owner. In event paving cracks on either side of pipe line or is otherwise disturbed or broken due to construction operations, Contractor shall repair or replace disturbed or broken area at no additional expense to the Owner.
- E. Clean and prime interior and exterior of casing pipe; and line with two coats of asphalt in accordance with area specifications and governing authorities.
- F. Butt weld steel casing. Welds shall be full penetration single butt-welds in accordance with AWWA C-205 and AWS D7-0-62.
- G. Install casing and utility pipe with end seals, vent pipe, and other special equipment in accordance with area specifications and governing authorities.

### **3.06 FIELD QUALITY CONTROL**

- A. At least three tests will be performed initially for moisture-density relationships initially for backfill materials in accordance with ASTM D 698, and for cement stabilized sand in accordance with ASTM D 558.
- B. In place density tests of compacted pipe foundation, embedment and trench backfill soil materials will be performed accordance with ASTM D 1556, or ASTM D 2922 and ASTM D 3017.
- C. For open cut construction projects compaction to be measured by one test per one hundred feet, and a minimum of three tests per length of trench.
- D. Density tests will be distributed among placement areas. Placement areas are: foundation, bedding, haunching, and backfill.
- E. Verification tests will be performed adjacent to in-place tests showing density less than the acceptable criteria.
- F. Recomacted placement shall be tested at the same frequency as first test series.
- G. Recondition, recompact, and retest at contractor's expense if testing indicates work does not meet specified compaction requirements.
- H. Acceptability of rock compaction will be determined by inspection.

**END OF SECTION 02315**



## **SECTION 02317**

### **EXCAVATING AND BACKFILLING FOR PAVEMENT**

#### **PART 1 GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Excavation and compaction of materials for pavement.
- B. Excavation and compaction of materials for side ditches and swales.

##### **1.02 RELATED SECTIONS**

- A. Section 02300 Earthwork
- B. Section 02710 Paving Base Course
- C. Section 2741 Asphaltic Concrete Paving
- D. Section 2750 Concrete Paving
- E. Section 2770 concrete Walks, Curbs and Gutters
- F. Geotechnical Report for findings of Subsurface Materials and Conditions.
- G. Construction Drawings

##### **1.03 REFERENCE STANDARDS**

- A. American Society for Testing and Materials (ASTM) latest edition.
  - D 422 Method for Particle Size Analysis of Soils
  - D 698 Test for Moisture-Density Relations of Soils Using 5.5 lb (2.5 kg) Rammer and 12-inch (304.8 mm ) Drop (Standard Proctor)
  - D 1556 Test for Density of soil in Place by the Sand Cone Method
  - D 1557 Test for Moisture-Density Relations of Soils Using 10-lb (4.5 kg) Rammer and 18-inch (457 mm) Drop (Modified Proctor)
  - D 1559 Test Method for Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus
  - D 2167 Test for Density of Soil in Place by the Rubber Balloon Method
  - D 2216 Laboratory Determination of Moisture content of Soil
  - D 2487 Classification of Soils for Engineering Purposes
  - D 2922 Tests for Density of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)
  - D 3017 Test for Moisture Content of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)
  - D 4318 Test for Plastic Limit, Liquid, Limit, and Plasticity Index of Soils
  - C 25 Chemical Analysis of Limestone, Quicklime and Hydrated Lime
  - C 110 Physical Testing for Quicklime and Hydrated Lime, Wet Sieve Method
  - C 618 Specification for Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Portland Cement Concrete
  - C 977 Quicklime and Hydrated Lime for Soil Stabilization
- B. American Association of State Highway and Transportation Officials (AASHTO) latest edition
  - T 88 Mechanical Analysis of Soils

#### **1.04 QUALITY ASSURANCE**

Independent testing laboratory selected and paid by owner shall be retained to perform construction on Excavation and backfilling operations and subgrade analysis as specified in Section 02300 and as stated herein.

#### **1.05 SUBMITTAL**

- A. Shop Drawings or details pertaining to excavating and backfilling for Pavement are not required unless otherwise shown on the drawings or if contrary procedures to the project document are proposed.
- B. Submit a sample of each type of off-site fill material that is to be used in backfilling in an air-tight, 10 lb container for the testing laboratory or submit a gradation and certification of the aggregate material that is to be used to the testing laboratory for review.

#### **PART 2 PRODUCTS**

- A. On-site fill material as specified in Section 02300 and Geotechnical Exploration Report, and as approved by the owner or owner's representative.
- B. Off-site fill material as specified in Section 02300 and Geotechnical Exploration Report, and as approved by the owner or owner's representative.
- C. Acceptable stabilization fabrics and geogrids:
  - 1. Mirafi 500X or 600X
  - 2. Phillips 66 Supac 6WS
  - 3. Dupont Tytar 3401 and 3601
  - 4. Trevira S1114 and S1120
  - 5. Tensar SS-1 and SS-2
  - 6. Exxon GTF-200 OR 350

#### **PART 3 EXECUTION**

##### **3.01 PREPARATION**

- A. Identify all datum, lines, elevations and grades necessary to construct pavements, curb and gutter, bases, walkways and roadways as shown in the plans and specifications. Comply with Section 01725.
- B. Carefully protect benchmarks, property corners, monuments or other reference points.
- C. Notify all utility companies that are to remove or relocate utilities.
- D. Locate and identify all site utilities that have previously been installed and may be in danger of damage by grading operations.

- E. Locate and identify all existing utilities that are to remain and protect them from damage.
- F. Over excavate and properly prepare areas of subgrade that are not capable of supporting the proposed systems. These areas shall be stabilized by using acceptable filter fabrics and/or aggregate material placed and compacted as specified.

### 3.02 EXCAVATION

- A. Protect structures, trees, shrubs, and other features outside of grading area that are to remain.
- B. Excavate roadway and pavement areas to line and grade as shown in the plans and specifications.
- C. Engage all suitable material into the project fill areas as specified in Section 02300.
- D. Unsuitable excavated material is to be disposed of in a manner and location that is acceptable to the owner and local governing agencies.
- E. Perform excavation using capable, well maintained equipment and methods acceptable to the owner and the project document requirements.

### 3.03 FILLING AND SUBGRADE PREPARATION

- A. Areas exposed by excavation or stripping and on which subgrade preparations for paving are to be performed, including future pavement areas, shall be scarified to minimum depth of 8" and compacted to minimum of 95% of optimum density, in accordance with ASTM D 698 (or 92% of optimum density, in accordance with ASTM D 1557), at a moisture content of not less than 1% below and not more than 3% above the optimum moisture content. These areas shall then be proofrolled to detect any areas of insufficient compaction. Proofrolling shall be accomplished by making a minimum of two (2) complete passes with a fully-loaded tandem-axle dump truck, or approved equivalent, in each of the two perpendicular directions under the supervision and direction of a field geotechnical engineer. Areas of failure shall be excavated and recompacted as stated above.
- B. Fill materials used in preparation of subgrade shall be place in lifts or layers not to exceed 8" loose measure and compacted to a minimum density of 95% of optimum density, in accordance with ASTM D 698, (or 92% of the optimum density, in accordance with ASTM D 1557) at a moisture content of not less than 1% below and not more than 3% above the optimum moisture content.
- C. The following table stipulates maximum allowable values for plasticity index (PI) and liquid limit (LL) of suitable fill materials to be used in the specified areas, unless specifically stated otherwise on the Geotechnical Report for the area being excavated:

| Description                        | PI | LL |
|------------------------------------|----|----|
| *Paving Area, below upper two feet | 20 | 50 |
| *Paving Area, upper two feet       | 15 | 40 |

(\*References to Depth are to Proposed Subgrade Elevations)

- D. Material imported from off-site shall have a CBR (California Bearing Ratio) or LBR (Limerock Bearing Ratio) value equal to or above the pavement design subgrade CBR or LBR value indicated on the Geotechnical Report.
- E. Remove material not qualified for reuse and any excess soil not being reused from the site in accordance with approved waste disposal methods.

### **3.04 COMPACTION**

- A. Maintain optimum moisture content of fill materials to attain required compaction density.
- B. All materials shall be tested in accordance with Section 02300.
- C. An independent testing laboratory selected and paid by the owner, shall be retained to perform testing on-site.
- D. Compaction test will be as specified in Section 02300 together with the following for paving areas:
  - a. In cut areas not less than one compaction test for every 10,000 square feet.
  - b. In fill areas, same rate of testing for each 8" lift (measured loose).
- E. If compaction requirements are not complied with at any time during construction process, remove, recompact and retest deficient areas until proper compaction is obtained at no additional expense to owner.
- F. Maintain a plus or minus ½ inch tolerance in top of compacted surface in cross section or in a 16 foot length.

### **3.05 MAINTENANCE OF SUBGRADE**

- A. Prevent erosion at all times. Maintain ditches and cut temporary swales to allow natural drainage. Do not allow water to pond.
- B. Repair and recompact slides, washouts, settlements, or areas with loss of density.
- C. Finished subgrades shall be verified to ensure proper elevation and conditions for construction above subgrade.
- D. Protect subgrade from excessive wheel loading during construction including concrete trucks and dump trucks. Protect exposed areas having high moisture content from wheel loads that may cause rutting.
- E. Remove areas of finished subgrade found to have insufficient compaction density to depth necessary and replace in a manner that will comply with compaction requirements by use of material equal to or better than best subgrade material on-site. Surface of subgrade after compaction shall be hard, uniform, smooth, stable, and true to grade and cross-section.

### **3.06 FINISH GRADING**

- A. Finish grading shall be in accordance with Section 02300 and as more specifically stated herein.
- B. Grading of paving areas shall be checked by string line from grade stakes (blue tops) set at not more than 50' centers. Maximum tolerances of 0.10 feet, will be permitted. Contractor to provide engineering and field staking necessary for verification of lines, grades, and elevations and per Section 01725.

**END OF SECTION 02317**

## **SECTION 02341**

### **SOIL LIME STABILIZATION**

#### **Part 1 GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Application of lime slurry to subgrade soils.
- B. Mixing, compaction, and curing of lime slurry, water and subgrade into a stabilized foundation.

##### **1.02 REFERENCES**

- A. ASTM D 698 Standard Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort
- B. ASTM D 2922 Standard Test Methods for Density of Soil and Soil Aggregate in Place by Nuclear Methods (shallow depths)
- C. ASTM D 4318 Standard Test Method for liquid Limit, Plastic Limit, and Plasticity Index of Soils
- D. Geotechnical Exploration Report for site specific soil characteristics and requirements.

##### **1.03 SUBMITTALS**

- A. Submit certification that hydrated lime, quicklime, or commercial lime slurry complies with specification.
- B. Submit weigh tickets, certified by supplier, including each bulk delivery to work site if requested by Project Manager or Owner.

#### **Part 2 Products**

##### **2.01 WATER**

- A. Use clean, clear water, free from oil, acids, alkali or vegetation.

##### **2.02 LIME**

- A. Type A Hydrated Lime: Dry material consisting essentially of calcium hydroxide or mixture of calcium hydroxide and an allowable percentage of calcium oxide as listed in chemical composition chart.
- B. Type B Commercial Lime Slurry: Liquid mixture consisting essentially of lime solids and water in slurry form. Water or liquid portion shall not contain dissolved material in sufficient quantity to be injurious or objectionable for purpose intended.
- C. Type C Quicklime: Dry material consisting essentially of calcium oxide. Furnish quicklime in either of the following grades:
  - a. Grade DS: Pebble quicklime of gradation suitable for use in preparation of slurry for wet placing.

- b. Grade S: Finely graded quicklime for use in preparation of slurry for wet placing. Do not use grade S quicklime for dry placing.
- D. Conform to the following requirements:

| CHEMICAL COMPOSITION                                                                                                                                                                                                                                                           | TYPE                     |                       |                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------|----------------------|
|                                                                                                                                                                                                                                                                                | A                        | B                     | C                    |
| Active lime content % by weight<br>$\text{Ca(OH)}_2 + \text{CaO}$                                                                                                                                                                                                              | 90.0<br>min <sup>1</sup> | 87.0 min <sup>2</sup> | -                    |
| Unhydrated lime content, % by weight<br>CaO                                                                                                                                                                                                                                    | 5.0<br>max               | -                     | 87.0 min             |
| Free water content, % by weight<br>H <sub>2</sub> O                                                                                                                                                                                                                            | 5.0<br>max               | -                     | -                    |
| SIZING                                                                                                                                                                                                                                                                         |                          |                       |                      |
| Wet sieve, as % by weight<br>residue retained:                                                                                                                                                                                                                                 |                          |                       |                      |
| No. 6                                                                                                                                                                                                                                                                          | 0.2<br>max               | 0.2 max <sup>2</sup>  | 8.0 max <sup>3</sup> |
| No. 30                                                                                                                                                                                                                                                                         | 4.0<br>max               | 4.0 max <sup>2</sup>  | -                    |
| Dry Sieve, as % by weight<br>residue retained:                                                                                                                                                                                                                                 |                          |                       |                      |
| 1-inch                                                                                                                                                                                                                                                                         | -                        | -                     | 0.0                  |
| ½ inch                                                                                                                                                                                                                                                                         | -                        | -                     | 10.0 max             |
| Notes:<br>1. Maximum 5.0% by weight CaO shall be allowed in determining total active lime content.<br>2. Maximum solids content of slurry.<br>3. Total active lime content as CaO in material retained on No. 6 sieve shall not exceed 2.0% by weight of original Type C lime. |                          |                       |                      |

- E. Deliver lime slurry to the job site as commercial lime or prepare at job site using hydrated lime or quicklime. Provide slurry free of liquids other than water and of consistency that can be handled and uniformly applied without difficulty.
- F. Lime containing magnesium hydroxide is prohibited.

## 2.03 SOIL

- A. Soil to receive lime treatment may include borrow or existing subgrade material, existing pavement or combination of all three. Where existing pavement or base material is encountered, pulverize or scarify material so that 100 percent of sampled material passes 2 inch sieve.

## Part 3 Execution

### 3.01 PREPARATION

- A. Complete backfill of utilities prior to stabilization.
- B. Cut material to bottom of subgrade using an approved cutting and pulverizing machine meeting the following requirements:
- Cutters accurately provide smooth surface over entire width of cut to plane of secondary

- grade.
- b. Provide cut to depth as specified or shown in the drawings.
- C. Alternatively, scarify or excavate to bottom of stabilized subgrade. Remove material or window to expose secondary grade. Obtain uniform stability.
- D. Correct wet or unstable material below secondary grade by scarifying, adding lime, and compacting as directed by Project Manager or Engineer.
- E. Pulverize existing material so that 100 percent passes a 1-3/4 inch sieve.

### **3.02 LIME SLURRY APPLICATION**

- A. Apply slurry with distributor truck equipped with an agitator to keep lime and water in consistent mixture. Make successive passes over measured section of roadway to attain proper moisture and lime content. Limit spreading to an area where preliminary mixing operations can be completed on the same working day.

### **3.03 PRELIMINARY MIXING**

- A. Use approved single pass or multiple pass rotary speed mixers to mix soil, lime, and water to required depth. Obtain homogenous friable mixture free of clods and lumps.
- B. Shape mixed subgrade to final lines and grades.
- C. Eliminate the following operations and final mixing if pulverization requirements of paragraph 3.04C can be met during preliminary mixing:
  - a. Seal subgrade as precaution against rainfall by rolling lightly with light pneumatic rollers.
  - b. Cure soil lime material for 24 to 72 hours or as required to obtain optimum hydration. Keep subgrade moist during cure.

### **3.04 FINAL MIXING**

- A. Use approved single-pass or multiple-pass rotary speed mixers to uniformly mix cured soil and lime to required depth.
- B. Add water to bring moisture content of soil mixture to optimum or above.
- C. Mix and pulverize until all material passes a 13/4 inch sieve; minimum of 85 percent, excluding non-slacking fractions passes 3/4 inch sieve; and minimum of 60 percent excluding non-slacking fractions passes no. 4 sieve. Test according to TxDOT Tex-101 E, Part III using dry method.
- D. Shape mixed subgrade to final lines and grades.
- E. Do not expose hydrated lime to open air for 6 hours or more during interval between application and mixing. Avoid excessive hydrated lime loss due to washing or blowing.

### **3.05 COMPACTION**

- A. Aerate or sprinkle to attain optimum moisture content to 3 percent above optimum as determined by ASTM 698 on material sample after mixing with lime.

- B. Start compaction immediately after final mixing.
- C. Spread and compact in two or more equal layers where total compacted thickness is greater than equipment manufacturer's recommended range of mixing and compaction.
- D. Compact with approved heavy pneumatic or vibrating rollers, or combination of tamping rollers and light pneumatic rollers. Begin compaction at bottom and continue until entire depth is uniformly compacted.
- E. Do not allow stabilized subgrade to mix with underlying material. Correct irregularities or weak spots immediately by replacing material and recompacting.
- F. Compact subgrade to a minimum density of 95 percent of maximum dry density, according to ASTM D 698, at moisture content of optimum to 3 percent above optimum, unless otherwise indicated on drawings or geotechnical report.
- G. Seal with approved light pneumatic tired rollers. Prevent surface hair line cracking. Rework and recompact areas where hairline cracking develops.

### **3.06 CURING**

- A. Moist cure for minimum of 3 days before placing base or surface course, or opening to traffic. Subgrade may be opened to traffic after 2 days when adequate strength has been attained to prevent damage. Restrict traffic to light pneumatic rollers or vehicles weighing less than 10 tons.
- B. Keep subgrade surface damp by sprinkling. Roll with light pneumatic roller to keep surface knit together. Place base or surface within 14 days after final mixing and compaction. Restart compaction and moisture content of base material when time is exceeded.

### **3.07 TOLERANCES**

- A. Completed surface: smooth and conforming to typical section and established lines and grades.
- B. Top of compacted surface: plus or minus ¼ inch cross section or 16 foot length.
- C. Depth of lime stabilization shall be plus or minus one inch of specified depth for each 1000 foot roadway section.

### **3.08 FIELD QUALITY CONTROL**

- A. Testing will be performed by a qualified Testing Laboratory.
- B. Test soils, lime, and mixtures as follows:
  - a. Tests and analysis of soil materials will be performed in accordance with ASTM D 4318, using the wet preparation method.
  - b. Sampling and testing of lime shall be in accordance with TxDOT Tex 600 J, except using a lime slurry cup.
  - c. Sample mixtures of hydrated lime or quicklime in slurry form will be tested to establish compliance with specifications.
  - d. Moisture-density relationship will be established on material sampled from roadway, after stabilization with lime and final mixing in accordance with ASTM 698.



- C. In place depth will be evaluated for each 1000 foot roadway section and determined in accordance with TxDOT Tex 140 E in hand excavated holes. For each 1000 foot section, 3 phenolphthalein tests will be performed. Average stabilization depth for 1000 foot section will be based on average depth for three tests.
- D. Perform compaction testing in accordance with ASTM D 2922. Three test will be performed for each 1000 foot roadway section.

### **3.09 REWORK OF FAILED SECTIONS**

- A. Rework sections that do not meet specified thickness.
- B. Perform the following steps when more than 72 hours have lapsed since completion of compaction:
  - a. Moist cure for minimum of 3 days after compaction to required density.
  - b. Add lime at a rate of 25 percent of specified rate at no additional cost to the Owner.
  - c. Moisture density test of reworked material must be complete by laboratory before field compaction testing can be completed.

### **3.10 PROTECTION**

- A. Maintain stabilized subgrade to lines and grades and in good condition until placement of base or surface course.
- B. Repair defects immediately by replacing materials to full depth.

**END OF SECTION 02341**

**SECTION 02505**

**PAVING BASE COURSE**

**Part 1            GENERAL**

**1.01    SECTION INCLUDES**

- A. Granular Base
- B. Caliche Base
- C. Full Depth Asphalt Base
- D. Hot-Mix Sand Asphalt Base
- E. Soil Cement Stabilized Base

**1.02    RELATED SECTIONS**

- A. Section 02100            - Site Preparation
- B. Section 02200            - Earthwork
- C. Section 02227            - Aggregate Materials
- D. Section 02245            - Soil Stabilization
- E. Section 02511            - Asphaltic Concrete Paving
- F. Section 02520            - Portland Cement Concrete Paving
- G. Section 02525            -Curbs and Sidewalks
- H. Section 02900            - Landscaping
- 1. Construction Drawings

**1.03    REFERENCES**

- A. ANSI/ASTM D698 - Test Methods for Moisture-Density Relations of Soils and Soil-Aggregate Mixtures, Using 5.5 lb (2.49 kg) Rammer and 12 inch (304.8 mm) Drop.
- B. ANSI/ASTM D1557 - Test Methods for Moisture-Density Relations of Soils and Soil-Aggregate Mixtures using 10 lb (4.4 kg) Rammer and 18 inch (457 mm) Drop.
- C. ASTM D2167 - Test Method for Density and Unit Weight of Soil in-place by the Rubber Balloon Method.
- D. ASTM D1556 - Test Method for Density of Soil in-place by the Sand-Cone Method.
- E. ASTM D2922 - Test Methods for Density of Soil and Soil-Aggregate in-

place by Nuclear Methods (Shallow Depth), Method B (Direct Transmission).

F. ASTM D3017 - Test Methods for Moisture Content of Soil and Soil-Aggregate Mixtures.

## **PART 2 PRODUCTS**

### **2.01 FILL MATERIALS**

- A. Submit materials certificate to on-site independent testing laboratory which is signed by material producer and Contractor, certifying that material comply with, or exceed, the requirements herein.

## **PART 3 EXECUTION**

### **3.01 EXAMINATION**

Contractor shall verify that the subgrade has been inspected, tested and the gradients and elevations are correct, dry and properly prepared.

### **3.02 CONSTRUCTION**

- A. Perform base course construction in a manner that will drain surface properly at all times and at the same time prevent runoff from adjacent areas from draining onto base course construction.
- B. Compact base material to not less than 98% of optimum density as determined by ASTM D 698 or 95% of optimum density, as determined by ASTM D 1557, unless otherwise indicated on the Drawings.
- C. Granular Base: Construct to thickness indicated on Drawings. Apply in lifts or layers not exceeding 8", measured loose.
- D. Caliche Base: Construct to thickness indicated on Drawings. Use Type A or B, Grades 1,2,or 3 caliche per TXDOT Spec. Item 247.
- E. Asphalt Institute Type IV mix for Full Depth Asphalt Base: Construct to thickness indicated on Drawings in lifts or layers not exceeding 3", measured loose.
- F. Asphalt Institute Type VI, VII, or VIII Mixes for Hot Mix Sand Asphalt Bases: Construct to thickness indicated on Drawings. Apply in lifts or layers not exceeding 3", measured loose.
- G. Soil Cement Stabilized Base: Construct to thickness and strength as indicated on Drawings and in accordance with applicable state highway specifications. If not indicated on the Drawings, the minimum compressive strength shall be 500 p.s.i., tested at 28 days.

### **3.03 FIELD QUALITY CONTROL**

- A. An Independent Testing Laboratory, selected and paid by Owner, shall be retained to perform construction testing of in-place base courses for compliance with requirements for thickness, compaction, density and tolerance. Paving base course tolerances shall be verified (by rod and level readings on not more than fifty-foot centers) to be not more than 0.05 feet above design elevation which will allow for paving thicknesses as shown in the Drawings. Contractor shall provide instruments and suitable benchmark.

1. The following tests shall be performed on each type of material used as base course material:
  1. Moisture and Density Relationship: ASTM D 698 or ASTM D 1557.
  2. Mechanical Analysis: AASHTO T-88.
  3. Plasticity Index: ASTM D-4318.
  4. Base material thickness: Perform one test for each 20,000 square foot of in-place base material area.
  5. Base material compaction: Perform one test in each lift for each 20,000 square feet of in-place base material area.
  6. Test each course of base material for compliance with applicable state highway specifications.
- B. Field density tests for in-place materials shall be performed according to one of the following standards as part of construction testing requirements:
  1. San-Cone Method: ASTM D 1556.
  2. Balloon Method: ASTM D 2167.
  3. Nuclear Method: ASTM D 2922, Method B (Direct Transmission).
- D. Independent Testing Laboratory shall prepare test reports that indicate test location, elevation data, and test results. The Owner, Engineer, and Contractor shall be provided with copies of reports within 96 hours of time test was performed. In event that any test performed fails to meet these Specifications, the Owner, Engineer and Contractor shall be notified immediately by Independent Testing Laboratory. The Owner reserves right to employ Independent Testing Laboratory and to direct any testing that is deemed by them to be necessary. Contractor shall provide free access to site for testing activities.

**END OF SECTION 02505**

## **SECTION 02511**

### **ASPHALTIC CONCRETE PAVING**

#### **Part 1 General**

- A. Section 02223 - Excavation, Backfill and Compacting for Pavement
- B. Section 02500 - Paving and Surfacing
- C. Section 02505 - Paving Base Course
- D. Section 02520 - Portland Cement Concrete Paving
- E. Section 02525 - Curbs
- F. Section 02584 - Parking Lot and Roadway Marking
- G. Construction Drawings
- H. State Highway Department Standard Specifications

#### **1.03 SUBMITTALS**

- A. Design Mix: Before any asphaltic concrete paving is constructed, submit actual design mix to the Owner's Construction Department for review and/or approval. Design mix submittal shall follow the format as indicated in the Asphalt Institute Manual MS-2, Marshall Stability Method; and shall include the type/name of the mix gradation analysis, grade of asphalt cement used, Marshall Stability (lbs), flow, effective asphalt content (percent), and direct references to the applicable highway department specifications sections for each material. The design shall be for a mixture listed in the current edition of the applicable state roadway specifications. Mix designs over three (3) years old will not be accepted by the owner.
- B. Material Certificates: Submit materials certificate to on-site independent testing laboratory which is signed by material producer and Contractor, certifying that materials comply with, or exceed, the requirements herein.

#### **1.04 JOB CONDITIONS**

- A. Weather Limitations:
- B. Apply prime and tack coats when ambient temperature is above 40°F, and when temperature has been above 35° F for 12 hours immediately prior to application. Do not apply when base is wet, contains excess moisture, or during rain.
- D. Construct asphaltic concrete paving when atmospheric temperature is above 40°F.

#### **1.05 REFERENCES**

- A. MS-2 Mix design methods for asphaltic concrete and other hot Mix types per The Asphalt Institute (AI)
- B. MS-3-Asphalt Plant Manual per The Asphalt Institute (AI)
- C. Hot Mix Asphalt Paving Handbook per US Army Corp of Engineers, UN-13 (CE MP-ET)

- D. MS-10 Basic Asphalt Emulsion Manual per The Asphalt Institute (AI)
- E. ASTM D946 - Penetration - Graded Asphalt Cement for use in Pavement Construction
- F. AASHTO M-226/ASTM D3381 Asphalt Cement
- G. AASHTO M-140/ASTM D997 or AASHTO M-208/ASTM D-2397 Tack Coat
- H. AASHTO M-117/ASTM D242 Mineral Filler
- I. AASHTO T-245/ASTM D1559 Mashall Mix Design

## PART 2 PRODUCTS

### 2.01 Materials

- A. Provide asphalt-aggregate mixture as recommended by local or state paving authorities to suit project conditions. Use locally available materials and gradations which meet state highway Specifications and exhibit satisfactory records of previous installations.
- B. Asphalt Cement: Comply with AASHTO M-226/ASTM D 3381; Table 2 AC-10, AC-20, or AC-30, viscosity grade, depending on local mean annual air temperature. (See chart below):

| <u>Temperature Condition</u>                                                                               | <u>Asphalt Grades</u> |
|------------------------------------------------------------------------------------------------------------|-----------------------|
| Cold, mean annual air temperature<br>At 7 degrees C (45 degrees F) or lower                                | AC-10<br>85/100 pen.  |
| Warm, mean annual air temperature<br>between 7 degrees C (45 degrees F) and<br>24 degrees C (75 degrees F) | AC-20<br>60/70 pen.   |
| Hot, mean annual air temperature<br>At 24 degrees C (75 degrees F) or higher                               | AC-30                 |

- C. Prime Coat: A medium curing cut-back asphalt or an asphalt penetrating prime coat consisting of either MC-30 or SS-1h.
- D. Tack Coat: Emulsified asphalt; AASHTO M-140/ASTM D 997 or AASHTO M 208/ASTM D 2397, SS-1, or CSS-1h, diluted with one part water to one part emulsified asphalt
- E. Mineral Filler: Rock or slag dust, hydraulic cement, or other inert material complying with AASHTO M-17/ASTM D 242, if recommended by applicable state highway standards.
- F. Asphalt-Aggregate Mixture: Unless otherwise noted on the Drawings the design Mix shall have a minimum stability based on a 50-blow Mashall complying with ASTM D 1559 of 1000 lb with a flow between 8 and 16. The Design Mix shall be within sieve analysis and bitumen ranges flow:

## SIEVE ANALYSIS OF MIX

| <u>Square Sieve</u> | <u>Total Percent Passign</u> | <u>Percent Tolerance</u> |
|---------------------|------------------------------|--------------------------|
| 1/4"                | 100.....                     | 7%                       |
| 1/2"                | 80 - 100%.....               | 5%                       |
| 3/8"                | 65 - 93%.....                | 4%                       |
| #8                  | 40 - 55%.....                | 4%                       |
| #50                 | 12 - 27%.....                | 2%                       |
| #200                | 0 - 10%.....                 | 0%                       |

Percent bitumen by weight of total mix: 5.0 -8.35

Air voids: 3-6%

Percent aggregate voids filled with asphalt cement: 70-82%

Allowable variance of percent bitumen by weight of total  
mix = 0.4

### 2.02 Equipment

Maintain equipment in satisfactory operating condition and correct breakdown in manner that will not delay or be detrimental to progress of paving operations.

## PART 3EXECUTION

### 3.01 Preparation

- A. Remove loose material from compacted base material surface immediately before applying prime coat.
- B. Proof roll prepared base material surface to check for areas requiring additional compaction and areas requiring removal and recompaction.
- C. Do not begin paving work until deficient base material areas have been corrected and are ready to receive paving.

### 3.02 Applications

- A. Prime Coat:
  1. Apply bituminous prime coat to all base material surfaces where asphaltic concrete paving will be constructed .
  2. Apply bituminous prime coat in accordance with APWA Section 2204 and applicable state highway specifications.
  3. Apply at minimum rate of 0.25 gallon per square yard over compacted base material. Apply to penetrate and seal, but not flood surface.
  4. Make necessary precautions to protect adjacent areas from overspray.
  5. Cure and dry as long as necessary to attain penetration of compacted base and evaporation of volatile substances.

B. Tack Coat:

1. Apply to contact surfaces of previously constructed asphaltic concrete base courses or portland cement concrete and surfaces abutting or projecting into asphaltic concrete or into asphaltic concrete pavement.
2. Apply tack coat to asphaltic concrete base course or sand asphalt base course. Apply emulsified asphalt tack coat between each lift or layer of full depth asphaltic concrete and sand asphalt base and on surface of all such bases where asphaltic concrete paving will be constructed.
3. Apply emulsified asphalt tack coat in accordance with APWA Section 2204 and applicable state highway specifications.
4. Apply at minimum rate of 0.05 gallon per square yard of surface.
5. Allow to dry until at proper condition to receive paving.

**3.03 Asphaltic Concrete Placement**

- A. Place asphaltic concrete mixture on completed compacted subgrade surface, spread, and strike off. Spread mixture at following minimum temperatures:
1. When ambient temperature is between 40°F and 50°F, mixture temp. = 285°F
  2. When ambient temperature is between 50°F and 60°F, mixture temp. = 280°F
  3. When ambient temperature is higher than 60°F, mixture temp. = 275°F
- B. Whenever possible, all pavement shall be spread by a finishing machine; however Inaccessible or irregular areas may be placed by hand methods. The hot mixture shall spread uniformly to the required depth with hot shovels and rakes. After spreading, the hot mixtures shall be carefully smoothed to remove all segregated course aggregate and rake marks. Rakes and lutes used for hand spreading shall be of the type designed for use on asphalt mixtures Loads shall not be dumped faster that they can be properly spread. Workers shall not stand on the loose mixture while spreading.
- C. Paving Machine Placement: Apply successive lifts of asphaltic concrete in transverse directions with the surface course placed in the direction of surface-water flow. Place in typical strips not less than 10'-0" wide.
- D. Joints: Make joints between old and new pavements, or between successive days and work in a manner that will provide a continuous bond between adjoining work. Construction joints shall have same texture, density, and smoothness as other sections of asphaltic concrete course. Clean contact surfaces of all joints and apply tack coat.

**3.04 Rolling and Compaction**

- A. The mixture, after being spread, shall be thoroughly compacted by rolling as soon as it will bear the weight of the rollers without undue displacement. The number, weight and types of rollers and sequences of rolling operations shall be such that the required density and surface are consistently attained while the mixture is in a workable condition.



- B. Compact mixture with hot hand tampers or vibrating plate compactors in areas inaccessible to rollers.
- C. Breakdown Rolling: Accomplish breakdown or initial rolling immediately following rolling of joints and outside edge. Check surface after breakdown rolling, and repair displaced areas by loosening and filling with hot material.
- D. Second Rolling: Follow breakdown rolling as soon as possible, while mixture is hot. Continue second rolling until mixture has been thoroughly compacted.
- E. Finish Rolling: Perform finish rolling while mixture is still warm enough for removal of roller marks. Continue rolling until roller marks are eliminated and course has attained maximum density.
- F. Patching: Remove and replace paving areas mixed with foreign materials and defective areas. Cut out such areas and fill with fresh, hot asphaltic concrete. Compact by rolling to maximum surface density and smoothness.
- G. Protection: After final rolling, do not permit vehicular traffic on pavement until it has cooled and hardened. Erect barricades to protect paving from traffic until mixture has cooled not to become marked.

### **3.05 Field Quality Control**

- A. Independent Testing Laboratory, selected and paid by Owner, shall be retained to perform construction testing on in-placed asphaltic concrete courses for compliance with requirements for thickness, compaction and surface smoothness. Asphaltic surface and base courses shall be randomly cored at a minimum rate of one core for every 20,000 square feet of paving. However, no less than three cores in light duty areas and three cores in heavy duty areas shall be obtained. Coring holes shall be immediately filled with full-depth asphalt or with concrete. Asphaltic Concrete pavement samples shall be tested for conformance with the mix design.
- B. Grade Control: Establish and maintain required lines and elevations.
- C. Thickness: In-place compacted thickness shall be not less than thickness specified on the drawings. Areas of deficient paving thickness shall receive a tack coat and a minimum 1" overlay; or shall be removed and replaced to the proper thickness, at the discretion of the Owner; until specified thickness of the course is met or exceed at no additional expense to the Owner.
- D. Surface Smoothness: Testing shall be performed on the finished surface of each asphalt concrete course for smoothness, using 10'-0" straightedge applied parallel with, and at right angles to centerline of paved area. The results of these tests shall be made available to the owner upon request. Surfaces will not be acceptable if the following 10' straightedge tolerances for Smoothness are exceeded.

|                         |       |
|-------------------------|-------|
| Base Course Surface:    | 1/4"  |
| Wearing Course Surface: | 3/16" |
- E. Check surface areas at intervals necessary to eliminate ponding areas. Remove and replace unacceptable paving as directed by Owner.
- F. Compaction: Field density test for in place material shall be performed by examination of field cores in accordance with one of the following standards:

- 1. Bulk specific gravity of paraffing-coated specimens: ASTM D-1188.

2. Bulk specific gravity using saturated surface-dry Specimens: ASTM D-2726.

Rate of testing shall be one core per 20,000 square feet of pavement, with a minimum of 3 cores from heavy duty areas and 3 cores from standard-duty areas. Cores shall be cut from areas representative of the project.

Areas of insufficient compaction shall be delineated, removed, and replaced in compliance with the specifications at no expenses to the Owner.

**END OF SECTION 02511**

**SECTION 02520**

**PORTLAND CEMENT CONCRETE PAVING**

**Part 1 General**

**1.01 SECTION INCLUDES**

Concrete, integral curbs, median barriers, parking areas and roads.

**1.02 RELATED SECTIONS**

- A. Section 02100 - Site Preparation.
- B. Section 02227 - Aggregate Material.
- C. Section 02505 - Paving Base Course.
- D. Section 02525 - Curbs and Sidewalk.
- E. Section 02584 - Parking Lot and Roadway Marking.
- F. Section 03300 - Cast-In-Place Concrete  
(See Architectural/Building Specifications)
- G. State Highway Department Standard Specifications.
- H. Construction Drawings.

**1.03 REFERENCES**

- A. ACI 301 - Specifications for Structural Concrete for Buildings.
- B. ACI 304 - Recommended Practice for Measuring, Mixing,  
Transporting and Placing Concrete.
- C. ANSI/ASTM A185 - Welded Steel Wire Fabric for Concrete.
- D. ANSI/ASTM A497 - Welded Deformed Steel Wire Fabric Concrete  
Reinforcement.
- E. ANSI/ASTM D1751 - Preformed Expansion Joint Fillers for Concrete  
Paving and Structural Construction.
- F. ANSI/ASTM D1752 - Preformed Sponge Rubber and Cork Expansion Joint  
Fillers for Concrete Paving and Structural Construction.
- G. ASTM A615 - Deformed and Plain Billet-Steel for Concrete  
Reinforcement.

- A. Concrete mix design and testing shall comply with requirements of applicable Section 03300.
- B. Design mix to produce normal weight concrete consisting of Portland cement, aggregate, water
- PORTLAND CEMENT CONCRETE PAVING 9/16/15 02520-2

reducing admixture, air-entraining admixture, and water to produce the following properties:

1. Compressive Strength: 3,500 psi, minimum at 28 days, unless otherwise indicated on the Drawings.
2. Slump Range: 3" - 5" at time of placement.
3. Air Entrainment: 5% to 8%.

### **PART 3 EXECUTION**

#### **3.01 PREPARATION**

- A. Proof-roll prepared base material surface to check for unstable areas. The paving work shall begin after the unsuitable areas have been corrected and are ready to receive paving. Compaction testing for the base material shall be completed prior to the placement of the paving.

Surface Preparation: Remove loose material from compacted base material surface to produce a firm, smooth surface immediately before placing concrete.

#### **3.02 INSTALLATION**

A. Form Construction

1. Set forms to required grades and lines, rigidly braced and secured.
2. Install sufficient quantity of forms to allow continuance of work and so that forms remain in place a minimum of 24 hours after concrete placement.
3. Check completed formwork for grade and alignment to following tolerances:  
  
Top of forms not more than 1/8" in 10' - 0"  
Vertical face on longitudinal axis, not more than 1/4" in 10'-0".

- B. Reinforcement: Locate, place and support reinforcement per applicable Section 03300.

C. Concrete Placement

1. Comply with applicable requirements of Section 03300.
2. Do not place concrete until base material and forms have been checked for line and grade. Moisten base material if required to provide uniform dampened condition at time concrete is placed. Concrete shall not be placed around manholes or other structures until they are at the required finish elevation and alignment.
3. Place concrete using methods which prevent segregation of mix. Consolidate concrete along face of forms and adjacent to transverse joints with internal

vibrator. Keep vibrator away from joint assemblies, reinforcement, or side forms. Consolidate with care to prevent dislocation of reinforcement, or side forms. Consolidated with care to prevent dislocation of reinforcing dowels, and joint devices.

4. Deposit and spread concrete in continuous operation between transverse joints, as far as possible. If interrupted for more than ½ hour, place construction joint.

D. Joint Construction: Construct expansion, weakened-plane Control (contraction), and construction joints straight with face perpendicular to concrete surface. Construct transverse joints perpendicular to centerline, unless otherwise detailed.

1. Weakened-Plane Control (Contraction) Joints: Provide joints at a spacing of 15'-0" o.c. maximum each way. Construct control joints for depth equal to at least 1/4 concrete thickness, as follows:

1. Form tooled joints in fresh concrete by grooving top portion with recommended tool and finishing with jointer.
2. Form sawed joint using powered saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut joints into hardened concrete as soon as surface will not be torn, abraded, or otherwise damaged by cutting action.

2. Construction Joints: Place concrete joints at end of placements and at locations where placement operations are stopped for period of more than ½ hours, except where such placements terminate at expansion joints. Construction joints using standard metal keyway-section forms.

3. Expansion Joints: Located expansion joints at 180'-0" o.c. maximum each way. Provide premolded joint filler for expansion joints abutting concrete curbs, catch basins, manholes, inlets, structures, walks, and other fixed objects.

E. Joint Fillers: Extend joint fillers full-width and depth of joint, and not less than ½" or more than 1" below finished surface where joint sealer is indicated. Furnish joint fillers in one-piece lengths for full width being placed, wherever possible. Where more than one length is required, lace or clip joint filler sections together.

F. Joint Sealants: All joints shall be sealed with approved exterior pavement joint sealants and shall be installed per manufacturer's recommendations.

### **3.03 CONCRETE FINISHING**

A. After striking off and consolidating concrete, smooth surface by screeding and floating to compact surface and produce uniform texture. After floating, test surface for trueness with 10'-0" straightedge. Distribute concrete as required to remove surface irregularities, and refloat repaired areas to provide continuous smooth finish.

B. Work edges of slabs, gutters, back top edge of integral curb, and formed joints with a edging tool, and round to ½" radius. Eliminate tool marks on concrete surface. After completion of floating and troweling when excess moisture or surface sheen has disappeared, complete surface finishing, as follow:

1. Inclined slab surfaces: Provide coarse, nonslip finish by scoring surface with stiff-bristled broom perpendicular to line of traffic.
  2. Paving: Provide coarse, nonslip finish by scoring surface with stiff-bristled broom perpendicular of traffic.
- C. Do not remove forms for 24 hours after concrete has been placed. After form removal, clean ends of joints and point up any minor honeycombed areas. Remove and replace areas or sections with major defects, as directed.
- D. Protect and cure finished concrete paving using acceptable moint-curing method, more particularly described in the "water-curing" section of ACT 308-81.

### **3.04 CLEANING AND ADJUSTING**

- A. Sweep concrete pavement and wash free of stains, discolorations, dirt, and other foreign material just prior to final inspection.
- B. Protect concrete form from damage until acceptance of work. Exclude traffic from payment for at least 14 days after placement. When construction traffic is permitted, maintain pavement as clean as possible by removing surface stains and spillage of materials.

### **3.05 FIELD QUALITY CONTROL**

An independent testing laboratory shall randomly core the pavement at a minimum rate of one core per 20,000 square feet of pavement, with a minimum of 3 cores from heavy-duty areas and 3 cores from standard duty areas. Core shall be tested for thickness and quality of aggregate distribution. Core holes shall be patched immediately with Portland cement concrete conforming to Section 2.02 and shall be finished to provide a level surface conforming to Section 3.03 A & 3.03 B.

**END OF SECTION 02520**

## **SECTION 02525**

### **CURB AND SIDEWALKS**

#### **PART I GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Combination concrete curb and gutter
- B. Concrete Curb
- C. Concrete Flume
- D. Concrete Sidewalk

##### **1.02 RELATED SECTIONS**

- A. Section 02100 - Site Preparation.
- B. Section 02584 - Parking Lot and Roadway Marking.
- C. Section 03300 - Cast-in-place Concrete (See Architectural/Building Specifications).
- D. State Highway Department Standard Specifications.
- E. Construction Drawings.

##### **1.03 REFERENCES**

- A. ACI 304 - Recommended Practice for Measuring, Mixing, Transporting and Placing Concrete.
- B. ANSI/ASTM D1751 - Performed Expansion Joint Fillers for Concrete Paving and Structural Construction.
- C. ANSI/ASTM D1752 - Performed Sponge Rubber and Cork Expansion Joint Fillers for Concrete Paving and Structural Construction.
- D. ASTM C33 - Concrete Aggregates.
- E. ASTM C94 - Ready Mix Concrete.
- F. ASTM C150 - Portland Cement.
- G. ASTM C260 - Air-Entraining Admixtures for Concrete.
- H. ASTM C309 - Liquid Membrane-Forming Compounds for Curing Concrete.
- I. ASTM C494 -Chemical Admixtures for Concrete.
- J. FS TT-C-800 - Curing Compound, Concrete, for New and Existing Surfaces.

##### **1.05 PERFORMANCE REQUIREMENTS**

- A. Contractor shall maintain access for vehicular and pedestrian traffic as required for other construction activities. Utilized temporary striping, flagmen, barricades, warning signs, and warning lights as required.



## **PART 2 PRODUCTS**

### **2.01 MATERIALS**

- A. Forms: Steel, wood, or other suitable material of size and strength to resist movement during concrete placement and to retain horizontal and vertical alignment until removal. Use straight forms, free of distortion and defects. Use flexible spring steel forms or laminated boards to form radius bends as required. The forms shall be of a depth equal to the depth of curbing or sidewalk, and so designed as to permit secure fastening together at the tops. Coat forms with non-staining type coating that will not discolor or deface surface of concrete.
- B. Concrete Materials: Comply with requirements of applicable Section 03300 for concrete materials, admixtures, bonding materials, curing materials, and others as required.
- C. Joint Fillers: Resilient premolded bituminous impregnated fiberboard units complying with ASTM D 1751 FS HH-F-341, Type II, Class A; or AASHTO M 153, Type I.
- D. Joint Sealers: Non-priming, pourable, self-leveling polyurethane. Acceptable sealants are Sonneborn "Sonolastic Paving Joint Sealant" Sonneborn "Sonomeric CT 1 Sealant", Sonneborn "Sonomeric CT 2 Sealant", Mameco "Vulken 45", or Woodmont Products "Chem-Caulk".

### **2.02 MIX DESIGN AND TESTING**

- A. Concrete mix design and testing shall comply with requirements of applicable Section 03300.
- B. Design mix to produce normal weight concrete consisting of Portland cement, aggregate, water-reducing admixture, air-entraining admixture, and water to produce the following properties:
  - 1. Compressive Strength: 3,500 psi, minimum at 28 days, unless otherwise indicated on the Drawings.
  - 2. Slump Range: 2"-5" at time of placement.
  - 3. Air Entrainment: 5% to 8%.

## **PART 3 EXECUTION**

### **3.01 PREPARATION**

- A. Proof-roll prepared base material surface to check for unstable areas. The paving work shall begin after any unsuitable areas have been corrected and are ready to receive paving. Compaction testing for the base material shall be completed prior to the placement of the paving.
- B. Surface Preparation: Remove loose material from compacted base material surface to produce a firm, smooth surface immediately before placing concrete.

### **3.02 INSTALLATION**

- A. Form Construction

1. Set forms to require grades and lines, rigidly braced and secured.
2. Install sufficient quantity of forms to allow continuance of work and so that forms remain in place a minimum of 24 hours after concrete placement.
3. Check completed formwork for grade and alignment to following tolerances:

Top of forms not more than 1/8" in 10'-0". Vertical face on longitudinal axis, not more than 1/4" in 10'-0".

- A. Clean forms after each use, and coat with form release agent as often as required to ensure separation from concrete without damage.

B. Concrete Placement

1. Comply with applicable requirements of Section 03300.
2. Do not place concrete until base material and forms have been checked for line and grade. Moisten base material if required to provide uniform dampened conditions at time concrete is place. Concrete shall not be placed around manholes or other structures until they are at the required finish elevation and alignment.
3. Place concrete using methods which prevent segregation of mix. Consolidate concrete along face of forms and adjacent to transverse joints with internal vibrator. Keep vibrator away from joint assemblies, reinforcement, or side forms. Consolidate with care to prevent dislocation of dowels, and joint devices.
4. Deposit and spread concrete in continuous operation between transverse joints, as fast as possible. If interrupted for more than 1/2 hours, place construction joints. Automatic machine may be used for curb and gutter placement at Contractor's option. Machine placement must produce curbs and gutters to required cross section, lines, grades, finish, and jointing as specified for formed concrete. If results are not acceptable, remove and replace with formed concrete as specified.

C. Joint Construction

1. Contraction Joints: Concrete curb, concrete gutter or concrete curb and gutter, where specified on the plans, shall be constructed in uniform sections of the length specified on the plans. The joints between sections shall be formed either by steel templates 1/8 inch in thickness, of a length equal to the width of the gutter and/or curb, and with a depth which will penetrate at least 2 inches below the surface of the curb an/or gutter; or with 3/4-inch thick preformed expansion joint filler cut to the exact cross section of the curb and/or gutter; or by sawing to a depth of at least 2 inches while the concrete is between 4 to 24 hours old. If steel templates are used, they shall be left in place until the concrete has set sufficiently to hold its shape, but shall be removed while the forms are still in place.
2. Longitudinal Construction Joints: Concrete curb, concrete gutter or combination concrete curb and gutter, where specified on the plans, shall be tied to concrete pavement with 1/2 inch round deformed reinforcement bars of the length and spacing shown on the plans.
3. Transverse Expansion Joints: Transverse expansion joint in curb, curb and gutter, gutters or sidewalk shall have the filler cut to the exact cross section of the curb, curb and gutter, gutter or sidewalk. The joints shall be similar to the type of expansion joint used in the adjacent pavement.

- D. Joint Fillers: Extend joint fillers full-width and depth of joint, and not less than 1/2" or more than 1" below finished surface where joint sealer is indicated. Furnish joint fillers in one-piece lengths for full

width being placed, wherever possible. Where more than one length is required, lace or clip joint filler section together.

D. Joint Sealants: All joints shall be sealed with approved exterior pavement joint sealants and shall be installed per manufacturer's recommendations.

### **3.03 CONCRETE FINISHING**

- A. After striking off and consolidating concrete, smooth surface by screeding and floating. Adjust floating to compact surface and produce uniform texture. After floating, test surface for trueness with 10'-0" straightedge. Distribute concrete as required to remove surface irregularities, and refloat repaired areas to provide continuous smooth finish.
- B. Work edges of sidewalks, gutters, back top edge of integral curb, and formed joints with and edging tool, and round to ½" radius. Eliminate tool marks on concrete surface. After completion of floating and troweling when excess moisture or surface sheen has disappeared, complete surface finishing, as follows:
  - 1. Inclined Slab Surfaces: Provide coarse, nonslip finish by scoring surface with stiff-bristled broom perpendicular to line of traffic.
  - 2. Curbs, gutters, and walks: Broom finish by drawing fine-hair broom across surface perpendicular to line of traffic. Repeat operation as necessary to produce a fine lien texture.
- C. Do not remove forms for 24 hours after concrete has been placed. After form removal, clean ends of joints and point up any minor honeycombed areas. Remove and replace areas or sections with major defects, as directed.
- D. Protect and cure finished concrete paving using acceptable moist- curing methods, more particularly described in the "water-curing" section of ACI 308-81.

### **3.04 BACKFILL**

After the concrete has set sufficiently, the spaces in front and back of the curb and gutter or sidewalk shall be refilled to the required elevation with suitable material which shall be compacted until firm and solid and neatly graded.

### **3.05 CLEANING AND ADJUSTING**

- A. Sweep concrete pavement and wash free of stains, discolorations, dirt, and other foreign material just prior to final inspection.
- B. Protect concrete from damage until acceptance of work. Exclude traffic from pavement for at least 14 days after placement. When construction traffic is permitted, maintain pavement as clean as possible by removing surface stains and spillage of materials.

**END OF SECTION 02525**

## **SECTION 02584**

### **PAVEMENT MARKINGS**

#### **PART 1 GENERAL**

##### **1.01 SECTION INCLUDES**

1. Painted pavement marking.
2. Painted curbs, guard posts and light pole bases.

##### **1.02 RELATED SECTIONS**

1. Section 02200 - Earthwork
2. Construction Drawings

##### **1.03 PROJECT CONDITIONS**

1. Maintain access for vehicular and pedestrian traffic as required for other construction activities. Utilize flagmen, barricades, warning signs and warning light as required.

#### **PART 2 PRODUCTS**

##### **2.01 MATERIALS**

1. The paint shall be a non-bleeding, quick-drying, alkyd petroleum base paint suitable for traffic-bearing surface and shall meet FS TTP-85E and mixed in accordance with manufacturer's instructions before application.

#### **PART 3 EXECUTION**

##### **3.01 PREPARATION**

1. Sweep and clean surface to eliminate loose material and dust.
2. Where existing pavement markings are indicated on the drawings to be removed or would interfere with the adhesion of new paint, a motorized abrasive device shall be used to remove the markings. The equipment employed shall not damage the existing paving or create a surface hazardous to vehicle or pedestrian traffic. In all areas within public rights-of-way, the method of marking removal shall be approved by governing authority.

##### **3.02 APPLICATION**

1. Apply two (2) coats of paint at manufacturer recommended rate without the addition of thinner, with a maximum of 100 square feet per gallon. Apply with mechanical equipment to produce uniform straight edges. At sidewalk curbs and crosswalks, use a straightedge to ensure a uniform, clean and straight stripe.
2. The following items shall be painted with the colors noted below:
  1. Pedestrian Crosswalks: Yellow
  2. Exterior Sidewalk Curbs, Light Pole Bases and Guardposts: Yellow

3. Fire Lanes: Red or per local code
4. Lane Striping where separating traffic in opposite directions: Yellow
5. Lane Striping where separating traffic in the same direction: White
6. Handicap Symbols: Per local code
7. Parking Stall Striping: Yellow, unless otherwise noted on plans

**END OF SECTION 02584**

## **SECTION 02605**

### **SEWER STRUCTURES**

#### **Part 1 GENERAL**

##### **1.01 SECTION INCLUDES**

- A. Monolithic concrete manhole barrel with the option of monolithic concrete or masonry transition to lid frame, covers, anchorage and accessories.
- B. Modular precast concrete manhole barrel with tongue-and-groove joints and with the option of precast concrete or masonry transition to lid frame, covers, anchorage and accessories.
- C. Masonry manhole barrel with masonry transition to lid frame, covers, anchorage and accessories.
- D. Precast Polyethylene manhole assemblies.

##### **1.02 RELATED SECTIONS**

- A. Section 02315 - Excavation, Backfill, and Compacting for utilities
- B. Section 02720 - Storm Sewer Systems
- C. Section 02730 - Sanitary Sewer Systems
- D. Section 03300 -Cast-In-Place Concrete (See Architectural/Building Specifications)
- E. Construction Drawings

##### **1.03 REFERENCES**

- A. ANSI/ASTM C55 - Concrete Building Brick.
- B. ASTM A48 - Gray Iron Castings.
- C. ASTM C478 - Precast Reinforced Concrete Manhole Sections.
- D. ASTM C923 - Resilient Connectors Between Reinforced Concrete Manhole Structures and Pipes.
- E. ASTM D1248 - Precast Polyethylene Manholes.
- F. International Masonry Industry All-Weather Council (IMIAC):  
Recommended Practices and Guide Specifications for Cold Weather  
Masonry Construction.

##### **1.04 SUBMITTALS**

- A. Shop Drawings: Indicate references to drawings of manhole locations, elevations, piping with sizes, locations and elevations of penetrations.
- B. Product Data: Provide data for manhole covers, manhole steps, component construction, features, configuration, and dimensions.

## **PART 2 PRODUCTS**

### **2.01 MATERIALS**

- A. Manhole Barrel: Reinforced precast concrete, in accordance with ASTM C478 with gaskets in accordance with ASTM C923.
  - 1. Construct manholes of precast concrete sections as required by Drawings to size, shape, and depth indicated, but never less than 4'-0" inside diameter.
- B. Manhole Barrel: Non-reinforced cast-in-place concrete as specified in Section 03300.
  - 1. Cast-in place manholes shall be constructed of 3500 psi concrete.
  - 2. Forms shall be made of steel sheets accurately shaped and fabricated of sufficient strength to form dense watertight walls to true dimensions.
  - 3. Concrete shall be deposited in evenly distributed layers of about 18 inches, with each layer vibrated to bond it to the preceding layer.
- C. Manhole Barrel: Precast Polyethylene in accordance with ASTM D 1248. Manholes shall be manufactured with factory-molded steps. The nominal cylinder internal diameter shall be 48" and shall be designed to accept concrete filled polyethylene manhole lids and standard cast iron frames with lid or grate. Manholes shall have compressive strength which meets ASTM D 2412 standards. Acceptable Manufacturers: Advanced Drainage Systems (ADS) or Owner-Approved equivalent manufacturer.
- D. Concrete Brick Units: ANSI/ASTM C55, Grade N Type I-Moisture Controlled, normal weight, of same Grade, Type and weight as block units, nominal modular size of 3 5/8 x 7 7/8 x 2 1/4 inches.
- E. Mortar and Grout: Mortar for finishing and sealing shall be Class "C". Honeycombing less than two (2) inches deep shall be repaired using Class "D" mortar.
- F. Brick Transition Reinforcement: Formed steel 8 gage wire with galvanized finish.

### **2.02 COMPONENTS**

- A. Lid and Frame: ASTM A48, Class 30B Heavy Duty Cast iron construction, machined flat bearing surface, removable lid, closed or open as specified on plans; sealing gasket; manufactured by Neenah Foundry Company or approved equal.
- B. Manhole Steps: Neenah Foundry Company catalog No. R-1982-F for precast or catalog No. R-1980-0 for brick/cast-in-place manholes or M.A. Industries PS-1 or approved equal.
- C. Base Pad: Cast-in-place concrete as specified in Section 03300.

### **2.03 CONFIGURATION**

- A. Barrel Construction: Concentric with eccentric cone top section.
- B. Shape: Cylindrical.
- C. Clear Inside Dimensions: 48 inch diameter or as indicated on plans.
- D. Design Depth: As indicated on plans.
- E. Clear Lid Opening: 22 inches minimum.
- F. Pipe Entry: Provide openings as indicated on plans.
- G. Main and Lateral Pipes: Neatly cut off main and lateral pipes flush with inside of manhole or inlet where they enter structure walls and point up irregularities and rough edges with nonshrinking grout.
- H. Inverts: Shape inverts for smooth flow across structure floor as shown on Drawings. Use concrete and mortar to obtain proper grade and contour and finish surface with fine textured wood float.

## **PART 3 EXECUTION**

### **3.01 EXAMINATION**

- A. Verify items provided by other sections of Work are properly sized and located.
- B. Verify that built-in items are in proper location, and ready for roughing into work.
- C. Verify excavation for manholes is correct.

### **3.02 PREPARATION**

Coordinate placement of inlet and outlet pipe or duct sleeve as indicated on plans.

### **3.03 PLACING PRE-CAST MANHOLE BARREL SECTIONS**

- A. Place base pad to proper elevation and location and trowel top surface level for placement of manhole barrel.
- B. Place manhole barrel plumb and level to correct elevations and anchor to base pad.
  - 1. After completion of slab foundation the first joint of manhole barrel shall be lowered into position-grooved end first, and set level and plumb on concrete base. Align and adjust to proper grade prior to placing and forming invert which shall be poured immediately after setting of first section of manhole barrel.
  - 2. Prior to setting subsequent manhole barrel sections, apply primer to tongue and groove ends and allow to set in accordance with manufacturer recommendations. Place "Ram-neck", or equivalent, plastic rope on tongue end. Lower next section into position, and remove excess material from interior of structure. Add additional material on exterior of joint, if necessary, for completely watertight joint.

### **3.04 MASONRY MANHOLE BARREL CONSTRUCTION**

- A. Maintain masonry courses to uniform dimension. Form vertical and horizontal joints of uniform thickness.
- B. Lay masonry units in running bond. Course 3 brick units and 3 mortar joints to equal 8



inches.

- C. Form flush mortar joints.
- D Lay masonry units in full bed of mortar, with full head joints, uniformly jointed with other work.
- E Install joint reinforcement 16 inches o.c.
- F.Place joint reinforcement in first and second horizontal joints above base pad and below lid frame opening.
- G. As work progresses, build-in fabricated metal items.
- H. Cut and fit masonry for pipes in accordance with 2.03-6 of this Section.
- I. Set cover frames and covers level without tipping, to correct elevations.

**END OF SECTION 02605**

## **SECTION 02660**

### **WATER DISTRIBUTION SYSTEMS**

#### **Part 1 GENERAL**

##### **1.01 SECTION INCLUDES**

Providing labor, materials, services, equipment, and other necessary items required for the construction of water systems. This shall include, but not be limited to the following: pipe and fittings for site water line including domestic water line and fire sprinkler system water line, valves and fire hydrants, setting line locations, elevations, and grade for water distribution systems work and control system for duration of work including careful maintenance of benchmarks, property corners, monuments, or other reference points.

##### **1.02 RELATED SECTIONS**

- A. Section 02315 - Excavating, Backfill and Compacting for Utilities.
- B. Section 02227 - Aggregate Materials.
- C. Section 02605 - Sewer Structures.
- D. Section 15300 - Fire Protection . (See Architectural/Building Specifications)
- E. Local Governing Authority and Code Requirements.
- F. All Necessary Construction Permits.
- G. Construction Drawings.

##### **1.03 REFERENCES**

- A. AASHTO T180 - Moisture-Density Relations of Soils Using a 10-lb (4.54 kg.) Rammer and an 18-in (457 mm) Drop.
- B. ANSI/ASME B16.18 - Cast Copper Alloy Solder Joint Pressure Fittings.
- C. ANSI/ASTM B16.22 - Wrought Copper and Cooper Alloy Solder Joint Pressure Fittings.
- D. ANSI/ASTM D1577 - Test Methods for Moisture-Density Relations of Soils and Soil-Aggregate Mixtures Using 10 lb (4.54 kg.) Rammer and 18 inch (457 mm) Drop.
- E. ANSI/ASTM D2466 - Polyvinyl Chloride (PVC) Plastic Pipe Fittings, Schedule 40.
- F. ANSI/AWS A5.8 - Brazing Filler Metal.
- G. ANSI/AWWA C104 - Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water.
- H. ANSI/AWWA C105 - Polyethylene Encasement for Ductile Iron Piping for Water and Other Liquids.

- I. ANSI/AWWA C111 - Rubber-Gasket Joints for Ductile Iron and Grey-Iron Pressure Pipe and Fittings.
- J. ANSI/AWWA C151 - Ductile-Iron Pipe, Centrifugally Cast in Metal Molds or Sand-Lined Molds, for Water or Other Liquids.
- K. ANSI/AWWA C500 - Gate Valves, 3 inch through 48 inch NPS, for Water and Sewage Systems.
- L. ANSI/AWWA C502 - Dry Barrel Fire Hydrants.
- M. ANSI/AWWA C504 - Rubber Seated Butterfly Valves.
- N. ANSI/AWWA C508 - Swing-Check Valves for Waterworks Service, 2 inch through 24 inch NPS.
- O. ANSI/AWWA C509 - Resilient seated Gate Valves 3 inch through 12 inch NPS, for Water and Sewage Systems.
- P. ANSI/AWWA C600 - Installation of Ductile-Iron Water Mains and Appurtenances.
- Q. ANSI/AWWA C606 - Grooved and Shouldered Type Joints.
- R. ANSI/AWWA C651 - Disinfecting Water Mains.
- S. ANSI/AWWA C900 - Standard for Polyvinyl Chloride (PVC) Pressure Pipe, 4 inch through 12 inch, for Water.
- T. ASTM D1785 - Seamless Copper water Tube.
- U. ASTM B88 - Polyvinyl Chloride (PVC) Plastic Pipe, Schedules 40, 80, and 120.
- V. ASTM D2241 - Polyvinyl Chloride (PVC) Plastic Pipe (SDR-PR).
- W. ASTM D2855 - Making Solvent-Cemented Joints with Polyvinyl Chloride (PVC) Pipe and Fittings.
- X. ASTM D2922 - Test Methods for Density of Soil and Soil - Aggregate in Place by Nuclear Methods (Shallow Depth).
- Y. ASTM D3017 - Test Methods for Moisture Content of Soil and Soil-Aggregate Mixtures.
- Z. ASTM D3139 - Joints for Plastic Pressure Pipes using Flexible Elastomeric Seals.
- AA. ASTM D3035 - Polyethylene (PE) Plastic Pipe (SDR-PR) Based on Controlled Outside Diameter.
- BB. AWWA C901 - Polyethylene (PE) Pressure Pipe, Tubing, and Fittings, 1½ inch through 3 inch, for water.
- CC. AWWA C600-82 - Hydrostatic Testing
- DD. UL 246 - Hydrants for Fire Protection Service.

#### **1.04 SUBMITTALS**

- A. Product Data: Provide Engineer with data on pipe materials, pipe fittings, hydrants, valves and accessories.
- B. Manufacturer's Certificate: Certify that products meet or exceed state or local requirements.

#### **1.05 PROJECT RECORD DOCUMENTS**

- A. Accurately record actual locations of piping mains, valves, connections, and top of pipe elevations.
- B. Identify and describe unexpected variations to subsoil conditions and location of uncharted utilities.

#### **1.06 QUALITY ASSURANCE**

- A. Perform work in accordance with utility company and/or municipality requirements.
- B. Valves: Manufacturer's name and pressure rating marked on valve body.

### **PART 2 PRODUCTS**

#### **2.01 PIPE**

- A. Pipe sizes less than 3 inches that are installed below grade and outside building shall comply with one or combination of following:
  - 1. Seamless Copper Tubing: Type "K" soft copper to comply with ASTM B 88-62 and installed with wrought copper (95-5 Tin Antimony solder joint) fittings in accordance with ASTM B16.22.
  - 2. Polyvinyl Chloride (PVC) Water Pipe: Pipe shall conform to ASTM D-2241 with an SDR 21 rating and shall be continually marked with manufacturer's name, pipe size, cell classification, SDR rating, and ASTM D 1784 classification. Pipe joints shall be integrally molded bell ends in accordance with ASTM D 3139 with factory supplied elastomeric gaskets and lubricant.
- B. Pipe sizes 3 inch and larger than are installed below grade and outside building shall comply with one of the following:
  - 1. Gray Cast Iron Water Pipe: In accordance with ANSI A21.6, thickness class 22, and pressure class 150. Fittings shall be either mechanical joint or push-on joint and shall comply with ANSI A21.10 or ANSI A21.11.
  - 2. Ductile Iron Water Pipe: In accordance with ANSI A21.51, Fittings shall be either mechanical joint or push-on joint complying with ANSI A21.10 or ANSI 21.11 (AWWA C-151) (CLASS 50).
  - 3. Polyvinyl Chloride (PVC) Water Pipe: Pipe shall meet the

requirements of AWWA C-900 and comply with ASTM D 2241, rated SDR 21 (Class 150). Pipe shall be continually marked as for smaller pipes. Pipe joints shall be integrally molded bell ends in accordance with ASTM D 3034, Table 2, with factory supplied elastomeric gaskets and lubricant.

#### **2.04 GATE VALVES - 2 Inches and Larger**

- A. Manufacturers:  
Mueller Resilient Seat Gate Valves or approved equal.
- B. ANSI/AWWA C509, Iron body, bronze mounted double disc, parallel seat type, non-rising stem with square nut, single wedge.

Resilient seat, flanged or mechanical joint ends, control rod, post indicator where indicated on plans, extension box and valve key.

#### **2.05 BALL VALVES - 2 Inches and Smaller**

- A. Manufacturers: Mueller Oriseal or approved equal.
- B. Brass body, teflon coated brass ball, rubber seats and stem seals, Tee stem pre-drilled for control rod, AWWA compression inlet end, compression outlet with electrical ground connector, with control rod, extension box and valve key.

#### **2.06 BUTTERFLY VALVES - From 2 inches to 24 inches**

- A. ANSI/AWWA C504, iron body, bronze disc, resilient replaceable seat, water or lug ends, infinite position lever handle.
- B. Check valves, post indicator valves and backflow preventors - Refer to Section 15300 (Fire Protection) in the Architectural/Building Specifications

#### **2.08 HYDRANT**

- A. Hydrant: Type as required by utility company and as shown on plans.
- B. Hydrant Extensions: Fabricate in multiples of 6 inches with rod and coupling to increase barrel length.
- C. Hose and Steamer Connection: Match sizes with utility company, two hose nozzles, one pumper nozzles.
- D. Finish: Primer and two coats of enamel or special coating to color as required by utility company.

#### **2.09 ACCESSORIES**

- A. Concrete for Thrust Blocks: Place thrust blocking consisting of 2,500 psi concrete to provide sufficient bearing area to transmit unbalanced thrust from bends, tees, caps, or plugs to undisturbed soil without loading undisturbed soil in excess of 2,500 lbs/sq ft when water main pressure is 100 psi.

#### MINIMUM THRUST BLOCKING BEARING AREAS

| Pipe<br>Diameter | Tees<br>Sq.Ft. | Sq.Ft. | 90 Bend<br>Sq.Ft. | 45 Bend<br>Sq.Ft. | Sq.Ft. | 22 ½ Bend | 11 ¾ Bend |
|------------------|----------------|--------|-------------------|-------------------|--------|-----------|-----------|
| 3"               | 1.0            |        | 1.0               | 1.0               |        | 1.0       | 1.0       |
| 4"               | 1.0            |        | 1.0               | 1.0               |        | 1.0       | 1.0       |
| 6"               | 1.5            |        | 2.0               | 1.0               |        | 1.0       | 1.0       |
| 8"               | 2.5            |        | 3.5               | 1.8               |        | 1.0       | 1.0       |
| 10" 4.0          | 5.5            |        | 2.8               | 1.5               |        | 1.0       |           |
| 12" 6.0          | 8.0            |        | 4.0               | 2.0               |        | 1.5       |           |
| 14" 8.0          | 11.0           |        | 5.5               | 3.0               |        | 2.0       |           |
| 16" 10.0         | 14.2           |        | 7.0               | 4.0               |        | 3.0       |           |
| 18" 21.0         | 21.0           |        | 12.0              | 6.0               |        | 4.0       |           |

### PART 3 EXECUTION

#### 3.01 EXAMINATION

- A. Verify existing conditions.
- B. Verify that building service connection and municipal utility water main size, location and depth are as indicated.

#### 3.02 PREPARATION

- A. Ream pipe and tube ends and remove burrs.
- B. Remove scale and dirt, on inside and outside, before assembly.
- C. Prepare pipe for connections to equipment with flanges or unions.

#### 3.03 BEDDING

- A. Excavate pipe trench and place bedding material in accordance with Section 02222 for work of this Section.

#### 3.04 INSTALLATION - PIPE AND FITTINGS

- A. Maintain separation of water main from sanitary and storm sewer piping in accordance with state or local code.
- B. Install pipe and fittings in accordance with ANSI/AWWA C600.
- C. Install pipe to allow for expansion and contraction without stressing pipe or joints or as specified by pipe manufacturer.
- D. Install access fittings in accordance with local codes to permit disinfection of water system performed under this section.
- E. Connection with Existing Pipelines: Where connections are made between new work and existing piping, make connection using suitable fittings for conditions encountered. Make each connection with existing pipe at time and under conditions which least interfere with operation of existing pipeline and in compliance with the local utility company.

- F. Form and place concrete for thrust blocks or other specified methods of retainage at each change of direction or end of pipe main.
- G. Establish elevations of buried piping in accordance with Section 02222 for work in this section.
- H. Backfill trench in accordance with Section 02222.

### **3.05 INSTALLATION - VALVES AND HYDRANTS**

- A. Install gate valves as indicated on Drawings and supported on concrete pads with valve stem vertical and plumb. Install valve boxes in a manner that will not transmit loads, stress, or shock to valve body. Center valve box over operating nut of valve vertical and plumb. Securely fit valve box together leaving cover flush with finished surface.
- B. Install fire hydrant assemblies as indicated on Drawings in vertical and plum position with steamer/pumper nozzle pointed perpendicular to traffic where hydrant is adjacent to a street, roadway or parking lot drive or toward the protected building unless otherwise directed by local authorities. Support hydrant assembly on concrete pad and firmly braced on side opposite inlet pipe against undisturbed soil and concrete blocking. Place minimum of 6 cu. ft. Of crushed stone or gravel around hydrant base and barrel after thrust blocking has cured at least 24 hours. Exercise care when backfilling and compacting so proper vertical position will not be altered.

### **3.06 DISINFECTION OF DOMESTIC WATER PIPING SYSTEM**

Disinfect distribution system with chlorine before acceptance for domestic operation. Amount of chlorine shall be such as to provide dosage of not less than 50 parts/million. Thoroughly flush lines before introduction of chlorinating materials and after contact period of not less than 24 hours, system shall Be flushed with clean water until residual chlorine content is not greater than 1.0 part/million. Open and close valves in lines being disinfected several times during contact period. After disinfection, take water sample and bacteriologically test in accordance with AWWA specifications. Do not place distribution system in service until approval is obtained from applicable governing authorities.

### **3.07 SERVICE CONNECTIONS**

Provide water service connection in compliance with utility company requirements including reduced pressure backflow preventer if required and water meter with by-pass valves and sand strainer.

### **3.08 FIELD QUALITY CONTROL**

- 1. Compaction testing of trench backfill shall be performed in accordance with Section 02315.
- 2. Water distribution system pipe installed below grade and outside building shall be tested in accordance with following procedures:
- 3. The Contractor shall perform the testing of pipe materials, joints, and/or other materials incorporated into the construction of water mains and force mains to determine leakage and watertightness. All pressure pipeline shall be tested in accordance with Section 4 of AWWA C600-82. In the event any state or local code requires a more stringent test, the more stringent shall apply.
- 4. Pressure Test:

After the pipe has been laid, all newly laid pipe or any valved section thereof shall be subjected to a hydrostatic pressure of at least 1.5 times the working pressure at the point of testing and not less than 1.25 times the working pressure at the highest point along the test section.

1. Leakage Test:

The leakage test shall be conducted concurrently with the pressure test. Leakage is defined as the quantity of water that must be supplied into the newly laid pipeline, or any valved section thereof, to maintain pressure within 5 psi of the specified test pressure after the air in the pipeline has been expelled and the pipeline has been filled with water. Leakage shall not be measured by a drop in pressure in a test section over a period of time.

No pipeline installation will be accepted if the leakage is greater than that determined by the following formula:

$$L = \frac{SDP}{133200}$$

L = allowable leakage, (gallons per hour)

S = length of pipe tested, (feet)

D = nominal diameter of pipe, (inches)

P = average test pressure during test, (psig)

2. Visible Leakage:

All visible leaks shall be repaired regardless of the amount of leakage.

3. Acceptance of Installation:

If any test of pipe laid in place discloses leakage greater than that specified, the Contractor shall, at his own expense, locate the leak and make repairs as necessary until the leakage is within the specified allowance.

Contractor shall supply all water for testing at no expense to the Owner.

Contractor shall furnish one copy of results of meter test and hydrostatic pressure test to the Owner and utility company upon completion of water distribution backfilling operations.

**END OF SECTION 02660**



## **SECTION 02720**

### **STORM SEWER SYSTEMS**

#### **Part 1 GENERAL**

##### **1.01 SECTION INCLUDES**

1. Site storm sewerage drainage piping, fittings and accessories, and bedding.
2. Connection of building storm water drainage system to municipal sewers.
3. Catch basins, paved area drainage, site surface drainage, and storm water detention facilities.

##### **1.02 RELATED REQUIREMENTS**

1. Section 02315 - Excavation, Backfill, and Compacting for Utilities
2. Section 01570 - Erosion and Sediment Control
3. Section 02605 - Sewer Structures
4. Section 02730 - Sanitary Sewerage System
5. Section 03300 -Cast-In-Place Concrete: (See  
Architectural/Building Specifications)
6. Local governing authority and code requirements.
7. All necessary construction permits.
8. Construction Drawings

##### **1.03 REFERENCES**

1. AASTHO M294 and M252 - Corrugated Polyethylene pipe smooth interior.
2. AASHTO M36 - Metallic (Zinc or Aluminum) Coated Corrugated Steel Culverts and Underdrains.
3. AASHTO T180 - Moisture-Density Relations of Soils Using a 10-lb (4.54 kg) Rammer and an 18-in. (457 mm) drop.
4. ANSI/ASTM A74 - Cast Iron Soil Pipe and Fittings.
5. ANSI/ASTM C12 - Practice for Installing Vitrified Clay Pipe Lines.
6. ANSI/ASTM C14 - Concrete Sewer, Storm Drain, and Culvert Pipe.
7. ANSI/ASTM C76 - Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe.
8. ANSI/ASTM C425 - Compression Joints for Vitrified Clay Pipe and Fittings.
9. ANSI/ASTM C443 - Joints for Circular Concrete Sewer and Culvert Pipe, Using Rubber Gaskets.
10. ASTM C564 - Rubber Gaskets for Cast Iron Soil Pipe and Fittings

11. ANSI/ASTM D1557 - Test Methods for Moisture-Density Relations of Soils and Soil-Aggregate Mixtures Using 10 lb (4.54 kg) Rammer and 18 inch (457 mm) Drop.
12. ANSI/ASTM D3034 - Type PSM Polyvinyl Chloride (PVC) Sewer Pipe and Fittings.
13. ASTM C700 - Vitrified Clay Pipe, Extra Strength, Standard Strength and Perforated.
14. ASTM D2922 - Test Methods for Density of Soil and Soil – Aggregate in Place by Nuclear Methods (Shallow Depth)
15. ASTM D3017 - Test Methods for Moisture Content of Soil and Soil-Aggregate

#### **1.04 DEFINITIONS**

Bedding: Fill placed under, beside and directly of pipe runs, prior to subsequent backfill operations.

#### **1.05 PROJECT RECORD DOCUMENTS**

1. Accurately record actual locations of pipe runs, connections, catch Basins, cleanouts, and invert elevations.
2. Identify and describe unexpected variations to subsoil conditions and location of uncharted utilities.

#### **1.06 COORDINATION**

Coordinate the Work with termination of storm sewer connection outside building including connection to municipal storm sewer system.

### **Part 2 PRODUCTS**

#### **2.01 SEWER PIPE MATERIALS AND ACCESSORIES**

1. Reinforced Concrete Pipe: Comply with requirements of ASTM C 76, Class III unless another class type is indicated on Drawings, installed with flexible plastic (Bitumen) gaskets at all joints. Gaskets shall comply with AASHTO M-198 75I, Type B, and shall be installed in strict accordance with pipe manufacturer's recommendations.
2. Corrugated Steel Pipe: Galvanized aluminized or bituminous coated as specified on the drawings. Only permitted when specifically indicated on Drawings and shall comply with requirements of ASTM A 760; 16 gage unless another gage is indicated on Drawings. Install with matching band connectors. Install sleeve gaskets in accordance with pipe manufacturer's recommendations. Corrugated steel pipe may be round pipe, arch pipe, or slotted drain pipe as indicated on Drawings. Slotted drain pipe shall have 1.75" wide drain guide waterway openings and 6" minimum height drain guide.
3. Spiral Rib Metal Pipe Type 1R: Galvanized, aluminized or bituminous coated as specified on Drawings. Only permitted when specifically indicated on Drawings. Pipe and shall be re-corrugated and installed with semi-corrugated Hugger-type bands and "O" ring gaskets in accordance pipe manufacturers installation requirements Spiral Rib metal pipe must comply with ASTM A 760 Type 1R. Acceptable manufacturer: CONTECH, INC. "ULTRA FLO or ULTRA FLO II" and Caldwell Culvert Co. "Smooth Cor" or approved equal.
4. Polyvinyl Chloride (PVC) Pipe: Only permitted when specifically indicated on Drawings. Pipe and fittings shall comply with ASTM D 3034, rated SDR 35. Pipe shall be continually marked with manufacturer's name, pipe size, cell classification, SDR rating,

and ASTM D 3034 classification. Pipe joints shall be integrally molded bell ends in accordance with ASTM D 3034, Table 2, with factory supplied elastomeric gaskets and lubricant.

5. Corrugated Polyethylene Pipe (CPP) Smooth Interior: Only permitted when specifically indicated on Drawings and shall conform with AASHTO Designation M294 and M252. Pipe must be installed in accordance with pipe manufacturers installation Guidelines for Culvert Storm Drainage Applications. Acceptable manufacturers: Advanced Drainage Systems, Inc. "ADS N-12" & HANCOR, INC. "Hi-G" or approved equal.
6. Polyvinyl Chloride (PVC) large diameter closed profile gravity sewer pipe, UNL-B-9: Only permitted when specifically indicated on drawings. Pipe and fittings shall comply with ASTM F-794. Pipe must be installed in accordance with pipe manufacturers installation guidelines. Acceptable manufacturer: Extrusion Technologies, Inc. "Ultra-Rib" or approved equal.
7. Polyvinyl Chloride (PVC) 8" through 30" diameter, smooth interior, open profile gravity sewer pipe. Only permitted when specifically indicated on drawings. Pipe and fittings shall comply with ASTM F-94 and Uni-Bell Uni-B-9. Pipe must be installed in accordance with pipe manufacturers installation guidelines. Acceptable manufacturer: Extrusion Technologies, Inc. "Ultra-Rib" or approved equal.

## **2.02 INLETS, CATCH BASINS AND JUNCTION BOXES**

1. Lid and frame per details shown on plans.
2. Structure construction in accordance with details shown on plans.

## **PART 3 EXECUTION**

### **3.01 EXAMINATION**

Verify that trench cut and excavation is ready to receive work and excavations, dimensions, and elevations are as indicated on civil engineering drawings.

### **3.02 PREPARATION**

1. Had trim excavations to required elevations. Correct over excavation with fine aggregate.
2. Remove large stones or other hard matter which could damage piping or impede consistent backfilling or compaction.

### **3.03 BEDDING**

1. Excavate pipe trench and place bedding material in accordance with Section 02222 for work of this Section.

### **3.04 INSTALLATION - PIPE**

1. Install pipe, fittings, and accessories in accordance with ASTM C12, ASTM D2321 or manufacturer's instructions and state or local requirements.
2. Install pipe on bedding in accordance with Section 02222 for work in this Section.
3. Lay pipe to slope gradients noted on construction drawings.
4. Refer to Section 02222 for trenching requirements. Do not displace or damage pipe when compacting.

5. Refer to Section 02605 for manhole requirements.

**3.05 INSTALLATION - CATCH BASINS, INLETS AND JUNCTION BOXES**

1. Form bottom of excavation clean and smooth to correct elevation.
2. Form and place cast-in-place concrete base pad, with provision for storm sewer pipe to placed at proper elevation.
3. Form and place cast-in-place concrete walls, sleeved at proper elevation to receive storm sewer pipe in accordance with details shown on the plans.
4. Form and place cast-in-place top of structure in accordance with details shown on the plans.

**END OF SECTION 02720**

## SECTION 02730

### SANITARY SEWER SYSTEMS

#### 1.1 RELATED DOCUMENTS

Construction drawings, specifications, and general provisions of Contract, including General and Supplementary Conditions and Specification sections, apply to work of this section.

#### 1 PART 1 - GENERAL

##### 1.2 DESCRIPTION OF WORK

1. Extent of sanitary sewage systems work is indicated on drawings and schedules, and by requirements of this section.
2. Refer to Division-2 section "Earthwork" for excavation and backfill required for sanitary sewage systems; not work of this section.
3. Refer to Division-15 section 15400 for interior building systems including drain, waste, and vent piping; not work of this section.

##### 1.3 QUALITY ASSURANCE

1. Manufacturer's Qualifications: Firms regularly engaged in manufacture of sanitary sewage system's products of types, materials, and sizes required, whose products have been in satisfactory use in similar service for not less than 5 years.
2. Plumbing Code Compliance: Comply with applicable portions of National Standard Plumbing Code pertaining to selection and installation of sanitary sewage system materials and products.

##### 1.3.3 SUBMITTALS

1. Record Drawings: At project closeout, submit record drawings of installed sanitary sewage piping and products, in accordance with requirements of Division 1.
2. Maintenance Data: Submit maintenance data and parts lists for sanitary sewage system materials and products. Include this data, product data, shop drawings, and record drawings in maintenance manual; in accordance with requirements of Division 1.

#### PART 2 - PRODUCTS

##### 2.1 IDENTIFICATION

1. Underground-Type Plastic Line Markers: Manufacturer's standard permanent, bright-colored, continuous-printed plastic tape, intended for direct-burial service; not less than 6" wide x 4 mils thick. Provide green tape with black printing reading "CAUTION SEWER LINE BURIED BELOW".
2. Manufacturer: Subject to compliance with requirements, provide identification markers of one of the following:
  - Allen Systems, Inc.
  - Emed Co., Inc.

Seton Name Plate Corp.

## **2.2 PIPES AND PIPE FITTINGS**

1. General: Provide pipes of one of the following materials, of weight/class indicated. Provide pipe fittings and accessories of same material and weight/class as pipes, with joining method as indicated.
2. Cast-Iron Soil Pipe: ASTM A 74, hub and spigot ends, service weight unless otherwise indicated.
3. Fittings: Cast-iron hub and spigot complying with ASTM A 74; lead/oakum calked joints, or compression joints with rubber gaskets complying with ASTM C 564.
4. Copper Tube: ASTM B 306, hard temper, Type DWV.
5. Fittings: Cast-copper alloy solder-joint drainage fittings, Type DWV, ANSI B16.23; or wrought-copper and wrought-copper-alloy solder-joint drainage fittings, ANSI B16.29; solder joints.
6. Polyvinyl Chloride (PVC) DWV Pipe: Schedule 40, ASTM D 2665.
7. Fittings: PVC Schedule 40, ASTM D 2665; solvent-cement joints, ASTM D 2664; or threaded joints.

### **2.2.8 CLEANOUTS**

1. General: Provide as indicated, pipe extension to grade with furrule and countersunk cleanout plug. Provide round cast-iron access frame over cleanout, with heavy-duty secured scoriated cover with lifting device.

## **PART 3 - EXECUTION**

### **3.1 INSTALLATION OF IDENTIFICATION**

1. General: During back-filling/top-soiling of sanitary sewage systems, install continuous underground-type plastic line marker, located directly over buried line at 6" to 8" below finished grade.

### **3.2 INSTALLATION OF PIPE AND FITTINGS**

1. General: Install piping in accordance with governing authorities having jurisdiction, except where more stringent requirements are indicated.
2. Inspect piping before installation to detect apparent defects. Mark defective materials with white paint and promptly remove from site.
3. Lay piping beginning at low point of system, true to grades and alignment indicated, with unbroken continuity of invert.

4. Place bell ends or groove ends of piping facing upstream.
5. Install gaskets in accordance with manufacturer's recommendations for use of lubricants, cements, and other special installation requirements.
6. Cast-Iron Soil Pipe: Install in accordance with applicable provisions of CISPI "Cast Iron Soil Pipe & Fittings Handbook".
7. Copper Tube: Install in accordance with applicable provisions of CDA "Copper Tube Handbook".
8. Plastic Pipe: Install in accordance with manufacturer's installation recommendations, and in accordance with ASTM D 2321.
9. Cleaning Piping: Clear interior of piping of dirt and other superfluous material as work progresses. Maintain swab or drag in line and pull past each joint as it is completed. Place plugs in ends of uncompleted conduit at end of day or whenever work stops. Flush lines between manholes if required to remove collected debris.
10. Joint Adapters: Make joints between different types of pipe with standard manufactured adapters and fittings intended for that purpose.

### **3.3 TAP CONNECTIONS**

1. Make connections to existing piping and underground structures, so that finished work will conform as nearly as practicable to requirements specified for new work.
2. Use commercially manufactured wyes for branch connections. Field cutting into piping will not be permitted. Spring wyes into existing line and encase entire wye, plus 6" overlap, with not less than 6" of 3,000 psi 28-day compressive strength concrete.
3. Branch connections made from side into existing 4" to 21" piping shall have wye sprung into existing line, and entire wye encased with not less than 6" of 3,000 psi 28-day compressive strength concrete.
4. For branch connections from side into existing 24" or larger piping or to underground structures, cut opening into unit sufficiently large to allow 3" of concrete to be packed around entering connection. Cut ends of connection passing through pipe or structure wall to conform to shape of, and be flush with, inside wall, unless otherwise indicated. On outside of pipe or structure wall, encase entering connection in 6" of concrete for a minimum length of 12" to provide additional support or collar from connection to undisturbed ground.

Provide concrete which will attain minimum 28-day compressive strength of 3,000 psi, unless otherwise indicated. Use epoxy bonding compound as interface between new and existing concrete and piping materials.

5. Take care while making tap connections to prevent concrete or debris from entering existing piping or structure. Remove debris, concrete, or other extraneous material which may accumulate.

### **3.4 BACKFILLING**

1. General: Conduct backfilling operations of open-cut trenches closely following laying, jointing, and bedding of pipe, and after initial inspection and testing are completed.
2. To minimize local area traffic interruptions, allow no more than 100' between pipe laying and point of complete backfilling.

#### **3.4.3 FIELD QUALITY CONTROL**

1. Testing: Perform testing of completed piping in accordance with local authorities having jurisdiction.

**END OF SECTION 02730**



## **SECTION 02846**

### **SIGNAGE**

#### **PART 1 GENERAL**

##### **1.01 SECTION INCLUDES**

Traffic control signs complying with U.S. Department of Transportation, Federal Highway Administration's Manual "Uniform Traffic Control Devices" and as specified. See Construction Drawings for type, location, and quantity of sign required.

##### **1.02 RELATED REQUIREMENTS:**

1. Construction Drawings
2. Manufacturer's mounting instructions

#### **PART 2 PRODUCTS**

##### **2.01 ACCEPTABLE MANUFACTURER**

Signs to be equivalent to those manufactured by SA-SO, Inc., Grand Prairie, Tx.

##### **2.02 SIGNS**

1. "STOPS" Signs: 24" x 24", Octagon, reflectorized copy and border.
2. "SPEED LIMIT 15" Signs: 12" x 24" , black legend on white reflective baked enamel background.
3. "HANDICAPPED SYMBOL" Signs: 12" x 18", White legend on blue reflective baked enamel background.
4. "NO PARKING, FIRE LANES" Signs: 12" x 18", red letters on white reflective or baked enamel background. (R7-8)
5. "KEEP RIGHT" Signs: 18" x 24", black letters and symbol on white reflective or baked enamel background. (R7-1)
6. "DO NOT ENTER" Signs: Highway Dept. Standard red and white sign except 24" x 24" size with reflective baked enamel finish.

##### **2.03 POSTS**

Round galvanized posts with galvanized sign-mounting hardware for each sign.

#### **PART 3 EXECUTION**

Install posts in 18" round x 24" deep concrete foundations. Set posts vertical and plumb with bottom of sign at 6'-5" above finish grade. Mount signs in accordance with manufacturer's instructions.

**END OF SECTION 02846**

**SECTION 02110**

**SITE CLEARING AND GRUBBING**

**PART 1 - GENERAL**

**RELATED DOCUMENTS**

Drawings and general provisions of Contract, including General and Supplementary Conditions and Divisions 1 Specifications Sections, apply to this section.

**SUMMARY**

This Section Includes the Following:

**Clearing and grubbing.**

**PROJECT CONDITIONS**

Protection of Existing Improvements: Provide protections necessary to prevent damage to existing improvements indicated to remain in place.

Protect improvements on adjoining properties and on Owner's property.

Restore damaged improvements to their original condition.

**PART 2 - PRODUCTS**

Not applicable to this section

**PART 3 - EXECUTION**

**SITE CLEARING**

General: Remove grass and other vegetation, improvements, or obstructions as required to permit installation of new construction. Remove similar items elsewhere on site or premises as specifically indicated. "Removal" includes digging out and off-site disposing of stumps and roots.

Fill depressions caused by clearing and grubbing operations with satisfactory soil material, unless further excavation or earthwork is indicated. Refer to other sections in this division relating to fill requirements.

**DISPOSAL OF WASTE MATERIALS**

Removal from Owner's Property: Remove waste materials and unsuitable or excess topsoil from Owner's properties.

**END OF SECTION 02110**

## SECTION 02200

### EARTHWORK

#### PART 1 - GENERAL

##### RELATED DOCUMENTS

Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this section.

##### SUMMARY

This Section includes the following:

Preparing of subgrade for building slabs, walks, and pavements.

Excavating and backfilling of trenches within building lines.

Excavating and backfilling for underground mechanical and electrical utilities and buried mechanical and electrical appurtenances within building lines.

##### SUBMITTALS

Test Reports: Submit the following reports directly to Architect from the testing services, with copy to Contractor:

Test reports on borrow material.

Field Reports; in-place soil density test.

One optimum moisture-maximum density curve for each type of soil encountered.

Report of percent compaction and moisture content of each strata tested.

##### QUALITY ASSURANCE

Codes and Standards: Perform excavation work in compliance with applicable requirements of authorities having jurisdiction.

Testing and Inspection Service: Owner will employ and pay for a qualified independent geotechnical testing and inspection laboratory to perform soil testing and inspection service during earthwork operations.

Testing Laboratory Qualifications: To qualify for acceptance, the geotechnical testing laboratory must demonstrate to Architect's satisfaction, based on evaluation of laboratory-submitted criteria conforming to ASTM E 699, that it has the experience and capability to conduct required field and laboratory geo-technical testing without delaying the progress of the work.

##### PROJECT CONDITIONS

Existing Utilities: Locate existing underground utilities in areas of excavation work. If utilities are indicated to remain in place, provide adequate means of support and protection during earthwork operations.

Should uncharted, or incorrectly charted, piping or other utilities be encountered during excavation, consult utility owner immediately for directions. Cooperate with Owner and utility companies in keeping respective services and facilities in operation. Repair damaged utilities to satisfaction of utility owner.

Use of Explosives: Use of explosives is not permitted.

Protection of Person and Property: Barricade open excavations occurring as part of this work and post with warning lights.

Operate warning lights as recommended by authorities having jurisdiction.

Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.

Perform excavation by hand within drip line of large trees to remain. Protect root systems from damage or dryout to the greatest extent possible. Maintain moist condition for root system and cover exposed roots with moistened burlap.

## **PART 2- PRODUCTS**

### **SOIL MATERIALS**

Select fill: Fill under all floor slabs and walks when properly slaked and tested by a qualified testing laboratory shall meet the following requirements:

1. Refer to Structural Plans

Samples of proposed select fill shall be furnished to the testing laboratory a minimum of 7 days prior to installation to permit time for specification compliance, inspection and approval.

Backfill and Fill Materials: Satisfactory soil materials free of clay, rock or gravel larger than 2 inches in any dimension, debris, waste, frozen materials, vegetation and other deleterious matter.

## **PART 3 - EXECUTION**

### **EXCAVATION**

Excavation is unclassified and includes excavation to subgrade elevations indicated, regardless of character of materials and obstructions encountered.

### **STABILITY OF EXCAVATIONS**

General: Comply with local codes, ordinances, and requirements of agencies having jurisdiction.

Slope sides of excavations to comply with local codes, ordinances, and requirements of agencies having jurisdiction. Shore and brace where sloping is not possible because of space restrictions or stability of material excavated. Maintain sides and slopes of excavations in safe condition until completion of backfilling.

### **DEWATERING**

Prevent surface water and subsurface or ground water from flowing into excavations and from flooding project site and surrounding area.

Do not allow water to accumulate in excavations. Remove water to prevent softening of foundation bottoms, undercutting footings, and soil changes detrimental to stability of subgrades and foundations. Provide and maintain pumps, well points, sumps, suction and discharge lines, and other dewatering system components necessary to convey water away from excavations.

Establish and maintain temporary drainage ditches and other diversions outside excavation limits to convey rain water and water removed from excavations to collecting or runoff areas. Do not use trench excavations as temporary drainage ditches.

### **STORAGE OF EXCAVATED MATERIALS**

Stockpile excavated materials acceptable for backfill and fill where directed. Place, grade, and shape stockpiles for proper drainage.

Located and retain soil materials away from edge of excavations. Do not store within drip line of trees indicated to remain.

Dispose of excess excavated soil material and materials not acceptable for use as backfill or fill.

### **EXCAVATION FOR STRUCTURES**

Conform to elevations and dimensions shown within a tolerance of plus or minus 0.10 foot, and extending a sufficient distance from footings and foundations to permit placing and removal of concrete formwork, installation of services, and other construction and for inspection.

Excavations for footing and foundations: Do not disturb bottom of excavation. Excavate by hand to final grade just before concrete reinforcement is placed. Trim bottoms to required lines and grades to leave solid base to receive other work.

Excavation for Grade Beams: Cut grade beams to the cross sections and grades as shown on the drawings. Deposit excavated materials away from the proposed building areas.

Excavation for Underground Tanks, Basins, and Mechanical or Electrical Structures: Conform to elevations and dimensions indicated within a tolerance of plus or minus 0.10 foot; plus a sufficient distance to permit placing and removal of concrete formwork, installation of services, and other construction and for inspection. Do not disturb bottom of excavations, intended for bearing surface.

### **TRENCH EXCAVATION FOR PIPES AND CONDUIT**

Excavate trenches to uniform width, sufficiently wide to provide ample working room and minimum of 6 to 9 inches of clearance on both sides of pipe or conduit.

Excavate trenches and conduit to depth indicated or required to establish indicated slope and invert elevations and to support bottom of pipe or conduit on undisturbed soil.

For pipes or conduit less than 6 inches in nominal size, and for flat-bottomed, multiple-duct conduit units, do not excavate beyond indicated depths. Hand-excavate bottom cut to accurate elevations and support pipe or conduit on undisturbed soil.

For pipes and equipment 6 inches or larger in nominal size, shape bottom of trench to fit bottom of pipe for 90 degrees (bottom 1/4 of the circumference). Fill depressions with tamped select fill. At each pipe joint, dig bell holes to relieve pipe bell of loads ensure continuous bearing of pipe barrel on bearing surface.

### **BACKFILL AND FILL**

General: Place soil material in layers to required subgrade elevations, for each area classification listed below, using materials specified in Part 2 of this Section.

Under grassed areas, use satisfactory excavated or borrow material.

Under walks and pavements, use select fill material.

Under steps, use select fill material.

Under building slabs, use select fill material.

### **PLACEMENT AND COMPACTION**

Remove at least 8 inches of top soil, vegetation, debris, etc., from the proposed building and paved areas to a distance of 3' - 0" outside the construction line.

Rework and compact the exposed subgrade under all floors per Plans.

Select Fill: Refer to Structural Plans.

Laboratory moisture-density curve or curves as required, and results of field density checks shall be submitted to the Architect or Engineer. A minimum of one (1) in place density test per 2000 square feet of slab are shall be taken on alternating lifts during placement of select fill.

No backfilling will be permitted where excavations for bearing under footings has been carried too deep. Such excavations shall be filled with concrete of the same class as the member it is to support.

### **GRADING**

General: Uniformly grade areas within limits of grading under this section, including adjacent transition areas. Smooth finished surface within specified tolerances, compact with uniform levels or slopes between points where elevations are indicated or between such points and existing grades.

Grading Outside Building Lines: Grades areas adjacent to building lines to drain away from structures and to prevent pounding. Finish surfaces free from irregular surface changes and as follows:

Lawn or Unpaved Areas: Finish areas to receive topsoil to within not more than 0.10 foot above or below required subgrade elevations.

Walks: Shape surface of areas under walks to line, grade, and cross-section, with finish surface not more than 0.10 foot above or below required subgrade elevation.

Pavements: Shape surface of areas under pavement to line, grade, and cross-section, with finish surface not more than 1/2 inch above or below required subgrade elevation.

### **FIELD QUALITY CONTROL**

Quality Control Testing During Construction: Allow testing service to inspect and approve each subgrade, base and fill layer before further backfill or construction work is performed.

If in opinion of Architect, based on testing service reports and inspection, subgrade or fills that have been placed are below specified density, perform additional compaction and testing until specified density is obtained. Contractor shall pay for such retesting.

**EROSION CONTROL:**

Provide erosion control methods in accordance with requirements of authorities having jurisdiction.

**MAINTENANCE**

Protection of Graded Areas: Protect newly graded areas from traffic and erosion. Keep free of trash and debris.

Repair and reestablish grades in settled, eroded, and rutted areas to specified tolerances.

Reconditioning Compacted Areas: Where completed compacted areas are disturbed by subsequent construction operations or adverse weather, scarify surface, reshape, and compact to required density prior to further constructions.

Settling: Where settling is measurable or observable at excavated areas during general project warranty period, remove surface (pavement, lawn, or other finish), add backfill material, compact, and replace surface treatment. Restore appearance, quality, and condition of surface or finish to match adjacent work, and eliminate evidence of restoration to greatest extent possible.

**DISPOSAL OF EXCESS AND WASTE MATERIALS**

Removal from Owner's Property: Remove waste materials, including unacceptable excavated material, trash, and debris, and dispose of it off Owner's property.

**END OF SECTION 02200**

**SECTION 02282**

**TERMITE CONTROL**

**PART 1 - GENERAL**

**RELATED DOCUMENTS**

Drawings and general provisions of Contract, including General and Supplementary Conditions and Division-1 Specifications sections, apply to work of this section.

Chemical soil treatment.

**SUMMARY**

Provide soil treatment for termite control, as herein specified, prior to placement of vapor barrier under concrete work.

**REFERENCES**

Title 7, United States Code, 136 through 136y – Federal Insecticide, Fungicide and Rodenticide Act; United States Code; 1947 (Revised 2001).

**SUBMITTALS**

See Section 01300 – Administrative Requirements, for submittal procedures.

Product Data: Indicate toxicants to be used, composition by percentage, dilution schedule, intended application rate.

Test Reports: Indicate regulatory agency approval reports when required.

Manufacturer's Application Instructions: Indicate caution requirements.

Manufacturer's Certificate: Certify that toxicants meet or exceed specified requirements.

Warranty: Submit warranty and ensure that forms have been completed.

Product Data: Submit manufacturer's technical data and application instructions.

**QUALITY ASSURANCE**

In addition to requirements of these specifications, comply with manufacturer's instructions and recommendations for work, including preparation of substrate and application.

Engage a professional pest control operator, licensed in accordance with regulations of governing authorities for application of soil treatment solution.

Use only termiticides which bear a Federal registration number of the U.S. Environmental Protection Agency.



### **JOB CONDITIONS**

**Restrictions:** Do not apply soil treatment solution until excavating, filling and grading operations are completed, except as otherwise required in construction operations.

To insure penetration, do not apply soil treatment to frozen or excessively wet soils or during inclement weather. Comply with handling and application instructions of the soil toxicant manufacturer.

### **SPECIFIC PRODUCT WARRANTY**

Furnish written warranty certifying that applied soil termiticide treatment will prevent infestation of subterranean termites and, that if subterranean terminate activity is discovered during warranty period, Contractor will retreat soil and repair or replace damage caused by termite infestation. Provide warranty for a period of 1 year from date of treatment, signed by Applicator and Contractor. This contract shall be renewable annually at the option of the Owner.

Warranty for repair or replacement of all termite damaged materials up to the sum of \$5,000.00.

### **PART 2 - PRODUCTS**

#### **SOIL TREATMENT SOLUTION**

Use an emulsible concentrate termiticide for dilution with water, especially formulated to prevent infestation by termites. Fuel oil will not be permitted as a diluent. Provide a solution consisting of one of following chemical elements and concentrations:

Chloropyrifos ("Dursban TC"); 1.0 percent is water emulsion.

Permethrin ("Dragnet", "Torpedo"); 0.5 percent in water emulsion.

Other solutions may be used as recommended by Applicator if also acceptable to Architect and approved for intended application by jurisdictional authorities. Use only soil treatment solutions which are not injurious to planting.

### **PART 3 - EXECUTION**

#### **APPLICATION**

**Surface Preparation:** Remove foreign matter which could decrease effectiveness of treatment on areas to be treated. Loosen, rake and level soil to be treated, except previously compacted areas under slabs and foundations. Toxicants may be applied before placement of compacted fill under slabs, if recommended by toxicant manufacturer.

**Application Rates:** Apply soil treatment solutions as follows:

**Under slab-on-grade** structures, treat soil before concrete slabs are placed, using the following rates of application:

Apply 4 gallons of chemical solution per 10 lin. ft. to soil in critical areas under slab, including entire inside perimeter inside of foundation walls, along both sides of interior partition walls, around plumbing pipes and electric conduit penetrating slab, and around interior column footers.

Apply one gallon of chemical solution per 10 sq. ft. as an overall treatment under slab and attached slab areas where fill is soil or unwashed gravel. Apply 1-1/2 gallons of chemical solution to areas where fill is washed gravel or other coarse absorbent material.

Apply 4 gallons of chemical solution per 10 lin. ft. of trench, for each foot of depth from grade to footing, along outside edge of building. Dig a trench 6" to 8" wide along outside of foundation to a depth of not less than 12". Punch holes to top of footing at not more than 12" o.c. and apply chemical solution. Mix chemical solution with the soil as it is being replaced in trench.

At hollow masonry foundations or grade beams, treat voids at rate of 2 gal. per 10 lin. ft., poured directly into the hollow spaces.

At expansion joints, control joints, and areas where slabs will be penetrated, apply at rate of 4 gals. per 10 lin. ft. of penetration.

Post signs in areas of application to warn workers that soil termiticide treatment has been applied. Remove signs when areas are covered by other construction.

Reapply soil treatment solution to areas disturbed by subsequent excavation, landscape grading, or other construction activities following application.

Outside building perimeter in a strip at least 2' wide, 1 gallon per 5 square feet.

**END OF SECTION 02282**

**SECTION 03200**

**CONCRETE REINFORCEMENT**

**PART 1 – GENERAL**

**SECTION INCLUDES**

Reinforcing steel for cast-in-place concrete.

Supports and accessories for steel reinforcement.

**RELATED SECTIONS**

Section 03310 – Concrete Work.

**REFERENCES**

ACI 301 – Specifications for Structural Concrete for Buildings; American Concrete Institute International; 2005.

ACI SP-66 – ACI Detailing Manual; American Concrete Institute International; 2004.

ASTM A 185/A 185M – Standard Specification for Steel Welded Wire Reinforcement, Plain, for Concrete; 2006.

ASTM A 615/A 615 M - Standard Specification for Deformed and Plain Billet-Steel Bars for Concrete Reinforcement; 2006.

CRSI (DA4) – Manual of Standard Practice; Concrete Reinforcing Steel Institute; 2001.

**SUBMITTALS**

See Section 01300 – Administrative Requirements, for submittal procedures.

Shop Drawings: Comply with requirements of ACI SP-66. Include bar schedules, shapes of bent bars, spacing of bars, and location of splices.

**QUALITY ASSURANCE**

Perform work of this section in accordance with ACI 301.

**PART 2 PRODUCTS**

**REINFORCEMENT**

Reinforcement Steel: ASTM A 615/A 615M Grade 60 (420),

- Deformed billet-steel bars.
- Unfinished

Steel Welded Wire Reinforcement: ASTM A 185/A 185M, plain type.

- Flat Sheets.
- Mesh Size: 6x6
- Wire Gage: W2.9 X W2.9 or as indicated on the plans.

Reinforcement Accessories:

- Tie Wire: Annealed, minimum 16 gage.
- Chairs, Bolster, Bar Supports, Spacers: Sized and shaped for adequate support of reinforcement during concrete placement.
- Provide stainless steel or plastic components for placement with 1-1/2 inches of weathering surfaces.

**FABRICATION**

Fabricate concrete reinforcing in accordance with CRSI (DA4) – Manual of Standard Practice.

Welding of reinforcement is not permitted.

**PART 3 EXECUTION**

**PLACEMENT**

Place, support and secure reinforcement against displacement. Do not deviate from required position.

Do not displace or damage vapor barrier.

Accommodate placement of formed openings.

Maintain concrete cover around reinforcing as follows:

- Footings and Concrete Formed Against Earth: 3 inch.
- Slabs on Fill: 2 inch.

Conform to applicable code for concrete cover over reinforcement.

**END OF SECTION 03200**

**SECTION 03310**

**CONCRETE WORK**

**PART 1 - GENERAL**

**RELATED DOCUMENTS:**

Drawings and general provisions of Contract, including General and Supplementary Conditions and Division-1 Specification sections, apply to work of this section.

**REFERENCES:**

ACI 211.1 – Standard Practice for Selectiong Proportions for Normal, Heavyweight, and Mass Concrete; American Concrete Institute International; 1991 (Reapproved 2002).

ACI 301 – Specifications for Structural Concrete for Buildings; American Concrete Institute International; 2005.

ACI 302.1R – Guide for Concrete Floor and Slab Construction; American Concrete Institute International; 2004.

ACI 304R – Guide for Measuring, Mixing, Transporting, and Placing Concrete; American Concrete Institute International; 2000.

ACI 305R – Hot Weather Concreting; American Concrete Institute International; 1999.

ACI 308R – Guide to Curing Concrete; American Concrete Institute International; 2001.

ACI 318 – Building Code Requirements for Structural Concrete and Commentary; American Concrete Institute International; 2005.

ASTM C 33 – Standard Specification for Concrete Aggregates; 2003.

ASTM C 39/C 39M – Standard Test Method for Compressive Strength of Cylindrical Concrete Sepcimens; 2005.

ASTM C 94/C 94M – Standard Specification for Ready-Mixed Cocrete; 2005.

ASTM C 150 – Standard Specification for Portaland Cement; 2005.

ASTM C 494/C 494M – Standard Specification for Chemical Admixtures for Concrete; 2005a.

ASTM C 618 – Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete; 2005

ASTM C 881/C 881M – Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete; 2002.

ASTM E 115 – Standard Test Method for Determining F (F) Floor Flatness and F (L) Floor Levelness Numbers; 1996 (Reapproved 2001).

ASTM E 1155M – Standard Test Method for Determining F (F) Floor Flatness and F (L) Floor Levelness Numbers [Metric]; 1996 (Reapproved 2001).

**SUMMARY:**

Extent of concrete work is shown on drawings.

**SUBMITTALS:**

Product Data: Submit data for non-proprietary materials and items, including admixtures, patching compounds, waterstops, joint systems, curing compounds, dry-shake finish materials, and others as requested by Architects.

Shop Drawings; Reinforcement: Submit original shop drawings prepared by registered Professional Engineer for fabrication, bending, and placement of concrete reinforcement. Comply with ACI 315 "Manual of Standard Practice for Detailing Reinforced Concrete Structures" showing bar schedules, stirrup spacing, diagrams of bent bars, and arrangement of concrete reinforcement. Include special reinforcement required for openings through concrete structures.

**QUALITY ASSURANCE:**

Codes and Standards: Comply with provisions of the following codes, specifications, and standards, except where more stringent requirements are shown or specified:

ACI 301 "Specifications for Structural Concrete for Buildings".  
ACI 318 "Building Code Requirements for Reinforced Concrete".  
Concrete Reinforcing Steel Institute (CRSI), "Manual of Standard Practice".

Materials and installed work may require testing and retesting at anytime during progress of work. Retesting of rejected materials for installed work, shall be done at Contractor's expense.

**PROJECT CONDITIONS:**

Protect adjacent finish materials against spatter during concrete placement.

**PART 2 - PRODUCTS**

**FORM MATERIALS:**

Forms for Exposed Finish Concrete: Plywood, metal, metal-framed plywood faced, or other acceptable panel-type materials, to provide continuous, straight, smooth, exposed surfaces. Furnish in largest practicable sizes to minimize number of joints and to conform to joint system shown on drawings.

Use plywood complying with U.S. Product Standard PS-1 "B-B (Concrete Form) Plywood", Class I, Exterior Grade or better, mill-oiled and edge-sealed, with each piece bearing legible inspection trademark.

Forms for Unexposed Finish Concrete: Plywood, lumber, metal, or other acceptable material. Provide lumber dressed on at least 2 edges and one side for tight fit.

Form Coatings: Provide commercial formulation form-coating compounds that will not bond with, stain, nor adversely affect concrete surfaces, and will not impair subsequent treatments of concrete surfaces.

Provide ties which, when removed, will leave holes not larger than 1" diameter in concrete surface.

**REINFORCING MATERIALS:**

Reinforcing Bars: ASTM A 615, Grade 60 deformed.

Steel Wire: ASTM A 82, plain, cold-drawn steel.

Welded Wire Fabric: ASTM A 185, welded steel wire fabric.

Supports for Reinforcement: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire fabric in place. Use wire bar type supports complying with CRSI specifications.

For slabs-on-grade, use supports with sand plates or horizontal runners where base material will not support chair legs. Brick scrap is acceptable.

**CONCRETE MATERIALS:**

Portland Cement: ASTM C 150, Type I, "Alamo Cement" or equal.

Use one brand of cement throughout project, unless otherwise acceptable to Architect.

Normal Weight Aggregates: ASTM C 33, and as herein specified. Provide aggregates from a single source for exposed concrete.

For exterior exposed surfaces, do not use fine or coarse aggregates containing spalling-causing deleterious substances.

Local aggregates not complying with ASTM C 33 but which have shown by special test or actual service to produce concrete of adequate strength and durability may be used when acceptable to Architect.

Water: Clean and Potable.

Water-reducing Admixture: ASTM C 194, Type A, and containing not more than 0.1 percent chloride ions.

Available Products: Subject to compliance with requirements, products which may be incorporated in the work include, but are not limited to, the following:

"WRDA Hycol"; W.R. Grace.  
"PSI N"; Gifford-Hill/American Admixtures  
"Eucon WR-75"; Euclid Chemical Co.  
"Pozzolith Normal"; Master Builders.  
"Plastocrete 160"; Sika Chemical Corp.  
"Chemtard"; Chem-Masters Corp.  
"Pro-Kete-N"; Protex Industries, Inc.

Water-Reducing, Non-Chloride Accelerator Admixture: ASTM C 494, Type E, and containing not more than 0.1 percent chloride ions.

Available Products: Subject to compliance with requirements, products which may be incorporated in the work include, but are not limited to, the following:

"Accelguard 80"; Euclid Chemical Co.  
"Pozzolith High Early"; Master Builders.  
"Gilco Accelerator"; Gifford-Hill/America Admixtures

Water-Reducing, Retarding Admixture: ASTM C 494, Type D, and containing not more than 0.1 percent chloride ions.

Available Products: Subject to compliance with requirements, products which may be incorporated in the work include, but are not limited to, the following:

"Edoco 20006"; Edoco Technical Products.  
"Pozzolith Retarder"; Master Builders.  
"Eucon Retarder 75"; Euclid Chemical Co.  
"Daratard"; W.R. Grace.  
"PSI R"; Gifford-Hill/American Admixtures.  
"Plastiment"; Sika Chemical Co.  
"Protard"; Protex Industries, Inc.

Prohibited Admixtures: Calcium chloride thycyanates or admixtures containing more than 0.1 percent chlorine ions are not permitted.

**RELATED MATERIALS:**

Vapor Retarder: Provide vapor retarder cover over prepared base material where indicated below slabs on grade. Vapor Retarder membrane must have the following qualities:

- |                                    |                |                          |
|------------------------------------|----------------|--------------------------|
| a) Water Vapor Transmission Rate   | ASTM E 96      | 0.04 Perms or lower      |
| b) Water Vapor Retarder            | ASTM E 1745    | Meets or exceeds Class C |
| c) Thickness of Retarder (plastic) | ACI 302.1 R-04 | Not less than 15 mils    |

Available Products: Subject to compliance with requirements, products which may be incorporated in the work include, but are not limited to, the following:

- a) Stego Wrap 15-mil Vapor Retarder by STEGO INDUSTRIES LLC, San Juan Capistrano, CA (877) 464-7834 [www.stegoindustries.com](http://www.stegoindustries.com).  
b) Griffolyn T-85 by Reef Industries  
c) Moistop Ultra by Fortifiber Industries

Liquid Membrane-Forming Curing Compound for **Interior use only**: Liquid type membrane-forming curing compound (must be dissipating cure) complying with ASTM C-309. Moisture loss not more than 0.055 gr./sq.cm. when applied at 300 sq.ft./gal.

Products: Provide the following:

"Kurez DR VOX" as manufactured by Euclid Chemical Company  
"Tamms Cure WB 30C" as manufactured by Tamms Industries

Liquid Membrane-Forming Curing Compound for **Exterior use only**: Liquid type membrane-forming curing & sealing compound complying with ASTM C-1315. Moisture loss not more than 0.045 gr./sq. cm. when applied at 200 sq. ft./gal.

Products: Provide the following:

"Super Rez Seal" as manufactured by Euclid Chemical Company  
"Master Cure & Seal 30" as manufactured by Master Builders

Bonding Compound: Polyvinyl acetate or acrylic base.

Available Products: Subject to compliance with requirements, products which may be incorporated in the work include, but are not limited to, the following:



Polyvinyl Acetate (Interior Only):

"Euco Weld"; Euclid Chemical Co.  
"Weldcrete"; Larsen Products Corp.

Acrylic or Styrene Butadiene:

"J-40 Bonding Agent"; Dayton Superior Corp.  
"Everbond"; L & M Construction Chemicals.  
"Hornweld"; A.C. Horn, Inc.  
"Sonocrete"; Sonneborn-Rexnord.  
"Acrylic Bondcrete"; The Burke Co.  
"SBR Latex"; Euclid Chemical Co.  
"Daraweld C"; W.R. Grace

**PROPORTIONING AND DESIGN OF MIXES:**

Prepare design mixes for each type and strength of concrete by either laboratory trial batch or field experience methods as specified in ACI 301. If trial batch method used, use an independent testing facility acceptable to Architect for preparing and reporting proposed mix designs. The testing facility shall not be the same as used for field quality control testing.

Submit written reports to Architect and Structural Engineer of each proposed mix for each class of concrete at least 15 days prior to start of work. Do not begin concrete production until mixes have been reviewed by Architect.

Design mixes to provide normal weight concrete with the following properties, as indicated on drawings and schedules:

3000 psi 28-day compressive strength; W/C ratio, 0.58 maximum (non-air-entrained), 0.46 maximum (air-entrained). For structural slabs and pavements. 2500 psi 28-day compressive strength; W/C ratio, 0.67 maximum (non-air-entrained), 0.54 maximum (air-entrained). For sidewalks and curbs.

Adjustment to Concrete Mixes: Mix design adjustments may be requested by Contractor when characteristics of materials, job conditions, weather, test results, or other circumstances warrant; at no additional cost to Owner and as accepted by Architect. Laboratory test data for revised mix design and strength results must be submitted to and accepted by Architect before using in work.

Admixtures:

Use water-reducing admixture in concrete as required for placement and workability.

Use non-chloride accelerating admixture in concrete slabs placed at ambient temperatures below 50 °F (10°C).

Slump Limits: Proportion and design mixes to result in concrete slump at point of placement as follows:

Ramps, slabs, and sloping surfaces: Not more than 5".

Reinforced foundation systems: Not less than 3" and not more than 5".

Other concrete: Not less than 3" nor more than 5".

**CONCRETE MIXING:**

Ready-Mix Concrete: Comply with requirements of ASTM C 94, and as herein specified.

During hot weather, or under conditions contributing to rapid setting of concrete, a shorter mixing time than specified in ASTM C 94 may be required.

### **PART 3 - EXECUTION**

#### **GENERAL:**

Coordinate the installation of joint materials and vapor retarders with placement of forms and reinforcing steel.

#### **FORMS:**

Design, erect, support, brace, and maintain formwork to support vertical and lateral, static, and dynamic loads that might be applied until such loads can be supported by concrete structure. Construct formwork so concrete members and structures are of correct size, shape, alignment, elevation, and position. Maintain formwork construction tolerances complying with ACI 347.

Design formwork to be readily removable without impact, shock, or damage to cast-in-place concrete surfaces and adjacent materials.

Construct forms to sizes, shapes, lines, and dimensions shown, and to obtain accurate alignment, location, grades level and plumb work in finished structures. Provide for openings, offsets, sinkages, keyways, recesses, moldings, rustications, reglets, chamfers, blocking, screeds, bulkheads, anchorages and insets, and other features required in work. Use selected materials to obtain required finishes. Solidly but joints and provide back-up at joints to prevent leakage of cement paste.

Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush plates or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces where slope is too steep to place concrete with bottom forms only. Kerf wood inserts for forming keyways, reglets, recesses, and the like, to prevent swelling and for easy removal.

Provide temporary openings where interior are of formwork is inaccessible for cleanout, for inspection before concrete placement, and for placement of concrete. Securely brace temporary openings and set tightly to forms to prevent loss of concrete mortar. Locate temporary openings on forms at inconspicuous locations.

Chamfer expose corners and edges as indicate, using wood, metal, PVC, or rubber chamfer strips fabricated to produce uniform smooth lines and tight edge joints.

Provisions for Other Trades: Provide an opening is concrete formwork to accommodate work of other trades. Determine size and location of openings, recesses, and chases from trades providing such items. Accurately place and securely support items built into forms.

Cleaning and Tightening: Thoroughly clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, or other debris just before concrete is placed. Retightening forms and bracing after concrete placement is required to eliminate mortar leaks and maintain proper alignment.

#### **VAPOR RETARDER INSTALLATION:**

##### **Preparation**

- Ensure that subsoil is approved by architect or geotechnical firm
- 1. Level and tamp or roll aggregate, sand or tamped earth base.
- 2. Insure termite treatment has been done.

**Installation**

Install Vapor Barrier/Retarder:

1. Installation shall be in accordance with manufacturer's instructions and ASTM E 1643-98.
  - a. Unroll Vapor Barrier/Retarder with the longest dimension parallel with the direction of the pour.
  - b. Lap Vapor Barrier/Retarder over footings and seal to foundation walls.
  - c. Over lap joints 6 inches and seal with manufacturer's tape.
  - d. Seal all penetrations (including pipes) per manufacturer's instructions.
  - e. No penetration of the Vapor Barrier/Retarder is allowed except for reinforcing steel and permanent utilities.
  - f. Repair damaged areas by cutting patches of Vapor Barrier/Retarder, overlapping damaged area 6 inches and taping all four sides with tape.

**PLACING REINFORCEMENT:**

Comply with Concrete Reinforcing Steel Institute's recommended practice for "Placing Reinforcing Bars", for details and methods of reinforcement placement and supports, and as herein specified.

Avoid cutting or puncturing vapor retarder during reinforcement placement and concreting operations.

Clean reinforcement of loose rust and mill scale, earth, ice, and other materials which reduce or destroy bond with concrete.

Accurately position, support and secure reinforcement against displacement by formwork, construction, or concrete placement operations. Locate and support reinforcing by metal chairs, runners, bolsters, spacers, and hangers, as required.

Place reinforcement to obtain at least minimum coverages for concrete protection. Arrange, space, and securely tie bars and bar supports to hold reinforcement in position during concrete placement operations. Set wire ties so ends are directed into concrete, not toward exposed concrete surfaces.

Install welded wire fabric in as long lengths as practicable. Lap adjoining pieces at least one full mesh and lace splices with wire. Offset end laps in adjacent widths to prevent continuous laps in either direction.

**JOINTS:**

Construction Joints: Locate and install construction joints as indicated or, if not indicated, located so as not to impair strength and appearance of the structure, as acceptable to Architect.

Place construction joints perpendicular to main reinforcement. Continue reinforcement across construction joints, except as otherwise indicated.

**INSTALLATION OF EMBEDDED ITEMS:**

General: Set and build into work anchorage devices and other embedded items required for other work that is attached to, or supported by, cast-in-place concrete. Use setting drawings, diagrams, instructions, and directions provided by suppliers of items to be attached thereto.

Edge Forms and Screed Strips for Slabs: Set edge forms or bulkheads and intermediate screed strips for slabs to obtain required elevations and contours in finished slab surface. Provide and secure units sufficiently strong to support types of screed strips by use of strike-off templates or accepted compacting type screeds.

**PREPARATION OF FORM SURFACES:**

Clean re-used forms of concrete matrix residue, repair and patch as required to return forms to acceptable surface condition.

Coat contact surfaces of forms with a form-coating compound before reinforcement is placed.

Thin form-coating compounds only with thinning agent of type, amount, and under conditions of form-coating compound manufacturer's directions. Do not allow excess form-coating material to accumulate in forms or to come into contact with in-place concrete surfaces against which fresh concrete will be placed. Apply in compliance with manufacturer's instructions.

Coat steel forms with, if used, a non-staining, rust-preventative form oil or otherwise protect against rusting. Rust-stained steel formwork is not acceptable.

**CONCRETE PLACEMENT:**

Replacement Inspection: Before placing concrete, inspect and complete formwork installation, reinforcing steel, and items to be embedded or casting. Notify other crafts to permit installation of their work; cooperate with other trades in setting such work. Moisten wood forms immediately before placing concrete where form coatings are not used.

Apply temporary protective covering to lower 2' of finished walls adjacent to poured floor slabs and similar conditions, and guard against spattering during placement.

General: Comply with ACI 304 "Recommended Practice for Measuring, Mixing, Transporting, and Placing Concrete", and as herein specified.

Deposit concrete continuously or in layers of such thickness that no concrete will be placed on concrete which has hardened sufficiently to cause the formation of seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as herein specified. Deposit concrete as nearly as practicable to its final location to avoid segregation.

Placing Concrete in Forms: Deposit concrete in forms in horizontal layers not deeper than 24" and in a manner to avoid inclined construction joints. Where placement consists of several layers, place each layer while preceding layer is still plastic to avoid cold joints.

Consolidate placed concrete by mechanical vibrating equipment supplemented by hand-spading, rodding, or tamping. Use equipment and procedures for consolidation of concrete in accordance with ACI 309.

Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly space locations not farther than visible effectiveness of machine. Place vibrators to rapidly penetrate placed layer and at least 6" into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to set. At each insertion limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing segregation of mix.

Placing Concrete Slabs: Deposit and consolidate concrete slabs in a continuous operation, within limits of construction joints, until the placing of a panel or section is completed.

Consolidate concrete during placing operations so that concrete is thoroughly worked around reinforcement and other embedded items and into corners.

Bring slab surfaces to correct level within straightedge and strike off. Use bull floats or darbies to smooth surface, free of humps or hollows. Do not disturb slab surfaces prior to beginning finishing operations.

Maintain reinforcing in proper position during concrete placement operations.

Cold Weather Placing: Protect concrete work from physical damage or reduced strength which could be caused by frost, freezing actions, or low temperatures, in compliance with ACT 306 and as herein specified.

When air temperature has fallen to or is expected to fall below 40 deg F (4deg C), uniformly heat water and aggregates before mixing to obtain a concrete mixture temperature of not less than 50 deg F (10 deg C), and not more than 80 deg F (27 deg C) at point of placement.

Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.

Do not use calcium chloride, salt, and other materials containing antifreeze agents or chemical accelerators, unless otherwise accepted in mix design.

Hot Weather Placing: When hot weather conditions exist that would seriously impair quality and strength of concrete, place concrete in compliance with ACI 305 and as herein specified.

Cool ingredients before mixing to maintain concrete temperature at time of placement below 90 deg F (32 deg C). Mixing water may be chilled, or chopped ice may be used to control temperature provided water equivalent of ice is calculated to total amount of mixing water. Use of liquid nitrogen to cool concrete is Contractor's option.

Cover reinforcing steel with water-soaked burlap if it becomes too hot, so that steel temperature will not exceed the ambient air temperature immediately before embedment in concrete.

Fog spray forms, reinforcing steel, and subgrade just before concrete is placed.

Used water-reducing retarding admixture (Type D) when required by high temperatures, low humidity, or other adverse placing conditions.

#### **FINISH OF FORMED SURFACES:**

Rough Form Finish: For formed concrete surfaces not exposed- to- view in the finish work or by other construction, unless otherwise indicated. This is the concrete surface having texture imparted by form facing material used, with tie holes and defective areas repaired and patched and fin and other projections exceeding 1/4" in height rubbed down or chipped off.

#### **MONOLITHIC SLAB FINISHES:**

Scratch Finish: Apply scratch finish to monolithic slab surfaces that are to receive concrete floor topping or mortar setting beds for tile, portland cement terrazzo, and other bonded applied cementitious finish flooring material, and as otherwise indicated.

After placing slabs, plane surface to tolerances for floor flatness (F) of 15 and floor levelness (F1) of 13. Slope surfaces uniformly to drains where requires. After leveling, roughen surface before final set, with stiff brushes, brooms, or rakes.

Float Finish: Apply float finish to monolithic slab surfaces to receive trowel finish and other finishes as hereinafter specified, and slab surfaces which are to be covered with membrane or elastic waterproofing, membrane or elastic roofing, or sand-bed terrazzo, and as otherwise indicated.

After screeding, consolidating, and leveling concrete slabs, do not work surface until ready for floating. Begin floating when surface water has disappeared or when concrete has stiffened sufficiently to permit operation of power-driven floats, or both, Consolidate surface with power-driven floats, or by hand-floating

if area is small or inaccessible to power units. Check and level surface plane to tolerances of F 18 - F 15. Cut down high spots and fill low spots. Uniformly slope surfaces to drains. Immediately after leveling, refloat surface to a uniform, smooth, granular texture.

Trowel Finish: Apply trowel finish to monolithic slab surfaces to be exposed-to-view, and slab surfaces to be covered with resilient flooring, carpet, ceramic or quarry tile, paint, or other thin firm finish coating system.

After floating, begin first trowel finish operation using a power-driven trowel. Begin final troweling when surface procedures a ringing sound as trowel is moved over surface. Consolidate concrete surface by final hand-troweling operation, free of trowel marks, uniform in texture and appearance, and with surface leveled to tolerances of F 20 - F 17. Grind smooth surface defects which would telegraph through applied floor covering system.

Trowel and Fine Broom Finish: Where ceramic or quarry tile is to be installed with thin-set mortar, apply trowel finish as specified, then immediately follow with slightly scarifying surface by fine brooming.

Non-Slip Broom Finish: Apply non-slip broom finish to exterior concrete platforms, steps, and ramps, and elsewhere as indicated.

Immediately after float finishing, slightly roughen concrete surface by brooming with fiber bristly broom perpendicular to main traffic route. Coordinate required final finish with Architect before application.

#### **CONCRETE CURING AND PROTECTION:**

General: Protect freshly placed concrete from premature drying and excessive cold or hot temperature.

Start initial curing as soon as free water has disappeared from concrete surface after placing and finishing. Weather permitting; keep continuously moist for not less than 7 days.

Begin final curing procedures immediately following initial curing and before concrete has dried. Continue final curing for at least 7 days in accordance with ACI 301 procedures. Avoid rapid drying at end of final curing period.

Provide curing and sealing compound to exposed interior slabs and to exterior slabs, walks, and curbs, as follows:

Apply specified curing and sealing compound to concrete slabs as soon as final finishing operations are complete (within 2 hours). Apply uniformly in continuous operation by power-spray or roller in accordance with manufacturer's directions. Recoat areas subjected to heave rainfall within 3 hours after initial application. Maintain continuity of coating and repair damage during curing period.

Curing Unformed Surfaces: Cure unformed surfaces, such as slabs, floor topping, and other flat surfaces by application of appropriate curing method.

Sealer and Dustproofer: Apply a second coat of specified curing and sealing compound only to surfaces given a first coat.

#### **REMOVAL OF FORMS:**

Formwork not supporting weight of concrete, such as sides of beams and similar parts of the work, may be removed after cumulatively curing at not less than 50 deg F (10 deg C) for 24 hours after placing concrete, provided concrete is sufficiently hard to not be damaged by form removal operations, and provided curing and protection operations are maintained.

**RE-USED OF FORMS:**

Clean and repair surfaces of forms to be re-used in work. Split, frayed, delaminated, or otherwise damaged form facing material will not be acceptable for exposed surfaces. Apply new form coating compound as specified for new formwork.

When forms are extended for successive concrete placement, thoroughly clean surfaces, remove fins and laitance, and tighten forms to close joints. Align and secure joint to avoid offsets. Do not use "patched" forms for exposed concrete surfaces, except as acceptable to Architect.

**MISCELLANEOUS CONCRETE ITEMS:**

Filling-In: Fill-in holes and openings left in concrete structures for passage of work by other trades, unless otherwise shown or directed, after work of other trades is in place. Mix, place, and cure concrete as herein specified, to blend with in-place construction. Provide other miscellaneous concrete filling shown or required to complete work.

Equipment Bases and Foundations: Provide machine and equipment bases and foundations, as shown on drawings. Set anchor bolts for machines and equipment to template at correct elevations, complying with certified diagrams or templates of manufacturer furnishing machines and equipment.

**CONCRETE SURFACE REPAIRS:**

Patching Defective Areas: Repair and patch defective areas with cement mortar immediately after removal of forms, when acceptable to Architect.

Cut out honeycomb, rock pockets, voids over 1/4" in any dimension, and holes left by tie rods and bolts, down to solid concrete but, in no case to a depth of less than 1". Make edges of cuts perpendicular to the concrete surface. Thoroughly clean, dampen with water, and brush-coat the area to be patched with specified bonding agent. Place patching mortar after bonding compound has dried.

For exposed-to-view surfaces blend white portland cement and standard portland cement so that, when dry, patching mortar will match color surrounding. Provide test areas at inconspicuous location to verify mixture and color match before proceeding with patching. Compact mortar in place and strike-off slightly higher than surrounding surface.

Repair of Formed Surfaces: Removed and replaced concrete having defective surfaces if defects cannot be repaired to satisfaction of Architect. Surface defects, as such, include color and texture irregularities, cracks, spalls, air bubbles, honeycomb, rock pockets; fins

and other projections on surface; and stains and other discolorations that cannot be removed by cleaning. Flush out form tie holes, fill with dry pack mortar, or precast cement cone plugs secured in place with bonding agent.

Repair concealed formed surfaces, where possible, that contain defects that affect the durability of concrete. If defects cannot be repaired, remove and replace concrete.

Repair of Unformed Surfaces: Test unformed surfaces, such as monolithic slabs, for smoothness and verify surface plane to tolerances specified for each surface and finish. Correct low and high areas as herein specified. Test unformed surfaces sloped to drain for trueness of slope, in addition to smoothness using a template having required slope.

Correct high area in unformed surfaces by grinding, after concrete has cured at least 4 days.

Correct low areas in unformed surfaces during or immediately after completion of surface finishing operations by cutting out low areas and replacing with fresh concrete. Finish repaired areas to blend into adjacent concrete. Proprietary patching compounds may be used when acceptable to Architect.

Repair defective areas, except random cracks and single holes not exceeding 1" diameter, by cutting out and replacing with fresh concrete. Remove defective areas to sound concrete with clean, square cuts and expose reinforcing steel with at least 3/4" clearance all around. Dampen concrete surfaces in contact with patching concrete and apply bonding compound. Mix patching concrete of same materials to provide concrete of same type or class as original concrete. Place, compact, and finish to blend with adjacent finished concrete. Cure in same manner as adjacent concrete.

Repair isolated random cracks and single holes not over 1" in diameter by dry-pack method. Groove top of cracks and cut-out holes to sound concrete and clean of dust, dirt, and loose particles. Dampen cleaned concrete surfaces and apply bonding compound. Mix dry-pack, consisting of one part portland cement to 2-1/2" parts fine aggregate passing a No. 16 mesh sieve, using only enough water as required for handling and placing. Place dry pack after bonding compound has dried. Compact dry-pack mixture in place and finish to match adjacent concrete. Keep patched area continuously moist for not less than 72 hours.

Perform structural repairs with prior approval of Architect or Structural Engineer for method and procedure, using specified epoxy adhesive and mortar.

Repair methods not specified above may be used, subject to acceptance of Architect.

#### **QUALITY CONTROL TESTING DURING CONSTRUCTION:**

The Owner will employ a testing laboratory to perform test and to submit test reports.

Sampling and testing for quality control during placement of concrete may include the following, as directed by Architect.

Sampling Fresh Concrete: ASTM C 172, except modified for slump to comply with ASTM C 94.

Slump: ASTM C 143; one test at point of discharge for each day's pour of each type of concrete; additional tests when concrete consistency seems to have changed.

Air Content: ASTM C 173, volumetric method for lightweight or normal weight concrete; ASTM C 231 pressure method for normal weight concrete; one for each day's pour of each type of air-entrained concrete.

Concrete Temperature: Test hourly when air temperature is 40 deg F (4 deg C) and below, and when 80 deg F (27 deg C) and above; and each time a set of compression test specimens are required.

Compression Test Specimen: ASTM C 31, one set of 4 standard cylinders for each compressive strength test, unless otherwise directed. Mold and store cylinders for laboratory cured test specimens except when field-cure test specimens are required.

Compressive Strength Tests: ASTM C 39; one set for each day's pour exceeding 5 cu. yds. plus additional sets for each 50 cu. yds. over and above the first 25 cu. yds. of each concrete class placed in any one day; one specimen tested at 7 days, two specimens tested at 28 days, and one specimen retained in reserve for later testing if required. When frequency of testing will provide less than 5 strength test for a given class of concrete, conduct testing from at least 5 randomly selected batches or from each batch if fewer than 5 are used.



When strength of field-cured cylinders is less than 85 percent of companion laboratory-cured cylinders, evaluate current operations and provide corrective procedures for protecting and curing the in-place concrete.

Strength level of concrete will be considered satisfactory if averages of sets of three consecutive strength test results equal or exceed specified compressive strength, and no individual strength test result falls below specified compressive strength by more than 500 psi.

Test results will be reported in writing to Architect, Structural Engineer, and Contractor within 24 hours after tests. Reports of compressive strength test shall contain the project identification name and number, date of concrete placement, name of concrete testing service, concrete type and class, location of concrete batch in structure, design compressive strength at 28 days, concrete mix proportions and materials; compressive breaking strength and type of break for both 7-day tests and 28-day tests.

Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted but shall not be used as the sole basis for acceptance or rejection.

Additional Tests: The testing service will make additional tests of in-place concrete when test results indicate specified concrete strengths and other characteristics have not been attained in the structure, as directed by Architect. Testing service may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C 42, or by other methods as directed. Contractor shall pay for such tests when unacceptable concrete is verified.

**END OF SECTION 03310**

**SECTION 04065**

**MORTAR AND MASONRY GROUT**

**PART 1 – GENERAL**

**SECTIONS INCLUDES**

Mortar for masonry.

Grout for masonry.

**RELATED SECTIONS**

**REFERENCES**

ASTM C 91 – Standard Specification for Masonry Cement; 2005.

ASTM C 94/C 94M – Standard Specification for Ready-Mixed Concrete; 2005.

ASTM C 144 – Standard Specification for Aggregate for Masonry Mortar; 2004.

ASTM C 270 – Standard Specification for Mortar for Unit Masonry; 2005a.

ASTM C 404 – Standard Specification for Aggregates for Masonry Grout; 2004.

ASTM C 476 – Standard Specification for Grout for Masonry; 2002.

ASTM C 780 – Standard Test Method for Preconstruction and Construction Evaluation of Mortars for Plain and Reinforced Unit Masonry; 2006.

ASTM C 979 – Standard Specification for Pigments for Integrally Colored Concrete; 2005.

ASTM C 1019 – Standard Test Method for Sampling and Testing Grout; 2005.

ASTM C 1072 – Standard Test Method for Measurement of Masonry Flexural bond Strength; 2005b.

ASTM C 1314 – Standard Test Method for Compressive Strength of Masonry Prisms; 2003b.

ASTM E 518 – Standard Test Methods for Flexural Bond Strength of Masonry; 2003.

IMAIWC (CW) – Recommended Practices & Guide Specifications for Cold Weather Masonry Construction; International Masonry Industry All-Weather Council; 1993.

IMIAWC (HW) – Recommended Practices & Guide Specifications for Cold Weather Masonry Construction; International Masonry Industry All-Weather Council; current edition.

**SUBMITTALS**

See Section 01300 – Administrative Requirements, for submittal procedures.

**QUALITY ASSURANCE**

**ENVIRONMENTAL REQUIREMENTS**

Cold Weather Requirements: Comply with recommendations of IMIAWC (CW).

Hot Weather Requirements: Comply with IMIAWC (HW).

**PART 2 – PRODUCTS**

**MATERIALS**

Masonry Cement: ASTM C 91, types as scheduled in this section.

- Colored mortar: Premixed cement as required to match Architect's color sample.

Mortar Aggregate: ASTM C 144.

Grout Aggregate: ASTM C 404.

Pigments for Colored Mortar: Pure, concentrated mineral pigments specifically intended for mixing into mortar and complying with ASTM C 979.

Water: Clean and potable.

**MORTAR MIXES**

Mortar for Unit Masonry: ASTM C 270, Property Specification.

- Exterior, loadbearing masonry: Type S.
- Exterior, non-loadbearing masonry: Type S.
- Interior, loadbearing masonry: Type S.
- Interior, non-loadbearing masonry: Type S.
- Glass unit masonry: Type S.

Colored Mortar: Proportion selected pigments and other ingredients to match Architect's sample, without exceeding manufacturer's recommended pigment-to-cement ratio.

**MORTAR MIXING**

Thoroughly mix mortar ingredients using mechanical batch mixer, in accordance with ASTM C 270 and in quantities needed for immediate use.

Maintain sand uniformly damp immediately before the mixing process.

Add mortar color in accordance with manufacturer's instructions. Provide uniformity of mix and coloration.

Do not use ant-freeze compounds to lower the freezing point of mortar.

If water is lost by evaporation, re-temper only within two hours of mixing.

Use mortar within two hours after mixing at temperatures of 90 degrees F, or two-and-one-half hours at temperatures under 40 degrees F.

### **GROUT MIXES**

Bond Beams and Lintels: 3,000 psi strength at 28 days; 8-10 inches slump; provide premixed type in accordance with ASTM C 94/C 94M.

- Fine grout for spaces with smallest horizontal dimension of 2 inches or less.
- Coarse grout for spaces with smallest horizontal dimension greater than 2 inches.

### **GROUT MIXING**

Mix grout in accordance with ASTM C 94/ 94M

Thoroughly mix grout ingredients in quantities needed for immediate use in accordance with ASTM C 476 for fine and coarse grout.

### **PRECONSTRUCTION TESTING**

Testing will be conducted by an independent test agency, in accordance with provisions of Section 01400.

Mortar Mixes: Test mortars pre batched by weight in accordance with ASTM C 780 recommendations for preconstruction testing.

- Test result will be used to establish optimum mortar proportions and establish quality control values for construction testing.

Grout Mixes: Test grout batches in accordance with ASTM C 1019 procedures.

- Test results will be used to establish optimum grout proportions and establish quality control values for construction testing.

## **PART 3 – EXECUTION**

### **PREPARATION**

Plug clean-out holes for grouted masonry. Brace masonry to resist wet grout pressure.

### **INSTALLATION**

Install mortar and grout to requirements of section (s) in which masonry is specified.

Work grout into masonry cores and cavities to eliminate voids.

Do not install grout in lifts greater than 48 inches without consolidating grout by rodding.

Do not displace reinforcement while placing grout.

Remove excess mortar from grout spaces.

### **GROUTING**

Grouting Technique:

- At the contractor's option, use either low-lift grouting techniques subject to the requirements which follow.

Low-Lift Grouting:

- Limit height of pours to 48 inches.
- Limit height of masonry to 8 inches above each pour.

- Pour grout only after vertical reinforcing and horizontal reinforcing is in place. Prevent displacement of bars as grout is poured.
- Place grout for each pour continuously and consolidated immediately; do not interrupt pours for more than 1-1/2 hours.

**High-Lift Grouting:**

- Verify that horizontal and vertical reinforcement is in proper position and adequately secured before beginning pours.
- Hollow Masonry: Limit lifts to maximum 4 feet and pours to maximum height of 8 feet.
- Place grout for spanning elements in single, continuous pour.
- Provide cleanout holes in first course at all vertical cells which are to be filled with grout.

**FIELD QUALITY CONTROL**

An independent testing agency will perform field test.

Test and evaluate mortar in accordance with ASTM C 780 procedures.

- Test with same frequency as specified for masonry units.

Test and evaluate grout in accordance with ASTM C 1019 procedures.

- Test with same frequency as specified for masonry units.

Prism Tests: Test masonry and mortar panels for compressive strength in accordance with ASTM C 1314, and for flexural bonds strength in accordance with ASTM C 1072 or ASTM 8 518; perform tests and evaluate results as specified in individual masonry sections

**END OF SECTION 04065**

**SECTION 04200**

**UNIT MASONRY**

**PART 1 - GENERAL**

**RELATED DOCUMENTS**

Drawings and general provisions of Contract, including General and Supplementary Conditions, and Division-1 Specification sections, apply to work of this section.

Extent of each type of masonry work is indicated on drawings and schedule.

Types of masonry work required include:

Brick masonry with pigmented mortar.

Brick vents.

**QUALITY ASSURANCE:**

Single Source Responsibility for Masonry Units: Obtain exposed masonry units of uniform texture and color, or uniform blend within the ranges accepted for these characteristics, from one manufacturer for each different product required for each surface or visually related surfaces.

Single Source Responsibility for Mortar Materials: Obtain exposed masonry ingredients of uniform quality, including color for exposed masonry, from one manufacturer for each cementitious component and from one source and producer for each aggregate.

Masonry Preconstruction Testing Service: The Owner Employ and pay for the services of a testing laboratory acceptable to Architect and experienced in performing types of preconstruction masonry test indicated.

Owner shall Engage a testing laboratory complying with ASTM E 329.

Preconstruction Tests by Unit Test Methods: Test as directed by Architect.

Mortar Tests: Test each mortar type per ASTM C 780.

**SUBMITTALS**

Product Data: Submit manufacturer's product data for each type of masonry unit, accessory, and other manufactured products, including certifications that each type complies with specified requirements.

Samples for Initial Selection Purposes: Submit samples of the following materials:

Colored masonry mortar samples showing full extent of colors available.

**DELIVERY, STORAGE, AND HANDLING:**

Deliver masonry materials to project in undamaged condition.

Store and handle masonry units to prevent their deterioration or damage due to moisture, temperature changes, contaminants, corrosion, or other causes.

Limit moisture absorption of concrete masonry units during delivery and until time of installation to the maximum percentage specified for Type I units for the average annual-relative humidity as reported by the U.S. Weather Bureau Station nearest project size.

Store cementations materials off the ground, under cover and in dry location.

Store masonry accessories including metal items to prevent deterioration by corrosion and accumulation of dirt.

### **PROJECT CONDITIONS:**

Protection of Work: During erection, cover top of walls with waterproof sheeting at end of each day's work. Cover partially completed structures when work is not in progress.

Extend Cover a minimum of 24 inches down both sides and hold cover securely in place.

Do not apply uniform floor or roof loading for at least 12 hours after building masonry walls or columns.

Do not apply concentrated loads for at least 3 days after building masonry walls or columns.

Staining: Prevent grout, mortar, or soil from staining the face of masonry to be left exposed or painted. Removed immediately grout or mortar in contact with such masonry.

Protect base of walls from rain-splashed mud and mortar splatter by means of coverings spread on ground and over wall surface.

Protect sills, ledges, and projections from droppings of mortar.

### **Cold Weather Protection:**

Do not lay masonry units which are wet or frozen.

Remove any ice or snow formed on masonry bed by carefully applying heat until top surface is dry to the touch.

Remove masonry damaged by freezing conditions.

For clay masonry units with initial rates of absorption (suction) which require them to be wetted before laying, comply with the following requirements.

For units with surface temperatures above 32 deg. F (0 deg. C.), wet with water heated to above 70 deg. F (21 deg. C).

For units with surface temperatures below 32 deg. F (0 deg. C), wet with water heated to above 130 deg. F (54 deg. C).

### **PART 2 - PRODUCTS**

**Facing Brick:** ASTM C216, Type FBS, Grade MW.

Color and Texture: As selected by the Architect.

Nominal Size: As selected by the Architect.

Compressive Strength: As selected by the Architect.

Manufacturer: As selected by the Architects.

Cornice Brick: ASTM C216, Type FBS, Grade MW.

Color and Texture As selected by the Architect.

Nominal Size: As selected by the Architect.

Compressive Strength: As selected by the Architect.

Manufacturer: As selected by the Architect.

**MORTAR AND GROUT MATERIALS:**

Portland Cement: ASTM C 150, Type I, Except Type III may be used for cold weather construction. Provide natural color or white cement as required to produce required mortar color.

Hydrate Lime: ASTM C 207, Type S.

Aggregate for Grout: ASTM C 404.

Available Products: Subject to compliance with requirements, colored mortar pigments which may be incorporated in the work include, but are not limited to, the following:  
"SGS Mortar Colors"; Solomon Grind-Chem Services, Inc.  
"True Tone Mortar Colors"; Davis Colors, A Subsidiary of Rockwood Industries, Inc.

Water: Clean and potable.

**JOINT REINFORCEMENT, TIES, AND ANCHORING DEVICES:**

Materials: Comply with requirements indicated below for basic materials and with requirements indicated under each form of joint reinforcement, tie, and anchor for size and other characteristics:

Zinc-Coated (galvanized) Steel Wire: ASTM A 82 for uncoated wire and with ASTM A 641 for zinc coating of class indicated below:

Class 1 (0.40 oz. per sq. ft. of wire surface).

Application: Use for masonry not exposed to exterior or earth.

Hot-Dip Galvanized Steel Wire: ASTM A 82 for uncoated wire and with ASTM A 123, Class B-2 (1.5 oz per sq. ft. of wire surface) for zinc coating applied after prefabrication into units.

Application: Use for masonry exposed to exterior and in contact with earth.

Zinc-Coated (Galvanized) Steel Sheet: Carbon steel with zinc coating complying with ASTM A 525, Coating Designation G90.

Joint Reinforcement: Provide welded-wire units prefabricated with deformed continuous side rods and plain cross rods into straight lengths of not less than 10', with prefabricated corner and tee units, and complying with requirements indicated below:

Width: Fabricate joint reinforcement in units with widths of approximately 2" less than nominal width of walls and partitions as required to provide mortar coverage of not less than 5/8" on joint faces exposed to exterior and 2" elsewhere.



Wire Size for Side Rods: 0.1875" diameter.

Wire Size for Cross Rods: 0.1483" diameter

For single-wythe masonry provide type as follows with single pair of side rods:

Truss design with continuous diagonal cross rods spaced not more than 16@ o.c.

For multi-wythe masonry provide type as follows:

Tab design with single pair of side rods and truss type diagonal cross rods spaced not more than 16" o.c.; with side rods spaced for embedment within each face shell of back-up wythe and rectangular ties extended to within 1" of exterior face of facing wythe and spaced not more than 24" o.c. Use units with adjustable 2-piece rectangular ties.  
Flexible Anchors: Where flexible anchors are indicated for connecting masonry to structural framework, provide 2-piece anchors as described below which permit vertical or horizontal differential movement between wall and framework parallel to, but resist tension and compression forces perpendicular to, plane of wall.

For anchorage to concrete framework, provide manufacturer's standard anchors with dovetail anchor section formed from 0.1046" (12 gage) thick sheet metal and triangular-shaped wire tie section sized to extend within 1" of masonry face.

For anchorage to steel framework provide manufacturer's standard anchors with crimped 1/4" diameter wire anchor section for welding to steel and triangular-shaped wire tie section sized to extend within 1" of masonry face.

Wire Size: 0.1875" diameter.

For anchorage to metal studs provide hot dip galvanized No. D/A 213 Veneer Anchor with DA807 screws as manufactured by DUR-O-WAL or equal.

#### **MISCELLANEOUS MASONRY ACCESSORIES:**

Reinforcing Bars: Deformed steel, ASTM A 615, Grade 60 for bars No. 3 to No. 18.

Non-Metallic Expansion Joint Strips: Premolded, flexible cellular neoprene rubber filler strips complying with ASTM D 1056, Grade RE41E1, capable of compression up to 35% of width and thickness indicated.

Premolded Control Joint Strips: Material as indicate below, designed to fit standard sash block and to maintain lateral stability in masonry wall; size and configuration as indicated.

Styrene-butadiene rubber compound complying with ASTM D 2000, Designation 2AA-508.

#### **MORTAR AND GROUT MIXES:**

General: Do not add admixtures including coloring pigments, air-entraining agents, accelerators, retarders, water repellent agents, anti-freeze compounds or other admixtures, unless otherwise indicated.

Do not use calcium chloride in mortar or grout.

Mixing: Combine and thoroughly mix cementitious, water and aggregates in a mechanical batch mixer; with referenced ASTM standards for mixing time and water content.

Mortar for Unit Masonry: Comply with ASTM C 270, Proportion Specification, for types of mortar required, unless otherwise indicated.

Limit cementitious materials in mortar to Portland cement-lime.

Use Type S mortar for all masonry.

Grout for Unit Masonry: Comply with ASTM C 476 for grout for use in construction of reinforced and nonreinforced unit masonry. Use grout of consistency indicated or if not otherwise indicated, of consistency (fine or coarse) at time of placement which will completely fill all spaces intended to receive grout.

Use fine grout in grout spaces less than 2" in horizontal direction, unless otherwise indicated.

Use coarse grout in grout spaces 2" or more in least horizontal dimension, unless otherwise indicated.

### **PART 3 - EXECUTION**

#### **INSTALLATION, GENERAL:**

Do not wet concrete masonry units.

Cleaning Reinforcing: Before placing, remove loose rust, ice, and other coatings from reinforcing.

Thickness: Build cavity composite walls, floors, and other masonry construction to the full thickness shown. Build single-wythe walls (if any) to the actual thickness of the masonry units using of nominal thickness indicated.

Build chases and recesses as shown or required for the work of other trades. Provide not less than 8" of masonry between chase or recess and jamb of openings, and between adjacent chases and recesses.

Leave openings for equipment to be installed before completion of masonry work. After installation of equipment, complete masonry work to match work immediately adjacent to the opening.

Cut masonry units using motor-driven saws to provide clean, sharp, unchipped edges. Cut units as required to provide continuous pattern and to fit adjoining work. Use full-size units without cutting where possible.

Use dry cutting saws to cut concrete masonry units.

#### **CONSTRUCTION TOLERANCES:**

Variation from Plumb: For vertical lines and surfaces of columns, walls, and arises do not exceed 1/4" in 10' or 3/8" in a story height not to exceed 20', nor 2" in 40' or more. For external corners, expansion joints, control joints and other conspicuous lines, do not exceed 1/4" in any story or 20' maximum, nor 2" in 40' or more. For vertical alignment of head joints, do not exceed plus or minus 1/4" in 10', 2" maximum.

Variation from level: For bed joints and lines of exposed lintels, sills, parapets, horizontal grooves and other conspicuous lines, do not exceed 1/4" in any bay or 20' maximum, nor 2" in 40' or more. For top surface of bearing walls, do not exceed 1/8" between adjacent floor elements in 10' or 1/16" within width of a single unit.

Variation of Linear Building Line: For position shown in plan and related portion of columns, walls, and partitions, do not exceed 2" in any bay or 20' maximum, nor 3 1/4" in 40' or more.

Variation in Cross-Sectional Dimensions: For columns and thickness of walls, from dimensions shown, do not exceed minus 1/4" nor plus 2".

**EDINBURG FIRE DEPARTMENT  
FIRE STATION #5**

**Hinojosa Engineering, Inc.  
Project No. 15-118**

Variation in Mortar Joint Thickness: Do not exceed bed joint thickness indicated by more than plus or minus 1/8", with a maximum thickness limited to 1/2". Do not exceed head joint thickness indicated by more than plus or minus 1/8".

**LAYING MASONRY WALLS:**

Layout walls in advance for accurate spacing of surface bond patterns with uniform joint widths and to accurately locate openings, movement-type joints, returns and offsets. Avoid the use of less-than-half-size units at corners, jambs, and wherever possible at other locations.

Lay-up walls to comply with specified construction tolerances, with courses accurately spaced and coordinated with other work.

Pattern Bond: Lay exposed masonry in the bond pattern shown or, if not shown, lay in running bond with vertical joint in each course centered on units in courses above and below. Lay concealed masonry with all units in a wythe in running bond or bonded by lapping not less than 2". Bond and interlock each course of each wythe at corners. Do not use units with less than nominal 4" horizontal face dimensions at corners or jambs.

Stopping and Resuming Work Rack back 1/2-unit length in each course; do not tooth. Clean exposed surfaces of set masonry, wet units lightly (if required) and remove loose masonry units and mortar prior to laying fresh masonry.

Built-in Work: As the work progresses, built-in items specified under this and other sections of these specifications. Fill in solidly with masonry around built-in items.

Fill space between hollow metal frames and masonry solidly with mortar, unless otherwise indicated.

**MORTAR BEDDING AND JOINTING:**

Lay hollow concrete masonry units with full mortar coverage on horizontal and vertical face shells. Bed webs in mortar in starting course on footings and in all courses of piers, columns, and pilasters, and where adjacent to cells or cavities to be reinforced or filled with concrete or grout. For starting course on footings where cells are not grouted, spread out full mortar bed including areas under cells.

Maintain joint widths shown except for minor variations required to maintain bond alignment. If not shown, lay walls with 3/8" joints.

Concave mortar joints for masonry walls which are to be concealed or to be covered by other materials, unless otherwise indicated.

Remove masonry units disturbed after laying; clean and reset in fresh mortar. Do not pound corners or jambs to shift adjacent stretcher units which have been set in position. If adjustments are required, remove units, clean off mortar and reset in fresh mortar.

**STRUCTURAL BONDING OF MULTI-WYTHER MASONRY:**

Use continuous horizontal joint reinforcement installed in horizontal mortar joints for bond tie between wythes. Install at not more than 16" o.c. vertically.

Corners: Provide interlocking masonry unit bond in each course at corners, unless otherwise shown.

For horizontally reinforced masonry, provide continuity at corners with prefabricated "L" units, in addition to masonry bonding.

Intersecting and Abutting Walls: Unless vertical expansion or control joints are shown at juncture, provide same type of bonding specified for structural bonding between wythes and spaces as follows:

Provide continuity with horizontal joint reinforcement using prefabricated "T" units.

**CAVITY WALLS:**

Keep cavity clean of mortar droppings and other materials during construction. Strike joints facing cavity flush. Tie exterior wythe to back-up with individual metal ties spaced not more than 16" o.c. vertically and 24" o.c. horizontally.

Provide weep holes in exterior wythe of cavity wall located immediately above ledges and flashing, spaced 2'-0" o.c. unless otherwise indicated.

**HORIZONTAL JOINT REINFORCEMENT:**

General: Provide continuous horizontal joint reinforcement as indicated. Install longitudinal side rods in mortar for their entire length with a minimum cover of 5/8" on exterior side of walls, 2" elsewhere. Lap reinforcing a minimum of 6".

Cut or interrupt joint reinforcement at control and expansion joints, unless otherwise indicated.

Reinforcement walls with continuous horizontal joint reinforcing unless specifically noted to be omitted. Provide continuity at corners and wall intersections by use of prefabricated "L" and "T" sections. Cut and bend reinforcement units as directed by manufacturer for continuity at returns, offsets, column fireproofing, pipe enclosures and other special conditions.

Space continuous horizontal reinforcement as follows:

For multi-wythe walls (solid or cavity) where continuous horizontal reinforcement acts as structural bond or tie between wythes, space reinforcement as required by code but not more than 16" o.c. vertically.

For single-wythe walls, space reinforcement at 16" o.c. vertically, unless otherwise indicated.

For parapets, space reinforcement at 8" o.c. vertically, unless otherwise indicated.

Reinforcement masonry openings greater than 1'-0" wide, with horizontal joint reinforcement placed in 2 horizontal joints approximately 8" apart, immediately above the lintel and immediately below the sill. Extend reinforcement a minimum of 2'-0" beyond jambs of the opening except at control joints.

In addition to wall reinforcement, provide additional reinforcement at openings as required to comply with the above.

**ANCHORING MASONRY WORK:**

General: Provide anchor devices of type indicated.

Anchor masonry to structural members where masonry abuts or faces structural members to comply with the following.

Provide an open space not less than 1" in width between masonry and structural member, unless otherwise indicated. Keep open space free of mortar or other rigid materials.

Anchor masonry to structural members with flexible anchors embedded in masonry joints and attached to structure.

Space anchors as indicated, but not more than 24" o.c. vertically and 36" o.c. horizontally.

**CONTROL AND EXPANSION JOINTS:**

General: Provide vertical and horizontal expansion, control and isolation joints in masonry where shown. Build-in related items as the masonry work progresses.

Build flanges of metal expansion strips into masonry. Lap each joint 4" in direction of water flow. Seal joints below grade and at junctures with horizontal expansion joints, if any.

Build flanges of factory-fabricated expansion joint units into masonry. See Division-7 section "Elastic Expansion Joints".

Build-in non-metallic joint fillers where indicated.

Build in horizontal pressure relieving joints where indicated; construct joints by either leaving an air space or inserting non-metallic compressible joint filler of width required to permit installation of sealant and backer rod.

Located horizontal pressure relieving joints beneath shelf angles supporting masonry veneer and attached to structure behind masonry veneer.

#### **LINTELS:**

Install steel lintels where indicated.

For hollow concrete masonry unit walls, use specially formed U-shaped lintel units with reinforcement bars placed as shown filled with coarse grout.

Provide minimum bearing of 8" at each jamb, unless otherwise indicated.

#### **FLASHING OF MASONRY WORK:**

General: Provide concealed flashing in masonry work at, or above, shelf angles, lintels, ledges, and other obstructions to the downward flow of water in the wall so as to divert such water to the exterior. Prepare masonry surfaces smooth and free from projections which could puncture flashing. Place through-wall flashing on sloping bed of mortar and cover with mortar. Seal penetrations in flashing with mastic before covering with mortar.

Extend flashing through exterior face of masonry and turn down to form drip.

Extend flashing the full length of lintels and shelf angles and minimum of 4" into masonry each end. Extend flashing from exterior face of outer wythe of masonry, through the outer wythe, turned up a minimum of 4", and through the inner wythe to within 1/2" of the interior face of the wall in exposed work. Where interior surface of inner wythe is concealed by furring, carry flashing completely through the inner wythe and turn up approximately 2". At heads and sills, turn up ends not less than 2" to form a pan.

Interlock end joints of deformed metal flashings by over-lapping deformations not less than 1-1/2" and seal lap with elastic sealant.

Provide weep holes in the head joints of the first course of masonry immediately above concealed flashings. Space 24" o.c., unless otherwise indicated.

Install reglets and nailers for flashing and other related work where shown to be built into masonry work.

#### **INSTALLATION OF REINFORCEMENT UNIT MASONRY:**

Refer to Division-4 sections "Reinforced Unit Masonry" for installation requirements applicable to reinforced unit masonry.

#### **FIELD QUALITY CONTROL:**

Owner will employ separate testing laboratory to perform field quality control testing.

**Unit Test Method:**

Concrete Masonry Unit Test: For each type, class, and grade of concrete masonry unit indicated, test units by method of sampling and testing of ASTM C 140.

Mortar Tests: For each type indicated, test mortar by methods of sampling and testing of ASTM C 780. Conduct tests no less frequently than that required to evaluate mortar used to install each increment of masonry units indicated above from which samples are taken for testing.

Report test results in writing and in form specified under each test method, to Architect and Contractor, on same day tests are made.

Evaluation of Quality Control Tests: Masonry work, in absence of other indications of noncompliance with requirements, will be considered satisfactory if results from construction quality control tests comply with minimum requirements indicated.

**REPAIR, POINTING, AND CLEANING:**

Remove and replace masonry units which are loose, chipped, broken, stained, or otherwise damaged, or if units do not match adjoining units as intended. Provide new units to match adjoining units and install in fresh mortar or grout, pointed to eliminate evidence of replacement.

Pointing: During the tooling of joints, enlarge any voids or holes, except weep holes, and completely fill with mortar. Point-up all joints including corners, openings, and adjacent work to provide a neat, uniform appearance, prepared for application of sealants.

Final Cleaning: After mortar is thoroughly set and cured, clean masonry as follows:

Remove large mortar particles by hand with wooden paddles and non-metallic scrape hoes or chisels.

Test cleaning methods on sample wall panel; leave 2 panel uncleaned for comparison purposes. Obtain Architect's approval of sample cleaning before proceeding with cleaning of masonry.

Protect adjacent stone and non-masonry surfaces from contact with cleaner by covering them with liquid strippable masking agent, polyethylene film or waterproof masking tape.

Saturate wall surfaces with water prior to application of cleaners; remove cleaners promptly by rinsing thoroughly with clear water.

Use bucket and brush hand cleaning method described in BIA "Technical Note No. 20 Revised" to clean brick masonry made from clay or shale, except use masonry cleaner indicated below.

Detergent.

Acidic cleaner; apply in compliance with directions of cleaner manufacturer.

Clean concrete unit masonry to comply with masonry manufacturer's directions and applicable NCMA "Tek" bulletins.

Protection: Provide final protection and maintain conditions in a manner acceptable to Installer, which ensures unit masonry work being without damage and deterioration at time of substantial completion.

**END OF SECTION 04200**

**SECTION 04230**

**REINFORCED UNIT MASONRY**

**PART 1 - GENERAL**

**RELATED DOCUMENTS:**

Drawings and general provisions of Contract, including General and Supplementary Conditions and Division-1 Specification sections, apply to work of this section.

Requirements of Section "Unit Masonry" apply to work of this section.

**DESCRIPTION OF WORK:**

Extent of each type of reinforced unit masonry work is indicated on drawings and in schedules.

**SUBMITTALS:**

General: Comply with ACI 530.1/ASCE6/TMS 602

Shop Drawings: Submit shop drawings for fabrication, bending, and placement of reinforcement bars. Comply with ACI 315 "Manual of Standard Practice for Detailing Reinforced Concrete Structures". Show bar schedules, diagrams of bent bars, stirrup spacing, lateral ties and other arrangements and assemblies as required for fabrication and placement of reinforcement for unit masonry work.

**PART 2 - PRODUCTS**

**MATERIALS:**

General: Refer to Section "Unit Masonry" for masonry materials and accessories not included in this section.

Reinforcement Bars: Provide deformed bars of following grades complying with ASTM A 615, Except as otherwise indicated.

Provide Grade 60 for bars No. 3 to No. 18, except as otherwise indicated.

Shop-fabricate reinforcement bars which are shown to be bent or hooked.

**PART 3- EXECUTION**

**PLACING REINFORCEMENT:**

General: Clean reinforcement of loose rust, mill scale, earth, ice, or other materials which will reduce bond to mortar or grout. Do not use reinforcement bars with kinks or bends not shown on drawings or final shop drawings, or bars with reduced cross-section due to excessive rusting or other causes.

Position reinforcement accurately at the spacing indicated. Support and secure vertical bars against displacement. Horizontal reinforcement may be placed as the masonry work progresses. Where vertical bars are shown in close proximity, provide a clear distance between bars of not less than the nominal bar diameter or 1" (whichever is greater).

For columns, piers, and pilasters, provide a clear distance between vertical bars as indicated, but not less than 1-1/2 times the nominal bar diameter or 1-1/2", whichever is greater. Provide lateral ties as indicated.

Splice reinforcement bars where shown; do not splice at other points unless acceptable to the Architect. Provide lapped splices, unless otherwise indicated. In splicing vertical bars of attaching to dowels, lap ends, place in contact and wire tie.

Provide not less than minimum lap indicated, or if not indicated, as required by governing code.

Embed metal ties in mortar joints as work progresses, with a minimum mortar cover of 5/8" on exterior face of walls and 1/2" at other locations.

Embed prefabricated horizontal joint reinforcement as the work progresses, with a minimum cover of 5/8" on exterior face of walls and 1/2" at other locations. Lap units not less than 6" at ends. Use prefabricated "L" and "T" units to provide continuity at corners and intersections. Cut and bend units as recommended by manufacturer for continuity at returns, offsets, column fire-proofing, pipe enclosures, and other special conditions.

Anchoring: Anchor reinforced masonry work to supporting structure they indicated.

Anchor reinforced masonry walls to non-reinforced masonry where they intersect.

#### **INSTALLATION, GENERAL:**

Refer to Section "Unit Masonry" for general installation requirements of unit masonry.

Temporary Formwork: Provide formwork and shores as required for temporary support of reinforced masonry elements.

Construct formwork to conform to shape, line, and dimensions shown. Make sufficiently tight to prevent leakage of mortar, grout, or concrete (if any). Brace, tie, and support as required maintaining position and shaping during construction and curing of reinforced masonry.

Do not remove forms and shores until reinforced masonry members have hardened sufficiently to carry their own weight and all other reasonable temporary loads that may be placed on them during construction.

Allow not less than the following minimum time to elapse after completion of members before removing shores or forms, provided suitable curing conditions have been obtained during the curing period.

- 10 days for girders and beams.
- 7 days for slabs.
- 7 days for reinforced masonry soffits.

#### **INSTALLATION OF REINFORCED CONCRETE UNIT MASONRY:**

##### **General:**

Do not wet concrete masonry units (CMU).

Lay CMU units with full-face shell mortar beds. Fill vertical head joints (end joints between units) solidly with mortar from face of unit to a distance behind face equal to not less than the thickness of longitudinal face shells. Solidly bed cross-webs of starting courses in mortar. Maintain head and bed joint widths shown, or if not shown, provide 3/8" joints.



**Walls:**

**Pattern Bond:** Lay CMU wall units in 1/2-running bond with vertical joints in each course centered on units in courses above and below, unless otherwise indicated. Bond and interlock each course at corners and intersections. Use special-shaped units where shown, and as required for corners, jambs, sash, control joints, lintels, bond beams, and other special conditions.

Maintain vertical continuity of core or cell cavities, which are to be reinforced and grouted, to provide minimum clear dimension indicated and to provide minimum clearance and grout coverage for vertical reinforcement bars. Keep cavities free of mortar. Solidly bed webs in mortar where adjacent to reinforced cores or cells.

Where horizontal reinforced beams (bond beams) are shown, use special units or modify regular units to allow for placement of continuous horizontal reinforcement bars. Place small mesh expanded metal lath or wire screening in mortar joints under bond beam courses over cores or cells of non-reinforced vertical cells, or provide units with solid bottoms.

**Grouting Technique:** At the Contractor's option, use either low-lift or high-lift grouting techniques subject to requirements which follow.

**Low-Lift Grouting:**

Provide minimum clear dimension of 2" and clear area of 8 sq. in. in vertical cores to be grouted.

Place vertical reinforcement prior to laying of CMU. Extend above elevation of maximum pour height as required for splicing. Support in position at vertical intervals not exceeding 192 bar diameters or 10 ft.

Lay CMU to maximum pour height. Do not exceed 4' height, or if bond beam occurs below 4' height stop pour at course below bond beam.

Pour grout using chute or container with spout. Rod or vibrate grout during placing. Place grout continuously; do not interrupt pouring of grout for more than one hour. Terminate grout pours 1-1/2" below top course of pour.

**Bond Beams:** Stop grout in vertical cells 1-1/2" below bond beam course. Place horizontal reinforcement in bond beam; lap at corners and intersections as shown. Place grout in bond beam course before filling vertical cores above bond beam.

**High-Lift Grouting:**

Do not use high-lift grouting technique for grouting of CMU unless minimum cavity dimension and area is 3" and 10 sq. in., respectively.

Provide cleanout holes in first course at all vertical cells which are to be filled with grout.

Use units with one face shell removed and provide temporary supports for units above, or use header units with concrete brick supports, or cut openings in one face shell.

Construct masonry to full height of maximum grout pour specified, prior to placing grout.

Limit grout lifts to a maximum height of 4' and grout pour to a full height of wall, unless otherwise indicated. Place vertical reinforcement before grouting. Place before or after laying masonry units, as required by job conditions. Tie vertical reinforcement to dowels at base of masonry where shown and thread CMU over or around reinforcement. Support vertical reinforcement at intervals not exceeding 192 bar diameters or 10'.

Where individual bars are placed after laying masonry, place wire loops extending into cells as masonry is laid and loosen before mortar sets. After insertion of reinforcement bar, pull loops and bar to proper position and tie free ends.

Place horizontal beam reinforcement as the masonry units are laid.

**Preparation of Grout Spaces:** Prior to grouting, inspect and clean grout spaces. Remove dust, dirt, mortar droppings, loose pieces of masonry and other foreign materials from grout spaces. Clean reinforcement and adjust to proper position. Clean top surface of structural members supporting masonry to ensure bond. After final cleaning and inspection, close cleanout holes and brace closures to resist grout pressures.

Do not place grout until entire height of masonry to be grouted has attained sufficient strength to resist displacement of masonry units and breaking of mortar bond. Install shores and bracing, if required, before starting grouting operations.

Place grout by pumping into grout spaces unless alternate methods are acceptable to the Architect.

Limit grout pours to sections which can be completed in one working day with not more than one hour interruption of pouring operation. Place grout in lifts which do not exceed 4'. Allow not less than 30 minutes, nor more than one hour between lifts of a given pour. Rod or vibrate each grout lift during pouring operation.

Place grout in lintels or beams over openings in one continuous pour.

Where bond beam occurs more than one course below top of pour, fill bond beam course to within 1" of vertically reinforced cavities, during construction of masonry.

When more than one pour is required to complete a given section of masonry, extend reinforcement beyond masonry as required for splicing. Pour grout to within 1-1/2" of top course of first pour. After grouted masonry is cured, lay masonry units and place reinforcement for second pour section before grouting. Repeat sequence if more pours are required.

**END OF SECTION 04230**

**SECTION 05100**

**STRUCTURAL STEEL**

**SUMMARY**

**GENERAL**

**RELATED DOCUMENTS**

Drawings and general provisions of Contract, including General and Supplementary Conditions and Division-1 Specification Sections, apply to work of this section.

**SUMMARY**

Extent of structural steel work is shown on drawings, including schedules, notes and details to show size and location of members, typical connections, and type of steel required.

Structural steel is that work defined in American Institute of Steel Construction (AISC) "Code of Standard Practice" and as otherwise shown on drawings.

Miscellaneous Metal Fabrications are specified elsewhere in Division 5.

Refer to Division 3 for anchor bolt installation in concrete; Division 4 for masonry.

Source Quality Control: Materials and fabrication procedures are subject to inspection and tests in mill, shop, and field, conducted by a qualified inspection agency. Such inspections and tests will not relieve Contractor of responsibility for providing materials and fabrication procedures in compliance with specified requirements.

Promptly remove and replace materials or fabricated components which do not comply.

Design of Members and Connections: Details shown are typical; similar details apply to similar conditions, unless otherwise indicated. Verify dimensions at site whenever possible without causing delay in the work. Promptly notify Architect whenever design of members and connections for any portion of structure are not clearly indicated.

**SUBMITTALS**

Shop Drawings: Submit shop drawings including complete details and schedules for fabrication and assembly of structural steel members, procedures and diagrams.

Include details of cuts, connections, camber, holes, and other pertinent data. Indicate welds by standard AWS A2.1 and A2.4 symbols, and show size, length, and type of each weld.

Provide setting drawings, templates, and directions for installation of anchor bolts and other anchorages to be installed as work of other sections.

**QUALITY ASSURANCE**

Code and Standards: Comply with provisions of following, except as otherwise indicated:

AISC "Code of Standard Practice for Steel Buildings and Bridges".

Paragraph 4.2.1 of the above code is hereby modified by deletion of the following sentence:  
"This approval constitutes the owner's acceptance of all responsibility for the design adequacy of any connections designed by the fabricator as a part of his preparation of these shop drawings".

AISC "Specifications for the Design, Fabrication, and Erection of Structural Steel for Buildings", including "Commentary" and Supplements thereto as issued.

AISC "Specifications for Architecturally Exposed Structural Steel".

AISC "Specifications for Structural Joints using ASTM A 325 or A 490 Bolts" approved by the Research Council on Riveted and Bolted Structural Joints of the Engineering Foundation.

American Welding Society (AWS) D1. "Structural Welding Code - Steel".

ASTM A 6 "General Requirements for Delivery of Rolled Steel Plates, Shapes, Sheet Piling and Bars for Structural Use".

Qualifications for Welding Work: Qualify welding processes and welding operators in accordance with AWS "Standard Qualification Procedure".

Provide certification that welders to be employed in work have satisfactory passed AWS qualification tests.

If recertification of welders is required, retesting will be Contractor's responsibility.

### **DELIVERY, STORAGE, AND HANDLING**

Deliver materials to site at such intervals to insure uninterrupted progress of work.

Deliver anchor bolts and anchorage devices, which are to be embedded in cast-in-place concrete or masonry, in ample time to not to delay work.

Store materials to permit easy access for inspection and identification. Keep steel members off ground, using pallets, platforms, or other supports. Protect steel members and packaged materials from erosion and deterioration.

Do not store materials on structure in a manner that might cause distortion or damage to members or supporting structures. Repair or replace damaged materials or structures as directed.

### **PRODUCTS**

#### **MATERIALS**

Metal Surfaces, General: For fabrication of work which will be exposed to view, use only materials which are smooth and free of surface blemishes including pitting, rust and scale seam marks, roller marks, rolled trade names and roughness. Remove such blemishes by grinding, or by welding and grinding, prior to cleaning, treating, and application of surface finishes.

Structural Steel Shapes, Plates and Bars: As shown on drawings.

Cold-Formed Steel Tubing: ASTM A 500, Grade B.

Headed Stud-Type Shear Connectors: ASTM A 108, Grade 1015 or 1020, cold finished carbon steel; with dimensions complying with AISC Specifications.

Anchor Bolts: ASTM A 307, nonheaded type unless otherwise indicated.

Adhesive Anchors: Hilti "HVA Adhesive Anchor System"

High-Strength Threaded Fasteners: Heavy hexagon structural bolts, heavy hexagon nuts, and hardened washers, as follows:

Quenched and tempered medium-carbon steel bolts, nuts and washers, complying with ASTM A 325.

Direct tension indicator washers may be used at Contractor's option.

Electrodes for Welding: Comply with AWS Code.

Structural Steel Primer Paint: GPA - 313.

Non-metallic Shrinkage-Resistant Grout: Pre-mixed, non-metallic, non-corrosive, non-staining product containing selected silica sands, portland cement, shrinkage compensating agents, plasticizing and water reducing agents, complying with CE-CRD-C621.

Available Products: Subject to compliance with requirements, products which may be incorporated in the work include, but are not limited to, the following:

Euco N.S.; Euclid Chemical Co.  
Crystex; L & M Construction Chemicals  
Masterflow 713; Master Builders  
Five Star Grout; U.S. Grout Corp.  
Upcon; Upco Chem. Div., USM Corp.  
Propak; Protex Industries, Inc.  
Set Non-Shrink; Set Products, Inc.

## **FABRICATION**

Shop Fabrication and Assembly: Fabricate and assemble structural assemblies in shop to greatest extent possible. Fabricate items of structural steel in accordance with AISC Specifications and as indicated on final shop drawings. Provide camber in structural members where indicated.

Properly mark and match-mark materials for field assembly. Fabricate for delivery sequence which will expedite erection and minimize field handling of materials.

Where finishing is required, complete assembly, including welding of units, before start of finishing operations. Provide finish surfaces of members exposed in final structure free of markings, burrs, and other defects.

Connections: Weld or bolt shop connections, as indicated.

Bolted field connections, except where welded connections or other connections are indicated.

Provide high-strength threaded fasteners for all bolted connections, except where unfinished bolts are indicated.

Provide unfinished threaded fasteners for only bolted connections of secondary framing members to primary members (including purlins, girts, and other framing members taking only nominal stresses) and for temporary bracing to facilitate erection.

High-Strength Bolted Construction: Install high-strength threaded fasteners in accordance with AISC "Specifications for Structural Joints using ASTM A 325 or ASTM A 490 bolts" (RCRBSJ).

Welded Construction: Comply with AWS Code for procedures, appearance and quality of welds, and methods used in correcting welding work.

Assemble and weld built-up sections by methods which will produce true alignment of axes without warp.

Holes for Other Work: Provide holes required for securing other work to structural steel framing, and for passage of other work through steel framing members, as shown on final shop drawings. Provide threaded nuts welded to framing, and other specialty items as indicated to receive other work.

Cut, drill, or punch holes perpendicular to metal surfaces. Do not flame cut holes or enlarge holes by burning. Drill holes in bearing plates.

### **SHOP PAINTING**

General: Shop paint structural steel, except those members or portions of members to be embedded in concrete or mortar. Paint embedded steel which is partially exposed on exposed portions and initial 2" of embedded areas only.

Do not paint surfaces which are to be welded.

Apply 2 coats of paint to surfaces which are inaccessible after assembly or erection. Change color of second coat to distinguish it from first.

Surface Preparation: After inspection and before shipping, clean steel work to be painted. Remove loose rust, loose mill scale, and spatter, slag or flux deposits. Clean steel in accordance with Steel Structures Painting Council (SSPC) as follows:

Painting: Immediately after surface preparation, apply structural steel primer paint in accordance with manufacturer's instructions and at a rate to provide dry film thickness of not less than 1.5 mils. Use painting methods which result in full coverage of joints, corners, edges, and exposed surfaces.

### **PART 3 EXECUTION**

#### **ERECTION**

Surveys. Employ a land surveyor for accurate erection of structural steel. Check elevations of concrete and masonry bearing surfaces, and locations of anchor bolts and similar devices, before erection work proceeds, and report discrepancies to Architect. Do not proceed with erection until corrections have been made, or until compensating adjustments to structural steel work have been agreed upon with Architect.

Temporary Shoring and Bracing: Provide temporary shoring and bracing members with connections of sufficient strength to bear imposed loads. Remove temporary members and connections when permanent members are in place and final connections are made. Provide temporary guy lines to achieve proper alignment of structures as erection proceeds.

Temporary Planking: AOSHA approved at temporary planking and working platforms as necessary to effectively complete work.

Setting Bases and Bearing Plates: Clean concrete and masonry bearing surfaces of bond-reducing materials and roughen to improve bond to surfaces. Clean bottom surface of base and bearing plates.

Set loose and attached base plates and bearing plates for structural members on wedges or other adjusting devices.

Tighten anchor bolts after supported members have been positioned and plumbed. Do not remove wedges or shims, but if protruding, cut off flush with edge of base or bearing plate prior to packing with grout.

Pack grout solidly between bearing surfaces and bases or plates to ensure that no voids remain. Finish exposed surfaces, protect installed materials, and allow to cure.

Field Assembly: Set structural frames accurately to lines and elevation indicated. Align and adjust various members forming part of complete frame or structure before permanently fastening. Clean bearing surfaces and other surfaces which will be in permanent contact before assembly. Perform necessary adjustments to compensate for discrepancies in elevations and alignment.

Level and plumb individual members of structure within specified AISC tolerances.

Splice members only where indicated.

Erection Bolts: On exposed welded construction, remove erection bolts, fill holes with plug welds and grind smooth at exposed surfaces.

Comply with AISC Specifications for bearing, adequacy of temporary connections, alignment, and removal of paint on surfaces adjacent to field welds.

Do not enlarge unfair holes in members by burning or by use of drift pins, except in secondary bracing members. Ream holes that must be enlarged to admit bolts.

Gas Cutting: Do not use gas cutting torches in field for correcting fabrication errors in primary structural framing. Cutting will be permitted only on secondary members which are not under stress, as acceptable to Architect. Finish gas-cut sections equal to a sheared appearance when permitted.

Touch-Up Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint. Apply paint to exposed areas using same material as used for shop painting.

Apply by brush or spray to provide minimum dry film thickness of 1.5 mils.

### **QUALITY CONTROL**

Owner will engage an independent testing and inspection agency to inspect high-strength bolted connections and welded connections and to perform tests and prepare test reports. Contractor shall pay for such services from the testing allowance specified in Division 1, Section 01020.

Testing agency shall conduct and interpret tests and state in each report whether test specimens comply with requirements, and specifically state any deviations there from.

Provide access for testing agency to places where structural work is being fabricated or produced so that required inspection and testing can be accomplished.

Testing agency may inspect structural steel at plant before shipment; however, Architect reserves the right, at any time before final acceptance, to reject material not complying with specified requirements.

Correct deficiencies in structural steel work which inspections and laboratory test reports have indicated to be not compliance with requirements. Perform additional tests, at Contractor's expense, as may be necessary to reconfirm any non-compliance of original work, and as may be necessary to show compliance of corrected work.

**END OF SECTION 05100**

**SECTION 05500**

**METAL FABRICATIONS**

**GENERAL**

**RELATED DOCUMENTS**

Drawings and general provisions of Contract, including General and Supplementary Conditions and Division-1 Specifications Sections, apply to work of this section.

**DESCRIPTION OF WORK**

Definition: Metal fabrications include items made from iron and steel shapes, plates, bars, strips, tubes, pipes and castings which are not a part of structural steel or other metal systems specified elsewhere.

Extent of metal fabrications is indicated on drawings.

Types of work in this section include metal fabrications for:

Rough hardware.

Steel Ladder

Loose steel lintels

Miscellaneous framing and supports.

Floor plates as applicable.

Steel pipe railings.

Structural steel is specified in another section within Division 5.

**SYSTEM PERFORMANCES**

Structural Performances: Engineer and provide assemblies which, when installed, comply with the following minimum requirements for structural performance, unless otherwise indicated.

Handrails and Toprails: Capable of withstanding the following loads applied as indicated when tested per ASTM E 935.

Concentrated loads of 250 lbs. applied at any point in any direction.

Uniform load of 50 lbs. per linear ft. applied simultaneously in both vertical and horizontal directions.

Concentrated and uniform loads above need not be assumed to act concurrently.

Guards: Intermediate rails, capable of withstanding a uniform load of 25 lb. per sq. ft. of gross areas of guard, including any open areas, of which they are a part.

Above load need not be assumed to be acting concurrently with uniform horizontal loads on top rails of railing assembly in determining stress on guard supporting members.



Steel Ladder at Pressbox: The tread and riser of the closed riser steel stair assembly shall be diamond plate with factory enamel paint to resist foot traffic. Color to be selected by Architect.

### **QUALITY ASSURANCE**

Shop Assembly: Preassemble items in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation.

### **SUBMITTALS**

Product Data: Submit manufacturer's specifications, anchor details, and installation instructions for products used in miscellaneous metal fabrications, including paint products and grout.

Shop Drawings: Submit shop drawings for fabrication and erection of miscellaneous metal fabrications. Include plans, elevations, and details of sections and connections. Show anchorage and accessory items. Provide templates for anchor and bolt installation by others.

Where materials or fabrications are indicated to comply with certain requirements for design loadings, include structural computations and letter certifying compliance.

### **PRODUCTS**

#### **MATERIALS**

Ferrous Metals:

Metal Surfaces, General: For fabrication of miscellaneous metal work which will be exposed to view, use only materials which are smooth and free of surface blemishes including pitting, seam marks, roller marks, rolled trade names and roughness.

Steel Plates, Shapes, and Bars: ASTM A 36.

Steel Tubing: Cold-formed, ASTM A 500; or hot-rolled, ASTM A 50.

Steel Pipe: ASTM A 53; Type and grade (if applicable) as selected by fabricator and as required for design loading; galvanized; standard weight (schedule 40), unless otherwise indicated.

Brackets, Flanges, and Anchors: Cast or formed metal of the same type material and finish as supported rails, unless otherwise indicated.

Concrete Inserts: Threaded or wedge type; galvanized ferrous castings, either malleable iron, ASTM A 27. Provide bolts, washers, and shims as required, hot-dip galvanized, ASTM A 153.

Grout:

Non-Shrink Non-Metallic Grout: Pre-mixed, factory-packaged, non-staining, non-corrosive, non-gaseous grout complying with CE CRD-C621. Provide grout specifically recommended by manufacturer for interior and exterior applications of type specified in this section.

Fasteners: As recommended by fabricator's engineer.

Galvanizing Repair Paint: High zinc dust content paint for regalvanizing welds in galvanized steel, complying with the Military Specifications MIL-P-21035 (Ships) or SSPC-Paint-20.

Concrete Fill:

Concrete Materials and Properties: Comply with requirements of Division-3 section "Concrete Work" for normal weight, ready-mix concrete.

Non-Slip Aggregate Finish: Factory-graded, packaged material containing fused aluminum oxide grits or crushed emery as abrasive aggregate; rust-proof and non-glazing; unaffected by freezing, moisture or cleaning materials.

### **FABRICATION, GENERAL**

Workmanship: Use materials of size and thickness indicated, or if not indicated, as required to produce strength and durability in finished product for use intended. Work to dimensions indicated or accepted on shop drawings, using proven details of fabrication and support. Use type of materials indicated or specified for various components of work.

For Exposed Work true to line and level with accurate angles and surfaces and straight sharp edges. Ease exposed edges to a radius of approximately 1/8". Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing work.

Weld corners and seams continuously, complying with AWS recommendations. At exposed connections, grid exposed welds smooth and flush to match and blend with adjoining surfaces.

Form exposed connections with hairline joints, flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of type indicated or, if not indicated, Phillips flat-head (countersunk) screws or bolts.

Provide for anchorage of type indicated, coordinated with supporting structure. Fabricate and space anchoring devices to provide adequate support for intended use.

Cut, reinforce, drill, and tap miscellaneous metal work as indicated to receive finish hardware and similar items.

Galvanizing: Provide a zinc coating for those items indicated or specified to be galvanized, as follows:

ASTM A 123 for galvanized rolled, pressed and forged steel shapes, plates, bars, and strip 1/8" thick and heavier.

ASTM A 386 for galvanizing assembled steel products.

Fabricate joints which will be exposed to weather in a manner to exclude water or provide weep holes where water may accumulate.

### **ROUGH HARDWARE**

Furnish bent or otherwise custom fabricated bolts, plates, anchors, hangers, dowels, and other miscellaneous steel and iron shapes as required for framing and supporting woodwork, and for anchoring or securing woodwork to concrete or other structures. Straight bolts and other stock rough hardware items as specified in Division-6 sections.

Fabricate items to sizes, shapes, and dimensions required. Furnish malleable-iron washers for heads and nuts which bear on wood structural connections; elsewhere, furnish steel washers.

### **LOOSE STEEL LINTELS:**

Provide loose structural steel lintels for openings and recesses in masonry walls and partitions as shown. Weld adjoining members together to form a single unit where indicated. Provide not less than 8" bearing at each side of openings, unless otherwise indicated.

Galvanize loose steel lintels to be installed in exterior walls.

### **MISCELLANEOUS FRAMING AND SUPPORTS**

Provide miscellaneous steel framing and supports which are not a part of structural steel framework, as required to complete work.

Fabricate miscellaneous units to sizes, shapes, and profiles indicated or, if not indicated, of required dimensions to receive adjacent other work to be retained by framing. Except as otherwise indicated, fabricated from structural steel shapes, plates, and steel bars of welded construction using mitered joints for field connection. Cut, drill, and tap units to receive hardware and similar items.

Equip units with integrally welded anchors for casting into concrete or building into masonry. Furnish insets if units must be installed after concrete is placed.

### **EXECUTION**

#### **PREPARATION**

Field Measurements: Take field measurements prior to preparation of shop drawings and fabrication, where possible. Do not delay job progress; allow for trimming and fitting where taking field measurements before fabrication might delay work.

Coordinate and furnish anchorages, setting drawings, diagrams, templates, instructions, and directions for installation of anchorages, such as concrete inserts, sleeves, anchor bolts, and miscellaneous items having integral anchors, which are to be embedded in concrete or masonry construction. Coordinate delivery of such items to project site.

#### **INSTALLATION**

General: Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary for securing miscellaneous metal fabrication to in-place construction; including, threaded fasteners for concrete and masonry inserts, toggle bolts, through-bolts, lag bolts wood screws, and other connectors as required.

Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installation, of miscellaneous metal fabrications. Set work accurately in location, alignment and elevation, plus, level, true and free of rack, measured from established lines and levels. Provide temporary bracing or anchors in form work for items which are to be built into concrete masonry or similar construction.

Fit exposes connections accurately together to form tight hairline joints. Weld connections which are not to be left as exposes joint, but cannot be shop welded because of shipping size limitations. Grind exposes joints smooth and touch-up shop paint coat. Do not weld, cut, or abrade the surfaces of exterior units which have been hot-dip galvanized after fabrication.

Field Welding: Comply with AWS Code for procedures of manual shielded metal-arc welding, appearance and quality of welds made, and methods used in correcting welding work.

Pack grout solidly between bearing surfaces and plates to ensure that no voids remain.

#### **ADJUST AND CLEAN**

For galvanize surfaces: Clean field welds, bolted connections and abraded areas and apply of galvanizing repair paint to comply with ASTM A 780.

**END OF SECTION 05500**

**SECTION 13120**

**METAL BUILDING SYSTEM**

**PART 1 - GENERAL**

**RELATED DOCUMENTS:**

Drawing and general provisions of the Contract, including General and Supplementary Conditions and Division-1 Specifications sections apply to work specified in this section.

**SUMMARY:**

Extend of metal building system work is shown on drawings.

**SUBMITTALS:**

Product Data: Submit manufacturer's product information, specifications, and installation instructions for building components and accessories.

Shop Drawings: Submit complete erection drawings showing anchor bolts settings, sidewall, end wall, and roof framing, transverse cross sections, covering and trim details, and accessory installation details to clearly indicate proper assembly of building components. Provide frame profile drawings indicating frame dimensions and clearances from finish floor.

Certification: Submit written Certification prepared and signed by a Professional Engineer, registered to practice in the State of Texas, verifying that building design meets indicated loading requirements and codes of authorities having jurisdiction.

**QUALITY ASSURANCE:**

**Design Criteria:**

Structural Framing: Design primary and secondary structural members and exterior covering materials for applicable loads and combinations of loads in accordance with the Metal Building Association's (MBMA) "Design Practices Manual".

Structural Steel: For design of structural steel members, comply with requirements of the American Institute of Steel Construction's (AISC) "Specifications for the Design, Fabrication, and Erection of Structural Steel for Buildings" for design requirements and allowable stresses.

Light Gage Steel: For design of light gage steel members, comply with requirements of the American Iron and Steel Institute's (AISI) "Specifications for the Design of Cold Formed Steel Structural Members" and "Design of Light Gage Steel Diaphragms" for design requirements and allowable stresses.

Welding Connections: Comply with requirements of the American Welding Society's (AWS) "Standard Code for Arc and Gas Welding in Building Construction" for welding procedures.

Design Loads: Basic design loads are indicated on the drawings. Collateral loads and Dead loads are not indicated in the drawings.

Basic design loads include live load and wind load in addition to the dead load and collateral loads.

Collateral loads include additional dead loads over and above the weight of the metal building system such as sprinkler systems and mechanical systems.

Design each member to withstand stresses resulting from combinations of loads that produce the maximum allowable stresses in that member as prescribed in MBMA's "Design Practices Manual".

Manufacturer's Qualifications: Provide metal building system as produced by a manufacturer with not less than 5 years successful experience in the fabrication of metal buildings of the type and quality required.

#### **DELIVERED, STORAGE, AND HANDLING:**

Deliver and store prefabricated components, sheets, panels, and other manufactured items so they will not be damaged or deformed.

Stack materials on platforms or pallets, covered with tarpaulins or other suitable weather tight ventilated covering. Store metal sheets or panels so that water accumulations will drain freely. Do not store sheets or panels in contact with other materials which might cause staining.

Maintenance Stock: Furnish at least 5% excess over required amount of nuts, bolts, screws, washers, and other required fasteners for each building. Pack in cartons and store on site where directed.

#### **PART 2 - PRODUCT**

##### **MANUFACTURERS:**

Manufacturer: Subject to compliance with specified requirements, provide the metal building systems provided by one of the following:

- A. Alliance Steel, Inc.
- B. Butler Manufacturing, Co.
- C. Ceko Building Division
- D. United Structures of America
- E. Varco-Pruden Buildings, AMCA International
- F. Whirlwind Steel Buildings, Inc.
- G. Fabricator/manufacturer that has AISC Certification.

##### **MATERIALS:**

###### **Metals:**

Hot-Rolled Structural Shapes: Comply with requirements of ASTM A36 or A529.

Tubing or Pipe: Comply with requirements of ASTM A500, Grade B, ASTM A501, or A53.

Paint and Coating Materials: Unless otherwise indicated, paint and coating materials shall comply with performance requirements of the federal specifications indicated. Unless specifically indicated otherwise, compliance with compositional requirements of the federal specifications indicated is not required.

Primers:

Shop Primer for Ferrous Metal: Provide fast-curing, lead-free, "universal" primer, as selected by the manufacturer for resistance to normal atmospheric corrosion, compatibility with finish paint systems indicated and capability to provide a sound foundation for field-applied topcoats despite prolonged exposure. Comply with performance requirements of FS TT-P-645.

Finish Coats:

Shop-Primed Metal Surfaces: Provide semi-gloss alkyd enamel finish coat as recommended by the manufacturer for use over primed ferrous metal surfaces. Comply with performance requirements of FS TT-E-529, Class A.

**STRUCTURAL FRAMING:**

Rigid Frames shall be fabricated from hot-rolled structural steel. Provide built-up "I-beam" shape consisting of either tapered or parallel flange beams and parallel columns. Provide frames factory welded and shop painted. Furnish frame complete with attachment plates, bearing plates and splice members. Factory drill frames for bolted field assembly.

Provide length of span and spacing of frames indicated. Slight variations in length of span and frame spacing may be acceptable if necessary to meet manufacturer's standard.

Maximum and/or minimum dimensions of Rigid Frames are indicated on the drawings.

Horizontal Wind Bracing: Provide adjustable wind bracing using not less than 1/2" diameter threaded steel rods; comply with requirements of ASTM A36 or A572, Grade D. Locate interior end bay bracing only where indicated.

Vertical Wind Bracing: Refer drawings for acceptable methods

Secondary Framing:

Provide not less than 16-ga. Shop painted rolled formed sections for the followings secondary framing members:

- Purlins.
- Eave struts.
- Endwall beams.
- Flange bracing.
- Sag bracing.

Provide not less than 14-ga. Cold-formed galvanized steel sections for the following secondary framing members:

Base channels.  
Sill angles.  
Endwall structural members (except columns and beams).  
Purlins spacers.

Bolts: Provide shop painted bolts, except when structural framing components are in direct contact with roofing and siding panels. Provide zinc-plated or cadmium-plated bolts when structural framing components are in direct contact with roofing and siding panels.

Shop Painting: Clean surfaces to be primed of loose mill scale, rust, dirt, oil, grease, and other matter precluding paint bond. Follow procedures of SSPC-SP3 for power tool cleaning, SSPC-SP7 for brush-off blast cleaning, and SSPC-SP1 for solvent cleaning.

Prime structural steel primary and secondary framing members with the manufacturer's standard rust-inhibitive primer.

Prime galvanized members, after phosphoric acid pretreatment with manufacturer's standard dust-zinc oxide primer.

Finish: Provide shop-applied Dexstar 850 finish to galvanized steel trim and accessory elements.

Clean galvanized steel with an alkaline compound, then treat with a zinc phosphate conversion coating, and seal with a chromic acid rinse.

Interior finish: Apply .2 mil thick wash coat.

Colors as selected by Architect from the manufacturer's standards.

## **FABRICATION:**

General: Design prefabricated components and necessary field connections required for erection to permit easy assembly and disassembly. Fabricate components in such a manner that once assembled, they may be disassembled, repackaged, and reassembled with a minimum amount of labor.

Clearly and legibly mark each piece and part of the assembly to correspond with previously prepared erection drawings, diagrams, and instruction manuals.

Structural Framing: Shop fabricate structural framing components to the indicated size and section complete with base plates, bearing plates and other plates required for erection, welded in place. Provide required holes for anchoring or connections either shop drilled or punched to template dimensions.

Shop Connections: Provide welded shop connections.

Field Connections: Provide bolted field connections.

**PART 3 - EXECUTION**

**ERECTION:**

Bracing: Provide diagonal rod and moment resisting frame bracing in both roof and sidewalls as indicated.

**ROOF AND SIDING:**

General: Arrange and nest sidelap joints so that prevailing winds blow over, not into, lapped joints. Lap ribbed or fluted sheets one full rib corrugation. Apply panels and associated items for neat and weathertight enclosure. Avoid "panel creep" or application not true to line. Protect factory finishes from damage.

Align bottoms of wall panels with blind rivets, bolts or self-tapping screws. Fasten flashings, trim around openings, and similar elements with self-tapping screws. When building height requires two rows of panels at gable ends, align lap of gable panels over wall panels at eave height.

Install screw fasteners with power tool having controlled torque adjusted to prevent damage to screw threads, or panels. Install screws in predrilled holes.

**END OF SECTION 13120**



SECTION 00 61 00 — PERFORMANCE BOND AND LABOR AND MATERIAL PAYMENT BOND

PART 1 - GENERAL

1.1 RELATED DOCUMENTS: PERFORMANCE BOND AND LABOR AND MATERIAL PAYMENT BOND:

- A. The Contractor shall, prior to the execution of the Contract, furnish bonds covering the faithful performance of the Contract and the payment of all obligations arising thereunder in the amount of 100% of the Contract Price covering 100% performance and 100% payment, and with such sureties secured through the contractor's usual sources as may be agreeable to the parties.
- B. The Contractor shall deliver the required bonds to the Owner not later than the date of execution of the Contract, or if the work is commenced prior thereto in response to a letter of intent, the Contractor shall, prior to commencement of the Work, submit evidence satisfactory to the Owner that such bonds will be finished.
- C. The Contractor shall require the Attorney-In-Fact who executes the required bonds on behalf of the surety to affix thereto a certificate and current copy of his Power of Attorney.
- D. Any Payment Bond and Performance Bond furnished pursuant to the provisions of Art. 5160, Vernon's Texas Civil Statutes, connected with this project, shall be furnished by a corporate surety or corporate or corporate sureties in accordance with Article 7.19-1, Vernon's Texas Insurance Code, that has a stated capital and surplus (as reported by it to the Texas Insurance Commission in its most recent report) that is in excess of ten times the stated amount of the Payment Bond or the Performance Bond. Provided however, that if any Payment Bond or any Performance Bond is in an amount in excess of ten percent (10%) of the surety company's capital and surplus (as reported to the Texas Insurance Commission in its most recent report), as a condition to accepting the bond, the Owner must receive written certification and information, satisfactory in form and substance to the Owner, that the surety company has reinsured the portion of the risk that exceeds ten percent (10%) of the surety company's capital and surplus, with one or more reinsurers who are duly authorized, accredited or trusted to do business in the State of Texas. For the purpose of this requirement, any amount reinsured by any reinsurer may not exceed ten percent (10%) of the reinsurer's capital and surplus (as reported to the Texas Insurance Commission by the reinsurer in its most recent report). In the event there is one or more reinsurer, the surety company must provide all necessary information and certification related to the current financial condition of the surety company and any and all reinsurers required by the Owner, together with copies of all reinsurance contracts with the surety company, before any such Payment Bond and Performance Bond is eligible to be considered acceptable by the Owner.
- E. ALL CONTRACTORS SHALL SUBMIT THE NAME, ADDRESS AND TELEPHONE NUMBER OF THE CORPORATED SURETIES PROVIDING THE PAYMENT BOND AND PERFORMANCE BOND AND THE LOCAL AGENT.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 00 62 76 — TAX EXEMPT ORGANIZATION CERTIFICATE

PART 1 - GENERAL

1.1 DEFINITION

- A. This Contract is to be performed for an exempt organization as defined by Title 2; Subtitle E; Chapter 150 of the Texas Limited Sales, Excise and Use Tax Act and Section 151.311 of the State Statutes. The Owner will furnish the Contractor proof or Certificate of Exemption upon award of contract.
- B. Proposer shall not include sales tax in their Proposal.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 00 73 00 — SUPPLEMENTARY CONDITIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS: SUPPLEMENTARY CONDITIONS

- A. The Supplementary Conditions modify, change, delete from or add to the General Conditions and shall apply to each and every Section of the Work as though written in full therein.
- B. The following paragraphs and subparagraphs take precedence over the General Conditions. Where any part of the General Conditions is modified or deleted by the Supplementary Conditions, the unaltered provisions remain in effect.
- C. Paragraph numbers and titles refer to like numbers and titles in the General Conditions.

1.2 EXECUTION, CORRELATION AND INTENT

1.3 Add the following subparagraphs.

- 1.4 1.2.6 Scope paragraphs placed at the beginning of the SECTIONS present a brief indication of the principal Work included in that SECTION, but do not limit Work to subject mentioned nor purport to itemize Work that may be included.

- 1.5 The Relation of Specifications and Drawings shall be equal in authority and priority. Should they disagree in themselves, or with each other, bids shall be based on the most expensive combination of quality and quantity of work indicated. The appropriate Work, in the event of the above mentioned disagreements, shall be determined by the Architect, at no additional cost to the Owner.

- 1.6 1.2.8 Failure to report a conflict in the Contract Documents, prior to opening of Proposal, shall be deemed evidence that the Contractor has elected to proceed in the more expensive manner, at no additional cost to the Owner.

- A. 1.2.9 The Specifications have been partially “streamlined” and some words and phrases have been intentionally omitted. Missing portions shall be supplied by inference as with notes on drawings.
- B. 1.2.10 The words “approved”, “inspected”, “directed”, “selected”, and similar words and phrases shall be presumed to be followed by “by Architect”. The words “satisfactory”, “submitted”, “reported”, and similar words and phrases shall be presumed to be followed by “to Architect”. Words like “install”, “provide”, “locate”, “furnish”, and “supply” shall be construed to include complete furnishing and installing of construction. Words like “Bids”, “Bidders”, shall be construed to be “Proposals”, “Proposers”, or “offers”, “offerors”, respectively.

C. INFORMATION AND SERVICES REQUIRED OF THE OWNER

- D. Delete 2.2.5 and replace with the following subparagraph.

- E. 2.2.5 The General Contractor will be furnished, free of charge, fifteen (15) sets of Drawings and Specifications for use in construction of this Project. Additional Drawings and Specifications will be

furnished the General Contractor at the Contractor's expense, but shall remain the property of the Architect. Cost of additional sets will be the cost of reproduction.

F. LABOR AND MATERIALS

G. Add the following subparagraphs 3.4.3 and 3.4.4 to 3.4:

H. After the Contract has been executed, the Owner and the Architect will consider a formal request for the substitution of products in place of those specified only under the conditions set forth in the General Requirements (Division 1 of the Specifications).

I. By making requests for substitutions based on subparagraph 3.4.3 above, the Contractor:

1. Represents that the Contractor has personally investigated the proposed substitute product and determined that it is equal or superior in all respects to that specified;
2. Represents that the Contractor will provide the same warranty for the substitution that the Contractor would for that specified;
3. Will coordinate the installation of the accepted substitute, making such changes as may be required for the Work to be complete in all respects, all at no additional cost to the Owner.

7.3 CONSTRUCTION CHANGE DIRECTIVES

7.3.3.1 CHANGE TO READ:

Mutual acceptance of a lump sum properly itemized in accordance with 7.3.6.1, 7.3.6.2 and 7.3.6.3. Items listed in 7.3.6.4 and 7.3.6.5 shall be a part of the overhead scheduled 7.3.10 following. Items shall be supported by sufficient substantiating data to permit evaluation;

7.3.6 In the first sentence, delete the words "a reasonable allowance for overhead and profit" and substitute "an allowance for overhead and profit in accordance with Clauses 7.3.10.1 through 7.3.10.6 following:

7.3.6.4 DELETE the final "and" then add the following to the sentence: are a part of overhead schedule in 7.3.10 following".

7.3.6.5 ADD the following to the sentence: "are apart of overhead schedule in 7.3.10 following".

ADD the following subparagraph 7.3.10 to 7.3:

7.3.10 In subparagraph 7.3.6, the allowance for the combined and profit included in the total cost to the Owner shall be based on the following schedule:

1. For the Contractor, for Work performance by the Contractor's own forces, 10 percent of the cost.
2. For the Contractor, for Work performance by the Contractor's contractor, 6 percent of the amount due to the Sub-subcontractor.
3. For each Subcontractor or Sub-subcontractor involved, for Work performed by that Subcontractor's or Sub-subcontractor's own forces, 10 percent of the cost.
4. For each Subcontractor, for Work performed by the Subcontractor's, Sub-subcontractor's, 6 percent of the amount due the Sub-subcontractor.
5. Cost to which overhead and profit is to be applied shall be determined in accordance with Subparagraph 7.3.6.
6. In order to facilitate checking of quotations for extras or credits, all proposals, except those so minor that their propriety can be seen by inspection, shall be accompanied by a complete itemization of costs including labor, materials and Subcontracts. Labor and materials shall be itemized in the manner prescribed above. Where major cost items are Subcontracts, they shall be itemized also. In no case will a change involving over \$500.00 be approved without such itemization.

8.1 DEFINITIONS

Add the following subparagraph.

8.1.5 The term working Day as used in the Contract Documents for extensions of time shall mean normal working day excluding weekends and legal holidays.

8.3 DELAYS AND EXTENSIONS OF TIME

Delete paragraph 8.3.2 and replace with the following subparagraph.

8.3.2 Any claim for extension of time shall be made in writing to the Architect not more than ten (10) days after the commencement of the delay; otherwise, it shall be waived. In case of a continuing delay only one claim is necessary. In case of claims for extensions of time because of adverse weather, such extensions of time shall be granted only when such adverse weather prevented the execution of major items of Work on normal working days

and exceeds the number of days included in the Contract time. The Contractor shall provide an estimate of the probable effect of such delay on the progress of the Work. In the event an extension of time is granted such extension shall be the complete claim allowed. Contractor shall not be entitled to additional compensation such as, but not limited to, compensable extended overhead or lost profit.

9.6 PROGRESS PAYMENTS

Add the following subparagraph to 9.6.1

1. Unless otherwise indicated in the Agreement, the Owner will pay ninety-five (95%) percent of the amount due the Contractor on account of progress payment until final payment.

Add the following paragraphs to 9.11 to Article 9:

9.11 LIQUIDATED DAMAGES:

9.11.1 If the Contractor neglects, fails or refuses to complete the Work within the time specified in the Contract, or any proper extension thereof granted by the Owner, then the Contractor does hereby agree, as part consideration of the awarding of this Contract, to pay the Owner the amount of ***FIVE HUNDRED DOLLARS (\$500.00)*** not as a penalty but as a liquidated damages for such breach of Contract as hereinafter set forth, for each and every calendar day that the Contractor shall be in default after the time stipulated in the Contract for completing the Work.

9.11.2 The said amount is fixed and agreed upon by and between the Contractor and the Owner because of the impracticability and extreme difficulty of fixing and ascertaining the actual damages the Owner would, in such event, sustain.

9.11.3 TIME SPECIFIED IN CONTRACT IS AS FOLLOWS:

*The Undersigned agrees to commence work within ten (10) days of Notice to Proceed and to substantially complete the work on or before six (6) months after notice to proceed.*

11.1 Article 11.1 Modify to include the following:

The Contractor shall furnish three (3) copies of insurance certificates to the Architect's office two (2) days after award of the project and before signing of the contract. The Certificate of Insurance shall include thirty (30) Day Notice of Cancellation; Architect and Owner shall receive the same notice in regard to any policy changes.

Owner and Architect shall be named as additional insured by the Contractor but not with respect to payment of premiums due under Contractor's policies. Coverage shall include any off site-work on adjacent public or private property.

Insurance Company/Carrier issuing the certificates must be listed by A.M. Best and have an "A" rating or better and based in the United States Mainland.

The insurance as required in Article 11.1 shall have "Minimum Limits" as follows:

A. WORKER'S COMPENSATION INSURANCE: Statutory Requirements-

1. All States Endorsements (Broad)
2. Voluntary Compensation
3. Waiver of Subrogation Endorsement

B. MINIMUM EMPLOYER'S LIABILITY: \$100,000/\$100,000/\$500,000

C. COMPREHENSIVE GENERAL LIABILITY INSURANCE MINIMUM LIABILITY AND COVERAGE:

1. Bodily Injury \$500,000 each person/\$500,000 each occurrence
2. Property Damage \$100,000 each occurrence/\$100,000 aggregate

OR-

3. \$500,000 Combined Single Limit Per Occurrence Bodily Injury and Property Damage.

- a. Premises and operations coverage
- b. Explosion and collapse hazard coverage
- c. Underground hazard coverage
- d. Products/completed operation hazard coverage with limits and coverage continuing one (1) year after job completion.
- e. Broad Form property damage coverage
- f. Personal injury coverage
- g. Waiver of subrogation endorsement
- h. Contractual liability (Broad Form) coverage
- i. Independent contractors coverage (Owners, Architects, and Contractors protective)

NOTE: If General Liability coverage is written on a "Claims Made" basis, the Certificate of Insurance should so indicate. If so written, Contractor agrees that coverage so certified beyond job completion and that coverage written will apply to claims made DURING CONSTRUCTION AND FOR ONE (1) YEAR THEREAFTER.

D. AUTOMOBILE LIABILITY INSURANCE with minimum limits of:

1. Bodily Injury: \$250,000 each person/\$500,000 each occurrence
  2. Property Damage: \$250,000 each occurrence/\$500,000 Combined Single Limit per Occurrence Bodily Injury and Property Damage.
  3. Automobile Liability Insurance shall include coverage for owned, non-owned, and hired vehicles with limits not less than shown above.
- E. OWNER'S AND CONTRACTOR'S PROTECTIVE LIABILITY:
1. Bodily Injury \$500,000 Single limit each occurrence
  2. Property Damage \$250,000 each occurrence/\$250,000 aggregate
- F. UMBRELLA LIABILITY:  
Minimum combined single limits \$100,000 with same inception and expiration dates as underlying liability policies and with coverage no less broad than in primary program.
- G. BUILDER'S RISK INSURANCE:  
The Contractor shall FURNISH AND PAY FOR and issue a Certificate of Builder's Risk Coverage to the Owner/Architect in accordance with the General Conditions and Conditions of the Contract.
- H. ARTICLE 11.4: PERFORMANCE BOND AND PAYMENT BOND:  
Delete in its entirety and substitute the following:  
11.4.1: Prior to signing of the Contract, the CONTRACTOR, at HIS/HER OWN EXPENSE, shall furnish a Performance Bond, and a Labor and Materials Payment Bond for one hundred (100%) percent of the Contract price on such form and with such sureties as the Owner may approve. ***Surety company furnishing the Bond must be listed by A.M. BEST and have an "A" rating or better and be based in the United States Mainland and authorized to provide such bonds on public work in the State of Texas.***
- J. Any Payment Bond and Performance Bond furnished pursuant to the provisions of Art. 5160, Vernon's Texas Civil Statutes, connected with this project, shall be furnished by a corporate surety or corporate or corporate sureties in accordance with Article 7.19-1, Vernon's Texas Insurance Code, that has a stated capital and surplus (as reported by it to the Texas Insurance Commission in its most recent report) that is in excess of ten times the stated amount of the Payment Bond or the Performance Bond. Provided however, that if any Payment Bond or any Performance Bond is in an amount in excess of ten percent (10%) of the surety company's capital and surplus (as reported to the Texas Insurance Commission in its most recent report), as a condition to accepting the bond, the Owner must receive written certification and information, satisfactory in form and substance to the Owner, that the surety company has reinsured the portion of the risk that exceeds ten percent (10%) of the surety company's capital and surplus, with one or more reinsurers who are duly authorized, accredited or trusted to do business in the State of Texas. For the purpose of this requirement, any amount reinsured by any reinsurer may not exceed ten percent (10%) of the reinsurer's capital and surplus (as reported to the Texas Insurance Commission by the reinsurer in its most recent report). In the event there is one or more reinsurer, the surety company must provide all necessary information and certification related to the current financial condition of the surety company and any and all reinsurers required by the Owner, together with copies of all reinsurance contracts with the surety company, before any such Payment Bond and Performance Bond is eligible to be considered acceptable by district.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

---

# **MEG GEOTECHNICAL ENGINEERING REPORT**

**PROPOSED  
FIRE STATION NO. 5**

**EDINBURG, HIDALGO COUNTY, TEXAS**

---



**Geotechnical Engineering • Construction Materials Engineering & Testing  
Environmental • Consulting • Forensics**

**GEOTECHNICAL ENGINEERING REPORT  
FOUNDATION AND PAVEMENT RECOMMENDATIONS  
PROPOSED FIRE STATION NO. 5  
EDINBURG, HIDALGO COUNTY, TEXAS**

**Prepared For  
Mardoqueo Hinojosa, P.E., CFM, CPM  
City of Edinburg**

**MEG Report No. 01-19-29113**

**March 1, 2019**



**MILLENNIUM ENGINEERS GROUP, INC.**  
TBPE FIRM NO. F-3913  
5804 N. GUMWOOD AVENUE  
PHARR, TEXAS 78577  
TEL:956-702-8500  
FAX:956-702-8140  
[WWW.MEGENGINEERS.COM](http://WWW.MEGENGINEERS.COM)



March 1, 2019

Mardoqueo Hinojosa, P.E., CFM, CPM  
City Engineer  
City of Edinburg  
415 W. University Drive  
Edinburg, Texas, 78541  
956-388-8955  
[mhinojosa@cityofedinburg.com](mailto:mhinojosa@cityofedinburg.com)

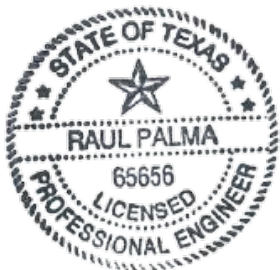
**Subject: Geotechnical Engineering Report  
MEG Report No. 01-19-29113  
Foundation and Pavement Recommendations  
Proposed Fire Station No. 5  
Edinburg, Hidalgo County, Texas**

Dear Mr. Hinojosa:

Millennium Engineers Group, Inc. is pleased to submit the enclosed geotechnical engineering report that was prepared for the above subject project. This report addresses the procedures and findings of our geotechnical engineering study. Our recommendations should be incorporated into the design and construction documents for the proposed development.

We want to emphasize the importance that all our recommendations presented in this report and/or addendums to this report be followed. We look forward to continuing our involvement in the project by providing construction monitoring in accordance with the report recommendations and materials testing services during construction. We strongly recommend that we be a part of the preconstruction meeting to address any specific issues that are pertinent to this project.

Thank you for the opportunity to be of service to you in this phase of the project and we would like the opportunity to assist you in the upcoming phases of the project. If you have any questions, please contact our office at the address, telephone, fax or electronic address listed below.



Cordially,  
**Millennium Engineers Group, Inc.**  
TBPE Firm No. F-3913

  
Raul Palma, P.E.  
President/ Chief Geotechnical Engineer

  
Amos Emerson, E.I.T.  
Geotechnical Engineer in Training

*The seal appearing on this document was authorized by Raul Palma, P.E. 65656 on March 1, 2019. Alteration of a sealed document without proper notification to the responsible engineer is an offence under the Texas Engineering Practice Act*

Cc: 1 Original and PDF Document

Millennium Engineers Group, Inc.  
5804 N. Gumwood Avenue  
Pharr, Texas 78577  
[www.megengineers.com](http://www.megengineers.com) Tel:956-702-8500 Fax:956-702-8140

MEG Project No.: 01-19-29113

Page II

## TABLE OF CONTENTS

|                                                                | Page |
|----------------------------------------------------------------|------|
| 1.0 INTRODUCTION .....                                         | 1    |
| 2.0 PROJECT DESCRIPTION .....                                  | 1    |
| 3.0 SCOPE AND LIMITATIONS OF STUDY .....                       | 1    |
| 4.0 FIELD EXPLORATION PROCEDURES .....                         | 2    |
| 5.0 GENERAL SITE CONDITIONS .....                              | 3    |
| 5.1 Site Description .....                                     | 3    |
| 5.2 Site Geology .....                                         | 3    |
| 5.3 Subsurface Conditions .....                                | 4    |
| 5.4 Groundwater Conditions .....                               | 4    |
| 6.0 ENGINEERING ANALYSIS AND RECOMMENDATIONS .....             | 5    |
| 6.1 General .....                                              | 5    |
| 6.2 Soil-Related Movements .....                               | 6    |
| 6.3 Conventional Shallow Slab Foundation Design Criteria ..... | 8    |
| 6.4 BRAB Design Criteria for Slab Foundations .....            | 9    |
| 7.0 CONSIDERATIONS DURING CONSTRUCTION .....                   | 10   |
| 7.1 Site Grading Recommendations .....                         | 10   |
| 7.2 Site Drainage Recommendations .....                        | 10   |
| 7.3 Site Preparation Recommendations .....                     | 10   |
| 7.4 Select Fill Recommendations .....                          | 11   |
| 7.5 Site Fill Recommendations .....                            | 12   |
| 7.6 Back Fill Recommendations .....                            | 12   |
| 7.7 Utility Considerations .....                               | 13   |
| 7.8 Utility Trench Recommendations .....                       | 13   |
| 7.9 Excavation, Sloping and Benching Considerations .....      | 13   |
| 7.10 Shallow Foundation Excavation Considerations .....        | 13   |
| 7.11 Landscaping Considerations .....                          | 14   |
| 7.12 Perimeter Foundation Cap .....                            | 14   |
| 8.0 PAVEMENT SECTION RECOMMENDATIONS .....                     | 15   |
| 8.1 General Information .....                                  | 15   |
| 8.2 Soil Stabilization Recommendations .....                   | 15   |
| 8.3 Design Methodology and Traffic Information .....           | 16   |
| 8.4 Recommend Pavement Sections .....                          | 17   |
| 8.5 Garbage Dumpster Considerations .....                      | 19   |
| 9.0 PAVEMENT MATERIAL SPECIFICATIONS SECTIONS .....            | 19   |
| 9.1 Pavement Preparation Recommendations .....                 | 19   |
| 9.2 Hot Mix Asphaltic Concrete Surface .....                   | 20   |
| 9.3 Reinforced Concrete .....                                  | 20   |
| 9.4 Untreated Caliche Base .....                               | 20   |
| 9.5 Lime Treated Caliche Base .....                            | 21   |
| 9.6 Lime Treated Subgrade .....                                | 21   |
| 9.7 Cement Treated Subgrade .....                              | 21   |
| 9.8 Moisture Conditioned Subgrade .....                        | 22   |

|      |                                          |    |
|------|------------------------------------------|----|
| 10.0 | OTHER CONSIDERATIONS.....                | 22 |
| 10.1 | Drainage .....                           | 22 |
| 10.2 | Curb and Gutters .....                   | 23 |
| 10.3 | Maintenance .....                        | 23 |
| 11.0 | PROJECT REVIEW AND QUALITY CONTROL ..... | 24 |

## APPENDIX

|                                                                          |  |
|--------------------------------------------------------------------------|--|
| APPENDIX A - CUSTOM SOIL RESOURCE REPORT .....                           |  |
| APPENDIX B - PROJECT LOCATION, TOPOGRAPHIC AND BOREHOLE MAPS.....        |  |
| APPENDIX C - BORING LOGS AND PROFILE .....                               |  |
| APPENDIX D - SUMMARY OF SOIL SAMPLE ANALYSIS .....                       |  |
| APPENDIX E - LABORATORY AND FIELD PROCEDURES .....                       |  |
| APPENDIX F - TENSAR SPECTRA PAVE TX5 CALCULATIONS AND SPECIFICATION..... |  |

## **1.0 INTRODUCTION**

Millennium Engineers Group, Inc. (MEG) has completed and is pleased to submit this document that presents our findings as a result of a geotechnical engineering study of this project to our client. The project site is located on Davis Road near Expressway 281 approximately at Latitude: 26°21'23.13"N, Longitude: 98° 8'30.37"W in Edinburg, Hidalgo County, Texas. The project location is shown on the Project Location Map, found in the Appendix section of this report. This report briefly describes the procedures utilized during this study and presents our findings along with our recommendation, for foundation design and construction considerations.

Our scope of services for the project was outlined in MEG proposal No. 01-18-309G, dated December 21, 2018 and approved by Mr. Juan G. Guerra, CPA, City Manager on January 23, 2018 through job task order number one (1) to the agreement between the City of Edinburg and Millennium Engineers Group.

## **2.0 PROJECT DESCRIPTION**

It is our understanding that the proposed site will accommodate the construction of a new Fire Station. It is also our understanding that the proposed Fire Station will consist of a one (1) story structure. The site construction for the proposed structure is anticipated to be on a slab-on-grade or on-fill foundation provided expansive, soil-related movements will not impair the performance of the structure.

The site will also accommodate the construction of either a rigid or flexible pavement that will serve for light duty and/or heavy duty traffic. Both flexible and rigid pavement types are viable and the selection on the type to be used will depend on the specific needs and criteria of this project site.

## **3.0 SCOPE AND LIMITATIONS OF STUDY**

This engineering report has been prepared in accordance with accepted geotechnical engineering practices currently exercised by geotechnical engineers in this area. No warranty, expressed or implied, is made or intended. This report is intended for the exclusive use by the client and client's authorized project team for use in preparing design and construction documents for this project only. This report may only be reproduced in its entirety for inclusion in construction documents. This report in its entirety shall not be reproduced or used for any other purposes without the written consent of our firm. This report may not contain sufficient information for purposes of other parties or other uses and is not intended for use in determining construction means and methods.

The recommendations presented in this report are based on data obtained from the soil borings drilled at this site and our understanding of the project information provided to us by our client and other project team members, and the assumption that site grading will result in only minor changes in the existing topography. Subsurface soil conditions have been observed and interpreted at the boring locations only.

This report may not reflect the actual variations of the subsurface conditions across the subject site. It is important to understand that variations may occur due to real geologic conditions or previous uses of the site. The nature and extent of variations across the subject site may not become evident until specific design locations are identified and/or construction commences. The construction process itself may also alter subsurface conditions. If variations appear evident at the time during the design phase and/or construction phase, we should be notified immediately to determine if our opinions, conclusions and recommendations need to be reevaluated. It may be necessary to perform additional field and laboratory tests and engineering analyses to establish the engineering impact of such variations. These services are additional and are not a part of our project scope.

The engineering report was conducted for the proposed project site described in this report. The conclusions and recommendations contained in this report are not valid for any other project sites. If the project information described in this report is incorrect, is altered, or if new information becomes available, we should be retained to review and modify our recommendations. These services are additional and are not a part of our project scope.

Our scope of services was limited to the proposed work described in this report, and did not address other items or areas. The scope of our geotechnical engineering study does not include environmental assessment of the air, soil, rock or water conditions on or adjacent to the site. No environmental opinions are presented in this report. If the client is concerned with environmental risk at this project site, the client should perform an environmental site assessment.

If final grade elevations are significantly different from existing grades at the time of our field activities (more than plus or minus one (1) foot), our office should be informed about these changes. If desired, we will reexamine our analyses and make supplemental recommendations.

#### **4.0 FIELD EXPLORATION PROCEDURES**

Subsurface conditions at the subject site were evaluated by five (5) 25-foot soil borings and nine (9) 10-foot soil borings. The Borings were drilled at the locations shown on the Borings Location Map, found in the Appendix section of this report. This location is approximate and distances were measured using a measuring wheel, tape, angles, and/or pacing from existing references. The structural soil borings were drilled in general accordance with American Society of Testing Materials (ASTM) D 420 procedures.

As part of our sampling procedures, the samples were collected in general conformance with ASTM D 1586 procedures. Representative portions of the samples were sealed in containers to reduce moisture loss, identified, packaged, and transported to our laboratory for subsequent testing. In the laboratory, each sample was evaluated and visually classified by a member of our Geotechnical Engineering staff. The geotechnical engineering properties of the strata were evaluated by a series of laboratory tests. The

results of the laboratory and field-testing are tabulated on the boring logs and Summary of Soil Sample Analyses which are found in the Attachments section of this report.

Standard penetration test results are noted on the boring logs as blows per 12 inches of penetration. Two 6 inch increments are performed for each standard penetration test. The sum of the blows for the two 6 inch increments is considered the “standard penetration resistance value” or “N-value.” Where hard or very dense materials were encountered, the tests are terminated as follows: (1) when a total of 50 blows have been applied in any of the 6 inch increments, or (2) when a total of 100 blows have been applied, or (3) when there is no observed advance of the sampler in the application of 10 successive blows. The boring logs in the case of hard or very dense materials will be noted as follows: 50/3”, where 50 is the number of blows applied in 3 inches of penetration, or 100/7½” where 100 is the number of blows applied in a total of 7 ½ inches of penetration, or 10/0”, where 10 is the number of blows applied in 0 inches of penetration.

Samples will be retained in our laboratory for 30 days after submittal of this report. Other arrangements may be provided at the request of the Client.

## **5.0 GENERAL SITE CONDITIONS**

### **5.1 Site Description**

The project site is located on Davis Road near Expressway 281 approximately at Latitude: 26°21'23.13"N, Longitude: 98° 8'30.37"W in Edinburg, Hidalgo County, Texas. The project location is shown on the Project Location Map, found in the Appendix section of this report. At the time of our field operations, the subject site can be described as an undeveloped tract of land. The general topography of the site is relatively flat sloping down to the south with a visually estimated vertical relief of less than 2 feet. Surface drainage is visually estimated to be poor to fair.

### **5.2 Site Geology**

According to the Soil Survey of Hidalgo County, Texas, published by the United States Department of Agriculture – Soil Conservation Service, the project site appears to be located within the Comitas, and Hidalgo soil association.

- The Comitas series consist of deep, well drained, sandy soils on uplands. These soils formed in sandy and loamy sediments that have been reworked by wind. Surface runoff is very slow and permeability is moderate rapid. Slopes range from 0 to 3 percent. Areas are elongated and have the long axis parallel to the prevailing southeasterly winds. These areas range from 40 to 200 acres. The corresponding soil symbol is 8, Comitas loamy fine sand, 0 to 3 percent slopes.



- The Hidalgo series consist of deep, well drained, nearly level soil is on convex uplands. These soils formed in calcareous loamy and clayey sediments. Surface runoff is slow and permeability is moderate. Slopes range from 0 to 1 percent. Areas are mostly broad and irregular in shape and range from 15 to 900 acres. The corresponding soil symbol is 25, Hidalgo - fine sandy loam, 0 to 1 percent slope.

### 5.3 Subsurface Conditions

On the basis of our borings, two (2) generalized strata that possess similar physical and engineering characteristics can describe the subsurface stratigraphy at this site. Table 5.1 summarizes the approximate strata range in our boring logs. These were prepared by visual classification and were aided by laboratory analyses of selected soil samples. The lines designating the interfaces between strata on the boring logs represent approximate boundaries. Transitions between strata may be gradual details for each of the borings can be found on the boring logs in the appendix of this report.

**Table 5.1. Approximate Subsurface Stratigraphy Depths.**

| Stratum | Range in Depth, ft <sup>1</sup> | Stratum Description <sup>1</sup>                                                           |
|---------|---------------------------------|--------------------------------------------------------------------------------------------|
| I       | 0 – 10                          | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff |
| II      | 10 – 25                         | Clayey SAND, brown to light brown, moist to wet, loose to dense                            |

Note 1: The stratum thickness and depths to strata interfaces are approximate. Our measurements are rounded off to the nearest foot increment and are referenced from ground surface at the time of our drilling activities. Subsurface conditions may vary between the boring locations.

### 5.4 Groundwater Conditions

The dry auger drilling technique was used to complete the soil borings in an attempt to observe the presence of subsurface water. During our drilling operations we encountered the groundwater table to be at approximately between fifteen (15) to twenty (20) feet below natural ground elevation for short term conditions. Moisture content testing exhibited high moisture contents at a depth of thirteen (13) feet below natural ground elevation. Table 5.2 summarizes the approximate groundwater and cave in depths measured in our explorations. It should be noted that the groundwater level measurements recorded are accurate only for the specific dates on which measurement were obtained and does not show fluctuations throughout the year.

Fluctuations in Groundwater levels are influenced by variations in rainfall and surface water run-off from season to season. The construction process itself may also cause variations in the groundwater level. If the subsurface water elevation is critical to the construction process the contractor should check the subsurface water conditions just prior to construction excavation activities.

**Table 5.2. Approximate Groundwater and Cave-in Depths.**

| Boring No. | Depth to Subsurface Water, Ft <sup>1</sup> |                | Depth to Cave-In, Ft <sup>1</sup> |                |
|------------|--------------------------------------------|----------------|-----------------------------------|----------------|
|            | Time of Drilling                           | 24 Hr. Reading | Time of Drilling                  | 24 Hr. Reading |
| B-1        | 20                                         | None           | None                              | 16             |
| B-2        | 15                                         | None           | None                              | 16             |
| B-3        | 19                                         | None           | None                              | 16             |
| B-4        | 18                                         | None           | None                              | 16             |
| B-5        | 20                                         | None           | None                              | 16             |
| P-1        | None                                       | None           | None                              | None           |
| P-2        | None                                       | None           | None                              | None           |
| P-3        | None                                       | None           | None                              | None           |
| P-4        | None                                       | None           | None                              | None           |
| P-5        | None                                       | None           | None                              | None           |
| P-6        | None                                       | None           | None                              | None           |
| P-7        | None                                       | None           | None                              | None           |
| P-8        | None                                       | None           | None                              | None           |
| P-9        | None                                       | None           | None                              | None           |

Note 1: Subsurface water levels and cave-in depths have been rounded to the nearest foot.

Based on the findings in our borings and on our experience in this region, we believe that groundwater seepage may not be encountered during site earthwork activities. If groundwater seepage is encountered during site earthwork activities, it may be controlled using temporary earthen berms and/or conventional sump-and-pump dewatering methods.

## 6.0 ENGINEERING ANALYSIS AND RECOMMENDATIONS

### 6.1 General

The analysis and recommendations presented in this report are applicable specifically to the proposed foundation structure. The data gathered from both the field and laboratory testing programs on soil samples obtained from the borings was utilized to establish geotechnical engineering parameters to develop recommendations for the proposed structure. The foundation system(s) considered in this report to provide support for the proposed structure must meet two independent criteria. One of the criteria is that the movement below the foundation structure due to compression (consolidation) or expansion (swell) of the underlying soils must be within tolerable limits. This criterion is addressed in the Soil Related Movements section of this report. The other criterion is that the dead and live loads must be distributed appropriately and the foundation structure



designed with an acceptable factor of safety to minimize the potential for bearing capacity failure of the underlying soils.

Geotechnical and structural engineers in this general area consider soil movements or Potential Vertical Rise (PVR) of approximately one (1) inch or less to be within acceptable structural design tolerances for most structures but may be different depending on structure use and the desired performance of the foundation. Therefore, movements of the underlying soils are not eliminated and thus one should expect a slab foundation structure to exhibit differential vertical movements. However, structural engineers design slab foundations for the expected magnitude of soil movements without failure of the structure. More stringent soil movement criteria may be established but the owner should consider the exponential increase in cost required to design and construct a structure for such soil movements. Data obtained in this study indicate that the soils at this site have strength characteristics capable of supporting the foundation and structure if designed appropriately. Stratum I is composed of Sandy lean CLAY to Clayey SAND and has a low to moderate potential to exhibit volumetric changes (contraction and expansion). Stratum II is composed of Clayey SAND and has a low potential to exhibit volumetric changes. The potential for soil volumetric changes is dependent on variations in moisture contents of the underlying soils. Based on this data, this site is suitable for a slab foundation provided the subgrade is modified in accordance with the recommendations established in this report to reduce the potential for these soil volumetric changes.

## 6.2 Soil-Related Movements

The anticipated ground movements due to swelling of the underlying soils at this site were estimated for slab foundation construction using the Texas Department of Transportation (TxDOT) procedures of test method TEX-124-E for determining Potential Vertical Rise (PVR). A PVR value of one (1) inch was estimated for the stratigraphic conditions encountered in our subsurface borings. A surcharge of 1 pound per square inch for the concrete slab, an active zone of 15 feet, and dry subsurface moisture conditions were assumed in estimating the above PVR values.

The following methods are generally acceptable for use in modifying the subgrade to reduce the potential for soil movements and volumetric changes below the foundation structure.

Excavate expansive clay soils and replace with select fill.  
Chemical injection of expansive clay soils.  
A combination of methods 1 and 2.

The method to be used is dependent on specific site conditions. At this site the grade will most likely need to be raised to obtain the proposed Finished Floor Elevation (FFE). **As of the date of this report the CLIENT/OWNER has not provided the proposed FFE. We recommend that the project civil engineer evaluate the proposed FFE with our recommendations to ensure that the subgrade modifications presented in the report are not diminished or compromised.** Adding select fill is generally the most cost effective method for reducing the potential for soil related movements. Therefore,

we only discuss this method in this report but we can provide details for the other methods if requested.

Based on the data obtained, the assumed FFE of two (2) feet above natural ground elevation, information provided by our client and our analysis of the site, we recommend the following modification (Table 6.1. Subgrade Modifications) of the subgrade within the building footprint to accomplish finish floor elevation of the subgrade at this site. While this method is not needed to maintain the potential for soil related movements to an approximate PVR value of less than one (1) inch, which is generally desired for projects of this type. It is recommended the Subgrade Modifications in Table 6.1 Subgrade Modifications be followed to produce provided bearing capacities. This method will maintain the potential for soil related movements to an approximate PVR value of less than one (1) inch, which is generally desired for projects of this type.

**Table 6.1. Subgrade Modifications**

| Item | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | See and adhere to the Site Preparation Recommendations section of this report.                                                                                                                                                                                                                                                                                                                             |
| 2    | Excavate existing soils to a depth of two (2) feet below natural ground elevation in accordance with the Site Preparation Recommendations section of this report.                                                                                                                                                                                                                                          |
| 3    | Condition and compact twelve (12) inches of subgrade below excavated soils in accordance with the Site Preparation Recommendations section of this report.                                                                                                                                                                                                                                                 |
| 4    | Place <b>select fill</b> , (a minimum of two (2) feet above natural ground, for a total of four (4) feet of select fill) condition and compact up to the proposed FFE in accordance with the Select Fill Recommendations section of this report. <b>The surface soils at this project site appear to be suitable soils (CL) for use as select fill. MEG needs to verify soils at time of construction.</b> |

The PVR method of estimating expansive, soil-related movements is based on empirical correlations utilizing the measured plasticity indices and assuming typical seasonal fluctuations in moisture content. If desired, other methods of estimating expansive, soil-related movements are available, such as estimations based on swell tests and/or soil-suction analyses. However, the performance of these tests and the detailed analyses of expansive, soil-related movements were beyond the scope of the current study. It should also be noted that actual movements can exceed the calculated PVR values as a result of isolated changes in moisture content (such as leaks, landscape watering, etc.) or if water seeps into the soils to greater depths than the assumed active zone depth due to deep trenching and/or excavations.

### 6.3 Conventional Shallow Slab-On-Grade Foundation Design Criteria

As indicated previously a slab foundation may be used at this site in conjunction with the subgrade modifications listed under the Soils Related Movements section. We recommend the following soil bearing pressures, and dimensional criteria for the slab grade beams. These recommendations ensure proper utilization of soil bearing capacity of continuous beam sections in the slab-on-grade foundation and reduce the potential of water migration from the outside to beneath the slab foundation. For structural considerations the beams may need to be greater and should be evaluated and designed by the structural engineer. Where concentrated load areas are present the grade beams or slab may be thickened and widened to serve as spread footings. Soil bearing pressures and beam dimensional criteria are as follows:

**Table 6.2. Bearing Criteria**

| <b>Grade Beams and Continuous Footings</b> |           |
|--------------------------------------------|-----------|
| Minimum depth below finished grade:        | 24 inches |
| Maximum depth below finished grade:        | 36 inches |
| Maximum width:                             | 30 inches |
| Maximum allowable bearing pressure:        | 1,500 psf |
| <b>Spread Footings (square)</b>            |           |
| Minimum depth below finished grade:        | 24 inches |
| Maximum depth below finished grade:        | 36 inches |
| Maximum width:                             | 60 inches |
| Maximum allowable bearing pressure:        | 1,800 psf |

The above-presented maximum allowable bearing pressures will provide a factor of safety of 3 with respect to the design soil strengths. For a slab foundation structure designed and constructed in accordance with the recommendations of this report, it is anticipated that total settlements will be in the order of one (1) inch or less. If lower anticipated total settlements are required for this project further mitigation may be required and MEG must be consulted for further recommendations.

Furthermore, the above design parameters are contingent upon the fill materials (if utilized) being selected and placed in accordance with the recommendations presented in the Select Fill Recommendations section of this report. Should select fill selection and placement differ from the recommendations presented herein, MEG should be informed of the deviations in order to reevaluate our recommendations and design criteria.

Excavations for slab on grade and spread footing foundations should be performed relatively clean and with an undisturbed bearing area. The bottom 6 inches of the excavation should be performed using a flat plate excavation bucket. The excavations

should be neatly excavated. No foreign debris or undisturbed soil should be left in the footing bottom. Should there be any abundance of foreign debris or disturbed soil found, it may be necessary to re-assess the fill site of its bearing capacity suitability. If the bearing area is found to be disturbed, the bearing area will require preparation and compaction for the entire depth of the disturbance in accordance with the Site Preparation and/or the Select Fill sections of this report.

The bearing surface of the grade beams and spread footings should be evaluated after excavation and immediately prior to concrete placement. We recommend that footing inspections be performed by a representative of MEG. The required inspections shall include inspecting for clean, dry (The moisture content should be within limits specified by the appropriate section in this report.) and undisturbed footing bottom, depth of footing, clearances from sides and size and spacing of reinforcing steel. Test results shall comply with the recommendations of this geotechnical report and shall be verified by an on-site representative of MEG.

Over excavation, if necessary, for compacted backfill placement below footings should extend laterally beyond all edges of the footings at least 8 inches per foot of over excavation depth below footing base elevation. The over excavation should then be backfilled up to the footing base elevation select fill placed in lifts of 8 inches or less in loose thickness and prepared and compacted in accordance with the Site Preparation and/or the Select Fill sections of this report. Equipment should not be operated and materials should not be placed or stockpiled within a horizontal distance equal to the excavation depth from the edge of the excavation. Excavations should not be placed next to existing structures or buried utilities/structures closer than a horizontal distance equal to the excavation depth unless some form of protection for the facilities is provided.

Water should not be allowed to accumulate at the bottom of the foundation excavation. Proper barriers such as berms or swales should be placed to divert any surface runoff away from excavations. To reduce the potential for groundwater seepage into the excavations and to minimize disturbance to the bearing area, we recommend that steel and concrete be placed as soon as possible after the excavations are completed, properly prepared and cleaned. Excavations should not be left open overnight.

#### **6.4 BRAB Design Criteria for Slab-on-Grade Foundations**

Table 6.3 list the values for criteria developed by the Building Research Advisory Board (BRAB) for the design of shallow slab-on-grade foundations. On the basis of stratigraphy encountered and the anticipated site modifications discussed earlier, the design criteria are as follows:

**Table 6.3. BRAB Values**

| <b>For Existing Conditions</b> |      |
|--------------------------------|------|
| Effective Plasticity Index     | 17   |
| Climatic Rating Cw.            | 15   |
| Soil Support Index, (c)        | 0.98 |
| <b>For Proposed Conditions</b> |      |
| Effective Plasticity Index     | 16   |
| Climatic Rating Cw.            | 15   |
| Soil Support Index, (c)        | 0.99 |

Note 1: Subgrade Modifications as outlined in the recommendations of this report;

## **7.0 CONSIDERATIONS DURING CONSTRUCTION**

### **7.1 Site Grading Recommendations**

Site grading plans can result in changes in almost all aspects of foundation recommendations. We have prepared the foundation recommendations based on the existing ground surface; there is a two (2) feet surcharge addition for the stratigraphic conditions encountered at the time of our study. If site grading plans differ from existing grades by more than plus or minus 1 foot, we must be retained to review the site grading plans prior to bidding the project for construction. This will enable us to provide input for any changes in our original recommendations that may be required as a result of site grading operations or other considerations.

### **7.2 Site Drainage Recommendations**

Drainage is one of the most important aspects to be addressed to ensure the successful performance of any foundation. Positive surface drainage should be implemented prior to, during and maintained after construction to prevent water ponding at or adjacent to the building facilities. It is recommended that the building and site design include rain gutters, downspouts and concrete gutters to channel runoff to paving or storm drains.

### **7.3 Site Preparation Recommendations**

Building areas and all area to support select fill should be stripped of all vegetation and organic topsoil up to a minimum of 5 ft. beyond the building perimeters. After stripping, remove at least six (6) inches of on-site soil as measured from existing grade when excavation of existing subgrade is not recommended in other sections of this report. The excavated material, if free of organic and/or deleterious material, may be stockpiled for use in the non-structural areas of the site. Where excavation of the subgrade is recommended in this report, the bottom of the excavation will extend at least five (5) feet

beyond the limits of the planned building perimeter including canopies and sidewalks. Exposed subgrades should be thoroughly proof rolled in order to locate and compact any weak, compressible and soft spots. Proof rolling shall be in accordance with TxDOT 2014 Specification Item 216. Proof rolling operations should be observed by the Geotechnical Engineer or his representative to document subgrade condition and preparation. Weak or soft areas identified during proof rolling or areas where large tree roots have been removed within the limits of excavation should be removed and replaced with a suitable, compacted select fill in accordance with the recommendations presented under the Select Fill Recommendations section of this report. Proof rolling operations and any excavation/backfill activities should be observed by **MEG** representatives to document subgrade preparation.

Prior to fill placement, the exposed subgrade shall be prepared based on what option is selected from the foundation and pavement recommendations. The exposed subgrade should be prepared, moisture-conditioned by scarifying to a minimum depth as recommended in the foundation and pavement recommendations and recompact to a minimum 98 percent of the maximum dry density as determined in accordance with ASTM D 698, moisture-density relationship. The moisture content of the subgrade should be maintained within the range of minus two (-2) percentage points below optimum to plus two (+2) percentage points above the optimum moisture content until the fill is permanently covered. The soil should be properly compacted in accordance with these recommendations and tested by **MEG** personnel for compaction as specified.

#### 7.4 Select Fill Recommendations

Materials used for select fill shall meet the following requirements:

1. Material shall conform to TxDOT 2014 Specification Item 247, Flexible Base; Type A, Grades 1 through 3.
2. Material shall conform to TxDOT 2014 Specification Item 247, Flexible Base, Types B or C, Grades 1 through 5 with a minimum plasticity index of 7.
3. Material shall conform to TxDOT 2014 Specification Item 247, Flexible Base, Type E, Grade 4 with a plasticity index between and inclusive of 7 and 15. Type E material shall be defined as Caliche (argillaceous limestone, calcareous or calcareous clay particles) and may contain stone, conglomerate, gravel, sand or granular materials when these materials are in situ with the caliche. Flexible Base (Type E, Grade 4) shall conform to the following requirements:

**Table 7.1. Type D, Grade 6 Requirements**

| Retained on Sq. Sieve | Percent Retained |
|-----------------------|------------------|
| 2"                    | 0                |
| 1/2"                  | 20-60            |
| No. 4                 | 40-75            |
| No. 40                | 70-90            |



|                                             |    |
|---------------------------------------------|----|
| Max. PI:                                    | 15 |
| Max. Wet Ball PI:                           | 15 |
| Wet Ball Mill Max Amount:                   | 50 |
| Wet Ball Increase, Max Passing No. 40 sieve | 20 |

4. Soils classified according to USCS as SM, SC, GM, GC, CL, ML and combinations of these soils. The soils shall be relatively free of organic matter. In addition to the USCS classification, select materials shall have a liquid limit of less than 40 and a plasticity index between and inclusive of 7 to 17.
5. Soils classified, as CH, MH, OH, OL and PT, under the USCS are not considered suitable for use as select fill materials at this site.

Select fill shall be placed in loose lifts not to exceed 8 inches (6 inches compacted) and compacted to a minimum 98 percent of the maximum dry density as determined in accordance with ASTM D 698. The moisture content of the fill shall be maintained within the range of minus two (-2) percentage points below optimum to plus two (+2) percentage points above the optimum moisture content until the fill is permanently covered. The select fill should be properly compacted in accordance with these recommendations and tested by **MEG** personnel for compaction as specified.

## 7.5 Site Fill Recommendations

Site fill shall be placed in loose lifts not to exceed 8 inches (6 inches compacted) and compacted to a minimum 98 percent of the maximum dry density as determined in accordance with ASTM D 698. The moisture content of the fill shall be maintained within the range of zero (0) percentage points below optimum to plus four (+4) percentage points above the optimum moisture content until the fill is permanently covered. The site fill should be properly compacted in accordance with these recommendations and tested by **MEG** personnel for compaction as specified.

## 7.6 Back Fill Recommendations

Back fill shall be placed in loose lifts not to exceed 8 inches (6 inches compacted) and compacted to a minimum 98 percent of the maximum dry density as determined in accordance with ASTM D 698. The moisture content of the fill shall be maintained within the range of zero (0) percentage points below optimum to plus four (+4) percentage points above the optimum moisture content until the fill is permanently covered. The back fill should be properly compacted in accordance with these recommendations and tested by **MEG** personnel for compaction as specified.

## **7.7 Utility Considerations**

Utilities that project through the slab-on-grade, slab-on-fill, floating floor slabs, or any other rigid unit should be designed with some degree of flexibility or with sleeves. Such features will help reduce the risk of damage to utility facilities from soil movements related to shrinkage and expansion.

## **7.8 Utility Trench Recommendations**

Bedding and initial backfill are buried around utility lines to support and protect the utility. The secondary backfill above the initial backfill also helps protect and support the foundation and/or pavement above. To ensure that settlement is not excessive in this secondary backfill we recommend the following:

- 1) If possible, trench and install utilities prior to work such as lime treatment and/or compaction of subgrade or placement of other fills or bases.
- 2) Place, moisture condition and compact the secondary backfill in accordance with the pertinent project requirements. Within the footprint of a building pad the secondary backfill should meet the same compaction requirements for select fill. Within the footprint of a pavement structure the secondary backfill should meet the same compaction requirements for the subgrade. When compaction of the subgrade is not specified it should meet the same compaction level of the adjacent natural ground. An alternative to compaction of secondary backfill is the use of flowable fill where secondary backfill is to be placed. If properly designed, the flowable fill can be excavated easily at a later date if necessary. No compaction and no testing is required when properly designed flowable fill is used.

## **7.9 Excavation, Sloping and Benching Considerations**

If trenches are to extend to or below a depth of five (5) ft., the contractor or persons doing the trenching should adhere to the current Occupational Health and Safety Administration (OSHA) guidelines on trench excavation safety and protection measures. Other industry standards may be applicable. The collection of specific geotechnical data and development of a plan for trench safety, sloping, benching or various types of temporary shoring, is beyond the scope of this study.

## **7.10 Shallow Foundation Excavation Considerations**

The Geotechnical Engineer or his representative prior to the placement of reinforcing steel and concrete should observe shallow foundation excavations. This is necessary to verify that the bearing soils at the bottom of the excavations are similar to those encountered during the subsurface soil exploration phase and that excessive loose materials and water are not present in the excavations. If soft pockets of soil are encountered in the foundation excavations, they should be removed and replaced with a compacted non-expansive fill material or lean concrete up to the design foundation bearing elevation.



## 7.11 Landscaping Considerations

Even though landscaping is a vital aesthetic component of any project, the owner, client and design team should be aware that placing trees or large bushes adjacent to any structure may distress the structure in the future. It is recommended that if any landscaping is to be placed adjacent to the structure in this project, it should be limited to small plants and shrubs. Trees and large bushes should be placed at a distance such that at their mature height, their canopy or “drip line” does not extend over the structures. The owner, client and design team should also be aware that if any watering is to be done in connection with the landscaping for this project it should be controlled, consistent and timely. Excessive or prolonged watering is not recommended. If watering is part of the landscaping plan, termination of watering for any extended period of time may also be detrimental to the structure. It is important that the moisture level in the subsurface soils remain constant so that shrinking and swelling of soils may be mitigated.

## 7.12 Perimeter Foundation Cap

We recommend that a cap of impervious fill be placed around the perimeter of the foundation to mitigate the intrusion of moisture into the soils surrounding the foundation. The top eighteen inches of fill around the foundation structure should be a low permeance clay cap to keep surface water away from the foundation. The low permeance clay cap should be sloped away from the foundation at a minimum slope of 2% and the surrounding areas should have positive drainage. The low permeance clay shall meet the USCS classification of CL and meeting the requirements in Tables 7.2 Gradation Requirements and Table 7.3 Atterberg Limits Requirements. The low permeance clay shall be compacted to minimum of 95 percent of the maximum dry density as determined in accordance with ASTM D 698. The moisture content of the subgrade should be maintained within the range of optimum to four (4) percentage points above the optimum moisture. If plantings are intended, add 4 to 6 inches of loam on top of the clay cap.

**Table 7.2. Gradation Requirements**

| Sieve Size | Percent Passing<br>(by dry weight) |
|------------|------------------------------------|
| 1/2 inch   | 100                                |
| No 4       | 70-100                             |
| No. 200    | 50 – 100                           |

**Table 7.3. Atterberg Limits Requirements**

| Test / ASTM               | Requirement                            |
|---------------------------|----------------------------------------|
| Atterberg Limits<br>D4318 | LL $\leq$ 45<br>20 $\leq$ PI $\leq$ 30 |

## **8.0 PAVEMENT SECTION RECOMMENDATIONS**

### **8.1 General Information**

The study was also performed to determine recommendations for the construction of a flexible pavement and these recommendations are presented in this report. The pavement recommendations are limited to samples taken from the existing soils within the roadway areas present at the site. The pavement design implemented for this project should be evaluated the civil engineer based on a traffic and design analysis for this project.

Recommendations for both flexible and rigid pavements are presented in this report for further evaluation by the project civil engineer. Both pavement types are viable and the selection on the type to be used will depend on the specific needs and criteria of this project site. Generally, flexible pavements have a lower initial construction cost when compared to rigid pavements. On the other hand a rigid pavement has lower maintenance cost throughout the life of the pavement structure as compared to flexible pavements. Flexible pavements generally require more frequent repairs and overlays at intervals from 6 to 10 years to meet the structural and functional requirements of the pavement during the design life. All pavements are very dependent on the condition of the soil platform on which they are supported and thus moisture conditions will play an important role in the performance of the pavement during its structure life. Proper consideration to drainage of the pavement structure and the surrounding areas is essential to the successful performance of a pavement structure.

After proof rolling and repairing deep subgrade deficiencies, the entire subgrade should be scarified and developed as recommended in the Site Preparation section of this report to provide a uniform subgrade for pavement construction. Areas that appear severely desiccated following site stripping may require further undercutting and moisture conditioning. If a significant precipitation event occurs after the evaluation or if the surface becomes disturbed, the subgrade should be reviewed by qualified personnel immediately prior to paving. The subgrade should be in its finished form at the time of the final review.

### **8.2 Soil Stabilization Recommendations**

The plasticity index of the surface soils at this site is approximately 7 to 15 percent. The existing subgrade soils may not require lime stabilization for soil shrink and swell mitigation. The existing subgrade soils may require cement stabilization for increased structural capacity. We recommend the addition of two (2) percent cement by weight to the existing surface soils for increased structural capacity. We recommend that the existing soils be tested after the pavement areas have been excavated or filled to the top of the subgrade elevation to verify the soil stabilization requirements. The natural ground should then be prepared as stated in the Site Preparation Recommendations of the pavement section. Proper consideration to drainage of the pavement structure and the surrounding areas is essential to the successful performance of a pavement structure.

### 8.3 Design Methodology and Traffic Information

Our engineering analysis of the pavement type considered the information obtained from our soil borings, the field and laboratory testing, our past experience with similar soils and site conditions. Design of new pavements for the project has been based on the procedures outlined in the 1993 Guide for Design of Pavement Structures by the American Association of State Highway and Transportation Officials (AASHTO). The client did not provide projected ESALs. We recommend that the project Civil Engineer or a Traffic Engineer review the project for the appropriate traffic levels and design periods to ensure that they are appropriate and consistent with the specific project site requirements. The assumptions for traffic used in the pavement design analysis are as follows:

**Table 8.1. Flexible Pavement Traffic Criteria Utilized**

| Criteria                      | Value <sup>1</sup> |
|-------------------------------|--------------------|
| Reliability Level             | 80%                |
| Standard Deviation            | 0.45               |
| Initial Serviceability level  | 4.0                |
| Terminal Serviceability level | 2.0                |

Note 1: The above traffic criteria as per 1993 AASHTO Pavement Design Guidelines.

**Table 8.2. Rigid Pavement Traffic Criteria Utilized**

| Criteria                      | Value <sup>1</sup> |
|-------------------------------|--------------------|
| Reliability Level             | 95%                |
| Standard Deviation            | 0.35               |
| Initial Serviceability level  | 4.0                |
| Terminal Serviceability level | 2.0                |

Note 1: The above traffic criteria as per 1993 AASHTO Pavement Design Guidelines.

## 8.4 Recommended Pavement Sections

The proposed pavement section for entrances, drives and parking areas may be chosen from one of the following options in the tables below.

***At the time of this study traffic volume information was not provided by the client and or their representatives. As such the following options provide a range of anticipated traffic levels. The chosen option should be evaluated against actual traffic levels by the civil engineer of record.***

The selection process of the appropriate pavement option should consider the following:

1. The client should consider the options presented as minimum pavement sections for the appropriate use and expected traffic levels. The civil engineer of record should evaluate the minimum pavement sections recommended with the appropriate class of pavement required for this project. The local government requirements should be met when they are more stringent than the minimum pavement sections recommended in our report.

**Table 8.3. Flexible Pavement Options**

| Materials                          | Light Duty Option One <sup>1</sup> | Light Duty Option Two <sup>3</sup> | Heavy Duty Option One <sup>2</sup> | Heavy Duty Option Two <sup>4</sup> | Heavy Duty Option Three <sup>5</sup> |
|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|--------------------------------------|
| Asphaltic Concrete (In)            | 2.0                                | 2                                  | 3.0                                | 4.0                                | 4.0                                  |
| Untreated Caliche Base (In)        | 10.0                               | 8.0                                | 8.0                                | 8.0                                | 8.0                                  |
| Mechanically Stabilized Base TX5   | -                                  | -                                  | -                                  | -                                  | Yes                                  |
| Moisture Conditioned Subgrade (In) | 8.0                                | -                                  | -                                  | -                                  | 8.0                                  |
| Cement Treated Subgrade (IN)       | -                                  | 6.0                                | 10.0                               | 10.0                               | -                                    |

Note 1: Light Duty Pavement Option One (ESALS = 9,000)

Note 2: Heavy Duty Pavement Option One (ESALS = 195,000)

Note 3: Light Duty Pavement Option Two (ESALS = 25,000)

Note 4: Heavy Duty Pavement Option Two (ESALS = 480,000)

Note 5: Heavy Duty Pavement Option Three (ESALS = 220,000)

**Table 8.4. Rigid Concrete Pavement Options**

| Materials                                                        | Light Duty<br>Option<br>One <sup>1</sup> | Light Duty<br>Option<br>Two <sup>3</sup> | Heavy<br>Duty<br>Option<br>One <sup>2</sup> | Heavy<br>Duty<br>Option<br>Two <sup>4</sup> |
|------------------------------------------------------------------|------------------------------------------|------------------------------------------|---------------------------------------------|---------------------------------------------|
| <b>Reinforced Concrete Pavement (In)</b>                         | 5.0                                      | 6.0                                      | 8.5                                         | 10.0                                        |
| <b>Untreated Caliche Base or<br/>Limed Treated Subgrade (In)</b> | 6.0                                      | 8.0                                      | 10.0                                        | 12.0                                        |

Note 1: Light Duty Pavement Option One (ESALS = 9,000)

Note 2: Heavy Duty Pavement Option One (ESALS = 195,000)

Note 3: Light Duty Pavement Option Two (ESALS = 25,000)

Note 4: Heavy Duty Pavement Option Two (ESALS = 480,000)

It is recommended that concrete pavements be reinforced. At a minimum, the reinforcing bars should be placed as follows:

**Table 8.5. Longitudinal Drives and Entrances**

| Thickness<br>(in) | Bar<br>Size | Longitudal Spacing<br>(in) | Transverse Spacing<br>(in) |
|-------------------|-------------|----------------------------|----------------------------|
| <=6               | 3           | 12                         | 24                         |
| <=7               | 4           | 12                         | 24                         |
| <=9               | 5           | 8.5                        | 17                         |
| <=10              | 6           | 11                         | 22                         |

**Table 8.6. Parking Areas**

| Thickness<br>(in) | Bar<br>Size | Longitudal Spacing<br>(in) | Transverse Spacing<br>(in) |
|-------------------|-------------|----------------------------|----------------------------|
| <=6               | 3           | 16                         | 16                         |
| <=7               | 4           | 16                         | 16                         |
| <=9               | 5           | 11                         | 11                         |
| <=10              | 6           | 14.5                       | 14.5                       |

Longitudinal reinforcement should be placed at  $\frac{1}{2}$  the slab depth  $\pm \frac{1}{2}$  inch from the surface. At a longitudinal edge, the first two spacing's for longitudinal reinforcement shall be at  $\frac{1}{2}$  the normal longitudinal spacing. At transverse construction joints, additional longitudinal reinforcement shall be placed at a spacing one half the normal longitudinal spacing for a length of 42 inches. At transverse joints, the first two spacing's for transverse reinforcement shall be at  $\frac{1}{2}$  the normal transverse spacing. All reinforcement should be specified as deformed steel meeting the requirement of ASTM A-615 (Grade 60) or ASTM A-616 (Grade 60). Splices should be a minimum of 33 nominal bar diameters. Reinforcing should not extend across longitudinal and expansion joints. Dowels across longitudinal and expansion joints are recommended to be 7/8-inch diameter, smooth bars with a length of 42 inches and spaced at a maximum 24 inches on

center for pavement thickness of seven (7) inches or less. Dowels across longitudinal and expansion joints are recommended to be 1 1/4-inch diameter, smooth bars with a length of 42 inches and spaced at a maximum 24 inches on center for pavement thickness of eleven (11) inches or less.

Longitudinal and transverse joints are recommended at a maximum spacing of 10 feet for pavements with a thickness of less than 6 inches and at a maximum spacing of 15 feet for pavements with a thickness of 6 inches or greater. The longitudinal and transverse joints should be formed or saw cut to a depth of 1/3 of the slab depth for concrete containing siliceous coarse aggregate and 1/4 of the slab depth for limestone aggregate. Sawing of joints should begin as soon as the concrete will not chip and ravel. It is recommended that longitudinal and expansion joints be doweled to promote load transfer. Expansion joint spacings are not to exceed a maximum of 75 feet and no expansion or contraction joints should be located within a swale or drainage collection area. Expansion joints are also needed to separate the concrete slab from fixed objects such as inlets, light standards and buildings.

It is recommended that the concrete pavement surface have a minimum slope of 0.015 ft/ft to provide adequate surface drainage. It is recommended that the concrete pavement should cure a minimum 7 days before allowing any traffic provided that adequate concrete strength has been attained as determined by the project Civil Engineer.

The curb shall be constructed in lengths equal to the adjoining pavement slab lengths, and expansion joints shall be provided in the curb opposite each transverse expansion joint in the pavement. Expansion joint material shall be of the same thickness, type and quality as specified for the pavement and shall be of the section as shown for the curb. All expansion joints shall be carried through the curb. Transverse contraction joints shall be sawed across the curb at the same time as sawing of the transverse contraction joints in the pavement are sawed. The curb shall be placed monolithically with the pavement for edge support and reinforced with a minimum one (1) #5 rebar. A finish coat of mortar shall be applied on the exposed surfaces of monolithic curbs.

## **8.5 Garbage Dumpster Considerations**

Within flexible pavement areas, it is recommended that reinforced concrete pads be provided in front of and beneath garbage dumpsters. Concrete paving is also recommended in areas where the dumpster trucks make turns with small radii to access the dumpsters. The pads should be a thickened concrete slab and reinforced similar to the concrete pavement recommendations or a minimum 10 inches thick.

## **9.0 PAVEMENT MATERIAL SPECIFICATION SECTIONS**

### **9.1 Pavement Preparation Recommendations**

Pavement areas should be stripped of all vegetation and organic topsoil up to a minimum of two (2) feet beyond the pavement perimeters. After stripping, remove at least six (6) inches of on-site soil as measured from existing grade when excavation of existing



subgrade is not recommended in other sections of this report. The excavated material, if free of organic and/or deleterious material, may be stockpiled for use in the non-pavement areas of the site. Where excavation of the subgrade is recommended in this report, the bottom of the excavation will extend at least two (2) feet beyond the limits of the planned pavement perimeter including canopies and sidewalks. Exposed subgrades should be thoroughly proof rolled in order to locate and compact any weak, compressible and soft spots. Proof rolling shall be in accordance with TxDOT 2014 Specification Item 216. Proof rolling operations should be observed by the Geotechnical Engineer or his representative to document subgrade condition and preparation. Weak or soft areas identified during proof rolling or areas where large tree roots have been removed within the limits of excavation should be removed and replaced with a suitable, compacted fill in accordance with the recommendations presented in TxDOT 2014 Specification Item 132 for density control and material requirements for Types A and B. If the fill is a clay it shall meet USCS Classification CL. Proof rolling operations and any excavation/backfill activities should be observed by MEG representatives to document subgrade preparation.

The exposed subgrade shall then be prepared based on what option is selected from the pavement recommendations. The soil should be worked in accordance with the recommendations and tested by MEG personnel for compaction as specified.

## 9.2 Hot Mix Asphaltic Concrete Surface

The asphaltic surface shall meet the requirements of the current TxDOT 2014 Specification Item 340 for Dense Graded Hot Mix Asphalt (small quantity) for projects with total production of less than 5,000 tons and TxDOT 2014 Specifications Item 341 Dense Graded Hot Mix Asphalt for projects with total production of 5,000 tons or greater. The hot mix asphaltic surface will be compacted to between 3.0 and 8.5 percent in place air voids in conformance with the specification. It is recommended that the testing required by this specification be performed during production.

**Table 9.1. Dense Grade Hot Mix Asphalt Additional Requirements**

| Test Procedure                      | Requirement                 |
|-------------------------------------|-----------------------------|
| Target Design Laboratory Density, % | 97.0 (for light duty areas) |
|                                     | 96.5 (for heavy duty areas) |

## 9.3 Reinforced Concrete

Concrete shall meet a minimum 28-day compressive strength of 3200 psi for light duty pavements and 4000 psi for medium and heavy duty pavements and a maximum water/cement ratio of 0.45. The concrete for concrete paving shall meet the requirements for Class P Concrete of TxDOT 2014 Specification Items 360 and 421. Aggregates used in the concrete design should meet the requirements of TxDOT 2014 Specification Item 421 or ASTM C33.

## 9.4 Untreated Caliche Base

The base shall be caliche base and meet the requirements of TxDOT 2014 Specification Item 247 Type E, Grade 3 or better and including the requirements of Table 9.2 shown below. The base shall be compacted to a minimum 98 percent of the maximum dry density as determined by the standard moisture density relation (ASTM D 698) at moisture contents ranging between minus two (-2) and plus two (+2) percentage points of the optimum moisture content. The base shall be placed in loose lifts not to exceed 8 inches (6 inches compacted) and not less than 5 ½ inches (4 inches compacted).

**Table 9.2. Flexible Base Type E, Additional Requirements**

| Retained Procedure                          | Requirements             |
|---------------------------------------------|--------------------------|
| Max. PI:                                    | 12 (15 for treated base) |
| Max. Wet Ball PI:                           | 12 (15 for treated base) |
| Wet Ball Mill Max Amount:                   | 50                       |
| Wet Ball Increase, Max Passing No. 40 sieve | 20                       |

## 9.5 Lime Treated Caliche Base

The base shall meet all the requirements of untreated caliche base. In addition the base shall be treated with a minimum one and a half (1 ½) percent lime or Portland cement by weight. Lime shall meet the requirements of TxDOT 2014 Specification Item 260. Portland cement shall be Type I meeting ASTM C150 or Type IP meeting ASTM C595. Lime treatment will be in accordance with TxDOT 2014 Specification Item 260 or 263 for base courses and Item 260 for subgrades. Cement treatment shall be in accordance with TxDOT 2014 Specification Item 275 or 276. The base shall be compacted to a minimum 98 percent of the maximum dry density as determined by the standard moisture density relation (ASTM D 698) at moisture contents ranging between minus two (-2) and plus two (+2) percentage points of the optimum moisture content. The base shall be placed in loose lifts not to exceed 8 inches (6 inches compacted) and not less than 5 ½ inches (4 inches compacted).

## 9.6 Lime Treated Subgrade

Lime stabilization of the soils may be used for soil shrink and swell mitigation and to increase the structural capacity of the subgrade as platform for the pavement. It is recommended that the subgrade plasticity index be verified after the site is stripped of the loose topsoil and the subgrade is exposed up to the subgrade elevation. Stabilization treatment of the subgrade soils may also be utilized to provide a platform for the base for resistance to moisture intrusion and changes. Prevention of moisture intrusion is extremely important to the life of a pavement. Increases in moisture content of the subgrade and the base will significantly reduce the material's structural ability to carry loads. The optimum lime content should reduce the PI of the soil to 20 or less and should



result in a soil lime mixture with a pH of at least 12.4 when tested in accordance with ASTM C 977, Appendix XI. The lime should be blended with a mixing device such as a pulvermixer to produce a soil mixture passing water added and be allowed to cure for at least 48 hours. After curing the lime soil mixture it shall pass the grading requirements of TxDOT 2014 Specification Item 260 and compacted to a minimum 98 percent of the maximum dry density determined in accordance with ASTM D 698 at moisture contents ranging from minus two (-2) percentage points below optimum to plus two (+2) percentage points above optimum moisture content. If the in place gradation requirements can be achieved during the initial mixing, the remixing after the curing period can be eliminated. Please note that there is a relationship between the time of mixing of the lime and soils with the maximum dry density. Any mixture older than three (3) days will require determination of new moisture density relationships.

Portland cement may be considered instead of lime to treat the subgrade. See the cement treated subgrade section of this report.

### **9.7 Cement Treated Subgrade**

Cement treatment of the on-site subgrade soils may be used for soil shrink and swell mitigation and to increase the structural capacity of the subgrade as a platform for the pavement. It is recommended that the subgrade plasticity index be verified after the site is stripped of the loose topsoil and the subgrade is exposed up to the subgrade elevation. Treatment of the subgrade soils may be utilized to provide a uniform platform for the base for resistance to moisture intrusion and changes. Prevention of moisture intrusion is extremely important to the life of a pavement. Increases in moisture content of the subgrade and the base will significantly reduce the material's structural ability to carry loads. The optimum cement content should be determined in accordance with TxDOT 2014 Specifications Item 275 and with a minimum dry strength requirement of 175 psi. It is anticipated that the soils will require about 2 to 3 percent cement by dry weight of the soils to meet the specification requirements. The cement should be blended into the soil and preferably with a mixing device such as a pulvermixer to produce a uniform soil-cement mixture. The soil-cement mixture shall be placed in accordance with TxDOT 2014 Specification Item 275 and compacted to a minimum 98 percent of the maximum dry density determined in accordance with ASTM D 698 at moisture contents ranging from minus two (-2) percentage points below optimum to plus two (+2) percentage points above the optimum moisture content.

Cracking of the subgrade can be expected when using cement treatment due to the increased rigidity of the soil. Maintenance of any reflected cracks will require immediate sealing to keep moisture from penetrating to the pavement layers below the surface. If reflective cracking on the pavement from the treated subgrade is a concern, we recommend that microcracking of the subgrade be performed and the procedure outlined in TxDOT 2014 Specification Item 275 be used to mitigate reflective cracking. Microcracking should be performed on treated subgrade layers not to exceed 8 inches compacted thickness.

### **9.8 Moisture Conditioned Subgrade**

The subgrade shall be scarified to a depth of eight (8) inches and moisture conditioned to within the range of minus two (-2) percentage points below optimum to plus two (+2) percentage points above the optimum moisture content. The subgrade shall be compacted to a minimum 98 percent of the maximum dry density determined in accordance with ASTM D 698.

## **10.0 OTHER CONSIDERATIONS**

### **10.1 Drainage**

Adequate perimeter drainage is essential for long-term performance of any pavement structure. Infiltration of surface water from unpaved areas surrounding the pavement should be minimized. We do not recommend the placement of landscape beds on the paved areas. Such design features provide a potential for water to enter into the pavement section and the underlying soil subgrade. This is especially true with time in paved areas that have limited traffic and lead to accelerated asphalt oxidation and thus cracking. Water intrusion will result in the deterioration of the pavement materials with time as vehicular traffic passes over affected areas. Above grade planter boxes with drainage discharge onto the top of the pavement or directed into storm sewers should be considered if landscape features are to be used.

### **10.2 Curb & Gutters**

Base material under curb and gutters and curbs shall be compacted to the same requirements as other areas. It is recommended that the base shall be placed below the curb & gutter and extended past the back of the curb & gutter a minimum of one and a half (1 ½) feet. The thickness of the base below the curb & gutter shall be the required pavement base thickness less six (6) inches. An adequate seal should be provided at all concrete- asphalt interfaces. It is recommended that a crack sealant compatible to both asphalt and concrete be used.

### **10.3 Maintenance**

A maintenance plan is recommended for the long-term performance of the paved areas. Asphaltic pavements have a tendency to strip and become oxidized with exposure to the elements. Thus cracks may become present in the pavement. It is recommended that a maintenance schedule of crack sealing, fog seals and overlays be used over the life of the pavement.

## 11.0 PROJECT REVIEW AND QUALITY CONTROL

Each project site is unique and it is important that the appropriate design data, construction drawings, specifications, change orders and related documents be reviewed by the respective design and construction professionals participating in this project. The performance of foundations, construction building pads and/or parking areas for this project will depend on correct interpretation of our geotechnical engineering report and proper compliance of and adherence to our geotechnical recommendations and to the construction drawings and specifications.

It is important that **MEG** be provided the opportunity to review the final design and construction documents to check that our geotechnical recommendations are properly interpreted and incorporated in the design and construction documents. We cannot be responsible for misinterpretations of our geotechnical recommendations if we have not had the opportunity to review these documents. This review is an additional service and not part of our project scope.

**MEG** should be retained to provide construction materials testing and observation services during all phases of the construction process of this project. As the Geotechnical Engineer of Record, it is important to let our technical personnel provide these services to make certain that our recommendations are interpreted properly and to ensure that actual field conditions are those described in our geotechnical report. Since our personnel are familiar with this project, **MEG's** participation during the construction phase of this project would help mitigate any problems resulting from variations or anomalies in subsurface conditions, which are among the most prevalent on construction projects and often lead to delays, changes, costs overruns, and disputes. If the client does not follow all of our recommendations presented in this report and/or addendums to this report, the client assumes the responsibility and liability of such actions and will hold our firm harmless and without responsibility and liability for client's actions.

A construction testing frequency plan and budget needs to be developed for the required construction materials engineering and testing services for this project. Before construction, we recommend that **MEG**, the project design team members and the project general contractor meet and jointly develop the testing plan and budget, as well as review the testing specifications as it pertains to this project. **A failure to implement a complete testing plan will negate the recommendations provided in this report.**

**MEG** looks forward to the opportunity to provide continued support on this project.

## APPENDIX

## **APPENDIX A**

### **CUSTOM SOIL RESOURCE REPORT**

Soil Map—Hidalgo County, Texas  
(Fire Station No. 5 (01-19-29113))





## MAP LEGEND

### Area of Interest (AOI)

 Area of Interest (AOI)

### Soils

 Soil Map Unit Polygons

 Soil Map Unit Lines

 Soil Map Unit Points

### Special Point Features



Blowout



Borrow Pit



Clay Spot



Closed Depression



Gravel Pit



Gravelly Spot



Landfill



Lava Flow



Marsh or swamp



Mine or Quarry



Miscellaneous Water



Perennial Water



Rock Outcrop



Saline Spot



Sandy Spot



Severely Eroded Spot



Sinkhole



Slide or Slip



Sodic Spot



Spoil Area



Stony Spot



Very Stony Spot



Wet Spot



Other



Special Line Features

### Water Features



Streams and Canals

### Transportation



Rails



Interstate Highways



US Routes



Major Roads



Local Roads

### Background



Aerial Photography

## MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:20,000.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service

Web Soil Survey URL:

Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Hidalgo County, Texas

Survey Area Data: Version 17, Sep 15, 2018

Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: Sep 20, 2015—Nov 5, 2017

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

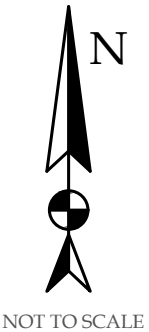
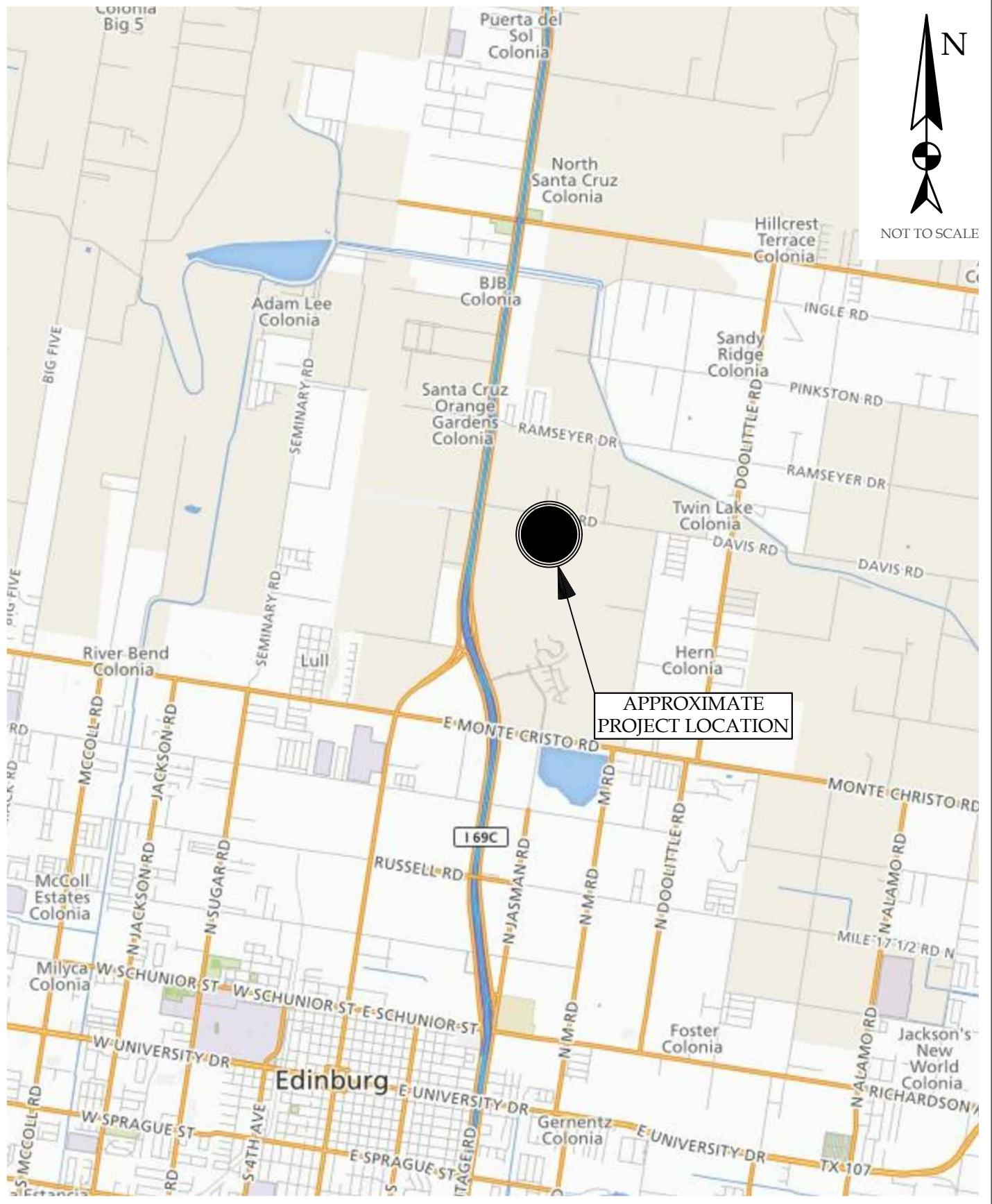
## Map Unit Legend

| Map Unit Symbol                    | Map Unit Name                                  | Acres in AOI | Percent of AOI |
|------------------------------------|------------------------------------------------|--------------|----------------|
| 8                                  | Comitas loamy fine sand, 0 to 3 percent slopes | 0.5          | 5.3%           |
| 25                                 | Hidalgo fine sandy loam, 0 to 1 percent slopes | 8.5          | 94.7%          |
| <b>Totals for Area of Interest</b> |                                                | <b>9.0</b>   | <b>100.0%</b>  |



## **APPENDIX B**

### **PROJECT LOCATION, TOPOGRAPHIC AND BOREHOLE LOCATION MAPS**



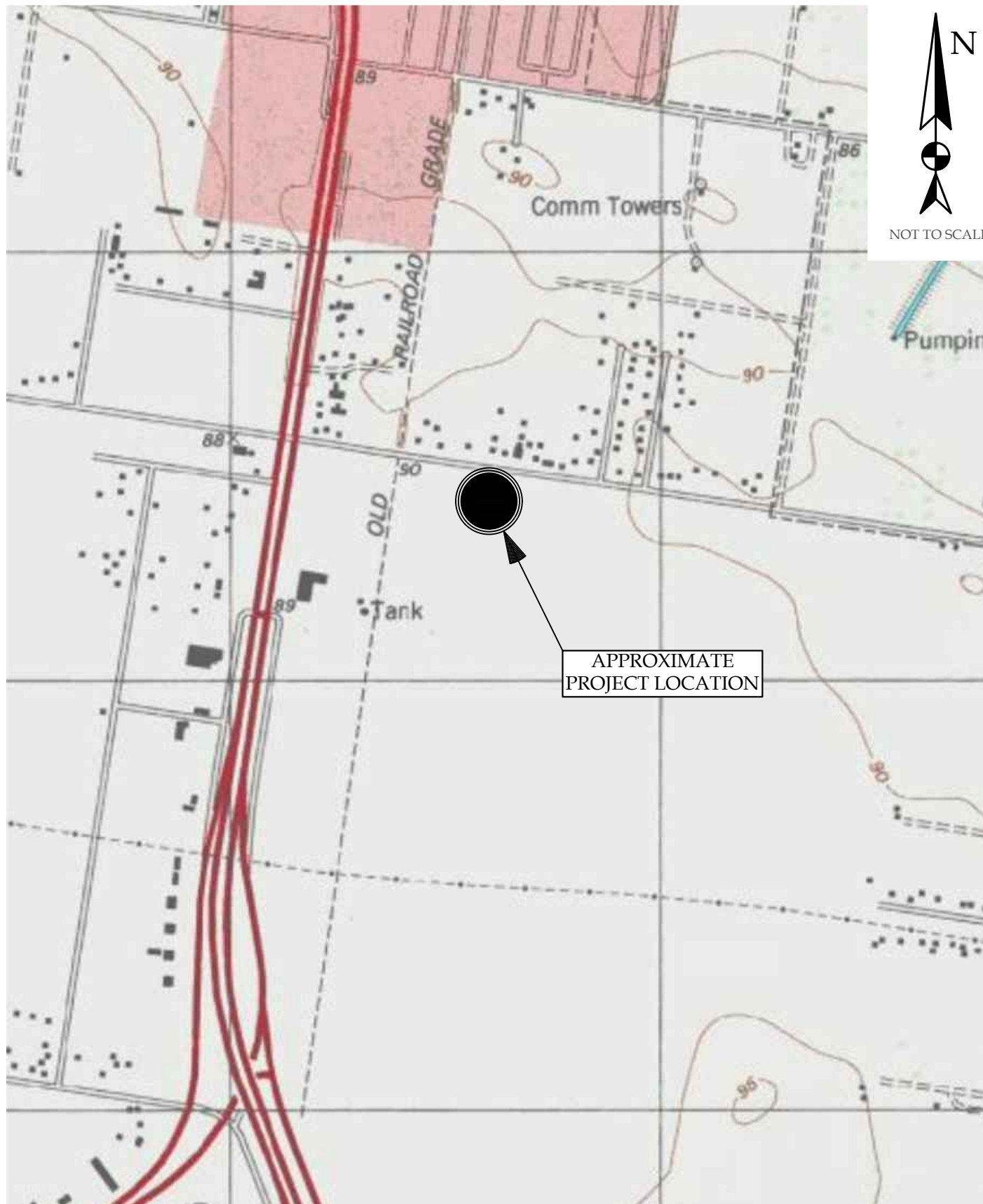
### PROJECT SITE LOCATION MAP

PROPOSED  
FIRE STATION No. 5

EDINBURG, HIDALGO COUNTY, TEXAS



MILLENNIUM ENGINEERS GROUP, INC.  
5804 N. GUMWOOD AVENUE  
PHARR, TEXAS 78577  
WWW.MEGENGINEERS.COM  
TEL: 956-702-8500  
FAX: 956-702-8140



## PROJECT TOPOGRAPHY MAP

PROPOSED  
FIRE STATION No. 5

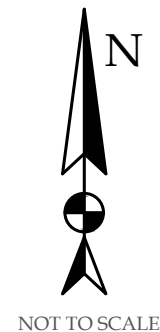
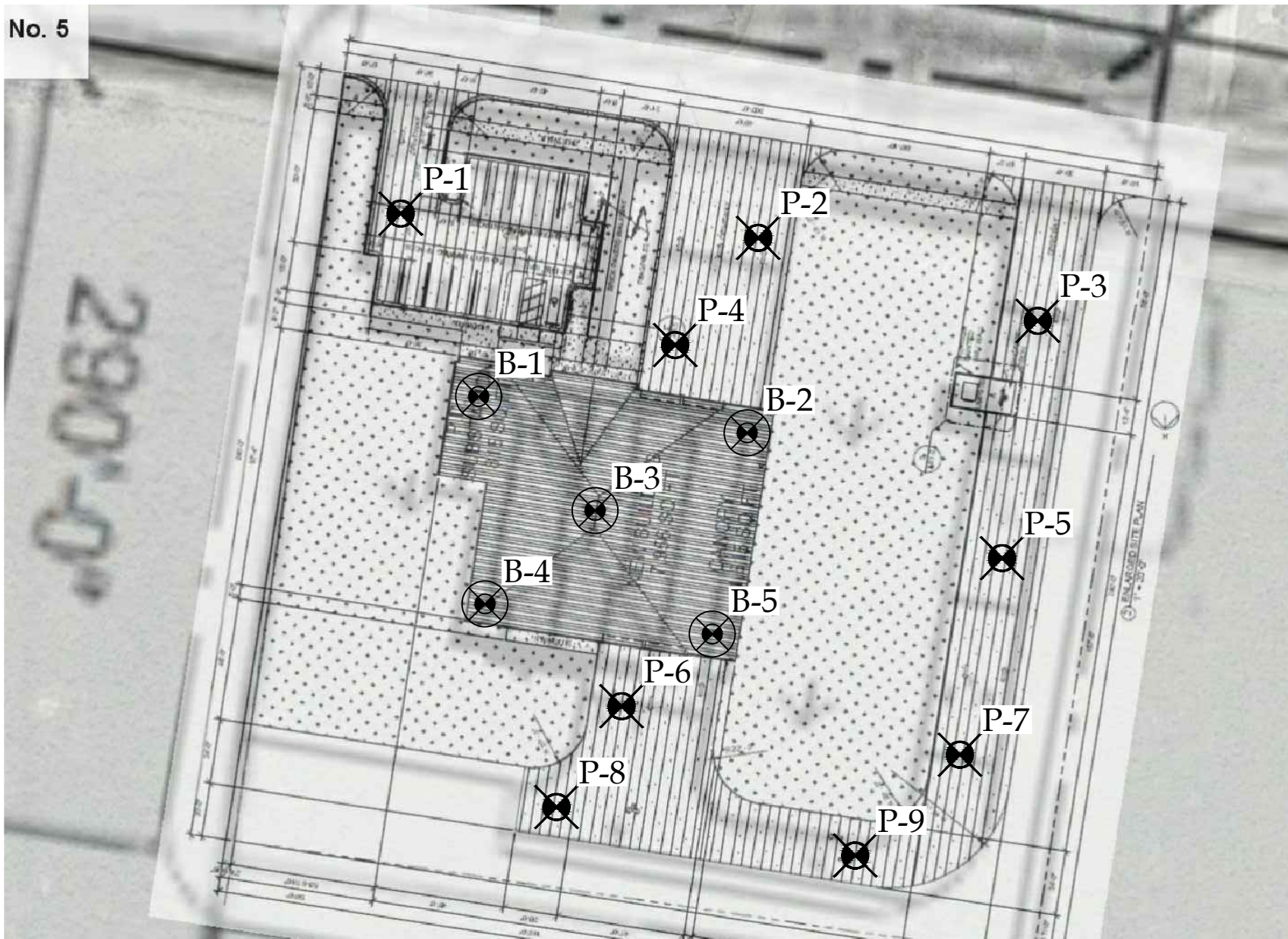
EDINBURG, HIDALGO COUNTY, TEXAS



MILLENNIUM ENGINEERS GROUP, INC.  
5804 N. GUMWOOD AVENUE  
PHARR, TEXAS 78577  
WWW.MEGENGINEERS.COM  
TEL: 956-702-8500  
FAX: 956-702-8140



No. 5



# PROJECT BOREHOLE LOCATION MAP

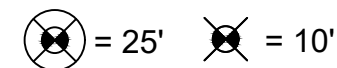
PROPOSED  
FIRE STATION No. 5

EDINBURG, HIDALGO COUNTY, TEXAS



MILLENNIUM ENGINEERS GROUP, INC.  
5804 N. GUMWOOD AVENUE  
PHARR, TEXAS 78577  
WWW.MEGENGINEERS.COM  
TEL: 956-702-8500  
FAX: 956-702-8140

## BOREHOLE DEPTH



## **APPENDIX C**

### **PROJECT BORING LOGS AND PROFILE**

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring B-1

Sheet 1 of 1

|                                                           |                                            |                                                       |
|-----------------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                        | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>                 | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>25 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                           | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>20 feet BGS</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>             | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             |               |                               | SC-CL         |             | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 10               | 22    | 12    |               |                         |
|                  |              |             | 1             | 6                             |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 2             | 5                             |               |             |                                                                                            | 12               |       |       | 48            |                         |
| 83               | 5            |             | 3             | 6                             |               |             |                                                                                            | 12               | 28    | 11    |               |                         |
|                  |              |             | 4             | 6                             |               |             |                                                                                            | 12               |       |       | 55            |                         |
|                  |              |             | 5             | 14                            |               |             |                                                                                            | 14               |       |       |               |                         |
| 78               | 10           |             |               |                               | SC            |             | Clayey SAND, brown to light brown, moist to wet, loose to dense                            |                  |       |       |               |                         |
|                  |              |             | 6             | 17                            |               |             |                                                                                            | 21               | 20    | 9     |               |                         |
| 73               | 15           |             |               |                               |               |             |                                                                                            |                  |       |       |               | ▼ Cave In 24 Hr         |
|                  |              |             | 7             | 13                            |               |             |                                                                                            | 25               |       |       |               | ▽ ATD                   |
| 68               | 20           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 8             | 21                            |               |             |                                                                                            | 18               |       |       | 26            |                         |
| 63               | 25           |             |               |                               |               |             | Bore Termination                                                                           |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 3**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring B-2

Sheet 1 of 1

|                                                           |                                            |                                                       |
|-----------------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                        | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>                 | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>25 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                           | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>15 feet BGS</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>             | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS  |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|--------------------------|
| 88               | 0            |             |               |                               | SC-CL         |             | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 7                | 20    | 7     |               |                          |
|                  |              |             | 1             | 12                            |               |             |                                                                                            |                  |       |       |               |                          |
|                  |              |             | 2             | 6                             |               |             |                                                                                            | 13               |       |       | 55            |                          |
| 83               | 5            |             | 3             | 9                             |               |             |                                                                                            | 14               |       |       |               |                          |
|                  |              |             | 4             | 12                            |               |             |                                                                                            | 11               |       |       |               |                          |
|                  |              |             | 5             | 18                            |               |             |                                                                                            | 12               |       |       | 51            |                          |
| 78               | 10           |             |               |                               | SC            |             | Clayey SAND, brown to light brown, moist to wet, loose to dense                            |                  |       |       |               |                          |
|                  |              |             | 6             | 16                            |               |             |                                                                                            | 23               | 21    | 14    |               |                          |
| 73               | 15           |             |               |                               |               |             |                                                                                            |                  |       |       |               | ▽ ATD<br>▼ Cave In 24 Hr |
|                  |              |             | 7             | 8                             |               |             |                                                                                            | 24               |       |       | 23            |                          |
| 68               | 20           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                          |
|                  |              |             | 8             | 30                            |               |             |                                                                                            | 19               |       |       |               |                          |
| 63               | 25           |             |               |                               |               |             | Bore Termination                                                                           |                  |       |       |               |                          |

Project: **Proposed Fire Station No. 3**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113****Log of Boring B-3****Sheet 1 of 1**

|                                                           |                                            |                                                       |
|-----------------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                        | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>                 | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>25 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                           | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>19 feet BGS</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>             | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             |               |                               | SC-CL         |             | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 9                |       |       |               |                         |
|                  |              |             | 1             | 8                             |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 2             | 5                             |               |             |                                                                                            | 12               | 26    | 18    |               |                         |
| 83               | 5            |             | 3             | 5                             |               |             |                                                                                            | 12               |       |       | 49            |                         |
|                  |              |             | 4             | 7                             |               |             |                                                                                            | 15               |       |       | 58            |                         |
|                  |              |             | 5             | 13                            |               |             |                                                                                            | 13               |       |       |               |                         |
| 78               | 10           |             |               |                               | SC            |             | Clayey SAND, brown to light brown, moist to wet, loose to dense                            |                  |       |       |               |                         |
|                  |              |             | 6             |                               |               |             |                                                                                            | 9                | 25    | 12    |               |                         |
| 73               | 15           |             |               |                               |               |             |                                                                                            |                  |       |       |               | ▼ Cave In 24 Hr         |
|                  |              |             | 7             |                               |               |             |                                                                                            | 24               |       |       |               | ▽ ATD                   |
| 68               | 20           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 8             |                               |               |             |                                                                                            | 23               |       |       | 30            |                         |
| 63               | 25           |             |               |                               |               |             | Bore Termination                                                                           |                  |       |       |               |                         |



Project: **Proposed Fire Station No. 3**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113****Log of Boring B-4****Sheet 1 of 1**

|                                                           |                                            |                                                       |
|-----------------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                        | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>                 | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>25 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                           | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>18 feet BGS</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>             | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             |               |                               | SC-CL         |             | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 9                |       |       | 54            |                         |
|                  |              |             | 1             | 9                             |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 2             | 6                             |               |             |                                                                                            | 12               | 26    | 11    |               |                         |
| 83               | 5            |             | 3             | 5                             |               |             |                                                                                            | 13               |       |       |               |                         |
|                  |              |             | 4             | 17                            |               |             |                                                                                            | 12               |       |       | 37            |                         |
|                  |              |             | 5             | 23                            |               |             |                                                                                            | 14               | 28    | 11    |               |                         |
| 78               | 10           |             |               |                               | SC            |             | Clayey SAND, brown to light brown, moist to wet, loose to dense                            |                  |       |       |               |                         |
|                  |              |             | 6             | 7                             |               |             |                                                                                            | 18               |       |       | 46            |                         |
| 73               | 15           |             |               |                               |               |             |                                                                                            |                  |       |       |               | ▼ Cave In 24 Hr         |
|                  |              |             | 7             | 24                            |               |             |                                                                                            | 22               |       |       |               | ▽ ATD                   |
| 68               | 20           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 8             | 27                            |               |             |                                                                                            | 21               |       |       |               |                         |
| 63               | 25           |             |               |                               |               |             | Bore Termination                                                                           |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113****Log of Boring B-5****Sheet 1 of 1**

|                                                           |                                            |                                                       |
|-----------------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                        | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>                 | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>25 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                           | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>20 feet BGS</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>             | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             | 1             | 4                             | SC-CL         |             | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 9                |       |       |               |                         |
|                  |              |             | 2             | 5                             |               |             |                                                                                            | 10               | 27    | 12    |               |                         |
| 83               | 5            |             | 3             | 6                             |               |             |                                                                                            | 13               | 31    | 21    |               |                         |
|                  |              |             | 4             | 8                             |               |             |                                                                                            | 14               |       |       |               |                         |
|                  |              |             | 5             | 12                            |               |             |                                                                                            | 15               |       |       | 59            |                         |
| 78               | 10           |             |               |                               | SC            |             | Clayey SAND, brown to light brown, moist to wet, loose to dense                            |                  |       |       |               |                         |
|                  |              |             | 6             | 8                             |               |             |                                                                                            | 20               |       |       | 49            |                         |
| 73               | 15           |             |               |                               |               |             |                                                                                            |                  |       |       |               | ▼ Cave In 24 Hr         |
|                  |              |             | 7             | 12                            |               |             |                                                                                            | 20               |       |       |               | ▽ ATD                   |
| 68               | 20           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 8             | 29                            |               |             |                                                                                            | 23               |       |       | 13            |                         |
| 63               | 25           |             |               |                               |               |             | Bore Termination                                                                           |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring P-1

Sheet 1 of 1

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             |               |                               | SC-CL         |             | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 8                | 17    | 7     |               |                         |
|                  |              |             | 1             | 9                             |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 2             | 6                             |               |             |                                                                                            | 10               |       |       |               |                         |
| 83               | 5            |             | 3             | 5                             |               |             |                                                                                            | 11               |       |       | 49            |                         |
|                  |              |             | 4             | 16                            |               |             |                                                                                            | 12               | 26    | 14    |               |                         |
|                  |              |             | 5             | 29                            |               |             |                                                                                            | 14               |       |       |               |                         |
| 78               | 10           |             |               |                               |               |             | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
| 73               | 15           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
| 68               | 20           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
| 63               | 25           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113****Log of Boring P-2****Sheet 1 of 1**

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>89</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 89               | 0            |             |               |                               | SC-CL         |             | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 8                | 22    | 9     |               |                         |
|                  |              |             | 1             | 5                             |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             | 2             | 4                             |               |             |                                                                                            | 11               |       |       |               |                         |
| 84               | 5            |             | 3             | 3                             |               |             |                                                                                            | 13               | 26    | 10    |               |                         |
|                  |              |             | 4             | 14                            |               |             |                                                                                            | 12               |       |       |               |                         |
|                  |              |             | 5             | 32                            |               |             |                                                                                            | 13               |       |       | 63            |                         |
| 79               | 10           |             |               |                               |               |             | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
| 74               | 15           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
| 69               | 20           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |
| 64               | 25           |             |               |                               |               |             |                                                                                            |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring P-3

Sheet 1 of 1

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>89</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log                                                                        | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 89               | 0            |             | 1             | 7                             | SC-CL         |  | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 7                |       |       | 34            |                         |
|                  |              |             | 2             | 3                             |               |                                                                                    |                                                                                            | 10               |       |       |               |                         |
| 84               | 5            |             | 3             | 6                             |               |                                                                                    |                                                                                            | 12               |       |       |               |                         |
|                  |              |             | 4             | 22                            |               |                                                                                    |                                                                                            | 14               |       |       |               |                         |
|                  |              |             | 5             | 26                            |               |                                                                                    |                                                                                            | 12               | 28    | 8     |               |                         |
| 79               | 10           |             |               |                               |               |                                                                                    | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 74               | 15           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 69               | 20           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 64               | 25           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring P-4

Sheet 1 of 1

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log                                                                        | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             | 1             | 5                             | SC-CL         |  | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 6                | 20    | 2     |               |                         |
|                  |              |             | 2             | 5                             |               |                                                                                    |                                                                                            | 10               |       |       | 40            |                         |
| 83               | 5            |             | 3             | 7                             |               |                                                                                    |                                                                                            | 13               |       |       |               |                         |
|                  |              |             | 4             | 13                            |               |                                                                                    |                                                                                            | 11               | 25    | 10    |               |                         |
|                  |              |             | 5             | 27                            |               |                                                                                    |                                                                                            | 13               |       |       |               |                         |
| 78               | 10           |             |               |                               |               |                                                                                    | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 73               | 15           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 68               | 20           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 63               | 25           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring P-5

Sheet 1 of 1

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log                                                                        | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             | 1             | 6                             | SC-CL         |  | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 10               | 23    | 9     |               |                         |
|                  |              |             | 2             | 4                             |               |                                                                                    |                                                                                            | 13               |       |       |               |                         |
| 83               | 5            |             | 3             | 9                             |               |                                                                                    |                                                                                            | 13               |       |       |               |                         |
|                  |              |             | 4             | 24                            |               |                                                                                    |                                                                                            | 13               |       |       | 60            |                         |
|                  |              |             | 5             | 31                            |               |                                                                                    |                                                                                            | 13               | 31    | 18    |               |                         |
| 78               | 10           |             |               |                               |               |                                                                                    | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 73               | 15           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 68               | 20           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 63               | 25           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring P-6

Sheet 1 of 1

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log                                                                        | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             | 1             | 6                             | SC-CL         |  | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 7                |       |       |               |                         |
|                  |              |             | 2             | 4                             |               |                                                                                    |                                                                                            | 12               | 27    | 15    |               |                         |
| 83               | 5            |             | 3             | 5                             |               |                                                                                    |                                                                                            | 12               |       |       | 50            |                         |
|                  |              |             | 4             | 10                            |               |                                                                                    |                                                                                            | 13               | 25    | 14    |               |                         |
|                  |              |             | 5             | 45                            |               |                                                                                    |                                                                                            | 13               |       |       |               |                         |
| 78               | 10           |             |               |                               |               |                                                                                    | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 73               | 15           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 68               | 20           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 63               | 25           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |



Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring P-7

Sheet 1 of 1

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>87</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log                                                                        | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 87               | 0            |             | 1             | 7                             | SC-CL         |  | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 8                |       |       | 42            |                         |
|                  |              |             | 2             | 7                             |               |                                                                                    |                                                                                            | 14               |       |       |               |                         |
| 82               | 5            |             | 3             | 5                             |               |                                                                                    |                                                                                            | 12               | 26    | 9     |               |                         |
|                  |              |             | 4             | 7                             |               |                                                                                    |                                                                                            | 12               |       |       |               |                         |
|                  |              |             | 5             | 19                            |               |                                                                                    |                                                                                            | 15               | 24    | 4     |               |                         |
| 77               | 10           |             |               |                               |               |                                                                                    | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 72               | 15           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 67               | 20           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 62               | 25           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |

Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring P-8

Sheet 1 of 1

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log                                                                         | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             | 1             | 8                             | SC-CL         |    | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 8                | 23    | 8     |               |                         |
|                  |              |             | 2             | 4                             |               |    |                                                                                            | 9                |       |       |               |                         |
| 83               | 5            |             | 3             | 5                             |               |  |                                                                                            | 12               |       |       | 58            |                         |
|                  |              |             | 4             | 22                            |               |  |                                                                                            | 14               | 26    | 9     |               |                         |
|                  |              |             | 5             | 26                            |               |  |                                                                                            | 12               |       |       |               |                         |
| 78               | 10           |             |               |                               |               |                                                                                     | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                     |                                                                                            |                  |       |       |               |                         |
| 73               | 15           |             |               |                               |               |                                                                                     |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                     |                                                                                            |                  |       |       |               |                         |
| 68               | 20           |             |               |                               |               |                                                                                     |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                     |                                                                                            |                  |       |       |               |                         |
| 63               | 25           |             |               |                               |               |                                                                                     |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                     |                                                                                            |                  |       |       |               |                         |

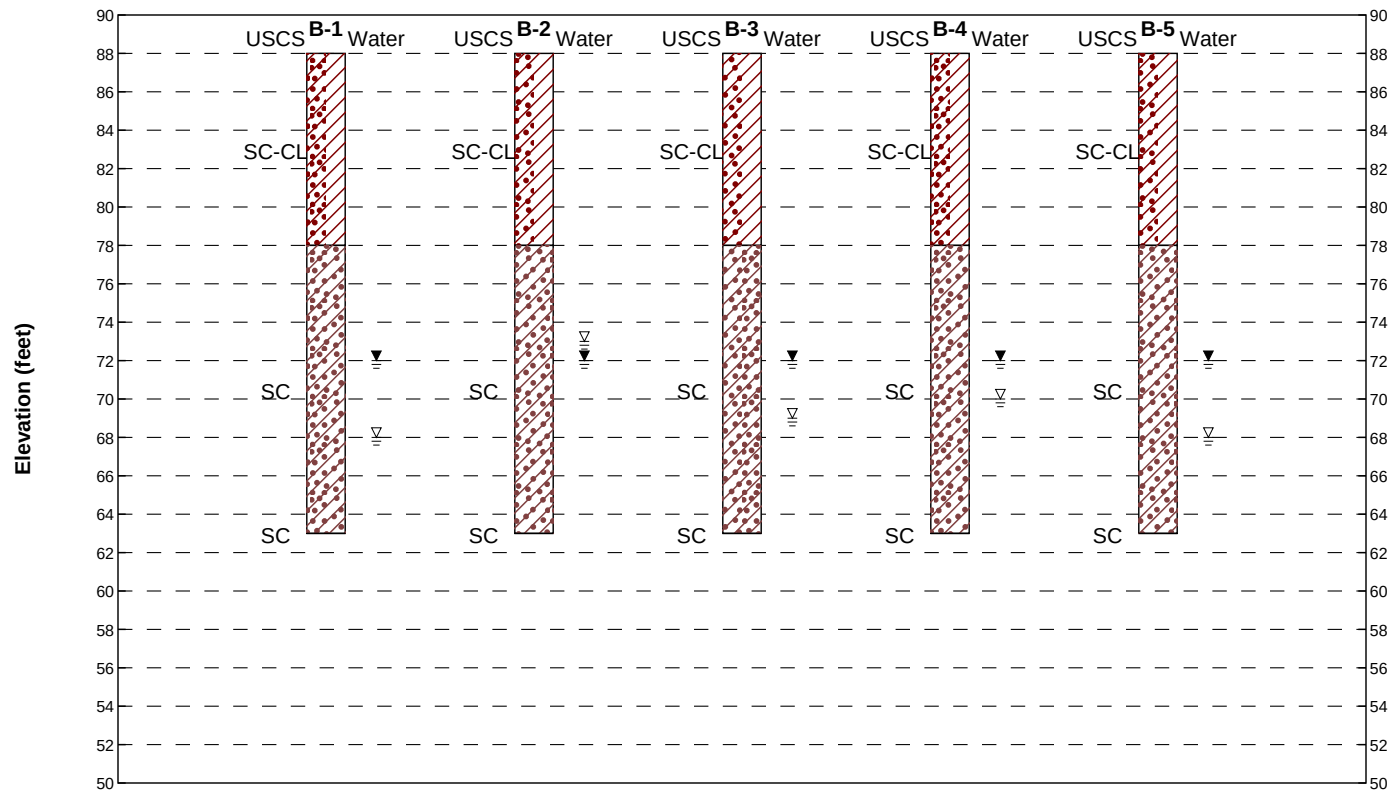
Project: **Proposed Fire Station No. 5**Project Location: **Edinburg, Hidalgo County, Texas**Project Number: **01-19-29113**

## Log of Boring P-9

Sheet 1 of 1

|                                                    |                                            |                                                       |
|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| Date(s) Drilled<br><b>2/6/2019</b>                 | Logged By<br><b>D. Juarez</b>              | Checked By<br><b>Raul Palma</b>                       |
| Drilling Method<br><b>Straight Flight</b>          | Drill Bit Size/Type<br><b>4" soil bit</b>  | Total Depth of Borehole<br><b>10 feet bgs</b>         |
| Drill Rig Type<br><b>CME 55</b>                    | Drilling Contractor<br><b>JEDI</b>         | Approximate Surface Elevation<br><b>88</b>            |
| Groundwater Level and Date Measured<br><b>None</b> | Sampling Method(s)<br><b>SPT</b>           | Hammer Data<br><b>140 lb., 30 in. drop, auto trip</b> |
| Borehole Backfill<br><b>Subgrade Cuttings</b>      | Location<br><b>See Boring Location Map</b> |                                                       |

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log                                                                        | MATERIAL DESCRIPTION                                                                       | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------|-------|-------|---------------|-------------------------|
| 88               | 0            |             | 1             | 8                             | SC-CL         |  | Sandy lean CLAY to Clayey SAND, dark brown to light brown, moist, med. stiff to very stiff | 9                | 25    | 9     | 49            |                         |
|                  |              |             | 2             | 5                             |               |                                                                                    |                                                                                            | 12               |       |       |               |                         |
| 83               | 5            |             | 3             | 8                             |               |                                                                                    |                                                                                            | 14               |       |       |               |                         |
|                  |              |             | 4             | 25                            |               |                                                                                    |                                                                                            | 12               |       |       |               |                         |
|                  |              |             | 5             | 20                            |               |                                                                                    |                                                                                            | 12               | 25    | 10    |               |                         |
| 78               | 10           |             |               |                               |               |                                                                                    | Bore Termination                                                                           |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 73               | 15           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 68               | 20           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
|                  |              |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |
| 63               | 25           |             |               |                               |               |                                                                                    |                                                                                            |                  |       |       |               |                         |



**MATERIAL GRAPHIC SYMBOLS**



Clayey SAND (SC)



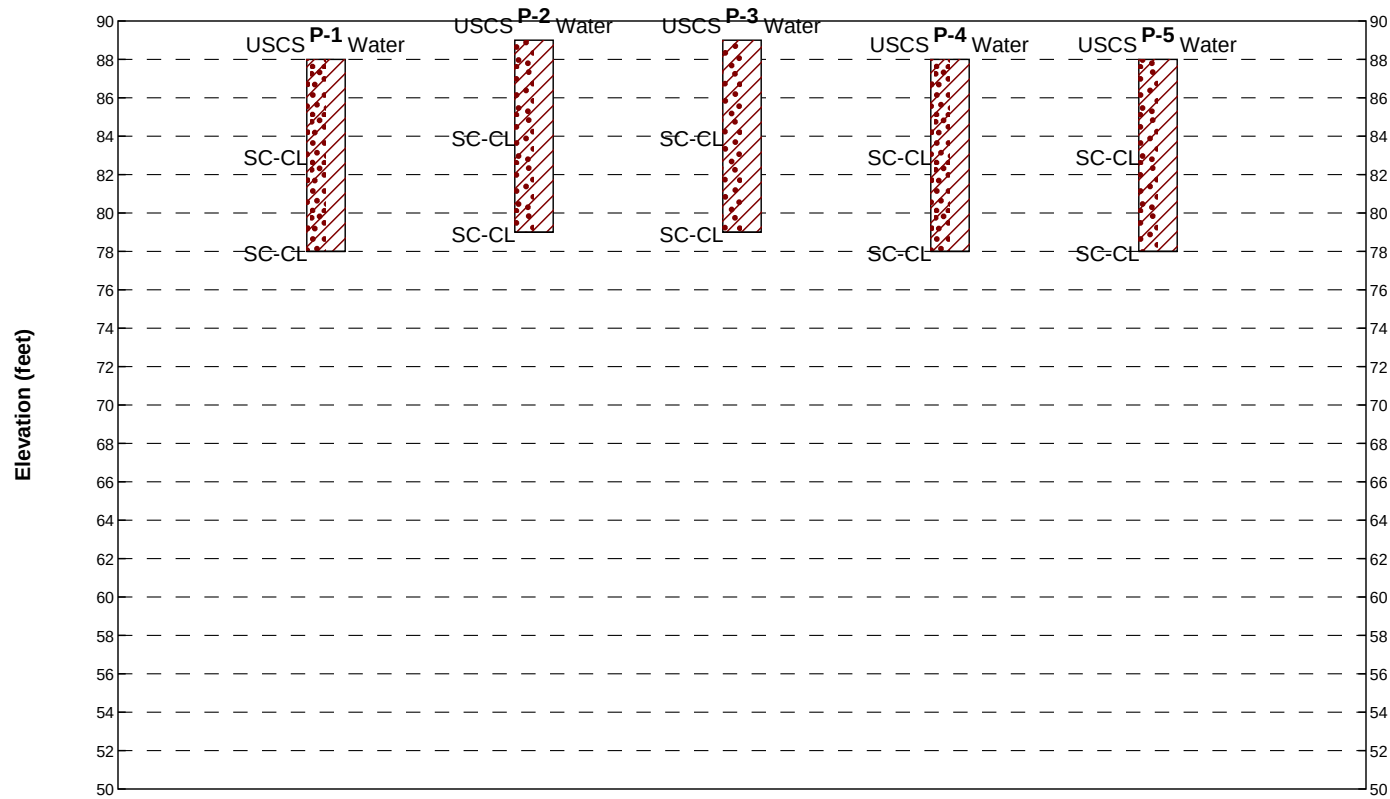
Clayey SAND to Sandy CLAY (SC-CL)

**Millennium Engineers Group**  
Pharr, Texas 78577

**Fire Station No.5**

**Project No.**  
01-19-29113

**Figure No.**  
C-1

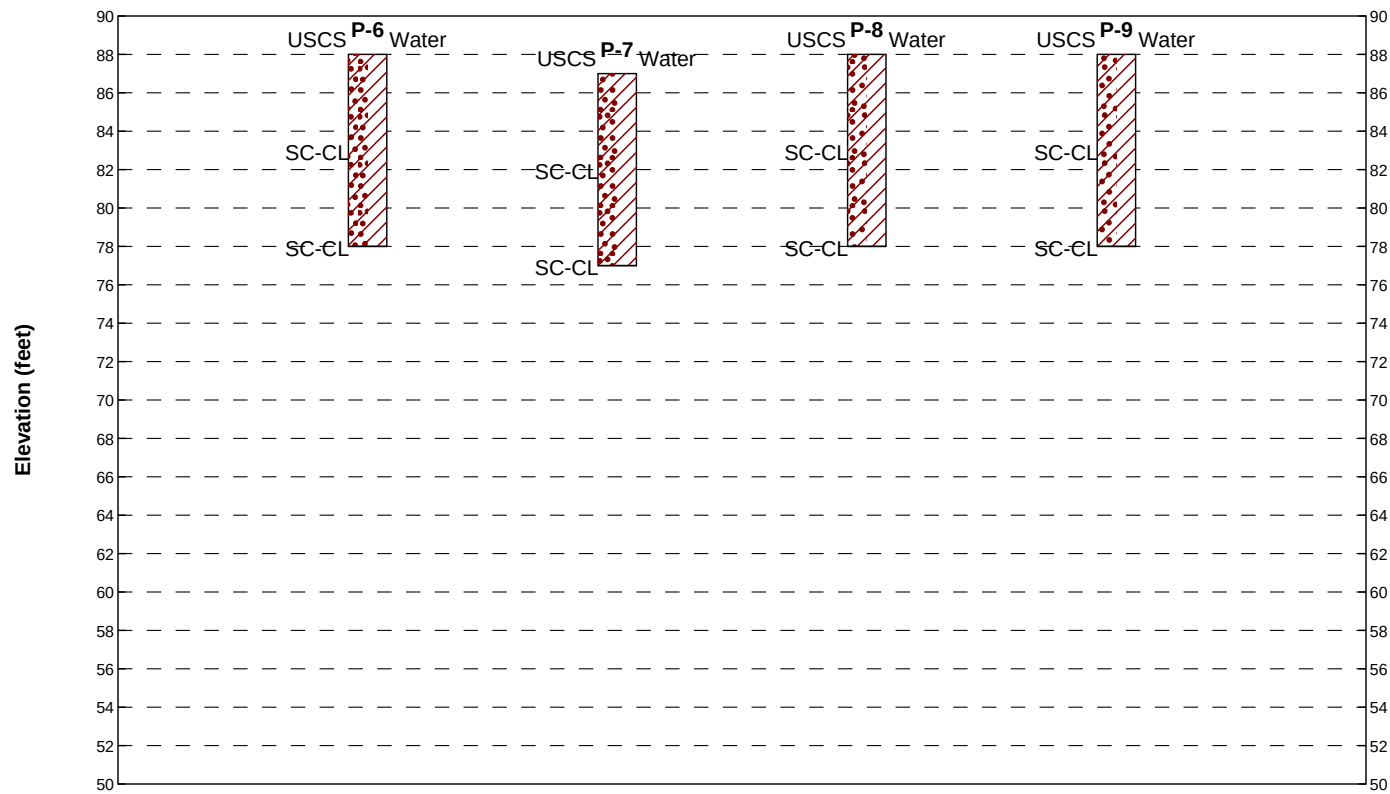


**MATERIAL GRAPHIC SYMBOLS**



Clayey SAND to Sandy CLAY (SC-CL)

|                                                         |                   |
|---------------------------------------------------------|-------------------|
| <b>Millennium Engineers Group</b><br>Pharr, Texas 78577 |                   |
| <b>Fire Station No.5</b>                                |                   |
| <b>Project No.</b>                                      | <b>Figure No.</b> |
| 01-19-29113                                             | C-1               |



#### MATERIAL GRAPHIC SYMBOLS



Clayey SAND to Sandy CLAY (SC-CL)

**Millennium Engineers Group**  
Pharr, Texas 78577

**Fire Station No.5**

**Project No.**

01-19-29113

**Figure No.**

C-1

Project: **Proposed Fire Station No. 5**

Project Location: **Edinburg, Hidalgo County, Texas**

Project Number: **01-19-29113**

## Key to Log of Boring

**Sheet 1 of 1**

| Elevation (feet) | Depth (feet) | Sample Type | Sample Number | Sampling Resistance, blows/ft | Material Type | Graphic Log | MATERIAL DESCRIPTION | Water Content, % | LL, % | PI, % | Percent Fines | REMARKS AND OTHER TESTS |
|------------------|--------------|-------------|---------------|-------------------------------|---------------|-------------|----------------------|------------------|-------|-------|---------------|-------------------------|
| 1                | 2            | 3           | 4             | 5                             | 6             | 7           | 8                    | 9                | 10    | 11    | 12            | 13                      |

### COLUMN DESCRIPTIONS

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>1</b> Elevation (feet): Elevation (MSL, feet).</p> <p><b>2</b> Depth (feet): Depth in feet below the ground surface.</p> <p><b>3</b> Sample Type: Type of soil sample collected at the depth interval shown.</p> <p><b>4</b> Sample Number: Sample identification number.</p> <p><b>5</b> Sampling Resistance, blows/ft: Number of blows to advance driven sampler one foot (or distance shown) beyond seating interval using the hammer identified on the boring log.</p> <p><b>6</b> Material Type: Type of material encountered.</p> <p><b>7</b> Graphic Log: Graphic depiction of the subsurface material encountered.</p> <p><b>8</b> MATERIAL DESCRIPTION: Description of material encountered. May include consistency, moisture, color, and other descriptive text.</p> | <p><b>9</b> Water Content, %: Water content of the soil sample, expressed as percentage of dry weight of sample.</p> <p><b>10</b> LL, %: Liquid Limit, expressed as a water content.</p> <p><b>11</b> PI, %: Plasticity Index, expressed as a water content.</p> <p><b>12</b> Percent Fines: The percent fines (soil passing the No. 200 Sieve) in the sample. WA indicates a Wash Sieve, SA indicates a Sieve Analysis.</p> <p><b>13</b> REMARKS AND OTHER TESTS: Comments and observations regarding drilling or sampling made by driller or field personnel.</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### FIELD AND LABORATORY TEST ABBREVIATIONS

CHEM: Chemical tests to assess corrosivity  
 COMP: Compaction test  
 CONS: One-dimensional consolidation test  
 LL: Liquid Limit, percent

PI: Plasticity Index, percent  
 SA: Sieve analysis (percent passing No. 200 Sieve)  
 UC: Unconfined compressive strength test, Qu, in ksf  
 WA: Wash sieve (percent passing No. 200 Sieve)

### MATERIAL GRAPHIC SYMBOLS



Clayey SAND (SC)



Clayey SAND to Sandy CLAY (SC-CL)

### TYPICAL SAMPLER GRAPHIC SYMBOLS



Auger sampler



CME Sampler



Bulk Sample



Grab Sample



3-inch-OD California w/ brass rings



2.5-inch-OD Modified California w/ brass liners



Pitcher Sample



2-inch-OD unlined split spoon (SPT)



Shelby Tube (Thin-walled, fixed head)

### OTHER GRAPHIC SYMBOLS



Water level (at time of drilling, ATD)



Water level (after waiting)



Minor change in material properties within a stratum



Inferred/gradational contact between strata



Queried contact between strata

### GENERAL NOTES

- Soil classifications are based on the Unified Soil Classification System. Descriptions and stratum lines are interpretive, and actual lithologic changes may be gradual. Field descriptions may have been modified to reflect results of lab tests.
- Descriptions on these logs apply only at the specific boring locations and at the time the borings were advanced. They are not warranted to be representative of subsurface conditions at other locations or times.



**Figure B-1**

## **APPENDIX D**

### **SUMMARY OF SOIL SAMPLE ANALYSIS**



## Summary of Soil Sample Analyses

**Project Name: Proposed Fire Station No.5**

| Boring No. | Sample Depth (ft) | Blows Per (ft) | Moisture Content | Liquid Limit | Plastic Limit | Plasticity Index | -200% Sieve | Shear Strength (tsf) | Dry Unit Weight (pcf) | USCS |
|------------|-------------------|----------------|------------------|--------------|---------------|------------------|-------------|----------------------|-----------------------|------|
| B-1        | .5 - 2            | 6              | 10               | 22           | 10            | 12               |             |                      |                       | CL   |
|            | 2.5 - 4           | 5              | 12               |              |               |                  | 48          |                      |                       |      |
|            | 4.5 - 6           | 6              | 12               | 28           | 16            | 11               |             |                      |                       | CL   |
|            | 6.5 - 8           | 6              | 12               |              |               |                  | 55          |                      |                       |      |
|            | 8.5 - 10          | 14             | 14               |              |               |                  |             |                      |                       |      |
|            | 13.5 - 15         | 17             | 21               | 20           | 11            | 9                |             |                      |                       | SC   |
|            | 18.5 - 20         | 13             | 25               |              |               |                  |             |                      |                       |      |
|            | 23.5 - 25         | 21             | 18               |              |               |                  | 26          |                      |                       |      |
| B-2        | .5 - 2            | 12             | 7                | 20           | 13            | 7                |             |                      |                       | CL   |
|            | 2.5 - 4           | 6              | 13               |              |               |                  | 55          |                      |                       |      |
|            | 4.5 - 6           | 9              | 14               |              |               |                  |             |                      |                       |      |
|            | 6.5 - 8           | 12             | 11               |              |               |                  |             |                      |                       |      |
|            | 8.5 - 10          | 18             | 12               |              |               |                  | 51          |                      |                       |      |
|            | 13.5 - 15         | 16             | 23               | 21           | 7             | 14               |             |                      |                       | SC   |
|            | 18.5 - 20         | 8              | 24               |              |               |                  | 23          |                      |                       |      |
|            | 23.5 - 25         | 30             | 19               |              |               |                  |             |                      |                       |      |
| B-3        | .5 - 2            | 8              | 9                |              |               |                  |             |                      |                       |      |
|            | 2.5 - 4           | 5              | 12               | 26           | 8             | 18               |             |                      |                       | CL   |
|            | 4.5 - 6           | 5              | 12               |              |               |                  | 49          |                      |                       |      |
|            | 6.5 - 8           | 7              | 15               |              |               |                  | 58          |                      |                       |      |
|            | 8.5 - 10          | 13             | 13               |              |               |                  |             |                      |                       |      |
|            | 13.5 - 15         |                | 9                | 25           | 14            | 12               |             |                      |                       | SC   |
|            | 18.5 - 20         |                | 24               |              |               |                  |             |                      |                       |      |
|            | 23.5 - 25         |                | 23               |              |               |                  | 30          |                      |                       |      |
| B-4        | .5 - 2            | 9              | 9                |              |               |                  | 54          |                      |                       |      |
|            | 2.5 - 4           | 6              | 12               | 26           | 15            | 11               |             |                      |                       | CL   |
|            | 4.5 - 6           | 5              | 13               |              |               |                  |             |                      |                       |      |
|            | 6.5 - 8           | 17             | 12               |              |               |                  | 37          |                      |                       |      |
|            | 8.5 - 10          | 23             | 14               | 28           | 16            | 11               |             |                      |                       | CL   |
|            | 13.5 - 15         | 7              | 18               |              |               |                  | 46          |                      |                       |      |
|            | 18.5 - 20         | 24             | 22               |              |               |                  |             |                      |                       |      |
|            | 23.5 - 25         | 27             | 21               |              |               |                  |             |                      |                       |      |

| Boring No. | Sample Depth (ft) | Blows Per (ft) | Moisture Content | Liquid Limit | Plastic Limit | Plasticity Index | -200% Sieve | Shear Strength (tsf) | Dry Unit Weight (pcf) | USCS |
|------------|-------------------|----------------|------------------|--------------|---------------|------------------|-------------|----------------------|-----------------------|------|
| B-5        | .5 - 2            | 4              | 9                |              |               |                  |             |                      |                       |      |
|            | 2.5 - 4           | 5              | 10               | 27           | 14            | 12               |             |                      |                       | CL   |
|            | 4.5 - 6           | 6              | 13               | 31           | 11            | 21               |             |                      |                       | CL   |
|            | 6.5 - 8           | 8              | 14               |              |               |                  |             |                      |                       |      |
|            | 8.5 - 10          | 12             | 15               |              |               |                  | 59          |                      |                       |      |
|            | 13.5 - 15         | 8              | 20               |              |               |                  | 49          |                      |                       |      |
|            | 18.5 - 20         | 12             | 20               |              |               |                  |             |                      |                       |      |
|            | 23.5 - 25         | 29             | 23               |              |               |                  | 13          |                      |                       |      |

| Boring No. | Sample Depth (ft) | Blows Per (ft) | Moisture Content | Liquid Limit | Plastic Limit | Plasticity Index | -200% Sieve | Shear Strength (tsf) | Dry Unit Weight (pcf) | USCS |
|------------|-------------------|----------------|------------------|--------------|---------------|------------------|-------------|----------------------|-----------------------|------|
| P-1        | .5 - 2            | 9              | 8                | 17           | 10            | 7                |             |                      |                       | CL   |
|            | 2.5 - 4           | 6              | 10               |              |               |                  |             |                      |                       |      |
|            | 4.5 - 6           | 5              | 11               |              |               |                  | 49          |                      |                       |      |
|            | 6.5 - 8           | 16             | 12               | 26           | 12            | 14               |             |                      |                       | CL   |
|            | 8.5 - 10          | 29             | 14               |              |               |                  |             |                      |                       |      |

|     |          |    |    |    |    |    |    |  |  |    |
|-----|----------|----|----|----|----|----|----|--|--|----|
| P-2 | .5 - 2   | 5  | 8  | 22 | 12 | 9  |    |  |  | CL |
|     | 2.5 - 4  | 4  | 11 |    |    |    |    |  |  |    |
|     | 4.5 - 6  | 3  | 13 | 26 | 16 | 10 |    |  |  | CL |
|     | 6.5 - 8  | 14 | 12 |    |    |    |    |  |  |    |
|     | 8.5 - 10 | 32 | 13 |    |    |    | 63 |  |  |    |

|     |          |    |    |    |    |   |    |  |  |    |
|-----|----------|----|----|----|----|---|----|--|--|----|
| P-3 | .5 - 2   | 7  | 7  |    |    |   | 34 |  |  |    |
|     | 2.5 - 4  | 3  | 10 |    |    |   |    |  |  |    |
|     | 4.5 - 6  | 6  | 12 |    |    |   |    |  |  |    |
|     | 6.5 - 8  | 22 | 14 |    |    |   |    |  |  |    |
|     | 8.5 - 10 | 26 | 12 | 28 | 20 | 8 |    |  |  | CL |

|     |          |    |    |    |    |    |    |  |  |    |
|-----|----------|----|----|----|----|----|----|--|--|----|
| P-4 | .5 - 2   | 5  | 6  | 20 | 18 | 2  |    |  |  | CL |
|     | 2.5 - 4  | 5  | 10 |    |    |    | 40 |  |  |    |
|     | 4.5 - 6  | 7  | 13 |    |    |    |    |  |  |    |
|     | 6.5 - 8  | 13 | 11 | 25 | 14 | 10 |    |  |  | CL |
|     | 8.5 - 10 | 27 | 13 |    |    |    |    |  |  |    |

|     |          |    |    |    |    |    |    |  |  |    |
|-----|----------|----|----|----|----|----|----|--|--|----|
| P-5 | .5 - 2   | 6  | 10 | 23 | 14 | 9  |    |  |  | CL |
|     | 2.5 - 4  | 4  | 13 |    |    |    |    |  |  |    |
|     | 4.5 - 6  | 9  | 13 |    |    |    |    |  |  |    |
|     | 6.5 - 8  | 24 | 13 |    |    |    | 60 |  |  |    |
|     | 8.5 - 10 | 31 | 13 | 31 | 12 | 18 |    |  |  | CL |

| Boring No. | Sample Depth (ft) | Blows Per (ft) | Moisture Content | Liquid Limit | Plastic Limit | Plasticity Index | -200% Sieve | Shear Strength (tsf) | Dry Unit Weight (pcf) | USCS |
|------------|-------------------|----------------|------------------|--------------|---------------|------------------|-------------|----------------------|-----------------------|------|
| P-6        | .5 - 2            | 6              | 7                |              |               |                  |             |                      |                       |      |
|            | 2.5 - 4           | 4              | 12               | 27           | 12            | 15               |             |                      |                       | CL   |
|            | 4.5 - 6           | 5              | 12               |              |               |                  | 50          |                      |                       |      |
|            | 6.5 - 8           | 10             | 13               | 25           | 11            | 14               |             |                      |                       | CL   |
|            | 8.5 - 10          | 45             | 13               |              |               |                  |             |                      |                       |      |
| P-7        | .5 - 2            | 7              | 8                |              |               |                  | 42          |                      |                       |      |
|            | 2.5 - 4           | 7              | 14               |              |               |                  |             |                      |                       |      |
|            | 4.5 - 6           | 5              | 12               | 26           | 17            | 9                |             |                      |                       | CL   |
|            | 6.5 - 8           | 7              | 12               |              |               |                  |             |                      |                       |      |
|            | 8.5 - 10          | 19             | 15               | 24           | 20            | 4                |             |                      |                       | CL   |
| P-8        | .5 - 2            | 8              | 8                | 23           | 14            | 8                |             |                      |                       | CL   |
|            | 2.5 - 4           | 4              | 9                |              |               |                  |             |                      |                       |      |
|            | 4.5 - 6           | 5              | 12               |              |               |                  | 58          |                      |                       |      |
|            | 6.5 - 8           | 22             | 14               | 26           | 17            | 9                |             |                      |                       | CL   |
|            | 8.5 - 10          | 26             | 12               |              |               |                  |             |                      |                       |      |
| P-9        | .5 - 2            | 8              | 9                | 25           | 16            | 9                |             |                      |                       | CL   |
|            | 2.5 - 4           | 5              | 12               |              |               |                  | 49          |                      |                       |      |
|            | 4.5 - 6           | 8              | 14               |              |               |                  |             |                      |                       |      |
|            | 6.5 - 8           | 25             | 12               |              |               |                  |             |                      |                       |      |
|            | 8.5 - 10          | 20             | 12               | 25           | 15            | 10               |             |                      |                       | CL   |

## **APPENDIX E**

### **LABORATORY AND FIELD PROCEDURES**

## Laboratory and Field Test Procedures

### **Soil Classification Per ASTM D2487-93:**

This soil-testing standard was used for classifying soils according to the Unified Soil Classification System. The soil classifications of the earth materials encountered are as noted in the attached boring logs.

### **Soil Water Content Per ASTM D2216-92:**

This test determines the water content of soil or rock expressed as a percentage of the solid mass of the soil. The test results are listed under **MC** in the attached boring logs.

### **Soil Liquid Limit Per ASTM D4318-93:**

The soil Liquid Limit identifies the upper limit soil water content at which the soil changes from a moldable (plastic) physical state to a liquid state. The Liquid Limit water content is expressed as a percentage of the solid mass of the soil. The test results are listed under **LL** in the attached boring logs.

### **Soil Plastic Limit Per ASTM D4318-93:**

The soil Plastic Limit identifies lower limit soil water content at which the soil changes from a moldable (plastic) physical state to a non-moldable (semi-solid) physical state. The Plastic Limit water content is expressed as a percentage of the solid mass of the soil. The test results are listed under **PL** in the attached boring logs.

### **Plasticity Index Per ASTM D4318-93:**

This is the numeric difference between the Liquid Limit and Plastic Limit. This index also defines the range of water content over which the soil-water system acts as a moldable (plastic) material. Higher Plasticity Index (PI) values indicate that the soil has a greater ability to change in soil volume or shrink and swell with lower or higher water contents, respectively. The test results are listed under **PI** in the attached boring logs.

### **Standard Penetration Test (SPT) and Split Spoon Sampler (SS) per ASTM D 1586:**

This is the standard test method for both the penetration test and split-barrel (spoon) sampling of soils. This sampling method is used for soils or rock too hard for sampling using Shelby Tubes. The method involves penetration of a split spoon sampler into the soil or rock through successive blows of a 140-pound hammer in a prescribed manner.

### **Blow Counts (N) per ASTM D 1586:**

This is the number of blows required to drive a Split Spoon Sampler by means of a 140 pound hammer for a distance of 12 inches in accordance with the variables stated in the test procedures.

**Shelby Tube (ST) per ASTM D 1587:**

This procedure is for using a thin-walled metal tube to recover relatively undisturbed soil samples suitable for laboratory tests of physical properties.

**Dry Density (DD) per ASTM D 2937:**

This procedure is for the determination of in-place density of soil. The test results are measured in pounds per cubic foot, pcf.

**Unconfined Compression Test (Uc) per ASTM D 2166:**

This test method covers the determination of the unconfined compressive strength of cohesive soil in the undisturbed, remolded, or compacted condition, using strain-controlled application of the axial load.

**Minus No. 200 Sieve per ASTM D 1140:**

This test method covers determination of the amount of material finer than a Number 200 sieve by washing. The results are stated as a percent of the total dry weight of the sample.

**Pocket Penetrometer (PP):**

This test method is an accepted modification of ASTM D 1558 test method for establishing the moisture-penetration resistance relationships of fine-grained soils. The test results are measured in tons per square foot, tsf. The strength values provided by this method should be considered qualitatively.

**Rock Quality Designation (RQD):**

The measure of the quality of a rock mass defined by adding intact rock core pieces greater than four inches in length by the total length of core advance.

**Recovery Ratio (REC):**

The Recovery Ratio is equal to the total length of core recovered divided by the total length of core advance.

**Boring Logs:**

This is a summary of the above-described information at each boring location.

## **APPENDIX F**

### **TENSAR SPECTRA PAVE TX5 CALCULATIONS AND SPECIFICATION**



# SpectraPave4 PRO™ Pavement Optimization Design Analysis



## Design Parameters for AASHTO (1993) Equation

|                         |          |                          |       |
|-------------------------|----------|--------------------------|-------|
| Reliability (%)         | = 80     | Initial Serviceability   | = 4.0 |
| Standard Normal Deviate | = -0.842 | Terminal Serviceability  | = 2.0 |
| Standard Deviation      | = 0.45   | Change in Serviceability | = 2   |

Aggregate fill shall conform to following requirement:

D50 ≤ 27mm (Base course)

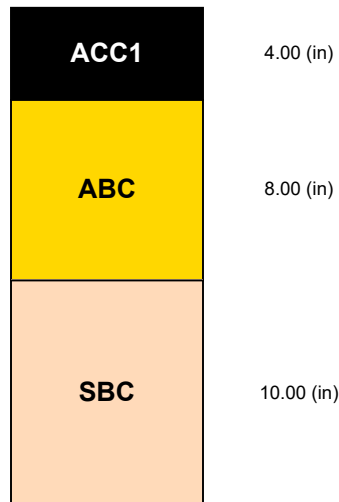
## Unstabilized Section Material Properties

| Layer | Description            | Cost (\$/ton) | Layer coefficient | Drainage factor |
|-------|------------------------|---------------|-------------------|-----------------|
| ACC1  | Asphalt Wearing Course | 70            | 0.400             | N/A             |
| ABC   | Aggregate Base Course  | 20            | 0.080             | 1.0             |
| SBC   | Subbase Course         | 16            | 0.080             | 1.0             |

## Stabilized Section Material Properties

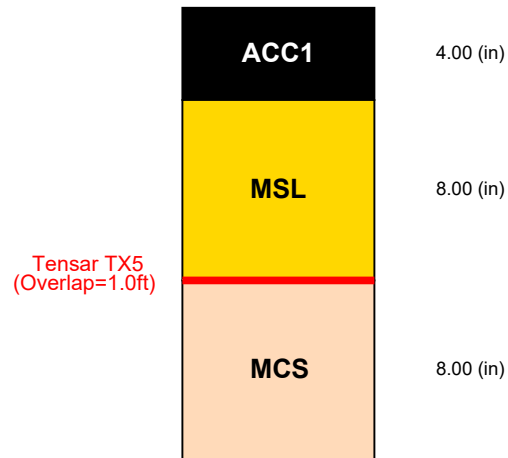
| Layer | Description                         | Cost (\$/ton) | Layer coefficient | Drainage factor |
|-------|-------------------------------------|---------------|-------------------|-----------------|
| ACC1  | Asphalt Wearing Course              | 70            | 0.400             | N/A             |
| MSL   | Mechanically Stabilized Base Course | 20            | 0.136             | 1.0             |
| MCS   | Moisture Conditioned Subgrade       | 16            | 0.000             | 1.0             |

## Unstabilized Pavement



Subgrade Modulus = 6,000 (psi)  
Structural Number = 3.040  
Calculated Traffic (ESALs) = 481,000

## Stabilized Pavement



Subgrade Modulus = 6,000 (psi)  
Structural Number = 2.688  
Calculated Traffic (ESALs) = 220,000

## LIMITATIONS OF THE REPORT

The designs, illustrations, information and other content included in this report are necessarily general and conceptual in nature, and do not constitute engineering advice or any design intended for actual construction. Specific design recommendations can be provided as the project develops.

|              |        |      |     |
|--------------|--------|------|-----|
| Project Name | N/A    |      |     |
| Company Name | Tensar |      |     |
| Designer     | N/A    | Date | N/A |



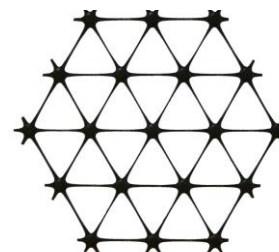
## Product Specification - TriAx® TX5 Geogrid

Tensor International Corporation reserves the right to change its product specifications at any time. It is the responsibility of the person specifying the use of this product and of the purchaser to ensure that product specifications relied upon for design or procurement purposes are current and that the product is suitable for its intended use in each instance.

### General

1. The geogrid is manufactured from a punched polypropylene sheet, which is then oriented in three substantially equilateral directions so that the resulting ribs shall have a high degree of molecular orientation, which continues at least in part through the mass of the integral node.
2. The properties contributing to the performance of a mechanically stabilized layer include the following:

**Tensor TriAx® Geogrid**



| Index Properties                         | Longitudinal | Diagonal   | Transverse | General     |
|------------------------------------------|--------------|------------|------------|-------------|
| ▪ Rib pitch <sup>(1)</sup> , mm (in)     | 40 (1.60)    | 40 (1.60)  | -          |             |
| ▪ Mid-rib depth <sup>(1)</sup> , mm (in) | -            | 1.3 (0.05) | 1.2 (0.05) |             |
| ▪ Mid-rib width <sup>(1)</sup> , mm (in) | -            | 0.9 (0.04) | 1.2 (0.05) |             |
| ▪ Rib shape                              |              |            |            | Rectangular |
| ▪ Aperture shape                         |              |            |            | Triangular  |

### Dimensions and Delivery

The TX geogrid shall be delivered to the jobsite in roll form with each roll individually identified and nominally measuring 3.0 meters (9.8 feet) and/or 4.0 meters (13.1feet) in width and 75 meters (246 feet) in length.

### Notes

1. Nominal dimensions.

**Tensor International Corporation**  
2500 Northwinds Pkwy.  
Atlanta, Georgia 30009  
Phone: 800-TENSAR-1  
[www.tensorcorp.com](http://www.tensorcorp.com)

This specification supersedes any and all prior specifications for the product designated above and is not applicable to any product shipped prior to February 1, 2011. Tensor and TriAx are trademarks of Tensor International Corporation or its affiliates in the US and many other countries. TriAx® geogrid and the use thereof are protected by U.S. Patent No. 7,001,112. Patents or patent applications also exist in other countries. Final determination of the suitability of the above-mentioned information or product for the use contemplated, and its manner of use are the sole responsibility of the user. Tensor International Corporation disclaims any and all express, implied or statutory warranties, including but not limited to, any warranty of merchantability or fitness for a particular purpose regarding this product or the Company's other products, technologies or services. The information contained herein does not constitute engineering advice.

SECTION 01 11 00 - SUMMARY

PART 1 - GENERAL

1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.2 SUMMARY OF WORK

- A. Project Identification: As follows:
  - 1. Project: Edinburg Fire Department – New Fire Station #5.
  - 2. Owner: City of Edinburg
  - 3. Location: 415 W. University Dr., Edinburg, Texas 78540.
- B. Contract Documents, dated February 28, 2019 were prepared by Milnet Architectural Services, 608 S. 12<sup>th</sup> St. McAllen, TX. 78501.
- C. This project consists of the 8,771 Sq. Ft. of new CMU walls, concrete slab & Insulated battenlok roof panel construction for the new Fire Station #5.

1.3 WORK RESTRICTIONS

- A. Contractor's Use of Premises: During construction, Contractor shall have **limited** use of **site** indicated. Contractor's use of premises is limited only by Owner's right to perform work or employ other contractors on portions of Project.
- B. Assume full responsibility for the protection and safekeeping of Products under this Contract, stored on the site.
- C. Move any stored Products, under Contractor's control, which interfere with operations of the Owner and separate contractor.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

## SECTION 01 20 00 - PRICE AND PAYMENT PROCEDURES

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 UNIT PRICES

- A. Changes to the Work incorporating Unit Prices will be made by Change Order.

#### 1.3 CONTRACT MODIFICATION PROCEDURES

- A. On Owner's approval of a proposal from Contractor, Architect will issue a Change Order on AIA Document G701, for all changes to Contract Sum or Contract Time.
- B. When Owner and Contractor disagree on the terms of a proposal, Architect may issue a Construction Change Directive on AIA Document G714, instructing Contractor to proceed with the change. Construction Change Directive will contain a description of the change and designate the method to be followed to determine changes to Contract Sum or Contract Time.

#### 1.4 PAYMENT PROCEDURES

- A. Submit a Schedule of Values **at least 10 days before** the first Application for Payment. In Schedule of Values, break down Contract Sum into at least one line item for each Specification Section. Correlate the Schedule of Values with Contractor's Construction Schedule.
- B. Submit 3 copies of each application for payment on AIA Document G702/703, according to the schedule established in Owner/Contractor Agreement.
  - 1. For the second Application for Payment through the Application for Payment submitted at Substantial Completion, submit partial releases of liens from each subcontractor or supplier for whom amounts were requisitioned in the previous Application for Payment.
  - 2. Submit final Application for Payment after completion of Project closeout procedures with release of liens and supporting documentation. Include consent of surety to final payment and insurance certificates.

Edinburg Fire Department  
Fire Station #5

MAS Proj. No. 219003

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 01 21 00 — ALLOWANCES

PART 1 - GENERAL

1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.2 RELATED DOCUMENTS

- A. Section 01028 – Modification Requirements.

1.3 CONTINGENCY ALLOWANCE

- A. Include in the Contract, a stipulated sum of **Sixty Thousand Dollars, (\$60,000.00)** for use upon Architect's instruction.

1.4 PROCEDURES FOR MANAGING ALLOWANCES

- A. Contractor's costs for Products, delivery, installation, labor, payroll, taxes, bonding, and equipment rental will be included in Change Orders authorizing expenditure of funds from Allowances.
- B. Funds will be drawn from Allowances only by Change Order.
- C. At closeout of Contract, funds remaining in Allowances will be credited to Owner by Change Order.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

## SECTION 01 25 00 – SUBSTITUTION PROCEDURES

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 SUBSTITUTION REQUIREMENTS

- A. When material, article, or method is specified using name of proprietary product manufacturer, vendor, or method followed by phrase "or equal," specific item mentioned establishes basis upon which projects are to be built.
  - 1. Other manufacturers' materials, articles, and methods not named will be considered as substitutions provided required information is submitted on "SUBSTITUTION REQUEST FORM" and will not require substantial revisions of Contract Documents.
  - 2. This applies to specific construction methods when required by Contract Documents.
  - 3. Substitution Requests must be filled out on enclosed "Substitution Request Form".
- B. Whenever material, article, or method is specified or described without phrase "or equal," no substitutions will be allowed.
- C. Costs for redesigns due to substituted items are responsibility of Applicant.
- D. In making request for substitution, Applicant/Contractor represents that he:
  - 1. Has personally investigated proposed product or method and determined that it is equal in all respects to that specified.
  - 2. Will provide same guarantee for substitution as for product or method specified.
  - 3. Will coordinate installation of accepted substitution into work, making design and construction changes to complete work in all respects following the Contract Documents.

#### 1.3 SUBMITTAL OF DATA FOR PROPOSED SUBSTITUTIONS

- A. In order for substitutions that do not change design intent to be considered, submit no later than 10 days prior to bid date deadline, 3 copies of complete data set forth herein to permit complete analysis of proposed substitutions listed on submitted "SUBSTITUTION REQUEST FORM".

1. For Products:
  - a. Identification including manufacturer's name and address.
  - b. Manufacturer's literature, including but not necessarily limited to:
    - 1) Product description, performance, and test data.
    - 2) Reference standards.
  - c. Samples where appropriate.
  - d. Name and address of similar projects on which product was used and dates of installation with contact name and telephone number.
2. For Construction Methods:
  - a. Detailed description of proposed method.
  - b. Drawings illustrating methods.
  - c. Name and address of similar projects on which method was used and dates of use with contact name and telephone number.
3. Comparison of proposed substitution with product or method specified
4. Data relating to impact on construction schedule by proposed substitution.
5. Impact on other contracts.

#### 1.4 APPROVAL OF SUBSTITUTION

- A. Architect's decision regarding evaluation of substitutions will be final and binding.
- B. All approved substitutions will be incorporated into the Contract Documents by Addendum.

PART 2 - PRODUCTS  
NOT USED

PART 3 - EXECUTION  
NOT USED

## SUBSTITUTION REQUEST FORM

Project: \_\_\_\_\_ Substitution Request Number: \_\_\_\_\_  
\_\_\_\_\_  
From: \_\_\_\_\_  
To: \_\_\_\_\_ Date: \_\_\_\_\_  
\_\_\_\_\_  
A/E Project Number: \_\_\_\_\_  
Re: \_\_\_\_\_ Contract For: \_\_\_\_\_  
\_\_\_\_\_  
Specification Title: \_\_\_\_\_ Description: \_\_\_\_\_  
Section: \_\_\_\_\_ Page: \_\_\_\_\_ Article/Paragraph: \_\_\_\_\_  
\_\_\_\_\_

Proposed Substitution: \_\_\_\_\_  
Manufacturer: \_\_\_\_\_ Address: \_\_\_\_\_ Phone: \_\_\_\_\_  
Trade Name: \_\_\_\_\_ Model No.: \_\_\_\_\_

Attached data includes product description, specifications, drawings, photographs, and performance and test data adequate for evaluation of the request; applicable portions of the data are clearly identified.

Attached data also includes a description of changes to the Contract Documents that the proposed substitution will require for its proper installation.

The Undersigned certifies:

- Proposed substitution has been fully investigated and determined to be equal or superior in all respects to specified product.
- Same warranty will be furnished for proposed substitution as for specified product.
- Same maintenance service and source of replacement parts, as applicable, is available.
- Proposed substitution will have no adverse effect on other trades and will not affect or delay progress schedule.
- Proposed substitution does not affect dimensions and functional clearances.
- Payment will be made for changes to building design, including A/E design, detailing, and construction costs caused by the substitution.

Submitted by: \_\_\_\_\_  
Signed by: \_\_\_\_\_  
Firm: \_\_\_\_\_  
Address: \_\_\_\_\_  
\_\_\_\_\_  
Telephone: \_\_\_\_\_  
\_\_\_\_\_

### A/E's REVIEW AND ACTION

- ☐ Substitution approved - Make submittals in accordance with Specification Section 01340 Submittals  
☐ Substitution approved as noted - Make submittals in accordance with Specification Section 01340 Submittals  
☐ Substitution rejected - Use specified materials.  
☐ Substitution Request received too late - Use specified materials.

Signed by: \_\_\_\_\_

Date: \_\_\_\_\_

Supporting Data Attached: ☐ Drawings ☐ Product Data ☐ Samples ☐ Tests ☐ Reports ☐ \_\_\_\_\_  
\_\_\_\_\_

END OF SECTION



## SECTION 01 30 00 - ADMINISTRATIVE REQUIREMENTS

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 PROJECT MANAGEMENT AND COORDINATION

- A. Verify layout information shown on Drawings, in relation to property survey and existing benchmarks, before laying out the Work.
- B. Coordinate construction to ensure efficient and orderly execution of each part of the Work.
- C. Progress meetings will be held at Project site every two weeks. Notify Owner and Architect of meeting dates. Each subcontractor or other entity concerned with current progress or involved with planning or coordination of future activities, shall attend.
  - 1. Prepare a progress meeting agenda.**
  - 2. Prepare a sign in sheet for each progress meeting.**
  - 3. Prepare minutes of each meeting and distribute to parties present.**

#### 1.3 CONSTRUCTION SCHEDULE

- A. Prepare a horizontal bar-chart construction schedule. Provide a separate time bar for each activity and a vertical line to identify the first workday of each week. Use same breakdown of Work indicated in the Schedule of Values. As Work progresses, mark each bar to indicate actual completion.
  - 1. Submit within twenty (20) days after date established for Commencement of the Work.
  - 2. Coordinate each element with other activities. Show each activity in proper sequence. Indicate sequences necessary for completion of related Work.
  - 3. Indicate Substantial Completion and allow time for Architect's procedures necessary for certifying Substantial Completion.
  - 4. Schedule Distribution: Distribute copies to Owner, Architect, subcontractors, and parties required to comply with dates.
  - 5. Updating: Revise the schedule after each meeting or activity where revisions have been made. Distribute revised copies to Owner, Architect, subcontractors, and parties required to comply with dates.

#

#### 1.4 SUBMITTAL PROCEDURES

- A. Coordinate submittal preparation with construction schedule, fabrication lead-times, other submittals, and activities that require sequential operations.
  - 1. No extension of Contract Time will be authorized due to failure to transmit submittals in time to permit processing sufficiently in advance of when materials are required in the Work.
  - 2. Architect will not accept submittals from sources other than Contractor.
- B. Prepare submittals by placing a permanent label on each for identification. Provide a 4- by 5-inch (100-by 125-mm) space on the label or beside title block to record review and approval markings and action taken. Include the following information on the label:
  - 1. Project name.
  - 2. Date.
  - 3. Name and address of Contractor.
  - 4. Name and address of subcontractor or supplier.
  - 5. Number and title of appropriate Specification Section.
  - 6. Contractor's certification that materials comply with specified requirements.
- C. Coordinate each submittal with other submittals and with work that does not require submittals.
- D. Product Data: Mark each copy to show applicable choices and options. Include the following:
  - 1. Data indicating compliance with specified standards and requirements.
  - 2. Notation of coordination requirements.
  - 3. For equipment data, include rated capacities, dimensions, weights, required clearances, and furnished specialties and accessories.
- E. Shop Drawings: Submit newly prepared information drawn to scale. Do not reproduce Contract Documents or copy standard information. Submit 1 reproducible print and 1 blue- or black-line print on sheets at least 8-1/2 by 11 inches (215 by 280 mm) but no larger than 30 by 42 inches (762 by 1067 mm). Architect will return the reproducible print. Include the following:
  - 1. Dimensions, profiles, methods of attachment, coordination with adjoining work, large scale details, and other information, as appropriate for the Work.
  - 2. Identification of products and materials.
  - 3. Notation of coordination requirements.
  - 4. Notation of dimensions established by field measurement.
  - 5. Identification of deviations from Contract Documents.
- F. Samples: Submit Samples finished as specified and identical with the material proposed. Where variations are inherent in the material, submit sufficient units to show limits of the variations. Include product name or name of the manufacturer.
- G. Architect will review each submittal, mark as appropriate to indicate action taken, and return copies less those retained. Compliance with specified requirements remains Contractor's responsibility.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION 01300

SECTION 01 33 00 — SUBMITTALS

PART 1 - GENERAL

1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.2 WORK INCLUDED

- A. Provide shop drawings, product data, physical samples and color samples as indicated herein and in each technical section of these specifications.

1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Additional submittal requirements specific to the particular section of the specifications.

PART 2 - PRODUCTS

2.1 SHOP DRAWINGS

- A. Prepare shop drawings using competent draftsmen, clearly and precisely showing the following:
  - 1. The size and gage of members.
  - 2. The method of anchoring and securing members of parts together.
  - 3. The quantity and location of each item.
  - 4. Other pertinent data necessary to show the Work to be done and where and how it is to be done.
- B. Prepare Drawings to scale, including full size details as required to fix and illustrate the Work required. Do not use Contract Documents or reproductions thereof as shop drawing submittals.
- C. Each sheet of Drawings shall be 30 x 40 inches maximum size with borders. Provide a title block in the lower right hand corner with the following information:
  - 1. Title of the sheet.
  - 2. Name and location of Project.
  - 3. Names of:
    - a. Architect/Engineer.
    - b. General Contractor.

- c. Manufacturer of the specified materials and equipment.
  4. The date of the Submittal.
  5. The date of each correction or revision.
  6. **Submittal number including Division No.** (such as submittal no. 3 under Division 11 is numbered "11-03").
- D. Fold drawings to 8-1/2x11 inch dimensions with title block exposed to top.
- E. Check the Drawings and add any corrections of field measurements needed. Stamp and sign the Contractor's approval, checker's signature, and date of approval before submitting to the Architect. Shop Drawings which do not bear the Contractor's stamp or have not been reviewed by the Contractor, will be returned by the Architect without review or approval.
- F. Number Shop Drawings consecutively. Indicate working and erection dimensions, arrangements, sectional views, necessary details including complete information for making connections with other Work, kinds of materials, and finishes.
- G. Provide a transmittal letter in duplicate, pointing out any deviations from items, methods or named manufacturers included in the Specifications or on the Drawings. Note submittal file number including Division.
- H. Submit one reproducible (sepia) print and minimum three blue-line prints of each Shop Drawing sheet.
- I. Make such corrections, changes, resubmit bound sets of Shop Drawings prints, as required herein, until approved is obtained. Any corrections or changes indicated on Shop Drawings shall not be considered as an extra work order.

## 2.2 PHYSICAL SAMPLES

- A. Provide duplicate samples of items as specified. Samples shall be 12 inches square or 12 inches long unless noted otherwise. Minimum liquid samples shall be 1 pint. Installed materials shall match approved samples.
- B. For Architect's permanent files provide one (1) 6" x 6" sample of all interior finishes, colors and materials (aluminum finish, glazing, plastic laminate, paint finish flooring materials, ceiling finish, etc.
- C. Provide a transmittal letter with each sample, listing the following:
1. Specification section title and paragraph specifying the material.
  2. Name and location of Project.
  3. Names of:
    - a. Architect/Engineer.
    - b. General Contractor.
    - c. Manufacturer of the specified materials and equipment.
  4. The date of the Submittal.
  5. Submittal file number including Division.
- D. If samples are not acceptable they will be returned directly to the Contractor for modification and resubmission.
- E. If samples are acceptable, notification will be sent directly to the Contractor, and the sample retained for comparison with the complete Work.

## 2.3 MANUFACTURER'S PRODUCT DATA

- A. Provide six copies of pre-printed Product Data of items as specified. Carefully mark out all items not applicable to the specified item.
- B. Standard catalogs, brochures, etc. including information not applicable to the project and not marked through, will be returned without review or approval.
- C. Provide a transmittal letter with the Product Data from each manufacturer, listing the following information:
  - 1. Name and location of Project.
  - 2. Names of:
    - a. Architect/Engineer.
    - b. General Contractor.
    - c. Manufacturer of the specified materials and equipment.
  - 3. The date of the Submittal.
  - 4. Submittal file number including Division.
- D. If Product Data is not approved, one copy will be marked and returned directly to the Contractor for modification and resubmission.
- E. If Product Data is approved, notification and one copy of the acceptable Product Data will be sent directly to the Contractor.
- F. When requested by the Architect, provide four copies of each ASTM Federal Specification, or other applicable documents referenced in the material Section.

## PART 3 - EXECUTION

### 3.1 REVIEW PROCEDURE

- A. Submittals will be reviewed with reasonable promptness so as to cause no delay, but only for conformance with the design concept of the project and with the information given in the Contract Documents. The review of a separate item shall not indicate a review of an assembly in which the item functions. Submittals that contain excessive errors or that are incomplete will be returned without review and approval and any delay caused thereby shall be the responsibility of the Contractor.
- B. If any submittals are not approved as submitted, all copies will be returned directly to the Contractor for revision. The reviewed submittals will be returned to the Contractor as soon as practicable.
- C. The Contractor shall make all revisions as noted and shall resubmit the required number of corrected copies of submittals, until no exceptions are taken. The Contractor shall direct specific attention, in writing or on resubmitted Shop Drawings, to revisions other than those requested on previous submissions.
- D. The review of submittals shall not relieve the Contractor of responsibility for deviations from the requirements of the Contract Documents unless the Contractor has submitted, in writing, such deviations and written approval has been given to each specific deviation. The review shall not relieve the Contractor from responsibility for errors and omissions in the Shop Drawings and samples.
- E. No portion of the Work requiring a submittal shall commence until the submittal has been approved as designated in the Conditions of the Contract. All such portions of the Work shall be in accordance with the submittal that has been stamped with final "Reviewed Without Exceptions" note, or "Approved" note.

- F. Materials and equipment specified or approved prior to beginning the Work are required to be used on the Project. Any proposed substitution resulting from no availability of specified items must be proven "better than" by the Contractor and approved in writing by the Architect. Substitutions included in submittals shall be so noted and brought to the Architect's attention in the submittal and on the transmittal. Failure to follow this procedure will render the substitution as not acceptable whether or not reviewed by the Architect.
- G. The Contractor shall have the approved shop drawings at the site at all times for use in the construction of the Work. Failure of the Contractor to supply such drawings will be deemed sufficient cause to delay the Work until such drawings are available for field use and reference.
- H. For submittals that will be reviewed by one of the Architect's consultants, these submittals shall be delivered directly to the Consultant with a copy of the transmittal to the Architect.
- I. For submittals that will be reviewed by one of the Architect's consultants, do not send to the Consultant as part of the package any items which will be reviewed by the Architect. As an example, do not provide a single submittal package combining Structural Steel and Miscellaneous Metal Fabrications.

END OF SECTION

## SECTION 01 40 00 - QUALITY REQUIREMENTS

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 SECTION REQUIREMENTS

- A. Quality-control services include inspections, tests, and related actions including reports. Quality-control services are further specified in other Sections of these Specifications and shall be performed by independent testing agencies provided by Contractor or Owner, as specified.
  - 1. Unless otherwise indicated, quality-control services required by authorities having jurisdiction will be provided by Owner.
- B. Contractor is responsible for scheduling inspections and tests.
- C. **Retesting: Contractor shall pay for retesting where results of inspections and tests prove unsatisfactory and indicate noncompliance with requirements.**
- D. Auxiliary Services: Cooperate with agencies performing inspections and tests. Provide auxiliary services as requested. Notify agency in advance of operations requiring tests or inspections, to permit assignment of personnel. Auxiliary services include the following:
  - 1. Access to the Work.
  - 2. Incidental labor and facilities to assist inspections and tests.
  - 3. Adequate quantities of materials that require testing, and assisting in taking samples.
  - 4. Facilities for storage and curing of test samples.
  - 5. Security and protection of samples and test equipment.
- E. Duties of Testing Agency: Testing agency shall cooperate with Architect and Contractor in performing its duties. Agency shall provide qualified personnel to perform inspections and tests.
  - 1. Agency shall promptly notify Architect and Contractor of irregularities or deficiencies observed in the Work during performance of its services.
  - 2. Agency shall not release, revoke, alter, or enlarge requirements of the Contract Documents or approve or accept any portion of the Work.
  - 3. Agency shall not perform duties of Contractor.

- F. Submittals: Testing agency shall submit a certified written report of each inspection and test to the following:
1. Owner.
  2. Architect.
  3. Contractor.
  4. Structural engineer.
  5. Authorities having jurisdiction, when authorities so direct.
- G. Report Data: Reports of each inspection, test, or similar service shall include at least the following:
1. Name, address, and telephone number of testing agency.
  2. Project title and testing agency's project number.
  3. Designation (number) and date of report.
  4. Dates and locations where samples were taken or inspections and field tests made.
  5. Names of individuals taking the sample or making the inspection or test.
  6. Designation of the product and test method.
  7. Complete inspection or test data including an interpretation of test results.
  8. Ambient conditions at the time of sample taking and testing.
  9. Comments or professional opinion on whether inspected or tested Work complies with requirements.
  10. Recommendations on retesting or reinspection.
  11. Name and signature of laboratory inspector.
- H. Testing Agency Qualifications: Engage inspection and testing agencies that are prequalified as complying with the American Council of Independent Laboratories' "Quality Assurance Manual" and that specialize in the types of inspections and tests to be performed.
1. Each testing agency shall be authorized by authorities having jurisdiction to operate in the state where Project is located.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION



## SECTION 01 50 00 — TEMPORARY FACILITIES AND CONTROLS

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 SECTION INCLUDES

- A. Temporary Utilities: Electricity, lighting, heat, ventilation, telephone and fax service, water, and sanitary facilities.
- B. Temporary Controls: Barriers, enclosures and fencing, protection of the Work, and water control.
- C. Construction Facilities: Access roads, parking, progress cleaning, project signage and temporary buildings.

#### 1.3 TEMPORARY ELECTRICITY

- A. Cost: By General Contractor. Utilize existing power service if approved by Owner. Extend temporary outlets in NEC and OSHA approved manner to facilitate construction.
- B. Provide power outlets for construction operations, with branch wiring and distribution boxes located as required. Provide flexible power cords as required.
- C. Provide main service disconnect and over correct protection at convenient location.
- D. Provide sufficient and adequate distribution equipment, wiring, and outlets to ensure unimpeded progress of the Work.
- E. Permanent convenience receptacles may be utilized during construction.

#### 1.4 TEMPORARY LIGHTING

- A. Provide and maintain lighting for construction operations to achieve a minimum lighting level of 2 watt/sq ft.

- B. Provide and maintain 1 watt/sq ft lighting to exterior staging and storage areas after dark for security purposes.
- C. Provide branch wiring from power source to distribution boxes with lighting conductors, pigtails, and lamps as required.
- D. Permanent building lighting may be utilized during construction.
- E. Maintain lighting and provide routine repairs.

1.5 TEMPORARY HEAT

- A. Provide and pay for heating devices and heat as needed to maintain specified conditions for construction operations.
- B. Maintain minimum ambient temperature of 50 degrees F (10 degrees C) in areas where construction is in progress, unless indicated otherwise in product sections.

1.6 TEMPORARY COOLING

- A. If required for the proper installation of particular materials, systems, or equipment, provide and pay for cooling devices and cooling as needed to maintain specified conditions.

1.7 TEMPORARY VENTILATION

- A. Ventilate enclosed areas to assist cure of materials, to dissipate humidity, and to prevent accumulation of dust, fumes, vapors, or gases.
- B. Utilize existing ventilation equipment if approved by Owner. Extend and supplement equipment with temporary fan units as required to maintain clear air for construction operations.

1.8 TELEPHONE SERVICE

- A. Provide, maintain and pay for telephone service to field office.

1.9 FACSIMILE SERVICE

- A. Provide, maintain and pay for separate telephone line to be used solely for fax service to field office.

1.10 TEMPORARY WATER SERVICE

- A. Utilize existing water service if approved by Owner for construction operations.
- B. Extend branch piping with outlets located so water is available by hoses with threaded connections. Provide temporary pipe insulation to prevent freezing as required.

1.11 TEMPORARY SANITARY

- A. Provide and maintain required facilities and enclosures. Existing facility use is **not** permitted. Provide at time of project mobilization.

#### 1.12 BARRIERS

- A. Provide barriers to prevent unauthorized entry to construction areas to protect existing facilities and adjacent properties from damage from construction operations and demolition. Barriers must isolate occupied use from construction activities. If and when needed, barriers must be capable of attenuating sound.
- B. Provide protection for existing plant life and landscaped. Maintain plant life and landscaped areas as necessary during construction operations. Replace damaged plant life.
- C. Protect non-owned vehicular traffic, stored materials, site, and structures from damage.
- D. Barrier plan and method subject to approval by the Architect and the Owner.

#### 1.13 FENCING

- A. Construction: Commercial grade chain link fence.
- B. Provide 6 foot high fence around construction site, equip with vehicular and pedestrian gates with locks. Fence must be capable of restricting entry by on-site facility users.

#### 1.14 WATER CONTROL

- A. Grade site to drain where additions are undertaken. Maintain excavations free of water. Provide, operate, and maintain pumping equipment and/or any other means, methods or techniques necessary to maintain excavation and site free of water.
- B. Protect site from puddling or running water. Provide water barriers as required to protect site from soil erosion.

#### 1.15 EXTERIOR ENCLOSURES

- A. Provide temporary insulated weather tight closure of exterior openings to accommodate acceptable working conditions and protect for products, to allow for temporary heating and maintenance of required ambient temperatures identified in individual specification sections, and to prevent entry of unauthorized persons. Provide access doors with self-closing hardware and locks.
- B. Provide temporary protection of existing wall cavities, substrates, and surfaces exposed to weather during cutting and minor demolition operations to prevent entrapment of moisture and development of mildew.

#### 1.16 PROTECTION OF INSTALLED WORK

- A. Protect installed Work and provide special protection to prohibit damage and where specified in individual specification sections.
- B. Provide temporary and removable protection for installed Products. Control activity in immediate work area to minimize damage.

- C. Provide protective coverings at walls, projections, jambs, sills, and soffits of openings.
- D. Protect finished floors, stairs, and other surfaces from traffic, dirt, wear, damage, or movement of heavy objects, by protecting with durable sheet materials.
- E. Prohibit traffic or storage upon waterproofed or roofed surfaces. If traffic or activity is necessary, obtain recommendations for protection from waterproofing or roofing material manufacturer.
- F. Prohibit traffic in all landscaped areas.

#### 1.17 SECURITY

- A. Provide security and facilities to protect Work and existing facilities from unauthorized entry, vandalism, or theft.
- B. Coordinate project security program with Owner's existing security operations at project mobilization.
- C. Maintain program throughout construction period until Owner acceptance precludes the need for Contractor security.
- D. Restrict entrance of persons and vehicles into Project site and existing facilities, allowing entrance only to authorized persons and persons identified by the Contract Document and/or the Architect or Owner as authorized to visit Project site.

#### 1.18 ACCESS

- A. Provide and maintain temporary roads accessing public thoroughfares to serve construction area.
- B. Extend and relocate as work progress requires. Provide detours necessary for unimpeded traffic flow.
- C. Provide and maintain access to fire hydrants, free of obstructions.
- D. Provide means of removing mud from vehicle wheels before entering streets.
- E. Existing on-site roads may be used for construction traffic.

#### 1.19 PARKING

- A. Provide temporary surface parking areas to accommodate construction personnel. Existing site areas may be used if approved in advance by the Owner.
- B. Contractor to propose plan for Owner concurrence and approval.

#### 1.20 PROGRESS CLEANING AND WASTE REMOVAL

- A. Maintain areas free of waste materials, debris, and rubbish. Maintain site in a clean and orderly condition.
- B. Remove debris and rubbish from pipe chases, plenums, attics, crawl spaces, and other closed or remote spaces, prior to enclosing the space.

- C. Broom and vacuum clean interior areas prior to start of surface finishing, and continue cleaning to eliminate dust.
- D. Collect and remove waste materials, debris, and rubbish from site weekly and dispose off-site.

1.21 PROJECT IDENTIFICATION

- A. Provide project sign. Refer to drawings for size and content.
- B. Erect on site at location established by Architect.
- C. No other signs are allowed without Owner permission except those required by law.

1.22 FIELD OFFICES AND SHEDS

- A. Office: Weather tight with lighting, electrical outlets, heating, cooling and ventilating equipment, and equipped with sturdy furniture drawing rack, and drawing display table, phone and fax.
- B. Provide space for Project meetings, with table and chairs to accommodate 6 persons.
- C. Provide storage sheds and facilities to accommodate Work. Size to storage requirements for products of individual Sections, allowing for access and orderly provision for maintenance and for inspection of products to requirements of Section 01 25 00.
- D. Designated existing covered and uncovered hard paved areas and facilities may be used for field storage areas. Protect and secure existing areas used for storage. Upon completion of Work, clean, repair, and restore all existing areas used for storage and restore to acceptable condition.

1.23 REMOVAL OF UTILITIES, FACILITIES, AND CONTROLS

- A. Remove temporary utilities, equipment, facilities, materials prior to Substantial Completion.
- B. Remove underground installation to a minimum depth of 2 feet. Grade site to drain.
- C. Clean and repair damage caused by installation or use of temporary work.
- D. Restore existing and permanent facilities used during construction to original condition. Restore permanent facilities used during construction to specified condition.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 04 05 13 — MORTAR

PART 1 - GENERAL

1.01 COORDINATION

- A. The General Conditions of the Contractor for Construction and the Supplementary Conditions to the General Conditions of the Contract for the Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addendum issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the stringent requirements and the greater quantity shall apply.

1.02 SCOPE:

- A. Perform all work required to furnish the Masonry Mortar indicated by the Contract Documents and furnish all supplementary items necessary for its proper installation.
- B. The requirements of Division 0 – “Bidding and Contract Requirements” and Division 1 – “General Requirements” of this Project Manual shall apply to all Work required for this Section.
- C. Application of Mortar used in the installation of masonry units is specified in each respective Unit Masonry Section and is not included in the work required for this Section.

1.03 SUBMITTALS:

- A. Submit product data on all mortar and admixtures.
- B. Submit certification that mortar and grout material meet ASTM standards.

1.04 PRODUCT DELIVERY AND STORAGE:

- A. Delivery: Delivery materials to Project site dry and in unbroken containers.
- B. Storage: Store materials above ground in waterproof shelters.

PART 2 - PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS:

- A. Material manufactured by any of the following manufacturers is acceptable, provided it complies with the Contract Documents.
  - 1. PORTLAND CEMENT:
    - a. Capitol Lone Star
    - b. Trinity Texas Industries
    - c. Universal Atlas Cement
  - 2. LIME:
    - a. Gibsonburg Lime Products Co., Tiger Limes
    - b. Texas Lime Company
    - c. United States Gypsum Company
    - d. National Gypsum Company
  - 3. WATER PROOFING ADMIXTURE:
    - a. Master Builders-Omicron Mortarproofing
    - b. Sonneborn Building Products-Hydracide
    - c. W.R. Grace-Hydratite Plus
  - 4. MORTAR COLOR:
    - a. Gray-
  - 5. ***DRY BLOCK-One pound per cubic foot of cementitious material, ½ sack per sack of 2 sacks of cement fluted, split –face CMU for warranty purposes***
- B. Refer to Section 01 25 00-Substitutions and Product Options for manufacturers not listed above.

2.02 MATERIALS:

- A. PORTLAND CEMENT: ASTM C150, TYPE I.
- B. HYDRATED LIME: ASTM C207, TYPE S.
- C. FINE AGGREGATE: ASTM C144,
- D. COARSE AGGREGATE: ASTM C404, Size No. 8
- E. WATER: Clean and free of deleterious acids, alkalis, or organic matter.
- F. WATERPROOFING ADMIXTURE: Omicron Mortarproofing, manufactured by Master Builders.
- G. GROUT ADMIXTURE: “Fluidifier” by Master Builders.
- H. SEALER: “DEFY” Block Water Repellant

2.03 PROPORTIONS AND MIXING:

- A. Meet requirements of ASTM C270 and proportion mortar types as specified.
- B. Meet requirements of ASTM C476 for masonry grout and proportion grout type as specified.
- C. Proportion material accurately and mix thoroughly by machine to a uniform consistency and color. Mix mortars with the maximum amount of water consistent with workability.

- D. Do not use mortar that has begun to set. Retemper mortar by adding water if mortar begins to stiffen from evaporation or absorption of a part of the mixing water. Use and place mortar in final position within 2-1/2 hours after mixing.

### PART 3 - EXECUTION

#### 3.01 INSTALLATION:

- A. See specific section of Masonry Materials for installation instructions.

#### 3.02 MORTAR SCHEDULE:

- A. Exterior Masonry Walls:
  - 1. Mortar-Type S, ASTM C270.
  - 2. Waterproofing Admixture-***dry block required to provide warranty.***
- B. Interior Masonry Partitions:
  - 1. Mortar-Type N, ASTM C270.
- C. Interior Paving Tile:
  - 1. Mortar-Type S, ASTM C270.
- D. Exterior Paving Tile:
  - 1. Mortar-Type M, ASTM C270.

#### 3.03 GROUT SCHEDULE:

- A. Paving Tile:
  - 1. Portland Cement-one part.
  - 2. Fine Aggregate-three parts.
  - 3. No lime.
  - 4. Sealer

END OF SECTION



SECTION 04 21 13 — BRICK MASONRY

PART 1 - GENERAL

1.1 COORDINATION:

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.2 SUBMITTALS:

- A. Within fifteen (15) calendar days after awarded of contract submit the following:
  - (1) Submit technical data for each type of masonry wall reinforcement, anchors, and ties.
  - (2) Submit sample panel of brick specified showing full color range.

1.3 MOCKUP:

- A. Lay up, where directed at Site, mock-up panel 4 feet high by 6 feet wide using range of brick selected for facing and mortar specified.
- B. **Do not proceed** with brickwork on Project until sample panel has been approved for color and shading. Approved panel shall be the standard of comparison for workmanship and materials. Do not destroy, alter, or move panel until brickwork is completed.

1.4 STORAGE AND HANDLING:

- A. Handle materials in manner to prevent breakage and chipping.
- B. Store materials on platforms raised free of the ground and protect materials with tarpaulin covers.

1.5 ENVIRONMENTAL CONDITIONS:

- A. Do not lay brick when the temperature of the outside air is 40°F or less, or will fall below 40°F twenty – four (24) hours after laying.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS:

- A. Material manufactured by any of the following manufacturers is acceptable provided it complies with the Contract Documents.

REINFORCEMENT, ANCHORS AND TIES:

1. AA Wire Products Company
2. Duro-O-Wall
3. Heckmann Building Products, Inc.
4. Hohmann and Barnard, Inc.
5. Masonry Reinforcing Corp. of America
6. National Wire Products Corp.

2.2 MATERIALS

A. MANUFACTURERS:

1. Hanson Brick Co.
2. Acme Brick Co.
3. Brick Selections

- B. Type: Comply with ASTM C-216, Grade SW, Type FBS for face brick. Comply with ASTM C902-87 for brick pavers. Provide letter for compliance from brick manufacturer or supplier.

- C. Shapes: Provide any special molded shapes required or as may be indicated in the drawings. All outside face brick corners other than 90° shall require shaped brick. All outside corners for soldier courses shall require shaped brick. Job cutting face brick shall not be acceptable where cut face is exposed. Provide shop drawings for Architect's approval.

- D. Solids: Provide solids as required to ensure against exposed cores or unfinished face on ends.

- E. Size: Modular (2-1/4" x 7-5/8" x 3-5/8") and as indicated in the drawings.

F. Selections:

- Brick Color, Field Color: Color to be selected by Architect.
- Brick Color, Accent #1: Color to be selected by Architect.
- Capstone, Accent #2: Color to be selected by Architect.

- G. CORRUGATED METAL TIES: DUR-O-WAL D/A 518 adjustable wall tie. Tie number as recommended by manufacturer for overall wall thickness. Zinc coating – 1.50 oz. per sq. ft. of uncoated wire surface – ASTM A 153- Class B-2.

- H. CLEANING AGENT: TRISODIUM PHOSPHATE (Calgon) and household detergent.

- I. WATER: Clean and free of deleterious acids, alkalies, or organic materials.

- J. CONTROL JOINT SPACER: As specified in the CONCRETE MASONRY Section.
- K. PREMOLDED NEOPRENE PAD: ASTM D – 1056SCE-43, R431-N by RUBTEX CORP., 3” wide, by 1/8” thick.
- L. WEEPS: Plastics tubes, 3/8” diameter, AA Wire Products Co., AA223.
- M. REINFORCING BARS: ASTM A615, Grade 60 deformation per ASTM A305, number 4 bars unless otherwise indicated.
- N. MORTAR SCREENS: #84 Weep thru mortar deflector by Heckmann Building Products.

### PART 3 - EXECUTION

#### 3.1 PREPARATION:

- A. Provide, install and maintain all scaffolding, staging and forms of protection necessary for execution of the work; substantially constructed, maintained, moved and dismantled as required to properly follow the sequence of the operation.
- B. Provide and install all shores and centering for the work, constructed true to required shape, size and form, well braced and made rigid in all parts, and capable of supporting and sustaining the loads to which subjected.
- C. Leave all shores and centering in place until the masonry has sufficiently set to safety carry its own weight and added loads of construction. Shore all free standing wall until protected from damage by windstorm.
- D. Examine surfaces to receive masonry and report and discrepancies before commencing work. Accept no former measurements, but layout work according to the plans and figures thereon.

#### 3.2 ADJUSTING

- A. All brick having absorption rates determined in accordance with ASTM C67 in excess of 0.025 oz. per sq. ft. inch per minute shall be wetted sufficiently so that the rate of absorption when laid does not exceed this amount. Wetting shall be such as to insure that each unit is nearly saturated, surface dry when laid.
- B. Lay brick plumb, level and true to a line running bond or as indicated. Lay 3 courses to 8” vertically with uniform horizontal and vertical joints. Glaze brick and tile shall be laid stack bond
- C. Lay brick in full bed or mortar with head and edge joints completely filled. Spread mortar for bed joint only so far ahead of laying units that the mortar will be plastic when units are laid. Butter end of brick for head joints with ample mortar so the vertical joint is completely filled with mortar when brick is shoved into place.
- D. Rock closures into place with both head joints and closure space spread with ample mortar. Shove against the two adjacent bricks in place so that both horizontal and vertical joints are completely filled. Do not disturb previously laid brick.
- E. Avoid over-plumbing and pounding of the corner and jambs to fit stretcher units after setting in place. Where adjustments must be made after initial setting, remove mortar and replace with fresh mortar.

- F. Fill the vertical, longitudinal joint between brick and backup as the course is laid, by pouring or slushing the vertical joint full of grout with the same mortar as used for setting.
- G. Keep cavity clear by laving a board,  $\frac{3}{4}$ " x cavity width, across a level of the ties to catch the droppings. As masonry reaches the next level for placing ties, raise the board, clean it and lay it on ties at that level.
- H. Finish joints that will remain exposed with a tool slightly larger than the width of the joint, to form a concave surface. Tool joints after the mortar has taken its initial set and in such manner as to squeeze the mortar back into the joint. Tool vertical joint first.
- I. Fill all nail or line-pipe holes with fresh mortar immediately upon removal. Provide weeps in head joints 16" on centers, in first horizontal course above flashings. Keep cavity side of weep free from mortar or accumulated materials.
- J. Cut brick, where necessary for fitting or bonding, with a power saw to insure straight, evenly cut edges.
- K. Cover tops of walls at end of day's work and when rain or is imminent with waterproof membrane. Overhang two feet on each side of wall and anchor securely. Protect masonry from weather or construction damage.
- L. Stop off longitudinal run of masonry where absolutely necessary by racking one-half length in each course. Remove loose mortar before new work is started and slightly wet old work.

### 3.2 ANCHORING:

- A. Space anchors not over 16" on center vertically and 24" on centers horizontally. Locate anchors at vertical and horizontal supports, and at other locations as indicated.
- B. Bond brick to backup with adjustable wall ties and joint reinforcing installed in adjacent masonry.
- C. Maintain a minimum of  $\frac{1}{2}$ " clearance between masonry and concrete or steel structure. Keep space free of mortar and other rigid material.

### 3.3 CONTROL AND EXPANSION JOINTS:

- A. Locate expansion joints where they occur in backup material. Locate control joints no to exceed 25'-0" on center. Keep vertical joints straight, true and continuous from top to bottom of masonry. Verify locations of control joints with Architect.
- B. Keep joints clean of mortar as work progresses. Build-in control joint spacer at control joints and flashings at expansion joints

### 3.4 EMBEDDED ITEMS:

- A. Build in flashings, sleeves, anchors, clips and accessories as work progresses. Install loose lintels, as indicated, in full beds of mortar.

### 3.5 CLEANING:

- A. Keep face of brickwork free excess mortar while laying brick. Brush with dry fiber brush prior to wet cleaning.

- B. Clean brickwork that will remain exposed promptly with fiber brushes, clean water and cleaning agent. Use or wire brushes, commercial cleaner or acid permitted only with specific approval.
- C. Repair and repoint defective work. Replace broken, damaged or discolored brick.
- D. Pressure wash all exterior brick work prior to Owner occupying new building.

END OF SECTION

SECTION 05 41 00 — LIGHT GAGE METAL FRAMING SYSTEMS AND GYPSUM SHEATHING

PART 1 - GENERAL

1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.2 WORK INCLUDED

- A. Furnish and install exterior metal stud framing as shown on the drawings and specified herein.
- B. Furnish and install water resistant gypsum board sheathing at exterior face of exterior metal studs.

1.3 RELATED WORK SPECIFIED ELSEWHERE

- A. Masonry.
- B. Interior drywall systems.
- C. Wall Insulation.
- D. Dampproofing and Waterproofing.
- E. Exterior plaster (stucco).

1.4 SUBMITTALS

- A. Submit manufacturer's product data describing all materials.
- B. Submit manufacturer's certification of structural properties, only for products to be used in the project.

1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.

1.6 DELIVERY, STORAGE AND HANDLING

- A. All materials shall be delivered in manufacturer's original packaging and stored flat in a covered, dry area providing protection from damage and exposure to the elements.
- B. Damaged or deteriorated materials shall be removed from the premises.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. STUDS AND FRAMING: Unimast, Dietrich Metals, Maverick Steel Co., Dale Industries, Delta Metals, Bostwick, American Studco Inc.
- B. GYPSUM BOARD SHEATHING: United States Gypsum Co., National Gypsum Co., Domtar Gypsum, Inc. Georgia Pacific, Temple Inland.

2.2 MATERIALS

- A. STRUCTURAL STUDS AND RUNNERS: Galvanized "Cee" studs in sizes and gauges as indicated in the drawings. Unless otherwise indicated in the drawings, minimum gauge shall be 16 gauge and the following structural properties shall apply:

| SIZE   | ABOUT MAJOR AXIS X-X |       |       | ABOUT MINOR AXIS Y-Y |      |      |
|--------|----------------------|-------|-------|----------------------|------|------|
|        | lx                   | Sx    | rx    | ly                   | Sy   | ry   |
| 3-5/8" | .906                 | .500  | 1.430 | .139                 | .142 | .614 |
| 4"     | 1.145                | .572  | 1.566 | .147                 | .143 | .615 |
| 6"     | 3.016                | 1.005 | 2.262 | .180                 | .149 | .595 |
| 8"     | 6.071                | 1.518 | 2.923 | .201                 | .152 | .565 |

- B. SHEATHING FASTENERS: Unimast self-drilling screw fasteners (bugle head).
- C. SHEATHING: Fire resistant gypsum board with treated water resistant gypsum core surfaced with water repellant paper both faces -1/2" x 4' x 8' with tongue and groove joint design at long edges. Meet requirements of ASTM C-79. Provide 5/8" thick rated X core where specifically indicated on the drawings.
- D. All metal studs, track, and bridging shall be formed from ASTM A-446 commercial grade steel having a minimum yield of 33,000 psi for 18 gauge and lighter members and 50,000 psi for 16 gauge and heavier members.
- E. All framing components shall be galvanized. Tracks, runners, bridging and bracing shall match grade and gauge of studs.

PART 3 - EXECUTION

3.1 GENERAL

- A. Install studs plumb and in plane, without twist. System installation shall be in accordance with AISI Design Manual for "Light Gauge Cold Formed Steel".
- B. All framing components shall be cut tight against abutting members. Members shall be held firmly in position until properly fastened.
- C. All attachments of axial loaded framing components shall be welded in accordance with the American Welding Society's "Recommended Practices for Resistance Welding" and shall transfer the imposed load into the adjoining member. Use no splices in axial loaded members.
- D. Attachments of framing components not subject to axial loads may be welded or screw fastened.
- E. Members shall be braced as required to resist all wind loads and construction loading for which the system has been designed. System shall be braced as erected and shall not be left overnight without adequate bracing.
- F. Framing components used to frame openings shall be of a size and type to transfer any load imposed on the opening into the members adjacent to the opening. Additional framing shall be provided adjacent to the opening to carry the load imposed.
- G. Welds in galvanized material shall be coated with "ZRC" cold galvanizing after wire brushing.

3.2 ERECTION

- A. TRACK FASTENING: Secure metal floor track to concrete floor slab with Type "A" or "B" fasteners spaced as scheduled in the table below. For determining unbraced wall height, ceiling does not qualify as bracing.
  - 1. Type "A" fastener – minimum 5/32" diameter x 1-1/4" long powder actuated fasteners. Hilti #DS32P10 or Ramset #2335.
  - 2. Type "B" fastener – minimum 1/4" diameter x 2" long drilled sleeve anchor. Hilti sleeve anchor or Ramset "Thunder Nail".
  - 3. Demonstrate to the Architect that fasteners can be driven full length into concrete slab tight to stud track.
  - 4. Use similar fasteners (and spacing) suitable for steel at overhead track or weld track to overhead steel at 12" o.c.
  - 5. At track splices use anchored channel inserts or fully weld.

Spacing Schedule for Type A & B Fasteners

| MAX. SPACING OF FASTENERS | *MAX. UNBRACED WALL HEIGHT |          |
|---------------------------|----------------------------|----------|
|                           | TYPE A                     | TYPE B   |
| 24"                       | 7.4 FT.                    | 8.3 FT.  |
| 16"                       | 11.1 FT.                   | 12.4 FT. |
| 12"                       | 14.8 FT.                   | 16.5 FT. |
| 8"                        | 24.9 FT.                   | 24.9 FT. |
| 6"                        | 29.7 FT.                   | 33.2 FT. |

\*NOTE: Ceiling at wall does not reduce unbraced wall height.



- B. STUD FASTENING: Each stud shall be fastened to top and bottom track (prior to gypsum board sheathing or interior wall finish) using one of the following two methods:
  - 1. Screw fastening: One self-drilling screw at the front and back faces of the top and bottom tracks for each stud (4 fasteners per stud.)
  - 2. Welding: One weld at the front face of the top and bottom tracks for each stud (2 welds per stud).
  - 3. Additional: The above minimum fasteners are required regardless of any additional bracing or intermediate fastening which may be indicated in the drawings or required.
- C. BRIDGING: Provide bridging at all exterior stud walls whether or not indicated in the drawings. Unless more stringent requirements are indicated in the drawings provide the following:
  - 1. Wind loading resistance only: Provide multiple bridging rows spaced 5'-0" o.c. vertically maximum.
  - 2. Axial loaded members: For stud lengths less than 10 feet, provide 2 rows of bridging at third points. For stud lengths 10 feet and greater, provide multiple bridging rows spaced 42" o.c. vertically maximum.
- D. SHEATHING INSTALLATION: Apply sheathing panels horizontally with the "v" edge turned up. Install with joints and penetrations tight and neatly fit. Stagger end joints over studs with screws spaced at maximum 12" centers at each stud and at 12" o.c. along top and bottom runners.

END OF SECTION

## SECTION 06 10 00 - ROUGH CARPENTRY

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 WORK INCLUDED

- A. Provide and install all rough carpentry, formwork, wood framing, blocking, wood furring, hardboard and related fasteners as indicated in the drawings or as required to complete the indicated construction.
- B. Install all related hardware and fasteners. Provide and install wood furring and/or trim for acoustical panels.

#### 1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Cast-in place concrete
- B. Painting
- C. Finish hardware

#### 1.4 SECTION REQUIREMENTS

- A. Submittals manufacturer's printed literature describing wood preservatives treatment system and certifying that system meets all current requirements for applicable Federal, State and local governing agencies.
- B. Submittals manufacturer's printed literature describing fire retardant treatment system, any structural or usage limitations, and certifying that system meets all current requirements for applicable Federal, State and local governing agencies.

#### 1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.

1.6 DELIVERY AND STORAGE

- A. Deliver and store lumber, plywood and hardwood on sills and cover for protection.

1.7 QUALITY ASSURANCE

- A. All lumber and plywood shall be grade marked by Southern Pine Inspection Bureau, West Coast Lumber Inspection Bureau, American Plywood Association, or Western Wood Products Association.
- B. All lumber and plywood shall be marked with producing manufacturer's trademark.
- C. Certificate of inspection issued by grading association for bundled lumber and plywood may substitute for individual piece marking.

PART 2 - PRODUCTS

2.1 LUMBER, GENERAL

- A. Dressed lumber, S4S, [19] [15] percent maximum moisture content for 2-inch (38-mm) thickness or less, marked with grade stamp of inspection agency.

2.2 TREATED MATERIALS

- A. Preservative-Treated Materials: AWPAC2 lumber and AWPAC9 plywood, labeled by an inspection agency approved by ALSC's Board of Review. After treatment, kiln-dry lumber and plywood to 19 and 15 percent moisture content, respectively. Treat indicated items and the following:
  - 1. Wood members in connection with roofing, flashing, vapor barriers, and waterproofing.
  - 2. Concealed members in contact with masonry or concrete.
  - 3. Wood framing members less than 18 inches (460 mm) above grade.
  - 4. Wood floor plates installed over concrete slabs directly in contact with earth.
- B. Fire-Retardant-Treated Materials: AWPAC20 lumber and AWPAC27 plywood, interior Type A treatment, labeled by a testing and inspecting agency acceptable to authorities having jurisdiction.
  - 1. Use treated lumber and plywood with bending strength, stiffness, and fastener-holding capacities that are not reduced below values published by manufacturer of chemical formulation under elevated temperature and humidity conditions.

2.3 LUMBER

- A. Miscellaneous Lumber: No. 3 or Standard grade of any species for nailers, blocking, and similar members as indicated on drawings.

2.4 MISCELLANEOUS PRODUCTS

- A. Fasteners: Size and type indicated. Where rough carpentry is exposed to weather, in ground contact, or in area of high relative humidity, provide fasteners with a hot-dip zinc coating per ASTM A 153 or of Type 304 stainless steel.
  - 1. Power-Driven Fasteners: CABO NER-272.

2. Bolts: Steel bolts complying with ASTM A 307, Grade A (ASTM F 568, Property Class 4.6); with ASTM A 563 (ASTM A 563M) hex nuts and, where indicated, flat washers.
- B. Metal Framing Anchors: Hot-dip galvanized steel of structural capacity, type, and size indicated.

### PART 3 - EXECUTION

#### 3.1 INSTALLATION

- A. COORDINATION: Coordinate work with other trades and provide cutting and patching required to accommodate the work. Verify all dimensions by taking field measurements to ensure proper fit. Accurately cut framing and blocking, and fit true to line and level, avoiding shims and wedges.
- B. Fit rough carpentry to other construction; scribe and cope for accurate fit. Correlate location of furring, blocking, and similar supports to allow attachment of other construction.
- C. ANCHORING AND FASTENTING: Use largest practicable fasteners for each type of work. Bolt nailers and blocking to steel, masonry or concrete members using bolts of proportionate strength to members attached. Unless otherwise noted in the drawings use 3/4" diameter bolts at maximum 4'-0" centers. Use concealed fasteners in finish work, set nails and use flathead countersunk screws.
- D. WOOD BLOCKING: Install fire-retardant tread wood blocking between metal studs where wall-supported drinking fountains, casework, railings, and other equipment is attached. Install between studs for toilet partitions systems and toilet accessories where anchored to wall. Use minimum 2 x 4 dimension where not indicated otherwise in the drawings.

END OF SECTION

## SECTION 06 40 23 - INTERIOR ARCHITECTURAL WOODWORK

### PART 1- GENERAL

#### 1.01 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

#### 1.02 SUMMARY

- A. This Section includes the following:
  - 1. Interior standing and running trim and rails.
  - 2. Wood cabinets (casework).
  - 3. Laminate clad cabinets (plastic-covered casework).
  - 4. Cabinet tops (countertops) and plastic-covered chair rails.
  - 5. Flush wood paneling.
  - 6. Interior door frames (jambs).
- B. Related Sections: The following sections contain requirements that relate to this section:
  - 1. Division 6 Section 'Rough Carpentry' for furring, blocking, and other carpentry work that is not exposed to view.
  - 2. Division 6 Section 'Finish Carpentry' for carpentry exposed to view that is not specified in this section.
  - 3. Division 6 Section 'Exterior Architectural Woodwork' for exterior woodwork.
  - 4. Division 8 Section "Flush Wood Doors" for doors specified by reference to architectural woodwork standards.
  - 5. Division 9 Section "Painting" for final finishing of installed painted finish architectural woodwork.

#### 1.03 SUBMITTALS

- A. General: Submit the following in accordance with Conditions of Contract and Division 1 Specification Sections.
- B. Product data for each type of product and process specified in this section and incorporated into items of architectural woodwork during fabrication, finishing, and installation.
- C. Fire-retardant treatment data for material impregnated by pressure process to reduce combustibility. Include certification by treating plant that treated materials comply with requirements.
- D. Shop drawings showing location of each item, dimensioned plans and elevations, large-scale details, attachment devices, and other components.
- E. Samples for initial selection purposes of the following in form of manufacturer's color charts consisting of actual units or sections of units showing full range of colors, textures, and patterns available for each type of material indicated.
  - 1. Plastic laminate.
- F. Samples for verification purposes of the following:
  - 1. Lumber with or for transparent finish, 50 square inches, for each species and cut, finished on one side and one edge.

2. Veneer leaves representative of and selected from flitches to be used for transparent finished woodwork.
  3. Wood veneer faced panel products, with or for transparent finish, 8-1/2 inches by 11 inches, for each species and cut with one half of exposed surface finished, with separate samples of unfaced panel product used for core.
  4. Lumber and panel products with factory-applied opaque finish, 8- 1/2 inches by 11 inches for panels and 50 square inches for lumber, for each finish system and color, with one half of exposed surface finished.
  5. Laminate clad panel products, 8-1/2 inches, by 11 inches for each type, color, pattern, and surface finish, with separate samples of unfaced panel product used for core.
  6. Corner pieces as follows:
    - a. Cabinet front frame joints between stiles and rail as well as exposed end pieces, 18 inches high by 18 inches wide by 6 inches deep.
    - b. Miter joints for standing trim.
  7. Exposed cabinet hardware, one unit of each type and finish.
- G. Product certificates signed by woodwork manufacturer certifying that products comply with specified requirements.
- H. Qualification data for firms and persons specified in "Quality Assurance" article to demonstrate their capabilities and experience. Include list of completed projects with project names, addresses, names of Architects and Owners, and other information specified.

#### 1.04 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Firm experienced in successfully producing architectural woodwork similar to that indicated for this Project, with sufficient production capacity to produce required units without causing delay in the Work.
- B. Single-Source Responsibility: Arrange for production by a single firm of architectural woodwork with sequence matched wood veneers.
1. Include the veneering of wood doors in the single-firm production, where veneer matching extends across wood doors.
- C. Single-Source Manufacturing and Installation Responsibility: Engage a qualified Manufacturer to assume undivided responsibility for woodwork specified in this section, including fabrication, finishing, and installation.
- D. Installer Qualifications: Arrange for installation of architectural woodwork by a firm that can demonstrate successful experience in installing architectural woodwork items similar in type and quality to those required for this project.
- E. AWI Quality Standard Comply with applicable requirements of "Architectural Woodwork Quality Standards" published by the Architectural Woodwork Institute (AWI) except as otherwise indicated.
- F. Hardware Coordination Distribute copies of approved schedule for cabinet hardware specified in Division 8 Section "Door Hardware" to manufacturer of architectural woodwork; coordinate cabinet shop drawings and fabrication with hardware requirements.

#### 1.05 DELIVERY, STORAGE, AND HANDLING

- A. Protect woodwork during transit, delivery, storage, and handling to prevent damage, soilage, and deterioration.
- B. Do not deliver woodwork until painting, wet work, grinding, and similar operations that could damage, soil, or deteriorate woodwork have been completed in installation areas. If woodwork must be stored in other than installation areas, store only in areas whose environmental conditions meet requirements specified in 'Project Conditions.'

#### 1.06 PROJECT CONDITIONS

- A. Environments<sup>1</sup> Conditions: Obtain and comply with Woodwork Manufacturer's and Installer's coordinated advice for optimum temperature and humidity conditions for woodwork during its storage and installation. Do not install woodwork until these conditions have been attained and stabilized so that woodwork is within plus or minus 1.0 percent of optimum moisture content from date of installation through remainder of construction period.
- B. Field Measurements: Where woodwork is indicated to be fitted to other construction, check actual dimensions of other construction by accurate field measurements before manufacturing woodwork; show recorded measurements on final shop drawings. Coordinate manufacturing schedule with construction progress to avoid delay of Work.
  - 1. Where field measurements cannot be made without delaying the Work, guarantee dimensions and proceed with manufacture of woodwork without field measurements. Coordinate other construction to ensure that actual dimensions correspond to guaranteed dimensions.

### PART 2- PRODUCTS

#### 2.01 HIGH PRESSURE DECORATIVE LAMINATE MANUFACTURERS

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering high pressure decorative laminates which may be incorporated in the work include:
  - 1. Formica Corp.
  - 2. Nevamar Corp.
  - 3. WilsonArt

#### 2.02 MATERIALS

- A. General: Provide materials that comply with requirements of the AWI woodworking standard for each type of woodwork and quality grade indicated and, where the following products are part of woodwork, with requirements of the referenced product standards, that apply to product characteristics indicated:
  - 1. Hardboard ANSI/AHA A135.4
  - 2. High Pressure Laminate: NEMA LD 3.
  - 3. Medium Density Fiberboard: ANSI A208.2.
  - 4. Particleboard ANSI A208.1
  - 5. Softwood Plywood PS 1.
- B. Formaldehyde Emission Levels: Comply with formaldehyde emission requirements of each voluntary standard referenced below:
  - 1. Particleboard: NPA 8.
  - 2. Medium Density Fiberboard: NPA 9.
  - 3. Hardwood Plywood: HPMA FE.

- C. Fire-Retardant Particleboard: Where indicated, provide panels complying with the following requirements that have fire-retardant chemicals bonded to softwood particles at time of panel manufacture to achieve products identical to those tested for flame spread of 20 or less and for smoke developed of 25 or less per ASTM B 84 by UL or other testing and inspecting organization acceptable to authorities having jurisdiction. Identify products with appropriate markings of applicable testing and inspecting organization.
  - 1. For 45-lb-density panels and thicknesses of 3/4 inch and less, comply with ANSI A208.1 for Grade 1-M-1 except that minimums for modulus of elasticity and screw-holding capacity on face and edge shall be 300,000 psi, 250 lb, and 225 lb, respectively.
  - 2. For 44-lb-density panels and thicknesses of 13/16 inch to 1-1/4 inch, comply with ANSI A208.1 for Grade 1-M-1 except that minimums for modulus of rupture, modulus of elasticity, internal bond, linear expansion, and screw-holding capacity on face and edge shall be 1300 psi, 250,000 psi, 60 psi, 0.50 percent, 250 lb, and 175 lb, respectively.
  - 3. Product: Subject to compliance with requirements, provide "Duraflake FR" by Duraflake Div.; Willamette Industries, Inc.

## 2.03 FABRICATION, GENERAL

- A. Wood Moisture Content: Comply with requirements of referenced quality standard for moisture content of lumber in relation to relative humidity conditions existing during time of fabrication and in installation areas.
- B. Fabricate woodwork to dimensions, profiles, and details indicated. Ease edges to radius indicated for the following:
  - 1. Corners of cabinets and edges of solid wood (lumber) members less than 1 inch in nominal thickness: 1/16 inch.
  - 2. Edges of rails and similar members more than 1 inch in nominal thickness: 1/8 inch.
- C. Complete fabrication, including assembly, finishing, and hardware application, before shipment to project site to minimum extent possible. Disassemble components only as necessary for shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting.
- D. Factory-cut openings, to maximum extent possible, to receive hardware, appliances, plumbing fixtures, electrical work, and similar items. Locate openings accurately and use templates or roughing-in diagrams to produce accurately sized and shaped openings. Smooth edges of cutouts and, where located in countertops and similar exposures, seal edges of cutouts with a water-resistant coating.

## 2.04 FIRE-RETARDANT-TREATED LUMBER

- A. General: Where indicated, pressure impregnate lumber with fire-retardant chemicals of formulation indicated to produce materials with fire performance characteristics specified.
- B. Fire-Retardant Chemicals: Use chemical formulations specified that do not bleed through or otherwise adversely affect finishes. Do not use colorants in solution to distinguish treated lumber from untreated lumber.
  - 1. Organic Resin-Based Formulation: Exterior type per AWPA C20 consisting of organic-resin solution, relatively insoluble in water, thermally set in wood by kiln drying.
  - 2. Low-Hygroscopic Formulation: Interior Type A per AWPA C20.
- C. Fire Performance Characteristics: Provide materials identical to those tested for the following fire performance characteristics per ASTM test methods indicated by UL or other testing and inspecting organizations acceptable to authorities having jurisdiction. Identify treated lumber with classification



marking of inspecting and testing organization in the form of separable paper label or, where required by authorities having jurisdiction, of imprint on lumber surfaces that will be concealed from view after installation.

1. Surface Burning Characteristics: Not exceeding values indicated below, tested per ASTM E 84 for 30 minutes with no evidence of significant combustion.
  - a. Flame Spread: 25.
  - b. Smoke Developed: 50.
- D. Mill lumber after treatment, within limits set for wood removal that does not affect listed fire performance characteristics, using a woodworking plant certified by testing and inspecting organization.
- E. Mill lumber before treatment and implement special procedures during treatment and drying processes that are needed to prevent lumber from warping and developing discolorations from drying sticks or other causes, marring, and other defects affecting appearance of treated woodwork.
- F. Kiln-dry woodwork after treatment to levels required for untreated woodwork. Maintain moisture content required by kiln drying before and after treatment.
- G. Discard treated lumber that does not comply with requirements of referenced woodworking standard. Do not use twisted, warped, bowed, discolored, or otherwise damaged or defective lumber.
- H. Products: Subject to compliance with requirements, provide one of the following:
  1. Low-Hygroscopic Formulation (Type A):
    - a. "Flameproof LHC"; Osmose Wood Preserving, Inc.
    - b. "Dricon"; Hickson Corporation.

## 2.05 STANDING AND RUNNING TRIM AND RAILS FOR TRANSPARENT FINISH

- A. Quality Standard: Comply with AWI Section 300.
- B. Backout or groove backs of flat trim members and kerf backs of other wide flat members, except for members with ends exposed in finished work.
- C. Assemble casings in plant except where limitations of access to place of installation require field assembly.
- D. Grade: Premium.
- E. Lumber Species: Birdseye Maple, half round; and American Black Cherry, rift sawn.

## 2.06 STANDING AND RUNNING TRIM AND RAILS FOR OPAQUE FINISH

- A. Quality Standard: Comply with AWI Section 300.
- B. Grade: Premium.
- C. Backout or groove backs of flat trim members and kerf backs of other wide flat members, except for members with ends exposed in finished work.
- D. Assemble casing in plant except where limitations of access to place of installation require field assembly.

- E. Lumber Species: Any dosed-grain hardwood listed in referenced woodworking standard.

## 2.07 WOOD CABINETS (CASEWORK) FOR TRANSPARENT FINISH

- A. Quality Standard: Comply with AWI Section 400 and its Division 400A Wood Cabinets.”
- B. Grade: Premium.
- C. AWI Type of Cabinet Construction: Flush overlay.
- D. Wood Species for Exposed Surfaces: American Black Cherry, rift sawn.
  - 1. Grain Matching: Run and match grain vertically for drawer fronts, doors, and fixed panels.
  - 2. Matching of Veneer Leaves: Slip match.
  - 3. Veneer Matching Within Panel Face: Balance match.
- E. Wood Species for Semiexposed Surfaces: Match species and cut indicated for exposed surfaces.

## 2.08 LAMINATE CLAD CABINETS (PLASTLC.COVERED CASEWORK)

- A. Quality Standard. Comply with AWI Section 400 and its Division 400B ‘Laminate Clad Cabinets.’
- B. Grade: Premium.
- C. AWI Type of Cabinet Construction: Flush overlay, unless otherwise indicated.
- D. Laminate Cladding High pressure decorative laminate complying with the following requirements:
  - 1. Colors, Patterns, and Finishes: Provide materials and products that result in colors and textures of exposed laminate surfaces complying with the following requirements:
    - a. Provide selections made by Architect from laminate manufacturer’s full range of standard colors and finishes in the following categories:
      - (1) Solid colors.
      - (2) Patterns.
  - 2. Laminate Grade for Exposed Surfaces: Provide laminate cladding complying with the following requirements for type of surface and grade.
    - a. Horizontal Surfaces Other Than Tops: GP-50 (0.050-inch nominal thickness).
    - b. Vertical Surfaces: GP-50 (0.050-inch nominal thickness).
    - c. Edges: GP-50 (0.050-inch nominal thickness).
  - 3. Semiexposed Surfaces: Provide surface materials indicated below:
    - a. High pressure laminate, GP-28.

## 2.09 CABINET HARDWARE AND ACCESSORY MATERIAL.8

- A. General: Provide cabinet hardware and accessory materials associated with architectural cabinets, except for items specified in Division 8 Section ‘Door Hardware.’
- B. Cabinet Hardware and Miscellaneous Item Schedule:

1. Adjustable Shelf Standard.
  - a. Manufacturer/Model No.: Knape & Vogt/No. 255.
  - b. Size/Type: 5/8" wide x 3/16" deep, recessed.
  - c. Finish: Bright zinc plate.
  - d. Remarks: 1/2" vertical adjustment.
2. Adjustable Shelf Support:
  - a. Manufacturer/Model No.: Knape & Vogt/No. 256.
  - b. Finish: Bright zinc plate.
  - c. Remarks: For use with No. 255 standard.
3. Slotted Shelf Standard:
  - a. Manufacturer/Model No.: Knape & Vogt/No. 51.
  - b. Size/Type: 3/4" x 3/8" x length shown, surface mount, heavy duty.
  - c. Finish: Bright nickel plate.
  - d. Remarks: 1-5/8" vertical adjustment.
4. Adjustable Shelf Bracket:
  - a. Manufacturer/Model No.: Knape & Vogt/No. 52.
  - b. Size/Type: 1 1/2" wide x shelf depth.
  - c. Finish: Bright nickel plate.
  - d. Remarks: For use with No. 51 standard.
5. Drawer Slide: (Typical)
  - a. Manufacturer/Model No.: Grant/No. 329.
  - b. Size/Type: Full extension, length to suit drawer.
  - c. Finish: Zinc plate.
  - d. Remarks: 100 lb. rating.
6. Drawer Slide: (To 4-1/2" drawer depth)
  - a. Manufacturer/Model No.: Grant/No. 328.
  - b. Size/Type: Full extension, length to suit drawer.
  - c. Finish: Zinc plate.
  - d. Remarks: 50 lb. rating.
7. Knobs (at Suites Level):
  - a. Forms + Surfaces Model No. HC430 Series, sizes as selected by Architect from manufacturers standards.
  - b. Finish: Anodized black matte.
8. Wire Pulls:
  - a. Manufacturer/Model No.: Stanley/No. 4483112
  - b. Size/Type: 3-1/2" center wire pulls.
  - c. Finish: US 28D.
9. Concealed Hinges:

- a. Manufacturer/Model No.: Stanley/No. 1510.

10. Continuous Hinges:

- a. Manufacturer/Model No.: Stanley/No. STS311-1/4.
- b. Size/Type: 1-1/2" wide x height of door.
- c. Finish: US 32.
- d. Remarks: Provide matching countersunk screws, 2" o.c., both sides.

11. Door Catch (Magnetic type)

- a. Manufacturer/Model No.: Stanley/No. SP4L
- b. Size/Type: 2" x 1-1/4" case size.
- c. Finish: Aluminum.
- d. Remarks: One per leaf to 48", two per leaf to 84".

12. Cabinet Lock

- a. Manufacturer/Model No.: National/No. C-8053.
- b. Size/Type: Disc tumbler cam lock.
- c. Finish: US 26D or US 32D.
- d. Remarks: Furnish two keys per lock; keyed to Building Standard.

13. Sliding Glass Door Locks: K&V 965NP, keyed to building system.

14. Track, Upper Guide & Sheaves: Stylmark Model No. 810005 Assembly, 204-Ri clear anodized finish.

- C. Hardware Standard Comply with ANSI/BEMA A156.9 "American National Standard for Cabinet Hardware" for items indicated by reference to BHMA numbers or referenced to this standard.

- D. Exposed Hardware Finishes: For exposed hardware, provide finish that complies with ANSI/BHMA A156.18 for BHMA code number indicated.

- 1. Dark Oxidized Satin Bronze, Oil Rubbed, on Bronze Base: BHMA 613 and matching Architect's sample.
- 2. Satin Chromium Plated, Brass or Bronze Base: BHMA 626.
- 3. Satin Chromium Plated, Steel Base: BHMA 652.
- 4. Satin Stainless Steel, Stainless Steel Base: BHMA 630.

- E. For concealed hardware provide manufacturer's standard finish that complies with product class requirements of ANSI/BHMA A156.9.

- F. Clear Tempered Float Glass for Shelves: ASTM C 1048, Condition A, style I, type I, quality q3, class 1, seamed at edges before tempering, 1/4-inch thick unless otherwise indicated.

2.10 ARCHITECTURAL CABINET TOPS (COUNTERTOPS) AND CHAIR RAILS:

- A. Quality Standard: Comply with AWI Section 400 and its Division 400C.

- B. Type of Top and Chair Rail: High pressure decorative laminate complying with the following:

- 1. Grade: Custom.
- 2. Laminate Cladding for Horizontal Surface: High pressure decorative laminate as follows:

- a. Colors, Patterns, and Finishes: Provide materials and products that result in colors and textures of exposed laminate surfaces complying with the following requirements:
  - (1) Provide selections made by Architect from manufacturer's full range of standard colors and finishes in the following categories:
    - (a) Solid colors.
    - (b) Patterns.
- b. Grade: GP-50 (0.050-inch nominal thickness).
- c. Edge Treatments:
  - (1) Plastic Laminate Edge Treatment: Same as laminate cladding on horizontal surfaces.
  - (2) Wood Edge Treatment: Lumber edge for transparent finish, with matching wood species and cut to be determined.

#### 2.11 FLUSH WOOD PANELING FOR TRANSPARENT FINISH

- A. Quality Standard: Comply with AWI Section 500 and its Division 500A.
- B. Grade: Premium.
- C. Veneer Species: Birdseye Maple 1 half round; and American Black Cherry, rift sawn.
- D. Matching of Adjacent Veneer Leaves: Slip match.
- E. Veneer Matching Within Panel Face: Book match.
- F. Fire Performance Characteristics: Provide paneling composed of panels of wood veneer density and fire-retardant particleboard that are identical in construction to units tested for the following surface burning characteristics per ASTM E 84 by UL or other testing and inspecting organization acceptable to authorities having jurisdiction. Identify panels with appropriate markings of applicable testing and inspecting organization on surfaces that will be concealed from view after installation.
  - 1. Flame Spread: 75 or less.
  - 2. Smoke Developed: 40 or less.

#### 2.12 INTERIOR DOOR FRAMES FOR TRANSPARENT FINISH

- A. Quality Standard: Comply with AWI Section 900B.
- B. Grade: Premium.
- C. Lumber Species: American Black Cherry, rift sawn.

#### 2.13 CLOSET AND UTILITY SHELVING:

- A. Quality Standard: Comply with Section 600.
- B. Shelving for Painted Finish (By Section 09900): Comply with the following requirements:
  - 1. Grade: Economy.
  - 2. Shelving Material: Birch faced veneer core plywood.
  - 3. Lumber: Ponderosa Pine or Poplar.

#### 2.14 FASTENERS AND ANCHORS

- A. Screws: Select material, type, size, and finish required for each use. Comply with FS FF-S-111 for applicable requirements.
  - 1. For metal framing supports, provide screws as recommended by metal framing manufacturer.
- B. Nails: Select material, type, size, and finish required for each use. Comply with FS FF-N-105 for applicable requirements.
- C. Anchors: Select material, type, size, and finish required by each substrate for secure anchorage. Provide nonferrous metal or hot-dip galvanized anchors and inserts on inside face of exterior walls and elsewhere as required for corrosion resistance. Provide toothed steel or lead expansion bolt devices for drilled-in-place anchors. Furnish inserts and anchors, as required, to be set into concrete or masonry work for subsequent woodwork anchorage.

#### 2.15 FACTORY FINISHING OF INTERIOR ARCHITECTURAL WOODWORK

- A. Quality Standard: Comply with AWI Section 1500 unless otherwise indicated.
- B. General: The entire finish of interior architectural woodwork is specified in this section, regardless of whether factory applied or applied after installation.
  - 1. Factory Finishing: To the greatest extent possible, finish architectural woodwork at factory. Defer only final touch-up, cleaning, and polishing until after installation. Painted finish by Section 09900 except prime coat.
- C. Preparations for Finishing: Comply with referenced quality standard for sanding, filling countersunk fasteners, sealing concealed surfaces and similar preparations for finishing of architectural woodwork, as applicable to each unit of work.
- D. Transparent Finish for Closed-Grain Woods: Comply with requirements indicated below for grade, finish system, staining, effect, and sheen.
  - 1. Grade: Premium.
  - 2. AWI Finish System #5: Catalyzed polyurethane.
  - 3. Staining for Cherry Only: Match approved sample for color.
  - 4. Effect: Open grain (not filled).
  - 5. Sheen: Dull satin 15-20 deg.
- E. Opaque Finish: Comply with requirements indicated below for grade, finish system, color, effect, and sheen:
  - 1. Grade: Premium.
  - 2. AWI Finish System #11: Catalyzed polyurethane.
  - 3. Color: Match Architect's sample.
  - 4. Sheen: Medium-gloss rubbed effect 35-45 deg.

#### 2.16 MISCELLANEOUS ACCESSORIES

- a. Steel Countertop Support Bracket: provide prefinished steel bracket supports at locations as shown on drawings. Brackets shall be by A&M Hardware (888) 647-0200 [info@aandmhardware.com](mailto:info@aandmhardware.com) Other equal products may be provided if and as specifically approved by Architect by substitution request during bidding period.

## PART 3- EXECUTION

### 3.01 PREPARATION

- A. Condition woodwork to average prevailing humidity conditions in installation areas before installing.
- B. Deliver concrete inserts and similar anchoring devices to be built into substrates well in advance of time substrates are to be built.
- C. Before installing architectural woodwork, examine shop-fabricated work for completion and complete work as required, including back priming and removal of packing.

### 3.02 INSTALLATION

- D. Quality Standard. Install woodwork to comply with AWI Section 1700 for same grade specified in Part 2 of this section for type of woodwork involved.
- E. Install woodwork plumb, level, true, and straight with no distortions. Shim as required with concealed shims. Install to a tolerance of 1/8 inch in 8'-0" for plumb and level (including tops) and with no variations in flushness of adjoining surfaces.
- F. Scribe and cut woodwork to fit adjoining work and refinish cut surfaces or repair damaged finish at cuts.
- G. Fire-Retardant-Treated Wood. Handle, store, and install fire- retardant-treated wood to comply with recommendations of chemical treatment manufacturer including those for adhesives where are used to install woodwork.
- H. Anchor woodwork to anchors or blocking built in or directly attached to substrates. Secure to grounds, stripping and blocking with countersunk, concealed fasteners and blind nailing as required for a complete installation. Except where prefinished matching fastener heads are required, use fine finishing nails for exposed nailing, countersunk and filled flush with woodwork and matching final finish where transparent finish is indicated.
- I. Standing and Running Trim and Rails: Install with minimum number of joints possible, using full-length pieces (from maximum length of lumber available) to the greatest extent possible. Stagger joints in adjacent and related members. Cope at returns and miter at corners.
- J. Cabinets: Install without distortion so that doors and drawers fit openings properly and are accurately aligned. Adjust hardware to center doors and drawers in openings and to provide unencumbered operation. Complete the installation of hardware and accessory items as indicated. Maintain veneer sequence matching (if any) of cabinets with transparent finish
- K. Tops: Anchor securely to base units and other support systems as indicated.
- L. Paneling: Anchor paneling to supporting substrate with concealed panel-hanger clips and by blind nailing on backup strips, splined-connection strips, and similar associated trim and framing. Do not face nail unless otherwise indicated.
- M. Complete the finishing work specified in this section to whatever extent not completed at shop or before installation of woodwork.
- N. Refer to the Division 9 sections for finishing of painted architectural woodwork.

### 3.03 ADJUSTMENT AND CLEANING

- A. Repair damaged and defective woodwork where possible to eliminate defects functionally and visually; where not possible to repair, replace woodwork. Adjust joinery for uniform appearance.
- B. Clean, lubricate, and adjust hardware.
- C. Clean woodwork on exposed and semiexposed surfaces. Touch up factory-applied finishes to restore damaged or soiled areas.

3.04 PROTECTION

- A. Provide final protection and maintain conditions, in a manner acceptable to manufacturer and Installer, that ensures that woodwork is being without damage or deterioration at time of Substantial Completion.

END OF SECTION 06402



## SECTION 06 61 16 - SOLID SURFACING

### PART 1 – GENERAL

#### 1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.1 SUMMARY

- A. Section Includes:
  - 1. Gibraltar Solid Surface Material.
  - 2. Earthstone Solid Surface Material.
  - 3. Solid Surface Material Shaped Goods (Wilsonart Sinks).
- B. Related Sections:
  - 1. Finish Carpentry: Section 06100.
  - 2. Architectural Woodwork: Section 06402.
  - 3. Sealants: Section 07920.
  - 4. Door Thresholds at Ceramic Tile: Section 09300.
  - 5. Plumbing: Division 15.

#### 1.2 SYSTEM DESCRIPTION

- A. Gibraltar Solid Surface Sheet: Homogenous sheet material composed of acrylic resins, fire-retardant filler materials, and coloring agents.
- B. Earthstone Solid Surface Sheet: Homogenous sheet material composed of acrylic resins, fire-retardant filler materials, and coloring agents.
- C. Solid Surface Shaped Goods (Wilsonart Sinks): Cast items of homogenous material composed of polyester and acrylic resins, fire-retardant filler materials, and coloring agents.

#### 1.3 SUBMITTALS

- A. Comply with Section 01 33 00, unless otherwise indicated.
- B. Product Data:
  - 1. Detailed specification of construction and fabrication.
  - 2. Manufacturer's installation instructions.
  - 3. Manufacturer's detailed recommendations for handling, storage, installation, protection, and maintenance.

- C. Shop Drawings: Installation details including location and layout of each type of fabrication and accessory.
- D. Samples: Full range of standard colors and patterns.
- E. Contract Closeout Submittals: Comply with Contract Documents.

#### 1.4 QUALITY ASSURANCE

- A. Fabricator Qualifications: Wilsonart certified solid surface fabricator/installer.
- B. Installer Qualifications: Firm experienced in installation or application of systems similar in complexity to those required for this Project, including specific requirements indicated.
  - 1. Acceptable to or licensed by manufacturer.
- C. Source Limitations: Obtain materials and products from single source.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver fabrications appropriately wrapped in protective materials.
- B. Protect fabrications from damage.

#### 1.6 PROJECT CONDITIONS

- A. Maintain relative humidity planned for building occupants and an ambient temperature between 65 and 75°F for 48 hours prior to and during installation. After installation, maintain relative humidity and ambient temperature planned for building occupants.

#### 1.7 WARRANTY

- A. Furnish manufacturer's limited 10 year warranty.

### PART 2 - PRODUCTS

#### 2.1 MANUFACTURER

- A. Wilsonart International, (800) 433-3222, [www.wilsonart.com](http://www.wilsonart.com).
  - 1. Gibraltar Solid Surface, Type 051.
  - 2. Earthstone Solid Surface, Type 051.
- B. Substitutions: Permitted, in accordance with Section 01 25 00 Substitution Procedures.

#### 2.2 GIBRALTAR SOLID SURFACE SHEET

- A. Nominal sheet thickness: 0.50 inch (13 mm)
- B. Surface burning characteristics in accordance with ASTM E 84: Class I or A, and as follows:
  - 1. Flame spread: < 25.
  - 2. Smoke developed: < 25.
- C. Liquid Absorption, ISO 4586-2, for 1/2 inch material thickness: 0.4 percent after 2 hour period.
- D. Izod Impact, ASTM D 256, Method A: 0.3 foot pounds per inch.

- E. Tensile Modulus, ASTM D 638 Nominal: 1.2 million pounds per square inch.
- F. Thermal Expansion, ASTM D 696: 0.000018 inch per inch per degree F, maximum.
- G. Hardness, ASTM D 2583, Barcol Impressor: 57.
- H. Flexural Toughness, ASTM D 790: 3 (in.-lb./in<sup>3</sup>).
- I. Deflection Temperature under load, ASTM D 648: 90 degrees C.
- J. Stain Resistance, ANSI Z-124.3 Modified; 3.4: No effect.
- K. Boiling Water Resistance, NEMA LD 3-3.05: No effect.
- L. High Temperature Resistance, NEMA LD 3-3.06: No effect.
- M. Radiant Heat Resistance, NEMA LD 3-3.10: No effect.
- N. Light Resistance, NEMA LD 3-3.03: No effect.
- O. Ball Impact Resistance, NEMA LD 3-3.08, one half pound ball, unsupported: 125 inches.
- P. Specific Gravity (Density ASTM D792): 1.60 grams per cubic centimeter.
- Q. Approximate weight: 4.20 pounds per square foot.
- R. Weatherability, ASTM D 2565: Pass.
- S. Fungus Resistance, ASTM G 21: Pass.
- T. Bacterial Resistance, ASTM G 22: Pass.
- U. Pittsburgh Protocol Toxicity: 66.9 grams.
- V. Patterns and Finishes: Selected from manufacturer's full range of available selections (*standard*) by Architect.

## 2.3 EARTHSTONE SOLID SURFACE SHEET

- A. Nominal sheet thickness: 0.50 inch (13 mm).
- B. Surface burning characteristics in accordance with ASTM E 84: Class II or B, and as follows:
  - 1. Flame spread: < 26.
  - 2. Smoke developed: < 35.
- C. Liquid Absorption, ISO 4586-2, for 1/2 inch material thickness: 0.4 percent after 2 hour period.
- D. Izod Impact, ASTM D 256, Method A: 0.3 foot pounds per inch.
- E. Tensile Modulus, ASTM D 638 Nominal: 1.1 million pounds per square inch.
- F. Thermal Expansion, ASTM D 696: 0.00002 inch per inch per degree F, maximum.
- G. Hardness, ASTM D 2583, Barcol Impressor: 57.
- H. Flexural Toughness, ASTM D 790: 5 (in.-lb./in<sup>3</sup>).
- I. Deflection Temperature under load, ASTM D 648: 90 degrees C.
- J. Stain Resistance, ANSI Z-124.3 Modified; 3.4: No effect.
- K. Boiling Water Resistance, NEMA LD 3-3.05: No effect.
- L. High Temperature Resistance, NEMA LD 3-3.06: No effect.
- M. Radiant Heat Resistance, NEMA LD 3-3.10: No effect.
- N. Light Resistance, NEMA LD 3-3.03: No effect.
- O. Ball Impact Resistance, NEMA LD 3-3.08, one half pound ball, unsupported: 125 inches.
- P. Specific Gravity (Density ASTM D792): 1.56 grams per cubic centimeter.
- Q. Approximate weight: 4.10 pounds per square foot.
- R. Fungus Resistance, ASTM G 21: Pass.
- S. Bacterial Resistance, ASTM G 22: Pass.
- T. Pittsburgh Protocol Toxicity: 65.4 grams.
- U. Patterns and Finish: Selected from manufacturer's full range of available selections (*standard*) by Architect.

## 2.4 ACCESSORY MATERIALS

- A. Joint adhesive: Manufacturer's standard adhesive to create inconspicuous, nonporous joints, with a chemical bond (WA8215).
- B. Sealant: Standard mildew resistant, FDA/UL recognized silicone sealant in color matched or clear

formulations.

- C. Sink/bowl mounting hardware: Manufacturer's approved bowl clips, brass inserts and fasteners for attachment of undermount sinks/bowls.

## 2.5 FABRICATION

- A. Fabrication to be performed by a Wilsonart certified solid surface fabricator/installer.
- B. Fabricate components in shop to greatest extent practical to size and shape indicated, in accordance with approved shop drawing and Wilsonart published requirements.
- C. Wilsonart Solid Surface Fabrication Manual (SS0319)
- D. Form joints between components using manufacture's standard joint adhesive. Joints shall be inconspicuous in appearance and without voids. Attach 4" (100mm) wide Gibraltar/Earthstone reinforcing strip under joints required by Deck Seam Section of the Wilsonart Solid Surface Fabrication Manual (SS0319).
- E. Provide holes and cutouts for plumbing and bath accessories as indicated on shop drawings.
- F. Rout and finish component edges to a smooth, uniform finish. Rout all cutouts then sand all edges smooth. Repair or reject defective or inaccurate work.
- G. Finish: Surfaces shall have a uniform finish.
  - 1. Matte: Standard finish for high traffic areas, requires the least amount of maintenance.
  - 2. Satin: Standard finish for darker Gibraltar and Earthstone patterns, requires minimal maintenance.
  - 3. Semi-gloss: Higher sheen with greater reflectance, suggested for lower traffic areas, requires increased maintenance
  - 4. Gloss: Maximum sheen and reflectance, recommended for light traffic areas or vertical applications.
- H. Thermoforming (optional): Comply with forming data from manufacturer.
  - 1. Construct matching molds to form components shape.
  - 2. Form pieces to shape prior to seaming and joining.
  - 3. Cut pieces larger than finished dimensions, sand edges, remove all nicks and scratches.
  - 4. Heat entire component uniformly between 280°–325°F during forming.
  - 5. Prevent blistering, whitening or cracking of Gibraltar/Earthstone during forming.

## PART 3 – EXECUTION

### 3.1 EXAMINATION

- A. Examine substrates to receive solid surfacing. Identify conditions detrimental to proper or timely installation. Do not commence installation until conditions have been corrected.

### 3.2 PREPARATION

- A. Precondition Wilsonart Solid Surfacing in accordance with manufacturer's printed installation instructions.

### 3.3 INSTALLATION

- A. Install components plumb and level, in accordance with approved shop drawings, project installation details and manufacturer's printed instructions.
- B. Form joints using manufacturer's approved adhesive, with joints inconspicuous in finished work.
- C. Adhere undermount sinks/bowls to countertop using manufacturer's recommended joint adhesive.

- D. Adhere topmount sinks/bowls to countertop using manufacturer's recommended adhesive/silicone sealant.
- E. Provide backsplashes and endsplashes as indicated on the drawings. Adhere to countertops using manufacturer's recommended silicone sealant.
- F. Remove excessive adhesive and sealants. Components shall be clean on Date of Substantial Completion.
- G. Coordinate plumbing installation with Division 15.

#### 3.4 INSTALLATION OF WINDOW STOOLS

- A. Install window stools full length of window, set securely into place using only concealed fasteners and manufacturer's approved adhesive.
- B. Window stools shall be plumb, true and level.
- C. Provide minimum 1/8" expansion gaps on both sides of window stools, sealed with Manufacturer's approved sealant.
- D. Ease edges and sand smooth.

#### 3.5 INSTALLATION OF VANITIES

- A. Install plumb, level, true and straight. Shim as necessary using concealed shims.
- B. Attach top securely to base unit or support brackets in accordance with manufacturer's printed instructions.
- C. Seal between wall and component with manufacturer's recommended silicone sealant.
- D. Attach backsplashes and endsplashes to countertops using manufacturer's recommended silicone sealant.

#### 3.6 PROTECTION

- A. Protect surfaces from damage until Date of Substantial Completion. Repair or replace damaged components that cannot be repaired to architect's satisfaction.
- B. Fabricator/Installer to provide the Wilsonart® Care and Maintenance kit, review maintenance procedures and the Wilsonart warranty with the head of maintenance upon completion of project.

END OF SECTION

SECTION 07 10 00 — DAMPPROOFING AND WATERPROOFING

PART 1 - GENERAL

1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.2 WORK INCLUDED

- A. Provide and install below-grade waterproofing.
- B. Provide and apply dampproofing on weather side of inside wythe of all exterior masonry cavity walls.
- C. Provide and apply dampproofing and joint taping on weather side of gypsum board sheathing.
- D. Provide and install membrane waterproofing (flashing) at exterior walls as indicated in the drawings and specified herein.

1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Unit masonry.
- B. Gypsum sheathing.
- C. Flashing at roof.
- D. Plastic membrane under slab-on-grade.
- E. Waterstops.
- F. Metal thru-wall flashing.

1.4 SUBMITTALS

- A. Submit manufacturer's printed literature describing each material, restrictions, and manufacturer's recommended procedures. Submit samples of each material.

- B. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

## 1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the Work under this section for a period of one year after the date of Substantial Completion of the Project.

## 1.6 QUALITY ASSURANCE

- A. Waterproofing company shall have a minimum of 3 years experience in the dampproofing and waterproofing of building structures of similar size and scope as this project.
- B. Retain at the job site a properly calibrated gauge for use by the Architect to verify applied thickness of materials.

## PART 2 - PRODUCTS

### 2.1 WALL MATERIALS

- A. MEMBRANE FLASHING: 40 mil thick polyethylene backed SBS modified bitumen self-adhering black membrane; "Protecto Flash" as manufactured by Protecto Wrap Co. or "Perm-A-Barrier" as manufactured by W.R. Grace and Co. Membrane shall comply with the following:
  - 1. Tensile Strength: ASTM D412; 46 psi.
  - 2. Elongation: ASTM D412; 300% min.
  - 3. Water Vapor Permeability: ASTM E96; 0.1 perms max.
- B. DAMPPROOFING: Non-asbestos emulsion type coating No. 352 over No. 207 adhesive primer, as manufactured by Gulf States Asphalt or approved equivalent by Monsey, Karnak, W.R. Meadows, Celotex, or Sonneborn. Comply with ASTM D1227, Type 1.
- C. SHEATHING TAPE: 4" wide glass fabric scrim complying with ASTM D1668 or 40 mil thick polyethylene backed SBS modified bitumen self-adhering tape as manufactured by Protecto Wrap Co. or equivalent by W.R. Grace and Co. Verify compatibility of tape with proposed dampproofing.

### 2.2 BELOW GRADE WATERPROOFING:

- A. WALLS: "Hydrocide Liquid Membrane 5000T", one part cold applied elastomeric, modified urethane. Trowel applied, non-sag, as manufactured by Sonneborn or approved equivalent by Toch Bros. or Tremco.
- B. SLABS: "Hydrocide Liquid Membrane, HLM 5000" Cold Applied Seamless Elastomeric, Modified Urethane for use between concrete seal slab and concrete slab-on-grade as manufactured by Sonneborn or approved equivalent by Toch Bros. or Tremco.
- C. PROTECTION BOARD: Water-resistant, semi-rigid panel composed of a core of asphalt and inorganic mineral filler particles, bottom reinforcing cover of asphalt-saturated felt and top cover of fiber glass mat weather-coated with a bond-breaking film, as manufactured by W.R. Meadows, Inc.

- D. WATERSTOPS: Reference concrete section.

## 2.3 SHOWER PANS:

- A. MEMBRANE SHOWER PAN: 30 mil thick synthetic, heavy-duty, flexible membrane PVC sheet, Nervastral 300.

## PART 3 - EXECUTION

### 3.1 INSPECTION

- A. Contractor shall inspect exterior face of all masonry cavity walls to ensure that all penetrations and joints are completely filled prior to dampproofing operations beginning.

### 3.2 MEMBRANE FLASHING

- A. Prime concrete and masonry surfaces scheduled to receive membrane flashing using flashing manufacturer's recommended primer to ensure good adhesion.
- B. WALL FLASHINGS: Shall be installed above all openings occurring in an exterior wall, at base of exterior wall, and at wall interruptions by columns, beams, slabs, spandrels and other locations as indicated in the drawings. Flashing shall extend to within 1" of outside face of wall, shall be continuous and shall extend through cavity and be turned up to the top first course above finish floor on face of inner wythe, and to extend 1" minimum into back up or inner wythe. End laps to be 9" and side laps 6".
- C. STEEL STRUCTURE: Cover all steel columns or beams in exterior walls not protected by dampproofed concrete block or sheathing. Cover steel completely with membrane flashing lap 6" on to masonry on each side of columns. Conform and adhere to steel shapes not fireproofed. Cover all protruding angles or miscellaneous steel.
- D. FRAMES: Install at exterior window and door frames and other locations as indicated in the drawings.
- E. SHEATHING: Wrap all corners of gypsum board sheathing. See drawings for other details.

### 3.3 SHEATHING TAPE: Use one of the following systems:

- A. Imbed and cover glass fabric scrim tape in dampproofing mastic at all joints, cracks and penetrations at gypsum board sheathing.
- B. Apply specified self-adhering tape continuously over all joints, cracks and penetrations prior to beginning dampproofing operations.

### 3.4 DAMPPROOFING

- A. Spray or brush apply dampproofing coating to weather side of all gypsum sheathing and primed concrete block back-up at exterior masonry cavity walls in accordance with the following:
  - 1. Primer: Minimum ½ gallon material per 100 sq. ft. of wall surface.



2. Coating: Minimum 1/32" dry film thickness and minimum 5 gallons material per 100 sq. ft.
- B. Cover all corners and work thoroughly into all joints, cracks, or crevices. Finished coating shall be monolithic and free of pin holes or cracks. Seal cracks, voids and joints at dissimilar materials with glass fabric embedded in dampproofing coating.
- C. Seal around penetrations including all masonry anchors.
- D. Dampproofing shall be applied only when temperature is at 40 degrees F. and rising or above, and when no rain is forecast for the 24 hour period following application. No dampproofing shall be covered by masonry prior to observation by the Architect. All dampproofing shall dry for a minimum of 24 hours prior to being covered by finish masonry.

### 3.5 BELOW GRADE WATERPROOFING

- A. LIQUID MEMBRANE:
  1. Install liquid membrane systems at earth side of all below grade walls, between sub-slab ("mud-slab") and structural slab, and all outside surfaces of elevator pit. Allow concrete work to cure a minimum of 14 days. All surfaces shall be smooth, dry, sound and free of honeycombs. Concrete shall be free of curing and parting compounds, wax or other foreign materials.
  2. Static joints or cracks less than 1/8" wide shall be sealed with "HLM" as manufactured by waterproofing manufacturer. Material shall fill and over-lap the edges of the joint to a width of 4" on both sides and shall have a minimum surface thickness of 55 (+5) mils.
  3. Immediately prior to application of membrane, remove all dust and dirt by use of high-pressure air, by brushing with a soft broom or vacuum cleaning.
  4. Apply material at a rate of 4 gallons per 100 square feet of surface to produce a membrane of 55 (+5) mil thick. Carefully control application to avoid runs and sags of fresh material.
  5. Apply membrane to prestripped areas at cracks, joints, intersections, penetrations, etc., to provide a minimum total thickness of 110 mils over these areas. Mask any membrane edge exposed to view to provide a straight clean edge.
  6. Before the membrane attains a final set, verify the applied thickness by use of a mil-thickness gauge. Where readings indicate a thickness less than specified, immediately apply additional membrane to produce required thickness.
  7. Following the application of the membrane, place protection boards over the membrane waterproofing at walls receiving backfill. Use membrane material as required to adhere protection boards. Boards shall be firmly in place with joints closely butted and sealed with gusset tape before backfilling is started.
  8. Protect membrane during construction. Any punctures or cuts in the membrane shall be patched and sealed in the manner described above for sealing joints in the sheeting.

### 3.6 SHOWER PANS

1. Ensure that surfaces receiving shower pan are clean, thoroughly dry and free from rough surfaces and sharp projections.
2. One-ply of 30 mil sheet shall be applied over concrete surface by embedding it in a coat of Nerva-Plast mastic trowel-applied at a rate of 40 sq. ft. per gallon. Turn up perimeter a minimum of 4".
3. Seal joints with 3" and final 2" wide strips of Nervastral tape in accordance with manufacturer's recommendations. Preform all corners and make without joints.
4. Roll entire horizontal area with 50 to 100 lb. Roller. Set corners and turn-ups with rubber roller.

END OF SECTION 07100

## SECTION 07 21 00 - BUILDING INSULATION

### PART 1 - GENERAL

#### 1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.1 SUMMARY

- A. This Section includes the following:
  - 1. Polystyrene foam insulation
  - 2. Open cell spray foam insulation
  - 3. Chicken Wire
  - 4. Fiberglass roll or batt insulation
  - 5. Polyencapsulated Batt Insulation
  - 6. Fiberboard ceiling insulation underlayment
- B. Related Sections include the following:
  - 1. Section 09 21 16.

#### 1.2 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Samples for Verification: Full-size units for each type of exposed insulation indicated.
- C. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, for insulation products.
- D. Research/Evaluation Reports: For foam-plastic insulation.

#### 1.3 QUALITY ASSURANCE

- A. Source Limitations: Obtain each type of building insulation through one source.
- B. Fire-Test-Response Characteristics: Provide insulation and related materials with the fire-test-response characteristics indicated, as determined by testing identical products per test method indicated below by UL or another testing and inspecting agency acceptable to authorities having jurisdiction. Identify materials with appropriate markings of applicable testing and inspecting agency.
  - 1. Surface-Burning Characteristics: ASTM E 84.
  - 2. Fire-Resistance Ratings: ASTM E 119.
  - 3. Combustion Characteristics: ASTM E 136.

#### 1.4 DELIVERY, STORAGE AND HANDLING

- A. Protect insulation materials from physical damage and from deterioration by moisture, soiling, and other sources. Store inside and in a dry location. Comply with manufacturer's written instructions for handling, storing, and protecting during installation.
- B. Protect plastic insulation as follows:
  - 1. Do not expose to sunlight, except to extent necessary for period of installation and concealment.
  - 2. Protect against ignition at all times. Do not deliver plastic insulating materials to Project site before installation time.
  - 3. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

### PART 2 - PRODUCTS

#### 2.1 MANUFACTURERS

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
- B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
  - 1. Extruded-Polystyrene Board Insulation:
    - a. DiversiFoam Products.
    - b. Dow Chemical Company.
    - c. Owens Corning.
    - d. Tenneco Building Products.

#### 2.2 INSULATING MATERIALS

- A. General: Provide insulating materials that comply with requirements and with referenced standards.

1. Preformed Units: Sizes to fit applications indicated; selected from manufacturer's standard thicknesses, widths, and lengths.
- B. Extruded-Polystyrene Board Insulation: ASTM C 578, of type and density indicated below, with maximum flame-spread and smoke-developed indices of 75 and 450, respectively:
  1. Type X, 1.30 lb/cu. ft.
- C. Open Cell Spray Foam Insulation:
  1. Icynene LD-R-50
  2. Demilec Sealection 500
  3. Application: Exterior Walls and other locations as indicated on plans.
- D. Polyencapsulated Batt Insulation
  1. Johns Manville
  2. Owens Corning
  3. Certainteed

Encapsulated, Glass-Fiber Blanket Insulation: ASTM C 665, Type II (non-reflective faced), Class A (faced surface with a flame-spread index of 25 or less); Category 1 (membrane is a vapor barrier).

  1. Roof/Ceiling Cavity: R-19
  2. Exterior Walls: R-19
- E. Batt or Roll Insulation:
  1. Johns Manville
  2. Owens Corning
  3. Certainteed

General: Insulation shall be fine fiber, flexible, resilient glass fiber blanket. Moisture absorption shall be less than .2% by volume.

1. Interior Stud Walls: 3 5/8" x 16" wide x 96" sound attenuation batts "R" factor 11. Unfaced.
2. Interior Stud Walls: 6" x 16" wide x 96" sound attenuation batts "R" factor 19. Unfaced
3. Above Acoustical Ceilings: 6" x 24" wide x 96" thermal batt insulation kraft faced fiberglass. "R" factor 19

#### AUXILIARY INSULATING MATERIALS

- A. Adhesive for Bonding Insulation: Product with demonstrated capability to bond insulation securely to substrates indicated without damaging insulation and substrates.
- B. Chicken Wire: Provide as support for encapsulated batt insulation attached to the underside of metal building roof z girts.
- C. Fiberboard ceiling insulation underlayment: Provide over scheduled ceilings as substrate to apply sprayed foam insulation. Provide Celotex or equivalent product.

### PART 3 - EXECUTION

#### 3.1 EXAMINATION

- A. Examine substrates and conditions, with Installer present, for compliance with requirements for Sections in which substrates and related work are specified and other conditions affecting performance.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
- C. Review and insure chemical compatibility of cavity wall dampproofing membrane and cavity rigid insulation board prior to installation.

#### 3.2 PREPARATION

- A. Clean substrates of substances harmful to insulations or vapor retarders, including removing projections capable of puncturing vapor retarders or of interfering with insulation attachment.
- B. Close off openings in cavities receiving poured-in-place insulation to prevent escape of insulation. Provide bronze or stainless steel screens (inside) where openings must be maintained for drainage or ventilation.

#### 3.3 INSTALLATION, GENERAL

- A. Comply with insulation manufacturers written instructions applicable to products and application indicated.
- B. Install insulation that is undamaged, dry, and unsoiled and that has not been left exposed at any time to ice and snow.
- C. Extend insulation in thickness indicated to envelop entire area to be insulated. Cut and fit tightly around obstructions and fill voids with insulation. Remove projections that interfere with placement.
- D. Water-Piping Coordination: If water piping is located on inside of insulated exterior walls, coordinate location of piping to ensure that it is placed on warm side of insulation and insulation encapsulates piping.
- E. Apply single layer of insulation to produce thickness indicated, unless multiple layers are required to make up total thickness.

#### 3.4 INSTALLATION OF FOAM INSULATION

- A. Per manufacturer's instructions. Installation by approved applicator only.

#### 3.5 INSTALLATION OF CAVITY WALL INSULATION

- A. On units of plastic insulation, install small pads of adhesive spaced approximately 24 inches o.c. both ways on inside face, as recommended by manufacturer. Fit courses of insulation between confining obstructions in cavity, with edges butted tightly both ways. Press units firmly against sheathing.

### 3.6 INSTALLATION OF POLYENCAPSULATED BATTS

- A. Encapsulated batts at vertical wall surfaces are to be attached with self tapping screws where attached at z girts. Batts at metal stud wall shall form fit to cavity.

### 3.7 INSTALLATION OF GENERAL BUILDING INSULATION

- A. Apply insulation units to substrates by method complying with manufacturer's written instructions. If no specific method is indicated, bond units to substrate with adhesive or use mechanical anchorage to provide permanent placement and support of units.
- B. Set vapor-retarder-faced units with vapor retarder to warm side of construction. Do not obstruct ventilation spaces, except for firestopping.
  - 1. Tape joints and ruptures in vapor retarder, and seal each continuous area of insulation to surrounding construction to ensure airtight installation.
- C. Apply spray foam insulation in strict compliance with insulation manufacturers' written recommendations by manufacturer approved applicator only. Do not apply insulation until installation of pipes, ducts, conduits, wiring, and electrical outlets in walls is completed and windows, electrical boxes, and other items not indicated to receive insulation are masked. After insulation is applied, make it even with studs by using method recommended by insulation manufacturer.

END OF SECTION

SECTION 07 21 19 - FOAMED-IN-PLACE MASONRY WALL INSULATION

PART 1 - GENERAL

1.00 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.01 SUMMARY

- A. Extent of insulation work is shown on drawings and indicated by provisions of this section.
- B. Applications of insulation specified in this section include the following:
  - 1. Foamed-In-Place masonry insulation for thermal, sound and fire resistance values.

1.02 SUBMITTALS

- A. Product and technical presentation as provided by the manufacturer.
- B. Certified Test Reports: With product data, submit copies of certified test reports showing compliance with specified performance values, including R-values, fire performance and sound abatement characteristics.
- C. Material Safety Data Sheet: Submit Material Safety Data Sheet complying with OSHA Hazard Communication Standard, 29 CFR 1910.1200.

1.03 QUALITY ASSURANCE

- A. Manufacturing Standards: Provide insulation produced by a single and approved manufacturer. The product must come from the manufacturer pre-mixed to ensure consistency.
- B. Installer Qualifications for Foamed-In-Place Masonry Insulation: Engage an experienced dealer/applicator who has been trained and licensed by the product manufacturer and which has not less than three years direct experience in the installation of the product used.
- C. Warranty: Upon request, a one year product and installation warranty will be issued by both the manufacturer and installer.

- D. Fire Performance Characteristics: Provide insulation materials which are identical to those whose fire performance characteristics, as listed for each material or assembly of which insulation is a part, have been determined by testing, per methods indicated below, by a testing agency acceptable to authorities having jurisdiction.

**Product must be classified by Underwriters Laboratory ® ("UL") as to Surface Burning Characteristics**

|                                  |            |
|----------------------------------|------------|
| Fire Resistance Ratings:         | ASTM E-119 |
| Surface Burning Characteristics: | ASTM E-84  |
| Combustion Characteristics:      | ASTM E-136 |

**PART 2 - PRODUCTS**

2.01 ACCEPTABLE MANUFACTURERS

- A. Manufacturers of Foamed-In-Place Masonry Insulation: Subject to compliance with requirements, provide products from the following:
- "Core-Fill 500™"; Tailored Chemical Products, P.O. Drawer 4186, Hickory, N.C. 28603, (800) 627-1687
  - Or approved equal.

2.02 INSULATING MATERIALS

- A. General: Provide insulating materials which comply with requirements indicated for materials, compliance with referenced standards, and other characteristics.
- B. Foamed-In-Place Masonry Insulation: Two component thermal insulation produced by combining a plastic resin and catalyst foaming agent surfactant which, when properly ratioed and mixed, together with compressed air produce a cold-setting foam insulation in the hollow cores of hollow unit masonry walls.
- Fire-Resistance Ratings: Minimum four (4) hour fire resistance wall rating (ASTM E-119) for 8" and 12" concrete masonry units when used in standard two (2) hour rated CMUs.
  - Surface Burning Characteristics: Maximum flame spread, smoke developed and fuel contributed of 0, 5 and 0 respectively.
  - Combustion Characteristics: Must be noncombustible, Class A building material.
  - Thermal Values: "R" Value of 4.91/inch @ 32 degrees F mean; ASTM C-177.
  - Sound Abatement: Minimum Sound Transmission Class ("STC") rating of 53 and a minimum Outdoor Indoor Transmission Class ("OITC") rating of 44 for 8" wall assembly (ASTM E 90-90).

**PART 3 - EXECUTION**

3.01 INSPECTION AND PREPARATION

- A. Application Assemblies:
- |               |                                           |
|---------------|-------------------------------------------|
| Block Walls:  | 6", 8", 10" or 12" concrete masonry units |
| Cavity Walls: | 2" cavity or greater                      |



3.02 INSTALLATION OF FOAMED-IN-PLACE INSULATION

- A. General: Install foamed-in-place insulation from interior, or as specified, prior to installation of interior finish work and after all masonry and structural concrete work is in place; comply with manufacturer's instructions.
- B. Installation: Fill all open cells and voids in hollow concrete masonry walls where shown on drawings. The foam insulation shall be pressure injected through a series of 5/8" to 7/8" holes drilled into every vertical column of block cells (every 8" on center) beginning at an approximate height of four (4) feet from finished floor level. Repeat this procedure at an approximate height of ten (10) feet above the first horizontal row of holes (or as needed) until the void is completely filled. Patch holes with mortar and score to resemble existing surface.

END OF SECTION

SECTION 07 22 00 - ROOF AND DECK INSULATION

PART 1. GENERAL

1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.1 Scope of Work:

- A. Provide all labor, equipment, and materials to install rigid thermal insulation and Dens Deck Prime recovery board over vented base sheets, where indicated on the drawings. Install cants, edge strips where indicated on drawings.

1.2 Related Sections:

- A. Division 6 "Rough Carpentry"
- B. Division 7 "Coal Tar Modified Bitumen Roofing"

1.3 Submittals:

- A. Samples and product literature for all products listed.

1.4 Delivery Storage and Handling:

- A. Deliver materials in manufacturer's original unopened packages, dry, undamaged, seals and labels intact.
- B. Store all insulation delivered to the site in enclosed trailers.

1.5 Environmental Requirements:

- A. Apply insulation only when the weather conditions are in compliance with the roof system limitations.
- B. Protect the installed insulation from water penetration at the end of each day's work.
- C. Application of the roof system shall immediately follow the installation of the roof insulation as it

is installed.

## PART 2. PRODUCTS

### 2.1 General: (Provide as listed or approved equal)

- A. When a particular make or trade name is specified, it shall be indicative of a standard required.

### 2.2 Materials:

- A. Polyisocyanurate Insulation Board: 2-inch with fiberglass reinforced facers. Johns Manville, E'NRG'2, LTTR-value: 11.2. Size: Four feet by four feet(4'x4')
- B. Recovery board: 1/4-inch DensDeckPrime.
- C. Insulation Adhesive: Cold applied single component urethane adhesive Insul-Lock II.
- D. Roof Board Joint Tape: Six (6) inches wide glass fiber mat with adhesive compatible with insulation board facers.
- E. Cant Strips: Fiberglass, Glass Cant.
- G. As required by the membrane manufacturer.

## PART 3. EXECUTION

### 3.1 Examination:

- A. Examine substrate surfaces to receive roof and deck insulation and associated work and conditions under which insulation will be installed. Do not proceed with roofing until unsatisfactory conditions have been corrected in a manner acceptable to Installer.
- B. Verify deck and surfaces are clean, smooth, dry, free of depressions or irregularities prior to beginning installation of materials.
- C. Verify roof openings, curbs, pipes, sleeves, ducts, penetrations or vents through roof are solidly set, wood nailing strips are in place.
- D. Verify all specifications related to Carpentry, have been followed prior to beginning installation of insulation. Beginning installation means acceptance of substrate.

### 3.2 Protection:

- A. During execution of work covered by this Section, the Contractor shall provide protection for roof insulation from water and wind penetration at the end of each day's work.
- B. Protect the roof insulation in areas that will receive excessive traffic with a surface protection such as plywood.
- C. All workmen shall wear clean, soft rubber-soled shoes for any application work where they may be walking on the in-place insulation.

3.3 Installation:

- A. Ensure all surfaces are clean, dry, free of dirt, debris, oils, loose or embedded gravel and other contaminants that may inhibit adhesion.
- B. Over the 40 mil SBS base sheet, apply insulation adhesive directly to the substrate using a ribbon pattern with one half (1/2) inch to three quarter (3/4) inch wide beads, using either the pail or an automatic applicator, placed every 1 ft. on center to achieve proper coverage rates.
- C. Immediately place insulation boards into wet adhesive. Do not slide boards into place. Do not allow the adhesive to skin over before installing insulation boards.
- D. Briefly step each board into place to ensure contact with the adhesive. Substrates with irregular surfaces may prevent the insulation board from making positive contact with the adhesive. Relief cuts or temporary weights may be required to ensure proper contact.
- E. All boards shall be cut and fitted where the roof deck intersects a vertical surface. The boards shall be cut to fit a minimum of one quarter (1/4) inch away from the vertical surface.
- F. Embed recovery board in adhesive after first layer has been attached as insulation was adhered. Stagger end joints of boards so all open joints will be eliminated. Walk in each piece of insulation and leave boards completely adhered to insulation. Each insulation board shall be butt firmly against adjoining panels. All open joints shall be eliminated.
- G. Cant Strips/Tapered Edge/Crickets: Install preformed 45-degree fiberglass cant strips at junctures of vertical surface. Provide preformed, tapered edge/crickets where indicated on the drawings.

END OF SECTION

## SECTION 07 26 16 - UNDER-SLAB VAPOR BARRIER

### PART 1 – GENERAL

#### 1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.1 SUMMARY

- A. Products supplied under this section:
  - 1. Vapor barrier and installation accessories for installation under concrete slabs.
- B. Related sections:
  - 1. Section 03 30 00 Cast-in-Place Concrete
  - 2. Section 07 26 00 Vapor Retarders

#### 1.2 REFERENCES

- A. American Society for Testing and Materials (ASTM):
  - 1. ASTM E1745- 11Standard Specification for Plastic Water Vapor Retarders Used in Contact with Soil or Granular Fill Under Concrete Slabs.
  - 2. ASTM E1643- 11Selection, Design, Installation, and Inspection of Water Vapor Retarders Used in Contact with Earth or Granular Fill Under Concrete Slabs.
- B. Technical Reference - American Concrete Institute (ACI):
  - 1. ACI 302.2R-06 Guide for Concrete Slabs that Receive Moisture-Sensitive Flooring Materials.

#### 1.3 SUBMITTALS

- A. Quality control/assurance:
  - 1. Summary of test results per paragraph 9.3 of ASTM E 1745.
  - 2. Manufacturer's samples and literature.
  - 3. Manufacturer's installation instructions for placement, seaming and penetration repair instructions.
  - 4. All mandatory ASTM E1745 testing must be performed on a single production roll per ASTM E1745 Section 8.1.

## PART 2 – PRODUCTS

### 2.1 MATERIALS

- A. Vapor barrier shall have all of the following qualities:
  - 1. Maintain permeance of less than 0.01 Perms [grains/(ft<sup>2</sup> · hr · inHg)] as tested in accordance with mandatory conditioning tests per ASTM E1745 Section 7.1 (7.1.1-7.1.5).
  - 2. Other performance criteria:
    - a. Strength: ASTM E1745 Class A.
    - b. Thickness: 15 mils minimum
- B. Vapor barrier products:
  - 1. Basis of Design: Stego Wrap Vapor Barrier (15-mil) by Stego Industries LLC., (877) 464-7834 [www.stegoindustries.com](http://www.stegoindustries.com).
  - 2. Substitutions will be in accordance with Section 01 25 00.

### 2.2 ACCESSORIES

- A. Seams :
  - 1. Stego Tape by Stego Industries LLC, (877) 464-7834 [www.stegoindustries.com](http://www.stegoindustries.com).
- B. Penetrations of Vapor barrier:
  - 1. Stego Mastic by Stego Industries LLC, (877) 464-7834 [www.stegoindustries.com](http://www.stegoindustries.com).
  - 2. Stego Tape by Stego Industries LLC, (877) 464-7834 [www.stegoindustries.com](http://www.stegoindustries.com).
- C. Perimeter/edge seal:
  - 1. Stego Crete Claw by Stego Industries LLC, (887) 464-7834 [www.stegoindustries.com](http://www.stegoindustries.com).
  - 2. Stego Term Bar by Stego Industries LLC, (877) 464-7834 [www.stegoindustries.com](http://www.stegoindustries.com).
  - 3. StegoTack Tape (double sided) by Stego Industries LLC, (877) 464-7834 [www.stegoindustries.com](http://www.stegoindustries.com).

## PART 3 – EXECUTION

### 3.1 PREPARATION

- A. Ensure that subsoil is approved by Architect or Geotechnical Engineer.
  - 1. Level and compact base material.

### 3.2 INSTALLATION

- A. Install vapor barrier in accordance ASTM E1643.
  - 1. Unroll vapor barrier with the longest dimension parallel with the direction of the concrete placement and face laps away from the expected direction of the placement whenever possible.
  - 2. Extend vapor barrier over footings and grade beams to a distance acceptable to the structural engineer or stop at impediments such as dowels and waterstops.
  - 3a. Seal vapor barrier to slab perimeter/edge using Stego Crete Claw and remove dirt, debris, and mud from Crete Claw prior to concrete placement.  
OR
  - 3b. Seal vapor barrier to footing/grade beam with double sided tape, termination bar, or both.
  - 4. Overlap joints 6 inches and seal with manufacturer's tape.
  - 5. Apply tape/Crete Claw to a clean and dry vapor barrier.
  - 6. Seal all penetrations (including pipes) per manufacturer's instructions.

7. No penetration of the vapor barrier is allowed except for reinforcing steel and permanent utilities.
8. Repair damaged areas by cutting patches of vapor barrier, overlapping damaged area 6 inches and taping all sides with tape.

END OF SECTION

## SECTION 07 41 13 - INSULATED METAL ROOF PANEL

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 SECTION INCLUDES

- A. Foamed-insulation-core standing seam metal roof panels, with related metal trim and accessories.

#### 1.3 RELATED REQUIREMENTS

- A. Division 05 Section "Structural Steel Framing" for steel framing supporting metal panels.
- B. Division 05 Section "Cold-Formed Metal Framing" for cold-formed metal framing supporting metal panels.
- C. Division 07 Section "Metal Wall Panels" for factory-formed metal wall panels.
- D. Division 07 Section "Sheet Metal Flashing and Trim" for sheet metal copings, flashings, reglets, and roof drainage items in addition to items specified in this Section.
- E. Division 07 Section "[Joint Sealants](#)" for field-applied [Joint Sealants](#).
- F. Division 13 Section "Metal Building Systems" for steel framing supporting metal panels.

#### 1.4 REFERENCES

- A. American Society of Civil Engineers (ASCE): [www.asce.org/codes-standards](http://www.asce.org/codes-standards):
  - 1. ASCE 7 - Minimum Design Loads for Buildings and Other Structures.
- B. ASTM International (ASTM): [www.astm.org](http://www.astm.org):
  - 1. ASTM A 653 - Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
  - 2. ASTM A 755 - Specification for Steel Sheet, Metallic Coated by the Hot-Dip Process and Prepainted by the Coil-Coating Process for Exterior Exposed Building Products.
  - 3. ASTM A 792 - Standard Specification for Steel Sheet, 55 % Aluminum-Zinc Alloy-Coated by the Hot-Dip Process.
  - 4. ASTM C 1363 - Standard Test Method for Thermal Performance of Building Materials and Envelope Assemblies by Means of a Hot Box Apparatus



5. ASTM D 1621 – Compressive Properties of Rigid Cellular Plastics.
6. ASTM D 1622 – Apparent Density of Rigid Cellular Plastics.
7. ASTM C 518 - Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.
8. ASTM D 2244 - Test Method for Calculation of Color Differences from Instrumentally Measured Color Coordinates.
9. ASTM D 4214 - Test Methods for Evaluating Degree of Chalking of Exterior Paint Films.
10. ASTM D 6226 - Standard Test Method for Open Cell Content of Rigid Cellular Plastics
11. ASTM E 72 - Standard Test Methods of Conducting Strength Tests of Panels for Building Construction.
12. ASTM E 84 - Test Methods for Surface Burning Characteristics of Building Materials.
13. ASTM E 1592 - Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference.
14. ASTM E 1646 - Standard Test Method for Water Penetration of Exterior Metal Roof Panel Systems by Uniform Static Air Pressure Difference.
15. ASTM E 1680 - Standard Test Method for Rate of Air Leakage Through Exterior Metal Roof Panel Systems.
16. ASTM E 1980 - Practice for Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces.

C. FM Global (FM): [www.fmglobal.com](http://www.fmglobal.com):

1. ANSI/FM 4471 Approval Standard for Class 1 Panel Roofs.
2. ANSI/FM 4880 American National Standard for Evaluating Insulated Wall and Roof/Ceiling Assemblies.

D. Underwriters Laboratories, Inc. (UL): [www.ul.com](http://www.ul.com):

1. UL 580 - Tests for Uplift Resistance of Roof Assemblies

1.5 QUALITY ASSURANCE

- A. Manufacturer/Source: Provide metal roof panel assembly and accessories from a single manufacturer approved under an accredited third-party quality control program.
- B. Manufacturer Qualifications: Approved manufacturer listed in this Section with minimum five years experience in manufacture of similar products in successful use in similar applications.

1.6 ADMINISTRATIVE REQUIREMENTS

- A. Preinstallation Meeting: Prior to erection of framing, conduct preinstallation meeting at site attended by Owner, Architect, manufacturer's technical representative, inspection agency and related trade contractors.
  1. Coordinate building framing in relation to metal panel system.
  2. Coordinate openings and penetrations of metal panel system.
  3. Coordinate work of Division 07 Sections "Roof Specialties" and "Roof Accessories" and openings and penetrations and manufacturer's accessories with installation of metal panels.

1.7 ACTION SUBMITTALS

- A. Product Data: Manufacturer's data sheets for specified products.
- B. Shop Drawings: Show layouts of metal panels. Include details of each condition of installation, panel profiles, and attachment to building. Provide details at a minimum scale 1-1/2-inch per foot of edge conditions, joints, fastener and sealant placement, flashings, openings, penetrations, curbs, vents, snow

guards, lightning arresting equipment, and special details. Make distinctions between factory and field assembled work.

1. Include data indicating compliance with performance requirements.
  2. Indicate points of supporting structure that must coordinate with metal panel system installation.
  3. Include structural data indicating compliance with performance requirements and requirements of local authorities having jurisdiction.
- C. Samples for Initial Selection: For each exposed product specified including sealants. Provide representative color charts of manufacturer's full range of colors.
- D. Samples for Verification:
1. Provide ~~12-inch-~~ (305 mm-) long section of each metal panel profile.
  2. Provide color chip verifying color selection.

#### 1.8 INFORMATIONAL SUBMITTALS

- A. Product Test Reports: Indicating compliance of products with requirements.
- B. Warranty:
1. Submit manufacturer's written two (2) year limited warranty providing panels to be free from defects in materials and workmanship, beginning from the date of substantial completion excluding coil coatings (paint finishes) that are covered under a separate warranty.
  2. The installation contractor shall issue a separate one (1) year warranty against defects in installed materials and workmanship, beginning from the date of substantial completion of the installation.

#### 1.9 CLOSEOUT SUBMITTALS

- A. Maintenance data.
- B. Manufacturer's Warranty: Executed copy of manufacturer's standard warranty.

#### 1.10 DELIVERY, STORAGE, AND HANDLING

- A. Protect products of metal panel system during shipping, handling, and storage to prevent staining, denting, deterioration of components or other damage. Protect panels and trim bundles during shipping. Protect painted surfaces with a protective covering before shipping.
1. Deliver, unload, store, and erect metal panel system and accessory items without misshaping panels or exposing panels to surface damage from weather or construction operations.
  2. Store in accordance with Manufacturer's written instruction. Provide wood collars for stacking and handling in the field.
  3. Shield foam insulated metal panels from direct sunlight until installation.

#### 1.11 WARRANTY

- A. Special Manufacturer's Warranty: On manufacturer's standard form, in which manufacturer agrees to repair or replace metal panel assemblies that fail in materials and workmanship within two years from date of Substantial Completion.
- B. The installation contractor shall issue a separate one (1) year warranty against defects in installed materials and workmanship, beginning from the date of substantial completion of the installation.
- C. Manufacturer's Weathertight Warranty: On manufacturer's standard form, in which manufacturer agrees to repair or replace metal panel assemblies that fail to remain weathertight, including leaks, without monetary limitation within 20 years from date of Substantial Completion.

- D. Special Panel Finish Warranty: Submit Manufacturer's twenty-five (25) year limited warranty on the exterior paint finish for adhesion to the metal substrate and twenty-five (25) year limited warranty on the exterior paint finish for chalk and fade.
  - 1. Fluoropolymer Two- Coat System:
    - a. Color fading in excess of 5 Hunter units per ASTM D 2244.
    - b. Chalking in excess of No. 6 rating per ASTM D 4214.
    - c. Failure of adhesion, peeling, checking, or cracking.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURER

- A. Basis of Design Manufacturer: MBCI Metal Roof and Wall Systems, Division of NCI Group, Inc.; Houston TX. Tel: (877)713-6224; Email: [info@ecoficientseries.com](mailto:info@ecoficientseries.com); Web: [www.mbc.com](http://www.mbc.com).
  - 1. Provide basis of design product , or comparable product in accordance with Section 01 25 00 substitution Procedures.

### 2.2 PERFORMANCE REQUIREMENTS

- A. General: Provide metal roof panel system meeting performance requirements as determined by application of specified tests by a qualified testing facility on manufacturer's standard assemblies.
- B. Structural Performance: Provide metal panel assemblies capable of withstanding the effects of indicated loads and stresses within limits and under conditions indicated, as determined by ASTM E 72 or ASTM E 1592 applied in accordance with IES AC 04, Section 4, Panel Load Test Option or Section 5, Panel Analysis Option:
  - 1. Wind Loads: Determine loads based on uniform pressure, importance factor, exposure category, and basic wind speed indicated on drawings.
    - a. Wind Uplift Testing: Certify capacity of metal panels by testing of proposed assembly per ASTM E 72 or ASTM E 1592.
  - 2. Deflection Limits: Withstand inward and outward wind-load design pressures in accordance with applicable building code with maximum deflection of 1/240 of the span with no evidence of failure. Reference structural drawings for addt. information.
  - 3. Seismic Performance: Comply with ASCE 7, Section 9, "Earthquake Loads."
- C. Wind Uplift Resistance: Comply with UL 580 for wind-uplift class UL-90.
- D. FM Approvals Listing: Comply with FM Approvals 4471 as part of a panel roofing system, and that are listed in FM Approvals' "RoofNav" for Class 1 construction. Identify materials with FM Approvals markings.
  - 1. Fire/Windstorm Classification: Class 1A-120.
  - 2. Hail Resistance Rating: Class 1-SH.
- E. Fire Performance Characteristics: Provide metal panel systems with the following fire-test characteristics determined by indicated test standard as applied by UL or other testing and inspection agency acceptable to authorities having jurisdiction.
  - 1. Surface-Burning Characteristics: Provide metal panel systems with the following characteristics when tested per ASTM E 84. The core shall have:

- a. Flame spread index: 25 or less.
  - b. Smoke developed index: 450 or less.
2. Fire Performance of Insulated Roof: Class 1 roof panel per ANSI/FM 4880.
- F. Air Infiltration, ASTM E 1680:
  1. Maximum **0.06 cfm/sq. ft. (0.30 L/s per sq. m)** at static-air-pressure difference of **6.24 lbf/sq. ft. (300 Pa)**.
- G. Water Penetration Static Pressure, ASTM E 1646: No uncontrolled water penetration at a static pressure of **6.24 lbf/sq. ft. (300 Pa)**.
- H. Thermal Movements: Allow for thermal movements from variations in both ambient and internal temperatures. Accommodate movement of support structure caused by thermal expansion and contraction. Allow for deflection and design for thermal stresses caused by temperature differences from one side of the panel to the other.
- I. Thermal Performance: When tested in accordance with ASTM C 518, "measurement of steady state thermal transmission," the panels shall provide a K-factor of **0.14 btu/sf/hr/deg F at a 75° F (24 C)** mean temperature.

## 2.3 INSULATED METAL ROOF PANELS

- A. Mechanically Seamed, Concealed Fastener, Foamed-Insulation-Core Metal Roof Panels: Structural metal roof panel consisting of ribbed exterior metal sheet and interior metal sheet, with factory foamed-in-place polyurethane core in thermally-separated profile, with tongue-and-groove panel edges, mechanically seamed, attached to supports using concealed clips and fasteners.
  1. Basis of Design: MBCI, Insulated BattenLok.
  2. ASTM A 792/A 792M, structural quality, Grade 50, Coating Class AZ50 (Grade 340, Coating Class AZM150), prepainted by the coil-coating process per ASTM A 755/A 755M.
    - a. Exterior Face Sheet: 22 gauge with stucco embossed surface and planked pan profile.
      - 1) Finish: Fluoropolymer two-coat system.
      - 2) Color: As selected by Architect from manufacturer's standard colors.
    - b. Interior Face Sheet: 24 gauge with stucco embossed surface and planked profile.
      - 1) Finish: Exposed Galvalume Plus coating.
      - 2) Color: As selected by Architect from manufacturer's standard colors.
  3. Panel Width: 36 inches (914 mm).
  4. Panel Thickness: 4 inch (102 mm).
  5. Insulating Core: Polyurethane with zero ozone depletion potential blowing agent
    - a. Closed Cell Content: 95% or more as determined by ASTM D 6226
    - b. Compressive Strength: As required to meet structural performance requirements and with a minimum of 15 psi as determined by ASTM D 1621
    - c. Minimum Density: 2.0 pcf (32 kg/m3) as determined by ASTM D 1622
    - d. Thermal Resistance (R-Value): 8 per inch of panel thickness. Minimum required R-Value of 30 as determined by ASTM C 518 at 75 degrees Fahrenheit mean temperature.

## 2.4 METAL ROOF PANEL ACCESSORIES

- A. General: Provide complete metal roof panel assembly incorporating trim, copings, fasciae, gutters and downspouts, and miscellaneous flashings. Provide required fasteners, closure strips, and sealants as indicated in manufacturer's written instructions.
- B. Flashing and Trim: Match material, thickness, and finish of metal panel face sheet.
- C. Panel Clips: ASTM C 645, with ASTM A 653/A 653M, **G90 (Z180)** hot-dip galvanized zinc coating, two-piece, configured for concealment in panel joints, and identical to clips utilized in tests demonstrating compliance with performance requirements.
- D. Panel Fasteners: Self-tapping screws and other acceptable fasteners recommended by roof panel manufacturer. Where exposed fasteners cannot be avoided, supply corrosion-resistant fasteners with heads matching color of metal panels by means of factory-applied coating.
- E. Joint Sealers: Provide Tape Mastic Sealants, Concealed Joint Sealant, and Urethane [Joint Sealants](#) per Section 07 92 00, "[Joint Sealants](#)".
- F. Steel Sheet Miscellaneous Framing Components: ASTM C 645, with ASTM A 653/A 653M, **G60 (Z180)** hot-dip galvanized zinc coating.
- G. Roof Accessories: Approved by metal roof panel manufacturer. Refer to Section 07 72 00 "Roof Accessories" for requirements for curbs, equipment supports, roof hatches, heat and smoke vents, ventilators, and preformed flashing sleeves.

## 2.5 FABRICATION

- A. General: Provide factory fabricated and finished metal panels and accessories meeting performance requirements, indicated profiles, and structural requirements.
- B. Fabricate metal panel joints configured to accept sealant tape providing weathertight seal and preventing metal-to-metal contact and minimizing noise resulting from thermal movement.
- C. Sheet Metal Flashing and Trim: Fabricate flashing and trim to comply with manufacturer's written instructions, approved shop drawings, and project drawings.

## 2.6 FINISHES

- A. Finishes, General: Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
- B. Exterior Face Sheet Coil-Coated Finish System
  - 1. Fluoropolymer Two-Coat System: 0.2 – 0.3 mil primer with 0.7 - 0.8 mil 70 percent PVDF fluoropolymer color coat, AAMA 621, meeting solar reflectance index requirements.
    - a. Basis of Design: MBCI, Signature 300.
- C. Interior Face Sheet Coil-Coated Finish System:
  - 1. Polyester Two-Coat System: 0.20 – 0.25 mil primer with 0.7 – 0.8 mil color coat.
    - a. Basis of Design: MBCI, Igloo White.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

- A. Examine metal panel system substrate with Installer present. Inspect for erection tolerances and other conditions that would adversely affect installation of metal panel installation.
  - 1. Inspect framing that will support insulated metal panels to determine if support components are installed as indicated on approved shop drawings and are within tolerances acceptable to metal panel manufacturer and installer. Confirm presence of acceptable framing members at recommended spacing to match installation requirements of insulated metal panels.
  - 2. Panel Support Tolerances: Confirm that panel supports are within tolerances acceptable to insulated metal panel system manufacturer but not greater than the following:
    - a. 1/8 inch (3 mm) in 5 foot (152 cm) in any direction.
    - b. 1/4 inch (6 mm) in 20 foot (610 cm) in any direction.
    - c. 1/2 inch (9 mm) over any single roof plane.
- B. Correct out-of-tolerance work and other deficient conditions prior to proceeding with insulated metal roof panel system installation.

### 3.2 METAL PANEL INSTALLATION

- A. Mechanically-Seamed, Foamed-Insulation-Core Metal Roof Panels: Install insulated metal panel system in accordance with manufacturer's written instructions, approved shop drawings, and project drawings. Install insulated metal roof panels in orientation, sizes, and locations indicated. Anchor panels and other components securely in place. Provide for thermal and structural movement.
- B. Attach panels to metal framing using clips, screws, fasteners, sealants, and adhesives recommended by manufacturer and indicated on approved shop drawings.
  - 1. Fasten metal panels to supports with concealed clips at each location indicated on approved shop drawings, with spacing and fasteners recommended by manufacturer.
  - 2. Provide weatherproof jacks for pipe and conduit penetrating metal panels.
  - 3. Dissimilar Materials: Where elements of metal panel system will come into contact with dissimilar materials, treat faces and edges in contact with dissimilar materials as recommended by manufacturer.
- C. Attach panel flashing trim pieces to supports using recommended fasteners and joint sealers.
- D. Joint Sealers: Install tape sealers and liquid sealants where indicated and where required for weatherproof performance of metal panel assemblies.
  - 1. Seal panels in accordance with insulated panel manufacturer's instructions, and project design drawings.
  - 2. Seal panel joints utilizing tape sealer and vapor seal bead of non-curing butyl; apply continuously without gaps in accordance with manufacturer's written instructions, approved shop drawings, and project drawings.
  - 3. Prepare joints and apply sealants per requirements of Division 07 Section "[Joint Sealants](#)."

### 3.3 ACCESSORY INSTALLATION

- A. General: Install metal panel accessories with positive anchorage to building and weather tight mounting and provide for thermal expansion. Coordinate installation with flashings and other components.

1. Install components required for a complete metal panel assembly, including trim, copings, flashings, sealants, closure strips, and similar items.
2. Comply with details of assemblies utilized to establish compliance with performance requirements and manufacturer's written installation instructions.
3. Provide concealed fasteners except where noted on approved shop drawings.
4. Set units true to line and level as indicated. Install work with laps, joints, and seams that will be permanently weather resistant.

#### 3.4 FIELD QUALITY CONTROL

- A. Testing Agency: Engage an independent testing and inspecting agency acceptable to Architect to perform field tests and inspections and to prepare test reports.
- B. Water-Spray Test: After completing portion of metal roof panel assembly including accessories and trim, test 2-bay area selected by Architect for water penetration, according to AAMA 501.2.

#### 3.5 CLEANING AND PROTECTION

- A. Remove temporary protective films immediately in accordance with metal roof panel manufacturer's instructions. Clean finished surfaces as recommended by metal roof panel manufacturer.
- B. Replace damaged panels and accessories that cannot be repaired to the satisfaction of the Architect.

END OF SECTION

SECTION 07 60 00 - SHEET METAL AND MISCELLANEOUS ACCESSORIES

PART 1 - GENERAL

1.00 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.01 SUMMARY

- A. Section Includes:
  - 1. Provide flashing and sheet metal components for moisture protection.
  - 2. Related accessories.

1.02 SUBMITTALS

- A. Product Data:
  - 1. Submit shop drawings, product data and mockups of all sheet metal.

1.03 QUALITY ASSURANCE

- A. Comply with governing local, state, and federal regulations, safety standards, and codes. Provide products of acceptable manufacturers in satisfactory use in similar service for five years. Use experienced installers. Deliver, handle and store materials in accordance with manufacturer's instructions.
- B. Reference Standards: Applicable portions of SMACNA, ASTM and NAAMM publications.

1.04 WARRANTIES

- A. Manufacturer's Product Warranty: Submit manufacturer's standard limited product warranty signed by the manufacturer's authorized official, guaranteeing to correct failures in product which may occur during the warranty period, without reducing or otherwise limiting any other rights to correction which the Owner/Project Consultant may have under the contract documents. Failure is defined to include product failure which leads to interruption of a watertight installation. Correction may include repair or replacement of failed product.



- B. Contractor's Warranty period: For roofing flashing and sheet metal, provide a written warranty which shall warrant work to be free of leaks and defects in materials and workmanship for two (2) years, starting from date of substantial completion.
- C. Defects of the sheet metal occurring during the warranty period shall be promptly corrected by the contractor, and defects of the roofing shall be promptly corrected by the manufacturer at no additional cost to the Owner. Upon notification from the Owner or the Owner's representative that evidence of a defect exists, the responsible party shall immediately inform the Owner's representative of the date on which corrective work will be scheduled, and shall notify the Owner's representative when the corrective work has been completed.

## PART 2 - PRODUCTS

### 2.01 SHEET METAL MATERIAL

- A. Hot-dipped Galvanized Steel for use as counterflashings (where not visible from the ground), pitch pans and expansion joints: Minimum 24-gauge, G-90, hot-dipped galvanized metal, commercial quality, ASTM A 525.
- B. Hot-dipped Galvanized Steel for use as continuous clips: Minimum 22-gauge, G-90, hot-dipped galvanized metal, commercial quality, ASTM A 525.
- C. Prefinished Galvanized Sheet Steel (where visible from the ground): Shall be 24-gauge flat stock, prefinished with Kynar finish meeting ASTM A 446, forty-five and one-half inches to forty-eight inches width by one hundred twenty inches in length (45-1/2" - 48" x 120") for use as new metal edge gravel guard, cover plates, downspouts, gutters, coping and miscellaneous metal. Standard color to be selected by Owner/Project Consultant.
- D. Stainless Steel: QQ-S-766, Class 301, 302, 304, or 316; or ASTM A 167, Type 301, 302, 304, or 316; form and condition most suitable for the purpose.
- E. Aluminum and Aluminum Alloy Plate and Sheet: QQ-A-250; form, alloy, and temper shall be that most suitable for the purpose.
- F. Sheet Lead: QQ-L-201, Grade B.
- G. All existing sheet metal shall be replaced with new metal of like gauge and type, or as specified on drawings.

### 2.02 FASTENERS

- A. Fasteners shall be same metal as flashing/sheet metal, or other non-corrosive metal as recommended by sheet manufacturer for the specific application. Match finish of exposed heads with material being fastened.
- B. Fasteners and fastening plates or bars shall be listed in the FM Global Approval Guide.
- C. Fastener for Brick: Shall be one-fourth inch by two inches (1/4" x 2"), zinc with plated steel or stainless steel nail, one piece unit, flat head.
- D. Screws: Self-taping sheet metal type with neoprene washer, as appropriate.
- E. Pop Rivets: Full stainless steel Series 42 or 44, as appropriate.

- F. Continuous Clip: Concealed hold-down clip type; of same materials as coping, gravel guard, sized to suit application. Use a continuous clip, minimum 22-gauge G-90 galvanized.

## 2.03 RELATED MATERIAL

- A. Bituminous Paint: Acid and alkali resistant, black color.
- B. Plastic Cement: FS SS-C-153, cutback asphalt type.
- C. Solder: QQ-S-571 composition best suited for purpose; use high tin content, minimum 60/40, for stainless steel and monel alloy.
- D. Copper, Sheet, and Strip: QQ-C-576, ASTM B 370, light cold-rolled temper.
- E. Lead-coated Copper: ASTM B 101, Type I or II, Class A.
- F. Sealant (for Sheet Metal): One-component polyurethane, conforming to requirements of FS TT-S-230C, non-staining and non-bleeding.

## 2.04 FABRICATION - GENERAL

- A. Shop-fabricate work to greatest extent possible. Comply with details shown, and with applicable requirements of SMACNA "Architectural Sheet Metal Manual" and other recognized industry practices. Fabricate for waterproof and weather-resistant performance; with expansion provisions for running work, sufficient to permanently prevent leakage, damage or deterioration of the work. Form work to fit substrates. Comply with material manufacturer's instructions and recommendations. Form exposed sheet metal work without excessive oil-canning, buckling, and tool marks, true to line and levels as indicated, with exposed edges folded back to form hems.
- B. Fabricate gravel stops/fascia, gutters/downspouts, counterflashings, pitch pans, expansion joints, and copings with new galvanized sheet metal as specified. Fabricate gravel guard and fascia to size and dimensions as indicated on the drawings. Fabricate light metal coping, gutters and downspouts as indicated.
- C. Form sheet metal on bending brake.
- D. Shape, trim and hand seam metal on bench insofar as practicable.
- E. Form materials with straight lines, sharp angles and smooth curves.
- F. Fold back edges on concealed side of exposed edge to form hem (1/2" minimum).
- G. Weld or solder joints on parts that are to be permanently and rigidly assembled.
- H. Submit sheet metal models for approval by the Owner/Project Consultant.
- I. Limit single-piece lengths to ten feet (10').
- J. Fabricate corner pieces with eighteen inch (18") extensions, metered and sealed by forming as one piece.
- K. Surface sand flange prior to applying any primers on Kynar metal.
- L. Backpaint flashing in contact with masonry or dissimilar materials with bituminous paint.

- M. All sheet metal shall be sealed and watertight.
  - N. Metal work should be secured so as to prevent damage from buckling or wind. Where clips are shown, these are to be continuous.
  - O. All metal to receive bitumen or adhesive shall be first primed with asphalt primer.
  - P. All prefinished metal shall be sanded and/or abraded prior to receiving primer.
  - Q. Seams: Fabricate non-moving seams in sheet metal with flat-lock seams. For metal other than aluminum, tin edges are to be seamed, form seams, and soldered.
  - R. Expansion Provisions: Form expansion joints of intermeshing hooked flanges, not less than one inch (1") deep, filled with mastic sealant (concealed within joints).
  - S. Sealant Joints: Where movable, non-expansion type joints are indicated or required for proper performance of work, form metal to provide for proper installation of elastomeric sealant, in compliance with industry standards.
  - T. Separations: Provide for separation of metal from non-compatible metal or corrosive substrates by coating concealed surfaces at locations of contact, with bituminous coating or other permanent separation as recommended by manufacturer/fabricator.
  - U. Bed flanges of work in a thick coat of bituminous roofing cement where required for waterproof performance.
- 2.05 FINISH
- A. Backpaint concealed metal surfaces with bituminous paint where expected to be in contact with cementitious materials or dissimilar metals. Exposed surfaces to be provided with a factory applied fluorocarbon Kynar finish meeting ASTM A 446 and AAMA specification 605.2 for high performance coating.
  - B. New 24-gauge hot-dipped galvanized metal shall be painted on all locations visible from the ground with an industrial grade paint to match existing, or standard color selected by Owner/Project Consultant. Galvanized metal surface must be properly prepared by removing all oil, grease, and/or protective mill coatings by solvent cleaning surface in accordance with SSPC-SP1, and according to paint manufacturer's recommendation, to ensure proper adhesion of paint to metal.

### PART 3 - EXECUTION

#### 3.01 INSPECTION

- A. Verify roof openings, curbs, pipes, sleeves, ducts or vents through roof are solidly set, cant strips and reglets in place, substrates are smooth and clean and nailing strips located.
- B. Verify membrane termination and base flashings are in place, sealed and secure.
- C. Beginning of installation means acceptance of conditions.

3.02 PREPARATION

- A. Field measure site conditions prior to fabricating work. Provide all shop drawings and mock-ups one month prior to installation to the Owner/Project Consultant for approval.
- B. Install starter and edge strips and cleats before starting installation.

3.03 INSTALLATION

- A. General: All sheet metal termination to vertical wall shall have a through-wall with receiver installed on masonry walls or prefabricated "Z" bar flashing pre-installed to fluid applied wall finished prior to installation of sheet metal termination. This applies to edge metal, base flashing closures and all vertical surface intersections. Refer to NRCA, SMACNA, and metal manufacturer's guidelines.
- B. Gravel Guard/Fascia:
  - 1. Shall be installed with expansion joints, ten feet (10') on center, one-fourth inch (1/4") expansion leeway, with a cover plate.
  - 2. Secure metal flashings per specifications.
  - 3. Lock seams and end joints.
  - 4. Form sections identical to profiles as shown or approved similar, to match existing building.
  - 5. Fabricate corner pieces with minimum eighteen inch (18"), maximum forty-eight inch (48") extensions, formed and sealed with rivets and sealant, as one piece.
  - 6. Hem exposed edges three-fourths inch (3/4") minimum.
  - 7. Backpaint flashing in contact with masonry or dissimilar materials with bituminous paint. Surface sand before applying primers.
  - 8. Integrate flashing in a manner consistent with detailing.
  - 9. Provide and install continuous clip around perimeter.
  - 10. Shall be fabricated in accordance with all SMACNA provisions.
- C. Coping:
  - 1. All coping shall be manufactured with low profile standing seam metal.
  - 2. Shall be minimum 24-gauge prefinished Kynar installed in ten foot (10') sections maximum with cover plates.
  - 4. Vertical fascia shall extend minimum two and one-half inches (2-1/2") or be minimum one and one-half inches (1-1/2") below bottom of nailer, whichever is greater.
  - 5. Secure metal flashings per specifications.
  - 6. Lock seams and end joints.
  - 7. Form sections identical to profiles as shown or approved similar, to match existing building.
  - 8. Fabricate corner pieces with minimum eighteen inch (18"), maximum forty-eight inch (48") extensions, formed and sealed with rivets and sealant, as one piece.
  - 9. Hem exposed edges three-fourths inch (3/4") minimum.
  - 10. Backpaint flashing in contact with masonry or dissimilar materials with bituminous paint. Surface sand before applying primers.
  - 11. Integrate flashing in a manner consistent with detailing.
  - 12. Provide and install continuous clip, minimum 22-gauge.
  - 13. Shall be fabricated in accordance with all SMACNA provisions.
- D. Expansion Joint Field and at Wall:
  - 1. Shall be as outlined by details, and be in full compliance with all provisions of SMACNA and FM Global requirements for attachment, installation and recommendations.
  - 2. Secure metal flashings per specifications.
  - 3. Lock seams and end joints.

4. Form sections identical to profiles as shown or approved similar, to match existing building.
  5. Fabricate corner pieces with minimum eighteen inch (18"), maximum forty-eight inch (48") extensions, formed and sealed with rivets and sealant, as one piece.
  6. Hem exposed edges three-fourths inch (3/4") minimum.
  7. Backpaint flashing in contact with masonry or dissimilar materials with bituminous paint. Surface sand before applying primers.
  8. Integrate flashing in a manner consistent with detailing.
  9. Provide and install continuous clip around perimeter.
  10. Shall be fabricated in accordance with all SMACNA provisions.
- E. Counterflashing:
1. Saw cut brick mortar joint to receive friction fit reglet and removable counterflashing as detailed and SMACNA Figure 4-3E.
- F. Gutter and Downspout:
1. Fabrication:
    - a) Fabricate gutter and downspout of profile and size indicated.
    - b) Field measure site conditions prior to fabricating work.
    - c) Fabricate with required connection pieces.
    - d) Fabricate section square, true, and accurate in size, in maximum possible lengths and free of distortion or defects detrimental to appearance or performance.
    - e) Hem exposed edges of metal.
    - f) Form and seal all metal joints; provide for expansion joints per SMACNA.
  2. Installation:
    - a) Install collector head, downspout, and accessories.
    - b) Join lengths with seams pop riveted and sealed watertight. Flash and seal collector head to downspouts and accessories.
    - c) Seal all metal joints watertight for full metal surface contact.
    - d) Collector Head: SMACNA style profile; submit detail for approval.
    - e) Downspouts: Rectangular profile. Seal all joints, four inches by six inches (4" x 6").
    - f) Support Brackets, Joint Fasteners: Profiled to suit gutters and downspouts.
    - g) Anchorage Devices: SMACNA requirements. Type recommended by fabricator.
    - h) Collector Head Support: Kynar. Color and Finish to match, as recommended by SMACNA.
    - i) Downspout Supports: Straps, Kynar. Color and Finish to match.
- G. Overflow Scupper, Collector Head and Downspout:
1. Fabrication:
    - a) Fabricate overflow scupper, collector head and downspout of profile and size indicated, taking care that the roof drain leader fits properly into the back of the collector head. Seal the pipe to the collector head for watertightness.
    - b) Field measure site conditions prior to fabricating work.
    - c) Fabricate with required connection pieces.
    - d) Fabricate section square, true, and accurate in size, in maximum possible lengths and free of distortion or defects detrimental to appearance or performance.
    - e) Hem exposed edges of metal.
    - f) Form and seal all metal joints; provide for expansion joints per SMACNA.
  2. Installation:

Install collector head, downspout, and accessories.

    - a) Join lengths with seams pop riveted and sealed watertight. Flash and seal collector head to downspouts and accessories.
    - b) Seal all metal joints watertight for full metal surface contact.
    - c) Collector Head: SMACNA style profile; submit detail for approval.

- d) Downspouts: Rectangular profile. Seal all joints, four inches by six inches (4" x 6").
- e) Support Brackets, Joint Fasteners: Profiled to suit gutters and downspouts.
- f) Anchorage Devices: SMACNA requirements. Type recommended by fabricator.
- g) Collector Head Support: Kynar. Color and Finish to match, as recommended by SMACNA.
- h) Downspout Supports: Straps, Kynar. Color and Finish to match.

H. Pitch Pans:

- 1. Install pitch pans of 24-gauge galvanized steel according to NRCA standards, minimum of six inches by six inches (6" x 6").
- 2. Pitch pans shall be fabricated to minimum of four inches (4") above the finished roof membrane. Seams of pitch pans shall be soldered.
- 3. Mastic shall be applied under pitch pan flange a minimum of one-half pound (1/2#) per linear foot.
- 4. All metal flanges shall be primed with asphalt primer prior to flashing installation. Inside of pitch pan shall be cleaned and primed as required by pitch pan sealant manufacturer.
- 5. All projections enclosed in pitch pans shall be cleaned in any manner suitable and coated with a rust inhibitive coating as approved by the Owner/Project Consultant. Coating shall be allowed to dry prior to pitch pan fill.
- 6. Base of pitch pans shall be filled with grout or cementitious binder to proper height and allowed to cure.
- 7. Top finish fill shall be self-leveling, one-part urethane, with maximum fill to within three-eighths inch (3/8") of top of pitch pan sides.
- 8. Strip metal flange of pitch pan with one strip of Type IV fiberglass felt set in hot bitumen extending from the outer edge of the flange a minimum of three inches (3") inward to base of pitch pan.
- 9. Strip in fiberglass felt with .060 inch coal-tar elastomeric membrane flashing set in hot asphalt extending from the outer edge of the Type IV fiberglass underlayment a minimum of three inches (3") inward to the base of the pitch pan.

I. Bonnets/Hoods:

- 1. Fabricate and install above all pitch pans, where necessary, or reinstall as applicable, metal bonnets over all pitch pans, NO EXCEPTIONS.
- 2. Bonnets/Hoods shall be manufactured with metal compatible with metal to which bonnet is to be attached.
- 3. On beams and other steel, weld in place bonnets fabricated from one-fourth inch (1/4") steel plate.
- 4. Draw band bonnets fabricated from 22-gauge galvanized steel may be used on circular projections.

3.04 FINISH

- A. Backpaint concealed metal surfaces with bituminous paint where expected to be in contact with cementitious materials or dissimilar metals. Exposed surfaces to be provided with a factory applied fluorocarbon Kynar finish meeting ASTM A 446 and AAMA specification 605.2 for high performance coating.

END OF SECTION

## SECTION 07 92 00 - JOINT SEALANTS

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 SECTION REQUIREMENTS

- A. Submittals: Product Data and color Samples.

### PART 2 - PRODUCTS

#### 2.1 JOINT SEALANTS

- A. Compatibility: Provide joint sealants, joint fillers, and other related materials that are compatible with one another and with joint substrates under service and application conditions.
- B. Elastomeric Sealants: Comply with ASTM C 920.
  - 1. Single-component, neutral-curing silicone sealant, Type S; Grade NS; Class 25; Uses T, M, and O, with the additional capability to withstand **[50 percent movement in both extension and compression for a total of 100 percent movement] [100 percent movement in extension and 50 percent movement in compression for a total of 150 percent movement]**. Use for building expansion joints.
  - 2. Single-component, nonsag polysulfide sealant, Type S; Grade NS; Class 12-1/2; Uses NT, M, G, A, and O. For general exterior use.
  - 3. Single-component, neutral-curing silicone sealant, Type S; Grade NS; Class 25; Uses T, NT, M, G, A, and O. For general exterior use.
  - 4. Single-component, nonsag urethane sealant, Type S; Grade NS; Class 25; and Uses NT, M, A, and O. For general exterior use.
  - 5. Single-component, nonsag urethane sealant, Type S; Grade NS; Class 25; Uses T, NT, M, G, A, and O. Use for exterior traffic-bearing joints, where slope precludes use of pourable sealant.
  - 6. Single-component, pourable urethane sealant, Type S; Grade P; Class 25; Uses T, M, G, A, and O. Use for exterior traffic-bearing joints.

- 7. Single-component, mildew-resistant silicone sealant, Type S; Grade NS; Class 25; Uses NT, G, A, and O; formulated with fungicide. Use for interior sealant joints in ceramic tile, stone, and other hard surfaces in kitchens and toilet rooms and around plumbing fixtures.
- C. Latex Sealant: Single-component, nonsag, mildew-resistant, paintable, acrylic-emulsion sealant complying with ASTM C 834. For interior use only at perimeters of door and window frames.
- D. Acoustical Sealant for Exposed Joints: Nonsag, paintable, nonstaining, latex sealant complying with ASTM C 834. For interior use only at acoustical assemblies.
- E. Acoustical Sealant for Concealed Joints: Nondrying, nonhardening, nonskinning, nonstaining, gunnable, synthetic-rubber sealant recommended for sealing interior concealed joints to reduce transmission of air-borne sound. For interior use only at acoustical assemblies.

### PART 3 - EXECUTION

#### 3.1 INSTALLATION

- A. Comply with ASTM C 1193.
- B. Comply with ASTM C 919 for use of joint sealants in acoustical applications.

END OF SECTION 07920



## SECTION 081113 - HOLLOW METAL DOORS AND FRAMES

### PART 1 - GENERAL

#### 1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

#### 1.2 SUMMARY

- A. Section Includes:

1. Standard and custom hollow metal doors and frames.
2. Steel sidelight, borrowed lite and transom frames.
3. Louvers installed in hollow metal doors.
4. Light frames and glazing installed in hollow metal doors.

- B. Related Sections:

1. Division 04 Section "Unit Masonry" for embedding anchors for hollow metal work into masonry construction.
2. Division 08 Section "Flush Wood Doors".
3. Division 08 Section "Glazing" for glass view panels in hollow metal doors.
4. Division 08 Section "Door Hardware".
5. Division 09 Sections "Exterior Painting" and "Interior Painting" for field painting hollow metal doors and frames.

- C. Codes and References: Comply with the version year adopted by the Authority Having Jurisdiction.

1. ANSI/SDI A250.8 - Recommended Specifications for Standard Steel Doors and Frames.
2. ANSI/SDI A250.4 - Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors, Frames, Frames Anchors and Hardware Reinforcing.
3. ANSI/SDI A250.6 - Recommended Practice for Hardware Reinforcing on Standard Steel Doors and Frames.
4. ANSI/SDI A250.10 - Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames.
5. ANSI/SDI A250.11 - Recommended Erection Instructions for Steel Frames.
6. ASTM A1008 - Standard Specification for Steel Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability.
7. ASTM A653 - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
8. ASTM A924 - Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process.
9. ASTM C 1363 - Standard Test Method for Thermal Performance of Building Assemblies by Means of a Hot Box Apparatus.
10. ANSI/BHMA A156.115 - Hardware Preparation in Steel Doors and Frames.

11. ANSI/SDI 122 - Installation and Troubleshooting Guide for Standard Steel Doors and Frames.
12. ANSI/NFPA 80 - Standard for Fire Doors and Fire Windows; National Fire Protection Association.
13. ANSI/NFPA 105: Standard for the Installation of Smoke Door Assemblies.
14. NFPA 252 - Standard Methods of Fire Tests of Door Assemblies; National Fire Protection Association.
15. UL 10C - Positive Pressure Fire Tests of Door Assemblies.
16. UL 1784 - Standard for Air Leakage Tests of Door Assemblies.

### 1.3 SUBMITTALS

- A. Product Data: For each type of product indicated. Include construction details, material descriptions, core descriptions, hardware reinforcements, profiles, anchors, fire-resistance rating, and finishes.
- B. Door hardware supplier is to furnish templates, template reference number and/or physical hardware to the steel door and frame supplier in order to prepare the doors and frames to receive the finish hardware items.
- C. Shop Drawings: Include the following:
  1. Elevations of each door design.
  2. Details of doors, including vertical and horizontal edge details and metal thicknesses.
  3. Frame details for each frame type, including dimensioned profiles and metal thicknesses.
  4. Locations of reinforcement and preparations for hardware.
  5. Details of anchorages, joints, field splices, and connections.
  6. Details of accessories.
  7. Details of moldings, removable stops, and glazing.
  8. Details of conduit and preparations for power, signal, and control systems.
- D. Samples for Verification:
  1. Samples are only required by request of the architect and for manufacturers that are not current members of the Steel Door Institute.

### 1.4 QUALITY ASSURANCE

- A. Source Limitations: Obtain hollow metal doors and frames through one source from a single manufacturer wherever possible.
- B. Quality Standard: In addition to requirements specified, furnish SDI-Certified manufacturer products that comply with ANSI/SDI A250.8, latest edition, "Recommended Specifications for Standard Steel Doors and Frames".
- C. Fire-Rated Door Assemblies: Assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency, for fire-protection ratings indicated, based on testing at positive pressure according to UL10C (neutral pressure at 40" above sill) or UL 10C.

1. Oversize Fire-Rated Door Assemblies Construction: For units exceeding sizes of tested assemblies, attach construction label certifying doors are built to standard construction requirements for tested and labeled fire rated door assemblies except for size.
2. Temperature-Rise Limit: Where indicated and at vertical exit enclosures (stairwell openings) and exit passageways, provide doors that have a maximum transmitted temperature end point of not more than 450 deg F (250 deg C) above ambient after 30 minutes of standard fire-test exposure.
3. Smoke Control Door Assemblies: Comply with NFPA 105.
  - a. Smoke "S" Label: Doors to bear "S" label, and include smoke and draft control gasketing applied to frame and on meeting stiles of pair doors.

- D. Fire-Rated, Borrowed-Light Frame Assemblies: Assemblies complying with NFPA 80 that are listed and labeled, by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire-protection ratings indicated, based on testing according to NFPA 257. Provide labeled glazing material.
- E. Pre-Submittal Conference: Conduct conference in compliance with requirements in Division 01 Section "Project Meetings" with attendance by representatives of Supplier, Installer, and Contractor to review proper methods and procedures for installing hollow metal doors and frames and to verify installation of electrical knockout boxes and conduit at frames with electrified or access control hardware.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver hollow metal work palletized, wrapped, or crated to provide protection during transit and Project site storage. Do not use non-vented plastic.
- B. Deliver welded frames with two removable spreader bars across bottom of frames, tack welded to jambs and mullions.
- C. Store hollow metal work under cover at Project site. Place in stacks of five units maximum in a vertical position with heads up, spaced by blocking, on minimum 4-inch high wood blocking. Do not store in a manner that traps excess humidity.
  1. Provide minimum 1/4-inch space between each stacked door to permit air circulation. Door and frames to be stacked in a vertical upright position.

#### 1.6 PROJECT CONDITIONS

- A. Field Measurements: Verify actual dimensions of openings by field measurements before fabrication.

#### 1.7 COORDINATION

- A. Coordinate installation of anchorages for hollow metal frames. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors. Deliver such items to Project site in time for installation.

## 1.8 WARRANTY

- A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace doors that fail in materials or workmanship within specified warranty period.
- B. Warranty includes installation and finishing that may be required due to repair or replacement of defective doors.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide steel doors and frames from a SDI Certified manufacturer:
  - 1. CECO Door Products (C).
  - 2. Curries Company (CU).
  - 3. Pioneer Industries (PI).
  - 4. Steelcraft (S).

### 2.2 MATERIALS

- A. Cold-Rolled Steel Sheet: ASTM A 1008/A 1008M, Commercial Steel (CS), Type B; suitable for exposed applications.
- B. Metallic-Coated Steel Sheet: ASTM A 653/A 653M, Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.
- C. Frame Anchors: ASTM A 653/A 653M, Commercial Steel (CS), Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

### 2.3 HOLLOW METAL DOORS

- A. General: Provide 1-3/4 inch doors of design indicated, not less than thickness indicated; fabricated with smooth surfaces, without visible joints or seams on exposed faces unless otherwise indicated. Comply with ANSI/SDI A250.8 and ANSI/NAAMM HMMA 867.
- B. Exterior Doors (Energy Efficient): Face sheets fabricated of commercial quality hot-dipped zinc coated steel that complies with ASTM A924 A60. Provide doors complying with requirements indicated below by referencing ANSI/SDI A250.8 for level and model, ANSI/SDI A250.4 for physical performance level, and HMMA 867 for door construction.
  - 1. Design: Flush panel.
  - 2. Core Construction: Foamed in place polyurethane and steel stiffened laminated core with no stiffener face welds, in compliance with HMMA 867 "Laminated Core".
    - a. Provide 22 gauge steel stiffeners at 6 inches on-center internally welded at 5" on-center to integral core assembly, foamed in place polyurethane core chemically bonded to all interior surfaces. No stiffener face welding is permitted.

- b. Thermal properties to rate at a fully operable minimum U-Factor 0.29 and R-Value 3.4, including insulated door, thermal-break frame and threshold.
    - c. Kerf Type Frames: Thermal properties to rate at a fully operable minimum U-Factor 0.36 and R-Value 2.7, including insulated door, kerf type frame, and threshold.
  3. Level/Model: Level 3 and Physical Performance Level A (Extra Heavy Duty), Minimum 16 gauge (0.053 inch - 1.3-mm) thick steel, Model 2.
  4. Vertical Edges: Vertical edges to be mechanically interlocked with hairline seam. Beveled Lock Edge, 1/8 inch in 2 inches (3 mm in 50 mm).
  5. Top and Bottom Edges: Reinforce tops and bottoms of doors with a continuous steel channel not less than 16 gauge, extending the full width of the door and welded to the face sheet. Doors with an inverted top channel to include a steel closure channel, screw attached, with the web of the channel flush with the face sheets of the door. Plastic or composite channel fillers are not acceptable.
  6. Hinge Reinforcement: Minimum 7 gauge (3/16") plate 1-1/4" x 9".
  7. Hardware Reinforcements: Fabricate according to ANSI/SDI A250.6 with reinforcing plates from same material as door face sheets.
- C. Interior Doors: Face sheets fabricated of commercial quality cold rolled steel that complies with ASTM A 1008/A 1008M. Provide doors complying with requirements indicated below by referencing ANSI/SDI A250.8 for level and model and ANSI/SDI A250.4 for physical performance level:
  1. Design: Flush panel.
  2. Core Construction: Manufacturer's standard kraft-paper honeycomb, or one-piece polystyrene core, securely bonded to both faces.
    - a. Fire Door Core: As required to provide fire-protection and temperature-rise ratings indicated.
  3. Level/Model: Level 2 and Physical Performance Level B (Heavy Duty), Minimum 18 gauge (0.042-inch - 1.0-mm) thick steel, Model 2.
  4. Top and Bottom Edges: Reinforce tops and bottoms of doors with a continuous steel channel not less than 16 gauge, extending the full width of the door and welded to the face sheet.
  5. Hinge Reinforcement: Minimum 7 gauge (3/16") plate 1-1/4" x 9" or minimum 14 gauge continuous channel with pierced holes, drilled and tapped.
  6. Hardware Reinforcements: Fabricate according to ANSI/SDI A250.6 with reinforcing plates from same material as door face sheets.
- D. Manufacturers Basis of Design:
  1. CECO Door Products (C) Honeycomb Core - Regent Series.
  2. CECO Door Products (C) Energy Efficient - Trio-E Series.
  3. Curries Company (CU) - Polystyrene Core - 707 Series.
  4. Curries Company (CU) - Energy Efficient - 777 Trio-E Series.

## 2.4 HOLLOW METAL FRAMES

- A. General: Comply with ANSI/SDI A250.8 and with details indicated for type and profile.
- B. Exterior Frames: Fabricated of hot-dipped zinc coated steel that complies with ASTM A 653/A 653M, Coating Designation A60.
  - 1. Fabricate frames with mitered or coped corners. Profile as indicated on drawings.
  - 2. Frames: Minimum 14 gauge (0.067-inch -1.7-mm) thick steel sheet.
  - 3. Manufacturers Basis of Design:
    - a. CECO Door Products (C) – SU SR Series.
    - b. Curries Company (CU) – M CM Series.
- C. Interior Frames: Fabricated from cold-rolled steel sheet that complies with ASTM A 1008/A 1008M.
  - 1. Fabricate frames with mitered or coped corners. Profile as indicated on drawings.
  - 2. Frames: Minimum 16 gauge (0.053-inch -1.3-mm) thick steel sheet.
  - 3. Manufacturers Basis of Design:
    - a. CECO Door Products (C) - SU Series.
    - b. Curries Company (CU) - M Series.
- D. Fire rated frames: Fabricate frames in accordance with NFPA 80, listed and labeled by a qualified testing agency, for fire-protection ratings indicated.
- E. Hardware Reinforcement: Fabricate according to ANSI/SDI A250.6 Table 4 with reinforcement plates from same material as frames.

## 2.5 FRAME ANCHORS

- A. Jamb Anchors:
  - 1. Masonry Type: Adjustable strap-and-stirrup or T-shaped anchors to suit frame size, formed from A60 metallic coated material, not less than 0.042 inch thick, with corrugated or perforated straps not less than 2 inches wide by 10 inches long; or wire anchors not less than 0.177 inch thick.
  - 2. Stud Wall Type: Designed to engage stud and not less than 0.042 inch thick.
  - 3. Compression Type for Drywall Slip-on (Knock-Down) Frames: Adjustable compression anchors.
- B. Floor Anchors: Floor anchors to be provided at each jamb, formed from A60 metallic coated material, not less than 0.042 inches thick.
- C. Mortar Guards: Formed from same material as frames, not less than 0.016 inches thick.

## 2.6 LOUVERS

- A. Metal Louvers: Door manufacturer's standard metal louvers unless otherwise indicated.

1. Blade Type: Vision proof inverted V or inverted Y.
  2. Metal and Finish: Galvanized steel, 0.040 inch thick, factory primed for paint finish with baked enamel or powder coated finish. Match pre-finished door paint color where applicable.
- B. Louvers for Fire Rated Doors: Metal louvers with fusible link and closing device, listed and labeled for use in doors with fire protection rating of 1-1/2 hours and less.
1. Manufacturers: Subject to compliance with requirements, provide door manufacturers standard louver to meet rating indicated.
  2. Metal and Finish: Galvanized steel, 0.040 inch thick, factory primed for paint finish with baked enamel or powder coated finish. Match pre-finished door paint color where applicable.

## 2.7 LIGHT OPENINGS AND GLAZING

- A. Stops and Moldings: Provide stops and moldings around glazed lites where indicated. Form corners of stops and moldings with butted or mitered hairline joints at fabricator's shop. Fixed and removable stops to allow multiple glazed lites each to be removed independently. Coordinate frame rabbet widths between fixed and removable stops with the type of glazing and installation indicated.
- B. Moldings for Glazed Lites in Doors and Loose Stops for Glazed Lites in Frames: Minimum 20 gauge thick, fabricated from same material as door face sheet in which they are installed.
- C. Fixed Frame Moldings: Formed integral with hollow metal frames, a minimum of 5/8 inch (16 mm) high unless otherwise indicated. Provide fixed frame moldings and stops on outside of exterior and on secure side of interior doors and frames.
- D. Preformed Metal Frames for Light Openings: Manufacturer's standard frame formed of 0.048-inch-thick, cold rolled steel sheet; with baked enamel or powder coated finish; and approved for use in doors of fire protection rating indicated. Match pre-finished door paint color where applicable.

## 2.8 ACCESSORIES

- A. Mullions and Transom Bars: Join to adjacent members by welding or rigid mechanical anchors.
- B. Grout Guards: Formed from same material as frames, not less than 0.016 inches thick.

## 2.9 FABRICATION

- A. Fabricate hollow metal work to be rigid and free of defects, warp, or buckle. Accurately form metal to required sizes and profiles, with minimum radius for thickness of metal. Where practical, fit and assemble units in manufacturer's plant. When shipping limitations so dictate, frames for large openings are to be fabricated in sections for splicing or splining in the field by others.
- B. Tolerances: Fabricate hollow metal work to tolerances indicated in ANSI/SDI A250.8.

C. Hollow Metal Doors:

1. Exterior Doors: Provide optional weep-hole openings in bottom of exterior doors to permit moisture to escape where specified.
2. Glazed Lites: Factory cut openings in doors with applied trim or kits to fit. Factory install glazing where indicated.
3. Astragals: Provide overlapping astragals as noted in door hardware sets in Division 08 Section "Door Hardware" on one leaf of pairs of doors where required by NFPA 80 for fire-performance rating or where indicated. Extend minimum 3/4 inch beyond edge of door on which astragal is mounted.
4. Continuous Hinge Reinforcement: Provide welded continuous 12 gauge strap for continuous hinges specified in hardware sets in Division 08 Section "Door Hardware".

D. Hollow Metal Frames:

1. Shipping Limitations: Where frames are fabricated in sections due to shipping or handling limitations, provide alignment plates or angles at each joint, fabricated of same thickness metal as frames.
2. Welded Frames: Weld flush face joints continuously; grind, fill, dress, and make smooth, flush, and invisible.
  - a. Welded frames are to be provided with two steel spreaders temporarily attached to the bottom of both jambs to serve as a brace during shipping and handling. Spreader bars are for bracing only and are not to be used to size the frame opening.
3. Sidelight and Transom Bar Frames: Provide closed tubular members with no visible face seams or joints, fabricated from same material as door frame. Fasten members at crossings and to jambs by butt welding.
4. High Frequency Hinge Reinforcement: Provide high frequency hinge reinforcements at door openings 48-inches and wider with mortise butt type hinges at top hinge locations.
5. Continuous Hinge Reinforcement: Provide welded continuous 12 gauge straps for continuous hinges specified in hardware sets in Division 08 Section "Door Hardware".
6. Provide countersunk, flat- or oval-head exposed screws and bolts for exposed fasteners unless otherwise indicated for removable stops, provide security screws at exterior locations.
7. Mortar Guards: Provide guard boxes at back of hardware mortises in frames at all hinges and strike preps regardless of grouting requirements.
8. Floor Anchors: Weld anchors to bottom of jambs and mullions with at least four spot welds per anchor.
9. Jamb Anchors: Provide number and spacing of anchors as follows:
  - a. Masonry Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:
    - 1) Two anchors per jamb up to 60 inches high.
    - 2) Three anchors per jamb from 60 to 90 inches high.
    - 3) Four anchors per jamb from 90 to 120 inches high.
    - 4) Four anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 120 inches high.



- b. Stud Wall Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:
  - 1) Three anchors per jamb up to 60 inches high.
  - 2) Four anchors per jamb from 60 to 90 inches high.
  - 3) Five anchors per jamb from 90 to 96 inches high.
  - 4) Five anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 96 inches high.
  - 5) Two anchors per head for frames above 42 inches wide and mounted in metal stud partitions.
- 10. Door Silencers: Except on weatherstripped or gasketed doors, drill stops to receive door silencers. Silencers to be supplied by frame manufacturer regardless if specified in Division 08 Section "Door Hardware".
- E. Hardware Preparation: Factory prepare hollow metal work to receive template mortised hardware; include cutouts, reinforcement, mortising, drilling, and tapping according to the Door Hardware Schedule and templates furnished as specified in Division 08 Section "Door Hardware".
  - 1. Locate hardware as indicated, or if not indicated, according to ANSI/SDI A250.8.
  - 2. Reinforce doors and frames to receive non-template, mortised and surface mounted door hardware.
  - 3. Comply with applicable requirements in ANSI/SDI A250.6 and ANSI/DHI A115 Series specifications for preparation of hollow metal work for hardware.
  - 4. Coordinate locations of conduit and wiring boxes for electrical connections with Division 26 Sections.

## 2.10 STEEL FINISHES

- A. Prime Finishes: Doors and frames to be cleaned, and chemically treated to insure maximum finish paint adhesion. Surfaces of the door and frame exposed to view to receive a factory applied coat of rust inhibiting shop primer.
  - 1. Shop Primer: Manufacturer's standard, fast-curing, lead and chromate free primer complying with ANSI/SDI A250.10 acceptance criteria; recommended by primer manufacturer for substrate; and compatible with substrate and field-applied coatings.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. General Contractor to verify the accuracy of dimensions given to the steel door and frame manufacturer for existing openings or existing frames (strike height, hinge spacing, hinge back set, etc.).
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 PREPARATION

- A. Remove welded in shipping spreaders installed at factory. Restore exposed finish by grinding, filling, and dressing, as required to make repaired area smooth, flush, and invisible on exposed faces.
- B. Prior to installation, adjust and securely brace welded hollow metal frames for square, level, twist, and plumb condition.
- C. Tolerances shall comply with SDI-117 "Manufacturing Tolerances Standard Steel Doors and Frames."
- D. Drill and tap doors and frames to receive non-template, mortised, and surface-mounted door hardware.

### 3.3 INSTALLATION

- A. General: Install hollow metal work plumb, rigid, properly aligned, and securely fastened in place; comply with Drawings and manufacturer's written instructions.
- B. Hollow Metal Frames: Install hollow metal frames of size and profile indicated. Comply with ANSI/SDI A250.11 and NFPA 80 at fire rated openings.
  - 1. Set frames accurately in position, plumbed, leveled, aligned, and braced securely until permanent anchors are set. After wall construction is complete and frames properly set and secured, remove temporary braces, leaving surfaces smooth and undamaged. Shim as necessary to comply with installation tolerances.
  - 2. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor, and secure with post-installed expansion anchors.
  - 3. Masonry Walls: Coordinate installation of frames to allow for solidly filling space between frames and masonry with mortar.
  - 4. Grout Requirements: Do not grout head of frames unless reinforcing has been installed in head of frame. Do not grout vertical or horizontal closed mullion members.
- C. Hollow Metal Doors: Fit hollow metal doors accurately in frames, within clearances specified below. Shim as necessary.
  - 1. Non-Fire-Rated Standard Steel Doors:
    - a. Jambs and Head: 1/8 inch plus or minus 1/16 inch.
    - b. Between Edges of Pairs of Doors: 1/8 inch plus or minus 1/16 inch.
    - c. Between Bottom of Door and Top of Threshold: Maximum 3/8 inch.
    - d. Between Bottom of Door and Top of Finish Floor (No Threshold): Maximum 3/4 inch.
  - 2. Fire-Rated Doors: Install doors with clearances according to NFPA 80.
- D. Field Glazing: Comply with installation requirements in Division 08 Section "Glazing" and with hollow metal manufacturer's written instructions.

### 3.4 ADJUSTING AND CLEANING

- A. Final Adjustments: Check and readjust operating hardware items immediately before final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including hollow metal work that is warped, bowed, or otherwise unacceptable.
- B. Remove grout and other bonding material from hollow metal work immediately after installation.
- C. Prime-Coat and Painted Finish Touchup: Immediately after erection, sand smooth rusted or damaged areas of prime coat, or painted finishes, and apply touchup of compatible air drying, rust-inhibitive primer, zinc rich primer (exterior and galvanized openings) or finish paint.

END OF SECTION 081113

## SECTION 08 14 16 - FLUSH WOOD DOORS

### PART 1 - GENERAL

#### 1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

#### 1.2 SUMMARY

- A. Section Includes:

- 1. Solid core doors with wood veneer faces.
  - 2. Factory finishing wood doors.
  - 3. Factory fitting wood doors to frames and factory machining for hardware.
  - 4. Light frames and glazing installed in wood doors.

- B. Related Sections:

- 1. Division 08 Section "Door Schedule".
  - 2. Division 08 Section "Hollow Metal Doors and Frames".
  - 3. Division 08 Section "Glazing".
  - 4. Division 08 Section "Door Hardware".
  - 5. Division 08 Section "Access Control Hardware".

- C. Standards and References: Comply with the version year adopted by the Authority Having Jurisdiction.

- 1. ANSI A117.1 - Accessible and Usable Buildings and Facilities.
  - 2. ANSI A208.1 – Wood Particleboard.
  - 3. Intertek Testing Service (ITS Warnock Hersey) - Certification Listings for Fire Doors.
  - 4. NFPA 80 - Standard for Fire Doors and Fire Windows; National Fire Protection Association.
  - 5. NFPA 252 - Standard Methods of Fire Tests of Door Assemblies; National Fire Protection Association.
  - 6. UL 10C - Positive Pressure Fire Tests of Door Assemblies; UL 1784 - Standard for Air Leakage Tests of Door Assemblies.
  - 7. Window and Door Manufacturers Association - WDMA I.S.1-A Architectural Wood Flush Doors.

#### 1.3 SUBMITTALS

- A. Product Data: For each type of door indicated. Include details of core and edge construction, louvers, trim for openings, and WDMA I.S.1-A classifications. Include factory finishing specifications.

- B. Shop Drawings shall include:

1. Indicate location, size, and hand of each door.
2. Indicate dimensions and locations of mortises and holes for hardware.
3. Indicate dimensions and locations of cutouts.
4. Indicate requirements for veneer matching.
5. Indicate location and extent of hardware blocking.
6. Indicate construction details not covered in Product Data.
7. Indicate doors to be factory finished and finish requirements.
8. Indicate fire protection ratings for fire rated doors.

C. Samples for Initial Selection: For factory finished doors.

1. Factory finishes applied to actual door face materials, approximately 8 by 10 inches, for each material and finish. For each wood species and transparent finish, provide set of three samples showing typical range of color and grain to be expected in the finished work.
2. Corner sections of doors, 8 by 10 inches, with door faces and edges representing actual materials to be used.
  - a. Provide samples for each species of veneer and core material.
  - b. Finish veneer faced door samples with same materials proposed for factory finished doors.
3. Frames for light openings, 6 inches long, for each material, type, and finish required.

D. Warranty: Provide sample of manufacturer's warranty.

1.4 QUALITY ASSURANCE

- A. Source Limitations: Obtain flush wood doors through one source from a single manufacturer wherever possible.
- B. Quality Standard: In addition to requirements specified, comply with WDMA I.S.1-A, latest edition, "Industry Standard for Architectural Wood Flush Doors".
- C. Fire Rated Wood Doors: Doors complying with NFPA 80 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire ratings indicated, based on testing at positive pressure according to NFPA 252 (neutral pressure at 40" above sill) or UL10C.
  1. Smoke Control Door Assemblies: Comply with NFPA 105.
    - 1) Smoke "S" Label: Doors to bear "S" label, and include smoke and draft control gasketing applied to frame and on meeting stiles of pair doors.
- D. Pre-Submittal Conference: Conduct conference in compliance with requirements in Division 01 Section "Project Meetings" with attendance by representatives of Supplier, Installer, and Contractor to review proper methods and procedures for receiving, handling, and installing flush wood doors.

## 1.5 DELIVERY, STORAGE, AND HANDLING

- A. Comply with requirements of referenced standard and manufacturer's written instructions.
- B. Package pre-finished doors individually in plastic bags and wrap bundles of doors in plastic sheeting.
- C. Mark each door on top rail with opening number used on Shop Drawings.

## 1.6 PROJECT CONDITIONS

- A. Environmental Limitations: Do not deliver or install doors until spaces are enclosed and weather tight, wet work in spaces is complete and dry, and HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.

## 1.7 WARRANTY

- A. Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace doors that fail in materials or workmanship within specified warranty period.
  - 1. Failures include, but are not limited to, the following:
    - a. Warping (bow, cup, or twist) more than 1/4 inch in a 42-by-84-inch section.
    - b. Telegraphing of core construction in wood face veneers exceeding 0.01 inch in a 3-inch span.
  - 2. Warranty includes installation and finishing that may be required due to repair or replacement of defective doors.
  - 3. Warranty Period for Solid Core Interior Doors: Life of installation.

## PART 2 - PRODUCTS

### 2.1 DOOR CONSTRUCTION – GENERAL

- A. WDMA I.S.1-A Performance Grade: Extra Heavy Duty; Aesthetic Grade: Premium.
- B. Fire Rated Doors: Provide construction and core as needed to provide fire ratings indicated.
  - 1. Category A Edge Construction: Provide fire rated door edge construction with intumescent seals concealed by outer stile (Category A) at 45, 60, and 90 minute rated doors. Comply with specified requirements for exposed edges.
  - 2. Pairs: Provide stiles with concealed intumescent seals. Comply with specified requirements for exposed edges.
    - a. Provide fire retardant stiles that are listed and labeled for applications indicated without formed steel edges and astragals.

- b. Where required for concealed hardware, provide formed steel edges and astragals with intumescent seals. Finish steel edges and astragals with baked enamel.

## 2.2 CORE CONSTRUCTION

### A. Particleboard Core Doors:

1. Particleboard: Wood fiber based materials complying with ANSI A208.1 Particleboard standard. Grade LD-2.
2. Adhesive: Fully bonded construction using Polyurethane (PUR) glue.
3. Blocking: As indicated under article "Blocking".

## 2.3 BLOCKING

### A. Non-Fire-Rated Doors:

1. Provide blocking as indicated below:
  - a. HB3: 5 inch top and bottom rail blocking, in doors indicated to have closers and kick plates.

## 2.4 VENEERED DOORS FOR TRANSPARENT FINISH

### A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Algoma Hardwoods: Architectural Series.
2. Eggers Industries: Premium Series.
3. Graham: GPD Series.
4. Marshfield: Signature Series.

### B. Interior Solid Core Doors:

1. Grade: Premium.
2. Faces: Veneer grades as noted below; veneer minimum 1/50-inch (0.5mm) thickness at moisture content of 12% or less.
  - a. Plain Sliced Red Oak, A grade faces.
3. Match between Veneer Leaves: Book match.
4. Assembly of Veneer Leaves on Door Faces:
  - a. Running Match.

5. Pair and Set Match: Provide for doors hung in same opening or separated only by mullions.
6. Transom Match: Continuous match.
7. Vertical Edges: Matching same species as faces. Wood or composite material, one piece, laminated, or veneered. Minimum requirements per WDMA section P-1, Performance Standards for Architectural Wood Flush Doors.
8. Horizontal Edges: Solid wood or structural composite material meeting the minimum requirements per WDMA section P-1, Performance Standards for Architectural Wood Flush Doors
9. Construction: Five plies. Stiles and rails are bonded to core, then entire unit sanded before applying face veneers.
10. At doors over 40% of the face cut-out for lights and or louvers, furnish engineered composite lumber core.

## 2.5 LIGHT FRAMES AND GLAZING

- A. Wood Beads for Light Openings in Wood Doors up to and including 20-minute rating:
  1. Wood Species: Same species as door faces.
  2. Profile:
    - a. M1 Flush Bead.
    - b. At wood core doors with 20-minute fire protection ratings, provide wood beads and metal glazing clips approved for such use.
- B. Metal Frames for Light Openings in Fire Rated Doors over 20-minute Rating: Manufacturer's standard frame formed of 0.048-inch-thick, cold rolled steel sheet; with baked enamel or powder coated finish; and approved for use in doors of fire protection rating indicated.
  1. Manufacturers:
    - a. Air Louver.
    - b. All Metal Stamping.
    - c. Anemostat.
    - d. Pemko.
- C. Glazing: Comply with installation requirements in Division 08 Section "Glazing" and with the flush wood door manufacturer's written instructions.

## 2.6 FABRICATION

- A. Factory fit doors to suit frame opening sizes indicated.
  1. Comply with requirements in NFPA 80 for fire rated doors.
  2. Undercut: As required per manufacturer's templates and sill condition.
- B. Factory machine doors for hardware that is not surface applied. Comply with final hardware schedules, door frame Shop Drawings, DHI A115-W series standards, and hardware templates.



1. Coordinate with hardware mortises in metal frames to verify dimensions and alignment before factory machining.
  2. Metal Astragals: Factory machine astragals and formed steel edges for hardware for pairs of fire rated doors.
- C. Openings: Cut and trim openings through doors in factory.
1. Light Openings: Trim openings with moldings of material and profile indicated.
  2. Glazing: Comply with applicable requirements in Division 08 Section "Glazing."

## 2.7 FACTORY FINISHING

- A. General: Comply with referenced quality standard for factory finishing. Complete fabrication, including fitting doors for openings and machining for hardware that is not surface applied, before finishing.
1. Finish faces, all four edges, edges of cutouts, and mortises. Stains and fillers may be omitted on top and bottom edges, edges of cutouts, and mortises.
- B. Transparent Finish: Provide a clear protective coating over the wood veneer allowing the natural color and grain of the selected wood species to provide the appearance specified. Stain is applied to the wood surface underneath the transparent finish to add color and design flexibility.
1. Finish: Meet or exceed WDMA I.S. 1A TR8 UV Cured Acrylated Polyester finish performance requirements.
  2. Staining:
    - a. As selected by Architect from manufacturer's full range.
    - b. Custom stain to match architect's sample.
  3. Sheen: Satin.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

- A. Examine doors and installed door frames before hanging doors.
1. Verify that frames comply with indicated requirements for type, size, location, and swing characteristics and have been installed with level heads and plumb jambs.
  2. Reject doors with defects.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

### 3.2 INSTALLATION

- A. Hardware: For installation, see Division 8 Section "Door Hardware."
- B. Installation Instructions: Install doors and frames to comply with manufacturer's written instructions and the referenced quality standard, and as indicated.
1. Install fire rated doors in corresponding fire rated frames according to NFPA 80.

- C. Factory Fitted Doors: Align in frames for uniform clearance at each edge.
- D. Factory Finished Doors: Restore finish before installation if fitting or machining is required at Project site.
- E. Field modifications to doors shall not be permitted, except those specifically allowed by manufacturer or fire rating requirements.

### 3.3 ADJUSTING

- A. Operation: Re-hang or replace doors that do not swing or operate freely.
- B. Finished Doors: Replace doors that do not comply with requirements. Doors may be repaired or refinished if work complies with requirements and shows no evidence of repair or refinishing.

END OF SECTION

## SECTION 08 41 13 – ALUMINUM STOREFRONT

### PART 1 - GENERAL

#### 1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.1 SUMMARY

- A. Related Documents: Conditions of the Contract, Division 1 - General Requirements, and Drawings apply to Work of this Section.
- B. Section Includes:
  - 1. Entrance and storefront systems, complete with reinforcing, fasteners, anchors and attachment devices.
  - 2. Aluminum doors complete with hardware.
  - 3. Accessories necessary to complete work.
- C. Related Sections:
  - 1. Section 01 40 00 - Quality Requirements.
  - 2. Section 05 50 00 - Metal Fabrications.
  - 3. Section 06 10 00 - Rough Carpentry.
  - 4. Section 07 92 00 - Joint Sealants.
  - 5. Section 08 71 00 - Door Hardware.
  - 6. Section 08 81 00 - Glass and Glazing.

#### 1.2 REFERENCES

- A. Aluminum Association (AA):
  - 1. DAF-45 Designation System for Aluminum Finishes.
- B. American Architectural Manufacturers Association (AAMA):
  - 1. 503.1 Test Method for Condensation Resistance of Windows, Doors and Glazed Wall Systems.
  - 2. 701.2 Specifications for Pile Weatherstripping.
  - 3. Manual #10 Care and Handling of Architectural Aluminum From Shop to Site.
  - 4. SFM-1 Aluminum Storefront and Entrance Manual.
- C. American National Standards Institute (ANSI):
  - 1. A117.1 Safety Standards for the Handicapped.

- D. American Society for Testing and Materials (ASTM):
  - 1. A36 Structural Steel.
  - 2. B209 Aluminum and Aluminum - Alloy Sheet and Plate.
  - 3. B221 Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes and Tubes.
  - 4. B308 Aluminum-Alloy 6061-T6 Standard Structural Shapes, Rolled or Extruded.
  - 5. C509 Cellular Elastomeric Pre-formed Gasket and Sealing Material.
  - 6. C864 Dense Elastomeric Compression Seal Gaskets, Setting Blocks and Spacers.
  - 7. E283 Test Method for Rate of Air Leakage Through Exterior Windows, Curtain Walls and Doors.
  - 8. E330 Test Method for Structural Performance of Exterior Windows, Curtain Walls and Doors by Uniform Static Air Pressure Difference.
  - 9. E331 Test Method for Water Penetration of Exterior Windows, Curtain Walls and Doors by Uniform Static Air Pressure Difference.
- E. Federal Specifications (FS):
  - 1. TT-P-645A Primer, Paint, Zinc Chromate, Alkyd Type.
- F. Steel Structures Painting Council (SSPC):
  - 1. Paint 12 Cold-Applied Asphalt Mastic (Extra Thick Film).

### 1.3 SYSTEM REQUIREMENTS

- A. Design Requirements:
  - 1. Drawings are diagrammatic and do not purport to identify nor solve problems of thermal or structural movement, glazing, anchorage or moisture disposal.
  - 2. Requirements shown by details are intended to establish basic dimension of units, sight lines and profiles of members.
  - 3. Provide concealed fastening.
  - 4. Provide entrance and storefront systems, including necessary modifications, to meet specified requirements and maintaining visual design concepts.
  - 5. Attachment considerations are to take into account site peculiarities and expansion and contraction movements so there is no possibility of loosening, weakening or fracturing connection between units and building structure or between units themselves.
  - 6. Anchors, fasteners and braces shall be structurally stressed not more than 50% of allowable stress when maximum loads are applied.
  - 7. Provide for expansion and contraction without detriment to appearance or performance.
  - 8. Assemblies shall be free from rattles, wind whistles and noise due to thermal and structural movement and wind pressure.
- B. Performance Requirements:
  - 1. Air infiltration: Air leakage through fixed light areas of storefront shall not exceed 0.06 cfm per square foot (0.0003 m<sup>3</sup>/sm<sup>2</sup>) of surface area when tested in accordance with ASTM E283 at differential static pressure of 6.24 psf (300 Pa).
  - 2. Water infiltration: No uncontrolled water penetration when tested in accordance with ASTM E 331 at test pressure of 8.0 psf 380 Pa.

C. Thermal Requirements:

1. Framing systems shall accommodate expansion and contraction movement due to surface temperature differentials of 180 degrees Fahrenheit (82 degrees Celsius) without causing buckling, stress on glass, failure of joint seals, excessive stress on structural elements, reduction of performance, or other detrimental effects.
2. Ensure doors function normally within limits of specified temperature range.

D. Structural Requirements, as measured in accordance with ANSI/ASTM E330:

1. Wind loads for exterior assemblies:
  - a. Basic loading:
    - 1) [ ] psf acting inward.
    - 2) [ ] psf acting outward.
2. Deflection: Maximum calculated deflection of any framing member in direction normal to plane of wall when subjected to specified design pressures shall not exceed 1/175 of its clear span.

E. Testing Requirements: Provide components that have been previously tested by an independent testing laboratory.

1.4 SUBMITTALS

A. General: Submit in accordance with Section 01 33 00.

B. Product Data:

1. Submit manufacturer's descriptive literature and product specifications.
2. Include information for factory finishes, hardware, accessories and other required components.
3. Include color charts for finish indicating manufacturer's standard colors available for selection.

C. Shop Drawings:

1. Submit shop drawings covering fabrication, installation and finish of specified systems.
2. Include following:
  - a. Fully dimensioned plans and elevations with detail coordination keys.
  - b. Locations of exposed fasteners and joints.
3. Provide detailed drawings of:
  - a. Composite members.
  - b. Joint connections for framing systems and for entrance doors.
  - c. Anchorage.
  - d. System reinforcements.
  - e. Expansion and contraction provisions.
  - f. Hardware, including locations, mounting heights, reinforcements and special installation provisions.
  - g. Glazing methods and accessories.
  - h. Internal sealant requirements as recommended by sealant manufacturer.
4. Schedule of finishes.

D. Samples:

1. Submit samples indicating quality of finish, in required colors, on alloys used for work, in sizes as standard with manufacturer.

2. Where normal texture or color variations are expected, include additional samples illustrating range of variation.
  - E. Test Reports:
    1. Standard Systems: Submit certified copies of previous test reports substantiating performance of system in lieu of re-testing. Include other supportive data as necessary.
  - F. Certificates:
    1. Submit manufacturer's certification stating that systems are in compliance with specified requirements.
  - G. Qualification Data:
    1. Submit installer qualifications verifying years of experience.
    2. Include list of projects having similar scope of work identified by name, location, date, reference name and phone number.
  - H. Manufacturer's Instructions: Submit manufacturer's printed installation instructions.
- 1.5 QUALITY ASSURANCE
- A. Single Source Responsibility:
    1. To ensure quality of appearance and performance, obtain materials for each system from either a single manufacturer or from manufacturer approved by each system manufacturer.
  - B. Installer Qualifications: Certified in writing by Contractor as qualified for installation of specified systems.
  - C. Perform Work in accordance with AAMA SFM-1 and manufacturer's written instructions.
  - D. Conform to requirements of ANSI A117.1 and local amendments.
- 1.6 DELIVERY, STORAGE AND HANDLING
- A. Comply with requirements of Section 01 60 00.
  - B. Protect finished surfaces as necessary to prevent damage.
  - C. Do not use adhesive papers or sprayed coatings which become firmly bonded when exposed to sun.
  - D. Do not leave coating residue on any surfaces.
  - E. Replace damaged units.
- 1.7 WARRANTY
- A. Provide warranties in accordance with the Contract General Conditions.
  - B. Provide written manufacturer's warranty, executed by company official, warranting against defects in materials and products for two (2) years from date of Substantial Completion.
  - C. Provide written installer's warranty, warranting work to be watertight, free from defective materials, defective workmanship, glass breakage due to defective design, and agreeing to replace components which fail within 1 year from date of Substantial Completion.

1. Warranty shall cover following:
  - a. Complete watertight and airtight system installation within specified tolerances.
  - b. Completed installation will remain free from rattles, wind whistles and noise due to thermal and structural movement and wind pressure.
  - c. System is structurally sound and free from distortion.
  - d. Glass and glazing gaskets will not break or "pop" from frames due to design wind, expansion or contraction movement.
  - e. Glazing sealants and gaskets will remain free from abnormal deterioration or dislocation due to sunlight, weather or oxidation.

## PART 2 - PRODUCTS

### 2.1 MATERIALS

#### A. Manufacturers

1. YKK AP America, Inc.
2. Vista Wall Architectural Products
3. Kawneer
4. Oldcastle Building Envelope

B. Substitutions: Submit under provisions of Section 01 25 00, a minimum of 10 days prior to bid date.

#### C. Acceptable Entrance Doors:

1. Standard Duty Doors: Model 35D Medium Stile Door YKK AP with Mid-panel panic device system or equivalent by specified manufacturer.

#### D. Acceptable Storefront Framing Systems:

1. Framing System: YES 45 FI (2" x 4-1/2") YKK AP or equivalent by specified manufacturer.

### 2.2 FRAMING MATERIALS AND ACCESSORIES

#### A. Aluminum:

1. ASTM B221, alloy 6063-T5 for extrusions; ASTM B209, alloy 5005-H34 for sheets; or other alloys and temper recommended by manufacturer appropriate for specified finish.

#### B. Internal Reinforcing:

1. ASTM A36 for carbon steel; or ASTM B308 for structural aluminum.
2. Shapes and sizes to suit installation.
3. Shop coat steel components after fabrication with alkyd type zinc chromate primer complying with FS TT-P-645.

#### C. Anchorage Devices:

1. Manufacturer's standard formed or fabricated steel or aluminum assemblies of shapes, plates, bars or tubes.

#### D. Fasteners:

1. Aluminum, non-magnetic stainless steel or other materials warranted by manufacturer to be non-

- corrosive and compatible with components being fastened.
- 2. Do not use exposed fasteners, except where unavoidable for application of hardware.
- 3. For exposed locations, provide countersunk Phillips head screws with finish matching items fastened.
- 4. For concealed locations, provide manufacturer's standard fasteners.
- 5. Provide nuts or washers of design having means to prevent disengagement; deforming of fastener threads is unacceptable.
- E. Expansion Anchor Devices: Lead-shield or toothed-steel, drilled-in, expansion bolt anchors.
- F. Protective Coatings: Cold-applied asphalt mastic complying with SSPC-Paint 12, compounded for 30 mil (0.77 mm) thickness for each coat; or alkyd type zinc chromate primer complying with FS TT-P-645.
- G. Glazing Gaskets:
  - 1. Compression type design, replaceable, molded or extruded, of neoprene, or ethylene propylene diene monomer (EPDM).
  - 2. Conform to ASTM C509 or C864.
  - 3. Profile and hardness as required to maintain uniform pressure for watertight seal.
  - 4. Provide in manufacturer's standard black color.
- H. Weatherstripping:
  - 1. Wool pile conforming to AAMA 701.2; or extruded EPDM elastomeric conforming to ASTM C509 or C864.
  - 2. Provide EPDM or vinyl-blade gasket weatherstripping in bottom door rail, adjustable for contact with threshold.
- I. Internal Sealants: Types recommended by sealant manufacturer.
- J. "Anti-Walk" Edge Blocking: "W" shaped EPDM blocks for use in keeping glazing material stationary under vibration or seismic loading.
- K. Baffles (at weep holes): Type as recommended by system manufacturer and shown in published installation instructions.

## 2.3 GLASS AND GLAZING ACCESSORIES

- A. Refer to Section 08 81 00.

## 2.4 FABRICATION

- A. Coordination of Fabrication:
  - 1. Check actual frame or door openings required in construction work by accurate field measurements before fabrication.
  - 2. Fabricate units to withstand loads which will be applied when system is in place.
- B. General:
  - 1. Conceal fasteners wherever possible.
  - 2. Reinforce work as necessary for performance requirements and for support to structure.
  - 3. Separate dissimilar metals and aluminum in contact with concrete utilizing protective coating or pre-formed separators which will prevent contact and corrosion.
  - 4. Comply with Section 08 81 00 for glazing requirements.
- C. Aluminum Framing:
  - 1. Provide members of size, shape and profile indicated, designed to provide for glazing from



- interior.
- 2. Fabricate frame assemblies with joints straight and tight fitting.
- 3. Reinforce internally with structural members as necessary to support design loads.
- 4. Maintain accurate relation of planes and angles, with hairline fit of contacting members.
- 5. Seal horizontals and direct moisture accumulation to exterior.
- 6. Provide flashings and other materials used internally or externally that are corrosive resistant, non-staining, non-bleeding and compatible with adjoining materials.
- 7. Provide manufacturer's extrusions and accessories to accommodate expansion and contraction due to temperature changes without being detrimental to appearance or performance.
- 8. Make provisions in framing for minimum edge clearance, nominal edge cover and nominal pocket width for thickness and type of glazing or infill used in accordance with recommendations of manufacturer and FGMA Glazing Manual.
- 9. Provide tight fitting, injection molded, plastic water deflectors at all intermediate horizontals.

D. Entrance Doors:

- 1. Fabricate with mechanical joints using internal reinforcing plates and shear blocks attached with fasteners and by welding.
- 2. Provide extruded aluminum glazing stops of [square] [beveled and mitered (for single glazing only)] design, [permanently anchored on security side and removable on opposite side.]

E. Hardware:

- 1. Receive hardware supplied in accordance with Section 08 71 00 and install in accordance with requirements of this Section.
- 2. Cut, reinforce, drill and tap frames and doors as required to receive hardware.
- 3. Comply with hardware manufacturer's templates and instructions.
- 4. Use concealed fasteners wherever possible.
- 5. Coordinate mid-panel panic device system with hardware at storefront doors to ensure compatibility.

F. Welding:

- 1. Comply with recommendations of the American Welding Society.
- 2. Use recommended electrodes and methods to avoid distortion and discoloration.
- 3. Grind exposed welds smooth and flush with adjacent surfaces; restore mechanical finish.

G. Flashings:

- 1. Form from sheet aluminum with same finish as extruded sections. Apply finish after fabrication. Material thickness as required to suit condition without deflection or "oil-canning".

2.5 FINISH

- A. Manufacturer's standard colors as selected by Architect.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions and proceed with Work in accordance with Section 01 40 00.

- B. Verify dimensions, tolerances and method of attachment with other Work.

### 3.2 INSTALLATION

#### A. Erection Tolerances:

1. Limit variations from plumb and level:
  - a. 1/8 inch (3 mm) in 10 feet (3 M) vertically.
  - b. 1/8 inch (3 mm) in 20 feet (6 M) horizontally.
2. Limit variations from theoretical locations: 1/4 inch (6 mm) for any member at any location.
3. Limit offsets in theoretical end-to-end and edge-to-edge alignment: 1/16 inch (2 mm) from flush surfaces not more than 2 inches (51 mm) apart or out-of-flush by more than 1/4 inch (6 mm).

- B. Install doors and hardware in accordance with manufacturer's printed instructions.

- C. Set units plumb, level and true to line, without warp or rack of frame.

- D. Anchor securely in place, allowing for required movement, including expansion and contraction.

- E. Separate dissimilar materials at contact points, including metal in contact with masonry or concrete surfaces, with bituminous paint or pre-formed separators to prevent contact and corrosion.

- F. Seal perimeter members as shown on manufacturer's installation instructions or as required for unique job conditions. Set other members with internal sealants and baffles as called for in manufacturer's installation instructions. Use sealants as recommended by sealant manufacturer.

- G. Coordinate installation of perimeter sealant and backing materials between assemblies and adjacent construction in accordance with requirements of Section 07 92 00.

- H. Glazing: Refer to requirements of Section 08 81 00. Utilize "anti-walk" edge blocking on all vertical edges of glazing.

### 3.3 ADJUSTING

- A. Test door operating functions. Adjust closing and latching speeds and other hardware in accordance with manufacturer's instructions to ensure smooth operation.

### 3.4 CLEANING

- A. Clean surfaces in compliance with manufacturer's recommendations; remove excess mastic, mastic smears, foreign materials and other unsightly marks.

- B. Clean metal surfaces exercising care to avoid damage.

END OF SECTION

## SECTION 08 71 00 – DOOR HARDWARE

### PART 1 - GENERAL

#### 1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

#### 1.2 SUMMARY

- A. This Section includes commercial door hardware for the following:
  - 1. Swinging doors.
  - 2. Other doors to the extent indicated.
- B. Door hardware includes, but is not necessarily limited to, the following:
  - 1. Mechanical door hardware.
  - 2. Cylinders specified for doors in other sections.
- C. Related Sections:
  - 1. Division 08 Section “Door Hardware Schedule”.
  - 2. Division 08 Section “Hollow Metal Doors and Frames”.
  - 3. Division 08 Section “Flush Wood Doors”.
  - 4. Division 08 Section “Aluminum-Framed Entrances and Storefronts”.
  - 5. Division 28 Section “Access Control”.
- D. Codes and References: Comply with the version year adopted by the Authority Having Jurisdiction.
  - 1. ANSI A117.1 - Accessible and Usable Buildings and Facilities.
  - 2. ICC/IBC - International Building Code.
  - 3. NFPA 70 - National Electrical Code.
  - 4. NFPA 80 - Fire Doors and Windows.
  - 5. NFPA 101 - Life Safety Code.
  - 6. NFPA 105 - Installation of Smoke Door Assemblies.
  - 7. State Building Codes, Local Amendments.
- E. Standards: All hardware specified herein shall comply with the following industry standards:
  - 1. ANSI/BHMA Certified Product Standards - A156 Series
  - 2. UL10C – Positive Pressure Fire Tests of Door Assemblies

### 1.3 SUBMITTALS

- A. Product Data: Manufacturer's product data sheets including installation details, material descriptions, dimensions of individual components and profiles, operational descriptions and finishes.
- B. Door Hardware Schedule: Prepared by or under the supervision of supplier, detailing fabrication and assembly of door hardware, as well as procedures and diagrams. Coordinate the final Door Hardware Schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.
  - 1. Format: Comply with scheduling sequence and vertical format in DHI's "Sequence and Format for the Hardware Schedule."
  - 2. Organization: Organize the Door Hardware Schedule into door hardware sets indicating complete designations of every item required for each door or opening. Organize door hardware sets in same order as in the Door Hardware Sets at the end of Part 3. Submittals that do not follow the same format and order as the Door Hardware Sets will be rejected and subject to resubmission.
  - 3. Content: Include the following information:
    - a. Type, style, function, size, label, hand, and finish of each door hardware item.
    - b. Manufacturer of each item.
    - c. Fastenings and other pertinent information.
    - d. Location of door hardware set, cross-referenced to Drawings, both on floor plans and in door and frame schedule.
    - e. Explanation of abbreviations, symbols, and codes contained in schedule.
    - f. Mounting locations for door hardware.
    - g. Door and frame sizes and materials.
    - h. Warranty information for each product.
  - 4. Submittal Sequence: Submit the final Door Hardware Schedule at earliest possible date, particularly where approval of the Door Hardware Schedule must precede fabrication of other work that is critical in the Project construction schedule. Include Product Data, Samples, Shop Drawings of other work affected by door hardware, and other information essential to the coordinated review of the Door Hardware Schedule.
- C. Keying Schedule: After a keying meeting with the owner has taken place prepare a separate keying schedule detailing final instructions. Submit the keying schedule in electronic format. Include keying system explanation, door numbers, key set symbols, hardware set numbers and special instructions. Owner must approve submitted keying schedule prior to the ordering of permanent cylinders/cores.
- D. Informational Submittals:
  - 1. Product Test Reports: Indicating compliance with cycle testing requirements, based on evaluation of comprehensive tests performed by manufacturer and witnessed by a qualified independent testing agency.

- E. Operating and Maintenance Manuals: Provide manufacturers operating and maintenance manuals for each item comprising the complete door hardware installation in quantity as required in Division 01, Closeout Submittals.

#### 1.4 QUALITY ASSURANCE

- A. Manufacturers Qualifications: Engage qualified manufacturers with a minimum 5 years of documented experience in producing hardware and equipment similar to that indicated for this Project and that have a proven record of successful in-service performance.
- B. Installer Qualifications: A minimum 3 years documented experience installing both standard and electrified door hardware similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.
- C. Door Hardware Supplier Qualifications: Experienced commercial door hardware distributors with a minimum 5 years documented experience supplying both mechanical and electromechanical hardware installations comparable in material, design, and extent to that indicated for this Project. Supplier recognized as a factory direct distributor by the manufacturers of the primary materials with a warehousing facility in Project's vicinity. Supplier to have on staff a certified Architectural Hardware Consultant (AHC) available during the course of the Work to consult with Contractor, Architect, and Owner concerning both standard and electromechanical door hardware and keying.
- D. Source Limitations: Obtain each type and variety of door hardware specified in this section from a single source unless otherwise indicated.
  - 1. Electrified modifications or enhancements made to a source manufacturer's product line by a secondary or third party source will not be accepted.
- E. Each unit to bear third party permanent label demonstrating compliance with the referenced standards.
- F. Keying Conference: Conduct conference to comply with requirements in Division 01 Section "Project Meetings." Keying conference to incorporate the following criteria into the final keying schedule document:
  - 1. Function of building, purpose of each area and degree of security required.
  - 2. Plans for existing and future key system expansion.
  - 3. Requirements for key control storage and software.
  - 4. Installation of permanent keys, cylinder cores and software.
  - 5. Address and requirements for delivery of keys.
- G. Pre-Submittal Conference: Conduct coordination conference in compliance with requirements in Division 01 Section "Project Meetings" with attendance by representative(s) of Supplier(s), Installer(s), and Contractor(s) to review proper methods and the procedures for receiving, handling, and installing door hardware.

1. Prior to installation of door hardware, conduct a project specific training meeting to instruct the installing contractors' personnel on the proper installation and adjustment of their respective products. Product training to be attended by installers of door hardware (including electromechanical hardware) for aluminum, hollow metal and wood doors. Training will include the use of installation manuals, hardware schedules, templates and physical product samples as required.
  2. Inspect and discuss electrical roughing-in, power supply connections, and other preparatory work performed by other trades.
  3. Review sequence of operation narratives for each unique access controlled opening.
  4. Review and finalize construction schedule and verify availability of materials.
  5. Review the required inspecting, testing, commissioning, and demonstration procedures
- H. At completion of installation, provide written documentation that components were applied to manufacturer's instructions and recommendations and according to approved schedule.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

- A. Inventory door hardware on receipt and provide secure lock-up and shelving for door hardware delivered to Project site. Do not store electronic access control hardware, software or accessories at Project site without prior authorization.
- B. Tag each item or package separately with identification related to the final Door Hardware Schedule, and include basic installation instructions with each item or package.
- C. Deliver, as applicable, permanent keys, cylinders, cores, access control credentials, software and related accessories directly to Owner via registered mail or overnight package service. Instructions for delivery to the Owner shall be established at the "Keying Conference".

#### 1.6 COORDINATION

- A. Templates: Obtain and distribute to the parties involved templates for doors, frames, and other work specified to be factory prepared for installing standard and electrified hardware. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing hardware to comply with indicated requirements.
- B. Door and Frame Preparation: Doors and corresponding frames are to be prepared, reinforced and pre-wired (if applicable) to receive the installation of the specified electrified, monitoring, signaling and access control system hardware without additional in-field modifications.

#### 1.7 WARRANTY

- A. General Warranty: Reference Division 01, General Requirements. Special warranties specified in this Article shall not deprive Owner of other rights Owner may have under other provisions of the Contract Documents and shall be in addition to, and run concurrent with, other warranties made by Contractor under requirements of the Contract Documents.

- B. Warranty Period: Written warranty, executed by manufacturer(s), agreeing to repair or replace components of standard and electrified door hardware that fails in materials or workmanship within specified warranty period after final acceptance by the Owner. Failures include, but are not limited to, the following:
  - 1. Structural failures including excessive deflection, cracking, or breakage.
  - 2. Faulty operation of the hardware.
  - 3. Deterioration of metals, metal finishes, and other materials beyond normal weathering.
  - 4. Electrical component defects and failures within the systems operation.
- C. Standard Warranty Period: One year from date of Substantial Completion, unless otherwise indicated.
- D. Special Warranty Periods:
  - 1. Seven years for heavy duty cylindrical (bored) locks and latches.
  - 2. Five years for exit hardware.
  - 3. Twenty five years for manual surface door closer bodies.
  - 4. Two years for electromechanical door hardware.

## 1.8 MAINTENANCE SERVICE

- A. Maintenance Tools and Instructions: Furnish a complete set of specialized tools and maintenance instructions as needed for Owner's continued adjustment, maintenance, and removal and replacement of door hardware.

## PART 2 - PRODUCTS

### 2.1 SCHEDULED DOOR HARDWARE

- A. General: Provide door hardware for each door to comply with requirements in Door Hardware Sets and each referenced section that products are to be supplied under.
- B. Designations: Requirements for quantity, item, size, finish or color, grade, function, and other distinctive qualities of each type of door hardware are indicated in the Door Hardware Sets at the end of Part 3. Products are identified by using door hardware designations, as follows:
  - 1. Named Manufacturer's Products: Product designation and manufacturer are listed for each door hardware type required for the purpose of establishing requirements. Manufacturers' names are abbreviated in the Door Hardware Schedule.
- C. Substitutions: Requests for substitution and product approval for inclusive mechanical and electromechanical door hardware in compliance with the specifications must be submitted in writing and in accordance with the procedures and time frames outlined in Division 01, Substitution Procedures. Approval of requests is at the discretion of the architect, owner, and their designated consultants.

## 2.2 HANGING DEVICES

- A. Hinges: ANSI/BHMA A156.1 certified butt hinges with number of hinge knuckles and other options as specified in the Door Hardware Sets.

1. Quantity: Provide the following hinge quantity:
  - a. Two Hinges: For doors with heights up to 60 inches.
  - b. Three Hinges: For doors with heights 61 to 90 inches.
  - c. Four Hinges: For doors with heights 91 to 120 inches.
  - d. For doors with heights more than 120 inches, provide 4 hinges, plus 1 hinge for every 30 inches of door height greater than 120 inches.
2. Hinge Size: Provide the following, unless otherwise indicated, with hinge widths sized for door thickness and clearances required:
  - a. Widths up to 3'0": 4-1/2" standard or heavy weight as specified.
  - b. Sizes from 3'1" to 4'0": 5" standard or heavy weight as specified.
3. Hinge Weight and Base Material: Unless otherwise indicated, provide the following:
  - a. Exterior Doors: Heavy weight, non-ferrous, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate standard weight.
  - b. Interior Doors: Standard weight, steel, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate heavy weight.
4. Hinge Options: Comply with the following:
  - a. Non-removable Pins: Provide set screw in hinge barrel that, when tightened into a groove in hinge pin, prevents removal of pin while door is closed; for the all out-swinging lockable doors.
5. Manufacturers:
  - a. Bommer Industries (BO).
  - b. Hager Companies (HA).
  - c. McKinney Products; ASSA ABLOY Architectural Door Accessories (MK).

## 2.3 DOOR OPERATING TRIM

- A. Door Push Plates and Pulls: ANSI/BHMA A156.6 certified door pushes and pulls of type and design specified in the Hardware Sets. Coordinate and provide proper width and height as required where conflicting hardware dictates.
1. Push/Pull Plates: Minimum .050 inch thick, size as indicated in hardware sets, with beveled edges, secured with exposed screws unless otherwise indicated.
  2. Door Pull and Push Bar Design: Size, shape, and material as indicated in the hardware sets. Minimum clearance of 2 1/2-inches from face of door unless otherwise indicated.



3. Offset Pull Design: Size, shape, and material as indicated in the hardware sets. Minimum clearance of 2 1/2-inches from face of door and offset of 90 degrees unless otherwise indicated.
4. Fasteners: Provide manufacturer's designated fastener type as indicated in Hardware Sets.
5. Manufacturers:
  - a. Burns Manufacturing (BU).
  - b. Hiawatha, Inc. (HI).
  - c. Rockwood Products; ASSA ABLOY Architectural Door Accessories (RO).

## 2.4 CYLINDERS AND KEYING

- A. General: Cylinder manufacturer to have minimum (10) years experience designing secured master key systems and have on record a published security keying system policy.
- B. Source Limitations: Obtain each type of keyed cylinder and keys from the same source manufacturer as locksets and exit devices, unless otherwise indicated.
- C. Cylinders: Original manufacturer cylinders complying with the following:
  1. Mortise Type: Threaded cylinders with rings and cams to suit hardware application.
  2. Rim Type: Cylinders with back plate, flat-type vertical or horizontal tailpiece, and raised trim ring.
  3. Bored-Lock Type: Cylinders with tailpieces to suit locks.
  4. Mortise and rim cylinder collars to be solid and recessed to allow the cylinder face to be flush and be free spinning with matching finishes.
  5. Keyway:
- D. Patented Cylinders: ANSI/BHMA A156.5, Grade 1, certified cylinders employing a utility patented and restricted keyway requiring the use of patented controlled keys. Provide bump resistant, fixed core cylinders as standard with solid recessed cylinder collars. Cylinders are to be factory keyed where permanent keying records will be established and maintained.
  1. Provide a 6 pin multi-level master key system comprised of patented controlled keys and security and high security cylinders operated by one (1) key of the highest level. Geographical exclusivity to be provided for all security and high security cylinders and UL437 certification where specified.
    - a. Level 1 Cylinders: Provide utility patented controlled keyway cylinders that are furnished with patented keys available only from authorized distribution.
  2. Manufacturers:
    - a. Sargent Manufacturing (SA) - Degree Series.
    - b. Corbin Russwin (RU) – Access 3 Series.
    - c. No Substitution.
- E. Keying System: Each type of lock and cylinders to be factory keyed.

1. Conduct specified "Keying Conference" to define and document keying system instructions and requirements.
2. Furnish factory cut, nickel-silver large bow permanently inscribed with a visual key control number as directed by Owner.
3. New System: Key locks to a new key system as directed by the Owner.

F. Key Quantity: Provide the following minimum number of keys:

1. Change Keys per Cylinder: Three (3).
2. Master Keys (per Master Key Level/Group): Five (5).

G. Key Registration List (Bitting List):

1. Provide keying transcript list to Owner's representative in the proper format for importing into key control software.
2. Provide transcript list in writing or electronic file as directed by the Owner.

H. Key Control Cabinet: Provide a key control system including envelopes, labels, and tags with self-locking key clips, receipt forms, 3-way visible card index, temporary markers, permanent markers, and standard metal cabinet. Key control cabinet shall have expansion capacity of 150% of the number of locks required for the project.

1. Manufacturers:

- a. Lund Equipment (LU).
- b. MMF Industries (MM).
- c. Telkee (TK).

## 2.5 MECHANICAL LOCKS AND LATCHING DEVICES

A. Cylindrical Locksets, Grade 1 (Heavy Duty): ANSI/BHMA A156.2, Series 4000, Grade 1 certified.

1. Furnish with solid cast levers, standard 2 3/4" backset, and 1/2" (3/4" at rated paired openings) throw brass or stainless steel latchbolt.
2. Locks are to be non-handed and fully field reversible.
3. Extended cycle test: Locks to have been cycle tested in ordinance with ANSI/BHMA 156.2 requirements to 2 million cycles.
4. Manufacturers:
  - a. Corbin Russwin Hardware (RU) – CL3300 Series.
  - b. Sargent Manufacturing (SA) – 10 Line.
  - c. Schlage (SC) – ND Series.

## 2.6 LOCK AND LATCH STRIKES

- A. Strikes: Provide manufacturer's standard strike with strike box for each latch or lock bolt, with curved lip extended to protect frame, finished to match door hardware set, unless otherwise indicated, and as follows:
  - 1. Flat-Lip Strikes: For locks with three-piece antifriction latchbolts, as recommended by manufacturer.
  - 2. Extra-Long-Lip Strikes: For locks used on frames with applied wood casing trim.
  - 3. Aluminum-Frame Strike Box: Provide manufacturer's special strike box fabricated for aluminum framing.
  - 4. Double-lipped strikes: For locks at double acting doors. Furnish with retractable stop for rescue hardware applications.
- B. Standards: Comply with the following:
  - 1. Strikes for Mortise Locks and Latches: BHMA A156.13.
  - 2. Strikes for Bored Locks and Latches: BHMA A156.2.
  - 3. Strikes for Auxiliary Deadlocks: BHMA A156.36.
  - 4. Dustproof Strikes: BHMA A156.16.

## 2.7 ELECTRIC STRIKES

- A. Standard Electric Strikes: Heavy duty, cylindrical and mortise lock electric strikes conforming to ANSI/BHMA A156.31, Grade 1, UL listed for both Burglary Resistance and for use on fire rated door assemblies. Stainless steel construction with dual interlocking plunger design tested to exceed 3000 lbs. of static strength and 350 ft-lbs. of dynamic strength. Strikes tested for a minimum 1 million operating cycles. Provide strikes with 12 or 24 VDC capability and supplied standard as fail-secure unless otherwise specified. Provide latchbolt and latchbolt strike monitoring indicating both the position of the latchbolt and locked condition of the strike where specified.
  - 1. Manufacturers:
    - a. Folger Adam EDC (FO).
    - b. HES (HS).
    - c. Security Door Controls (SD).

## 2.8 CONVENTIONAL EXIT DEVICES

- A. General Requirements: All exit devices specified herein shall meet or exceed the following criteria:
  - 1. At doors not requiring a fire rating, provide devices complying with NFPA 101 and listed and labeled for "Panic Hardware" according to UL305. Provide proper fasteners as required by manufacturer including sex nuts and bolts at openings specified in the Hardware Sets.

2. Where exit devices are required on fire rated doors, provide devices complying with NFPA 80 and with UL labeling indicating "Fire Exit Hardware". Provide devices with the proper fasteners for installation as tested and listed by UL. Consult manufacturer's catalog and template book for specific requirements.
  3. Except on fire rated doors, provide exit devices with hex key dogging device to hold the pushbar and latch in a retracted position. Provide optional keyed cylinder dogging on devices where specified in Hardware Sets.
  4. Devices must fit flat against the door face with no gap that permits unauthorized dogging of the push bar. The addition of filler strips is required in any case where the door light extends behind the device as in a full glass configuration.
  5. Lever Operating Trim: Where exit devices require lever trim, furnish manufacturer's heavy duty escutcheon trim with threaded studs for thru-bolts.
    - a. Lock Trim Design: As indicated in Hardware Sets, provide finishes and designs to match that of the specified locksets.
    - b. Where function of exit device requires a cylinder, provide a cylinder (Rim or Mortise) as specified in Hardware Sets.
  6. Vertical Rod Exit Devices: Where surface or concealed vertical rod exit devices are used at interior openings, provide as less bottom rod (LBR) unless otherwise indicated. Provide dust proof strikes where thermal pins are required to project into the floor.
  7. Narrow Stile Applications: At doors constructed with narrow stiles, or as specified in Hardware Sets, provide devices designed for maximum 2" wide stiles.
  8. Dummy Push Bar: Nonfunctioning push bar matching functional push bar.
  9. Extended cycle test: Devices to have been cycle tested in ordinance with ANSI/BHMA 156.3 requirements to 9 million cycles.
  10. Rail Sizing: Provide exit device rails factory sized for proper door width application.
  11. Through Bolt Installation: For exit devices and trim as indicated in Door Hardware Sets.
- B. Conventional Push Rail Exit Devices (Heavy Duty): ANSI/BHMA A156.3, Grade 1 certified panic and fire exit hardware devices furnished in the functions specified in the Hardware Sets. Exit device latch to be stainless steel, pullman type, with deadlock feature.
1. Manufacturers:
    - a. Corbin Russwin Hardware (RU) - ED4000 / ED5000 Series.
    - b. Sargent Manufacturing (SA) - 80 Series.
    - c. Von Duprin (VD) - 35A/98 XP Series.
- C. Tube Steel Removable Mullions: ANSI/BHMA A156.3 removable steel mullions with malleable-iron top and bottom retainers and a primed paint finish.

1. Provide keyed removable feature where specified in the Hardware Sets.
2. Provide stabilizers and mounting brackets as required.
3. Provide electrical quick connection wiring options as specified in the hardware sets.
4. Manufacturers:
  - a. Corbin Russwin Hardware (RU) - 700/900 Series.
  - b. Sargent Manufacturing (SA) - 980S Series.
  - c. Von Duprin (VD) - 9954 Series.

## 2.9 DOOR CLOSERS

### A. All door closers specified herein shall meet or exceed the following criteria:

1. General: Door closers to be from one manufacturer, matching in design and style, with the same type door preparations and templates regardless of application or spring size. Closers to be non-handed with full sized covers including installation and adjusting information on inside of cover.
2. Standards: Closers to comply with UL-10C for Positive Pressure Fire Test and be U.L. listed for use of fire rated doors.
3. Cycle Testing: Provide closers which have surpassed 15 million cycles in a test witnessed and verified by UL.
4. Size of Units: Comply with manufacturer's written recommendations for sizing of door closers depending on size of door, exposure to weather, and anticipated frequency of use. Where closers are indicated for doors required to be accessible to the physically handicapped, provide units complying with ANSI ICC/A117.1.
5. Closer Arms: Provide heavy duty, forged steel closer arms unless otherwise indicated in Hardware Sets.
6. Closers shall not be installed on exterior or corridor side of doors; where possible install closers on door for optimum aesthetics.
7. Closer Accessories: Provide door closer accessories including custom templates, special mounting brackets, spacers and drop plates as required for proper installation. Provide through-bolt and security type fasteners as specified in the hardware sets.

### B. Door Closers, Surface Mounted (Large Body Cast Iron): ANSI/BHMA A156.4, Grade 1 surface mounted, heavy duty door closers with complete spring power adjustment, sizes 1 thru 6; and fully operational adjustable according to door size, frequency of use, and opening force. Closers to be rack and pinion type, one piece cast iron body construction, with adjustable backcheck and separate non-critical valves for closing sweep and latch speed control.

1. Manufacturers:

- a. Corbin Russwin Hardware (RU) - DC8000 Series.
- b. Norton Door Controls (NO) – 9500 Series.
- c. Sargent Manufacturing (SA) - 281 Series.

## 2.10 ARCHITECTURAL TRIM

### A. Door Protective Trim

1. General: Door protective trim units to be of type and design as specified below or in the Hardware Sets.
2. Size: Fabricate protection plates (kick, armor, or mop) not more than 2" less than door width (LDW) on stop side of single doors and 1" LDW on stop side of pairs of doors, and not more than 1" less than door width on pull side. Coordinate and provide proper width and height as required where conflicting hardware dictates. Height to be as specified in the Hardware Sets.
3. Where plates are applied to fire rated doors with the top of the plate more than 16" above the bottom of the door, provide plates complying with NFPA 80. Consult manufacturer's catalog and template book for specific requirements for size and applications.
4. Protection Plates: ANSI/BHMA A156.6 certified protection plates (kick, armor, or mop), fabricated from the following:
  - a. Stainless Steel: 300 grade, .050-inch thick.
5. Options and fasteners: Provide manufacturer's designated fastener type as specified in the Hardware Sets. Provide countersunk screw holes.
6. Manufacturers:
  - a. Burns Manufacturing (BU).
  - b. Hager Companies (HA).
  - c. Hiawatha, Inc. (HI).
  - d. Rockwood Products; ASSA ABLOY Architectural Door Accessories (RO).

## 2.11 DOOR STOPS AND HOLDERS

- A. General: Door stops and holders to be of type and design as specified below or in the Hardware Sets.
- B. Door Stops and Bumpers: ANSI/BHMA A156.16, Grade 1 certified door stops and wall bumpers. Provide wall bumpers, either convex or concave types with anchorage as indicated, unless floor or other types of door stops are specified in Hardware Sets. Do not mount floor stops where they will impede traffic. Where floor or wall bumpers are not appropriate, provide overhead type stops and holders.
  1. Manufacturers:

- a. Burns Manufacturing (BU).
- b. Hager Companies (HA).
- c. Hiawatha, Inc. (HI).
- d. Rockwood Products; ASSA ABLOY Architectural Door Accessories (RO).

## 2.12 ARCHITECTURAL SEALS

- A. General: Thresholds, weatherstripping, and gasket seals to be of type and design as specified below or in the Hardware Sets. Provide continuous weatherstrip gasketing on exterior doors and provide smoke, light, or sound gasketing on interior doors where indicated. At exterior applications provide non-corrosive fasteners and elsewhere where indicated.
- B. Smoke Labeled Gasketing: Assemblies complying with NFPA 105 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for smoke control ratings indicated, based on testing according to UL 1784.
  1. Provide smoke labeled perimeter gasketing at all smoke labeled openings.
- C. Fire Labeled Gasketing: Assemblies complying with NFPA 80 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire ratings indicated, based on testing according to UL-10C.
  1. Provide intumescent seals as indicated to meet UL10C Standard for Positive Pressure Fire Tests of Door Assemblies, and NFPA 252, Standard Methods of Fire Tests of Door Assemblies.
- D. Sound-Rated Gasketing: Assemblies that are listed and labeled by a testing and inspecting agency, for sound ratings indicated.
- E. Replaceable Seal Strips: Provide only those units where resilient or flexible seal strips are easily replaceable and readily available from stocks maintained by manufacturer.
- F. Manufacturers:
  1. National Guard Products (NG).
  2. Pemko Products; ASSA ABLOY Architectural Door Accessories (PE).
  3. Reese Enterprises, Inc. (RE).

## 2.13 ELECTRONIC ACCESSORIES

- A. Push-Button Switches: Industrial grade momentary or alternate contact, back-lighted push buttons with stainless-steel switch enclosures. 12/24 VDC bi-color illumination suitable for either flush or surface mounting.
  1. Manufacturers:
    - a. Security Door Controls (SD) - 400 Series.
    - b. Securitron (SU) - PB Series.

- B. Door Position Switches: Door position magnetic reed contact switches specifically designed for use in commercial door applications. On recessed models the contact and magnetic housing snap-lock into a 1" diameter hole. Surface mounted models include wide gap distance design complete with armored flex cabling. Provide SPDT, N/O switches with optional Rare Earth Magnet installation on steel doors with flush top channels.

1. Manufacturers:

- a. Sargent Manufacturing (SA) – 3280 Series.
- b. Security Door Controls (SD) - DPS Series.
- c. Securitron (SU) - DPS Series.

- C. Power Supplies: Provide Nationally Recognized Testing Laboratory Listed 12VDC or 24VDC (field selectable) filtered and regulated power supplies. Include battery backup option with integral battery charging capability in addition to operating the DC load in event of line voltage failure. Provide the least number of units, at the appropriate amperage level, sufficient to exceed the required total draw for the specified electrified hardware and access control equipment.

1. Manufacturers:

- a. Corbin Russwin Hardware (RU) - BPS Series.
- b. Sargent Manufacturing (SA) - 3500 Series.
- c. Security Door Controls (SD) - 630 Series.
- d. Securitron (SU) - BPS Series.

## 2.14 FABRICATION

- A. Fasteners: Provide door hardware manufactured to comply with published templates generally prepared for machine, wood, and sheet metal screws. Provide screws according to manufacturers recognized installation standards for application intended.

## 2.15 FINISHES

- A. Standard: Designations used in the Hardware Sets and elsewhere indicate hardware finishes complying with ANSI/BHMA A156.18, including coordination with traditional U.S. finishes indicated by certain manufacturers for their products.
- B. Provide quality of finish, including thickness of plating or coating (if any), composition, hardness, and other qualities complying with manufacturer's standards, but in no case less than specified by referenced standards for the applicable units of hardware
- C. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.



## PART 3 - EXECUTION

### 3.1 EXAMINATION

- A. Examine scheduled openings, with Installer present, for compliance with requirements for installation tolerances, labeled fire door assembly construction, wall and floor construction, and other conditions affecting performance.
- B. Notify architect of any discrepancies or conflicts between the door schedule, door types, drawings and scheduled hardware. Proceed only after such discrepancies or conflicts have been resolved in writing.

### 3.2 PREPARATION

- A. Hollow Metal Doors and Frames: Comply with ANSI/DHI A115 series.
- B. Wood Doors: Comply with ANSI/DHI A115-W series.

### 3.3 INSTALLATION

- A. Install each item of mechanical and electromechanical hardware and access control equipment to comply with manufacturer's written instructions and according to specifications.
  - 1. Installers are to be trained and certified by the manufacturer on the proper installation and adjustment of fire, life safety, and security products including: hanging devices; locking devices; closing devices; and seals.
- B. Mounting Heights: Mount door hardware units at heights indicated in following applicable publications, unless specifically indicated or required to comply with governing regulations:
  - 1. Standard Steel Doors and Frames: DHI's "Recommended Locations for Architectural Hardware for Standard Steel Doors and Frames."
  - 2. Wood Doors: DHI WDHS.3, "Recommended Locations for Architectural Hardware for Wood Flush Doors."
  - 3. Where indicated to comply with accessibility requirements, comply with ANSI A117.1 "Accessibility Guidelines for Buildings and Facilities."
  - 4. Provide blocking in drywall partitions where wall stops or other wall mounted hardware is located.
- C. Retrofitting: Install door hardware to comply with manufacturer's published templates and written instructions. Where cutting and fitting are required to install door hardware onto or into surfaces that are later to be painted or finished in another way, coordinate removal, storage, and reinstallation of surface protective trim units with finishing work specified in Division 9 Sections. Do not install surface-mounted items until finishes have been completed on substrates involved.

- D. Thresholds: Set thresholds for exterior and acoustical doors in full bed of sealant complying with requirements specified in Division 7 Section "Joint Sealants."
- E. Storage: Provide a secure lock up for hardware delivered to the project but not yet installed. Control the handling and installation of hardware items so that the completion of the work will not be delayed by hardware losses before and after installation.

### 3.4 FIELD QUALITY CONTROL

- A. Field Inspection: Supplier will perform a final inspection of installed door hardware and state in report whether work complies with or deviates from requirements, including whether door hardware is properly installed, operating and adjusted.

### 3.5 ADJUSTING

- A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.

### 3.6 CLEANING AND PROTECTION

- A. Protect all hardware stored on construction site in a covered and dry place. Protect exposed hardware installed on doors during the construction phase. Install any and all hardware at the latest possible time frame.
- B. Clean adjacent surfaces soiled by door hardware installation.
- C. Clean operating items as necessary to restore proper finish. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of owner occupancy.

### 3.7 DEMONSTRATION

- A. Instruct Owner's maintenance personnel to adjust, operate, and maintain mechanical and electromechanical door hardware.

### 3.8 DOOR HARDWARE SETS

- A. The hardware sets represent the design intent and direction of the owner and architect. They are a guideline only and should not be considered a detailed hardware schedule. Discrepancies, conflicting hardware and missing items should be brought to the attention of the architect with corrections made prior to the bidding process. Omitted items not included in a hardware set should be scheduled with the appropriate additional hardware required for proper application and functionality.

B. The supplier is responsible for handing and sizing all products and providing the correct option for the appropriate door type and material where more than one is presented in the hardware sets. Quantities listed are for each pair of doors, or for each single door.

C. Manufacturer's Abbreviations:

1. MK - McKinney
2. SA - Sargent
3. HS - HES
4. RO - Rockwood
5. PE - Pemko
6. SU - Securitron
7. LU - Lund Equipment Co

### Hardware Sets

#### Set: 1.0

Doors: 100, 200

Description: Exterior

|                           |                              |       |    |
|---------------------------|------------------------------|-------|----|
| 6 Hinge                   | T4A3386 NRP 4-1/2" x 4-1/2"  | US32D | MK |
| 1 Removable Mullion       | L980S                        | PC    | SA |
| 1 Exit Device (storeroom) | DG1 16 AD8504 862 MK         | US32D | SA |
| 1 Rim Exit Device         | DG1 16 AD8510 862 MK         | US32D | SA |
| 1 Cylinder                | DG1 41 MK                    | US32D | SA |
| 2 Door Closer             | 281 CPS                      | EN    | SA |
| 2 Drop Plate              | 281D                         | EN    | SA |
| 2 Kit                     | 581-2                        | EN    | SA |
| 2 Door Stop               | 462                          | US2C  | RO |
| 1 Threshold               | 279x292AFGPK x Opening Width |       | PE |
| 2 Sweep                   | 345ANB x Door Width          |       | PE |
| 2 Astragal                | 305CN x Door Height          |       | PE |

Notes: Weatherstripping furnished by Aluminum Door Supplier.

#### Set: 2.0

Doors: 112

Description: Exterior

|                  |                              |       |    |
|------------------|------------------------------|-------|----|
| 3 Hinge          | T4A3386 NRP 4-1/2" x 4-1/2"  | US32D | MK |
| 1 Storeroom Lock | DG1 28 10G04 LL MK           | US26D | SA |
| 1 Door Closer    | TB 281 CPS                   | EN    | SA |
| 1 Kick Plate     | K1050 10" x 2" LDW CSK       | US32D | RO |
| 1 Door Stop      | 462                          | US2C  | RO |
| 1 Threshold      | 279x292AFGPK x Opening Width |       | PE |
| 1 Gasketing      | S773D (Head & Jambs)         |       | PE |

|   |            |                            |    |
|---|------------|----------------------------|----|
| 1 | Rain Guard | 346C x Overall Frame Width | PE |
| 1 | Sweep      | 345ANB x Door Width        | PE |

**Set: 3.0**

Doors: 115, 115E

Description: Exterior

|   |                         |                              |       |    |
|---|-------------------------|------------------------------|-------|----|
| 3 | Hinge                   | T4A3386 NRP 4-1/2" x 4-1/2"  | US32D | MK |
| 1 | Exit Device (storeroom) | DG1 16 8804 PSB MK           | US32D | SA |
| 1 | Door Closer             | TB 281 CPS                   | EN    | SA |
| 1 | Kick Plate              | K1050 10" x 2" LDW CSK       | US32D | RO |
| 1 | Door Stop               | 462                          | US2C  | RO |
| 1 | Threshold               | 279x292AFGPK x Opening Width |       | PE |
| 1 | Gasketing               | S773D (Head & Jambs)         |       | PE |
| 1 | Rain Guard              | 346C x Overall Frame Width   |       | PE |
| 1 | Sweep                   | 345ANB x Door Width          |       | PE |

**Set: 4.0**

Doors: 115A, 115B, 115C, 115D

Description: Exterior

|   |          |                                    |       |    |
|---|----------|------------------------------------|-------|----|
| 1 | Cylinder | DG1 34 MK or DG1 41 MK as required | US32D | SA |
|---|----------|------------------------------------|-------|----|

Notes: Balance of hardware furnished by Overhead Door Supplier.

Hardware supplier to verify the correct cam or tailpiece for the cylinder required.

**Set: 5.0**

Doors: 101

Description: Kitchen

|   |             |                        |       |    |
|---|-------------|------------------------|-------|----|
| 3 | Hinge       | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Office Lock | DG1 28 10G05 LL MK     | US26D | SA |
| 1 | Door Closer | 281 O                  | EN    | SA |
| 1 | Kick Plate  | K1050 10" x 2" LDW CSK | US32D | RO |
| 1 | Wall Stop   | 409                    | US32D | RO |
| 1 | Gasketing   | S88D (Head & Jambs)    |       | PE |

**Set: 6.0**

Doors: 102, 108

Description: Dispatch (Rated)

|   |             |                        |       |    |
|---|-------------|------------------------|-------|----|
| 3 | Hinge       | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Office Lock | DG1 28 10G05 LL MK     | US26D | SA |
| 1 | Door Closer | TB 281 P10             | EN    | SA |
| 1 | Kick Plate  | K1050 10" x 2" LDW CSK | US32D | RO |
| 1 | Wall Stop   | 409                    | US32D | RO |
| 1 | Threshold   | 271A x Opening Width   |       | PE |
| 1 | Gasketing   | S773D (Head & Jambs)   |       | PE |
| 1 | Door Bottom | 420APKL x Door Width   |       | PE |

**Set: 7.0**

Doors: 104, 203, 204

Description: Office

|   |             |                        |       |    |
|---|-------------|------------------------|-------|----|
| 3 | Hinge       | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Office Lock | DG1 28 10G05 LL MK     | US26D | SA |
| 1 | Wall Stop   | 409                    | US32D | RO |
| 3 | Silencer    | 608                    |       | RO |

**Set: 8.0**

Doors: 105, 110

Description: Fitness, Men

|   |             |                        |       |    |
|---|-------------|------------------------|-------|----|
| 3 | Hinge       | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Push Plate  | 70C-RKW                | US32D | RO |
| 1 | Pull Plate  | BF 107x70C             | US32D | RO |
| 1 | Door Closer | 281 O                  | EN    | SA |
| 1 | Kick Plate  | K1050 10" x 2" LDW CSK | US32D | RO |
| 1 | Wall Stop   | 409                    | US32D | RO |
| 3 | Silencer    | 608                    |       | RO |

**Set: 9.0**

Doors: 106

Description: Laundry

|   |                |                        |       |    |
|---|----------------|------------------------|-------|----|
| 3 | Hinge          | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Classroom Lock | DG1 28 10G37 LL MK     | US26D | SA |
| 1 | Wall Stop      | 409                    | US32D | RO |
| 3 | Silencer       | 608                    |       | RO |

**Set: 10.0**

Doors: 107

Description: Dorm

|   |               |                        |       |    |
|---|---------------|------------------------|-------|----|
| 3 | Hinge         | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Passage Latch | 28 10U15 LL            | US26D | SA |
| 1 | Door Closer   | 281 CPS                | EN    | SA |
| 1 | Kick Plate    | K1050 10" x 2" LDW CSK | US32D | RO |
| 1 | Gasketing     | S88D (Head & Jambs)    |       | PE |

**Set: 11.0**

Doors: 109

Description: Women

|   |             |                        |       |    |
|---|-------------|------------------------|-------|----|
| 3 | Hinge       | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Push Plate  | 70C-RKW                | US32D | RO |
| 1 | Pull Plate  | BF 107x70C             | US32D | RO |
| 1 | Door Closer | 281 P10                | EN    | SA |

|   |            |                        |       |    |
|---|------------|------------------------|-------|----|
| 1 | Kick Plate | K1050 10" x 2" LDW CSK | US32D | RO |
| 1 | Wall Stop  | 409                    | US32D | RO |
| 3 | Silencer   | 608                    |       | RO |

**Set: 12.0**

Doors: 111

Description: Server

|   |                |                        |       |    |
|---|----------------|------------------------|-------|----|
| 3 | Hinge          | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Storeroom Lock | DG1 28 10G04 LL MK     | US26D | SA |
| 1 | Door Closer    | 281 O                  | EN    | SA |
| 1 | Kick Plate     | K1050 10" x 2" LDW CSK | US32D | RO |
| 1 | Wall Stop      | 409                    | US32D | RO |
| 3 | Silencer       | 608                    |       | RO |

**Set: 13.0**

Doors: 114, 114A

Description: Bunker Gear

|   |                |                        |       |    |
|---|----------------|------------------------|-------|----|
| 3 | Hinge          | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Classroom Lock | DG1 28 10G37 LL MK     | US26D | SA |
| 1 | Door Closer    | TB 281 P10             | EN    | SA |
| 1 | Kick Plate     | K1050 10" x 2" LDW CSK | US32D | RO |
| 1 | Wall Stop      | 409                    | US32D | RO |
| 1 | Threshold      | 271A x Opening Width   |       | PE |
| 1 | Gasketing      | S773D (Head & Jambs)   |       | PE |
| 1 | Door Sweep     | 315CN x Door Width     |       | PE |

**Set: 14.0**

Doors: 201, 205

Description: Toilet

|   |              |                        |       |    |
|---|--------------|------------------------|-------|----|
| 3 | Hinge        | TA2714 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Privacy Lock | 28 10U65 LL            | US26D | SA |
| 1 | Wall Stop    | 409                    | US32D | RO |
| 3 | Silencer     | 608                    |       | RO |

**Set: 15.0**

Doors: 202

Description: Hallway

|   |                            |                            |       |    |
|---|----------------------------|----------------------------|-------|----|
| 3 | Hinge                      | TA2714 NRP 4-1/2" x 4-1/2" | US26D | MK |
| 1 | Storeroom Lock             | DG1 28 10G04 LL MK         | US26D | SA |
| 1 | Electric Strike            | 1500C                      | 630   | HS |
| 1 | SMART Pac Bridge Rectifier | 2005M3                     |       | HS |
| 1 | Door Closer                | 281 CPS                    | EN    | SA |
| 1 | Kick Plate                 | K1050 10" x 2" LDW CSK     | US32D | RO |
| 3 | Silencer                   | 608                        |       | RO |
| 1 | Push Button                | PB3ER                      |       | SU |

|   |                 |                                  |    |
|---|-----------------|----------------------------------|----|
| 1 | Timer           | TM-9                             | SU |
| 1 | Motion Sensor   | XMS                              | SU |
| 1 | Position Switch | DPS-W-BK or DPS-M-BK as required | SU |
| 1 | Power Supply    | BPS-24-1                         | SU |

Notes: Card reader furnished by Owner.

PB3ER push button will be located per architect instructions.

Operation: Door is normally closed and locked. When a valid credential is presented to the wall mounted card reader the electric strike will release and you can pull the door open. When the door comes back closed the electric strike will relock. The receptionist will also have a push button PB3ER located around the desk that can be pushed to release the electric strike. The The XMS motion sensor will be used by the access control system for the REX or request to exit switch. You can always turn the inside lever and exit out of the space.

**Set: 16.0**

Doors: Misc.

Description: Extra's for the project.

|    |             |        |    |
|----|-------------|--------|----|
| 50 | Keyblank    | DG1    | SA |
| 1  | Key Cabinet | 1201-A | LU |

Notes: Key blanks will be given directly to the owner or the owners representative.

END OF SECTION 087100

## SECTION 08 81 00 — GLASS AND GLAZING

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 WORK INCLUDED

- A. Glazing for hollow metal doors and frames.
- B. Glazing for aluminum frames.

#### 1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Hollow metal doors and frames.
- B. Aluminum entrances and storefronts.
- C. Aluminum window systems.

#### 1.4 SUBMITTALS

- A. Submit manufacturer's literature with material and performance descriptions for each type of glass, sealant and glazing accessories.
- B. Submit detailed shop drawings indicating locations, installation and sealing methods.
- C. Submit 12" x 12" physical samples of each type of tinted or wire glass and panel.
- D. Obtain approved shop drawings from hollow metal supplier, aluminum frame supplier, plastic laminate door supplier.
- E. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

#### 1.5 WARRANTY



- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.
- B. Warranted defects shall include but not necessarily be limited to water infiltration, air infiltration, glass failure due to improper sizing or installation, sealant failure.

#### 1.6 QUALITY ASSURANCE

- A. Glazing contractor shall have a minimum of 3 years experience in the installation of glazing products for projects of similar size and scope as this project.
- B. Each piece of glass shall bear manufacturer's label indicating type.

#### 1.7 DELIVERY, STORAGE AND HANDLING

- A. Do not deliver glass or panels to the jobsite until openings are ready for glazing.
- B. Deliver glass and panels in manufacturer's original protective packaging. Store in a dry, well ventilated area and take care to prevent condensation on the materials. Keep glass faces separated.

#### 1.8 MINIMUM COMPLIANCE STANDARDS

- A. SAFETY: Contractor shall be responsible for meeting all Federal and applicable code requirements for types and locations of glazing regardless of drawing indications. Comply with the current standards of the Consumer Products Safety Commission and Federal Standard 16 CFR 1201 Federal Architectural Glazing Materials Safety Standard.
- B. INSTALLATION: Comply with recommendations of Flat Glass Marketing Association – FGMA Glazing Manual.

### PART 2 - PRODUCTS

#### 2.1 GENERAL

- A. GLAZING SHEETS: Glazing materials shall conform to the highest qualities as specified in the following standards:
  - 1. Float glass: FS DD-G-451d and ASTM C1036.
  - 2. Float glass, heat strengthened: ASTM C1036 and ASTM C1048.
  - 3. Float glass, tempered: FS DD-G-1403B and ASTM C1036, ASTM C1048, ANSI Z97.1, and Consumer Product Safety Commission 16 CFR 1201.
  - 4. Wired glass: FS DD-G-451, ASTM C1036 and ANSI Z97.1. Misco diamond pattern.
  - 5. Insulating glass: ASTM C1036. Meet industry standards set by the Sealed Insulating Glass Manufacturers Association (SIGMA).
- B. MISCELLANEOUS
  - 1. Glazing sealants: FS TT-S-1543A (silicone rubber); FS TT-S-230 (synthetic rubber); FS TT-S-001657 (butyl rubber).
  - 2. Glazing tape: Architectural Aluminum Manufacturer's Association.

## 2.2 MANUFACTURERS

- A. GLASS:
  - 1. Guardian
  - 2. PPG Industries
  - 3. Pilkington.
- B. TEMPERING, LAMINATING AND HEAT STRENGTHENING:
  - 1. Oldcastle
  - 2. Trulite
- C. WIRE GLASS:
  - 1. Pilkington
  - 2. PPG Industries
- D. GLAZING TAPE:
  - 1. TREMCO – tape, shims, setting blocks, edge blocking.
- E. GLAZING SEALANT:
  - 1. TREMCO,
  - 2. General Electric.

## 2.3 MATERIALS: Types as indicated in the drawings.

- A. TEMPERED GLASS: 1/4" clear and solar tint float glass tempered by the vertical or horizontal process and meeting requirements of FS DD-G-1403B.
- B. WIRE GLASS: Shall be 1/4" thick. Polish plate glass reinforced with diamond pattern wire mesh No. 24 gauge minimum, with a mesh not larger than 1".
- C. HOLLOW METAL FRAME AND DOOR GLAZING SYSTEM:
  - 1. Glazing: 1/4" Tempered.
  - 2. Glazing tape: 1/8" x 3/8" x continuous preshimmed butyl tape; Tremco 440.
  - 3. Setting blocks: Neoprene or EPDM in minimum 4" lengths.
  - 4. Edge blocking: Neoprene or EPDM in minimum 4" lengths and sized to allow for 1/8" clear expansion at both vertical edges.
  - 5. Add sealant at exterior glazing.

## 2.4 INSULATING GLASS

- A. Solar Control Tinted Insulated Units
  - 1. Conformance: ASTM C 1172 and complying with testing requirements in CPSC 16CFR-1201 for Category II materials.
  - 2. Overall Thickness: 1 inch (25 mm)
  - 3. Outboard Lite: Bronze float glass.
    - a. Tinted Float Glass: ASTM C 1036, Type I, Class 2, Quality q3.
    - b. Glass Thickness: 1/4 inch (6 mm).
    - c. Heat Treatment: Fully Tempered, ASTM C 1048, Kind FT
  - 4. Interspace: 1/2 inch (12 mm) hermetically sealed air
  - 5. Inboard Lite: Clear float glass.
    - a. Clear Float Glass: ASTM C 1036, Type I, Class 1, Quality q3.
    - b. Glass Thickness: 1/4 inch (6 mm).
    - c. Heat Treatment: Fully Tempered, ASTM C 1048, Kind FT
  - 6. Sealant: Approved by glass manufacturer.

7. Nominal shading coefficient: 0.53

### PART 3 - EXECUTION

#### 3.1 INSTALLATION

- A. GENERAL: Install glass without warping, binding or stress. Allow for expansion and contraction of glass due to temperature changes. Do not install sealant with surfaces or ambient temperature below 40 degrees F.
- B. HOLOW METAL FRAMES AND DOORS:
  1. Ensure that finish painting of doors and frames is complete.
  2. Cut glazing tape to length and install against permanent stop, flush with face of stop.
  3. Place setting blocks at 1/4 points.
  4. Rest glass on setting blocks and press against stop for full contact and adhesion at perimeter.
  5. Place continuous glazing tape on opposite-face perimeter of glass in same manner described above. Install removable stop; avoid displacement of tape; and exert pressure on tape for full continuous contact.
  6. Knife trim excess of protruding tape (leave recessed for sealant at exterior glazing).
  7. Do not touch glass to metal.
- C. PLASTIC LAMINATE DOORS:
  1. Follow procedures specified above for non-rated doors. Metal stops provided by door manufacturer.
  2. Follow recommendations of door manufacturer for rated doors. Metal stops provided by door manufacturer.
- D. ALUMINUM FRAMES: Follow door and frame manufacturer's printed instructions for glazing gasketed systems. Provide watertight installation at exterior systems.

#### 3.2 CLEANING AND PROTECTION

- A. During glazing operations, provide sufficient stick-on safety labels or hang streamers on new glazing.
- B. Prior to project closeout, thoroughly clean all glazing inside and out with commercial glass cleaner.
- C. Reglaze any openings where glass is chipped, broken, scratched, pitted or stained.

END OF SECTION

## SECTION 08 88 56 - BULLET RESISTANT HOLE AND BACKER WINDOW ASSEMBLY

### PART 1 GENERAL

#### 1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.1 REFERENCE

- A. Underwriters Laboratory UL 752-Standard for Bullet Resisting Equipment & ASTM E119-98-Standard Test Methods for Fire Tests of Building Construction and Materials, NIJ Standard 0108.01-(National Institute of Justice) Standard for Ballistic Resistant Protective Materials, ASTM B 209/B 209M- Standard Specification for Aluminum and Aluminum Alloy Sheet and Plate, ASTM A 666-Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar.

#### 1.2 SUBMITTALS

- A. The following shall be submitted by the manufacturer in accordance with Sections 08 88 00 and any Special Contract Requirements and coordinate with Sections 01 33 00: Submit for approval prior to fabrication: samples, product data (including preparation, storage and installation methods), cuts & anchor spacing, reinforcement & location , product specifications, shop drawings, test reports (current UL Listing Verification & UL 752 Test Results as provided by Underwriters Laboratories), and printed data in sufficient detail to indicate compliance with the contract documents.
- B. Manufacturer's Instructions for installation and cleaning of TSS Bullet Transaction Window Assemblies. All required submittals shall be approved prior to installation.

#### 1.3 DESIGN PERFORMANCE

- A. Through the design, manufacturing techniques and material application the TSS Bullet Hole and Backer Design Transaction Window shall be of the "non-ricochet" type. This design is intended to permit the encapture and retention of an attacking projectile lessening the potential of a random injury or lateral penetration. This assembly shall provide single transaction positions utilizing the "Acrylic" backer configuration. This design shall employ an acrylic voice ports in transaction glazing to complete the "Hole and Backer" design. Each transaction position shall have a stainless steel dip tray as shown on the drawings. Components must be manufactured in strict accordance with the specifications, design and details. All vision panels shall be cut to size with all exposed edges polished. Necessary holes shall be pre drilled and tapped where required. Stainless Steel assembly screws and acrylic spacers shall be provided. Clear anodized angles and channels shall be provided. Anchor screws shall be provided by the installer.

- B. No field alterations to the construction of the units fabricated under the acceptable standards shall be allowed unless approved by the manufacturer and the architect. Standard manufacturing tolerances shall be +/- 1/16".
- C. Materials shall meet or exceed UL 752 requirements.

#### 1.4 QUALITY ASSURANCE

- A. Manufacturer shall be a Company that specializes in manufacturing products of the specified type with a minimum of five years experience. Installer shall be a Company that specializes in product type specified and Certified for the installation by the manufacturer. Manufacturer shall provide a Mock-up, if required, for evaluation of surface preparation and application workmanship and color/finish to the Architect for approval prior to start of work.

#### 1.5 DELIVERY, STORAGE & HANDLING

- A. Delivery the materials to the project with the manufacturer's UL Listed Labels intact and legible. Handle the materials with care to prevent damage. Store materials inside and under cover, stack flat and off floor. Project conditions (temperature, humidity, and ventilation) shall be within the maximum limit recommendations set by manufacturer. Do not install products that are under conditions outside these limits.

#### 1.6 WARRANTY

- A. All materials shall be warranted against defects for a period of 1 year after Substantial Completion Date. Certificates of manufacturer's standard limited warranty shall be provided at project completion.

### PART 2 PRODUCTS

#### 2.1 ACCEPTABLE MANUFACTURERS

- A. Products shall be manufactured by: Total Security Solutions, Inc, 170 National Park Drive, Fowlerville, MI 48836, 866-930-7807. Jim Richards, [info@demandtss.com](mailto:info@demandtss.com) . Web: [www.tssbulletproof.com](http://www.tssbulletproof.com) .
- B. Substitutions shall be in accordance with Section 01 25 00 Substitution Procedures.

#### 2.2 BULLET RESISTANT ARCH TRANSACTION WINDOW

- A. Product shall be: TSS HB Transaction Window: The Hole and Backer System consists of custom prefabricated bullet resistant panels with secure air passage as required for voice transmission. Aluminum frame, with a stainless steel base and counter mounted cash tray. All accessories for installation are included.
- B. Glazing Panels shall be:  
*Bullet Resistant Level 1*  
1 1/4" AR Coated Acrylic
- C. Cash tray: Counter mounted, Stainless Steel.
- D. Aluminum sections to be manufactured in accordance with ASTM B209, Extruded aluminum alloy 6063 T5 Anodized to match the existing décor and be free of sharp edges or burrs when in place. Glazing Channel: U-Channel specifically designed for securing transparencies tightly in place.

Angles and stops are only acceptable for top attachment.

- 1) Frame to be 18 ga. stainless steel. The bottom of the glazing to be capped with corresponding material on the frame (ie: stainless steel on stainless steel). Provide a shelf 2" thick with a recessed deal tray. The shelf to be full width of window, 12" deep, centered under the glazing and covered with a stainless steel 18 ga. #4 finish.
  - 2). Deal tray to be 18 ga. stainless steel, # 4 finish 16" x 10" from the outside edge of flanges with a clear opening .
- E. Product size shall be: As noted on drawings.

## PART 3 EXECUTION

### 3.1 PREPARATION

- A. Prior to installing the bullet resistive material, the contractor shall verify that all supports have been installed as required by the contract documents and architectural drawings, and approved shop/CAD drawings, if required. Installer shall notify architect of any unsatisfactory preparation that is responsibility of another installer.
- B. Clean and prepare all surfaces per manufacturers recommendations for achieving the best results for the substrate under the project conditions.

### 3.2 INSTALLATION

- A. Do not begin installation until openings have been verified and surfaces properly prepared in accordance with Drawings. Install in accordance with manufacturer's instructions and UL 752. Set all equipment plumb. All products shall be installed per installation instructions provided by the manufacturer, if warranty is to be issued.
- B. HB Transaction Window: shall arrive on site as a completed unit. Unit shall be installed in provided opening (wall/door), secured to structure (anchors by others).

### 3.3 POST APPLICATION

- A. HB Transaction Window: shall be installed in accordance with manufacturer's printed recommendations, including adhering to anchoring and finishing details.
- B. Inspection and Cleaning: Verify installation is complete and complies with manufacturer's requirements. Clean product and accessories, removing excess sealant, labels and protective covers.
- C. Touch-up, repair or replace damaged products before Substantial Completion.
- D. Product Warranty: Applicable warranty shall be issued to owner upon final release of completed project.

END OF SECTION

SECTION 09 20 00 — LATH AND PLASTER

PART 1 - GENERAL

1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.2 WORK INCLUDED

- A. Provide and install plaster lathing and accessories, three coat stucco system with floated finish as indicated in the drawings and specified herein.

1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Metal studs and gypsum sheathing.
- B. Insulation
- C. Dampproofing and waterproofing.
- D. Painting

1.4 SUBMITTALS

- A. Submit manufacturer's product data describing masonry mix, waterproofing additive, oriental stucco, lath and metal accessories.
- B. Submit mix design.
- C. Submit a 12" x 12" lath and plaster, metal edged sample for each type of plaster and each finish texture for Architect's approval.
- D. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.
- B. Warranted defects shall include but not necessarily be limited to cracking, water infiltration, loss of adhesion, spalling or discoloration.

## 1.6 QUALITY ASSURANCE

- A. Plaster contractor shall have a minimum of 3 years experience in the installation of plaster systems for projects of similar size and scope as this project.

## PART 2 - PRODUCTS

### 2.1 LATHING MATERIALS

- A. CHANNELS: 16 gauge, cold rolled pressed steel, galvanized. Flanges minimum 7/16" wide. Minimum weight shall be 475 pounds per 1000 lineal feet for 1-1/2" channels and 300 pounds per 1000 lineal feet for 3/4" channels.
- B. METAL LATH: Copper - alloy steel as follows: 1. Interior dry areas: Flat expanded diamond mesh at ceilings and soffits. Self-furring type at sheathed walls. Galvanized or painted finish. Minimum 3.4 lbs. per square yard. 2. Exterior and interior wet areas: Flat expanded diamond mesh at ceilings and soffits. Self-furring type at sheathed walls. Galvanized finish meeting requirements of FS QQ-Z-325C, Type 1. Minimum 3.4 lbs. per square yard.
- C. WIRE: Annealed galvanized metal wire. Minimum 18 gauge tie wire; minimum 8 gauge hanger wire.
- D. LATHING ACCESSORIES: Galvanized for interior dry areas; solid zinc alloy for exterior work.
  - 1. Casino Beads: MUcor #66 with expanded metal flange, 26 gauge.
  - 2. Corner Beads: U.S.G. NO. 4-R, or approved equal, 26 gauge expansion type.
  - 3. Control Joints: No. 75 per U.S.G.
  - 4. Expansion Joints: No. 40 zinc expansion flange type per Keene.
- E. Wire clips for attachment of furring channels to runner channels shall be formed hairpin clips, 8 gauge galvanized soft steel wire.
- F. METAL STUDS: Provided and installed under another section of these specifications

### 2.2 PLASTERING MATERIALS

- A. REINFORCING: Alkaline resistant fiberglass strands, 1/2".
- B. PORTLAND CEMENT: ASTM C-150, Type I, white for finish coat.
- C. SAND: ASTM C-144, red torpedo sand for scratch and brown coats.
- D. MASONRY MIX: Pre-mixed dry masonry mortar mix meeting requirements of A.S.T.M. C-91, Type N, as manufactured by Trinity, Ideal, TXI or Lonestar.



- E. FINISH: "Oriental Exterior" stucco as manufactured by United States Gypsum Company, or approved equal. Colors shall be selected by the Architect. Deliver to job in manufacturer's original packages, with labels intact, seals unbroken. Prepare stucco finish coat for application by mixing with water only.
- F. WATERPROOFING: "Hvdrocide" as manufactured by Sonneborn.
- G. WATER: Clean, potable and free from any amounts of mineral and organic substances that would affect set of Plaster.
- H. No asbestos or admixtures.

## 2.3 MIXES

- A. SCRATCH COAT:
  - 1. 1 sack Portland cement.
  - 2. 2 sacks masonry mix.
  - 3. 9 cu. ft. sharp sand
  - 4. 1-1/2 lbs. fiberglass strands.
- B. BROWN COAT:
  - 1. 1 sack Portland cement.
  - 2. 2 sacks masonry mix.
  - 3. 10 cu. ft. sharp sand.
  - 4. 1-1/2 lbs. fiberglass strands
  - 5. Integral waterproofing per manufacturer's recommendations
- C. Plaster mixes shall comply with ASTM C926.

## PART 3 - EXECUTION

### 3.1 GENERAL

- A. Install furring, lathing, and all plaster work level and plumb, true and rigid. Ensure that all work to be concealed by plaster has been completed and inspected prior to beginning plaster work.
- B. Obtain access panels, frames, or other built-in items from the appropriate trades before beginning plaster work.
- C. Exercise precautions to prevent damage to work of other crafts. Plaster droppings on glass or aluminum surfaces shall be immediately removed with clean water and soft cloths.

### 3.2 EXTERIOR PLASTER (STUCCO)

- A. All exterior work and interior wet areas to have Portland Cement Finish "Oriental Exterior" sand finish 3/16" to 1/4" thick finished coat, texture to match existing as approved by Architect. Color and texture shall be uniform.

- B. Exterior plaster and interior wet areas to include integral waterproofing, galvanized lath, and pure zinc accessories.

### 3.3 SUSPENDED CEILING & SOFFIT INSTALLATION

- A. Install suspended ceilings or exterior soffits in indicated locations. Unless otherwise indicated, suspension system shall consist of 1-1/2" runner channels, 3/4" furring channels, suspended from structure above by galvanized hanger wires.
- B. Space hanger wire 48" maximum in either direction for interior ceilings. Maximum 36" in either direction for stucco soffits.
- C. Space runner channels 48" on center maximum for interior plaster ceilings, supported from resilient hangers; space 36" on center maximum for plaster soffits
- D. Wrap each hanger wire twice around channels; secure by at least 3 turns around itself. Space furring channels 12-1/2" on center maximum and at right angles to runners.
- E. Secure to runner channels with wire clips or saddle tied with 2 strands of 16 gauge tie wire giving wire ends 3 twists.
- F. Isolate penetrations (such as light fixtures) with control joints and reinforce with furring channels

### 3.4 METAL LATH & ACCESSORIES

- A. Apply metal lath to form true surfaces, straight, without sags or buckles, with long dimension at right angles to direction of supports. Secure lath to supports at 6" intervals. Secure side laps on ceilings to supports; tie at 6" intervals between supports. Lap lath at sides at least 1/2". Lap lath at ends at least 1", stagger laps; and locate only over supports. Break end joints of lath on alternate sheets of lath. Lath ties shall have a minimum of three complete turns.
- B. Provide corner beads on external plaster corners and where indicated. Corner beads shall be single lengths where length of corner does not exceed standard stock lengths. Miter or cope beads at corners; fasten securely with tie wire spaced 8" maximum; stagger on two wings.
- C. Install casing beads (stops) where plaster abuts other surfaces, at edges of plaster panels, and elsewhere as indicated. Set casing beads level, true to line. Install casing beads in lengths as long as practicable, with joints in straight runs aligned with suitable formed splices. Secure casing beads to metal lath with tie wire spaced 8" maximum.
- D. Provide expansion joints in exterior and interior plaster as shown. Expansion joints shall be in single lengths where possible. Secure expansion joints to metal lath with tie wire; space ties or nail anchors not over 8" apart.
- E. Provide control joints in exterior and interior plaster between expansion joints so that no panel dimension exceeds 12' or 120 square feet of area.

### 3.5 APPLICATION

- A. Maintain temperature of at least 40 degrees F. in building prior to plaster application, until it is dry. Plaster shall be three coat work on all bases. Plaster thickness from plaster base to finished plaster surface shall be as noted on drawings but shall be a minimum thickness of 3/4" at its thinnest point. Do not

combine scratch and brown coats. No irregularities shall show in finished surface, such as "cat faces", streaks, waviness, trowel, float or brush marks. Finished surfaces shall be true, uniform in texture and finish.

- B. Apply scratch coat with sufficient pressure to force mortar through mesh and key firmly to lath. Scratch to form rough surfaces. Apply brown coat 48 hours after scratch coat has set; bring out to grounds; straighten to true surface with rod, darby; leave rough, and ready for finish coat.
- C. Apply finish over base coat which has been wetted evenly by brushing or spraying. Apply finish coat not sooner than seven (7) days after brown coat. Provide light sand finish per approved sample.
- D. Keep plaster moist for the curing period between coats. Limit thickness *of* scratch, and brown coat to maximum 3/8" each.

### 3.6 PLASTER CUTTING & PATCHING

- A. Execute after other work is in place, and after painter has applied priming coat. Thoroughly rake out, or cut out, moisten and fill with finishing material. Float finish with adjoining work. Point up around fixtures, outlet boxes, switch plates, fittings, piping and other appliances abutting or extending into plastering.

### 3.7 FINISH PLASTER PROTECTION

- A. Provide protection against damage for finished plaster work. Protect plastering from freezing or premature drying. Execute no plastering work in cold weather, except where work is adequately protected and proper temperatures are maintained to prevent freezing.

END OF SECTION

## SECTION 09 21 16 — INTERIOR DRYWALL SYSTEMS

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 WORK INCLUDED

- A. Provide and install acoustical batt insulation within interior drywall partitions.
- B. Provide and install all interior drywall systems including light gauge metal studs and tracks, horizontal bridging, gypsum wall board and finishing systems, suspended gypsum board ceilings and soffits, furred gypsum board.
- C. Provide and install troweled firestopping system at drywall ceiling and wall penetrations at rated walls.
- D. Provide and install specified corner guards at each wall corner.

#### 1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Painting
- B. Door frames
- C. Carpentry (wood blocking)
- D. Plaster on metal studs
- E. Mechanical, electrical and plumbing penetrations in rated drywall systems.

#### 1.4 SUBMITTALS

- A. Submit manufacturer's product data describing all materials.
- B. Submit gypsum board finish schedule indicating level of finish proposed per each area. Finish levels shall be levels 1 through 4 as specified herein and defined by "Recommended Specification: Levels of

Gypsum Board Finish" as jointly published by AWCI, CISA, GA, and PDCA. Submit copy of publication with finish schedule.

- C. Submit manufacturers detail drawings and detailed installation methods for fire rated penetrations and filling of voids with specified firestopping system. Submit only those systems applicable to this project.
- D. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

## 1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.
- B. Warranted defects shall include but not necessarily be limited to cracking, joint tape delamination or tearing, dimpling at fastener heads, bowing or warping of wall board, cracking at metal accessories, acoustical sealant failure.

## 1.6 DELIVERY, STORAGE AND HANDLING

- A. All materials shall be delivered in manufacturer's original packaging and stored flat in a covered, dry area providing protection from damage and exposure to the elements.
- B. Damaged or deteriorated materials shall be removed from the premises.
- C. During cold weather installation of gypsum panels and joint finishing, temperatures within the building shall be maintained within the range of 50 degrees to 80 degrees F. Adequate ventilation shall be provided to carry off excess moisture.
- D. Steel framing and related accessories shall be stored and handled in accordance with AISI's "Code of Standard Practice"

## PART 2 - PRODUCTS

### 2.1 MANUFACTURERS

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>A. <u>Drywall Framing:</u><ul style="list-style-type: none"><li>1. ClarkDietrich Building Systems</li></ul></li><li>B. <u>Gypsum Board and Related Accessories:</u><ul style="list-style-type: none"><li>1. United States Gypsum Co.</li><li>2. National Gypsum Co.</li><li>3. Georgia Pacific</li><li>4. Temple Inland</li><li>5. James Hardie</li></ul></li><li>C. <u>Acoustical Batts:</u><ul style="list-style-type: none"><li>1. Owens-Corning</li><li>2. Certainteed</li><li>3. Manville</li></ul></li></ul> | <ul style="list-style-type: none"><li>D. <u>Acoustical Sealant:</u><ul style="list-style-type: none"><li>1. TREMCO</li><li>2. Ohio Sealants, Inc.</li></ul></li><li>E. <u>Specialty Trims:</u><ul style="list-style-type: none"><li>1. Fry Reglet Corp.</li><li>2. MM Systems Corp.</li></ul></li><li>F. <u>Corner Guards:</u><ul style="list-style-type: none"><li>1. WallProtex, (877) 880-8115</li></ul></li></ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 2.2 FRAMING: Comply with ASTM C645-09 for conditions indicated.

1. Steel Sheet Components: Comply with ASTM C645-09 requirements for metal unless otherwise indicated.
  2. Protective coating: Comply with ASTM C645-09; roll formed from hot dipped galvanized steel; complying with ASTM A1003/A1003M and ASTM A653/A653M G40 (Z120) or having a coating that provides equivalent corrosion resistance. A40 galvanized products are not acceptable.
- A. METAL STUDS: 25 gauge galvanized roll formed, screw channel type studs with minimum 5/16 inch flanges and 1-1/4 inch legs. Provide widths of 1-5/8 inch, 2-1/2 inch, 3-5/8 inch, 4 inches and 6 inches as indicated in the drawings. Provide conduit punchouts at 24" o.c.
1. "EQ" (Equivalent Gauge Thickness) Steel Studs and Runners: Members that can show certified third party testing with gypsum board in accordance with ICC ES AC86-2010 (approved February 2010 Effective March 1, 2010) need not meet the minimum thickness limitation or minimum section properties set forth in ASTM C645-09.
  2. Non-structural Studs: Cold-formed galvanized steel C-studs, ClarkDietrich Building Systems Pro STUD drywall studs as per ASTM C645-09 for conditions indicated below:
    - a. Flange Size: 1 1/4 inch (32mm)
    - b. Web Depth: As specified on drawings, 1-5/8 inches (41 mm) 2-1/2 inches (64 mm) 3-5/8 inches (92 mm) 4 inches (102 mm) 6 inches (152 mm).
    - c. Member Description: ProSTUD 25 (25ga equivalent drywall stud) 70ksi Minimum Thickness: 0.0150 inches (0.3810mm) Minimum Design Thickness: 0.0158 inches (0.4013mm)
    - d. Member Description: ProSTUD 22 (22ga equivalent drywall stud) 70ksi Minimum Thickness: 0.0179 inches (0.4547mm) Minimum Design Thickness: 0.0188 inches (0.4775mm)
    - e. Member Description: ProSTUD 20 (20ga equivalent drywall stud) 65ksi Minimum Thickness: 0.0220 inches (0.5588mm) Minimum Design Thickness: 0.0232 inches (0.5893mm)
- B. RUNNER CHANNELS: Provide 25 gauge galvanized channels with minimum 1-1/4 inch flanges with hemmed edges, in widths to accommodate stud sizes.
1. Non structural Track: Cold-Formed galvanized steel runner tracks, ClarkDietrich Building Systems ProTRAK drywall track in conformance with ASTM C645-09 for conditions indicated below:
    - a. Flange Size: 1 1/4 inch (32mm)
    - b. Web Depth: Track web to match stud web size.
    - c. Minimum Material Thickness: Track thickness to match wall stud thickness or as per design.
- C. FURRING CHANNELS: Provide 20 gauge galvanized "hat" channels with face width of 1-1/4 inches, depth of 7/8 inches, and back Width of 2-9/16 inches minimum, hemmed edges.
- D. CEILING SUSPENSION: Provide 16 gauge galvanized channels, 3/4" x 1/2" and 11/2" or 2" x 17/32".
1. Firestop tracks: Top runner manufactured to allow partition heads to expand and contract with movement of the structure while maintaining continuity of fire-resistance rated assembly

indicated; in thickness not less than indicated for studs and in width to accommodate depth of studs.

- a. Basis of Design Product: Subject to compliance with requirements, provide ClarkDietrich Building Systems; MaxTrak or an equivalent product.

## 2.3 ACCESSORIES

- A. CORNER BEADS: 26 gauge galvanized beaded angle with 1-1/4" legs.
- B. Channel Bridging and Bracing: Steel, 0.0538-inch (1.37mm) minimum base metal thickness, with minimum 1/2 inch (13mm) wide flanges.
  - a. Basis of Design Product: Subject to compliance with requirements, provide ClarkDietrich Building Systems; Spazzer 9200 Bridging and Spacing Bar, or an equivalent product.
  - b. Depth: As indicated on drawings, 7/8 inch by 7/8 inch by 50 inches.
  - c. Install at 48" o.c. horizontally.
2. Backing Plate: Proprietary fire-resistance treated blocking and bracing in width indicated.
  - a. Basis of Design Product: Subject to compliance with requirements, provide ClarkDietrich Building Systems; Danback Fire-treated wood backing plate or an equivalent product.
- C. EDGE TRIM: 26 gauge galvanized steel "J" mould and angle with continuous bead. ClarkDietrich Building Systems 200.A and 200.B.
- D. WIRE: 9 gauge galvanized hanger wire and 16 gauge galvanized be wire.
- E. SCREWS: Bugel head Type "S" self tapping drywall screws in lengths recommended by wallboard manufacturer. USG "Super-Tite".
- F. CONTROL JOINTS: Roll formed zinc with 1/4" open joint, and perforated flanges. Provide with fireseal backing at rated systems. ClarkDietrich Building Systems No. 093.
- G. JOINT ADHESIVE: Premixed water based compound. USG taping joint compound.
- H. LAMINATING ADHESIVE: Durabond sheetrock setting-type for double-layer application and column fireproofing.
- I. JOINT REINFORCING: Center creased paper tape equal to "Perf-A-Tape".
- J. TROWELED FIRESTOPPING
  1. System Type: A combination of glass fiber or mineral wool insulation packing material with troweled-on application of sealing compound.
  2. Sealing Compound: Red tinted compound job mixed with water providing protection from heat (to temperatures of 1850 degrees F), smoke, toxic gas, fire and water. "Sta-Smooth FS 90 Fire-Shield Compound Fire and Smoke Stop" as manufactured by National Gypsum Co. or approved equivalent by Domtar Gypsum, Inc.
  3. Approvals:
    - a. Rated as noncombustible as defined by NFPA Standard 220 when tested in accordance with ASTM E 136 at Underwriters Laboratories.
    - b. Meet all requirements of ASTM E 814 and UL 1479: Fire tests of through penetration fire stops.
- K. CORNER GUARDS: Textured Vinyl Corner Guards 3" by WallProtex. 4' lengths, taped. Color as selected by Architect.

## 2.4 WALLBOARD

- A. TYPICAL: 5/8" thick x 48" wide paper-faced gypsum panels, tapered long edges, lengths as required. U.L. listed and conforming to ASTM C-1396/C1396M-09a Standard Specification for Gypsum Board, Type X. USG fire code.
- B. WATER RESISTANT: 5/8" thick x 48" wide U.L. listed, Type X board with chemically treated face paper and water resistant gypsum core. Comply with ASTM C-1396/C1396M-09a Standard Specification for Gypsum Board.
- C. HIGH IMPACT: 5/8" thick x 48" wide, length as required. U.L. listed, "Fiberock Interior Panel Abuse Resistant" by USG or equal.

## 2.5 TILE BACKER BOARD

- A. 5/8" thick cement board formed of aggregated Portland cement slurry with polymer-coated, glass-fiber mesh. "Durock" as manufactured by United States Gypsum Co or approved equivalent.

# PART 3 - EXECUTION

## 3.1 PARTITION INSTALLATION

- A. STUD SYSTEM ERECTION: Attach metal runners at floor and to structural elements with suitable fasteners spaced maximum 24" o.c. Position studs vertically, engaging floor track and runner at ceiling or structure. Place studs in direct contact with all door frame jambs, abutting partitions, partition corners and existing construction elements.
- B. Anchor all studs adjacent to door and window frames, partition intersections, and corners to ceiling and floor runner flanges. Securely anchor studs to jamb and head anchor clips of door or side-light frames by screw attachment. Over door and side-light frames, install horizontal runner with a web-flange bend at each end, and secure with one positive attachment per flange.
- C. Install diagonal stud bracing above ceiling at strike side of door jambs and at other locations as indicated in the drawings. Secure to structure.
- D. Follow stud manufacturer's recommendations for all framing construction and fastening.

## 3.2 WALL PANEL ERECTION

- A. Apply gypsum panels vertically or horizontally. Position all edges over studs for vertical application; all ends over studs for horizontal application. Use maximum practical lengths to eliminate end joints. Fit ends and edges closely together. Stagger joints on opposite side of partition.
- B. For single-layer vertical application of gypsum panels, space screws 12" o.c. in field of panels and 8" o.c. staggered along vertical abutting edges. For horizontal panel application, space screws 12" o.c. in field and along abutting end joints.
- C. For double-layer screw attachment, space screws 16" o.c. for both layers. Apply both layers of gypsum panels vertically with joints in face layer offset from base layer joints. For 5/8" panels, use 1" screws for base layer and 1-5/8" screws for face layers. For 1/2" panels, use 7/8" screws for base layer and 1-5/16" screws for face layer.



### 3.3 CHASE WALL ERECTION

- A. Align two parallel rows of floor and ceiling runners spaced as indicated in the drawings. Attach to concrete slabs with powder actuated anchors 24" o.c. and to suspended ceiling tees or structure with suitable fasteners 24" o.c.
- B. Position metal studs vertically in runners, 16" o.c., with flanges in the same direction and with studs on opposite sides of chase directly across from each other. Anchor all studs to floor and ceiling runner flanges with U.S.G. Metal Lock Fastener tool.
- C. Cut gypsum panel bracing to be placed between rows of studs, 12" high by chase wall width. Space braces 48" o.c. vertically and attach to stud webs with screw fasteners. 2-1/2" metal studs may be used in lieu of gypsum panels. Anchor web at each end of metal brace to stud web with two 3/8" pan head screws.

### 3.4 CEILING FRAMING

- A. GRILLAGE ERECTION: Space 8 gauge hanger wires 48" o.c. along carrying channels and within 6" of ends of carrying-channel runs. Wrap hanger around and through beams or joists. Install 1-1/2" carrying channels at 24" o.c. Position channels for proper ceiling height, level and secure with hanger wire saddle-bed along channel. Provide 1" clearance between runners and abutting walls and partitions. Secure furring to carrying channels with clips or saddle-tie to support. Overlap splices at least 8" and securely wire-tie each end with double-strand 16 gauge tie wire.
- B. Erect metal furring channels at right angles to 1-1/2" carrying channels or main support members Space furring (16") o.c. and within 6" of walls. Provide 1" clearance between furring ends and abutting walls and partitions. Secure furring to carrying channels with clips or saddle-tie to supports with double strand 16 gauge tie wire. Overlap splices at least 8" and securely wire-tie each end with double-strand 16 gauge tie wire.
- C. At light troffers or any openings that interrupt the carrying or furring channels, install additional cross reinforcing to restore lateral stability of grillage.
- D. At rated ceilings meet all requirements of selected U.L. Design No.
- E. METAL STUD CEILING FRAMING OPTION: Attach runners at ceiling height through gypsum panels to each partition stud with two screws. Insert metal studs in runners and attach each end with one 3/8" pan head screw. Install 1-5/8" stud cross-bracing over stud framing, space 48" o.c. and attach to each framing stud with two 3/8" pan head screws. At hangers, install 12" long stud section for box reinforcing or lap studs 12" and secure each end with two 3/8" pan head screws. At light troffers or any openings that interrupt the ceiling, install additional cross reinforcing to maintain structural integrity of framing.
- F. GYPSUM PANEL ERECTION: Apply gypsum panels of maximum practical length with long dimension at right angles to furring channels. Position end joints over channel web and stagger in adjacent rows. Fit ends and edges closely. Fasten panels to channels with 1", Type S screws, spaced 8" o.c. in field of panels and 8" along ends and edges.

### 3.5 EXTERIOR WALLS: Reference Section 05 41 00.

### 3.6 ACOUSTICAL BATTS

- A. Install unfaced full thickness acoustical fiberglass batts between studs at partitions as scheduled on the drawings. Fit batts tight to studs, tight to floor and head tracks and tight to one another. Batts shall run full height of partition unless indicated otherwise in the drawings.

### 3.7 ACOUSTICAL SEALANT

- A. Install continuous bead of sealant at bottom tracks at drywall partitions.
- B. Install vinyl foam double stick tape and sealant where head track terminates at ceiling.
- C. See drawings for additional locations.

### 3.8 ACCESSORY APPLICATION

- A. JOINT SYSTEM: Finish all face panel joints and corners with U.S.G. Joint System installed according to manufacturer's directions.
  - 1. Mix joint cement in strict accordance with manufacturers directions.
  - 2. Butter cement into joints filling them evenly and fully.
  - 3. Center tape and press down into cement leaving sufficient cement under tape for proper bond. Cover with thin coat of cement to fill recess between tape and board to bring material flush with surface.
  - 4. Face panels shall be cut fit around all wall outlets and switch boxes, utility lines, etc. All voids and cracks, occurring around all openings in board shall be taped and covered with joint cement.
- B. LAMINATING ADHESIVE: Spread to provide 1/2" adhesive beads 4-1/2" o.c. for full sheet lamination. For strip lamination, apply adhesive in vertical strips of four 1/2" beads, 1-1/2" to 2" o.c. Space strips 24" o.c.
- C. CORNER BEAD: Reinforce all vertical and horizontal exterior corners with corner bead fastened with 9/16" rosin-coated staples 9" o.c. on both flanges along entire length of bead.
- D. METAL TRIM: At exposed edges of board or where board terminates against other materials, apply metal trim over panel edge and fasten with screws.
- E. SCREWS: Power-drive at least 3/8" from edges or ends of panel to provide uniform dimple of 1/32" deep.
- F. CONTROL JOINTS: Cut panel at joint and back with double framing members. Attach control joint to face layer with 9/16" rosin-coated staples spaced 6" o.c. on both flanges along entire length of joint. At rated walls, provide fireseal behind joint. Provide joints at 25' maximum or as otherwise indicated in the drawings.
- G. CORNER GUARDS: Install as per manufacturer's recommendations. Double sided adhesive tape factory applied to corner guard.

### 3.9 TROWELED FIRESTOPPING:

- A. General: Install systems in complete accordance with manufacturers printed instructions and approved submittal for the required fire rating of the particular condition. Install firestopping systems at all penetrations and voids in all rated drywall ceilings and walls.

- B. Through-penetrations. Ensure that pipe, conduit, duct, cables or other penetration element is rigidly supported by drywall framing on both sides of wall or ceiling assembly. Oversize opening in wall board to allow for required opening size and thickness of packing material in accordance with system and rating requirements. Install packing material in accordance with system requirements and compressed to allow for required thickness of sealing material. Trowel red-tint sealing material into void (same thickness as gypsum board) and smooth flush with both faces of drywall. Provide additional layer(s) of gypsum board around penetration where necessary to achieve required minimum thickness of sealing material.
  - C. Void-filling: For voids such as intersection of walls and smooth or corrugated deck, pack void with compressed packing material and trowel red-tint sealing material into void (same thickness as gypsum board) and smooth flush with both faces of drywall. Provide additional layer(s) of gypsum board around penetration where necessary to achieve required minimum thickness of sealing material.
- 3.10 WOOD BLOCKING: Coordinate with project carpenter to ensure installation of fire retardant wood blocking between studs for mounting casework, millwork, toilet partitions, drinking fountains and other equipment.
- 3.11 FINISHING SCHEDULE: Follow published "Recommended Specification: Levels of Gypsum Board Finish" as follows:
- A. LEVEL 1 FINISH: At concealed areas above ceiling.
  - B. LEVEL 2 FINISH: At gypsum backing board to be covered with file or panels thicker than 1/4".
  - C. LEVEL 3 FINISH: At mechanical rooms, storage rooms, custodial and maintenance rooms, electrical and telephone closets.
  - D. LEVEL 4 FINISH: All other drywall areas scheduled for paint, fabric or vinyl wall covering.

END SECTION

SECTION 09 22 00 - PLASTER ACCESSORIES

PART 1 – GENERAL

1.0 COORDINATION

- A. The General Conditions of the Contractor for Construction and the Supplementary Conditions to the General Conditions of the Contract for the Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addendum issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the stringent requirements and the greater quantity shall apply.

1.1 SUMMARY

- A. Related work specified elsewhere includes:
  - 1. Rough carpentry.
  - 2. Finish carpentry and millwork.
  - 3. Metal support systems.
  - 4. Furring and lathing.
  - 5. Plaster.
  - 6. Gypsum board systems.
  - 7. Painting.
  - 8. Wallcovering.

1.2 SUBMITTALS

- A. Product data: Indicate product description, including compliance with specified requirements and installation requirements. Mark manufacturer's brochures to include only those products proposed for use.

1.3 QUALITY ASSURANCE

- A. Applicable standards; standards of the following, as referenced herein:
  - 1. Aluminum Association (AA).
  - 2. American Society for Testing and Materials (ASTM).
- B. Allowable tolerances in horizontal planes:
  - 1. Variation from level: +1/8" in 12'-0".
  - 2. Variation in plane of adjacent wallboard panels prior to joint treatment: 1/16".
- C. Allowable tolerances in framed vertical construction.
  - 1. Position: +1/4" maximum variation from design position.
  - 2. Alignment: 1/8" in 8'-0"; 1/4" maximum in any continuous wall, line or surface.

3. Surface smoothness: No joint or fastener location, roughness or blemish discernible after application of finish when viewed at any angle from a distance of 5'-0" under occupancy lighting conditions, with surface preparation as specified in Painting section.

#### 1.4 DELIVERY, STORAGE AND HANDLING

##### A. Storage:

1. Stack accessories off floor on pallets or similar platforms providing continuous support for accessories to prevent sagging. Stack accessories so that long lengths are not over short lengths.
2. Do not overload floor systems.
3. Handle materials to prevent damage to surfaces, edges and ends of sheet metal items. Reject and remove damaged material from site.

#### 1.5 WARRANTY

- ##### A. Finish warranty:
- Warrant fluoropolymer coating to remain free, under normal atmospheric conditions, from peeling, checking, cracking, chalking in excess of numerical rating of 8 when measured in accord with ASTM D4214-98, of fading in excess of 5 N. B. S. Units during warranty period. Warranty period shall be 5 years, beginning at Date of Substantial Completion.

### PART 2 - PRODUCTS

#### 2.1 MANUFACTURER:

- ##### A. Acceptable manufacturer; subject to compliance with specified requirements:

Fry Reglet Corporation.  
625 S. Palm Ave.  
Alhambra, CA 91803  
Phone 800-237-9773 Fax 800-200-4397

- ##### B. Accessory systems of similar design and construction, as manufactured by other manufacturers, may be submitted for Architect's consideration. Acceptance is subject to compliance with specified design criteria, as evidenced by submittal of specified product data. Submittals shall comply with requirements of Product Options and Substitutions section.

#### 2.2 MATERIALS AND FINISH:

##### A. Anodized finish:

1. Architectural 200R1 medium etch (AA-M32c10A21), clear color.
2. Thickness of anodic coating shall be tested in accord with ASTM B244-97 and sealed to pass modified dye stain test ASTM B136-84(1998).

##### B. Color anodized finish:

1. Two-step impregnated color Class II Architectural 0.40-0.70 mils (AA-M12C22A33).
2. Thickness of anodic coating shall be tested in accord with ASTM B244-97 and sealed to pass modified dye stain test ASTM B136-84(1998).
3. Color: As selected by Architect from manufacturer's standard color selection.

2.3 REVEALS:

- A. Channel screed:
  - 1. Acceptable product: Number PCS. DCS.
  - 2. Characteristics: Screed shall provide reveal and control joint to plaster walls and Ceilings.
  - 3. Description: Sc
    - b. Material: Extruded aluminum.
    - c. Dimensions: As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
- B. 1/4" channel screed:
  - 1. Acceptable product: Number PCS-75-25/25.
  - 2. Characteristics:
  - 3. Description: Screed provides narrow profile control joint.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
- C. "V" reveal molding:
  - 1. Acceptable product: Number PCSV-75.PCSV-75-Short
  - 2. Characteristics:
    - a. Description: Molding shall create a "V" shaped reveal.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
- D. "F" reveal molding:
  - 1. Acceptable product: Number FDM. FPM.
  - 2. Characteristics:
    - a. Description: Reveal molding shall form a trim reveal around doors or between walls and floors.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
- E. "T" reveal molding:
  - 1. Acceptable product: Number TRM
  - 2. Characteristics:
    - a. Description: Molding shall provide a reveal at junction of dissimilar materials.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
- F. Plaster key reveal molding:
  - 1. Acceptable product: Number PRZ.

2. Characteristics:
  - a. Description: Molding shall provide termination and reveal at juncture of walls, plaster ceilings and soffits.
  - b. Material: Extruded aluminum.
  - c. Dimensions: \_\_\_ As indicated on drawings.
  - d. Radius: \_\_\_ As indicated on drawings.
  - e. Ventilation: Provide horizontal vents for air movement.

G. Reveal molding:

1. Acceptable product: Number PRM.
2. Characteristics:
  - a. Description: Reveal shall provide termination and terminate juncture of walls, plaster ceilings and soffits.
  - b. Material: Extruded aluminum.
  - c. Dimensions: \_\_\_ As indicated on drawings.
  - d. Radius: \_\_\_ As indicated on drawings.
  - e. Ventilation: Provide horizontal vents for air movement.

H. Contemporary reveal molding

1. Acceptable product: Number CDRMCVPR
2. Characteristics:
  - a. Description: Contemporary Reveal Molding provides an attractive vertical or horizontal recess in drywall or veneer plaster that creates the illusion the panels are "floating" in space.
  - b. Material: Extruded aluminum
  - c. Dimensions: As indicated on drawings.
  - d. Radius: \_\_\_ As indicated on drawings.

2.4 CONTROL JOINTS:

- A. 2-piece plaster control screed.
  1. Acceptable product: Number DCS.PCS
  2. Characteristics:
    - a. Description: Screed shall provide control joint in interior plaster and exterior stucco.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.

2.5 SOFFIT VENTS:

- A. Soffit vents:
  1. Acceptable product: DS.
  2. Characteristics:
    - a. Description: Vent shall act as drip screed and provide ventilation to areas above soffit.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.

B. 5/8" soffit vent:

1. Acceptable product: Number DS-875-5/8-V-300.

2. Characteristics:
    - a. Description: Vent shall function as drip screed and vent for areas above soffit and shall feature a 3/8" long bottom leg to support 5/8" soffit material.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
- C. Soffit vent:
1. Acceptable product: Number DS-375-V-875.
  2. Characteristics:
    - a. Description: Vent shall function as drip screed and vent for areas above soffit and shall separate soffit from fascia with reveal and add uniform edge to soffit/fascia joint.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
- D. Soffit vent:
1. Acceptable product: Number DCS.PCS
  2. Characteristics:
    - a. Description: Vent shall provide ventilation to areas above soffit, forming a reveal.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
- E. Vented soffit molding:
1. Acceptable product: Number WPM.
  2. Characteristics:
    - a. Description: Molding shall provide reveal and venting for soffits and screed point for applying stucco.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
- F. Soffit vent E.I.F.S.:
1. Acceptable product: Number VFS.
  2. Characteristics:
    - a. Description: Vent shall provide drip screed and ventilation to areas above soffit.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
- 2.6 E.I.F.S. MATERIALS:
- A. E.I.F.S. channel screed:
1. Acceptable product: PCS.
  2. Characteristics:



- a. Description: Screed shall act as a 1/4" ground channel screed, control joint, and reveal details on exterior walls and soffits.
  - b. Material: Extruded aluminum.
  - c. Dimensions: \_\_\_ As indicated on drawings.
  - d. Radius: \_\_\_ As indicated on drawings.
  - e. Ventilation: Provide horizontal vents for air movement.
  
- B. E.I.F.S. soffit vent:
  - 1. Acceptable product: Number PCS.
  - 2. Characteristics:
    - a. Description: Vent shall provide ventilation in soffits.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
  
- C. E.I.F.S. Clean Finish soffit vent:
  - 1. Acceptable product: Number SV-75-V-300/EIFS.
  - 2. Characteristics:
    - a. Description: Vent shall be applied after E.I.F.S. application is complete.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
  
- D. Direct application E.I.F.S. soffit vent:
  - 1. Acceptable product: Number DS.
  - 2. Characteristics:
    - a. Description: Vent shall provide drip screed in stucco or synthetic E.I.F.S. finishes.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
  
- E. E.I.F.S. soffit vent (Direct Applied):
  - 1. Acceptable product: Number DRM.
  - 2. Characteristics:
    - a. Description: Vent shall provide vent in direct applied E.I.F.S. substrates in soffits with an E.I.F.S. system.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.
  
- F. E.I.F.S. soffit vent:
  - 1. Acceptable product: Number VFS.
  - 2. Characteristics:
    - a. Description: Vent shall function as drip screed and provide ventilation to areas above soffit.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.

- G. E.I.F.S. drip screed:
  - 1. Acceptable product: Number DS.
  - 2. Characteristics:
    - a. Description: Screed shall join soffit and fascia while providing drip joint to prevent water stains on soffit and vertical wall of building.
    - b. Material: Extruded aluminum.
    - c. Dimensions: \_\_\_ As indicated on drawings.
    - d. Radius: \_\_\_ As indicated on drawings.
    - e. Ventilation: Provide horizontal vents for air movement.

## 2.7 FASTENERS:

- A. Fasteners: Exposed fasteners (provided by installer) shall be countersunk and shall match accessories in color.
  - 1. Aluminum to aluminum: Aluminum or Type 302 or 304 stainless steel.
  - 2. Aluminum to stainless steel or carbon steel: Type 302 or 304 stainless steel.

## 2.8 FABRICATION:

- A. Mounting holes: Provide mounting holes located at 8" o. c.
- B. Venting: Vent slots shall be 1" by 1/8" spaced 1-1/2" lengthwise. Each row of vent slots shall provide 1 square inch of free area per lineal foot.
- C. Make custom miters and intersections with welded corners or with high-strength industrial tape on backs.
- D. Radius shapes shall be as indicated on drawings.

## PART 3 - EXECUTION

### 3.1 INSTALLATION:

- A. Install plaster accessories in accord with manufacturer's product data and as follows:
  - 1. Corner trim: Install at designated corners.
  - 2. Metal trim shapes: Install at exposed edge of wallboard at door and window openings, at intersections with other materials and at intersection of walls with ceilings.
  - 3. Install corner beads and metal trim shapes to framing system with mechanical anchors spaced at 8" o. c.
  - 4. Joint treatment: Finish joints and attachment flanges as specified in Plaster Stucco section.
  - 5. Dust surfaces and leave ready for decoration. Joint and fastener treatment shall be indistinguishable in finished work.

### 3.2 PROTECTION

- A. Protect accessories from damage until date of Substantial Completion. Replace accessories which become damaged.

END OF SECTION

## SECTION 09 30 00 — WALL AND FLOOR TILE

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 WORK INCLUDED

- A. Provide and install all ceramic wall and floor tile and base as indicated in the drawings and specified herein.
- B. Provide and install all quarry tile flooring and base as indicated in the drawings and specified herein.

#### 1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Cast-in-place concrete.
- B. Drywall systems.
- C. Masonry.
- D. Waterproofing and dampproofing.

#### 1.4 SUBMITTALS

- A. Per SUPPLEMENTARY GENERAL CONDITIONS, submit samples, type of tile and color for Architect's approval. Mark with manufacturer's name and space where tile is to be installed.
- B. Submit manufacturer's printed literature describing products.
- C. Submit (2) boxes of tile chips showing full range of available colors.
- D. Submit (2) boxes of grout color samples.
- E. Submit 12" x 12" grouted sample board for each tile/grout combination selected.

- F. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

#### 1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.
- B. Warranted defects shall include but not necessarily be limited to cracking, crazing, staining, joint spalling or cracking, loosening of bond.

#### 1.6 QUALITY ASSURANCE

- A. Tile Contractor shall have a minimum of 3 years experience in tile installation for projects of similar size and scope as this project.
- B. Conform with all applicable requirements of the American Standards Association Specifications (A-108 Series) and the "Tile Handbook" of the Tile Council of America. Tile shall bear the seal of Tile Council of America, Inc., and be equal to or exceed Standard Grade.

#### 1.7 DELIVERY & STORAGE

- A. Deliver all manufactured materials in original, unbroken containers bearing name of manufacturer, brand and grade seal. Keep materials dry, clean and protected against deterioration in any form and at room temperature.
- B. Maintain room temperature between 70 and 80 degrees F. 24 hours prior, during and a minimum of 48 hours after installation.

### PART 2 - PRODUCTS

#### 2.1 MANUFACTURERS

- A. CERAMIC TILE:
  - 1. American Olean
  - 2. Dal-Tile
  - 3. United States Ceramic Tile Co.
- B. QUARRY TILE: American Olean, United States Ceramic Tile Co.
- C. GROUT:
  - 1. American Olean
  - 2. Laticrete
  - 3. Tex Rite

## 2.2 MATERIALS

### A. GENERAL:

1. Floor Tile: Unglazed porcelain ceramic with cushioned edges and sheet backing.
  - a. Water absorption: Classified "Impervious" per A.S.T.M. C-373. Less than 1/2 of 1% absorption.
  - b. Size: Nominal 12" x 12" x 1/4" thick.
  - c. Base: 4" high base. Bottom tile with integral cove (provide bullnose plastic edge strip at top of all ceramic tile base).
  - d. Type: Porcelain Tile in **groups 3 through 4** as selected by the Architect from one of the specified manufacturers.
  - e. Color(s): Bidders shall assume a different color scheme for each room unless colors and patterns are indicated in the drawings.

### B. TOILET/SHOWER ROOMS:

1. Floor Tile: Unglazed porcelain ceramic with cushioned edges and sheet backing.
  - a. Water absorption: Classified "Impervious" per A.S.T.M. C-373. Less than 1/2 of 1% absorption.
  - b. Size: Nominal 1" x 1" x 1/4" thick.
  - c. Base: 4" high base. Bottom tile with integral cove.
  - d. Type: Porcelain Tile in **groups 1 through 2** as selected by the Architect from one of the specified manufacturers.
  - e. Color(s): Bidders shall assume a different color scheme for each room unless colors and patterns are indicated in the drawings.
2. Wall Tile: Glazed ceramic with cushion edges.
  - a. Size: Nominal 4" x 4" x 1/4" thick.
  - b. Base: See floor base.
  - c. Type: **Groups 3 through 4** for field tile and for accent banding as selected by the Architect from one of the specified manufacturers.
  - d. Color(s): Bidders shall assume a different color scheme for each room unless colors and patterns are indicated in the drawings.
3. Trim: Terminate tile with bullnose edges and rounded outside corners. Provide square inside corners and at ceiling/wall joints.

C. **THINSET BOND COAT**: Latex/Portland Cement mortar mix meeting requirements of ANSI A118.4. Provide Portland cement and sand in a 1 to 1 mixture gauged with Laticrete 4237 latex additive. Use on dry cured mortar bed at slab recesses, where thinset on concrete slab, and where thinset on wall substrates.

D. **SEALANT**: One part silicone rubber meeting requirements of FS TT-S-001543, as manufactured by Dow Corning or General Electric.

### E. GROUT:

1. Walls: Portland Cement waterproof, dry set grout as manufactured by American Olean. Color(s) as selected by Architect.
2. Floor and base: Interior grout shall be epoxy type as manufactured by American Olean. Color(s) as selected by Architect.

F. **SEALANT**: One part silicone rubber meeting requirements of FS TT-S-001543, as manufactured by Dow Corning or General Electric.

## PART 3 - EXECUTION

### 3.1 INSPECTION AND PREPARATION

- A. Examine surfaces to receive tile and do not start work until defects that will adversely affect tile work have been corrected.
- B. Inspect all surfaces to see that they are dry, clean, free of oily or waxy film, firm, level and plumb. Report any unsatisfactory conditions to the Architect. Starting installation shall be deemed as acceptance of surfaces.
- C. Do not start until work of other trades, which goes through or in the space behind tile has been completed. Do not proceed with installation until adjoining work is satisfactory protected. Close off spaces in which tile is being set to traffic and other work during installation and for at least 48 hours after completion of tile work.
- D. Do not apply mortar and adhesives to surfaces covered by frost. Maintain minimum temperature-for installation of tile above 50 Deg. F. Prevent rapid evaporation of moisture from mortar bed. Do not set tile on dry bed.
- E. Install specified mortar bed at slab depressions. Slope mortar bed uniformly to drain(s).

### 3.2 INSTALLATION

- A. GENERAL: Tile shall be installed in accordance with current Tile Council of America's "Handbook for Ceramic Tile Installation", design numbers as indicated below.
- B. Center fields and patterns on applied areas so that no tile is less than half size. For heights stated in feet and inches, maintain full courses to nearest attainable height without cutting tile.
- C. Except where otherwise shown or specified, make joints in wall tile vertical and horizontal and joints in floor tile perpendicular and parallel to walls. Control joint widths of glazed tile by lugs on the sides of tile. Control joints widths between sheets of ceramic mosaic tile by supporting boards with metal spacing strips.
- D. Grind and fit tile carefully at intersections, against trim finish and at built-in fixtures and accessories. Fit tile closely around outlets, pipes, fixtures and fittings so that plates, escutcheons and collars will overlap cuts. Cut and drill tile and trim shapes accurately without damage. Rub all exposed cut edges smooth with abrasive stone.
- E. Coat trim with 1/32 to 1/16" pure coat paste. Set in same mortar mix as is recommended for setting flat tile on walls. Do not use pure coat as mortar to set trim and angles.
- F. FLOORS:
  - 1. Interior thinset on concrete floor slab:
    - a. Tile bonded with minimum 3/32" thick latex-Portland cement bond coat over cleavage membrane adhered to floor slab **(modified TCA F113)**.
    - b. Adhere cleavage membrane to slab in strict accordance with manufacturer's recommendations using specified latex-Portland cement bond coat. Increase typical curing time of bond coat by 50%.
  - 2. Interior thin-set on recessed mortar bed (where required at existing kitchens): Tile bonded with minimum 3/32" thick Latex-Portland Cement bond coat to reinforced mortar bed over loose bond breaker membrane over floor slab **(TCA F111)**.

G. WALLS:

1. Ceramic Tile at Drywall Toilets: Thinset to water resistant gypsum wallboard.
2. Ceramic Tile at Drywall Showers: Thinset to tile backer board.
3. Ceramic Tile at Masonry: Bonded to mortar bed at masonry. No. **W211**.
4. 12" x 12" Porcelain – Up to 3 ft. High Wainscot: Install with mastic over drywall.
5. 12" x 12" Porcelain – Over 3 ft. High Wainscot: Thinset over tile backer board.

H. EXPANSION JOINTS:

1. At floor tile provide 1/4" sealant expansion joints in accordance with TCA recommendations where tile abuts walls, curbs, columns and other restraining surfaces, where substrate material changes, at floor slab construction joints (cold joints), and each way in pattern approved by the Owner.
2. At walls install sealant expansion joints at inside corners, at maximum 30', and at other conditions subject to cracking or movement. Install specified sealant at expansion and control joints, at doorframe perimeters and similar conditions.

3.3 LAYOUT

- A. Layout all work so that no tiles less than half size occur. Align all joints vertically and horizontally.
- B. Cut and drill neatly without marring tile. Rub smooth any necessary cuts with a fine stone and set no cut edge against any fixture, cabinet, or other tile without a joint at least 1/16" wide.
- C. Maximum plane variation shall be 1/8" + or – in 10' when a straight edge is laid on the surface in any direction.

3.4 GROUTING AND SEALING:

- A. Follow grout manufacturer's recommendations for grouting procedures and precautions. Damp cure non-epoxy grout in accordance with manufacturer's recommendations.
- B. Grout Haze Removal:
  1. Unglazed Tile: For cement grout remove all grout haze following grout manufacturer's recommendations for use of acid and chemical cleaners. Rinse tilework thoroughly with clean water before and after chemical cleaners. Polish surface of tilework with soft cheesecloth.
  2. Glazed Tile: For cement grout remove all grout haze with cheesecloth rub.
  3. Take special care with epoxy grout to keep tiles clean as work progresses.

3.5 PROTECTION

- A. Protect tiled floors from foot and wheel traffic for at least 7 days after installation.
- B. Place plywood panels over traffic floors.
- C. In non-traffic areas, cover floors with heavy paper taped in place.
- D. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents. Furnish quantity not less than 5 percent for each color, pattern, and type of tile installed.

END OF SECTION

#

SECTION 09 51 00 - ACOUSTICAL TILE CEILINGS

PART 1 - GENERAL

COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.2 WORK INCLUDED

- A. Provide and install all lay-in acoustical ceiling panels and suspended grid system in accordance with the drawings and as specified herein.
- B. Provide and install light fixture protection at all rated ceilings.
- C. Provide and install hold-down clips where required for rated system.

1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Steel joists (spacing)
- B. Mechanical (air devices)
- C. Electrical (lighting fixtures)

1.4 DRAWING REFERENCES

- A. See drawings and finish schedule for ceiling types and ratings.

1.5 SUBMITTALS

- A. Submit manufacturer's product data describing all materials, finishes, ratings and installation requirements.
- B. Submit physical samples for each type of acoustical tile proposed.
- C. Submit physical samples for each type of grid proposed.
- D. Submit tile manufacturer's certification for whether hold-down clips are required for the selected tile(s) and rated system(s).
- E. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.



#  
1.6 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.
- B. Warranted defects shall include but not necessarily be limited to rusting or deflection of grid, deterioration or deflection of acoustical tiles.

1.7 QUALITY ASSURANCE

- A. Suspended acoustical ceiling contractor shall have a minimum of 3 years experience in the installation of specified systems for projects of similar size and scope of this project.
- B. Installation of acoustical tile and panels shall not begin until residual moisture from plaster, drywall, concrete or terrazzo work is dissipated. Before installation, the building shall be enclosed and permanent heating and cooling equipment in operation.

1.8 DELIVERY AND STORAGE OF MATERIALS

- A. Do not deliver materials to jobsite until spaces are ready for ceiling installation.
- B. All materials shall be delivered in manufacturer's original packaging and stored in an enclosed shelter providing protection from damage and exposure to the elements.
- C. Damaged, rusted or deteriorated materials shall be removed from the premises.

PART TWO - PRODUCTS

2.1 MANUFACTURERS

- A. TYPICAL CEILING PANELS:
  - 1. Armstrong World Industries, Inc.
  - 2. USG Interiors, Inc.
- B. SPECIALTY CEILING PANELS
  - 1. Acoustical Resources, Inc.
  - 2. Wenger
  - 3. U.S.G.
- C. GRID SYSTEMS:
  - 1. Armstrong World Industries, Inc.
  - 2. USG Interiors, Inc.
  - 3. Chicago Metallic Corp.

2.2 MATERIALS:

- A. TYPICAL CEILING PANELS:
  - 1. Type "C-1": 24" x 48" x 5/8" white "Cortega Square Lay-in" No. 747 as manufactured by Armstrong or equivalent (color, pattern, texture) by specified manufacturer. **Non-rated system.**

#

#

#

B. SUSPENSION SYSTEM:

1. Components shall be formed from commercial quality cold-rolled steel, electro-galvanized, 2'x4' module.
2. The suspension system shall support the ceiling assembly with a maximum deflection of 1/360 of the span per A.S.T.M. C-635-69.
3. Main tee with double web design 1-1/2" high and rectangular bulb; 15/16" exposed flange with rolled cap; cross tee holes at 6" o.c.
4. Four foot cross tee 1-1/2" high with double web design. Rectangular bulb joining main runners at 2' on center.
5. Two foot cross tees perpendicular to 4' cross tees. Two foot cross tees minimum of 1-1/2" high, No. CMC 222-41 or equivalent by specified manufacturer.
6. Wall molding - hemmed edge, electro-galvanized cold rolled steel with equal leg width, finish to match grid.
7. Finish: Typical finish, factory white painted steel. At high humidity areas including kitchens, dressing rooms, toilet rooms provide factory white painted aluminum cap.
8. Rating: Provide U.L. listed grid for scheduled system rating.

PART THREE - EXECUTION

3.1 COORDINATION

- A. Verify that above ceiling work, including fire dampers, ductwork, piping, wiring and insulation is complete and approved prior to beginning ceiling work.

3.2 INSTALLATION

- A. Ceiling systems shall be suspended from structural members by 12 gauge annealed wire; spacing as recommended by manufacturer. Provide additional support for light fixtures and grilles at each corner. Provide secondary support framing ("Unistrut") where spacing of structural members exceeds suspension system manufacturer's recommendations.
- B. Acoustical lay-in panels shall be installed in strict accordance with the manufacturer's instructions. Tile shall be installed with fissures or pattern all in same direction.
- C. Provide additional hangers at ceiling suspended items including projection screens, speakers, exit lights, air supply and return grilles.
- D. Space main runner hangers a maximum of 6 inches from wall. Do not support systems from wall.
- E. Adjust hangers to ensure level ceiling in plane.

3.3 RATED CEILINGS

- A. Provide specified ceilings in fire rated assembly. Protect light fixture protection in accordance with approved U.L. Design to meet required assembly rating. Provide additional hangers to meet the requirements of the particular U.L. rating.
- B. Ceiling system manufacturers not listed in the required U.L. design number (reference drawings) shall be responsible for determining whether their rated system is acceptable to the particular local code authority.
- C. For ceiling tiles weighing 1 lb. per square foot or more, verify no requirement for hold-down clips at rated systems.

3.4 CLEANING AND REPLACEMENT

- A. At completion, replace file unit and grid systems that are damaged. Clean or replace tile and grid systems that cannot be cleaned. Deliver three (3) full boxes of each type-ceiling tile to the Owner.

END OF SECTION 09512

## SECTION 09 65 00 – RESILIENT FLOORING AND BASE

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 WORK INCLUDED

- A. Provide and install all vinyl tile flooring as indicated in the drawings and specified herein.
- B. Provide and install all resilient base as scheduled throughout the project, regardless of floor finish.
- C. Provide and install all resilient transition strips at resilient flooring, steps, stairs and change of flooring materials.
- D. Install tapered resilient transition edge strip at any place where resilient floor is installed on concrete steps/stairs and terminates with risers.
- E. Provide five (5) coats of wax on all new resilient flooring.

#### 1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Cast in place concrete.
- B. Millwork.

#### 1.4 SUBMITTALS

- A. Submit manufacturer's product data describing all materials.
- B. Submit physical samples of all resilient materials to the Architect for approval. Color(s) to be selected by the Architect.
- C. Submit manufacturer's recommendations for finishing and maintenance of resilient flooring materials.
- D. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

#### 1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.

- B. Warranted defects shall include but not necessarily be limited to loss of adhesion, excessive surface wear, color change, curling or other deterioration.

## 1.6 DELIVERY, STORAGE AND HANDLING

- A. Do not deliver flooring materials to the jobsite until spaces are ready for installation of resilient flooring.
- B. Open material packages and acclimate flooring materials within the installation spaces for a minimum of 3 days prior to installation.

## 2 PART TWO – PRODUCTS

### 2.1 MANUFACTURERS

- A. VINYL TILE:
  - 1. Armstrong World Industries
  - 2. Azrock Industries, Inc.
  - 3. Johnsonite
  - 4. Tarkett
- B. RESILIENT BASE:
  - 1. Roppe
  - 2. Burke Flooring Products
  - 3. Mercer Products Co., Inc.

### 2.2 MATERIALS

- A. VINYL TILE: 12" x 12" x 1/8" Standard Excelon "Imperial Texture" vinyl composition tile, as manufactured by Armstrong World Industries, or equivalent in color and design by specified manufacturer.
- B. RESILIENT BASE: 4" high x 1/8" thick x 4' lengths, rubber cove base as manufactured by Roppe or equivalent by specified manufacturer. Color(s) to be selected by the Architect. Provide manufacturer's pre-molded outside corners.
- C. TRANSITION STRIPS: Vinyl transition strips as manufactured by Roppe or equivalent by specified manufacturer. Color(s) to be selected by the Architect.

## 3 PART THREE – EXECUTION

### 3.1 PREPARATION

- A. Inspect the completed floor slab for defects which may adversely affect the finished resilient tile work. Commencing resilient flooring operations indicates acceptance of the sub-floor.
- B. Subfloor depressions shall be brought to level with latex underlayment. Raised areas shall ground and smoothed prior to resilient flooring installation.
- C. Thoroughly clean subfloor of all wax, oil, dusting, dirt or other deleterious material.

### 3.2 INSTALLATION

- A. VINYL COMPOSITION TILE

1. Tile shall be installed in strict accordance with the manufacturer's recommendations using adhesive approved by tile manufacturer.
2. Unless otherwise indicated in the drawings, lay flooring with joints and seams aligned with building walls. Start laying tiles from the center of the room out for equal sized tiles at the perimeters. Avoid tiles of less than ½ size.
3. Spread adhesive using notched trowel. Apply only enough adhesive at one time to allow placing of tile prior to initial setting of adhesive.
4. Use heavy roller to smooth tile and ensure complete adhesion.
5. Install tapered resilient edge strip at any place where resilient floor meets concrete, carpet or other finish flooring material. Typically material changes should be made at the centerlines of doors. Color(s) as selected by the Architect.
6. Install tapered resilient transition edge strip at any place where resilient/wood flooring is installed on concrete steps/stairs and terminates with risers. Ensure a tight fit so resilient floor will not crack or be damaged by foot traffic. Color(s) and size as selected by the Architect.
7. Install tapered resilient transition edge strip at any place where there is a change of height and/or flooring materials.
8. A feature strip shall be used to divide any two areas where it is not possible to maintain alignment from one area to the adjoining area. Coordinate with Architect.
9. Scribe flooring to walls, columns, cabinets, floor outlets and other interruptions to ensure tight fitted joints.

B. RESILIENT BASE:

1. Install base using manufacturer's recommended adhesive applied with notched trowel. Install with contact cement within 6" of a job-formed outside corner.
2. Miter inside corners. Use factory-formed outside corners unless job-formed corners are specifically approved by the Architect.
3. Butt joints tight and scribe base to door frames, columns and other interruptions.

C. TRANSITION STRIP:

1. Subfloor must be smooth, sound, dry, clean, and free of dirt, wax, polish, paint, and all other foreign matter which may interfere in a good bond, including curing agents and sealers.
2. Carefully follow warnings on container of the Solvent-Based Contact Adhesive. Follow adhesive manufacturer's recommendations for the installation of TRANSITION STRIPS.
3. Roll TRANSITION STRIP with a hand roller.

3.3 CLEANING AND ADJUSTING

- A. After installation all resilient flooring shall be cleaned and contractor shall provide five (5) coats of wax on all new resilient flooring.
- B. Replace any damaged tile or tile that shows inconsistent shades of color/pattern. Remove glue stains or other marks.
- C. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents. Furnish quantity not less than 5 percent for each color, pattern, and type of resilient flooring installed.

END OF SECTION #

## SECTION 09 65 19 – SPECIALTY FLOORING

### PART 1 GENERAL

#### 1.00 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.01 SUMMARY

- A. Section Includes: Rubber Floor Tile.

#### 1.02 REFERENCED DOCUMENTS

- A. ASTM International
  - 1. D2859, Standard Test Method for Ignition Characteristics of Finished Textile Floor Covering Materials.
  - 2. E662, Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials.
  - 3. F373, Standard Test Method for Embossed Depth of Resilient Floor Coverings.
  - 4. F386, Standard Test Method for Thickness of Resilient Flooring Materials Having Flat Surfaces.
  - 5. F511, Standard Test Method for Quality of Cut (Joint Tightness) of Resilient Floor Tile.
  - 6. F710, Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring.
  - 7. F925, Standard Test Method for Resistance to Chemicals of Resilient Flooring.
  - 8. F1344, Standard Specification for Rubber Floor Tile.
  - 9. F1859, Standard Specification for Rubber Sheet Floor Covering Without Backing.
  - 10. F2055, Standard Test Method for Size and Squareness of Resilient Floor Tile by Dial Gage Method.
- B. Other Referenced Documents
  - 1. National Fire Protection Association (NFPA) 255, Test Method for Critical Radiant Flux of Floor Covering Systems Using a Radiant Energy Source.
  - 2. National Fire Protection Association (NFPA) 258, Test Method for Specific Density of Smoke Generated by Solid Materials.
  - 3. FMVSS 302, Title 49, Part 571, Flammability of Interior Materials.
  - 4. IMO Resolution A.653 (16), Surface Flammability of Deck Finish (Shipboard use).
  - 5. Title 16, CFR Chapter II, Subchapter D, Part 1630, Standard for the Surface Flammability of Carpets and Rugs (FF 70).
  - 6. California Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65).
  - 7. The Collaborative for High Performance Schools (CHPS).

#### 1.03 SUBMITTALS

- A. Product Data: Submit product data, including manufacturer's specification summary sheet for specified products.

- B. Shop Drawings: Submit shop drawings showing layout, finish colors, patterns and textures.
- C. Samples: Submit selection and verification samples for finishes, colors, and textures.
- D. Quality Assurance Submittals: Submit the following
  - 1. Test Reports: Certified test reports showing compliance with specified performance characteristics and physical properties.
  - 2. Manufacturer's Instructions: Manufacturer's installation and maintenance instructions.
- E. Maintenance Information: Maintenance information for installed products in accordance with Division 1 sections.
  - 1. Methods for maintaining installed products.
  - 2. Precautions against cleaning materials and methods detrimental to finishes and performance.
- F. Warranty: Warranty documents specified herein.

#### 1.04 QUALITY ASSURANCE

- A. Installer Qualifications: Installer experienced in performing work of this section who has specialized in installing work similar to that required for this project.
- B. Regulatory Requirements
  - 1. Fire Performance characteristics: Provide resilient rubber floor covering with the following Fire performance characteristics as determined by testing products in accordance with ASTM method (and NFPA method) indicated below by a certified testing laboratory or another testing and inspecting agency acceptable to authorities having jurisdiction.
    - a. ASTM E648 (NFPA 253), Critical Radiant Flux of Floor Covering Systems; Class 1, Greater than 0.45 W/cm<sup>2</sup>.
    - b. ASTM E662 (NFPA 258), Specific Optical Density of Smoke Generated by Solid Materials; < 450.
- C. Single-Source Responsibility: Obtain rubber floor tile and manufacturer's recommended adhesive from a single supplier.
- D. Pre-Installation Meetings: Conduct pre-installation meeting to verify project requirements, Manufacturer's conditions, recommended adhesive depending on product, substrate type and type of installation, manufacturer's installation instructions and manufacturer's warranty requirements. Comply with requirements in Division 1.

#### 1.05 DELIVERY, STORAGE AND HANDLING

- A. General: Comply with requirements in Division 1.
- B. Ordering: Comply with manufacturer's ordering instructions and lead-time requirements to avoid construction delays.
- C. Delivery: Deliver materials in manufacturer's original, unopened, undamaged containers with Identification labels intact.
- D. Storage and Protection: Store materials protected from exposure to harmful weather conditions and acclimated to site conditions at temperature and humidity conditions recommended by manufacturer.

#### 1.06 PROJECT CONDITIONS

- A. Environmental Requirements/Conditions: In accordance with manufacturer's recommendations, areas to receive rubber flooring shall be clean, fully enclosed, weather tight with the permanent HVAC set at a uniform temperature of 65° - 85° F for 48 hours prior to, during and thereafter installation of rubber flooring. Rubber flooring and adhesive shall be conditioned in the same manner. Rubber flooring/tile must be un-boxed at least 48 hours prior to installation in the areas in which it will be installed.
- B. Existing Conditions: (Specify existing conditions affecting product use and installation).

#### 1.07 SEQUENCING AND SCHEDULING

- A. Finishing Operations: Install rubber flooring after finishing operations, including floor covering, painting and ceiling operations, have been completed.



1.08 MAINTENANCE

- A. Extra Materials: Deliver to Owner five (5) percent extra materials from same production run as products installed. Package products with protective covering and identify with descriptive labels. Comply with Division 1, Closeout Submittals Section.
- B. Quantity: Furnish quantity of Rubber flooring equal to 5% of amount to be installed.
- C. Delivery, Storage and Protection: Comply with Owner's requirements for delivery, storage and protection of extra materials.
- D. Maintenance of finished floor covering to be conducted per Manufacturer's Maintenance Guide.

1.09 WARRANTY

- A. Manufacturer's Materials Warranty: Submit, for Owner's acceptance, manufacturer's standard warranty document. Manufacturer's warranty is in addition to, and not a limitation of, other rights Owner may have under Contract Documents.
  - 1. Warranty: 1 year limited warranty commencing on Date of Substantial Completion. Notice of any defect must be made in writing to manufacturer within 30 days after buyer learns of the defect.
  - 2. Limited Wear Warranty: 10 year limited wear warranty available depending on product type/design & type of use.

PART 2 PRODUCTS

2.01 RUBBER TILE FLOORING

- A. Manufacturer:
  - 1. Flexco Corporation, 1401 E. 6th Street, Tuscumbia, AL 35674. Phone: 800-633-3151, Fax: 800-346-9075, Web: [www.flexcofloors.com](http://www.flexcofloors.com) or Approved Equal.
  - 2. Tuflex Recycled Rubber Tile interlocking tiles from Roppe. Titan or Spartus or Approved Equal.
- B. Test results
  - 1. ASTM C1028, Static Coefficient of Friction (Slip Resistance); 0.88 dry
  - 2. ASTM D395, Compression Set; 12% or less
  - 3. ASTM D792, Density PCF; 65 lbs/cu/ft
  - 4. ASTM D2240, Hardness Shore A; 60 +/- 5
  - 5. ASTM D3389, (H-18 Wheels, 500 g/load, 1000 cycles) Taber Abrasion; < 0.50 grams weight loss
  - 6. ASTM D624, Die C, Tear Strength; 70 ppi
  - 7. ASTM F137, Flexibility; bends around a 6 mm mandrel with no cracks or breaks
  - 8. ASTM F925, Resistance to Chemicals; Passes, List available
  - 9. ASTM F970, Static Load Limit; 250 psi
  - 10. ASTM F1514, Resistance to Heat; Delta E: 2.24 and AATCC Gray Scale: 3.85 average
- C. Product Warranty
  - 1. Manufacturer's three year limited warranty for manufacturing defects
  - 2. Manufacturer's ten year limited wear warranty
- D. Products
  - 1. Flexco Prime Sports Flooring, homogeneous rubber product manufactured from Vulcanized post-industrial rubber color chips (EPDM) and vulcanized post-consumer tires, bonded with urethane binder.
    - a. Color: To be selected by Architect from Manufacturer's standard color selections.
    - b. EPDM Chip Content Percentage: To be Selected by Architect.
    - c. Size:

- 1) Inter-locking tile (measured from outside tabs): 36" x 36" (914.4 mm x 914.4 mm)  
nominal
- d. Thickness:
  - 1) 1/2" (12.00mm)

E. Miscellaneous

- 1. Provide transition strips at any change of flooring material and/or height.

2.06 PRODUCT SUBSTITUTIONS

- A. Substitutions: Permitted. Must submit proposed substitution 7 days prior to bid date. Manufacturer's literature and samples. Approval to be determined by Architect.

PART 3 EXECUTION

3.01 MANUFACTURER'S INSTRUCTIONS

- A. Compliance: Comply with manufacturer's requirements as published in manufacturer's installation instructions
- B. Adhesive: Manufacturer's brand adhesives, as required, based on product specified, job site conditions, subfloor type, and planned use of finished floor (select one)
  - 1. Section 2.01 Rubber Tile Flooring
    - a. Flexco 220 Sports Flooring Adhesive or approved equivalent by specified manufacturer.

3.02 EXAMINATION

- A. Site Verification of Conditions: Confirm substrate conditions (which have been previously addressed under other sections) are acceptable for product installing in accordance with manufacturer's instructions.
- B. Material Inspection: In accordance with manufacturer's installing requirements, visually inspect materials prior to installing. Material with visual defects shall not be installed.

3.03 PREPARATION

- A. Adjacent Surfaces Protection: Protect adjacent work areas and finish surfaces from damage while installing.
- B. Substrate Preparation: Prepare substrate to be free of paint, old adhesive, sealers, coatings, finishes, dirt, film-forming curing compounds, or other substances which may affect the adhesion of floor covering to the substrate.
  - 1. Concrete Substrate: Reference Standard ASTM F710 Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring. If the following test results exceed the floor covering manufacturer's limits, installing shall not commence until results conform to limits.
    - a. Concrete Moisture Test: Per ASTM F1869 Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride
      - 1) Flexco 220 Sports Flooring Adhesive, 5 lbs or less
    - b. Concrete Moisture Test: Per ASTM F2170 Determining Relative Humidity in Concrete Floor Slabs using *in situ* Probes
      - 1) Flexco 220 Sports Flooring Adhesive, 75% RH or less

- c. Concrete pH Test: Perform pH tests on concrete regardless of its age or grade level or history of use. Readings below 7.0 and above 10.0 can adversely affect resilient flooring or adhesives, or both.
2. Do not install over existing floor covering or over substrates not approved by manufacturer.

### 3.04 INSTALLING

- A. Refer to manufacturer's installation instructions for specific Rubber Flooring detailed specifications on installing.
  1. Flash Cove Procedure
    - a. Follow manufacturer's recommended procedures for flash cove installation.
    - b. Install Flexco® #195 Cove Stick Filler at the wall and floor junction.
    - c. Install Flexco® #197 Resilient Cove Cap on the wall.
    - d. Install metal inside and outside corners, to same height as cove cap.
    - e. Butterfly corners can also be installed based on installer expertise.
  2. Finish Floor Covering Designs: As selected by Architect.
  3. Accessories: Architect shall specify manufacturers' color coordinated accessories as required, including (but not limited to) resilient wall base, stair nosing, reducers or other edgings, welding rods for heat welded seams.

### 3.05 FIELD QUALITY REQUIREMENTS

- A. Manufacturer's Field Services: Upon Owner's request and with minimum 72 hours notice, provide manufacturer's field service consisting of product use recommendations and periodic site visits to confirm installing of product is in accordance with manufacturer's instructions.
- B. Site Visits: A minimum of two (2) site visits.

### 3.06 PROTECTION

- A. Protection: Protect installed product and finish surfaces from damage during construction. Remove and legally dispose of protective covering at time of substantial completion.
- B. No heat welding for first 48 hours.
- C. Restrict foot traffic from new floor covering for first 48 hours.
- D. Restrict furniture, fixtures and rolling traffic for first 72 hours.
- E. Restrict cleaning/buffing for first 72 hours.

### 3.07 INITIAL MAINTENANCE PROCEDURES

- A. General: Include in contract sum cost for initial maintenance procedures and execution by professional maintenance personnel after floor covering has been installed for 72 hours as specified in the manufacturer's maintenance instructions.

### 3.08 CLEANING

- A. Cleaning: See manufacturer's maintenance instructions. Remove temporary coverings and protection of adjacent work areas. Repair or replace damaged installed products. Clean installed products in accordance with manufacturer's instructions prior to owner's acceptance. Remove construction debris from project site and legally dispose of.
- B. Floor finishes: See manufacturer's product data information for information on floor finishes to be used.

END OF SECTION

#

SECTION 09 91 00 – PAINTING AND FINISHING

PART 1 - .

PART 2 - GENERAL

2.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

2.2 WORK INCLUDED

- A. Provide all labor, materials, and equipment required for all painting, staining and finishing as indicated in the drawings, the approved submittals, and as specified herein. Painted or stained systems include but are not necessarily limited to the items listed below:
- B. EXTERIOR SYSTEMS:
  - 1. All visible wood unless noted otherwise.
  - 2. All ferrous metal. All galvanized metal unless noted otherwise. Touch-up on welds or damaged finishes.
  - 3. Exposed conduit, piping, etc., except for roof mounted piping not visible.
  - 4. Exposed roof mounted equipment visible from ground level or from upper floors of the building.
  - 5. All exposed concrete masonry units.
  - 6. All items normally painted in accordance with good construction practice.
- C. INTERIOR SYSTEMS:
  - 1. All visible wood or behind cabinet doors unless noted otherwise.
  - 2. All ferrous metal. All galvanized metal unless noted otherwise. Touch-up on welds or damaged finishes. Structural steel, steel joists and deck exposed to view except in mechanical rooms.
  - 3. Exposed conduit, piping, outlet boxes, raceways, and panel boxes except galvanized or aluminum piping located in mechanical or electrical rooms.
  - 4. All exposed concrete masonry units, gypsum board and plaster unless otherwise noted.
  - 5. All factory-primed hardware. Back-priming of all wood trim, millwork or finished carpentry prior to installation.
  - 6. All hollow metal doors and frames.
  - 7. All items normally painted in accordance with good construction practice.
  - 8. All unfinished louvers and grilles.

2.3 WORK TYPICALLY EXCLUDED

#

- A. Shop applied primer on structural steel and miscellaneous metals items.
- B. Aluminum frames, doors, and windows.
- C. Plastic clad casework, millwork, and wall panels.
- D. Factory finished equipment unless noted otherwise (provide job touch-up).

2.4 DRAWING REFERENCE: Reference any paint or finish notes in the drawings for any pre-selected colors or other requirements.

#### 2.5 SUBMITTALS

- A. Submit manufacturer's product data describing each proposed type of paint, sealer, stain, or coating and its recommended use. Include viscosity and percent solids information. Where not the specified base manufacturer, list the specified brand name and type and the proposed substitute. The Architect shall be the sole judge as to equivalency of systems.
- B. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

#### 2.6 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of two years after the date of Substantial Completion of the project.
- B. Warranted defects shall include but not necessarily be limited to peeling, crazing, cracking, blistering, mildewing, chalking or dusting, pin holes, color fade or loss of hardness or sheen.

#### 2.7 QUALITY ASSURANCE

- A. Painting contractor shall have a minimum of 5 years experience in the application of the specified systems for projects of similar size and scope as this project.
- B. If requested by the Architect, provide system manufacturer's certification of the proposed painting contractor as approved for application of the product.

#### 2.8 DELIVERY, STORAGE AND HANDLING

- A. Do not deliver painting materials to the jobsite until spaces and surfaces are ready for painting.
- B. Deliver materials in manufacturer's original containers, unopened except for shop mixing of colors. Containers shall bear manufacturer's readable labels indicating brand and type of paint. Any additional containers with labels indicating products not approved shall be removed from the jobsite. Any applied material not previously approved by the Architect is subject to removal and reapplication with the appropriate approved product.
- C. Store materials in environmentally controlled area. Interior products shall be acclimated to a temperature range of 50-80 degrees F. at least 24 hours prior to application.

#

1.9 MOCK UP

- A. Prior to starting any painting work; contractor must paint an area 6' x 6' on any interior wall for Owner's approval. **Painting work must not begin until paint sample has been approved by Owner.**

### PART 3 - PRODUCTS

#### 3.1 MANUFACTURERS

- A. TYPICAL PAINTS: Systems are based on the first listed manufacturer. Only equivalent systems provided by specified manufacturers in accordance with attached Product Comparison sheet and as approved by the Architect are approved for use.
1. Sherwin Williams, Inc.
  2. Pittsburgh Paints
  3. Pratt & Lambert
  4. Benjamin Moore Co.
- B. SPECIALTY PAINTS:
1. Epoxies: Sherwin Williams, PPG, Pratt & Lambert.
  2. Projection Screen Paint: Sherwin Williams

#### 3.2 INTERIOR SYSTEMS

- A. SYSTEM TYPES FOR NEW WALLS (Unless indicated otherwise on Finish Schedule or drawings):
1. Drywall in toilet rooms, storage rooms, and mechanical/electrical/toilet rooms/ classrooms: **Semi Gloss Enamel** at walls and ceilings.
  2. Drywall soffits: **Eggshell Enamel**.
  3. Typical masonry (CMU): **Gloss Enamel**.
  4. Masonry (CMU) in toilet rooms: **Gloss Epoxy**.
  5. Steel railings: **Gloss Aliphatic Urethane**.
  6. Suspended rigging over stage: **Dry Fog**.
- B. SYSTEM DESCRIPTIONS (Reference item 3.3 for modifications and preparation required for these systems when applied to existing walls already painted):
1. Primer on gypsum board: SW PrepRite High Build Primer B28W601 – one coat over light to medium texture (submit texture sample for approval)
  2. Eggshell Enamel on Drywall: SW Pro Mar 400 Latex Eg-Shel B20W4400 – one coat over specified primer.
  3. Semi-Gloss Enamel on Drywall: SW Pro Mar 400 Latex Semi Gloss B31W4400 - one coat over specified primer.
  4. Epoxy Paint on Drywall: One coat SW PrepRite 200 Latex Primer B28W200 over specified primer.
  5. Gloss Enamel on Drywall: Two coats SW Water Based Catalyzed Epoxy B70 Series gloss acrylic over specified primer.

#

6. Semi-Gloss Enamel on shop-primed metals: SW Water Based Industrial Enamel B53-300 acrylic gloss Enamel – two coats.
7. Natural Finish on Wood: SW Sherwood BAC Wiping Stain (one coat) + SW Wood Classics Sanding Sealer B26V3 (one coat) + SW Wood Classics Satin Varnish A66.
8. Clear Finish on Wood: SW Wood Classics Polyurethane Varnish A67 (two coats). Sand lightly between all coats.
9. Block Filler: SW Prep Rite Block Filler B25W25 (for areas not subject to moisture); SW Heavy Duty Block Filler (for areas subject to moisture). Provide 2 coats as specified under “Execution”.
10. Gloss Enamel on CMU or concrete: Two coats block filler plus two coats SW Water based Industrial Enamel gloss acrylic latex over specified primer.
11. Semi-Gloss Enamel on CMU or concrete: Two coats block filler plus two coats SW Water Based Industrial Enamel semi-gloss acrylic latex over specified primer.
12. Semi-Gloss Epoxy Paint on concrete: One coat SW Water Based Epoxy semi-gloss over cured concrete plus finish coat of SW Water Based Epoxy semi-gloss. Minimum paint thickness 3.0 dry mils.
13. Gloss Epoxy Paint on CMU: Two coats block filler (unless surface-bonded) plus finish coat of gloss. Minimum paint thickness 3.0 dry mils.
14. Gloss Epoxy Paint on concrete: One coat SW Water Based Epoxy gloss over cured concrete plus finish coat of SW Water Based Epoxy gloss. Minimum paint thickness 3.0 dry mils.
15. Semi-Gloss Enamel on utility piping and galvanized metals: SW Pro-Cryl Universal Metal Primer – one coat + SW DTM Acrylic Semi Gloss – two coats.
16. Semi-Gloss Epoxy Paint on CMU: Two coats block filler plus finish coat of SW Water Based Epoxy semi-gloss. Minimum paint thickness 3.0 dry mils.
17. Gloss Aliphatic Urethane Enamel on primed steel railings: Over epoxy shop primer apply two coats SW Hydrogloss Single Component Water Based Urethane B65-181 Urethane Gloss Enamel using airless spray equipment.
18. Dry Fall Acrylic (exposed deck, structure and rigging): One coat SW Super Save Lite Acrylic Dry Fall Eggshell Primer & Finish. Black color. Overspray dries to non-adhering dust in a ten foot fall.

## 2.3 EXTERIOR SYSTEMS

### A. SYSTEM TYPES:

1. Exterior Metals: **Gloss Enamel.**
2. Field welds: **Zinc-Rich Coating.**

### B. SYSTEM DESCRIPTION:

1. Gloss Enamel on Galvanized Metals: SW Pro-Cryl Universal Metal Primer B66W310 (one coat) + SW Sher-Cryl HPA B66-300 enamel – two coats.
2. Block Filler on CMU: SW Heavy Duty Block Filler B24W46, one coat.
3. Gloss Enamel on Shop-Primed Metals: SW Sher-Cryl HPA B66-300 gloss enamel-two coats.
4. Gloss Enamel on Aluminum: SW Pro-Cryl Universal Metal Primer B66W310 – (one coat) + SW Sher-Cryl HPA B66-300 gloss enamel – two coats.
5. Field Welds: “ZRC” cold-applied galvanizing.

## PART 4 - EXECUTION

### 4.1 PREPARATION

- A. METALS: Remove grease, oil, and dirt. Touch-up any damaged primer with like material. Remove any welding tags and grind smooth before painting. Fill any open galvanizing ports.
- B. PLASTER, CMU, CONCRETE: Remove dusting and mortar residue. Remove any efflorescence and seal. Ensure that plaster, concrete and mortar joints are dry and fully cured.

#

#### 4.2 APPLICATION

- A. GENERAL: All paint and finishes be brushed or sprayed in even, uniform coats without runs or sags. Allow each coat to dry completely before applying succeeding coats. All surfaces shall be dry and no painting shall be done in damp conditions or when the ambient temperature is below 50 degrees F.
- B. WOOD DOORS: Factory sealed tops, bottoms, and edges of plastic laminate surfaced doors left undisturbed require no additional finishing. Reseal any job cuts. Paint metal glazing stops.
- C. MECHANICAL/ELECTRICAL EQUIPMENT: Painting contractor shall examine the mechanical and electrical drawings to determine quantities and locations of exposed piping, louvers not shown in Architectural drawings, electrical and telephone panels in finished areas, exposed electrical conduit in finished areas.
- D. BLOCK FILLER AT CMU: Apply **first coat** of filler to ensure penetration into voids and work into block texture with bristle brush. Follow with a **minimum of one additional coat**. Provide uniform finish with no pinholes.
- E. DRYWALL: Paint finish, sheen and texture shall be uniform and match the samples submitted to and approved by the Architect.

#### 4.3 PREPARATION OF EXISTING PAINTED SURFACES

Maintenance painting will frequently not permit or require complete removal of all old coatings prior to repainting. However, all surface contamination such as oil, grease. Loose paint, mill scale dirt, foreign matter, rust, mold, mildew, mortar, efflorescence, and sealers must be removed to assure sound bonding to the tightly adhering old paint. Glossy surfaces of old paint films must be clean and dull before repainting. Thoroughly washing with an abrasive cleanser will clean and dull in one operation, or, wash thoroughly and dull by sanding. Spot prime any bare areas with an appropriate primer. Recognize that any surface preparation short of total removal of the old coating may compromise the service length of the system. Check for compatibility by applying a test patch of the recommended coating system, covering at least 2 to 3 square feet. Allow to dry one week before testing adhesion per ASTM D3359. If the coating system is incompatible, complete removal is required.

### PART 4 – SCHEDULES

#### 4.1 COLOR SELECTIONS

- A. SCHEDULE: Unless colors are pre-selected in the Bidding Documents, the Architect shall prepare color schedule for the project using colors selected from the approved paint manufacturer(s). Where colors are pre-selected, the painting contractor shall use the colors selected or submit a schedule of proposed exact color matches by one of the specified paint manufacturers. **Provide 12” x 12” samples of actual paint for each color** whether pre-selected color or proposed color match.
- B. DOCUMENTATION: Upon completion of the Project, painting contractor shall furnish to the Architect a complete schedule of paint brands, types, and colors actually used for each room and area.

#### 4.2 EXTRA MATERIALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents. Furnish quantity not less than 5 percent for each color (field and accent) of paint used.



Edinburg Fire Department  
Fire Station #5  
#  
END OF SECTION - 09900

MAS Proj. No. 219003

## SECTION 09800 - ELASTOMERIC COATING

### PART 1 – GENERAL

#### 1.0 Coordination

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.1 Description

- A. Provide internally plasticized, elastomeric high-build waterproof emulsion coating with terpolymer acrylic resins coating for vertical applications for sealing exterior exposed masonry, stone, stucco, tile and concrete walls.

#### 1.2 Submittals

- A. Submit for approval samples, product data.

#### 1.3 Quality Assurance

- A. Comply with governing codes and regulations. Provide products of acceptable manufacturers which have been in satisfactory use in similar service for five years. Use experienced installers. Deliver, handle, and store materials in accordance with manufacturer's instructions.

### PART 2 - PRODUCTS

#### 2.1 Materials

- A. Elastomeric Coating: Concretite. Color: Architect to select from standard palette of colors.
- B. Sealant: One-part Urethane sealant, Tuff-Stuff.

### PART 3 - EXECUTION

#### 3.1 Installation

- A. Install materials and systems in accordance with manufacturer's instructions and approved submittals. Install materials with uniform appearance.
- B. Coordinate with work of other sections.

B. Surface Preparation:

1. Masonry: Masonry, concrete and stucco surfaces should be clean and free of oil and grease. Remove loose particles, laitance, efflorescence, and other foreign materials by wire brushing, power washing, or other effective means. Surface with heavy chalk face should be power-washed or sand blasted. Repair all cracks, openings, and imperfections with one-part urethane sealant.
2. Concrete Block: Concrete block should be clean and dry. Mortar joints should be free of voids and cracks. Fill all block with a quality exterior block filler.
3. Previously Coated Surfaces: All surface contamination- such as oil, grease, loose paint, mill scale, dirt, foreign matter, rust, mold, mildew, loose mortar and efflorescence- must be removed to assure sound bonding to the tightly adhering old paint.

C. Application:

Roller Application: Spread Rate: 1-1/2 gallons/100 sq. ft.

3.2 CLEANING

- A. As work proceeds and upon completion of areas, remove coating where spilled, splashed or spattered.
- B. During progress of work keep premises free from accumulation of tools, equipment, surplus materials.
- C. Upon completion of work, leave premises neat and clean, to the satisfaction of the Owner.

END OF SECTION

## SECTION 10 14 00 — GRAPHICS AND SIGNAGE

### PART 1 - GENERAL

#### 1.1 COORDINATION:

- A. The General Conditions of the Contractor for Construction and the Supplementary Conditions to the General Conditions of the Contract for the Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addendum issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the stringent requirements and the greater quantity shall apply.

#### 1.2 WORK INCLUDED

- A. Material and installation for the Plastic Room Identification Plaques.
- B. Material and Installation for Exterior/Interior Building Identification Letters.
- C. Material and Installation For Building Dedication Plaque.

#### 1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Interior wall materials and finishes.
- B. Exterior wall materials and finishes.
- C. Typical handicapped site signage provided under Section 02846.

#### 1.4 SUBMITTALS

- A. Submit manufacturer's product data describing materials, and mounting methods for Room Identification Plaques, Exterior/Interior Building Identification Letters, and Building Dedication Plaque.
- B. Submit color samples of actual material for color and finish selection by Architect.
- C. Submit finished sample of room identification plaque(s) with any required symbols other than text.
- D. Submit paper "rubbing" of final layout of Building Dedication Plaque for Architect's approval.
- E. Submit full size paper layout of Exterior Building Identification Letters for each line of text.
- F. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

## 1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.
- B. Warranted defects shall include but not necessarily be limited to color fading, delamination, failure of anchoring or fastening, cracking, breaking or tarnishing.
- C. Exterior signage or building letters contributing to streaking or staining of building shall be a defect to be corrected by the Contractor, with building materials cleaned or replaced as required.

## 1.6 QUALITY ASSURANCE

- A. Fabrication and installation company shall have a minimum of 3 years experience in the installation of similar systems for projects of similar size and scope.

## 1.7 DELIVERY, STORAGE AND HANDLING

- A. Do not deliver materials to the jobsite until surfaces are ready for installation of graphics.
- B. Store materials in covered, dry, temperature and humidity controlled space.

## 2 PART TWO – PRODUCTS

### 2.1 MANUFACTURERS

- A. ROOM IDENTIFICATION PLAQUES:
  - 1. South Texas Graphic Specialties, Inc.
  - 2. The Southwell Co.
  - 3. Cantrell Industries.
- B. EXTERIOR BUILDING IDENTIFICATION LETTERS:
  - 1. A.R.K. Ramos
  - 2. Gemini Inc.
  - 3. Matthews Bronze Div. – Architectural Products
  - 4. Metal Arts
- C. BUILDING DEDICATION PLAQUE:
  - 1. A.R.K. Ramos
  - 2. Matthews Bronze Div. – Architectural Products
  - 3. Metal Arts
  - 4. OMC Industries, Inc.

### 2.2 MATERIALS

- A. ROOM IDENTIFICATION PLAQUES:
  - 1. 6" X 6" X 1/4" thick two tone series:
  - 2. Fabrication: Constructed of Wilson Art face laminate (as selected by the Architect) laminated to a solid acrylic core. The raised 1/32" acrylic copy shall be cut through the laminate face color and chemically welded to the acrylic core to assure permanent attachment, including the symbols. Any lower and secondary copy shall be 5/8" high Sans Serif Medium (all caps) incised copy paint filled. Colors as selected by the Architect. Any secondary copy shall be 8-stroke computer

- engraved. Rounded corner letters will not be acceptable. The edge of the signs shall be finished to match the face laminate color-to-color as selected by the Architect.
3. At toilet rooms also provide with 2" high raised gender and wheelchair symbols when handicapped equipped noted on schedule. Symbols shall be chemically welded through the face laminate to the acrylic core. Edges painted a color as selected.
  4. The raised copy shall be accompanied with grade 2 Braille by means if Visi Touch DuraDot Braille manufacturing system. The clear Glass DuraDot shall have a 0.059 surface diameter and raised 1/32" above the face laminate and shall be unitized to the acrylic core through the face laminate. The edges of the sign shall be finished to match the face laminate color-to color as selected by the Architect. Any secondary copy shall be 8-stroke computer engraved. Rounded corner letters will not be acceptable.
  5. Installed plaques shall comply with all state, local, and federal requirements for compliance.

#### B. EXTERIOR BUILDING IDENTIFICATION LETTERS

1. Scope: The project shall include a cast letters as described below, to be provided and installed by contractor. Letterstyle, finish and mounting to be selected by Architect.
2. Fabrication of Letters: Fabricate letters to comply with requirements indicated below and as indicated on drawings.
  - A . Cut letters : Form letters by cutting from solid sheet material of thickness specified. Produce characters with smooth flat faces, sharp corners, precisely formed lines and profiles, free from pits, scale, sand holes and other defects. Supply anchoring devices on reverse side of individual letters as required.
3. Characteristics:
  - A. Metal: Aluminum
  - B. Size: As noted on drawings.
  - C. Thickness: 1 1/2 inches.
  - D. Letterstyle: Sans Serif
  - E. Finish: As selected by Architect from manufacturer's finish options (submit samples).
  - F. Mounting: Concealed (refer to drawings for wall type).
  - G. Text: As noted on drawings
4. Template: Provide full size paper mounting template showing hole placement and location of mounting holes.
5. Finishes: Colors and surface textures for exposed letters as selected by the architect from the manufacturer's standard and **premium** selections.

#### C. BUILDING DIRECTIONAL SIGNS:

1. 16" X 16" X 1/4" thick two-tone series.
2. Fabrication shall consists of Wilson Art plastic laminate with an incised demarcation line paint filled with a color as selected. Face laminate to be laminated to an acrylic core. Edges and backside of sign to match when applied to glass to include back-up plate. Characters shall be raised acrylic copy chemically welded through the laminate to the acrylic core to assure a permanent attachment of copy and symbols. Colors as selected by the Architect from laminate manufacturer's full range.
3. Include Braille notations.

4. Symbols shall be chemically welded through the face laminate to the acrylic core. Edges painted a color as selected.
5. Include two (2) directional signs as part of base bid.

C. BUILDING IDENTIFICATION PLAQUE:

1. 18" wide X 24" high cast bronze alloy plaque. Borders and raised text shall have satin finish. Background shall receive a dark oxidized leatherette finish. Faces and edges to be chemically cleaned and sprayed with two coats of clear acrylic lacquer.
2. Provide threaded stainless steel or brass studs on back for concealed mounting with epoxy. Letter style "Helvetica Medium" per A.R. Ramos or equivalent by specified manufacturer.
3. Layout, logos and letter sizes to be provided by the Architect. General contractor shall perform all conversions of architectural drawings & logos into useable vector line art format or any other type of format as required in order to produce the building plaque layout as provided by the Architect.

3 PART THREE- EXECUTION

3.1 INSPECTION AND PREPARATION

- A. Ensure that wall surfaces are completed and accepted by the Architect prior to installing wall-mounted items or painted wall graphics.
- B. Obtain approved location schedule for Room Identification Plaques prior to delivery of plaques to the jobsite.

3.2 INSTALLATION

A. ROOM IDENTIFICATION PLAQUES:

1. Apply top and bottom strips of 1/8" thick double stick vinyl foam tape and backs of each plaque. Apply liberal amount of clear silicone rubber adhesive to a minimum of 50% coverage of back of plaque.
2. Plaques shall be mounted to the strike side of the door on the wall within 5' of the floor and 2" from the jamb; when location is a glass side light, mount with a solid color back-up plate to cover reverse side of the glass. Attachment shall be with foam tape and silicone.

B. BUILDING DIRECTIONALS SIGNS:

1. Apply top and bottom strips 1/8" thick double stick vinyl foam tape on backs of each sign. Apply liberal amount of clear silicone rubber adhesive to a minimum of 50% coverage of back of sign.
2. Signs shall be mounted to the strike side of the door on the wall within 5' of the floor and 2" from the jamb; when location is a glass sidelight, mount with a solid color back-up plate to cover reverse side of the glass. Attachment shall be with foam tape and silicone.

C. EXTERIOR/INTERIOR BUILDING IDENTIFICATION LETTERS.

1. Pre-drill holes into masonry and insert threaded stud on back of letters into epoxy adhesive filled holes. Provide stainless steel spacers to set letters off wall 1/2" minimum 2 studs per letter.

D. BUILDING IDENTIFICATION PLAQUE:

1. Masonry Wall: Pre-drill holes into masonry walls and insert threaded studs on back of letters into epoxy adhesive filled holes. Mount plaque tight against wall.

2. Drywall: Mount plaque using a minimum of 4 moly type expansion screws and silicone adhesive.  
Mount plaque tight against wall.

END OF SECTION



## SECTION 10 21 13 - TOILET COMPARTMENTS

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 WORK INCLUDED

- A. Provide and install **solid phenolic toilet partition system and urinal screens** as indicated in the drawings, the approved shop drawings and as specified herein.
- B. Provide and install all toilet room and shower accessories as indicated in the drawings and as specified herein.

#### 1.3 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Wood blocking between studs.
- B. Wall and floor finishes.
- C. Electrical power provided.

#### 1.4 SUBMITTALS

- A. **SOLID PHENOLIC PARTITION SYSTEMS:**
  - 1. Submit shop drawings for solid phenolic partition system indicating plan and elevation dimensions and mounting details. Submit hardware samples and full chain of melamine samples for partition doors.
  - 2. Shop drawings indicating handicapped stall layouts not meeting State and Federal requirements will be returned and rejected without review.
- B. **ACCESSORIES:**
  - 1. Submit manufacturer's product data describing size, type, finish and installation requirements for each item.
  - 2. Indicate mounting heights for each item. Meet State and Federal requirements for the handicapped.
- C. Reference Section 01 33 00 SUBMITTALS for additional submittal requirements.

#### 1.5 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the work under this section for a period of one year after the date of Substantial Completion of the project.
- B. Warranted defects shall include but not necessarily be limited to delamination of facing or edging, swelling of core, change in alignment of parts, failure of anchorage or fasteners.
- C. Provide manufacturer's extended written warranty for systems and accessories where available.

#### 1.6 QUALITY ASSURANCE

- A. Partition system installation company shall have a minimum of 5 years experience in the installation of similar system for projects of similar size and scope.
- B. Partition system installation company shall be authorized by the system manufacturer for this installation.

#### 1.7 DELIVERY, STORAGE AND HANDLING

- A. Deliver partition system materials to the job site in manufacturer's original packaging.
- B. Store materials in covered, dry, temperature and humidity controlled space.

### PART 2 - PRODUCTS

#### 2.1 MANUFACTURERS

- A. SOLID PHENOLIC PARTITION SYSTEMS:
  - 1. Bobrick Washroom Equipment
  - 2. American Specialties, Inc.
  - 3. Ampco
  - 4. General Partitions
- B. ACCESSORY MANUFACTURERS:
  - 1. Bobrick Washroom Equipment
  - 2. Bradley Wash Fountain Co.
- C. PAPER TOWEL DISPENSERS:
  - 1. Bobrick Washroom Equipment
- D. ELECTRIC HAND DRYERS:
  - 1. Bobrick Washroom Equipment
  - 2. World Dryer Corporation

#

#### 2.2 TOILET ROOM ACCESSORIES:

- 1. Grab bars: Furnish and install grab bars at each handicapped toilet stall. Bars shall be Bobrick No. B-6806 series, 1-1/2" outside diameter, satin finish stainless steel, configuration as indicated on the drawings, 1-1/2" clear to wall. **Where bars are mounted over back of toilet, General Contractor shall hold flush valve low.**
- 2. Mirrors: Mirrors shall be 1/4" plate glass, mirror quality, with copper backs. Provide polished stainless steel or brass chrome plated frames in sizes indicated on the drawings.
  - a. Tilted mirrors shall be Bobrick No. B-293 or equivalent by specified manufacturer.

- b. Flat mirrors shall be Bobrick No. B-290 or equivalent by specified manufacturer.
- 3. Mop Holder: Bobrick B223X24 stainless steel. Furnished and installed by Contractor.
- 4. Soap dispensers: Bobrick Contura Series Surface Mounted Soap Dispenser Model B-4112. Furnished and installed by Contractor.
- 5. Tissue dispensers: ClassicSeries Surface-Mounted Toilet Tissue Dispenser for Two Rolls Model B-265 of Bobrick Washroom Equipment, Inc. Furnished and installed by Contractor.
- 6. Paper Towel Dispensers: Recessed Paper Towel Dispenser and Waster Receptacle shall be Model B-369 of Bobrick Washroom Equipment, Inc. Furnished and installed by Contractor.
- 7. Shower Seats: Right hand folding handicap shower seat Model B-517 of Bobrick Washroom Equipment, Inc. Left hand folding handicap shower seat Model B-518 of Bobrick Washroom Equipment, Inc.
- 8. Clothes hooks: Surface mounted clothes hooks Model B-981 of Bobrick Washroom Equipment, Inc.
- 9. Soap Dish: Surface mounted soap dish Model B-680 of Bobrick Washroom Equipment, Inc.
- 10. Towel Pin: Surface mounted towel pin Model B-677 of Bobrick Washroom Equipment, Inc.
- 11. Shower Rod: Model B-207 of Bobrick Washroom Equipment, Inc.
- 12. Shower Curtain & Hooks: Model B-204-1 and Model B-204-2 of Bobrick Washroom Equipment, Inc.

## 2.3 SOLID PHENOLIC PARTITION SYSTEMS:

### A. STILES, PANELS, DOORS, SCREENS, BENCHES

- 1. Solid phenolic material constructed of solidly fused plastic laminate with matte-finish melamine surfaces, colored face sheets, and black phenolic-resin core that are integrally bonded. Edges shall be black. Brown edges shall not be acceptable. Color and pattern as selected by architect from manufacturer's standard colors.
- 2. Solid phenolic material shall meet National Fire Protection Association and International Building Code Interior Wall and Ceiling Finish Class A, Uniform Building Code Class I, ASTM E-84 Fire Resistance Standards; flame spread 20, smoke density 95.
- 3. Finish Thickness
  - a. Stiles and doors shall be 3/4" (19mm).
  - b. Panels and benches shall be 1/2" (13mm).

### B. HARDWARE

- 1. All hardware to be 18-8, type-304 stainless steel with satin-finish.
- 2. All hardware shall be concealed inside compartments with the exception of outswing doors.
- 3. Hardware of chrome-plated "Zamak" is unacceptable.

### C. LATCH

1. Sliding door latch shall be 16-gauge (1.6mm).
2. Sliding door latch shall require less than 5-lb force to operate. Twisting latch operation will not be acceptable.
3. Latch track shall be attached to door by flathead machine screws into factory installed threaded brass inserts.
4. Latch handle shall have rubber bumper to act as door stop.
5. Latch shall allow door to be lifted over 16-gauge (1.6mm) keeper for emergency access.
6. Metal-to-metal connection shall withstand a direct pull of over 1,500 lb. per screw.

D. HINGES

1. Cam shall be adjustable in the field to permit door to be fully closed or partially open when compartment is unoccupied.
2. Hinges shall be attached to door and stile by theft-resistant, one-way stainless steel machine screws into factory-installed metal inserts. Fasteners secured directly into the core are not acceptable.
3. Metal-to-metal connection shall withstand a direct pull of over 1,500 lb. per screw.

E. Clothes Hook shall be constructed of stainless steel and shall project no more than 1-1/8" (29mm) from face of door. Clothes hook shall be secured by theft-resistant, one-way stainless steel screws.

F. Mounting Brackets shall be constructed of stainless steel and shall be mounted inside compartment. Mounting brackets exposed on the exterior of the compartment will not be acceptable. Wall mounted urinal screen brackets shall be 11-gauge (3mm) double thickness.

G. Leveling Device shall be 3/16" (5mm) hot rolled steel bar; chromate-treated and zinc-plated; through bolted to base of solid phenolic stile.

H. Stile Shoe shall be one-piece, 4" (102mm) high, type-304, 22-gauge (0.8mm) stainless steel with satinfinish. Top shall have 90° return to stile. Patented one-piece shoe capable of adapting to 3/4" or 1" stile thickness and capable of being fastened (by clip) to stiles starting at wall line.

I. Headrail (Overhead-Braced) shall be satin finish, extruded anodized aluminum (.065" / 1.65mm thick) with anti-grip profile. Type: Floor mounted, overhead braced continuously over entire system.

2.4 ELECTRIC HAND DRYERS:

1. Hand Dryers:
  - a. Provide and install No. B-750 115V Recessed Aircraft Automatic Hand Dryer, as manufactured by Bobrick or equivalent by specified manufacturer. Fixed outlet.
  - b. Provide units wired for 115 volt AC, 20 amps, unless electrical drawings provide other power.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Ensure that Contractor has properly installed solid wood blocking between studs at all mounting points.
- B. Ensure that Contractor has provided coordinated electrical power at each electric dryer location.

3.2 INSTALLATION

- A. Install accessories and partition systems in accordance with the project drawings, approved shop drawings and as specified herein. Use tamper proof stainless steel fasteners for all items.
- B. ACCESSORIES:
  - 1. Install through finished stud walls into solid wood blocking with stainless steel one-way screws. No plastic anchors.
  - 2. Attach to masonry walls using stainless steel machine screws in lead shield anchors.
- C. PARTITION SYSTEMS AND URINAL SCREENS:
  - 1. Mount channels using stainless steel one-way screws through finished stud walls into solid wood blocking.
  - 2. Mount channels to masonry walls using stainless steel machine screws in lead shield anchors.
  - 3. Job measure for proper fit and to ensure that the maximum space between edge of any pilaster or panel and its adjacent surface is one inch.
  - 4. Install pilaster, doors and panels plumb and square. Adjust doors for gravity closing.
- D. FRAMED MIRRORS:
  - 1. Mirrors shall be installed with theft-proof anchors at height shown on drawings. Furnish tilted mirrors where shown.
  - 2. Install mirrors at other locations in addition to toilet rooms as indicated in the drawings.
  - 3. Unframed mirrors are provided and installed under another section of these specifications.

END OF SECTION

## SECTION 10 26 41 – UL 752 LEVEL 1 BULLET-RESISTANT FIBERGLASS PANELS

### PART 1 – GENERAL

#### 1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions.

#### 1.2 REFERENCES

- A. Underwriters Laboratories:
  - 1. UL 752 Specifications and Ammunition, 11th Edition, Standard for Bullet Resisting Equipment published September 9, 2005, revised December 21, 2006, Level 1

#### 1.3 SUBMITTALS

The following shall be submitted in accordance with Section 10 26 00 and the SPECIAL CONTRACT REQUIREMENTS (Submit for approval prior to fabrication samples, brochures, specifications):

- A. UL LISTING Verification and UL752 Current Test Results as provided by Underwriters Laboratories.
- B. Printed data in sufficient detail to indicate compliance with the contract documents.
- C. ASTM E119-98 One-Hour Fire Rating of Building Construction and Materials.
- D. Manufacturer's Instructions for installation of Bullet Resistant Fiberglass Panels.

#### 1.4 DESIGN

- A. Through the design, manufacturing technique and material application the Bullet Resistant Fiberglass shall be of the "non-ricochet type." This design is intended to permit the encapture and retention of an attacking projectile lessening the potential of a random injury or lateral penetration.

#### 1.5 DELIVERY, HANDLING, AND STORAGE

- A. Deliver the materials to the project with the manufacturer's UL LISTED Labels intact and legible.
- B. Handle the material with care to prevent damage. Store the materials inside under cover, stack flat and off the floor.

#### 1.6 WARRANTY

- A. All materials and workmanship shall be warranted against defects for a period of two (2) years from the date of receipt at the project site.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURER

- A. Acceptable Manufacturers:

- 1. Waco Composites, Ltd., Waco, TX 76710, fax: 254-752-3634, 254-752-3622  
email: [sales@armorcore.com](mailto:sales@armorcore.com)

- B. Substitutions: Permitted in accordance with Section 01 25 00 Substitution Procedures.

### 2.2 MATERIAL

- A. The panels shall be made of multiple layers of woven roving ballistic grade fiberglass cloth impregnated with a thermoset polyester resin and compressed into flat rigid sheets. The production technique and materials used shall provide the controlled internal delamination to permit the encapture of a penetrating projectile.
- B. Bullet Resistant Fiberglass panels: 1/4" nominal thickness, and 2.6 lbs. per sq. ft. nominal weight.

### 2.3 SECURITY LEVEL

- A. The Bullet Resistant Fiberglass will be rated and tested for UL752 Level 1.

## PART 3 - EXECUTION

### 3.1 SUPPORTING MEMBERS

- A. Prior to installing the bullet resistive material the contractor shall verify that all supports have been installed as required by the contract documents and the architectural drawings.

### 3.2 JOINTS

- A. All joints shall be reinforced by a back-up layer of bullet resistive material. The bullet resistance of the joint, as reinforced, shall be at least equal to that of the panel. Minimum width of reinforcing layer at joint shall be 4-inches (2" on each panel or a 2" minimum overlap).

### 3.3 APPLICATION

- A. Armor shall be installed in accordance with the manufacturer's printed recommendations. Armor panels shall be adhered using an industrial adhesive, mastic, screws or bolts. Method of application shall maintain the bullet resistive rating at junctures with the concrete floor slab, the concrete roof slab, the bullet resistive door frames, the bullet resistive window frames, and all required penetrations.

END OF SECTION

## SECTION 10 28 13 - TOILET & BATH ACCESSORIES

### PART 1 GENERAL

#### 1.00 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.01 SECTION INCLUDES

- A. Vertical or Horizontal Baby Changing Stations and Child Protection Seat for use in commercial men's and women's toilet and bath facilities.

#### 1.02 QUALITY ASSURANCE

- A. Regulatory, Requirements: Conform to ASTM F2285 (formerly ASTM PS125) Standard Consumer Safety Performance Specification for Diaper Changing Tables for Commercial Use, ANSI A117.1 Accessible and Usable Buildings and Facilities, ANSI Z535.4 Product Safety Signs and Labels, German TUV Technical Inspection Association, or local code if more stringent installation requirements are applicable for barrier-free accessibility. FDA approved high-density polyethylene (HDPE) materials conform to ASTM G21 Antifungal, and ASTM G22 Antibacterial Standards.

#### 1.03 WARRANTY

- A. Submit manufacturer's 5-year limited warranty on materials and workmanship and 5-year replacement warranty against vandalism agreeing to repair or replace unit that fails to perform as intended from date of substantial completion.

### PART 2 PRODUCTS

#### 2.01 MANUFACTURERS

- A. Acceptable Manufacturers: Koala Kare Products, a Division of Bobrick, Englewood, CO, 877-284-3906



## 2.02 ACCEPTABLE PRODUCTS

- A. Baby Changing Stations
  1. Horizontal Design with molded Braille instructions (Model KB100-00)
  2. Vertical Design with molded Braille instructions (Model KB101-00)
  3. Stainless Steel Recessed Mounted Horizontal Design (Model KB110-SSRE)
  4. Stainless Steel Surface Mounted Horizontal Design (Model KB110-SSWM)
- B. Sanitary Liner Refills. Liners are for use with all Koala Baby Changing Station designs (Model KB150-99).
- C. Child Protection Seat: Koala Child Protection Seat (Model KB102-00)

## 2.03 MATERIALS

- A. Materials/finishes: {[Molded FDA approved high density polyethylene HDPE, cream color] (Models KB100-00, KB101-00, KB102-00)} {[18 gauge, type 304 satin stainless steel exterior finish with grey polyethylene interior (Models KB110-SSRE, KB110-SSWM)]}.
- B. Hinges: reinforced, full-length steel-on-steel.
- C. Mounting supports: multiple, 11-gauge steel.
- D. Operation: hidden pneumatic gas spring mechanism for safe open/close motions.

## 2.04 ACCESSORIES

- A. Integral, built-in Liner Dispenser for use with 3-ply chemical-free biodegradable 13" x 19" sanitary liners.
- B. Replaceable snap-lock protective holding straps.
- C. Molded graphic instructions and safety messages in 6 languages and Braille. Identifying door plaque.
- D. Optional antimicrobial polyethylene.
- E. Optional lock supplied with 2 keys (not available for stainless steel exterior finish models).

## PART 3 EXECUTION

### 3.01 PREPARATION

- A. Provide templates and rough-in measurements as required.

### 3.02 INSTALLATION

- A. Install in accordance with manufacturer's hardware and instructions.
- B. Locate products to eliminate interference with door swings or use of fixtures in compliance with ADA regulations

END OF SECTION

## SECTION 10 44 00 - FIRE EXTINGUISHERS AND CABINETS

### PART 1 GENERAL

#### 1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.01 SECTION INCLUDES

- A. Fire extinguishers.
- B. Fire extinguisher cabinets.
- C. Accessories.

#### 1.02 RELATED SECTIONS

- A. Section 04220 – Concrete Masonry Units; CMU walls to receive bracket mounted fire extinguisher.
- B. Section 06100 – Rough Carpentry: Wood blocking and framing to receive semi-recessed fire extinguisher cabinets.
- C. Section 09260 – Gypsum Drywall Assemblies: Finished openings in walls for semi-recessed fire extinguisher cabinets.

#### 1.03 REFERENCES

- A. NFPA 10 – Standard for Portable Fire Extinguishers; National Fire Protection Association; 2002.
- B. UL (FPED) – Fire Protection Equipment Directory; Underwriters Laboratories Inc.; current edition.

#### 1.04 PERFORMANCE REQUIREMENTS

- A. Conform to NFPA 10.
- B. Provide extinguishers classified and labeled by Underwriters Laboratories Inc. for the purpose specified and indicated.

## 1.05 SUBMITTALS

- A. See Section 01 33 00 – Submittals, procedures and requirements for shop drawings, product data and submittal requirements.
- B. Shop Drawings: Indicate cabinet physical dimensions.
- C. Product Data: Provide extinguisher operational features.
- D. Manufacturer's Installation Instructions: Indicate special criteria and wall opening coordination requirements.
- E. Manufacturer's Certificate: Certify that products meet or exceed specified requirements.
- F. Maintenance Data: Include test, refill or recharge schedules and re-certification requirements.

## 1.06 ENVIRONMENTAL REQUIREMENTS

- A. Do not install extinguishers when ambient temperature may cause freezing of extinguisher ingredients.

## PART 2 PRODUCTS

### 2.01 MANUFACTURERS

- A. Fire Extinguishers, Cabinets and Accessories:
  - 1. JL Industries, Inc; Product 1037B20 with Extinguisher: [www.jlindustires.com](http://www.jlindustires.com).
  - 2. Larsen's Manufacturing Co: [www.larsensmfg.com](http://www.larsensmfg.com).
  - 3. Potter-Roemer: [www.potterroemer.com](http://www.potterroemer.com).
  - 4. Substitutions: See Section 01 25 00 – Product Requirements.

### 2.02 FIRE EXTINGUISHERS

- A. Basis of Design: JL Industries, "Cosmic 10E".
- B. Type: Multipurpose dry chemical.
- C. Rating: Sized for project requirements.
- D. Mounting: Refer to floor plans for locations, annotated FEC for cabinets and FE extinguishers alone.
- E. Dry Chemical Type: Stainless steel tank, with pressure gage.
  - 1. Class A:B:C.
  - 2. Size 10.
  - 3. Finish: Baked enamel, Red color.
- G. ALL fire extinguishers shall be inspected and certified by the local authority having jurisdiction that they are charged and ready for use and shall be "tagged" identifying such.

### 2.03 FIRE EXTINGUISHER CABINETS

- A. Basis of Design:
  - 1. JL Industries, "Cosmopolitan – 1035B20 ADAC with Saf-T-Loc, TAS compliant.
  - 2. Designations: Refer to the floor plans, FEC for Extinguishers in cabinets and FE for surface mounted extinguishers secured to walls.

- B. Surface Mounted (Non-Cabinet, FE Type) – Bracket and Extinguisher (non-cabinet):  
Manufacturer's standard stainless steel strap with enamel finished bracket with locking band retainer.
  - 1. Bracket shall match the extinguisher type.
- C. Metal for Cabinets: Formed stainless steel sheet; 0.036 inch thick base metal; #4 finish stainless steel.
- D. Cabinet Configuration: Recessed type.
  - 1. Sized to accommodate accessories.
  - 2. Exterior nominal dimensions of 13 7/8 inch wide x 27 3/8 inch high x 6 inch deep.
  - 3. Trim: Returned to wall surface, with 3 inch projection, 1 1/2 inch wide face.
  - 4. Form cabinet enclosure with right angle inside corners and seams. Form perimeters trim and door stiles.
- E. Door: 0.036 inch thick, reinforced for flatness and rigidity; lock with full glass access. Hinge doors for 180 degree opening with two butt hinge. Provide nylon catch.
- F. Door Glazing: Glass, clear, 1/8 inch thick float. Set in resilient channel gasket glazing.
- G. Cabinet Mounting Hardware: Appropriate to cabinet. Pre-drill for anchors.
- H. Weld, fill, and grind components smooth.
- I. Finish of Cabinet Interior: Enamel, color to select from manufacturer's full color line.

#### 2.04 ACCESSORIES

- A. Extinguisher Brackets: Formed steel, chrome-plated.
- B. Cabinet Signage: FIRE EXTINGUISHER, vertical up face of cabinet to one side.

### PART 3 EXECUTION

#### 3.01 EXAMINATION

- A. Verify existing conditions before starting work.
- B. Verify rough openings for cabinet are correctly sized and located.

#### 3.02 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Install cabinets plumb and level in wall openings, 30 inches from finished floor to inside bottom of cabinet.
- C. Secure rigidly in place.
- D. Place extinguishers in cabinets.

### END OF SECTION

## SECTION 10 51 53 — HARDWOOD LOCKER BENCHES

### PART 1 - GENERAL

#### 1.1 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.2 DESCRIPTION

- A. Hardwood bench work includes the following, where indicated:
  - 1. Hardwood Locker Room Benches.
  - 2. ADA Compliant Hardwood Benches.
- B. Furnish all labor and materials necessary for the completion of work in this section as shown on the contract drawings and specified herein.
- C. Work in this section shall include, but is not limited to:
  - 1. Locker room benches and steel, or aluminum pedestals.
  - 2. Hardware for locker room benches and pedestals.
  - 3. Manufacturer's guarantee.
- D. Related work specified elsewhere shall include accessories and anchorage/blocking for attachment of locker room benches.

#### 1.3 PRODUCT

- A. Submit six (6) sets of shop drawings and details for architect's approval.
- B. Colors shall be selected from the manufacturer's full range of colors: REPUBLIC STANDARD COLORS; or approved equal.
- C. Color samples and hardware samples shall be submitted for approval by the architect upon request.

## PART 2 - PRODUCTS

### 2.1 MANUFACTURER

- A. Locker room benches and pedestals to be supplied by REPUBLIC STORAGE SYSTEM Co., or approved equal.

### 2.2 MATERIALS

- A. Locker room bench tops shall be 9-1/2" wide x 1-1/4" full finish thickness laminated maple. Benches and pedestals shall have an overall height of 17-1/2". All Bench tops are to be maple laminated. Locker room bench tops shall be fabricated from polymer resins compounded under high pressure, forming a single component section which is waterproof, nonabsorbent and has a self-lubricating surface that resists marks from pens, pencils, markers and other writing instruments. All plastic components shall be covered with a protective plastic masking.
- B. ADA Compliant Bench: ADA compliant bench tops are to be 48" wide and 24" deep max, laminated maple, 1-1/4" full finished thickness. All corners are to be rounded and sanded. Top and edges have two coats of a clear finish with one coat on the bottom. Four pedestals required. Benches shall have an overall height of 17-1/2".

### 2.3 CONSTRUCTION

- A. Locker room bench tops shall be 1-1/2" thick with all edges rounded to a 1/4" radius. Standard bench top size is 9-1/2" wide by customer specified length not to exceed 96" for a single piece.

### 2.4 HARDWARE

- A. Standard pedestals shall be 16-1/4" high, and secured to bench tops with stainless steel, torx screws and secured to the floor using lead expansion shields and stainless steel, phillips head screws. Pedestal consist of sturdy 1-1/4" outside diameter tubing with 10 gauge steel flanges.
- B. Moveable Pedestal shall be a freestanding trapezoidal-shaped 16" high, with a 1/4" x 3" aluminum bar stock and feature a black anodized finish. Non-skid pads shall be provided for each pedestal.
- C. Pedestal spacing not to exceed 4' on center.

## PART 3 - EXECUTION

### 3.1 PREPARATION

- A. Examine areas to receive locker room benches for anchorage/blocking that may affect installation of benches. Report any discrepancies to the architect.
- B. Take complete and accurate measurements of locations to receive locker room benches.
- C. Start of work constitutes acceptance of job.

3.2 INSTALLATION

- A. Install locker room benches in a rigid, straight, plumb and level manner, with plastic laid out as shown on shop drawings and manufacturer's installation instructions.
- B. No evidence of cutting, drilling and/or patching shall be visible on the finished work.
- C. Finished surfaces shall be cleaned after installation and be left free of all imperfections.

3.3 WARRANTY

- A. Provide written warranty against defects in materials and workmanship for the Work under this section for a period of one year after the date of Substantial Completion of the Project.
- B. Warranted defects shall include but not necessarily be limited to deterioration of finish, corrosion, faulty latch or lock operation, loosening of anchorage, failure of rivets or other fastenings.

END OF SECTION

## SECTION 10 75 00 - FLAGPOLES

### PART 1 GENERAL

#### 1.0 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division-1 specification section, apply to work of this section.

#### 1.2 SESSION INCLUDES:

- 1.3 Aluminum flagpoles, ground mounted.

#### 1.4 REFERENCES

- A. AASHTO M246/M246M Standard Specification for steel sheet, metallic-coated and polymer pre-coated for corrugated steel pipes.
- B. ASTM A53/A53M-02 Pipe, Steel, Black and Hot-Dipped, Zinc Coated, Welded and Seamless.
- C. ASTM A123/A123M-02 - Zinc (Hot Dipped Galvanized) Coatings on Iron and Steel Products.
- D. ASTM A790/A790M-02 Standard Specification for Seamless and Welded Ferrite/Austenitic Stainless Steel Pipe
- E. ASTM B221-02 Aluminum Alloy Extruded Bar, Rod, Wire, Shape, and Tube.
- F. ASTM B241/B241M-02 Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube
- G. CDA (Copper Development Association) - Handbook.

#### 1.5 PERFORMANCE REQUIREMENTS

- A. The General Contractor shall furnish and install a tapered aluminum flagpole, complete with a 50-star U.S. Flag, a State of Texas flag and a Mexico Flag, and all fittings, top ornament, lighting, ground spike with plate, foundation, grounding facilities, and all appurtenant work, all in accordance with the requirements of the Contract Documents.

#### 1.6 SUBMITTALS

- A. Submit under provisions of Section 01 33 00.
- B. Shop Drawings: Indicate detailed dimensions, base details, anchor requirements and imposed loads.
- C. Product Data: Provide product data on pole, accessories and configurations.



## 1.7 OPERATION AND MAINTENANCE DATA

- A. Maintenance Data: Provide lubrication and periodic maintenance requirement schedules.

## 1.8 QUALIFICATIONS

- A. Design flagpole foundation under direct supervision of a Professional Structural Engineer experienced in design of this work and licensed in the State of Texas. Provide calculations.

## 1.9 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, protect and handle products to site under provisions of Section 01 25 00.
- B. Spiral wrap flagpole with protective covering and pack in protective shipping tubes or containers.
- C. Protect flagpole and accessories on site from damage or moisture.

## PART 2 PRODUCTS

### 2.1 POLE MATERIALS

- A. Aluminum: ASTM B241; 6063 alloy, T6 temper.

### 2.2 POLE CONFIGURATION

- A. Nominal Height: 25 ft measured from nominal ground elevation.
- B. Flagpole: Ground mounted type.
- C. Flagpole Design: Cone tapered, seamless.
- D. Halyard: External type.

### 2.3 COMPONENTS AND ACCESSORIES

- A. Finial Ball: 6" diameter, spun aluminum.
- B. Truck Assembly: Cast aluminum; revolving, stainless steel ball-bearings, non-fouling.
- C. Flags: (1) U.S.A. design, 8 x 5 feet size, nylon fabric, brass grommets, hemmed edges.  
(1) Fire Department flag, 8 x 5 feet size, nylon fabric, brass grommets, hemmed edges.
- D. Cleats: Aluminum with stainless steel fastenings, two per halyard.
- E. Cleat Box: Aluminum with built-in hinge and hasp assembly, attached to pole with tamper proof screws inside box.
- F. Halyard: 3/8" diameter polypropylene, braided, white.

### 2.4 MOUNTING COMPONENTS

- A. Foundation Tube Sleeve: AASHTO M-246, corrugated 16-gage steel, galvanized.
- B. Pole Base Attachment: Flush, aluminum base with base cover.
- C. Lightning Ground Rod: Copper rod, 3/4" diameter; design length.

### 2.5 FINISHES

- A. Metal Surfaces in Contact with Concrete: Asphaltic paint.
- B. Concealed Steel Surfaces: Galvanized to ASTM A123 1.25 oz/sq ft.
- C. Aluminum: Anodized to color as selected.
- D. Finial: Spun and Lacquered.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

- A. Verify site conditions.
- B. Verify that concrete foundation is ready to receive work and dimensions are as indicated on shop drawings.

#### 3.2 PREPARATION

- A. Coat metal sleeve surfaces below grade and surfaces in contact with dissimilar materials with asphaltic paint.

#### 3.3 INSTALLATION

- A. Install flagpole, base assembly, and fittings in accordance with manufacturer's instructions.
- B. Electrically ground flagpole installation.
- C. Fill foundation tube sleeve with sand specified in Section 02 22 30 and compact.
- D. Install foundation plate and centering wedges, welded base assembly for flagpoles base set on concrete base and fasten.

#### 3.4 ERECTION TOLERANCES

- A. Maximum Variation From Plumb: One inch

#### 3.5 ADJUSTING

- A. Adjust work under provisions of Section 01 70 00.
- B. Adjust operating devices so that halyard and flag function smoothly.

END OF SECTION

SECTION 12 21 13 - HORIZONTAL LOUVER BLINDS

PART 1 – GENERAL

1.00 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

1.01 DESCRIPTION

- A. Material and installation of 2" Horizontal Louver Blinds With Aluminum Slats at each window location.

1.02 SUBMITTALS

- A. Manufacturer's complete CSI 3- part specification sheet.
- B. Submit working hand sample or mock up blind as required.
- C. Submit two 6" samples of aluminum slat indicating color and dimensions.
- D. Approval of submittals by Architect shall not relieve contractor from installing blinds with adequate clearance to permit smooth operation of the blinds and demonstrating blinds to be in smooth, uniform working order. Contractor must field verify all dimensions.

1.03 DELIVERY, STORAGE AND HANDLING

- A. Product to be delivered in manufacturer's original packaging.
- B. Products to be handled and stored to prevent damage to materials, finishes and operating mechanisms. Store in a clean, dry area, laid flat to prevent sagging and twisting of packaging.

1.04 EXTRA STOCK

- A. Describe extra attic stock as required: One (1) additional horizontal louver blind.

PART 2 – PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

- A. Springs Window Fashions LLC or approved architect equivalent.
- B. Substitutions Request: Submit for approval under provisions of section 01 25 00.

2.02 HORIZONTAL BLINDS

- A. Product: Bali® 2" Aluminum School Blinds

- B. Color Name: As Selected by Architect; Color Number: As Selected by Architect.
- C. SureClose® Headrail shall be 1 5/8" high x 2 1/4" wide x .022" thick U-shaped steel with 1/8" light blocking lip on the bottom centerline. The steel finishing process includes phosphate treatment for corrosion resistance, a chrome-free sealer, a low HAP urethane primer and a topcoat with low HAP polyester baked enamel.
- D. Cord tilter shall be a snap-in component incorporating a worm and pulley of low-friction thermoplastic and a nylon gear. Standard tilt cords shall measure 2.2 mm in diameter.  
Select One:  
☐ Cord tilter (standard)
- E. Cord lock shall be metal of a snap-in design incorporating a floating, shaft-type locking pin and shall incorporate a crash proof safety feature that will lock blind automatically upon release of cord  
Options:  
☐ Ring pull provides a single plated steel ring in lieu of tassels with a nominal 4" cord length.
- F. Lift cord shall be made of braided polyester measuring 2.2mm in diameter.
- G. Vinyl ladder tape shall be 1 1/2" wide reinforced vinyl. Standard ladder spacing shall be 42mm.  
Select One Vinyl Tape Color:  
☐ 005 White  
☐ 670 Creamy Beige  
☐ 904 Lamplight  
☐ 983 Gray Haze  
☐ 062 Char Brown
- H. Slats shall be 5000 series cold-rolled aluminum containing the maximum allowable recycled content to produce a high strength and corrosion resistant flexible product. Slats shall be nominally 2" wide x .008" thick and treated with Advanced Finishing Technology (AFT), providing a smooth, hard, less porous surface. AFT delivers anti-static performance to repel dust and anti-microbial qualities to resist fungal and bacterial growth. Slats shall be treated with a chrome-free sealer and a topcoat of low HAP polyester baked enamel.
- I. Bottom rail shall be "C" shaped 9/16" high x 2" wide x .040 thick anodized aluminum. It is fully enclosed with a dust cover slat and finished with a polyester baked enamel to match headrail.

## 2.03 FABRICATION

- A. Blinds shall be fabricated according to specifications and accurate to tolerance established by SWF engineering standards

## PART 3 – EXECUTION

### 3.01 INSPECTION

- A. Installer shall be responsible for inspection of jobsite, approval of mounting surfaces, verification of field measurements and installation conditions. Installation shall commence when satisfactory conditions are met.

- B. Do not dimension the drawings. Any questions concerning dimensions should be directed to the Architect for clarification.

### 3.02 INSTALLATION

- A. Install blinds in accordance with manufacturer's instructions including recommended support brackets and fasteners.
- B. Install blinds with adequate clearance to permit smooth operation of the blinds. Demonstrate blinds to be in smooth, uniform working order.

### 3.03 MAINTENANCE AND CLEANING

- A. Maintain and clean blinds in accordance with manufacturer's instructions.

END OF SECTION

## SECTION 13 34 19 — PRE-ENGINEERED METAL BUILDINGS

### PART 1 - GENERAL

#### 1.00 COORDINATION

- A. The General Conditions of the Contract for Construction and the Supplementary Conditions to the General Conditions of the Contract for Construction shall be considered as part of this section of the specifications.
- B. Each Bidder shall be responsible for determining during the bidding period the extent that any addenda issued during the bidding period may affect this section of the specifications.
- C. Reference Instructions to Bidders for requirements regarding substitutions of materials and products.
- D. Where conflicts occur between the drawings and specifications, between different drawings, between different portions of this section of the specifications, or between different sections of the specifications, the more stringent requirements and the greater quantity shall apply.

#### 1.01 SCOPE:

- A. Provide all pre-engineered metal buildings, complete, as shown on the Drawings, specified herein, or needed for a complete and proper installation and not specifically called for under other Sections of these specifications.
- B. The requirements of Division 0 – “Bidding and Contract Requirements” and Division 1 – “General Requirements” of this Project Manual apply to all work required for this section.

### PART 2 - PRE-ENGINEERED METAL BUILDING SYSTEM:

#### 2.01 GENERAL:

- A. The intent of these specifications and drawings is to establish a quality and performance level for structural design, material, durability, and workmanship.
- B. All bidders must conform strictly to these specifications in their bid.
- C. The building shall be the design of a manufacturer who is regularly engaged in the fabrication of pre-engineered structures. All materials shall be new, unused, free from defects and of American manufacture.
- D. The following standards and criteria (of most recent issue) shall be used where applicable in the structural design of the building covered by this specification:
  - “MANUAL OF STEEL CONSTRUCTION”-American Institute of Steel Construction
  - “COLD FORMED STEEL DESIGN MANUAL”-American Iron and Steel Institute
  - “ALUMINUM CONSTRUCTION MANUAL”-The Aluminum Association
  - “CODE FOR WELDING IN BUILDING CONSTRUCTION”-American Welding SocietyThe following criteria shall also be applicable in other phases of design: latest edition of the INTERNATIONAL BUILDING CODE.

- E. Listing by:  
Underwriters' Laboratories Inc.  
Factory Mutual System or other recognized testing laboratories

2.02 DESIGN LOADS:

A. GENERAL:

- 1. The basic design loads shall include live and wind, in addition to dead load. All other design loads, whether they be of static or dynamic nature, shall be considered as auxiliary loads.

B. VERTICAL LIVE LOAD:

- 1. Roof covering shall be designed for either 50 psf uniformly distributed or a 200-pound concentrated (point) load (over a 1' x 1' area) located at center of maximum roof (panel) span. The most severe conditions shall govern.
- 2. Purlins shall be designed for 20 psf uniformly distributed over the roof area which they support.
- 3. Primary framing (frames) shall be designed for 20 psf uniformly distributed over the roof area which it supports.
- 4. All the above loads to be in addition to the applicable dead loads and shall be applied to the horizontal projection of the roof.

C. WIND LOADS:

- 1. The wind load on the structure shall be proportioned and applied as horizontal and uplift forces according to and as recommended by the latest edition of the INTERNATIONAL BUILDING CODE.
- 2. The roof construction shall carry a U.L. Construction (Uplift) Listing of not less than Class 90.
- 3. Wind load may be proportioned as allowed by the latest edition of the INTERNATIONAL BUILDING CODE. However, such proportioning shall not compromise the UL-Class 90 listing.

D. AUXILIARY (ADDITIONAL COLLATERAL) LOADS:

- 1. Other superimposed dynamic and/or static loads shall be considered as part of the design requirements and combined with normal design (live and/or wind) loads as prescribed hereafter:

DYNAMIC LOADS: VARIOUS HVAC EQUIPMENT (REFER TO DRAWINGS FOR LOCATIONS).

STATIC LOADS: THE ROOF FRAMING AT 1:12 PITCH SHALL BE DESIGNED FOR AN AUXILIAR LOAD OF 5 PSI.

E. COMBINATION OF LOADS:

- 1. The combining of normal loads and auxiliary loads for design purposes shall be as prescribed and recommended by the latest edition of the INTERNATIONAL BUILDING CODE.

F. CERTIFICATION:

- 1. After the awarding of the Contract, complete structural analysis shall be submitted by the Metal Building Manufacturer to the Architect upon request for same. Structural design must be sealed by a **Texas Registered Professional Structural Engineer.**

2.03 DESCRIPTION:

- A. The pre-engineered metal buildings covered in this specification are to be rigid frame structure of steel (frames) rafter beams.
- B. The roof slope shall be not less than 1": 12" as indicated on the drawings.
- C. Column spacing shall be as indicated on the drawings.
- D. Nominal eave height shall be as indicated on the drawings.

2.04 ROOF COVERING AND SUPPORTS:

- A. ROOF PANELS – “Standing Seam Insulated Metal Roof” Panel:
  - 1. The exposed metal roof covering shall be 24-gauge (minimum) “Standing Seam Metal Roof” Panel as by approved manufacturer. Refer to specification section 07 41 13 for additional information.
- B. WARRANTY:
  - 1. Refer to specification section 07 41 13 for warranty information.
- C. PURLINS (ROOF COVER SUPPORT MEMBERS):
  - 1. The configuration, thickness and spacing of the purlins shall be the Building Manufacturer’s standard. The allowance design capacity of cold-formed purlin members shall be calculated in accordance with the provisions of the AISI Specification for the Design of Cold-Formed Steel Structural Members.
  - 2. The deflection of the purlin or secondary member shall not exceed L/180 of its span when supporting the applicable vertical live loads previously prescribed and any collateral loads required.
  - 3. The standing seam roof does not provide a diaphragm or purlin bracing function. Brace purlins as required to conform with A.I.S.C. and A.I.S.I. specifications.
- D. ROOF JACKS AND CURBS:
  - 1. At roof penetrations for plumbing vents, install roof jacks (rubber) DEKTITE type where standing seam roofing is installed. Refer to specification section 07412 for additional information.
  - 2. At roof penetrations for mechanical equipment skylights requires curbs; provide custom fabricated roof curbs as manufactured by CUSTOM CURB, INC., LCM INC. Refer to specification section 07 41 13 for additional information.

2.05 RAKE, TRIM, GUTTERS, ROOF DOWNSPOUTS, TRIMS AND SOFFIT LINERS:

- A. The closures, flashings, fascias, gutters, and trim shall be the Building Manufacturer’s standard, compatible with the material furnished as roof panels.
- B. Buildings shall have continuous gutters with downspouts where shown on the drawings.
- C. Gutters, downspouts, rake trim, ridge panels, and trim associated with standing seam roof panels shall be a color to be selected by Architect from manufacturer’s KYNAR 500 custom and to be part of base bid colors.
- D. Refer to specification section 07 41 13 for additional information.
- E. WARRANTY:



1. Refer to specification section 07 41 13 for additional information.

F. GIRTS:

1. The girt's configuration and thickness shall be the Building Manufacturer's standard provided all design criteria, including deflection and girt spacing is met.
2. Based on a simple span, the deflection of the girts (supporting the wall covering) shall be proportioned with due regard to that produced by the previously prescribed design (wind) load.

2.06 STRUCTURAL STEEL PRIMER:

- A. All uncoated structural steel shall be given one (1) coat of rust inhibitive (primer) paint which meets or exceeds Federal Specifications TT-P-664, or certification shall be submitted that it conforms to a recognized authoritative specification, such as from a Federal or Military authority or the Structural Steel Painting Council.
- B. Exposed pre-engineered metal building must be painted with at least one (1) coat of primer and two (2) coats of finish paint.

2.07 INSULATION AND INTERIOR FINISH:

A. ROOF

1. Thermal Resistance (R-Value): 8 per inch of panel thickness. Minimum required R-Value of 30 as determined by ASTM C 518 at 75 degrees Fahrenheit mean temperature.

PART 3 - EXECUTION

3.01 ERECTION:

- A. Erection of metal building, accessories, and insulation shall be performed by one of the following:
  1. Authorized systems contractors or builders of the manufacturer.
  2. Building manufacturer's crews.
  3. Other erectors authorized by the manufacturer as trained and qualified to erect that manufacturer's product. In this case, the manufacturer shall inspect the work and certify its correctness.

END OF SECTION

## **SECTION 21 00 00 - FIRE PROTECTION**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 22 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

The scope and work shall include providing treatment for MIC (Microbiologically Influenced Corrosion).

The Fire Protection Contractor shall provide the Owner with a periodic plan for the testing of the system's water.

#### **1.3 WORK INCLUDED:** Include the following Work in addition to items normally part of this Section: Stand Pipes, Fire Hose Valves, Siamese Connections, Test Header and Roof Manifolds.

#### **1.4 REGULATORY CODES**

- A. WORK IN ACCORDANCE WITH:
  - 1. NFPA 13 and 14.
  - 2. Local municipal codes that have jurisdiction.

#### **1.5 CERTIFICATE OF TESTING**

- A. Furnish Owner with test certificate certifying the system approved by:
  - 1. Fire Marshall.
  - 2. Insurance Services Officials.

### **PART 2 - PRODUCTS**

#### **2.1 STANDPIPE SYSTEM**

- A. VALVES:
  - 1. GATE VALVES: Provide OS & Y gate valves with bronze seat and steel stem with iron body, flanged similar to Jenkins Figure "825A".
  - 2. CHECK VALVES: Provide flanged, bronze mounted check valves similar to Jenkins Figure "629".
  - 3. Valves shall be U.L. listed.
- B. PIPING:
  - 1. All piping associated with the standpipe system shall be schedule 40 black steel pipe with Victaulic couplings.
  - 2. All fitting shall be cast iron rated for 300 psi and in accordance with NFPA 14.
- C. PRODUCTS:

1. All Piping, fittings, equipment and valves shall be by a domestic manufacturer.

## 2.2 FIRE HOSE EQUIPMENT

- A. APPROVED MANUFACTURERS: This specification is based on the first named, other listed below are approved.
  1. Potter-Roemer, Inc.
  2. Allen.
  3. Elkhart.
  4. Croker.
- B. STANDPIPE HOSE VALVES: Potter-Roemer, Inc., 2-1/2" pressure regulating valve #4054 complete with brass cap, chain and visual indicator.
- C. SIAMESE CONNECTIONS: shall be a Potter-Roemer #5761, polished chrome plated, equipped with chrome plated ball drip and with required lettering per N.F.P.A.
- D. TEST HEADER: Potter-Roemer #5865 polished chrome plated on all exposed parts.
- E. CONNECTION THREADS: Conforming to Fire Department requirements.
- F. ROOF MANIFOLD: Potter-Roemer wall mounted or free standing rough chrome plated 3-way.

## PART 3 - EXECUTION

### 3.1 DESIGN

- A. The standpipe system shall conform to the requirement of NFPA.
- B. The Contractor shall conform to the National Fire Protection Association's Fire Code No. 14, latest edition. It shall be the Contractor's responsibility to determine if any deficiency or deviations, such as an inadequate water supply, or any other item which would materially affect the acceptability of the system.

### 3.2 INSTALLATION

- A. Install all items in accordance with applicable codes.
- B. Run piping concealed above furred ceilings to minimize obstructions. Exact routing of piping shall be approved by Architect, and relocated as required at no additional cost to Owner.

### 3.3 REPLACEMENT

Upon receipt of written notice of failure of any part of the guaranteed equipment during the guaranteed period, the Contractor will replace the affected part or parts promptly at no additional cost.

### 3.4 TESTING

Upon completion of the installation, test the system and obtain approval of the local fire insurance rating organization having jurisdiction. Particular attention is called to the requirements of NFPA 14 pamphlet.

### 3.5 TRAINING

- A. Owner's people shall be fully briefed in the normal start-up of the system, operation, normal and emergency shutdown, and maintenance of the system.
- B. Routine maintenance, yearly maintenance, winterization, and spring start-up shall be fully discussed and documented.
- C. Names of those instructed and dates, as well as a list of information handed over to the owner, shall be included in the final report.

**END OF SECTION**

## SECTION 21 02 01 - COORDINATION DRAWINGS

### PART 1 - GENERAL

#### 1.1 GENERAL REQUIREMENTS

- A. The requirements of the General Conditions 013100 and Supplementary Conditions apply to all Work herein.

#### 1.2 COORDINATION DRAWINGS

- A. The Contractor shall take the lead in coordinating the Mechanical, Electrical, Plumbing, Communications, Electronic Safety/Security and Fire Protection systems within the building.
- B. The Mechanical Contractor shall coordinate a three-dimensional (3D) model of the building which includes the Mechanical, Electrical, Plumbing, and Fire Protection systems. The Electrical, Plumbing, and Fire Protection Contractors shall prepare their work and generate 3D models which will be given to the Mechanical Contractor for coordination. The Contractor will be provided with the REVIT model that was used to generate the contract documents, this file may be used as the background file. The Contractor shall replace the systems drawn with the actual shop drawing models. The Contractor is not limited to using REVIT, but may use any 3-D software in generating and combining the coordination model.
- C. Submitting the contract drawings as coordination drawings will not be acceptable.
- D. The model shall include detailed and accurate representations of all equipment to be installed based upon the reviewed equipment submittals.
- E. The Mechanical Contractor shall hold a 3-D coordination meeting with all sub-contractors present to review the model and discuss coordination of the installation of the building systems.
- F. Upon completion of the coordination meeting, the Contractor shall submit the 3-D model and 1/4" scale drawings for review.
- G. The model shall detail major elements, components, and systems in relationship with other systems, installations, and building components. Indicate locations where space is limited for installation and access and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:
  - 1. Indicate the proposed locations of pipe, duct, equipment, and other materials. Include the following:
    - a. Wall and type locations.
    - b. Clearances for installing and maintaining insulation.
    - c. Locations of light fixtures and sprinkler heads.
    - d. Clearances for servicing and maintaining equipment, including tube removal, filter removal, and space for equipment disassembly required for periodic maintenance.
    - e. Equipment connections and support details.
    - f. Exterior wall and foundation penetrations.
    - g. Routing of storm and sanitary sewer piping.
    - h. Fire-rated wall and floor penetrations.
    - i. Sizes and location of required concrete pads and bases.

- j. Valve stem movement.
    - k. Structural floor, wall and roof opening sizes and details.
  - 2. Indicate scheduling, sequencing, movement, and positioning of large equipment into the building during construction.
  - 3. Prepare floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
  - 4. Prepare reflected ceiling plans to coordinate and integrate installations, air distribution devices, light fixtures, communication systems components, and other ceiling-mounted items.
- A. Sequence of Coordination
- Below is hierarchy of model elements and the sequencing by which the models will be coordinated.
- 1. Structural and Architectural model
  - 2. Miscellaneous steel
  - 3. Perform preliminary space allocation
  - 4. Identify hard constraints (locations of access panels, lights, A/V space requirements, etc.)
  - 5. Main and medium pressure ducts from the shaft out
  - 6. Main graded plumbing lines and vents
  - 7. Sprinkler mains and branches
  - 8. Cold and hot water mains and branches
  - 9. Lighting fixtures and plumbing fixtures
  - 10. Smaller sized ducts and flex ducts
  - 11. Smaller size cold water and hot water piping, flex ducts, etc.
- B. The Contractor and Sub-Contractors shall not install any item until the coordination has been completed and reviewed by the Construction Manager, Owner, and A/E team.
- C. This Contractor shall be responsible for coordination of all items that will affect the installation of the work of this Division. This coordination shall include, but not be limited to: voltage, ampacity, capacity, electrical and piping connections, space requirements, sequence of construction, building requirements and special conditions.
- D. By submitting shop drawings on the project, this Contractor is indicating that all necessary coordination has been completed and that the systems, products and equipment submitted can be installed in the building and will operate as specified and intended, in full coordination with all other Contractors and Subcontractors.

#### **END OF SECTION**

## **SECTION 21 05 29 – HANGERS AND SUPPORTS FOR FIRE SUPPRESSION PIPING AND EQUIPMENT**

### **PART 1 - GENERAL**

#### **1.1 WORK INCLUDED**

- A. Pipe, and equipment hangers, supports and associated anchors.
- B. Sleeves and seals.
- C. Flashing and sealing equipment and pipe stacks.

#### **1.2 RELATED WORK**

- A. Section 21 00 00 – Fire Protection
- B. Section 21 13 13 – Wet Pipe Sprinkler System.
- C. Section 21 13 16 – Dry Pipe Sprinkler System.

#### **1.3 REFERENCES**

- A. ANSI/ASME B31.1 - Power Piping.
- B. NFPA 13 - Standard for the Installation of Sprinkler Systems.
- C. NFPA 14 - Standard for the Installation of Standpipe and Hose Systems.

#### **1.4 QUALITY ASSURANCE**

- A. Supports for Sprinkler Piping: In conformance with NFPA 13.
- B. Supports for Standpipes: In conformance with NFPA 14.

#### **1.5 SUBMITTALS**

- A. Submit shop drawings and product data under provisions of Division One.
- B. Indicate hanger and support framing and attachment methods.

### **PART 2 - PRODUCTS**

#### **2.1 PIPE HANGERS AND SUPPORTS**

- A. Hangers for Pipe Sizes 1/2 to 1-1/2 Inch Malleable iron, adjustable swivel, split ring.
- B. Hangers for Pipe Sizes 2 to 4 Inches Carbon steel, adjustable, clevis.
- C. Hangers for Pipe Sizes 6 Inches and Over: Adjustable steel yoke, cast iron roll, double hanger.
- D. Multiple or Trapeze Hangers: Steel channels with welded spacers and hanger rods; cast iron roll and stand for pipe sizes 6 inches and over.
- E. Wall Support for Pipe Sizes to 3 Inches: Cast iron hook.

- F. Wall Support for Pipe Sizes 4 Inches and Over: adjustable steel yoke and cast iron roll.
- G. Vertical Support: Steel riser clamp.
- H. Floor Support for Pipe Sizes to 4 Inches: Cast iron adjustable pipe saddle, locknut nipple, floor flange, and concrete pier or steel support.
- I. Floor Support for Pipe Sizes 6 Inches and Over: Adjustable cast iron roll and stand, steel screws, and concrete pier or steel support.
- J. Roof Pipe Supports and Hangers: Galvanized Steel Channel System as manufactured by Portable Pipe Hangers, Inc. or approved equal.  
  
For pipes 2-1/2" and smaller – Type PP10 with roller  
For pipes 3" through 8" – Type PS  
For multiple pipes – Type PSE - Custom
- K. For installation of protective shields refer to specification section 22 05 29 -3.03.

## 2.2 HANGER RODS

- A. Galvanized Hanger Rods: Threaded both ends, threaded one end, or continuous threaded.

## 2.3 INSERTS

- A. Inserts: Malleable iron case of galvanized steel shell and expander plug for threaded connection with lateral adjustment, top slot for reinforcing rods, lugs for attaching to forms; size inserts to suit threaded hanger rods.

## 2.4 FLASHING

- A. Metal Flashing: 20 gage galvanized steel.
- B. Lead Flashing: 4 lb./sq. ft. sheet lead for waterproofing; 1 lb./sq. ft. sheet lead for soundproofing.
- C. Caps: Steel, 20 gage minimum; 16 gage at fire resistant elements.
- D. Coordinate with roofing contractor/architect for type of flashing on metal roofs.

## 2.5 EQUIPMENT CURBS

- A. Fabricate curbs of hot dipped galvanized steel.

## 2.6 SLEEVES

- A. Sleeves for Pipes through Non-fire Rated Floors: Form with 18 gage galvanized steel, tack welded to form a uniform sleeve.
- B. Sleeves for Pipes through Non-fire Rated Beams, Walls, Footings, and Potentially Wet Floors: Form with steel pipe, schedule 40.
- C. Sleeves for Pipes through Fire Rated and Fire Resistive Floors and Walls, and Fireproofing: Prefabricated fire rated steel sleeves including seals, UL listed.



- D. Fire Stopping Insulation: Glass fiber type, non-combustible, U.L. listed.
- E. Caulk: Paintable 25-year acrylic sealant.
- F. Pipe Alignment Guides: Factory fabricated, of cast semi-steel or heavy fabricated steel, consisting of bolted, two-section outer cylinder and base with two-section guiding spider that bolts tightly to pipe. Length of guides shall be as recommended by manufacturer to allow indicated travel.

## 2.7 FABRICATION

- A. Size sleeves large enough to allow for movement due to expansion and contraction. Provide for continuous insulation wrapping.
- B. Design hangers without disengagement of supported pipe.
- C. Design roof supports without roof penetrations, flashing or damage to the roofing material.

## 2.8 FINISH

- A. Prime coat exposed steel hangers and supports. Hangers and supports located in crawl spaces, pipe shafts, and suspended ceiling spaces are not considered exposed.

## PART 3 - EXECUTION

### 3.1 INSERTS

- A. Provide inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams. Coordinate with structural engineer for placement of inserts.
- B. Provide hooked rod to concrete reinforcement section for inserts carrying pipe over 4 inches.
- C. Where concrete slabs form finished ceiling, provide inserts to be flush with slab surface.
- D. Where inserts are omitted, drill through concrete slab from below and provide thru-bolt with recessed square steel plate and nut recessed into and grouted flush with slab. Verify with structural engineer prior to start of work.

### 3.2 PIPE HANGERS AND SUPPORTS

- A. Support horizontal piping as follows:

| <u>PIPE SIZE</u>  | <u>MAX. HANGER SPACING</u> | <u>HANGER DIAMETER</u> |
|-------------------|----------------------------|------------------------|
| (Steel Pipe)      |                            |                        |
| 1/2 to 1-1/4 inch | 7'-0"                      | 3/8"                   |
| 1-1/2 to 3 inch   | 10'-0"                     | 3/8"                   |
| 4 to 6 inch       | 10'-0"                     | 1/2"                   |
| 8 to 10 inch      | 10'-0"                     | 5/8"                   |
| 12 to 14 inch     | 10'-0"                     | 3/4"                   |
| 15 inch and over  | 10'-0"                     | 7/8"                   |
| (Copper Pipe)     |                            |                        |
| 1/2 to 1-1/4 inch | 5'-0"                      | 3/8"                   |

|                     |        |      |
|---------------------|--------|------|
| 1-1/2 to 2-1/2 inch | 8'-0"  | 3/8" |
| 3 to 4 inch         | 10'-0" | 3/8" |
| 6 to 8 inch         | 10'-0" | 1/2" |
| (PVC Pipe)          |        |      |
| 1-1/2 to 4 inch     | 4'-0"  | 3/8" |
| 6 to 8 inch         | 4'-0"  | 1/2" |
| 10 and over         | 4'-0"  | 5/8" |

- B. Install hangers to provide minimum 1/2 inch space between finished covering and adjacent work.
- C. Place a hanger within 12 inches of each horizontal elbow and at the vertical horizontal transition.
- D. Use hangers with 1-1/2 inch minimum vertical adjustment.
- E. Support horizontal cast iron pipe adjacent to each hub, with 5 feet maximum spacing between hangers.
- F. Support vertical piping at every floor. Support vertical cast iron pipe at each floor at hub.
- G. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers.
- H. Support riser piping independently of connected horizontal piping.
- I. Install hangers with nut at base and above hanger; tighten upper nut to hanger after final installation adjustments.
- J. Portable pipe hanger systems shall be installed per manufactures instructions.

### 3.3 Insulated Piping: Comply with the following installation requirements.

- A. Clamps: Attach galvanized clamps, including spacers (if any), to piping with clamps projecting through insulation; do not exceed pipe stresses allowed by ASME B31.9.
- B. Saddles: Install galvanized protection saddles MSS Type 39 where insulation without vapor barrier is indicated. Fill interior voids with segments of insulation that match adjoining pipe insulation.
- C. Shields: Install protective shields MSS Type 40 on cold and chilled water piping that has vapor barrier. Shields shall span an arc of 180 degrees and shall have dimensions in inches not less than the following:

| <u>NPS</u>        | <u>LENGTH</u> | <u>THICKNESS</u> |
|-------------------|---------------|------------------|
| 1/4 THROUGH 3-1/2 | 12            | 0.048            |
| 4                 | 12            | 0.060            |
| 5 & 6             | 18            | 0.060            |
| 8 THROUGH 14      | 24            | 0.075            |
| 16 THROUGH 24     | 24            | 0.105            |

- D. Piping 2" and larger provide galvanized sheet metal shields with calcium silicate at hangers/supports.

- E. Insert material shall be at least as long as the protective shield.
- F. Thermal Hanger Shields: Install where indicated, with insulation of same thickness as piping.

### 3.4 EQUIPMENT BASES AND SUPPORTS

- A. Provide equipment bases of concrete.
- B. Provide templates, anchor bolts, and accessories for mounting and anchoring equipment.
- C. Construct support of steel members. Brace and fasten with flanges bolted to structure.
- D. Provide rigid anchors for pipes after vibration isolation components are installed.

### 3.5 FLASHING

- A. Provide flexible flashing and metal counter flashing where piping and ductwork penetrate weather or waterproofed walls, floors, and roofs.
- B. Flash vent and soil pipes projecting 8 inches minimum above finished roof surface with lead worked one inch minimum into hub, 8 inches minimum clear on sides with 24 x 24 inches sheet size. For pipes through outside walls, turn flanges back into wall and caulk, metal counter flash and seal.
- C. Flash floor drains in floors with topping over finished areas with lead, 10 inches clear on sides with minimum 36 x 36 inch sheet size. Fasten flashing to drain clamp device.
- D. Seal floor shower mop sink and all other drains watertight to adjacent materials.
- E. Provide curbs for mechanical roof installations 8 inches minimum high above roofing surface. Contact architect for all flashing details and roof construction. Seal penetrations watertight.

### 3.6 SLEEVES

- A. Set sleeves in position in formwork. Provide reinforcing around sleeves.
- B. Extend sleeves through floors minimum one inch above finished floor level. Caulk sleeves full depth with fire rated thermfiber and 3M caulking and provide floor plate.
- C. Where piping penetrates floor, ceiling, or wall, close off space between pipe or duct and adjacent work with U.L. listed fire stopping insulation and caulk seal air tight. Provide close fitting metal collar or escutcheon covers at both sides of penetration.
- D. Fire protection sleeves may be flush with floor of stairways.

## END OF SECTION

## **SECTION 21 13 13 - WET PIPE SPRINKLER SYSTEM**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 22 02 00, are included as a part of this Section as though written in full in this document.
- C. Refer to Section 21 00 00 for integral requirements.

#### **1.2 SCOPE**

- A. Scope of the work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for owner's use.
- B. The scope and work shall include providing treatment for MIC (Microbiologically Influenced Corrosion).
- C. The Fire Protection Contractor shall provide the Owner with a periodic plan for the testing of the system's water.

#### **1.3 REGULATORY CODES**

- A. Work in accordance with:
  - 1. NFPA.
  - 2. Local municipal codes that have jurisdiction.
- B. Products in accordance with:
  - 1. United Laboratories (UL) listed.
  - 2. Factory Mutual (FM) approved.

#### **1.4 CERTIFICATE OF TESTING**

- A. Furnish Owner with test certificate certifying the system approved by:
  - 1. City Fire Marshall.
  - 2. Insurance Services Officials

### **PART 2 - PRODUCTS**

#### **2.1 FIRE SPRINKLER SYSTEM**

- A. GENERAL:
  - 1. Work Included:
    - a. Design, coordination and installation of inside and outside piping, including sprinkler heads, valves, hangers and supports sleeves.
    - b. The sprinkler system is a wet type and is designed to provide coverage for entire building. The Contract Drawings indicate the extent and general arrangement, and the various occupancy classifications.

- c. Sprinkler heads are not shown.
  - d. The plans provide a preliminary layout with riser assembly location, flow switch locations, valve locations, and fire department Siamese connections. These are a guide for subsequent preparation of the Contractor's detailed working drawings.
  - e. Interface system with building fire and smoke alarm system.
2. Quality Assurance: Equipment and installation to meet requirements of NFPA Number 13, latest edition and local authority having jurisdiction. All components of the completed system shall be UL listed for the intended service.

## 2.2 SUBMITTALS:

- A. Submit shop drawings in accordance with Section 22 02 00.
- B. Submit preliminary layout showing only head locations for review by Architect/Engineer. Furnish additional heads which may be required for coordinated ceiling pattern without added cost, even though number of heads may exceed minimum code requirements.
- C. Submit shop drawings of entire sprinkler system including hydraulic calculations to Architect/Engineer.
- D. Provide Architect with six complete sets of final approved shop drawings before starting the installation. Include details of the sprinkler system showing sections, light fixtures, air conditioning, ducts, and a plan giving fire department connections, location of all exposed structures within twenty feet of this structure, and other equipment to be used. Drawings shall bear the stamp of review of the local fire insurance rating organization having jurisdiction.
- E. Service Utility Diagram: Furnish Architect with an accurately marked print showing location of underground pipes and valves as installed upon completion of underground Work.
- F. Provide a printed sheet giving brief instructions relative to all necessary aspects of sprinkler controls and emergency procedures next to sprinkler riser mains. Instruction sheet to be protected by glass or a transparent plastic cover.
- G. Materials:
  - 1. Piping:
    - a. All piping above grade shall be:
      - schedule 10 black steel pipe with a rolled groove ends, joined with mechanical coupling and cut groove cast iron fittings for pipe 2-1/2" and greater.
      - schedule 40 black steel threaded pipe and fittings for pipe 2" and smaller.
    - b. Acceptable manufacturer:
      - American Tube
      - Wheatland Tube
      - Gem Sprinkler
    - c. Acceptable mechanical coupling manufacturer:
      - Victaulic
      - Grinnell
      - Reliable
      - Gruvlok
    - d. In all locations subject to corrosive conditions, ensure to provide a suitable epoxy (spray) coating to all exposed surfaces of pipe and

fittings. Such locations shall include, but not necessarily be limited to, the following: natatoriums, pool equipment rooms, chemical and metal process areas, and animal pens. All job-applied protective coatings shall be provided as approved and only after verification that the proper piping has been installed, per the markings along the length of the pipe.

2. Sprinkler Heads:
  - a. Suspended Ceiling Type: Standard Concealed pendant type with white cover plate.
  - b. Exposed Area Type: Standard upright type with brass finish.
  - c. Sidewall Type: Chrome plated finish with matching escutcheon.
  - d. Temperature rating on fusible links to suit specific hazard area with minimum margin or safety 50 degrees F.
  - e. Sprinkler heads of the "O"-ring seal type are not acceptable.
  - f. In all locations subject to corrosive conditions provide heads entirely constructed of stainless steel or heads with polyester coated finish and dipped in wax. Such locations shall include, but not necessarily be limited to, the following: natatoriums, pool equipment rooms, chemical and metal process areas, and animal pens.
  - g. In all locations subject to occupant abuse or vandalism provide institutional type sprinkler heads with tamper-resistant construction and suitable "break-away" weight test documentation from the manufacturer. Such locations shall include, but not necessarily be limited to, the following: patient areas of institutional mental health occupancies, prisoner areas (cells, etc.) of correctional facilities – jails, prisons, juvenile detention facilities.
  - h. Flexible type sprinkler head connection systems are not acceptable.
  - i. Acceptable manufacturer:
    - Reliable
    - Grinnell
    - Viking
3. Sprinkler Alarm Valve:
  - a. Provide approved automatic sprinkler valve with one or two pole (as required) flow detectors, pressure switch, outside water motor gong and inside electric gong and circuit breaker.
  - b. Acceptable manufacturer:
    - Reliable
    - Grinnell
    - Viking
4. Valves:
  - a. 2" and smaller: bronze, rising stem, inside screw, solid wedge, U.L. listed valve.
  - b. 2-1/2" and larger: iron body, bronze trim, rising stem, OS&Y, solid wedge, U.L. listed valve.
  - c. Check valve: cast iron flanged body, bronze fitted, non-slam type.
  - d. Install valves with stems upright or horizontal, not inverted.
  - e. Acceptable manufacturer:
    - Nibco
    - Grinnell
    - Stockham
    - Victaulic
5. Fire Department Connection: Refer to Section 21 00 00.
6. Insulation:
  - a. All piping and valves exposed to the weather or within building and exposed to the weather shall be insulated with Phenolic foam with ASJ and all joints sealed. Insulation density shall not be less than 1.5 pounds per cubic foot, and conductivity (K) not higher than 0.25 and 75°F mean

- temperature difference, with factory applied all weather vapor barrier jacket.
  - b. All insulated pipe and valves subject to damage shall be protected with an aluminum jacket with sealed joints.
  - c. Refer to Section 22 07 19 for detailed specification.
- 7. Products:
  - a. All piping, fitting, equipment, sprinkler heads, and valves shall be by a domestic manufacturer.

### PART 3 - EXECUTION

- 3.1 All equipment shall be installed in accordance with the manufacturer's recommendations and printed installation instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Provide all items as required by NFPA and installed as per manufacturer's recommendations.

#### 3.3 DESIGN

- A. Design spacing of sprinkler heads and selection sizes shall conform to the requirement of NFPA 13 for the indicated occupancy.
- B. Uniform discharge density design shall be based on hydraulic calculations utilizing the method outlined in NFPA 13. Density of discharge from sprinkler heads shall conform to NFPA 13.
- C. Friction losses in pipe will be based on a value of "C" =120 in the Hazen - Williams formula.
- D. Design and install the system so that no part will interfere with doors, windows, heating, plumbing, or electrical equipment. Do not locate sprinkler heads within 6 inches of lighting fixtures, HVAC diffusers and other obstructions. Sprinkler piping cannot penetrate ductwork or lighting fixtures.
- E. The Contractor shall conform to the National Fire Protection Association's Fire Code No. 13, latest edition. Special attention shall be given to Article 1-9, working plans. It shall be the Registered Fire Protection Engineer's responsibility to determine if any deficiency or deviations, such as an inadequate water supply, or any other item which would materially affect the acceptability of the system.

#### 3.4 INSTALLATION

- A. Install all items in accordance with applicable codes.
- B. Install piping so that mains and branches are not located directly underneath HVAC equipment or other items needing access.
- C. All sprinkler heads shall be located as near the center of ceiling tiles as is practical ( $\pm 1/2"$ ). Location shall present a uniform pattern with all heads aligned when completely installed.
- D. Run piping concealed above furred ceilings and in joists to minimize obstructions. Expose only heads. Exact routing of piping shall be approved by Architect or relocated as required at no additional cost to Owner.

- E. Wire guards on all pendant or upright sprinklers heads in mechanical rooms, gymnasiums, athletic areas, wood and metal shops.
- F. Protect sprinkler heads against mechanical injury with standard guards.
- G. Locate outside alarms on wall of building adjacent to siamese fire department connection.
- H. Provide on wall near sprinkler valve, cabinet containing four extra sprinkler heads of each type and wrench suitable for each head type.
- I. Provide 1 inch diameter nipple and 1 inch x 1/2 inch reducing fitting for each upright head.
- J. Painting shall be as follows:
  - 1. Exposed sprinkler riser, alarm valve and all related piping shall be painted red.
  - 2. Exposed sprinkler piping in finished areas shall be painted as directed by Architect.

### 3.5 REPLACEMENT

Upon receipt of written notice of failure of any part of the guaranteed equipment during the guaranteed period, the Contractor will replace the affected part or parts promptly at no additional cost.

### 3.6 TESTING

- A. Prior to testing, the entire sprinkler system shall be thoroughly flushed clean.
- B. Upon completion of the installation and flushing, test the system and obtain approval of the local fire insurance rating organization having jurisdiction. Particular attention is called to the requirements of NFPA 13 pamphlet.

### 3.7 TRAINING

- A. Owner's people shall be fully briefed in the normal start-up of the system, operation, normal and emergency shutdown, and maintenance of the system.
- B. Routine maintenance, yearly maintenance, winterization, and spring start-up shall be fully discussed and documented.
- C. Names of those instructed and dates, as well as a list of information handed over to the owner, shall be included in the final report.

## END OF SECTION



## **SECTION 22 02 00 - BASIC MATERIALS AND METHODS**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all Work herein.
- B. The Contract Drawings indicate the extent and general arrangement of the systems. If any departure from the Contract Drawings is deemed necessary by the Contractor, details of such departure and the reasons therefore shall be submitted to the Architect/Engineer for approval as soon as reasonably practicable. No such departures shall be made without the prior written approval of the Architect/Engineer.
- C. Notwithstanding any reference in the Specifications to any article, device, product, material, fixture, form or type of construction by name, make or catalog number, such reference shall not be construed as limiting competition; and the Contractor, in such cases, may at his option use any article, device, product, material, fixture, form or type of construction which in the judgment of the Architect/Engineer, expressed in writing, is equal to that specified.

#### **1.2 SCOPE OF WORK**

- A. The Work included under this Contract consists of the furnishing and installation of all equipment and material necessary and required to form the complete and functioning systems in all of their various phases, all as shown on the accompanying Drawings and/or described in these Specifications. The Contractor shall review all pertinent Drawings, including those of other contracts, prior to commencement of Work.
- B. This Division requires the furnishing and installing of all items as specified herein, indicated on the Drawings, or reasonably inferred as necessary for safe and proper operation; including every article, device or accessory (whether or not specifically called for by item) reasonably necessary to facilitate each system's functioning as indicated by the design and the equipment specified. Elements of the work include, but are not limited to: materials, labor, supervision, transportation, storage, equipment, utilities, all required permits, licenses and inspections. All work performed under this Section shall be in accordance with the Project Manual, Drawings and Specifications and is subject to the terms and conditions of the Contract.
- C. The approximate locations of Plumbing and Fire Protection items are indicated on the Drawings. These Drawings are not intended to give complete and accurate details with regards to location of outlets, apparatus, etc. Exact locations are to be determined by actual measurements at the building/job-site, and will in all cases be subject to the Review of the Owner or Engineer, who reserves the right to make any reasonable changes in the locations indicated without additional cost to the Owner.
- D. Items specifically mentioned in the Specifications but not shown on the Drawings and/or items shown on Drawings but not specifically mentioned in the Specifications shall be installed by the Contractor under the appropriate section of work as if they were both specified and shown.
- E. All discrepancies between the Contract Documents and actual job-site conditions shall be reported to the Owner or Engineer so that they will be resolved prior to bidding. Where this cannot be done at least seven (7) working days prior to bid; the greater or more

costly of the discrepancy shall be bid. All labor and materials required to perform the work described shall be included as part of this Contract.

- F. It is the intention of this Section of the specifications to outline minimum requirements to furnish the Owner with a turn-key and fully operating system in cooperation with other trades.
- G. It is the intent of the above "Scope" to give the Contractor a general outline of the extent of the Work involved; however, it is not intended to include each and every item required for the Work. Anything omitted from the "Scope" but shown on the Drawings, or specified elsewhere, or necessary for complete and functioning plumbing systems shall be considered a part of the overall "Scope".
- H. The Contractor shall rough-in fixtures and equipment furnished by others from rough-in and placement drawings furnished by others. The Contractor shall make final connection to fixtures and equipment furnished by others.
- I. The Contractor shall participate in the Commissioning process as required; including, but not necessarily limited to: meeting attendance, completion of checklists, and participation in functional testing.

### 1.3 SCHEMATIC NATURE OF CONTRACT DOCUMENTS

- A. The contract documents are schematic in nature in that they are only to establish scope and a minimum level of quality. They are not to be used as actual working construction drawings. The actual working construction drawings shall be the reviewed Shop Drawings.
- B. All piping, fixture, or equipment locations as indicated on the documents do not indicate every transition, offset, or exact location. All transitions, offsets, clearances and exact locations shall be established by actual field measurements, coordination with the structural, architectural and reflected ceiling plans, and other trades. Submit Shop Drawings for review.
- C. All transitions, offsets and relocations as required by actual field conditions shall be performed by the Contractor at no additional cost to the owner.
- D. Additional coordination with Electrical Subcontractor may be required to allow adequate clearances of electrical equipment, fixtures, and associated appurtenances. Contractor to notify Architect and Engineer of unresolved clearances, conflicts, or equipment locations.

### 1.4 SITE VISIT AND FAMILIARIZATION

- A. Before submitting a bid, it will be necessary for each Contractor whose work is involved to visit the site and ascertain for himself the conditions to be met therein in installing his work and make due provision for same in his bid. It will be assumed that this Contractor in submitting his bid has visited the premises and that his bid covers all work necessary to properly install the piping, fixtures and equipment shown. Failure on the part of the Contractor to comply with this requirement shall not be considered justification for the omission or faulty installation of any work covered by these Specifications and Drawings.
- B. Understand the existing utilities from which services will be supplied; verify locations of utility services, and determine requirements for connections.
- C. Determine in advance that equipment and materials proposed for installation fit into the confines indicated.

#### 1.5 WORK SPECIFIED IN OTHER SECTIONS

- A. Finish painting is specified. Prime and protective painting is included in the work of this Division.
- B. Owner and General Contractor furnished equipment shall be properly connected to plumbing systems.
- C. Furnishing and installing all required plumbing equipment, control relays and electrical interlock devices, conduit, wire and junction boxes are included in the Work of this Division.

#### 1.6 PERMITS, TESTS, INSPECTIONS

- A. Arrange and pay for all permits, fees, tests, and all inspections as required by governmental authorities.

#### 1.7 DATE OF FINAL ACCEPTANCE

- A. The date of final acceptance shall be the date of Owner occupancy, or the date all punch list items have been completed, or the date final payment has been received. Refer to Division 01 for additional requirements.
- B. The date of final acceptance shall be documented in writing and signed by the Architect, Owner and Contractor.

#### 1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to the project properly identified with names, model numbers, types, grades, compliance labels, and other information needed for identification.
- B. Deliver products to the project at such times as the project is ready to receive the equipment, pipe or valves - properly protected from incidental damage and weather damage.
- C. Damaged equipment, valves or pipe shall be promptly removed from the site and new, undamaged equipment, valves and pipe shall be provided in its place promptly with no additional charge to the Owner.

#### 1.9 NOISE AND VIBRATION

- A. The plumbing systems and the component parts thereof shall be guaranteed to operate without objectionable noise and vibration.
- B. Provide foundations, supports and isolators as specified or indicated, properly adjusted to prevent transmission of vibration to the building structure, piping and other items.
- C. Carefully fabricate pipe and fittings with smooth interior finish to prevent turbulence and generation or regeneration of noise.
- D. All equipment shall be selected to operate with minimum of noise and vibration. If, in the opinion of the Architect/Engineer, objectionable noise or vibration is produced or transmitted to or through the building structure by equipment, piping or other parts of the Work, the Contractor shall rectify such conditions without extra cost to the Owner.

#### 1.10 APPLICABLE CODES

- A. Obtain all required permits and inspections for all work required by the Contract Documents and pay all required fees in connection thereof.
- B. Arrange with the serving utility companies for the connection, relocation, and upgrade of all required utilities and pay all charges, meter charges, connection fees and inspection fees, if required.
- C. Comply with all applicable codes, specifications, local ordinances, industry standards, utility company regulations and the applicable requirements of the following nationally accepted codes and standards, including, but not necessarily limited to:
  - 1. American Standards Association, ASA.
  - 2. American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc., ASHRAE.
  - 3. American Society of Mechanical Engineers, ASME.
  - 4. American Society of Plumbing Engineers, ASPE.
  - 5. American Society of Testing Materials, ASTM.
  - 6. American Water Works Association, AWWA.
  - 7. National Bureau of Standards, NBS.
  - 8. National Fire Protection Association, NFPA.
  - 9. UL, LLC (formerly Underwriters Laboratories).
  - 10. FM Global.
  - 11. International Energy Conservation Code, IECC.
  - 12. International Fire Code.
  - 13. International Gas Code.
- D. Where differences exist between the Contract Documents and applicable state or city building codes, state and local ordinances, industry standards, utility company regulations and the applicable requirements of the above listed nationally accepted codes and standards, the more stringent or costly application shall govern. Promptly notify the Architect/Engineer in writing of all differences.
- E. When directed in writing by the Architect/Engineer, remove all work installed that does not comply with the Contract Documents and applicable state or city building codes, state and local ordinances, industry standards, utility company regulations and the applicable requirements of the above listed nationally accepted codes and standards. Correct the deficiencies and complete the work at no additional cost to the Owner.

#### 1.11 DEFINITIONS AND SYMBOLS

- A. General Explanation: A substantial amount of construction and Specification language constitutes definitions for terms found in other Contract Documents, including Drawings which must be recognized as diagrammatic and schematic in nature and not completely descriptive of requirements indicated thereon. Certain terms used in Contract Documents are defined generally in this article, unless defined otherwise in Division 01.
- B. Definitions and explanations of this Section are not necessarily either complete or exclusive, but are general for work to the extent not stated more explicitly in another provision of the Contract Documents.
- C. Indicated: The term "Indicated" is a cross-reference to details, notes or schedules on the Drawings, to other paragraphs or schedules in the Specifications and to similar means of recording requirements in Contract Documents. Where such terms as "Shown", "Noted", "Scheduled", "Specified" and "Detailed" are used in lieu of "Indicated", it is for the

purpose of helping the reader locate cross-reference material, and no limitation of location is intended except as specifically shown.

- D. Directed: Where not otherwise explained, terms such as "Directed", "Requested", "Accepted", and "Permitted" mean by the Architect or Engineer. However, no such implied meaning will be interpreted to extend the Architect's or Engineer's responsibility into the Contractor's area of construction supervision.
- E. Reviewed: Where used in conjunction with the Engineer's response to submittals, requests for information, applications, inquiries, reports and claims by the Contractor the meaning of the term "Reviewed" will be held to limitations of Architect's and Engineer's responsibilities and duties as specified in the General and Supplemental Conditions. In no case will "Reviewed" by Engineer be interpreted as a release of the Contractor from responsibility to fulfill the terms and requirements of the Contract Documents.
- F. Furnish: Except as otherwise defined in greater detail, the term "Furnish" is used to mean supply and deliver new to the project site, ready for unloading, unpacking, assembly, installation, etc., as applicable in each instance.
- G. Install: Except as otherwise defined in greater detail, the term "Install" is used to describe operations at the project site including unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimension, finishing, curing, protection, cleaning and similar operations, as applicable in each instance.
- H. Provide: Except as otherwise defined in greater detail, the term "Provide" is used to mean "Furnish and Install", complete and ready for intended use, as applicable in each instance.
- I. Installer: Entity (person or firm) engaged by the Contractor or its Subcontractor for performance of a particular unit of work at the project site, including unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimension, finishing, curing, protection, cleaning and similar operations, as applicable in each instance. It is a general requirement that such entities (Installers) be expert in the operations they are engaged to perform.
- J. Imperative Language: Used generally in Specifications. Except as otherwise indicated, requirements expressed imperatively are to be performed by the Contractor. For clarity of reading at certain locations, contrasting subjective language is used to describe responsibilities that must be fulfilled indirectly by the Contractor, or when so noted by other identified installers or entities.
- K. Minimum Quality/Quantity: In every instance, the quality level or quantity shown or specified is intended as minimum quality level or quantity of work to be performed or provided. Except as otherwise specifically indicated, the actual work may either comply exactly with that minimum (within specified tolerances), or may exceed that minimum within reasonable tolerance limits. In complying with requirements, indicated or scheduled numeric values are either minimums or maximums as noted or as appropriate for the context of the requirements. Refer instances of uncertainty to Owner or Engineer via a request for information (RFI) for decision before proceeding.
- L. Abbreviations and Symbols: The language of Specifications and other Contract Documents including Drawings is of an abbreviated type in certain instances, and implies words and meanings which will be appropriately interpreted. Actual word abbreviations of a self-explanatory nature have been included in text of Specifications and Drawings. Specific abbreviations and symbols have been established, principally for lengthy technical terminology and primarily in conjunction with coordination of Specification

requirements with notations on Drawings and in Schedules. These are frequently defined in Section at first instance of use or on a Legend and Symbol Drawing. Trade and industry association names and titles of generally recognized industry standards are frequently abbreviated. Singular words will be interpreted as plural and plural words will be interpreted as singular where applicable and where full context of Contract Documents so indicate. Except as otherwise indicated, graphic symbols and abbreviations used on Drawings and in Specifications are those recognized in construction industry for indicated purposes. Where not otherwise noted symbols and abbreviations are defined by 2009 ASHRAE Fundamentals Handbook, chapter 34 "Abbreviations and Symbols", ASME and ASPE published standards.

#### 1.12 DRAWINGS AND SPECIFICATIONS

- A. These Specifications are intended to supplement the Drawings. It will not be the province of the Specifications to mention any part of the work which the Drawings are competent to fully explain in every particular and such omission shall not to relieve the Contractor from carrying out portions indicated on the Drawings only.
- B. Should items be required by these Specifications and not indicated on the Drawings, they are to be supplied even if of such nature that they could have been indicated thereon. In case of disagreement between Drawings and Specifications, or within either Drawings or Specifications, the better quality or greater quantity of work shall be estimated and the matter referred to the Architect or Engineer for review with a request for information and clarification at least seven (7) working days prior to bid opening date for issuance of an addendum.
- C. The listing of product manufacturers, materials and methods in the various sections of the Specifications, and indicated on the Drawings, is intended to establish a standard of quality only. It is not the intention of the Owner or Engineer to discriminate against any product, material or method that is equal to the standards as indicated and/or specified, nor is it intended to preclude open, competitive bidding. The fact that a specific manufacturer is listed as an acceptable manufacturer should not be interpreted to mean that the manufacturers' standard product will meet the requirements of the project design, Drawings, Specifications and space constraints.
- D. The Architect or Engineer and Owner shall be the sole judge of quality and equivalence of equipment, materials and methods.
- E. Products by other reliable manufacturers, other materials, and other methods, will be accepted as outlined, provided they have equal capacity, construction, and performance. However, under no circumstances shall any substitution be made without the written permission of the Architect or Engineer and Owner. **Request for prior approval must be made in writing at least ten (10) days prior to the bid date without fail.**
- F. Wherever a definite product, material or method is specified and there is not a statement that another product, material or method will be acceptable, it is the intention of the Owner or Engineer that the specified product, material or method is the only one that shall be used without prior approval.
- G. Wherever a definite material or manufacturer's product is specified and the Specification states that products of similar design and equal construction from the specified list of manufacturers may be substituted, it is the intention of the Owner or Engineer that products of manufacturers that are specified are the only products that will be acceptable and that products of other manufacturers will not be considered for substitution without approval.

- H. Wherever a definite product, material or method is specified and there is a statement that "OR EQUAL" product, material or method will be acceptable, it is the intention of the Owner or Engineer that the specified product, material or method or an "OR EQUAL" product, material or method may be used if it complies with the specifications and is submitted for review to the Engineer as outline herein.
- I. Where permission to use substituted or alternative equipment on the project is granted by the Owner or Engineer in writing, it shall be the responsibility of the Contractor or Subcontractor involved to verify that the equipment will fit in the space available which includes allowances for all required Code and maintenance clearances, and to coordinate all equipment structural support, plumbing and electrical requirements and provisions with the Mechanical and Plumbing Design Documents and all other trades, including Division 26.
- J. Changes in architectural, structural, electrical, mechanical, and plumbing requirements for the substitution shall be the responsibility of the bidder wishing to make the substitution. This shall include the cost of redesign by the affected designer(s). Any additional cost incurred by affected Subcontractors shall be the responsibility of this bidder and not the Owner.
- K. If any request for a substitution of product, material or method is rejected, the Contractor will automatically be required to furnish the product, material or method named in the Specifications. Repetitive requests for substitutions will not be considered.
- L. The Owner or Engineer will investigate all requests for substitutions when submitted in accordance with above and if accepted, will issue a written acceptance allowing the substitutions.
- M. Where equipment other than that used in the design as specified or shown on the Drawings is substituted (either from an approved manufacturers list or by submittal review), it shall be the responsibility of the substituting Contractor to coordinate space requirements, building provisions and connection requirements with his trades and all other trades and pay all additional costs to other trades, the Owner, the Architect or Engineer, if any, due to the substitutions.

#### 1.13 SUBMITTALS

- A. Coordinate with Division 01 for submittal timetable requirements, unless noted otherwise within thirty (30) days after the Contract is awarded. The Contractor shall submit an electronic copy of a complete set of Shop Drawings and complete data covering each item of equipment or material. The submittal of each item requiring a submittal must be received by the Architect or Engineer within the above thirty (30) day period. The Architect or Engineer shall not be responsible for any delays or costs incurred due to excessive Shop Drawing review time for submittals received after the thirty (30) day time limit. The Architect and Engineer will retain a copy of all Shop Drawings for their files. All literature pertaining to items subject to Shop Drawing submittal shall be submitted at one time. Submittals shall be placed in one electronic file in PDF 8.0 format and bookmarked for individual specification sections. Individual electronic files of submittals for individual specifications shall not be permitted. Each submittal shall include the following items:
  - 1. A cover sheet with the names and addresses of the Project, Architect, MEP Engineer, General Contractor and the Subcontractor making the submittal. The cover sheet shall also contain the section number covering the item or items submitted and the item nomenclature or description.
  - 2. An index page with a listing of all data included in the Submittal.

3. A list of variations page with a listing all variations, including unfurnished or additional required accessories, items or other features, between the submitted equipment and the specified equipment. If there are no variations, then this page shall state "NO VARIATIONS". Where variations affect the work of other Contractors, then the Contractor shall certify on this page that these variations have been fully coordinated with the affected Contractors and that all expenses associated with the variations will be paid by the submitting Contractor. This page will be signed by the submitting Contractor.
  4. Equipment information including manufacturer's name and designation, size, performance and capacity data as applicable. All applicable Listings, Labels, Approvals and Standards shall be clearly indicated.
  5. Dimensional data and scaled drawings as applicable to show that the submitted equipment will fit the space available with all required Code and maintenance clearances clearly indicated and labeled at a minimum scale of 1/4" = 1'-0", as required to demonstrate that the alternate or substituted product will fit in the space available.
  6. Identification of each item of material or equipment matching that indicated on the Drawings.
  7. Sufficient pictorial, descriptive and diagrammatic data on each item to show its conformance with the Drawings and Specifications. Any options or special requirements or accessories shall be so indicated. All applicable information shall be clearly indicated with arrows or another approved method.
  8. Additional information as required in other Sections of this Division.
  9. Certification by the General Contractor and Subcontractor that the material submitted is in accordance with the Drawings and Specifications, signed and dated in long hand. Submittals that do not comply with the above requirements shall be returned to the Contractor and shall be marked "REVISE AND RESUBMIT".
- B. Refer to Division 00 and Division 01 for additional information on Shop Drawings and submittals.
- C. Equipment and materials submittals and Shop Drawings will be reviewed for compliance with design concept only. It will be assumed that the submitting Contractor has verified that all items submitted can be installed in the space allotted. Review of Shop Drawings and submittals shall not be considered as a verification or guarantee of measurements or building conditions.
- D. Where Shop Drawings and submittals are marked "REVIEWED", the review of the submittal does not indicate that submittals have been checked in detail nor does it in any way relieve the Contractor from his responsibility to furnish material and perform work as required by the Contract Documents.
- E. Shop Drawings shall be reviewed and returned to the Contractor with one of the following categories indicated:
1. REVIEWED: Contractor need take no further submittal action, shall include this submittal in the O&M manual and may order the equipment submitted on.
  2. REVIEWED AS NOTED: Contractor shall submit a letter verifying that required exceptions to the submittal have been received and complied with including additional accessories or coordination action as noted, and shall include this submittal and compliance letter in the O&M manual. The Contractor may order the equipment submitted on at the time of the returned submittal providing the Contractor complies with the exceptions noted.
  3. NOT APPROVED: Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is not



- approved, the Contractor will automatically be required to furnish the product, material or method named in the Specifications and/or Drawings. Contractor shall not order equipment that is not approved. Repetitive requests for substitutions will not be considered.
4. **REVISE AND RESUBMIT:** Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is marked revise and resubmit, the Contractor will automatically be required to furnish the product, material or method named in the Specifications and/or provide as noted on previous Shop Drawings. Contractor shall not order equipment marked revise and resubmit. Repetitive requests for substitutions will not be considered.
  5. **CONTRACTOR'S CERTIFICATION REQUIRED:** Contractor shall resubmit submittal on material, equipment or method of installation. The Contractor's stamp is required stating the submittal meets all conditions of the contract documents. The stamp shall be signed by the General Contractor. The submittal will not be reviewed if the stamp is not placed and signed on all Shop Drawings.
  6. **MANUFACTURER NOT AS SPECIFIED:** Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is marked manufacturer not as specified, the Contractor will automatically be required to furnish the product, material or method named in the specifications. Contractor shall not order equipment where submittal is marked manufacturer not as specified. Repetitive requests for substitutions will not be considered.
- F. Materials and equipment which are purchased or installed without Shop Drawing review shall be at the risk of the Contractor and the cost for removal and replacement of such materials and equipment and related work which is judged unsatisfactory by the Owner or Engineer for any reason shall be at the expense of the Contractor. The responsible Contractor shall remove the material and equipment noted above and replace with specified equipment or material at his own expense when directed in writing by the Architect or Engineer.
- G. Shop Drawing Submittals shall be complete and checked prior to submission to the Engineer for review.
- H. Submittals are required for, but not necessarily limited to, the following items:
1. Basic Materials.
  2. Plumbing Fixtures and Valves.
  3. Supports and Carriers.
  4. Floor Drains and Cleanouts.
  5. Interceptors/Traps (All Types).
  6. Water Heaters.
  7. Expansion Tanks.
  8. Water Filters.
  9. Backflow Preventers.
  10. Plumbing Piping.
  11. Piping and Equipment Insulation.
  12. Expansion Fittings and Devices.
  13. Noise and Vibration Controls.
  14. Pipe and Equipment Hangers and Supports.
  15. Plumbing Specialties.
  16. Coordination Drawings.
- I. Refer to other Division 22 sections for additional Shop Drawing and submittal requirements. Provide samples of actual materials and/or equipment to be used on the Project upon request of the Owner or Engineer.

#### 1.14 COORDINATION DRAWINGS

- A. Prepare coordination drawings to a scale of 1/4"=1'-0" or larger; detailing major elements, components, and systems of plumbing equipment and materials in relationship with other systems, installations, and building components. Indicate locations where space is limited for installation and access and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:
  - 1. Indicate the proposed locations of pipe, equipment, and other materials. Include the following:
    - a. Wall locations and types.
    - b. Clearances for installing and maintaining insulation.
    - c. Locations of light fixtures and sprinkler heads.
    - d. Clearances for servicing and maintaining equipment, including tube removal and space for equipment disassembly required for periodic maintenance.
    - e. Equipment connections and support details.
    - f. Exterior wall and foundation penetrations.
    - g. Routing of storm, sanitary sewer piping and plumbing piping.
    - h. Fire-rated wall and floor penetrations.
    - i. Sizes and location of required concrete pads and bases.
    - j. Valve stem movement.
    - k. Structural floor, wall and roof opening sizes and details.
  - 2. Indicate scheduling, sequencing, movement, and positioning of large equipment into the building during construction.
  - 3. Prepare floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
- B. This Contractor shall be responsible for coordination of all items that will affect the installation of the work of this Division. This coordination shall include, but not be limited to: voltage, ampacity, capacity, electrical and piping connections, space requirements, sequence of construction, building requirements and special conditions.
- C. By submitting Shop Drawings on the project, this Contractor is indicating that all necessary coordination has been completed and that the systems, products and equipment submitted can be installed in the building and will operate as specified and intended, in full coordination with all other Contractors and Subcontractors.

#### 1.15 RECORD DOCUMENTS

- A. Prepare Record Documents in accordance with the requirements of Division 00 and Division 01, in addition to the requirements specified in Division 22.
- B. The Contractor shall maintain a separate set of clearly and legibly marked Record Drawings on the job site to record all changes and modifications, including, but not limited to the following: work details, alterations to meet site conditions, and changes made by "Change Order" notices. Mark the drawings with colored pencil(s). These shall be available for review by the Owner, Architect or Engineer during the entire construction stage.
- C. The Record Drawings shall be updated concurrently as construction progresses, and in no case less frequently than a daily basis. They shall indicate accurate dimensions for all buried or concealed work; precise locations of all concealed pipe; locations of all valves, controls and operable devices; and any deviations from the work shown on the

Construction Documents. All dimensions shall include at least two dimensions to permanent structure points.

- D. Record Drawings shall indicate, at a minimum, the following installed conditions:
1. Mains and branches of piping systems, with valves and control devices located and numbered, unions located, and with items requiring maintenance located (i.e., traps, strainers, expansion fittings, tanks, etc.). Valve location diagrams, complete with valve tag chart. Indicate actual inverts and horizontal locations of underground piping.
  2. Equipment locations (exposed and concealed), dimensioned from prominent building lines.
  3. Approved substitutions, Contract Modifications, and actual equipment and materials installed.
  4. Contract Modifications, actual equipment and materials installed.
- E. Engage the services of a Land Surveyor or Professional Engineer registered in the state in which the project is located as specified herein to record the locations and invert elevations of underground installations.
- F. If the Contractor does not keep an accurate set of Record Documents, the pay request may be altered or delayed at the request of the Architect. Delivery of Record Documents is a condition of final acceptance. Record Drawings shall be furnished in addition to Shop Drawings.
- G. Upon completion of the Work, the Contractor shall submit three (3) full size sets of Record Drawing prints to the Architect or Engineer for review prior to scheduling the final inspection at the completion of the work. The drawings shall have the name(s) and seal(s) of the Engineer(s) removed or blanked out and shall be clearly marked and signed on each sheet as follows:

CERTIFIED RECORD DRAWINGS

DATE:

(NAME OF GENERAL CONTRACTOR)

BY: \_\_\_\_\_  
(SIGNATURE)

(NAME OF SUBCONTRACTOR)

BY: \_\_\_\_\_  
(SIGNATURE)

1.16 CERTIFICATIONS AND TEST REPORTS

- A. Submit a detailed schedule for completion and testing of each system indicating scheduled dates for completion of system installation and outlining tests to be performed and scheduled dates for each test. This detailed completion and test schedule shall be submittal at least ninety (90) days before the projected Project completion date.
- B. Test result reporting forms shall be submitted for review no later than the date of the detailed schedule submitted.

- C. Submit four (4) copies of all certifications and test reports to the Architect or Engineer for review adequately in advance of completion of the Work to allow for remedial action as required to correct deficiencies discovered in equipment and systems.
- D. Certifications and test reports to be submitted shall include, but not be limited to those items outlined in other Sections of Division 22.

#### 1.17 OPERATIONS AND MAINTENANCE MANUALS

- A. Prepare Operations and Maintenance manuals in accordance with the requirements of Division 01 and Division 22. In addition to the requirements of other Sections, this shall include the following information for fixtures, specialties, and equipment items:
  - 1. Identifying names, name tags designations and locations for all equipment.
  - 2. Valve tag lists with valve number, type, color coding, location and function.
  - 3. Reviewed Shop Drawing submittals with exceptions noted compliance letter.
  - 4. Fabrication drawings.
  - 5. Equipment and device bulletins and data sheets clearly highlighted to show equipment installed on the project and including performance curves and data as applicable, i.e., description of function, normal operating characteristics and limitations, performance curves, engineering data and tests, and complete nomenclature and model numbers of replacement parts.
  - 6. Manufacturer's printed operating procedures to include start-up, break-in, and routine and normal operating instructions; regulation, control, stopping, shutdown, and emergency instructions; and summer and winter operating instructions.
  - 7. Maintenance procedures for routine preventative maintenance and troubleshooting; disassembly, repair, and reassembly; aligning and adjusting instructions.
  - 8. Servicing instructions and lubrication charts and schedules.
  - 9. Equipment and motor name plate data.
  - 10. Wiring diagrams.
  - 11. Exploded parts views and parts lists for all equipment and devices.
  - 12. Color coding charts for all painted equipment and conduit.
  - 13. Location and listing of all spare parts and special keys and tools furnished to the Owner.
  - 14. Furnish recommended lubrication schedule for all required lubrication points with listing of type and approximate amount of lubricant required.
- B. Coordinate with Division 01 for Operations and Maintenance manual requirements. Unless noted otherwise, bind together in "D ring" style three-ring binders (National model no. 79-883 or equivalent). Binders shall be large enough to allow 1/4" of spare capacity. Include three (3) sets with all approved Shop Drawing submittals, fabrication drawings, bulletins, maintenance instructions, operating instructions and parts exploded views and lists for each and every piece of equipment furnished under this Specification. All sections shall be typed and indexed into sections with tabbed insertable dividers, labeled for easy reference. Utilize the individual specification section numbers shown in the Plumbing Specifications as an organization guideline. Bulletins containing information about equipment that is not installed on the project shall be properly marked up or stripped and reassembled. All pertinent information required by the Owner for proper operation and maintenance of equipment supplied by Division 22 shall be clearly and legibly set forth in memoranda that shall, likewise, be bound with bulletins.
- C. In addition to the bound "hard-copy" Operation and Maintenance manuals referenced above, provide an identical electronic copy in searchable PDF format, with all sections

bookmarked within the file for easy reference. Provide a USB flash drive with the final manual to the Owner.

- D. Operating and Maintenance Manuals shall be turned over to the Owner or Engineer for review a minimum of fourteen (14) working days prior to the beginning of the operator training period.
- E. **Operating and Maintenance Manuals which the Engineer deems incomplete, poorly organized, or otherwise unacceptable will be rejected in writing.** The Contractor will subsequently be required to again turn over Operating and Maintenance Manuals, with all deficiencies corrected, until deemed acceptable by the Engineer.

#### 1.18 OPERATOR TRAINING

- A. The Contractor shall furnish the services of factory trained specialists to instruct the Owner's operating personnel. The Owner's operator training shall include a minimum of 12 hours of on-site training in three (3) shifts of four (4) hours each.
- B. Before proceeding with the instruction of Owner's Personnel, prepare a typed outline in triplicate, listing the subjects that will be covered in this instruction, and submit the outline for review by the Owner. At the conclusion of the instruction period, obtain the signature of each person being instructed on each copy of the reviewed outline to signify that he or she has a proper understanding of the operation and maintenance of the systems and then resubmit the signed outlines.
- C. Refer to other Sections of Division 22 for additional Operator Training requirements.

#### 1.19 FINAL COMPLETION

- A. At the completion of the work, all equipment, operable appurtenances, and systems shall be tested. All faulty equipment and material shall be repaired or replaced. Refer to other Sections of Division 22 for additional requirements.
- B. Clean and adjust all fixtures, flushometers, valves and operable devices. Replace faulty or otherwise damaged parts immediately prior to final acceptance.
- C. Touch up and/or refinish any scratched equipment and devices immediately prior to final acceptance. This shall be acceptable **only for minor superficial scratches**, the determination of which rests solely on the judgment of the Architect or Engineer.

#### 1.20 CONTRACTOR'S GUARANTEE

- A. Use of the Plumbing systems to provide temporary service during the construction period shall not be allowed without written permission from the Owner, and, if granted, shall not be cause for the warranty period to start, except as defined below.
- B. Contractor shall guarantee to keep the entire installation in repair and perfect working order for a period of one (1) year after its completion and final acceptance, and shall furnish free of additional cost to the Owner all materials and labor necessary to comply with the above guarantee throughout the year beginning from the date of issue of Substantial Completion, Beneficial Occupancy by the Owner, or the Certificate of Final Payment as agreed upon by all parties.
- C. This guarantee shall not include cleaning or changing filters except as required by testing, adjusting and balancing.

## 1.21 TRANSFER OF ELECTRONIC FILES

- A. Project documents are not intended or represented to be suitable for re-use by Architect/Owner or others on extensions of this project or on any other project. Any such re-use or modification without written verification or adaptation by Engineer, as appropriate for the specific purpose intended, will be at Architect/Owner's risk and without liability or legal exposure to Engineer or its consultants from all claims, damages, losses and expense, including attorney's fees arising out of or resulting thereof.
- B. Because data stored in electric media format can deteriorate or be modified inadvertently, or otherwise without authorization of the data's creator, the party receiving the electronic files agrees that it will perform acceptance tests or procedures within sixty (60) days of receipt, after which time the receiving party shall be deemed to have accepted the data thus transferred to be acceptable. Any errors detected within the sixty (60) day acceptance period will be corrected by the party delivering the electronic files. Engineer is not responsible for maintaining documents stored in electronic media format after acceptance by the Architect/Owner.
- C. When transferring documents in electronic media format, Engineer makes no representations as to the long-term compatibility, usability or readability of documents resulting from the use of software application packages, operating systems, or computer hardware differing from those used by Engineer at the beginning of the Project.
- D. Any re-use or modifications will be Contractor's sole risk and without liability or legal exposure to Architect, Engineer or any consultant.
- E. The Texas Board of Architectural Examiners (TBAE) has stated that it is in violation of Texas law for persons other than the Architect of record to revise the Architectural drawings without the Architect's written consent.
  - 1. It is agreed that "MEP" hard copy or computer-generated documents will not be issued to any other party except directly to the Architect/Owner. The Contract Documents are contractually copyrighted and cannot be used for any other project or purpose except as specifically indicated in AIA B-141 Standard Form of Agreement Between Architect and Owner.
  - 2. If the client, Architect/Owner, or developer of the project requires electronic media for "record purposes", then an AutoCAD based compact disc ("CD") will be prepared. The "CD" will be submitted with all title block references intact and will be formatted in a "plot" format to permit the end user to only view and plot the drawings. Revisions will not be permitted in this configuration.
- D. At the Architect/Owner's request, Engineer will prepare one "CD" of electronic media to assist the Contractor in the preparation of submittals. The Engineer will prepare and submit the "CD" to the Architect/Owner for distribution to the Contractor.
  - 1. The "CD" will be prepared and all title blocks, names and dates will be removed. The "CD" will be prepared in a ".dwg" format to permit the end user to revise the drawings.

## PART 2 - PRODUCTS

### 2.1 MATERIALS

- A. Provide materials and equipment manufactured by a domestic United States manufacturer and assembled in the United States for all local and Federal Government

projects. These materials and equipment shall comply with "Buy American Act."

- B. Access Doors: Provide access doors as required for access to equipment, valves, controls, cleanouts and other apparatus where concealed. Access doors shall have concealed hinges and screw driver cam locks unless indicated otherwise.
- C. All access panels located in wet areas such as toilet rooms, locker rooms, shower rooms, natatoriums, kitchens, and any other wet areas shall be constructed of stainless steel.
- D. Access doors shall be as follows:
  - 1. Plastic Surfaces: Milcor Style K.
  - 2. Ceramic Tile Surfaces: Milcor Style M.
  - 3. Drywall Surfaces: Milcor Style DW.
  - 4. Install panels only in locations approved by the Architect.

## 2.2 EQUIPMENT PADS

- A. Provide four (4) inch high concrete pads for indoor floor mounted equipment. Pads shall conform to the shape of the equipment with a minimum extension of six (6) inches beyond the equipment. Top and sides of pads shall be troweled to a smooth finish, equivalent to the floor. External corners shall be bull-nosed to a 3/4" radius, unless shown otherwise.
- B. Provide six (6) inch high concrete pads for all exterior mounted equipment. Pads shall conform to the shape of the equipment with a minimum extension of six (6) inches beyond the equipment. Provide a four (4) foot monolithic extension to the pad in front of the equipment for service when mounted on a non-finished area (i.e. landscape, gravel, clay, etc.) Top and sides of pads shall be troweled to a smooth finish. External corners shall be bull-nosed to a 3/4" radius, unless shown otherwise.

## PART 3 - EXECUTION

### 3.1 ROUGH-IN

- A. Verify final locations for rough-ins with field measurements and with the requirements of the actual equipment to be connected via reviewed submittals.
- B. Refer to equipment specifications in other Divisions (10, 11, 12, 13, 21, 22, etc.) for additional rough-in requirements as necessary and provide accordingly.

### 3.2 PLUMBING INSTALLATIONS

- A. General: Sequence, coordinate, and integrate the various elements of plumbing and fire systems, materials, and equipment. Comply with the following requirements:
  - 1. Coordinate plumbing and fire protection systems, equipment, and materials installation with other building components.
  - 2. Verify all dimensions by field measurements.
  - 3. Arrange for chases, slots, leave-outs, and other openings in building components during progress of construction to allow for plumbing installations.
  - 4. Coordinate the installation of required supporting devices, sleeves, and pathways to be set in poured-in-place concrete and other structural components, as they are constructed.

5. Sequence, coordinate, and integrate installations of plumbing materials and equipment for efficient flow of the Work. Give particular attention to large equipment requiring positioning prior to closing in the building.
6. Where mounting heights are not detailed or dimensioned, install systems, materials, and equipment to provide the maximum headroom possible.
7. Coordinate connection of plumbing and fire protection systems with exterior underground and overhead utilities and services. Comply with requirements of governing regulations, franchised service companies, and controlling agencies. Provide required connection for each service.
8. Install systems, materials, and equipment to conform with architectural action markings on submittal, including coordination drawings, to greatest extent possible. Conform to arrangements indicated by the Contract Documents, recognizing that portions of the Work are shown only in diagrammatic form. Where coordination requirements conflict with individual system requirements, resolve conflicts and submit proposed solution to the Architect for review.
9. Install systems, materials, and equipment level and plumb, parallel and perpendicular to other building systems and components, where installed exposed in finished spaces.
10. Install equipment to facilitate servicing, maintenance, and repair or replacement of equipment components. As much as practical, connect equipment for ease of disconnecting, with minimum of interference with other installations. Extend grease fittings to an accessible location and label.
11. Install access panels or doors where valves, operable devices, and equipment are concealed behind finished surfaces. Refer to Article 2.1 of this Section and to Architectural documents for specifications and locations.
12. Install systems, materials, and equipment giving right-of-way priority to systems required to be installed at a specified slope.
13. Provide roof curbs for all roof mounted equipment. Coordinate with roof construction for pitched roof. Provide roof curb to match roof slope. Refer to architectural drawings and details.
14. The equipment to be furnished under this Specification shall be essentially the standard product of the manufacturer. Where two or more units of the same class of equipment are required, these units shall be products of a single manufacturer; however, the component parts of the system need not be the product of the same manufacturer.
15. The architectural and structural features of the building and the space limitations shall be considered in selection of all equipment. No equipment shall be furnished which will not suit the arrangement and space limitations indicated.
16. Lubrication: Prior to start-up, check and properly lubricate all bearings as recommended by the manufacturer.
17. Where the word "Concealed" is used in these Specifications in connection with insulating, painting, piping, valves, etc., it shall be understood to mean hidden from sight as in chases, furred spaces or suspended ceilings. "Exposed" shall be understood to mean the opposite of concealed.
18. Identification of Plumbing Equipment:
  - a. Plumbing equipment shall be identified by means of nameplates permanently attached to the equipment. Nameplates shall be engraved laminated plastic or etched metal. Shop Drawings shall include dimensions and lettering format for approval. Attachments shall be with escutcheon pins, self-tapping screws, or machine screws.
  - b. Tags shall be attached to all valves, including control valves, with nonferrous chains. Tags shall be brass and at least 1-1/2 inches in diameter. Nameplate and tag symbols shall correspond to the identification symbols on the Record Drawings.

### 3.3 CUTTING AND PATCHING



- A. Protection of Installed Work: During cutting and patching operations, protect adjacent installations.
- B. Perform cutting, fitting, and patching of plumbing equipment and materials required to:
  - 1. Uncover Work to provide for installation of ill-timed Work.
  - 2. Remove and replace defective Work.
  - 3. Remove and replace Work not conforming to requirements of the Contract Documents.
  - 4. Remove samples of installed Work as specified for testing.
  - 5. Install equipment and materials in existing structures.
  - 6. Upon written instructions from the Engineer, uncover and restore Work to provide for Engineer/Owner's observation of concealed Work, without additional cost to the Owner.
  - 7. Patch existing finished surfaces and building components using new materials matching existing materials and experienced Installers. Patch finished surfaces and building components using new materials specified for the original installation and experienced Installers; refer to the materials and methods required for the surface and building components being patched; Refer to Article 1.11 DEFINITIONS AND SYMBOLS for definition of "Installer."
- C. Cut, remove and legally dispose of selected plumbing equipment, components, and materials as indicated, including but not limited to removal of plumbing piping, equipment, plumbing fixtures and trim, and other plumbing items made obsolete by the new Work.
- D. Protect the structure, furnishings, finishes, and adjacent materials not indicated or scheduled to be removed.
- E. Provide and maintain temporary partitions or dust barriers adequate to prevent the spread of dust and dirt to adjacent areas.

### 3.4 WORK SEQUENCE, TIMING, COORDINATION WITH OWNER, ARCHITECT AND ENGINEER

- A. The Owner will cooperate with the Contractor, however, the following provisions must be observed:
  - 1. A meeting will be held at the project site, prior to any construction, between the Owner's Representative, the General Contractor, the Subcontractors and the Architect/Engineer to discuss Contractor's employee parking space, access, storage of equipment or materials, and use of the Owner's facilities or utilities. The Owner's decisions regarding such matters shall be final.
  - 2. During the construction of this project, normal facility activities will continue in existing buildings until renovated areas are completed. Plumbing, fire protection, lighting, electrical, communications, heating, air conditioning, and ventilation systems will have to be maintained in service within the occupied spaces of the existing building.
- B. Start-up for major plumbing and fire protection equipment shall be performed by a factory authorized technician. Such equipment shall include, but not necessarily be limited to, the following: domestic water boilers and packaged water heating systems, water softeners, ultra-pure water equipment systems, domestic water booster pumps, fire pumps, and break tank level alarm systems. Refer to other Sections of Divisions 21 and 22 for additional requirements.

### 3.5 DEMOLITION AND WORK WITHIN EXISTING BUILDINGS

- A. In the preparation of these documents every effort has been made to show the approximate locations of, and connections to the existing piping, utilities, equipment and other apparatus related to this phase of the work. However, this Contractor shall be responsible for verifying all of the above information. This Contractor shall visit the existing site to inspect the facilities and related areas. This Contractor shall inspect and verify all details and requirements of all the Contract Documents, prior to the submission of a proposal. All discrepancies between the Contract Documents and actual job-site conditions shall be resolved by this Contractor, who shall produce drawings that shall be submitted to the Architect/Engineer for review. All labor and materials required to perform the work described shall be a part of this Contract.
- B. All equipment and/or systems noted on the Drawings "To Remain" shall be inspected and tested on site to certify its working condition. A written report on the condition of all equipment to remain, including a copy of the test results and recommended remedial actions and costs shall be made by this Contractor to the Architect/Engineer for review.
- C. All equipment and/or systems noted on the Drawings "To Be Removed" shall be removed including, associated pipe, supports, and hangers. Where pipe is to be capped for future or end of line use, it shall be properly tagged with its function or service appropriately identified. Where existing equipment is to be removed or relocated and has an electric motor or connection, the Electrical Contractor shall disconnect motor or connection, remove wiring to a safe point and this Contractor shall remove or relocate motor or connection along with the equipment.
- D. Ensure existing piping and equipment to remain that is adjacent to and impacted by the scope of Work is properly supported, fastened, and secure.
- E. During the construction and remodeling, portions of the Project shall remain in service. Construction equipment, material tools, extension cords, etc., shall be arranged so as to present minimum hazard or interruption to the occupants of the building. None of the construction work shall interfere with the proper operation of the existing facility or be so conducted as to cause harm or danger to persons on the premises. All fire exits, stairs or corridors required for proper access, circulation or exit shall remain clear of equipment, materials or debris. The General Contractor shall maintain barricades, other separations in corridors and other spaces where work is conducted.
- F. Certain work during the demolition phase of construction may require overtime, night time, or weekend shifts or temporary evacuation of the occupants. Coordinate and schedule all proposed down time with the Owner at least seventy-two (72) hours in advance in writing.
- G. Any salvageable equipment as determined by the Owner, shall be delivered to the Owner, and placed in storage at the location of his choice. All other debris shall be removed from the site immediately and disposed of lawfully.
- H. Equipment, piping or other potential hazards to the working occupants of the building or the general public shall not be left overnight outside of the designated working or construction area.
- I. Make every effort to minimize damage to the existing building and the Owner's property. Repair, patch or replace as required any damage that occurs as a result of work at the site. Care shall be taken to minimize interference with the Owner's activities during construction and to keep construction disrupted areas to a minimum. Coordinate with the Owner and other trades in scheduling and performance of the work.

- J. Include in the contract price all rerouting of existing pipe, utilities, etc., and the reconnecting of the existing equipment and plumbing fixtures as necessitated by field conditions to allow the installation of the new systems regardless of whether or not such rerouting, reconnecting or relocating is shown on the Drawings. Provide all temporary pipe, utilities, controls, etc., as required to maintain heating, cooling, ventilation and plumbing services for the existing areas with a minimum of interruption.
- K. All existing plumbing fixtures, pipe, utilities, materials, equipment, controls and appurtenances not included in the remodel or alteration areas are to remain in place.
- L. Pipe, utilities, equipment and controls serving mechanical, plumbing and owner's equipment, etc., which is to remain but which is served by pipe, utilities, equipment and controls that are disturbed by the remodeling work, shall be reconnected in such a manner as to leave this equipment in proper operating condition.
- M. No portion of the **fire protection systems** shall be turned off, modified or changed in any way without the express knowledge and written permission of the Owner's representative in order to protect systems that shall remain in service.
- N. It is the intention of this Section of the Specifications to outline minimum requirements to furnish the Owner with a turn-key and operating system in cooperation with other trades with a minimum of disruption or downtime.
- O. Refer to Architectural Demolition and/or Alteration plans for actual locations of walls, ceiling, etc., being removed and/or remodeled.

**END OF SECTION**

## SECTION 22 02 01 - COORDINATION DRAWINGS

### PART 1 - GENERAL

#### 1.1 GENERAL REQUIREMENTS

- A. The requirements of the General Conditions 013100 and Supplementary Conditions apply to all Work herein.

#### 1.2 COORDINATION DRAWINGS

- A. The Contractor shall take the lead in coordinating the Mechanical, Electrical, Plumbing, Communications, Electronic Safety/Security and Fire Protection systems within the building.
- B. The Mechanical Contractor shall coordinate a three-dimensional (3D) model of the building which includes the Mechanical, Electrical, Plumbing, and Fire Protection systems. The Electrical, Plumbing, and Fire Protection Contractors shall prepare their work and generate 3D models which will be given to the Mechanical Contractor for coordination. The Contractor will be provided with the REVIT model that was used to generate the contract documents, this file may be used as the background file. The Contractor shall replace the systems drawn with the actual shop drawing models. The Contractor is not limited to using REVIT, but may use any 3-D software in generating and combining the coordination model.
- C. Submitting the contract drawings as coordination drawings will not be acceptable.
- D. The model shall include detailed and accurate representations of all equipment to be installed based upon the reviewed equipment submittals.
- E. The Mechanical Contractor shall hold a 3-D coordination meeting with all sub-contractors present to review the model and discuss coordination of the installation of the building systems.
- F. Upon completion of the coordination meeting, the Contractor shall submit the 3-D model and 1/4" scale drawings for review.
- G. The model shall detail major elements, components, and systems in relationship with other systems, installations, and building components. Indicate locations where space is limited for installation and access and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:
  - 1. Indicate the proposed locations of pipe, duct, equipment, and other materials. Include the following:
    - a. Wall and type locations.
    - b. Clearances for installing and maintaining insulation.
    - c. Locations of light fixtures and sprinkler heads.
    - d. Clearances for servicing and maintaining equipment, including tube removal, filter removal, and space for equipment disassembly required for periodic maintenance.
    - e. Equipment connections and support details.
    - f. Exterior wall and foundation penetrations.
    - g. Routing of storm and sanitary sewer piping.
    - h. Fire-rated wall and floor penetrations.
    - i. Sizes and location of required concrete pads and bases.

- j. Valve stem movement.
  - k. Structural floor, wall and roof opening sizes and details.
- 2. Indicate scheduling, sequencing, movement, and positioning of large equipment into the building during construction.
  - 3. Prepare floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
  - 4. Prepare reflected ceiling plans to coordinate and integrate installations, air distribution devices, light fixtures, communication systems components, and other ceiling-mounted items.
- A. Sequence of Coordination
- Below is hierarchy of model elements and the sequencing by which the models will be coordinated.
- 1. Structural and Architectural model
  - 2. Miscellaneous steel
  - 3. Perform preliminary space allocation
  - 4. Identify hard constraints (locations of access panels, lights, A/V space requirements, etc.)
  - 5. Main and medium pressure ducts from the shaft out
  - 6. Main graded plumbing lines and vents
  - 7. Sprinkler mains and branches
  - 8. Cold and hot water mains and branches
  - 9. Lighting fixtures and plumbing fixtures
  - 10. Smaller sized ducts and flex ducts
  - 11. Smaller size cold water and hot water piping, flex ducts, etc.
- B. The Contractor and Sub-Contractors shall not install any item until the coordination has been completed and reviewed by the Construction Manager, Owner, and A/E team.
- C. This Contractor shall be responsible for coordination of all items that will affect the installation of the work of this Division. This coordination shall include, but not be limited to: voltage, ampacity, capacity, electrical and piping connections, space requirements, sequence of construction, building requirements and special conditions.
- D. By submitting shop drawings on the project, this Contractor is indicating that all necessary coordination has been completed and that the systems, products and equipment submitted can be installed in the building and will operate as specified and intended, in full coordination with all other Contractors and Subcontractors.

#### **END OF SECTION**

## **SECTION 22 05 16 – EXPANSION FITTINGS AND LOOPS FOR PLUMBING PIPING**

### **PART 1 - GENERAL**

#### **1.1 WORK INCLUDED**

- A. Flexible pipe connections.
- B. Expansion joints and compensators
- C. Pipe loops, offsets, and swing joints.

#### **1.2 RELATED WORK**

- A. Section 22 05 29 – Hangers and Support for Plumbing Piping and Equipment.
- B. Section 22 10 00 – Plumbing Piping.

#### **1.3 PERFORMANCE REQUIREMENTS**

- A. Provide structural work and equipment required to control expansion and contraction of piping. Verify that anchors, guides, and expansion joints provided, adequately protect system.
- B. Expansion Calculations:
  - 1. Installation Temperature: 50 degrees F (10 degrees C).
  - 2. Hot Water Heating: 210 degrees F (99 degrees C).
  - 3. Domestic Hot Water: 140 degrees F (60 degrees C).
  - 4. Safety Factor: 30 percent.
- C. Pipe sizes indicated are to establish a minimum quality of compensator. Refer to manufacturers' literature for model series for different pipe sizes.

#### **1.4 SUBMITTALS**

- A. Submit shop drawings under provisions of Division One.
- B. Product Data:
  - 1. Flexible Pipe Connectors: Indicate maximum temperature and pressure rating, face-to-face length, live length, hose wall thickness, hose convolutions per foot (meter) and per assembly, fundamental frequency of assembly, braid structure, and total number of wires in braid.
  - 2. Expansion Joints: Indicate maximum temperature and pressure rating, and maximum expansion compensation.
- C. Design Data: Indicate selection calculations.
- D. Manufacturer's Installation Instructions: Indicate special procedures, and external controls.

#### **1.5 PROJECT RECORD DOCUMENTS**

- A. Submit under provisions of Division One.

- B. Record actual locations of flexible pipe connectors, expansion joints, anchor, and guides.

#### 1.6 OPERATION AND MAINTENANCE DATA

- A. Submit under provisions of Division One.
- B. Maintenance Data: Include adjustment instructions.

#### 1.7 QUALIFICATIONS

- A. Manufacturer: Company specializing in manufacturing the products specified in this section with minimum five years documented experience.
- B. Design expansion compensation system under direct supervision of a Professional Engineer experienced in design of this work and licensed in the state where the project is located.

#### 1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, project and handle products to site under provisions of Division One.
- B. Accept expansion joints on site in factory packing with shipping bars and positioning devices intact. Inspect for damage.
- C. Protect equipment from exposure by leaving factory coverings, pipe end protection, and packaging in place until installation.

#### 1.9 WARRANTY

- A. Provide five year warranty under provisions of Division One.
- B. Warranty: Include coverage for leak free performance of packed expansion joints.

#### 1.10 EXTRA MATERIALS

- A. Furnish under provisions of Division One.

### PART 2 - PRODUCTS

#### 2.1 FLEXIBLE PIPE CONNECTORS

- A. Steel Piping (Based on 2" Pipe):
  - 1. Manufacturers:
    - a. Amber/Booth Metal-Flex, Model Type SS-PM or FW
    - b. Triplex, Model Flexonics Series 400M
    - c. Mercer Rubber Company, Model BSS-EM (Mason Industries)
  - 2. Inner Hose: Type 321, stainless steel, corrugated metal.
  - 3. Exterior Sleeve: Type 321, single braided stainless steel.
  - 4. Pressure Rating: 350 psig WOG and 70 degrees F. For 4 inch pipe - 200 psig WOG and 70 degrees F.
  - 5. Joint: Schedule 40 steel, threaded with male nipple and hex boss each end and Union. Flanged joints for pipe sizes 2½ inch and larger.
  - 6. Size: Use pipe sized units.
  - 7. Maximum offset: 1/2 inch on each side of installed center line.
  - 8. Application: Air handling units cooling and heating coils.

B. Copper Piping (Based on 2" Pipe):

1. Manufacturers:
  - a. Amber/Booth Metal-Flex, Model Type BR-SM
  - b. Triplex, Model Flexonics Series 300
  - c. Mercer Rubber Company, Type BFF (Mason Industries)
2. Inner Hose: Corrugated Bronze
3. Exterior Sleeve: Braided bronze.
4. Pressure Rating: 250 psig WOG and 70 degrees F.
5. Joint: Threaded with male nipple and hex boss each end with Union. Flanged joints for pipe sizes 2½ inch and larger.
6. Size: Use pipe sized units.
7. Maximum offset: 1/2 inch on each side of installed center line.
8. Application: Air handling units cooling and heating coils.

2.2 EXPANSION JOINTS

A. Bellows Type (Based on 4" Pipe):

1. Manufacturers:
  - a. Amber/Booth, Style EB
  - b. Triplex, Model Resistoflex R6905
  - c. Mercer Rubber Company, Style 803 or 805 (Mason Industries)
2. Body: Monel wire reinforced molded TFE teflon bellows, multiple arch.
3. Pressure Rating: 70 psig WSP and 250 degrees F (66 degrees C).
4. Maximum Compression: 1 inch.
5. Maximum Extension: 1 inch.
6. Maximum Offset: 1/2 inch.
7. Joint: ASA standard ductile iron flanges, integral molded gasket.
8. Size: Use pipe sized units.
9. Accessories: Control rod limit bolts.
10. Application: Steel piping 8 inch and under.

2.3 ACCESSORIES

A. Pipe Alignment Guides to Direct Axial Movement:

1. Manufacturers:
  - a. Triplex, Model Flexonics
  - b. Metraflex, Style II
2. Two piece welded steel with shop paint, bolted, with spider to fit standard pipe, frame with four mounting holes, clearance for minimum 1 inch thick insulation, minimum 3 inch travel.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Construct spool pieces to exact size of flexible connection for future insertion.
- C. Install flexible pipe connectors on pipes connected to equipment supported by vibration isolation. Provided line size flexible connectors.
- D. Install flexible connectors at right angles to displacement. Install one end immediately



adjacent to isolated equipment and anchor other end. Install in horizontal plane unless indicated otherwise.

- E. Provide miscellaneous metals to rigidly anchor pipe to building structure. Provide pipe guides so that movement takes place along axis of pipe only. Erect piping such that strain and weight is not on cast connections or apparatus.
- F. Provide support and equipment required to control expansion and contraction of piping. Provide loops, pipe offsets, and swing joints, or expansion joints where required.

### 3.2 MANUFACTURER'S FIELD SERVICES

- A. Prepare and start systems under provisions of Division One.
- B. Provide inspection services by flexible pipe manufacturer's representative for final installing and certify installation is in accordance with manufacturer's recommendations and connectors are performing satisfactorily.

**END OF SECTION**

## **SECTION 22 05 29 – HANGERS AND SUPPORT FOR PLUMBING PIPING AND EQUIPMENT**

### **PART 1 - GENERAL**

#### **1.1 WORK INCLUDED**

- A. Pipe, and equipment hangers, supports, and associated anchors.
- B. Sleeves and seals.
- C. Flashing and sealing equipment and pipe stacks.

#### **1.2 RELATED WORK**

- A. Section 22 05 29 – Hangers and Support for Plumbing Piping and Equipment.
- B. Section 22 07 19 – Plumbing Piping Insulation.
- C. Section 22 07 16 – Plumbing Equipment Insulation.
- D. Section 21 00 00 - Fire Protection and 21 13 13 Wet Pipe Sprinkler System.
- E. Section 22 10 00 - Plumbing System.
- F. Section 23 21 13 – Above Ground Hydronic Piping.
- G. Section 23 23 00 – Refrigerant Piping

#### **1.3 REFERENCES**

- A. ANSI/ASME B31.1 - Power Piping.
- B. NFPA 13 - Standard for the Installation of Sprinkler Systems.
- C. NFPA 14 - Standard for the Installation of Standpipe and Hose Systems.

#### **1.4 QUALITY ASSURANCE**

- A. Supports for Sprinkler Piping: In conformance with NFPA 13.
- B. Supports for Standpipes: In conformance with NFPA 14.

#### **1.5 SUBMITTALS**

- A. Submit shop drawings and product data under provisions of Division One.
- B. Indicate hanger and support framing and attachment methods.

### **PART 2 - PRODUCTS**

#### **2.1 PIPE HANGERS AND SUPPORTS**

- A. Hangers for Pipe Sizes 1/2 to 1-1/2 Inch Malleable iron, adjustable swivel, split ring.
- B. Hangers for Pipe Sizes 2 to 4 Inches Carbon steel, adjustable, clevis.

- C. Hangers for Pipe Sizes 6 Inches and Over: Adjustable steel yoke, cast iron roll, double hanger.
- D. Multiple or Trapeze Hangers: Steel channels with welded spacers and hanger rods; cast iron roll and stand for pipe sizes 6 inches and over.
- E. Wall Support for Pipe Sizes to 3 Inches: Cast iron hook.
- F. Wall Support for Pipe Sizes 4 Inches and Over: adjustable steel yoke and cast iron roll.
- G. Vertical Support: Steel riser clamp.
- H. Floor Support for Pipe Sizes to 4 Inches: Cast iron adjustable pipe saddle, locknut nipple, floor flange, and concrete pier or steel support.
- I. Floor Support for Pipe Sizes 6 Inches and Over: Adjustable cast iron roll and stand, steel screws, and concrete pier or steel support.
- J. Roof Pipe Supports and Hangers: Galvanized Steel Channel System as manufactured by Portable Pipe Hangers, Inc. or approved equal.  
  
 For pipes 2-1/2" and smaller – Type PP10 with roller  
 For pipes 3" through 8" – Type PS  
 For multiple pipes – Type PSE - Custom
- K. Copper Pipe Support and Hangers: Electro-galvanized with thermoplastic elastomer cushions; Unistrut "Cush-A-Clamp" or equal. Hangers: Plastic coated; Unistrut or equal.
- L. For installation of protective shields refer to specification section 22 07 19 -3.3.
- M. Shields for Vertical Copper Pipe Risers: Sheet lead.
- N. Pipe Rough-In Supports in Walls/Chases: Provide preformed plastic pipe supports, Sioux Chief "Pipe Titan" hold rite or equal.

## 2.2 HANGER RODS

- A. Galvanized Hanger Rods: Threaded both ends, threaded one end, or continuous threaded.

## 2.3 INSERTS

- A. Inserts: Malleable iron case of galvanized steel shell and expander plug for threaded connection with lateral adjustment, top slot for reinforcing rods, lugs for attaching to forms; size inserts to suit threaded hanger rods.

## 2.4 FLASHING

- A. Metal Flashing: 20 gage galvanized steel.
- B. Lead Flashing: 4 lb./sq. ft. sheet lead for waterproofing; 1 lb./sq. ft. sheet lead for soundproofing.
- C. Caps: Steel, 20 gage minimum; 16 gage at fire resistant elements.
- D. Coordinate with roofing contractor/architect for type of flashing on metal roofs.

## 2.5 EQUIPMENT CURBS

- A. Fabricate curbs of hot dipped galvanized steel.

## 2.6 SLEEVES

- A. Sleeves for Pipes Through Non-fire Rated Floors: Form with 18 gage galvanized steel, tack welded to form a uniform sleeve.
- B. Sleeves for Pipes Through Non-fire Rated Beams, Walls, Footings, and Potentially Wet Floors: Form with steel pipe, schedule 40.
- C. Sleeves for Pipes Through Fire Rated and Fire Resistive Floors and Walls, and Fireproofing: Prefabricated fire rated steel sleeves including seals, UL listed.
- D. Sleeves for Round Ductwork: Form with galvanized steel.
- E. Sleeves for Rectangular Ductwork: Form with galvanized steel.
- F. Fire Stopping Insulation: Glass fiber type, non-combustible, U.L. listed.
- G. Caulk: Paintable 25-year acrylic sealant.
- H. Pipe Alignment Guides: Factory fabricated, of cast semi-steel or heavy fabricated steel, consisting of bolted, two-section outer cylinder and base with two-section guiding spider that bolts tightly to pipe. Length of guides shall be as recommended by manufacturer to allow indicated travel.

## 2.7 FABRICATION

- A. Size sleeves large enough to allow for movement due to expansion and contraction. Provide for continuous insulation wrapping.
- B. Design hangers without disengagement of supported pipe.
- C. Design roof supports without roof penetrations, flashing or damage to the roofing material.

## 2.8 FINISH

- A. Prime coat exposed steel hangers and supports. Hangers and supports located in crawl spaces, pipe shafts, and suspended ceiling spaces are not considered exposed.

## PART 3 - EXECUTION

### 3.1 INSERTS

- A. Provide inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams. Coordinate with structural engineer for placement of inserts.
- B. Provide hooked rod to concrete reinforcement section for inserts carrying pipe over 4 inches.
- C. Where concrete slabs form finished ceiling, provide inserts to be flush with slab surface.
- D. Where inserts are omitted, drill through concrete slab from below and provide thru-bolt

with recessed square steel plate and nut recessed into and grouted flush with slab. Verify with structural engineer prior to start of work.

### 3.2 PIPE HANGERS AND SUPPORTS

A. Support horizontal piping as follows:

| <u>PIPE SIZE</u>    | <u>MAX. HANGER SPACING</u> | <u>HANGER DIAMETER</u> |
|---------------------|----------------------------|------------------------|
| (Steel Pipe)        |                            |                        |
| 1/2 to 1-1/4 inch   | 7'-0"                      | 3/8"                   |
| 1-1/2 to 3 inch     | 10'-0"                     | 3/8"                   |
| 4 to 6 inch         | 10'-0"                     | 1/2"                   |
| 8 to 10 inch        | 10'-0"                     | 5/8"                   |
| 12 to 14 inch       | 10'-0"                     | 3/4"                   |
| 15 inch and over    | 10'-0"                     | 7/8"                   |
| (Copper Pipe)       |                            |                        |
| 1/2 to 1-1/4 inch   | 5'-0"                      | 3/8"                   |
| 1-1/2 to 2-1/2 inch | 8'-0"                      | 3/8"                   |
| 3 to 4 inch         | 10'-0"                     | 3/8"                   |
| 6 to 8 inch         | 10'-0"                     | 1/2"                   |
| (Cast Iron)         |                            |                        |
| 2 to 3 inch         | 5'-0"                      | 3/8"                   |
| 4 to 6 inch         | 10'-0"                     | 1/2"                   |
| 8 to 10 inch        | 10'-0"                     | 5/8"                   |
| 12 to 14 inch       | 10'-0"                     | 3/4"                   |
| 15 inch and over    | 10'-0"                     | 7/8"                   |
| (PVC Pipe)          |                            |                        |
| 1-1/2 to 4 inch     | 4'-0"                      | 3/8"                   |
| 6 to 8 inch         | 4'-0"                      | 1/2"                   |
| 10 and over         | 4'-0"                      | 5/8"                   |

- B. Install hangers to provide minimum 1/2 inch space between finished covering and adjacent work.
- C. Place a hanger within 12 inches of each horizontal elbow and at the vertical horizontal transition.
- D. Use hangers with 1-1/2 inch minimum vertical adjustment.
- E. Support horizontal cast iron pipe adjacent to each hub, with 5 feet maximum spacing between hangers.
- F. Support vertical piping at every floor. Support vertical cast iron pipe at each floor at hub.
- G. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers.
- H. Support riser piping independently of connected horizontal piping.
- I. Install hangers with nut at base and above hanger; tighten upper nut to hanger after final installation adjustments.
- J. Portable pipe hanger systems shall be installed per manufactures instructions.

3.3 Insulated Piping: Comply with the following installation requirements.

- A. Clamps: Attach galvanized clamps, including spacers (if any), to piping with clamps projecting through insulation; do not exceed pipe stresses allowed by ASME B31.9.
- B. Saddles: Install galvanized protection saddles MSS Type 39 where insulation without vapor barrier is indicated. Fill interior voids with segments of insulation that match adjoining pipe insulation.
- C. Shields: Install protective shields MSS Type 40 on cold and chilled water piping that has vapor barrier. Shields shall span an arc of 180 degrees and shall have dimensions in inches not less than the following:

| <u>NPS</u>        | <u>LENGTH</u> | <u>THICKNESS</u> |
|-------------------|---------------|------------------|
| 1/4 THROUGH 3-1/2 | 12            | 0.048            |
| 4                 | 12            | 0.060            |
| 5 & 6             | 18            | 0.060            |
| 8 THROUGH 14      | 24            | 0.075            |
| 16 THROUGH 24     | 24            | 0.105            |

- D. Piping 2" and larger provide galvanized sheet metal shields with calcium silicate at hangers/supports.
- E. Insert material shall be at least as long as the protective shield.
- F. Thermal Hanger Shields: Install where indicated, with insulation of same thickness as piping.

3.4 EQUIPMENT BASES AND SUPPORTS

- A. Provide equipment bases of concrete.
- B. Provide templates, anchor bolts, and accessories for mounting and anchoring equipment.
- C. Construct support of steel members. Brace and fasten with flanges bolted to structure.
- D. Provide rigid anchors for pipes after vibration isolation components are installed.

3.5 FLASHING

- A. Provide flexible flashing and metal counter flashing where piping and ductwork penetrate weather or waterproofed walls, floors, and roofs.
- B. Flash vent and soil pipes projecting 8 inches minimum above finished roof surface with lead worked one inch minimum into hub, 8 inches minimum clear on sides with 24 x 24 inches sheet size. For pipes through outside walls, turn flanges back into wall and caulk, metal counter flash and seal.
- C. Flash floor drains in floors with topping over finished areas with lead, 10 inches clear on sides with minimum 36 x 36 inch sheet size. Fasten flashing to drain clamp device.
- D. Seal floor shower mop sink and all other drains watertight to adjacent materials.
- E. Provide curbs for mechanical roof installations 8 inches minimum high above roofing surface. Contact architect for all flashing details and roof construction. Seal penetrations

watertight.

### 3.6 SLEEVES

- A. Set sleeves in position in formwork. Provide reinforcing around sleeves.
- B. Extend sleeves through floors minimum one inch above finished floor level. Caulk sleeves full depth with fire rated thermfiber and 3M caulking and provide floor plate.
- C. Where piping or ductwork penetrates floor, ceiling, or wall, close off space between pipe or duct and adjacent work with U.L. listed fire stopping insulation and caulk seal air tight. Provide close fitting metal collar or escutcheon covers at both sides of penetration.
- D. Fire protection sleeves may be flush with floor of stairways.

**END OF SECTION**

## **SECTION 22 05 53 – IDENTIFICATION FOR PLUMBING PIPING AND EQUIPMENT**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 22 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for owner's use.

- 1.3 Refer to Architectural Sections for additional requirements.

### **PART 2 - PRODUCTS**

#### **2.1 VALVE AND PIPE IDENTIFICATION**

##### **A. Valves:**

- 1. All valves shall be identified with a 1-1/2" diameter brass disc wired onto the handle. The disc shall be stamped with 1/2" high depressed black filled identifying numbers. These numbers shall be numerically sequenced for all valves on the job.
- 2. The number and description indicating make, size, model number and service of each valve shall be listed in proper operational sequence, properly typewritten. Three copies to be turned over to Owner at completion.
- 3. Tags shall be fastened with approved meter seal and 4 ply 0.018 smooth copper wire. Tags and fastenings shall be manufactured by the Seton Name Plate Company or approved equal.
- 4. All valves shall be numbered serially with all valves of any one system and/or trade grouped together.

##### **B. Pipe Marking:**

- 1. All interior visible piping located in accessible spaces such as above accessible ceilings, equipment rooms, attic space, under floor spaces, etc., shall be identified with all temperature pipe markers as manufactured by W.H. Brady Company, 431 West Rock Ave., New Haven, Connecticut, or approved equal.
- 2. All exterior visible piping shall be identified with UV and acid resistant outdoor grade acrylic plastic markers as manufactured by Set Mark distributed by Seton nameplate company. Factory location 20 Thompson Road, Branford, Connecticut, or approved equal.
- 3. Generally, markers shall be located on each side of each partition, on each side of each tee, on each side of each valve and/or valve group, on each side of each piece of equipment, and, for straight runs, at equally spaced intervals not to exceed 75 feet. In congested area, marks shall be placed on each pipe at the points where it enters and leaves the area and at the point of connection of each piece of equipment and automatic control valve. All markers shall have directional arrows.
- 4. Markers shall be installed after final painting of all piping and equipment and in



such a manner that they are visible from the normal maintenance position. Manufacturer's installation instructions shall be closely followed.

5. Markers shall be colored as indicated below per ANSI/OSHA Standards:

| <u>SYSTEM</u>      | <u>COLOR</u> | <u>LEGEND</u>   |
|--------------------|--------------|-----------------|
| Sanitary Sewer     | Green        | Vent            |
|                    |              | Sanitary Sewer  |
| Storm Drain        | Green        | Storm Drain     |
| Domestic Water     | Green        | Domestic Water  |
| Domestic Hot Water | Yellow       | Domestic Hot    |
| Supply             |              | Water Supply    |
| Domestic Hot Water | Yellow       | Domestic Hot    |
| Recirculating      |              | Water Return    |
| Fire Protection    | Red          | Fire Protection |
| Automatic          | Red          | Fire            |
| Sprinkler          |              | Sprinkler       |
| Gas                | Yellow       | Natural Gas     |
| Compressed Air     | Blue         | Compressed Air  |
| Oxygen             | Yellow       | Oxygen          |
| Nitrogen           | Green        | Nitrogen        |
| Deionized Water    | Green        | Deionized Water |

C. Pipe Painting:

1. All piping exposed to view shall be painted as indicated or as directed by the Architect in the field. Confirm all color selections with Architect prior to installation.
2. The entire fire protection piping system shall be painted red.
3. All piping located in mechanical rooms and exterior piping shall be painted as indicated below:

| <u>System</u>                        | <u>Color</u> |
|--------------------------------------|--------------|
| Storm Sewer                          | White        |
| Sanitary Sewer Waste and Vent        | Light Gray   |
| Domestic Cold Water                  | Dark Blue    |
| Domestic Hot Water Supply and Return | Orange       |

PART 3 - EXECUTION

- 3.1 All labeling equipment shall be installed as per manufacturers printed installation instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Contractor's price shall include all items required as per manufacturers' requirements.
- 3.3 All piping shall be cleaned of rust, dirt, oil and all other contaminants prior to painting. Install primer and a quality latex paint over all surfaces of pipe.

**END OF SECTION**

## **SECTION 22 07 19 – PLUMBING PIPING INSULATION**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 22 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

- A. Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for owner's use.
- B. Furnish and install piping insulation to:
  - 1. Interior domestic hot water and hot water return piping.
  - 2. Interior domestic cold water piping.
  - 3. Exterior domestic cold water piping.
  - 4. Drain bodies and associated piping.
  - 5. Condensate drainage piping.
  - 6. All pipes subject to freezing conditions shall be insulated.
- C. Work specified elsewhere.
  - 1. Painting.
  - 2. Pipe hangers and supports.
- D. For insulation purposes, piping is defined as the complete piping system including supplies and returns, pipes, valves, automatic control valve bodies, fittings, flanges, strainers, thermometer wells, unions, pressure reducing stations, and orifice assemblies.

#### **1.3 WARRANTY**

- A. Warrant the Work specified herein for one year against becoming unserviceable or causing an objectionable appearance resulting from either defective or nonconforming materials or workmanship.
- B. Defects shall include, but not be limited to, the following:
  - 1. Mildewing.
  - 2. Peeling, cracking, and blistering.
  - 3. Condensation on exterior surfaces.

#### **1.4 SUBMITTALS**

- A. **SHOP DRAWINGS:** Indicate size, material, and finish. Show locations and installation procedures. Include details of joints, attachments, and clearances.
- B. **PRODUCT DATA:** Submit schedules, charts, literature, and illustrations to indicate the performance, fabrication procedures, project variations, and accessories.

#### **1.5 DELIVERY AND STORAGE**

- A. DELIVERY: Deliver undamaged materials in the manufacturer's unopened containers. Containers shall be clearly labeled with the insulation's flame and smoke ratings.

## PART 2 - PRODUCTS

- 2.1 It is the intent of these specifications to secure superior quality workmanship resulting in an absolutely satisfactory installation of insulation from the standpoint of both function and appearance. Particular attention shall be given to valves, fittings, pumps, etc., requiring low temperature insulation to insure full thickness of insulation and proper application of the vapor seal. All flaps of vapor barrier jackets and/or canvas covering must be neatly and securely smoothed and sealed down.
- 2.2 The type of insulation and its installation shall be in strict accordance with these specifications for each service, and the application technique shall be as recommended by the manufacturer. All insulation types, together with adhesives and finishes shall be submitted and approved prior to installation.
- 2.3 A sample quantity of each type of insulation and each type application shall be installed and approval secured prior to proceeding with the main body of the work. Condensation caused by improper installation of insulation shall be corrected by Installing Contractor. Any damage caused by condensation shall be made good at no cost to the Owner or Architect/Engineer.
- 2.4 All insulation shall have composite (insulation, jacket or facing, and adhesive used to adhere the facing or jacket to insulation) fire and smoke hazard as tested by Procedure ASTM E084, NFPA 255 and UL 723 not exceeding:

Flame Spread 25  
Smoke Developed 50

- 2.5 Accessories, such as adhesives, mastics and cements shall have the same component ratings as listed above.
- 2.6 All products or their shipping cartons shall have a label affixed, indicating flame and smoke ratings do not exceed the above requirements.
- 2.7 APPROVED MANUFACTURERS

- A. Calcium silicate materials shall be as manufactured by Johns Manville.
- B. Glass fiber materials shall be as manufactured by Johns Manville or Owens-Corning and shall have the same thermal properties, density, fire rating, vapor barrier, etc., as the types specified herein, subject to review by the Engineer.
- C. Adhesives shall be as manufactured by Childers, Foster, HB Fuller or Armstrong, and shall have the same adhesive properties, fire rating, vapor seal, etc., as the types specified herein, subject to review by the Engineer.
- D. Armaflex elastomeric cellular thermal insulation by Armstrong.
- E. Phenolic foam insulation shall be as manufactured by Kooltherm Insulation (Koolphen).
- F. Metal jacketing and fitting covers shall be as manufactured by Childers or RPR Products, Inc.

## 2.8 MATERIALS

- A. INTERIOR DOMESTIC WATER PIPE: provide fiberglass pipe insulation with all service jackets with self sealing lap joint.
- B. EXTERIOR DOMESTIC WATER PIPE: Provide elastomeric cellular thermal, or preformed phenolic foam pipe insulation with secured aluminum jacketing.
- C. DRAIN BODIES AND DOWNSPOUTS: Insulate underside of roof and overflow drain bodies, associated horizontal piping, including first turn down to vertical conductor. Insulate chilled water waste lines from drinking fountain to junction with main waste stacks. Insulate branch lines including traps and exposed underside of floor drains receiving cooling coil condensate, same as water piping where exposed to building occupant view. When concealed, insulation may be same as specified for external duct wrap.
- D. CONDENSATE DRAINAGE PIPING: Fire resistant fiberglass insulation; insulation not required when piping is exposed on roof.
- E. ALUMINUM OR STAINLESS STEEL JACKETING: Utilize strap-on type jacketing, banding, and accessories. Provide pre-formed fitting covers for all elbows and tees.

### PART 3 - EXECUTION

- 3.1 All insulation shall be installed in accordance with the manufacturers' recommendations and printed installation instructions, including high density inserts at all hangers and pipe supports to prevent compression of insulation.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Provide all items required as per manufacturers requirements.
- 3.3 Pipes located outdoors or in tunnels shall be insulated same as concealed piping; and in addition shall have a jacket of 0.016 inch thick, smooth aluminum with longitudinal modified Pittsburgh Z-Lock seam and 2 inch overlap. Jacketing shall be easily removed and replaced without damage. All butt joints shall be sealed with gray silicone. Galvanized banding is not acceptable.
- 3.4 All insulated piping located over driveways shall have an aluminum shield permanently banded over insulation to protect it from damage from car antennas.

### 3.5 WATER PIPE INSULATION INSTALLATION

- A. The insulation shall be applied to clean, dry pipes with all joints firmly butted together. Where piping is interrupted by fittings, flanges, valves or hangers and at intervals not to exceed 25 feet on straight runs, an isolating seal shall be formed between the vapor barrier jacket and the bare pipe. The seal shall be by the applications of adhesive to the exposed insulation joint faces, carried continuously down to and along 4 inches of pipe and up to and along 2 inches of jacket.
- B. Pipe fittings and valves shall be insulated with pre-molded or shop fabricated glass fiber covers finished with two brush coats of vapor barrier mastic reinforced with glass fabric.
- C. All under lap surfaces shall be clean and free of dust, etc. before the SSL is sealed. These laps shall be firmly rubbed to insure a positive seal. A brush coat of vapor retarder shall be applied to all edges of the vapor barrier jacket.
- D. At hangers and supports, provide a high density foam insulation insert that extends 2" beyond the shield on each side and a protective shield/saddle to prevent

compression/damage. Secure shield/saddle to insulation using mastic or strapping tape.

### 3.6 FIRE RATED INSULATION

- A. All pipe penetrations through walls and concrete floors shall be fire rated by applying USG Thermafiber in the space between the concrete and the pipe.
- B. The fire rating shall be additionally sealed by using 3M brand model CP 25 or 303 fire barrier caulk and putty.
- C. All fire rating material shall be insulated in accordance with manufacturer's printed instructions.

## PART 4 - SCHEDULES

### 4.1 LOW TEMPERATURE SURFACES

#### MINIMUM INSULATION THICKNESS BASED ON FIBERGLASS

- A. Exposed exterior domestic water pipe: 1½ inch
- B. Interior domestic cold water pipe: 1 inch
- C. Condensate drain lines: ¾ inch
- D. Drains receiving condensate: 1 inch
- E. Concealed piping from roof drains: 1½ inch blanket wrap
- F. Exposed piping from roof drains: 1 inch thick rigid with all service jacket

### 4.2 HIGH TEMPERATURE SURFACES

#### MINIMUM INSULATION THICKNESS

- A. Domestic Hot Water and Domestic Hot Water Return Piping
  - 1. Pipe sizes 1-1/4 inch and smaller with Operating temperatures of 140°F or less 1 inch
  - 2. Pipe sizes 1-1/2 inch and larger with Operating temperatures of 140°F or less 1-1/2 inch
  - 3. Pipe sizes 1-1/4 inch and smaller with Operating temperatures greater than 140°F 1-1/2 inch
  - 4. Pipe sizes 1-1/2 inch and larger with Operating temperatures greater than 140°F 2 inch

**END OF SECTION**

## **SECTION 22 10 00 - PLUMBING PIPING**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 22 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

- A. The scope of the work shall include the furnishing and complete installation of the piping covered by this Section, with all appurtenances, ready for the Owner's use.
- B. Include the following work in addition to items normally part of this Section:
  - 1. Pipe and pipe fittings:
    - a. Sanitary drainage piping system.
    - b. Domestic water piping system.
  - 2. Adapters, Transitions, Unions, Couplings, Flanges, Connectors
  - 3. Valves
  - 4. Excavation, Bedding, and Backfill

#### **1.3 RELATED WORK**

- A. Section 22 05 29 – Hangers and Support for Plumbing Piping and Equipment.
- B. Section 22 05 48 – Vibration and Seismic Controls for Plumbing Piping.
- C. Section 22 05 53 – Identification for Plumbing Piping and Equipment.
- D. Section 22 07 19 – Plumbing Piping Insulation.
- E. Section 22 11 19 – Plumbing Specialties.
- F. Section 22 30 00 – Plumbing Equipment.
- G. Section 22 40 00 – Plumbing Fixtures.

#### **1.4 REFERENCES**

- A. ASME – Boiler and Pressure Vessel Code.
- B. ASME Section IX – Welding and Brazing Qualifications.
- C. ASME B1.20.1 – Pipe Threads, General Purpose.
- D. ASME B16.1 – Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125, and 250.

- E. ASME B16.3 - Malleable Iron Threaded Fittings: Classes 150 and 300.
- F. ASME B16.4 – Gray Iron Threaded Fittings: Classes 125 and 250.
- G. ASME B16.5 – Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard.
- H. ASME B16.14 – Ferrous Pipe Plugs, Bushings, and Locknuts with Pipe Threads.
- I. ASME B16.22 - Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings.
- J. ASME B16.23 – Cast Copper Alloy Solder Joint Drainage Fittings: DWV.
- K. ASME B16.51 – Copper and Copper Alloy Press-Connect Pressure Fittings.
- L. ASME B31.3 – Process Piping.
- M. ASME B31.9 – Building Services Piping.
- N. ASTM A47 – Standard Specification for Ferritic Malleable Iron Castings.
- O. ASTM A53 – Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc Coated, Welded and Seamless.
- P. ASTM A74 – Standard Specification for Cast Iron Soil Pipe and Fittings.
- Q. ASTM A126 – Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
- R. ASTM A153 – Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
- S. ASTM A197 – Standard Specification for Cupola Malleable Iron.
- T. ASTM A395 – Standard Specification for Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures.
- U. ASTM A536 – Standard Specification for Ductile Iron Castings.
- V. ASTM A582 – Standard Specification for Free-Machining Stainless Steel Bars.
- W. ASTM B32 – Standard Specification for Solder Metal.
- X. ASTM B42 – Standard Specification for Seamless Copper Pipe, Standard Sizes.
- Y. ASTM B43 – Standard Specification for Seamless Red Brass Pipe, Standard Sizes.
- Z. ASTM B62 – Standard Specification for Composition Bronze or Ounce Metal Castings.
- AA. ASTM B75 – Standard Specification for Seamless Copper Tube.
- BB. ASTM B148 – Standard Specification for Aluminum-Bronze Sand Castings.
- CC. ASTM B306 – Standard Specification for Copper Drainage Tube (DWV).

- DD. ASTM B584 – Standard Specification for Copper Alloy Sand Castings for General Applications.
- EE. ASTM B828 – Standard Practice for Making Capillary Joints by Soldering of Copper and Copper Alloy Tube and Fittings.
- FF. ASTM C33/C33M – Standard Specification for Concrete Aggregates.
- GG. ASTM C94 – Standard Specification for Ready-Mix Concrete.
- HH. ASTM C150 – Standard Specification for Portland Cement.
- II. ASTM C564 – Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings.
- JJ. ASTM C1053 – Standard Specification for Borosilicate Glass Pipe and Fittings for Drain, Waste, and Vent (DWV) Applications.
- KK. ASTM C1173 – Standard Specification for Flexible Transition Couplings for Underground Piping Systems.
- LL. ASTM C1277 – Standard Specification for Shielded Couplings Joining Hubless Cast Iron Soil Pipe and Fittings.
- MM. ASTM C1540 – Standard Specification for Heavy Duty Shielded Couplings Joining Hubless Cast Iron Soil Pipe and Fittings.
- NN. ASTM D635 – Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position.
- OO. ASTM D1784 – Standard Specification for Rigid Poly (Vinyl Chloride) (PVC) Compounds and Chlorinated Poly (Vinyl Chloride) (CPVC) Compounds.
- PP. ASTM D1785 – Standard Specification for Poly (Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120.
- QQ. ASTM D2241 – Standard Specification for Poly (Vinyl Chloride) (PVC) Pressure-Rated Pipe (SDR Series).
- RR. ASTM D2321 – Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications.
- SS. ASTM D2464 – Standard Specification for Threaded Poly (Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80.
- TT. ASTM D2466 – Standard Specification for Poly (Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40.
- UU. ASTM D2467 – Standard Specification for Poly (Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80.
- VV. ASTM D2564 – Standard Specification for Solvent Cements for Poly (Vinyl Chloride) (PVC) Plastic Piping Systems.
- WW. ASTM D2665 – Standard Specification for Poly (Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings.



- XX. ASTM D2672 – Standard Specification for Joints for IPS PVC Pipe Using Solvent Cement.
- YY. ASTM D2729 – Standard Specification for Poly (Vinyl Chloride) (PVC) Sewer Pipe and Fittings.
- ZZ. ASTM D2774 – Standard Practice for Underground Installation of Thermoplastic Pressure Piping.
- AAA. ASTM D2843 – Standard Test Method for Density of Smoke from the Burning or Decomposition of Plastics.
- BBB. ASTM D2846 – Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Hot and Cold Water Distribution Systems.
- CCC. ASTM D2855 – Standard Practice for the Two-Step (Primer and Solvent Cement) Method of Joining Poly (Vinyl Chloride) (PVC) or Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Piping Components with Tapered Sockets.
- DDD. ASTM D3034 – Standard Specification for Type PSM Poly (Vinyl Chloride) (PVC) Sewer Pipe and Fittings.
- EEE. ASTM D3139 – Standard Specification for Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals.
- FFF. ASTM D3212 – Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals.
- GGG. ASTM D3311 – Standard Specification for Drain, Waste, and Vent (DWV) Plastic Fittings Patterns.
- HHH. ASTM D4976 – Standard Specification for Polyethylene Plastics Molding and Extrusion Materials.
- III. ASTM D5926 – Standard Specification for Poly (Vinyl Chloride) (PVC) Gaskets for Drain, Waste, and Vent (DWV), Sewer, Sanitary, and Storm Plumbing Systems.
- JJJ. ASTM D6707 – Standard Specification for Circular-Knit Geotextile for Use in Subsurface Drainage Applications.
- KKK. ASTM E84 – Standard Test Method for Surface Burning Characteristics of Building Materials.
- LLL. ASTM F439 – Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80.
- MMM. ASTM F441 – Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80.
- NNN. ASTM F477 – Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe.
- OOO. ASTM F493 – Standard Specification for Solvent Cements for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe and Fittings.

- PPP. ASTM F656 – Standard Specification for Primers for Use in Solvent Cement Joints of Poly (Vinyl Chloride) (PVC) Plastic Pipe and Fittings.
- QQQ. ASTM F913 – Standard Specification for Thermoplastic Elastomeric Seals (Gaskets) for Joining Plastic Pipe.
- RRR. ASTM F1336 – Standard Specification for Poly (Vinyl Chloride) (PVC) Gasketed Sewer Fittings.
- SSS. ASTM F1476 – Standard Specification for Performance of Gasketed Mechanical Couplings for Use in Piping Applications.
- TTT. ASTM F1548 – Standard Specification for Performance of Fittings for Use with Gasketed Mechanical Couplings Used in Piping Applications.
- UUU. AWS A5.8 – Specification for Filler Metals for Brazing and Braze Welding.
- VVV. AWS 5.31 – Specification for Fluxes for Brazing and Braze Welding.
- WWW. AWWA C105 – Standard for Polyethylene Encasement for Ductile-Iron Pipe Systems.
- XXX. AWWA C111 – Rubber-Gasket Joints for Ductile Iron Pressure Pipe and Fittings.
- YYY. AWWA C209 – Standard for Cold-Applied Tape Coatings for the Exterior of Special Sections, Connections, and Fittings for Steel Water Pipelines.
- ZZZ. AWWA C219 – Bolted, Sleeve-Type Couplings for Plain-End Pipe.
- AAAA. AWWA C509 – Resilient-Seated Gate Valves for Water Supply Service.
- BBBB. AWWA C515 – Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Services.
- CCCC. AWWA C651 – Disinfecting Water Mains.
- DDDD. CISPI 301 – Standard Specification for Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications.
- EEEE. CISPI 310 – Specification for Coupling for Use in Connection with Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications.
- FFFF. ASSE 1003 Performance Requirements for Water Pressure Reducing Valves for Domestic Water Distribution Systems.
- GGGG. ASSE 1079 – Performance Requirements for Dielectric Pipe Unions.
- HHHH. UL 94 Standard for Tests for Flammability of Plastic Materials for Parts in Devices and Appliances.
- IIII. UL 1285 Standard for Safety Pipe and Couplings, PVC and PVCO for Underground Fire Service.
- JJJJ. NSF/ANSI 61 – Drinking Water System Components – Health Effects.
- KKKK. NSF/ANSI 372 – Drinking Water System Components – Lead Content.
- LLLL. Federal Specifications and Standards DD-G-541B – Glass (Laboratory).

## 1.5 QUALITY ASSURANCE

- A. Manufacturer: For each product specified, provide components by the same manufacturer throughout.
- B. Valves: Manufacturer's name, size, and pressure rating shall be cast or marked on valve body or handle.
- C. Piping shall be labeled along its entire length indicating size, class, material specification, manufacturer's name and **country of origin**.
- D. Foreign pipe, fittings or valves are unacceptable.
- E. All cast iron soil pipe and fittings shall be marked with the collective trademark of the Cast Iron Soil Pipe Institute (CISPI) and shall be listed by NSF International.
- F. Welding Materials and Procedures: Conform to ASME Code and applicable state labor regulations.
- G. Welders Certification: In accordance with ASME Section IX.

## 1.6 SUBMITTALS

- A. Submit under provisions of Division One.
- B. Submit product data under provisions of Division One.
- C. Include pipe materials, pipe fittings, valves, and accessories. Provide manufacturer's catalog information, product certifications, and **country of origin**. Indicate valve data and ratings.

## 1.7 PROJECT RECORD DOCUMENTS

- A. Submit under provisions of Division One.
- B. Record actual locations of valves.

## 1.8 OPERATION AND MAINTENANCE DATA

- A. Submit under provisions of Division One.
- B. Maintenance Data: Include installation instructions, spare parts lists, exploded assembly views.

## 1.9 QUALIFICATIONS

- A. Manufacturer: Company specializing in manufacturing the Products specified in this section with a minimum of 5 years documented experience and must be a domestic manufacturer.
- B. Installer: Company specializing in performing the work of this section with a minimum of 5 years documented experience.

## 1.10 DELIVERY, STORAGE, AND HANDLING

- A. DELIVERY: Deliver clearly labeled piping and valves to; and store, protect and handle products on site in accordance with the provisions of Division One.
- B. TIMING AND COORDINATION: Arrange for delivery of materials to allow for minimum storage time at the project site. Coordinate with the scheduled time of installation.
- C. ACCEPTANCE: Accept product on site in original factory packaging. Receive valves on site in shipping containers with labeling in place. Inspect for damage. Damaged valves shall not be acceptable.
- D. STORAGE: Store materials in a clean, dry location, protected from weather and damage.
- E. Provide temporary protective coating on cast iron and steel valves.
- F. Provide temporary end caps and closures on piping and fittings. Maintain in place until installation.
- G. Protect installed piping systems from entry of foreign materials by providing temporary covers, as completing sections of the work, and isolating parts of completed systems. Tape will not be allowed as an acceptable end cover.

#### 1.11 EXTRA MATERIALS

- A. Furnish under provisions of Division One.

#### 1.12 REGULATORY REQUIREMENTS

- A. Perform work in accordance with plumbing and building codes having jurisdiction.
- B. **No PVC pipe or fittings, or similar un-rated material, will be allowed in any areas where pipe is to penetrate a fire rated assembly or is to be installed in a return air plenum unless the entire length of all such piping is encased within a minimum two (2) hour fire rated enclosure.**
- C. Provide a water pressure regulating valve assembly at the service entry where incoming water supply pressure is greater than 70 psi.

### PART 2 - PRODUCTS

#### 2.1 SANITARY SOIL, WASTE AND VENT PIPING, BURIED BEYOND 5 FEET OUTSIDE OF BUILDING

- A. PVC Pipe: ASTM D1785/D2665 schedule 40 solid wall; installed per ASTM D2321.
  - 1. Fittings: PVC, ASTM D3311/D2665 drainage pattern, with bell and spigot ends. Furnished by the same manufacturer as pipe or approved equal.
  - 2. Joints: solvent weld with ASTM D2564 solvent cement, installed per the requirements of ASTM D2855.

#### 2.2 SANITARY SOIL, WASTE AND VENT PIPING, BURIED WITHIN 5 FEET OF BUILDING, BELOW GRADE

- A. PVC Pipe: ASTM D1785/D2665 schedule 40 solid wall; installed per ASTM D2321.

1. Fittings: PVC, ASTM D3311/D2665 drainage pattern, with bell and spigot ends. Furnished by the same manufacturer as pipe or approved equal.
2. Joints: solvent weld with ASTM D2564 solvent cement, clear, medium bodied, for sizes 3" and smaller and gray, heavy bodied, for sizes 4" and larger. Mating surfaces shall be prepared with ASTM F656 purple primer immediately prior to cement application.
3. Acceptable except for kitchen waste and vent systems including piping associated with the grease trap, which shall be as specified below.

## 2.3 SANITARY SOIL, WASTE AND VENT PIPING, WITHIN BUILDING, ABOVE GRADE

### A. Cast Iron Pipe: ASTM A74 service weight.

1. Fittings: Cast iron, ASTM A74 drainage pattern.
2. Joints: Hub and spigot, ASTM C564 neoprene, compression type gaskets or lead and oakum.

## 2.4 DOMESTIC WATER PIPE, BURIED WITHIN 5 FEET OF BUILDING EDGE, BELOW GRADE

### A. Copper Tubing: ASTM B88, Type K, soft annealed. Provide for pipe sizes up to and including 2-1/2".

1. Fittings: ASTM B16.22 wrought copper pressure fittings.
2. Joints shall be as follows:
  - a. **No joints shall be permitted for pipe sizes 2" and smaller. All such piping must be run continuous below slab on grade and brought up to no less than 12" above the finished floor before any joint is provided.**
  - b. For sizes larger than 2", joints between copper pipe and fittings shall be brazed and shall be made in accordance with all the applicable portions of ASTM B828, manufacturer's recommendations, and AWS requirements. Brazing filler metal shall be in accordance with AWS A5.8 and any required flux shall meet AWS A5.31, Type FB3-A or FB3-C.
2. Beginning at no closer than the 5'-0" mark from the building, all piping buried or in contact with concrete shall be provided with one of the following, which shall also extend to a minimum of 6" above the finished floor:
  - a. AWWA C209 cold-applied, integrated primer type, elastomeric adhesive, laminate polymeric tape coating, minimum 35 mil nominal thickness, in accordance with manufacturer's installation guidelines. Chase Construction Products Tapecoat H35 or approved equivalent.
  - b. Continuous polyethylene lining, minimum 60 mil nominal thickness.

### B. Stainless Steel Pre-Fabricated In-Building Riser (acceptable for sizes 2" through 10")

1. Corrosion resistant Type 304 stainless steel construction single, extended 90 degree fitting.
2. UL listed, FM approved and NFPA 24 compliant.
3. Lead free and NSF/ANSI 61 (372) certified.
4. Acceptable manufacturers:
  - a. Ames Fire & Waterworks Series IBR (4" through 10") and IBR2 (2", 2-1/2", and 3")
  - b. Zurn Wilkins Model WBR (4" through 10")
5. Note: For this application, the inlet joint for larger diameter (4" through 10") piping (which **shall not** be located below a building slab or foundation) can be rubber

gasketed push-on type, ANSI/AWWA C111/A21.11. Installation shall be in accordance with ANSI/AWWA C600.

6. Provide continuous polyethylene encasement for all piping buried or in contact with concrete in accordance with ANSI/AWWA C105/A21.5, beginning at no closer than the 5'-0" mark from the building and to a minimum of 6" above the finished floor.

## 2.5 DOMESTIC WATER PIPING, WITHIN BUILDING, BELOW GRADE

### A. Copper Tubing: ASTM B88, Type K, soft annealed.

1. No joints allowed below slab, run tubing continuous.
2. Provide AWWA C209 cold-applied, integrated primer type, elastomeric adhesive, laminate polymeric tape coating, minimum 35 mil nominal thickness, in accordance with manufacturer's installation guidelines, for all piping buried or in contact with concrete, to a minimum of 6" above finished floor. Chase Construction Products Tapecoat H35 or approved equivalent.
3. Applies to installations including services to island sinks and trap primer lines.

## 2.6 DOMESTIC WATER PIPING, WITHIN BUILDING, ABOVE GRADE

### A. Copper Tubing: ASTM B88, Type L, hard drawn.

1. Fittings: ASME B16.18, cast bronze or ASTM B16.22 wrought copper alloy solder joint pressure fittings.
2. Joints between copper pipe and fittings shall be made in accordance with ASTM B828 using ASTM B32 Alloy HB lead-free solder.
3. Fittings and joints for pipe sizes 1/2" through 4" may be mechanical press-connect system joints with ASME B16.51 lead-free copper bodied fittings with integral ethylene-propylene diene monomer rubber (EPDM) sealing gaskets. All fittings, couplings, and adapters shall be the product of a single system manufacturer and only that manufacturer's approved press tools, kits, and jaws shall be used.
  - a. EPDM o-rings shall be pre-installed and lubricated with ANSI/NSF 61 listed lubricant.
  - b. All installers of copper press-connect fittings shall be trained by the fitting manufacturer's appointed representative and carry such credentials for the duration of the project.
  - c. The fitting manufacturer's representative shall conduct periodic inspections of the installation and shall provide written reports of such inspections to the Contractor and Engineer, including any observed deviations from the manufacturer's recommended installation practices.
  - d. Acceptable system manufacturers: Viega or pre-approved equal.
4. Fittings and joints for pipe sizes 2-1/2 inch and larger shall be rolled groove type for copper tubing with all tools, couplings, adapters, fittings, gaskets, and **valves** the product of a single system manufacturer.
  - a. Fittings shall be cast bronze using lead-free alloys per ASTM B584 or copper wrought copper constructed to ASTM B75, compliant with NSF/ANSI 61 for potable water service applications, and meet ASTM F1548.
  - b. Couplings shall be epoxy/enamel (rust-inhibiting) coated ductile iron housings conforming to ASTM A536.
  - c. Gaskets shall be EPDM for potable water, meeting ASTM F1476, and NSF 61/NSF 372 certified for potable water service from 30 degrees to 180 degrees F.
  - d. Acceptable system manufacturers: Victaulic, Grinnell, Anvil Gruvlok.

## 2.7 ADAPTERS, TRANSITIONS, UNIONS, COUPLINGS, FLANGES, CONNECTORS

A. (Non-Acid Waste) Drainage Applications:

1. Provide approved listed adapter and transition fittings appropriate to the specific pipe transition and in accordance with code requirements.
2. For dissimilar piping above ground, provide stainless steel shielded, molded elastomeric couplings and adapters meeting ASTM C564 and ASTM C1460. Applies to installations including cast iron to PVC transitions immediately adjacent to building slabs on grade.
3. For dissimilar underground piping not below building slab, provide shear resistant .012" thick 300 series stainless steel shielded, PVC gasketed flexible couplings and adapters meeting ASTM D5926 and ASTM C1173. For direct-bury applications, provide AWWA C209 cold-applied, integrated primer type, elastomeric adhesive, laminate polymeric tape coating, minimum 35 mil nominal thickness, in accordance with manufacturer's installation guidelines, to completely wrap the shield, banding, and screws. Chase Construction Products Tapecoat H35 or approved equivalent.
4. Acceptable manufacturers:
  - a. Anaco-Husky/Cremco
  - b. Mission Rubber Company LLC
  - c. Fernco, Inc.
  - d. Fernco, Inc. Strong Back RC 1000 Series (underground piping, not below building slab; or readily accessible underground piping transitions in backwater valve pits, etc.)
5. Adapters, couplings, bushings for copper DWV pipe shall be cast bronze or wrought copper, ASME B16.23/B16.29.

B. Domestic Water Applications:

1. Provide joints between various materials with approved adapter and transition fittings appropriate to the specific pipe transition and in accordance with code requirements and the manufacturer's instructions.
2. For copper tube and pipe: adapters, bushings, plugs, caps, and couplings shall be wrought copper or cast bronze; flanges (minimum class 150) and unions shall be cast bronze. Provide with solder or threaded connections as necessary and as produced to applicable ASME standards B16.15, B16.18, B16.22, B16.24, B16.50, B1.20.1. All such appurtenances shall be for use in above ground potable water systems.
3. Above slab transitions for water service entries:
  - e. 100% fusion bonded epoxy coated ASTM A536 cast ductile iron construction coupling with acrylonitrile butadiene rubber (NBR) gaskets and EPDM insulating boot for water service. 5/8 inch high strength stainless steel bolts and nuts. Coupling shall meet AWWA C219. Romac Industries, Inc. IC501 or pre-approved equivalent.
  - f. 100% fusion bonded 14 mil epoxy coated coupling with ASTM A536 cast ductile iron rings. Complete with acrylonitrile butadiene rubber (NBR) gaskets and type 304 stainless steel bridge, spacers, nuts, and bolts. Coupling shall meet AWWA C219, NSF 61, and NSF 372. Krausz USA Hymax Grip Coupling Restraint or pre-approved equivalent.
4. Dielectric connections:
  - a. For pipe sizes 2 inch and smaller, provide lead-free dielectric unions, rated to 180 F at 250 psi and compliant to ASSE 1079.
  - b. For pipe sizes larger than 2 inches, provide lead-free dielectric flanged pipe fittings, rated to 180 F at 175 psi and meeting ASME B16.1.

- c. For grooved copper joining systems, provide grooved end dielectric transition fitting from system manufacturer, with virgin polypropylene internal lining, meeting NSF 61.

C. General:

1. Unions for ferrous pipe shall be ASTM B16.39 galvanized malleable iron, threaded, minimum pressure class 150.
2. Plugs and bushings for ferrous pipe shall be ASME B16.14 galvanized malleable iron, threaded.
3. Nipples for ferrous pipe shall be schedule 40, galvanized, ASTM A53 welded steel pipe nipples, threaded, meeting ASTM A733.
4. Couplings for ferrous pipe shall be galvanized steel, threaded, manufactured in accordance with ASTM A865.
5. Flanges for ferrous pipe shall be galvanized forged steel construction, either socket weld or slip-on weld type, minimum pressure class 150, manufactured to ASME B16.5.
6. Bolts, nuts, and gaskets for flanged connections shall be appropriate to the pipe material, fluid type, temperature, and pressure. 1/16" thick pre-formed neoprene, typical.
7. Provide flexible stainless steel connectors at pumps and other such equipment, in accordance with manufacturer's recommendations. Connectors shall have corrugated hose and braided 300 series stainless steel jacketing. Carbon steel flanged or grooved ends as appropriate. NSF 372 lead-free for all potable water applications. Metraflex Company or pre-approved equivalent.

2.8 BUTTERFLY VALVES (GROOVED)

- A. Grooved end, lead-free, copper alloy bodied, 300 psi butterfly valve with EPDM encapsulated ductile iron or aluminum bronze disc, EPDM seat/seal, stainless steel stem and trim, and extended neck. ANSI/NSF 61 certified for potable water systems. Lever handle operated for sizes 2-1/2" through 4" and manual gear operated with handwheel for sizes 6" and larger.
- B. Acceptable manufacturers:
  1. Victaulic 608N (cast brass body with aluminum bronze construction disc).
  2. Grinnell B680 (ASTM B584 bronze bodied).
  3. Anvil Gruvlok Series 6700 CTS (ASTM B584 bronze bodied).
- C. Applies only to domestic water system installations employing grooved copper joining systems, as specified elsewhere in this section. Manufacturers shall only be acceptable where their grooved systems are provided.

2.9 CHECK VALVES (BRONZE)

- A. ASTM B62/ASTM B584 bronze body and disc, minimum 200 psi (cold working pressure) Y-pattern horizontal swing type check valve with removable bronze bonnet, Type 300 series stainless steel nuts and hinge pin, and PTFE disc seat. Threaded connections. Certified lead-free to NSF 61/NSF 372 and suited to 180 degrees F.
- B. ASTM A126 cast iron bodied, (minimum) class 125 globe style spring loaded (silent) check valve with ASTM B584 bronze disc and seat. Flanged connections. Certified lead-free to NSF 61/NSF 372 and suited to 200 degrees F.



- C. Basis of design:
  - 1. NIBCO T-413-Y-LF (Y-pattern swing type) for sizes up through 2".
  - 2. NIBCO F-910-B-LF (globe style spring loaded type) for sizes 2-1/2" and larger.
- D. Acceptable alternate manufacturers:
  - 1. Apollo (for sizes up through 2")
- E. Applies to domestic water system installations including associated pump discharge lines. Valves shall be suited for installation in both horizontal lines and vertical lines with upward flow, in accordance with manufacturer's recommendations.

#### 2.10 PRESSURE REGULATING VALVES (PRV's)

- A. ASTM B62/ASTM B584 bronze bodied direct acting, ASSE 1003 single diaphragm type pressure regulating valve with removable bronze bonnet, in-line stainless steel strainer and spring, and FDA approved EPDM seat disc and Buna-N diaphragm. Threaded connections. Certified lead-free to NSF 61/NSF 372 and suited to 180 degrees F.
- B. NSF 61 epoxy coated ductile iron bodied pilot-operated globe style pressure regulating valve assembly. Complete with low-flow bypass and stainless steel, bronze, and copper trim and fittings. NSF 61 EPDM seat disc and diaphragm. Threaded or flanged connections. Suited to 180 degrees F.
- C. Basis of design:
  - 1. Apollo PRH-T-Y-LF (36HLF series) for direct acting valves, sizes up through 3".
  - 2. Apollo A127-LF series for pilot operated valves, sizes 1-1/4" through 4".
- D. Acceptable alternate manufacturers:
  - 1. Cla-Val
  - 2. Victaulic (pilot-operated valves)
- E. PRV's shall automatically reduce inlet pressure to a steady lower downstream pressure, regardless of changing flow rate. Provide complete with inlet strainer, inlet and outlet pressure gauges, isolation valves, and unions. Provide bypass line around assembly with normally closed valve.

#### 2.11 BALANCING VALVES

- A. Self-contained, fully automatic thermally actuated balancing valve shall continuously adjust flow to maintain the desired domestic hot water temperature within the branch line, regardless of system operating pressure. Valve shall modulate between open and closed position within a 10 degrees F range. Valve body and all internal components shall be constructed of stainless steel with major components constructed of Type 303 stainless. Rated for 200 psi maximum working pressure and no less than 250 degrees F maximum working temperature. Lead-free and ANSI/NSF 61 compliant. Threaded connections.
- B. Basis of design:
  - 1. ThermOmegaTech Circuit Solver, sizes 1/2" through 2". Provide a union and ball type shutoff valve on both sides of the balancing valve.

2. ThermOmegaTech Circuit Solver with integrated union (CSU) assembly, sizes 1/2" and 3/4". Balancing valve assembly shall come complete with union body and ball type shutoff valves on both sides.
- C. Applies to circulated domestic hot water system installations including multi-branch parallel piping circuits and single-loop piping circuits.
1. Provide balancing valve at end of **each** domestic hot water supply line (after last fixture served) just prior to the hot water return line, as indicated on Drawings and in accordance with manufacturer's installation recommendations.
  2. Provide a pipe tee or elbow with bushing as appropriate, 3/4" threaded thermowell, and bi-metal adjustable angle 3 inch dial thermometer upstream of each balancing valve. Thermowell stem length and thermometer temperature probe length to be suited for pipe size, insulation thickness, and to ensure clearance for maintenance access and easy viewing of thermometer. Terice bimetal/sensor, threaded-stepped shank thermowell (style 76) of lead-free brass (PBF) material. Terice Model B836 thermometer with 300 stainless steel case and stem, hermetically sealed, double strength glass windowed, aluminum white-faced dial, complete with external reset and 0 to 200 degrees F range. Thermowell and thermometer face to be oriented upright for readability.

## PART 3 - EXECUTION

### 3.1 EXCAVATION, BEDDING AND BACKFILL

- A. This section shall apply for the excavation, bedding, and backfill of all buried piping unless specifically noted otherwise. All work shall be coordinated with any job site subsurface drainage/dewatering and adjusted accordingly.
- B. Establish elevations of buried piping outside the building to ensure the following:
1. Not less than 2 feet of cover, or not less than maximum depth of frost penetration, whichever is the greater.
  2. For water lines intended for fire protection service, the depth of cover shall be:
    - a. Not less than 2'-6" in those locations where frost is not a factor.
    - b. Not less than 1'-0" below the frost line for the locality.
    - c. Not less than 3'-0" for piping under driveways.
    - d. Not less than 1'-0" below the bottom of the building foundation/footers.
    - e. In full compliance with the requirements of NFPA 13 and NFPA 24.
- C. Excavation:
1. Excavate trenches for underground piping to the required depths.
  2. The bottom of the trench or excavation shall be cut to a uniform grade.
  3. Should rock be encountered, excavate 6 inches below grade, fill with bedding material and tamp to existing density.
  4. Coordinate alignment of pipe trenches to avoid obstructions. Ensure that proposed routing of pipe will not interfere with building foundation before any trenching has begun. Should conflicts occur, contact Architect/Engineer before proceeding.
  5. Should any sleeving of the building foundation be required, this shall be provided as directed by the structural engineer of record AND in accordance with the prevailing code, but in no case shall the sleeve be any less than two (2) pipe sizes greater than the pipe it serves.
- D. Bedding and Backfill:

1. Backfill shall not be placed until the piping has been inspected, tested and approved. Complete backfill to the surface of natural ground or to the lines and grades indicated on drawings. **Provide 6 inch stabilized sand bed with 4 inch stabilized sand cover around each pipe.** Provide select fill up to finished surface or grade, unless indicated otherwise by project geotechnical report or specified otherwise in Division 02.
2. Compacting Backfill: Place material in uniform layers of 8 inches maximum, loose measure and compact to not less than 95% of maximum soil density as determined by ASTM D-698 Standard Proctor.
3. Restoration: Compact backfill, where trenching or excavation is required in improved areas such as pavements, walks and similar areas, to a condition equal to the adjacent undisturbed earth and restore surface of the area to the condition existing prior to trenching or excavating operation.
4. A clay fill "trench plug" extending 3 feet inside the building line and 5 feet outside the building line shall be placed to completely surround utility lines passing beneath the foundation and grade beam. The materials shall consist of on-site soils with a plasticity index (PI) between 30 and 40 percent compacted to at least 95 percent of the Standard Proctor and maximum dry density as determined by ASTM D-698.

E. Cement Stabilized Sand:

1. Materials:
  - a. Cement shall be Type I Portland cement conforming to ASTM C150.
  - b. Sand shall be clean, durable sand meeting grading requirements for fine aggregates of ASTM C33 and free of organic matter and deleterious substances.
  - c. Water shall be potable and free of oils, acids, alkalis, organic matter, or other deleterious substances, meeting requirements of ASTM C94.
2. Mixture:
  - a. Product shall consist of not less than 1.5 sacks of Portland cement per ton of dry sand.
  - b. Mixture shall contain sufficient water to hydrate the cement and be thoroughly mixed in a pugmill type mixer.

F. For water lines (including In-Building Risers) intended for fire protection service, provide joint restraints by way of concrete thrust blocks in accordance with the requirements of NFPA 13 and NFPA 24.

G. Aggressive Soil Conditions: Soil shall be considered aggressive and protection of buried metallic piping shall be provided as specified if any of the following situations exist:

1. Conditions are identified as such by the project geotechnical report or project geotechnical engineer.
2. The soil environment is a landfill area, swamp, marsh, polluted river bottom, cinder bed, or has alkaline soils.
3. A score of ten or higher is tallied when applying the soil assessment tool detailed in Appendix A of AWWA C105. An excerpt of this evaluation procedure is provided below for reference but is not intended as a substitute for the complete and latest Standard:

### Numerical Corrosivity Scale

| Soil Parameter       | Assigned Points |
|----------------------|-----------------|
| Resistivity (ohm-cm) |                 |

|                                |     |
|--------------------------------|-----|
| <700                           | 10  |
| 700 – 1,000                    | 8   |
| 1,000 – 1,200                  | 5   |
| 1,200 – 1,500                  | 2   |
| 1,500 – 2,000                  | 1   |
| >2,000                         | 0   |
| <b>pH</b>                      |     |
| 0 – 2                          | 5   |
| 2 – 4                          | 3   |
| 4 – 6.5                        | 0   |
| 6.5 – 7.5                      | 0   |
| 7.5 – 8.5                      | 0   |
| >8.5                           | 3   |
| <b>Redox Potential (mV)</b>    |     |
| >100                           | 0   |
| 50 – 100                       | 3.5 |
| 0 – 50                         | 4   |
| <0                             | 5   |
| <b>Sulfides</b>                |     |
| Positive                       | 3.5 |
| Trace                          | 2   |
| Negative                       | 0   |
| <b>Moisture</b>                |     |
| Poor drainage continuously wet | 2   |
| Fair drainage generally moist  | 1   |
| Good drainage generally dry    | 0   |

### 3.2 INSTALLATION

#### A. General requirements for piping:

1. Ream pipe and tube ends. Remove burrs. Bevel plain end ferrous pipe.
2. Remove any scale, oil and dirt, on inside and outside, before assembly.
3. Prepare piping connections to equipment with flanges or unions.
4. Confirm pipe placement, depth/elevation, and flow lines prior to any installation.

#### B. General requirements for valves:

1. Install valves with stems upright or horizontal, not inverted.
2. Valves shall be line-sized unless specifically noted otherwise.
3. Provide clearance for installation of insulation and access to valves and operable fittings. Valves installed beyond reasonable reach shall be provided with a chain operator.
4. Provide access doors where valves and operable fittings are not otherwise accessible. Access doors shall be of approved types set in locations pre-approved by submittal to the Architect.
5. Gate valves installed below grade shall be covered with an adjustable cast iron roadway box extended to grade. Cover shall be cast iron with 'water' cast on top of cover and shall be set flush to finished paving or 2" above finished earthen

grade. Box shall be supported from undisturbed soil or concrete base and shall not introduce any stress to piping under all traffic conditions.

- C. Install all materials in accordance with the manufacturer's published instructions.
- D. All exposed sewer and water pipe in toilet rooms or other finished areas of the building shall be chrome plated.
- E. Provide non-conducting dielectric connections wherever joining dissimilar metals.
- F. Route piping in an orderly manner, parallel and perpendicular to building column grid lines, unless indicated otherwise on drawings, and maintain gradients.
- G. Install piping to conserve building space and not conflict with other trades or interfere with intended use of space.
- H. Group piping whenever practical at common elevations.
- I. Install piping to allow for expansion and contraction without stressing pipe, joints, or connected equipment.
- J. Where pipe support members are welded to structural building framing, scrape, brush clean, and apply one coat of zinc rich primer to welding.
- K. Provide encasement for and support for utility meters in accordance with the requirements of utility companies.
- L. Prepare pipe, fittings, supports, and accessories not pre-finished, ready for finish painting.
- M. Maintain uniformity in the installation of piping materials and joining methods. Do not mix material types.
- N. Where connecting new underground sanitary, storm, or vent piping to existing piping of dissimilar material, provide suitable mechanical transition fittings complete with corrosion protection for metallic elements. Chase Construction Products Tapecoat H35 or approved equivalent and a final coat of coal tar to completely cover the transition.
- O. Solder joints shall be wiped clean at each joint, remove excess metal while molten and flux residue when cooled.
- P. Waste nipple from wall to tapped tee shall be schedule 40 threaded galvanized steel pipe or brass or copper with threaded adapter.
- Q. General requirements for cast iron piping installation:
  - 1. Install all pipe and fittings in accordance with published recommendations from the manufacturer and the Cast Iron Soil Pipe Institute (CISPI). Specific items referenced below are not intended as a substitute for the complete and latest recommendations.
  - 2. Install bell and spigot type pipe with bell end upstream.
  - 3. Above ground horizontal pipe (suspended) shall:
    - a. Be supported at no less than at every joint, and within 18" of the hub or coupling.
    - b. Be maintained in alignment. Sagging or grade reversal shall be unacceptable.

- c. Be supported at terminal ends of all runs or branches and at each change of direction or alignment.
    - d. Have all closet bends, traps, trap arms, and similar branches firmly secured.
    - e. Be braced to prevent movement or joint separation.
    - f. Be provided with suitable sway bracing (such as clamps, rods, and hardware) where pipe and fittings are suspended in excess of 18" by means of non-rigid hangers.
  - 4. Above ground vertical pipe shall:
    - a. Be secured at each stack base.
    - b. Be secured at each floor and riser clamps shall be provided on no greater than 15'-0" intervals.
    - c. Be adequately supported to keep the system (pipe and contents) in alignment.
  - 5. Provide seismic restraints in seismically active areas, whether specifically required by the prevailing code or not.
- R. For all underground non-metallic piping outside the building, provide minimum 14 AWG solid copper tracer wire (ASTM B-1, B-3) with high molecular weight polyethylene insulation (HMWPE) per ASTM D-1248. Wire shall be suited for direct bury applications to facilitate the detection and tracing of underground piping systems. THHN wire and other such nylon jacketing shall **not** be allowed. Insulation color shall be provided per the particular utility, in accordance with the American Public Works Association (APWA) uniform color code. Provide corrosion proof wire connectors with twist locking design and protective dielectric sealant. Copperhead Industries, LLC Snakebite or pre-approved equivalent. Tracer wire shall be placed in the same orientation as the installed pipe and laid six inches directly above the piping. One end of the tracer wire shall be brought aboveground at a building wall or riser for easy identification.
- S. **No PVC pipe or fittings, or similar un-rated material, will be allowed in any areas where pipe is to penetrate a fire rated assembly or is to be installed in a return air plenum unless the entire length of all such piping is encased within a minimum two (2) hour fire rated enclosure**
- T. Installations of underground thermoplastic piping systems shall be in strict conformity with the manufacturer's published instructions and the requirements of ASTM D2321 (gravity pipe) and ASTM D2774 (pressure pipe).
- U. Installation of above ground thermoplastic piping systems shall be in accordance with the manufacturer's recommendations. The specific items indicated below are not intended as a substitute for the complete and latest manufacturer's recommendations.
- 1. Hangers and supports shall not compress, distort, cut, or abrade the piping. Nor shall they force the pipe and fittings into position.
  - 2. Piping shall be supported at intervals sufficiently close to maintain pipe alignment and to prevent any sagging or grade reversal. System maximum operating temperature will determine support spacing.
  - 3. Piping shall be supported at all branch ends and at all changes of direction, as close as practical to the fitting to avoid introducing excessive torsional stresses into the system.
  - 4. Directly support (or if need be, immediately adjacent to) concentrated loads in the system, such as valves and other appurtenances.
  - 5. Allowances must be made for thermal expansion and contraction of the piping system where temperature fluctuations can reasonably be expected to produce such movement. Provide and place hangers accordingly so as not to restrict.

6. Plastic piping systems shall not be placed alongside steam or other high temperature pipe lines or other high temperature objects.
  7. Drainage piping shall be supported at trap arms as close as possible to the trap and all closet bends shall be supported and braced.
- V. Installation of solvent cement joints for PVC and CPVC piping shall be in strict conformity with the requirements of ASTM D2855 and manufacturer's published instructions.
- W. Provide approved heavy duty transition coupling at each transition from above ground cast iron pipe to underground PVC pipe as specified elsewhere in this section. Transition shall be made as close as possible to the floor for sanitary DWV piping systems and at test tee "minimum 12 inches A.F.F." for storm drainage piping. Support vertical cast iron pipe from floor anchors using riser clamp and galvanized all thread rod as specified in Section 22 05 29.
- X. All grooved system tools and components (couplings, adapters, fittings, gaskets, **valves**, and specialties) shall be the product of a single domestic system manufacturer.
- Y. Grooved pipe system manufacturer shall provide on-site training for contractor's field personnel by a factory trained representative in the proper use of grooving tools, application of groove, and product installation. Factory trained representative shall periodically visit the job site and inspect installation. Contractor shall remove and replace any improperly installed products at no additional cost to the owner.

### 3.3 APPLICATION

- A. Provide union downstream of all valves at equipment or apparatus connections.
- B. Provide male adapters each side of threaded valves in copper piped system. Sweat solder adapters to tube prior to make-up of threaded connections.
- C. Provide approved isolation valves for shut-off and to isolate all equipment items and distinct parts of systems. Isolation valves shall be provided for both hot and cold water in locations including, but not necessarily limited to, the following:
1. At each floor for each domestic water tap branching off from a vertical riser.
  2. At each domestic water branch line capped for future use.
  3. At each restroom or restroom group.
  4. At each hose bibb, wall hydrant, hose reel, and trap primer device (except for flush valve or tailpiece type trap primer devices).
  5. At each domestic water branch line within 24" of the corresponding main.
  6. At each plumbing fixture not otherwise served by a localized fixture group isolation valve.
- D. Each plumbing water rough-in stub out shall be fitted with a supply stop.
- E. Valves installed in insulated piping shall be fitted with extended lever operators of sufficient length to raise handle above the insulation jacket material. Where valve is used for throttling service, the valve handle shall be equipped with adjustable memory stop device.
- F. Provide non-slam type check valves on discharge lines from all water pumps. Install at a minimum length of 5 times the pipe diameter from the pump and in accordance with manufacturer's installation recommendations.

### 3.4 ERECTION TOLERANCES

- A. All gravity drainage lines in the building shall have 1/4 inch per foot fall where possible and not less than 1/8 inch per foot fall toward the main sewer. Pipe must be laid so that the slope will be uniform and continuous. Permission shall be secured from the Architect and Engineer before proceeding with any Work where existing conditions prevent the installation at the minimum grade specified.
- B. All vent and branch vent pipes shall be graded and connected as to drip back by gravity to the drainage pipe it serves. A slope of 1 inch per 40 feet will suffice for this requirement, subject to the approval of the local Authority Having Jurisdiction.
- C. Slope all horizontal water piping with uniform pitch of 1/8 inch per 10 feet to low points to allow for complete system drainage. For long runs, where constant pitch cannot be maintained, provide intermediate low points and rise up again from such locations. Slope horizontal branches back to mains or risers. Provide clearly identified supplementary drain valves where hose bibbs, hydrants, or sill cocks will not suffice for this requirement.

### 3.5 DISINFECTION OF DOMESTIC WATER PIPING SYSTEM

- A. Prior to starting work, all domestic water systems shall be complete, thoroughly flushed clean and free of all foreign matter or erection residue.
- B. Ensure PH of water to be treated is between 7.4 and 7.6 by adding alkali (caustic soda or soda ash) or acid (hydrochloric).
- C. On building side of the main shut off valve, provide a 3/4" connection through which chlorine can be introduced into the water piping
- D. Inject disinfectant, free chlorine in liquid, powder, tablet or gas form, in sufficient quantity to obtain 50 to 80 mg/L residual free chlorine solution throughout the entire domestic water piping systems.
- E. Bleed water from outlets as required to ensure complete distribution and test for disinfectant residual at a minimum 15 percent of total outlets.
- F. Maintain disinfectant in system for 24 hours.
- G. If final disinfectant residual tests less than 25 mg/L, repeat treatment.
- H. Flush disinfectant from system until residual equal to that of incoming water or 1.0 mg/L.
- I. Take samples no sooner than 24 hours after flushing, from 5 percent of outlets and from water entry, and analyze in accordance with AWWA C651.

### 3.6 SERVICE CONNECTIONS

- A. Provide new sanitary and storm sewer services connecting to existing building services or utility lines as shown on the drawings.
- B. Before commencing work, field verify invert elevations required for sewer connections, confirm inverts and ensure that these can be properly connected with slope for drainage and cover as required.
- C. Provide new domestic water service connecting to existing building services or utility lines as shown on plans. Assure connections are in compliance with requirements of the jurisdiction having authority.



- D. Extension of services to the building shall be fabricated from the same materials as the utility service lines or those materials specified herein.
- E. Should points of connection vary from those indicated on the drawings contractor shall properly allow for this in the actual connections field fabricated.

### 3.7 RODDING SEWERS

- A. All sanitary soil and waste lines, both in the building and out, shall be rodded out after completion of the installation.
- B. This Work shall be done, as part of the contract, to make certain that all lines are clear, and any obstruction that may be discovered shall be removed immediately. Rodding shall be accomplished by utilizing a rotary cutter, which shall be full size of pipe being cleaned.

### 3.8 TESTING OF PLUMBING PIPING SYSTEMS

- A. During the progress of the work and upon completion, tests shall be made as specified herein and as required by Authorities Having Jurisdiction, including Inspectors, Owner or Architect. The Architect or duly authorized Construction Inspector shall be notified in writing at least 2 working days prior to each test or other Specification requirement which requires action on the part of the Construction Inspector.
- B. Tests shall be conducted as part of this work and shall include all necessary instruments, equipment, apparatus, and service as required to perform the tests with qualified personnel. Submit proposed test procedures, recording forms, and test equipment for approval prior to the execution of testing.
- C. Tests shall be performed before piping of various systems have been covered or furred-in. For insulated piping systems testing shall be accomplished prior to the application of insulation.
- D. All piping systems shall be tested and proved absolutely tight for a period of not less than 24 hours. Tests shall be witnessed by the Architect or an authorized representative and pronounced satisfactory before pressure is removed or any water drawn off.
- E. Leaks, damage or defects discovered or resulting from test shall be repaired or replaced to a like new condition. Leaking pipe joints, or defective pipe, shall be removed and replaced with acceptable materials. Test shall be repeated after repairs are completed and shall continue until such time as the entire test period expires without the discovery of any leaks.
- F. Wherever conditions permit, each piping system shall thereafter be subjected to its normal operating pressure and temperature for a period of no less than five 5 days. During that period, it shall be kept under the most careful observation. The piping systems must demonstrate the propriety of their installation by remaining absolutely tight during this period.
- G. Domestic Water: Pressure test at one and one half times the normal working pressure or 125 psig, whichever is the greater, for 24 hours.
- H. Sanitary Soil, Waste and Vents and Storm Sewer:
  - 1. After the rough-in soil, waste and vent and other parts of the sanitary sewer including branch laterals have been set from the lowest level, at point of connection to existing utility lines, to above the floor line, all outlets shall be temporarily

plugged or capped, except as are required for testing as described herein. Ground work shall not permit the backfill of trenches to cover any joints until the completion of testing. Back fill shall be limited to mid sections of full joints of piping only. For pipe in ground the piping shall be readied as described herein and filled with water to a verifiable and visible level to 10' above the lowest portions of the system being tested.

2. On multi-level buildings only one floor level shall be tested at a time. Each floor shall be tested from a level below the structure of the floor, or the outlet of the building in the case of the lowest level, to a level of 12 inches above the floor immediately above the floor being tested, or the top of the highest vent in the case of the highest building level. The pipes for the level being tested shall be filled with water to a verifiable and visible level as described above and be allowed to remain so for 24 hours. If after 24 hours the level of the water has been lowered by leakage, the leaks must be found and stopped, and the water level shall again be raised to the level described, and the test repeated until, after a 24 hour retention period, there shall be no perceptible lowering of the water level in the system being tested.
3. Should the completion of these tests leave any reasonable question or doubt of the integrity of the installation, additional tests including peppermint smoke, or other measures shall be performed to demonstrate the reliability of these systems to the complete satisfaction of the Owner's duly authorized representative. Such tests shall be conducted and completed before any joints in plumbing are concealed or made inaccessible.

### 3.9 COMPLETE FUNCTIONING OF WORK

- A. All work reasonably implied as essential to the complete functioning of the systems shown on the Drawings and Specification shall be completed as part of the work of this Division, unless specifically stated otherwise. It is the intention of the Drawings and Specification to establish the type and function of systems but not to set forth each item essential to the functioning of any system. In case of doubt as to the work intended or in the event of amplification or clarification thereof, the Contractor shall call upon the Architect for Supplementary Instructions and Drawings, etc.

**END OF SECTION**

## **SECTION 22 11 19 - PLUMBING SPECIALTIES**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 22 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

- A. The scope of the work shall include the furnishing and complete installation of the specialties covered by this Section, with all appurtenances, ready for the Owner's use.
- B. Include the following work in addition to items normally part of this Section:
  - 1. Hose Bibbs and Hydrants
  - 2. Backflow Preventers
  - 3. Water Hammer Arresters
  - 4. Strainers and Filters
  - 5. Thermostatic Mixing Valves
  - 6. Floor Drains and Floor Sinks
  - 7. Cleanouts
  - 8. Trap Primers
  - 9. Interceptors and Separators

#### **1.3 RELATED WORK**

- A. Section 22 05 29 – Hangers and Support for Plumbing Piping and Equipment
- B. Section 22 10 00 – Plumbing Piping
- C. Section 22 30 00 – Plumbing Equipment
- D. Section 22 40 00 – Plumbing Fixtures

#### **1.4 REFERENCES**

- A. ANSI/ASSE 1010 – Performance Requirements for Water Hammer Arresters
- B. ANSI/ASSE 1011 – Performance Requirements for Hose Connection Vacuum Breakers
- C. ANSI/ASSE 1012 – Performance Requirements for Backflow Preventers with an Intermediate Atmospheric Vent

- D. ANSI/ASSE 1013 — Performance Requirements for Reduced Pressure Principle Backflow Preventers and Reduced Pressure Principle Fire Protection Backflow Preventers
- E. ANSI/ASSE 1015 – Performance Requirements for Double Check Backflow Prevention Assemblies and Double Check Fire Protection Backflow Prevention Assemblies
- F. ANSI/ASSE 1019 – Performance Requirements for Wall Hydrants with Backflow Protection and Freeze Resistance
- G. ANSI/ASSE 1057 – Performance Requirements for Freeze Resistant Sanitary Yard Hydrants with Backflow Protection
- H. ASME A112.6.3 – Floor Drains and Trench Drains
- I. ASME A112.6.7 – Sanitary Floor Sinks
- J. ASME A112.6.4 – Roof, Deck, and Balcony Drains
- K. ASME A112.14.1 – Backwater Valves
- L. ASME A112.14.3 – Grease Interceptors
- M. ASME/ANSI A112.26.1 – Water Hammer Arresters
- N. PDI WH-201 Water Hammer Arresters
- O. AWWA C506 – Standard for Backflow Prevention Devices – Reduced Pressure Principle and Double Check Valve Types
- P. AWWA C510 – Standard for Double Check Valve Backflow Prevention Assembly
- Q. ASSE 1069 – Performance Requirements for Automatic Temperature Control Mixing Valves
- R. ASSE 1070 – Performance Requirements for Water Temperature Limiting Devices
- S. PDI G-101 – Testing and Rating Procedure for Hydro Mechanical Grease Interceptors
- T. NSF/ANSI Standard 61: Drinking Water System Components – Health Effects
- U. NSF/ANSI 372: Drinking Water System Components – Lead Content

#### 1.5 QUALITY ASSURANCE

- A. Manufacturer: For each product specified, provide components by the same manufacturer throughout.

#### 1.6 SUBMITTALS

- A. Submit under provisions of Division One.
- B. Submit shop drawings and product data under provisions of Division One.
- C. Include component sizes, rough-in requirements, service sizes, and finishes.

- D. Manufacturer's Installation Instructions: Indicate assembly and support requirements.

#### 1.7 PROJECT RECORD DOCUMENTS

- A. Submit under provisions of Division One.
- B. Record actual locations of equipment and backflow preventers.

#### 1.8 OPERATION AND MAINTENANCE DATA

- A. Submit under provisions of Division One.
- B. Operation Data: Indicate frequency of treatment required for interceptors and separators.
- C. Maintenance Data: Include installation instructions, spare parts lists, exploded assembly views.

#### 1.9 DELIVERY, STORAGE, AND HANDLING

- A. DELIVERY: Deliver clearly labeled specialties to; and store, protect and handle products on site in accordance with the provisions of Division One.
- B. TIMING AND COORDINATION: Arrange for delivery of materials to allow for minimum storage time at the project site. Coordinate with the scheduled time of installation.
- C. ACCEPTANCE: Accept specialties on site in original factory packaging. Inspect for damage. Damaged specialties shall not be acceptable.
- D. STORAGE: Store materials in a clean, dry location, protected from weather and damage.

#### 1.10 EXTRA MATERIALS

- A. Furnish under provisions of Division One.
- B. Provide two loose keys for hose bibbs and hydrants and spare hose end vacuum breakers.

#### 1.11 REGULATORY REQUIREMENTS

- A. Conform to applicable codes for the provision and installation of all required backflow prevention devices.
- B. Provide certificate of compliance from authority having jurisdiction indicating approval of installation of backflow prevention devices.
- C. Provide backflow prevention assembly test and maintenance report for all devices. A printed and signed form by the licensed tester that performed the work shall be provided both to the Owner and to the Public Water System in accordance with TCEQ (Texas Commission on Environmental Quality) requirements.

### PART 2 - PRODUCTS

#### 2.1 HOSE BIBBS/HYDRANTS

- A. Hose Bibb: ANSI/ASSE 1011; bronze or brass construction, replaceable hexagonal disc, hose thread spout, chrome plated where exposed, with lockshield and removable key, integral vacuum breaker.
- B. Wall Hydrant: ANSI/ASSE 1019; non-freeze, self-draining type with chrome plated lockable recessed box hose thread spout, lockshield and removable key, and vacuum breaker.
- C. Floor Hydrant: ANSI/ASSE 1019; chrome plated lockable recessed box, hose thread spout, lockshield and removable key, and vacuum breaker.
- D. Roof/Post Hydrant; ANSI/ASSE 1057; bronze non-freeze post hydrant with cast iron casing guard, vacuum breaker, and removable tee handle key.
- E. ACCEPTABLE MANUFACTURERS:
  - 1. J.R. Smith
  - 2. Zurn
  - 3. Mifab
  - 4. Watts
  - 5. Wade
  - 6. Josam
  - 7. Chicago
  - 8. Woodford

## 2.2 RECESSED VALVE BOX

- A. Washing Machine: Pre-formed galvanized rough-in box with brass long shank valves with wheel handles, threaded drain fitting for waste, and matching secured faceplate.
- B. Refrigerator: Pre-formed galvanized rough-in box with brass long shank valve with wheel or quarter-turn handle and matching secured faceplate.
- C. ACCEPTABLE MANUFACTURERS:
  - 1. Guy Gray
  - 2. Mifab
  - 3. Sioux Chief
  - 4. Oatey

## 2.3 BACKFLOW PREVENTERS

- A. General Requirements: All assemblies shall be suited for the system anticipated working pressure and temperature as well as the intended orientation (vertical or horizontal) of the installation. All assemblies shall be lead-free per NSF/ANSI 372, unless specifically noted otherwise on the Drawings.
- B. Reduced Pressure Backflow Preventers: ANSI/ASSE 1013; Bronze or FDA approved epoxy coated cast iron body with corrosion resistant internal parts and stainless steel springs; two independently operating, spring loaded check valves; intermediate internal pressure intermediate relief valve with water outlet; test cocks and isolation valves.
- C. Double Check Valve Assemblies: ANSI/ASSE 1015; Cast copper alloy or FDA approved epoxy coated cast iron body with corrosion resistant internal parts and stainless steel springs; two independently operating check valves, test cocks and isolation valves.

- D. Dual Check Valve with Intermediate Atmospheric Vent: ANSI/ASSE 1012; Brass body with corrosion resistant internal parts and stainless steel springs; two independently operating check valves with intermediate atmospheric vent.

- E. ACCEPTABLE MANUFACTURERS:

- 1. Watts
  - 2. Wilkins
  - 3. Ames
  - 4. Febco
  - 5. Conbraco

## 2.4 WATER HAMMER ARRESTERS

- A. Engineered water hammer arresters: ASSE 1010 listed, lead-free, pre-charged, permanently sealed, maintenance- free, suited for concealed installation, with a working temperature range of 33 to no less than 212 degrees F and a maximum working pressure of no less than 250 psi during pressure surges. Stainless steel or copper body construction. Shall be sized and located in accordance with Plumbing Drainage Institute standard PDI-WH 201.

- B. ACCEPTABLE MANUFACTURERS:

- 1. J.R. Smith
  - 2. Zurn
  - 3. Mifab
  - 4. Wade
  - 5. Josam
  - 6. P.P.P.
  - 7. Sioux Chief

## 2.5 THERMOSTATIC MIXING VALVES

- A. Provide thermostatic mixing valves in accordance with manufacturer's recommendations and as indicated and scheduled on Drawings.
  - 1. Unless scheduled otherwise, all units other than under-counter point of use units shall be provided complete in lockable cabinet of 16 gage (1.5 mm) prime coated steel when located in finished areas.
  - 2. All under-counter point of use units shall be provided complete with integral checks and dual stainless steel strainers on inlets for protection against fouling.

- B. ACCEPTABLE MANUFACTURERS:

- 1. Bradley
  - 2. Powers
  - 3. Symmons
  - 4. Acorn

## 2.6 FLOOR DRAINS AND FLOOR SINKS

- A. Provide floor drains and floor sinks in accordance with manufacturer's recommendations, as appropriate for floor construction, and as indicated and scheduled on Drawings.
- B. Provide clamping devices for all drains in membrane floor areas.
- C. ACCEPTABLE MANUFACTURERS:

1. J.R. Smith
2. Zurn
3. Mifab
4. Watts
5. Wade
6. Josam

- D. Provide drains of suitable and compatible material for specialized piping systems conveying acid waste.

## 2.7 CLEANOUTS

- A. General: Provide cleanouts as indicated and scheduled on Drawings and also as required by the prevailing code, whether shown on the Drawings or not.

- B. Construction: All cleanouts shall have tapered bronze plugs.

- C. Provide clamping devices for all cleanouts in membrane floor areas.

- D. Provide cleanouts of suitable and compatible material for specialized piping systems conveying acid waste.

- E. Types:

1. Finished floor cleanouts: Provide cast iron body, with adjustable floor level assembly, and round nickel bronze scoriated top.
2. Resilient or tile finished floor cleanouts: Provide cast iron body, with adjustable floor level assembly, and round nickel-bronze top with gasketed water tight cover and depressed top to receive flooring finish material.
3. Interior finished wall cleanouts: Provide cast iron tee body or cleanout ferrule as required for wall construction and provide counter-sunk bronze plug with stainless steel access cover and securing screw(s).
4. Interior unfinished accessible cleanouts: Calked or threaded type. Provide bolted stack cleanouts on vertical rainwater leaders.

- F. ACCEPTABLE MANUFACTURERS:

1. J.R. Smith
2. Zurn
3. Mifab
4. Watts
5. Wade
6. Josam

## 2.8 TRAP PRIMERS

- A. General: Provide trap primers as indicated and scheduled on Drawings and in accordance with manufacturer's recommendations.

- B. ACCEPTABLE MANUFACTURERS:

1. J.R. Smith
2. Zurn
3. Mifab
4. Watts



5. Wade
6. Josam
7. P.P.P.
8. Sioux Chief

## 2.9 INTERCEPTORS

### A. Grease Interceptor – Precast Concrete

1. Construction: Refer to detail(s) on Drawings.
2. Provide buried installation in accordance with manufacturer's requirements. Provide complete with traffic rated access covers at finished elevation, complete with lid liners for odor control.
3. Unit Rating: Refer to Drawings for connection sizes and grease capacity.
4. ACCEPTABLE MANUFACTURERS:
  - a. Park-USA
  - b. Old Castle Precast

### B. Oil Interceptor – Precast Concrete

1. Construction: Refer to detail(s) on Drawings.
2. Provide buried installation in accordance with manufacturer's requirements. Provide complete with traffic rated access covers at finished elevation, complete with lid liners for odor control.
3. Unit Rating: Refer to Drawings for flow rating, connection sizes, and oil capacity.
4. ACCEPTABLE MANUFACTURERS:
  - a. Park-USA
  - b. Old Castle Precast

## 2.10 BACKWATER VALVES

- A. ASME A112.14.1; Lacquered cast iron body and cover, brass valve, access cover, extension sleeve as required and cover at finished elevation.

### B. ACCEPTABLE MANUFACTURERS:

1. J.R. Smith
2. Zurn
3. Mifab
4. Watts
5. Wade
6. Josam

## PART 3 - EXECUTION

### 3.1 INSTALLATION AND APPLICATION

- A. Install specialties in accordance with manufacturer's instructions to provide intended performance.
- B. The contractor shall provide water hammer arresters as shown on Drawings and also in accordance with PDI Standard WH-201, whether shown on Drawings or not. Water hammer arresters shall be PDI certified and sized and placed as recommended by manufacturer. Provide above lay-in ceiling, within chase or wall or above solid ceiling

complete with access panel, or otherwise accessible location complete with isolation valve to facilitate replacement.

- C. The use of air chambers for the control of water hammer shock shall not be acceptable.
- D. Provide strainers at all backflow preventers.
- E. Contractor shall certify all newly installed backflow preventers and provide proof of certification to the Owner.
- F. Pipe relief line from backflow preventer via manufacturer's air gap assembly, full size to nearest drain. Such routing shall not pose a trip hazard.
- G. All backflow preventers shall be securely supported with wall supports and/or pipe stands as appropriate for the size and weight of the unit and shall be installed with sufficient access and clearance for testing and maintenance. Unless specifically noted otherwise on Drawings, all backflow preventers shall be installed at 48"-60" above finished floor.
- H. Extend cleanouts to finished floor or wall surface. Lubricate threaded cleanout plugs with mixture of graphite and linseed oil. Ensure clearance at cleanouts for rodding of drainage system.
- I. All cleanouts outside of building on grade shall be set in an 18" x 18" x 4" thick concrete pad, flush with final grade/paving.
- J. All cleanouts shall be the same nominal size as the pipe they serve, up to 4 inches. For pipes larger than 4 inches, provide a 4 inch cleanout.
- K. Coordinate with casework to ensure that all interceptors are readily accessible and removable for servicing and cleaning.
- L. Coordinate with casework to ensure that all point of use interceptors do not interfere with required accessibility requirements. Provide offset(s) as required and in accordance with code.
- M. Provide approved sampling well downstream of centralized interceptors and separators.

#### **END OF SECTION**

## **SECTION 22 11 21 - NATURAL GAS PIPING SYSTEMS**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 22 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

- A. Scope of the Work shall include the furnishing, complete installation and testing of the gas piping system, with all metering, valves, piping and auxiliaries, ready for owner's use.
- B. Coordinate with the gas company and pay all fees and permits required for a complete and operating gas service to the project.

### **PART 2 - PRODUCTS**

- 2.1 All gas piping above ground shall be Schedule 40 black steel as manufactured by National Tube, Republic, Youngstown, or approved equal domestic manufacturer.
- 2.2 All gas piping larger than 2" shall be of welded construction. Screwed fittings will only be permitted for size 2" and smaller. Unions and valves will not be permitted above furred ceiling areas or in walls or chases.
- 2.3 All pipe fittings shall be of materials as follows:
  - A. All welding fittings shall be factory-made and shall be full line size, for each tee, branch, elbow, etc., with reducers after fittings, if required.
  - B. All screwed fittings shall be Crane, or approved equal, Class 150 malleable iron. Screw joints shall be made up with graphite and oil or Teflon tape. Screwed threads shall be in accordance with American Pipe Thread Standards.
  - C. All piping and fittings shall be from a domestic manufacturer.
- 2.4 All underground gas piping with 5 pound working pressure or less shall be as follows:
  - A. The pipe shall be yellow polyethylene with socket heat fusion joints and fittings. Pipe sizes 1-1/2" and 2" shall be SDR 11, (PE 2406) and pipe sizes 3" and 4" shall be SDR 11.5 (PE 2406).
  - B. All socket heat fusion fittings shall be D.O.T. approved and meet ASTM D-2513 and ANSI B31.8 codes.
  - C. All gas valves shall be polyethylene ball type, doubled union, rated for natural gas use. All valves shall be placed in a cast-iron valve box of an adequate size for accessibility and maintenance.
  - D. All transition meter risers shall be D.O.T. approved anode-less service type, fusion coupled and PE 2406 rated.

- E. The contractor shall take thermal expansion under consideration during installation. The contractor shall follow all requirements set by the manufacturer to protect the system from damage due to thermal expansion.
  - F. The contractor shall provide detector tape approximately 12" above all gas piping.
  - G. Wrap pipe with 18 gauge minimum copper tracer wire.
- 2.5 Gas piping installed in unventilated spaces shall be routed in properly vented continuous sleeve where required by the building code.
- 2.6 Gas valves shall be U.L. listed as follows:
- A. Ball Valves: Nibco T585-70-UL for ¼" to 1" and T580-70-UL for 1-¼" to 3".
  - B. Plug Valves: DeZurick Series 425 or 435 Eccentric valves with RS 49 plug seals.
- 2.7 Gas pressure regulators shall be capable of reducing 75 psi pressure gas to 0.5 psi gas at capacities required by Gas Demand. Install per A.G.A. Bulletin 90. Regulators shall be as manufactured by Rockwell, Fisher-Governor or approved equal.
- 2.8 All gas regulators located inside the building shall be vented to atmosphere with schedule 40 black steel pipe. This includes all regulators provided with mechanical and plumbing equipment and all other regulators provided under this contract. Vent piping shall be the full size of regulatory port opening, or as recommended by regulator manufacturer, and shall run independent of any other regulator vent through to point of termination.

### PART 3 - EXECUTION

- 3.1 All piping shall be installed in accordance with the manufacturer's recommendations and printed installation instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Provide all items required as per manufacturer's requirements.
- 3.3 All underground gas piping shall be laid on 6" of wet compact banksand approximately 24" below grade. Backfill trench with wet compacted banksand to 6" above pipe. The remainder of backfill shall be selected backfill and shall meet all compaction requirements set forth by the general trenching and backfill requirements.
- 3.4 Provide lever handle gas valve, drip leg and union to each piece of equipment and where indicated.
- 3.5 All gas lines entering building shall be valved on the exterior of the building above grade.

### PART 4 - TESTING

#### 4.1 TESTING OF GAS PIPING SYSTEMS

- A. All gas system testing shall be in compliance with local codes or as required in NFPA 54 National Fuel Gas Code whichever is the more stringent requirement.
- B. All work shall be performed by a Journeyman Plumber holding current State and local licenses.

- C. All tests shall be accomplished during normal working hours and after having given due notification to building owner, construction manager or designee, of tests to be performed. All tests shall be performed in the presence of and witnessed by the building owners representative or designee
- D. All gas system piping shall be subjected to a pneumatic test pressure of 60 psig for not less than 2 hours upon completion of all rough-in work and prior to covering. While the systems are subjected to this air pressure test, all joints shall have a soapy water solution applied and shall be observed for leaks. During test period there shall be no perceptible drop in test gage pressure
- E. A final test shall be performed after all portions of the piping system are completely installed and covered. The entire system shall be tested, with all system outlets plugged or capped, before any equipment or appliances are connected to the piping.
  - 1. Final test shall be with mercury, measured with a manometer or slope gage. Test pressures shall in no case be less than one and one half times the normal operating pressure or as listed below; which ever is the greater:
    - a. 10.5 inches mercury (5 psig) for 4 ounce system.
    - b. 21.0 inches mercury (10 psig) for 8 ounce system.
  - 2. Tests shall be for a period of not less than 30 minutes and shall prove absolutely tight, showing no perceptible drop, for the entire test period.
- F. Purge air from test piping before connecting equipment or appliances. Purge air to outdoors or to ventilated space of sufficient volume to prevent accumulation of flammable mixtures.

**END OF SECTION**

## **SECTION 22 30 00 - PLUMBING EQUIPMENT**

### **PART 1 - GENERAL**

#### **1.1 SECTION INCLUDES**

- A. Water Heaters.
- B. In-line circulator pumps.

#### **1.2 RELATED SECTIONS**

- A. Section 22 05 29 – Hangers and Support for Plumbing Piping and Equipment.
- B. Section 22 05 48 – Vibration and Seismic Controls for Plumbing Equipment.
- C. Section 22 10 00 - Plumbing Piping.
- D. Section 22 11 19 - Plumbing Specialties.
- E. Section 26 05 19 - Wire, Cable, and Related Materials.

#### **1.3 REFERENCES**

- A. ANSI/ASHRAE 90A - Energy Conservation in New Building Design.
- B. ASME Section VIII - Pressure Vessels; Boiler and Pressure Vessel Codes.
- C. ANSI/NFPA 54 - National Fuel Gas Code.
- D. ANSI/NFPA 70 - National Electrical Code.
- E. ANSI/UL 1453 - Electric Booster and Commercial Storage Tank Water Heaters.

#### **1.4 SUBMITTALS**

- A. Submit under provisions of Division One.
- B. Shop Drawings:
  - 1. Include water heater dimensions. size of tapings, and performance data.
  - 2. Include dimensions of tanks, tank lining methods, anchors, attachments, lifting points, tapings, and drains.
- C. Product Data:
  - 1. Include dimension drawings of water heaters indicating components and connections to other equipment and piping.
  - 2. Indicate pump type, capacity, power requirements, and affected adjacent construction.
  - 3. Submit certified pump curves showing pump performance characteristics with pump and system operating point plotted. Include NPSH curve when applicable.
  - 4. Provide electrical characteristics and connection requirements.
- D. Manufacturer's Installation Instructions.

## 1.5 OPERATION AND MAINTENANCE DATA

- A. Submit under provisions of Division 22.
- B. Include operation, maintenance, and inspection data, replacement part numbers and availability, and service depot location and telephone number.

## 1.6 QUALITY ASSURANCE

- A. Perform Work in accordance with authorities having jurisdiction.
- B. Provide pumps with manufacturer's name, model number, and rating/capacity identified.
- C. Ensure products and installation of specified products are in conformance with recommendations and requirements of the following organizations:
  - 1. American Gas Association (AGA).
  - 2. National Sanitation Foundation (NSF).
  - 3. American Society of Mechanical Engineers (ASME).
  - 4. National Board of Boiler and Pressure Vessel Inspectors (NBBPVI).
  - 5. National Electrical Manufacturers' Association (NEMA).
  - 6. Underwriters Laboratories (UL).
  - 7. American Society of Plumbing Engineers (ASPE)
- D. Ensure pumps operate at specified system fluid temperatures without vapor binding and cavitation, are non-overloading in parallel or individual operation, operate within 25 percent of midpoint of published maximum efficiency curve.

## 1.7 REGULATORY REQUIREMENTS

- A. Conform to AGA NSF ANSI/NFPA 54 ANSI/NFPA 70 ANSI/UL 1453 requirements for water heaters.
- B. Conform to ASME Section VIII for manufacture of pressure vessels for heat exchangers.

## 1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, protect and handle products to site under provisions of Section Division One.
- B. Provide temporary inlet and outlet caps. Maintain caps in place until installation.

## 1.9 WARRANTY

- A. Provide five year warranty under provisions of Division One.
- B. Warranty: Include coverage of domestic water heaters, water storage tanks, and packaged water heating systems.

## 1.10 EXTRA MATERIALS

- A. Furnish under provisions of Division One.

- B. Provide two sets of electric heater elements.

#### 1.11 OPERATIONS PERSONNEL TRAINING

- A. Provide a training session for the owner's operations personnel. Training session shall be performed by a qualified person who is knowledgeable in the subject system/equipment. Submit a training agenda two weeks prior to the proposed training session for review and approval. Training session shall include at the minimum:
  - 1. Purpose of equipment.
  - 2. Principle of how the equipment works
  - 3. Important parts and assemblies
  - 4. How the equipment achieves its purpose and necessary operating conditions
  - 5. Most likely failure modes, causes and corrections
  - 6. On site demonstration

### PART 2 - PRODUCTS

#### 2.1 COMMERCIAL ELECTRIC WATER HEATERS

- A. Manufacturers:
  - 1. A.O. Smith
  - 2. Other acceptable manufacturers offering equivalent products.
    - a. State
    - b. Rheem.
    - c. Bradford White.
    - d. Bock.
- B. Type: Factory-assembled and wired, electric, vertical storage.
- C. Tank: Glass lined welded steel; 4 inch diameter inspection port (when applicable), thermally insulated with minimum 2 inches glass fiber encased in corrosion-resistant steel jacket; baked-on enamel finish.
- D. Controls: Automatic immersion water thermostat; externally adjustable temperature range from 60 to 180 degrees F, flanged or screw-in nichrome elements, high temperature limit thermostat.
- E. Accessories: Brass water connections and dip tube, drain valve, high-density magnesium anode, and ASME rated temperature and pressure relief valve.
- F. Provide training per 1.11.

#### 2.2 IN-LINE CIRCULATOR PUMPS

- A. Manufacturers:
  - 1. Bell & Gossett.
  - 2. Other acceptable manufacturers offering equivalent products.
    - a. TACO.
    - b. Grundfos.
- B. Casing: Bronze, rated for 125 psig working pressure.



- C. Impeller: Bronze.
- D. Shaft: Alloy steel with integral thrust collar and two oil lubricated bronze sleeve bearings.
- E. Seal: Carbon rotating against a stationary ceramic seat.
- F. Drive: Flexible coupling.

### PART 3 - EXECUTION

#### 3.1 WATER HEATER INSTALLATION

- A. Install water heaters in accordance with manufacturer's instructions and to AGA NSF ANSI/NFPA 54 UL requirements.
- B. Coordinate with plumbing piping and related work to achieve operating system.

#### 3.2 PUMP INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Provide line sized isolating valve and strainer on suction and line sized soft seated check valve and balancing valve on discharge.

**END OF SECTION**

## **SECTION 22 40 00 - PLUMBING FIXTURES**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 22 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

- A. The scope of the work shall include the furnishing and complete installation of the fixtures covered by this Section, with all appurtenances, ready for the Owner's use.
- B. Include the following work in addition to items normally part of this Section:
  - 1. Plumbing Fixtures
  - 2. Fixture Carriers
  - 3. Faucets, Supplies, and Trim
  - 4. Flushometers

#### **1.3 RELATED WORK**

- A. Section 22 05 29 – Hangers and Support for Plumbing Piping and Equipment
- B. Section 22 10 00 – Plumbing Piping
- C. Section 22 11 19 – Plumbing Specialties
- D. Section 22 30 00 – Plumbing Equipment

#### **1.4 REFERENCES**

- A. ASME A112.4.3 – Plastic Fittings for Connecting Water Closets to the Sanitary Drainage System
- B. ASME A112.6.1M – Floor Affixed Supports for Off-the-Floor Plumbing Fixtures for Public Use
- C. ASME A112.18.1 – Plumbing Supply Fittings
- D. ASME A112.18.2 – Plumbing Waste Fittings
- E. ASME A112.18.9 – Protectors/Insulators for Exposed Waste and Supplies on Accessible Fixtures
- F. ASME A112.19.1 – Enameled Cast Iron and Enameled Steel Plumbing Fixtures
- G. ASME A112.19.2 – Ceramic Plumbing Fixtures

- H. ASME A112.19.3 – Stainless Steel Plumbing Fixtures
- I. ASME A112.19.7 – Hydromassage Bathtub Systems
- J. NSF/ANSI 61 – Drinking Water System Components – Health Effects
- K. ANSI Z358.1 – Emergency Eyewash and Shower Equipment
- L. ASSE 1016 – Performance Requirements for Individual Thermostatic, Pressure Balancing, and Combination Pressure Balancing and Thermostatic Control Valves for Individual Fixture Fittings.
- M. ASSE 1037 – Performance Requirements for Pressurized Flushing Devices for Plumbing Fixtures
- N. ADA (Americans with Disabilities Act)
- O. TAS (Texas Accessibility Standards)

#### 1.5 QUALITY ASSURANCE

- A. Manufacturer: For each product specified, provide components by the same manufacturer throughout.
- B. Warranty: Warrant the work specified herein for one year against becoming unserviceable or causing an objectionable appearance resulting from defective or non-conforming materials and workmanship.
- C. Defects shall include, but not necessarily be limited to, the following:
  - 1. Noisy operation.
  - 2. Noticeable deterioration of finish.
  - 3. Leakage of water.

#### 1.6 SUBMITTALS

- A. Submit under provisions of Division One.
- B. Submit product data under provisions of Division One.
- C. Include component sizes, rough-in requirements, service sizes, finishes, materials, dimensions, performance information, and accessories.
- D. Manufacturer's Installation Instructions: Indicate assembly and support requirements.

#### 1.7 OPERATION AND MAINTENANCE DATA

- A. Submit under provisions of Division One.
- B. Provide pre-printed operating and maintenance instructions for each item specified. Instruct and demonstrate the proper operation and maintenance to the Owner's designated representative.

#### 1.8 DELIVERY, STORAGE, AND HANDLING

- A. DELIVERY: Deliver clearly labeled specialties to; and store, protect and handle products on site in accordance with the provisions of Division One.
- B. TIMING AND COORDINATION: Arrange for delivery of materials to allow for minimum storage time at the project site. Coordinate with the scheduled time of installation.
- C. ACCEPTANCE: Accept specialties on site in original factory packaging. Inspect for damage. Damaged specialties shall not be acceptable.
- D. STORAGE: Store materials in a clean, dry location, protected from weather and damage.

#### 1.9 FIELD MEASUREMENTS

- A. Verify that field measurements are as indicated on the Contract Documents.
- B. Confirm and field coordinate that millwork is constructed with adequate provisions for the installation of counter top lavatories and sinks.

### PART 2 - PRODUCTS

#### 2.1 PLUMBING FIXTURES

- A. GENERAL: Provide plumbing fixtures in accordance with manufacturer's recommendations and as indicated and scheduled on Drawings. Acceptable manufacturers of each fixture type are as indicated below.
  - 1. Provide floor-affixed fixture carriers as appropriate for all wall-hung plumbing fixtures unless specifically noted otherwise.
  - 2. Fixture drilling shall match faucet spread and match any related trim and accessories.
- B. WATER CLOSETS, URINALS, LAVATORIES (Vitreous China)
  - 1. American Standard
  - 2. Kohler
  - 3. Zurn
  - 4. Sloan
  - 5. Toto
- C. SINKS – COUNTER MOUNTED (Stainless Steel)
  - 1. Elkay
  - 2. Just
  - 3. Moen Commercial
- D. SHOWER
  - 1. Acorn
  - 2. Bradley
  - 3. Symmons Commercial
- E. MOP SINKS
  - 1. Stern-Williams
  - 2. Fiat
  - 3. E.L. Mustee & Sons

F. EMERGENCY SAFETY FIXTURES

1. Bradley
2. Guardian
3. Chicago
4. Haws
5. Speakman

G. DRINKING FOUNTAINS AND WATER COOLERS

1. Halsey Taylor
2. Elkay
3. Haws
4. Oasis

2.2 FAUCETS, SUPPLIES, AND TRIM

A. GENERAL: Provide faucets, supplies, and trim in accordance with manufacturer's recommendations, as appropriate for fixtures to be served, and as indicated and scheduled on Drawings. Acceptable manufacturers for each type of appurtenance are as indicated below.

1. Flushometer flush rate shall match gallon-per-flush criteria of fixtures served.
2. Strainers shall be heavy cast brass chrome plated with matching grid type strainer, with or without overflow as required, 17 gauge seamless brass tailpiece of length determined by installation requirements. Provide complete with washers and brass locknut.
3. P-traps shall be 17 gauge seamless chrome plated brass, adjustable type. Provide complete with cleanout plug, chrome plated brass slip nuts, wall bend, and wrought brass escutcheon of depth determined by installation requirements.
4. Angle stops shall be lead-free commercial pattern chrome plated brass, quarter turn ball type with loose key handles. Provide complete with chrome plated copper supply risers and wrought brass escutcheon of depth determined by installation requirements.
5. Toilet seats shall be commercial grade and provided complete with stainless steel posts and self-sustaining check hinges.
6. Pipe trim insulation shall be compliant, white molded vinyl, fade/discoloration-resistant, bacteria/fungal-resistant insulation.

B. FAUCETS

1. Chicago
2. T&S Brass
3. Zurn
4. Moen Commercial
5. Delta Commercial
6. American Standard
7. Kohler
8. Symmons Commercial
9. Speakman

C. SHOWER VALVES

1. Acorn
2. Bradley
3. Symmons Commercial
4. Chicago

5. Powers
6. Zurn
7. Speakman

D. FLUSHOMETERS

1. Sloan
2. Zurn
3. Moen Commercial
4. Delta Commercial
5. American Standard
6. Toto

E. SUPPLY STOPS

1. McGuire
2. Zurn
3. Chicago

F. CHROME PLATED TUBULAR BRASS

1. McGuire
2. Zurn
3. Kohler

G. TOILET SEATS

1. Church
2. Bemis
3. American Standard
4. Zurn
5. Toto

H. PIPE TRIM INSULATION

1. Truebro
2. McGuire
3. Plumberex

2.3 FIXTURE CARRIERS

- A. GENERAL: ANSI/ ASME A112.6.1M; Provide floor-affixed fixture carriers as appropriate for all wall-hung plumbing fixtures unless specifically noted otherwise. Fixture carrier foot supports shall be securely anchored to the floor with 1/2" bolts and anchors at all locations.
1. Chair type carriers shall be adjustable, with coated cast iron body with integral no hub waste and vent connections, complete with gasketed adjustable faceplate assembly, adjustable nipple with test cap, neoprene bowl gasket, lugs for floor and wall attachment, threaded fixture studs, and hardware. Provide single or double type of vertical or horizontal configuration as required and with auxiliary inlet as required.
  2. Lavatory carriers shall be adjustable, with steel uprights and welded base feet, coated cast iron support brackets, cast or ductile iron concealed support arms, alignment rod, complete with leveling and support hardware. Provide single or back to back configuration as required.

3. Drinking fountain and urinal carriers shall be adjustable, with steel uprights and welded base feet, upper and lower bearing plates, threaded rods, and mounting hardware. Provide single or side-by-side configuration as required

B. ACCEPTABLE MANUFACTURERS

1. J.R. Smith
2. Zurn
3. Mifab
4. Watts
5. Wade
6. Josam

PART 3 – EXECUTION

3.1 PREPARATION

- A. EXAMINATION OF CONDITIONS: Examine conditions affecting this work. Report unsatisfactory conditions to the proper authority and do not proceed until those conditions have been corrected. Commencing work implies acceptance of existing conditions as satisfactory to the outcome of this work.
- B. Coordinate forming of floor construction to receive drains to required invert elevations.

3.2 INSTALLATION

- A. Install fixtures in locations and heights as shown on Drawings and as directed by the Architect.
- B. Install materials plumb, level, securely, and in accordance with manufacturer's recommendations.
- C. All rough-in pipe openings for final connections with supply, waste, vent, and storm systems shall be closed with caps or plugs during early stages of construction and installation. Tape shall not be considered sufficient protection.
- D. Rough-in fixture piping connections in accordance with minimum sizes indicated in fixture rough-in schedule for particular fixtures.
- E. Provide ball valves in piping serving batteries of fixtures. Label stops "Hot" and "Cold." Valves shall be located above accessible ceilings. If ceilings are not accessible, provide access panels of adequate size to ensure valves are fully accessible and can be fully operated.
- F. Provide lockable ball valves in piping serving emergency safety fixtures and clearly label such valves as to the fixtures served.
- G. Plumbing fixtures shall be supported by a concealed carrier where required to properly support the fixture specified. All carriers to be securely mounted, bolted and checked prior to concealment.
- H. Caulk around fixtures with best grade white silicone caulking. Do not use grout.
- I. All handles on supply and drainage fittings or other brass items shall be properly lined up and adjusted. Fittings shall not be left in any haphazard manner.

- J. All fixtures shall have individual chrome plated heavy pattern loose key quarter-turn cutoff stops on supply lines, complete with escutcheons. Where same are not specified as a part of the fixture trim, they shall be installed as close to fixtures as possible in the hot and cold water supply.
- K. Install each fixture with trap, easily removable for servicing and cleaning.
- L. All showers and similar installations shall be installed with type "L" copper pipe between shower valve and shower head rough-in. The termination point shall have a brass drop ear elbow for shower head arm connection. Contractor shall provide proper anchoring support.

### 3.3 INTERFACE WITH OTHER PRODUCTS

- A. Review millwork shop drawings. Confirm location and size of fixtures and openings before rough-in and installation.
- B. Review architectural drawings. Confirm configuration and orientation of shower controls and trim prior to rough-in and installation.

### 3.4 ADJUSTING

- A. Adjust work under provisions of Division One.
- B. Adjust stops or valves for intended water flow rate to fixtures without splashing, noise, or overflow.

### 3.5 CLEANING

- A. Clean work under provisions of Division One.
- B. At completion clean plumbing fixtures and appurtenances.

### 3.6 PROTECTION OF FINISHED WORK

- A. Protect finished Work under provisions of Division One.
- B. Do not permit use of fixtures.

### 3.7 ADA ACCESSIBLE FIXTURES

- A. At all locations required to be accessible, such fixtures, controls, and final installations shall comply with the requirements of ADA and any applicable state accessibility standards. Install fixtures to heights, indicated on architectural drawings.
- B. All exposed water supply and drain pipes under accessible lavatories and sinks shall be insulated with securely fastened pipe trim insulation kits of the proper model for the fixtures specified.
- C. Wall mounted drinking fountains and coolers which protrude into passages or corridor space, whether single or paired with an adjacent accessible fixture, shall be supplied with a matching skirt or apron to lower the underside clearance of the non-accessible fixture equal to that required for accessible fixture.

## END OF SECTION



## **SECTION 22 61 00 - COMPRESSED AIR SYSTEMS**

### **PART 1 - GENERAL**

#### **1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division-1 Specification sections, apply to work of this section.
- B. Division-15 Basic Mechanical Materials and Methods sections apply to work of this section.

#### **1.2 DESCRIPTION OF WORK**

- A. Extent of compressed air systems work, is indicated on drawings and schedules, and by requirements of this section.
- B. Refer to Division-16 sections for the following work; not work of this section.
  - 1. Power supply wiring from power source to power connection on compressed air equipment. Include starters, disconnects, and required electrical devices, except where specified as furnished, or factory-installed, by manufacturer.
  - 2. Interlock wiring between electrically-operated compressed air equipment units; and between equipment and field-installed control devices.
    - a. Interlock wiring specified as factory-installed is work of this section.
- C. Provide the following electrical work as work of this section, complying with requirements of Division-16 sections:
  - 1. Control wiring between field-installed controls, indicating devices, and unit control panels.

#### **1.3 QUALITY ASSURANCE**

- A. Manufacturer's Qualifications: Firms regularly engaged in manufacture of compressed air systems equipment and products, of types, materials, and sizes required, whose products have been in satisfactory use in similar service for not less than 5 years.
- B. Installer Qualifications: Firm with at least 3 years of successful installation experience on projects with compressed air systems work similar to that required for project.
- C. Codes and Standards:
  - 1. ASME Compliance: Provide compressed air pressure vessels and relief valves in accordance with ASME "Boiler and Pressure Vessel Code"; provide ASME Code Symbol Stamp.
  - 2. ASME Compliance: Fabricate and install compressed air systems in accordance with ASME B31.9 "Building Services Piping".
  - 3. CGA Compliance: Fabricate and install compressed air systems in accordance with CGA Standards (Compressed Gas Association).
  - 4. UL Compliance: Provide electrical components which are UL listed and have UL label affixed.

#### 1.4 SUBMITTALS

- A. Product Data: Submit manufacturer's technical product data and installation instructions for compressed air systems materials and products.
- B. Record Drawings: At project closeout, submit record drawings of installed piping, and compressed air systems products; in accordance with requirements of Division 1.
- C. Wiring Diagrams: Submit manufacturer's electrical requirements for power supply wiring to units. Submit manufacturer's ladder-type wiring diagrams for interlock and control wiring. Clearly differentiate between portions of wiring that are factory-installed and portions to be field-installed.
- D. Maintenance Data: Submit maintenance data and parts lists for compressed air systems materials and products. Include this data, product data, shop drawings, record drawings, and wiring diagrams in maintenance manual; in accordance with requirements of Division 1.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver compressed air equipment, including compressed air dryers and receiver tanks with factory-installed skid; with accessories as scheduled on the plans.

### PART 2 - PRODUCTS

#### 2.1 ACCEPTABLE MANUFACTURERS

- A. Saylor – Beall
- B. Ingersoll – Ran
- C. Champion

#### 2.2 BASIC IDENTIFICATION

- A. General: Provide identification complying with Division-15 Basic Mechanical Materials and Methods section "Mechanical Identification", in accordance with the following listing:
  - 1. Piping Above Ground: Plastic pipe markers.
  - 2. Valves: Plastic valve tags.

#### 2.3 BASIC PIPES AND PIPE FITTINGS

- A. General: Provide pipes and pipe fittings complying with Division-15 Basic Mechanical Materials and Methods.
- B. All Sizes: Copper tube, hard-temper for exposed locations, soft-temper for underground and concealed locations; Type "L"; wrought-copper, solder-joint fittings

#### 2.4 BASIC SUPPORTS AND ANCHORS

- A. General: Provide supports and anchors, complying with Division-15 Basic Mechanical Materials and Methods section "Supports and Anchors", in accordance with the following listing:

1. Extension split pipe clamp, copper plated, hinged or 2-bolt for pipe support from any substrate.

## 2.5 VALVES

- A. Shutoff Valves: Provide compressed air shutoff valves, bronze-body, double seal, full port, union ball type, with seat seals and stem seals. Design so quarter turn of lever-type valve handle is maximum travel between open and closed positions. Design for working pressure of 300 psi minimum. Provide valves with color-coded gas identification labels.

## 2.6 BASIC VIBRATION CONTROL

- A. General: Provide vibration control products complying with Division-15 Basic Mechanical Materials and Methods section "Vibration Control", in accordance with the following listing:

## 2.7 COMPRESSED AIR EQUIPMENT AND ACCESSORIES

- A. General: Provide air compressor system, size as specified on drawings.
- B. The air compressor system shall be of a single point connection, base mounted design consisting of two compressor, two motors, and regulator, integral pre-wired control panel and corrosion resistant air receiver. The complete package shall be pre-wired, pre-piped, and assembled on one common base with single point connection for electrical.
- C. Compressor Module: The compressor shall be a continuous duty rated. The design shall be single stage, air-cooled, rated for 120 psig discharge pressure. The compressor shall be v-belt driven with totally enclosed belt guard. The drive motor shall be a NEMA rated, open drip proof, 3600 RPM, with 1.15 service factor suitable for 208 or 230/460V electrical service.
- D. Isolation System: The compressor and motor shall be fully isolated from the air receiver by means of a four point spring isolation system.
- E. Air Receiver: The vertical air receiver shall be corrosion resistant, ASME coded, National Board Certified, rated for a minimum 150 psig design pressure, and include a liquid level gauge glass, safety relief valve, manual drain valve, and a timed automatic solenoid drain valve.
- F. Dryer System: The air drying system shall consist of a hi – temp refrigerated type. The dryer shall be individually sized for peak calculated demand and capable as scheduled on the plans.
- G. Control System: The duplex mounted and wired control system shall be NEMA 12 and U.L. labeled. This control system shall provide automatic lead/lag sequencing with circuit breaker disconnects for each compressor with external operators, full voltage motor starters with overload protection, redundant 120V control circuit transformers, visual and audible unit alarm with isolated contacts for remote alarm, hand-off-auto lighted selector switches, pressure gauge, and duplexed runtime hour meters. Automatic alternation of both compressors based on a first-on/first-off principle, with provisions for simultaneous operation if required, automatic activation of reserve unit if required, visual and audible alarm indication for high discharge air temperature shutdown with isolated contacts for remote alarm shall be included.

## PART 1 - EXECUTION

### 3.1 INSPECTION

- A. Examine areas and conditions under which compressed air systems and equipment are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to Installer.

### 3.2 INSTALLATION OF BASIC IDENTIFICATION

- A. General: Install mechanical identification in accordance with Division-15 Basic Mechanical Materials and Methods section "Mechanical Identification".

### 3.3 INSTALLATION OF COMPRESSED AIR PIPING

- A. General: Install piping in accordance with Division- 15 Basic Mechanical Materials and Methods section "Pipes and Pipe Fittings".

### 3.4 INSTALLATION OF SUPPORTS AND ANCHORS

- A. Install supports and anchors, in accordance with Division-15 Basic Mechanical Materials and Methods section "Supports and Anchors". Locate spacing for supports and anchors in accordance with piping manufacturer's recommendations.
- B. Support Spacing: Install piping supports at the following maximum intervals:
  - 1. 1/2" Pipe Size: 6' o.c.
  - 2. 3/4" to 1" Pipe Size: 8' o.c.
  - 3. 1-1/4" or Larger Pipe Size: 10' o.c.
  - 4. Vertical Piping: Every floor level.

### 3.5 INSTALLATION OF VALVES

- A. Shutoff Valves: Provide shutoff valves where indicated, on plans.

### 3.6 INSTALLATION OF EQUIPMENT AND ACCESSORIES

- A. Install equipment and accessories where indicated, in accordance with equipment manufacturer's written instructions, and with recognized industry practices, to ensure that compressed air equipment and accessories comply with requirements and serve intended purposes.
- B. Coordinate with other work including plumbing, as necessary to interface installation of compressed air piping and equipment with other work.
- C. Support: Install equipment on 4" high reinforced concrete pads, 4" larger on each side than equipment base. Cast anchor bolt inserts into pad.
- D. Electrical Wiring: Install electrical devices furnished by manufacturer but not specified to be factory-mounted. Furnish copy of manufacturer's wiring diagram submittal to Electrical Installer.
  - 1. Verify that electrical wiring installation is in accordance with manufacturer's submittal and installation requirements of Division-16 sections. Do not proceed with equipment start-up until wiring installation is acceptable to equipment installer.

3.7 EQUIPMENT CONNECTIONS

- A. General: Connect compressed air systems to mechanical equipment as indicated, and comply with equipment manufacturer's instructions where not otherwise indicated.

3.8 FIELD QUALITY CONTROL

- A. Test compressed air alarms for proper operation at high pressure, low pressure.

**END OF SECTION**

## **SECTION 23 02 00 - BASIC MATERIALS AND METHODS**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all Work herein.
- B. The Contract Drawings indicate the extent and general arrangement of the systems. If any departure from the Contract Drawings is deemed necessary by the Contractor, details of such departures and the reasons therefore, shall be submitted to the Architect/Engineer for review as soon as practicable. No such departures shall be made without the prior written approval of the Architect/Engineer.
- C. Notwithstanding any reference in the Specifications to any article, device, product, material, fixture, form or type of construction by name, make or catalog number, such reference shall not be construed as limiting competition; and the Contractor, in such cases, may at his option use any article, device, product, material, fixture, form or type of construction which in the judgment of the Architect/Engineer, expressed in writing, is the equivalent of that specified.

#### **1.2 SCOPE OF WORK**

- A. The Work included under this Contract consists of the furnishing and installation of all equipment and material necessary and required to form complete and functioning systems in all of their various phases, all as shown on the accompanying Drawings and/or described in these Specifications. The Contractor shall review all pertinent drawings, including those of other contracts, prior to commencement of Work.
- B. This Division requires the furnishing and installing of all items as specified herein, indicated on the Drawings or reasonably inferred as necessary for safe and proper operation; including every article, device or accessory (whether or not specifically called for by item) reasonably necessary to facilitate each system's functioning as indicated by the design and the equipment specified. Elements of the work include, but are not limited to, materials, labor, supervision, transportation, storage, equipment, utilities, all required permits, licenses and inspections. All work performed under this Section shall be in accordance with the Project Manual, Drawings and Specifications and is subject to the terms and conditions of the Contract.
- C. The approximate locations of Mechanical (HVAC) items are indicated on the Drawings. These Drawings are not intended to give complete and accurate details in regard to location of outlets, apparatus, etc. Exact locations are to be determined by actual measurements at the building, and will in all cases be subject to the review of the Owner or Engineer, who reserves the right to make any reasonable changes in the locations indicated without additional cost to the Owner.
- D. Items specifically mentioned in the Specifications but not shown on the Drawings and/or items shown on Drawings but not specifically mentioned in the Specifications shall be installed by the Contractor under the appropriate section of work as if they were both specified and shown.
- E. All discrepancies between the Contract Documents and actual job-site conditions shall be reported to the Owner or Engineer so that they will be resolved prior to bidding. Where this cannot be done at least 7 working days prior to bid; the greater or more costly of the

discrepancy shall be bid. All labor and materials required to perform the work described shall be included as part of this Contract.

- F. It is the intention of this Section of the Specifications to outline minimum requirements to furnish the Owner with a turn-key and fully operating system in cooperation with other trades.
- G. It is the intent of the above "Scope" to give the Contractor a general outline of the extent of the Work involved; however, it is not intended to include each and every item required for the Work. Anything omitted from the "Scope" but shown on the Drawings, or specified later, or necessary for a complete and functioning heating, ventilating and air conditioning system shall be considered a part of the overall "Scope".
- H. The Contractor shall rough-in fixtures and equipment furnished by others from rough-in and placement drawings furnished by others. The Contractor shall make final connection to fixtures and equipment furnished by others.
- I. The Contractor shall participate in the commissioning process as required; including, but not limited to, meeting attendance, completion of checklists, and participation in functional testing.

### 1.3 SCHEMATIC NATURE OF CONTRACT DOCUMENTS

- A. The Contract Documents are schematic in nature in that they are only to establish scope and a minimum level of quality. They are not to be used as actual working construction drawings. The actual working construction drawings shall be the reviewed shop drawings.
- B. All duct or pipe or equipment locations as indicated on the documents do not indicate every transition, offset, or exact location. All transitions, offsets, clearances and exact locations shall be established by actual field measurements, coordination with the structural, architectural and reflected ceiling plans, and other trades. Submit shop drawings for review.
- C. All transitions, offsets and relocations as required by actual field conditions shall be performed by the Contractor at no additional cost to the Owner.
- D. Additional coordination with electrical contractor may be required to allow adequate clearances of electrical equipment, fixtures and associated appurtenances. Contractor to notify Architect and Engineer of unresolved clearances, conflicts or equipment locations.

### 1.4 SITE VISIT AND FAMILIARIZATION

- A. Before submitting a bid, it will be necessary for each Contractor whose work is involved to visit the site and ascertain for himself the conditions to be met therein in installing his work and make due provision for same in his bid. It will be assumed that this Contractor in submitting his bid has visited the premises and that his bid covers all work necessary to properly install the equipment shown. Failure on the part of the Contractor to comply with this requirement shall not be considered justification for the omission or faulty installation of any work covered by these Specifications and Drawings.
- B. Understand the existing utilities from which services will be supplied; verify locations of utility services, and determine requirements for connections.
- C. Determine in advance that equipment and materials proposed for installation fit into the confines indicated.

#### 1.5 WORK SPECIFIED IN OTHER SECTIONS

- A. Finish painting is specified. Prime and protective painting are included in the work of this Division.
- B. Owner and General Contractor furnished equipment shall be properly connected to Mechanical (HVAC) systems.
- C. Furnishing and installing all required Mechanical (HVAC) equipment control relays and electrical interlock devices, conduit, wire and J-boxes are included in the Work of this Division.

#### 1.6 PERMITS, TESTS, INSPECTIONS

- A. Arrange and pay for all permits, fees, tests, and all inspections as required by governmental authorities.

#### 1.7 DATE OF SUBSTANTIAL COMPLETION

- A. The date of final acceptance shall be the date of substantial completion. Refer to Division One for additional requirements.
- B. The date of final acceptance shall be documented in writing and signed by the Architect, Owner and Contractor.

#### 1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to the project properly identified with names, model numbers, types, grades, compliance labels, and other information needed for identification.
- B. Deliver products to the project at such time as the project is ready to receive the equipment, pipe or duct - properly protected from incidental damage and weather damage.
- C. Damaged equipment, duct or pipe shall be promptly removed from the site and new, undamaged equipment, pipe or duct shall be installed in its place promptly with no additional charge to the Owner.

#### 1.9 NOISE AND VIBRATION

- A. The heating, ventilating and air conditioning systems, and the component parts thereof, shall be guaranteed to operate without objectionable noise and vibration.
- B. Provide foundations, supports and isolators as specified or indicated, properly adjusted to prevent transmission of vibration to the building structure, piping and other items.
- C. Carefully fabricate ductwork and fittings with smooth interior finish to prevent turbulence and generation or regeneration of noise.
- D. All equipment shall be selected to operate with minimum of noise and vibration. If, in the opinion of the Architect, objectionable noise or vibration is produced or transmitted to or through the building structure by equipment, piping, ducts or other parts of the Work, the Contractor shall rectify such conditions without extra cost to the Owner.



#### 1.10 APPLICABLE CODES AND STANDARDS

- A. Obtain all required permits and inspections for all work required by the Contract Documents and pay all required fees in connection thereof.
- B. Arrange with the serving utility companies for the connection of all required utilities and pay all charges, meter charges, connection fees and inspection fees, if required.
- C. Comply with all applicable codes, specifications, local ordinances, industry standards, utility company regulations and the applicable requirements which includes and is not limited to the following nationally accepted codes and standards:
  - 1. Air Moving & Conditioning Association, AMCA.
  - 2. American Standards Association, ASA.
  - 3. American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc., ASHRAE.
  - 4. American Society of Mechanical Engineers, ASME.
  - 5. American Society of Plumbing Engineers, ASPE.
  - 6. American Society of Testing Materials, ASTM.
  - 7. American Water Works Association, AWWA.
  - 8. National Bureau of Standards, NBS.
  - 9. National Fire Protection Association, NFPA.
  - 10. Sheet Metal & Air Conditioning Contractors' National Association, SMACNA.
  - 11. Underwriters' Laboratories, Inc., UL.
  - 12. International Energy Conservation Code, IECC.
  - 13. International Building Code.
  - 14. International Mechanical Code.
  - 15. International Fire Code.
  - 16. International Gas Code.
- D. Where differences existing between the Contract Documents and applicable state or city building codes, state and local ordinances, industry standards, utility company regulations and the applicable requirements of the nationally accepted codes and standards, the more stringent or costly application shall govern. Promptly notify the Engineer in writing of all differences.
- E. When directed in writing by the Engineer, remove all work installed that does not comply with the Contract Documents and applicable state or city building codes, state and local ordinances, industry standards, utility company regulations and the applicable requirements of the above listed nationally accepted codes and standards, correct the deficiencies, and complete the work at no additional cost to the Owner.

#### 1.11 DEFINITIONS AND SYMBOLS

- A. General Explanation: A substantial amount of construction and Specification language constitutes definitions for terms found in other Contract Documents, including Drawings which must be recognized as diagrammatic and schematic in nature and not completely descriptive of requirements indicated thereon. Certain terms used in Contract Documents are defined generally in this article, unless defined otherwise in Division 01.
- B. Definitions and explanations of this Section are not necessarily either complete or exclusive, but are general for work to the extent not stated more explicitly in another provision of the Contract Documents.
- C. Indicated: The term "Indicated" is a cross-reference to details, notes or schedules on the

Drawings, to other paragraphs or schedules in the Specifications and to similar means of recording requirements in Contract Documents. Where such terms as "Shown", "Noted", "Scheduled", "Specified" and "Detailed" are used in lieu of "Indicated", it is for the purpose of helping the reader locate cross-reference material, and no limitation of location is intended except as specifically shown.

- D. Directed: Where not otherwise explained, terms such as "Directed", "Requested", "Accepted", and "Permitted" mean by the Architect or Engineer. However, no such implied meaning will be interpreted to extend the Architect's or Engineer's responsibility into the Contractor's area of construction supervision.
- E. Reviewed: Where used in conjunction with the Engineer's response to submittals, requests for information, applications, inquiries, reports and claims by the Contractor the meaning of the term "Reviewed" will be held to limitations of Architect's and Engineer's responsibilities and duties as specified in the General and Supplemental Conditions. In no case will "Reviewed" by Engineer be interpreted as a release of the Contractor from responsibility to fulfill the terms and requirements of the Contract Documents.
- F. Furnish: Except as otherwise defined in greater detail, the term "Furnish" is used to mean supply and deliver to the project site, ready for unloading, unpacking, assembly, installation, etc., as applicable in each instance.
- G. Install: Except as otherwise defined in greater detail, the term "Install" is used to describe operations at the project site including unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimension, finishing, curing, protection, cleaning and similar operations, as applicable in each instance.
- H. Provide: Except as otherwise defined in greater detail, the term "Provide" is used to mean "Furnish and Install", complete and ready for intended use, as applicable in each instance.
- I. Installer: Entity (person or firm) engaged by the Contractor, or its Subcontractor or Sub-subcontractor for performance of a particular unit of work at the project site, including unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimension, finishing, curing, protection, cleaning and similar operations, as applicable in each instance. It is a general requirement that such entities (Installers) be expert in the operations they are engaged to perform.
- J. Imperative Language: Used generally in Specifications. Except as otherwise indicated, requirements expressed imperatively are to be performed by the Contractor. For clarity of reading at certain locations, contrasting subjective language is used to describe responsibilities that must be fulfilled indirectly by the Contractor or, when so noted, by other identified installers or entities.
- K. Minimum Quality/Quantity: In every instance, the quality level or quantity shown or specified is intended as minimum quality level or quantity of work to be performed or provided. Except as otherwise specifically indicated, the actual work may either comply exactly with that minimum (within specified tolerances), or may exceed that minimum within reasonable tolerance limits. In complying with requirements, indicated or scheduled numeric values are either minimums or maximums as noted or as appropriate for the context of the requirements. Refer instances of uncertainty to Owner or Engineer via a request for information (RFI) for decision before proceeding.
- L. Abbreviations and Symbols: The language of Specifications and other Contract Documents including Drawings is of an abbreviated type in certain instances, and implies

words and meanings which will be appropriately interpreted. Actual word abbreviations of a self-explanatory nature have been included in text of Specifications and Drawings. Specific abbreviations and symbols have been established, principally for lengthy technical terminology and primarily in conjunction with coordination of Specification requirements with notations on Drawings and in Schedules. These are frequently defined in Section at first instance of use or on a Legend and Symbol Drawing. Trade and industry association names and titles of generally recognized industry standards are frequently abbreviated. Singular words will be interpreted as plural and plural words will be interpreted as singular where applicable and where full context of Contract Documents so indicate. Except as otherwise indicated, graphic symbols and abbreviations used on Drawings and in Specifications are those recognized in construction industry for indicated purposes. Where not otherwise noted symbols and abbreviations are defined by the latest ASHRAE Fundamentals Handbook, chapter 34 "Abbreviations and Symbols", ASME and ASPE published standards.

#### 1.12 DRAWINGS AND SPECIFICATIONS

- A. These Specifications are intended to supplement the Drawings and it will not be the province of the Specifications to mention any part of the Work which the Drawings are competent to fully explain in every particular and such omission is not to relieve the Contractor from carrying out portions indicated on the Drawings only.
- B. Should items be required by these Specifications and not indicated on the Drawings, they are to be supplied even if of such nature that they could have been indicated thereon. In case of disagreement between Drawings and Specifications, or within either Drawings or Specifications, the better quality or greater quantity of work shall be estimated and the matter referred to the Architect or Engineer for review with a request for information and clarification at least 7 working days prior to bid opening date for issuance of an addendum.
- C. The listing of product manufacturers, materials and methods in the various sections of the Specifications, and indicated on the Drawings, is intended to establish a standard of quality only. It is not the intention of the Owner or Engineer to discriminate against any product, material or method that is the equivalent of the standards as indicated and/or specified, nor is it intended to preclude open, competitive bidding. The fact that a specific manufacturer is listed as an acceptable manufacturer should not be interpreted to mean that the manufacturer's standard product will meet the requirements of the project design, Drawings, Specifications and space constraints.
- D. The Architect or Engineer and Owner shall be the sole judge of quality and equivalence of equipment, materials and methods.
- E. Products by other reliable manufacturers, other materials, and other methods, will be accepted as outlined, provided they have equivalent capacity, construction, and performance. However, under no circumstances shall any substitution be made without the written permission of the Architect or Engineer and Owner. Request for prior approval must be made in writing 10 calendar days prior to the bid date without fail.
- F. Wherever a definite product, material or method is specified and there is not a statement that another product, material or method will be acceptable, it is the intention of the Owner or Engineer that the specified product, material or method is the only one that shall be used without prior approval.
- G. Wherever a definite material or manufacturer's product is specified and the Specification states that products of similar design and equivalent construction from the specified list of manufacturers may be substituted, it is the intention of the Owner or Engineer that

products of manufacturers that are specified are the only products that will be acceptable and that products of other manufacturers will not be considered for substitution without approval.

- H. Wherever a definite product, material or method is specified and there is a statement that "OR EQUIVALENT" product, material or method will be acceptable, it is the intention of the Owner or Engineer that the specified product, material or method or an "OR EQUIVALENT" product, material or method may be used if it complies with the Specifications and is submitted for review to the Engineer as outline herein.
- I. Where permission to use substituted or alternative equipment on the project is granted by the Owner or Engineer in writing, it shall be the responsibility of the Contractor or Subcontractor involved to verify that the equipment will fit in the space available which includes allowances for all required Code and maintenance clearances, and to coordinate all equipment structural support, plumbing and electrical requirements and provisions with the Mechanical (HVAC) Design Documents and all other trades, including Division 26.
- J. Changes in architectural, structural, electrical, mechanical, and plumbing requirements for the substitution shall be the responsibility of the bidder wishing to make the substitution. This shall include the cost of redesign by the affected designer(s). Any additional cost incurred by affected Subcontractors shall be the responsibility of this bidder and not the Owner.
- K. If any request for a substitution of product, material or method is rejected, the Contractor will automatically be required to furnish the product, material or method named in the Specifications. Repetitive requests for substitutions will not be considered.
- L. The Owner or Engineer will investigate all requests for substitutions when submitted in accordance with the requirements listed above; and if accepted, will issue a letter allowing the substitutions.
- M. Where equipment other than that used in the design as specified or shown on the Drawings is substituted (either from an approved manufacturers list or by submittal review), it shall be the responsibility of the substituting Contractor to coordinate space requirements, building provisions and connection requirements with his trades and all other trades; and to pay all additional costs to other trades, the Owner, the Architect or Engineer, if any, due to the substitutions.

#### 1.13 SUBMITTALS

- A. Coordinate with Division 01 for submittal timetable requirements, unless noted otherwise within thirty (30) days after the Contract is awarded. The Contractor shall submit an electronic copy of a complete set of shop drawings and complete data covering each item of equipment or material. The submittal of each item requiring a submittal must be received by the Architect or Engineer within the above thirty-day period. The Architect or Engineer shall not be responsible for any delays or costs incurred due to excessive shop drawing review time for submittals received after the thirty (30) day time limit. The Architect and Engineer will retain a copy of all shop drawings for their files. All literature pertaining to items subject to Shop Drawing submittal shall be submitted at one time. Submittals shall be placed in one electronic file in PDF 8.0 format and bookmarked for individual specification sections. Individual electronic files of submittals for individual specifications shall not be permitted. Each submittal shall include the following items:
  - 1. A cover sheet with the names and addresses of the Project, Architect, MEP

- Engineer, General Contractor and the Subcontractor making the submittal. The cover sheet shall also contain the section number covering the item or items submitted and the item nomenclature or description.
2. An index page with a listing of all data included in the Submittal.
  3. A list of variations page with a listing of all variations, including unfurnished or additional required accessories, items or other features, between the submitted equipment and the specified equipment. If there are no variations, then this page shall state "NO VARIATIONS". Where variations affect the work of other Contractors, then the Contractor shall certify on this page that these variations have been fully coordinated with the affected Contractors and that all expenses associated with the variations will be paid by the submitting Contractor. This page will be signed by the submitting Contractor.
  4. Equipment information including manufacturer's name and designation, size, performance and capacity data as applicable. All applicable Listings, Labels, Approvals and Standards shall be clearly indicated.
  5. Dimensional data and scaled drawings as applicable to show that the submitted equipment will fit the space available with all required Code and maintenance clearances clearly indicated and labeled at a minimum scale of 1/4" = 1'-0", as required to demonstrate that the alternate or substituted product will fit in the space available.
  6. Identification of each item of material or equipment matching that indicated on the Drawings.
  7. Sufficient pictorial, descriptive and diagrammatic data on each item to show its conformance with the Drawings and Specifications. Any options or special requirements or accessories shall be so indicated. All applicable information shall be clearly indicated with arrows or another approved method.
  8. Additional information as required in other Sections of this Division.
  9. Certification by the General Contractor and Subcontractor that the material submitted is in accordance with the Drawings and Specifications, signed and dated in long hand. Submittals that do not comply with the above requirements shall be returned to the Contractor and shall be marked "REVISE AND RESUBMIT".
- B. Refer to Division 00 and Division 01 for additional information on shop drawings and submittals.
- C. Equipment and materials submittals and shop drawings will be reviewed for compliance with design concept only. It will be assumed that the submitting Contractor has verified that all items submitted can be installed in the space allotted. Review of shop drawings and submittals shall not be considered as a verification or guarantee of measurements or building conditions.
- D. Where shop drawings and submittals are marked "REVIEWED", the review of the submittal does not indicate that submittals have been checked in detail nor does it in any way relieve the Contractor from his responsibility to furnish material and perform work as required by the Contract Documents.
- E. Shop drawings shall be reviewed and returned to the Contractor with one of the following categories indicated:
1. REVIEWED: Contractor need take no further submittal action, shall include this submittal in the O&M manual and may order the equipment submitted on.
  2. REVIEWED AS NOTED: Contractor shall submit a letter verifying that required exceptions to the submittal have been received and complied with including additional accessories or coordination action as noted, and shall include this submittal and compliance letter in the O&M manual. The contractor may order

- the equipment submitted on at the time of the returned submittal providing the Contractor complies with the exceptions noted.
3. NOT APPROVED: Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is not approved. The Contractor will automatically be required to furnish the product, material or method named in the Specifications and/or Drawings. Contractor shall not order equipment that is not approved. Repetitive requests for substitutions will not be considered.
  4. REVISE AND RESUBMIT: Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is marked revise and resubmit. The Contractor will automatically be required to furnish the product, material or method named in the Specifications and/or provide as noted on previous shop drawings. Contractor shall not order equipment marked revise and resubmit. Repetitive requests for substitutions will not be considered.
  5. CONTRACTOR'S CERTIFICATION REQUIRED: Contractor shall resubmit submittal on material, equipment or method of installation. The Contractor's stamp is required stating that the submittal meets all conditions of the Contract Documents. The stamp shall be signed by the General Contractor. The submittal will not be reviewed if the stamp is not placed and signed on all shop drawings.
  6. MANUFACTURER NOT AS SPECIFIED: Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is marked manufacturer not as specified. The Contractor will automatically be required to furnish the product, material or method named in the Specifications. Contractor shall not order equipment when submittal is marked manufacturer not as specified. Repetitive requests for substitutions will not be considered.
- F. Materials and equipment which are purchased or installed without submittal review shall be at the risk of the Contractor and the cost for removal and replacement of such materials and equipment and related work which is judged unsatisfactory by the Owner or Engineer for any reason shall be at the expense of the Contractor. The responsible Contractor shall remove the material and equipment noted above and replace with specified equipment or material at his own expense when directed in writing by the Architect or Engineer.
- G. Shop Drawing Submittals shall be complete and checked prior to submission to the Engineer for review.
- H. Submittals are required for, but not limited to, the following items subject to project requirements:
1. Coordination Drawings
  2. Common Motor Requirements for HVAC Equipment
  3. Expansion Fittings and Loops for HVAC Piping
  4. Variable Frequency Motor Speed Control for HVAC Equipment
  5. Hangers and Support for Piping and Equipment HVAC
  6. Vibration and Seismic Controls for HVAC Piping and Equipment
  7. Testing, Adjusting, and Balancing
  8. Duct Insulation
  9. HVAC Equipment Insulation
  10. HVAC Piping Insulation
  11. Refrigerant Monitor System
  12. Energy Management and Control System
  13. Above Ground Hydronic Piping
  14. Hydronic Specialties

15. Hydronic Pumps
16. Refrigerant Piping
17. Metal Ductwork
18. Ductwork Accessories
19. HVAC Fans
20. Series Fan Powered Terminal Units
21. Single Duct VAV Terminal Box
22. Parallel Fan Powered Terminal Unit
23. Dual Duct Air Terminal Units
24. Air Distribution Devices
25. Air Filters
26. Flue Pipe Systems
27. Non-Condensing Boiler-Gas Fired (Forced Draft)
28. Condensing Boiler – Gas Fired
29. Finned Water-Tube Boilers
30. Steel Water-Tube Boilers
31. Gas Fired Furnaces
32. Gas Fired Roof Mounted Make-up Air Unit Heaters
33. Shell and Tube Heat Exchanger
34. Centrifugal Liquid Chiller
35. Rotary Screw Water Chillers
36. Air Cooled Rotary Liquid Chiller
37. Induced Draft Cooling Tower
38. Energy Recovery Ventilator
39. Modular Indoor Central Station Air Handling Units
40. Packaged Air Handling Unit
41. Modular Outdoor Central Station Air Handling Units
42. 100% Outside Air Rooftop Unit with Gas Heat
43. Self-Contained Air Conditioners
44. Rooftop Heating and Cooling Units Electric Cooling-Gas Heating
45. Rooftop Heating and Cooling Units Electric Cooling-Electric Heat
46. Variable Air Volume Rooftop Units
47. Variable Refrigerant Flow (VRF) for HVAC System
48. Water Source Heat Pump Unit
49. Fan Coil Unit
50. Unit Ventilators
51. Electric Duct Heaters
52. Radiant Heating Electric Cables
53. Air Conditioning Unit for Swimming Pool Enclosures

- I. Refer to other Division 23 sections for additional submittal requirements. Provide samples of actual materials and/or equipment to be used on the Project upon request of the Owner or Engineer.

#### 1.14 COORDINATION DRAWINGS

- A. Prepare coordination drawings to a scale of 1/4"=1'-0" or larger; detailing major elements, components, and systems of mechanical equipment and materials in relationship with other systems, installations, and building components. Indicate locations where space is limited for installation and access, and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:
  1. Indicate the proposed locations of pipe, duct, equipment, and other materials. Include the following:

- a. Wall and type locations.
    - b. Clearances for installing and maintaining insulation.
    - c. Locations of light fixtures and sprinkler heads.
    - d. Clearances for servicing and maintaining equipment, including tube removal, filter removal, and space for equipment disassembly required for periodic maintenance.
    - e. Equipment connections and support details.
    - f. Exterior wall and foundation penetrations.
    - g. Routing of storm and sanitary sewer piping.
    - h. Fire-rated wall and floor penetrations.
    - i. Sizes and location of required concrete pads and bases.
    - j. Valve stem movement.
    - k. Structural floor, wall and roof opening sizes and details.
  - 2. Indicate scheduling, sequencing, movement, and positioning of large equipment into the building during construction.
  - 3. Prepare floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
  - 4. Prepare reflected ceiling plans to coordinate and integrate installations, air distribution devices, light fixtures, communication systems components, and other ceiling-mounted items.
- B. This Contractor shall be responsible for coordination of all items that will affect the installation of the work of this Division. This coordination shall include, but not be limited to: voltage, ampacity, capacity, electrical and piping connections, space requirements, sequence of construction, building requirements and special conditions.
- C. By submitting coordination drawings on the project, this Contractor is indicating that all necessary coordination has been completed and that the systems, products and equipment submitted can be installed in the building and will operate as specified and intended, in full coordination with all other Contractors and Subcontractors.

#### 1.15 RECORD DOCUMENTS

- A. Prepare Record Documents in accordance with the requirements in Special Project Requirements, in addition to the requirements specified in Division 23, indicate the following installed conditions:
- 1. Duct mains and branches, size and location, for both exterior and interior; locations of dampers, fire dampers, duct access panels, and other control devices; filters, fuel fired heaters, fan coils, condensing units, and roof-top A/C units requiring periodic maintenance or repair.
  - 2. Mains and branches of piping systems, with valves and control devices located and numbered, concealed unions located, and with items requiring maintenance located (i.e., traps, strainers, expansion compensators, tanks, etc.). Valve location diagrams, complete with valve tag chart. Indicate actual inverts and horizontal locations of underground piping.
  - 3. Equipment locations (exposed and concealed), dimensioned from prominent building lines.
  - 4. Approved substitutions, Contract Modifications, and actual equipment and materials installed.
  - 5. Contract Modifications, actual equipment and materials installed.
- B. Engage the services of a Land Surveyor or Professional Engineer registered in the state in which the project is located as specified herein to record the locations and invert elevations of underground installations.



- C. The Contractor shall maintain a set of clearly marked black line record "AS-BUILT" prints on the job site on which he shall mark all work details, alterations to meet site conditions and changes made by "Change Order" notices. These shall be kept available for inspection by the Owner, Architect or Engineer at all times.
- D. Refer to Division 00 and Division 01 for additional requirements concerning Record Drawings. If the Contractor does not keep an accurate set of as-built drawings, the pay request may be altered or delayed at the request of the Architect. Mark the drawings with a colored pencil. Delivery of as-built prints and re-producibles is a condition of substantial completion.
- E. The record prints shall be updated on a daily basis and shall indicate accurate dimensions for all buried or concealed work, precise locations of all concealed pipe or duct, locations of all concealed valves, controls and devices and any deviations from the work shown on the Construction Documents which are required for coordination. All dimensions shall include at least two dimensions to permanent structure points.
- F. Submit three prints of the tracings for review. Make corrections to tracings as directed and deliver "Auto Positive Tracings" to the Architect. "As-Built" drawings shall be furnished in addition to submittals.
- G. When the option described in paragraph F above is not exercised, then upon completion of the Work, the Contractor shall transfer all marks from the tracings and submit a set of clear concise reproducible record "AS-BUILT" drawings and shall submit the reproducible drawings with corrections made by a competent draftsman and three (3) sets of black line prints to the Architect or Engineer for review prior to scheduling the final inspection at the completion of the Work. The reproducible record "AS-BUILT" drawings shall have the Engineer's Name and Seal removed or blanked out and shall be clearly marked and signed on each sheet as follows:

CERTIFIED RECORD DRAWINGS

DATE:

(NAME OF GENERAL CONTRACTOR)

BY: \_\_\_\_\_  
(SIGNATURE)

(NAME OF SUBCONTRACTOR)

BY: \_\_\_\_\_  
(SIGNATURE)

1.16 OPERATING AND MAINTENANCE MANUALS

- A. Prepare operating and maintenance manuals in accordance with Division 00 and Division 01 and, in addition to the requirements specified in those Divisions, include the following information for equipment items:
  - 1. Description of function, normal operating characteristics and limitations, performance curves, engineering data and tests, and complete nomenclature and commercial numbers of replacement parts.
  - 2. Manufacturer's printed operating procedures to include start-up, break-in, and routine and normal operating instructions; regulation, control, stopping, shutdown, and emergency instructions; and summer and winter operating

- instructions.
- 3. Maintenance procedures for routine preventative maintenance and troubleshooting; disassembly, repair, and reassembly; aligning and adjusting instructions.
- 4. Servicing instructions and lubrication charts and schedules.

#### 1.17 CERTIFICATIONS AND TEST REPORTS

- A. Submit a detailed schedule for completion and testing of each system indicating scheduled dates for completion of system installation and outlining tests to be performed and scheduled date for each test. This detailed completion and test schedule shall be submitted at least 90 days before the projected substantial completion date.
- B. Test result reporting forms shall be submitted for review no later than the date of the detailed schedule.
- C. Submit 4 copies of all certifications and test reports to the Architect or Engineer for review adequately in advance of substantial completion of the Work to allow for remedial action as required to correct deficiencies discovered in equipment and systems.
- D. Certifications and test reports to be submitted shall include, but not be limited to, those items outlined in Section 23 02 00.

#### 1.18 OPERATING AND MAINTENANCE MANUALS

- A. Coordinate with Division 00 and Division 01 for operating and maintenance manual requirements. Unless noted otherwise, bind together in "D ring type" binders (National model no. 79-883 or equal). Binders shall be large enough to allow ¼" of spare capacity. Three (3) sets of all reviewed submittals, fabrication drawings, bulletins, maintenance instructions, operating instructions and parts exploded views and lists for each and every piece of equipment furnished under these Specifications. All sections shall be typed and indexed into sections and labeled for easy reference and shall utilize the individual specification section numbers shown in the Mechanical Specifications as an organization guideline. Bulletins containing information about equipment that is not installed on the project shall be properly marked up or stripped and reassembled. All pertinent information required by the Owner for proper operation and maintenance of equipment supplied by Division 23 shall be clearly and legibly set forth in memoranda that shall, likewise, be bound with bulletins.
- B. Prepare maintenance manuals in accordance with Special Project Conditions. In addition to the requirements specified in Division 23, include the following information for equipment items:
  - 1. Identifying names, name tag designations and locations for all equipment.
  - 2. Valve tag lists with valve number, type, color coding, location and function.
  - 3. Reviewed submittals with exceptions noted compliance letter.
  - 4. Fabrication drawings.
  - 5. Equipment and device bulletins and data sheets clearly highlighted to show equipment installed on the project and including performance curves and data as applicable (i.e., description of function, normal operating characteristics and limitations, performance curves, engineering data and tests, and complete nomenclature and model numbers of replacement parts).
  - 6. Manufacturer's printed operating procedures to include start-up, break-in, and routine and normal operating instructions; regulation, control, stopping, shutdown, and emergency instructions; and summer and winter operating

- instructions.
  - 7. Maintenance procedures for routine preventative maintenance and troubleshooting; disassembly, repair, and reassembly; aligning and adjusting instructions, servicing instructions and lubrication charts and schedules.
  - 8. Equipment and motor name plate data.
  - 9. Wiring diagrams.
  - 10. Exploded parts views and parts lists for all equipment and devices.
  - 11. Color coding charts for all painted equipment and piping.
  - 12. Location and listing of all spare parts and special keys and tools furnished to the Owner.
  - 13. Furnish recommended lubrication schedule for all required lubrication points with listing of type and approximate amount of lubricant required.
- C. Refer to Division 00 and Division 01 for additional information on Operating and Maintenance Manuals.
  - D. Operating and Maintenance Manuals shall be turned over to the Owner or Engineer for review a minimum of 14 working days prior to the beginning of the operator training period.

#### 1.19 OPERATOR TRAINING

- A. The Contractor shall furnish the services of factory trained specialists to instruct the Owner's operating personnel. The Owner's operator training shall include a minimum of 12 hours of onsite training in three 4 hour shifts.
- B. Before proceeding with the instruction of Owner Personnel, prepare a typed outline in triplicate, listing the subjects that will be covered in this instruction, and submit the outline for review by the Owner. At the conclusion of the instruction period, obtain the signature of each person being instructed on each copy of the reviewed outline to signify that he has a proper understanding of the operation and maintenance of the systems and resubmit the signed outlines.
- C. Refer to other Division 23 Sections for additional Operator Training requirements.

#### 1.20 FINAL COMPLETION

- A. At the completion of the Work, all equipment and systems shall be tested and faulty equipment and material shall be repaired or replaced. Refer to Sections of Division 23 for additional requirements.
- B. Clean and adjust all air distribution devices and replace all air filters immediately prior to Substantial Completion.
- C. Touch up and/or refinish all scratched equipment and devices immediately prior to Substantial Completion.

#### 1.21 CONTRACTOR'S GUARANTEE

- A. Use of the HVAC systems to provide temporary service during construction period will not be allowed without permission from the Owner in writing; and, if granted, shall not cause the warranty period to start, except as defined below.
- B. Contractor shall guarantee to keep the entire installation in repair and perfect working order for a period of one year after the date of the Substantial Completion, and shall

furnish (free of additional cost to the Owner) all materials and labor necessary to comply with the above guarantee throughout the year beginning from the date of Substantial Completion, Beneficial Occupancy by the Owner, or the Certificate of Final Payment as agreed upon by all parties.

- C. This guarantee shall not include cleaning or changing filters except as required by testing, adjusting and balancing.
- D. All air conditioning compressors shall have parts and labor guarantees for a period of not less than 5 years beyond the date of Substantial Completion.
- E. Refer to Sections in Division 23 for additional guarantee or warranty requirements.

#### 1.22 TRANSFER OF ELECTRONIC FILES

- A. Project documents are not intended or represented to be suitable for reuse by Architect/Owner or others on extensions of this project or on any other project. Any such reuse or modification without written verification or adaptation by Engineer, as appropriate for the specific purpose intended, will be at Architect/Owner's risk and without liability or legal exposure to Engineer or its consultants from all claims, damages, losses and expense, including attorney's fees arising out of or resulting thereof.
- B. Because data stored in electronic media format can deteriorate or be modified inadvertently, or otherwise, without authorization of the data's creator, the party receiving the electronic files agrees that it will perform acceptance tests or procedures within sixty (60) days of receipt, after which time the receiving party shall be deemed to have accepted the data thus transferred to be acceptable. Any errors detected within the sixty (60) day acceptance period will be corrected by the party delivering the electronic files. Engineer is not responsible for maintaining documents stored in electronic media format after acceptance by the Architect/Owner.
- C. When transferring documents in electronic media format, Engineer makes no representations as to the long term compatibility, usability or readability of documents resulting from the use of software application packages, operating systems, or computer hardware differing from those used by Engineer at the beginning of the Project.
- D. Any reuse or modifications will be at the Contractor's sole risk and without liability or legal exposure to Architect, Engineer or any consultant.
- E. The Texas Board of Architectural Examiners (TBAE) has stated that it is in violation of Texas law for persons other than the Architect of record to revise the Architectural drawings without the Architect's written consent.
  - 1. It is agreed that "MEP" hard copy or computer-generated documents will not be issued to any other party except directly to the Architect/Owner. The Contract Documents are contractually copyrighted and cannot be used for any other project or purpose except as specifically indicated in AIA B-141 Standard Form of Agreement Between Architect and Owner.
  - 2. If the client, Architect or Owner of the project requires electronic media for "record purposes", then AutoCAD/ Revit documents will be prepared by Engineer on electronic media such as removable memory devices, flash drives or CD's. These documents can also be submitted via file transfer protocols. AutoCAD/ Revit files will be submitted with all title block references intact to permit the end user to only view and plot the drawings. Revisions will not be permitted in this configuration.

3. At the Architect/Owner's request, Engineer will assist the Contractor in the preparation of the submittals and prepare one copy of AutoCAD/ Revit files on electronic media or submit through file transfer protocols. The electronic media will be prepared with all indicia of documents ownership removed. The electronic media will be prepared in a ".rvt" or ".dwg" format to permit the end user to revise the drawings.

## PART 2 - PRODUCTS

### 2.1 MATERIALS

- A. Provide materials and equipment manufactured by a domestic United States manufacturer and assembled in the United States for all local and Federal Government projects. These materials and equipment shall comply with "Buy American Act."
- B. Access Doors: Provide access doors as required for access to equipment, valves, controls, cleanouts and other apparatus where concealed. Access doors shall have concealed hinges and screw driver cam locks.
- C. All access doors located in wet areas such as restrooms, locker rooms, shower rooms, kitchen and any other wet areas shall be constructed of stainless steel.
- D. Access Doors: shall be as follows:
  1. Plaster Surfaces: Milcor Style K.
  2. Ceramic Tile Surface: Milcor Style M.
  3. Drywall Surfaces: Milcor Style DW.
  4. Install doors only in locations approved by the Architect.

### 2.2 EQUIPMENT PADS (See 2.4 in Section 26 02 00)

## PART 3 - EXECUTION

### 3.1 ROUGH-IN

- A. Verify final locations for rough-ins with field measurements and with the requirements of the actual equipment to be connected via reviewed submittals.
- B. Refer to equipment specifications in Divisions 2 through 48 for additional rough-in requirements.

### 3.2 MECHANICAL INSTALLATIONS

- A. General: Sequence, coordinate, and integrate the various elements of mechanical systems, materials, and equipment. Comply with the following requirements:
  1. Coordinate mechanical systems, equipment, and materials installation with other building components.
  2. Verify all dimensions by field measurements.
  3. Arrange for chases, slots, and openings in other building components during progress of construction, to allow for mechanical installations.
  4. Coordinate the installation of required supporting devices and sleeves to be set in poured-in-place concrete and other structural components, as they are constructed.
  5. Sequence, coordinate, and integrate installations of mechanical materials and

- equipment for efficient flow of the Work. Give particular attention to large equipment requiring positioning prior to closing in the building.
6. Where mounting heights are not detailed or dimensioned, install systems, materials, and equipment to provide the maximum headroom possible.
  7. Coordinate connection of mechanical systems with exterior underground and overhead utilities and services. Comply with requirements of governing regulations, franchised service companies, and controlling agencies. Provide required connection for each service.
  8. Install systems, materials, and equipment to conform with architectural action markings on submittal, including coordination drawings, to greatest extent possible. Conform to arrangements indicated by the Contract Documents, recognizing that portions of the Work are shown only in diagrammatic form. Where coordination requirements conflict with individual system requirements, resolve conflicts and submit proposed solution to the Architect for review.
  9. Install systems, materials, and equipment level and plumb, parallel and perpendicular to other building systems and components, where installed exposed in finished spaces.
  10. Install mechanical equipment to facilitate servicing, maintenance, and repair or replacement of equipment components. As much as possible, connect equipment for ease of disconnecting, with minimum of interference with other installations. Extend grease fittings to an accessible location and label.
  11. Install access doors where units are concealed behind finished surfaces. Refer to paragraph 2.1 in this section and architect for access doors specifications and location.
  12. Install systems, materials, and equipment giving right-of-way priority to systems required to be installed at a specified slope.
  13. Provide roof curbs for all roof mounted equipment. Coordinate with roof construction for pitched roof. Provide roof curbs which match the roof slope and provides a level top for equipment installation. Refer to Architectural drawings and details.
  14. The equipment to be furnished under these Specifications shall be essentially the standard product of the manufacturer. Where two or more units of the same class of equipment are required, these units shall be products of a single manufacturer; however, the component parts of the system need not be the product of the same manufacturer.
  15. The Architectural and Structural features of the building and the space limitations shall be considered in selection of all equipment. No equipment shall be furnished which will not suit the arrangement and space limitations indicated.
  16. Lubrication: Prior to start-up, check and properly lubricate all bearings as recommended by the manufacturer.
  17. Where the word "Concealed" is used in these Specifications in connection with insulating, painting, piping, ducts, etc., it shall be understood to mean hidden from sight as in chases, furred spaces or suspended ceilings. "Exposed" shall be understood to mean the opposite of concealed.
  18. Identification of Mechanical Equipment:
    - a. Mechanical equipment shall be identified by means of nameplates permanently attached to the equipment. Nameplates shall be engraved laminated plastic or etched metal. Submittals shall include dimensions and lettering format for approval. Attachment shall be with escutcheon pins, self-tapping screws, or machine screws.
    - b. Tags shall be attached to all valves, including control valves, with nonferrous chain. Tags shall be brass and at least 1-1/2 inches in diameter. Nameplate and tag symbols shall correspond to the identification symbols on the temperature control submittal and the "as-built" drawings.

19. Provide construction filters for all air handling units, fan coil unit, VAV boxes, and all other air handling equipment during the entire construction period.
20. Provide temporary construction strains for all strainers in the hydronic systems during the initial flushing of the systems.

### 3.3 CUTTING AND PATCHING

- A. Protection of Installed Work: During cutting and patching operations, protect adjacent installations.
- B. Perform cutting, fitting, and patching of mechanical equipment and materials required to:
  1. Uncover Work to provide for installation of ill-timed Work.
  2. Remove and replace defective Work.
  3. Remove and replace Work not conforming to requirements of the Contract Documents.
  4. Remove samples of installed Work as specified for testing.
  5. Install equipment and materials in existing structures.
  6. Upon written instructions from the Engineer, uncover and restore Work to provide for Engineer/Owner's observation of concealed Work, without additional cost to the Owner.
  7. Patch existing finished surfaces and building components using new materials matching existing materials and experienced Installers. Patch finished surfaces and building components using new materials specified for the original installation and experienced Installers; refer to the materials and methods required for the surface and building components being patched; Refer to Paragraph 1.11 I for definition of "Installer."
- C. Cut, remove and legally dispose of selected mechanical equipment, components, and materials as indicated, including but not limited to removal of mechanical piping, mechanical ducts and HVAC units, and other mechanical items made obsolete by the new Work.
- D. Protect the structure, furnishings, finishes, and adjacent materials not indicated or scheduled to be removed.
- E. Provide and maintain temporary partitions or dust barriers adequate to prevent the spread of dust and dirt to adjacent areas.

### 3.4 WORK SEQUENCE, TIMING, COORDINATION WITH OWNER, ARCHITECT AND ENGINEER

- A. The Owner will cooperate with the Contractor, however, the following provisions must be observed:
  1. A meeting will be held at the project site, prior to any construction, between the Owner's Representative, the General Contractor, the Sub-Contractors and the Engineer to discuss Contractor's employee parking space, access, storage of equipment or materials, and use of the Owner's facilities or utilities. The Owner's decisions regarding such matters shall be final.
  2. During the construction of this project, normal facility activities will continue in existing buildings until renovated areas are completed. Plumbing, fire protection, lighting, electrical, communications, heating, air conditioning, and ventilation systems shall be maintained in service within the occupied spaces of the existing building.
  3. Contractor shall not start-up any of the HVAC equipment unless the Owner,

- Architect and Engineer are signed off.
4. Start-up for major HVAC equipment such as chillers, cooling towers, variable frequency drives and hot water boilers shall be performed by a factory technician. The start-up shall include a written report signed off by Contractor, Engineer and Owner.

### 3.5 DEMOLITION AND WORK WITHIN EXISTING BUILDINGS

- A. In the preparation of these documents every effort has been made to show the approximate locations of, and connections to, the existing piping, duct, equipment and other apparatus related to this phase of the Work. However, this Contractor shall be responsible for verifying all of the above information. This Contractor shall visit the existing site to inspect the facilities and related areas. This Contractor shall inspect and verify all details and requirements of all the Contract Documents, prior to the submission of a proposal. All discrepancies between the Contract Documents and actual job-site conditions shall be resolved by the contractor, who shall produce drawings that shall be submitted to the Architect/Engineer for review. All labor and materials required to perform the work described shall be a part of this Contract.
- B. All equipment and/or systems noted on the Drawings "To Remain" shall be inspected and tested on site to certify its working condition. A written report on the condition of all equipment to remain, including a copy of the test results and recommended remedial actions and costs shall be made by this Contractor to the Architect/Engineer for review.
- C. All equipment and/or systems noted on the Drawings "To Be Removed" shall be removed including, associated pipe and duct, pipe and duct hangers and/or line supports. Where duct or pipe is to be capped for future or end of line use, it shall be properly tagged with its function or service appropriately identified. Where existing equipment is to be removed or relocated and has an electric motor or connection, the Electrical Contractor shall disconnect motor or connection, remove wiring to a safe point and this Contractor shall remove or relocate motor or connection along with the equipment.
- D. During construction and remodeling, portions of the Project shall remain in service. Construction equipment, material, tools, extension cords, etc., shall be arranged so as to present minimum hazard or interruption to the occupants of the building. None of the construction work shall interfere with the proper operation of the existing facility; or be so conducted as to cause harm or danger to persons on the premises. All fire exits, stairs or corridors required for proper access, circulation or exit shall remain clear of equipment, materials or debris. The General Contractor shall maintain barricades, other separations in corridors and other spaces where work is conducted.
- E. Certain work during the demolition and construction phases may require overtime or night time shifts or temporary evacuation of the occupants. Coordinate and schedule all proposed down time with the Owner at least seventy-two (72) hours in advance in writing.
- F. Any salvageable equipment as determined by the Owner, shall be delivered to the Owner, and placed in storage at the location of his choice. All other debris shall be removed from the site immediately.
- G. Equipment, piping or other potential hazards to the occupants of the building shall not be left overnight outside of the designated working or construction area.
- H. Make every effort to minimize damage to the existing building and the Owner's property. Repair, patch or replace as required any damage that occurs as a result of work at the site. Care shall be taken to minimize interference with the Owner's activities during



construction and to keep construction disrupted areas to a minimum. Coordinate with the Owner and other trades in scheduling and performance of the work.

- I. Include in the contract price all rerouting of existing pipe, duct, etc., and the reconnecting of the existing equipment as necessitated by field conditions to allow the installation of the new systems regardless of whether or not such rerouting, reconnecting or relocating is shown on the Drawings. Furnish all temporary pipe, duct, controls, etc., as required to maintain heating, cooling, and ventilation services for the existing areas with a minimum of interruption.
- J. All existing pipe, duct, materials, equipment, controls and appurtenances not included in the remodel or alteration areas are to remain in place.
- K. Pipe, duct, equipment and controls serving mechanical and other Owner's equipment, etc., which is to remain but is served by pipe, duct, equipment and controls that are disturbed by the remodeling work, shall be reconnected in such a manner as to leave this equipment in proper operating condition.
- L. No portion of the **fire protection systems** shall be turned off, modified or changed in any way without the express knowledge and written permission of the Owner's representative in order to protect systems that shall remain in service.
- M. It is the intention of this Section of the Specifications to outline minimum requirements to furnish the Owner with a turn-key and operating system in cooperation with other trades with a minimum of disruption or downtime.
- N. Refer to Architectural Demolition and/or Alteration plans for actual location of walls, ceilings, etc., being removed and/or remodeled.

#### END OF SECTION

## SECTION 23 02 01 - COORDINATION DRAWINGS

### PART 1 - GENERAL

#### 1.1 GENERAL REQUIREMENTS

- A. The requirements of the General Conditions 013100 and Supplementary Conditions apply to all Work herein.

#### 1.2 COORDINATION DRAWINGS

- A. The Contractor shall take the lead in coordinating the Mechanical, Electrical, Plumbing, Communications, Electronic Safety/Security and Fire Protection systems within the building.
- B. The Mechanical Contractor shall coordinate a three-dimensional (3D) model of the building which includes the Mechanical, Electrical, Plumbing, and Fire Protection systems. The Electrical, Plumbing, and Fire Protection Contractors shall prepare their work and generate 3D models which will be given to the Mechanical Contractor for coordination. The Contractor will be provided with the REVIT model that was used to generate the contract documents, this file may be used as the background file. The Contractor shall replace the systems drawn with the actual shop drawing models. The Contractor is not limited to using REVIT, but may use any 3-D software in generating and combining the coordination model.
- C. Submitting the contract drawings as coordination drawings will not be acceptable.
- D. The model shall include detailed and accurate representations of all equipment to be installed based upon the reviewed equipment submittals.
- E. The Mechanical Contractor shall hold a 3-D coordination meeting with all sub-contractors present to review the model and discuss coordination of the installation of the building systems.
- F. Upon completion of the coordination meeting, the Contractor shall submit the 3-D model and 1/4" scale drawings for review.
- G. The model shall detail major elements, components, and systems in relationship with other systems, installations, and building components. Indicate locations where space is limited for installation and access and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:
  - 1. Indicate the proposed locations of pipe, duct, equipment, and other materials. Include the following:
    - a. Wall and type locations.
    - b. Clearances for installing and maintaining insulation.
    - c. Locations of light fixtures and sprinkler heads.
    - d. Clearances for servicing and maintaining equipment, including tube removal, filter removal, and space for equipment disassembly required for periodic maintenance.
    - e. Equipment connections and support details.
    - f. Exterior wall and foundation penetrations.
    - g. Routing of storm and sanitary sewer piping.
    - h. Fire-rated wall and floor penetrations.
    - i. Sizes and location of required concrete pads and bases.
    - j. Valve stem movement.
    - k. Structural floor, wall and roof opening sizes and details.

2. Indicate scheduling, sequencing, movement, and positioning of large equipment into the building during construction.
  3. Prepare floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
  4. Prepare reflected ceiling plans to coordinate and integrate installations, air distribution devices, light fixtures, communication systems components, and other ceiling-mounted items.
- H. Sequence of Coordination
- Below is hierarchy of model elements and the sequencing by which the models will be coordinated.
1. Structural and Architectural model
  2. Miscellaneous steel
  3. Perform preliminary space allocation
  4. Identify hard constraints (locations of access panels, lights, A/V space requirements, etc.)
  5. Main and medium pressure ducts from the shaft out
  6. Main graded plumbing lines and vents
  7. Sprinkler mains and branches
  8. Cold and hot water mains and branches
  9. Lighting fixtures and plumbing fixtures
  10. Smaller sized ducts and flex ducts
  11. Smaller size cold water and hot water piping, flex ducts, etc.
- I. The Contractor and Sub-Contractors shall not install any item until the coordination has been completed and reviewed by the Construction Manager, Owner, and A/E team.
- J. This Contractor shall be responsible for coordination of all items that will affect the installation of the work of this Division. This coordination shall include, but not be limited to: voltage, ampacity, capacity, electrical and piping connections, space requirements, sequence of construction, building requirements and special conditions.
- K. By submitting shop drawings on the project, this Contractor is indicating that all necessary coordination has been completed and that the systems, products and equipment submitted can be installed in the building and will operate as specified and intended, in full coordination with all other Contractors and Subcontractors

#### **END OF SECTION**

## **SECTION 23 05 29 – HANGERS AND SUPPORTS FOR PIPING AND EQUIPMENT - HVAC**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 - Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 WORK INCLUDED**

- A. Pipe, and equipment hangers, supports and associated anchors.
- B. Sleeves and seals.
- C. Flashing and sealing equipment and pipe stacks.

#### **1.3 RELATED WORK**

- A. Section 21 00 00 – Fire Suppression.
- B. Section 22 10 00 – Plumbing Piping and Pumps.
- C. Section 23 05 48 – Vibration and Seismic Controls for HVAC Piping and Equipment.
- D. Section 23 07 16 – HVAC Equipment Insulation.
- E. Section 23 07 19 – HVAC Piping Insulation.
- F. Section 23 21 13 – Above Ground Hydronic Piping.
- G. Section 23 21 16 – Underground Hydronic Piping.

#### **1.4 REFERENCES**

- A. ANSI/ASME B31.1 - Power Piping.
- B. NFPA 13 - Standard for the Installation of Sprinkler Systems.
- C. NFPA 14 - Standard for the Installation of Standpipe and Hose Systems.

#### **1.5 QUALITY ASSURANCE**

- A. Supports for Sprinkler Piping: In conformance with NFPA 13.
- B. Supports for Standpipes: In conformance with NFPA 14.

#### **1.6 SUBMITTALS**

- A. Submit shop drawings and product data under provisions of Division One.
- B. Indicate hanger and support framing and attachment methods.

## PART 2 - PRODUCTS

### 2.1 PIPE HANGERS AND SUPPORTS

- A. Hangers for Pipes Sizes 1/2 to 1-1/2 Inch: Malleable iron, adjustable swivel, split ring.
- B. Hangers for Pipes Sizes 2 to 4 Inch: Carbon steel, adjustable clevis.
- C. Hangers for Pipes Sizes 6 Inches and Over: Adjustable steel yoke, cast iron roller, double hanger.
- D. Multiple or Trapeze Hangers: Steel channels with welded spacers, pre-formed manufactured saddles and hanger rods; cast iron roller and stand for pipe sizes 6 inches and over.
- E. Wall Support for Pipe Sizes to 3 Inches: Cast iron hook.
- F. Wall Support for Pipe Sizes 4 Inches and over: adjustable steel yoke and cast iron roller.
- G. Vertical Support: Steel riser clamp.
- H. Floor Support for Pipe Sizes to 4 Inches: Cast iron adjustable pipe saddle, locknut nipple, floor flange, and concrete pier or steel support.
- I. Floor Support for Pipe Sizes 6 Inches and Over: Adjustable cast iron roller and stand, steel screws, and concrete pier or steel support.
- J. Roof Pipe Supports and Hangers: Galvanized Steel Channel System as manufactured by Portable Pipe Hangers, Inc. or approved equal.
  - 1. For pipes 2-1/2" and smaller – Type PP10 with roller
  - 2. For pipes 3" through 8" – Type PS
  - 3. For multiple pipes – Type PSE - Custom
- K. Copper Pipe Support and Hangers: Electro-galvanized with thermoplastic elastomer cushions; Unistrut "Cush-A-Clamp" or equal. Hangers: Plastic coated; Unistrut or equal.
- L. Shields for Vertical Copper Pipe Risers: Sheet lead.
- M. Pipe Rough-In Supports in Walls/Chases: Provide preformed plastic pipe supports, Sioux Chief "Pipe Titan" or equal.

### 2.2 HANGER RODS

- A. Galvanized Hanger Rods: Threaded both ends, threaded one end, or continuous threaded.

### 2.3 INSERTS

- A. Inserts: Malleable iron case with galvanized steel shell and expander plug for threaded connection with lateral adjustment, top slot for reinforcing rods, lugs for attaching to forms; size inserts to suit threaded hanger rods.

### 2.4 FLASHING

- A. Metal Flashing: 20 gage galvanized steel.

- B. Lead Flashing: 4 lb. /sq. ft. sheet lead for waterproofing; 1 lb. /sq. ft. sheet lead for soundproofing.
- C. Caps: Steel, 20 gage minimum; 16 gage at fire resistant elements.
- D. Coordinate with roofing contractor/Architect for type of flashing on metal roofs.

## 2.5 EQUIPMENT CURBS

- A. Fabricate curbs of hot dipped galvanized steel.
- B. For metal roof construction, roof curbs shall be made of aluminum or stainless steel. Coordinate with Architectural Drawings and details.

## 2.6 SLEEVES

- A. Sleeves for Pipes through Non-fire Rated Floors: Form with 18 gage galvanized steel, tack welded to form a uniform sleeve.
- B. Sleeves for Pipes through Non-fire Rated Beams, Walls, Footings, and Potentially Wet Floors: Form with steel pipe, Schedule 40.
- C. Sleeves for Pipes through Fire Rated and Fire Resistive Floors and Walls, and Fireproofing: Prefabricated fire rated steel sleeves including seals, UL listed.
- D. Sleeves for Round Ductwork: Form with galvanized steel.
- E. Sleeves for Rectangular Ductwork: Form with galvanized steel.
- F. Fire Stopping Insulation: Glass fiber type, non-combustible, U.L. listed.
- G. Caulk: Paintable 25-year acrylic sealant.
- H. Pipe Alignment Guides: Factory fabricated, of cast semi-steel or heavy fabricated steel, consisting of bolted, two-section outer cylinder and base with two-section guiding spider that bolts tightly to pipe. Length of guides shall be as recommended by manufacturer to allow indicated travel.

## 2.7 FABRICATION

- A. Size sleeves large enough to allow for movement due to expansion and contraction. Provide for continuous insulation wrapping.
- B. Design hangers without disengagement of supported pipe.
- C. Design roof supports without roof penetrations, flashing or damage to the roofing material.

## 2.8 FINISH

- A. Prime coat exposed steel hangers and supports. Hangers and supports located in crawl spaces, pipe shafts, and suspended ceiling spaces are not considered exposed.

## PART 3 - EXECUTION

### 3.1 INSERTS

- A. Provide inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams. Coordinate with Structural Engineer for placement of inserts.
- B. Provide hooked rod to concrete reinforcement section for inserts carrying pipe over 4 inches.
- C. Where concrete slabs form finished ceiling, provide inserts to be flush with slab surface.
- D. Where inserts are omitted, drill through concrete slab from below and provide thru-bolt with recessed square steel plate and nut recessed into and grouted flush with slab. Verify with Structural Engineer prior to start of work.

### 3.2 PIPE HANGERS AND SUPPORTS

- A. Support horizontal piping as follows:

| <u>PIPE SIZE</u>    | <u>MAX. HANGER SPACING</u> | <u>HANGER DIAMETER</u> |
|---------------------|----------------------------|------------------------|
| (Steel Pipe)        |                            |                        |
| 1/2 to 1-1/4 inch   | 7'-0"                      | 3/8"                   |
| 1-1/2 to 3 inch     | 10'-0"                     | 3/8"                   |
| 4 to 6 inch         | 10'-0"                     | 1/2"                   |
| 8 to 10 inch        | 10'-0"                     | 5/8"                   |
| 12 to 14 inch       | 10'-0"                     | 3/4"                   |
| 15 inch and over    | 10'-0"                     | 7/8"                   |
| (Copper Pipe)       |                            |                        |
| 1/2 to 1-1/4 inch   | 5'-0"                      | 3/8"                   |
| 1-1/2 to 2-1/2 inch | 8'-0"                      | 3/8"                   |
| 3 to 4 inch         | 10'-0"                     | 3/8"                   |
| 6 to 8 inch         | 10'-0"                     | 1/2"                   |
| (Cast Iron)         |                            |                        |
| 2 to 3 inch         | 5'-0"                      | 3/8"                   |
| 4 to 6 inch         | 10'-0"                     | 1/2"                   |
| 8 to 10 inch        | 10'-0"                     | 5/8"                   |
| 12 to 14 inch       | 10'-0"                     | 3/4"                   |
| 15 inch and over    | 10'-0"                     | 7/8"                   |
| (PVC Pipe)          |                            |                        |
| 1-1/2 to 4 inch     | 4'-0"                      | 3/8"                   |
| 6 to 8 inch         | 4'-0"                      | 1/2"                   |

10 and over                      4'-0"                      5/8"

- B. Install hangers to provide minimum 1/2 inch space between finished covering and adjacent work.
- C. Place a hanger within 12 inches of each horizontal elbow, and at the vertical to horizontal transition.
- D. Use hangers with 1-1/2 inch minimum vertical adjustment.
- E. Support horizontal cast iron pipe adjacent to each hub, with 5 feet maximum spacing between hangers.
- F. Support vertical piping at every floor. Support vertical cast iron pipe at each floor at hub.
- G. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers.
- H. Support riser piping independently of connected horizontal piping.
- I. Install hangers with nut at base and above hanger; tighten upper nut to hanger after final installation adjustments.
- J. Portable pipe hanger systems shall be installed per manufacturer's instructions.
- K. Distances between supports are maximum distance. Supports shall be provided to carry the pipe/equipment load.

3.3 INSULATED PIPING: Comply with the following installation requirements.

- A. Clamps: Attach galvanized clamps, including spacers (if any), to piping with clamps projecting through insulation; do not exceed pipe stresses allowed by ASME B31.9.
- B. Saddles: Install galvanized protection saddles MSS Type 39 where insulation without vapor barrier is indicated. Fill interior voids with segments of insulation that match adjoining pipe insulation. Secure the full contact area of the saddle to the pipe insulation with 1/8" thick coat of mastic.
- C. Shields: Install protective shields MSS Type 40 on cold and chilled water piping that has vapor barrier. Secure the full contact area of the shield to the pipe insulation with 1/8" thick coat of mastic.
- D. Shields shall span an arc of 180 degrees and shall have dimensions in inches not less than the following:

| NPS               | LENGTH | THICKNESS |
|-------------------|--------|-----------|
| 1/4 THROUGH 3-1/2 | 12     | 0.048     |
| 4                 | 12     | 0.060     |
| 5 & 6             | 18     | 0.060     |
| 8 THROUGH 14      | 24     | 0.075     |



- E. Piping 2" and larger: provide galvanized sheet metal shields with calcium silicate insulation at hangers/supports.
- F. Insert material shall be at least as long as the protective shield.
- G. Thermal Hanger Shields: Install where indicated, with insulation of same thickness as piping.

### 3.4 EQUIPMENT BASES AND SUPPORTS

- A. Provide equipment bases of concrete.
- B. Provide templates, anchor bolts, and accessories for mounting and anchoring equipment.
- C. Construct support of steel members. Brace and fasten with flanges bolted to structure.
- D. Provide rigid anchors for pipes after vibration isolation components are installed.

### 3.5 FLASHING

- A. Provide flexible flashing and metal counter flashing where piping and ductwork penetrate weather or waterproofed walls, floors, and roofs.
- B. Flash vent and soil pipes projecting 8 inches minimum above finished roof surface with lead worked one inch minimum into hub, 8 inches minimum clear on sides with 24 x 24 inches sheet size. For pipes through outside walls, turn flanges back into wall and caulk. Provide metal counter flash and seal.
- C. Flash floor drains in floors with topping over finished areas with lead, 10 inches clear on sides with minimum 36 x 36 inch sheet size. Fasten flashing to drain clamp device.
- D. Seal floor, shower, mop sink, and all other drains watertight to adjacent materials.
- E. Provide curbs for mechanical roof installations 8 inches minimum high above roofing surface. Contact Architect for all flashing details and roof construction. Seal penetrations watertight.

### 3.6 SLEEVES

- A. Set sleeves in position in formwork. Provide reinforcing around sleeves.
- B. Extend sleeves through floors minimum one inch above finished floor level. Caulk sleeves full depth with fire rated thermafiber and 3M caulking and provide floor plate.
- C. Where piping or ductwork penetrates floor, ceiling, or wall, close off space between pipe or duct and adjacent work with U.L. listed fire stopping insulation and caulk seal air tight. Provide close fitting metal collar or escutcheon covers at both sides of penetration.
- D. Fire protection sleeves may be flush with floor of stairways.

### END OF SECTION

## **SECTION 23 05 48 – VIBRATION AND SEISMIC CONTROLS FOR HVAC PIPING AND EQUIPMENT**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. This Section and Section 23 02 00 – Basic Materials and Methods are part of each Division 23 Section which references the vibration control products specified herein.

#### **1.2 WORK INCLUDED**

- A. Vibration and sound control products.

#### **1.3 QUALITY ASSURANCE**

- A. Manufacturer's Qualifications: Firms regularly engaged in manufacture of vibration control products of type, size, and capacity required, whose products have been in satisfactory use in similar service for not less than 5 years.
- B. Vibration and sound control products shall conform to ASHRAE criteria for average noise criteria curves for all equipment at full load conditions.
- C. Unless otherwise indicated, sound and vibration control products shall be provided by a single manufacturer.

#### **1.4 SUBMITTALS**

- A. SHOP DRAWINGS: Indicate size, material, and finish. Show locations and installation procedures. Include details of joints, attachments, and clearances.
- B. PRODUCT DATA: Submit schedules, charts, literature, and illustrations to indicate the performance, fabrication procedures, product variations, and accessories.

### **PART 2 - PRODUCTS**

#### **2.1 ACCEPTABLE MANUFACTURERS**

- A. Amber/Booth Company, Inc.
- B. Mason Industries, Inc.
- C. Kinetics Noise Control, Inc.

#### **2.2 GENERAL**

- A. Provide vibration isolation supports for equipment, piping and ductwork, to prevent transmission of vibration and noise to the building structure that may cause discomfort to the occupants.
- B. Model numbers of Amber/Booth products are included for identification. Products of the additional manufacturers will be acceptable provided they comply with all the requirements of this specification.

## 2.3 FLOOR MOUNTED AIR HANDLING UNITS

- A. Provide Amber/Booth CAL-2 aluminum housed isolators sized for 2" static deflection. Cast iron or steel housings may be used provided they are hot-dip galvanized after fabrication
- B. If floor mounted air handling units are furnished with internal vibration isolation option, provide 2" thick Amber/Booth NRC ribbed neoprene pads to address high frequency breakout and afford additional unit elevation for condensate drains. Ribbed neoprene pads shall be located in accordance with the air handling unit manufacturer's recommendations.

## 2.4 SUSPENDED AIR HANDLING UNITS

- A. Provide Amber/Booth HRS-2 combination spring and rubber-in-shear isolation hanger sized for 2" static deflection.
- B. If suspended air handling units are furnished with internal vibration isolation option, furnish Amber/Booth HR rubber-in-shear isolation hangers sized for approximately ½" deflection to address high frequency break-out.

## 2.5 SUSPENDED FANS AND FAN COIL UNITS

- A. Provide Amber/Booth HS spring hangers sized for 1" static deflection.

## 2.6 PIPING

- A. Provide spring and rubber-in-shear hangers, Amber/Booth HRS in mechanical equipment rooms, for a minimum distance of 50 feet from isolated equipment for all chilled water and hot water piping 1-1/2" diameter and larger. Springs shall be sized for 1" deflection.
- B. Floor supported piping is required to be isolated with Amber/Booth AW-1 open springs sized for 1" deflection.
- C. All condenser water piping shall be supported with Amber/Booth AW-1 with 1" deflection for floor or roof mounted piping and Amber/Booth HRS isolators with 1" deflectors for suspended piping.
- D. Furnish line size flexible connectors at supply and return of pumps, Amber/Booth style 2800 single sphere EPDM construction, connector shall include 150 lb. cadmium plated carbon steel floating flanges.

## 2.09 CORROSION PROTECTION

- A. All vibration isolators shall be designed and treated for resistance to corrosion.
- B. Steel components: PVC coated or phosphate coated and painted with industrial grade enamel. Nuts, bolts, and washers: zinc-electroplated.

## PART 3 - EXECUTION

- 3.1 All equipment shall be installed in accordance with the manufacturer's recommendations and printed installation instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Provide all items required as per manufacturer's requirements.

- 3.3 If internal isolation option is used on air handling units, the mechanical contractor shall verify proper adjustment and operation of isolators prior to start-up. All shipping brackets and temporary restraint devices shall be removed.
- 3.4 The vibration isolation supplier shall certify in writing that he has inspected the installation and that all external isolation materials and devices are installed correctly and functioning properly.

**END OF SECTION**

## **SECTION 23 05 53 – IDENTIFICATION FOR HVAC PIPING AND EQUIPMENT**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 – Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for owner's use.

- 1.3 Refer to Architectural Sections for additional requirements.

### **PART 2 - PRODUCTS**

#### **2.1 VALVE AND PIPE IDENTIFICATION**

##### **A. Valves:**

- 1. All valves shall be identified with a 1-1/2" diameter brass disc wired onto the handle. The disc shall be stamped with 1/2" high depressed black filled identifying numbers. These numbers shall be numerically sequenced for all valves on the job.
- 2. The number and description indicating make, size, model number and service of each valve shall be listed in proper operational sequence, properly typewritten. Three copies to be turned over to Owner at completion.
- 3. Tags shall be fastened with approved meter seal and 4 ply 0.018 smooth copper wire. Tags and fastenings shall be manufactured by the Seton Name Plate Company or approved equal.
- 4. All valves shall be numbered serially with all valves of any one system and/or trade grouped together.

##### **B. Pipe Marking:**

- 1. All interior visible piping located in accessible spaces such as above accessible ceilings, equipment rooms, attic space, under floor spaces, etc., shall be identified with all temperature pipe markers as manufactured by W.H. Brady Company, 431 West Rock Ave., New Haven, Connecticut, or approved equal.
- 2. All exterior visible piping shall be identified with UV and acid resistant outdoor grade acrylic plastic markers as manufactured by Set Mark distributed by Seton (Name plate Company Factory location 20 Thompson Road, Branford, Connecticut) or approved equal.
- 3. Generally, markers shall be located on each side of each and every partition, on each side of every tee, on each side of every valve and/or valve group, on each side of every piece of equipment, and, for straight runs, at equally spaced intervals not to exceed 75 feet. In congested area, marks shall be placed on each pipe at the points where it enters and leaves the area and at the point of connection of each piece of equipment and automatic control valve. All markers shall have directional arrows.
- 4. Markers shall be installed after final painting of all piping and equipment and in

such a manner that they are visible from the normal maintenance position. Manufacturer's installation instructions shall be closely followed.

5. Markers shall be colored as indicated below per ANSI/OSHA Standards:

| <u>SYSTEM</u>     | <u>COLOR</u>   | <u>LEGEND</u>                                          |
|-------------------|----------------|--------------------------------------------------------|
| Chilled Water     | Green          | Chilled Water Supply<br>Chilled Water Return           |
| Hot Water         | Reddish Orange | Hot Water Supply<br>Return                             |
| Condenser Water   | Green          | Condenser Water<br>Supply<br>Condenser Water<br>Return |
| Compressed Air    | Blue           | Compressed Air                                         |
| Pneumatic Control | Yellow         | Pneumatic Controls                                     |
| Oxygen            | Yellow         | Oxygen                                                 |
| Nitrogen          | Green          | Nitrogen                                               |
| Deionized Water   | Green          | Deionized Water                                        |
| Steam             | Yellow         | Steam Supply<br>Steam Return                           |

C. Pipe Painting:

1. All piping exposed to view shall be painted as indicated or as directed by the Architect in the field. Confirm all color selections with Architect prior to installation.
2. All piping located in mechanical rooms and exterior piping shall be painted as indicated below:

| <u>System</u>                       | <u>Color</u>   |
|-------------------------------------|----------------|
| Condenser Water Supply and Return   | Light Green    |
| Chilled Water Supply and Return     | Light Blue     |
| Heating Hot Water Supply and Return | Reddish Orange |

## 2.2 EQUIPMENT IDENTIFICATION

- A. Mechanical equipment shall be identified by means of nameplates permanently attached to the equipment. Nameplates shall be engraved laminated plastic or etched metal. Submittals shall include dimensions and lettering format for approval. Attachment shall be with escutcheon pins, self-tapping screws, or machine screws.

## PART 3 - EXECUTION

- 3.1 All labeling equipment shall be installed as per manufacturer's printed installation instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Contractors price shall include all items required as per manufacturer's requirements.
- 3.3 All piping shall be cleaned of rust, dirt, oil and all other contaminants prior to painting. Refer to

Division 9 for Architect's required paint system(s).

**END OF SECTION**

## **SECTION 23 05 93 - TESTING, ADJUSTING, AND BALANCING**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 – Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 RELATED DOCUMENTS**

Approved submittal date on equipment installed, to accomplish the test procedures, outlined under paragraph 3.1 of this Section, will be provided by the Contractor.

#### **1.3 DESCRIPTION**

- A. The TAB of the air conditioning systems shall be performed by an impartial technical firm hired by Owner whose operations are limited only to the field of professional TAB. The TAB work will be done under the direct supervision of a qualified engineer employed by the TAB firm.
- B. The TAB firm will be responsible for inspecting, adjusting, balancing, and logging the data on the performance of fans, dampers in the duct system, and air distribution devices. The Contractor and the various Subcontractors of the equipment installed shall cooperate with the TAB firm to furnish necessary data on the design and proper applications of the system components and provide labor and material required to eliminate deficiencies or malperformance.

#### **1.4 QUALITY ASSURANCE**

- A. **QUALIFICATIONS OF CONTRACTOR PERSONNEL:** Submit evidence to show that the personnel who shall be in charge of correcting deficiencies for balancing the systems are qualified. The Owner and Engineer reserve the right to require that the originally approved personnel be replaced with other qualified personnel if, in the Owner and Engineer's opinion, the original personnel are not qualified to properly place the system in condition for balancing.
- B. **QUALIFICATIONS OF TAB FIRM PERSONNEL:**
  - 1. A minimum of one registered Professional Engineer licensed in the State, is required to be in permanent employment of the firm.
  - 2. Personnel used on the jobsite shall be either Professional Engineers or technicians, who shall have been permanent, full time employees of the firm for a minimum of six months prior to the start of Work for that specified project.
  - 3. Evidence shall be submitted to show that the personnel who actually balance the systems are qualified. Evidence showing that the personnel have passed the tests required by the Associated Air Balance Council (AABC) shall be required.
- C. **CALIBRATION LIST:** Submit to the Engineer for approval, a list of the gauges, thermometers, velometer, and other balancing devices to be used in balancing the system. Submit evidence to show that the balancing devices are properly calibrated before proceeding with system balancing.



## 1.5 OPERATIONS PERSONNEL TRAINING

- A. Provide a training session for the owner's operations personnel. Training session shall be performed by a qualified person who is knowledgeable in the subject system/equipment. Submit a training agenda two (2) weeks prior to the proposed training session for review and approval. Training session shall include at the minimum:
  - 1. Purpose of equipment.
  - 2. Principle of how the equipment works.
  - 3. Important parts and assemblies.
  - 4. How the equipment achieves its purpose and necessary operating conditions.
  - 5. Most likely failure modes, causes and corrections.
  - 6. On site demonstration.

## PART 2 - PRODUCTS (NOT USED)

## PART 3 - EXECUTION

### 3.1 SERVICES OF THE CONTRACTOR

- A. The Drawings and Specifications have indicated valves, dampers, and miscellaneous adjustment devices for the purpose of adjustment to obtain optimum operating conditions. Install these devices in a manner that leaves them accessible, and provide access as requested by the TAB firm.
- B. Have systems complete and in operational readiness prior to notifying the TAB firm that the project is ready for their services, and certify in writing to the Architect and Owner that such a condition exists.
- C. As a part of the Work of this Section, make changes in the sheaves, belts, and dampers or the addition of dampers required for correct balance of the new work as required by the TAB firm, at no additional cost to the Owner.
- D. Fully examine the existing system to be balanced, to determine whether or not sufficient volume dampers, balancing valves, thermometers, gauges, pressure and temperature taps, means of reading static pressure and total pressure in duct systems, means of determining water flow, and other means of taking data needed for proper water and air balancing are existing. Submit to the Engineer in writing a listing of omitted items considered necessary to balance existing systems. Submit the list and proposal as a cost add item.
- E. Verify that fresh air louvers are free of blockage, coils are clean and fresh air ducts to each air handling unit have individually adjustable volume regulating dampers.
- F. Provide, correct, repair, or replace deficient items or conditions found during the testing, adjusting, and balancing period.
- G. In order that systems may be properly tested, balanced, and adjusted as specified, operate the systems at no expense to the Owner for the length of time necessary to properly verify their completion and readiness for TAB period.
- H. Project construction schedules shall provide time to permit the successful completion of TAB services prior to Substantial Completion. Complete, operational readiness, prior to commencement of TAB services, shall include the following services of the Contractor:

1. Construction status of building shall permit the closing of doors, windows, ceilings installed and penetrations complete, to obtain project operating conditions.
2. AIR DISTRIBUTION SYSTEMS:
  - a. Verify installation for conformity to design. Supply, return, and exhaust ducts terminated and pressure tested for leakage as specified.
  - b. Volume and fire dampers properly located and functional. Dampers serving requirements of minimum and maximum outside air, return and relief shall provide tight closure and full opening, smooth and free operation.
  - c. Supply, return, exhaust and transfer grilles, registers and diffusers shall be installed.
  - d. Air handling systems, units and associated apparatus, such as heating and cooling coils, filter sections, access doors, etc., shall be blanked and sealed to eliminate excessive bypass or leakage of air.
  - e. Fans (supply and exhaust) operating and verified for freedom from vibrations, proper fan rotation and belt tension; overload heater elements shall be of proper size and rating; record motor amperage and voltage and verify that these functions do not exceed nameplate ratings.
  - f. Furnish or revise fan drives or motors as necessary to attain the specified air volumes.
3. WATER CIRCULATING SYSTEMS:
  - a. Position valves pertinent to system design and require operation to permit full flow of water through system components. Operate hydronic systems under full flow conditions until circulating water is clean. Remove and clean strainers as required during this cycle of operation.
  - b. For retrofit projects, record each existing pump motor amperage and voltage. Readings shall not exceed nameplate rating.
  - c. Verify, on new equipment, electrical starter overload heater elements to be of proper size and rating.
  - d. Ensure that water circulating systems shall be full of water and free of air; expansion tanks set for proper water level, and air vents installed at high points of systems and operating freely. Advise Engineer of deficiencies.
  - e. Check and set operating temperatures of heat exchangers to design requirements.
  - f. The various existing water circulating systems shall be cleaned, filled, purged of air, and put into operation before hydronic balancing.
4. AUTOMATIC CONTROLS:
  - a. Verify that control components are installed in accordance with project documents and functional, electrical interlocks, damper sequences, air and water resets, fire and freeze stats.
  - b. Controlling instruments shall be functional and set for design operating conditions. Factory precalibration of room thermostats and pneumatic equipment will not be acceptable.
  - c. The temperature regulation shall be adjusted for proper relationship between the controlling instruments and calibrated by the TAB Contractor. Advise Engineer of deficiencies or malfunctions.

- I. Contractor shall repair any insulation removed from piping system by TAB Contractor during water balancing.

### 3.2 SERVICES OF THE TAB FIRM

- A. The TAB firm will act as liaison between the Owner, Engineer, and the Contractor and inspect the installation of mechanical piping system, sheet metal work, temperature

controls and other component parts of the heating, air conditioning and ventilating systems being retrofitted, repaired, or added under this Contract. The reinspection of the Work will cover that part related to proper arrangement and adequate provision for the testing and balancing and will be done when the Work is 80 percent complete.

- B. Upon completion of the installation and start-up of the mechanical equipment, to check, adjust, and balance system components to obtain optimum conditions in each conditioned space in the building. Prepare and submit to the Engineer complete reports on the balance and operations of the systems.
- C. Measurements and recorded readings of air, water, and electricity that appear in the reports will be done by the permanently employed technicians or engineers of the TAB firm.
- D. Make an inspection in the building during the opposite season from that in which the initial adjustments were made. At the time, make necessary modifications to the initial adjustments required to produce optimum operation of system components to affect the proper conditions as indicated on the Drawings. At time of opposite season check-out, the Owner's representative will be notified before readings or adjustments are made.
- E. In fan systems, the air quantities indicated on the Drawings may be varied as required to secure a maximum temperature variation of two degrees within each separately controlled space, but the total air quantity indicated for each zone must be obtained. It shall be the obligation of the Contractor to furnish or revise fan drive and motors if necessary, without cost to the Owner, to attain the specified air volumes.
- F. Contractor shall utilize ultrasonic flow meter to balance water flow of existing water system if the original pressure drop data is not available. Contractor shall remove insulation as necessary to use flow meter.

### 3.3 PROFESSIONAL REPORT

- A. Before the final acceptance of the report is made, the TAB firm will furnish the Engineer the following data to be approved by the Owner and Engineer:
  - 1. Summary of main supply, return and exhaust duct pitot tube traverses and fan settings indicating minimum value required to achieve specified air volumes.
  - 2. A listing of the measured air quantities at each outlet corresponding to the temperature tabulation as developed by the Engineer and TAB firm.
  - 3. Air quantities at each return and exhaust air handling device.
  - 4. Static pressure readings entering and leaving each supply fan, exhaust fan, filter, coil, balancing dampers and other components of the systems. Including the retrofit Work. These readings will be related to performance curves in terms of the CFM handled if available.
  - 5. Motor current readings at each equipment motor on load side of capacitors. The voltages at the time of the reading shall be listed.
  - 6. The final report shall certify test methods and instrumentation used, final velocity reading obtained, temperatures, pressure drops, RPM of equipment, amperage of motors, air balancing problems encountered, recommendations and uncompleted punch list items. The test results will be recorded on standard forms.
  - 7. A summary of actual operating conditions shall be included with each system outlining normal and ventilation cycles of operation. the final report will act as a reference of actual operating conditions for the Owner's operating personnel.

### 3.4 BALANCING AIR CONDITIONING SYSTEM

A. GENERAL:

1. Place all equipment into full operation, and continue operating during each working day of balancing and testing. If the air conditioning system is balanced during Off-Peak cooling season Contractor shall return to rebalance air side system as required to put system in proper balance at that season.
2. The Contractor shall submit detailed balancing and recording forms for approval. After approval by the Engineer, prepare complete set of forms for recording test data on each system. All Work shall be done under the supervision of a Registered Professional Engineer. All instruments used shall be accurately calibrated to within 1% of scale and maintained in good working order.
3. Upon completion of the balancing and testing, the TAB Contractor shall compile the test data in report forms, and forward five copies to the Engineer for evaluation.
4. The final report shall contain logged results of all tests, including such data as:
  - a. Tabulation of air volume at each outlet.
  - b. Outside dry bulb and wet bulb temperature.
  - c. Inside dry bulb and wet bulb temperatures in each conditioned space room or area.
  - d. Actual fan capacities and static pressures. Motor current and voltage readings at each fan.

B. AIR SYSTEMS: Perform the following operations as applicable to balance and test systems:

1. Check fan rotation.
2. Check filters (balancing shall be done with clean filters).
3. Test and adjust blower rpm to design requirements.
4. Test and record motor full load amperes.
5. Test and record system static pressures, suction and discharge.
6. Test and adjust system for design cfm, return air and outside air ( $\pm 2\%$ ). Change-out fan sheaves as required to balance system.
7. Test and record entering air temperatures, db and wb.
8. Test and record leaving air temperatures, db and wb.
9. Adjust all zones to design cfm ( $\pm 2\%$ ).
10. Test and adjust each diffuser, grille, and register to within 5% of design.

C. AIR DUCT LEAKAGE: (From SMACNA Duct Standards latest edition) Test all ductwork (designed to handle over 1000 CFM) as follows:

1. Test apparatus  
The test apparatus shall consist of:
  - a. A source of high pressure air--a portable rotary blower or a tank type vacuum cleaner.
  - b. A flow measuring device consisting of straightening vanes and an orifice plate mounted in a straight tube with properly located pressure taps. Each orifice assembly shall be accurately calibrated with its own calibration curve. Pressure and flow readings shall be taken with U-tube manometers.
2. Test Procedures
  - a. Test for audible leaks as follows:
    - 1) Close off and seal all openings in the duct section to be tested. Connect the test apparatus to the duct by means of a section of flexible duct.
    - 2) Start the blower with its control damper closed.

- 3) Gradually open the inlet damper until the duct pressure reaches 1.2 times the standard designed duct operating pressure.
  - 4) Survey all joints for audible leaks. Mark each leak and repair after shutting down blower. Do not apply a retest until sealants have set.
- b. After all audible leaks have been sealed, the remaining leakage should be measured with the orifice section of the test apparatus as follows:
- 1) Start blower and open damper until pressure in duct reaches 25% in excess of designed duct operating pressure.
  - 2) Read the pressure differential across the orifice on manometer No. 2. If there is no leakage, the pressure differential will be zero.
  - 3) Total allowable leakage shall not exceed one (1) percent of the total system design air flow rate. When partial sections of the duct system are tested, the summation of the leakage for all sections shall not exceed the total allowable leakage.
  - 4) Even though a system may pass the measured leakage test, a concentration of leakage at one point may result in a noisy leak which must be corrected.

D. DX SYSTEMS:

1. Test and record suction and discharge pressures at each compressor and record ambient air temperature entering the condensing coils.
2. Test and record unit full load amps and voltage.
3. Test and record staging and unloading of unit required by sequence of operation or drawing schedule.

E. Automatic temperature controls shall be calibrated; and all thermostats and dampers adjusted so that the control system is in proper operating condition, subject to the approval of the Engineer/Owner.

F. The TAB Contractor shall report to Engineer all air distribution devices or other equipment that operate noisily so that corrective measures may be implemented by the Contractor at no additional cost to the Owner or Architect/Engineer.

**END OF SECTION**

## **SECTION 23 07 13 - DUCT INSULATION**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 - Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 WORK INCLUDED**

- A. Ductwork system insulation.

#### **1.3 RELATED SECTIONS**

- A. Section 23 05 29 - Hangers and Support for HVAC Piping and Equipment
- B. Section 23 05 53 – Identification for HVAC Piping and Equipment
- C. Section 23 31 13 – Metal Ductwork

#### **1.4 REFERENCE STANDARDS**

- A. ASTM International. (ASTM)
- B. American Society of Heating, Refrigerating, and Air Conditioning Engineers, Inc. (ASHRAE).
- C. North American Insulation Manufacturers Association (NAIMA).
- D. National Fire Protection Association (NFPA).
- E. Sheet Metal and Air Conditioning Contractors National Association, Inc. (SMACNA).
- F. Underwriter's Laboratories (UL).
- G. Underwriter's Laboratories Environmental (UL Environment).

#### **1.5 QUALITY ASSURANCE**

- A. Installer's Qualifications: Firm with at least 5 years successful installation experience on projects with mechanical insulations similar to that required for this project.
- B. Flame/Smoke Ratings: Provide composite mechanical insulation (insulation, jackets, coverings, sealers, mastics and adhesives) that is UL Classified per UL 723 or with flame-spread index of 25 or less, and smoke-developed index of 50 or less, as tested by ASTM E 84 (NFPA 255) method.
  - 1. Exception: Outdoor mechanical insulation may have flame spread index of 75 and smoke developed index of 150.
- C. Duct and plenum insulation shall comply with minimum R-value requirements of 2015 International Energy Conservation Code and ASHRAE 90.1 - 2013.

- D. Adhesive and other material shall comply with NFPA and NBFU Standards No. 90A and 90B.

## 1.6 WARRANTY

- A. Warrant the Work specified herein for one year against becoming unserviceable or causing an objectionable appearance resulting from either defective, or nonconforming materials and workmanship.
- B. Defects shall include, but not be limited to, the following:
  - 1. Mildewing.
  - 2. Peeling, cracking, and blistering.
  - 3. Condensation on exterior surfaces.

## 1.7 SUBMITTALS

- A. SHOP DRAWINGS: Indicate size, material, and finish. Show locations and installation procedures. Include details of joints, attachments, and clearances.
- B. PRODUCT DATA: Submit schedules, charts, literature, and illustrations to indicate the performance, fabrication procedures, product variations, and accessories.

## 1.8 DELIVERY, STORAGE AND HANDLING

- A. Deliver insulation, coverings, cements, adhesives, and coatings to site in unopened containers with manufacturer's stamp, clearly labeled with flame and smoke rating, affixed showing fire hazard indexes of products.
- B. Protect insulation against dirt, water and chemical and mechanical damage. Do not install damaged or wet insulation; remove such from project site.

# PART 2 - PRODUCTS

## 2.1 GENERAL DESCRIPTION

- A. The type of insulation and its installation shall be in strict accordance with these specifications for each service, and the application technique shall be as recommended by the manufacturer. All insulation types, together with adhesives and finishes shall be submitted and approved before any insulation is installed.
- B. A sample quantity of each type of insulation and each type of application shall be installed and approval secured prior to proceeding with the main body of the Work.

## 2.2 ACCEPTABLE MANUFACTURERS

- A. Glass mineral wool materials shall be as manufactured by Knauf Insulation, Certain-Teed, Johns-Manville or Owens-Corning and shall have the same thermal properties, density, fire rating, vapor barrier, etc., as the types specified herein, subject to review by the Engineer.
- B. Adhesives shall be as manufactured by Minnesota Mining, Arabol, Benjamin-Foster, Armstrong or Insulmastic, Inc., and shall have the same adhesive properties, fire rating, vapor seal, etc., as the types specified herein, subject to review by the Engineer.
- C. Ceramic fiber materials shall be as manufactured by Primer Refractories, A.P. Green

Refractories or approved equal.

## PART 3 - EXECUTION

### 3.1 GENERAL

- A. All insulation shall be installed in accordance with the manufacturer's recommendations and printed installation instructions.
- B. All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Provide all items required as per manufacturer's requirements.

### 3.2 EXTERNAL DUCT INSULATION

- A. Fasten all longitudinal and circumferential laps with outward clinching staples 3" on center. On rectangular ducts over 24" wide apply as above and hold insulation in place on bottom side with mechanical pins and clips on 12" centers.
- B. Seal all joints, fastener penetrations and other breaks in vapor barrier with 3-inch wide strips of white glass fabric embedded between two coats of vapor barrier mastic, Childers CP-30 or approved equal.
- C. All external duct insulation shall be Knauf Insulation Atmosphere Duct Wrap with ECOSE Technology, Johns Manville Microlite EQ duct wrap insulation with reinforced aluminum facing or approved equal.
- D. External duct wrap is required on all outside air ducts, supply and return air ducts that are not internally insulated. External duct wrap is also required on all exhaust and relief air ducts that are used in airside energy recovery systems. Any exhaust ductwork located in an unconditioned space shall also be provided with external duct wrap. Duct wrap shall be provided as follows:
  - 1. 1½" thick, 1.0 PCF density minimum; minimum installed R-value of 4.2 when ducts are located in conditioned spaces.
  - 2. 2" thick with a minimum installed R-value of 6 when ducts are located in unconditioned spaces, such as ceiling plenum space.
- E. Any ductwork located in an air plenum that is comprised of materials that do not comply with the 25/50 flame and smoke rating per ASTM E 84 testing requirements shall be provided with a single layer of duct wrap to establish a noncombustible rating per ASTM E 136. Duct wrap products which are approved for such non-compliant combustible duct materials located in air plenums shall be 3M Fire Barrier Plenum Wrap 5A+ or Unifrax FyreWrap 0.5 Plenum. Insulation products for this application shall be installed in strict accordance with the manufacturer's instructions.

### 3.3 DUCT LINER

- A. Duct liner shall be kept clean and dry during transportation, storage, installation, and throughout the construction process care should be taken to protect the liner from exposure to the elements or damage from mechanical abuse.
- B. All portions of duct designed to receive duct liner shall be completely covered with liner as specified. The smooth, black, mat facing or acrylic-coated surfaces with flexible glass cloth reinforcement shall face the airstream. All duct liner shall be cut to assure tight, overlapped corner joints. The top pieces shall be supported by the sidepieces. Duct liner



shall be installed following the guidelines in the NAIMA "Duct Liner Installation Standard".

- C. The duct liner shall be tested according to erosion test method in ASTM C 1071 and shall be guaranteed to withstand velocities in the duct system up to 6000 fpm without surface erosion.
- D. Duct liner shall be adhered to the sheet metal with full coverage of an approved adhesive that conforms to ASTM C 916, and all exposed leading edges and transverse joints shall be coated with Permacote factory-applied or field-applied edge coating and shall be neatly butted without gaps. Shop or field cuts shall be liberally coated with Johns Manville SuperSeal® duct butter and Edge Treatment or approved adhesive.
- E. Metal nosings shall be securely installed over transversely oriented liner edges facing the airstream at forward discharge and at any point where lined duct is preceded by unlined duct.
- F. When velocity exceeds 4000 fpm (20.3 m/sec), use metal nosing on every leading edge. Nosing may be formed on duct or be channel or zee attached by screws, rivets or welds.
- G. The liner shall further be secured with Graham welding pins and washers on not more than 18 inch centers both vertical and horizontal surfaces, and the pins and washers shall be pointed up with adhesive.
- H. Duct liner shall be Knauf Insulation Atmosphere Duct Liner with ECOSE Technology, Johns Manville Linacoustic RC duct liner with factory-applied edge coating and acrylic coating on the mat surface of airstream side or approved equal. The liner shall meet the Life Safety Standards as established by NFPA 90A and 90B, FHC 25/50 and Limited Combustibility and the air stream surface coating should contain an immobilized, EPA-registered, anti-microbial agent so it will not support microbial growth as tested in accordance with ASTM G21 and G22. The duct liner shall conform to the requirements of ASTM C 1071, UL 2824, with an NRC not less than .70 as tested per ASTM C 423 using a Type "A" mounting, and a thermal conductivity no higher than 0.24 BTU•in/(hr•ft<sup>2</sup>•°F) at 75°F mean temperature.
- I. Line supply and return ductwork at connection of HVAC unit to a point of 15 feet upstream and downstream of the equipment and in return air boots. Attach with full cover coat of cement, duct dimensions up to 16 inches; provide stick clips or screws and cap for dimensions over 16 inches, spaced 16 inches o.c. maximum. Provide sheet metal liner cap over all leading edges of internal insulation exposed to air stream.
- J. Duct liner shall be provided as follows:
  - 1. 1" Thick, 1.5 PCF density minimum; minimum installed R-value of 4.2 when ducts are located in conditioned spaces.
  - 2. 1 ½" Thick with a minimum installed R-value of 6 when ducts are located in unconditioned spaces, such as ceiling plenum space.
  - 3. 2" Thick with a minimum installed R-value of 8 when ducts are located outdoors.

### 3.4 EXPOSED DUCTWORK LOCATED INDOORS

- A. Duct routed exposed in occupied spaces shall be double wall.
- B. Round and flat oval duct routed exposed shall be double wall with perforated inner liner and 1" thick layer of glass mineral wool insulation as manufactured by United McGill Company model no. Acousti-27 or approved equal. Insulation density shall be a minimum of 1.0 PCF.

### 3.5 AIR DEVICE AND MISCELLANEOUS DUCT INSULATION

- A. The backside of all supply air devices shall be insulated with taped and sealed 1½ inch thick external duct wrap.
- B. The contractor shall install an additional layer of 1-½ inch thick external glass mineral wool duct wrap on any portion of the supply air, return air, outside air, or exhaust air system that has condensation forming during any period of operation. The insulation shall be taped and vapor-sealed and located until all evidence of the condensation has been eliminated, at no additional cost to the Owner.

### 3.6 KITCHEN GREASE HOOD EXHAUST DUCT

- A. All type I kitchen range hood exhaust duct shall be enclosed with 2-hour fire rated enclosure.
- B. The duct enclosure shall be sealed around the duct at the points of penetration with an approved fire barrier sealant per ASTM E 1966. Sealant shall be 3M CP 25WB+ or 303 fire barrier caulk and putty.
- C. The enclosure shall be separated from the duct by at least 3 inches and not more than 12 inches.
- D. Cleanout openings at exhaust duct with access openings at the fire rated enclosure and access doors shall be provided at each duct offset and as required for proper operation and maintenance.
- E. As an alternate method, the contractor may use two layers of 2-hour fire rated duct wrap such as 3M Fire Barrier Duct Wrap 615+ or Unifrax FyreWrap Elite 1.5 in lieu of the fire rated enclosure, provided that all the following constraints are satisfied:
  - a. Duct wrap system shall be tested per ASTM E 2336 internal fire testing and have an achieved minimum fire resistance rating of 2 hours.
  - b. Product shall be approved by the local Authority Having Jurisdiction (AHJ).
  - c. Duct wrap system shall be mechanically attached to the duct using steel banding and/or weld pins per manufacturer's instructions.
  - d. Duct wrap system shall be installed in strict accordance with the manufacturer's instructions, including but not limited to zero clearance to combustibles at all locations on the wrap surface.
- F. Insulation and all other requirements shall be provided per local codes.

### 3.7 DRYER VENT DUCT

- A. All dryer vent duct routed within an air plenum shall be enclosed within a 1-hour fire rated enclosure.
- B. The duct enclosure shall be sealed around the duct at the points of penetration with an approved fire barrier sealant per ASTM E 1966. Sealant shall be 3M CP 25WB+ or 303 fire barrier caulk and putty.
- C. The enclosure shall be separated from the duct by at least 3 inches and not more than 12 inches.

- D. Cleanout openings at exhaust duct with access openings at the fire rated enclosure and access doors shall be provided at each duct offset and as required for proper operation and maintenance.
- E. As an alternate method, the contractor may use a single layer of 1-hour fire rated duct wrap such as 3M Fire Barrier Duct Wrap 615+ or Unifrax FyreWrap Elite 1.5 in lieu of the fire rated enclosure, provided that all the following constraints are satisfied:
  - a. Duct wrap system shall be tested per ASTM E 2336 internal fire testing and have an achieved minimum fire resistance rating of 1 hour.
  - b. Product shall be approved by the local Authority Having Jurisdiction (AHJ).
  - c. Duct wrap system shall be mechanically attached to the duct using steel banding and/or weld pins per manufacturer's instructions.
  - d. Duct wrap system shall be installed in strict accordance with the manufacturer's instructions, including but not limited to zero clearance to combustibles at all locations on the wrap surface.
- F. Insulation and all other requirements shall be provided per local codes.

**END OF SECTION**

## **SECTION 23 07 19 – HVAC PIPING INSULATION**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 – Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

- A. Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for Owner's use.
- B. Furnish and install piping insulation to:
  - 1. Chilled water and heating hot water piping.
  - 2. Condensate drain piping.
  - 3. Refrigerant piping.
  - 4. All pipes subject to freezing conditions shall be insulated.
- C. Work specified elsewhere.
  - 1. Painting.
  - 2. Pipe hangers and supports.
- D. For insulation purpose piping is defined as the complete piping system including supplies and returns, pipes, valves, automatic control valve bodies, fittings, flanges, strainers, thermometer well, unions, reducing stations, and orifice assemblies.

#### **1.3 RELATED SECTIONS**

- A. Section 23 05 29 - Hangers and Support for HVAC Piping and Equipment
- B. Section 23 05 53 – Identification for HVAC Piping and Equipment
- C. Section 23 21 13 - Above Ground Hydronic Piping
- D. Section 23 21 16 - Underground Hydronic Piping
- E. Section 23 22 13 - Steam and Condensate Heating Piping
- F. Section 23 23 00 - Refrigerant Piping

#### **1.4 REFERENCE STANDARDS**

- A. ASTM International. (ASTM)
- B. American Society of Heating, Refrigerating, and Air Conditioning Engineers, Inc. (ASHRAE).
- C. North American Insulation Manufacturers Association (NAIMA).

- D. National Fire Protection Association (NFPA).
- E. Sheet Metal and Air Conditioning Contractors National Association, Inc. (SMACNA).
- F. Underwriter's Laboratories (UL).
- G. Underwriter's Laboratories Environmental (UL Environment).

#### 1.5 WARRANTY

- A. Warrant the Work specified herein for one year against becoming unserviceable or causing an objectionable appearance resulting from either defective or nonconforming materials or workmanship.
- B. Defects shall include, but not be limited to, the following:
  - 1. Mildewing.
  - 2. Peeling, cracking, and blistering.
  - 3. Condensation on exterior surfaces.

#### 1.6 SUBMITTALS

- A. SHOP DRAWINGS: Indicate size, material, and finish. Show locations and installation procedures. Include details of joints, attachments, and clearances.
- B. PRODUCT DATA: Submit schedules, charts, literature, and illustrations to indicate the performance, fabrication procedures, project variations, and accessories.

#### 1.7 DELIVERY AND STORAGE

- A. Deliver insulation, coverings, cements, adhesives, and coatings to site in unopened containers with manufacturer's stamp, clearly labeled with flame and smoke rating, affixed showing fire hazard indexes of products.
- B. Protect insulation against dirt, water and chemical and mechanical damage. Do not install damaged or wet insulation; remove such from project site.

### PART 2 - PRODUCTS

- 2.1 It is the intent of these specifications to secure superior quality workmanship resulting in an absolutely satisfactory installation of insulation from the standpoint of both function and appearance. Particular attention shall be given to valves, fittings, pumps, etc., requiring low temperature insulation to insure full thickness of insulation and proper application of the vapor seal. All flaps of vapor barrier jackets and/or canvas covering must be neatly and securely smoothed and sealed down.
- 2.2 The type of insulation and its installation shall be in strict accordance with these specifications for each service, and the application technique shall be as recommended by the manufacturer. All insulation types, together with adhesives and finishes shall be submitted and reviewed prior to installation.
- 2.3 A sample quantity of each type of insulation and each type application shall be installed and accepted prior to proceeding with the main body of the work. Condensation caused by improper installation of insulation shall be corrected by Installing Contractor. Any damage caused by condensation shall be made good at no cost to the Owner or Architect/Engineer.

- 2.4 All insulation shall have composite (insulation, jacket or facing, and adhesive used to adhere the facing or jacket to insulation) fire and smoke hazard as tested by Procedure ASTM E 84, NFPA 255 and UL 723 not exceeding:

Flame Spread 25  
Smoke Developed 50

- 2.5 All HVAC piping insulation shall comply with minimum requirements of 2015 International Energy Conservation Code and ASHRAE 90.1 - 2013.
- 2.6 Accessories, such as adhesives, mastics and cements shall have the same component ratings as listed above.
- 2.7 All products or their shipping cartons shall have a label affixed, indicating flame and smoke ratings do not exceed the above requirements.
- 2.8 Any piping located in an air plenum that is comprised of materials that do not comply with the 25/50 flame and smoke rating per ASTM E 84 testing requirements shall be provided with a single layer of high-temperature insulation to establish a noncombustible rating per ASTM E 136. Insulation products which are approved for such non-compliant combustible piping materials located air plenums shall be 3M Fire Barrier Plenum Wrap 5A+ or Unifrax FyreWrap 0.5 Plenum. Insulation products for this application shall be installed in strict accordance with the manufacturer's instructions.

2.9 APPROVED MANUFACTURERS

- A. Calcium silicate materials shall be as manufactured by Johns Manville.
- B. Glass mineral wool materials shall be as manufactured by Knauf Insulation, Johns Manville or Owens-Corning and shall have the same thermal properties, density, fire rating, vapor barrier, etc., as the types specified herein, subject to review by the Engineer.
- C. Adhesives shall be as manufactured by Childers, Foster, HB Fuller or Armacell, and shall have the same adhesive properties, fire rating, vapor seal, etc., as the types specified herein, subject to review by the Engineer.
- D. Flexible elastomeric cellular thermal insulation by Armacell.
- E. Phenolic foam insulation shall be as manufactured by Kooltherm Insulation (Koolphen).
- F. Metal jacketing and fitting covers shall be as manufactured by Childers or RPR Products.

2.10 MATERIALS

- A. CHILLED WATER: Provide Phenolic foam with ASJ jacket and all joints sealed
- B. HEATING HOT WATER PIPE: Provide glass mineral wool pipe insulation with ASJ-SSL jacket or phenolic foam with ASJ and all joints sealed.
- C. CONDENSATE DRAINAGE PIPING: Flexible elastomeric cellular thermal insulation, model "AP-2000", fire rated for use in environmental air plenums; insulation not required when piping is exposed on roof.
- D. REFRIGERANT PIPING: Flexible elastomeric cellular thermal insulation, model "AP-2000", fire rated for use in environmental air plenums. Apply manufacturers

recommended finish and sealant for exterior applications.

- E. METAL JACKETING: Utilize Childers "Strap-On" jacketing. Provide preformed fitting covers for all elbows and tees.

### PART 3 - EXECUTION

- 3.1 All insulation shall be installed in accordance with the manufacturer's recommendations and printed installation instructions, including high density inserts at all hangers and pipe supports to prevent compression of insulation.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Provide all items required as per manufacturer's requirements.
- 3.3 All heat recovery piping between air conditioning equipment and hydronic or domestic hot water piping shall be insulated per the High Temperature Surfaces Schedule in Part 4.
- 3.4 All condenser water piping located in a ceiling plenum shall be insulated per the Low Temperature Surfaces Schedule in Part 4.
- 3.5 Pipes located outdoors or in tunnels shall be insulated same as concealed piping and shall have a jacket of 0.016 inch thick, smooth aluminum with longitudinal modified Pittsburgh Z-Lock seam and 2 inch overlap. Jacketing shall be easily removed and replaced without damage. All insulation butt joints shall be sealed with gray silicone. Galvanized banding is not acceptable.
- 3.6 All insulated piping located over driveways shall have an aluminum shield permanently banded over insulation to protect it from damage from car antennas.
- 3.7 WATER PIPE INSULATION INSTALLATION
  - A. The insulation shall be applied to clean, dry pipes with all joints firmly butted together. Where piping is interrupted by fittings, flanges, valves or hangers and at intervals not to exceed 25 feet on straight runs, an isolating seal shall be formed between the vapor barrier jacket and the bare pipe. The seal shall be by the applications of adhesive to the exposed insulation joint faces, carried continuously down to and along 4 inches of pipe and up to and along 2 inches of jacket.
  - B. Pipe fittings and valves shall be insulated with pre-molded or shop fabricated glass mineral wool covers finished with two brush coats of vapor barrier mastic reinforced with glass fabric.
  - C. All under lap surfaces shall be clean and free of dust, etc. before the Joint is sealed. These laps shall be firmly rubbed to insure a positive seal. A brush coat of vapor retarder shall be applied to all edges of the vapor barrier jacket.
- 3.8 STANDBY-GENERATOR ENGINE EXHAUST PIPING
  - A. Entire engine exhaust pipe from exhaust manifold to outside terminal shall be enclosed in a 1" thick layer of calcium silicate insulation.
  - B. A second insulating layer of 1" thick calcium silicate shall cover the first layer.
  - C. Joints for the first and second layer shall be staggered.
  - D. Apply aluminum jacket over outer layer of insulation.

- E. Insulate exhaust muffler in the same manner as the exhaust piping.

### 3.9 FIRE RATED INSULATION

- A. All pipe penetrations through walls and concrete floors shall be fire rated by applying USG Thermafiber in the space between the concrete and the pipe.
- B. The penetration shall be additionally sealed by using 3M brand model CP 25 or 303 fire barrier caulk and putty.
- C. All fire rating material shall be insulated in accordance with manufacturer's printed instructions.

## PART 4 - SCHEDULES

- 4.1 Provide all HVAC piping insulation to comply with the ASHRAE 90.1 Minimum Thickness Schedule.

### 4.2 LOW TEMPERATURE SURFACES MINIMUM INSULATION THICKNESS

- A. Condensate drain lines: 1 inch
- B. Chilled Water Piping:
  - 1. Located outdoors: 2 inch
  - 2. Located indoors:
    - a. 4 inch and smaller: 1 ½ inch
    - b. Larger than 4 inch: 2 inch
- C. Refrigerant Piping
  - 1. 1½" and smaller: 1 inch
  - 2. Larger than 1½ inch: 1 ½ inch

### 4.3 HIGH TEMPERATURE SURFACES MINIMUM INSULATION THICKNESS

- A. Hot Water Piping:
  - 1. Operating temperature 105°F or less: 1 inch
  - 2. Operating temperature higher than 105°F and pipe size 1¼ inch or smaller: 1 ½ inch
  - 3. Operating temperature higher than 105°F and pipe size greater than 1¼ inch: 2 inch
- B. Steam Piping:
  - 1. Pipe size 1½ inch and smaller: 2 ½ inch
  - 2. Pipe size more than 1½ inch: 3 inch

**END OF SECTION**



## SECTION 23 09 63 - ENERGY MANAGEMENT AND CONTROL SYSTEM (EMCS)

### PART 1 - GENERAL

#### 1.1 GENERAL

- A. The Energy Management and Control System (**EMCS**) shall be comprised of a Local Area Network (**LAN**) infrastructure, Operator Workstations (**OWS**), Engineering Workstations (**EWS**), a Primary Network Server (**PNS**), Network Area Controllers (**NAC**), Application Specific Controllers (**ASC**), Unitary System Controllers (**USC**), and Field Devices installed within the facility.
- B. The Workstations, Primary Network Server, and Network Area Controllers shall be connected by a EMCS Contractor supplied and installed Local Area Network. The LAN shall comply with all IEEE Standards as outlined in: IEEE STD 802-1990: IEEE Standards for Local and Metropolitan Networks, Overview and Architecture.
- C. If the EMCS contractor wishes to connect to the Owner's Wide Area/Local Area Network as part of the control system network, the EMCS contractor shall acquire permission in writing and include the letter in the submittal. Any system that requires connection to the owner's network for communication between NAC, ASC, USC and/or field devices that is submitted without the written permission from the owner shall be rejected. The EMCS Contractor shall coordinate with the Owner and supply all required information.
- D. Access to the system, either locally in the building, or remotely from a central site or sites, shall be accomplished through standard web browsers, via the Internet and/or a local area network.
- E. All EMCS controllers and workstations shall communicate using the protocols and network standards as defined by ANSI/ASHRAE Standard 135-2010, latest revision. Management level TCP/IP Ethernet network speeds shall be 1 Gbps minimum and the Automation Level MS/TP network speeds shall be 76.8 Kbps minimum.
- F. The Server shall gather data from the system and generate HTML pages accessible through a conventional web browser from all personal computers (PCs) connected to the network. System shall include any and all software and hardware to support at least 50 simultaneous users. The EMCS shall be compatible with all common web browsers.
- G. Facility Operators shall be able to view and configure systems through the standard web browser and all graphical/data representations shall appear identical, whether the user is on site or viewing via the Internet at a remote location. Standard operator functions such as control point manipulation, configuration and viewing of trends, schedules and alarms shall be performed through the standard browser. Each mechanical system and building floor plan shall be depicted on the operator workstation by point-and-click graphics.
- H. The EMCS shall directly control HVAC equipment as specified in the Sequence of Operations. Furnish Energy Conservation features such as Optimal Start/Stop, Night Setback, Setpoint Reset logic, and Demand Control Ventilation.
- I. The EMCS vendor shall provide the following additional services as part of this specification: warranty and service during the warranty period; submittals, samples and record documentation; comprehensive startup and testing of the EMCS with documentation; training services for the owner and facility operators; coordination with other contractors and suppliers; operator and technician training program, and shall cooperate fully with the Project Commissioning Agent.

- J. Products furnished under this specification but installed by other.
1. Mechanical devices installed under Division 23 by the mechanical contractor or other suppliers: temperature sensing thermowells; automatic control valves; pipe taps for flowmeters; water pressure sensors and switches; automatic control dampers not installed in air handling unit mixing boxes or louver schedules; damper actuators for variable air volume (VAV) terminal units; mounting cost of controller and actuator for VAV terminal units.
  2. Electrical devices installed under Division 26 by the electrical contractor:
    - a. 120 VAC power to controllers and control panels at locations indicated on the drawings. Review and verify that these locations are adequate for the proposed EMCS.
    - b. Interlock wiring to duct mounted smoke detector or fire alarm shutdown relays to HVAC equipment motor starters and variable frequency drives (VFD).
- K. Provide and install all interconnecting cables between all operator's terminals and peripheral devices (such as printers, etc.) supplied under this section.

## 1.2 RELATED DOCUMENTS & REFERENCES

- A. Drawings and general provisions of the contract documents, apply to this section including:
1. Division 01 for General Conditions and Supplementary Conditions.
  2. Division 21 for fire protection equipment.
  3. Division 22 for plumbing equipment and domestic water systems.
  4. Division 23 for mechanical equipment, ductwork, and piping systems.
  5. Division 26 for electrical equipment, lighting control, and fire alarm systems.
- B. The latest edition of the following standards and codes in effect as approved by the authority having jurisdiction and amended as of supplier's proposal date, and any applicable subsections thereof, shall govern design and selection of equipment and material supplied:
1. ANSI MC85.1 - Terminology for Automatic Control.
  2. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE).
  3. ANSI/ASHRAE Standard 135-2010, BACnet.
  4. BTL Mark by the BACnet Testing Laboratories.
  5. Uniform Building Code (UBC), including local amendments.
  6. UL 916 Underwriters Laboratories Standard for Energy Management Equipment. Canada and the US.
  7. National Electrical Code (NEC).
  8. FCC Part 15, Subpart J, Class A.
  9. National Institute of Standards and Technology (NIST).
  10. IEEE STD 802-1990: IEEE Standards for Local and Metropolitan Networks, Overview and Architecture.

## 1.3 RELATED WORK IN OTHER SECTIONS

- A. Refer to Division 00 and Division 01 for allowances and related contractual requirements.
- B. Refer to Division 21 for General Fire Protection Provisions and fire suppression pump.

1. The EMCS contractor shall provide communications integration via BACnet/IP interface to each installed system listed above. BACnet MS/TP acceptable if IP interface is not available from equipment manufacturer.
  2. The EMCS contractor shall coordinate with all vendors providing above systems to obtain communications requirements and points lists. Map all available points to EMCS.
- C. Refer to Division 22 for General Plumbing Provisions, domestic water heating systems, domestic water pumping systems, and domestic water metering.
1. The EMCS contractor shall provide communications integration via BACnet/IP interface to each installed system listed above. BACnet MS/TP acceptable if IP interface is not available from equipment manufacturer.
  2. The EMCS contractor shall coordinate with all vendors providing above systems to obtain communications requirements and points lists. Map all available points to EMCS.
- D. Refer to Division 23 for General Mechanical Provisions for equipment such as chillers, cooling towers, boilers, pumps, air-handling units, terminal units, ventilation fans, variable frequency drives, unitary AC units, etc.
1. The EMCS contractor shall provide communications integration via BACnet/IP interface to each installed system listed above. BACnet MS/TP acceptable if IP interface is not available from equipment manufacturer.
  2. The EMCS contractor shall coordinate with all vendors providing above systems to obtain communications requirements and points lists. Map all available points to EMCS.
- E. Refer to Section 26 for General Electrical Provisions for equipment such as electrical switchgear control, electrical power monitoring, emergency generators, lighting control system, etc.
1. The EMCS contractor shall provide communications integration via BACnet/IP interface to each installed system listed above. BACnet MS/TP acceptable if IP interface is not available from equipment manufacturer.
  2. The EMCS contractor shall coordinate with all vendors providing above systems to obtain communications requirements and points lists. Map all available points to EMCS.

#### 1.4 ELECTRICAL POWER PROVISIONS

- A. Primary power will be provided under Division 26 by the electrical contractor to the panel locations indicated on the mechanical & electrical drawings. Provide step down transformers within panel enclosures. Provide all necessary fuses and circuit protection devices
- B. Power will be provided to the controllers serving fan powered terminal units with electric heat via the control transformer provided with the unit.
- C. All components of the EMCS shall be powered from the sources above. Provide final terminations from the locations indicated on the Division 23 Drawings.
- D. The EMCS Contractor shall provide any additional control power that is required as part of this contract and not indicated by other. This shall include all conduit, cabling, circuit breakers, etc.

## 1.5 CONTRACTOR QUALIFICATIONS

### A. The EMCS Contractor shall:

1. Have a local staff of trained personnel capable of giving instructions and providing routine and emergency maintenance on the EMCS, all components and software/firmware and all other elements of the EMCS.
2. Have a proven record of experience in the supply and installation of equivalent BACnet systems over a minimum period of five years. Provide documentation of at least three equal and complexity, if so requested by the Owner's Representative.
3. Be a factory certified representative of the native BACnet EMCS manufacturer for design, installation, and service of the proposed system.
4. Have comprehensive local service, training and support facilities for the total EMCS as provided. Maintain local, supplies of essential expendable parts.

## 1.6 SUBMITTALS

### A. ALL DOCUMENTS SUBMITTED SHALL BE IN NATIVE PDF FORMAT. NO SCANS.

### B. Shop Drawings:

1. The following information shall be included on the cover page for each shop drawing and equipment documentation submittal:
  - a. Project name with date. Refer to the applicable specifications by name and number.
  - b. Provide submittal number and re-submittal number and date as applicable.
  - c. Provided name and address of Consulting Engineer, Mechanical Contractor, General Contractor
2. Shop drawings shall be CAD generated, plot size of 8-1/2" x 11" or 11" x 17". Drawings shall include diagrams, mounting instructions, installation procedures, equipment details and software descriptions for all aspects of the system to be installed.
3. Provide schematic of systems indicating instrumentation locations, all interconnecting cables between supplied cabinets on a mechanical floor plan.
4. Software specifications and descriptions including operating sequences.
5. Provide a bill of material that indicates specific manufacturer, part number, part description and quantity of each device for all system components.
6. Provide a list of the wire labels to be installed on each end of the control wiring, at the device and the control panel terminal. Labels shall be machine generated, typed and legible with a maximum of 17 characters. The label description "AHU-1 SAT" shall indicate the supply air temperature of AHU-1.
7. Equipment Schematic: Provide an electronic equipment schematic for each piece of mechanical equipment. The schematic shall display all mechanical equipment characteristics including fans, dampers, valves, sensors and other applicable control devices. The schematic shall show wiring terminations to each control device as shown in the submittal and as-build documentation. Control devices shall be labeled by a symbol that can easily be identified in a bill of material that is shown on this graphic. The bill of material shall show the device symbol, description, manufacture and part number.
8. Sequence of Operations: The control sequences shall be viewable for each piece of mechanical equipment and be in a text format as shown in the as built documentation. The sequence of operations shall be selectable at the applicable location for the control program.

- C. Control component submittals:
1. Component technical data sheets with mounting and installation details.
  2. The documentation shall include comprehensive and complete details of the BIBB and automation level documentation including address, associated controller type, etc. as required and for the interface to the EMCS.
  3. Details of networks/communications equipment, cabling and protocols proposed. Provide schedule of cabling including details of proposed cable types.
  4. Module Drawing: Provide an electronic wiring diagram of each control module (as shown in submittal documentation). Diagram shall display wiring schematic and terminations to end devices. Diagram shall display each input and output terminals and label those that are used for the control application. Diagram shall display module type/name and network address.
  5. Field sensor and instrumentation specification sheets. Provide complete manufacturer's specifications for all items that are supplied. Include vendor name of every item supplied.
  6. Schedule and specification sheets for dampers, valves and actuators.
  7. Design and provide layout of all components of panel mounted control devices, terminal strips and power supplies.
- D. Colorgraphics: Provide sample layout of color graphic representations of the systems for review. The submittal shall indicate the quality of the graphic to be provided with the system with a sample of the specific control points to be included. Control points shall as a minimum include points indicated in the input/output summary, control schematic and primary controlling points defined in the sequences of operation. Provide a sample of a floor plan layout, typical AHU, terminal unit, outside air pretreatment unit, variable frequency drive, exhaust/supply fan, chiller plant and hot water plant. For control points to be provided by equipment BACnet integration provide sample of the control points, up to 25 total.
- E. Verification Reports. The submittal shall include a sample of the verification reports to be utilized during the verification section of this specification. Sample reports shall be approved as submitted or be modified by the engineer or owner's representative. The verification reports shall be included in the final Operation & Maintenance Manuals. Reports shall be provided in electronic PDF format.
1. Project Systems Verification Form for each controller.
    - a. General information for each form shall include: project name; associated equipment with mark number; control panel number and location; controller number and model number; controller device instance number (address); MS/TP LAN segment number; verifying technician and date.
    - b. Each connected control point and device shall contain the following columns with a separate line for each connected physical point: point description (same as device label); input/output number for each connected control device (AI-XX, AO-XX, DI-XX, or DO-XX).
    - c. Check boxes confirming that the verification tasks have been completed: device location, proper termination at device; proper termination at control panel; sequence is verified; point trend is enabled.
    - d. Data entry boxes indicating measured/confirmed values: preliminary control point value on the graphic; observed control point value; calibration or adjustment value to correct offset; final displayed point value on the color-graphic; date of verification; engineer or owner's representative verification.
  2. Control Panel Verification Form for each control panel.

- a. General information: panel location and identification number; panel dimensions and NEMA rating; panel properly installed; Class 1 and Class 2 wiring are properly separated; correct voltage to the panel; no shorts or grounds in panel; no induce voltages in panel wiring; point to point termination match submittal; devices are mounted in the correct location; controller software revision number; address of controllers; panel device checkout is complete; panel startup is complete.
  3. **Sequence of Operation Verification Form** per piece of equipment (AHU, VAV, chiller, boiler, etc.).
    - a. General information: project name; system identifier; building area served; control panel and controller numbers; controller model number and instance number (address); MS/TP LAN segment number; name of verifying technician and date.
    - b. Each step of the sequence of operation for each piece of equipment shall be documented shall include a "description of test", "input to trigger test" and "expected outcome". A pass/fail checkbox shall indicate each of these actions. Provide space for technician approval with associated date.
- F. Operating and Maintenance (O&M) manuals: Provide O&M manual with full information to allow the owner to operate, maintain and repair installed products. Include trade names with model numbers, color, dimensions and other physical characteristics.
  1. Format: Produce on 8-1/2 x 11-inch pages, and bind in 3-ring/binders with durable plastic covers. Label binder covers with printed title "OPERATION AND MAINTENANCE MANUAL", title of project, and subject matter and "Number \_ of \_" of binder. Provide substantial dividers tabbed and titled by section/component number.
  2. Table of Contents for each volume:
    - a. Part 1: Directory with name, address and telephone number of Designer, Contractor and Subcontractors and Suppliers for each Project Manual section.
    - b. Part 2: Operation and maintenance instructions, arranged by Project Manual Section number where practical and where not, by system. Include:
  3. Product design criteria, functions, normal operating characteristic and limiting conditions. Installation, alignment, adjustment, checking instructions and troubleshooting guide. Operating instructions for start-up, normal operation, regulation and control, normal shutdown and emergency shutdown. Test data and performance curves.
  4. Spare parts list for operating products, prepared by manufacturers including detailed drawings giving location of each maintainable part, lists of spares recommended for user- service inventory and nearest source of in-stock spares.
- G. Record Documentation:
  1. Details of all alarm, diagnostic, error and other messages. Detail the Operator action to be taken for each instance.
  2. Detail special programs provided and provide a complete programming instruction manual. Detail operation of all software applications.
  3. Detailed list of the database for all installed devices.
  4. Record drawings shall be CAD generated and shall include final locations and point ID for each monitored and controlled device.
  5. In addition to the required hard-copies, provide a CD-ROM with all of the record

documentation in PDF format and a CD-ROM containing backup copies of all installed software and graphics.

6. Online as-built documentation: provide digital replications of as-builts that shall be accessible from each equipment graphic controlled or monitored by the EMCS.

## 1.7 WARRANTY

- A. Warranty work and the equipment provided under this contract shall be for a period of one year from the date of Substantial Completion. Warranty shall cover all components, system software, parts and assemblies supplied by this contractor and shall be guaranteed against defects in materials and workmanship for one (1) year from the date of Substantial Completion. Labor to troubleshoot, repair, reprogram or replace system components that have failed due to defects in materials and workmanship shall be provided by this contractor at no charge to the owner during the warranty period. All corrective software modifications made during warranty service periods shall be updated on all user documentation and on user and manufacturer archived software disks. All warranty work shall be performed by the EMCS contractor's local service group.
- B. Warranty shall not include routine maintenance, e.g., equipment cleaning, mechanical parts lubrication, pilot lamp replacement, operational testing, etc. Warranty shall not cover repair or replacement of equipment damaged by under- or over-voltage, misuse, lack of proper maintenance, lightning, water damage from weather or piping failure.
- C. Hardware and software personnel supporting this warranty agreement shall provide on-site or off-site service in a timely manner after failure notification to the EMCS contractor. The maximum acceptable response time to provide this service at the site shall be 24 hours, during normal working hours.

## 1.8 OPERATIONS PERSONNEL TRAINING

- A. Provide a training session for the owner's operations personnel. Training session shall be performed by a qualified person who is knowledgeable in the subject system/equipment. Submit a training agenda two (2) weeks prior to the proposed training session for review and approval. Training session shall include at the minimum:
  1. Purpose of equipment.
  2. Principle of how the equipment works.
  3. Important parts and assemblies.
  4. How the equipment achieves its purpose and necessary operating conditions
  5. Most likely failure modes, causes and corrections.
  6. On site demonstration that includes hands-on demonstration of the manipulation of setpoints, schedules and other adjustable elements of the system.
  7. The demonstration shall be on the actual, completed graphic interface pages for the specific project.
- B. Provide a second training session 3 months after initial session for any follow-up or additional training requested by owner's personnel. Allow 3 hours for the second training session.

## 1.9 OPERATOR WORKSTATION (OWS)

- A. The Operator Workstation shall be any personal computer, connected to the LAN, with appropriate web browser software installed.

#### 1.10 ENGINEERING WORKSTATION (EWS)

- A. The Engineering Workstation shall be any personal computer, connected to the LAN, with a registered copy of the EMCS contractor supplied engineering and/or programming software installed. The EMCS contractor shall provide at least one copy of all required software(s), to enable the Owner complete editing/programming functions of all controllers, graphics, and control logic.
- B. The EMCS shall provide one personal computer (PC) which is compatible with the performance required by the EMCS Engineering Software if an engineering workstation is specified for the system.

### PART 2 - PRODUCTS

#### 2.1 THE ACCEPTABLE EMCS VENDORS ARE;

- A. ALERTON - Climatec
- B. AUTOMATED LOGIC - Branch Office
- C. RELIABLE CONTROLS - Unify Energy Solutions
- D. TRANE - Branch Office

#### 2.2 PRIMARY NETWORK SERVER (PNS)

- A. The EMCS Contractor shall provide and install the Primary Network Server as part of this system. The PNS shall utilize the Internet and provide efficient integration of standard open protocols. The PNS shall maintain comprehensive database management, alarm management and messaging services, and graphical user interface as follows:
  - 1. Support an unlimited number of users over the Internet/intranet with a standard web browser to access alarms, trend logs, graphics, schedules and configuration data. Access to the PNS shall be password protected utilizing authentication and encryption techniques. An audit trail of database changes indicating user, time stamp, and audit action shall be provided.
  - 2. Enterprise level information exchange using an SQL database and HTTP/HTML/XML text formats.
  - 3. Synchronize controller databases, database storage scheduling, control and energy management routines
  - 4. Alarm processing and routing which includes email, SMS text messages and paging.
  - 5. HTML based help system that includes comprehensive online system documentation.
  - 6. Support of multiple Network Area Controllers (NAC) connected to a Local Area Network.
- B. Server Functions
  - 1. It shall be possible to access all Network Area Controllers (NAC) via a single connection to the server through the Ethernet LAN. In this configuration, each Network Area Controller can be accessed from a single user login.
  - 2. The PNS shall provide the following functions, at a minimum:
    - a. The server shall provide complete access to distributed global data. The



server shall provide the ability to execute global control strategies based on control and data objects in any NAC in the network, local or remote.

- b. The server shall include a master clock service for its subsystems and provide time synchronization for all NACs.
- c. The server shall provide scheduling for all NACs and their underlying field control devices.
- d. The server shall provide demand limiting control that operates across all NACs. The network server shall be capable of multiple demand limiting programs for sites with multiple meters and or multiple sources of energy. Each demand program shall be capable of supporting separate demand shed lists for effective demand control.
- e. The server shall implement the BACnet Command Prioritization scheme (16 levels) for safe and effective contention resolution of all commands issued to NACs. Each Network Area Controller supported by the server shall have the ability to archive its log data, alarm data and database to the server, automatically. Archiving options shall be user-defined including archive time and archive frequency.
- f. The server shall provide central alarm management for all NACs supported by the server. Alarm management shall include: routing of alarms to a video display, a printer, an email and pager; view and acknowledge alarms; query alarm logs based on user-defined parameters
- g. The server shall provide central management of logged data for all NACs supported by the server. Logged data shall include process logs, runtime and event counter logs, audit logs and error logs. Log data management shall include: viewing and printing log data; exporting log data to other software applications; query log data based on user-defined parameters

- 3. The Primary Network Server shall be capable of supporting the following open system drivers;

- a. BACnet/IP
- b. Modbus TCP

C. Network Server Platform Requirements

- 1. Rack-Mounted Server Computer Hardware: DELL PowerEdge R220 or equal, Intel Pentium Core 2 Duo 2.0 GHz or higher, 8GB RAM, 500GB harddrive, video card, 22" color monitor, and Ethernet adapter 1Gbps or higher.
- 2. Operating system software shall be Microsoft Windows 7 Professional or higher.

2.3 NETWORK AREA CONTROLLER (NAC)

- A. Provide one or more Network Area Controllers (NAC) to meet the sequence of operations and the type and quantity of devices being integrated into the system. The NAC shall provide the interface between the local area network and the field controllers. The NAC shall provide global supervisory control functions over the associated controllers and shall be capable of executing application control programs to provide: calendar functions; scheduling; trending; alarm monitoring and routing; time synchronization; integration of controller data for each applicable protocol; network management functions for all network devices. The user may view real-time information via web-based data.
- B. The Network Area Controller shall provide the following hardware features as a minimum: Ethernet Ports 100Mbps or higher, BACnet MS/TP ports, battery backup, DDR RAM memory, flash memory for long term data backup.
- C. Provide an uninterruptible power source (UPS) per network controller to maintain

operation for 1 hours.

- D. The NAC shall be capable of operation over a temperature range of 32 to 122 °F and operation over a humidity range of 5 to 95% RH, non-condensing; storage temperatures of between 32 and 158 °F.
- E. The NAC shall provide multiple user access to the system and support for ODBC or SQL. A database resident on the NAC shall be an ODBC-compliant database or must provide an ODBC data access mechanism to read and write data stored within it.
- F. The NAC shall be capable of supporting the following open system drivers;
  - 1. BACnet/IP
  - 2. BACnet MS/TP
  - 3. Modbus TCP
  - 4. Modbus RTU
- G. Event Alarm Notification and actions: The NAC shall provide alarm recognition, storage; routing, management, and analysis to supplement distributed capabilities of equipment or application specific controllers. Alarm conditions shall be routed to any defined user location whether connected to a local or wide-area network.
  - 1. Alarm generation shall be selectable for annunciation type and acknowledgement requirements including but limited to: alarm; return to normal; fault.
  - 2. Provide for the creation of a minimum of eight alarm classes for the purpose of routing types and or classes of alarms, i.e.: security, HVAC, Fire, etc. Allow timed routing of alarms by class, object, group, or node.
  - 3. Provide alarm generation from binary object "runtime" and/or event counts for equipment maintenance (i.e. filter status, fan run status). Authorized users shall be able to reset runtime or event count values with appropriate password control.
  - 4. Control equipment and network failures shall be treated as alarms and annunciated.
  - 5. Alarms shall be annunciated in any of the following manners as defined by the user: screen message text; e-mail of the complete alarm message to multiple recipients. Provide the ability to route and email alarms based on: day of the week, time of day and recipient.
  - 6. Color-graphic shall have flashing alarm object(s). Printed message may be routed directly to a dedicated alarm printer.
  - 7. The following shall be recorded by the NAC for each alarm (at a minimum): time and date; location (building, floor, zone, office number, etc.); associated equipment. Upon acknowledgement of the alarm the NAC shall document the time, date and authorized user. The number of alarm occurrences since the last acknowledgement shall be recorded.
  - 8. Defined users shall be given proper access to acknowledge any alarm, or specific types or classes of alarms defined by the user. Alarm actions may be initiated by user defined programmable objects created for that purpose.
  - 9. Alarm archiving: A log of all alarms shall be maintained by the NAC and/or a server and shall be available for review by the user. Provide a "query" feature to allow review of specific alarms by user defined parameters. A separate log for system alerts (controller failures, network failures, etc.) shall be provided and available for review by the user.
- H. Data Collection and Storage: The NAC shall have the ability to collect data for any property of any object and store this data for future use.
  - 1. The user shall designate the log as an interval log or deviation log. For an

- interval log, the object shall be configured for time of day, day of week and the sample collection interval. For deviation log, the object shall be configured for the deviation of a variable to a fixed value. This value, when reached, will initiate logging of the object. For all logs, provide the ability to set the maximum number of data stores for the log and to set whether the log will stop collecting when full, or rollover the data on a first-in, first-out basis. Each log shall have the ability to have its data cleared on a time-based event or by a user-defined event or action.
2. All log data shall be stored in a relational database in the NAC and the data shall be accessed from the server or a standard web browser. All log data, when accessed from the server, shall be capable of being manipulated using standard SQL statements.
  3. All log data shall be available to the user in the following data formats: HTML, XML, plain text, comma separated values, as a minimum.
  4. The NAC shall have the ability to archive its log data either locally or remotely to the server or other NAC on the network.
- I. Local Access: The NAC shall provide redundancy of system access to the local controllers at the remote building if the Primary Network Server should lose communication or be off-line. The NAC shall maintain setpoint and scheduling features, access to the color-graphic displays, maintain trend logs and reports. Upon restoration of communication with the PNS the archived information shall be transmitted to the server for archiving.

## 2.4 SOFTWARE FOR THE NAC

- A. The distributed architecture of the operating system for the PNS and NACs shall provide the operator a comprehensive interface to allow the operator to configure and customize the EMCS to optimize the HVAC system to save energy, schedule and maintain equipment and provide occupant comfort. The provided graphical toolset shall allow the operator to create applications in a drag and drop environment.
1. Input/output capability shall allow the operator to request the current value or status of the control point; command/override equipment to a specific state; add, change or delete control points, alarm limits and controllers; change descriptors to control points and equipment; modify parameters; create or modify DDC loops.
- B. Operator System Access: Via software password with five access levels at workstations and at each control unit.
- C. Color graphic tools shall allow the user to create equipment and floor plan graphics from a standard library of symbols; allow custom generation of symbols; utilize over 64 or more colors; create real-time dynamic data for the graphics. Up to 60 control points may be displayed on each graphic.
1. Provide a link between compatible graphics to minimize the paths to additional information. For example provide the link from the zone sensor to the VAV terminal to the air handling unit and to the central plant. Web pages shall be provided to allow the operator to zoom into specific areas of the facility and then link the space to the floor plan to the overall building and then to the facility site plan.
  2. Graphical tools shall allow the creation of bar graphs, pie graphs and other tools to visualize control information such as run time hours, energy consumed and occupant comfort.
- D. Alarm processing tools shall allow the operator to create alarm messages that include as a minimum: time of alarm, point descriptor, alarm condition and remote annunciation.

Critical alarms shall be displayed, archived to a storage device or printed on a alarm printer. Alarms shall be displayed in order of occurrence and have an optional audible alarm indicator.

1. Print alarm messages, up to 60 characters in length, for each alarm point specified.
  2. Alarms may be routed to other devices including web-enabled cell phones, pagers, tablet PCs and designated personal computers on the network or Internet.
  3. Operator specifies when alarm requires acknowledgment. Continue to indicate unacknowledged alarms after return to normal. An alarm log shall be maintained to archive alarms for future reference with the above specified parameters as well as indicating the person acknowledging the alarm.
  4. The graphical display shall indicate the number of the current unacknowledged alarms by individual building site or by sum of all campus-wide facilities.
  5. The operator may create and forward an e-mail message to another user directly from the graphical interface so that the message can be read when the second user logs on to the system.
- E. Upon a power failure to equipment in the facility, the EMCS shall automatically start equipment upon the restoration of power. Program a time delay between individual equipment restart on a schedule to minimize demand charges from the utility company.
- F. Custom reports may be created by the operator with a requested time and date manually or automatically. All reports may be logged to a storage device for future reference. The data reports shall allow customization and scaling of the X-Y coordinates; plotting of tabular reports; provide multi-point graphical reports with not less than eight variables on the same report. Print reports on daily, weekly, monthly, yearly or scheduled basis as scheduled.
- G. The network server current operating system, database, color-graphics, custom reports shall be backed up automatically to a remote server or storage device as directed by the owner's representative.
- H. Maintenance Management capability shall allow the system to monitor and log the run-time for HVAC equipment; schedule maintenance reports that include recommended material and labor for the assigned task.

## 2.5 APPLICATION SPECIFIC CONTROLLERS (ASC)

- A. All devices required for single loop control shall be terminated on a single controller. (for example, CHW loop pressure control. The differential pressure sensor and the pump VFD ramp signal.)
- B. ASCs shall be capable of implementing control strategies for the system based on information from any or all connected inputs. The AC shall utilize factory pre-programmed global strategies that may be modified by field personnel on-site. Global control algorithms and automated control functions should execute via a 32-bit processor
- C. Programming shall be object-oriented using control program blocks that will support a minimum of 500 Analog Values and 500 Binary Values. Analog and binary values shall support standard BACnet priority arrays. Provide means to graphically view inputs and outputs to each program block in real-time as program is executing.
- D. Controller shall have adequate data storage to ensure high performance and data

reliability. Battery shall retain static RAM memory and real-time clock functions for a minimum of 1 year (cumulative). Battery shall be a field-replaceable (non-rechargeable) lithium type. The onboard, battery-backed real time clock must support schedule operations and trend logs.

- E. The base unit of the ASC shall host various I/O combinations including universal inputs, binary outputs, and switch selectable analog outputs (0-10V or 0-20 mA). Inputs shall support thermistors, 0-5VDC, 0-10VDC, 4-20mA, dry contacts and pulse inputs directly.
- F. All binary outputs shall have onboard Hand-Off-Auto switches and a status indicator light. HOA switch position shall be monitored. The position of each HOA switch shall be available system wide as a BACnet object.
- G. Controller shall be capable of BACnet communication. BACnet Conformance:
  - 1. Standard BACnet object types supported shall include as a minimum: Analog Input, Binary Input, Analog Output, Binary Output, Analog Value, Binary Value, Device, File, Group, Event Enrollment, Notification Class, Program and Schedule object types. All necessary tools shall be supplied for working with proprietary information.
- H. Schedules: Each ASC shall support a minimum of 10 BACnet schedule objects.
- I. Logging Capabilities: Each controller shall support a minimum of 100 trend logs. Sample time interval shall be adjustable at the operator's workstation. Controller shall periodically upload trended data to system server for long term archiving if desired. Archived data stored in database format shall be available for use in third-party spreadsheet or database programs.
- J. Alarm Generation: Alarms may be generated within the system for any object change of value or state either real or calculated. This includes things such as analog object value changes, binary object state changes, and various controller communication failures. Alarm logs shall be provided for alarm viewing. Log may be viewed on-site at the operator's terminal or off-site via remote communications. Controller must be able to handle up to 200 alarm setups stored as BACnet event enrollment objects – system destination and actions individually configurable.

## 2.6 UNITARY SYSTEM CONTROLLERS (USC)

- A. All devices required for single loop control shall be terminated on a single controller. (for example, cooling coil control valve control. The temperature sensor and the valve control signal.)
- B. The EMCS Contractor shall provide all Unitary System Controllers. USCs shall be fully programmable or applications specific controllers with pre-packaged operating sequences maintained in Flash RAM.
- C. The USC shall be a node on the automation network and shall control its own communications so that the failure of any one node, shall not inhibit communications on the network between the remaining nodes. USCs shall be totally independent of other network nodes for their monitoring and control functions.
- D. Provide each USC with a battery back-up for the protection of volatile memory for a minimum of 72 hours. Batteries shall be rated for a seven year life.

- E. All associated applications programs shall reside at the USC. The USC shall not require communication to any other panel for normal operating sequences other than time scheduled base commands.
- F. Control shall be based on algorithms, i.e. proportional plus integral plus derivative (PID), proportional plus integral (PI), or proportional to comply with the sequences of operation PID algorithms shall maintain the system operation within +/- 2% of setpoint.
- G. The USC shall be configured with sufficient input/output capacity to achieve the required control points to meet the sequence of operations.

## 2.7 VAV TERMINAL UNIT CONTROLLER (TUC)

- A. All devices required for single loop control shall be terminated on a single controller. (for example, terminal unit air valve control. The flow sensor and the actuator control signal.)
- B. The EMCS Contractor shall provide all controllers required for all variable air volume (VAV) terminal units. The number and location of terminal units and airflow rates shall be as indicated on the mechanical drawings.
- C. The TUC shall be capable of monitoring and controlling the following parameters for VAV terminal units per the sequences of operation and input/output summary: space temperature; primary air flow rate; damper modulation; heating coil stage control, heating valve control, heating SCR control (as applicable); fan on/off control; supply air sensor; occupancy sensor; carbon dioxide sensor or humidity sensor.
- D. Furnish primary damper actuators, for factory mounting, meeting the following requirements: direct shaft mounting; adequate torque, to properly operate the damper from fully open to fully closed without binding; locking "V" groove or similar means to prevent slippage between actuator and shaft.
- E. The EMCS Contractor shall field install the following components for each terminal unit: space temperature sensor; supply air temperature sensor; occupancy sensor, and carbon dioxide sensor as indicated on the Mechanical Drawings.
- F. The EMCS Contractor shall furnish to the terminal unit manufacturer the following components for factory installation and wiring for each terminal unit: VAV controller with integral differential pressure transducer and damper actuator.
- G. The terminal unit manufacturer may provide the following components for each terminal unit for interface and mounting of the TUC: primary air dampers; enclosure to house the TUC and associated components including suitable mounting brackets shall be NEMA 1 rating and located outside the terminal unit; multi-point averaging type flow sensor at the primary air inlet to the terminal unit; 24 VAC control transformer; 24 VAC fan control relay interface; 24 VAC heater control relay interface (up to two stages); 24 volt SCR heater input as scheduled (0-10 Vdc or 4-20 mA).
- H. Any items required for proper operation but not provided by TU vendor, shall be provided under this section.

## 2.8 AIR HANDLING UNIT CONTROLLER

- A. All devices required for single loop control shall be terminated on a single controller. (for example, AHU static pressure control. The differential pressure sensor and the VFD ramp signal.)

- B. The EMCS Contractor shall provide controllers required for chilled/hot water and DX/electric heat air handling units and fan coil units. Provide an enclosure to house the controller and associated components including suitable mounting brackets shall be NEMA 1 rated and located outside the FCUs.
- C. The controller shall be capable of monitoring and controlling the following parameters per the sequences of operation and input/output summary; space temperature; space relative humidity sensor ; cooling/heating stage control or modulating valve control; fan on/off control and status; supply air sensor; occupancy sensor; carbon dioxide sensor; VFD control and monitoring.

## 2.9 EMCS CONTROLLER LEVEL NETWORK

- A. EMCS Automation Level Network shall consist of BACnet MS/TP (76.8 Kbps minimum). Data transfer rate and data throughput as required to meet the alarm annunciation requirements.

## 2.10 SOFTWARE OVERVIEW

- A. **Dynamic Colored Floor plans:** Dynamic colored floor plans that compare actual space conditions to setpoints shall be provided on all floorplan graphics displayed on the front-end. Floorplan enlargements shall also use the thermographs to display space conditions. Zones within the set point range shall appear transparent white. As the space gets warmer the zone color shall gradually modulate from transparent white to transparent red to identify a hot zone. As the space conditions get cooler the zone color shall gradually modulate from transparent white to transparent blue to identify a cold zone. Each zone shall indicate the current actual zone temperature within the zone. The floor plans shall use a dynamic scheduling icon to indicate schedule occupancy for each zone and provide direct one-click access to that zones unique schedule. Provide a designated icon or symbol indicating that the zone is in the occupied/unoccupied condition. From the floorplan graphic, the operator shall be able to click on any zone and go directly to the graphic for the piece of equipment controlling that zone. All dynamic floor plans shall be visible via web interface as well as on the LAN. The authorized system operator shall be able to change the zone or system identifier (or name) on the graphic and that change shall be distributed to other associated graphics and to the equipment controller.
- B. **Pop up Trends:** Provide trend logs that automatically pop up when the operator mouse clicks on the point from the graphic. Provide pop up trends for all dampers, control valves, temperature sensors, carbon dioxide sensors, humidity sensors, airflows, static pressures, flow meters, VFD speeds, etc. The EMCS contractor shall set up all trends for the owner. The pop-up trend shall include a trend tool that allows the operator to modify the trend time scale and sample interval for up to 10 sample values. The trends shall be graphical on the computer screen but shall provide an output as an .xls, .csv, .pdf, HTML, or text file.
- C. **Interactive Maps:** Implement JAVA SCRIPT API 3.0 or newer, such as Google Interactive maps depicting the facility location to indicate the site plan. This is not a static image and must be completely interactive.
- D. **Custom User HTML applications:** The EMCS shall utilize HTML applications as an extra feature. At minimum, provide 7-day forecast, weather radar, traffic map and hurricane tracker. All of these features shall be imbedded into the EMCS system.
- E. Provided a web-based EMCS platform; contractor shall provide an Open License

software. Licenses that are not open are not acceptable. There shall be no per seat or per user licensing fee charged to the owner by the contractor.

- F. System shall use the **BACnet** protocol for communication to the operator workstation or web server and for communication between control modules. Schedules, setpoints, trends, and alarms shall be BACnet Objects.
- G. **User access** shall include 50 assigned operators that shall include five levels of access within the web system. Each operator log-in shall have an expiration date to allow for temporary access to the system. The operator's access description shall include his e-mail address and cell/phone numbers. The operator access can be limited from 5 minutes to permanent access. The user shall be limited to eight bad login attempts before being locked out of the system.
- H. **Global modification:** Provide the capability for global modification of user definable parameters of all points shall be provided. Global modification is defined as the mass adjustment of user definable parameters across a defined group, area, facility, campus, or network. Parameters shall include, but not be limited to temperature set point (VAV boxes, AHU Discharge, VAV AHU Static Pressure Setpoints etc.), equipment start/stop, equipment status, valve output signal, VFD speed control signal, and damper position signal. User shall be able to lock the definable parameter to a set value, or adjust a set point to an operator adjustable value. This function shall be accomplished through the standard graphical user interface/workstation and is to be selectively applicable by the user to all controllers on the network, all controllers in a specific facility or all controllers in a specific zone within a specific facility.
- I. The system operator shall be able to override the output signal to the valves, dampers, variable frequency drives, etc. with the use of the PC mouse click on the device. The system override shall include a **Hand-Off-Auto (HOA)** capability. If the output is commanded to the hand position the operator shall designate an output value of 0-100% in 1% increments. The hand override position shall be permanent or expire after a designated time period and revert to the auto position. The color-graphic shall indicate the device that has been overridden by a color change of the output value.
- J. For non-emergency in-warranty events the system operator may submit a **Service Request** directly from the floor plan or system graphic. The web interface shall include the EMCS suppliers contact information including phone numbers and e-mail address. The service request will be logged into the EMCS suppliers service department. A non-response by the assigned technician shall elevate the request to the next highest manager or supervisor until the system operator receives an response that their request has been received and is scheduled for a resolution. All requests for service shall be maintained in the customer's database for future reference. The service request capability may be extended after the expiration of the warranty as part of a service agreement.
- K. The web-based system shall be accessible from **Tablet PCs** and provide the same functionality that is available from personal computers connected through the LAN or WAN to the system operator. The tablet PCs as a minimum shall include an Apple iPad and Google Android based tablet PC. Operation shall include touch screen capability and use of the tablet keyboard screen. The operator shall be able to view color-graphics, system trends, override setpoints, change time schedules, and override damper and valve positions.

## 2.11 ENERGY SAVING PROGRAMS

- A. Demand Limiting: Monitor total power consumption for each power meter and shed



associated loads automatically to reduce power consumption to an operator set maximum demand level.

- B. Duty Cycling: Periodically stop and start loads, based on space temperature, and according to various on/off patterns.
- C. Automatic Time Scheduling: Self-contained programs for automatic start/stop/scheduling of building loads. Support up to seven (7) normal day schedules, seven (7) "special day" schedules and two (2) temporary schedules.
- D. Optimal Start/Stop: Perform optimized start/stop as function of outside conditions, inside conditions, or both. Optimization shall be adaptive and self-tuning, adjusting to changing conditions by modifying occupancy period based upon the desired temperature at beginning and end of the occupancy period. Base optimization on occupancy schedules, outside air temperature, seasonal requirements, and interior room temperature. Employ adaptive model prediction for how long building takes to warm up or cool down under different conditions.
- E. Night-Setback Program: Reduce heating space temperature setpoint or raise cooling space temperature setpoint during unoccupied hours in conjunction with scheduled start/stop and optimum start/stop programs.
- F. Setpoint Reset: Setpoints for control of variable load systems shall be reset based on load demand, as described in the Sequence of Operations.
- G. Calculated Points: Define calculations and totals computed from monitored points (analog/digital points), constants, or other calculated points.
- H. Event Initiated Programming: Any data point capable of initiating event, causing series of controls in a sequence.
- I. Holiday Scheduling
- J. Direct Digital Control: Furnish software so operator is capable of customizing control strategies and sequences of operation by defining appropriate control loop algorithms and choosing optimum loop parameters.
- K. Trend logging shall be provided for all points per the input/output summary where there is a change in the analog or binary signal. Each controller shall be capable of storing trend values and then automatically transfer data to the NAC or the NS hard disk. Trend data shall be updated continuously per the operator assigned interval at intervals as low as one minute. Collect samples at intervals specified in minutes, hours, days, or month. Output trend logs as line-graphs or bar graphs. Binary points (input and output) shall only be logged upon a change of value (COV). Display trend samples on workstation in graphic format. Automatically scale trend graph with minimum 60 samples of data in plot of time versus data.

## 2.12 FIELD INSTRUMENTATION

- A. Temperature Sensors: All temperature sensors shall be thermistor type, factory-calibrated to within 0.5 °F, interchangeable with housing appropriate for application. Sensors shall have a temperature curve rated for the application. Sensor wiring terminations shall be in a galvanized box.
  - 1. Outside air temperature sensors shall be installed in weather proof enclosure

- with ventilated sun-shield
  2. Duct mounted temperature sensors shall be averaging type for supply air, mixed air and low temperature applications for air handling units. Duct probe temperature sensor shall be acceptable for terminal units.
  3. Space temperature sensors shall contain a backlit LCD digital display and user function keys along with temperature sensor, setpoint adjustment and after-hours override use. Override time may be set in one-hour increments.
  4. Thermowell temperature sensors shall be stainless steel probe of length that is equivalent to a minimum of 50% of the pipe diameter. End-to-end accuracy shall be  $\pm 0.5$  deg. F. Connection box shall be moisture/water proof with conduit fitting. Furnish the stainless steel thermowell to the mechanical contractor for installation. A thermal conducting grease shall be installed in the thermowell to provide uniform temperature sensing.
  5. Provide flat plate stainless steel space temperature sensors with no local setpoint adjustment as indicated on the drawings.
- B. Carbon Dioxide Sensors: The sensor shall be capable of monitoring carbon dioxide concentration with an accuracy of  $\pm 30$  parts per million (PPM). The sensor shall produce a linear 0-10 VDC or 4-20 mA signal over the range of 0 to 2000 PPM. The sensor shall measure using non-dispersed infrared (NDIR) technology to measure carbon dioxide gas and shall be;
1. Wall mounted carbon dioxide sensors shall be Veris CWE series or equivalent.
  2. Duct mounted carbon dioxide sensor shall be Veris CWD series or equivalent.
  3. The EMCS contractor shall utilize the required calibration devices to properly commission and calibrate the sensors per the manufacturer's requirements.
- C. Relative Humidity Sensors: relative humidity sensors shall be a two-wire type, 4-20 mA output proportional to the relative humidity range of 0-100%. The accuracy of the sensors shall be  $\pm 3\%$  over a range of 5-95% RH.
1. Outdoor relative humidity sensors: provide non-corroding outdoor shield to minimize wind effects and solar heating. Install wall-mount weather proof enclosure with conduit fitting. Sensor shall be Veris HO series, or equivalent.
  2. Wall-mounted relative humidity sensor: sensor shall be installed in a wall-mounted enclosure with white cover. Sensor shall be Veris HEW series or equivalent.
  3. Duct-mounted relative humidity sensor: sensor shall be provided with a moisture resistant enclosure with conduit fitting. The probe length shall be 8" minimum. Sensor shall be Veris HED series or equivalent.
- D. Pressure Transducers:
1. Air pressure sensor: The pressure sensors shall have an input range compatible with the medium being measured. The proportional output signal shall be 0-10 VDC or 4-20 mA. Sensor shall be SETRA Model 264 or equivalent.
  2. Water pressure sensors: The pressure sensors shall have an input range compatible with the medium being measured. The proportional output signal shall be 0-10 VDC or 4-20 mA. Sensor shall be SETRA Model 230 or equivalent. Sensor shall be installed with a valved piping bypass and bleed off for each port.
- E. Freezestat: Provide freezestats for all chilled water air handling systems that receive more than 10% untreated outside air. Freezestats shall provide vapor tension elements, which shall serpentine the inlet face on all coils. Provide additional sensors, wired in series, to provide one linear foot per square foot of coil surface area. Freezestat shall be manually reset at the switch. Interlock to the associated fan so that fan will shut down

when HOA switch is in hand or auto position. Provide time delay relays with a 0-10 minute time delay relay duration to minimize nuisance freezestat trips. Time delay relay shall be adjustable at the associated control panel.

- F. Air differential pressure switch: For fan shutdown, provide air differential pressure switches for all fans controlled by a variable frequency drive (VFD) to shut down the associated fan in the event of sensing high differential pressure. Air differential pressure switches shall have an adjustable setpoint with a range of 0-10 inches w.g. with manual reset at the switch. Provide ¼ inch copper tubing with compression fittings to mount to the side of the duct. Sensor shall be DWYER Series 1620 or equivalent.
- G. Momentary control relays: Provide momentary control relays as indicated. . Relays shall have coil ratings of 120 VAC, 50 mA or 10-30 VAC/VDC, 40 mA as suitable for the application. Contact ratings shall be 10 amp. Provide complete isolation between the control circuit and the digital output. Relays shall be located in the UC or other local enclosures and have pin-type terminals. Relays shall have LED indication of status.
- H. Current sensing relay: Current sensing relays shall be rated for the applicable load. The output relay shall have an accessible trip adjustment over its complete operating range. Enclosure shall have an LED to indicate relay status.
- I. Photocell: Ambient light level shall by a photocell in a non-corroding in a weatherproof housing with sun shield suitable for exterior installation. The control signal output shall be 4-20 ma or binary contact closure as specified in the sequences of operation. Mount the photocell on the north side of the building on the roof. The sensor reading shall be 0-750 foot-candles.
- J. Occupancy Sensors
  - 1. The dual-technology occupancy ceiling mounted sensor shall be capable of detecting presence in the control area by via Doppler shifts in transmitted ultrasound and passive infrared (PIR) heat changes. Sensor shall utilize Dual Sensing Verification Principle for coordination between ultrasonic and PIR technologies. Detection verification of both technologies must occur in order to activate lighting systems. Sensor shall have a retrigger feature in which detection by either technology shall retrigger the lighting system on within 5 seconds of being switched off. The sensor shall operate at 24 VDC/VAC. WattStopper DT-300 or approve equal.
  - 2. Sensors shall have a time delay that is adjustable with configuration software or shall have a fixed time delay of 5 to 30 minutes, set by a DIP switch. Sensors shall feature a walk-through mode, where lights turn off 3 minutes after the area is initially occupied if no motion is detected after the first 30 seconds.
  - 3. The sensor shall have an additional single-pole, double throw isolated relay with normally open, normally closed and common outputs. The isolated relay is for use with HVAC control, data logging, and other control options. The sensor shall have an LED indicator that remains active at all times in order to verify detection within the area to be controlled.

## 2.13 WATER FLOW METERS

- A. Insertion Turbine Flow Meters shall be provided for HVAC applications in piping larger than 2 inches. The flow meter shall have a stainless steel insertion probe with non-metallic rotors; 2.0 % accuracy of actual reading from 0.4 to 20 ft/s; turndown ratio of 30:1; pulse outputs proportional to flow rate. The flow meter shall be single turbine type on applications with 20 diameters of pipe upstream and 5 diameters of pipe downstream. The flow meter shall be a dual turbine type on applications with less than 20 diameters of

pipe upstream and 5 diameters of pipe downstream. Provide full port valve to allow for removal and re-insertion without disruption to the water service, to be installed by Division 23. Meter shall be ONICON F-1200 series or pre-approved substitution.

- B. Inline nutating-disk type flow meters shall be provided for domestic water and cooling tower metering applications. The meter shall include a pulse output for monitoring by the EMCS. Provide meter to be installed by Division 23. Meter shall be Badger RCDL series or approved substitution.

## 2.14 AIRFLOW MEASURING STATIONS (AFMS)

- A. Duct mounted airflow measuring stations with combination airflow and air temperature measurement devices shall have the following features:
  - 1. Multi-point sensors in one or more probe assemblies with a maximum of one to sixteen sensor nodes per location, and a single remotely mounted microprocessor-based transmitter for each measurement location. Each sensor node shall consist of two hermetically sealed bead-in-glass thermistors. Each sensing point shall independently determine the airflow rate and temperature at each node, which shall be equally weighted in calculations by the transmitter prior to output as the cross-sectional average. Each ducted sensor probe shall have an integral, U.L. Listed, plenum rated cable. Each independent temperature sensor shall have a calibrated accuracy of  $\pm 0.14^{\circ}\text{F}$  ( $0.08^{\circ}\text{C}$ ) over the entire operating temperature range of  $-20^{\circ}\text{F}$  to  $160^{\circ}\text{F}$  ( $-28.9^{\circ}\text{C}$  to  $71^{\circ}\text{C}$ ), and be calibrated at 3 temperatures against standards that are traceable to NIST. Acceptable manufacturer shall be EBTRON, Inc. GTx116-PC.
  - 2. Each transmitter shall have a display capable of simultaneously displaying both airflow and temperature. Airflow rate shall be field configurable to be displayed as velocity or volumetric rates, selectable as IP or SI units. Each transmitter shall operate on 24 VAC and be fused and protected from over voltage, over current and power surges.
  - 3. Each independent airflow sensor shall have a laboratory accuracy of  $\pm 2\%$  of Reading over the entire calibrated airflow range of 0 to 5,000 fpm (25.4 m/s), and be wind tunnel calibrated at 16 points against air velocity standards that are traceable to NIST.

## 2.15 DAMPERS:

- A. Provide motorized volume control and shutoff dampers as detailed in 23 33 00 Ductwork Accessories.

## 2.16 DAMPER ACTUATORS:

- A. Outside and exhaust air damper actuators shall be mechanical spring return. The actuator mounting arrangement and spring return feature shall permit normally open or normally closed positions of the damper as required.
- B. Outside and return air modulating actuators shall utilize analog (proportional) control 0-10 VDC. Actuators shall be driven in both the open and closed directions.
- C. Electric damper actuators shall be direct shaft mounted and use a V-bolt and toothed V-clamp causing a cold weld effect for positive gripping. Single bolt or setscrew type fasteners are not acceptable.
- D. Single section dampers shall have one electronic actuator direct shaft mounted.

- E. Multi-section dampers with electric actuators shall be arranged so that each damper section operates individually. One electronic actuator shall be direct shaft mounted per damper section.
- F. Damper actuators shall be BELIMO or equivalent.

## 2.17 CONTROL VALVES

- A. Furnish all valves controlled by the EMCS as shown on the Mechanical Drawings. Furnish all automated isolation valves as shown on the Mechanical Drawings. Control valves shall be factory fabricated of type, body material, and pressure class based on maximum pressure and temperature rating of piping system, unless otherwise indicated. EMCS contractor to size control valve with a maximum of three psi pressure drop. 2-position isolation valves shall be full-line size.
  - 1. All chilled water, condenser water, and hot water valves shall meet, at minimum, the following ANSI Class 150 ratings. Valves 0.5 inch to 2 inches shall have NPT female screwed ends. Valves 2.5 inches and larger shall have flanged ends.
  - 2. Equal Percentage control characteristic shall be provided for all water coil control valves.
- B. Pressure Independent Characterized Control Ball Valves ½" to 6", for **two-way modulating applications** shall have equal percentage characteristics and control the flow from 0 to 100% full rated flow with an operating pressure differential range of 5 to 50 PSI across the valve. The pressure independent control valve shall be provided and delivered from a single manufacturer as a complete assembly. The actuator shall be integrally mounted to the valve at the factory with a single screw on a direct coupled DIN mounting-base. All valve actuators shall be capable of being electronically programmed in the field by use of external computer software or a dedicated handheld tool for the adjustment of flow and/or temperature control. Programming using actuator mounted switches or multi-turn actuators are not acceptable. The control valves shall be sized for the scheduled flow and not pressure drop. Calibrated Balancing Valves and Automatic Flow-Control Valves shall be prohibited from use at coil circuit piping where pressure independent control valves are installed. Contractor shall provide a section of straight pipe five times the pipe diameter with respect to the nominal valve size upstream of the control valve assembly where utilizing integral flow sensor to guarantee sensor accuracy.
  - 1. NPS 3/4" and Smaller: Belimo PIQCV or equal. Forged brass body rated at no less than 360 PSI, stainless steel ball and blowout-proof stem, characterizing disc integral to ball, PTFE ball seat, dual EPDM lubricated O-rings, and female NPT union ends. Close off pressure rating of 100 psi. Integral pressure regulator located upstream of ball to maintain a constant pressure differential. Replaceable cartridge type regulators are not permitted.
  - 2. NPS 1" through 2": Belimo ePIV or equal. Forged brass, nickel-plated body rated at no less than 360 PSI, stainless steel ball and blowout-proof stem, PTFE ball seat, dual EPDM lubricated O-rings, stainless steel or TEFZEL characterizing disc, and female NPT union ends. Close off pressure rating of 200 psi. Valve shall be integrated with an electronic (ultra-sonic or electromagnetic) flow sensor (accuracy +/- 2%) providing analog flow feedback. The valve shall reposition to maintain the required flow with a +/- 5% accuracy over a pressure differential range of 1 to 50 psi.
  - 3. NPS 2-1/2" through 6": Belimo ePIV or equal. GG25 cast iron body according to ANSI 125, Class B, stainless steel ball and blowout-proof stem, PTFE ball seat, with a dual EPDM lubricated O-rings and a stainless steel flow characterizing disc. End connection pattern to match ANSI 125 flange. Close off pressure rating of 100 psi. Valve shall be integrated with an electronic (ultra-sonic or

electromagnetic) flow sensor (accuracy +/- 2%) providing analog flow feedback. The valve shall reposition to maintain the required flow with a +/- 5% accuracy over a pressure differential range of 1 to 50 psi.

- C. Characterized Control Ball Valves (CCV) for ½' to 2": for **3-way modulating applications** shall have equal percentage characteristics. Manufacturer shall be Belimo or approved equal.
  - 1. Valve housing shall consist of forged brass rated at no less than 400 psi at 250 °F. Three-way valves shall have EPDM O-rings behind ball seals to allow for a minimum close-off pressure of 40 psi with an actuator that provides 35 in-lbs torque for ½ to 2 in. sizes. Three-way valves shall be installed in a "tee" configuration with actuator perpendicular to the shaft. Confirm mixing or diverting application for correct valve selection.
- D. Globe Valves 2-1/2" to 6": for **3-way modulating applications** shall have equal percentage characteristics. Manufacturer shall be Belimo G7 series or approved equal.
  - 1. Valve housing shall consist of cast iron rated at no less than 125 psi at 300 °F. Valve shall have stainless steel stem, plug and seat. Three-way valves shall be installed in a "tee" configuration with actuator perpendicular to the shaft. Confirm mixing or diverting application for correct valve selection.
- E. Butterfly valves: For chiller and cooling tower isolation control valves, butterfly control valves may be provided.
  - 1. Butterfly Isolation valves shall be line-size. Design velocity shall be less than 12 feet per second when used with standard EPDM seats. Butterfly valves shall have ductile iron body, 304 stainless steel disc and EPDM seat. The valve body close-off pressure rating shall be 150 psi over a range of -20 F to 250 F. The flange shall be ANSI 125/250. Belimo F6 and F7 series or approved manufacturer.
- F. Actuators for characterized control valves and globe valves: Provide electric actuators for all control valves that are furnished as part of the EMCS contract. Two-way and three-way control valve actuators shall meet, at minimum, the following requirements:
  - 1. Motor driven type with gear assembly made of hardened steel. Actuator shall have an input voltage of 24 VAC. Interior actuator housings shall be NEMA-2 rated. Exterior housings shall require a weather shield or shall be NEMA-4 rated. Provide visual mechanical position indication
  - 2. Valves shall be sized to meet the shut-off requirements when operating at the maximum system differential pressure and with the installed system pump operating at shut-off head. Actuators shall control against system maximum working pressures.
  - 3. Normal and failure positions shall be as indicated in the operating sequences. Provide spring return action per the sequences.
  - 4. Manual declutch lever to enable manual operation of the valve. It shall be possible for an operator to manually modulate valves located in mechanical rooms in the event of loss of power.
  - 5. Overload Protection: Actuators shall provide protection against actuator burnout by using an internal current limiting circuit or digital motor rotation sensing circuit. Circuit shall insure that actuators cannot burn out due to stalled damper or mechanical and electrical paralleling. End switches to deactivate the actuator at the end of rotation are acceptable only for butterfly valve actuators.
  - 6. All actuators shall be capable of being electronically programmed in the field by

use of external computer software or a dedicated handheld tool for the adjustment of flow. Programming using actuator mounted switches or multi-turn actuators are not acceptable.

7. Electric actuators shall be Belimo, compatible with the valves furnished.

G. Butterfly Valve Industrial Actuators

1. Enclosure shall be NEMA 4 (weatherproof) enclosure and will have an industrial quality coating.
2. Actuator shall have a motor rated for continuous duty. The motor shall be fractional horsepower; permanent split capacitor type designed to operate on a 120 VAC, 1 phase, 60 Hz supply. Two adjustable cam actuated end travel limit switches shall be provided to control direction of travel. A self-resetting thermal switch shall be imbedded in the motor for overload protection.
3. Reduction gearing shall be designed to withstand the actual motor stall torque. Gears shall be hardened alloy steel, permanently lubricated. A self-locking gear assembly or a brake shall be supplied.
4. Actuator shall have a 6 ft wiring harness provided for ease in field wiring (above 1500 in-lbs). Two adjustable SPDT cam-actuated auxiliary switches, rated at 250 VAC shall be provided for indication of open and closed position. Actuator shall have heater and thermostat to minimize condensation within the actuator housing.
5. Actuator shall be equipped with a hand wheel for manual override to permit operation of the valve in the event of electrical power failure or system malfunction. Hand wheel must be permanently attached to the actuator and when in manual operation electrical power to the actuator will be permanently interrupted. The hand wheel will not rotate while the actuator is electrically driven.
6. The actuator shall be analog, floating, or two position as called out in the control sequence of operation. All analog valves shall be positive positioning, and respond to a 2-10 VDC, 4-20 mA, or adjustable signal as required. Analog actuators shall have a digital control card allowing any voltage input for control and any DC voltage feedback signal for position indication.
7. Butterfly valve actuators shall be Belimo furnished with specified butterfly valves.

2.18 REFRIGERANT LEAK DETECTION:

A. Refrigerant leak detection monitors shall be provided for the refrigerant and number of chillers installed.

1. Power consumption: AC - 325 mA, DC - 250 mA. Volt free contacts to indicate an alarm condition.
2. Operating temperature range of 32 °F to 105 °F. Operating humidity range of 0 to 95% non-condensing.
3. Measuring range of 0-1000 ppm proportional to 4 to 20mA output range for each sampling point.
4. System shall detect the presence of the following types of refrigerants regardless of refrigerant type using sequential sampling and multi-point monitoring method: CFC, HCFC and HFC as provided with the chillers.
5. System shall annunciate to the EMCS through a contact closure and have a local alarm (audible and visual) Control panel shall have a silencing alarm button. Signage at all entry points to the chiller room shall be installed. Initial alarm shall comply with recommended Allowable Exposure Level (AEL). Adjustable 3 level alarm for each point shall be supplied with common alarm output contacts. Provide local digital indication of ppm level for a minimum of 1 sample point per chiller. A sample point shall be located close to each chiller and the refrigerant pump out unit location. Location to be approved by the engineer. Sample point if

in alarm shall flash the associated LED. Provide local alarm horns and visual (stroboscopic) beacons at the following locations to activate upon alarm to an approved detail:

- a. Outside of entrance doors to chiller machine room.
  - b. Inside rooms without an escape route other than through the chiller room.
  - c. At each chiller location.
  - d. At any other location in the chiller room as necessary to ensure that a person at any location in the chiller room and room that can be entered from the chiller room can see the visual alarm and hear the audible alarm and at any other location required to meet the applicable codes.
6. System shall shut down all electrical equipment (chiller systems and associated pumps, AHU, FCU, etc.) and sequence emergency extract equipment as required to meet regulations. Where combustion equipment is employed, refrigerant vapor monitoring system shall automatically shut down the combustion process in event of refrigerant leakage if other alternative acceptable conditions are not applied. Ventilation system, chiller and associated pumps and other equipment shut down as a result of the refrigerant leak alarm shall return to normal operation when the refrigerant monitoring system is no longer detecting refrigerant levels above set points and alarms have been silenced.
  7. System shall have self-diagnostics and supply common malfunction output. Loss of sample flow at either sample or ZERO line and electrical malfunction shall annunciate to the EMCS.
  8. Provide two (2) additional particulate filters and zero gas filter cartridges.
  9. Provide an emergency shut-off control button outside each chiller plant room entrance/exit door. Button shall be mounted at 48 inches above finished floor adjacent to refrigerant leak detection alarm light. Activation of any one of the buttons shall de-energize all chillers and other electrical equipment within the chiller plant room. Button shall be manually reset.
  10. Provide BACnet MS/TP interface to EMCS. Provide Strobe/Horns and Emergency Push Buttons.
  11. Maximum System Maintenance Requirements - The system shall require no periodic maintenance other than periodic checking. Periodic checking or adjustments of the unit shall be capable of being accomplished by one person at the unit location.
  12. Manufacturer Capability Requirements - As a minimum, the Gas Monitoring Equipment manufacturer must meet the following requirements:
    - a. Be capable of supplying all equipment used to check or calibrate the unit
    - b. Be capable of providing onsite service with factory trained personnel
    - c. Be capable of providing start-up assistance and training for the owner/operator
  13. Gas Monitoring System shall be a Mine Safety Appliances Company Chillgard RT Refrigerant Monitor or equal.

## 2.19 PANELS AND ENCLOSURES

- A. Provide panels and enclosures for all components of the EMCS, which are susceptible to physical or environmental damage.
- B. Interior panels and enclosures shall meet be NEMA 1 rated painted steel panels with locking door.
- C. Exterior mounted panels and enclosures shall be NEMA 4 painted steel panels with



locking door.

- D. Panels for USCs shall be mounted on the outside of all unit ventilators and fan coil units with three feet of wall clearance in front of them and no higher than 7 feet to the bottom of the panel.

## 2.20 LABELING and WARNING NOTICES

- A. Provide labeling for all control panels and enclosures.
- B. Provide labeling of all control wires and input/output points at the controller and at the control device; the label at each end of the wire shall be the same. Labels shall be machine generated, typed and clearly legible with a maximum of 17 characters. Hand written labels or labels written on the control wire jacket will not be acceptable. Each label shall be unique to its function and shall reference the applicable system. For example "AHU-1 SAT" will indicate the supply air temperature sensor for AHU-1. Improper labeling shall be removed and shall require re-commissioning of the control device and controller to document correct functionality.
- C. Provide high voltage warning notices at all equipment controlled by the EMCS and at all associated motor starters when used by equipment controller.

## 2.21 TUBING AND PIPING

- A. Provide tubing and piping as required for the field instrumentation.
- B. Tubing within equipment rooms, vertical risers, and penetrations to ductwork shall be either copper pipe or shall be plastic tubing within conduit. Tubing for all water-based instrumentation shall be copper pipe. Identify the type of tubing proposed in the shop drawing submittal.
- C. Provide suitable bulk head fittings for duct and panel penetrations.
- D. Tubing in plenum rated areas may be plastic tubing. Polyethylene tubing shall meet, at minimum, the following requirements: flame retardant; crack resistant; 300 psi burst pressure.

## 2.22 CONDUIT AND FITTINGS

- A. Provide all conduits, raceways and fittings for the EMCS monitoring, communication and control cabling. All work shall meet all applicable codes.
- B. Conduit, where required, shall meet, the requirements specified within Division 26.
- C. EMCS monitoring and control cable shall not share conduit with cable carrying voltages in excess of 90 VAC.

## 2.23 CABLING

- A. Provide all cables for the EMCS. Cable shall meet, at minimum, the following requirements:
  - 1. Minimum 98% conductivity stranded copper.
  - 2. Proper impedance for the application as recommended by the EMCS component manufacturer.

3. Monitoring and control cable shall be #18 AWG or larger, dependent on the application. Analog input and output cabling shall be shielded.
4. Management Level Network cable shall be CAT 6, 24 gauge unshielded.
5. Automation Level Network cable shall be #24 AWG shielded.
6. Shield shall be grounded at the CCP, UC, or control panel. Ground at one end only to avoid ground loops.
7. Identification of each end at the termination point. Identification should be indicated on and correspond to the record drawings.

B. 120 VAC power wiring shall be of #12 AWG solid conductor or larger as required.

## PART 3 - EXECUTION

### 3.1 PRE-CONSTRUCTION

- A. The EMCS supplier shall provide a pre-construction coordination meeting with the affected trades to ensure a cooperative efficient process of installation. The invited trades shall include the general contractor, mechanical contractor, electrical contractor, test and balance contractor, owner's representative, consulting engineer and others with a direct interest in the coordination of the affected systems. The EMCS contractor shall provide an outline of the meeting agenda highlighting the construction schedule, coordination with mechanical and electrical trades. Provide a sign-in sheet and submit it through the attendees along with a summary of the meeting notes for future reference.

### 3.2 INSPECTION DURING INSTALLATION

- A. Provide a technician to assist the Engineer or Owner's Representative with inspections made during the installation period that are required to review the progress and quality of ongoing work. The engineer/owner's representative shall generate field observation reports on the findings of the inspection. The engineer or owner's representative shall advise the EMCS contractor during the inspection of any concerns noted with respect to the installation and shall repeat the concerns in writing as soon as possible after the inspection is completed. The EMCS contractor shall take corrective action to meet the requirements of the specifications. Upon correction, the EMCS contractor shall submit written documentation through the contractors to the engineer.

### 3.3 INSTALLATION OF COMPONENTS

- A. Provide all interlock and control wiring. All wiring shall be installed in a neat and professional manner in accordance with specification Division 26 and all national, state and local electrical codes.
- B. Provide wire and wiring techniques recommended by equipment manufacturers. Control wiring shall not be installed in power circuit raceways. Magnetic starters and disconnect switches shall not be used as junction boxes. Provide auxiliary junction boxes as required. Coordinate location and arrangement of all control equipment with the Owner's Representative prior to rough-in. Provide auxiliary pilot duty relays on motor starters as required for control function.
- C. Electrical Contractor shall provide 120 or 277 volt power at a junction box within 48" of the controller. The BAS Contractor shall coordinate with the Electrical Contractor to identify locations of power requirements prior to the installation of the controls.
- D. Conduit for control wiring shall be provided whenever one of the following conditions exists:

1. Conduit is indicated on the drawings or specifically required by the specifications.
  2. Cabling runs through inaccessible areas such as within partitions/walls, above closed in ceilings, under floor; within trenches and underground; on the exterior of the building; exposed on the surface of the building; when encased in concrete or other material that makes the cable inaccessible or when located such that access to the cable is not readily obtained.
  3. Cable within mechanical, telecommunications and electrical equipment rooms and control rooms.
  4. Conduit shall be installed, inside wall from sensor box to above the wall, for all wall mounted temperature, humidity and CO2 sensors.
- E. Control wiring located above an accessible ceiling space may be plenum rated cable. Plenum rated wire shall be bundled and routed at right angles to the building lines and secured to the building structure every 15 feet.
- F. Control wiring located in underground conduits shall be provided with direct-burial-rated insulation.
- G. When communication bus enters or exits a building, a surge suppressor shall be installed. The surge suppressor shall be installed according to the controls manufacturer's instructions.
- H. Provide sleeves for all cable and conduit passing through walls, partitions, structural components, floors and roof
- I. All sensor wiring shall be labeled to indicate the origination (at the device) and destination of data (at the control panel). The description shall indicate the type and location of the control device such as "AHU-1 SA temp" or "VAV 1-1 space temp".
- J. Wall temp sensors at 48" above the finished floor to comply with ADA requirements and to match the height of the light switches. Mount humidity sensor at equal height to wall temperature sensor.

### 3.4 VERIFICATION REQUIREMENTS

- A. Verification shall be provided by the EMCS contractor to demonstrate and confirm that the installed system complies with the specifications and the control sequences of operation herein specified. upon completion of the verification process the EMCS contractor shall demonstrate to the engineer or owner's representative the functionality of the control system devices are in compliance with the contract documents.
- B. Technicians provided by the EMCS contractor shall be factory trained and qualified in the operation of the provided control system. The EMCS contractor shall provide, if requested, the factory training certificates of the individuals providing the verification services on this project.
- C. Verification tools, applicable to the system provided, shall be utilized by the factory-trained technicians for proper verification of system operation and functionality. Temperature verification sensors shall be NIST certified within the last 12 months. Meters such as Fluke 52 series or better shall be utilized. Use of non-certified meters may require the system to be re-verified with certified meters at no cost to the owner.
- D. Documentation of the verification process shall be provided per the project general conditions in electronic PDF format as required. Documentation shall include the following forms:

1. **Project System Verification Forms** for each controller provided on the project to verify the proper function of each controller, control device and system component provided.
  2. **Panel Verification Forms** for each control panel to document the proper installation and function of each control panel provided.
  3. **Sequence of Operation Verification Forms** for each piece of controlled equipment to confirm compliance of the control system with the specified sequences of operation.
  4. Not providing proper documentation for each control devices, panel, or system, upon request by the engineer or owner's representative, may require the EMCS contractor to re-verify the applicable systems at no additional cost to the owner.
- E. After completion of the verification, the EMCS contractor shall be able to demonstrate the sequence of operations for each system to the engineer and the owner's representative.
- F. Equipment checkout sheets are to be produced by this contractor showing checkboxes and compliance with the following procedures for each piece of equipment and turned over to the owner and/or mechanical engineer.

### 3.5 COLORGRAPHICS

- A. The colorgraphics shall be provided for the EMCS system prior to system acceptance and owner training.
- B. The colorgraphics provided shall include the following as a template. Provide forward and backward links on the graphic.
1. Site plan with link to overall building plan including detached buildings. The site plan shall be referenced to an automatically updated aerial view or map view of the area such as Google Maps or Bing Maps. Provide link to proceed to the overall building floor plan.
  2. The overall building plan shall indicate space temperature conditions referenced by the color of the zone. Specific details of the zone temperatures and equipment are not required. Provide a link to the floor plan wings, upper floors and remote buildings.
  3. The floor plan colorgraphics shall indicate the space temperatures by color references. Additional information shall indicate the space temperature, the occupancy of the zone, air handling units, VAV terminals and ductwork with diffusers. A link at each terminal unit or AHU shall automatically connect the system operator to the equipment colorgraphic.
  4. The colorgraphics for the equipment shall as a minimum be equal to the points from the input/output summary or control schematic. Primary control devices as required by the sequences of operation shall also be provided.
  5. Control points from equipment that are integrated into the EMCS via BACnet shall be provided to convey the operating conditions of the attached equipment. Coordination of the integration points shall be accomplished during the submittal phase. The EMCS contractor shall provide a list of all integrated points on their submittal.

### 3.6 CONTROL SYSTEM DEMONSTRATION AND ACCEPTANCE

- A. Startup testing documentation: Prepare the checklist documenting startup testing of each input and output device, with technician's initials and date certifying each device has been tested and calibrated prior to acceptance testing. This document shall indicate proof that the following functions have been commissioned and shall be included in the as-built documentation: short to ground check, configuration of trends, confirmation that color-

graphics are accurately representing actual systems, point to point checkout, all damper and valve actuators respond to input change, control modules are addressed and have functional descriptors, specified interlocks are functional, calibration report of all sensors, discrete outputs respond to time schedule or manual enable command.

B. Demonstration. Prior to acceptance, demonstrate the following performance tests to demonstrate system operation and compliance with specifications.

1. Engineer, owner's representative and mechanical contractor shall be invited to observe and review system demonstration. Provide attendees at least 10 days notice.
2. Demonstration shall follow process approved as part of the submittal and shall include complete checklists and forms for each system as part of system demonstration.
3. Demonstrate actual field operation of each sequence of operation as specified. Demonstrate calibration and response of any input and output points requested by engineer or owner's representative.
4. Demonstrate complete operation of operator interface including review of color-graphics, time schedules, trend logs, alarm notification, functionality of tablet PC operation.
  - a. PID loop response. Supply graphical trend data output showing each PID loop's response to a set point change representing an actuator position change of at least 25% of full range. Trend sampling rate shall be selectable from 10 seconds to 3 minutes, depending on loop speed. Each sample's trend data shall show set point, actuator position, and controlled variable values.
  - b. Demand limiting. Supply trend data output showing demand-limiting algorithm action. Trend data shall document action sampled each minute over at least a 30-minute period and shall show building kW, demand limiting setpoint, and status of set points and other affected equipment parameters.
  - c. Trend logs for each system. Trend data shall indicate set points, operating points, valve positions, and other data as specified. Logs shall be accessible through system's operator interface and shall be retrievable for use in other software programs.
5. Alarms and Interlocks. Check each alarm with an appropriate signal at a value that will trip the alarm. Trip interlocks using field contacts to check logic and to ensure that actuators fail in the proper direction. Alarm verification shall include temperatures exceeding alarm threshold (high and low), fan failure safety, duct high static pressure switch, freezestat, and smoke detector shutdown.
6. Tests that fail to demonstrate proper system operation to the engineer shall be repeated after contractor makes necessary repairs or revisions to hardware or software to successfully complete each test.

C. Owner Acceptance.

1. After tests described in this specification are performed to the satisfaction of both engineer and owner's representative, the engineer shall accept the control system as meeting completion requirements. Engineer may exempt tests from completion requirements that cannot be performed due to circumstances beyond EMCS contractor's control. Engineer shall provide written statement of each exempted test. Exempted tests shall be performed as part of warranty.
2. System shall not be accepted until completed demonstration forms and checklists are submitted and approved by the engineer.

### 3.7 DEMONSTRATION AND OWNER TRAINING

- A. Furnish basic operator training for multiple persons on data display, alarm and status descriptors, requesting data, execution commands and log requests. Include a minimum of 16 hours: 8 hours instructor time for onsite training and 8 hours of hands on class environment training. Training sessions may be provided in 4-hour increments as approved by the owner's representative.
  - 1. Change/modify temperature setpoints.
  - 2. Change/modify time of day, holiday and override schedules.
  - 3. Display, create, and modify trends of system points.
  - 4. Update room numbers on the color-graphics.
- B. Demonstrate complete and operating system to Owner. Provide written documentation listing the attendees of the specified training with sign-in sheet and training time and date.

## PART 4 - SEQUENCE OF OPERATIONS

### 4.01 DX SPLIT SYSTEM UNIT CONTROLS (FCU-1 thru FCU-4)

- A. The system control panel shall be capable of communicating with each individual split system. The control manufacturer shall provide one controller per DX system that communicates back to the main control panel.
- B. Space mounted sensor shall modulate stages of heating or cooling as required to maintain space setpoint.

### 4.02 EXHAUST FANS

- A. Fans EF-2 thru EF-5:
  - 1. Exhaust fans shall be interlocked with the respective fan coil unit.
- B. Fans EF-1:
  - 1. Exhaust fans shall be interlocked with Carbon monoxide sensors in bay area. Interlock with motorized damper on louvers to open when fan is activated. Provide motor speed control. Exhaust fan shall energize when carbon monoxide detects a concentration of 50PPM.

**END OF SECTION 23 09 63**

## **SECTION 23 23 00 - REFRIGERANT PIPING**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 – Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for Owner's use.

### **PART 2 - PRODUCTS**

#### **2.1 GENERAL**

Provide for the systems as shown. Submit shop drawings of piping systems showing all traps, pipe sizes, and accessories; drawing to be marked "Approved", and signed by a representative of the Application Engineering Department of the condensing unit manufacturer. Pipe sizes shall be as recommended by unit manufacturer. Refer to piping schematic on Drawings.

#### **2.2 MATERIAL**

- A. PIPE: Copper ACR hard-drawn tubing.
- B. FITTINGS: Wrought copper streamlined sweat fitting.
- C. SOLDER: Sil-Fos; except on valves use solder recommended by valve manufacturer.

#### **2.3 ACCESSORIES**

All accessories shall be UL listed and rated in accordance with ARI Standard 710.

- A. On systems 7-1/2 tons and larger, each separate refrigerant circuit shall have a separate filter drier. Each filter drier shall have a replaceable core and a three valve bypass. The filter drier shall be full line size and installed in the refrigerant liquid line. The filter shall have a minimum 4-3/4 inches diameter shell with removable flange and gasket. Flange shall be tapped for 1/4 inch FPT access valve. Size filter-drier for maximum 2.0 psi pressure drop at evaporator operating temperature. Similar to Mueller Brass Company model Drymaster micro-guard refillable filter series SD-485 through SD19217 or Sporlan catch-all.
- B. On systems less than 7-1/2 tons, the filter drier shall be the sealed type; sizes as above. One drier per refrigerant circuit.
- C. Liquid-Moisture Indicator shall be installed in liquid refrigerant line; full line size similar to Mueller Brass Company model "Vuemaster" with soldered ends.
- D. Thermostatic expansion valve shall have adjustable super heat and be as manufactured by Sporlan.

## 2.4 EVACUATION

Evacuate moisture completely by applying a commercial vacuum pump for a minimum of 24 hours. Moisture indicator shall indicate a completely moisture-free condition at time of final inspection. The vacuum pump shall run until the system indicates a maximum of 35 degrees FDB. The system shall be flushed with the operating refrigerant and the vacuum pump connected and rerun to repeat the evacuation. Evaluation shall be performed under supervision of the Engineer.

## 2.5 REFRIGERANT AND OIL

- A. Contractor shall leave the refrigeration system with a full charge of refrigerant and oil and shall be responsible for the maintenance of a full charge of refrigerant and oil in the systems for a period of one year from date of Substantial Completion.
- B. Should any leaks in the refrigeration system occur during the guarantee period, the Contractor shall eliminate such leaks and recharge system to a full charge of refrigerant and oil at no cost to the Owner.

## PART 3 - EXECUTION

- 3.1 All equipment and piping shall be installed in accordance with the manufacturer's recommendations and printed installation instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the Drawings or in the Specifications. Provide all items required as per manufacturer's requirements.

## END OF SECTION



## **SECTION 23 31 13 - METAL DUCTWORK**

### **PART 1 - GENERAL**

#### **1.1 WORK INCLUDED**

- A. Low pressure ductwork.
- B. Medium and high pressure ductwork.
- C. Casings.
- D. Kitchen hood ductwork.
- E. Commercial dryer vent.
- F. Duct cleaning.

#### **1.2 RELATED SECTIONS**

Division 9 - Finishes: Weld priming, weather resistant, paint or coating.

- A. Section 23 02 00 - Basic Material and Methods.
- B. Section 23 05 29 – Hangers and Support for Piping and Equipment HVAC.
- C. Section 23 05 93 - Testing, Adjusting and Balancing.
- D. Section 23 07 13 - Duct Insulation.
- E. Section 23 33 00 - Ductwork Accessories.
- F. Section 23 37 13 - Air Distribution Devices.

#### **1.3 QUALITY ASSURANCE**

- A. Manufacturer's Qualifications: Firms regularly engaged in manufacture of metal ductwork products of types, materials and sizes required, whose products have been in satisfactory use in similar service for not less than 5 years.
- B. Installer's Qualifications: Firms with least 3 years of successful installation experience on projects with metal ductwork systems similar to that required for project.
- C. Codes and Standards:
  - 1. SMACNA Standards: Comply with latest SMACNA's "HVAC Duct Construction Standards, Metal and Flexible" for fabrication and installation of metal ductwork.
  - 2. ASHRAE90.1 Standards: Comply with ASHRAE Handbook, Equipment Volume, Chapter 1 "Duct Construction", for fabrication and installation of metal ductwork.
  - 3. NFPA Compliance: Comply with NFPA 90A "Standard for the Installation of Air Conditioning and Ventilating Systems", NFPA 90B "Standard for the Installation of Warm Air Heating and Air Conditioning Systems", and NFPA 96 Standard.
  - 4. IECC 2015: Comply with 2015 International Energy Conservation Code.

#### **1.4 GENERAL DESCRIPTION**

- A. Extent of metal ductwork is indicated on drawings and in schedules, and by requirements of this section.

## 1.5 SUBMITTALS

- A. Submit shop drawings, duct fabrication standards and product data under provisions of Division One.
- B. Indicate duct fittings, particulars such as gauges, sizes, welds, and configuration prior to start of work.
- C. The contract documents are schematic in nature and are to be used only for design intent. The contractor shall prepare sheet metal shop drawings, fully detailed and drawn to scale, indicating all structural conditions, all plumbing pipe and light fixture coordination, and all offsets and transitions as required to permit the duct to fit in the space allocated and built. All duct revisions required as a result of the contractor not preparing fully detailed shop drawings will be performed at no additional cost.

## 1.6 DEFINITIONS

- A. Duct Sizes: Inside clear dimensions. For lined ducts, maintain indicated clear size inside lining. Where offsets or transitions are required, the duct shall be the equivalent size based on constant friction rate.
- B. Low Pressure: Low pressure ductwork shall be rated for an operating pressure of 2". Low pressure ductwork shall be defined as all return, exhaust, and outside air ducts, all supply ductwork associated with constant volume air handling units with a scheduled external static pressure of less than 2", and all supply ductwork downstream of terminal units in variable volume systems.
- C. Medium Pressure: Medium pressure ductwork shall be rated for an operating pressure of 4". Medium pressure ductwork shall be defined as all supply ductwork extending from variable volume air handling units to terminal units in variable volume systems with air handling units having a scheduled external static pressure of less than 4". The supply ductwork of constant volume air handling units having a scheduled external static pressure greater than 2" and less than 4" shall be rated for medium pressure.
- D. High Pressure: High pressure ductwork shall be rated for an operating pressure of 6", or the scheduled external pressure of the equipment it is connected to, whichever is greater. The supply ductwork of air handling units having a scheduled external static pressure greater than 4" shall be high pressure.

## 1.7 DELIVERY, STORAGE, AND HANDLING

- A. Protection: Protect shop-fabricated and factory-fabricated ductwork, accessories and purchased products from damage during shipping, storage and handling. Prevent end damage and prevent dirt and moisture from entering ducts and fittings, use sheet metal end caps on any lined duct exposed to the weather.
- B. Storage: Where possible, store ductwork inside and protect from weather. Where necessary to store outside, store above grade and enclose with waterproof wrapping.

## PART 2 - PRODUCTS

### 2.1 DUCTWORK MATERIALS

- A. Exposed Ductwork Materials: Where ductwork is indicated to be exposed to view in occupied spaces, provide materials which are free from visual imperfections including pitting, seam marks, roller marks, stains and discolorations, and other imperfections, including those which would impair painting.
- B. Sheet Metal.: Except as otherwise indicated, fabricate ductwork from galvanized sheet steel complying with ASTM A 527, lockforming quality, with G 90 zinc coating in accordance with ASTM A 525; and mill phosphatized for exposed locations.
- C. Stainless Steel Sheet: Where indicated, provide stainless steel complying with ASTM A167; Type 316; with No. 4 finish where exposed to view in occupied spaces, No. 1 finish elsewhere. Protect finished surfaces with mill-applied adhesive protective paper, maintained through fabrication and installation.
- D. Aluminum Sheet: Where indicated, provide aluminum sheet complying with ASTM B 209, Alloy 3003, Temper H14.

## 2.2 MISCELLANEOUS DUCTWORK MATERIALS

- A. General: Non-combustible and conforming to UL 181, Class 1 air duct materials.
- B. Flexible Ducts: Flexmaster U.S.A., Inc. Type 3M or approved equal, corrosive resistant galvanized steel formed and mechanically locked to inner fabric with 1" thick insulation when flexible ducts are located in conditioned spaces and with R-5 insulation when located in unconditioned spaces. Flexible duct shall have reinforced metalized outer jacket comply with UL 181, Class 1 air duct.
- C. Sealants: Hard-Cast "iron grip" or approved equal, non-hardening, water resistant, fire resistive and shall not be a solvent curing product. Sealants shall be compatible with mating materials, liquid used alone or with tape or heavy mastic.
- D. Ductwork Support Materials: Except as otherwise indicated, provide hot-dipped galvanized steel fasteners, anchors, rods, straps, trim and angles for support of ductwork.
  - 1. For exposed stainless steel ductwork, provide matching stainless steel support materials.
  - 2. For aluminum ductwork, provide aluminum support materials.

## 2.3 LOW PRESSURE DUCTWORK

- A. Fabricate and support in accordance with latest SMACNA Duct Construction Standards and ASHRAE handbooks, except as indicated. Provide duct material, gauges, reinforcing, and sealing for operating pressures indicated.
- B. Size round ducts installed in place of rectangular ducts in accordance with ASHRAE table of equivalent rectangular and round ducts. No variation of duct configuration or sizes permitted except by approved shop drawings. Obtain engineer's approval prior to using round duct in lieu of rectangular duct.
- C. Construct T's, bends, and elbows with radius of not less than 1-1/2 times width of duct on centerline. Where not possible and where rectangular elbows are used, provide airfoil-turning vanes. Where acoustical lining is indicated, provide turning vanes of perforated metal with glass fiber insulation.

- D. Increase duct sizes gradually, not exceeding 15 degrees divergence wherever possible. Divergence upstream of equipment shall not exceed 30 degrees; convergence downstream shall not exceed 45 degrees.
- E. Use crimp joints with bead for joining round duct sizes 6 inch smaller with crimp in direction of airflow.
- F. Use double nuts and lock washers on threaded rod supports.

## 2.4 MEDIUM AND HIGH PRESSURE DUCTS

- A. Fabricate and support in accordance with SMACNA Duct Construction Standards and ASHRAE handbooks, except as indicated. Provide duct material, gauges, reinforcing, and sealing for operating pressures indicated.
- B. Construct T's, bends, and elbows with radius of not less than 1½ times width of duct on centerline. Where not possible and where rectangular elbows are used, provide airfoil-turning vanes. Where acoustical lining is required, provide turning vanes of perforated metal with glass fiber insulation. Weld in place.
- C. Transform duct sizes gradually, not exceeding 15 degrees divergence and 30 degrees convergence.
- D. Fabricate continuously welded medium and high pressure round and oval duct fittings two gauges heavier than duct gauges indicated in SMACNA Standard. Joints shall be minimum 4 inch cemented slip joint, brazed or electric welded. Prime coat welded joints.
- E. Provide standard 45 degree lateral wye takeoffs unless otherwise indicated where 90 degree conical tee connections may be used.

## 2.5 CASINGS

- A. Fabricate casings in accordance with SMACNA Duct Construction Standards and SMACNA High Pressure Duct Construction Standards and construct for operating pressures indicated.
- B. Mount floor mounted casings on 4 inch high concrete curbs. At floor, rivet panels on 8 inch centers to angles. Where floors are acoustically insulated, provide liner of 18 gauge galvanized expanded metal mesh supported at 12 inch centers, turned up 12 inches at sides with sheet metal shields.
- C. Reinforce doorframes with steel angles tied to horizontal and vertical plenum supporting angles. Install hinged access doors where indicated or required for access to equipment for cleaning and inspection. Provide clear wire glass observation ports, minimum 6 X 6 inch size.
- D. Fabricate acoustic casings with reinforcing turned inward. Provide 16 gauge back facing and 22 gauge perforated front facing with 3/32 inch diameter holes on 5/32 inch centers. Construct panels 3 inches thick packed with 4.5 lb./cubic foot minimum glass fiber media, on inverted channels of 16 gauge.

## 2.6 BURIED UNDERGROUND DUCTS

- A. Buried ducts may be concrete encased sheet metal or fiberglass reinforced plastic as indicated.

## 2.7 KITCHEN HOOD EXHAUST DUCTWORK

- A. Fabricate in accordance with SMACNA Duct Construction Standards, and NFPA 96.
- B. Construct of 16 gauge carbon steel or 18 gauge stainless steel, using continuous external welded joints.

## 2.8 DISHWASHER/SHOWER/LOCKER ROOM EXHAUST DUCTWORK

- A. All ductwork shall be stainless steel, one gauge heavier than that required for galvanized steel duct.
- B. Slope all duct to drain out grilles or provide drain line to floor drain.

## 2.9 COMMERCIAL DRYER VENT

- A. Construct of 18 gauge, Type 316 stainless steel.
- B. All welded construction.
- C. Provide hard duct connection to dryer.

# PART 3 - EXECUTION

## 3.1 GENERAL INSTALLATION REQUIREMENTS

- A. Obtain manufacturer's inspection and acceptance of fabrication and installation of ductwork at beginning of installation.
- B. Provide openings in ductwork where required to accommodate thermometers and controllers. Provide pitot tube openings where required for testing of systems, complete with metal can with spring device or screw to ensure against air leakage. Where openings are provided in insulated ductwork, install insulation material inside a metal ring.
- C. Locate ducts with sufficient space around equipment to allow normal operating and maintenance activities.
- D. Connect terminal units to medium or high pressure ducts with four feet maximum length of flexible duct. Do not use flexible duct to change direction.
- E. Connect diffusers or troffer boots to low pressure ducts with 6 feet maximum, 4 feet minimum, length of flexible duct. Hold in place with strap or clamp.
- F. During construction provide temporary closures of metal or taped polyethylene on open ductwork to prevent construction dust from entering ductwork system.
- G. The interior surface of all ductwork shall be smooth. No sheet metal parts, tabs, angles, or anything else may project into the ducts for any reason, except as specified to be so. All seams and joints shall be external.
- H. All ductwork located exposed on roof shall be "crowned" to prevent water from ponding. Ref: Insulation for additional requirements.
- I. Where ducts pass through floors, provide structural angles for duct support. Where ducts pass through walls in exposed areas, install suitable sheet metal escutcheons as closers.

- J. All angles shall be carried around all four sides of the duct or group of ducts. Angles shall overlap corners and be welded or riveted.
- K. All ductwork shall be fabricated in a manner to prevent the seams or joints being cut for the installation of grilles, registers, or ceiling outlets.
- L. All duct hangers shall be attached to building structure. Cutting slots in roof or floor decking for hanger straps to be cast in concrete is not acceptable.

### 3.2 INSTALLATION OF FLEXIBLE DUCTS

- A. Maximum Length: For any duct run using flexible ductwork, do not exceed 6'-0" extended length.
- B. Installation: Install in accordance with Section III of SMACNA's, "HVAC Duct Construction Standards, Metal and Flexible".

### 3.3 REQUIREMENTS FOR DUCTS BURIED UNDERGROUND

- A. Slope underground ducts to plenums or low pump-out points at 1:500. Provide access doors for inspection.
- B. Coat buried, metal ductwork without factory jacket with one coat and seams and joints with additional coat of asphalt base protective coating.
- C. Insulate buried supply duct runs over 50 feet long with one inch thick insulation covered with plastic vapor barrier.
- D. Encase buried metal ductwork in 3 inch minimum of concrete. Provide adequate tie-down points to prevent ducts from floating during concrete placement. Introduce no heat into ducts for 20 days following placement of concrete.

### 3.4 REQUIREMENTS FOR UNIT CASINGS

- A. Set plenum doors 6 to 12 inches above floor. Arrange door swings so that fan static pressure holds door in closed position.

### 3.5 REQUIREMENTS FOR KITCHEN HOOD EXHAUST DUCT

- A. Provide residue traps in kitchen hood exhaust ducts at base of vertical risers with provisions for cleanout.
- B. Provide access openings in each change in direction, located on sides of duct 1½" minimum from bottom, and fitted with grease-tight covers of same material as duct
- C. Use stainless steel for ductwork exposed to view.

### 3.6 DUCTWORK APPLICATION SCHEDULE

#### AIR SYSTEM

#### MATERIAL

Low Pressure Supply

Galvanized Steel, Aluminum

Buried Supply or Return

Concrete, Glass Fiber Reinforced Plastic

|                                 |                                                                       |
|---------------------------------|-----------------------------------------------------------------------|
| Medium and High Pressure Supply | Galvanized Steel                                                      |
| Return and Relief               | Galvanized Steel, Aluminum                                            |
| General Exhaust                 | Galvanized Steel, Aluminum                                            |
| Kitchen Hood Exhaust            | Carbon Steel, Stainless Steel                                         |
| Dishwasher/Shower/Locker Room   | Stainless Steel                                                       |
| Dryer Vent/Paint Hood Exhaust   |                                                                       |
| Fume Hood Exhaust               | Stainless Steel, Glass Fiber Reinforced Plastic if not in air plenum. |
| Outside Air Intake              | Steel                                                                 |
| Combustion Air                  | Steel                                                                 |
| Emergency Generator Ventilation | Steel                                                                 |

### 3.7 DUCTWORK HANGERS AND SUPPORTS

- A. All ductwork shall be properly suspended or supported from the building structure. Hangers shall be galvanized steel straps or hot-dipped galvanized rod with threads pointed after installation. Strap hanger shall be attached to the bottom of the ductwork, provide a minimum of two screws one at the bottom and one in the side of each strap on metal ductwork. The spacing, size and installation of hangers shall be in accordance with the recommendations of the latest SMACNA edition.
- B. All duct risers shall be supported by angles or channels secured to the sides of the ducts at each floor with sheet metal screws or rivets. The floor supports may also be secured to ducts by rods, angles or flat bar to the duct joint or reinforcing. Structural steel supports for duct risers shall be provided under this Division.

### 3.8 AIR DUCT LEAKAGE: (From SMACNA Duct Standards Latest Edition) Test all ductwork (designed to handle over 1000 CFM) as follows:

#### A. Test apparatus

The test apparatus shall consist of:

1. A source of high pressure air--a portable rotary blower or a tank type vacuum cleaner.
2. A flow measuring device consisting of straightening vanes and an orifice plate mounted in a straight tube with properly located pressure taps. Each orifice assembly shall be accurately calibrated with its own calibration curve. Pressure and flow readings shall be taken with U-tube manometers.

#### B. Test Procedures

1. Test for audible leaks as follows:
2. Close off and seal all openings in the duct section to be tested. Connect the test apparatus to the duct by means of a section of flexible duct.
  - a. Start the blower with its control damper closed.
  - b. Gradually open the inlet damper until the duct pressure reaches 1.5

- c. times the standard designed duct operating pressure.
  - c. Survey all joint for audible leaks. Mark each leak and repair after shutting down blower. Do not apply a retest until sealants have set.
- 3. After all audible leaks have been sealed, the remaining leakage should be measured with the orifice section of the test apparatus as follows:
  - a. Start blower and open damper until pressure in duct reaches 50% in excess of designed duct operating pressure.
  - b. Read the pressure differential across the orifice on manometer No. 2. If there is no leakage, the pressure differential will be zero.
  - c. Total allowable leakage shall not exceed one (1) percent of the total system design air flow rate. When partial sections of the duct system are tested, the summation of the leakage for all sections shall not exceed the total allowable leakage.
  - d. Even though a system may pass the measured leakage test, a concentration of leakage at one point may result in a noisy leak which, must be corrected.
- 4. Testing Report
  - a. Contractor shall provide a testing report for each air system to the engineer. The report shall indicate the completion of testing and compliance with testing specification.
  - b. All duct testing reports shall be included in the final close out documents.

### 3.9 DUCT SYSTEM CLEANING

- A. Duct system cleaning shall be performed in accordance with the current published standards of ASHRAE and NADCA.
- B. Duct system cleaning method used shall incorporate the use of vacuum collection devices that are operated continuously during cleaning. A vacuum device shall be connected to the downstream end of the section being cleaned through a predetermined opening. The vacuum collection device must be of sufficient power to render all areas being cleaned under negative pressure, such that containment of debris and the protection of the indoor environment is assured.
- C. All vacuum devices exhausting air inside the building shall be equipped with HEPA filters (minimum efficiency), including hand-held vacuums and wet-vacuums.
- D. All vacuum devices exhausting air outside the facility shall be equipped with Particulate Collection including adequate filtration to contain debris removed from the HVAC system. Such devices shall exhaust in a manner that will not allow contaminants to re-enter the facility. Release of debris outdoors must not violate any outdoor environmental standards, codes or regulations.
- E. Fibrous glass thermal or acoustical insulation elements present in any equipment or ductwork shall be thoroughly cleaned with HEPA vacuuming equipment, while the HVAC system is under constant negative pressure, and not permitted to get wet in accordance with applicable NADCA and NAIMA standards and recommendations.
- F. Duct cleaning method used shall not damage the integrity of the ductwork, nor damage porous surface materials such as liners inside the ductwork or system components.
- G. Replace the fiberglass material if there is any evidence of damage, deterioration, delamination, friable material, mold or fungus growth, or moisture such that fibrous glass materials cannot be restored by cleaning or resurfacing with an acceptable insulation repair coating.



- H. Clean external surfaces of foreign substances which might cause corrosive deterioration of metal or, where ductwork is to be painted, might interfere with painting or cause paint deterioration.
- I. Strip protective paper from stainless ductwork surfaces, and repair finish wherever it has been damaged.
- J. Temporary Closure: At ends of ducts which are not connected to equipment or air distribution devices at time of ductwork installation, provide temporary closure of polyethylene film or other covering which will prevent entrance of dust and debris until time connections are to be completed.
- K. Cleaning Report: Contractor shall provide a report to the Owner indicating the completion of duct cleaning per specification and areas of the duct system found to be damaged and/or in need of repair.

### 3.10 DUCT JOINTS AND SEAMS

- A. Seal all non-welded duct joints with duct sealant as indicated.

**END OF SECTION**

## **SECTION 23 33 00 – DUCTWORK ACCESSORIES**

### **PART 1 – GENERAL**

#### **1.1 WORK INCLUDED**

- A. Volume control dampers.
- B. Shutoff Dampers.
- C. Round Duct Taps.
- D. Conical Duct Taps.
- E. Fire dampers.
- F. Combination fire and smoke dampers.
- G. Back draft dampers.
- H. Air turning devices.
- I. Flexible duct connections.
- J. Duct access doors.
- K. Duct test holes.

#### **1.2 RELATED WORK**

- A. Section 23 05 48 - Vibration and Seismic Controls for HVAC Piping and Equipment.
- B. Section 23 31 13 - Metal Ductwork.

#### **1.3 REFERENCES**

- A. NFPA 90A - Installation of Air Conditioning and Ventilating Systems.
- B. SMACNA - Low Pressure Duct Construction Standards.
- C. UL 33 - Heat Responsive Links for Fire-Protection Service.
- D. UL 555 - Fire Dampers and Ceiling Dampers.

#### **1.4 SUBMITTALS**

- A. Submit shop drawings and product data under provisions of Division One.
- B. Provide shop drawings for shop fabricated assemblies indicated, including volume control dampers duct access doors duct test holes. Provide product data for hardware used.
- C. Submit manufacturer's installation instructions under provisions of Division 1, for fire dampers and combination fire and smoke dampers.

### **PART 2 - PRODUCTS**

## 2.1 VOLUME CONTROL DAMPERS

- A. Fabricate in accordance with SMACNA Low Pressure Duct Construction Standards, and as indicated.
- B. Fabricate splitter dampers of material same gauge as duct to 24 inches size in either direction, and two gauges heavier for sizes over 24 inches.
- C. Fabricate splitter dampers of double thickness sheet metal to streamline shape. Secure blade with continuous hinge or rod. Operate with minimum 1/2 inch diameter rod in self aligning, universal joint, action flanged bushing, with set screw.
- D. Fabricate single blade dampers for duct sizes to 9-1/2 x 24 inch.
- E. Fabricate multi-blade damper of opposed blade pattern with maximum blade sizes 12 x 72 inch.
  - 1. Assemble center and edge crimped blades in prime coated or galvanized channel frame with suitable hardware.
  - 2. On outside air, return air, and all other dampers required to be low leakage type, provide galvanized blades and frames, seven inches wide maximum, with replaceable vinyl, EPDM, silicone rubber seals on blade edges and stainless steel side seals. Provide blades in a double sheet corrugated type construction for extra strength. Provide hat channel shape frames for strength and blade linkage enclosure to keep linkage out of the air stream. Construction leakage not to exceed 1/2%, based on 2,000 fpm and 4 inch static pressure.
- F. Except in round ductwork 12 inches and smaller, provide end bearings. On multiple blade dampers, provide oil-impregnated nylon or sintered bronze bearings.
- G. Provide locking, indicating quadrant regulators on single and multi-blade dampers. Where rod lengths exceed 30 inches provide regulator at both ends.
- H. On insulated ducts mount quadrant regulators on stand-off mounting brackets, bases, or adapters.

## 2.2 SHUTOFF DAMPERS

- A. Fabricate in accordance with SMACNA Low Pressure Duct Construction Standards, and as indicated.
- B. Provide Class I multi-blade damper of parallel blade pattern for all ductwork systems which penetrate the building thermal envelope.
  - 1. Damper shall be constructed of one-piece 16 ga. roll-formed galvanized steel hat-shaped channel frame. Blades shall be 14 ga. roll-formed galvanized steel, airfoil type. Blade edge seals shall be neoprene gaskets mechanically locked to blade edge. Bearings shall be 304 stainless steel, oil-impregnated and self-lubricating sleeve type, turning in extruded holes in damper frame.
- C. Shutoff dampers shall have an air leakage rate not greater than 4 cfm/ft<sup>2</sup> of damper surface area at 1.0 in.w.g. and shall be labeled by an approved agency when tested in accordance with AMCA 500D for such purpose.

## 2.3 ROUND DUCT TAPS

- A. Taps to trunk duct for round flexible duct shall be spin-in fitting with locking quadrant butterfly damper, model no. FLD-B03 by Flexmaster or approved equal.

#### 2.4 CONICAL DUCT TAPS

- A. Taps to trunk duct for primary air inlet to all VAV terminal units shall be conical fitting, model no. CB by Flexmaster or approved equal.

#### 2.5 ACCEPTABLE MANUFACTURERS - FIRE DAMPERS AND COMBINATION FIRE AND SMOKE DAMPERS

- A. Greenheck.
- B. Louvers and Dampers Inc.
- C. Ruskin.
- D. Nailor Industries.
- E. Pottorff

#### 2.6 FIRE DAMPERS

- A. Fabricate in accordance with NFPA 90A and UL 555, and as indicated.
- B. Provide curtain type dampers of galvanized steel with interlocking blades. Provide stainless steel closure springs and latches for horizontal installations. Configure with blades out of air stream.
- C. Fabricate multiple blade fire dampers per U.L. with 16 gauge minimum galvanized steel frame and blades, oil-impregnated bronze or stainless steel sleeve bearings and plated steel axles, 1/8 x 1/2 inch plated steel concealed linkage, stainless steel closure spring, blade stops, and lock.
- D. Fusible links, UL 33, shall separate at 160 degrees F. Provide adjustable link straps for combination fire/balancing dampers.

#### 2.7 COMBINATION FIRE AND SMOKE DAMPERS

- A. Fabricate in accordance with NFPA 90A and UL 555, and as indicated.
- B. Provide factory sleeve for each damper. Install damper operator on exterior of sleeve and link to damper operating shaft.
- C. Fabricate with multiple blades with 16 gauge galvanized steel frame and blades, oil-impregnated bronze or stainless steel sleeve bearings and plated steel axles, stainless steel jamb seals, 1/8 x 1/2 inch plated steel concealed linkage, stainless steel closure spring, blade stops, and lock, and 1/2 inch actuator shaft.
  - 2. Operators shall be spring return electric type suitable to operate on 120 V AC, 60 cycle.
  - 3. Operators shall be UL listed and labeled.

#### 2.8 SMOKE DAMPERS

- A. Fabricate in accordance with NFPA 90A and UL 555, and as indicated.

- B. Motorized Smoke Dampers: multi-blade type, normally open with power on, close automatically when power is interrupted, UL-listed and labeled damper and damper operator.

## 2.9 ACCEPTABLE MANUFACTURERS – BACKDRAFT DAMPERS

- A. Greenheck
- B. American Warming and Vent.
- C. Louvers and Dampers Inc.
- D. Ruskin.
- E. Pottorff
- F. Substitutions: Under provisions of Division One.

## 2.10 BACKDRAFT DAMPERS

- A. Gravity back draft dampers, size 18 x 18 inches or smaller, furnished with air moving equipment, may be air moving equipment manufacturers standard construction.
- B. Fabricate multi-blade, parallel action gravity balanced back draft dampers of 16 gauge galvanized steel, or extruded aluminum, with blades of maximum 6 inch width, with felt or flexible vinyl sealed edges, linked together in rattle-free manner with 90 degree stop, steel ball bearings, and plated steel pivot pin; adjustment device to permit setting for varying differential static pressure.
- C. Gravity backdraft dampers shall have an air leakage not greater than 20 cfm/ft<sup>2</sup> where not less than 24 inches in either dimension and 40 cfm/ft<sup>2</sup> where less than 24 inches in either dimension. The rate of air leakage shall be determined at 1.0 in.w.g. when tested in accordance with AMCA 500D for such purpose.

## 2.11 ACCEPTABLE MANUFACTURERS – AIR TURNING DEVICES

- A. Young Regulator.
- B. Titus.
- C. Tuttle and Bailey.
- D. Substitutions: Under provisions of Division One.

## 2.12 AIR TURNING DEVICES

- A. On duct sizes less than 12 x 12, multi-blade device with blades aligned in short dimension; steel or aluminum construction; with individually adjustable blades, mounting straps.
- B. Multi-blade device with radius blades attached to pivoting frame and bracket, steel or aluminum construction, with worm drive mechanism with 18 inch long removable key operator.

## 2.13 ACCEPTABLE MANUFACTURERS – FLEXIBLE DUCT CONNECTIONS

- A. Metaledge.
- B. Ventglass.
- C. Substitutions: Under provisions of Division One.

#### 2.14 FLEXIBLE DUCT CONNECTIONS TO AIR MOVING EQUIPMENT

- A. Fabricate in accordance with SMACNA Low Pressure Duct Construction Standards, and as indicated.
- B. UL listed fire-retardant neoprene coated woven glass fiber fabric to NFPA 90A, minimum density 20 oz per sq yd, approximately 6 inches wide, crimped into metal edging strip.

#### 2.15 ACCEPTABLE MANUFACTURERS – DUCT ACCESS DOORS

- A. Greenheck.
- B. American Warming and Vent.
- C. Ruskin.
- D. Titus.
- E. Substitutions: Under provisions of Division One.

#### 2.16 DUCT ACCESS DOORS

- A. Fabricate in accordance with SMACNA Low Pressure Duct Construction Standards and as indicated.
- B. Review locations prior to fabrication.
- C. Fabricate rigid and close-fitting doors of galvanized steel with sealing gaskets and quick fastening locking devices. For insulated ductwork, install minimum one inch thick insulation with sheet metal cover. Insulation shall be replaceable without field cutting or patching.
- D. Access doors smaller than 12 inches square may be secured with sash locks.
- E. Provide two hinges and two sash locks for sizes up to 18 inches square, three hinges and two compression latches with outside and inside handles for sizes up to 24 x 48 inches. Provide an additional hinge for larger sizes.
- F. Access doors with sheet metal screw fasteners are not acceptable.

#### 2.17 DUCT TEST HOLES

- A. Cut or drill temporary test holes in ducts as required. Cap with neat patches, neoprene plugs, threaded plugs, or threaded or twist-on metal caps.
- B. Permanent test holes shall be factory fabricated, air tight flanged fittings with screw cap. Provide extended neck fittings to clear insulation.

### PART 3 – EXECUTION

### 3.1 INSTALLATION

- A. Install accessories in accordance with manufacturer's instructions.
- B. Balancing Dampers
  - 1. Provide at points on low pressure supply, return, and exhaust systems where branches are taken from larger ducts and as required for air balancing. Use splitter dampers only where indicated.
  - 2. All regulators mounted on externally insulated ductwork shall have 16 gauge elevated platforms at least 1/8 inch higher than the thickness of the insulation. Damper shaft shall have Ventlock No. 607 bearing mounted on ductwork within elevated platform. If duct is inaccessible the operating handle shall be extended and the regulator installed on the face of the wall or ceiling. Where regulators are exposed in finished parts of the building, they shall be flush type, Ventlock No. 666. All regulators shall be manufactured by Ventlock, or approved equal.
  - 3. All dampers in lined ductwork shall have bushing to prevent damper damage to liner.
- C. Provide fire dampers at locations indicated, where ducts and outlets pass through fire rated components, and where required by authorities having jurisdiction. Install with required perimeter mounting angles, sleeves, breakaway duct connections, corrosion resistant springs, bearings, bushings and hinges.
- D. Demonstrate re-setting of fire dampers to authorities having jurisdiction and Owner's representative.
- E. Provide gravity backdraft dampers or motorized shutoff dampers in accessible location nearest to exterior wall/roof penetrations and where indicated for all outdoor air intake and exhaust systems to automatically shut when the associated systems or spaces served are not in use.
- F. Provide flexible duct connections immediately adjacent to equipment in ducts associated with fans and motorized equipment. Provide at least one inch slack at all flexible duct connections.
- G. Provide duct access doors for inspection and cleaning before and after filters, coils, fans, automatic dampers, at fire dampers, and elsewhere as indicated. Provide minimum 8 x 8 inch size for hand access, 18 x 18 inch size for shoulder access, and as indicated.
- H. Provide duct test holes where indicated and required for testing and balancing purposes.

**END OF SECTION**

## **SECTION 23 34 00 – HVAC FANS**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 - Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 WORK INCLUDED**

- A. Centrifugal roof ventilators.
- B. Ceiling and inline ventilators.

#### **1.3 RELATED SECTIONS**

- A. Section 23 05 13 – Common Motor Requirements for HVAC Equipment
- B. Section 23 05 48 – Vibration and Seismic Controls for HVAC Piping and Equipment
- C. Section 23 09 63 – Energy Management and Control System
- D. Section 23 05 93 – Testing, Adjusting and Balancing

#### **1.4 QUALITY ASSURANCE**

- A. UL Compliance: Fans shall be designed, manufactured, and tested in accordance with UL 705 "Power Ventilators."
- B. UL Compliance: Fans and components shall be UL listed and labeled.
- C. Nationally Recognized Testing Laboratory Compliance (NRTL): Fans and components shall be NRTL listed and labeled. The term "NRTL" shall be as defined in OSHA Regulation 1910.7.
- D. NEMA Compliance: Motors and electrical accessories shall comply with NEMA standards.
- E. Electrical Component Standard: Components and installation shall comply with NFPA 70 "National Electrical Code."
- F. Sound Power Level Ratings: Comply with AMCA Standard 301 "Method for Calculating Fan Sound Ratings From Laboratory Test Data." Test fans in accordance with AMCA Standard 300 "Test Code for Sound Rating." Fans shall be licensed to bear the AMCA Certified Sound Ratings Seal.
- G. Fan Performance Ratings: Establish flow rate, pressure, power, air density, speed of rotation, and efficiency by factory tests and ratings in accordance with AMCA Standard 210/ASHRAE Standard 51 - Laboratory Methods of Testing Fans for Rating.

#### **1.5 SUBMITTALS**



- A. General: Submit the following in accordance with Conditions of Contract and Division 1 Specification Sections:
- B. Product data for selected models, including specialties, accessories, and the following:
  - 1. Certified fan performance curves with system operating conditions indicated.
  - 2. Certified fan sound power ratings.
  - 3. Motor ratings and electrical characteristics plus motor and fan accessories.
  - 4. Materials, gages and finishes, include color charts.
  - 5. Dampers, including housings, linkages, and operators.
  - 6. Full color paint samples.
- C. Shop drawings from manufacturer detailing equipment assemblies and indicating dimensions, weights, required clearances, components, and location and size of field connections.
- D. Coordination drawings, in accordance with Division 23, Section "Basic Materials and Methods", for roof penetration requirements and for reflected ceiling plans drawn accurately to scale and coordinating penetrations and units mounted above ceiling. Show the following:
  - 1. Roof framing and support members relative to duct penetrations.
  - 2. Ceiling suspension members.
  - 3. Method of attaching hangers to building structure.
  - 4. Size and location of initial access modules for acoustical tile.
  - 5. Ceiling-mounted items including light fixtures, diffusers, grilles, speakers, sprinkler heads, access panels, and special moldings.
- F. Wiring diagrams that detail power, signal, and control wiring. Differentiate between manufacturer installed wiring and field installed wiring.
- G. Product certificates, signed by manufacturer, certifying that their products comply with specified requirements.
- H. Maintenance data for inclusion in Operating and Maintenance Manual specified in Division 1 and Division 23, Section "Basic Materials and Methods".

#### 1.6 DELIVERY, STORAGE, AND HANDLING

- A. Fans shall be stored and handled in accordance with the unit manufacturer's instructions.
- B. Lift and support units with the manufacturer's designated lifting or supporting points.
- C. Disassemble and reassemble units as required for movement into the final location following manufacturer's written instructions.
- D. Deliver fan units as a factory-assembled unit to the extent allowable by shipping limitations, with protective crating and covering.

#### 1.7 ENVIRONMENTAL REQUIREMENTS

- A. Do not operate units for any purpose, temporary or permanent, until ductwork is clean, filters are in place, bearings lubricated, and fan has been test run under observation.

#### 1.8 OPERATIONS PERSONNEL TRAINING

- A. Provide a training session for the owner's operations personnel. Training session shall be performed by a qualified person who is knowledgeable in the subject system/equipment. Submit a training agenda two (2) weeks prior to the proposed training session for review and approval. Training session shall include at the minimum:
  - 1. Purpose of equipment.
  - 2. Principle of how the equipment works.
  - 3. Important parts and assemblies.
  - 4. How the equipment achieves its purpose and necessary operating conditions.
  - 5. Most likely failure modes, causes and corrections.
  - 6. On site demonstration.

## PART 2 - PRODUCTS

### 2.1 ACCEPTABLE MANUFACTURERS

- A. PennBarry
- B. Loren Cook Company
- C. Greenheck Fan Corporation
- D. ACME
- E. Twin City Fan and Blower

### 2.2 GENERAL DESCRIPTION

- A. Provide fans that are factory fabricated and assembled, factory tested, and factory finished with indicated capacities and characteristics.
- B. Fans and Shafts shall be statically and dynamically balanced and designed for continuous operation at the maximum rated fan speed and motor horsepower.
- C. Provide factory baked-enamel finish coat after assembly. Color for roof mounted fans shall be chosen by Architect during the submittal process.

### 2.3 CENTRIFUGAL ROOF VENTILATORS

- A. Fan shall be a spun aluminum, centrifugal, roof mounted, direct driven or belt driven as indicated.
- B. Fan shall be listed by Underwriters Laboratories (UL 705). Fan shall bear the AMCA certified ratings seal for sound and air performance.
- C. The fan shall be of bolted and welded construction utilizing corrosion resistant fasteners. The spun aluminum structural components shall be constructed of minimum 16 gauge marine alloy aluminum, bolted to a rigid aluminum support structure.
- D. The aluminum base shall have continuously welded curb cap corners for maximum leak protection. A discharge baffle conduit chase shall be provided through the curb cap and into the motor compartment to facilitate wiring connections.
- E. The motor, bearings and drives shall be mounted on a minimum 14 gauge steel power assembly, isolated from the unit structure with rubber vibration isolators. These

components shall be enclosed in a weather-tight compartment, separated from the exhaust airstream. Unit shall bear an engraved aluminum nameplate and shall be shipped in transit tested packaging.

- F. Wheel shall be centrifugal backward inclined, constructed of 100% aluminum, including a precision machined cast aluminum hub. Wheel inlet shall overlap an aerodynamic aluminum inlet cone to provide maximum performance and efficiency. Wheel shall be balanced in accordance with AMCA standard 204-96, balance quality and vibration levels for fans.
- G. Motor shall be heavy duty type with permanently lubricated sealed ball bearings.
- H. Bearings shall be designed and individually tested specifically for use in air handling applications. Construction shall be heavy duty re-greaseable ball type in a cast iron housing selected for a minimum L50 life in excess of 200,000 hours at maximum cataloged operating speed.
- I. Accessories: The following accessories are required.
  - 1. Disconnect Switch: Non-fusible type, with thermal overload protection, mounted inside fan housing, factory-wired through an internal aluminum conduit.
  - 2. Bird Screens: Removable ½ inch mesh, 16 gauge, aluminum or brass wire.
  - 3. Dampers: Gravity backdraft damper or motorized shutoff damper mounted in accessible location. Refer to 23 33 00 – Ductwork Accessories.
  - 4. Roof Curbs: Prefabricated, 12 inch high, heavy-gauge, galvanized steel; mitered and welded corners; 2 inch thick, rigid, fiberglass insulation adhered to inside walls; built-in cant and mounting flange for flat roof decks; and 2 inch wood nailer. Size as required to suit roof opening and fan base. Roof curb shall match roof slope so that the curb is level.

## 2.4 CEILING AND INLINE VENTILATORS

- A. Ceiling and inline ventilators shall be direct drive or belt drive as indicated, centrifugal blower type. Fan wheel shall be constructed of galvanized steel and shall be dynamically balanced. The housing shall be constructed of minimum 20 gauge corrosion resistant galvanized steel and acoustically insulated for quiet operation. Blower and motor assembly shall be easily removable from the housing without disturbing the ductwork. The motor shall be permanently lubricated with built-in thermal overload protection and shall be factory tested prior to shipment. The ceiling ventilators shall be furnished standard with a powder-painted white steel grille.
- B. Ventilators shall be certified and licensed to bear the AMCA Seal for Air and Sound Performance. Ventilator performance shall be based on tests and procedures performed in accordance with AMCA publication 211 and comply with the requirements of the AMCA Certified Ratings Program. Fan sound power level ratings shall be based on tests and procedures performed in accordance with AMCA publication 311 and comply with the requirements of the AMCA Certified Ratings Program. Ventilators shall be UL listed and CSA certified.
- C. Accessories: The following accessories are required.
  - 1. Dampers:
    - a. Aluminum backdraft damper.
    - b. Motor-operated volume control damper.

- c. U.L. listed ceiling radiation damper for ceiling fans comply with NFPA Standard 90A rated for 3 hours.
- 2. Disconnect Switch: Nonfusible type with thermal overload protection.
- 3. Speed Controls: Fan mounted, solid state speed controller.

### PART 3 – EXECUTION

- 3.1 Install in accordance with manufacturer's instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Provide all items required as per manufacturer's requirements.

### END OF SECTION

## **SECTION 23 37 13 - AIR DISTRIBUTION DEVICES**

### **PART 1 - GENERAL**

#### **1.1 WORK INCLUDED**

- A. Ceiling air diffusers.
- B. Wall registers and grilles.
- C. Louvers.
- D. Other air devices indicated on drawings and schedules.

#### **1.2 RELATED SECTIONS**

- A. Section 23 02 00 – Basic Materials and Methods
- B. Section 23 05 93 – Testing, Adjusting and Balancing
- C. Section 23 31 13 – Metal Ductwork
- D. Section 23 31 16 – Fibrous Glass Ductwork
- E. Section 23 31 19 – Ductwork Accessories

#### **1.3 QUALITY ASSURANCE**

- A. Manufacturer's Qualifications: Firms regularly engaged in manufacture of air distribution devices of types and capacities required, whose products have been in satisfactory use in similar service for not less than 5 years.
- B. Codes and Standards:
  - 1. ARI Compliance: Test and rate air distribution devices in accordance with ARI 650 "Standard for Air Outlets and Inlets".
  - 2. ASHRAE Compliance: Test and rate air distribution devices in accordance with ASHRAE 70 "Method of Testing for Rating the Air Flow Performance of Outlets and Inlets".
  - 3. AMCA Compliance: Test and rate louvers in accordance with AMCA 500 "Test Method for Louvers, Dampers and Shutters".
  - 4. AMCA Seal: Provide louvers bearing AMCA Certified Rating Seal.
  - 5. NFPA Compliance: Install air distribution devices in accordance with NFPA 90A "Standard for the Installation of Air Conditioning and Ventilating Systems".

#### **1.4 SUBMITTALS**

- A. Product Data: Submit manufacturer's technical product data for air distribution devices including the following:
  - 1. Schedule of air distribution devices indicating drawing designation, room location, number furnished, model number, size, and accessories furnished.
  - 2. Data sheet for each type of air distribution devices, and accessory furnished; indicating construction, finish, and mounting details.

3. Performance data for each type of air distribution devices furnished, including aspiration ability, temperature and velocity traverses; throw and drop; and noise criteria ratings. Indicate selections on data.
- B. Shop Drawings: Submit manufacturer's assembly-type shop drawing for each type of air distribution devices, indicating materials and methods of assembly of components.
- C. Maintenance Data: Submit maintenance data, including cleaning instructions for finishes, and spare parts lists. Include this data, product data, and shop drawings in maintenance manuals; in accordance with requirements of Division 1.

#### 1.5 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. Deliver air distribution devices wrapped in factory-fabricated fiber-board type containers. Identify on outside of container type of outlet or inlet and location to be installed. Avoid crushing or bending and prevent dirt and debris from entering and settling in devices.
- B. Store air distribution devices in original cartons and protect from weather and construction work traffic. Where possible, store indoors; when necessary to store outdoors, store above grade and enclose with waterproof wrapping.

#### 1.6 WARRANTY

- A. Warrant the installation of the Work specified herein for one year against becoming unserviceable or causing an objectionable appearance resulting from defective or nonconforming workmanship.

### PART 2 – PRODUCTS

#### 2.1 ACCEPTABLE MANUFACTURERS

- A. Titus Company
- B. Metalaire Industries, Inc.
- C. Nailor Industries
- D. Krueger
- E. Price
- F. Substitutions under provisions of Division One.

#### 2.2 GENERAL DESCRIPTION

- A. Unless otherwise indicated, provide manufacturer's standard air devices when shown of size, shape, capacity, type and accessories indicated on drawings and schedules, constructed of materials and components as indicated and as required for complete installation and proper air distribution.
- B. Provide air devices that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device and listed in manufacturer's current data.
- C. Unless noted otherwise on drawings, the finish shall be #26 white. The finish shall be an anodic acrylic paint, baked at 315°F for 30 minutes. The pencil hardness must be HB to

H. The paint must pass a 100 hour ASTM D117 Corrosive Environments Salt Spray Test without creepage, blistering, or deterioration of film. The paint must pass a 250 hour ASTM-870 Water Immersion Test. The paint must also pass the ASTM D-2794 Reverse Impact Cracking Test with a 50 inch pound force applied.

- D. Provide air device with border styles that are compatible with adjacent ceiling or wall system, and that are specially manufactured to fit into the wall construction or ceiling module with accurate fit and adequate support. Refer to architectural construction drawings and specifications for types of wall construction and ceiling systems.
- E. Provide integral volume damper with roll formed steel blades where indicated on drawings or schedules. Dampers shall be opposed blade design with a screw driver slot or a concealed lever operator for adjustment through the face of the air device.
- F. Air devices designated for fire rated systems shall be pre-assembled with UL classified radiation damper and thermal blanket. Fire rated air devices shall be shipped completely assembled; one assembly per carton, Each assembly shall be enclosed in plastic shrink wrap with installation instructions.

## 2.3 LOUVERS

- A. Except as otherwise indicated, provide manufacturer's standard louvers where shown; of size, shape, capacity and type indicated; constructed of materials and components as indicated, and as required for complete installation.
- B. Provide louvers that have minimum free area, and maximum pressure drop of each type as listed in manufacturer's current data, complying with louver schedule.
- C. Provide louvers with frame and sill styles that are compatible with adjacent substrate, and that are specifically manufactured to fit into construction openings with accurate fit and adequate support, for weatherproof installation. Refer to architectural construction drawings and specifications for types of substrate.
- D. Louvers shall be constructed of aluminum extrusions, ASTM B 221, Alloy 6063-T5. Weld units or use stainless steel fasteners.
- E. Louver Screens: On inside face of exterior louvers, provide 1/2" square mesh anodized aluminum wire bird screens mounted in removable extruded aluminum frames.
- F. Acceptable Manufacturers:
  - 1. Ruskin Manufacturing Company
  - 2. Greenheck Company
  - 3. Louvers and Dampers, Inc.
  - 4. Pottorff
  - 5. Arrow
  - 6. Substitutions under provisions of Division One.

## PART 3 – EXECUTION

- 3.1 All interior surfaces of all air devices shall be painted flat black.
- 3.2 See floor plans for type, neck size and CFM of air for all air distribution devices.
- 3.3 Install all air distribution devices as detailed on plans and in accordance with manufacturer's recommendations.

**END OF SECTION**



## **SECTION 23 41 00 - AIR FILTERS**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 23 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for owner's use.

### **PART 2 - PRODUCTS**

#### **2.1 FILTERS**

- A. Air filters shall be medium efficiency ASHRAE pleated panels consisting of cotton and synthetic media, welded wire media support grid, and beverage board enclosing frame, Camfil Farr 30/30, 2-inch thick or approved equal.
- B. **APPROVED MANUFACTURERS:** The following manufacturers are approved subject to specification compliance.
  - 1. American Air Filter.
  - 2. Camfil.
  - 3. Airguard Industries, Inc.
  - 4. Cambridge.
  - 5. Filtration Group

#### **2.2 LOW VELOCITY FILTER SECTION**

- A. Filters shall be of the throwaway cartridge type in 24 inches X 24 inches X 2 inch frames. When installing multiple filters into slide-in frames tape adjacent filters together with duct tape to prevent bypassing of air around the filter. Media shall be rated at 500 feet per minute.
- B. Filtering media shall be formed of non-woven reinforced cotton fabric type filtering media bonded to 96% open area media support grid folded into a non-creased radial pleat design. The filter pack shall be bonded to the inclosing frame to prevent air bypass. Minimum Efficiency Reporting Value of MERV 8 when evaluated under the guidelines of ASHRAE Standard 52.2-2017. Initial resistance shall not exceed 0.23 inches water gauge at 350 FPM face velocity.

### **PART 3 - EXECUTION**

#### **3.1 INSTALLATION**

- A. Install differential pressure switch to activate "Filter Dirty" light when pressure difference across filters reaches 0.5 inch W.G. (adjustable). Locate "filter dirty" lights in mechanical rooms with identifying label

- B. Install and relocate filters in the mechanical or the storage room in accordance with manufacturer's recommendations.
- C. Refer to Section 23 02 00 for additional filter information.

**END OF SECTION**

## **SECTION 23 62 13 - AIR COOLED CONDENSING UNITS**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 – Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for Owner's use.

#### **1.3 OPERATIONS PERSONNEL TRAINING**

- A. Provide a training session for the owner's operations personnel. Training session shall be performed by a qualified person who is knowledgeable in the subject system/equipment. Submit a training agenda two (2) weeks prior to the proposed training session for review and approval. Training session shall include at the minimum:
  - 1. Purpose of equipment.
  - 2. Principle of how the equipment works.
  - 3. Important parts and assemblies.
  - 4. How the equipment achieves its purpose and necessary operating conditions.
  - 5. Most likely failure modes, causes and corrections.
  - 6. On site demonstration.

### **PART 2 - PRODUCTS**

#### **2.1 AIR-COOLED CONDENSING UNITS**

- A. Air-cooled condensing unit shall be designed for use with split system having a remote direct-expansion (DX) cooling coil mounted in evaporator fan unit. Capacity shall be as called for on the Drawings when matched to the appropriate evaporator coil.
- B. Condensing unit shall consist of high-efficiency hermetic compressor, air-cooled condenser with quiet fan, factory wired controls, R410A or R407C refrigerant and refrigeration circuit and valves.
- C. Cabinet shall be heavy-gauge galvanized steel with bonding primer and baked-enamel finish coat. The entire cabinet shall be protected from rust.
- D. Compressor shall be protected from excessive current and temperatures and shall be provided with a thermostatically controlled crankcase heater to operate only when needed for protection of the compressor. Compressor shall be spring-mounted on rubber isolators. Compressor shall be located in compartment isolated from condenser fan and coil. Provide a high-capacity dryer in the system to remove moisture and dirt.
- E. Condenser fan shall be directly connected to a weather-protected, quiet, high-efficiency motor. Fan guard shall be provided and shall be protected from rust by PVC finish. Condenser coil shall be aluminum fin with copper tube.

- F. Connections for refrigerant suction and liquid lines shall be extended outside the cabinet and provided with service valves with gauge connections.
- G. Power connections shall be made to the connectors located inside the electrical connection box.
- H. Standard operating and safety controls shall include high-pressure switch, low pressure switch, compressor overload service, and solid-state timed-off control.
- I. All components (parts and labor) of the sealed refrigeration circuit shall be warranted by the manufacturer for five years.

## 2.2 AUXILIARY EQUIPMENT

- A. Auxiliary equipment shall consist of refrigerant lines prepared for the unit involved. These lines shall be cleaned, dried, and pressurized at the factory.
- B. Low ambient kit to allow operation at outside temperature below 35 deg. F (2 deg. C) shall be provided.
- C. Expansion valve shall be provided with the evaporator coil.
- D. Provide thermostat to match the requirements of the job. Thermostat shall provide subbase with Heat-Cool-Off and Fan On-Auto switch. See section on controls for other related requirements.
- E. Provide polyethylene structural base designed for that service, and intended to support the unit and eliminate vibration transmission.
- F. Provide hard-start kit with unit.
- G. Provide guards for condenser coils.

## 2.3 ACCEPTABLE MANUFACTURERS

- A. Condensing unit shall be the make and model number shown on the drawings or acceptable equivalents by Lennox, Carrier, York, Trane, or Daikin.

## PART 3 - EXECUTION

- 3.1 All HVAC equipment shall be installed as per manufacturer's printed installation instructions.

- 3.2 All items required for a complete and proper installation are not necessarily indicated on the Drawings or in the Specifications. Provide all items required as per manufacturer's requirements.

## 3.3 INSTALLATION

- A. Install the condensing unit on proper foundation as shown on the Drawings, and in location that will not restrict the air entry or discharge from the unit.
- B. Install refrigerant lines as recommended by the manufacturer, taking care not to lose the refrigerant charge contained in the lines, or allow air to enter the lines or equipment. Locate the lines in such a way as to not obstruct access to the condensing unit or other equipment. Lines located underground or under concrete shall be installed in a PVC sleeve for protection.

- C. Provide electrical connections as required by the applicable codes. Provide control wiring required. All power wiring and control wiring shall be in conduit and located so as not to obstruct access to the unit or other equipment.

#### 3.4 TESTING

- A. Operate the condensing unit and the system to assure that unit is operating properly and without excessive noise and vibration.
- B. Read and record the power draw and the refrigeration suction and liquid pressures as required by Section 23 05 93 – Testing, Adjusting, and Balancing.

**END OF SECTION**

## **SECTION 23 82 19 - FAN COIL UNIT**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Section 23 02 00 - Basic Materials and Methods is included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for owner's use.

#### **1.3 OPERATIONS PERSONNEL TRAINING**

- A. Provide a training session for the owner's operations personnel. Training session shall be performed by a qualified person who is knowledgeable in the subject system/equipment. Submit a training agenda two (2) weeks prior to the proposed training session for review and approval. Training session shall include at the minimum:
  - 1. Purpose of equipment.
  - 2. Principle of how the equipment works.
  - 3. Important parts and assemblies.
  - 4. How the equipment achieves its purpose and necessary operating conditions.
  - 5. Most likely failure modes, causes and corrections.
  - 6. On site demonstration.

### **PART 2 - PRODUCTS**

#### **2.1 FAN COIL UNITS**

Fan coil units shall be factory built, manufactured as scheduled on Drawings. Carrier, York, Trane or Daikin shall be considered as equal, if they comply with the specifications and schedule. Special Note: Contractor shall field verify exact clearances required for fan coil units. Units shall be field located as required and shop drawings shall indicate final location for approval by Architect/Engineer.

- A. Furnish and install fan coil units of the type, capacities, ratings and drive motor horsepower shown on the Drawings.
- B. Units shall be factory fabricated, draw-thru type, and shall have fan section, cooling coil section, condensate drain pan, adjustable blower drive with motor on resilient mounted base, vee-belts with guard, filter section, and mixing box (if scheduled) assembled as integrated fan coil units.
- C. REQUIREMENTS:
  - 1. Mill-galvanized steel, rigidly framed, braced, and reinforced; access panels each side of unit; minimum panel ga. - 18; minimum weight formed framing member 14 ga.
  - 2. Fan section, cooling coil section, and outlet frame throats shall be internally insulated at the factory with 1" thick, 3/4 PCF density, Neoprene coated

- fiberglass cemented in place with waterproof adhesive, having fire-retardant characteristics in accordance with NFPA 90A.
3. Drain pan shall not be lighter than 14 ga.; extend completely under the coil section and be all-galvanized, foam insulated pan with drain connections.
  4. When the fan coil unit is installed above an accessible ceiling, the unit shall incorporate a secondary drain pan. The secondary pan shall be fabricated from galvanized sheet metal, 16 gauge minimum with cross breaking sloped towards a drain. The sides shall be a minimum 2" tall and the corners shall be soldered watertight. The top edge shall have a 1/4" hem to provide additional rigidity and the secondary pan shall be supported at a minimum of six points. The pan shall extend on all sides a minimum of 3" beyond the sides of the unit casing. Route the secondary drain piping to a conspicuous location, or install a float switch at the low point in the secondary pan. The secondary pan should be sloped a minimum of 1/8" per foot and supported so that the unit is not in contact with the bottom of the secondary pan.
  5. The fan section, including wheels, shafts, bearings, drive, etc., shall be statically and dynamically balanced as an assembly, and the shaft shall not pass through the first critical speed, while accelerating from rest to operating speed. Submittal data shall state the first critical shaft speed. Shaft bearings shall be of vacuum de-gassed steel, and shall be selected for 200,000 hours average life.
  6. Coil shall be as hereinafter specified.

## 2.2 COILS

- A. Cooling coils shall be cartridge type and, when mounted in fan coil units, shall be removable from either end. Coils shall be constructed of copper tubes with aluminum fins and shall be designed for even distribution of air across the face of the coils; air shall not pass around coil frames. Coils shall have same end connection for DX or chilled water piping.
- B. DX refrigerant coils shall be counter-flow refrigerant to air; shall have inlet and outlet connections permanently marked; shall have thermostatic expansion valves with adjustable super heat.
- C. Maximum face velocity across cooling coils shall be 500 FPM, unless noted otherwise on schedule.

## PART 3 - EXECUTION

- 3.1 All HVAC equipment shall be installed as per manufacturers printed installation instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the Drawings or in the Specifications. Provide all items required as per manufacturer's requirements.

## END OF SECTION

## **SECTION 23 82 43 - ELECTRIC DUCT HEATERS**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. The Basic Materials and Methods, Section 23 02 00, are included as a part of this Section as though written in full in this document.

#### **1.2 SCOPE**

- A. Scope of the Work shall include the furnishing and complete installation of the equipment covered by this Section, with all auxiliaries, ready for owner's use.

#### **1.3 OPERATIONS PERSONNEL TRAINING**

- A. Provide a training session for the owner's operations personnel. Training session shall be performed by a qualified person who is knowledgeable in the subject system/equipment. Submit a training agenda two (2) weeks prior to the proposed training session for review and approval. Training session shall include at the minimum:
  - 1. Purpose of equipment.
  - 2. Principle of how the equipment works.
  - 3. Important parts and assemblies.
  - 4. How the equipment achieves its purpose and necessary operating conditions.
  - 5. Most likely failure modes, causes and corrections.
  - 6. On site demonstration.

### **PART 2 - PRODUCTS**

#### **2.1 ELECTRIC DUCT HEATER**

- A. Provide open coil, electric heating coils as listed in the schedule.
- B. Three phase heaters shall have balanced three phase steps unless specified otherwise.
- C. All heaters to be UL listed for zero clearance to combustible surfaces and bear the UL label.
- D. All heaters shall meet the requirements of the National Electrical Code.
- E. Standard terminal box, with 1/2 inch insulation, as well as element housing and racks wall to be made of heavy gauge galvanized steel. All contactors shall be silent type operation mercury contactors.
- F. All heating coils to be made of high grade nickel/chromium resistance wire and terminated by means of a loop of wire being sandwiched between stainless steel or nickel plated washers and terminal hardware. All terminal hardware to be insulated from the heater by a two piece ceramic bushing.
- G. Safety Controls:



1. Positive air pressure switch to prevent heater from energizing until air flow is proven.
2. Primary over temperature protection shall be provided by built-in disc type automatic reset thermal cutouts.
3. Secondary over-temperature protection shall consist of a sufficient number of load carrying manual reset controls to deenergize the elements if the primary system fails; one pilot duty manual reset and back up contactors.

H. Wiring Diagrams:

1. A separate, complete and specific wiring diagram shall be permanently attached to each heater. Typical wiring diagrams are not acceptable.
  2. Control and line terminals in each heater shall be marked identical to the wiring diagram.
  3. Additional diagrams instructions, etc., to be firmly held in position by a metallic snap clip or pocket inside the cover.
- I. Overcurrent protection incorporating fuses or circuit or breakers must be provided for all heaters rated more than 48 amperes, factory installed, within the heater enclosure, or provide as a separate assembly by the heater manufacturer. Heaters exceeding 48 amperes total line current must be divided into subcircuits (as allowed by stages) of less than 48 amperes and be protected at not more than 60 amperes. The main conductors supplying these overcurrent protective devices are considered branch circuit conductors and are subject to the 125% ampere rating rule of NEC.
- J. Units as manufactured by Nailor, Indeeco, Markel, Warren or Brasch Manufacturing shall be considered "as equal" provided they comply with the schedule and specifications.
- K. All units shall include an integral disconnect switch to meet the NEC requirements for a disconnecting means within sight of the heater. Disconnect switch shall be located inside of heater control cabinet and shall be interlocked with control cabinet door. Disconnect switch shall have labeled "on" and "off" positions. If any other external sources of control voltage are required, a separate toggle switch shall be provided.

PART 3 - EXECUTION

- 3.1 All HVAC equipment shall be installed as per manufacturers printed installation instructions.
- 3.2 All items required for a complete and proper installation are not necessarily indicated on the plans or in the specifications. Provide all items required as per manufacturer's requirements.

**END OF SECTION**

## **SECTION 26 02 00 - BASIC MATERIALS AND METHODS**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all Work herein.
- B. The Contract Drawings indicate the extent and general arrangement of the systems. If any departure from the Contract Drawings are deemed necessary by the Contractor, details of such departures and the reasons therefore, shall be submitted to the Architect for approval as soon as practicable. No such departures shall be made without the prior written approval of the Architect.

#### **1.2 SCOPE OF WORK**

- A. The Work included under this Contract consists of the furnishing and installation of all equipment and material necessary and required to form the complete and functioning systems in all of its various phases, all as shown on the accompanying Drawings and/or described in these Specifications. The contractor shall review all pertinent drawings, including those of other contracts prior to commencement of Work.
- B. This Division requires the furnishing and installing of all items Specified herein, indicated on the Drawings or reasonably inferred as necessary for safe and proper operation; including every article, device or accessory (whether or not specifically called for by item) reasonably necessary to facilitate each system's functioning as indicated by the design and the equipment specified. Elements of the work include, but are not limited to, materials, labor, supervision, transportation, storage, equipment, utilities, all required permits, licenses and inspections. All work performed under this Section shall be in accordance with the Project Manual, Drawings and Specifications and is subject to the terms and conditions of the Contract.
- C. The approximate locations of Electrical items are indicated on the Drawings. These Drawings are not intended to give complete and accurate details in regard to location of outlets, apparatus, etc. Exact locations are to be determined by actual measurements at the building, and will in all cases be subject to the Review of the Owner or Engineer, who reserves the right to make any reasonable changes in the locations indicated without additional cost to the Owner.
- D. Items specifically mentioned in the Specifications but not shown on the Drawings and/or items shown on Drawings but not specifically mentioned in the Specifications shall be installed by the Contractor under the appropriate section of work as if they were both specified and shown.
- E. All discrepancies within the Contract Documents discrepancies between the Contract Documents and actual job-site conditions shall be reported to the Owner or Engineer so that they will be resolved prior to the bidding, where this cannot be done at least 7 working days prior to bid; the greater or more costly of the discrepancy shall be bid. All labor and materials required to perform the work described shall be included as part of this Contract.
- F. It is the intention of this Section of the Specifications to outline minimum requirements to furnish the Owner with a turn-key and fully operating system in cooperation with other trades.

- G. It is the intent of the above "Scope" to give the Contractor a general outline of the extent of the Work involved; however, it is not intended to include each and every item required for the Work. Anything omitted from the "Scope" but shown on the Drawings, or specified later, or necessary for a complete and functioning heating, ventilating and air conditioning system shall be considered a part of the overall "Scope".
- H. The Contractor shall rough-in fixtures and equipment furnished by others from rough-in and placement drawings furnished by others. The Contractor shall make final connection to fixtures and equipment furnished by others.
- I. Contractor shall participate in the commissioning process; including but not limited to meeting attendance, completion of checklists and participation in functional testing.

### 1.3 RELATED SECTIONS

- A. General Conditions
- B. Supplementary Conditions
- C. Division One

### 1.4 COOPERATION WITH TRADES:

- A. Cooperation with trades of adjacent, related, or affected materials or operations shall be considered a part of this work in order to affect timely and accurate placing of work and bring together in proper and correct sequence, the work of such trades.

### 1.5 REFERENCES

- A. National Electrical Code (NEC)
- B. American Society for Testing and Materials (ASTM)
- C. Underwriter's Laboratories, Inc. (UL)
- D. Insulated Cable Engineer's Association (ICEA).
- E. National Electrical Manufacturer's Association (NEMA).
- F. Institute of Electrical and Electronic's Engineers (IEEE).
- G. American National Standards Institute (ANSI).
- H. National Fire Protection Association (NFPA).
- I. International Energy Conservation Code (IECC).

### 1.6 COMPLETE FUNCTIONING OF WORK:

- A. All work fairly implied as essential to the complete functioning of the electrical systems shown on the Drawings and Specifications shall be completed as part of the work of this Division unless specifically stated otherwise. It is the intention of the Drawings and Specifications to establish the types of the systems, but not set forth each item essential to the functioning of the system. In case of doubt as to the work intended, or in the event of amplification or clarification thereof, the Contractor shall call upon the Architect for supplementary instructions, Drawings, etc.

- B. Contractor shall review all pertinent Drawings and adjust his work to all conditions shown there on. Discrepancies between Plans, Specifications, and actual field conditions shall be brought to the prompt attention of the Architect.
  - 1. Approximate location of transformers, feeders, branch circuits, outlets, lighting and power panels, outlets for special systems, etc., are indicated on the Drawings. However, the Drawings, do not give complete and accurate detailed locations of such outlets, conduit runs, etc., and exact locations must be determined by actual field measurement. Such locations will, at all times, be subject to the approval of the Architect.
  - 2. Communicate with the Architect and secure his approval of any outlet (light fixture, receptacle, switch, etc.) location about which there may be the least question. Outlets obviously placed in a location not suitable to the finished room or without specific approval, shall be removed and relocated when so directed by the Architect. Location of light fixtures shall be coordinated with reflected ceiling plans.
- C. Additional coordination with mechanical contractor may be required to allow adequate clearances of mechanical equipment, fixtures and associated appurtenances. Contractor to notify Architect and Engineer of unresolved clearances, conflicts or equipment locations.

#### 1.7 SCHEMATIC NATURE OF CONTRACT DOCUMENTS

- A. The contract documents are schematic in nature in that they are only to establish scope and a minimum level of quality. They are not to be used as actual working construction drawings. The actual working construction drawings shall be the approved shop drawings.

#### 1.8 CONTRACTOR'S QUALIFICATIONS

- A. An approved contractor for the work under this division shall be:
  - 1. A specialist in this field and have the personnel, experience, training, and skill, and the organization to provide a practical working system.
  - 2. Able to furnish evidence of having contracted for and installed not less than 3 systems of comparable size and type that have served their Owners satisfactorily for not less than 3 years.
  - 3. Perform work by persons qualified to produce workmanship of specified quality. Persons performing electrical work shall be required to be licensed. Onsite supervision, journeyman shall have minimum of journeyman license. Helpers, apprentices shall have minimum of apprentice license.

#### 1.9 DATE OF FINAL ACCEPTANCE

- A. The date of final acceptance shall be the date of owner occupancy, or the date all punch list items have been completed or final payment has been received. Refer to Division One for additional requirements.
- B. The date of final acceptance shall be documented in writing and signed by the architect, owner and contractor.

## 1.10 DEFINITIONS AND SYMBOLS

- A. General Explanation: A substantial amount of construction and Specification language constitutes definitions for terms found in other Contract Documents, including Drawings which must be recognized as diagrammatic and schematic in nature and not completely descriptive of requirements indicated thereon. Certain terms used in Contract Documents are defined generally in this article, unless defined otherwise in Division 1.
- B. Definitions and explanations of this Section are not necessarily either complete or exclusive, but are general for work to the extent not stated more explicitly in another provision of the Contract Documents.
- C. Indicated: The term "Indicated" is a cross-reference to details, notes or schedules on the Drawings, to other paragraphs or schedules in the Specifications and to similar means of recording requirements in Contract Documents. Where such terms as "Shown", "Noted", "Scheduled", "Specified" and "Detailed" are used in lieu of "Indicated", it is for the purpose of helping the reader locate cross-reference material, and no limitation of location is intended except as specifically shown.
- D. Directed: Where not otherwise explained, terms such as "Directed", "Requested", "Accepted", and "Permitted" mean by the Architect or Engineer. However, no such implied meaning will be interpreted to extend the Architect's or Engineer's responsibility into the Contractor's area of construction supervision.
- E. Reviewed: Where used in conjunction with the Engineer's response to submittals, requests for information, applications, inquiries, reports and claims by the Contractor the meaning of the term "Reviewed" will be held to limitations of Architect's and Engineer's responsibilities and duties as specified in the General and Supplemental Conditions. In no case will "Reviewed" by Engineer be interpreted as a release of the Contractor from responsibility to fulfill the terms and requirements of the Contract Documents.
- F. Furnish: Except as otherwise defined in greater detail, the term "Furnish" is used to mean supply and deliver to the project site, ready for unloading, unpacking, assembly, installation, etc., as applicable in each instance.
- G. Install: Except as otherwise defined in greater detail, the term "Install" is used to describe operations at the project site including unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimension, finishing, curing, protection, cleaning and similar operations, as applicable in each instance.
- H. Provide: Except as otherwise defined in greater detail, the term "Provide" is used to mean "Furnish and Install", complete and ready for intended use, as applicable in each instance.
- I. Installer: Entity (person or firm) engaged by the Contractor or its subcontractor or Sub-contractor for performance of a particular unit of work at the project site, including unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimension, finishing, curing, protection, cleaning and similar operations, as applicable in each instance. It is a general requirement that such entities (Installers) be expert in the operations they are engaged to perform.

- J. Imperative Language: Used generally in Specifications. Except as otherwise indicated, requirements expressed imperatively are to be performed by the Contractor. For clarity of reading at certain locations, contrasting subjective language is used to describe responsibilities that must be fulfilled indirectly by the Contractor, or when so noted by other identified installers or entities.
- K. Minimum Quality/Quantity: In every instance, the quality level or quantity shown or specified is intended as minimum quality level or quantity of work to be performed or provided. Except as otherwise specifically indicated, the actual work may either comply exactly with that minimum (within specified tolerances), or may exceed that minimum within reasonable tolerance limits. In complying with requirements, indicated or scheduled numeric values are either minimums or maximums as noted or as appropriate for the context of the requirements. Refer instances of uncertainty to Owner or Engineer via a request for information (RFI) for decision before proceeding.
- L. Abbreviations and Symbols: The language of Specifications and other Contract Documents including Drawings is of an abbreviated type in certain instances, and implies words and meanings which will be appropriately interpreted. Actual word abbreviations of a self-explanatory nature have been included in text of Specifications and Drawings. Specific abbreviations and symbols have been established, principally for lengthy technical terminology and primarily in conjunction with coordination of Specification requirements with notations on Drawings and in Schedules. These are frequently defined in Section at first instance of use or on a Legend and Symbol Drawing. Trade and industry association names and titles of generally recognized industry standards are frequently abbreviated. Singular words will be interpreted as plural and plural words will be interpreted as singular where applicable and where full context of Contract Documents so indicate. Except as otherwise indicated, graphic symbols and abbreviations used on Drawings and in Specifications are those recognized in construction industry for indicated purposes. Where not otherwise noted symbols and abbreviations are defined by 1993 ASHRAE Fundamentals Handbook, chapter 34 "Abbreviations and Symbols", ASME and ASPE published standards.

#### 1.11 DELIVERY, STORAGE, AND HANDLING

- A. Deliver products to the project properly identified with names, model numbers, types, grades, compliance labels, and other information needed for identification.
- B. Deliver products to the project at such time as the project is ready to receive the equipment, pipe or duct properly protected from incidental damage and weather damage.
- C. Damaged equipment shall be promptly removed from the site and new, undamaged equipment shall be installed in its place promptly with no additional charge to the Owner.

#### 1.12 SUBMITTALS

- A. Coordinate with Division 01 for submittal timetable requirements, unless noted otherwise within thirty (30) days after the Contract is awarded. The Contractor shall submit an electronic copy of a complete set of shop drawings and complete data covering each item of equipment or material. The submittal of each item requiring a submittal must be received by the Architect or Engineer within the above thirty day period. The Architect or Engineer shall not be responsible for any delays or costs incurred due to excessive shop drawing review time for submittals received after the thirty (30) day time limit. The Architect and Engineer will retain a copy of all shop drawings for their files. All literature pertaining to items subject to Shop Drawing submittal shall be submitted at one time.

Submittals shall be placed in one electronic file in PDF 8.0 format and bookmarked for individual specification sections. Individual electronic files of submittals for individual specifications shall not be permitted. Each submittal shall include the following items:

1. A cover sheet with the names and addresses of the Project, Architect, MEP Engineer, General Contractor and the Subcontractor making the submittal. The cover sheet shall also contain the section number covering the item or items submitted and the item nomenclature or description.
  2. An index page with a listing of all data included in the Submittal.
  3. A list of variations page with a listing all variations, including unfurnished or additional required accessories, items or other features, between the submitted equipment and the specified equipment. If there are no variations, then this page shall state "NO VARIATIONS". Where variations affect the work of other Contractors, then the Contractor shall certify on this page that these variations have been fully coordinated with the affected Contractors and that all expenses associated with the variations will be paid by the submitting Contractor. This page will be signed by the submitting Contractor.
  4. Equipment information including manufacturer's name and designation, size, performance and capacity data as applicable. All applicable Listings, Labels, Approvals and Standards shall be clearly indicated.
  5. Dimensional data and scaled drawings as applicable to show that the submitted equipment will fit the space available with all required Code and maintenance clearances clearly indicated and labeled at a minimum scale of 1/4" = 1'-0", as required to demonstrate that the alternate or substituted product will fit in the space available.
  6. Identification of each item of material or equipment matching that indicated on the Drawings.
  7. Sufficient pictorial, descriptive and diagrammatic data on each item to show its conformance with the Drawings and Specifications. Any options or special requirements or accessories shall be so indicated. All applicable information shall be clearly indicated with arrows or another approved method.
  8. Additional information as required in other Sections of this Division.
  9. Certification by the General Contractor and Subcontractor that the material submitted is in accordance with the Drawings and Specifications, signed and dated in long hand. Submittals that do not comply with the above requirements shall be returned to the Contractor and shall be marked **"REVISE AND RESUBMIT"**.
- B. Refer to Division 1 for additional information on shop drawings and submittals.
- C. Equipment and materials submittals and shop drawings will be reviewed for compliance with design concept only. It will be assumed that the submitting Contractor has verified that all items submitted can be installed in the space allotted. Review of shop drawings and submittals shall not be considered as a verification or guarantee of measurements or building conditions.
- D. Where shop drawings and submittals are marked **"REVIEWED"**, the review of the submittal does not indicate that submittals have been checked in detail nor does it in any

way relieve the Contractor from his responsibility to furnish material and perform work as required by the Contract Documents.

- E. Shop drawings shall be reviewed and returned to the Contractor with one of the following categories indicated:
1. **REVIEWED:** Contractor need take no further submittal action, shall include this submittal in the O&M manual and may order the equipment submitted on.
  2. **REVIEWED AS NOTED:** Contractor shall submit a letter verifying that required exceptions to the submittal have been received and complied with including additional accessories or coordination action as noted, and shall include this submittal and compliance letter in the O&M manual. The contractor may order the equipment submitted on at the time of the returned submittal providing the Contractor complies with the exceptions noted.
  3. **NOT APPROVED:** Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is not approved, the Contractor will automatically be required to furnish the product, material or method named in the Specifications and/or drawings. Contractor shall not order equipment that is not approved. Repetitive requests for substitutions will not be considered.
  4. **REVISE AND RESUBMIT:** Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is marked revise and resubmit, the Contractor will automatically be required to furnish the product, material or method named in the Specifications and/or provide as noted on previous shop drawings. Contractor shall not order equipment marked revise and resubmit. Repetitive requests for substitutions will not be considered.
  5. **CONTRACTOR'S CERTIFICATION REQUIRED:** Contractor shall resubmit submittal on material, equipment or method of installation. The Contractor's stamp is required stating the submittal meets all conditions of the contract documents. The stamp shall be signed by the General Contractor. The submittal will not be reviewed if the stamp is not placed and signed on all shop drawings.
  6. **MANUFACTURER NOT AS SPECIFIED:** Contractor shall resubmit new submittal on material, equipment or method of installation when the alternate or substitute is marked manufacturer not as specified, the Contractor will automatically be required to furnish the product, material or method named in the specifications. Contractor shall not order equipment where submittal is marked manufacturer not as specified. Repetitive requests for substitutions will not be considered.
- F. Materials and equipment which are purchased or installed without shop drawing review shall be at the risk of the Contractor and the cost for removal and replacement of such materials and equipment and related work which is judged unsatisfactory by the Owner or Engineer for any reason shall be at the expense of the Contractor. The responsible Contractor shall remove the material and equipment noted above and replace with specified equipment or material at his own expense when directed in writing by the Architect or Engineer.
- G. Shop Drawing Submittals shall be complete and checked prior to submission to the Engineer for review.



- H. Furnish detailed shop drawings, descriptive literature, **table of contents listing all items being submitted at the beginning of each submittal package**, physical data and a specification critique for each section indicating "compliance" and/or "variations" for the following items:

Distribution Panelboards  
Panelboards  
Wiring Gutters  
Heavy Duty Disconnect Switches  
Lighting Fixtures  
Lighting Contactors  
Time Clocks  
Lighting Control System  
Photocells  
Wiring Devices and Plates  
Conduit and Fittings  
Wire  
Switchboards  
General Purpose Dry Type Transformers  
Harmonic Mitigating Type Transformers  
Emergency Generator  
Automatic Transfer Switches  
Sound Reinforcing System  
Fire Alarm System  
Surge Protection Device (SPD)  
Lightning Protection

- I. Refer to each specification section for additional requirements.

#### 1.13 OPERATION AND MAINTENANCE MANUALS

- A. Prepare maintenance manuals in accordance with Division 1 and in addition to the requirements specified in Division 1, include the following information for equipment items:
1. Description of function, normal operating characteristics and limitations, performance curves, engineering data and tests, and complete nomenclature and commercial numbers of replacement parts.
  2. Manufacturer's printed operating procedures to include start-up, break-in, and routine and normal operating instructions; regulation, control, stopping, shutdown, and emergency instructions; and summer and winter operating instructions.
  3. Maintenance procedures for routine preventative maintenance and troubleshooting; disassembly, repair, and reassembly; aligning and adjusting instructions.
  4. Servicing instructions and lubrication charts and schedules.

#### 1.14 COORDINATION DRAWINGS

- A. Prepare coordination drawings to a scale of 1/4"=1'-0" or larger; detailing major elements, components, and systems of mechanical equipment and materials in relationship with other systems, installations, and building components. Indicate locations where space is

limited for installation and access and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:

1. Indicate the proposed locations of pipe, duct, equipment, and other materials. Include the following:
    - a. Wall and type locations.
    - b. Clearances for installing and maintaining insulation.
    - c. Locations of light fixtures and sprinkler heads.
    - d. Clearances for servicing and maintaining equipment, including tube removal, filter removal, and space for equipment disassembly required for periodic maintenance.
    - e. Equipment connections and support details.
    - f. Exterior wall and foundation penetrations.
    - g. Routing of storm and sanitary sewer piping.
    - h. Fire-rated wall and floor penetrations.
    - i. Sizes and location of required concrete pads and bases.
    - j. Valve stem movement.
    - k. Structural floor, wall and roof opening sizes and details.
  2. Indicate scheduling, sequencing, movement, and positioning of large equipment into the building during construction.
  3. Prepare floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
  4. Prepare reflected ceiling plans to coordinate and integrate installations, air distribution devices, light fixtures, communication systems components, and other ceiling-mounted items.
- B. This Contractor shall be responsible for coordination of all items that will affect the installation of the work of this Division. This coordination shall include, but not be limited to: voltage, ampacity, capacity, electrical and piping connections, space requirements, sequence of construction, building requirements and special conditions.
- C. By submitting shop drawings on the project, this Contractor is indicating that all necessary coordination has been completed and that the systems, products and equipment submitted can be installed in the building and will operate as specified and intended, in full coordination with all other Contractors and Subcontractors.

#### 1.15 RECORD DRAWINGS

- A. Maintain a continuous record during the course of construction of all changes and deviations in the work from the contract drawings. Upon completion of the work, purchase a set of "Auto Positive Tracings" on vellum and make corrections as required to reflect the electrical systems as installed. Location and size of all conduit shall be accurately shown to dimension. Submit three prints of the tracings for approval. Make corrections to tracings as directed and deliver "Auto Positive Tracings" to the Architect. Record drawings shall be furnished in addition to shop drawings. Symbols on the Record drawings shall correspond to the identification symbols on the contract drawings and equipment identification plates and tags.
- B. The Contractor shall maintain a set of clearly marked black line record "AS-BUILT" prints on the job site on which he shall mark all work details, alterations to meet site conditions and changes made by "Change Order" notices. These shall be kept available for inspection by the Owner, Architect or Engineer at all times.

- C. Refer to Division 1 for additional requirements concerning record drawings. If the Contractor does not keep an accurate set of as-built drawings, the pay request may be altered or delayed at the request of the Architect. Mark the drawings with a colored pencil. Delivery of as-built prints and reproducibles is a condition of final acceptance.
- D. The record prints shall be updated on a daily basis and shall indicate accurate dimensions for all buried or concealed work, precise locations of all concealed pipe or duct, locations of all concealed valves, controls and devices and any deviations from the work shown on the Construction Documents which are required for coordination. All dimensions shall include at least two dimensions to permanent structure points.
- E. Submit three prints of the tracings for approval. Make corrections to tracings as directed and delivered "Auto Positive Tracings" to the architect. "As-Built" drawings shall be furnished in addition to shop drawings.
- F. When the option described in paragraph F., above is not exercised then upon completion of the work, the Contractor shall transfer all marks from the submit a set of clear concise set of reproducible record "AS-BUILT" drawings and shall submit the reproducible drawings with corrections made by a competent draftsman and three (3) sets of black line prints to the Architect or Engineer for review prior to scheduling the final inspection at the completion of the work. The reproducible record "AS-BUILT" drawings shall have the Engineers Name and Seal removed or blanked out and shall be clearly marked and signed on each sheet as follows:

CERTIFIED RECORD DRAWINGS

DATE:

(NAME OF GENERAL CONTRACTOR)

BY: \_\_\_\_\_  
(SIGNATURE)

(NAME OF SUBCONTRACTOR)

BY: \_\_\_\_\_  
(SIGNATURE)

#### 1.16 CERTIFICATIONS AND TEST REPORTS

- A. Submit a detailed schedule for completion and testing of each system indicating scheduled dates for completion of system installation and outlining tests to be performed and schedule date for each test. This detailed completion and test schedule shall be submittal at least 90 days before the projected Project completion date.
- B. Test result reporting forms shall be submitted for review no later than the date of the detailed schedule submitted.
- C. Submit 4 copies of all certifications and test reports to the Architect or Engineer for review adequately in advance of completion of the Work to allow for remedial action as required to correct deficiencies discovered in equipment and systems.
- D. Certifications and test reports to be submitted shall include, but not be limited to those items outlined in Section of Division 26.

#### 1.17 MAINTENANCE MANUALS

- A. Coordinate with Division 1 for maintenance manual requirements, unless noted otherwise bind together in "D ring type" binders by National model no. 79-883 or equal, binders shall be large enough to allow 1/4" of spare capacity. Three (3) sets of all approved shop drawing submittals, fabrication drawings, bulletins, maintenance instructions, operating instructions and parts exploded views and lists for each and every piece of equipment furnished under this Specification. All sections shall be typed and indexed into sections and labeled for easy reference and shall utilize the individual specification section numbers shown in the Electrical Specifications as an organization guideline. Bulletins containing information about equipment that is not installed on the project shall be properly marked up or stripped and reassembled. All pertinent information required by the Owner for proper operation and maintenance of equipment supplied by Division 26 shall be clearly and legibly set forth in memoranda that shall, likewise, be bound with bulletins.
- B. Prepare maintenance manuals in accordance with Special Project Conditions, in addition to the requirements specified in Division 26, include the following information for equipment items:
1. Identifying names, name tags designations and locations for all equipment.
  2. Fault Current calculations and Coordination Study.
  3. Reviewed shop drawing submittals with exceptions noted compliance letter.
  4. Fabrication drawings.
  5. Equipment and device bulletins and data sheets clearly highlighted to show equipment installed on the project and including performance curves and data as applicable, i.e., description of function, normal operating characteristics and limitations, performance curves, engineering data and tests, and complete nomenclature and model numbers of replacement parts.
  6. Manufacturer's printed operating procedures to include start-up, break-in, and routine and normal operating instructions; regulation, control, stopping, shutdown, and emergency instructions; and summer and winter operating instructions.
  7. Maintenance procedures for routine preventative maintenance and troubleshooting; disassembly, repair, and reassembly; aligning and adjusting instructions, servicing instructions and lubrication charts and schedules.
  8. Equipment name plate data.
  9. Wiring diagrams.
  10. Exploded parts views and parts lists for all equipment and devices.
  11. Color coding charts for all painted equipment and conduit.
  12. Location and listing of all spare parts and special keys and tools furnished to the Owner.
  13. Furnish recommended lubrication schedule for all required lubrication points with listing of type and approximate amount of lubricant required.

- C. Refer to Division 1 for additional information on Operating and Maintenance Manuals.
- D. Operating and Maintenance Manuals shall be turned over to the Owner or Engineer a minimum of 14 working days prior to the beginning of the operator training period.

#### 1.18 OPERATOR TRAINING

- A. The Contractor shall furnish the services of factory trained specialists to instruct the Owner's operating personnel. The Owner's operator training shall include 12 hours of onsite training in three 4 hour shifts.
- B. Before proceeding with the instruction of Owner Personnel, prepare a typed outline in triplicate, listing the subjects that will be covered in this instruction, and submit the outline for review by the Owner. At the conclusion of the instruction period obtain the signature of each person being instructed on each copy of the reviewed outline to signify that he has a proper understanding of the operation and maintenance of the systems and resubmit the signed outlines.
- C. Refer to other Division 26 Sections for additional Operator Training requirements.

#### 1.19 SITE VISITATION

- A. Visit the site of the proposed construction in order to fully understand the facilities, difficulties and restriction attending the execution of the work.
- B. Before submitting a bid, it will be necessary for each Contractor whose work is involved to visit the site and ascertain for himself the conditions to be met therein in installing his work and make due provision for same in his bid. It will be assumed that this Contractor in submitting his bid has visited the premises and that his bid covers all work necessary to properly install the equipment shown. Failure on the part of the Contractor to comply with this requirement shall not be considered justification for the omission or faulty installation of any work covered by these Specifications and Drawings.
- C. Understand the existing utilities from which services will be supplied; verify locations of utility services, and determine requirements for connections.
- D. Determine in advance that equipment and materials proposed for installation fit into the confines indicated.

#### 1.20 WARRANTY

- A. The undertaking of the work described in this Division shall be considered equivalent to the issuance, as part of this work, of a specific guarantee extending one year beyond the date of completion of work and acceptance by Owner, against defects in materials and workmanship. Materials, appliances and labor necessary to effect repairs and replacement so as to maintain said work in good functioning order shall be provided as required. Replacements necessitated by normal wear in use or by Owner's abuse are not included under this guarantee.
- B. All normal and extended warranties shall include parts, labor, miscellaneous materials, travel time, incidental expenses, freight/shipping, refrigerant, oils, lubricants, belts, filters and any expenses related to service call required to diagnose warranty problems.

#### 1.21 TRANSFER OF ELECTRONIC FILES

- A. Project documents are not intended or represented to be suitable for reuse by Architect/Owner or others on extensions of this project or on any other project. Any such reuse or modification without written verification or adaptation by Engineer, as appropriate for the specific purpose intended, will be at Architect/Owner's risk and without liability or legal exposure to Engineer or its consultants from all claims, damages, losses and expense, including attorney's fees arising out of or resulting thereof.
- B. Because data stored in electric media format can deteriorate or be modified inadvertently, or otherwise without authorization of the data's creator, the party receiving the electronic files agrees that it will perform acceptance tests or procedures within sixty (60) days of receipt, after which time the receiving party shall be deemed to have accepted the data thus transferred to be acceptable. Any errors detected within the sixty (60) day acceptance period will be corrected by the party delivering the electronic files. Engineer is not responsible for maintaining documents stored in electronic media format after acceptance by the Architect/Owner.
- C. When transferring documents in electronic media format, Engineer makes no representations as to the long term compatibility, usability or readability of documents resulting from the use of software application packages, operating systems, or computer hardware differing from those used by Engineer at the beginning of the Project.
- D. Any reuse or modifications will be Contractor's sole risk and without liability or legal exposure to Architect, Engineer or any consultant.
- E. The Texas Board of Architectural Examiners (TBAE) has stated that it is in violation of Texas law for persons other than the Architect of record to revise the Architectural drawings without the Architect's written consent.
  - 1. It is agreed that "MEP" hard copy or computer-generated documents will not be issued to any other party except directly to the Architect/Owner. The contract documents are contractually copyrighted and cannot be used for any other project or purpose except as specifically indicated in AIA B-141 Standard Form of Agreement Between Architect and Owner.
  - 2. If the client, Architect or Owner of the project requires electronic media for "record purposes", then AutoCAD/ Revit documents will be prepared by Engineer on electronic media such as removable memory devices, flash drives or CD's. These documents can also be submitted via file transfer protocols. AutoCAD/ Revit files will be submitted with all title block references intact to permit the end user to only view and plot the drawings. Revisions will not be permitted in this configuration.
  - 3. At the Architect/Owner's request, Engineer will assist the Contractor in the preparation of the submittals and prepare one copy of AutoCAD/ Revit files on electronic media or submit through file transfer protocols. The electronic media will be prepared with all indicia of documents ownership removed. The electronic media will be prepared in a ".rvt" or ".dwg" format to permit the end user to revise the drawings.

## PART 2 - PRODUCTS

### 2.1 SUBSTITUTIONS

- A. The names and manufacturers and model numbers have been used in the Contract documents to establish types of equipment and standards of quality. Where more than

one manufacturer is named for a specific item of equipment, only one of the specified manufacturers will be considered for approval. Where only one manufacturer is mentioned with the phrase "or approved equal", Contractor may submit an alternate manufacturer for consideration, provided the following conditions are met:

1. Submit alternate equipment with complete descriptive data in shop drawing form. Provide sample of equipment upon request for review by Architect. Samples will be returned if requested in writing.
  2. Alternate equipment must be equal from the standpoint of materials, construction and performance.
  3. Alternate submittal must be presented to the Engineer/Architect ten (10) days prior to bid date for approval.
- B. The Architect and Engineer shall be the sole judge of quality and equivalence of equipment, materials and methods.

2.2 All materials and products used on this project shall be listed by Underwriters' Laboratories.

### 2.3 ACCESS DOORS

- A. Wherever access is required in walls or ceilings to concealed junction boxes, pull boxes, equipment, etc., installed under this Division, furnish a hinged access door and frame with flush latch handle to another Division for installation. Doors shall be as follows:
1. Plaster Surfaces: Milcor Style K.
  2. Ceramic Tile Surfaces: Milcor Style M.
  3. Drywall Surfaces: Milcor Style DW.
  4. Install panels only in locations approved by the Architect.

### 2.4 EQUIPMENT PADS

- A. Provide 4-inch-high concrete pads for indoor floor mounted equipment. Pads shall conform to the shape of the equipment with a minimum of 6 inch beyond the equipment. Top and sides of pads shall be troweled to a smooth finish, equivalent to the floor. External corners shall be bullnosed to a 3/4" radius, unless shown otherwise.
- B. Provide 6-inch-high concrete pads for all exterior mounted equipment. Pads shall conform to the shape of the equipment with a minimum of 6 inch beyond the equipment. Provide a 4-foot monolithic extension to the pad in front of the equipment for service when mounted on a non-finished area (i.e. landscape, gravel, clay, etc.) Top and sides of pads shall be troweled to a smooth finish. External corners shall be bullnosed to a 3/4" radius, unless shown otherwise.
- C. Provide a minimum 6-inch-high, steel reinforced concrete pad for generators. Pads shall be sized 6" larger than the outside perimeter dimensions. Provide a 4-foot monolithic extension to the pad around the equipment for service when mounted on a non-finished area (i.e. landscape, gravel, clay, etc.). Refer to structural details. Top and sides of pads shall be troweled to a smooth finish. External corners shall be bullnosed to a 3/4" radius, unless shown otherwise. The generator shall be bolted to the concrete pad per the manufacturers details.

- D. Provide steel reinforced concrete pad for utility transformers. Pads shall comply with Utility Company Standards.

## 2.5 ESCUTCHEONS

- A. Provide heavy chrome or nickel plated plates, of approved pattern, on conduit passing through walls, floors and ceilings in finished areas. Where conduit passes through a sleeve, no point of the conduit shall touch the building construction. Caulk around such conduit with sufficient layers of two hour rated firesafing by Thermafiber 4.0 P.C.F. density, U.S.G. fire test 4/11/78 and seal off openings between conduit and sleeves with non-hardening mastic prior to application of escutcheon plate. Escutcheons shall be Gravler Sure-Lock, or approved equal.

## 2.6 SPACE LIMITATIONS

- A. Equipment shall be chosen which shall properly fit into the physical space provided and shown on the drawings, allowing ample room for access, servicing, removal and replacement of parts, etc. Adequate space shall be allowed for clearances in accordance with Code requirements. Physical dimensions and arrangement of equipment shall be subject to the approval of the Architect.

## 2.7 PAINTING

- A. All factory assembled equipment for electrical work, except light fixtures, that normally is delivered with a factory applied finish shall be delivered with a hard surface factory applied finish such as baked-on machinery enamel which will not require additional field painting. The finish shall consist of not less than 2 coats of medium gray color paint USA No. 61 Munsell Notation 8-3G, 6. 10/0.54 enamel. This Contractor shall protect this finish from damage due to construction operations until acceptance of the building. He shall be responsible for satisfactorily restoring any such finishes or replacing equipment that becomes stained or damaged.

## 2.8 ELECTRICAL SYSTEM IDENTIFICATION

- A. Conduit Systems: Provide adequate marking of major conduit which is exposed or concealed in accessible spaces to distinguish each run as either a power or signal/communication conduit. Except as otherwise indicated, use orange banding with black lettering. Provide self-adhesive or snap-on type plastic markers. Indicate voltage for that raceway. Locate markers at ends of conduit runs, on pull boxes, on junction boxes, near switches and other control devices, near items of equipment served by the conductors, at points where conduit passes through walls or floors, or enters non-accessible construction and at spacings of not more than 50 feet along each run of conduit. Switch-leg conduit and short branches for power connections do not have to be marked, except where conduit is larger than  $\frac{3}{4}$  inch. Branch circuit conduits, junction boxes and pull boxes shall be marked with a permanent marker indicating panel name and branch circuit numbers.
- B. Underground Cable Identification: Bury a continuous, preprinted, bright colored plastic ribbon cable marker with each underground cable (or group of cables), regardless of whether conductors are in conduit, duct bank, or direct buried. Locate each directly over cables, 6 to 8 inches below finished grade.
- C. Identification of Equipment:
  - 1. All major equipment shall have a manufacturer's label identifying the manufacturer's address, equipment model and serial numbers, equipment size,



and other pertinent data. Care shall be taken not to obliterate this nameplate in any way.

2. A black-white-black laminated plastic engraved identifying nameplate shall be secured by stainless steel screws to each automatic transfer switch, switchboard, distribution panel, motor control center, motor starter panels and panelboards.
  - a. Identifying nameplates shall have ¼ inch high engraved letters and shall contain the following information:
    - 1) Name
    - 2) Voltage
    - 3) Phase
    - 4) "3" or "4" wire, and
    - 5) Where it is fed from.
  - b. An example of a panelboard nameplate is:  
Center Panel – 1HB  
480/277 volt, 3 phase, 4 wire  
Center Fed from DP2
  - c. An example of an automatic transfer switch nameplate is:  
Center ATS #2  
480/277 volt, 3 phase, 4 wire, 4 pole  
Center Fed from MSB and DPE
3. Each feeder device in a switchboard, distribution panel, and motor control center device shall have a nameplate showing the load served in ½ inch high engraved letters.
4. A black-white-black laminated plastic engraved identifying nameplate shall be secured by screws to each safety switch, disconnect switch, individual motor starter, enclosed circuit breaker, wireway, and terminal cabinet.
  - a. Identifying nameplates shall have ¼ inch high engraved letters and shall indicate the equipment served.
  - b. An example if a disconnect switch is: AHU-1.
5. Prohibited Markings: Markings which are intended to identify the manufacturer, vendor, or other source from which the material has been obtained are prohibited for installation within public, tenant, or common areas within the project. Also, prohibited are materials or devices which bear evidence that markings or insignias have been removed. Certification, testing (example, Underwriters' Laboratories, Inc.), and approval labels are exceptions to this requirement.
6. Warning Signs: Provide warning signs where there is hazardous exposure associated with access to or operation of electrical facilities. Provide text of sufficient clarity and lettering of sufficient size to convey adequate information at each location; mount permanently in an appropriate and effective location. Comply with recognized industry standards for color and design.
7. Operational Tags: Where needed for proper and adequate information on operation and maintenance of electrical system, provide tags of plasticized card stock, either preprinted or hand printed. Tags shall convey the message, example: "DO NOT OPEN THIS SWITCH WHEN BURNER IS OPERATING."

### PART 3 - EXECUTION

#### 3.1 EXCAVATING AND BACKFILLING

- A. Trenching and backfilling and other earthwork operations required to install the facilities

specified herein shall conform to the applicable requirements of Division 2 (95% of maximum standard density). Where trenching or excavation is required in improved areas, the backfill shall be compacted to a condition equal to that of adjacent undisturbed earth and the surface of the area restored to the condition existing prior to trenching or excavating operations. Provide a minimum of 3" of sand underneath all conduits. The plans indicate information pertaining to surface and sub-surface obstructions; however, this information is not guaranteed. Should obstructions be encountered whether or not shown, the Contractor shall alter routing of new work, reroute existing lines, remove obstructions where permitted, or otherwise perform whatever work is necessary to satisfy the purpose of new work and leave existing surfaces and structures in a satisfactory and serviceable condition. **All work shall comply with OSHA Standards.**

### 3.2 WORKMANSHIP AND CONCEALMENT

- A. The work of this Section shall be performed by workman skilled in their trade. Installation shall be consistent in completeness whether concealed or exposed. Each item of electrical work shall be concealed in walls, chases, under floors and above ceilings except:
  - 1. Where shown to be exposed.
  - 2. Where exposure is necessary to the proper function.

### 3.3 SLEEVES, CUTTING AND PATCHING

- A. This section shall be responsible for placing sleeves for all conduit passing through walls, partitions, sound walls, beams, floors, roof, etc. Sleeves through below-grade walls shall use water-tight fitting manufactured by O-Z/Gedney.
- B. All cutting and patching will be done under another Division, but this Section will be responsible for timely performance of this work and layout of holes and setting sleeves.
- C. All un-used sleeves shall be sealed with 2 hour UL approved fire sealant manufactured by "3M" or approved equal.
- D. Refer to 26 05 33 for additional requirements.

### 3.4 ELECTRICAL GEAR

- A. Install all electrical equipment in accordance with the National Electrical Code and as shown on the drawings.
- B. Lighting contractors, time clocks, disconnect switches, etc. mounted in mechanical/electrical rooms shall be mounted at a working height not requiring a ladder, when wall space is available. Installation of these devices at greater elevations shall be approved by the Engineer. Contractor shall provide a coordination sketch of each mechanical/electrical room noting locations and mounting heights of all electrical devices (note bottom and top elevations) shown to be installed. Sketches shall be provided to the Engineer for review and the general contractor for coordination with other trades working in these rooms.

### 3.5 CLEANING

- A. Clean lighting fixtures and equipment.
- B. Touch-up and refinish scratches and marred surfaces on panels, switches, starters, and

transformers.

### 3.6 CORROSIVE AREAS

- A. In areas of a corrosive nature, which include but are not limited to the following: pool equipment rooms, cooling towers and areas subject to salt air, etc., provide NEMA 4 X stainless steel or fiberglass reinforced enclosures for contactors, panel boards, controllers, starters, disconnects and materials used as supporting means (i.e. plastibond unistrut, pipe, fittings). The use of spray on coating may be acceptable in some applications.

### 3.7 TESTS AND INSPECTIONS

- A. Tests and inspection requirements shall be coordinated with Division I.
- B. Date for final acceptance test shall be sufficiently in advance of completion date of contract to permit alterations or adjustments necessary to achieve proper functioning of equipment prior to contract completion date.
- C. Conduct re-tests as directed by Architect on portions of work or equipment altered or adjusted as determined to be necessary by final acceptance test. No resultant delay or consumption of time as a result of such necessary re-test beyond contract completion date shall relieve Contractor of his responsibility under contract.
- D. Put circuits and equipment into service under normal conditions, collectively and separately, as may be required to determine satisfactory operation. Demonstrate equipment to operate in accordance with requirements of these specifications. Perform tests in the presence of Architect. Furnish instruments and personnel required for tests.
- E. Final Inspection:
  - 1. At the time designated by the Architect, the entire system shall be inspected by the Architect and Engineer. The contractor or his representative shall be present at this inspection.
  - 2. Panelboards, switches, fixtures, etc., shall be cleaned and in operating condition.
  - 3. Certificates and documents required hereinbefore shall be in order and presented to the Architect prior to inspection.
  - 4. Panel covers, junction box covers, etc., shall be removed for visual inspection of the wire, bus bars, etc.
  - 5. After the inspection, any items which are noted as needing to be changed or corrected in order to comply with these specifications and the drawings shall be accomplished without delay.
- F. The contractor shall provide a thermographic test using an independent testing laboratory using an infrared scanning device. This test shall include but not limited to all switchboards, distribution panelboards, panelboards, automatic transfer switches and other electrical distribution devices. This test shall be conducted to locate high temperature levels. This test shall be conducted between 3 to 8 months after occupancy, but not beyond the one year warranty period. Submit test to the architect and engineer using test reporting forms. All unacceptable conditions shall be corrected prior to the end of the warranty period.

**END OF SECTION**

## SECTION 26 02 01 - COORDINATION DRAWINGS

### PART 1 - GENERAL

#### 1.1 GENERAL REQUIREMENTS

- A. The requirements of the General Conditions 013100 and Supplementary Conditions apply to all Work herein.

#### 1.2 COORDINATION DRAWINGS

- A. The Contractor shall take the lead in coordinating the Mechanical, Electrical, Plumbing, Communications, Electronic Safety/Security and Fire Protection systems within the building.
- B. The Contractor shall coordinate a three-dimensional (3D) model of the building which includes the Mechanical, Electrical, Plumbing, and Fire Protection systems. The Mechanical, Electrical, Plumbing, and Fire Protection Contractors shall prepare their work and generate 3D models which will be given to the Contractor for coordination. The Contractor will be provided with the REVIT model that was used to generate the contract documents, this file may be used as the background file. The Contractor shall replace the systems drawn with the actual shop drawing models. The Contractor is not limited to using REVIT, but may use any 3-D software in generating and combining the coordination model.
- C. Submitting the contract drawings as coordination drawings will not be acceptable.
- D. The model shall include detailed and accurate representations of all equipment to be installed based upon the reviewed equipment submittals.
- E. The Contractor shall hold a 3-D coordination meeting with all sub-contractors present to review the model and discuss coordination of the installation of the building systems.
- F. Upon completion of the coordination meeting, the Contractor shall submit the 3-D model and 1/4" scale drawings for review.
- G. The model shall detail major elements, components, and systems in relationship with other systems, installations, and building components. Indicate locations where space is limited for installation and access and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:
  - 1. Indicate the proposed locations of pipe, duct, equipment, and other materials. Include the following:
    - a. Wall and type locations.
    - b. Clearances for installing and maintaining insulation.
    - c. Locations of light fixtures and sprinkler heads.
    - d. Clearances for servicing and maintaining equipment, including tube removal, filter removal, and space for equipment disassembly required for periodic maintenance.
    - e. Equipment connections and support details.
    - f. Exterior wall and foundation penetrations.
    - g. Routing of storm and sanitary sewer piping.
    - h. Fire-rated wall and floor penetrations.
    - i. Sizes and location of required concrete pads and bases.
    - j. Valve stem movement.
    - k. Structural floor, wall and roof opening sizes and details.

2. Indicate scheduling, sequencing, movement, and positioning of large equipment into the building during construction.
3. Prepare floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
4. Prepare reflected ceiling plans to coordinate and integrate installations, air distribution devices, light fixtures, communication systems components, and other ceiling-mounted items.

H. Sequence of Coordination

1. Below is hierarchy of model elements and the sequencing by which the models will be coordinated:
  - a. Structural and Architectural model
  - b. Miscellaneous steel
  - c. Perform preliminary space allocation
  - d. Identify hard constraints (locations of access panels, lights, A/V space requirements, etc.)
  - e. Main and medium pressure ducts from the shaft out
  - f. Main graded plumbing lines and vents
  - g. Sprinkler mains and branches
  - h. Cold and hot water mains and branches
  - i. Lighting fixtures and plumbing fixtures
  - j. Smaller sized ducts and flex ducts
  - k. Smaller size cold water and hot water piping, flex ducts, etc.
- I. The Contractor shall not install any item until the coordination has been completed and reviewed by the Construction Manager, Owner, and A/E team.
- J. This Contractor shall be responsible for coordination of all items that will affect the installation of the work of this Division. This coordination shall include, but not be limited to: voltage, ampacity, capacity, electrical and piping connections, space requirements, sequence of construction, building requirements and special conditions.
- K. By submitting shop drawings on the project, this Contractor is indicating that all necessary coordination has been completed and that the systems, products and equipment submitted can be installed in the building and will operate as specified and intended, in full coordination with all other Contractors and Subcontractors.

**END OF SECTION**

## **SECTION 26 05 19 - WIRE, CABLE AND RELATED MATERIALS**

### **PART 1 - GENERAL**

#### **1.1 SCOPE**

- A. Provide 600 volt building wire, cable and connectors and 300 volt wire, cable and connectors.
- B. **WORK INCLUDED:** Include the following Work in addition to items normally part of this Section.
  - 1. Wiring for lighting and power.
  - 2. Automatic Control Wiring.
  - 3. Connection of equipment shown.
  - 4. Fire Alarm System.
- C. **WORK SPECIFIED ELSEWHERE:**
  - 1. Heating, ventilating, and air conditioning equipment.
  - 2. Structured cabling system.
  - 3. Coaxial cables

#### **1.2 STANDARDS**

- A. UL83
- B. ASTM B-3
- C. All wire cable and connectors shall be UL approved.

#### **1.3 ACCEPTABLE MANUFACTURERS**

- A. **600 VOLT WIRE AND CABLE**
  - 1. Southwire
  - 2. Encore
  - 3. Cerro
- B. **300 VOLT WIRE AND CABLE**
  - 1. Westpenn
  - 2. Beldon
  - 3. Alpha
  - 4. Tappan - Southwire

C. FLEXIBLE CABLE SYSTEMS

1. AFC Modular Cable Systems

D. CONNECTORS

1. IIsco
2. Cooper
3. AMP - TYCO
4. Burndy
5. Ideal
6. 3M
7. O.Z. Gedney
8. Thomas & Betts
9. Buchanan

1.4 SUBMITTALS

- A. Shop drawings shall include, but not limited to:

1. Cutsheets of wire, cable and connectors to indicate the performance, fabrication procedures, product variations, and accessories.

1.5 REQUIREMENTS OF REGULATORY AGENCIES WORK IN ACCORDANCE WITH:

- A. National Electrical Code.
- B. Local, municipal, or state codes that have jurisdiction.

PART 2 - PRODUCTS

2.1 WIRING

- A. All wire shall be new and continuous without weld, splice, or joints throughout its length. It must be uniform in cross-section, free from flaws, scales and other imperfections.
- B. WIRE MATERIAL: Conductors shall be soft drawn, annealed copper. Aluminum wiring is not acceptable unless otherwise noted on drawings.
- C. TYPES:
1. Provide type XHHW insulation for all buried feeders and service entrance conductors.
  2. Provide type "THHN/THWN-2" insulation for all branch circuits and above grade feeders.



3. All wire No. 8 and larger shall be stranded. All wire No. 10 and smaller shall be stranded or solid.
4. Provide type "XHHW" or other 90 degrees insulation wiring for branch circuit wiring installed through continuous rows of fluorescent fixture bodies.
5. All 300-volt cable including but not limited to telephone, fire alarm, data, CATV and security shall be UL listed for use in return air plenums.

D. CONDUCTOR SIZES

1. Feeder conductors shall be sized for a maximum of 2% drop in rated voltage at scheduled load.
2. Branch circuit conductors shall be sized for a maximum 3% drop in the rated voltage to the longest outlet on the circuit.
3. Minimum wire shall be No. 12, unless otherwise shown on Drawings or required by Code.

- E. COLOR CODING: No. 6 or larger shall use tape for color coding. No. 8 and smaller wire shall be color coded in accordance with the governing authority requirements or as follows:

120/208 Volt  
Neutral: White  
Phase A: Black  
Phase B: Red  
Phase C: Blue  
Ground: Green

2.2 GROUNDING

Permanently connect all conduit work, motors, starters, and other electrical equipment to grounding system in accordance with the National Electrical Code.

2.3 METAL CLAD CABLE - TYPE MC

Allowed for use as:  
Fixture whips per 26 51 00  
For receptacles in casework or millwork.  
No other applications accepted.

2.4 ARMORED CABLE - TYPE AC

Not allowed

PART 3 - EXECUTION

3.1 WIRE

- A. Do not pull wire into conduit until Work of an injurious nature is completed. Where two or more circuits run to a single outlet box, each circuit shall be properly tagged. Wyreze or approved equal may be used as a lubricant where necessary.
- B. Splices shall be fully made up in outlet boxes with compression crimp-on type splice

connectors.

- C. Joints and splices will not be permitted in service entrance or in feeders. Joints in branch circuits will be permitted where branch circuits divide, and then shall consist of one through-circuit to which the branch shall be spliced. Joints shall not be left for the fixture hanger to make. Connect joints and splices with Buchanan Series "2000" solderless connectors complete with insulating caps or properly sized twist on wire nuts. **"Wago" push-in connectors are not acceptable.**
- D. All stranded conductors shall be furnished with lugs or connectors.
- E. Connectors furnished with circuit breakers or switches shall be suitable for copper wire termination.
- F. "Sta-Cons" shall be used to terminate stranded conductors on all switches and receptacles.
- G. Metal Clad Cable - Type MC
  - 1. Metal clad cable shall be UL approved connectors and shall be used and installed per Article 334 of the National Electrical Code. The cable shall be supported at intervals not exceeding 6 feet and within 12 inches of every box.
  - 2. Provide anti-short bushing at cable ends.
  - 3. Metal clad cable shall not be installed in concrete.
- H. All stranded #10 and small conductors shall be terminated with an approved solderless terminal if the device or light fixture does not have provisions for clamp type securing of the conductor.
- I. The jacket for all travelers used on 3-way and 4-way switches shall be pink.
- J. Route conductors for 480Y/277 systems in a separate raceway. Do not combine with 208Y/120 volt or 120/240 volt systems.
- K. Emergency circuits shall not be routed with normal conductors.

### 3.2 BALANCING SYSTEM

The load on each distribution and lighting panel shall be balanced to within 10% by proper arrangement of branch circuits on the different phase legs. Provide written documentation showing results. Submit with O & M manuals.

### 3.3 LOW VOLTAGE WIRING

- A. Low voltage wiring shall be plenum rated. All wiring in mechanical rooms, electrical rooms, drywall ceiling, inaccessible areas, underground, plaster ceiling, inside concealed walls areas exposed to occupant view, and other areas subject to physical damage shall be run in conduit.
- B. Low voltage wiring shall be routed in separate raceways from power wiring systems.
- C. Sleeves shall be placed in the forms of concrete, masonry and fire rated walls, floor slabs and beams, for the passage of wiring. Sleeves should be set in place a sufficient time ahead of the concrete work so as not to delay the work. Sleeves shall be rigid galvanized

steel.

- D. Provide Caddy J-hooks supported independently from other system to support cable at 4-foot on center or closer if required by manufacture.

#### 3.4 CABLE SUPPORTS

- A. Provide cable supports in all vertical raceways in accordance with Article 300-19 of the NEC.

#### 3.5 DEFECTS

- A. Defects shall include, but are not to limited to, the following:
  - 1. Tripping circuit breakers under normal operation.
  - 2. Improperly connected equipment.
  - 3. Damaged, torn, or skinned insulation.

**END OF SECTION**

## **SECTION 26 05 26 - GROUNDING**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.

#### **1.2 SCOPE**

- A. **WORK COMBINED WITH OTHER SECTIONS:** Combine the work specified herein with the following Sections to form a single responsibility for the Work:
  - 1. Electrical.
  - 2. Basic materials and methods.
- B. Provide electrical service, equipment and wiring device grounding as shown, scheduled and as specified.
- C. The types of grounding include, but not limited to, the grounding bonding of all equipment devices, building steel piping, and as required by the National Electrical Code, Local Inspection Department and Power Company.

#### **1.3 STANDARDS**

- A. NATIONAL ELECTRICAL CODE (NFPA-70)
- B. Local municipal and State codes that have jurisdiction.
- C. NECA

#### **1.4 ACCEPTABLE MANUFACTURES**

- A. Provide grounding products manufactured by Copperweld and Cadweld.

#### **1.5 SUBMITTALS**

- A. Shop drawings shall include, but not limited to the following:
  - 1. Cut sheets of ground rods, clamps and connectors.
  - 2. Grounding system diagram.

### **PART 2 - PRODUCTS**

- A. **GENERAL:** Provide all materials required to construct a complete grounded electrical system.
- B. **GROUND RODS:** Ground rods shall be 3/4" inch diameter by 10 feet long construction with copper jacket and a steel core.
- C. **CLAMPS:** Ground clamps shall be copper except for steel or iron pipes in which the clamps shall be galvanized iron.

- D. CONDUCTORS: Conductors shall be connected by means of an approved pressure connector or clamp.

## PART 3 - EXECUTION

### 3.1 INSTALLATION

- A. GENERAL: Install grounding system as shown and specified to ensure a properly grounded system.
- B. SERVICE ENTRANCE GROUNDING SYSTEM: Provide a main bonding jumper between the neutral and ground bus of each switchboard. Route a separate grounding electrode conductor in conduit from each **main distribution panel** to the ground rod grid, incoming cold water piping system. Provide a bonding jumper around water meter. The grounding electrode conductor shall be stranded copper, 98% conductivity and shall be run continuous without splices or joints and installed at least 12" below grade.
- C. BUILDING STEEL AND PIPING SYSTEM: Install a bonding jumper between building steel and metallic piping systems to bond them to the electrical grounding system.
- D. NEUTRAL: The neutral shall be grounded only at the service entrance and other separately derived systems. The neutral shall be kept separate from the grounding system and shall not be used as a ground.
- E. GROUNDING SEPARATELY DERIVED ALTERNATING CURRENT SYSTEM
  - 1. TRANSFORMERS: The center point (neutral) of each wye connected transformer shall be bonded to the case and a grounding electrode conductor shall be connected to a ground rod or building steel.
  - 2. STANDBY EMERGENCY GENERATOR: The generator neutral shall be bonded to the generator when a 4 pole switched neutral automatic transfer switch is specified.
- F. GROUNDING CONDUCTOR: A grounding conductor and metallic conduit system shall bond all equipment served by the electrical system. Provide a flexible bonding jumper for isolated metallic piping and ductwork and around expansion fittings and joints.
- G. CONDUIT GROUNDING BUSHING:

Conduit terminating in equipment that has a ground bus such as switchboards, panelboards, etc., shall have grounding bushings installed. Ground each conduit by means of a grounding bushing and to the ground bus in the equipment.
- H. MOTORS: The frame of all motors shall be grounded.
- I. SPECIAL GROUNDING: Provide a #6 AWG copper grounding conductor for each telephone board, television system, etc. Terminate the grounding conductor on ground bus and to the building electrical grounding system. Refer to 800-40(d) and 820-40(d) of the NEC.
- J. REMOTE PANELBOARDS: Provide a grounding electrode conductor all remote panels as required by the NEC and shown on drawings.
- K. LIGHTING FIXTURES: Flexible fixture whips containing a green grounding conductor shall be used to connect light fixtures. Flexible fixture whips shall not exceed ten feet.

- L. RECEPTACLES: All receptacles shall be grounded using the branch circuit grounding conductor. Receptacles shall use an approved grounding yoke.
- 3.2 TESTING: Perform a ground resistance test using a biddle analog or digital portable earth/ground resistance tester. The system resistance shall not exceed 5 OHMS. Provide additional electrodes as required (refer to 250-84 of the NEC or the most current edition 250-56). Test shall not be conducted following wet weather. Provide personal instruments to conduct these tests and submit certified test for review. Test shall be verified by Engineer.

**END OF SECTION**

## **SECTION 26 05 33 - RACEWAYS**

### **PART 1 - GENERAL**

#### **1.1 SCOPE**

- A. Provide electrical raceways and fittings as shown, scheduled and specified.
- B. The types of raceways and fittings required are as follows:
  - 1. Rigid hot-dipped galvanized steel conduit (GRC) (RMC)
  - 2. Electrical metallic tubing (EMT)
  - 3. PVC (Sch. 40 & 80)
  - 4. Flexible metal conduit (FMC)
  - 5. PVC coated rigid galvanized steel conduit

#### **1.2 STANDARDS**

- A. ANSI, C80.1 & C80.3
- B. NEMA FB-1
- C. NEMA TC3
- D. UL, 6, 797 & 1242

#### **1.3 ACCEPTABLE MANUFACTURERS**

- A. Raceways
  - 1. Allied
  - 2. Republic
  - 3. Prime Conduit (Carlton)
  - 4. Wheatland Tube
  - 5. Cantex
  - 6. Western Tube
  - 7. Robroy Industries
- B. Fittings
  - 1. Appleton
  - 2. Crouse Hinds
  - 3. Steel City

4. O.Z. Gedney
5. Carlon
6. Raco, Inc.
7. Bridgeport

C. Boxes

1. RACO
2. Thomas and Betts
3. EATON
4. Crouse-Hinds
5. Appleton

D. Surface

1. Hubbell
2. Wiremold

1.4 SUBMITTALS

- A. Product data shall include but not be limited to:
1. Cutsheets for raceways, fitting, solvents, primers, etc.

1.5 REQUIREMENTS OF REGULATORY AGENCIES WORK IN ACCORDANCE WITH:

- A. National Electrical Code.
- B. Local, municipal, or state codes that have jurisdiction.

PART 2 – PRODUCTS

2.1 **CONDUIT AND FITTINGS:**

- A. Rigid Galvanized Steel Conduit.
1. Hot-dip galvanized rigid steel conduit, galvanized after fabrication. Products shall comply with UL6 and ANSI C80.1. All threads shall be galvanized after cutting. A uniform zinc coating shall be applied to the inner and outer walls.
  2. Fittings shall be threaded and shipped with thread protectors.
- B. Electrical Metallic Tubing (EMT).
1. EMT shall be made of hot-dip galvanized strip steel. The interior shall be coated



with a corrosion-resistant lubricant for ease of wiring pulling.

C. Rigid Nonmetallic Conduit (PVC).

1. Conduit shall be schedule 40 or 80 polyvinyl chloride (PVC), UV stabilized, rated for 90°C conductors.
2. Fittings shall be solvent weld socket type.

D. Flexible Metal Conduit (Greenfield).

1. Spirally wound continuously interlocked zinc coated strip steel.
2. Fittings shall be one screw for smaller than 1-1/2-inch, two screw for 1-1/2-inch and larger, double clamp steel or malleable iron, either cadmium plated or hot-dip galvanized.

## 2.2 PULL BOXES

- A. Exterior in-ground pull boxes shall be concrete or polymer as manufactured by Brooks, Dalworth, Hubbell Quazite, or approved equivalent. Covers shall include identification of systems contained.
- B. **Pull boxes in pole bases shall be as manufactured by Carlon.**

## 2.3 WIREWAYS

- A. Wireways shall be made of not less than 16-gauge sheet steel for 4 inch and 6 inch square sizes and 14 gauge steel for 8 inch and 12 inch square sizes. Couplings end plates, and knockouts shall be furnished as required. Each section of wireways shall be rigidly supported.
- B. The finish shall be ANSI-49 gray epoxy paint applied by a cathodic electrode position paint process over a corrosion resistant phosphate preparation for NEMA 1 wireways. Provide galvanized steel for NEMA 3R wireways. NEMA 3R wireways and auxiliary gutters are for horizontal mounting only.

## 2.4 FITTINGS

- A. Couplings for rigid steel or intermediate conduit shall be hot dipped galvanized steel. Set screw type is not acceptable.
- B. Steel or malleable iron fittings shall be used on all other raceway types except for PVC. Die-cast fittings are not allowed.
- C. EMT systems shall utilize steel insulated throat, set screw connectors and steel set screw couplings in all indoor conditioned spaces. EMT system shall utilize steel insulated throat, threadless, watertight compression type connectors and steel threadless watertight compression type coupling in all non-conditioned spaces.
- D. Coupling and connectors accessories and fittings for PVC coated rigid galvanized steel shall be PVC coated.
- E. Liquidtight Flexible Metal Conduit (LFMC) fittings shall be steel. Plastic is not acceptable.

- F. Provide nylon bushing on end of all low voltage cabling system conduits (sleeves, rough-ins, etc.).

### PART 3 - EXECUTION

#### 3.1 PROVIDE CONDUIT AS FOLLOWS:

##### A. GENERAL

The Drawings are diagrammatic, and are intended to show the general location of outlets, devices, fixtures, and arrangement and control of circuits. The Contractor shall determine exact locations by actual measurement of the building or by reference to the Architectural Drawings.

- B. Except as noted or otherwise specified, all wiring shall be installed in galvanized rigid steel, rigid aluminum conduit or electrical steel tube (EMT) of the proper size to contain the number of conductors required in accordance with the latest edition of the N.E.C. Where conduit sizes are shown on the drawings, these shall take preference. Contractor shall epoxy coat galvanized rigid steel conduit for use in natatoriums.
- C. Raceways shall not be routed below or within slab-on-grade, foundations, or below grade of suspended slab structures, unless specifically noted or indicated otherwise on plan.
- D. EMT in sizes up to 4 inches when concealed or not exposed to damage and located indoors only. (EMT is not acceptable in wet and damp location.)
- E. PVC coated rigid galvanized steel shall be used for all penetrations of slab on grade.
- F. Rigid galvanized steel where embedded in concrete or masonry construction, mechanical yard or in exterior/interior applications where subject to damage.
- G. Rigid aluminum shall be used in exterior applications. (i.e. roof, top of canopies)
- H. PVC schedule 40 and 80 may be utilized underground, in or below slab where shown on the construction documents.
- I. MINIMUM SIZE: 3/4 inch.
- J. PVC coated rigid galvanized steel conduit shall be coated inside and outside.
- K. PVC coated rigid galvanized steel conduit shall be used at cooling towers, corrosive areas and pool pump rooms.
- L. Fixture whips: Refer to 26 51 00 for additional information.
- M. Flexible metal shall be used for connecting rotating equipment installed in conditioned spaces.
- N. Of such size, and so installed that conductors may be drawn in without injury or excessive strain.
- O. Where entering panels, pull boxes, junction boxes, or outlet boxes, shall be secured in place with lock nuts inside and outside, and insulated bushings inside.
- P. Have Red seal type VCC or approved equal cable supports in risers, as required by N.E.C.

- Q. Have ends reamed after cutting and application of die.
- R. Keep conduit corked and dry during construction, and swab out before conductors are pulled.
- S. Have bends and offsets made with approved tools. Bends or offsets in which the pipe is crushed or deformed shall not be installed.
- T. Where not embedded in concrete or masonry, be firmly secured by approved clamps, half-straps or hangers.
- U. Have O.Z. Gedney or approved equal expansion fittings where crossing building expansion joints.
- V. Except in the mechanical equipment rooms, run conduit concealed, and by the shortest practicable route between outlets. Install risers, drops, and offsets necessary to avoid conflict with ductwork, piping, structural members, and similar items.
- W. Install exposed conduit in mechanical rooms, and elsewhere as indicated, parallel to horizontal and vertical lines of walls, ceilings, and floors.
- X. Fixtures in finished areas having suspended acoustical ceilings shall be connected to outlet boxes of lighting grid by flexible metal conduit; length not to exceed ten feet (six feet if using 3/8" manufactured fixture "whips").
- Y. Outlet boxes in partitions shall never be set back to back. They shall be offset to prevent undue noise transmission from room to room.
- Z. Concealed conduit shall run in as direct manner as possible using long bends. Exposed conduit shall be run parallel with or at right angles to the lines of the building; and all bends shall be made with standard conduit elbows or conduit benders. Not more than equivalent of four quarter bends shall be used in any run between terminals and cabinet, or between outlet or junction boxes. Approved condulets shall be used in lieu of conduit elbows where ease of installation and appearance warrants their use and approved by the engineer. Conduit joints shall be made with approved couplings and unions.
- AA. Conduits shall be continuous from outlet to outlet and from outlets to cabinets, junction or pull boxes and shall be electrically continuous throughout. Terminals of all conduits shall be provided with double lock nuts and bushing or terminated on conduit hubs. Use of running threads is prohibited.
- BB. Each entire conduit system shall be installed complete before any conductors are drawn in. Every run of conduit shall be finished before covering up to guard against obstructions and omissions.
- CC. Sleeves shall be placed in the forms of concrete, masonry and fire rated walls, floor slabs and beams, for the passage of conduits. Sleeves should be set in place a sufficient time ahead of the concrete work so as not to delay the work. Sleeves shall be rigid galvanized steel with a minimum thickness of 1.07MM and set to extend 4" above slab.
- DD. All pipe penetrations through walls and concrete floors shall be fire rated by applying USG Thermafiber in the space between the concrete and the pipe. The fire rating shall be additionally sealed by using 3M brand model CP 25 or 303 fire barrier caulk and putty. All fire rating material shall be installed in accordance with manufacturer's printed instructions.
- EE. All conduit shall be cleaned and swabbed to remove all foreign matter and moisture prior

to pulling wire and cable. All boxes in which conduits terminate shall be cleaned of all concrete mortar and other foreign matter.

- FF. Provide #30 nylon pulling line in all conduits in which permanent wiring is not installed.
- GG. All conduit shall be securely fastened and supported using hot galvanized malleable iron one-hole pipe straps, clamps, hanger or other means approved by the engineer. Supports shall be as required per NEC. Tie wire shall not be used as support or securing means. Support conduit independently of ceiling hanger wire. Use all thread rods to support outlet boxes, junction boxes and conduit.
- HH. When PVC conduit is routed underground, all stub-up's and bends 15° and greater shall be PVC coated rigid galvanized steel. Use PVC coated rigid galvanized steel when penetrating concrete on grade.
- II. Flexible and liquid-tight flexible steel conduit shall be used for final connections to utilization equipment. Liquid-tight flexible steel conduit shall be used for all exterior locations and all interior locations subject to moisture, vibrations, rotating equipment and dry-type transformers. Refer to Section 26 02 00 for additional information concerning flexible steel conduit.
- JJ. Contact the Architect and Engineer for an installation review before covering any below grade or above grade conduit.
- KK. All new outlets shall be flush mounted. In remodeled areas where wall construction prohibits flush mounting, provide Hubbell 2400 series, unless noted otherwise. Verify exact location and routing with architect before installation.
- LL. Contractor shall not penetrate water proof barriers without using proper fitting to maintain barriers. This shall include exterior walls and slabs. Coordinate with Architect for proper methods.

### 3.2 CONDUIT CORROSION PROTECTION

- A. Branch circuit conduits installed in concrete slabs on fill or grade shall be positioned in a manner to ensure complete concrete cover. In no case shall such conduits be exposed below or above the slab surfaces, or penetrate the waterproof membrane.
- B. At locations where metallic conduits pass through slabs on grade or transitions below grade, Contractor shall use 3M corrosive protective tap on rigid galvanized conduit in lieu of PVC coated rigid galvanized conduit.
- C. Conduit installed in the air gap between the water resistant barrier and finish brick shall not exceed 2ft in length.

### 3.3 EXPANSION JOINTS

- A. Install approved expansion fitting in all conduit runs in excess of 150 feet or when crossing building expansion joints.

### 3.4 OUTLET AND JUNCTION BOXES

- A. Provide an approved galvanized outlet box with adequate volume for number of conductors installed.
- B. Provide standard galvanized switch boxes of the required number of gangs. Switch boxes

where conduit is exposed shall be handy boxes or approved equal.

- C. Outlet boxes for receptacles shall be similar to Universal 52151 with suitable raised cover. Receptacle boxes where conduit is exposed shall be handy boxes or approved equal.
- D. Weatherproof boxes shall be FS or FD. Provide these boxes in all non-conditioned areas, exterior areas and natatoriums.
- E. Outdoor boxes shall be NEMA 3R, with conduit connections made by Myers Hubs.
- F. See notes and details on Drawings for special box requirements.
- G. Provide junction boxes required to facilitate installation of the various conduit systems. Provide support boxes required for risers, each complete with approved cable supports as described elsewhere in this Division.
- H. Outlet boxes for drywall shall be standard galvanized 4" square boxes with the appropriate device cover. Secure all outlet boxes with a backing brace connected to two adjacent studs. Mounting brackets with a single ear to rest against the backing sheet rock are not acceptable.
- I. Provide floor outlet fittings for telephone to match fittings for duplex floor receptacles.
- J. Provide 3-1/2" deep gangable masonry boxes in all masonry wall (CMU). Steel City GW-135-G or approved equal.
- K. Provide shallow 4"x4" boxes in all demountable partitions.
- L. Metallic boxes located in fire rated walls or partitions shall be separated by a minimum horizontal distance of 24 in. This minimum separation distance between metallic boxes may be reduced when "Wall Opening Protective Materials" (CLIV) are installed according to the requirements of their Classification. Metallic boxes shall not be installed on opposite side of walls or partitions of staggered stud construction unless "Wall Opening Protective Materials" are installed with the metallic boxes in accordance with Classification requirements for the protective materials.
- M. Junction, pull boxes, condulets, gutters, disconnects, contactors, etc., above 2-foot x 2-foot grid ceilings shall be mounted within 18-inches of ceiling grid. Above 2-foot x 4 – foot grid ceiling they shall be mounted within 30-inches of ceiling grid. All junction box, pull box, gutter openings shall be side or bottom accessible.

### 3.5 THRU-WALL SEALS

- A. Provide O.Z. Gedney "Thru-wall" seals for all conduits passing through concrete structure below grade, above grade, and floor penetrations below grade. These prevent moisture from entering the building.
- B. Straight sleeves are not acceptable.

### 3.6 PULL BOXES

- A. Interior Pull boxes shall be provided for conduit systems as required and shall be constructed of galvanized steel of not less than gauge and size specified by National Electrical Code. Size pull boxes per NEC 314.28.

- B. Where two or more feeders pass through a common pull box, they shall be tagged to indicate clearly their electrical characteristics, circuit number, and panel designation.
- C. Exterior in-ground pull boxes shall have open bottoms with sand and rock beds below box for drainage of water. Provide closed bottom boxes where specified. Closed bottom boxes shall be provided with sumps for portable pump to allow for extracting water. Refer to details on the drawings.
- D. Pull boxes mounted in pole bases shall be coordinated with the pour of the pole base and shall be flush with finished footing.

### 3.7 WIREWAYS

- A. Wireways shall be installed as indicated or required and locations shall be coordinated with architect.
- B. Wiring in wireways shall be neatly bundled, tied and suitably tagged.

### 3.8 UNDERGROUND DUCTBANK SYSTEM

#### A. DUCT SYSTEM

1. The duct system shall consist of Schedule 40 PVC or type 1-EB PVC conduits encased in concrete as detailed on the drawings. Use rigid conduit for stub-ups and the last ten feet at the end of each ductbank. Duct lines shall be laid to a minimum grade of 4 inches per 100 feet and shall be free from either horizontal or vertical waves. Duct lines shall be straight unless otherwise noted on the drawings. Duct lines shall be installed so that the top of concrete in encased duct lines is not less than 24 inches below finished grade or finished paving at any point. Changes in direction or runs exceeding a total of 10 degrees, either vertical or horizontal, shall be accomplished by long sweep bends having a minimum radius of curvature of 5 feet. The long sweep bends may be made up of one or more curved or straight sections and/or combinations thereof using five degree angle couplings. Conduit shall be thoroughly cleaned before using or laying. During construction and after the duct line is completed, the ends of the conduit shall be plugged to prevent water washing mud into the conduits. Particular care shall be taken to keep the conduits clean of concrete, dirt, and any other substance during the course of construction.
2. Each single conduit of the duct bank shall be completely encased in steel reinforced concrete as indicated. The thickness of concrete encasement indicated is the minimum thickness, and may be increased to fit the actual shape of trench.
3. Concrete for duct bank envelopes shall be standard 2000 psi concrete mix as described in Division 03,
  - a. Envelopes may be poured directly against sides of trenches if the "cut" is clean, even and free of loose material. All loose dirt and extraneous material shall be removed from the trenches before and during the pouring of concrete to ensure sound envelopes. Concrete shall be carefully spaded during pouring to eliminate all voids under and between the conduit and honeycombing of the exterior surfaces. Power driven tampers of agitators shall not be used, unless specifically designed for the application, in order to ensure that the water-tightness of the conduits is not destroyed.

- b. Generally, each run of envelopes shall be poured in one continuous operation. Where more than one pour is necessary, each pour shall terminate in a vertical plane. Partial pours shall not terminate in horizontal or angular planes.
- B. For normal underground installation see Section 26 02 00, paragraph 3.1 for Excavating and Backfilling.

**END OF SECTION**

## **SECTION 26 08 00 – COMMISSIONING OF ELECTRICAL SYSTEMS**

### **PART 1 – GENERAL**

#### **1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract Documents, including General and Supplementary Conditions and Division 01 Specifications, apply to this section.
- B. Related SECTIONS:
  - 1. SECTION 01 91 00 - GENERAL COMMISSIONING REQUIREMENTS

#### **1.2 SUMMARY**

- A. The commissioning of the lighting system and associated controls shall be performed by an impartial technical firm hired by the owner or shall be performed by the installing contractor if the owner has not hired a commissioning firm. The commissioning provider shall be certified under one or more of the following certifications:
  - 1. CxA – Certified Commissioning Authority – ACG
  - 2. CBCP – Certified Building Commissioning Professional – AEE
  - 3. CCP – Certified Commissioning Professional – BCA
  - 4. CPMP – Certified Process Management Professional – ASHRAE
  - 5. BSC – Building System Commissioning Certification – NEBB
- B. The commissioning provider (Commissioning authority) shall be responsible for leading the entire construction team through the commissioning process including, but not limited to, conducting the commissioning kick-off meeting, preparing the commissioning plan, preparing pre-functional checklists, preparing functional test scripts, participation in functional testing and preparation of required documentation and reports.

#### **1.3 RESPONSIBILITIES**

- A. Contractor: Responsibilities of the Contractor as relate to Commissioning Process include, but are not limited to the following:
  - 1. Facilitate coordination of Commissioning work by Commissioning authority.
  - 2. Attend Commissioning meetings or other meetings called by Commissioning authority to facilitate the Commissioning Process.
  - 3. Review Functional Performance Test procedures for feasibility, safety, and impact on warranty, and provide Commissioning authority with written comment on same.
  - 4. Provide all documentation relating to manufacturer's recommended performance testing of equipment and systems.
  - 5. Provide Operations & Maintenance data to Commissioning authority for preparation of checklists and training manuals.
  - 6. Provide As-built drawings and documentation to facilitate Testing.



7. Assure and facilitate participation and cooperation of Sub Contractors and equipment suppliers as required for the Commissioning Process.
8. Certify to Commissioning authority that installation work listed in Pre-Functional Checklists has been completed.
9. Install systems and equipment in strict conformance with project specifications, manufacturer's recommended installation procedures, and Pre-Functional Checklists.
10. Provide data concerning performance, installation, and start-up of systems.
11. Provide copy of manufacturers filled-out start-up forms for equipment and systems.
12. Ensure systems have been started and fully checked for proper operation prior to arranging for Testing with Commissioning authority. Prepare and submit to Commissioning authority written certification that each piece of equipment and/or system has been started according to manufacturer's recommended procedure, and that system has been tested for compliance with operational requirements.
  - a. Contractor shall carry out manufacturer's recommended start-up and testing procedures, regardless of whether or not they are specifically listed in Pre-Functional Checklists.
  - b. Contractor is not relieved of obligation for systems/equipment demonstration where performance testing is required by specifications, but a Functional Performance Test is not specifically designated by Commissioning authority.
13. Coordinate with Commissioning authority to determine mutually acceptable date of Functional Performance Tests.
14. Provide qualified personnel to assist and participate in Commissioning.
15. Provide test instruments and communications devices, as prescribed by Commissioning authority, required for carrying out Testing of systems.
16. Proprietary test equipment required by the manufacturer, whether specified or not, shall be provided by the manufacturer of the equipment. Manufacturer shall provide the test equipment, demonstrate its use, and assist in the commissioning process. Proprietary test equipment shall become the property of the Owner upon completion of commissioning.
17. Ensure deficiencies found in the Commissioning Issues Log are corrected within the time schedule shown in the Commissioning Plan.
18. Provide Commissioning authority with all submittals, start-up instructions manuals, operating parameters, and other pertinent information related to Commissioning Process. This information shall be routed through Architect.
19. Prepare and submit to Commissioning authority proposed Training Program outline for each system.
20. Coordinate and provide training of Owner's personnel.
21. Prepare Operation & Maintenance Manuals and As-Built drawings in accordance with specifications; submit copy to Commissioning authority in addition to other

contractually required submissions. Revise and resubmit manuals in accordance with Design Professionals and Commissioning authority's comments.

- 22. Commissioning requires participation of this Division Subcontractors to ensure that systems are operating in manner consistent with Contract Documents. All costs associated with the participation of Contractor, Sub-Contractors, Design Professionals, and Equipment Vendors in the Commissioning Process shall be included as part of the Construction Contract.
- B. Subcontractors and vendors shall prepare and submit to Commissioning Agent proposed Startup procedures to demonstrate proper installation of systems, according to these specifications and checklists prepared by Commissioning authority.
- C. Electrical contractor shall provide a letter certifying the installed lighting controls meet documented performance criteria specified in the commissioning plan within 90 days of substantial completion.

#### 1.4 COMMISSIONING PLAN

- A. Commissioning Process tasks and activities:
  - a. Commissioning kick-off meeting: Conducted by commissioning authority and attended by construction team and design team.
  - b. Pre-functional checklists: Prepared by the commissioning authority and filled out by subcontractors performing the work that is applicable.
  - c. Site visits to review installation of applicable systems and progress of checklist documentation performed and reported by commissioning authority.
  - d. Functional testing: Commissioning authority shall conduct functional testing with assistance of applicable subcontractors and document successful results as well as deficiencies (issues). Functional performance testing shall demonstrate the installation and operation of components, systems, and system-to-system interfacing in accordance with plans and specifications.
  - e. Preliminary commissioning report: Commissioning authority shall issue a preliminary commissioning report to the owner that has results of the first round of functional testing including deficiencies discovered.
  - f. Systems manual: Commissioning authority shall compile the systems manual using submittal data provided by the general contractor and applicable subcontractors.
  - g. Final commissioning report: Commissioning authority shall issue final commissioning report documenting the entire process and final results of functional testing. Report shall include final testing and balancing report.
- B. Electrical System Equipment to be tested
  - a. Occupancy sensors.
  - b. Time switch controls

- c. Daylighting controls.
- C. Testing functions and conditions
  - a. Daylighting control devices
    - a. Verify the devices have been calibrated, properly located and adjusted.
    - b. Loads adjust to light level set points in response to daylight.
    - c. Location of calibration equipment is accessible to authorized personnel only.
  - b. Time switches
    - a. Verify schedule, time, date and programming is accurate.
    - b. Verify override time limit is set, battery is installed and switch operates the lights that are specified in the design documents.
    - c. All specified lights can be turned on and off by area control switch.
    - d. Manual override switch allows only the lights in the space where the switch is located turn on or remain on until next scheduled shut off.
  - c. Occupant sensors:
    - a. Certify the sensor has been located and aimed in accordance with manufacturer recommendations.
    - b. For projects with fewer than seven sensors, each sensor shall be tested.
    - c. For projects with more than seven occupant sensors, testing shall be done for each unique combination of sensor type and space geometry. Where multiples of each combination are provided not less than 10 percent shall be tested.
    - d. Verify correct operation of status indicators.
    - e. Controlled lights turn off or down to the permitted level within the required time.
    - f. For auto-on sensor, the lights turn-on to the permitted level when an occupant enters space.
    - g. Verify the lights are not incorrectly turned-on by movement in adjacent areas or by HVAC operation.
- D. Performance criteria
  - a. Daylighting controls shall maintain specified light levels within 5% of design.
  - b. All time switches shall be accurate to time on cellular network devices.

## PART 2 – PRODUCTS

### 2.1 NO PRODUCTS SUPPLIED

## PART 3 – EXECUTION

### 3.1 GENERAL

- A. This Division has startup responsibilities and are required to complete sub-systems so COMPLETE SYSTEMS are fully functional. Insuring they meet design requirements of Contract Documents. Commissioning procedures and testing do not relieve or lessen this

responsibility or shift this responsibility, in whole or in part, to Commissioning Agent or Owner.

- B. Coordinate with other Sub-Contractors and equipment vendors to set aside adequate time to address Pre-Functional Checklists, Functional Performance Tests, Operations & Maintenance Manual creation, Owner Training, and associated coordination meetings.
- C. Commissioning authority will also conduct site inspections at critical times and issue Cx Field Reports with observations on installation deficiencies so that they may be issued by Architect as deemed appropriate.

### 3.2 WORK PRIOR TO COMMISSIONING

- A. Complete all phases of the work so the systems can be started, adjusted, balanced and otherwise tested.
- B. See pertinent specification sections in this Division, which outline responsibilities for start-up of equipment with obligations to complete systems, including all sub-systems so that they are fully functional.
- C. Assist Commissioning Agent with all information pertaining to actual equipment and installation as required complete the full commissioning scope.
- D. Contractor shall prepare startup procedures to demonstrate compliance with pre-functional checklists, and coordinate scheduling for completion of these checklists.
- E. A minimum of 7 days prior to date of system startup, submit to Commissioning Agent for review, detailed description of equipment start-up procedures which contractor proposes to perform to demonstrate conformance of systems to specifications and Checklists.

### 3.3 PARTICIPATION IN COMMISSIONING

- A. Attend meetings related to the Commissioning Process; arrange for attendance by personnel and vendors directly involved in the project, prior to testing of their systems.
- B. Provide skilled technicians to startup and test all systems, and place systems in complete and fully functioning service in accordance with Contract Documents.
- C. Provide skilled technicians, experienced and familiar with systems being commissioned, to assist Commissioning authority in commissioning process.

### 3.4 WORK TO RESOLVE DEFICIENCIES

- A. Complete corrective work in a timely manner to allow expeditious completion of Commissioning Process. If deadlines pass without resolution of identified problems, Owner reserves the right to obtain supplementary services and/or equipment to resolve the problem. Costs thus incurred will be Contractor's responsibility.

### 3.5 PRE-FUNCTIONAL CHECKLISTS (PFC)

- A. Contractor shall complete Pre-Functional Checklists to validate compliance with Contract Documents installation and start-up requirements, for this Division's systems.
- B. Refer to commissioning plan for detailed list of equipment to be commissioned.

### 3.6 FUNCTIONAL PERFORMANCE TESTING (FPT)

- A. Contractor, in cooperation with Commissioning Agent, shall conduct Functional Performance Testing to validate compliance with Contract Documents.
- C. Refer to commissioning plan for detailed list of equipment to be commissioned.
- B. Assist Commissioning authority in Functional Testing by removing equipment covers, opening access panels, etc. Furnish ladders, flashlights, meters, gauges, or other inspection equipment as necessary.

### 3.7 TRAINING

- A. The following requirements are in addition to Operations & Maintenance requirements specified elsewhere in this specifications manual.
- B. Contractor shall be responsible for training coordination and scheduling, and ultimately to ensure that training is completed.
- C. The training agenda (plan) shall include, at a minimum, the following elements:
  - 1. Purpose of equipment.
  - 2. Principle of how the equipment works.
  - 3. Important parts and assemblies.
  - 4. How the equipment achieves its purpose and necessary operating conditions.
  - 5. Most likely failure modes, causes and corrections.
  - 6. On site demonstration.
- D. Commissioning Agent shall be responsible for overseeing and approving content and adequacy of training of Owner personnel for all installed systems. Provide Commissioning Agent with training plan two weeks before planned training.

### 3.8 OPERATIONS & MAINTENANCE MANUALS

- A. The following requirements are in addition to Operations & Maintenance requirements specified elsewhere in this specifications manual.
- B. Contractor shall compile and prepare documentation for equipment and systems specified in this Division, and shall deliver documentation to Contractor for inclusion in Operation & Maintenance Manuals, in accordance with requirements of Division 01, prior to training Owner personnel.
- C. Provide Commissioning authority with a single, electronic copy of Operation & Maintenance Manuals for review. Commissioning authority's copy of O&M manuals shall be submitted through Architect.
- D. Operation and maintenance manuals shall include, service agency contact information, maintenance requirements, controls system settings and a narrative of how each system is intended to operate, including set points.

### 3.9 DOCUMENTATION

- A. Commissioning authority shall provide documentation of process as follows:
- a. Preliminary commissioning report including test procedures, results of testing, itemization of deficiencies, deferred tests and climatic conditions required for performance of deferred tests. Preliminary commissioning report shall be issued to owner to demonstrate the first pass of testing has occurred and to demonstrate compliance with applicable codes.
  - b. Final commissioning report shall include the final test and balance report, final results of functional testing, disposition of deficiencies discovered during testing, including the details of corrective measures used and functional testing procedures used for repeatability of testing in the future.

**END OF SECTION**

## **SECTION 26 21 13 - ELECTRICAL SERVICE ENTRANCE**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. Power company fees shall be paid by Owner.
- C. Provide electrical service to comply with contract documents and the power company standards.

#### **1.2 SCOPE**

##### **A. RELATED WORK:**

- 1. Basic Electrical Materials and Methods.
- 2. Earthwork
- 3. Division 1

- B. PERMANENT POWER: Contractor shall coordinate with the power company to provide electrical service to the building or campus as shown on the drawings. Contractor shall coordinate with power company and submit all required documentation; and shall establish the sequence of work to obtain permanent power.

- C. TEMPORARY POWER: Provided by the contractor.

#### **1.3 PERMITS, CODES, LAWS AND ORDINANCES**

- A. NFPA-70, NESC, State and local.

#### **1.4 MINIMUM COMPLIANCE STANDARDS**

- A. Easements required by power company.
- B. Duct bank required by power company.

### **PART 2 - PRODUCTS**

#### **2.1 ELECTRICAL SERVICE**

- A. SERVICE TRANSFORMER: Exterior Pad Mounted
- B. POWER COMPANY: AEP.
- C. SOURCE VOLTAGE: 208Y/120 volt, three phase, four wire system.
- D. TERMINAL

#### **2.2 CABLE TAP BOX**

- A. Provide when required by the power company for terminating additional conductors in which the power company transformers terminations are exceeded or for metering.

## 2.3 METERS

- A. Provide metering as required by Electric Utility Company Standards and requirements which may include self-contained meters and sockets, current transformers (CT's) and CT enclosures, bussed CT enclosures, transockets, meter modules, and bussed gutters and meter assemblies.

## PART 3 - EXECUTION

### 3.1 SERVICE LATERAL

- A. Route service lateral underground to main service disconnect.

### 3.2 COORDINATION

- A. Confirm with power company exact locations of service entry and other requirements.
- B. Provide all service laterals, whether above ground or below ground, in accordance with utility company engineered drawings. Information shown on contract documents may vary slightly from utility company engineered drawings and all utility company drawings and specifications shall supersede these contract drawings.
- C. All electrical service infrastructure, up to the point of delivery, shall be installed per utility company standards. Plans and details included in these contract documents do not depict exact installation details as they may be required by the utility company.

### 3.3 CONCRETE PADS

- A. Provide steel reinforced concrete pads with leave outs for conduits to comply with power company standards. Refer to Basic Materials and Methods.

### 3.4 INSTALLATION

- A. Provide concrete encased Primary ductbank, including any and all steel reinforcement, conduit spaces and straps as shown in the plans, utility company standards shall supersede where furnished by the utility company.
- B. Provide concrete encased secondary service lateral ductbank, including any and all steel reinforcement, sand-bed conduit spaces and straps as shown on the plans, utility company standards shall supersede where furnished by the utility company.
- C. All unmetered conductors shall be installed in rigid steel conduit, where installed above grade.

### 3.5 STEEL STRUCTURES

- A. Provide galvanized steel structure embedded in concrete to support metering equipment and disconnect switches.

**END OF SECTION**



## **SECTION 26 24 16 - PANELBOARDS**

### **PART 1 - GENERAL**

#### **1.1 SCOPE**

- A. Provide panelboards as shown, scheduled and as specified herein.
- B. The types of panelboards include:
  - 1. Panelboards.
  - 2. Power distribution panelboards.

#### **1.2 STANDARDS**

- A. Products shall be designed, manufactured, tested and installed in compliance with applicable standards.
- B. Products shall conform to all applicable UL standards and shall be UL-labeled.

#### **1.3 ACCEPTABLE MANUFACTURERS**

- A. Provide one of the following manufacturers:
  - 1. General Electric Company/ABB
  - 2. Square D Company
  - 3. Siemens
  - 4. Eaton

#### **1.4 SUBMITTALS**

- A. Shop drawings shall include, but not be limited to:
  - 1. Cutsheets of all enclosures, circuit breakers, fusible switches, bussing, rating, schedules and all accessories clearly labeled.

#### **1.5 REQUIREMENTS OF REGULATORY AGENCIES**

- A. WORK IN ACCORDANCE WITH:
  - 1. National Electrical Code.
  - 2. Local, municipal, or state codes that have jurisdiction.

### **PART 2 - PRODUCTS**

#### **2.1 MATERIALS AND COMPONENTS**

- A. General

Provide power distribution and panelboards as indicated in the panelboard schedule and as shown on the plans. . Power distribution panelboards shall be equipped with fusible

switches or circuit breakers as shown on the schedule. Panelboards shall be equipped with thermal-magnetic, molded case circuit breakers of frame and trip ratings as shown on the schedule.

**B. Busing Assembly and Temperature Rise**

Panelboard bus structure and main lugs or main breaker shall have current ratings as shown on the panelboard schedule. Such ratings shall be established by heat rise tests with maximum hot spot temperature on any connector or bus bar not to exceed 50°C. rise above 40°C ambient. Heat rise test shall be conducted in accordance with Underwriters Laboratories Standard UL 67. The use of conductor dimensions will not be accepted in lieu of actual heat tests. All current carrying parts of the bus shall be tin or silver plated copper.

1. Bus structure shall be isolated. Bus bar connections to the branch circuit breakers shall be distributed phase or phase sequence type and shall accept bolt-on circuit breakers for lighting and appliance panelboards.
2. The lugs for terminating conductors shall be rated at 75° C on all panel boards and circuit breakers.

Provide an extruded bare copper ground bus. Provide an isolated ground copper bus in each panel serving isolated ground circuits. Provide a full size copper neutral bus in each panelboard enclosure. Provide a double size neutral buss when served by a harmonic mitigating transformer.

**C. Distribution Panelboards**

Circuit breakers shall be equipped with individually insulated, braced and protected connectors. The front faces of all circuit breakers shall be flush with each other. Large, permanent, individual circuit numbers shall be affixed to each breaker in a uniform position. Tripped indication shall be clearly shown by the breaker handle taking a position between "ON" and "OFF". Provisions for additional breakers shall be such that no additional connectors will be required to add breakers. Circuit breakers shall be of the frame size, trip setting and interrupting capacity as indicated on the drawings. Circuit breakers shall be rated 65,000 AIC unless otherwise noted on plans.

1. Provide arc energy reduction switch for each breaker rated 1200 amps or larger to comply with 240.87 of the NEC. Switch shall be equipped with a pad lockable cover with a blue LED pilot light that illuminates when system is activated. Locate switch and cover recessed mounted adjacent to the breaker it serves or remote as indicated on the plans. Provide label and all required hardware. Remote switch(es) shall be flush mounted in wall near entry to the room.

**D. 240 Volt Panelboards**

Main breakers shall be vertically mounted. Horizontally mounted main breakers are not acceptable.

Circuit breakers shall be bolt-on thermal-magnetic, molded case circuit breakers. Breakers shall be 1, 2, or 3 pole with an integral crossbar to assure simultaneous opening of all poles in multiple circuit breakers. Breakers shall have an overcenter, trip-free, toggle-type operating mechanism with quick-make, quick-break action and positive handle indication.

Handles shall have "ON", "OFF" and "TRIPPED" positions.

Circuit breakers shall be UL listed in accordance with UL standard 489 and shall be rated 240 volts ac maximum with continuous current rating as noted on the plans.

Branch circuit breakers feeding convenience outlets shall have sensitive instantaneous trip settings of not more than 10 times the trip settings of the breaker to prevent repeated arcing short resulting from frayed appliance cords. Single pole 15 and 20 ampere circuit breakers shall be UL listed as "Switching Breakers" at 120V ac and carry the SWD marking.

UL Class A 5mA ground fault circuit protection shall be provided on all receptacle circuits serving wet areas and on all 120V ac branch circuits as specified on the plans or panelboard schedule. This protection shall be an integral part of the branch circuit breaker which also provides overload and short circuit protection for branch circuit wiring. Tripping of a branch circuit breaker containing ground fault circuit interruption shall not disturb the feeder circuit to the panelboard. A single pole circuit breaker with integral ground fault circuit interruption shall require no more panelboard branch circuit space than a conventional circuit breaker.

UL Class B 30mA ground fault circuit protection (GFEP) shall be provided on all equipment circuits requiring ground fault protection. This protection shall be an integral part of the branch circuit breaker which also provides overload and short circuit protection for branch circuit wiring.

Provide Shunt Trip Breakers including control power for circuits under cooking hoods and other equipment having this requirement.

Provide Breaker with Switched Neutral circuits with common trip for gasoline pumps and other equipment having this requirement.

Circuit breakers shall be rated 22,000 AIC at 240V unless otherwise noted on plans.

Provide double sized neutral bus with panels served from a non-linear transformer or when indicated on drawings. This shall be a UL approved assembly.

#### E. Cabinets and Fronts

The panelboard bus assembly shall be enclosed in a steel cabinet. The rigidity and gauge of steel to be as specified in UL Standard 50 for cabinets. Wiring gutter space shall be in accordance with UL Standard 67 for panelboards. The box shall be fabricated from galvanized steel or equivalent rust resistant steel. All NEMA-1 panels shall have hinged front covers. The front cover shall have a door with hinges, latch and a lock. The piano hinged front covers door-in-door shall allow full access to the circuit breaker gutter area without having to remove the entire front cover. All panelboard lock shall be keyed alike. Circuit breaker and fusible distribution panels shall have four-piece trims. A welded circuit directory frame and card with a clear plastic covering shall be provided on the inside of the door. Provide NEMA 1 enclosure where installed indoors unless otherwise noted. Provide NEMA 3R enclosure where installed outside or in a sprinkled area.

#### F. Safety Barrier

The distribution panelboard interior assembly shall be dead front with panelboard cover removed. Main lugs or main breakers shall have barriers on five sides. The barrier in front of the main lugs shall be hinged to a fixed part of the interior. The end of the bus structure opposite the mains shall have barriers.

G. Integrated Equipment Short Circuit Rating

Each panelboard, as a complete unit, shall have a short circuit current rating equal to or greater than the integrated equipment rating shown on the panelboard schedule or on the plans. This rating shall be established by testing with the over-current devices mounted in the panelboard. The short circuit tests on the over-current devices and on the panelboard structure shall be made simultaneously by connecting the fault to each over-current device with the panelboard connected to its rated voltage source. Method of testing shall be per Underwriters Laboratories Standard UL 67. The source shall be capable of supplying the specified panelboard short circuit current or greater. Testing of panelboard over-current devices for short circuit rating only while individually mounted is not acceptable. Also, testing of the bus structure alone is not acceptable. Panelboards shall be marked with their maximum short circuit current rating at the supply voltage and shall be UL listed.

H. Load Centers

1. Provide load centers where indicated.
2. Load centers shall have mains ratings and branch circuit breaker ratings of the size and number as indicated on drawings. Load centers shall be plug-on type construction. All current carrying parts of the bus assembly shall be plated copper. Terminals for feeder conductors to mains and branch neutral shall be UL listed as suitable for the type conductor specified. The load center bus assembly shall be enclosed in steel cabinet. The size of the wiring gutters and gauge steel shall be in accordance with UL Standards No. 67 and No. 50. Fronts shall include door and be provided with a directory for circuit identification. Load center boxes and fronts shall have corrosion resisting phosphate treatment and a gray baked enamel finish. Load centers shall be UL listed and meet Federal Specification U-P-115b as Type I, Class 2.
3. All breakers shall be full width, plug-on-type, toggle action with quick-make, quick-break mechanism. Provide Square D 'QO' breakers or approved equal. Piggy-back or half width breakers are not acceptable. Trip indication shall be clearly shown by the breaker handle taking a position between "ON" and "OFF" when the breaker is tripped. All multi-pole breakers shall be single operating handle, common trip variety. Branch circuit breakers feeding convenience outlets shall have sensitive, instantaneous trip in order to give "flash protection" for frayed, stranded wire cords. QWIK-GARD circuit breakers with ground fault circuit interrupters shall be used in accordance with the National Electrical Code. Circuit breakers shall be UL listed and meet the requirements of Federal Specification W-C-375B/GEN, Class 1. Provide ARC Fault circuit interrupter breaker to comply with 210.12 of the N.E.C.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General: Install panelboards, including electrical connections, in accordance with manufacturers written instructions, NEC and recognized industry practices.
- B. Housekeeping Pads: Mount floor mounted panelboards on 4 inch high concrete housekeeping pads.
- C. Fuses: Install fuses of the rating and class as shown in each fusible distribution panel scheduled on drawings.

- D. Conduits: Stub up three one inch conduits to an accessible location above the ceiling for each recessed panelboard.

### 3.2 IDENTIFICATION

- A. Nameplate: Each panelboard shall have an engraved bakelite nameplate. Nameplates shall be white with black letters and show panel designation. Nameplates shall be attached with stainless steel screws. Refer to Section 26 02 00, paragraph 2.8(C).
- B. Directory Card: Cardholders and directory cards shall be furnished for circuit identification in panelboards. Cardholder shall be located on inside of panel door and shall be in a metal frame with clear plastic front. Circuit lists shall be typewritten. Circuit descriptions shall include location and name of each item of equipment served. Spares and spaces shall be written in erasable pencil for future use. Circuit directory shall show the room served by each circuit. The final graphs/signage room numbers shall be used. Do not use Architectural numbering on plans.
- C. Replacement Components: Nameplate shall identify replacement components.

### END OF SECTION

## **SECTION 26 27 26 – WIRING DEVICES**

### **PART 1 – GENERAL**

#### **1.1 SCOPE**

- A. Provide wiring devices as shown; scheduled, required and as specified.
- B. The types of wiring devices required include:
  - 1. Receptacles
  - 2. Switches
  - 3. Coverplates

#### **1.2 STANDARDS**

- A. NEMAWD-1
- B. NEMA WD-5
- C. UL
- D. Federal Spec WC-596-F and WS-896

#### **1.3 ACCEPTABLE MANUFACTURERS**

- A. Hubbell
- B. Leviton
- C. Pass & Seymour

#### **1.4 SUBMITTALS**

- A. Shop drawings shall include but not be limited to:
  - 1. Cut sheets of all devices indicating NEMA configuration, rating, materials, color, and all accessories.
  - 2. Cut sheets of all coverplates indicating materials, color and any engraving specified on drawing or in the specifications.

#### **1.5 REQUIREMENTS OF REGULATORY AGENCIES WORK IN ACCORDANCE WITH:**

- A. National Electric Code.
- B. Local, municipal, or state codes that have jurisdiction.

### **PART 2 – PRODUCTS**

#### **2.1 MATERIALS AND COMPONENTS**

- A. GENERAL

1. Provide factory assemble wiring devices with the rating type and color as required and specified for the service indicated.
2. Provide matching one-piece multiple gang plates where switches are ganged.
3. Provide wall plates for each receptacle furnished.
4. Architect reserves the right to select wiring device styles and colors to match wall finish.
5. Wall plates shall be of same manufacturer as devices.

## 2.2 SWITCHES

- A. Provide specification grade White toggle switches where indicated on the Drawings. Provide "Red" switches for switching emergency lighting circuits where switching is indicated. Coordinate exact locations with architect.
- B. Wall switches shall be 20 amp, 120-277 volt and shall be Hubbell, Leviton or P&S as follows:
  1. SINGLE POLE SWITCHES: Hubbell HBL1221, Leviton 1221-2, P&S PS20AC1
  2. DOUBLE POLE SWITCHES: Hubbell HBL1222, Leviton 1222-2, P&S PS20AC2
  3. THREE WAY SWITCHES: Hubbell HBL1223, Leviton 1223-2, P&S PS20AC3
  4. FOUR WAY SWITCHES: Hubbell HBL1224, Leviton 1224-2, P&S PS20AC4
  5. MOMENTARY CONTACT SWITCHES: Hubbell HBL1557, Leviton 1257, P&S 1251
  6. THREE POSITION, TWO CIRCUIT MAINTAINED CONTACT SWITCHES: Leviton 1285, Hubbell HBL1385, P&S 1225
  7. KEY TYPE LOCKABLE BARREL KEY OR CORBIN STYLE: Leviton 1221-2KL with 2KL key or P&S PS20AC1-KL with 4609 key for each switch, Hubbell #HBL 1221-RKL.
  8. Dwelling units shall use Hubbell CS115I and CS120I.
- C. Dimmers: Provide Lutron DIVA or as shown on drawings. Wall box dimmers shall be sized to handle the load. Where fluorescent dimming ballasts are to be used, coordinate wall box dimmer with ballast manufacturer.
- D. Light Handle Switches: Provide Leviton 1221-LHC, Hubbell HBL1221-II, P&S PS20AC1-ISI lighted handles to switch emergency lights were noted on the drawings.

## 2.3 RECEPTACLES

- A. Provide specification grade White receptacles where indicated on the drawings. Provide "Red" receptacles for receptacles on emergency power. Coordinate exact location with architect.
- B. Receptacles shall be Hubbell, Leviton or Pass & Seymour as follows:
1. Duplex 20A-125V-self grounding:with Brass mounting yoke (NEMA configuration 5-20R):  
Hubbell HBL5352, Leviton 5362, P&S 5362A
  2. Simplex 20A-125V-Self Grounding with Brass mounting yoke (NEMA configuration 5-20R):  
Hubbell HBL5361, Leviton 5361, P&S 5361 with Brass mounting yoke.
  3. Isolated ground duplex, 20A-125V: (Orange, NEMA configuration 5-20R)  
Hubbell IG5352, Leviton 5362IG , P&S IG5362.
  4. Clock hanger receptacle 15A-125V: (Brown with stainless steel plate with hanger, NEMA configuration 5-15R):  
Leviton 5361-CH, Hubbell 5235, P&S S3733-SS
  5. Ground fault circuit interrupter (GFCI) receptacle 20A-125V; (NEMA Configuration 5-20R, shall incorporate self-test, auto monitoring technology and features which will lock-out or render the device incapable of being reset if ground fault protection is compromised, with "Feed through" connectors capable of protecting connected downstream receptacles on a single circuit, and of being installed in a 2-3/4" deep outlet box without adapter, Hubbell GFRST20, Leviton GFNT2 or P & S 2097.  
Install Hubbell GFTRST20, Leviton GFTR2 or P&S 2097TR Tamper Resistant type for locations requiring Tamper Resistant installations  
Install Hubbell GFTWRST20, Leviton GFWR2 or P&S 2097TRWR Weather Resistant type for installations in damp or wet locations
  6. Tamper resistant receptacles 20A-125V (NEMA configuration 5-20R):
  7. Hubbell HBL8300SGA, Leviton 8300-SG, P&S TR63-H.
  8. Surge Protection Duplex Receptacles 20A-125V, (NEMA 5-20R) Hospital grade to include **LED** light and audible alarm:  
Hubbell HBL8362SA, Leviton 8380, P&S 8300SP
  9. Equipment receptacles shall be coordinated with owner/manufacture requirements and the correct and appropriate receptacle and coverplate shall be installed.
  10. Receptacles for dwelling units shall be Hubbell CR15TR and CR20TR tamper resistant receptacles.
  11. USB Charger types receptacles shall be Hubbell, 20A, 125V AC Hospital Grade, Tamper Resistant, with two USB Type 2.0 Ports 5.0 Amp, 5V DC, Decorator Type duplex receptacle. Hubbell USB8300A5 or equal by other approved wiring device manufacturers.
  12. Plug load controlled receptacles shall be Hubbell DR20C2WHI, white, two controlled faces or equal by Leviton or P&S.
  13. ARC Fault circuit interrupter receptacles shall be Hubbell AFR20TR.
  14. Ground fault circuit interrupter/ARC Fault dual function receptacles shall be



Hubbell AFGF20TR.

## 2.4 OCCUPANCY SENSORS

- A. Provide [ivory white gray] dual technology wall mounted sensors, provide one of the following:
1. Single Pole:
    - Wattstopper #DSW301
    - Lutron #MSA102
  2. Double Pole:
    - Wattstopper # DSW302
    - Lutron #MSA202
  3. Dimmer:
    - Wattstopper #DW311
    - Lutron #MSZ101
- B. Provide dual technology ceiling sensor with low voltage controlling switch and power pack.
- A. Single Button:
- Wattstopper # DT300 Sensor, BZ150 Power Pack and LVSW101 Digital Switch
  - Lutron; LOSCDT-2000 Sensor, PP-DV-M Power Pack and NTRCS-1 Digital Switch
- C. Provide Ultra Sonic Ceiling sensor for restrooms.
- a. Wattstopper #UT3000, BZ150 Power Pack
  - b. Lutron #LOS-CUT-2000, PP-DV-M Power Pack

## 2.5 DIGITAL TIMER SWITCHES

- A. Provide wattstopper TS-400-G digital timer. Locate in mechanical, electrical, MDF, and IDF Rooms.
- B. The time switch shall provide audible notification and visual notification (blink the room lights) prior to turning lights off.
- C. The time switch shall have a 12 hour manual over ride setting.

## 2.6 PLATES

[SELECT PLATE TYPE]

- A. Furnish and install plates on all outlet boxes. Oversize (Jumbo) plates are not acceptable.
- B. Plates shall be 302/304 smooth stainless steel
- C. Provide Hubbell WP Series, Bell, Carlon or Leviton NEMA 3R weatherproof coverplates on all exterior wiring devices. Enclosure shall be suitable for wet locations when in use.
- D. Plates shall be Hubbell SS Series, Leviton, Pass & Seymour 302/304 smooth stainless steel on all receptacles 30 amps and larger.
- E. Stainless steel device plates shall be provided at locations with tile or stone walls.

2.7 Floor boxes with surface activation shall be cast iron as manufactured by Hubbell or equal by Wiremold and as indicated below:

- A. Slab at grade (dual level, fully adjustable type 1).
  - 1. Single gang: #B-2436 w/#SB-3083 carpet flange.
  - 2. Two gang: #B-4233 w/#SB-3084 carpet flange.
  - 3. Three gang: #B-4333 w/#SB-3085 carpet flange.
- B. Slab above grade (shallow, fully-adjustable, type II)
  - 1. Single gang: #B-2421 w/#SB-3083 carpet flange.
  - 2. Two gang: #B-2422 w/#SB-3084 carpet flange.
  - 3. Three gang: #B-2423 w/#SB-3085 carpet flange.
- C. Cover plates shall have brass finish as follows:
  - 1. #S-3825 for duplex flap for duplex receptacles.
  - 2. #S-3826 for data/communications.

2.8 PVC floor boxes manufactured by Hubbell or equal shall be as follows:

- A. Provide CFBS1R4CFB dual service cast iron body floor box with PVC riser. Provide CFBS1R4CUP adjustable mounting cup, S1R4SP2X2DUPLEX sub-plate for (1) Duplex and (2) RJ-45 Keystone jacks OR S1R4SP2X2STYLE for (1) GFCI duplex, USB or Surge Device & (2) Keystone jacks, OR S1R4SPQUAD sub-plate with (4) 20A simplex receptacles, single and dual circuit wiring capability. Provide with CFBS1R4CVR cover, Color to be chosen by Architect.  
NOTE TO SPECIFIER: Minimum depth of pour 5-inches, Maximum is 6-inches  
Maximum finished floor thickness (above top of box collar) with maximum adjustability is 1-1/2-inches at 5-inch, At 6-inches maximum adjustability is 1/2".

2.9 Floor boxes, recessed activation type, meet UL 514A scrub water requirements, shall be stamped steel with corrosion resistant finish, UL Listed for slab-on-grade installations, or stamped steel for above-grade installations as manufactured by Hubbell or equal by Wiremold and as indicated below:

- A. Recessed Activation Slab at grade:
  - 1. Two gang: #CFB2G30CR or CFB2G30RCR (provisions for round cover), capable of up to 2" entry per gang. Flush flange, Surface flange and Furniture Feed cover availability. Surface Type Covers shall not exceed 0.15" rise. Covers with provisions for cable egress, when in use, shall not exceed/extend past 0.15" rise
  - 2. Four Gang: #CFB4G30CR or CFB4G30RCR (provisions for round cover), capable of up to 2" entry per gang. Flush flange, Surface flange availability. Cover shall not exceed 0.15" rise. Surface Type Covers shall not exceed 0.15" rise. Covers with provisions for cable egress, when in use, shall not exceed/extend past the 0.15" rise.

3. Six Gang: #CFB6G30CR or CFB6G30RCR (provisions for round cover), capable of up to 2" entry per gang. Flush flange, Surface flange availability. Surface Type Covers shall not exceed 0.15" rise. Covers with provisions for cable egress, when in use, shall maintain the 0.15" rise. Covers with provisions for cable egress, when in use, shall not exceed/extend past the 0.15" rise.
4. Ten Gang AV: #CFB10G55CR or CFB10G55RCR (provisions for round cover), with minimum (2) 2" KO's, multiple front and back  $\frac{3}{4}$ " to 1-1/2" concentric KO's.. Flush flange, Surface flange availability. Cover shall not exceed 0.15" rise. Covers with provisions for cable egress, when in use, shall not exceed/extend past the 0.15" rise.

B. Recessed Activation Slab above grade

1. Two Gang: #CFB2G30 or CFB2G30R (provisions for round cover), capable of up to 2" entry per gang. Flush flange, Surface flange and Furniture Feed cover availability. Surface Type Covers shall not exceed 0.15" rise. Covers with provisions for cable egress, when in use, shall not exceed/extend past the 0.15" rise.
2. Four Gang: #CFB4G30 or CFB4G30R (provisions for round cover), capable of up to 2" entry per gang. Flush flange, Surface flange availability. Cover shall not exceed 0.15" rise. Surface Type Covers shall not exceed 0.15" rise. Covers with provisions for cable egress, when in use, shall not exceed/extend past 0.15" rise.
3. Six Gang: #CFB6G30 or CFB6G30R (provisions for round cover), capable of up to 2" entry per gang. Flush flange, Surface flange availability. Surface Type Covers shall not exceed 0.15" rise. Covers with provisions for cable egress, when in use, shall maintain the 0.15" rise. Covers with provisions for cable egress, when in use, shall not exceed/extend past the 0.15" rise
4. Ten Gang AV: #CFB10G55 or CFB10G55R (provisions for round cover), with minimum (2) 2" KO's, multiple front and back  $\frac{3}{4}$ " to 1-1/2" concentric KO's.. Flush flange, Surface flange availability. Cover shall not exceed 0.15" rise. Covers with provisions for cable egress, when in use, shall not exceed/extend past the 0.15" rise.

C. Service Fittings

1. Surface Style Rectangular for use with carpet, tile, VCT and other engineered floors, available with or without carpet insert and offer system's furniture feed type cover providing (1) 1-inch and (1) 2-inch threaded openings
2. Flush Style Rectangular for use with tile, finished concrete or Terrazzo floors, available with or without carpet insert and offer system's furniture feed type cover providing (1) 1-inch and (1) 2-inch threaded openings
- D. Rectangular covers shall be powder coated in variety of common finishes, Aluminum, Black, Brass, Bronze and Satin Nickel
4. Round Covers for use with all floor types. Shall provide cable egress doors and systems furniture feed type cover providing (1)  $\frac{3}{4}$ -inch and (1) 2-inch threaded openings. Round covers shall be plated metal in variety of finishes except Black (powder coated). Brushed Aluminum, Brass Plated, Bronze Plated, Satin Nickel Plated.

## PART 3 – EXECUTION

### 3.1 WIRING DEVICE MOUNTING HEIGHTS

- A. Unless noted to the contrary on plans, or directed otherwise during the progress of the Work, wiring devices shall be set as follows:
  - 1. Switches 42" above finished floor.
  - 2. Wall mounted receptacles shall be installed vertically at 15 inches to the bottom outlet above finished floor unless otherwise noted or as required by local codes.
  - 3. Wall telephone outlets shall be mounted 15 inches to the bottom above finished floor unless otherwise noted. Mount even with wall mounted receptacles.
  - 4. At locations above counters, set devices at 6 inches above to the centerline counter tops, verify exact mounting height with the architect.

### 3.2 INSTALLATION (Refer to 26 05 33 for outlet box specifications).

- A. Wall switches shall be set in a suitable steel box and shall be installed on the strike side of the door as finally hung, whether so indicated on the Drawings or not.
- B. Receptacles shall be installed in a suitable steel box.
- C. The Architect reserves the right to relocate wiring device up to a distance of 5 feet from the location shown, before rough-in, without additional cost.
- D. Provide multi-gang device covers at locations where devices gang together.
- E. Device locations are indicated schematically on the drawings along with the type and mounting height. Final locations and mounting heights shall be coordinated with the Architect on the jobsite, and with shop drawings of equipment; including equipment to be furnished and installed by the Owner. Devices installed in walls covered with vinyl, fabric wallpaper or other special finishes shall be coordinated and verified with the Architect on the job-site.
- F. Stranded wire termination to switches, receptacles, devices and miscellaneous control devices shall be with an approved solderless terminal if clamp type securing is not possible (i.e. Sta-Con crimp on fork tongue connectors; Burndy Type TP-F).
- G. Provide keyed switches in all common areas not monitored by the faculty (i.e. gym, corridors, cafeteria, commons natatoriums).
- H. Tamper-resistant type receptacles shall be installed in all classrooms, cafeterias, corridors, special education, ALE, computer labs, special use classroom and all spaces where children 7 years and younger may occupy. In Child-Care facilities, tamper resistant receptacles shall be provided for all spaces with exception to back-of-house spaces, such as kitchens, custodial closets, electrical and mechanical rooms.
- I. All 20A, 120V receptacles in food service areas shall be GFCI.
- J. All circuit breaker serving electric drinking fountains shall be GFCI.
- K. Provide ARC Fault circuit interrupters (AFCI) as required to comply with 210.12 of the N.E.C. This shall include but not limited to dwelling units and dormitory's. AFCI breakers may be used.

- L. Provide ground fault circuit interrupter (GFCI)/ARC Fault circuit interrupter (AFCI) dual function receptacles to comply with 210.8, 210.12 and 406.4 of the N.E.C.

END OF SECTION

## **SECTION 26 28 16 - SAFETY AND DISCONNECT SWITCHES**

### **PART 1 - GENERAL**

#### **1.1 SCOPE**

- A. Provide safety and disconnect switches as shown, scheduled and as specified herein.

#### **1.2 STANDARDS**

- A. Products shall be designed, manufactured, tested and installed in compliance with applicable standards.
  - 1. NEMA KS1 - Enclosed switches
  - 2. Federal specification W-S-865C-Heavy duty switches
- B. Products shall conform all applicable UL standards, including UL98 (standard for safety, enclosed and dead front switches) and shall be UL-labeled.

#### **1.3 ACCEPTABLE MANUFACTURERS**

- A. Provide one of the following manufacturers:
  - 1. General Electric Company
  - 2. Square D Company
  - 3. Siemens
  - 4. Eaton

#### **1.4 SUBMITTALS**

- A. Shop drawings shall include, but not be limited to:
  - 1. Cutsheets of switches with ratings, physical dimensions and all accessories clearly labeled.

#### **1.5 REQUIREMENTS OF REGULATORY AGENCIES**

- A. WORK IN ACCORDANCE WITH:
  - 1. National Electrical Code.
  - 2. Local, municipal, or state codes that have jurisdiction.

### **PART 2 - PRODUCTS**

#### **2.1 GENERAL**

- A. Furnish and install heavy duty type safety switches with the number of switched poles as indicated on the plans and specifications. All safety switches shall be NEMA Heavy Duty Type HD, and Underwriters Laboratories listed.

#### **2.2 MATERIALS AND COMPONENTS**

A. Switch Interior

All switches shall have switch blades that are fully visible in the "OFF" position when the door is open. Switches shall have removable arc suppressor where necessary, to permit easy access to line side lugs. Lugs shall be front removable and UL listed for 60°C and 75°C copper or aluminum cables. All switches blades and contacts shall be plated copper. Adjust fuse block to accept Class J fuses.

B. Switch Mechanism

Switches shall have a quick-make and quick-break operating handle and mechanism, which shall be an integral part of the box, not the cover. Padlocking provisions shall be provided for locking in the "OFF" position with at least three padlocks. Switches shall have a dual cover interlock to prevent unauthorized opening of the switch door when the handle is in the "ON" position, and to prevent closing of the switch mechanism with the door open. A means shall be provided to permit authorized personnel to release the interlock for inspection purposes. Handle position shall indicate if switch is "ON" or "OFF".

C. Neutral

Provide a solid neutral with the safety switch where a neutral is present in the circuit.

D. Ratings

Switches shall be horsepower rated for ac and/or dc as indicated by the plans. The fused switches shall have Class R rejection fuse clips or adjusted for Class J fuses. UL listed short circuit ratings of the switches, when equipped with Class R fuses, shall be 200,000 symmetrical amperes.

E. Enclosures

1. Indoor switches shall be furnished in NEMA 1 enclosures.
2. Outdoor switches, switches located in wet areas or sprinkled areas shall be furnished in NEMA 3R enclosures.
3. Switches installed in wet areas such as cooling tower areas shall be NEMA 4X stainless steel or fiberglass reinforced polyester.
4. Switches installed in kitchens shall be stainless steel.
5. Switches installed in areas of a corrosive nature and subjected to salt air shall be NEMA 4X stainless steel or fiberglass reinforced polyester.

F. Electrical Interlock Contacts

Provide electrical interlock contacts on all disconnect switches serving motors in which remote VFDs are serving the motor. Provide conductors from contacts to the safe circuit inside the VFD. De-energizing the disconnect switch shall signal VFD to stop.

G. Service Entrance

Switch shall be suitable for use as service entrance equipment when installed in accordance with the National Electrical Code.

## PART 3 - EXECUTION

### 3.1 GENERAL

- A. Install safety and disconnect switches, including electrical connections, and fuses in accordance with manufacturer's written instructions, NEC and recognized industry practices.
- B. Location: Install switches within sight of controllers.
- C. Hubs: Provide bolt-on hubs for rainproof or wet area applications.

### 3.2 IDENTIFICATION

- A. Nameplate: Each disconnect switch shall have an engraved bakelite nameplate. Nameplates shall be white with black letters and show equipment served. Nameplates shall be attached with stainless steel screws.

**END OF SECTION**



## **SECTION 26 29 26 - MISCELLANEOUS ELECTRICAL CONTROLS AND WIRING**

### **PART 1 - GENERAL**

#### **1.1 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.

#### **1.2 SCOPE**

- A. Provide the various miscellaneous control devices, wiring and additional branch circuits as required, shown and specified.
- B. The types of miscellaneous control devices and wiring include but not limited to the following.
  - 1. Contactors
  - 2. Relays
  - 3. Photocells
  - 4. Time switches
  - 5. Additional control wiring and safety devices as shown and specified.
  - 6. Connect power from fire alarm relays to starters to shut down air handling units.
- C. WORK SPECIFIED ELSEWHERE:
  - 1. Various control devices, of an electrical nature, for the safe operation and temperature control of the heating, ventilating, air conditioning and plumbing systems provided under Division 23.
  - 2. All control wiring and conduit shall be furnished under Division 23. All power wiring 120 volt or larger shall be provided by Division 26.
  - 3. Refer to building controls specification, Division 23 for scope of work required to be performed by Division 26 (electrical contractor).

#### **1.3 REQUIREMENTS OF REGULATORY AGENCIES**

- A. WORK IN ACCORDANCE WITH:
  - 1. National Electrical Code.
  - 2. Local municipal or state codes that have jurisdiction.

#### **1.4 ACCEPTABLE MANUFACTURERS**

- A. Provide one of the following manufacturers:
  - 1. LIGHTING CONTACTORS AND RELAYS
    - a. General Electric
    - b. Square D Company

- c. Automatic Switch Company
- 2. PHOTOCELLS AND TIME SWITCHES
  - a. Tork, Inc.
  - b. Intermatic time controls
  - c. AMF paragon

## PART 2 - PRODUCTS

### 2.1 MATERIAL

- A. GENERAL: This Section shall outline the basic installation of electric devices, conduit, boxes, fittings, and wiring required for complete interconnection of several systems, this may not reflect every required appurtenance. It does not cover integral parts of mechanical equipment.
- B. CONTACTORS AND RELAYS: Provide control wiring, contactors, and relays with the ampere-rating and number of poles as shown, specified, and required for a complete and functioning system:
  - 1. Rated at 600 volts, 60 hertz.
  - 2. Continuously rated contacts for all types of ballast and tungsten lighting, resistance and motor loads. Contacts shall be sized as scheduled or noted.
  - 3. Shall have totally enclosed, double-break silver-cadmium-oxide power contacts. Auxiliary arcing contacts are not acceptable. Contact inspection and replacement shall be possible without disturbing line or load wiring.
  - 4. The contactor shall have straight-through wiring with all terminals clearly marked.
  - 5. The contactor shall be approved per UL508 and/or CSA, and be designed in accordance with NEMA ICS2-21 1B.
  - 6. They shall be industrial-duty rated for applications to 600 volts maximum.
  - 7. The contactor shall have provisions for factory or field addition of:
    - a. Four (4) N.O. or N.C. auxiliary contacts rated 6 amperes continuous at 600 volts.
    - b. Single or double circuit, N.O. or N.C., 30 or 60 ampere 600 volt power-pole adder.
  - 8. The contactor shall have a NEMA type 1 enclosure unless otherwise noted.
  - 9. Control power to the contactor 120V control circuit shall be provided from the nearest panelboard 120V circuit. If the 120V control power circuit is not shown, provide a control power transformer for 120 volt control power and a 120 volt coil when required for control. Provide primary and secondary fuses on the control power transformer.
  - 10. Electrically Held Lighting - Contactor coils shall be continuously rated and encapsulated. Electrically held contactors are not to be used unless specifically shown on the plans.
  - 11. Mechanically Held Lighting Contactors - Coil-clearing contacts shall be supplied so that the contactor coils shall be energized only during the instance of

operation. Both latch and unlatch coils shall be encapsulated. All contactors shall be mechanically held unless noted otherwise on the plans.

12. Provide 2-wire or 3-wire control modules as required to operate lighting contactors.
  13. Provide hand-off-automatic controls (H-O-A) for each lighting contactor.
  14. Provide relays and contactors to shut down air handling units.
- C. Photocells: Provide a specification grade self contained, weatherproof, photoelectric control that shall be mounted on an FS type weatherproof junction box. The photocell shall:
1. Switch "ON" at dusk and "OFF" at dawn.
  2. Adjustable from 2 to 50 foot candles.
  3. Rated at 2000 watts.
  4. Use 1" diameter cadmium sulphide cell.
  5. Have a 2 minute delay to prevent false switching.
- D. TIME SWITCHES: Provide a 7-day digital time clock with battery back-up feature installed in a NEMA 3R enclosure.
- E. Control wiring shall be not less than #14 AWG type TW, and shall be color coded and labeled with Brady markers throughout. Bundle multiple conductors with Ty-Raps.

### PART 3 - EXECUTION

- 3.1 Install miscellaneous electrical controls and wiring to provide a functioning system.
- 3.2 DIVISION 22, 23, 27 AND 28 MISCELLANEOUS POWER AND CONTROLS
  - A. Install electrical devices not an integral part of system equipment providing conduit, boxes, fittings, wiring, circuit breakers, disconnecting means and other devices.
  - B. Contractor is responsible for providing all line voltage power to devices that require electrical power to operate. Contractor shall terminate line voltage power to termination points. Contractor shall coordinate between all trades to determine sizing and quantities of line voltage circuits to adequately power and control devices. Provide circuits from nearest low voltage panel using spare circuits provided, if device requires power not already available or indicated.
  - C. Provide GFCI receptacle with weather proof cover within 25 feet of all heating, air conditioning and refrigeration equipment.
- 3.3 Install contactor and relays in electrical/mechanical rooms unless otherwise noted.
- 3.4 Install photocells on the roof unless otherwise directed by the architect. Coordinate any roof penetrations with all other trades and shield from other light sources.
- 3.5 Provide miscellaneous connections for signs and other furnished equipment as shown on the Drawings.

**END OF SECTION**

**SECTION 26 32 00**  
**DIESEL ENGINE DRIVEN STANDBY**  
**ENGINE GENERATING SYSTEM**

**PART 1 - GENERAL**

**1.01 GENERAL REQUIREMENTS**

- A. The requirements of the General Conditions and Supplementary Conditions apply to all work herein.
- B. SCOPE
  - 1. Provide a standby electric generating system manufactured by Cummins C150D6D series electric set rated for continuous standby service at 150 kW, 187 kVA at 0.8 power factor, 3-phase, 208Y/120 VAC, 60 cycle. The system shall be a package of new and current equipment consisting of:
    - a. A diesel engine driven electric set installed outdoors to provide emergency electric power.
    - b. Automatic transfer switch(es) to provide automatic starting and stopping of the plant and switching of the load for separate transfer switch section.
    - c. Mounted accessories as specified.
    - d. Control wiring.
    - e. Alternator oversized as shown.

**1.02 PERMITS, TEST INSPECTIONS**

- A. This system shall be completely built, tested and shipped by a manufacturer who has been regularly engaged in the production of such equipment for the past ten years and who has parts and service facilities locally available so there is one source of supply and responsibility. The performance of the electric plant shall be certified by an independent testing laboratory as to the plant's full power rating and voltage and frequency regulation. The complete system shall bear a seal showing that it is prototype test supported.

**1.03 REQUIREMENTS**

- A. Level 1 applications are legally-required emergency systems.
- B. The electric generating system must meet all requirements of NFPA 110 - latest edition including design specification, prototype tests, one-step full-load pickup, and installation acceptance. Engine-generator system to provide source of power for Level 1 applications.

**1.04 STANDARDS**

- A. Equipment shall meet the latest versions of the following codes:
  - 1. N.E.C.
  - 2. NFPA 101, 110, 37, 99, 30
  - 3. IEEE - 446, 587
  - 4. NEMA - MG1, ICS
  - 5. ANSI
  - 6. UL – 1008

7. MIL-STD 461 C – Part 9, IEC 801.2, IEC 801.3, IEC 801.5, IEC 1000-4-2,3,6 – RFI and EMI Performance.
8. UL 2200

#### **1.05 SUBMITTALS**

A. Shop drawings shall include but not be limited to:

1. Catalog cut sheets with all equipment, fuel tank, accessories and devices including all ratings.
2. Interconnection wiring diagrams.
3. Complete bill of materials.
4. Certified performance tests.

B. Operation and Maintenance Data

1. Submit under provisions of Division One.
2. Furnish three copies of the manuals and books listed below in substantial three-ring binders for each unit:
  - a. Operating Instructions: Describe and illustrate all switchgear controls and indicators and engine and general controls. Include instructions for operating transfer switch equipment under normal and emergency conditions when engine generator is running.
  - b. Parts Books: Illustrate and list all assemblies, subassemblies and components, except standard fastening hardware (nuts, bolts, washers, etc.).
  - c. Preventative Maintenance Instructions: Describe the daily, weekly, monthly biannual and annual maintenance requirements and include a complete lubrication chart.
  - d. Routine Test Procedures: Describe procedure for engine, radiator, all electronic and electrical circuits, and the generator.
  - e. Troubleshooting Chart: Describe and list all troubles, probable causes, and suggested remedies.
  - f. Recommended Spare Parts List: List all consumables anticipated to be required during routine maintenance and testing. List special tools, maintenance materials and replacement parts.
  - g. Wiring Diagrams and Schematics: Show function of all electrical components.

#### **1.06 DELIVERY, STORAGE AND HANDLING**

- A. Deliver, store, protect and handle products to site under provisions of Division One.
- B. Accept unit on site on skids. Inspect for damage.
- C. Protect equipment from dirt and moisture by securely wrapping in heavy plastic.

#### **1.07 ACCEPTABLE MANUFACTURERS**

- A. Provide products complying with these specifications and produced by one of the following:
  1. Generator Manufacturers
    - a. Caterpillar
    - b. Cummins

2. Engines
  - a. Cummins
  - b. Caterpillar
3. Generators
  - a. IEC – Baylor
  - b. Kato
  - c. Marathon
  - d. Cummins Power Generation/Onan
4. Automatic Transfer Switches
  - a. ASCO
  - b. Russelectric
  - c. Cummins Power Generation/Onan
  - d. GE/Zenith

#### **1.08 ACCEPTABLE SUPPLIERS**

- A. All equipment provided shall be supplied by an authorized distributor of the manufacturer who has been continuously engaged in the distribution of industrial grade Power System products for a minimum of 15 years. The supplier shall provide initial start-up services, conduct field acceptance testing, and warranty service. The supplier be authorized to perform warranty service on all products provided.
- B. Within 50 mile of the job site, the supplier shall maintain; a minimum of 6 factory trained and qualified field technicians; a proper supply of spare parts for the supplied equipment; a shop with overhaul capabilities; and be able to provide 24 hour, 7 day per week, 365 day per year field service capability.

### **PART 2 - PRODUCTS**

#### **2.01 ENGINE**

- A. The engine shall be radiator cooled, diesel fueled, 4 cycle, 6 cylinder. It shall have a total piston displacement of not less than 543 cubic inches and develop not less than 464 brake horsepower at its operating speed. A radiator air discharge duct flange shall be provided for a connecting duct to allow all heated air and gases to be discharged out of the building or enclosure through one opening. The radiator cooling system shall be rated at 54 degrees C. ambient against an external restriction of 0.5 inch water column. Engine cooling air requirements shall not exceed 21,000 CFM.
- B. The engine shall be of 1-piece cast alloy iron construction with cast alloy iron heads. Valves shall be overhead and free to rotate. Valves shall be hard chrome-cobalt alloy faced with replaceable valve seat inserts of solid chrome-cobalt alloy. The crankshaft shall be forged steel. main bearings between all cylinders. The connecting rods shall be forged steel with connecting rod bearings. Fuel injection system with automatic fuel shut-off, automatic positive head maintained on injectors, and a reusable air element air cleaner; mechanical fuel transfer pump with filters. Provide full-flow, replaceable, oil filter with bypass; oil pressure gauge shall be included.
- C. 120/240 VAC thermostatically controlled water jacket heater system 1500 watts shall be provided. Contractor shall install normal power to the heater.
- D. Provide the following safety shutdown fault devices:
  1. Low oil pressure

2. Over-speed
3. Over-crank
- High temperature (with low water level)

E. Provide the following alarms:

1. Low engine temperature (indicating jacket heater malfunction)
2. Marginally high engine temperature
3. Marginally low oil pressure
4. Low fuel
5. Flashing for control switch in "Stop" position.

The engine shall be equipped with adjustable electronic governor isochronous (speed regulation 0.5 percent, no load to full load) with controls

F. Generator main circuit breaker: set-mounted and wired, UL listed, molded case type with electronic trip unit, (trip units less than 300 amp may be thermo-magnetic) rated at 400 amps, 3 pole, 48 volts. Submittals shall demonstrate that the circuit breaker provides proper protection for the alternator by a comparison of the trip characteristic of the breaker with the thermal damage characteristic of the alternator. Field circuit breakers shall not be acceptable for generator overcurrent protection. (Lugs on breaker shall match "ATS" lugs)

G. Provide 24 volt electrical system and starting shall be a 24 volt electric starter.

## 2.02 ALTERNATOR

- A. A. Rating 175 kW, 215 kVA, at 0.8 power factor 105 degrees Celsius, 480Y/277 volts, 60 Hz at 1800 rpm.
- B. The alternator shall be a single bearing revolving field type, 2/3 pitch, 4-pole and shall be completely brush less. No commutator or commutator brushes shall be allowed. The main alternator and exciter shall be vacuum impregnated. The alternator shall be directly connected to the engine through a rigid coupling to insure permanent alignment. Voltage regulation shall be within plus or minus 1% of rated voltage, from no load to full load. Voltage recovery to rated voltage after acceptance of 100% of rated load in one step shall occur within 10 seconds. Provide a permanent magnet generator (PMG) excitation system. Motor starting capability shall be a minimum of 1000 kVA. Rating for non-linear loads shall not be less than 320 kW at 0.8 power factor. The generator set shall be capable of sustaining a minimum of 90% of rated no load voltage with the specified kVA load at near zero power factor applied to the generator set. The instantaneous voltage dip shall be less than 10% of rated voltage when full load and rated power factor is applied to the alternator. Stable or steady-state operation is defined as operation with terminal voltage remaining constant within plus or minus 1% of rated voltage. Temperature rise shall be within rating as defined by NEMA MG1-22.40. Radio interference reduction shall exceed requirements for general civilian or commercial applications with TIF less than 50 and waveform deviation less than 0.06 line to line.
- C. Provide a 120 volt anti-condensation heater minimum (100 watts) to prevent condensation during non-operating periods. Heater shall be thermostatically controlled, or rated for continuous use for the frame. Provide normal power to the heater.
- D. Overload Rating: Capable of withstanding a three phase load of 300% rated current for 10 seconds, 150% of rated current for 2 minutes and 110% rated current for 60 minutes with field set for normal rated load excitation, and capable of withstanding an overspeed of 125%.
- E. Performance Criteria:
  1. Waveform Deviation: Less than 5%.



2. Crest Factor: 1.41 +/- 0.07.
  3. Form Factor: 1.11 +/-0.05.
  4. Total Harmonic Distortion: 5%.
  5. Single Harmonic Distortion: 3%.
  6. Telephone Interference Factor: 50% maximum.
  7. Dynamic Balance: Less than 1 mil displacement peak to peak.
- F. Enclosure: NEMA MG1, open drip proof.
- G. Neutral Ground: As shown on drawings.

### **2.03 CONTROLS AND INSTRUMENTS**

- A. Provide comprehensive monitoring and control system integral to the Generator Set control to guard the electrical integrity of the alternator and power system. Provide single and 3-phase fault current regulation, so that downstream protective devices have the maximum current available to quickly clear fault conditions, without subjecting the alternator to potentially catastrophic failure conditions. Include provisions to either prevent over voltage due to single phase faults, or to shut down the generator set if line to neutral voltage on any phase exceeds 115% for more than 0.5 seconds. Acceptable methods are a fully rated (100%) 600 volt Circuit Breaker, mounted in the generator enclosure, GE Programmable VersaTrip of size as indicated on drawings with handheld programmer or inherent protection provided by microprocessor-based GenSet AmpSentry protection. Submittals shall demonstrate that the protective device provides proper protection for the alternator by a comparison of the trip characteristic of the breaker with the thermal damage characteristic of the alternator. Field circuit breakers shall not be acceptable for generator overcurrent protection.
- B. An instrument panel mounted on top of the alternator shall contain the following:
1. Run-stop-remote switch
  2. Lighted charge rate ammeter
  3. Lighted oil pressure gauge
  4. Lighted coolant temperature gauge
  5. Remote start-stop terminals
  6. Running time meter
  7. Full A.C. instrument panel (A.C. ammeter, A.C. voltage, phase selector switch, frequency meter, and voltage adjusting rheostat.) All parameters shall have a readout that is not less than 2.5% accuracy.
  8. Red alarm lights shall be provided for each fault and alarm condition.
  9. Two sets of spare terminals shall be provided for customer selected faults.
  10. An Emergency Shutdown contact shall be provided through which customer's push button or other momentary-closing switch contacts shall shutdown the generator set engine.
  11. A fault reset switch contacts shall shutdown the generator set engine.
  12. A fault reset switch shall be provided to clear fault indications and allow restarting of the engine after shutdown faults.
  13. The control design shall be such that the fault indication shall remain until reset. The fault indicator memory shall not be dependent on the presence of either A-C

or D-C voltage and shall retain the fault status memory even through complete removal and replacement of the starting batteries.

14. A battery warning that included load testing the battery on each crank shall be provided.
15. The fault reset function shall operate only when the RUN-STOP-REMOTE switch is in the STOP position.
16. All devices for interconnection and compatibility with digital accuracy and response shall be provided. Digital panels shall comply with electromagnetic interference requirements of Minimum Standard 461C – Part 9, and IEC Standard 801.2, 801.3 and 801.4.indication of voltage level
17. Include a full wave rectified automatic digital voltage regulation system matched and prototype tested by the engine manufacturer with the governing system provided. It shall be immune from misoperation due to load-induced voltage waveform distortion and provide a pulse width modulated output to the alternator exciter. The voltage regulation system shall be equipped with three-phase RMS sensing and shall control buildup of AC generator voltage to provide a linear rise and limit overshoot. The system shall include a torque-matching characteristic, which shall reduce output voltage in proportion to frequency below an adjustable frequency threshold. Torque matching characteristic shall be adjustable for roll-off frequency and rate, and be capable of being curve-matched to the engine torque curve with adjustments in the field.
18. The automatic voltage regulator shall be temperature compensated, solid-state design and include overvoltage and overexcitation protection functions. The voltage regulator shall be equipped with three phase RMS sensing. The regulator shall control buildup of AC generator voltage to provide a linear rise and limit overshoot. Overvoltage protection shall sense the AC generator output voltage and in the event of regulator failure or loss of reference, shut down regulator output on a sustained overvoltage of one (1) second duration. Overexcitation protection shall sense regulator output and shutdown regulator output if overloads exceed ten (10) seconds in duration. Both overvoltage and overexcitation protection shutdowns shall be latched, requiring the AC generator to be stopped for reset.
19. The regulator shall include an under frequency rolloff torque-matching characteristics, which shall reduce output voltage in proportion to frequency below a threshold of 58-59 Hz). The torque-matching characteristic shall include differential rate of frequency change compensation to use maximum available engine torque and provide optimal transient load response. Regulators which use a fixed volts per hertz characteristic are not acceptable.
20. An electronic governor system shall provide automatic isochronous frequency regulation. The governing system dynamic capabilities shall be controlled as a function of engine coolant temperature to provide fast, stable operation at varying engine operating temperature conditions.
21. All analog and digital metering shall be true-RMS indicating, and shall not be disrupted by non-linear load generated waveform distortion.
22. Digital metering set shall indicate generator RMS voltage and current, frequency, output current, output kW, kW-hours, and power factor. Generator output voltage shall be available in line-to-line neutral voltages, and shall display all three phase voltages (line to neutral or line-to-line) simultaneously.

23. An under frequency sensing and protection system shall be provided which causes a shutdown of the generator set if true RMS frequency falls below 90% of rated for more than 20 seconds.
24. The control system provided shall withstand the surge voltage produced by a 70A DC battery charging alternator operating at full load when the battery bank is disconnected. The test shall be successfully completed without tripping protective circuit breakers or blowing fuse protective devices.
25. All switches, lamps and meters shall be oil-tight and dust-tight and the enclosure door shall be gasketed.
26. All switches shall be provided with fully illuminated back-lit labels and all metering shall be individually lighted to allow for easy reading of functions in a completely dark room.
27. The field connections shall be made on permanently labeled terminal blocks, which are designed and tested by the manufacturer of the generator set to be suitable for use without wire termination lugs. Provisions shall be made for future addition of DIN-rail mounted components.
28. Control panel and interconnection enclosures shall be UL508 listed as a unit assembly.
29. Communications:
  - a. Alarm Relay Mode: Provide Form C alarm contacts that can be individually linked to alarm or status outputs from the generator set to external devices.
  - b. Software: Provide software designed to operate on an IBM-compatible PC. Monitor and control the on-site power system either locally or from a remote location through a modem and dedicated telephone line or other communications link.

#### **2.04 ELECTRIC PLANT MOUNTING**

The plant shall be provided with shock or anti-vibration mounts with the plant. Provide Korfund LKD spring-type isolators or type EU pads. Vibration isolation may be integrally a part of the generator set to skid packaged from the set manufacturer. The plants integral base shall have forklift sockets. Battery rack shall be integral part of plant base.

#### **2.05 ACCESSORIES**

- A. All accessories needed for the proper operation of each plant shall be furnished. These shall include but not limited to the following:
  1. Critical rated side inlet silencers with installation attachments for mounting within the set housing, flexible exhaust connection. For sets over 500 kW mount silencer on top of set housing.
  2. Belt driven battery charging alternator.
  3. Lead acid starting batteries
  4. Battery cables.
  5. Fully automatic 10 amp battery charger 120 volt, 325 watts (Onan Series 305-0813-01).
  6. Fuel/water separator.

7. Integral sub base double wall fuel storage tank size with 12 hours gallons capacity. Tank shall have 7 gauge steel walls and installed underneath the generator. The tank shall include the following features:
  - a. Low level switch
  - b. Low level fuel alarm
  - c. Leak detection alarm
  - d. Vent, drain
  - e. Fuel gauge.
  - f. High level switch.
  - g. The tank shall be constructed of corrosion resistant steel and shall be UL 2085 listed. The equipment, as installed, shall meet all local and regional requirements for above ground integral tanks.

## **2.06 REMOTE ALARM ANNUNCIATOR**

Onan no. 541-0814-02 shall be provided for flush mounting at inside locations remote from the generator set located by the fire alarm control panel.

## **2.07 AUTOMATIC TRANSFER SWITCH**

### **A. Scope**

1. Furnish and install open transition automatic transfer switches (ATS) 4 poles, amperage, voltage and withstand current ratings as shown on the plans. Each automatic transfer shall consist of an inherently double throw power transfer switch unit and a control module interconnected to provide complete automatic operation.

### **B. Codes and Standards**

1. UL 1008 - Standard for Automatic Transfer Switches
2. NFPA 70 - National Electrical Code
3. NFPA 99 - Essential Electrical Systems for Health Care Facilities
4. NFPA 110 - Emergency and Standby Power Systems

### **C. Acceptable Manufacturers**

1. Cummins Power Generation/Onan
2. Russelectric
3. GE/Zenith
4. ASCO

### **D. Mechanically Held Transfer Switch**

1. The transfer switch unit shall be electrically operated and mechanically held. The switch shall be mechanically interlocked to ensure only one of two possible positions, normal or emergency.
2. All main contacts shall be silver composition. Switches rated 600 amperes and above shall have segmented, blow-on construction and be protected by separate arcing contacts.
3. Inspection of all contacts shall be possible from the front of the switch without disassembly of operating linkages and without disconnection of power conductors. A manual operating handle shall be provided for maintenance

purposes.

4. Designs utilizing components of molded-case circuit breakers, contactors, or parts thereof are not acceptable.

E. Microprocessor Control Panel

1. The control panel shall direct the operation of the transfer switch. The panel's sensing and logic shall be controlled by a built-in microprocessor.

F. Enclosure

1. The ATS shall be furnished in a NEMA type 1 enclosure.

G. Voltage and Frequency Sensing

1. The voltage of each phase of the normal source shall be monitored, with pickup adjustable from 85% to 100% of nominal and dropout adjustable from 75% to 98% of pickup setting.
2. Single-phase voltage sensing of the emergency source shall be provided, with pickup voltage adjustable from 85% to 100% of nominal and independent frequency sensing with pickup adjustable from 90% to 100% of nominal.

H. Time Delays

1. A time delay shall be provided to override momentary normal source outages and delay all transfer and engine starting signals. Adjustable from 0 to 6 seconds.
2. A time delay shall be provided on transfer to emergency, adjustable from 0 to 5 minutes for controlled timing of transfer of loads to emergency.
3. A time delay shall be provided on retransfer to normal, adjustable from 0 to 30 minutes. Time delay shall be automatically bypassed if emergency source fails and normal source is acceptable.
4. A time delay shall be provided on shutdown of engine generator for cool down, adjustable from 0 to 60 minutes.

I. Additional Features

1. A set of DPDT gold-flashed contacts rated 10 amps, 32 VDC shall be provided for a low-voltage engine start signal.
2. A momentary-type test switch shall be provided to simulate a normal source failure.
3. One set of auxiliary contacts, rated 10 amps, 250 VAC shall be provided.
4. Position indicating lights shall be provided.
5. An in-phase monitor or delayed transition shall be provided to allow decay of magnetic fields in transformers and motors.

J. Withstand and Closing Ratings

1. The ATS shall be UL listed in accordance with UL 1008 and be labeled in accordance with that standard's 3 cycle, long-time ratings. ATS's that are not tested and labeled with 3 cycle (any breaker) ratings and have series, or specific breaker ratings only, are not acceptable unless the switch performance is guaranteed and UL listed for molded case circuit breakers or current limiting fuses. If current limiting fuses are utilized, provide current limiting fuses and disconnect switch mounted in or on Automatic Transfer Switch.

K. The ATS manufacturer shall maintain a national service organization of company-employed  
(250KW) DIESEL ENGINE DRIVEN STANDBY Page 9 of 12  
ENGINE GENERATING SYSTEM

Edinburg Fire Station No. 5

personnel located throughout the contiguous United States. The service center's personnel must be factory trained and must be on call 24 hours a day, 365 days a year.

## **2.08 OUTDOOR WEATHER-PROTECTIVE HOUSING**

- A. Generator set housing shall be provided factory-assembled to generator set base and radiator cowlings. Housing shall provide ample airflow for generator set operation at rated load in the ambient conditions previously specified. The housing shall have hinged side-access doors and rear control door. All doors shall be lockable. All sheet metal shall be primed for corrosion protection and finish painted with the manufacturer's standard color using a two step electro coating paint process, or equal meeting the performance requirements specified below. All surfaces of all metal parts shall be primed and painted. The painting process shall result in a coating that meets the following requirements:
  - 1. Primer thickness, 0.5-2.0 mils. Top coat thickness, 0.8-1.2 mils.
  - 2. Gloss, per ASTM D523-89, 80% plus or minus 5%. Gloss retention after one year shall exceed 50%.
  - 3. Crosshatch adhesion, per ASTM D3359-93, 4B-5B.
  - 4. Impact resistance, per ASTM D2794-93, 120-160 inch-pounds.
  - 5. Salt spray, per ASTM B117-90, 1000+ hours.
  - 6. Humidity, per ASTM D2247-92, 1000+ hours.
  - 7. Water soak, per ASTM D2247-92, 1000+ hours.
- B. Painting of hoses, clamps, wiring harnesses, and other non-metallic service parts shall not be acceptable. Fasteners used shall be corrosion resistant, and designed to minimize marring of the painted surface when removed for normal installation or service work.
- C. A 120 VAC heater with thermostat shall be provided within the generator set control panel to eliminate condensation. Contractor shall provide 120 volt circuit from normal utility source as required.

## **PART 3 - EXECUTION**

### **3.01 INSTALLATION**

- A. Install generator on a 6 inch high reinforced concrete housekeeping pad. Provide blackouts as required.
- B. Provide all power wiring, control wiring, additional contacts and relays required for a complete installation. All conduit shall be in a two hour enclosure. Underground conduits shall be concrete encased. The wall mounted dual rate battery shall be taken off line during the starting of the generator. The generator belt driven alternator shall charge the batteries when the generator is running.
- C. Install generator set and transfer switches in accordance with manufacturer's instructions.
- D. Ground and bond transfer switches under provisions of Division 16.
- E. Provide engraved plastic nameplates under the provisions of Section 16050.

### **3.02 WARRANTY**

- A. The complete standby electric power system, including engine-generator set equipped with set exerciser, running time meter, and automatic transfer switch, shall be warranted for a period of five years from the date of initial start-up. This shall include parts and labor. Multiple warranties for individual components (engine, alternator, controls, etc.) will not be acceptable. Satisfactory warranty documents must be provided. This warranty shall be as detailed in available written documents. In the judgement of the specifying authority, the manufacturer

supplying the warranty for the complete system must have necessary financial strength and technical expertise with all components supplied to provide adequate warranty support. All items of the engine, generator, and controls that are warranted in the first year shall be covered for the full five year term of the warranty.

- B. Extensions of warranty term up to 10 years from start-up and inclusion of comprehensive terms shall be available for one year after start-up.

### 3.03 TESTS

- A. Factory production model tests: Before shipment of the equipment, the generator sets shall be tested under rated load and power factor for performance and proper functioning of control and interfacing circuits. Testing at unity power factor only (resistance banks only) is not acceptable, since kW output is affected by the higher generator efficiency at unity power factor, and the kVAR for motor starting and regulation loads is not correlatable between unity and rated power factor. Other tests shall include:

1. Single step load pickup per NFPA 110.
2. Transient response and steady state governing.
3. Safety Shutdowns.
4. Prototype tests in accordance with NFPA 110 level 1 have been done on a complete and functional set, component level type tests will not substitute for this requirement.

The engineer shall be notified in advance of these test, and shall have the option of witnessing these tests. Certified copies of test results shall be forwarded to the engineer for review.

- B. Field Test After Installation:

1. The complete installation shall be initially started and checked out for operational compliance by factory-trained representative(s) of the engine-generator set and transfer switch manufacturer. The engine lubrication oil and antifreeze, as recommended by the manufacturer for operation under environmental conditions specified shall be provided by the engine-generator set supplier.
2. Upon completion of initial start-up and system checkout, the supplier of the generator set shall perform a field test, with the engineer notified in advance, to demonstrate load carrying capability, stability, voltage, and frequency. The engineer shall be present during the field test.
3. The generator shall be run for four hours continuously with all available facilities emergency load connected to its output; in addition the generator set supplier must provide a portable load bank to supplement any existing load to enable full load testing. Generator shall be run with no load for first 1/2 hour, during first initial run for proper engine break-in. Records shall be maintained throughout this period to record water temperature, oil pressure, ambient air temperature, voltage, current, frequency, kilowatts, and power factor. The above data shall be recorded at 15 minute intervals throughout the test. [Generator set manufacturer shall confirm paralleling equipment is complete and performing properly.] There shall be a 10 minute unloaded run at the conclusion of the test to allow engine to cool before shutdown. Three copies of the field test data shall be furnished to the engineer. The contractor shall make all necessary hook-ups to accomplish field tests and shall furnish all fuel necessary for field test and start-up. The fuel oil tank shall be filled up at the completion of all testing and at the end of the project.

- C. Training

1. Provide training designed for a minimum of four persons, to include:
2. Training in the system operation possible configurations.
3. Training in the maintenance of the system.
4. Minimum of four hours of instruction, but sufficient to cover all items specified.
5. Four sets of instruction materials.
6. Provide videotape (DVD format) of all owner training and instruction.

D. Exercising

1. Schedule for weekly non load running and monthly load running. Coordinate times with owner

**END OF SECTION**



## **SECTION 26 51 00 - LIGHTING FIXTURES**

### **PART 1 - GENERAL**

#### **1.1 SCOPE**

- A. Furnish and install general and emergency lighting fixtures as noted on the drawings. Fixtures shall be completely wired with lamps installed and shall be in perfect operating condition at the time of substantial completion.
- B. The types of lighting fixtures required for this project include:
  - 1. Fluorescent
  - 2. LED

#### **1.2 STANDARDS**

- A. All fixtures shall conform to all applicable UL standards and shall be UL label including damp and wet location ratings. "ETL listed" is an acceptable listing.
- B. All fluorescent ballast shall comply with certified ballast manufacture (CBM) standard and CBM labeled.
- C. NFPA 101
- D. ANSI C82.1
- E. NEMA-LE
- F. IEEE Publication 587 Category "A" (Electronic Ballast)
- G. All LED drivers shall be UL recognized Class 2 per UL1310 or non-Class 2 per UL 1012 as applicable.
- H. All LED drivers shall comply with applicable requirements of the Federal Communications Commission (FCC) rules and regulations, Title 47 CFR part 15, for Non-Consumer Equipment.
- I. All LED drivers shall be RoHS compliant.
- J. TM-21
- K. LM-80
- L. LM-79
- M. L70
- N. DLC
- O. UL 1008

#### **1.3 ACCEPTABLE MANUFACTURERS**

- A. Provide lighting fixtures produced by manufacturers as shown and scheduled.

B. BALLAST:

1. Provide one of the following manufacturers
  - a. Advance Transformer Company
  - b. Universal Lighting Technologies
  - c. Osram Sylvania

C. LAMPS:

1. Provide one of the following manufacturers
  - a. General Electric Company
  - b. Osram Sylvania
  - c. North American Philips

1.4 SUBMITTALS

- A. Shop drawings shall include a brochure with a separate cut sheet for each fixture type arranged in alphabetical order with fixture and all accessories/options clearly labeled. Provide performance data for each fixture. Provide an independent test lab report for each fixture if requested by the Architect/Engineer.
- B. Provide ballast and lamp data brochures indicating which lamp and ballast (if required) will be used in each fixture type.
- C. Furnish air handling and heat removal data for light fixtures specified with these requirements.

1.5 REQUIREMENTS OF REGULATORY AGENCIES

A. WORK IN ACCORDANCE WITH:

1. National Electrical Code.
2. Local, municipal, or state codes that have jurisdiction.
3. UL fire resistance directory.

PART 2 - PRODUCTS

2.1 MATERIALS AND COMPONENTS

A. General:

Provide the size, type and rating of each light fixture shown and scheduled. All light fixtures shall complete with reflectors, lens, trim rings, flanges, lamps, lamp holders, ballast, starters, fuses, wiring, earthquake clips, etc. to provide a complete functioning light fixture.

B. Lighting Fixture Types:

1. Fluorescent Fixtures
  - a. Fixture ballast and lamp holders shall be pre-wired and installed. Fixture shall be equipped with a top access plate with knockouts for conduit entry. Fixture shall also include knockouts at each end plate for conduit entry.
  - b. Provide disconnect switch as required by the latest version of the National Electrical Code.

- c. Fixtures shall be cold roll steel finished using a multistage iron phosphate pretreatment to ensure maximum bonding and rust inhibitor. Finish shall be a lighting grade, baked white enamel finish with a minimum reflectance of 85%.
  - d. Door frame shall be heavy gauge flush white steel or aluminum and hinged from one side and use a positive spring action latch on the other side for latching.
  - e. Any lamp shall be easily replaced without removing another lamp.
  - f. Fixtures installed in continuous rows shall utilize nipples or other accessories such as snap together plug in connectors supplied by the fixture manufacturer.
  - g. Provide battery ballast for emergency light fixtures.
2. LED Fixtures
- a. Fixtures shall be pre-wired with frame-in kit and integral thermal protection required by UL for recessed fixtures. Driver shall be encased in metal-can construction for optimal thermal performance.
  - b. Total fixture lumen output is dependent on the chip, thermal management, driver current and optical system. LED fixtures shall be tested as a complete unit or system. Only DOE recognized CALiPER testing laboratory results shall be utilized.
  - c. LED fixtures shall have integral common mode and differential mode surge protection of 3kV(1.2/50µs, 2 ohm combination wave).
3. Exit signs
- a. Exit signs shall meet all federal, state and local codes.
  - b. Provide fire alarm interface relay when required to flash exit signs.
  - c. Provide battery packs for emergency operation when not connected to emergency generator power.

## 2.2 BALLASTS AND DRIVERS - COORDINATE WITH LIGHT FIXTURE SCHEDULE

### A. Fluorescent

- 1. Program start electronic ballast shall be high power factor 98% minimum), operate lamp at 40 KHz, less than 10% total harmonic content, normal ballast factor 50,000 switching cycles, universal voltage, crest factor less than 1.7, multi-lamp, class "P" thermally protected, sound "A" rating, encased and potted and 0°F minimum starting temperature. Provide 5 year warranty parts and labor. Ballast shall be Osram Sylvania PSN Series or approved equal by Universal Lighting Technologies or Advance Transformer Company.
- 2. All outdoor ballast unless otherwise noted shall be high power factor, rapid start, class P thermally protected, encased and potted, sound rating B and a 0°F temperature rating. Ballast shall be CBM certified by an ETL and UL approved.
- 3. Provide suitable dimming ballast where indicated.
- 4. Compact fluorescent ballast shall be electronic, shall have circuitry designed to shut down the system reliably and safely when lamps have reached their end of life, high power factor, sound rating "A" and UL approved. Provide 5-year warranty, parts and labor.

### B. LED

- 1. Driver manufacturer shall have a 10-year history producing electronic drivers for

the North American market.

2. Driver shall carry a five year limited warranty from date of manufacture against defects in material or workmanship (including replacement) for operation at a maximum case temperature of 80 degrees Celsius.
3. Drivers shall not contain any Polychlorinated Biphenyl (PCB).
4. Provide driver with integral color-coded leads.
5. Driver shall operate from 50/60 Hz input source of 120 Volt through 277 Volt or 347 Volt through 480 Volt with sustained variations of +/- 10% (voltage) with no damage to the driver.
6. Driver output shall be regulated to +/- 5% across published load range. And shall have a power factor greater than .90 for primary application to 50% of full load rating with an input current Total Harmonic Distortion (THD) of less than 20% to 50% of full load rating.
7. Provide drivers with a Class A sound rating.
8. Provide LED drivers for outdoor fixtures with a minimum operating temperature of -40 degrees Celsius (-40 F). Provide LED drivers for indoor fixtures with a minimum operating temperature of -20 degrees Celsius (-2F).
9. Drivers shall tolerate sustained open circuit and short circuit output conditions without fail and auto-resetting without need for external fuses or trip devices.
10. Driver output ripple current shall be less than 15% measured peak-to-average, with ripple frequency being greater than 100Hz.
11. Driver performance requirements shall be met when operated to 50% of full load rating.
12. Driver shall have integral thermal foldback to reduce driver power above rated case temperature to protect the driver if temperatures reach unacceptable levels.
13. Drivers shall comply with NEMA 410 for in-rush current limits.
14. Dimmable drivers shall be controlled by a Class 2 low voltage 0-10VDC controller with dimming range controlled between 1 and 8VDC with source current 150μA.

## 2.3 LAMPS – COORDINATE WITH LIGHT FIXTURE SCHEDULE

- A. F032T8 fluorescent lamps shall be 24,000 hours 4,000°K, 32 watt and low mercury. Life rating is based on 3 hours/start.
- B. Extended life F032 T8 Fluorescent lamps shall be 40,000 hours 4,000°K, 32 watt and low mercury. Life rating is based on 3 hours/start using programmed start ballast.
- C. All incandescent lamps shall be inside frosted, extended life rated for 2500 hours unless otherwise noted. 130 volt lamps may be used to provide extended life.
- D. All HID lamps shall be base up, base down, horizontal, or universal burn as indicated or specified.

- E. All compact fluorescent (T5 and smaller diameter) lamps shall be 3500°K and 4-pin

#### 2.4 EMERGENCY FLUORESCENT BATTERY BALLAST

- A. Provide Bodine #B50 emergency battery ballast for emergency light fixtures using T8 or T12 lamps in 9 or 10 foot ceiling.
- B. Provide Bodine #B30 Emergency Ballast for emergency light fixtures using T8 lamps in ceiling heights greater than 12 feet.
- C. Provide Bodine #B84C emergency battery ballast for emergency light fixtures using compact fluorescent lamps.
- D. Provide unswitched hot leg. Hot leg shall originate from the same branch circuit as required in NEC article 700.12 (F).

#### 2.4 BRANCH CIRCUIT EMERGENCY TRANSFER SWITCH (BCELTs)

- A. Provide 20 amp, 120-277 Volt, UL1008 listed Branch Circuit Emergency Transfer Switch to control emergency light fixtures ~~served by a generator~~ transferring from normal to emergency branch circuits. Provide Bodine GTD 20A or ETC. SC 1008. UL 924 Devices are not acceptable.

#### 2.5 AUTOMATIC LOAD CONTROL RELAY (ALCR)

- A. Provide 3 amp, 120-277 volt UL 924 listed. Relay to bypass switching controlling emergency branch circuit light fixtures Provide Bodine GTD or Wattstopper ELCLI.

#### 2.6 SPARE LAMPS

- B. Provide 5% spare lamps, minimum of 3 of each type.
- C. Ship lamps to the Owner in original cartons (loose lamps are not acceptable).

### PART 3 - EXECUTION

#### 3.1 INSTALLATIONS

##### A. General

1. Install the type of light fixture where shown and indicated in accordance with manufacturer's written instructions.
2. Provide earthquake clips on all recessed lay-in light fixtures as required by building code.
3. Adjust all adjustable light fixtures, as directed by the Architect.
4. Provide safety chains and wire guards for light fixtures located in gymnasium, multi-purpose rooms, play areas, etc.

##### B. Coordination

1. The contractor shall verify the type of fixtures with the ceiling types as indicated on

the drawings. Any discrepancies shall immediately be brought to the architect's attention before the contractor places his order and accepts delivery. Fixtures shall fit exact in the type of ceiling scheduled. Provide plaster frames, trim rings and other accessories required for a correct fit.

2. Provide supports attached to structural member to support fixtures when the ceiling system cannot maintain support. Provide separate supports for all recessed ceiling mounted HID fixtures.
3. Refer to architectural reflected ceiling plan for the exact location of all light fixtures. Notify the architect for any discrepancies or conflicts with structural, architectural, mechanical piping or ductwork before installation.

#### C. Mounting

1. Provide support channels to support outlet boxes used support surface mounted light fixtures such as exit signs or downlights.
2. Pendant or surface mounted fixture shall be provided with required mounting devices and accessories, including hickey and stud-extensions, ball-aligners, canopies and stems. Locations of fixtures in mechanical areas shall be coordinated with mechanical contractor. Mounting stems of pendant fixtures shall be of the correct length to uniformly maintain the fixture heights shown on the drawings or established in the field. The allowable variation tolerance in mounting individual fixtures shall not exceed 1/4 inch and shall not vary more than 1/2 inch from the floor mounting height shown on the Drawings. Fixtures hung in continuous runs shall be installed absolutely level and in line with each other. Hanging devices shall comply with Code requirements. Fixtures shall employ single - not twin - stem hangers unless otherwise noted.
3. All structure mounted fixtures (i.e. bracket mounted, pipe mounted and surface mounted) shall be provided with cables of suitable size and weight to support the weight of the fixture. Cables shall be fastened around or fastened to the housing of the fixture. On pendant fixtures, one safety cable of suitable size and weight to support the weight of the fixture assembly shall connect the top of the pendant to the supporting structure by means of welding or bolting, and one safety cable shall connect the housing of the fixture to the bottom of the pendant. Where more than one pendant per fixture occurs, only one pendant must be cabled. Track fixtures for pendant mounted track shall also be supplied with clip-on safety cables of suitable size and weight to support the weight of the fixture.
4. Provide secondary support wires from all four (4) corners of the lay-in fixtures to the structure above. Do not support fixtures from ceiling grid wire supports, piping, conduit, side walls, or mechanical equipment. Ceiling specifications do not supersede this requirement.

#### D. Electrical Connection

1. All light fixtures shall be connected from a branch circuit junction box using 1/2" flexible metal conduit or MC cable fixture pigtails not exceeding 8'- 0". Provide #12 AWG conductors. All fixtures must be grounded by using a grounding conductor. Fixture to fixture wiring of fixtures installed in an accessible ceiling is not permitted. Fixture whips shall not lay-on ceiling tile or grid. Provide caddy clips to provide additional support.

#### E. Fire Rated Ceiling

1. Provide fire rated canopy or enclosure for all fixtures recessed in a fire rated ceiling. The fire rated canopy or enclosure shall be as required by the UL design number listed in the UL fire resistance directory. Refer to architectural drawing for the UL design number. Coordinate with ceiling installer and manufacturer.

F. Air Handling Fixtures

1. Install all air handling light fixtures with return air slot in the open position, if it is to be as an air handling fixture. Coordinate with mechanical contractor.

3.2 FINAL INSPECTION

- A. Remove all plastic and protective coating from all fixtures. Fixtures shall be thoroughly cleaned. Replace any damaged fixture or fixture parts including reflectors, louvers, lens and metal parts that show signs of corrosion.
- B. All final incandescent lamps used during construction shall be replaced with new lamps. Replace all other defective ballast, lamps or discolored lamps, showing signs of excessive usage.
- C. Demonstrate proper operation of all fixtures and controls.

**END OF SECTION**

**SECTION 27 00 00**  
**COMMUNICATIONS**

**PART 1 - GENERAL**

**1.1 SUMMARY**

- A. This section includes general design requirements, administration topics, and installation for communications systems.

**1.2 SYSTEM DESCRIPTION**

- A. The objective of this project is to provide a complete communications cabling infrastructure system installation including, but not limited to: fiber backbone, riser system, horizontal data and voice cabling with attendant terminations, mounting equipment, cable pathway and management systems, testing and other items/materials, as specified in drawings, these specifications, and contract documents.
- B. Related Sections
1. Section 270526 Grounding and Bonding for Communications Systems
  2. Section 270528 Pathways for Communications
  3. Section 271100 Communications Equipment Room Fittings
  4. Section 271500 Communications Horizontal Cabling

**1.3 SCOPE OF WORK**

- A. This section establishes an infrastructure to be used as signal pathways for communications systems, but is not limited to the following:
1. Comply with all Project Contract documents and the following requirements for a complete project installation.
  2. Provide a structured cabling system as described hereafter that includes, but is not limited to, supplying, installing and testing of: backbone cabling, riser cabling; data and voice horizontal cabling, cable connectors, communications outlets and terminations, and equipment racks/cabinets for networking hardware and patch panels.
  3. Furnish all labor, materials, tools, equipment and services for the installation described herein.
  4. Follow industry standard installation procedures for communications cable to assure that the mechanical and electrical transmission characteristics of this cable plant and equipment are maintained.
- B. Work of this section covers complete installation of permanent links for a data and voice communications networks utilizing copper and fiber transmission media that includes, but is not limited to the following:
1. Provide, install, terminate, test, and document all fiber and copper backbone cables, riser cables, and horizontal cables.
  2. Provide and install all termination devices such as, but not limited to, modular patch panels, termination blocks, information outlets (jacks and plates), phone jacks, fiber distribution panels, bulkheads, connectors, and fiber fan out kits. Document all termination devices with proper labeling.
  3. Provide in quantities specified, interconnect components such as, but not limited to, fiber patch cables, copper patch cords, and station cables.



4. Provide and install specified Telecommunication Room equipment such as, but not limited to, racks, cabinets, horizontal and vertical cable support devices, cable trays and cable runway, and required mounting brackets/hardware.
  5. Provide and install UL-approved firestopping systems in all communication pass-through locations of rated ceiling, wall or floor penetrations involving, conduits, cable, and cable trays in coordination with General Contractor.
  6. Provide and install grounding and bonding connection to the bus (TGMB/TGB) provided by Division 26.
  7. Provide and install all appropriate consumable items required to complete the installation.
  8. Coordination with other trades.
  9. Provide complete documentation and demonstration of work.
  10. Provide indexed and organized complete Test Results of all copper and fiber cable and their components.
  11. Provide Submittals as outlined below.
  12. Provide a Manufacturer's Extended Product Warranty and System Assurance Warranty for this wiring system.
  13. Conduct a final document handover meeting with client, consultant, and PM to review, discuss and educate the Owner on the final product, test results, and As-Built Drawings.
- C. Changes to the Scope of Work
1. Owner changes to the scope of work shall be in writing.
  2. Change orders shall be submitted to the Owner/Project Manager complete with price breakdown and description for approval before any work is done.
  3. The Contractor shall respond to these changes with a complete material list, including pricing, labor, and taxes in writing to be presented to the Owner for approval.
  4. The Contractor shall not proceed with additional scope of work without signed approval by the Owner. Owner will not pay for additional work performed by the Contractor without written/signed approval of these changes.
  5. Contractor will attach a copy of the signed change order with billing information.

#### **1.4 PRODUCTS AND WORK BY OTHERS (NIC)**

- A. The Owner may separately procure and/or provide certain equipment and component that will be installed during the course of project. Such items may not be indicated in the documents.
- B. Contractor shall cooperate with the Owner and Owner's suppliers when considering:
1. The provision and installation of phone systems, related system equipment/software, and employee station equipment/software.
  2. The provision and installation of multi-port routers, switches, and other Layer 2 / Layer 3 networking components in communications rooms.
  3. The provision and installation of Uninterruptable Power Source (UPS) devices in communications rooms.
  4. Communications grounding busbars and grounding wires connecting to the main building electrode system.
  5. Dedicated power panels, ground busbars, circuits, and utility outlets.
  6. The installation and finishing of plywood backboards.

7. Building mechanical ductwork, cooling/heating system (HVAC), and environmental control sensors.
8. Communication pathway devices such as, but not limited to, cable tray and flex-tray in corridors, office spaces and open areas, outlet boxes and stub-ups, conduits, conduit sleeves, and penetrations in walls and floors.

## **1.5 SUBSTITUTION PROCEDURES**

- A. Substitution may be considered when a product becomes unavailable through no fault of the Contractor. An alternate product must be equal to or exceed specified requirements. The material substituted shall not void, alter or change manufacturers' structured cabling system warranty.
- B. Document substitution requests with complete data substantiating compliance of proposed substitution with Contract Documents. Include in each request for substitution:
  1. Product identification, manufacturer's name and address
  2. Product Data:
    - a) Description, performance and test data, reference standards, finishes and colors.
    - b) Samples: Finishes.
    - c) Complete and accurate drawings indicating construction revisions required (if any) to accommodate substitutions.
    - d) Data relating to changes required in construction schedule.
    - e) Cost comparison between specified and proposed substitution.
- C. Substitutions will not be considered when they are indicated or implied on shop drawing or product data submittals, without separate written request, or when acceptance will require revision to the Contract Documents.
- D. The Owner will be the final judge of acceptability, with review by DBR and the distribution of the acceptance by the Architect.
- E. No substitute shall be ordered, installed or utilized without the Architect's prior written verification of acceptance from the Owner.

## **1.6 REFERENCES AND RELATED DOCUMENTS**

- A. Drawings and General provisions of the contract, including Uniform General Conditions, Supplementary General Conditions, Architectural plans and specifications, requirements of Division 1, Electrical, Mechanical, Plumbing, Audio-Visual, Security and Communications specifications and plans, and the publications listed below apply to the Communications section, are incorporated into this specification by reference, and shall be considered a part of this section.
- B. Reference to codes, rules, regulations, standards, manufacturer's instructions, or requirements of regulatory agencies shall mean reference to the latest printed edition of each in effect at the date of contract.
- C. The Contractor shall read all sections in their entirety and apply them as appropriate for work in this section.
- D. Conflicts
  1. Drawings and specifications are to be used in conjunction with one another and to supplement one another.
  2. In general, the specifications determine the nature and quality of the materials and tests, and the drawings establish the quantities, details, and give characteristics of performance that should be adhered to during the installation of the communications system components.

3. If there is an apparent conflict between the drawings and specifications, or between specification sections, the items with the greater quantity and/or quality shall be estimated and installed.
  4. Clarification with the Owner and/or DBR about these items shall be made in writing prior to procurement and installation.
- E. Codes and Standards
1. American National Standards Institute/Telecommunications Industry Association (ANSI/TIA)
    - a) ANSI/TIA-568-C.0 "Generic Telecommunications Cabling for Customer Premises"
    - b) ANSI/TIA-568-C.1 "Commercial Building Telecommunications Cabling Standard"
    - c) ANSI/TIA-568-C.2 "Balanced Twisted-Pair Telecommunication Cabling and Components Standard"
    - d) ANSI/TIA-568-C.3 "Optical Fiber Cabling Components Standard"
    - e) ANSI/TIA-568-C.4 "Broadband Coaxial Cabling and Components Standard"
    - f) ANSI/TIA-569-C "Telecommunications Pathways and Spaces"
    - g) ANSI/TIA-606-B "Administration Standard for Commercial Telecommunications Infrastructure"
    - h) ANSI/TIA-607-B "Commercial Building Grounding (Earthing) and Bonding Requirements for Telecommunications"
    - i) ANSI/TIA-758-B "Customer-Owned Outside Plant Telecommunications Infrastructure Standard"
    - j) ANSI/TIA-862-A "Building Automation Systems Cabling Standard"
    - k) ANSI/TIA-942-A: "Telecommunications Infrastructure Standard for Data Centers"
    - l) ANSI/TIA-1152: "Requirements for Field Test Instruments and Measurements for Balanced Twisted-Pair Cabling"
  2. American National Standards Institute (ANSI)
    - a) ANSI C80.1 Electrical rigid steel conduit (ersc)
  3. American Society for Testing Materials (ASTM)
    - a) ASTM A123/A123M-13 Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
    - b) ASTM A510/A510M-13 Standard Specification for General Requirements for Wire Rods and Coarse Round Wire, Carbon Steel, and Alloy Steel
    - c) ASTM A653/A653M-13 Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
    - d) ASTM B3-13 Standard Specification for Soft or Annealed Copper Wire
    - e) ASTM B8-11 Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
    - f) ASTM B33-10 Standard Specification for Tin-Coated Soft or Annealed Copper Wire for Electrical Purposes
    - g) ASTM B633-13 Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
    - h) ASTM B653/B653M-11 Standard Specification for Seamless and Welded Zirconium and Zirconium Alloy Welding Fittings
  4. BICSI

- a) BICSI Outside Plant Design Reference Manual
  - b) BICSI Telecommunications Distribution Methods Manual (TDMM)
- 5. Federal Specifications (FS)
  - a) FS W-C-58C Conduit Outlet Boxes, Bodies Aluminum and Malleable Iron
  - b) FS W-C-1094 Conduit and Conduit Fittings Plastic, Rigid
  - c) FS WW-C-566C Flexible Metal Conduit
  - d) FS WW-C-581D Coatings on Steel Conduit
- 6. Institute of Electrical and Electronic Engineers (IEEE)
  - a) IEEE 142-1991 Recommended Practice for Grounding of Industrial and Commercial Power Systems
  - b) IEEE 1100-2005 IEEE Recommended Practice for Powering and Grounding Electronic Equipment
- 7. National Electrical Code (NEC)
  - a) NEC Article 250 - Grounding and Bonding
  - b) NEC Chapter 8 - Communications Systems
- 8. National Electrical Manufacturers Association (NEMA)
  - a) NEMA RN1 Polyvinyl Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit
  - b) NEMA TC2 Electrical Polyvinyl Chloride (PVC) Tubing and Conduit
  - c) NEMA TC3 Polyvinyl Chloride (PVC) Fittings for Use with Rigid PVC Conduit and Tubing
  - d) NEMA VE 1 - Metal Cable Tray Systems
  - e) NEMA VE 2 - Cable Tray Installation Guidelines
- 9. Underwriters' Laboratories (UL)
  - a) UL Cable Certification and Follow-Up Program
  - b) UL 6: Electrical Rigid Metal Conduit - Steel
  - c) UL 83: Thermoplastic-Insulated Wires and Cables
  - d) UL 467: Grounding and Bonding Equipment
  - e) UL 514B: Conduit, Tubing, and Cable Fittings
  - f) UL 651: Standard for Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
  - g) UL 651A: Schedule 40 and 80 High Density Polyethylene (HDPE) Conduit
  - h) UL 1666: Standard for Test for Flame Propagation Height of Electrical and Optical-Fiber Cables Installed Vertically in Shafts
- 10. Local, county, state and federal regulations and codes in effect as of date of installation.
- 11. Equipment of foreign manufacture must meet U.S. codes and standards.
  - a) It shall be indicated in the proposal the components that may be of foreign manufacture, if any, and the country of origin.

## 1.7 QUALITY ASSURANCE

- A. Communications Contractor shall have a complete working knowledge of low voltage communications cabling applications such as, but not limited to data, voice and video network systems.

- B. Communications Contractor shall have installed similar-sized systems in at least ten (10) other projects in the last five (5) years prior to this bid and be regularly engaged in the business of installation of the types of systems specified in this document.
- C. Communications Contractor and individual installation crew members shall be experienced and qualified to perform the work specified herein at time of bid submission. All onsite supervision personnel that will be assigned to this project shall be listed in the Pre-Installation Submittal.
  - 1. 80% shall have a minimum of three (3) years of experience in the installation of the types of systems, equipment, and cables specified in this document prior to this bid.
  - 2. All installation team members must demonstrate knowledge and compliance with all applicable methods, standards, and codes.
  - 3. All members of the installation team shall be certified by the Structured Cabling System Assurance Warranty provider as having completed the necessary training to complete their part of the installation and capable of an installation that falls under manufacturer's guidelines necessary to obtain the Manufacturer's System Assurance Warranty.
  - 4. Any personnel substitutions shall be noted in writing to the Owner.
- D. A BICSI RCDD shall supervise and approve all on-site work as a recognized member of the Contractor's installation team.
- E. Refer also to General Conditions.

## 1.8 CONTRACTOR REQUIREMENTS

- A. In order to accomplish the conditions of this agreement, the Contractor shall perform the specific duties listed herein.
- B. Contractor shall provide and pay for all labor, supervision, tools, equipment, test equipment, tests and services to provide and install a complete communications cabling infrastructure system. Pay all required sales, gross receipts, and other taxes.
- C. Insurance
  - 1. The Contractor shall procure, submit for review, and maintain for the duration of this agreement, insurance against claims for injuries to persons or damages to property which may arise from, or in connection with, the performance of work hereunder by the Contractor, his agents, representatives, employees or subcontractor. The Contractor shall pay the cost of such insurance.
  - 2. The Owner, its directors, officers, representatives, agents and employees, respectively, shall have no responsibility to the Contractor with respect to any insurance in accordance with the provisions set forth herein.
- D. Regulatory Requirements
  - 1. Communications Contractor shall supply all city, county, and state telecommunication cabling permits required by Authority Having Jurisdiction (AHJ).
  - 2. Communications Contractor shall be licensed and/or bonded as required for telecommunications/low voltage cabling systems.
- E. Privacy and Confidentiality
  - 1. The Contractor will respect and protect the privacy and confidentiality of Owner, its employees, processes, products, and intellectual property to extent necessary, consistent with the legal responsibilities of the Owner policies.
  - 2. Contractors shall sign a non-disclosure agreement and abide by the requirements to keep confidential all information concerning bid documents and this project.

- F. Use of Subcontractors
  - 1. Successful bidder shall inform the Owner's contact and General Contractor in writing about the intention to use Subcontractors and the scope of work for which they are being hired.
  - 2. The Owner or Owner's designated contact must approve the use of Subcontractors in writing prior to the Subcontractor's hiring and start of any work.
- G. The Contractor's designated Project Manager will be recognized as the single point of contact. The Project manager shall oversee all work performed to ensure compliance with specifications as outlined in bid documents (which includes all specifications, references, and drawings) to ensure a quality installation and attend project meetings with the telecommunication consultant, the Owner and others.
- H. Coordination
  - 1. Coordinate installation work with other trades (examples include ceiling grid contractors, HVAC and sheet metal contractors, etc.) to resolve procedures and installation placement for cable trays and cable bundle pathways.
  - 2. The goal of this coordination will be to establish priority pathways for critical data/voice network cable infrastructure, materials, associated hardware, as well as mitigate delays to the project and to allow service access for communications and HVAC components.
  - 3. Exchange information and agree on details of equipment arrangements and installation interfaces.
  - 4. Coordinate with electrical contractors and plan for the pathway routes used communications cabling to minimize cable lengths. Report any potential over distance cable runs for approval before pulling the cables.
  - 5. Record agreements with other trades and distribute record to other participants, Owner and telecommunication consultant.

## **1.9 PRE-INSTALLATION MEETINGS**

- A. Communications Contractor shall attend and/or arrange a scheduled pre-installation conference prior to beginning any work of this section. This venue is to ask and clarify questions in writing with consultant and/or project manager/Owner representative.
- B. Agenda
  - 1. Safety
  - 2. Work to be performed
  - 3. Scheduling
  - 4. Coordination
  - 5. Other topics as necessary
- C. Attendance
  - 1. Communications project manager/supervisor shall attend meetings arranged by General Contractor, Owner's representatives, and other parties affected by work of this document.
  - 2. All individuals who will serve in an on-site supervisory capacity, including project managers, site supervisors, and lead installers, shall be required to attend the pre-installation conference. Individuals who do not attend the conference will not be permitted to supervise the installation and testing of communications cables on the project.

## **1.10 CONTRACT ADMINISTRATION**

- A. DBR Inc may perform site visits and provide job field reports upon inspection of Contractor's installation, materials, supporting hardware, coordination with other trades and progress to schedule to the client.
- B. Job Field Report outline:
  - 1. General: The general installation progress in relation to scheduled work made by the Contractor up to that date.
  - 2. Deficiencies and/or Items of Note: Documents observations of the cable installation that may require corrective action by the Contractor.

#### **1.11 POST INSTALLATION MEETINGS**

- A. At the time of substantial completion the contractor shall call and arrange for a post installation meeting to present and review all submittal documents to include but not be limited to As-Built Drawings, Test reports, Warranty paperwork, etc.
- B. Attendees shall include
  - 1. Communications Contractor
  - 2. Project Manager/Owner Representative
  - 3. General Contractor
  - 4. DBR Inc.
  - 5. Other trades that the GC deems appropriate.
- C. At this meeting the Communications Contractor shall present and explain all documentation.
- D. Any discrepancies or deviations noted by and agreed to by participants shall be remedied by the Communications Contractor and resubmitted within one (1) week of the meeting.

#### **1.12 DELIVERY, STORAGE, AND HANDLING**

- A. Coordination with delivery companies, drivers, site address, and contact person(s) will be the responsibility of the Contractor.
- B. Communications Contractor requirements:
  - 1. Be responsible for prompt material deliveries to meet contracted completion date.
  - 2. Coordinate deliveries and submittals with the General Contractor to ensure a timely installation.
  - 3. No equipment materials shall be delivered to the job site more than three weeks prior to the commencement of its installation.
  - 4. Equipment shall be delivered in original packages with labels intact and identification clearly marked.
  - 5. Equipment shall not be damaged in any way and shall comply with manufacturer's operating specifications.
  - 6. Equipment and components shall be protected from the weather, humidity, temperature variations, dirt, dust, or other contaminants.
  - 7. Equipment damaged prior to system acceptance shall be replaced at no cost to the Owner.
  - 8. Contractor shall be responsible for all handling and control of equipment. Contractor is liable for any material loss due to delivery and storage problems.
- C. Owner/General Contractor shall provide the security requirements for Contractor to follow.

#### **1.13 PROJECT/SITE CONDITIONS**

- A. For all environmental recommendations, refer to master Architectural section.

- B. For all security recommendations, refer to related consultant sections.
- C. Contractor shall provide daily a clean work environment that is free from trash/rubbish accumulated during and after cabling installation.
- D. Contractor shall keep all liquids (drinks, sodas, etc.) away from finished spaces. If any liquid or other detriment (cuts, soils, stains, etc.) damages any finishes, Contractor shall provide professional services to clean or repair scratched/soiled finishes, at Contractor's expense.
- E. Damage by Communications Contractor to the work of others will be remedied at the Contractor's expense in a timely manner.

#### **1.14 WARRANTY**

- A. The Contractor shall be a certified Manufacturer's Value Added Reseller (VAR) and/or Authorized Installer and provide an end-to-end product warranty, adhere to the industry standard engineering, installation and testing procedures and utilize the authorized manufacturer components and distribution channels in provisioning this project.
- B. Contractor shall coordinate with manufacturer for warranty paperwork and procedures prior to the start of the project.
- C. Contractor shall provide a minimum one (1) year warranty on installation and workmanship PLUS an Extended Product Warranty and System Assurance Warranty for this wiring system and shall commit to make available local support for the product and system during the Warranty period.
  - 1. The Extended Product Warranty shall apply to all passive structured cabling system components and shall cover the replacement or repair of defective products and labor for the replacement or repair of such defective products for a minimum of one (1) year.
  - 2. The System Assurance Warranty provides a complete system and product warranty that will be extended to the end-user, ensuring the structured cabling system will be free of defects in materials and workmanship, will meet or exceed applicable performance requirements defined in the contract documents, and support all current and future network applications for a minimum of twenty (20) years.
- D. System Certification: Upon successful completion of the installation and subsequent inspection, the customer shall be provided with a numbered certificate, from the manufacturer, registering the installation.

#### **1.15 PAYMENT**

- A. Refer to the General Contractor contract documents and/or master specifications issued by Architect for project and cost payment details.

#### **1.16 SUBMITTALS**

- A. Refer to Requirements of Division 1
- B. Refer to Sections 271300 and 271500
- C. The Communications Contractor shall not perform any portion of the work requiring submittal and review of shop drawings, product data, or samples until Owner has approved the respective submittal in writing. Such work shall be in accordance with approved submittals.
- D. Pre-Installation Submittal Requirements
  - 1. Communications Contractor shall provide certificates for the appropriate insurance coverage as defined in contract documents.



2. City, county, and/or state telecommunication cabling permits as required by Authority Having Jurisdiction (AHJ).
3. Executed non-disclosure agreement.
4. Appoint a Project Manager and provide the name and contact information.
5. Shop Drawings
  - a) Communications Contractor shall submit, for approval, floor plans that identify all device locations, cable routes and quantities, cable types, riser locations, and references to installation details and diagrams.
    - 1) Communication Contractor shall notify Owner of cable routes exceeding standardized lengths.
  - b) Communications Contractor shall submit, for approval, diagrams that show room layouts, rack layouts (including elevations), riser layouts, etc.
  - c) The Contractor shall make any corrections as required by the consultant team and submit revised shop drawings to the team for approval.
  - d) Approval by the Consultant of such drawings or schedules shall not relieve the Contractor from responsibility for deviations from the drawings or specifications, nor shall it relieve the Contractor from responsibility for errors of any sort in shop drawings or schedules. Requests to deviate shall be submitted in writing to the Architect.
6. Product Data Cut-sheets
  - a) Communications Contractor shall submit catalogue cut-sheets that include manufacturer, trade name, and complete model number for each product specified. Model number shall be handwritten and/or highlighted to indicate exact selection.
  - b) Communications Contractor shall identify applicable specification section reference for each product performance for each component specified for approval prior to purchase and installation.
7. Warranty
  - a) The Communications Contractor shall submit appropriate documentation from the certifying manufacturer showing the project is registered and qualified for the System Assurance Warranty.
  - b) All subsequent work shall be in accordance with approved submittals. The Communications Contractor shall not perform any portion of the work requiring approval of the System Assurance Warranty manufacturer's warranty registration qualification procedures that would disqualify any part or all of the wiring system from that warranty qualification.
8. Qualifications
  - a) Communications Contractor shall submit a list of the Contractor's previous projects that demonstrate qualification for this project. This list shall include, but not be limited to:
    - 1) At least ten (10) other projects in the last five (5) years
    - 2) Name and location of project
    - 3) Project contacts, email addresses, and phone numbers
    - 4) Total square footage
    - 5) Total number of cables/drops
    - 6) Types of media

- b) Communications Contractor shall submit an up-to-date and valid statement of qualifications for those assigned to perform the work specified herein at time of bid submission.
      - 1) Communications Contractor Employees
      - 2) Subcontractors
    - c) Manufacturer certifications for Contractor and installers.
  - 9. Cable Testing Plan
    - a) The Contractor shall provide a complete and detailed test plan for approval of the cabling system specified herein, including a complete list of test equipment for copper and fiber components and accessories prior to beginning cable testing.
    - b) The following minimal items shall be submitted for review:
      - 1) A testing plan that clearly describes procedures and methods.
      - 2) Product data for test equipment.
      - 3) Certifications and qualifications of all persons conducting the testing.
      - 4) Calibration certificates indicating that equipment calibration meets National Institute of Standards and Technology (NIST) standards and has been calibrated at least once in the previous year of the testing date.
      - 5) Examples of test reports, including all graphs, tables, and charts necessary for display of testing results.
  - 10. Samples
    - a) For workstation outlet connectors, jack assemblies, housings and faceplates for color selection and evaluation of technical specifications and requirements. Confirm with Architect, interior designer, and Owner representative for color before purchasing materials.
  - 11. Unit Pricing
    - a) Provide add/deduct unit pricing for a permanent link, including all components, with an average cable length of a linear 150'-0". All requirements and specifications will be enforced.
- E. Closeout Submittal Requirements
  - 1. As-Built Drawings
    - a) Communications Design drawings are to be supplied to the Architect to prepare the master "As-Built" drawings.
    - b) Submit one electronic copy and one hard copy with project deliverables within three (3) weeks subsequent to substantial completion. Provide a laminated floorplan with drop designations in the respective serving Telecom Room.
    - c) As-Built drawings shall be in AutoCAD format, same version as used by Architect and consultant. Dimensions and scale of the drawing sheets submitted shall match the size of the drawing sheets used for the contract documents.
    - d) Utilize normal recognized drafting procedures that match AutoCAD standards, Architect and Consultant guidelines, and methodology.

- e) The As-Built drawings shall incorporate all changes made to the building identified in, but not limited to, addendum, change notices, site instructions or deviations resulting from site conditions.
  - 1) Contractor shall clearly identify any resubmitted drawing sheets, documents or cut sheets either by using a color to highlight or cloud around resubmitted information.
  - 2) Maintain drawing numbering or page/sheet scheme consistency as per previously issued drawings/documents.
- f) Provide dimensioned plan and elevation views of networking components, showing:
  - 1) All work area outlet locations complete with outlet/cable labeling.
  - 2) Rack and/or cabinet locations complete with labeling.
  - 3) One-line diagram of equipment/device interconnections with the cable plant.
  - 4) Standard or typical details of installations unique to Owner's requirements.
  - 5) Graphic symbols and component identification on detail drawing shall conform to the latest conventions:
    - i) ANSI/TIA-568-C.0 "Generic Telecommunications Cabling for Customer Premises"
    - ii) ANSI/TIA-569-C "Telecommunications Pathways and Spaces"
    - iii) ANSI/TIA-606-B "Administration Standard for Commercial Telecommunications Infrastructure"
    - iv) ANSI/TIA-607-B "Commercial Building Grounding (Earthing) and Bonding Requirements for Telecommunications"
- 2. The Communications Contractor shall deliver the Installer's Extended Product Warranty and Manufacturer's signed System Assurance Warranty of installed cabling system to include all components that comprise the complete cabling system.
  - a) Delivery shall be completed within two (2) weeks of the time of final punch list review.
  - b) Product Certificates shall be signed by manufacturers of cables, connectors, and terminal equipment certifying that products furnished comply with requirements.
- 3. Cable Testing Report Requirements
  - a) Submit certified test reports of Contractor-performed tests. Contractor shall submit the required Test Reports in the format and media specified, upon completion of testing the installed system.
  - b) The tests shall clearly demonstrate that the media and its components fully comply with the requirements specified herein.
  - c) Three (3) sets of electronic and hardcopy versions of test reports shall be submitted together and clearly identified with cable designations.
  - d) Cable inventory data shall be submitted for all fiber, copper, and coaxial cabling and termination components. Include products furnished:
    - 1) Manufacturer's name
    - 2) Manufacturer's part numbers
    - 3) Cable designations

- 4) Location and riser assignments
  - 5) Product Data
- 4. Supply Owner with training manuals with instructions on methods of adding or removing cabling to/from firestopped sleeves and chases.
- F. The Contractor's BICSI Registered Communications Distribution Designer (RCDD) supervisor shall review, approve and stamp all documents prior to submitting. The Contractor's RCDD shall warrant in writing that 100% of the installation meets the requirements specified herein upon completion of all work.

## **PART 2 - PRODUCTS**

### **2.1 SUMMARY**

- A. Equipment and materials shall be standard products of a manufacturer regularly engaged in the manufacture of telecommunications cabling products and shall be the manufacturer's latest standard design in satisfactory use for at least one year prior to bid opening.
- B. All material and equipment, as provided, should be the standard Commercial-Off-The-Shelf (COTS) products of a manufacturer engaged in the manufacturing of such products.
  - 1. All shall be typical commercial designs that comply with the requirements specified.
  - 2. All material and equipment shall be readily available through manufacturers and/or distributors.
- C. All equipment shall be standard catalogued items of the manufacturer and shall be supplied complete with any optional items required for proper installation.
- D. Coordinate the features of materials and equipment so they form an integrated system. Match components and interconnections for optimum future performance and backward compatibility.
- E. All materials shall be UL- and/or ETL-approved and labeled in accordance with NEC for all products where labeling service normally applies.
- F. Materials and equipment requiring UL 94, 149 or 1863 listing shall be so labeled. Modification of products that nullifies UL labels is not permitted.
- G. Backward Compatibility: The provided products shall be backward compatible with lower category ratings such that if higher category components are used with lower category components, the basic link and channel measures shall meet or exceed the lower category's specified parameters.
- H. Component Compliance: The provided products shall each meet the minimum transmission specifications listed herein such that no individual component will be less than specifications for permanent link and channel, regardless of the fact that tests for link and channel ultimately meet required specifications.

### **2.2 ACCEPTABLE MANUFACTURERS**

- A. Identification (Labeling) System
  - 1. Brady
  - 2. Dymo
  - 3. Hellerman-Tyton
  - 4. Acceptable alternate
- B. Fire-Stop Systems

1. Hilti
  2. SpecSeal
  3. 3M
  4. Acceptable alternate
- C. Other Products as Referenced in other Division 27 Specifications.

## **PART 3 - EXECUTION**

### **3.1 PREPARATION**

- A. Field Measurements
1. Verify dimensions in areas of installation by field measurements before fabrication and indicate measurements on shop drawings. Coordinate fabrication schedule with construction progress to avoid delaying the work.
- B. Established Dimensions
1. Where field measurements cannot be made without delaying the work, coordinate with the General Contractor to establish dimensions.
  2. When approved in writing, proceed with fabricating units without field measurements.
  3. Coordinate supports, adjacent construction, and fixture locations to ensure actual dimensions correspond to established dimensions.
- C. Pre-installation inspection
1. The Contractor shall visually inspect all cables, cable reels, and shipping cartons to detect possible cable damage incurred during shipping and transport.
  2. Visibly damaged goods are not acceptable and shall be replaced by the contractor at no additional cost to the Owner.

### **3.2 INSTALLATION**

- A. General
1. Contractor shall install work in accordance with specifications, drawings, manufacturer's instructions and approved submittal data.
- B. Allowable cable bend radius and pull tension:
- a) In general, communications cable cannot tolerate sharp bends or excessive pull tension during installation.
  - b) Refer to cable manufacturer's bend radius recommendations for the maximum allowable limits.
  - c) After installation, exposed cable and other surfaces must be cleaned free of lubricant residue. Use only lubricants specifically designed for cable installation.
- C. Pull Strings
1. Provide pull strings in all new conduits, including all conduits with cable installed (trailer strings) as part of this contract.
  2. Data and video cables can be pulled in tandem with pull strings.
  3. The pull strings must move freely to prevent cable jacket/cable damage during pulls.
- D. Labeling
1. Cable labels: Self-adhesive vinyl or vinyl-cloth wraparound tape markers, machine printed with alphanumeric cable designations.

2. Flat-surface labels: Self-adhesive vinyl or vinyl-cloth labels, machine printed with alphanumeric cable designations.
3. Provide transparent plastic label holders, and 4-pair marked colored labels.
4. In accordance with ANSI/TIA-606-B "Administration Standard for Commercial Telecommunications Infrastructure":
  - a) Install colored labels according to the type of field as per color code designations.
  - b) Use "designation strip color-code guidelines for voice, data, cross-connect, riser, and backbone fields".
5. Pathway Labels and Labeling System
  - a) Labeling system shall consist of a hand-held portable printer
  - b) Conduits: General-purpose label designed for powdered coated surfaces with an ultra-aggressive adhesive. Label size shall be appropriate for the conduit size. Font size shall be legible from the finished floor.
  - c) Inner duct: Polyethylene general-purpose tagging material attached using tie wraps.
  - d) Junction boxes: General-purpose label designed for powdered coated surfaces with an ultra-aggressive adhesive, trade name. Font size shall be easily visible from the finished floor.
  - e) All labels shall be permanent, i.e. will not fade, peel, or deteriorate due to environment or time.
  - f) Identification
    - 1) All conduits, junction boxes, gutters, and pull boxes shall have machine-generated labels easily visible from the finished floor.
    - 2) Conduits shall be labeled with the word "communications" and the conduit's origination room number and destination room number.
    - 3) The Contractor shall label conduit at each wall and floor penetration and at each conduit termination, such as outlet boxes, pull boxes, and junction boxes, or as otherwise specified in other sections.
    - 4) Junction boxes, gutters and pull boxes shall be labeled with identification name or number as determined by contractor and submitted for approval.
    - 5) The Contractor shall label conduit sleeves at each wall and floor penetration.

E. Firestop

1. Provide approved fire-resistant materials to restore originally-designed fire-ratings to all wall, floor, and ceiling penetrations used in the distribution and installation for communications cabling system.
2. Install and seal penetrations (conduit, sleeves, slots, chases) in fire-rated barriers created for communications infrastructure to prevent the passage of smoke, fire, toxic gas, or water through the penetrations.
3. The firestopping material shall maintain/establish the fire-rated integrity of the wall/barrier that has been penetrated.
4. All through penetrations in a fire rated surface require a sleeve, regardless of penetration diameter or penetrating cable count.

5. Using a "ring and string" method of installing cabling for membrane penetrations in a wall cavity is acceptable, provided the solution was accepted by the Owner in writing. Code-compliant firestopping rules still apply.
  6. Coordinate firestopping procedures and materials with General Contractor.
  7. Sharing the pathway of other trades/utilities through compliant and non-compliant penetrations does not remove the requirement to maintain code-compliant firestopping.
  8. Provide and install removable, intumescent mechanical systems in floor chases for all openings greater than 0'-4".
  9. Provide and install removable, intumescent, firestop bricks for all openings greater than 0'-4" where there are penetrations through walls.
  10. Bricks shall be listed for insertion in fire-rated openings and require restraining materials or apparatus as needed per manufacturers' specifications.
  11. Provide manufacturer recommended material for rated protection for any given barrier.
  12. Laminate and permanently affix adjacent to chases the following information:
    - a) Manufacturer of firestop system.
    - b) Date of installation/repair.
    - c) Part and model numbers of system and all components.
    - d) Name and phone numbers of local distributor and manufacturer's corporate headquarters.
  13. Solutions and shop drawings/submittals for firestop materials and systems shall be presented to the General Contractor for written approval of materials/systems prior to purchase and installation.
  14. Materials shall be installed per manufacturer instructions, be UL-listed for intended use, and meet NEC and locals codes for fire stopping measures.
  15. The material chosen shall be distinctively colored to be clearly distinguishable from other materials, adhere to itself, and maintain the characteristics for which it is designed to allow for the removal and/or addition of communication cables without the necessity of drilling holes in the material.
  16. Develop training manuals with instructions on methods of adding or removing cabling to/from firestopped sleeves and chases.
- F. Within the normal environment, the installed systems shall not generate nor be susceptible to any harmful electromagnetic emission, radiation, or induction that degrades, or obstructs any equipment.
- G. Expansion Capability: Unless otherwise indicated, provide spare conductor pairs in cables, positions in patch panels, cross connects, and terminal strips, and space in cable pathways and backboard layouts to accommodate 20% future increase in structure cable system capacity.
- H. In the event of a breach of the representations and warranties contained herein, the Contractor, at their own expense, shall take all measures necessary to make the cabling system work and comply with the applicable manufacturer written technical recommendations and standards.
- I. System Tests
1. Upon completion of the installation of the communications infrastructure systems, including all pathways and grounding, the Contractor shall test the system.

- a) Cables and termination modules shall be affixed, mounted or installed to the designed/specified permanent location prior to testing.
- b) Any removal and reinstallation of any component in a circuit, including faceplates, shall require retesting of that circuit and any other disturbed or affected circuits.
- c) Approved instruments, apparatus, services, and qualified personnel shall be utilized.
- d) The Contractor must verify that the requirements of the specifications are fully met through testing with an approved tester (rated for testing parameters listed elsewhere), and documentation as specified below.
- e) This includes confirmation of requirements by demonstration, testing and inspection. Demonstration shall be provided at final walk-through in soft copy and printed test data.

2. Non-Compliant Cabling

- a) Testing that shows some or all pairs of a cable do not comply with specifications, without written approval by the Owner, shall be replaced at Contractor's expense (including respective connectors).
- b) With the Owner's written approval, the over-length cable(s) shall be excluded from requirements to pass standardized tests and shall be explicitly identified.
  - 1) Testing is still required for non-compliant cabling.
  - 2) The tests shall be for wire-mapping, opens, cable-pair shorts, and shorts-to-ground.
  - 3) The test results must be within acceptable tolerances and shall be submitted with the Owner's acceptance document.

3. Failed Tests

- a) If tests fail, Contractor shall correct as required to produce a legitimate passing test.
- b) Manipulation of tester parameters on a failing test in order to achieve a passing test is unacceptable.
- c) If the Contractor is found to have manipulated or falsified any failing test result to show a "PASS" for any reason (without written notice and prior approval of the Owner), the Contractor shall be required to employ a Third-Party Testing Agent selected by the Owner to retest the complete cable plant and shall be required to pay all costs associated with this retesting.

4. Owner reserves the right to be present during any or all testing.

### 3.3 CLEANING

- A. The Contractor will clean all surfaces prior to final acceptance by Owner.

### 3.4 COMPLETION INSPECTION AND PUNCH LIST

- A. When the Contractor determines that the Scope of Work has been completed in accordance with the plans and specifications, Contractor shall schedule a Completion Inspection with the Owner.
- B. A Punch List will be generated during the Completion Inspection containing deficiencies in need of corrective action.
- C. Complete all punch list deficiencies within 10 working days. The work is not complete until all punch list deficiencies have been addressed.

### 3.5 ACCEPTANCE



- A. Once all work has been completed, test documentation has been submitted, and Owner is satisfied that all work is in accordance with contract documents, the Owner shall notify Contractor in writing of formal acceptance of the system.
- B. Contractor must warrant in writing that 100% of the installation meets the requirements specified herein (Standards Compliance & Test Requirements).
- C. Acceptance shall be subject to completion of all work, successful post-installation testing which yields 100% PASS rating, and receipt of full documentation soft and hard copies as described herein.

END OF SECTION 270000

**SECTION 27 05 26**  
**GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS**

**PART 1 - GENERAL**

**1.1 SUMMARY**

- A. This section includes grounding and bonding products, design requirements and installation for communications systems.
- B. Related Sections
  - 1. Section 270000 Communications
  - 2. Section 270528 Pathways for Communications Systems
  - 3. Section 271100 Communications Equipment Room Fittings
  - 4. Section 271500 Communications Horizontal Cabling

**1.2 REFERENCES**

- A. The publications referenced in Section 270000 form a part of this specification. The publications are referred to in the text by basic designation only.
- B. Specific reference in specifications to codes, rules, regulations, standards, manufacturer's instructions, or requirements of regulatory agencies shall mean the latest printed edition of each in effect at the date of contract unless the document is shown dated.
- C. Conflicts
  - 1. Refer to Section 270000.
- D. Codes and Standards
  - 1. Refer to Section 270000.

**1.3 SYSTEM REQUIREMENTS**

- A. General
  - 1. All conductor wire, busbars and conduit shall be UL listed.
  - 2. The communications ground system shall be independent from all power grounding except for the connection to the building's electrical service main grounding electrode system.
  - 3. Power grounding and/or bonding shall not be allowed to interfere or provide any back feed or be a conductor to the separate communications ground system source or to any communications bonded materials or equipment.
- B. Telecommunications Main Grounding Busbar (TMGB) and Bonding Conductor for Telecommunications (BCT)
  - 1. The main ground source feed for the Telecommunications Main Grounding Busbar (TMGB) in the MC (MDF) shall be an independent feed from the building's electrical service main grounding electrode system, known as the Bonding Conductor for Telecommunications (BCT).

2. The BCT shall be a stranded copper ground wire from the building ground system to the TMGB in the MC (MDF) sized at a minimum #4/0 unless otherwise sized by the Electrical Engineer of Record.
  3. The BCT connections shall be low emission exothermic welds at the connecting ends.
- C. Telecommunication Bonding Backbone (TBB) and Telecommunications Grounding Busbar (TGB)
1. The Telecommunication Bonding Backbone (TBB) originates at the TMGB and shall be extended from the TMGB within the MC (MDF) throughout the building along the same route as the telecommunications backbone pathways, to the Telecommunications Grounding Busbar(s) (TGBs) in each TR (IDF).
  2. The minimum TBB conductor size between busbars shall be a stranded copper ground wire one (1) AWG size smaller than the Bonding Conductor for Telecommunications (BCT).
- D. Grounding Equalizer (GE)
1. Whenever two or more TBBs are used in a multistory building, the TBBs shall be bonded together with a GE (by low emission exothermic welds) at the top floor and at a minimum of every third floor in between with a copper conductor equal to the gauge of the TBB.
- E. TEBC and RBC
1. All cabinets and racks shall be connected by the Telecommunications Equipment Bonding Conductor (TEBC). The TEBC is a stranded copper #4 conductor from the TMGB/TGB extending along each row of racks within the room. Bond each rack with a Rack Bonding Conductor (RBC). The RBC is a stranded copper #6 conductor connected to the vertical rack bonding terminal. All connections shall be irreversible crimp connections. Route conductor so as to minimize the quantity of sweeping bends.

#### **1.4 SUBMITTALS**

- A. Refer to Section 270000

#### **1.5 QUALITY ASSURANCE**

- A. Refer to Section 270000.

#### **1.6 DELIVERY, STORAGE, AND HANDLING**

- A. Refer to Section 270000.
- B. The Contractor shall ship on manufacturer's standard reel sizes of one continuous length. Where cut lengths are specified, mark reel quantity accordingly.

### **PART 2 - PRODUCTS**

#### **2.1 ACCEPTABLE MANUFACTURERS**

- A. Subject to compliance with requirements, manufacturers that may be incorporated in the work, include:
- B. Cable Manufacturers

1. Houston Wire and Cable Company
  2. Okonite Company
  3. General Cable
  4. Pirelli Cable Corporation
  5. Triangle Wire and Cable
  6. Owner Approved Alternate
- C. Electrical Service Entrance Bonding Conductor and Connector Manufacturers
1. Copperweld
  2. Thomas & Betts
  3. Blackburn
  4. Owner Approved Alternate
- D. Exothermic Connector Manufacturers
1. Erico Products (Cadweld)
  2. Continental Industries (thermOweld)
  3. Harger
  4. Owner Approved Alternate
- E. Crimp Connector Manufacturers
1. Thomas & Betts
  2. FCI Burndy Electrical
  3. O-Z/Gedney
  4. Owner Approved Alternate
- F. Telecommunication Grounding Busbars
1. Chatsworth
  2. Panduit
  3. Leviton
  4. Owner Approved Alternate
- G. Bonding Straps
1. Chatsworth
  2. Harger
  3. Brundy
  4. Owner Approved Alternate
- H. C-Type Compression Taps
1. Brundy
  2. Harger
  3. Owner Approved Alternate
- I. Antioxidant Joint Compound
1. Chatsworth
  2. Owner Approved Alternate
- J. Labeling

1. Refer to Section 270000.

K. Firestopping

1. Refer to Section 270000.

## 2.2 MATERIALS

A. Communications Grounding Conductors: Copper American Wire Gauge (AWG) wire of the following sizes:

1. Bonding Conductor for Telecommunications (BCT): #4/0 (unless otherwise sized by the Electrical Engineer of Record)
2. Telecommunication Bonding Backbone (TBB): #3/0 (unless otherwise sized by the Electrical Engineer of Record)
3. Grounding Equalizer (GE): equal AWG as the TBB (unless otherwise sized by the Electrical Engineer of Record)
4. Telecommunications Equipment Bonding Conductor (TEBC): #4
5. Rack Bonding Conductor (RBC): #6

B. Grounding Connectors

1. Connectors shall be a copper alloy material and two-hole, double-crimp compression lug type at the connecting ends.

C. Telecommunications Main Ground Busbar (TMGB)

1. Use pre-drilled copper busbar with standard NEMA bolt hole sizing and spacing for the type of connectors.
2. Sized for the immediate requirements and allow for 25% growth.
3. The minimum dimensions shall be 0'-1/4" thick X 0'-4" wide X 1'-8" long.
4. Contain (2) tiers of pre-drilled holes for use with standard sizes of two-hole copper compression lugs.
5. ASTM-B187-C11000 Copper bar suitable for use with two-hole compression-type copper lugs.

D. Telecommunications Ground Busbar (TGB)

1. Use pre-drilled copper busbar with standard NEMA bolt hole sizing and spacing for the type of connectors.
2. Sized for the immediate requirements and allow for 25% growth.
3. The minimum dimensions shall be 0'-1/4" thick X 0'-4" wide X 0'-10" long.
4. Contain (2) tiers of pre-drilled holes for use with standard sizes of two-hole copper compression lugs.
5. ASTM-B187-C11000 Copper bar suitable for use with two-hole compression type copper lugs.

## PART 3 - EXECUTION

### 3.1 EXAMINATION

- A. Refer to Section 270000

### 3.2 PREPARATION

- A. Refer to Section 270000.

- B. Copper and copper alloy connections should be cleaned prior to connection.

### **3.3 INSTALLATION**

- A. Refer to Section 270000.
- B. The Contractor shall install the work in accordance with the specifications, drawings, manufacturer's instructions and approved submittal data.
- C. All work shall be supervised and reviewed by the Contractor's on-site RCDD.
- D. Installation plans and Requests For Information (RFIs) shall be reviewed by the Contractor's RCDD.
- E. General
  - 1. Bonding and grounding procedures and components shall comply with ANSI/TIA-607-B "Commercial Building Grounding (Earthing) and Bonding Requirements for Telecommunications".
  - 2. Bonding should be accomplished such that the bonding system is integrated and compliant with NEC specifications.
  - 3. Bonding conductors shall be routed with minimum bends or changes in direction and should be made directly to the points being bonded.
  - 4. Bonding connections should be made by using compression copper lugs. However, for parts of the ground electrode system that are subject to corrosion, must carry high currents reliably, or for locations that require minimum maintenance, connections are made with low emission exothermic welding (see NEC Article 250).
  - 5. Make connections to dry surfaces only.
  - 6. Remove paint, rust, oxides, scales, grease and dirt from surfaces before making connection.
  - 7. Burnish clean a 0'-1" X 0'-1" area, drill, tap, apply an adequate amount of antioxidant joint compound mixed for the metal surface types affected, and bolt conductor and connector to burnished and compounded area. Ensure proper conductivity.
  - 8. Route bonding conductor(s) the shortest distance between bonding contact points.
  - 9. The ground-wire connecting ends shall have a minimum amount of insulation removed at the ground lug.
  - 10. Do not connect ground wire in power cable assemblies to the telecommunications ground system.
  - 11. All grounding and bonding conductors shall be copper and may be insulated. If bare-bonding conductors are used, isolate bonding conductors and prevent contact.
  - 12. Antioxidant material shall be installed to separate dissimilar metals and prevent corrosion.
  - 13. If multiple systems are involved (lightning protection systems, communications, radio and TV, CATV, etc.), those systems shall be bonded together to minimize potential differences between the systems, per NEC 250.94.
- F. Telecommunication Bonding Conductors
  - 1. Each telecommunications grounding and bonding conductor shall be labeled at each end detailing the function and room number of its opposite end. Labels shall be located on conductors as close as practicable to their point of termination in a readable position. Labels shall be nonmetallic and include the following text, "TELECOMMUNICATIONS GROUND - DO NOT REMOVE. IF THIS CONNECTOR OR CABLE IS LOOSE OR MUST BE REMOVED, PLEASE CALL THE BUILDING TELECOMMUNICATIONS MANAGER".

2. Furnish and install all required bonding material, hardware, and utilize tools manufactured for this purpose.
3. The connections of the BCT, TBB, GE, TEBC, and RBC shall be made using low emission exothermic welding or hydraulically crimped with a double crimp connector. Two-hole grounding lugs are preferred for connection to the grounding bus bars.
  - a) All low emission exothermic welding shall be by Division 26.
  - b) Coordinate with the building services personnel in occupied spaces to prevent the smoke from the exothermic weld process from potentially setting off smoke/fire alarms.
4. Grounding and bonding conductors should not be placed in ferrous metallic conduit. If it is necessary to place grounding and bonding conductors in a ferrous metallic conduit that exceeds 1m (3ft) in length, the conductors shall be bonded to each end of the conduit using a grounding bushing or a No. 6AWG conductor, minimum.
5. The bonding conductors should be installed without splices.
  - a) Where splices are necessary, the number of splices should be minimized, be accessible, and be located within the telecommunications spaces.
  - b) Joined segments of a bonding conductor shall be connected using exothermic welding, irreversible compression-type connectors, or equivalent. All joints shall be adequately supported and protected from damage.

#### G. Equipment Cabinets and Racks

1. The busbar shall be installed at the base and back of each cabinet/rack for floor fed cabinets/racks.
2. The busbar shall be installed at the top and back of each cabinet/rack for top fed cabinets/racks.
3. Each cabinet and rack shall be provided with a minimum # 6 AWG ground wire.
4. Do not loop from cabinet/rack to cabinet/rack.
5. Each cabinet or rack bay against the wall shall be bottom/side ground fed from the wall.
  - a) Wall ground feeds/raceways to racks shall not be exposed on the walls.
  - b) Exception: Some cabinet or rack bays will require the ground to be fed from the ceiling raceway.
6. All ground raceways within each cabinet/rack or cabinet base and adjacent-ganged cabinet base shall be an insulated metallic flex type raceway and shall not interfere with equipment mounting frames or equipment mounting brackets.

#### H. Cable Runway, Cable Raceway and Support System Grounding

1. The Contractor shall provide communications cable tray and cable runway systems with a communications dedicated ground from the TGB.
2. All cable tray needs to be electrically continuous per NEC 250.96.
  - a) Metal raceways, wire-mesh cable trays, cable armor, cable sheath, enclosures, frames, fittings, and other metal non-current-carrying parts that are to serve as an alternate grounding path, with or without the use of supplementary equipment grounding conductors, shall be effectively bonded where necessary to ensure

electrical continuity and the capacity to conduct safely any fault current plausibly to be imposed on them.

- b) Any nonconductive paint, enamel, or similar coating shall be removed at the threads, contact points, and contact surfaces.
- c) Grounding or bonding conductors shall be connected by fittings designed for that purpose to ensure adequate bonding.
- 3. The Contractor shall provide and install a #6 AWG ground wire to bond one end of each cable tray/runway system to the TGB.
- 4. For electrically non-continuous conduits that contain only grounding conductor, the Contractor shall bond the conduit and conductor together at both ends to ground to the nearest TGB with grounding bushings or ground clamps.

I. Shielded Backbone Cabling

- 1. The Contractor shall terminate and bond the shield to the nearest TGB or TMGB at both ends, following manufacturer's guidelines.

**3.4 FIELD QUALITY CONTROL**

A. Testing

- 1. Upon completion of the electrical system, including all grounding, the Electrical Contractor shall test the system for stray currents, ground shorts, etc.
- 2. Approved instruments, apparatus, services, and qualified personnel shall be utilized.
- 3. If stray currents, shorts, etc., are detected, eliminate or correct as required.

END OF SECTION 270526



SECTION 27 05 28  
PATHWAYS FOR COMMUNICATIONS SYSTEMS

**PART 1 - GENERAL**

**1.1 SUMMARY**

A. Section Includes

1. Hangers and Supports, including open-top supports (cable hooks) for communications systems.
2. Conduits and Pull Boxes for communications systems.
3. Cable Tray and Cable Runway with associated accessories and fittings for communications systems.

B. Related Sections

1. Section 270000 Communications Systems
2. Section 270526 Grounding and Bonding for Communications Systems
3. Section 271100 Communications Equipment Room Fittings
4. Section 271500 Communications Horizontal Cabling

**1.2 REFERENCES**

- A. The publications referenced in Section 270000 form a part of this specification. The publications are referred to in the text by basic designation only.
- B. Specific reference in specifications to codes, rules, regulations, standards, manufacturer's instructions, or requirements of regulatory agencies shall mean the latest printed edition of each in effect at the date of contract unless the document is shown dated.
- C. Conflicts
1. Refer to Section 270000.
- D. Codes and Standards
1. Refer to Section 270000.

**1.3 SUBMITTALS**

- A. Refer to Section 270000.

**1.4 QUALITY ASSURANCE**

- A. Refer to Section 270000.

**1.5 DELIVERY, STORAGE, and HANDLING**

- A. Refer to Section 270000.
- B. Conduit Storage
1. Package conduits in bundles maximum 10'-0" long, with conduit and coupling thread protectors for indoor/outdoor storage.
  2. Package fittings in manufacturer's standard quantities and packaging suitable for indoor storage.

3. Protect coating on plastic-coated rigid conduit, fittings, and bodies from damage during shipment and storage.
4. Store conduit above ground on horizontal racks to prevent corrosion and entrance of debris.
5. Equipment and components shall be protected from the weather, humidity, temperature variations, dirt, dust, or other contaminants. Protect plastic conduit and inner duct from sunlight. Equipment damaged prior to system acceptance shall be replaced at no cost to the Owner.

## **PART 2 - PRODUCTS**

### **2.1 ACCEPTABLE MANUFACTURERS**

- A. Subject to compliance with requirements, manufacturers that may be incorporated in the work, include:
  1. Cooper B-Line, Inc.
  2. Erico
  3. Caddy
  4. Owner approved alternate
- B. Cable Hooks
  1. Cooper B-Line, Inc.
  2. Erico
  3. Caddy
  4. Owner approved alternate
- C. Basket Type Cable Tray (for use in work areas)
  1. Cooper B-Line, Inc.
  2. Chatsworth (CPI)
  3. Owner approved alternate
- D. Wire Basket Cable Tray Cutting Tool
  1. Cooper B-Line, Inc.
  2. Chatsworth (CPI)
  3. Owner approved alternate
- E. Polyethylene Cable Support System
  1. Erico
  2. Owner approved alternate
- F. Innerduct
  1. Carlon Riser Guard Flexible Raceway (corrugated innerduct)
  2. Owner approved alternate
- G. Measured pull tape (pull tape printed with sequential footage markings)
  1. Fibertek
  2. Condux International
  3. Owner approved alternate
- H. Labeling
  1. Refer to Section 270000.

I. Firestopping

1. Refer to Section 270000.

**2.2 CABLE HOOKS**

- A. Cable hooks shall be factory assembled for direct attachment to walls, hanger rods, beam flanges, purlins, strut, floor posts, etc. to meet job conditions.
- B. Features
  1. Cable hooks shall have a flat bottom and provide a minimum of 0'-1.625" cable-bearing surface.
  2. Cable hooks shall have 90° radius edges to prevent damage while installing cables.
  3. Cable hooks shall be designed so that the mounting hardware is recessed to prevent cable damage.
  4. Cable hooks for non-corrosive areas shall be pre-galvanized steel. Where additional strength is required, cable hooks shall be spring steel with a zinc-plated finish.
  5. Cable hooks for corrosive areas shall be stainless steel.
  6. Cable hooks shall have a stainless steel cable latch retainer to provide containment of cables within the hook.
  7. The retainer shall be removable and reusable.
- C. Factory assembled multi-tiered cable hooks shall be used where required to provide separate cabling compartments, or where additional capacity is needed.
- D. Load cable hooks in accordance with manufacturer requirements and recommendations.
- E. Provide capacity for 25% growth, add additional hooks as needed.

**2.3 PULL BOXES, JUNCTION BOXES, AND GUTTERS**

- A. All junction boxes, gutters and pull boxes shall be UL listed and comply with NEC requirements.
- B. All junction boxes, gutters and pull boxes shall meet the following minimum material requirements:
  1. 16-gauge steel or heavier
  2. Seams shall be continuously welded and grounded smooth
  3. External screws and clamps
  4. External mounting feet (where applicable)
  5. Oil-resistant gasket and adhesive
  6. ANSI 61 gray polyester powder coating inside and out over phosphatized surface
- C. All junction boxes, gutters and pull boxes shall be provided with bushings for conduits and/or cabling.
- D. All junction boxes, gutters and pull boxes shall be securely installed.

**2.4 CONDUITS**

- A. All conduits shall be UL listed and comply with NEC requirements.
- B. Conduit Fittings
  1. All fittings shall be compression or threaded.

2. Fittings shall provide a secure connection for pulling communications cables.
  3. Setscrew fittings are not permitted.
  4. Conduit "condulets" are not permitted.
- C. Non-metallic conduits are not permitted in above ground installations. Conversion fittings are required for non-metallic (below ground) to metallic (above ground) transitions.
- D. Innerduct:
1. All fiber shall be installed in innerduct unless fiber cabling is armored.
  2. Shall be constructed of non-metallic material.
- E. Only manufacturer's fittings, transition adapters, terminators and fixed bends shall be used.
- F. Measured Pull Tape
1. Pre-lubricated, woven polyester, low friction, and high abrasion resistant yarn with footage markers printed on the tape.
  2. Minimum average tensile strength shall be 1130 lbs. for 0'-1.5" and smaller conduits and innerduct.
  3. Minimum average tensile strength shall be 1800 lbs. for conduits larger than 0'-1.5".
- G. Fill and Bend Radius
1. Conduit fill shall comply with NEC requirements.
  2. The minimum bend radius is 6 X the conduit inside diameter (ID) for 0'-2" conduit or less.
  3. The minimum bend radius is 10 X the conduit ID for a conduit greater than 0'-2".
  4. There shall be no more than two 90° bends (180° total) between conduit pull boxes.
  5. Changes in direction shall be accomplished with sweeping bends observing minimum bend radius requirements above.
  6. Do not use pull boxes for direction changes unless specifically designated otherwise in the drawings.
  7. Unless otherwise noted in the drawings, conduits entering pull boxes shall be aligned with exiting conduits.
- H. Routing
1. Conduits shall be routed in the most direct route possible, with the fewest number of bends possible.
  2. There shall be no continuous conduit sections longer than 100'-0" for premises conduits. For runs that total more than 100'-0", insert junction or pull boxes so that no continuous run between pull boxes is greater than 100'-0".
- I. Penetrations
1. All conduit penetrations shall comply with all applicable fire codes.
  2. All conduit penetrations in fire-rated walls or floors shall be sealed and fire-proofed to meet or exceed the designed rating of the penetration area.

## **2.5 CABLE TRAY**

- A. Cable tray systems are defined to include, but are not limited to, straight sections of cable trays, bends, tees, elbows, reducers, crosses, wyies, vertical bends, up/down tees, cable support fittings, drop-outs, supports and accessories.

- B. Install all tray types utilizing manufacturer recommended installation instructions and applicable standards.
- C. Load cable tray and cable runway in accordance with manufacturer requirements and applicable standards.
- D. Cable Tray Materials
  - 1. Aluminum
  - 2. Pre-galvanized Steel
  - 3. Hot-dip Galvanized Steel
  - 4. Stainless Steel
  - 5. Yellow Zinc Dichromate
  - 6. Pre-Galvanized Zinc
  - 7. Electro-Galvanized Zinc
- E. Cable Tray Systems
  - 1. Wire basket (mesh) of types and sizes indicated on the drawings; with connector assemblies, clamp assemblies, connector plates, splice plates, cable drop outs, bonding accessories, and splice bars. Construct units with rounded edges and smooth surfaces.
  - 2. Continuous mesh polyethylene cable-support system: with connector assemblies and appropriate support components. All parts shall be UL-listed. Plastic (non-metallic) parts shall have a zero detectable halogen content as substantiated by an independent test laboratory.
  - 3. Ladder type trays shall consist of two longitudinal members (side rails) with transverse members (rungs) welded to the side rails. Rungs shall be spaced 0'-9" on center. Spacing in radius fittings shall be 0'-9" as measured at the center of the tray's width. Rungs shall have a minimum cable-bearing surface of 0'-.875" with radius edges. No portion of the rungs shall protrude below the bottom plane of the side rails.
  - 4. Ventilated trough type trays shall consist of two longitudinal members (side rails) with a corrugated bottom welded to the side rails.
  - 5. Solid bottom trough type trays shall consist of two longitudinal members welded to the side rails.
- F. Cable trays shall have sufficient depth and width so as not to exceed a maximum 50% fill ratio, including 25% capacity for anticipated growth.
- G. All straight sections shall be supplied in minimum 8'-0" lengths, except where shorter lengths are permitted to facilitate tray assembly lengths.

## **2.6 HANGERS AND SUPPORT**

- A. Steel support brackets shall be galvanized steel and capable of supporting a minimum of 200 lbs with a safety factor of 3.
- B. Steel support brackets shall have a removable galvanized steel retaining strap.
- C. Steel support brackets shall accept 0'-3/8" (10mm) threaded rod for attachment to building structure or sub structure.

## **PART 3 - EXECUTION**

### **3.1 EXAMINATION**

- A. Refer to Section 270000.

### **3.2 PREPARATION**

- A. Refer to Section 270000.
- B. Verify system is properly sized for cables before installation.
- C. Verify that the manufacturer recommended loads are not exceeded.
- D. Verify general routing and coordinate locations with other trades before installation. Layout cable runs in advance to determine quantities of cable to be installed along pathways, and to ensure non-interference from other trade installations.

### **3.3 INSTALLATION**

- A. Refer to Section 270000.
- B. Cable Hooks
  - 1. Provide cable hook (j-hook) cable support system for horizontal and/or riser cabling in accessible ceiling space. Assemblies shall be complete with mounting hardware.
  - 2. Provide threaded rod for supporting hangers when hanging from floor deck and deck members.
  - 3. Follow manufacturers fill capacities.
  - 4. Locate cable hooks on 4' to 5' centers to adequately support and distribute the cable's weight.
  - 5. Suspended cables shall be installed with at least 0'-3" of clear vertical space above the ceiling tiles and support channels.
  - 6. For larger quantities of cables, provide special supports that are specifically designed to support the required cable weight and volume.
  - 7. Do not support pathways or cables with the ceiling suspension system or use electrical, plumbing, or other pipes for support.
  - 8. Cable supports shall be permanently anchored to building structure or substrates. Provide attachment hardware and anchors designed for the structure to which attached, and that are suitably sized to carry the weight of the cables to be supported.
  - 9. Secure and support exposed horizontal cable at intervals not exceeding 5'-0" and not less than 1'-4" (16") from cabinets, rack pole, boxes, fittings, outlets, racks, frames, and terminals.
  - 10. Cable sag between vertical supports for horizontal pathway shall not exceed 0'-6". Provide at least 0'-3" cable sag between supports.
  - 11. Painted J-hooks shall meet or exceed NEC requirements for the environment in which the product is installed.
- C. Conduit and Pull Boxes
  - 1. The Contractor shall route the conduit in approximate locations unless drawing is dimensioned for precise placement.
  - 2. Conduit cuts shall be square. Ream ends of burrs, and remove metal shavings and cutting lubricants before conduit is connected to the conduit system.

3. For conduit embedded in concrete, coat threaded connections in conduits with colloidal rust and corrosion inhibitor and sealant. Conduit must be clean and dry and must pass standard sizing test after concrete is poured.
  4. Cap unused conduits with watertight caps
  5. Make conduit connections with appropriate fittings and tighten securely.
  6. Use appropriate tools to install PVC coated conduit; avoid damage to exterior coating.
  7. Install liquid-tight flexible metal conduit where exposed to weather, water, or other liquids.
  8. Use IMC, PVC conduit, or rigid galvanized steel conduit in underground installations.
  9. The Contractor shall provide fabric innerduct in all underground conduits, as indicated on the drawings.
- a) The Contractor shall use pre-lubricated, woven polyester, low friction, and high abrasion resistant fabric.
  - b) The Contractor shall be trained for proper installation technique by the innerduct manufacturer. The Contractor shall coordinate with the owner to demonstrate that pull ropes in each inner duct cell move freely from end to end.

D. Cable Tray and Cable Runway

1. Cut wire basket tray members square with approved cable tray cutting tool as to not leave sharp edges at cut point. Remove burrs and smooth the ends before the cut is connected to wire-mesh tray system.
2. Ensure that standard splices are designed to have less than 1 milliohm (0.0001  $\Omega$ ) of resistance between connections and provide bonding between sections. Painted wire mesh tray requires the outer mask of the non-conductive surface be removed at each end of the tray prior to installing the splice to provide continuity between painted tray sections.
3. Threaded rod (minimum 0'-1/2" diameter) or equivalent and slotted channel shall be used for hanging cable tray between floor deck and deck members

E. Fiber Support:

1. Support vertical fiber optic cable with basket weave wire/cable grips. Support fiber riser with single weave support grip with a single offset eye.
2. Mount/attach pulling eye to a wall or ceiling deck secured hook to support/provide strain relief to riser cable. Provide a minimum 3'-0" loop of fiber prior to entering fire stopped floor sleeve.
3. Where required coil up slack fiber cable into pull box and secure with single weave support grip.

F. Clearances

1. A minimum of 1'-0" access headroom shall be provided above a cable tray. Ensure that other building components do not restrict access to the cable trays from the sides.
2. Power outlets shall not be installed in or mounted to cable tray or cable runway.
3. Provide 3'-0" of unencumbered space for every 10'-0" segment of tray.
4. Cable tray clearances
  - a) Motors or transformers: 4'-0"
  - b) Power cables and conduit: 1'-0"
  - c) Fluorescent lighting: 0'-5"
  - d) Halide lights: 1'-0"
  - e) Above the ceiling tiles: 0'-3"

- f) Access above and on one side of the cable tray: 1'-0"

### **3.4 FIELD QUALITY CONTROL**

- A. Test system to ensure electrical continuity of bonding and grounding connections.
- B. Ensure compliance with specified maximum ground resistance.
- C. Refer to NFPA 70B Chapter 18 for testing and test methods.

### **3.5 CLEANING**

- A. Remove all unnecessary tools and equipment, unused materials, packing materials, and debris from each area where Work has been completed unless designated for storage.
- B. Wipe clean all cable trays and apply appropriate manufacturer's paint to areas that have been scratched.

END OF SECTION 270528



## SECTION 27 11 00

### COMMUNICATIONS ROOM FITTINGS

#### **PART 1 - GENERAL**

##### **1.1 SUMMARY**

- A. This section includes basic communications and equipment room design requirements and fittings including:
  - 1. Equipment cabinets, racks, frames and enclosures
  - 2. Cable management and ladder racks
  - 3. Telecommunications service entrance pathways
  - 4. Rack mounted power protection and power strips
- B. Related Sections
  - 1. Section 270000 Communications
  - 2. Section 270526 Grounding and Bonding for Communications Systems
  - 3. Section 270528 Pathways for Communications
  - 4. Section 271500 Communications Horizontal Cabling

##### **1.2 REFERENCES**

- A. The publications listed below form a part of this specification. The publications are referred to in the text by basic designation only.
- B. Specific reference in specifications to codes, rules, regulations, standards, manufacturer's instructions, or requirements of regulatory agencies shall mean the latest printed edition of each in effect at the date of contract unless the document is shown dated.
- C. Conflicts
  - 1. Refer to section 270000
- D. Codes and Standards (Most recent editions or as required in contract)
  - 1. ANSI/TIA-568-C: Commercial Building Telecommunications Wiring Standard
  - 2. ANSI/TIA-569-B: Commercial Building Standard for Telecommunications Pathways and Spaces
  - 3. ANSI/TIA-606-A: Administration Standard for Commercial Telecommunications Infrastructure
  - 4. ANSI/NECA/BICSI-607-A: Commercial Building Grounding (Earthing) and Bonding Requirements for Telecommunications
  - 5. TIA-758-A: Customer-Owned Outside Plant Telecommunications Infrastructure Standard
  - 6. National Electrical Code (NEC), based upon year approval by local codes or AHJ
  - 7. BICSI Telecommunications Distribution Methods Manual (TDMM)
  - 8. Local, county, state and federal regulations and codes in effect as of date of purchase
  - 9. Equipment of foreign manufacture must meet U.S. codes and standards. It shall be indicated in the proposal the components that may be of foreign manufacture, if any, and the country of origin.

##### **1.3 COMMUNICATIONS ROOMS**

- A. Communications rooms must be dedicated to designated equipment and services:
  - 1. Space shall not be used for storage of equipment not related to designated equipment and services.
  - 2. Hazardous or corrosive materials shall not be stored in the space.

3. Piping, ductwork and distribution of power, not related to designated equipment and services shall not pass through or be located within the space.
  - a) Foreign piping such as water pipes, steam pipes, soil pipes, sanitary drains, storm drains, A/C ducts, and other unrelated systems utilized for or containing liquids, or gases shall not be installed or pass through communication rooms.
  - b) With the exception of fire sprinklers, all water pipes shall be routed around communications room.
- B. Each communication room shall be equipped with fire detection, fire-extinguishing system and prevention devices. Connect detection devices to base building fire alarm system. A minimum of one (1) smoke detector shall be installed in each communications room.
- C. Walls shall be covered with 0'- $\frac{3}{4}$ " X 4'-0" X 8'-0" AC-grade plywood backboard 1'-0" AFF (smooth side to interior of room mounted vertically), capable of supporting mounted hardware and equipment.
  1. Plywood shall be affixed to the studs in the walls with screws that penetrate the studs a minimum of 0'-1", are spaced not greater than 1'-6" (18") apart in each stud, and with screws 0'-0" from the top and bottom of plywood.
  2. Plywood shall be sealed against the wall and painted on all exposed sides with two coats of flat white non-reflective paint.
  3. If applicable fire-treatment verification stamps on plywood shall be left unpainted to be readable.
- D. Communications room walls shall extend from floor slab to ceiling deck, with no drop ceilings installed.
- E. Cable tray or ladder rack should be used to distribute cables between rooms through finished wall penetrations.
- F. Cable ladder rack should be used to distribute cables within rooms, complete with cable bend limiters (drop outs).
- G. To reduce static, floors should not have carpet, but be sealed concrete to prevent concrete dust from forming.
- H. Communications rooms shall have only one lockable entrance door, a minimum of 3'-0" wide and 7'-0" high, that opens towards the outside of the room, and does not open into another room.
  1. Doors shall be provided with a lockset for the appropriate technology key with pinned hinges and anti-pry guards.
  2. Doors should have no windows or door seals.
  3. Communications rooms should have no exterior identifying markings.
- I. Mechanical
  1. Install monitoring sensors with dedicated environmental controls operating 24 hours a day, 365 days a year in the communications rooms.
  2. Provide ventilation in the communications rooms to dissipate heat generated by active devices.
  3. Temperature and Humidity requirements:
    - a) Maintain communication rooms at an average of 60°F to 70°F, with a relative non-condensing humidity of 30% to 50%.
    - b) The temperature range should be maintained within  $\pm 9^\circ$
- J. Plumbing
  1. If "wet" fire suppression is used, install wire cages on sprinkler heads to prevent accidental operation.

2. Do not place sprinkler heads over equipment or cabling. In the event of a leak this will protect the equipment and cabling.
3. Drainage troughs are also recommended for leakage protection.

K. Electrical

1. One manufacturer's product is recommended for each type of installation. The mixing of different manufacturer products for one item is not acceptable.
2. No electrical feeders/branch circuits shall be placed in or run through any communications room except as required to service those rooms.
3. The Contractor shall install a slot (a UL-approved fire-rated assembly) to accommodate cable runway entry from corridor and a fire-retardant system (bricks, boards, mechanical, etc). The formed slot shall have no burrs or sharp edges. This opening in the wall will be used to pass data and voice cabling from the corridor cable tray into the communications room.
4. The Contractor shall provide uniform illumination of at least 50 foot-candles (fc) 3'-0" AFF for communications rooms located a minimum of 8'-6" AFF.
  - a) Light fixtures in communications rooms are to be positioned for maximum lighting. Do not install over cable tray, ladder rack, or 1'-7" (19") standing racks.
  - b) Provide enough power receptacles to support equipment and service. Coordinate power requirements of active equipment with electrical designer.

L. Relay Racks

1. 1'-7" (19") X 7'-0" relay racks are to be used for mounting and termination of inter-building and intra-building fiber optic/ copper cables and components.
  - a) The racks shall have adequate horizontal and vertical cable management for the 8P8C patch panels and switches.
  - b) Racks with active electronics shall have rack mounted power strips.

**1.4 SUBMITTALS**

- A. Refer to section 270000

**1.5 QUALITY ASSURANCE**

- A. Refer to section 270000
- B. Product Standards
  1. Equipment and materials shall be standard products of a manufacturer regularly engaged in the manufacture of telecommunications cabling products and shall be the manufacturer's latest standard design in satisfactory use for at least one year prior to bid opening.
  2. Items of the same classification shall be identical. This requirement includes equipment, modules, assemblies, parts, and components.

**1.6 DELIVERY, STORAGE, AND HANDLING**

- A. Refer to section 270000
- B. Coordinate layout and installation of equipment with owner's communications and LAN equipment and service suppliers. Coordinate service entrance arrangement with local exchange carrier.

**1.7 PROJECT/SITE CONDITIONS**

- A. Refer to section 270000

**1.8 WARRANTY**

- A. Refer to section 270000

- B. At the start of the project, contractor shall register the project with the manufacturer to help insure and facilitate manufacturer's warranty process.

## **PART 2 - PRODUCTS**

### **2.1 ACCEPTABLE MANUFACTURERS**

- A. 1'-7" (19") x 7" Floor-Mounted 2-post Equipment Racks and Support Components
  - 1. Chatsworth (CPI #55053-703)
  - 2. Owner approved alternate
- B. Horizontal Runway and Support Components
  - 1. Chatsworth (CPI #10250-712, 12" black)
  - 2. Owner approved alternate
- C. Horizontal Rack-Mount Cable Management
  - 1. Chatsworth (CPI, 1RU #30139-719, 2RU #30130-719)
  - 2. Owner approved alternate
- D. Vertical Rack-Mount Cable Management
  - 1. Chatsworth (CPI CCS, #30162-703; 7'-0"H x 6"W, #30163-703; 7'-0"H x 10"W)
  - 2. Owner approved alternate
- E. Labeling
  - 1. Refer to section 270000
- F. Firestopping
  - 1. Refer to section 270000

### **2.2 ACCESSORIES**

- A. Rack-mounted Uninterruptible Power Supply (UPS)
  - 1. Owner furnished owner installed.
- B. Rack-mounted Power Strips
  - 1. Owner furnished owner installed.

## **PART 3 - EXECUTION**

### **3.1 EXAMINATION**

- A. Refer to Section 270000

### **3.2 PREPARATION**

- A. Refer to section 270000
- B. Meet jointly with telecommunications and LAN equipment suppliers, local exchange carrier representatives, and Owner to exchange information and agree on details of equipment arrangements and installation interfaces.
- C. Adjust arrangements and locations of equipment with distribution frames, cross-connects, and patch panels of cabling systems of other communications, electronic safety and security, and related systems that share space in the equipment room.
- D. Coordinate location of power raceways and receptacles with locations of communications equipment requiring electrical power to operate.

### **3.3 INSTALLATION**

A. Refer to section 270000

**3.4 FIELD QUALITY CONTROL**

A. Refer to section 270000

**3.5 CLEANING**

A. Refer to section 270000

**3.6 ACCEPTANCE**

A. Refer to section 270000

END OF SECTION 271100

## **SECTION 27 15 00**

### **COMMUNICATIONS HORIZONTAL CABLING**

#### **PART 1 - GENERAL**

##### **1.1 SUMMARY**

- A. This section of the horizontal cabling portion of a structured cabling system includes:
  - 1. UTP Copper cabling
  - 2. Termination and patch cables
- B. Provide all horizontal cabling, terminating hardware, adapters, and cross-connecting hardware necessary to interconnect all system equipment including equipment located in communications rooms.
- C. Related Sections
  - 1. Section 270000 Communications
  - 2. Section 270526 Grounding and Bonding for Communications Systems
  - 3. Section 270528 Pathways for Communications
  - 4. Section 271100 Communications Equipment Room Fittings

##### **1.2 REFERENCES**

- A. The publications listed below form a part of this specification. The publications are referred to in the text by basic designation only.
- B. Specific reference in specifications to codes, rules, regulations, standards, manufacturer's instructions, or requirements of regulatory agencies shall mean the latest printed edition of each in effect at the date of contract unless the document is shown dated.
- C. Conflicts
  - 1. Refer to section 270000
- D. Codes and Standards
  - 1. Refer to section 270000

##### **1.3 SUBMITTALS**

- A. Refer to sections 270000 and 271300

##### **1.4 QUALITY ASSURANCE**

- A. Refer to section 270000

##### **1.5 DELIVERY, STORAGE, AND HANDLING**

- A. Refer to sections 270000 and 271300
- B. Storage temperature range: -40°F to 149°F (-40°C to 65°C)

##### **1.6 PROJECT/SITE CONDITIONS**

- A. Refer to section 270000

##### **1.7 WARRANTY**

- A. Refer to section 270000

##### **1.8 MAINTENANCE AND SUPPORT**

- A. Refer to section 271300

## **PART 2 - PRODUCTS**

### **2.1 ACCEPTABLE MANUFACTURERS**

- A. Labeling
  - 1. Refer to section 270000
- B. Firestopping
  - 1. Refer to section 270000

### **2.2 ACCEPTABLE COPPER MANUFACTURERS**

- A. UTP Plenum Rated Cable Cat6 (blue)
  - 1. Berk-tek
  - 2. General Cable
  - 3. Owner approved alternate
- B. Data/Voice Outlet Components Cat6
  - 1. Leviton
  - 2. Panduit
  - 3. Owner approved alternate
- C. Patch Panels (24 or 48 port)
  - 1. Leviton
  - 2. Panduit
  - 3. Owner approved alternate
- D. Copper Patch Cords Cat6 Pre-assembled
  - 1. Panduit
  - 2. Leviton
  - 3. Owner approved alternate
- E. Wall and/or Rack Mount 110 Termination Blocks
  - 1. Panduit
  - 2. Leviton
  - 3. Owner approved alternate
- F. Faceplate for wall-mount telephones
  - 1. Leviton 4108W-1SP
  - 2. Panduit KWP6
  - 3. Owner approved alternate

### **2.3 ACCESSORIES**

- A. Mount one laminated full-size hard copy in color of an as-built floor plan designating workstation locations, pathways, and communications room locations. Confirm hard copy size with Owner.
- B. Provide clear plastic lamination serving each communication room.
- C. Install the laminated drawings within a protective Plexiglas encasement on the wall of the servicing communications rooms. To ease accessibility the Plexiglas encasement shall be in either flip-down format or file folder format.

### **2.4 HORIZONTAL COPPER CABLING**

- A. Recognized cabling for providing the signal medium from the work area to the communications room shall include the following:
  - 1. Category 6 UTP cable
- B. Category 6 UTP Cable Requirements

1. 23/24 AWG solid bare copper
2. Cable jacket shall comply with NEC Article 800 for use as a plenum cable and shall be UL and c (UL) Listed Type CMP (communications multipurpose plenum)
3. Cable shall terminate on an eight-pin modular jack at each outlet. All horizontal cabling shall meet or exceed the ANSI/TIA-568-C.2 Commercial Building Telecommunications Cabling Standard, Part 2: Balanced Twisted Pair Cabling Components
4. Cables shall be marked as UL verified with a minimum of Category 6 rating
5. The cable shall support Voice, Analog Base band Video/Audio, Fax, Modem, Switched-56, T-1, ISDN, RS-232, RS-422, RS-485, 10BASE-T Ethernet, Token Ring, 100Mbps TP-PMD, 100BASE-T Ethernet, 155 Mbps ATM, AES/EBU Digital Audio, 270 Mbps Digital Video, 622 Mbps 64-CAP ATM and emerging high-bandwidth applications, including 1 Gbps Ethernet, gigabit ATM, as well as all 77 channels (550 Mhz) of analog broadband video
6. The maximum horizontal cable length for Category 6 copper cable from the termination of the cable in the communications room to the outlet is 295'-0".
7. Texas A&M ONLY - The maximum horizontal cable length for Category 6 copper cable from the termination of the cable in the communications room to the outlet is 270'-0".
8. Cable shall meet or exceed the following electrical characteristics:
9. Cable shall be specified to 250 MHz and shall meet the manufacturer's guaranteed electrical performance and physical specifications.

## 2.5 TERMINATION HARDWARE

- A. Patch panels
  1. Patch panels shall be rated to match installed cable plant
  2. The wiring block shall accommodate #23 AWG cable conductors.
  3. All modular cross connect panels shall be UL-listed.
- B. Work Area Outlet Cat6
  1. Universal eight-position jack pin/pair assignments
  2. Jack Color:
    - a) Data: White
- C. Work Area Outlet Faceplates:
  1. White or ivory to match electrical outlets.

## 2.6 PATCH CABLES

- A. Verify exact quantities and lengths with Owner prior to purchase
- B. Patch Cable requirements:
  1. Category 6, stranded UTP cable
  2. Standard modular non-keyed, 8-position 8-conductor plug
  3. 94V-0 rated
  4. UL listed
  5. Meets FCC Part 68
- C. Provide 3'-0", 5'-0", 7'-0", and 10'-0" Patch Cords at the communications room for each installed port.
  1. Coordinate with Owner on the active equipment layout prior to purchase to ensure correct sizing of patch cords from patch panels to switching equipment.
  2. When connecting voice ports to a copper riser, provide a one-pair stranded 8P8C connector on one end and 110GS on the other end and shall be of appropriate length for application.
- D. Provide a 10'-0" Station Cord for each work area outlet port.



- E. Place each size/length patch cord in a separate container, and mark the containers that hold the patch cords with the length of patch cords contained within.
- F. All cords shall conform to the requirements of ANSI/TIA-568-C.2 Commercial Building Telecommunications Cabling Standard, Horizontal Cabling Section, and be part of the UL LAN Certification and Follow-up Program.
- G. Cords shall be equipped with an eight-pin modular connector on each end, wired straight through and shall be of appropriate length for application.
- H. All rated patch cords shall be round, and consist of #23 AWG copper, stranded conductors, tightly twisted into individual pairs.
- I. Patch cords shall be made and warranted by the manufacturer of the cabling system installed in this project and shall meet or exceed patch cord specifications as outlined in TIA standards.

## **2.7 IDENTIFICATION (LABELING) SYSTEM**

- A. Refer to sections 270000 and 271300.

## **PART 3 - EXECUTION**

### **3.1 EXAMINATION**

- A. Refer to Section 270000 and 271300

### **3.2 PREPARATION**

- A. Refer to section 270000
- B. The Contractor shall check pathways, raceways, and other elements for compliance with space allocations, installation tolerances, debris, hazards to cable installation, and other conditions affecting installation prior to installation.

### **3.3 INSTALLATION REQUIREMENTS**

- A. Refer to section 270000
- B. All installation shall be done in conformance with ANSI/TIA-568-C standards, BICSI methods, industry standards and manufacturer's installation guidelines.
  - 1. The Contractor shall ensure that the maximum pulling tensions of the specified distribution cables are not exceeded and cable bends maintain the proper radius during the placement of the facilities.
  - 2. Failure to follow the appropriate guidelines shall require the Contractor to provide in a timely fashion the additional material and labor necessary to properly rectify the situation.
  - 3. This shall also apply to any and all damages sustained to the cables by the Contractor during the implementation.
- C. Install cable using techniques, practices, and methods that are consistent with specified data cabling and the installed components and that ensure specified performance levels of completed and linked signal paths, end to end.
  - 1. Pull cables in smooth and regular motions using methods that prevent cable kinking.
  - 2. Pull cables simultaneously if more than one is being installed in the same raceway/pathway.
  - 3. If necessary, use approved cable pulling lubricant
  - 4. Use fish tape, cable, rope, basket weave wire/cable grips, and other tools that will ensure no damage to the media or raceway.
  - 5. Install open cabling parallel and perpendicular to surfaces or structural members following surface contours where possible.

6. Do not bend cable greater than a bend radius of 0'-1".
- D. Provide a 10'-0" service loop at the communications room and shall provide a 3'-0" service loop above the access ceiling or cable trays unless specified otherwise.
  1. All service loops shall be a minimum of 1'-6" (18") in diameter and be accessible for maintenance.
- E. Coordinate loop placement and orientation with the technology consultant.
  1. This allows for future changes or expansion without installing new cables.
- F. Install cables in continuous "home run" lengths from work station outlet to specified patch panel.
  1. No intermediate punch down blocks or splices may be installed or utilized between the communications rooms and the workstation outlet without written Owner permission.
- G. All cable must be handled with care during installation so as not to change performance specifications.
  1. Factory twists of each individual pair must be maintained up to the connection points at both ends of the cable.
  2. There shall never be more than 0'-1/2" of unsheathed cable at either the wiring closet or the workstation termination locations.
- H. All cabling and associated hardware shall be placed so as to make efficient use of available space.
  1. All cabling and associated hardware shall be placed so as not to impair equipment's efficient use of their full capacity.

### 3.4 CABLING METHODS

- A. The Contractor shall provide cabling in accessible spaces, cable tray, (surface and/or enclosed raceway), conduits, and/or J-Hook cable support system.
  1. Within consoles, racks, cabinets, desks, and counters, in accessible ceilings spaces and in gypsum board partitions where open cable method may be used.
  2. Use UL or ETL listed plenum rated cable in all spaces.
  3. Provide all necessary installation materials, hardware, tools and equipment to perform insulation displacement type terminations at all data outlets, patch panels, and voice termination materials.
- B. Conceal raceway and cabling except in unfinished spaces as is practical.
- C. Exposed Cable
  1. All station cabling shall be installed inside walls or ceiling spaces whenever possible.
  2. Exposed station cable will only be run where indicated on the drawings and will only be allowed when no other options exist.
    - a) Owner must approve all exceptions.
- D. The Contractor shall utilize conduits/cable tray as indicated on the drawings.
- E. All cabling placed above drop ceilings must be supported by cable tray, J-hooks, caddy bags or conduit.
  1. The Contractor shall permanently affix cable supports to the building structure or substrates and provide attachment hardware and anchors designed for the structure to which attached and are suitably sized to sustain the weight of the cables to be supported.
    - a) Attaching cable to pipes or other mechanical items is not permitted.

- b) Cabling shall not be attached to ceiling grid wires.
  - 2. Multiple cables are to be dressed every 5'-0" to 7'-0".
    - a) Maximum cable sag between cable hooks is 3"-6".
- F. The Contractor shall route data and voice cables separately in a neat and orderly fashion.
  - 1. No cable ties or wraps shall be used to secure the cables in the runway outside of the communications rooms. Cable ties shall be rated for the environment.
- G. Keep all items protected before and after installation with dust and moisture proof barrier materials/envelopes.
- H. If wiring is terminated on patch panels, data, voice jacks prior to painting, carpet installation, and general finish clean up, these jacks shall be placed in a protective envelope to ensure dust, debris, moisture, and other foreign material do not settle onto jacks' contacts.
  - 1. Envelope will be removed on final trim out after other trades have completed their finish work.
  - 2. It shall be the Contractor's responsibility to ensure the integrity of these protective measures throughout the life/installation of the project.
    - a) Cable bundles brought into the communications rooms shall be routed and dressed in such a manner that prior to termination the cables are not subject to damage and misuse such as installers walking on the bundles that are on the floor.
    - b) Cable pulling force shall not exceed 25 lbs of pulling tension or cable manufacturer's recommended pulling tensions.
    - c) Do not leave cables on the floor unprotected or cable bundles hanging from the ceilings. Coil them up in a temporary manner and protect them from damage.
- I. Communications room cables shall be combed and dressed in a manner as to prevent twists, "braiding" and crossed cables in the cable bundle from the communication room entrance to the termination point at the rear of the patch panel.
  - 1. Behind the patch panel, the cable bundle shall be attached to the rear cable support bar, and shall drop out each cable in a neat, cascading manner to prevent crossed and/or interwoven cables to each patch panel port termination point.
    - a) Use Velcro wraps instead of cables ties for all bundling in the communications rooms.
    - b) Plastic/nylon tie-wraps are not allowed to permanently secure cables inside the communications room.

### **3.5 CABLING SEPARATION**

- A. Comply with TIA rules for separating unshielded copper communication and data-processing equipment cables from potential EMI sources, including electrical power lines and equipment.
- B. Maintain a minimum spacing of 1'-6" (18") from electrical feeders and/or branch circuit wiring including, but not limited to, light fixtures, sources of heat and EMI sources.
- C. Maintain a minimum spacing of 1'-0" from auxiliary systems cabling.
- D. Maintain a 1'-0" separation where cables must pass perpendicularly to electrical, plumbing, or other wiring, conduit, or piping systems.
  - 1. Use non-conduit bushings, if necessary to maintain separation, which allow for the addition of a reasonable number of cables in the future.

- E. Maintain communications pathways away from electrical apparatus such as motor driven equipment and transformers, minimum separation distance of 10'-0" is recommended.

### **3.6 CABLING TERMINATION**

- A. Terminate cables in consistent consecutive order.
- B. Terminate cables onto 8P8C modular patch panels without damaging twisted pairs or jacket.
- C. Arrange cables on patch panels and voice termination hardware in ascending order of room numbers and outlet numbers within rooms.
- D. Provide a 10'-0" service loop for horizontal cables at each rack in communications rooms.
  - 1. Locate loop at ceiling deck or on bottom of cable runway in minimum 1'-6" (18") diameter.
- E. Provide a 3'-6" service loop for horizontal cables at work area outlets. Locate service loop above or below data/voice outlet where vertical cable run transitions to horizontal run.
- F. Maintain twists in cable pairs to within 0'-1/2" of termination.
- G. Building Systems Cabling (BAS, FA, elevator line, etc)
  - 1. Coordinate exact placement and connectivity requirements with applicable trade prior to installation.
  - 2. Group all building systems cables in one group.
  - 3. Clearly label cable number and function, in the last positions on the horizontal cabling blocks in each communications room.
- H. Limit cable-bending radius to 20X the cable diameter during installation, and 15X the cable diameter after installation.
- I. Start numbering at the left of the main door to the room and continue in a clockwise direction around the room.
  - 1. The cables within the room will be terminated starting with the cables located to the left of the main door to the room and continue around the room in a clockwise direction.

### **3.7 TERMINATION HARDWARE**

- A. Station Hardware
  - 1. Flush mount jacks shall be mounted in a faceplate with back box.
  - 2. Outlets shall not be mounted on temporary, movable, or removable surfaces, doors, or access hatches without prior Owner approval.
  - 3. 8P8C Jack Pin Assignments for work area outlets shall match the T-568B wiring scheme.
- B. Patch panels
  - 1. Copper cables shall be terminated in eight position/eight conductor (8P8C) modular patch panels.
  - 2. All Modular jack panels shall match the T-568B wiring scheme.
- C. Work Area Outlet
  - 1. 8P8C non-keyed modular outlets for applications up to one Gbps and ANSI/TIA-568-C compliant for the specified transmission requirements
- D. Work Area Outlet Faceplates:
  - 1. Furnish and install blank plates in all unused ports.

### **3.8 SPECIAL CIRCUITS**

- A. The Contractor shall coordinate with the Owner on the cable termination plan for special circuits, including cables to wireless access point locations, security, elevators, fire alarms, etc.
- B. Wireless Access Points
  - 1. Install two (2) cable(s) from dedicated wireless patch panel(s) in communications room to outlets having 8P8C connectors within a BISCUIT box.
  - 2. Enclosures shall be NEMA rated for the environment to which they are exposed.
  - 3. 30'-0" of cable slack shall be coiled and hung on a "J"-hook at the enclosure location.

### 3.9 IDENTIFICATION AND LABELING

- A. Labeling system shall consist of a hand-held portable printer and labels appropriate to the application. Handwritten labels are not acceptable.
- B. Fiber termination hardware (designation strip) shall have a 0'-3/4" x 0'-1/4" thermal transfer printable label with a permanent acrylic adhesive
- C. All labels shall be permanent and shall not fade, peel, or deteriorate due to environment or time.
- D. The Contractor shall provide a copy of the finalized plan in writing to the Owner representative and DataCom Design Group for review and authorization to proceed.
  - 1. Coordinate with Owner for specifications on labeling of all hardware, cabling, and related equipment prior to any testing.
- E. Labeling requirements:
  - 1. Label cable terminations on designation strips
  - 2. Label all cable at each terminating point.
  - 3. Label each port of the work area outlet.
  - 4. Cable identification numbers shall not be duplicated.
  - 5. Label patch panels in the communications rooms to match those on the corresponding voice and data outlets.
    - a) The font shall be at least 0'-1/8" in height.
  - 6. Where a wireless access point is installed above an acoustical ceiling, label the ceiling grid frame below the access point, displaying the data port number and, if applicable, the access point identification number. Coordinate labeling of grid with Owner and Architect prior to application of labels.
  - 7. Label each distribution rack, block and other terminating equipment unit and field within that unit within 0'-4" from the block or patch panel termination. Keep labels in a neat and orderly lineup.
  - 8. Label each connector and each discrete unit of cable-terminating and connecting hardware within connector fields, in wiring closets and equipment rooms.
    - a) Where similar jacks and plugs are used for both communication and data-processing equipment, use a different color for jacks and plugs of each service.
  - 9. Post the cable schedule in a prominent location in each wiring closet and equipment room. List incoming and outgoing cables and their designations, origins, and destinations.
- F. Location and termination field description
  - 1. Room location
  - 2. Rack-mount or Wall mount
  - 3. Termination field type

- a) Specific patch panel ports versus a separate dedicated patch panel
- G. Unique identifiers
  - 1. Segregation and position on equipment rack
  - 2. Port color-coding
  - 3. Unique labeling
- H. Documentation
  - 1. Provide electronic copy of final comprehensive schedules for project in software and format selected by Owner.
    - a) All labels shall correspond to as-built drawings and to final test reports.
  - 2. All cable inventory data documentation shall be submitted in format coordinated with and approved by Owner so that data can be incorporated into existing databases.
  - 3. Documentation shall include cable identification number, source and destination, type of cable, length of cable and number of pairs or fibers.
  - 4. Complete cross connect documentation is required.

### **3.10 FIELD QUALITY CONTROL**

- A. Refer to section 270000

### **3.11 POST-INSTALLATION TESTING**

- A. Contractor shall test each pair or strand of every cable prior to acceptance. (100% PASS)
- B. Contractor shall submit acceptance documentation as defined below. No cabling installation is considered complete until test results have been completed, submitted and approved.
- C. Standards Compliance and Test Requirements:
  - 1. Cabling shall meet ANSI/TIA-568-C.2 Category 6 Horizontal cabling requirements.
- D. Attenuation, NEXT, PSNEXT, Return Loss, ELFEXT, and PSELFEXT data that indicate the worst case result, the frequency at which it occurs, the limit at that point, and the margin.
  - 1. These tests shall be performed in a swept frequency manner from 1 MHz to highest relevant frequency, using a swept frequency interval that is consistent with TIA and ISO requirements.
  - 2. Information shall be provided for all pairs or pair combinations and in both directions when required by the appropriate standards.
  - 3. Length, propagation delay, and delay skew relative to the relevant limit.
    - a) Length, propagation delay, and delay skew shall be tested relative to the relevant limit.
    - b) Test shall also include mutual capacitance and characteristic impedance.
      - 1) Any individual test that fails the relevant performance specification shall be marked as a 'FAIL'.
- E. Cable Test Documentation:
  - 1. Cable test documentation shall be submitted in hard copy and electronic formats.
    - a) If proprietary software is used, disk or CD shall contain any necessary software application required to view test results.

- b) Electronic reports shall be accompanied by a Certificate signed by an authorized representative of the Contractor warranting the truth and accuracy of the electronic report.
  - c) Certificate shall reference traceable circuit numbers that match the electronic record.
- 2. Each test record shall contain the cable ID as follows:
  - a) "MEDIA TYPE – SOURCE ROOM – DESTINATION ROOM – STRAND/PAIR #", e.g. MM-MC-HC23-001.
- 3. Test results saved within the field-test instrument shall be transferred into an accessible database utility that allows for the maintenance, inspection and archiving of the test records.
  - a) These test records shall be uploaded to the PC unaltered, i.e., "as saved in the field-test instrument".
  - b) The file format, CSV (comma separated value), does not provide adequate protection of these records and shall not be used.
- 4. Test reports shall include the following information for each cabling element:
  - a) Wire map results that indicate that 100% of the cabling has been tested for shorts, opens, miss-wires, splits, polarity reversals, transpositions, presence of AC voltage and end-to-end connectivity.
  - b) Length, propagation delay, and delay skew relative to the relevant limit. Any individual test that fails the relevant performance specification shall be marked as a FAIL.
  - c) Cable manufacturer, cable model number/type, and NVP
  - d) Tester make & model, serial number, hardware version, and software version.
  - e) Cable ID and project name
  - f) Auto-test specification used
  - g) Overall pass/fail indication
  - h) Date of test

F. Cable Test Equipment

- 1. Contractor shall supply all of the required test equipment used to conduct acceptance tests.
- 2. Test equipment used under this contract shall be from manufacturers that have a minimum of 5 years experience in producing field test equipment. Manufacturers shall be ISO 9001 certified.
- 3. Testing equipment shall be UL-verified to meet Level III accuracy.
  - a) The cable installers shall have a copy of this reference in their possession and be familiar with the contents.
- 4. Testing equipment shall be within the calibration period recommended by the manufacturer.
- 5. Testing equipment shall have the latest software and firmware installed.
- 6. Testing equipment of a given type shall be from the same manufacturer, and have compatible electronic results output.
- 7. Test adapter cables shall be approved by the manufacturer of the test equipment.
  - a) Adapter cables from other sources are not acceptable.
  - b) Adapter cables must be replaced after 1000 tests to ensure accuracy.

8. Test equipment must have a dynamic range of at least 100 dB to minimize measurement uncertainty.
9. Test equipment must be capable of storing full frequency sweep data for all tests and printing color graphical reports for all swept measurements.
10. Test equipment must include S-Band time domain diagnostics for NEXT and return loss (TDNXT and TDRL) for accurate and efficient troubleshooting.
11. Test equipment must be capable of running individual NEXT, return loss, etc measurements in addition to auto tests. Individual tests increase productivity when diagnosing faults.
12. Test equipment must include a library of cable types, sorted by major manufacturer.
13. Test equipment must be able to internally group auto tests and cables in project folders for good records management.
  - a) Test equipment must store at least 1000 auto tests in internal memory.
14. Test equipment must include DSP technology for support of advanced measurements.
15. Test equipment must make swept frequency measurements in compliance with TIA standards.
16. The measurement reference plane of the test equipment shall start immediately at the output of the test equipment interface connector.
17. There shall not be a time domain dead zone of any distance that excludes any part of the link from the measurement.
18. Acceptable testers:
  - a) Fluke DTX CableAnalyzer
  - b) Owner approved equivalent

### **3.12 CLEANING**

- A. Refer to section 270000

### **3.13 ACCEPTANCE**

- A. Once all work has been completed, test documentation has been submitted and approved, and the Owner is satisfied that all work has been completed in accordance with contract documents, the Owner will notify Contractor in writing of formal acceptance of the system.
- B. Contractor's RCDD shall warrant in writing that 100% of the installation meets the requirements specified herein.
- C. Acceptance shall be subject to completion of all work, successful post-installation testing which yields 100% PASS rating, and submittal and approval of full documentation as described above. Tests with the "\*" PASS" (asterisk) will not be acceptable.
  1. These circuits must be repaired to meet "PASS".

END OF SECTION 271500



**SECTION 28 00 00**  
**ELECTRONIC SECURITY**

**PART 1 - GENERAL**

**1.1 PROJECT SUMMARY/OVERVIEW**

- A. This document covers the general requirements for work to be performed to provide electronic security and surveillance.
- B. The contents of this document, along with related drawings and other documentary material, are critical to the security of this project and Owner and shall remain secure and confidential.
  - 1. Confidential information shall not be deliberately or inadvertently disclosed to anyone other than the Contractor's personnel and subcontractors who require disclosure to perform their portion of the work.
  - 2. This confidential information shall be tracked to ensure that copies are accounted for and properly destroyed when no longer needed to perform the work.
- C. The security systems shall consist of the following integrated subsystems as specified herein:
  - 1. Electronic Access Control and Intrusion Detection
  - 2. Video Surveillance
  - 3. Emergency Intercommunications and Duress
  - 4. Wire and Cable
- D. Provide complete turnkey systems with the exception of those items noted within this specification as being provided by others.
- E. Related Sections include:
  - 1. Section 087100 Door Hardware
  - 2. Section 260000 Electrical (including related sub-sections)
  - 3. Section 270000 Communications (including related sub-sections)
  - 4. Section 281000 Electronic Access Control and Intrusion Detection
  - 5. Section 282300 Video Surveillance
  - 6. Section 283100 Fire Alarm and Smoke Detection

**1.2 GENERAL REQUIREMENTS**

- A. Upon completion of commissioning testing and Owner acceptance, DBR bears no liability or responsibility for the continued proper operation of the installed systems.
- B. The Items described herein shall not be modified or substituted without consent of DBR and/or the Owner.
- C. Electronic security systems integrator (security subcontractor) manager/supervisor shall attend meetings arranged by the Contractor, Architect, Owner or other parties affected by the work of this Section 280000.
- D. If the manufacturer of security devices or connecting hardware has supplied post manufacture performance data, copies of such are to be kept for inclusion in the documentation and made available to the Owner upon request.
- E. All materials are to be new unused and of the latest series of model number, unless otherwise indicated by the Owner or security system designer.
- F. All security integrator personnel must be manufacturer certified and capable of an installation that falls under the manufacturer's guidelines necessary to obtain a manufacturer warranty.
  - 1. The integrator shall provide all components/materials essential for a complete and functional security access and surveillance system.

- G. Security integrator shall issue a two (2) year warranty on installation and workmanship.
- H. These Specifications and Drawings are intended for bidding purposes only, No part shall be copied or used for any purpose other than bidding on this project.
  - 1. This package shall be contractual upon bid award.
- I. Drawings and Specifications are to be used in conjunction with one another and to supplement one another.
  - 1. In general Specifications determine the nature and quality of the materials and tests, and drawings establish the quantities, details and give characteristics of performance that should be adhered to in the installation of the security system components.
  - 2. If there is an apparent conflict between the drawings and specifications, or within the specifications themselves, the items with greater quantity or quality shall be estimated and installed.
  - 3. Clarification with the Owner/Designer about these items shall be made prior to purchase and installation.
  - 4. Questions regarding the Specification or system requirements should be directed in writing to DBR or the Owner.
- J. Security integrator shall adhere to Division 1 general requirements and written security Specifications and Drawings within this construction package and shall be responsible for complying with all local, state and federal laws or regulations applicable to the work being performed, even though said law, rule or regulation is not identified herein.
- K. Security integrator shall arrange and pay for any inspections required by the public agencies having jurisdiction in the area.
- L. The security contractor shall procure and maintain for the duration of this agreement, insurance against claims for injuries to persons or damages to property which may arise from, or conjunction with, the performance of the work hereunder by the security integrator, his agents, representatives, or employees.
  - 1. The security integrator shall pay the cost of such insurance.
- M. The security integrator will respect and protect the privacy and confidentiality of the Owner, his employees, processes, products, and intellectual property to the extent necessary, consistent with the legal responsibilities of the State of Texas and the Owner.
- N. If required the security integrator shall sign a non-disclosure agreement and abide by its requirements to keep confidential all information concerning bid documents and this Project.
- O. Furnish submittals and manuals in accordance with Division 1.
- P. Furnish a detailed material list complete with suppliers (distributors) list of components and distributors name, address, and phone number.
- Q. Refer to Specifications issued by Architect, Division 1, for Project and cost payments.

### **1.3 REFERENCES**

- A. The publications listed below form a part of this Specification. The publications are referred to in the text by basic designation only.
- B. Specific reference in Specifications to codes, rules, regulations, standards, manufacturer's instructions, or requirements of regulatory agencies shall mean the latest printed edition of each in effect at the date of contract unless the document is shown dated.
- C. For conflicts between referenced requirements and contract documents comply with the one that is more stringent.

1. Federal, State, and Local codes, regulations and ordinances
  2. NFPA 101: Life Safety Code
  3. NFPA 72: National Fire Alarm Code
  4. NFPA 730: Guide for Premises Security
  5. NFPA 731: Standard for the Installation of Electronic Premises Security
  6. National Electric Code (NEC), latest edition
  7. Building Codes (UBC) (IBC), latest editions
  8. Occupational Health and Safety Act (OSHA)
  9. Americans with Disabilities Act (ADA)
  10. Local Governing Authorities Having Jurisdiction
  11. Underwriters Laboratory (UL) Applicable Standards for Safety and Security
  12. Institute of Electrical and Electronics Engineers (IEEE) Applicable Standards
  13. Telecommunications Industry Association (TIA) Applicable Standards
  14. Client standards for Security Systems
- D. Related Documents
1. Security Drawings
  2. General provisions of contract
  3. Uniform general conditions
  4. Supplementary general conditions
  5. Architectural plans & specifications
  6. Requirements of Division 1
  7. Electrical / Mechanical / Telecommunications specifications and plans.

#### **1.4 DESCRIPTION OF SYSTEM WORK**

- A. Furnish and install all materials, tools, equipment, and services for all electronic security/surveillance devices to provide functioning systems in accordance with performance requirements specified and any modifications resulting from reviewed shop and field coordinated drawings.
- a) Access Control System
  - b) This system replaces the typical mechanical key controlled door lock with a door locking system that uses an access card as the access credential.
  - c) The system includes an electric door-locking mechanisms, card reader located adjacent the door, door status sensor, door prop alarm and a request to exit device.
  - d) Typical system configuration is card or schedule controlled entry with free exiting.
2. Intrusion Detection System
- a) This system monitors areas for unauthorized entrance or intruder.
  - b) This system can consist of motion sensors, door status sensors, glass break sensors and one or more control keypads.
  - c) The keypad is used to arm/disarm system by entering a numeric code on the keypad.
3. Video Surveillance System
- a) This system is used to provide video surveillance through the use of cameras of security sensitive areas and target items.
  - b) The system shall allow for the viewing and recording of images.
4. Emergency Intercommunications and Duress Systems
- a) Duress Buttons
    - 1) These buttons, also known as panic buttons, are installed in locations where potential personal safety or security threats exist.

- 2) Depressing the button sends a silent priority alarm signal to assigned monitor with location and specific alarm information
- 3) The panic button is usually located in the knee space underneath a desk or service counter.

**B. RACKS AND ENCLOSURES**

1. Wall mounted enclosures, data gathering panels, and power supply panels shall be installed as per manufacturer's requirements.
  - a) Coordinate pathways and power with Electrical and Telecommunications Contractors
  - b) Furnish all labor, materials, tools, equipment, and services for all control consoles, equipment racks, cabinets, and enclosures not provided by others in accordance with contract documents.
  - c) Completely coordinate with work of other trades to avoid duplication in purchasing.
  - d) Although such work is not specifically indicated, furnish and install all supplementary or miscellaneous items, and devices incidental to or necessary for a sound, secure and complete installation.
2. The installation of the relay racks/cabinets for Electronic Surveillance shall be by the Telecommunications Contractor.
  - a) Coordinate locations with G.C.
3. The designated security space will provide an area reserved for rack and wall mounted security equipment.
  - a) The rack area allows for vertical relay rack(s).
  - b) Backboard wall area of 8'-0" X 8'-0" shall be reserved for wall-mounted components.
  - c) Cable tray/ ladder shall be by the telecommunications contractor and is provided to facilitate cable access into both wall and rack mounted equipment.

C. Provide all supplementary or miscellaneous items and devices incidental to or necessary for a sound and complete installation.

D. Drawings are representative and show general arrangement of systems and equipment, except when dimensioned or detailed.

1. For exact locations refer to dimensioned architectural drawings.
  - a) Field measurements take precedence over dimensioned drawings.
  - b) Field verify locations and arrangement of all systems and equipment.
  - c) Coordinate all work with other trades and Contractor.

**E. Circuit Supervision**

1. Supervise all signal and data transmission lines, links with other systems, and sensors.
  - a) Indicate circuit and detection device faults with both protected zone and trouble signals.
  - b) Initiate an alarm in response to opening, closing, or shorting of a signal or data transmission line.

F. Electronics systems work as specified in this Section and Sections 281000, 282300, 282600 shall include:

1. A project kick-off/pre-submittal meeting with the Architect, Designer, and Contractor to review security design package.
  - a) Additional participants shall include:

- 1) Division 8 subcontractors
- 2) Division 26 subcontractors
2. Preparation of pre-installation submittals, including point-to-point wiring information for security equipment to interface to work by others prior to start of any installation work. Include lock permit requests in submittals for review.
3. Furnishing and installation of all security devices, components and accessories.
4. The furnishing and coordination on installation of special back boxes for security equipment and field devices as required.
5. Furnishing, installation and termination of all copper wiring and cabling including any special purpose wire and cable for electronic security systems.
  - a) Coordinate all network and fiber optic cable interface provided by telecommunications subcontractor.
6. Coordinate raceway and power distribution systems provided by Division 26.
7. Provide and install 12/24 VAC/DC input power to all field devices as required.
8. Coordination with other trades and Owner required to facilitate the installation of the security equipment including:
  - a) Division 08 (doors)
  - b) Division 26 (power, raceways, and fire alarms)
  - c) Division 27 (telecommunications network interface).
9. Wiring and termination of electrified door hardware by security subcontractor shall be concurrent with the installation of these electrified components by the door hardware subcontractor.
10. Programming of all security control equipment and prior coordination with the Owner's security and telecommunications personnel.
11. Preparation of "As-Built" documentation.
12. Warranty service for completed work.

## 1.5 SUBMITTALS

- A. Refer to Requirements of Division 1.
- B. Pre-Installation Submittal Requirements
  1. Submittals for electronic security shall be complete and submitted at the same time.
    - a) No partial submittals will be accepted for review.
    - b) Allow 2 weeks for consultant review of submittals.
  2. General Requirements
    - a) A functional description of each system.
    - b) All cable and wiring types for each device type used.
    - c) Certification that lock wiring and access control systems requirements have been coordinated with electrified door hardware, fire alarm systems, automatic door controls, and overhead door controls specified in other sections and other packages.
    - d) Power supply points listing with devices and maximum loads to prevent overloading.
    - e) Battery backup calculations to show load and back-up times for UPS and power supplies with batteries.
    - f) Equipment schedules listing all system components, manufacturer, model number and quantities of each.
    - g) Qualifications and proof of work history (with references).
  3. Product Data Cut-sheets
    - a) Complete manufacturer's technical data including manufacturer warranty information, descriptive literature, illustrations, and installation

instructions for all components included within this project indicating compliance with applicable referenced standards, size, dimensions, model number, electrical characteristics, support requirements, connection requirements and all applicable information verifying that submitted components comply with Contract Documents.

4. Shop Drawings

- a) Floor plans necessary to identify specific device locations, cable routes and quantities, cable types, riser locations, and references to installation details and diagrams.
- b) Riser diagram showing routes between floors or other areas that are not easily identified on the floor plans.
- c) Security One-line diagrams showing all input and output points of the system.
  - 1) The Contractor shall make any corrections required by the consultant team, file with him two corrected copies and furnish such other copies as may be needed.
  - 2) The consultant's approval of such drawings or schedules shall not relieve the Contractor from responsibility for deviations from drawings or specifications, unless he has in writing called the Architect's attention to such deviations at the time of submission, nor shall it relieve him from responsibility for errors of any sort in shop drawings or schedules.

5. Warranty

- a) The Contractor shall provide the appropriate documentation to comply with the requirements described in the WARRANTY section.

6. Qualifications

- a) The Contractor shall provide the appropriate documentation to comply with the requirements described in the QUALITY ASSURANCE section.

C. As-Built drawings shall be in current AutoCAD format, same version as used by the Architect.

- 1. Dimensions and scale of the drawing sheets submitted shall match the size of the drawing used for the contract documents, and shall include the following.
  - a) Utilize normally recognized drafting procedures that match AutoCAD standards, Architect, and Designer guidelines and methodology.
  - b) The As-Built drawings shall incorporate all changes made to the building identified in, but not limited to, Addenda, contemplated change notices, Site Instructions or deviations resulting from site conditions.
  - c) Dimensioned plan and elevation views of all security components.
  - d) Cable routing paths of security cables to identified infrastructure pathways.
  - e) All rack, cabinet, and enclosure locations and labeling thereof.
  - f) One-line diagrams of equipment/device interconnecting cabling of the security systems.
  - g) Standard or typical installation details of installations unique to Owner's requirements.
  - h) Submit one soft and one hard copy with project deliverables within 30 days of project completion.

D. Security integrator shall provide three (3) paper copies and one (1) electronic copy (PDF format) of a properly indexed O&M Manual at the conclusion of the project, which will include, but not be limited to the following requirements:

1. Ring binder with project title, properly indexed, and contractor's name on cover and spine including:
  - a) Sequence of operations, design philosophy, and specific functions
  - b) System block diagram
  - c) Equipment list including:
    - 1) A brief description
    - 2) Model
    - 3) Total number of each item used in the project.
  - d) Camera schedule including:
    - 1) Number
    - 2) Location
    - 3) Camera model/manufacturer
    - 4) View
    - 5) Lens
    - 6) Power source
    - 7) Multiplexer/input
    - 8) Settings entered on site
  - e) Manufacturers' data sheet and O&M manual for associated equipment.
  - f) Maintenance requirements for equipment, inspections and preventative maintenance schedules.
  - g) As-built drawings for each floor plan layout and rack and wall elevation layouts. Each drawing shall show:
    - 1) Cable type and identifier
    - 2) Actual cable routing pathway
    - 3) Device number (camera, etc.),
    - 4) Device input/output number.
  - h) Final test data (measured video levels, day & night camera snapshots in JPEG format and other significant operating parameters).
  - i) List of system associated mechanical locking keys with key codes and tamper resistant hardware types.

## 1.6 QUALITY ASSURANCE

- A. Electronic security systems integrator (security subcontractor) shall meet the following minimum requirements.
  1. Maintain a valid Type B license from the Texas Private Security Bureau.
  2. Have successfully completed three (3) projects of similar size and complexity that have been in proper operation for a period of one (1) year.
  3. Technicians shall be factory trained and certified in specified systems.
  4. The Project manager and supervising/lead technician shall have been regularly engaged in the installation and testing of the products specified for not less than five (5) years and maintain manufacturer certification.
  5. The security integrator must maintain an operating facility in the local area (50 mile radius) of the Project location to provide service to the Owner for the warranty period.
    - a) At the Owners request for service, the security integrator shall dispatch a service technician to the location to affect the required repairs or adjustments.
  6. The contractor shall maintain a spare parts inventory necessary to resolve component failures of the system.

- a) Refer to individual specification section for a list of specifically required parts provided to the owner and stored on site. These parts will become the property of the owner.
      - 1) At the end of the warranty period the security integrator shall test the owner's spare parts and repair or replace as needed to bring the parts up to proper operation.
  - 7. A BICSI RCDD shall approve all on-site work as a recognized member of the Contractor's installation team.
    - a) All installation team members must demonstrate knowledge and compliance with all BICSI, TIA/EIA, UL, and NEC methods, standards and codes.
- B. Security integrators desiring approval must comply with Division 1 requirements.
- C. Security integrator must be cognizant of site conditions, verify locations of new and existing equipment, and determine exact requirements for connection and interface.

### **1.7 PRE-INSTALLATION MEETINGS**

- A. Attend and/or arrange a scheduled pre-installation conference prior to beginning any work of this section.
  - 1. Agenda
    - a) This venue is to ask and clarify questions in writing related to work to be performed, scheduling, and coordination with the Project manager/Owner representative and consultant.
  - 2. Attendance
    - a) The security project manager/supervisor shall attend meetings arranged by General Contractor, Owner's representatives, and other parties affected by work of this document.
    - b) All individuals who will be installers of the electronic security system and equipment in an on-site supervisory capacity, including project managers and lead installers, shall be required to attend the pre-installation conference.
    - c) Individuals who do not attend the conference will not be permitted to install, or supervise the installation of, any component of the security system.
      - 1) This includes supervisors, project managers, and lead installers of this project.

### **1.8 POST INSTALLATION MEETINGS**

- A. At the time of substantial completion the contractor shall call and arrange for a post installation meeting to present and review all submittal documents to include but not be limited to As-Built Drawings, Warranty paperwork, etc.
  - 1. Attendees to be invited shall include:
    - a) Project manager/Owner representative
    - b) DBR Inc.
    - c) General Contractor
    - d) Other trades that the GC deems appropriate.
  - 2. At this meeting the contractor shall present and explain all documentation, asking for feedback on its completeness.
  - 3. Any discrepancies or deviations noted by and agreed to by participants shall be remedied by the contractor and resubmitted within one week of the meeting.

### **1.9 DELIVERY, STORAGE AND HANDLING**



- A. Equipment and components shall be delivered properly protected and undamaged with original containers, packaging, and labels intact.
- B. Store, handle, and protect all related materials and equipment in accordance with Manufacturer's recommendations.
- C. Provide additional protection during handling as necessary to prevent breaking, scraping, marring, or otherwise damaging products or surrounding areas.
- D. Equipment and components shall be protected from the weather, humidity, temperature variations, dirt, dust, or other contaminants.
  - 1. Equipment damaged prior to system acceptance shall be replaced at no cost to the owner.
- E. Protect all equipment and components that are to be installed from theft, vandalism, or use by unauthorized persons.

#### **1.10 PROJECT/SITE CONDITIONS**

- A. Security integrator is responsible for conducting a site survey prior to the commencement of work to determine locations of all existing security devices and verify the proposed locations of the new components to be installed.
- B. Security integrator will coordinate all work through the Contractor and schedule work to cause as little interference or interruption of existing services as possible.
- C. Security integrator will arrange and pay for all necessary permits, licenses, and inspections.
  - 1. Security integrator shall prepare all information necessary to obtain a permit for Electronic Locking Mechanisms in compliance with the Owner requirements.
- D. Verify with Division 26 installer all conduits and special back box requirements in a timely manner.

#### **1.11 WARRANTY**

- A. See requirements in Division 1 Specifications.
- B. The Security Integrator shall warrant all completed work, including all materials and labor, to be free from defects in design, workmanship, and/or materials for a period of two (2) years from final acceptance date.
  - 1. System acceptance is defined as the completion of all functional performance testing and the resolution of all punch list items.
- C. Warranty Service
  - 1. In the event that defects in the materials and/or workmanship are identified during the warranty period, the contractor shall provide all labor and materials to correct the deficiency.
  - 2. All service work shall be performed by factory certified technicians.
  - 3. All warranty service shall include the replacement of all parts and or components as required to restore normal system operation.
    - a) If parts or components need to be repaired, a loaner will be supplied and installed until the part or component can be repaired and reinstalled.
  - 4. Immediately following a warranty service request, the Contractor shall provide written documentation to Owner which details the service work completed, cause of trouble, and any outstanding work required to restore a complete and normal system.
- D. Warranty service requests shall be responded to within 4 hours of notification with a qualified service technician on site.
- E. All repairs shall be completed within 48 hours upon site arrival.

1. If the failure exceeds 48 hours, the Owner reserves the right to require the contractor provide on-site manufacturer support at no additional cost to Owner.
- F. Extended warranties on equipment components offered by the manufacturer shall be passed through to the Owner.
  1. Warranty provisions shall be fully transferable only at the direction of the Owner, in the event that ownership of the installed security systems is transferred.

#### **1.12 SYSTEMS STARTUP AND TRAINING**

- A. After all systems have been tested, accepted and turned on for operation, the Security integrator shall provide "User Training" to Owner personnel.
  1. The onsite training shall cover all newly installed electronic security components, devices and systems. The training classes shall total a minimum of twenty (20) hours for up to eight (8) people of the Owner's choosing.
  2. Two (2) separate training sessions will be conducted, one for system operators and one for system administrators.
  3. The contents of the manuals will include:
    - a) Title page with subject, system name, owner's name, and an owner approved confidentiality notice.
    - b) Table of contents.
    - c) Manual that details system and sub-system operation.
    - d) Manuals that details system administration procedures and tasks.
    - e) Manuals that fully detail all programming commands.
  4. Provide ten (10) Bound hardcopy System Operation training manuals and one electronic copy (PDF format).
  5. Provide two (2) Bound hardcopy System Administration training manuals and one electronic copy (PDF format).

### **PART 2 - PRODUCTS**

#### **2.1 MANUFACTURERS**

- A. Acceptable Manufacturer's are shown in individual specification sections.
- B. Equipment manufacturers and model numbers indicated in individual specification sections are identified as minimum equipment requirements.
- C. All substitutions shall meet or exceed these minimum requirements and must be approved by the Owner/Architect prior to purchase.
- D. All manufacturers' equipment shall be available through a nationally recognized supplier network.

#### **2.2 EQUIPMENT**

- A. Provide security fasteners on all equipment, device plates, etc. within public areas.
  1. Allen head with center pin, hardened steel.
  2. Provide four (4) fastener tools to Owner.
- B. Equipment installed in exterior applications shall be fitted with fasteners and exposed surfaces of stainless steel or other corrosion resistant material.
- C. All materials and equipment used must be new and unused, prime quality products.
- D. All equipment or components installed on the exterior of a building where the equipment is subject to adverse weather/elements shall be enclosed in weatherproof enclosures.

#### **2.3 WIRE AND CABLE**

- A. All wire and cable shall be U.L. approved for its intended application and shall meet or exceed manufacturer's recommendations for the components connected.
- B. All conductors and cable shall meet individual security system manufacturer specifications.
  - 1. Provide shielded conductors and cable as required by the manufacturer or as required to provide for interference-free signals.
  - 2. Color coding shall be accomplished by using solidly colored insulation.
    - a) Grounding conductors, where insulated, shall be colored solid green or identified with green color as required by NEC.
- C. Increase conductor sizes on cables as required to be consistent with circuit current ratings, length of wire runs, and manufacturers' recommendations.
  - 1. Alarm device field wiring shall be in accordance with the equipment manufacturer's specifications.
  - 2. Low voltage power circuits shall use conductors as required by the equipment manufacturer's specifications.
  - 3. Plenum rated cable shall be used as required by code.
- D. UTP Structured Cabling Systems for IP cameras and intercoms (including pulling, terminating, and testing) by Division 27 Telecommunications contractor.
  - 1. Intra-building data communications circuits shall utilize UTP cable as specified in Telecommunications specifications.
- E. Patch Cables
  - 1. Provide pre-manufactured patch cables (cable, connectors, boots, etc.) as required to connect security systems to voice and data communication outlets.
  - 2. Patch cables shall be certified for their specific use to meet or exceed applicable industry specifications.
  - 3. Provide cable lengths as necessary to neatly route cables through cable management systems and other cable organization systems.
  - 4. Provide connectors as required for proper termination.
    - a) Provide boots for connectors where applicable to prevent snagging.
- F. The minimum conductor sizes are for distances as per the manufacturer's specifications from security device to security panel.
  - 1. The contractor shall size the conductor accordingly for longer runs.
  - 2. Minimum Conductor and Cable Types and Sizes.
    - a) Alarm device field wiring shall be 18/20 AWG stranded copper conductors.
    - b) Low voltage power circuits will use 18 AWG stranded copper conductors.
      - 1) Increase conductor gauge consistent with circuit current requirements.

### **PART 3 - EXECUTION**

#### **3.1 INSTALLATION**

- A. All personnel working on this project shall be experienced, highly skilled installers with a minimum of three (3) years work on similar type projects.
- B. Changes in location of any work require the written approval of the Architect/Owner prior to initiation.
- C. Changes in indicated sizes shall not be made without the written approval of the Owner/Architect.

- D. Install all equipment in accordance with manufacturer's recommendations.
- E. All systems shall be designed and installed to provide 24 hour a day, 7 days a week operation.
- F. Primary pathways
  - 1. All security cabling run from rack/enclosure head-end equipment to security devices shall follow primary telecom routing pathways.
  - 2. Security wire non-UTP cabling shall be kept separated from the data cabling
  - 3. Security wire non-UTP cabling shall be routed in bridge rings secured to the outside of the telecom tray where applicable.
    - a) Arlington loops or J hooks shall be used where telecom pathways are not present
  - 4. Provide all necessary anchoring devices and supports.
    - a) Use structural supports suitable for equipment, or as indicated.
    - b) Check loading and dimensions of equipment with shop drawings.
    - c) Do not cut or weld to, building structural members.
- G. Secondary pathways
  - 1. Arlington loops or J hooks shall be used for secondary pathways
  - 2. Security wire non-UTP cabling shall be kept separated from the data cabling
  - 3. Provide all necessary anchoring devices and supports.
    - a) Use structural supports suitable for equipment, or as indicated.
    - b) Check loading and dimensions of equipment with shop drawings.
    - c) Do not cut or weld to, building structural members.
- H. Coordinate extension and connection to commercial, emergency/UPS power circuits provided by Division 26.
  - 1. Make power connections in accordance with Division 26.
- I. Shielded and/or screened cables shall be grounded per the hardware manufacturer's instruction.
  - 1. Single point shield grounds shall be grounded at the field panel feeding the device or sub panel and insulated from ground at the termination end of the cable.
- J. All installation of security systems shall be complete at least thirty calendar days prior to occupancy.

### **3.2 RACK AND CABINET INSTALLATION**

- A. Coordinate rack/cabinet installation with Telecommunications contractor and follow provisions in Section 281000.
- B. Rack/cabinet installation by Telecommunications contractor.
- C. After racks are installed, install all required components to support rack mounted security equipment.
  - 1. Extend UPS/emergency power to rack mounted equipment as required.
- D. Install all conduits, back boxes, wire and cable management as required for interconnection of security equipment, data gathering panels, power supply enclosures, and distribution panels in the Security room.
- E. Extend commercial/emergency/UPS power circuits as required to security components as required.
- F. Neatly lace and dress all cables in each rack.
  - 1. All wiring and cable shall be properly supported.

2. Utilize suitable cable management devices, no tie-wraps for UTP structured cabling allowed.

### 3.3 GROUNDING AND BONDING

#### A. Equipment Cabinets and Racks

1. To provide electrical continuity between rack elements, paint-piercing grounding washers shall be used where rack sections bolt together, on both sides, under the head of the bolt and between the nut and rack.
2. A horizontal busbar shall be installed at the top and back of each rack for floor fed cabinets/racks.
3. A vertical busbar shall be installed to the rear of the right-hand side rail with thread-forming screws to ensure metal-to-metal contact.
4. Each rack shall be provided with a minimum # 6 AWG insulated ground wire.
5. Do not bond racks serially (loop from rack to rack).
6. Each rack bay against a wall shall be bottom/side ground feeds from the wall.
  - a) Wall ground feeds/raceways to racks shall not be exposed on the walls.
  - b) Exception
    - 1) Some rack bays will require the ground to be fed from the ceiling raceway. Refer to drawings for details.
7. The Contractor shall provide a ground strap for each equipment rack and bond to the nearest Telecommunications Bonding Backbone (TBB) connection, Furnish all required bonding materials and hardware manufactured for this purpose.
  - a) Follow NEC bonding procedures/specifications.
8. All ground raceways within each rack shall be an insulated metallic flex type raceway and shall not interfere with equipment mounting frames or equipment mounting brackets.
9. Each ground feed shall provide proper installation allowances and penetration depths to provide conversion fittings from solid metallic to insulated metallic flex conduit raceways.
10. To bond each rack to ground, burnish clean a one square inch area, drill, tap, apply an adequate amount of antioxidant joint compound mixed for the metal surface types affected, and bolt connectorized conductor to burnished and compounded area.
  - a) Ensure proper conductivity.

#### B. Cable Runway, Cable Raceway and Support System Grounding

1. The Contractor shall provide communications cable tray and cable runway systems with a communications isolated ground from the TBB.
2. All cable tray needs to be electrically continuous per NEC 250.96.
  - a) Metal raceways, wire-mesh cable trays, cable armor, cable sheath, enclosures, frames, fittings, and other metal non-current-carrying parts that are to serve as an alternate grounding path, with or without the use of supplementary equipment grounding conductors, shall be effectively bonded where necessary to ensure electrical continuity and the capacity to conduct safely any fault current likely to be imposed on them.
  - b) Any nonconductive paint, enamel, or similar coating shall be removed at threads, contact points, and contact surfaces, and be connected by means of fittings designed so as to make good bonding points.
3. The Contractor shall provide and install #6 AWG insulated ground wire to bond one end of each cable tray/runway system to the #2/0 TBB.

4. For electrically non-continuous conduits that contain only grounding conductor, the Contractor shall bond the conduit and conductor together at both ends to ground to nearest TGB with grounding bushings or ground clamps.

### **3.4 Labeling**

- A. Provide labeling for all security equipment components using waterproof, self-adhesive computer printed labels.
  1. Coordinate with Owner on numbering/labeling scheme.
- B. Provide labeling for all security cable/wiring using waterproof, self-adhesive computer printed labels.
  1. Coordinate with Owner on numbering/labeling scheme.
  2. Label all cables/wiring on both ends.
  3. At multi conductor cable terminations label each conductor.
  4. At a minimum, each cable/wire label shall designate:
    - a) Origination Point
    - b) Alarm point description
    - c) Opening description (if applicable)
- C. Provide a complete cable/wire identification plan/list with project completion submittal.
- D. Conduit and junction box exteriors may be identified with unique color paint, but shall not be identified with written words that easily identify the function of the conduit and boxes.

### **3.5 POWER REQUIREMENTS**

- A. 120 VAC emergency power dedicated to security will be provided. (By Electrical Contractor)
- B. Back-up power for all equipment and devices shall be for at least 4 hours unless otherwise specified.
  1. When generator backup power is available, provide a UPS, rated to maintain the load for a minimum of 15 minutes for all 120VAC equipment.
- C. Rack-mounted Uninterruptible Power Supply (UPS)
  1. Provide a UPS to support 120% of the required load to allow for future load expansion and age related deterioration of the battery performance.
  2. The UPS interface port shall have an RS-232 communications port and a 10 Base-T Ethernet for LAN management.
    - a) Provide the necessary data connection, hardware and software to remotely monitor the UPS
    - b) Provide user configurable computer operating system shutdown capability
  3. The control panel shall have a LED status display for load and battery bar graphs in addition to replace battery and overload indicators.
    - a) Rack-mounted surge suppression shall be vertically mounted and made for this orientation.
- D. All electronic locks shall be 12/24VDC (By Division 08)
- E. Connect to AC power and provide UL listed power supplies and transformers to distribute low voltage power to the system components as required.
  1. Provide uninterrupted battery backup power for the duration required above.
- F. All equipment connected to AC circuits shall be protected from power surges.
  1. The devices shall be installed and grounded per manufacturer instructions.
  2. Equipment protection shall meet requirements of ANSI C62.41.

3. Fuses shall not be used for surge protection.
- G. All non-fiber optic data circuits that serve devices exterior to the buildings will be protected by surge protectors at the device and the termination.
  1. The devices shall be installed and grounded per manufacturer instructions.
  2. Equipment protection shall meet requirements of ANSI C62.41.
  3. Fuses shall not be used for surge protection.

### **3.6 Testing**

- A. Ensure that all provisions and requirements of this specification are met.
  1. Verify through inspections, demonstrations and tests.
- B. Perform required tests to demonstrate workmanship, operation, and performance.
  1. Conduct tests with Architect/Owner and if required, inspectors of agencies having jurisdiction present.
  2. Arrange test dates in advance and give all parties a minimum of 48 hours notice.
- C. Repair or replace equipment or systems found defective or inoperative and re-test until 100% satisfactory results are obtained.
- D. Verification inspections will be made of all equipment components and installations for proper functioning of locking hardware and lock controls, mounting/placement of sensors, and cameras, etc. to guarantee requirements of the Contract Documents are complied with.
  1. The Owner's quality control representative shall have the opportunity to witness all inspections, or to conduct installation inspections of his own.

### **3.7 Functional Performance Test**

- A. The Functional Performance Test (FPT) will be conducted at the end of the project and prior to system acceptance by the Owner.
  1. The security integrator will provide all necessary staff and communications needed to fully test all functions of the system.
  2. The contractor will submit for approval by the Architect and Owner, a comprehensive test plan that will include testing of every function on every door and security device thirty (30) days prior to the scheduled start of the test.
  3. The system will not be considered for acceptance prior to the successful completion of the FPT and completion of punch list items.
- B. Pre-Testing
  1. Following installation and prior to the FPT, the security integrator shall individually test each component and field device and verify the proper functioning of each component within a particular sub-system.
    - a) The contractor shall also test each sub-system until all detection zones, alarm assessment components, alarm reporting, surveillance and display components; along with access control functions have been verified.
    - b) Prior to the FPT all deficiencies must be corrected.
    - c) After sub-system verification is complete, test the entire system to assure that all elements and subsystems are compatible and function properly as a complete system.
- C. Upon completion of the outlined tasks and tests the security integrator shall schedule the FPT with the Architect and Owner.
  1. The security contractor must demonstrate that the security system components and sub-systems operate together as a system and meet specification requirements in the "As-Installed" operating environment.

2. On conclusion of the FPT the test report document will be submitted to the architect for approval.
3. The FPT will be observed by the architect's and Owner's representatives.
4. The FPT may be stopped at any time by these representatives if they believe the failure rate is too high or the system is not performing to contract document requirements.
5. The FPT will only resume when all deficiencies have been corrected.
6. Retesting will be required of all failed tests.

### **3.8 System Operational Test**

- A. Upon completion of the FPT, conduct a formal test to be known as the System Operational Test (SOT), in which all components and sub-systems of the security system are demonstrated to operate error and failure free together as a system.
  1. This test is to be performed over a continuous seventy-two (72) hour period.
  2. A formal test plan and test procedures shall be prepared by the security subcontractor and submitted to the Owner/Architect for approval.
  3. The Security integrator must demonstrate that the system components and sub-systems meet specification requirements in the "As-Installed" operating environment and operate error and failure free for the duration of the test.
  4. If a system failure does occur, the failure must be documented and repaired, after which the seventy-two hour SOT period will restart.
- B. In the event that the Owner, Architect, or Contractor are required to witness a retest at a later date because the Security integrator is not properly prepared to conduct the acceptance tests or because the systems being tested have failed such tests, which shall be solely determined by the Architect or Owner witnessing the tests, the cost of witnessing additional tests shall be borne exclusively by the Security integrator.
  1. Costs are to be based on time and materials at the established rates of the Architect or Owner.

**END OF SECTION 28 00 00**



SECTION 28 10 00  
ELECTRONIC ACCESS CONTROL AND INTRUSION DETECTION

**PART 1 - GENERAL**

**1.1 SUMMARY/OVERVIEW**

- A. This section provides specifications for the installation of Electronic Access Control (AC), Intrusion Detection (ID), and related components.
- B. Related Sections
  - 1. Section 087100 Door Hardware
  - 2. Section 260000 Electrical (including related sub-sections)
  - 3. Section 270000 Communications (including related sub-sections)
  - 4. Section 280000 Electronic Security
  - 5. Section 282300 Video Surveillance
  - 6. Section 283100 Fire Alarm and Smoke Detection

**1.2 REFERENCES**

- A. See Section 280000 Electronic Security.

**1.3 SYSTEM COORDINATION**

- A. The Security Integrator shall completely coordinate all relevant work of other trades/systems including, but not limited to:
  - 1. Door Hardware
  - 2. Fire Alarm System
  - 3. Electrical Systems(s)
  - 4. Telecommunications System(s)
- B. Electric Locking Mechanisms
  - 1. The security integrator and door hardware contractor shall coordinate all door hardware, door and door frame design.
  - 2. The security contractor shall verify all specified door hardware is appropriate for the security application and verify the sequence of operations for each access controlled opening.
- C. Fire Alarm and Life Safety
  - 1. The security integrator shall coordinate the access control system design with the life safety consultant to insure compliance with applicable codes and requirements.
  - 2. This includes, but is not limited to:
    - a) Fire alarm interface
    - b) Fail safe/fail secure locking mechanisms
    - c) Delayed egress

**1.4 GENERAL SYSTEM DESCRIPTION**

- A. General Requirements
  - 1. Furnish all labor, materials, tools, equipment, and services for a complete security system as indicated and in accordance with provisions of the contract documents.
  - 2. Although such work is not specifically indicated, furnish and install all supplementary or miscellaneous items, and devices incidental to or necessary for a sound, secure and complete installation.
  - 3. Comply with the provisions of Division 1 for General Requirements.
    - a) In the event of a conflict between the provisions of this Section and Division 1, the more stringent provisions shall apply.

4. All system devices and components included shall be compatible.
- B. The project shall be equipped with new system that is maintained by the police department.
  1. All work required within the project for new AC/ID system head end shall be furnished and installed by the project security contractor.
- C. The AC/ID system will support the needs of the project in accordance with these specifications.
  1. The AC/ID system shall have the capability for future expansion to support the security needs of the completed complex.
- D. The AC/ID system shall be interfaced with the Fire Alarm system (by others) as required to comply with all building code requirements.
- E. Emergency/UPS power will be utilized to power the AC/ID system's computer workstation (client) at the Security head end equipment location.
- F. Emergency/APS power will be utilized to power the AC/ID system's Data Gathering Panels and control components as required throughout the facility.

## 1.5 ACCESS CONTROL SYSTEM

- A. The AC system will consist of card readers, door position switches, and request-to-exit sensors operating in conjunction with associated electric door hardware.
  1. Card readers and adjunct devices shall be provided as shown on the drawings.
    - a) Provide card readers, Data Gathering Panels <DGP>, and alarm input and output devices connected to the security management system via Local Area Network (LAN).
    - b) The security integrator shall coordinate network and IP address requirements with Owner to identify the Media Access Control (MAC) address (Layer 2) of each provided device, the location to be installed, and the port configuration needed for communication.
    - c) Furnish all labor, materials, tools, equipment, and services for a complete system as indicated and in accordance with provisions of the contract documents.
    - d) Although such work is not specifically indicated, furnish and install all supplementary or miscellaneous items, and devices incidental to or necessary for a sound, secure and complete installation.
- B. Card readers will work such that upon presentation of a valid AC card, the unique card data shall be transmitted to an associated control panel where the data is compared to an authorized user database and access is approved or rejected accordingly.
  1. A valid authorization will activate operation of the electric lock and shunt the door position switch. The alarm shunt will not affect supervision of the detection circuit.
  2. Coordinate with owner on card format and other pertinent details.
- C. Card readers shall support Magnetic Stripe, 125 KHz proximity, and 13.56 MHz smart card technologies
- D. Door position switches at card reader controlled location serve to indicate the open/closed status of the associated door and shall establish the basis for reporting a door-propped or unauthorized entry condition.
  1. Security contractor is responsible for coordinating the contact configuration (SPDT) (DPDT) and rating for door position switches, and for connection of switches with the AC.
  2. The Division 08 contractor shall be responsible for providing all flush mounted door position switches as indicated on drawings.

- E. Electrified door hardware for card reader controlled doors will include electrified locksets, electric exit devices, and electric power transfer as shown on the drawings.
  - 1. All electrified door hardware shall be provided under the work of Division 08 unless otherwise noted.
  - 2. Security subcontractor shall provide all security cables and, low voltage power supplies for operation of electrified door hardware associated with card reader controlled doors.
- F. Request-to-exit (REX) devices at designated card reader controlled doors shall cause the associated door position switches to be shunted.
  - 1. The alarm shunt shall not affect the supervision of the alarm detection circuit.
  - 2. Electrified Lockset shall have an integral REX switch.
  - 3. Electrified Exit devices shall have an integral REX switch.
  - 4. Magnetic Locks shall have a Passive InfraRed (PIR) motion sensor REX device.
    - a) Wire the PIR to the Door REX Input. The configuration on this motion shall be non-resettable and activate for only 2 seconds.
    - b) A second set of output contacts for the REX motion sensor shall be wired in series with the power to the lock, disconnecting power to the lock when motion is sensed.
    - c) The PIR REX shall be mounted and the sensor positioned to avoid detection more than three feet from the door and at the door bottom sweep.
      - 1) Deter under door spoofing attacks by pointing the sensor away from the door threshold. Position the sensor to detect motion at the door handle or door push plate.
    - d) Connect to REX switch in exit device (by Division 8).
      - 1) Coordinate with Division 8 to ensure proper REX switch configuration
      - 2) Wire the REX switch as described for the above motion sensor to disconnect power to the lock and activate the REX input on the DGP.
    - e) Doors with out exit devices, a UL listed double pole pushbutton exit switch shall be provided as a redundant REX device.
      - 1) It shall be wired as described for the above motion sensor to disconnect power to the lock and activate the REX input on the DGP.
      - 2) Locate within 6'-0" of the door push-plate/handle.

## 1.6 INTRUSION DETECTION SYSTEM

- A. A series of field installed alarm initiating devices shall be connected to the ID system so that status changes of the devices are transmitted to the security management system.
  - 1. Provide Data Gathering Panels <DGP>, alarm devices, and keypads to be connected to the security management system via Local Area Network (LAN).
  - 2. The security integrator shall coordinate network and IP address requirements with Owner to identify the Media Access Control (MAC) address (Layer 2) of each provided device, the location to be installed, and the port configuration needed for communication.
- B. Glass Break Detector
  - 1. Contractor will need to provide compatible glass break tester for device being installed.
  - 2. One glass break detector per zone, do not wire in series.

- C. Tamper Switches
  - 1. Typically closed tamper switches to monitor the secure status of all DGP's, power supplies, terminal cabinets, power distribution units, and other Security System cabinets and enclosures.
  - 2. Fasten tamper switches within the cabinet to provide no access to the switch and fasteners when the cabinet is closed.
  - 3. Provide independent monitoring of tamper conditions for each cabinet.
    - a) Include the number of tamper switches in the total alarm input figures.
- D. Provide ID keypads conveniently located near areas being protected so as to allow devices to arm and disarm.

## **1.7 SUBMITTALS**

- A. Follow provisions of Section 280000 additional requirements.
- B. Field Test Reports
  - 1. Upon completion and testing of the installed system, test reports shall be submitted in booklet form and electronic media showing all field tests performed on, and adjustments made to each/any component and all field tests performed to prove compliance with the specified performance criteria.
  - 2. Indicate and interpret test results in written form and verbally to owner/DataCom for compliance with performance requirements at a pre-scheduled meeting.
- C. Battery calculations to show the expected loads and backup duration for power supplies and UPS devices for all active AC/ID equipment.
- D. Security Contractor is responsible to prepare and submit as required to the Authority Having Jurisdiction (AHJ) any and all information to obtain an Electronic Locking Mechanisms permit.

## **1.8 QUALITY ASSURANCE**

- A. Follow provisions of Section 280000.
- B. Spare Parts:
  - 1. Provide two (2) spare components for every model and configuration of electronic components and devices used on the project as spare parts inventory.
    - a) The security integrator will turn over the new and unused components and devices to the owner at project closeout.

## **1.9 DELIVERY, STORAGE AND HANDLING**

- A. Follow provisions of Section 280000.

## **1.10 PROJECT/SITE CONDITIONS**

- A. Follow provisions of Section 280000.

## **1.11 WARRANTY**

- A. Follow provisions of Section 280000.
- B. All devices and components shall comply with applicable U.L. standards.

## **PART 2 - PRODUCTS**

### **2.1 ACCEPTABLE SYSTEM MANUFACTURERS**

- A. AC System Platform Software
  - 1. Tyco Software House
    - a) C-Cure 9000
  - 2. Genetec

- a) Unified Security Center – Synergis Access Control Management System
- B. System Platform Server (By Owner)
  - 1. Dell
  - 2. Hewlett Packard
  - 3. IBM
  - 4. Owner Approved Equivalent
- C. Workstation (refer to 282300)

## 2.2 ACCEPTABLE ACCESS CONTROL MANUFACTURERS

- A. Access Control Data Gathering Panels <DGP>
  - 1. AC/ID System compatible
  - 2. Owner Approved Equivalent
- B. Proximity Card Readers <CR>
  - 1. HID
  - 2. XceedID
  - 3. Owner Approved Equivalent
- C. ID credential printer (with card stock and consumables to produce 250 credentials)
  - 1. Fargo
  - 2. Nisca
  - 3. Datacard
  - 4. Magicard
  - 5. Owner Approved Equivalent
- D. Door Position Switches <DP> (By Division 08)
  - 1. Concealed Magnetic Door Position Switch
    - a) Sentrol 1076D
    - b) Magnasphere MSS-19C/L / MSS-25C/L
    - c) Detection Systems, Inc
    - d) Owner Approved Equivalent
- E. Electric Locking Mechanism Power Supply
  - 1. Altronix
  - 2. Alarm-Saf
  - 3. LifeSafety Power
  - 4. Owner Approved Equivalent
- F. Electric Locking Mechanisms (By Division 08)
  - 1. Sargent
  - 2. Schlage
  - 3. Von Duprin
  - 4. Locknetics
  - 5. Owner Approved Equivalent
- G. Electric Power Transfer (By Division 08)
  - 1. Security Door Controls (SDC)
  - 2. Schlage
  - 3. Von Duprin
  - 4. Owner Approved Equivalent
- H. Uninterruptible Power Supply <UPS>
  - 1. Eaton UPS
    - a) 5S series for workstations

- b) 9170 for rack mounted equipment
  - 2. APC Smart-UPS Series
    - a) SMT series for workstations
    - b) Smart-UPS on-Line series for rack mounted equipment
  - 3. MinuteMan
    - a) Pro series for workstations
    - b) Enterprise Plus series for rack mounted equipment
  - 4. Owner Approved Equivalent
- I. Wire & Cable
  - 1. Belden
  - 2. Windy City
  - 3. General Cable
  - 4. Owner Approved Equivalent

## **2.3 ACCEPTABLE INTRUSION DETECTION MANUFACTURERS**

- A. ID System Platform Software
  - 1. Bosch
- B. Intrusion Detection Data Gathering Panels <DGP>
  - 1. AC/ID System compatible
  - 2. Owner Approved Equivalent
- C. Keypads
  - 1. AC/ID System compatible
  - 2. Owner Approved Equivalent
- D. Tamper Switches
  - 1. Sentrol 3010
  - 2. Owner Approved Equivalent
- E. Dual Technology Motion Detectors
  - 1. Detection Systems
  - 2. GE/Sentrol
  - 3. AC/ID System compatible
  - 4. Owner Approved Equivalent
- F. Glass Break Detectors
  - 1. Sentrol
  - 2. C&K Systems IntelliSense
  - 3. AC/ID System compatible
  - 4. Owner Approved Equivalent

## **PART 3 - EXECUTION**

### **3.1 GENERAL REQUIREMENTS**

- A. Power Supplies
  - 1. Power supply requirements
    - a) A switch and on/off indicator within the power supply cabinet.
    - b) Four hours of sealed gel battery backup to provide continuous operation during power failure.
      - 1) Provide batteries as required to provide specified battery backup time for a fully loaded power supply, regardless of the connected load.

- c) A battery charger to maintain the battery.
  - d) Low battery and power fail contacts to monitor the status of the input power and the battery.
    - 1) Connect each power supply low battery and power fail alarm as a separate alarm input into DGP.
  - e) Key lockable wall mount metal enclosure with tamper switch.
- 2. Additional DGP Power Supply Requirements
  - a) The DGP power supply provides power only to DGP's and shall not provide power for locks or any other low voltage device.
- 3. Additional Electric Locking Mechanism Power Supply Requirements
  - a) Fail secure electric locking mechanisms shall remain locked during power failure and fire alarm conditions.
  - b) Connect fail safe locking devices in accordance with applicable life safety codes to unlock automatically under the following conditions:
    - 1) Loss of power to the power supply
    - 2) Failure of the power supply
    - 3) Fire alarm activation
  - c) Provide power distribution boards with independently fused output relays and fire alarm control panel interface.
- 4. Additional Device Power Supply Requirements
  - a) Provide device power supplies for other security system devices requiring power (e.g. card readers, local alarms, motion sensors, etc.)
  - b) Provide power distribution boards with independently fused outputs.
- B. Video Surveillance System Integration
  - 1. Automatic Video Call-up
    - a) All alarms shall call up all cameras in the area of alarm to the screen of the ACID alarm operator workstation to allow for operator assessment of the alarm.
  - 2. Pre and Post Alarm Video
    - a) The operator shall be able to view up to 10 seconds of video before the alarm and 30 seconds after the alarm for all cameras associated with the alarm.
    - b) This feature is to be integrated with the operator alarm notification to assist in alarm assessment.
    - c) This feature shall be displayed as an option on the alarm notification screen and will not require operator to make a manual video search.
  - 3. Recording
    - a) All cameras whose field of view that include images of the area affected by the alarm, shall be recorded when an alarm is detected for use in forensic analysis, including the pre and post alarm video.
  - 4. Duress and Emergency Intercommunications Integration
    - a) Calls from emergency intercoms/phones with cameras shall provide the above video call-up and the pre and post alarm video capabilities.
- C. Tamper Resistant Screws
  - 1. Provide appropriate screw heads for each application (e.g. countersunk heads for recessed cover plate screws, flat head screws for standard junction box covers, etc.).

2. The security integrator shall provide Torx® tamper resistant screws for:
  - a) Junction boxes located above doors
  - b) Junction boxes located below ceiling height and/or within reach of hatch ladders
  - c) Security device cover plates
  - d) Surface mounted door position switches and armored cable

### **3.2 ENCLOSURE INSTALLATION**

- A. Enclosures shall be lockable with a tamper switch and installed in a manner to be accessible with clearance to fully open enclosure door.
- B. All security panels shall be wired through a dedicated power supply with battery backup.
  1. Power to the data gathering panels is to be hardwired utilizing EMT or rigid conduit in accordance with the Electrical specifications.
  2. A circuit from the Fire Alarm panel must be installed to each lock power distribution panel.
- C. Enclosures shall be installed on designated wall fields in a neat and compact manner to allow for future growth.
- D. Enclosures shall be sized to allow for 20% growth in each panel.
- E. All panels and boards shall be installed in enclosure(s) suitable to their environment and have sufficient size and orientation to include all system components.
- F. Each panel shall be labeled accordance with Owner standards.

The label for each panel shall be posted on the exterior of the panel door.

- a) Each panel shall have a list of devices connected to it located on the inside cover.
- b) A detailed device layout drawing will be located on the inside of the panel door in an appropriate sleeve and keeper.

### **3.3 FURTHER REQUIREMENTS**

- A. Refer to provisions of Section 280000.
- B. Furnish and coordinate installation of all special device back boxes and ACID field devices as shown on the security drawings and as specified in this section.
- C. The exact installation locations of all equipment shall be coordinated and verified with the Contractor prior to installation.
  1. Subcontractor shall notify the Contractor if any location appears to be unsuitable.
- D. Provide low voltage power supplies for electric locking devices and ACID devices and components as shown on the security drawings and specified in this Section.
- E. Coordinate with the Telecommunications Subcontractor for data network connections, IP address requirements, and telephone circuits as required.
- F. Prepare all systems for user operation.
  1. The security system must be complete and ready to operate prior to Owner final acceptance of the system.
- G. Coordinate with the Owner for all system programming requirements.
- H. Perform database programming as required to support the card reader, alarm point, surveillance system integration, and control panel configuration as required.



END OF SECTION 28 10 00

**SECTION 28 23 00**  
**VIDEO SURVEILLANCE**

**PART 1 - GENERAL**

**1.1 SUMMARY/OVERVIEW**

- A. This section provides specifications for the installation of an IP based Video Surveillance System (VS) and related components.
- B. Related Sections
  - 1. Section 260000 Electrical (including related sub-sections)
  - 2. Section 270000 Communications (including related sub-sections)
  - 3. Section 280000 Electronic Security
  - 4. Section 281000 Electronic Access Control and Intrusion Detection

**1.2 REFERENCES**

- A. See Section 280000 Electronic Security.

**1.3 SYSTEM DESCRIPTION**

- A. The project will be equipped with a new Video Management System (VMS) as a Stand Alone System.
  - 1. Provide Network Video Recorder (NVR) and Network Attached Storage (NAS) located in relay rack in the MDF of the project site.
    - a) The new system shall provide for recording, local monitoring, and remote monitoring of IP cameras.
    - b) The system shall be maintained and monitored by Lamar police/security dept.
- B. The security integrator shall furnish and install the surveillance system, consisting of camera assemblies, NVR, wiring & cabling, and low voltage camera power supplies.
  - 1. All active surveillance equipment and communication devices shall be on emergency/UPS power.
- C. Camera assemblies include camera, lens, housing, and mount. Provide and install wiring and low voltage power from the security wall field/rack to the camera locations.
  - 1. Scope of work shall be complete from point of origin (camera) to point of termination (security rack).
- D. Provide rack mount 1U monitor/keyboard/track ball drawer with 20" LCD monitor for display and programming, keyboard, track ball, and integrated four-port KVM switch.
  - 1. Provide additional KVM ports as required plus two spare ports if Work includes more than two NVRs.
- E. Coordinate all work that must be performed in security head end spaces with the General Contractor, the Electrical Contractor, and the Telecommunications contractor. (if applicable)
- F. Camera images shall support H.264 compression formats.
- G. The NVR shall not be loaded to exceed 50% of the camera and/or storage capacity to allow room for expansion.
- H. Camera lenses for fixed cameras shall be varifocal and sized to provide the owner approved field of view. The lens shall be IR corrected and have megapixel resolution.
- I. Surveillance camera audio functions shall not be installed and/or disabled unless specifically requested by Owner.

**1.4 SUBMITTALS**

- A. Follow provisions of Section 280000 for additional requirements.
- B. Project Data
  - 1. Provide a description of system operation indicating the purpose and capability of each device/component of the system with a functional diagram indicating all interfaces to other systems.
- C. IP: Video Quality test reports shall be provided for all cameras to confirm an optimum high definition video signal.
- D. Shop drawings shall reflect all requirements associated with Owner provided or existing equipment and materials that will be used as part of this system.
- E. Video Storage calculations to show the system capacity can accommodate the specified video retention.
- F. Battery calculations to show the expected loads and backup duration for camera power supplies and UPS devices for all active surveillance equipment.
- G. System programming, camera titles, descriptions, camera images and database
  - 1. Camera titles and descriptions prior to system programming
  - 2. Programming/database prior to performance testing
  - 3. Provide a cross reference between specified camera numbers and programmed camera numbers
  - 4. Final programming, camera images and system documentation on electronic media to Owner
- H. Product Data
  - 1. Manufacturer's technical data sheets and specifications

## **1.5 QUALITY ASSURANCE**

- A. Follow provisions of Section 280000.
- B. Spare Parts:
  - 1. Provide two (2) spare components for every model and configuration of electronic components and devices used on the project as spare parts inventory.
    - a) The security integrator will turn over the new and unused components and devices to the owner at project closeout.

## **1.6 DELIVERY, STORAGE AND HANDLING**

- A. See Section 280000.

## **1.7 PROJECT/SITE CONDITIONS**

- A. See Section 280000.

## **1.8 WARRANTY**

- A. See Section 280000.

# **PART 2 - PRODUCTS**

## **2.1 CAMERA SPECIFICATIONS**

- A. All cameras shall be a Dome Camera unless otherwise specified
  - 1. Compatible with the VMS
  - 2. Vandal resistant with polycarbonate dome
  - 3. Wide Dynamic Range Feature: All exterior cameras and interior cameras that have exterior lighting or headlights in their field of view shall have a Wide Dynamic Range feature to improve picture quality in situations with strong backlighting.

4. Multi-stream so that recording and viewing can be at different frame rate and compression.
  5. Day-night Color/B&W camera with cut filter
  6. Exterior cameras:
    - a) Shall be outdoor rated
    - b) Include a heater to permit fog-free viewing in low temperatures
    - c) Fan to prevent overheating in high temperatures (as required)
- B. Interior Fixed Dome Cameras:
1. Shall be IP, PoE IEEE 802.3af
  2. Day-night Color/B&W camera with cut filter
  3. Smoked lower dome
  4. Include an SD card sized to allow for scheduled and event based storage of images
  5. Resolution shall be a minimum of 2MP (1080p) at 30 FPS
  6. Vari-focal auto-iris fixed lens sized to provide the owner approved field of view
  7. Shall have a minimum sensitivity of 0.05 Lux at:
    - a) 30 IRE
    - b) 75% reflectance
    - c) AGC off
    - d) 1.2 f-stop
    - e) Sense-Up off
- C. Interior/Exterior Fixed Wide Angle Lens Cameras:
1. Resolution shall be a minimum of 3.0 MP at 30 FPS
  2. Dome Camera is preferred over a Box Camera
  3. Include an SD card sized to allow for scheduled and event based storage of images
  4. Auto-iris wide angle lens
    - a) Shall be Rectilinear Lens to minimize barrel distortion
    - b) C/CS mount removable lens
    - c) Wide horizontal field of view minimum of 90 degrees
    - d) Size lens to provide the owner approved field of view
  5. Size camera and lens for facial recognition at a minimum of 60 pixels per foot at the target location
  6. Shall have a minimum sensitivity of 0.05 Lux at:
    - a) 30 IRE
    - b) 75% reflectance
    - c) AGC off
    - d) 1.2 f-stop
    - e) Sense-Up off
- D. Exterior Fixed Dome Cameras:
1. Shall be IP, PoE IEEE 802.3af
  2. Clear lower dome, unless otherwise specified
  3. Include an SD card sized to allow for scheduled and event based storage of images
  4. Resolution shall be a minimum of 2MP (1080p) at 30 FPS
  5. Vari-focal auto-iris fixed lens sized to provide the owner approved field of view.
  6. Shall include remote focus and zoom over the network
  7. Shall have a minimum sensitivity of 0.05 Lux at:
    - a) 30 IRE
    - b) 75% reflectance
    - c) AGC off

- d) 1.2 f-stop
  - e) Sense-Up off
- E. Exterior Pan Tilt Zoom (PTZ) Dome Cameras
  - 1. Shall be IP
  - 2. Clear lower dome
  - 3. Include an SD card sized to allow for scheduled and event based storage of images
  - 4. Resolution shall be a minimum of 1.3MP (720p) at 30 FPS
  - 5. Auto-iris
  - 6. Zoom - 18 x optical zoom lens and 12 x digital zoom.
  - 7. Operator controlled panning speed shall be at least 300 degrees per second and 360 degree continuous pan rotation.
  - 8. 50 presets minimum
  - 9. Shall have a minimum sensitivity of 0.04 Lux at:
    - a) 30 IRE
    - b) 75% reflectance
    - c) AGC off
    - d) 1.2 f-stop
    - e) Sense-Up off

## 2.2 NETWORK VIDEO RECORDER

- A. The Network Video Recorder shall be compatible with the existing video management system along with the following minimal requirements:
  - 1. Provide hot swappable storage drives in a RAID 5 array
  - 2. Support dual stream cameras
  - 3. Include a dual Network Interface Card (NIC) that supports:
    - a) Full duplex
    - b) Fault tolerance
    - c) Link aggregation
    - d) Load Balancing
    - e) Traffic prioritization
    - f) Hot swap
  - 4. Native support of H.264 compression that does not require the use of additional software or equipment
  - 5. Provide a 20 inch 1080P service monitor and keyboard at each NVR equipment rack
  - 6. Provide video storage capacity for 30 days using the following minimum criteria
    - a) All cameras using H.264 compression
    - b) Interior cameras: Use native camera resolution
    - c) Exterior cameras: Use native camera resolution
    - d) Wide angle lens cameras: Use native camera resolution
    - e) Motion triggered recording
      - 1) Assume that motion will be detected 50% of the day
      - 2) Motion detection will be configurable by camera and schedule to mitigate nuisance triggers
      - 3) Record video at 12 FPS when motion is detected
      - 4) Record video at 1 FPS when no motion is detected
- B. Coordinate with owner on the number of user licenses required
- C. Provide UPS for backup power to the NVR and peripheral equipment
  - 1. Follow provisions of Section 28000 for UPS power requirements
  - 2. Coordinate with Division 26 to provide a dedicated Emergency Power circuit

## **2.3 PoE NETWORK SWITCH**

- A. Provide a fully configured rack mounted PoE Managed 10/100/1000Mbps Layer 3 Switch.
  - 1. Coordinate switch make and model with Owner / Owner network representative prior to purchase to ensure compatibility with existing and non-surveillance related network appliances.
  - 2. Coordinate with Owner / Owner network representative on video surveillance subnet assignment.
  - 3. Switch port count shall allow for 10% growth.

## **2.4 ACCEPTABLE MANUFACTURERS**

- A. Video Management System (VMS) Platform Software
  - 1. Tyco Security
    - a) Exacq Vision
  - 2. Genetec
    - a) Unified Security Center – Omnicast Video Management System
  - 3. Owner Approved Equivalent
- B. NVR Server: Compatible with VMS Requirements
  - 1. NVR specification compliant hardware
- C. Workstation Hardware:
  - 1. Dell
  - 2. HP
  - 3. Owner Approved Equivalent
- D. Video Monitor:
  - 1. Viewsonic
  - 2. Samsung
  - 3. Optquest
  - 4. LG
  - 5. Owner Approved Equivalent
- E. Interior Color Camera Mini Dome
  - 1. Axis Communications
  - 2. American Dynamics
  - 3. Bosch
  - 4. Owner Approved Equivalent
- F. Exterior Color Camera Mini Dome Mini Dome
  - 1. Axis Communications
  - 2. American Dynamics
  - 3. Bosch
  - 4. Owner Approved Equivalent
- G. Exterior Color Camera PTZ Dome
  - 1. Axis Communications
  - 2. American Dynamics
  - 3. Bosch
  - 4. Owner Approved Equivalent
- H. Exterior Parapet/Roof Deck Camera Mount
  - 1. Axis
  - 2. American Dynamics
  - 3. Owner Approved Equivalent

- I. Wide Angle Megapixel Camera Lens
  - 1. Computar
  - 2. Theia
  - 3. Owner Approved Equivalent
- J. Camera Power Supply
  - 1. Altronix
  - 2. Alarm-Saf
  - 3. LifeSafety Power
  - 4. Owner Approved Equivalent
- K. Equipment Racks and Racks Components: (By Division 27)
  - 1. Chatsworth Products (CPI)
  - 2. Ortronics
  - 3. Newton Instruments
  - 4. Owner Approved Equivalent
- L. Lockable Equipment Cabinet and Components: (By Division 27)
  - 1. Chatsworth Products (CPI)
  - 2. Ortronics
  - 3. Newton Instruments
  - 4. Owner Approved Equivalent
- M. Media Converter - Copper-to-Fiber Outdoor rated (Hardened for extreme temperatures)
  - 1. Axis Communications
  - 2. Pelco
  - 3. IFS / UTC Security
  - 4. Owner Approved Equivalent
- N. PoE Network Switch
  - 1. Brocade
  - 2. Cisco
  - 3. Juniper
  - 4. Arista
  - 5. Owner Approved Equivalent
- O. Video Wire & Cable
  - 1. Windy City
  - 2. General Cable
  - 3. Belden
  - 4. CommScope
  - 5. Owner Approved Equivalent
- P. Uninterruptible Power Supply (UPS)
  - 1. Eaton UPS
    - a) 5S series for workstations
    - b) 9170 for rack mounted equipment
  - 2. APC Smart-UPS Series
    - a) SMT series for workstations
    - b) Smart-UPS on-Line series for rack mounted equipment
  - 3. MinuteMan
    - a) Pro series for workstations
    - b) Enterprise Plus series for rack mounted equipment
  - 4. Owner Approved Equivalent

## **PART 3 - EXECUTION**

### **3.1 CONFIGURATION**

- A. Video Cameras
  - 1. Provide day/night cameras in exterior locations
  - 2. Lenses shall be field tested with Owner present to verify clear, crisp images and desired field of view
    - a) Substitute camera lenses as necessary to obtain required field of view at no additional cost
    - b) Provide spot filters for exterior lenses as required to reduce picture washout caused by sunlight
- B. IP PoE Cameras
  - 1. The security integrator shall coordinate network and IP address requirements with Owner to identify the Media Access Control (MAC) address (Layer 2) of each provided camera, the location to be installed, and the port configuration needed for communication.
  - 2. Make all necessary adjustments to camera lenses to obtain clear, crisp images and desired field of view to the Owners satisfaction.
    - a) Substitute camera lenses as necessary to obtain required field of view at no additional cost.
      - 1) Adjust all cameras to produce high-definition images with no blooming, streaking or noticeable lag.
      - 2) Provide and install in-line PoE injectors as required when non PoE network switches are used or when manufacturer specified power is not available to the camera.
      - 3) All camera power shall comply with the specified power requirements.

### **3.2 POWER REQUIREMENTS**

- A. Provide uninterruptible power supplies for all active surveillance equipment
  - 1. Rack mounted components, including all active network communication hardware, shall be on an Uninterruptible Power Supply <UPS> system.
  - 2. Refer to Section 280000 for UPS and power requirements
  - 3. Camera power supplies shall be on an Auxiliary Power Supply <APS>, system as required, with a battery backup.
    - a) The Auxiliary power supply shall be furnished with a power distribution panel with each camera individually fused or protected with an over-current protector.
- B. Power supplies shall provide:
  - 1. 120 VAC input and output voltage as required
  - 2. UL Listed
  - 3. Power fail contacts to monitor the status of the input power
    - a) Connect each power supply power fail alarm as a separate alarm input into AC/ID system
  - 4. Key lockable wall mount metal enclosure with tamper switch
  - 5. Independently fused outputs

### **3.3 INSTALLATION**

- A. Refer to provisions of Section 280000
- B. All surveillance system devices and components shall be compatible.



- C. Camera Housings and Mounts
  - 1. Cameras shall include housings and mounts as indicated in the Drawings.
    - a) Provide the smallest available housing for each camera application.
      - 1) Integrated miniature dome cameras are preferred
  - 2. Wiring to cameras shall pass from the back-box through the mount and into the housing. Exposed wiring or conduit shall not be acceptable.
  - 3. Provide sun shields for camera housings in outdoor locations exposed directly to sunlight.
  - 4. Provide surge protection for power and copper video cables for exterior cameras at the camera and at the point of termination (security rack).
  - 5. Field verify the exact camera location, position, and mounting prior to installation.
  - 6. Roof mounted cameras shall use roof deck brackets.
- D. Video Management Control System
  - 1. System platform software shall be 'open architecture' allowing for compatibility and integration with other building automated systems.
  - 2. The system shall allow for secure remote viewing of live and recorded video as required.
- E. Provide labeling suitable to Owner for all major equipment components. Coordinate with Owner on numbering scheme to match existing. Major equipment components:
  - 1. IP Video monitors, IP camera Patch Panels, PoE Switches (or mid-span units), Network Video Recorders (NVR), and fiber mux units (if required).
- F. Coordinate with Telecommunication subcontractor for network and patch panel provisions for security connections in the IT room. (If applicable)
- G. Coordinate with Owner for all system programming and database requirements.
  - 1. Provide all programming, setup, camera and device titling and data entry
  - 2. Camera and device title and descriptions shall be consistent for all components
- H. Install all Point-to-Point wiring with appropriate terminal connections for every wire and component termination so that all connections are mechanically and electrically secure.
- I. Install field wiring in continuous lengths, without splices.
- J. Verify upon job completion that all wiring and terminations are clearly labeled to identify the wire and terminal.
- K. Testing of the surveillance system includes checkout of installed cameras back to the Security head end equipment to confirm proper operation of camera assemblies. Security integrator shall provide all necessary test equipment to fully demonstrate proper performance of field devices. Copies of test results shall be included in the project completion submittals given to the Owner.

**END OF SECTION 28 23 00**

## **SECTION 28 31 00 13 - FIRE ALARM AND SMOKE DETECTION SYSTEM**

### **PART 1 - GENERAL**

#### **1.1 SCOPE**

- A. The contractor shall design, furnish and install a complete microprocessor based 24VDC, electrically supervised, analog intelligent fire alarm system as specified herein and provide drawings. The system shall include, but not be limited to, all control equipment, power supplies, signal initiating and signaling devices, conduit, wire, fittings, and all other accessories required to provide a complete and operable system.
- B. The system shall operate as a non-coded, continuous sounding system, which will sound alarm devices until manually silenced, as herein specified.
- C. The system shall be wired as a style B and style 4 supervised system for all circuits.

#### **1.2 CODES AND STANDARDS**

- A. The system shall comply with the applicable Codes and Standards as follows:
  - 1. National Electrical Code - Article 760.
  - 2. National Fire Protection Association Standards: NFPA 70 NEC
    - a. NFPA 72 Protective Signaling Systems (current State adopted version)
    - b. NFPA 90A Air Conditioning
    - c. NFPA 101 Life Safety Code
    - d. UL 1971 Visual Devices
    - e. ANSI 117.1 Visual Devices
  - 3. Local & State Building Codes
  - 4. Requirements of Local Authorities having Jurisdiction. If local authorities design requirements differ substantially from contract drawings, the design engineer shall be notified no less than 10 days prior to bid date, to allow time for addendum to be provided to all contractors. Contractor to provide additional devices as required by local authorities in bid pricing.
  - 5. Underwriters Laboratory Requirements and Listings for use in Fire Protective
- B. Signaling Systems as follows:
  - 1. UL 864 Control Panels 9th Edition UL 268 Smoke Detectors - Systems UL 268A Duct Smoke Detectors
  - 2. UL 521 Heat Detectors
  - 3. UL 228 Door Holder-Closers
  - 4. UL 464 Audible Signaling Appliances UL 1971 Visual Signaling Appliances UL 38 Manual Alarm Stations

#### **1.3 ACCEPTABLE MANUFACTURERS**

- A. To establish the type, quality, and features of system required, the equipment specified is that of the Notifier Fire Systems.
- B. All equipment, materials, accessories, devices, etc. covered by the specifications and/or noted on the contract drawings shall be new and unused and be U.L. listed for their

intended use.

- C. All references to manufacturer or supplier's model numbers and other pertinent information herein is intended to establish a minimum standard of quality, performance and features required. All equipment proposed as an EQUAL to that specified shall COMPLETELY conform to the specifications herein.
- D. Equipment of other manufacturer's or supplier's may be considered as an equal to that specified provided that completely marked and identified catalog sheets of all proposed equipment is provided to the architect/ engineer for review ten (10) days prior to the date of bid for evaluation. In addition, a list of the contractor's qualifications and any exceptions to the specifications must be provided for review. Approval for any such substitution of equipment must be obtained in writing from the architect/engineer five (5) days prior to bid.
- E. Provide one of the following manufacturers:
  - 1. Notifier Fire Systems
  - 2. Siemens
  - 3. Edwards System Technology (EST)

#### 1.4 GENERAL REQUIREMENTS

- A. Contractor Qualifications:
  - 1. The equipment supplier shall be an authorized and designated representative of the Fire Alarm Manufacturer to sell, install, and service the proposed manufacturer's equipment.
  - 2. The equipment supplier and installing contractor shall be licensed by the State
    - a. Fire Marshall to sell, install, and service fire alarm systems as required by Article 5.43-2 of the Texas Insurance Code.
  - 3. The installing contractor and/or equipment supplier shall have on his staff a minimum of three (3) installation superintendents who are licensed by the State Fire Marshall's office for such purpose and under whose supervision installation, final connections, and check out will take place as required by the Texas Insurance Code.
  - 4. The installing contractor or equipment supplier shall have on staff a minimum of one (1) certified NICET Level III state licensed fire alarm planner under whose supervision system design shall take place.
  - 5. The installing contractor shall provide 24 hour, 365 days per year emergency service with qualified and state licensed service technicians.
  - 6. The installing contractor shall have been actively engaged in the business of selling, installing, and servicing fire alarm systems for at least ten (10) years.

#### 1.5 SUBMITTALS

- A. The installing contractor and/or equipment manufacturer shall provide complete and detailed shop drawings and include:
  - 1. Control panel configuration including wiring and interconnection schematics.
  - 2. Complete point to point wiring diagram showing terminal connections to all

- system devices.
3. Riser wiring diagram and associated zoning/addressing configurations with associated conduit sizes.
  4. Complete floor plan drawings locating all devices associated with the fire alarm system. Floor plan drawings shall include conduit and wiring routing complete with conduit sizing and number of conductors by type.
  5. Factory data sheets on each piece of equipment to be used and so marked as to model, dimensions, size, voltage, and configuration.
  6. Detailed system description in this specification format describing system functions and operation. All specification variations and deviations shall be clearly noted and marked.
  7. Complete Bill of Material for reference.
  8. Programming matrix defining all input/output functions and zoning.
  9. Power supply and battery calculations.
  10. A letter from the manufacturer stating that the fire alarm system contractor is authorized to sell, service and install the submitted equipment.
- B. Submittal shall include documentation confirming all qualifications in 1.04-A have been met. Submittals without qualification documentation shall be rejected.
- C. All submittal data will be in bound form with contractor's name, supplier's name, project name, and state fire alarm license number adequately identified.
- D. Only basic equipment devices have been shown on the contract drawings. Specific wiring between equipment/devices has not been shown. It is the contractor's responsibility to submit for approval the COMPLETE ENGINEERED system configuration and layout showing all devices, wiring, conduit, and locations along with other required information as specified herein.

## 1.6 COORDINATION

- A. It shall be the responsibility of the installing contractor to coordinate all requirements surrounding installation of the fire alarm system with all trades including, but, not exclusive of: electrical contractor, sprinkler contractor, and HVAC/controls contractor and intercom system. Adequate coordination shall be provided to insure proper installation and interface to all peripheral items required to interact with the fire alarm and communication system to provide a complete and functional life safety system.

## PART 2 - PRODUCTS

### 2.1 SYSTEM FUNCTIONAL OPERATION

#### A. Alarm Detection

1. When a fire alarm condition is detected by any of the system alarm initiating devices, the following functions shall occur:
  - a. The system common alarm LED on the CPU Module shall flash. The internal audible trouble device shall sound. Acknowledgement or silencing the alarm condition shall silence the alarm signals and cause flashing alarm LED's to illuminate steady.
  - b. An 80 character back-lit LCD display shall indicate all applicable information associated with the alarm condition including: zone, device

type, divide location, and time of alarm. Location and zoning messages shall be custom field programmed to respective premises.

- c. Any remote or local annunciator LED's associated with the alarm zone shall be illuminated as herein specified.
- d. A three-channel digital alarm communicator shall be integrally provided and transmit trouble and alarm signals to an approved remote station (remote station connection and service provided by Owner).
- e. All automatic events programmed to the alarm point shall be executed and the associated indicating devices and/or outputs activated.
- f. Activate all audible and visual alarm notification devices.
- g. De-activate HVAC systems over 2,000 CFM.
- h. Display system status changes on the remote annunciators.
- i. Release all smoke doors, fire doors, fire coiling doors, fire smoke dampers and fire shutters.

B. System Trouble Detection

1. When a trouble condition is detected by the CPU, one of the system initiating, alarm or SLC circuits, the following functions shall immediately occur:
  - a. The system trouble LED on the CPU module shall flash and the internal audible trouble device shall sound. Acknowledgement of the trouble condition shall silence the audible trouble device and cause all trouble LED's to illuminate steady.
  - b. The 80-character alphanumeric LCD annunciator shall display all applicable information via the alphanumeric display associated with the respective trouble condition and its location.

C. Auxiliary Control

1. All designated "non-silenceable" auxiliary control functions shall remain in operation (even upon silencing of audible alarms) until such time as the control panel is cleared and reset manually (i.e. fan control outputs, central station interface, elevator recall interface, etc.).
2. Activation of duct smoke detectors associated fans shall shutdown their respective units immediately in addition to identifying the condition as herein specified. Duct detectors shall be programmed as a supervisory condition per NFPA 72.

D. System Supervisory Detection

1. When a supervisory condition is detected by the fire alarm control panel, the following functions shall occur:
  - a. The fire alarm control panel supervisory indicator shall flash and the internal audible device shall sound. Acknowledgment of the supervisory condition shall silence the audible device and cause the supervisory indicator to illuminate steady.
  - b. The 80-character liquid crystal display shall display all applicable
  - c. information associated with the respective supervisory condition.
  - d. Activate a supervisory contact closure to interface with the owner provided central station monitoring service.
  - e. Print the status change messages on the system printer.

- f. Display the system status change on the remote annunciators.

#### E. Fire Drill Control

- 1. Provide a fire drill switch located on the Fire Alarm Control Panel. When activated, this switch will activate all horn/strobes and speakers for a fire drill. It shall not release fire shutter, shut down air handling equipment or recall elevators. If a fire alarm condition is detected, the system shall operate as defined in part 2.01A of this section.

### 2.2 ZONING

- A. The system shall have the inherent capability to employ 'Intelligent' smoke detectors and addressable interface devices capable of being recognized and annunciated at the main control panel on an individual basis. All zoning/device location information shall be totally field programmable to exact job requirements as approved by the Architect/Engineer.

### 2.3 FIRE ALARM CONTROL PANEL

- A. The fire alarm control panel shall be Notifier series NFS2-640. The control panel shall utilize DISTRIBUTED solid-state MICROPROCESSORS. The microprocessor based CPU shall be completely FIELD PROGRAMMABLE. CPU module shall provide for programmable non-volatile EEPROM memory. All circuitry shall be U.L. listed for power-limited application. System shall be sized to accommodate the capacity of the system specified and shown on the drawings. System shall be capable of being networked for future expansion.

#### B. Central Processing Unit Module (CPU)

- 1. The CPU shall contain and execute all custom time control functions or control-by-event programs for specified events including 'Holiday' exceptions. Time control event/programs shall be automatically overridden by priority fire alarm events. All programs shall be held in non-volatile programmable EEPROM memory, and shall be lost if both system primary and secondary power failure occurs
- 2. System CPU shall also provide for non-alarm points for non-fire, low priority building functions. The CPU shall provide capability of multi-stage signaling, tornado warning, positive alarm sequencing as well as remote control system operation.

#### C. Display

- 1. The DIA shall provide an 80-character backlit, supertwist Liquid Crystal Display (LCD). It shall provide Light-Emitting Diodes (LED's) for AC POWER; SYSTEM ALARM; SYSTEM TROUBLE; SUPERVISORY; CPU FAIL; and ALARM SILENCED.
- 2. The display shall provide power to a 21-key membrane keypad with control capability to command all system functions, status readouts, manual control action, and entry of any alphanumeric or numeric information. The keypad shall include means to enter multiple five-digit passwords to prevent unauthorized manual control programming.

#### D. Control Switches

1. Acknowledge/Step Switch
2. Signal Silence Switch
3. Evacuate
4. Lamp Test/Reset

E. System Outputs

1. The system shall provide the following outputs:
  - a. One port for CRT, modem, and/or printer (RS-232c)
  - b. One port for supervised remote LCD annunciators (RS-485)
  - c. Four notification appliance circuits (NAC) F. Loop Interface (SLC)
  - d. The CPU shall communicate and provide power to all devices on its loop over a single pair of wires. The CPU shall receive digital/ANALOG information from all "intelligent" detectors and shall process this information to determine normal, alarm, trouble, and sensitivity conditions. The analog information may be used for automatic test and determination of maintenance requirements, and be U.L. listed for such use. The CPU module shall individually monitor all "intelligent" detectors for sensitivity variation initiating a trouble condition should detector sensitivity "drift" become excessive. The system control unit shall have the capability to remotely read each detector's sensitivity in % obscuration, and if need be, electronically adjust the detector sensitivity as required for existing conditions within U.L. recommended limits. In addition, the system shall incorporate a "day/night" sensitivity feature. The system shall provide capability to program each individual detector for multiple 'pre-alarm' conditions. Each 'pre-alarm' level shall be field programmable as a function of the programmed alarm level. The system shall allow designated control-by-event actions to occur as may be required prior to any sensor reaching the designated alarm point.

F. Non-Lock Walk Test

1. The system shall include a special non-lock "walk test" mode. The walk test mode shall incorporate a one-hour time-out feature to return system to normal. Test results shall be capable of being generated and displayed on LCD annunciator or printed out on system printer.

G. Automatic Detector Test

1. The system shall include a special automatic detector test feature, which permits reading and adjustment of the sensitivity of all intelligent detectors from the main control panel. In addition, the automatic test feature shall also permit the functional testing of any "intelligent" detector or addressable interface device individually from the main control panel. An automatic detector test shall occur automatically a minimum of every two-hour period or be initiated manually from the FACP as desired. Automatic detector test sequencing shall be terminated upon receipt of a true alarm condition.

H. Special System Reports

1. The system shall have the ability to generate and print, upon command, system and point status reports. Selection of 'system' read status provides the operator

with global system programming information as well as providing the operator with all individual point programming data. The system shall also provide the capability to print out a detailed 'history' report from system history file upon command.

I. Field Programming

1. The system shall be 100% field programmable without the need for external computers or, PROM programmers, and shall NOT require replacement of memory IC's. All programs shall be stored in non-volatile EEPROM memory. Programming shall be accomplished only after entering an appropriate and pre-selected five-digit password security code. System programming mode shall NOT require the system to be taken off-line nor prohibit the system from performing its normal operations and routines. The system shall be capable of revising/changing programmed functions or system expansion at any time subsequent to initialization as described herein without factory modifications or factory programming. Field programming via the use of external computers may be considered provided programming can be accomplished on-site and the owner is permanently furnished with the required programming apparatus and software as part of this contract.

J. Event History

1. The main fire alarm panel shall have the resident ability to store a minimum of 600 system events in chronological order of occurrence. Event history shall include all system alarms, troubles, operator actions, unverified alarms, circuit/point alterations, and component failures. Events shall be time and date stamped. Events shall be stored in non-volatile buffer memory. Access to history buffer shall be secured via five-digit password security code. Systems not employing event history memory storage shall be required to furnish a printer/recorder for recording system events.

K. Power Supply

1. The power supply shall provide all control panel and peripheral power needs with filtered power as well as rectified 24VDC power for external audio-visual devices. All power supplies shall be designated to meet UL and NFPA requirements for POWER-LIMITED operation on all external signaling lines, including initiating circuits and indicating circuits.
2. Input power shall be 120VAC 60Hz. The power supply shall provide internal supervised batteries and automatic charger. The power supply shall provide both positive and negative ground fault supervision, battery/charger fail condition, A.C. power fail indicators. The power supply shall also provide supervision of modular expansion power supplies as may be required.

2.4 FIELD DEVICES

A. Multi sensor Detector (Smoke and Heat)

1. Provide Notifier FAPT-851 intelligent multi sensor smoke detectors. The multi sensor analog detector shall use a light scattering type photoelectric smoke sensor, a unipolar ionization smoke sensor and an ambient temperature sensor to sense changes in air samples from its surroundings. The integral



microprocessor shall employ time based algorithms to dynamically examine data. The Multi sensor shall be capable of adapting to ambient environmental conditions. The temperature sensor shall self-adjust to the ambient temperature of the surrounding air and input an alarm when there is a change of 65° F in ambient temperature. Systems using central intelligence for alarm decisions shall not be acceptable. The detector shall continually monitor any changes in sensitivity due to the environmental effects of dirt, smoke, temperature, age and humidity. The information shall be stored in the integral processor and transferred to the analog loop controller for retrieval using a laptop PC. Separately mounted photoelectric detectors, ionization detectors and heat detectors in the same location are not acceptable alternatives.

2. The Multi sensor smoke detector shall be rated for ceiling installation at minimum of 30 ft. (9.1m) centers and suitable for wall mount applications. The Multi sensor shall be suitable for direct insertion into air ducts up to 3 ft. (0.91m) high and 3 ft. (0.91m) wide and air velocities up to 500 ft./min. (0-2.54m/sec) without requiring specific duct detector housings or supply tubes. The percent smoke obscuration per foot alarm set point shall be field selectable to any of five sensitivity settings ranging from 1.0% to 3.5%. The integral heat sensor shall cause an alarm when it senses a change in ambient temperature of 65° F or reaches it fixed temperature alarm set point of 135° F nominal. The Multi sensor detector shall be suitable for operation in the following environment:

- a. Temperature: 32° F to 100° F (0° C to 38° C)
- b. Humidity 0-93% RH, non-condensing c. Elevation: Up to 6,000 ft. (1828m)

#### B. Intelligent Duct Detector

1. Notifier model DNR series duct mounted "intelligent" photoelectric smoke detectors shall be provided per applicable codes. Detectors shall operate on the same principles and exhibit the same basic characteristics as area type "intelligent" smoke sensors. The unit shall be capable of interchanging/accepting either photo-electronic or ionization type sensors. The detector shall operate in air velocities of 300 FPM to 4,000 FPM. Each detector shall interface directly to the system SLC loop without the use of zone modules.
2. The unit shall consist of a clear noryl molded plastic enclosure with integral conduit knockouts. The unit shall be provided with clear faceplate cover to provide visual viewing of detector/sensor for monitoring sensor operation and chamber condition. The duct housing shall be provided with gasket seals to insure proper seating of the housing to the associated ductwork. Each unit's sampling tubes shall extend the width of the duct and be provided with porosity filters to reduce sensor/chamber contamination. Detectors shall be installed per NFPA 90A, and be listed with the fire alarm control panel. A remote LED shall be located on the corridor ceiling adjacent to the respective detector where detectors are not plainly visible or concealed from view.

#### C. Intelligent Thermal Detectors

1. Notifier Model FST-851R analog, fixed temperature and rate of rise thermal detectors shall be provided where indicated on the drawings. The detectors shall use dual electronic thermostats to measure temperature levels in the chamber and shall, on command from the control panel, send data to the panel representing the analog temperature level.
2. The detectors shall provide address-setting means on the detector heat using

rotary decimal switches. No binary coding shall be required. Systems requiring separate detector programming apparatus will be unacceptable.

3. The detectors shall provide dual alarm and power/status LED's. Status LED's shall flash under normal conditions, indicating that the detector is operational and in regular communication with the control panel. Both LED's may be placed into steady illumination by the control panel, indicating that an alarm condition has been detected. An output connection shall also be provided in the base to connect an external remote alarm LED.
4. The detector shall be semi-flush ceiling mounted and be provided with modular detector head with twist-lock base.
5. Provide weatherproof heat detectors in the Garage Areas or other non-airconditioned areas where detection is required.

D. Addressable Manual Pull Stations

1. Notifier Model NBG-12LX manual stations shall be provided where indicated on the drawings. The manual station shall provide address-setting means using rotary decimal switches. No binary coding shall be required.
2. Manual stations shall be designed for semi-flush mounting on standard electrical box. The station shall be constructed of hi-impact red molded Lexan with instructions for station operation in raised white letters. Stations shall be of the dual action type.
3. All manual pull stations shall be provided with an STI-1100 series clear plastic cover with integral horn.

E. Monitor Module

1. Notifier model FMM-101 addressable monitor modules shall be provided where required to interface to contact alarm devices. The monitor module shall be used to connect a supervised zone of conventional initiating devices to an intelligent SLC loop.
2. The monitor module shall provide address-setting means using rotary decimal switches. No binary coding shall be required.

F. Control Module

1. Notifier model FCM-1 or FRM-1 control and relay modules shall be provided where required to provide audible alarm interface and/or relay control interface. The control module shall be used to connect a supervised zone of conventional indicating devices to an intelligent loop. The zone may be wired class A or class B - field selected. The control module may be optionally wired as dry contact (form C) relay.
2. The control module shall provide address-setting means using rotary decimal switches. No binary coding shall be required. A status LED shall be provided which shall flash under normal conditions, indicating that the control module is operational and in regular communication with the control panel. The LED shall illuminate steady when the device is actuated via the fire alarm control panel.

G. Electronic Audio Visual Devices

1. Audible/Visual alarm devices shall be Notifier "P" Series SpectrAlert Advance electronic horn/strobe units, to be located where indicated on the drawings. Devices shall be wall or ceiling mounted as indicated on the drawings. AV

devices shall be provided with the ability to provide multiple candela settings. Units shall operate at 24VDC and be polarized supervised. Each unit shall provide a choice of three different audible tones capable of being field selected. Preferred alarm signal shall be a temporal tone producing a sound pressure level of 84 dBA. The visual device shall use Xenon strobe type producing a minimum of 15 candela on a 24 VDC limited energy supervised circuit and meet the requirements of ADA and TAS. Strobe unit shall automatically flash upon operation of the horn. Horn/strobe unit shall be provided in textured white finish and be flush mounted. All visual devices shall be synchronized.

H. Electronic Alarm Horn

1. Provide Notifier H Series solid state electronic alarm device where indicated on the contract drawings. Units shall operate at 24 VDC and be polarized supervised. Each unit shall provide a choice of three different audible tones capable of being field selected. Preferred alarm signal shall be a temporal tone producing a sound pressure level of 84 dBA. Units shall be flush mounted and molded of high-impact white plastic.

I. Exterior Audio-Visual Devices

1. All audiovisual devices located outside or labeled weatherproof shall be weatherproof. Provide the following devices:
  - a. SpectrAlert Advance "PK" Series for audio/visual devices.
  - b. SpectrAlert Advance for "SK" Series for visual devices
  - c. All devices shall be provided with a weather proof type back box.

J. High Intensity Visual Signals

1. Provide a Notifier "S" Series SpectrAlert Advance visual signal device. High intensity visual signals shall be installed where shown on the drawings and as may be required by the Americans with Disabilities Act (Public Law 101-336) and TAS.
2. High intensity visual alarms shall be Xenon strobe type producing a minimum of 15 candela on a 24 VDC limited energy supervised circuit. Alarm devices shall be designated to be wall or ceiling mounted as indicated on the drawings. Signals shall operate in unison with audible alarm appliances. All visual devices shall be synchronized. Units shall be flush mounted and shall be provided in textured white.

K. Auxiliary AHU Relays

1. Notifier/Air Products model MR-101/C relays or approved equal shall be provided for HVAC and AHU control and interface. Relays shall be heavy duty type and rated up to 10 amps at 24 VDC, 60 HZ. Relays shall be provided with NEMA I dust cover assembly and be provided with SPDT contacts as well as (fail safe) so that if the cable is broken, disconnected etc., the AHU will automatically shut down.

L. Field Charging Power Supplies

1. Provide Notifier FCPS-24 power supplies with battery backup as required.

Provide 120 volts dedicated circuit to each power supply.

M. Remote LCD Alpha-Numeric Annunciators

1. Provide where indicated on the drawings, a Notifier FDU-80 remote LCD alpha-numeric annunciator to annunciate all system events and duplicate the displayed status at the main FACP. The annunciator shall be a backlit eighty-character LCD display and operate via the system RS485 and RS232 serial output terminal from main FACP. The LCD display shall automatically illuminate upon receipt of an alarm or trouble condition. The luminary source shall extinguish during normal/standby model to conserve power. The unit shall operate from FACP 24VDC power and function during system power failure while the system resides on standby batteries. The remote LCD annunciator shall include:
  - Integral time-date clock
  - Time-date select clock
  - Time-date/contrast adjust
  - Display/step switch
  - System reset
  - System silence
  - System acknowledge
  - Integral trouble buzzer
2. Annunciator shall upon command display the first system alarm, last alarm, and system alarm count. The unit shall be equipped with an integral lamp test feature. The unit shall be semi flush mounted where shown.

N. Protective Covers

1. Provide protective covers on all wall mounted fire alarm devices located in student restrooms, corridors and in the cafeteria. These protective covers shall be manufactured by Safety Technology International, Inc. (STI). These covers shall be provided on all devices including but not limited to smoke detectors, heat detectors, audible and visual devices, pull stations, etc. The mounting of a device shall be reinforced to enable the protective covers to protect the fire alarm devices.

## PART 3 - EXECUTION

### 3.1 DESIGN CRITERIA

- A. The contractor shall provide drawings for Owner, Engineer and Fire Marshall's approval.
- B. Drawings shall be prepared by a state licensed alarm planning superintendent.
- C. Drawings shall comply with all local, state and federal code. These include but not limited to N.E.C., U.L., NFPA 101, Etc.
- D. Locate the fire alarm control panel in the day room unless otherwise directed by Owner.
- E. Locate a remote annunciator in the lobby unless otherwise directed by Owner.
- F. Additional items required above minimum codes include the following:
  1. Pull Stations – All exits including exit stair wells on multi story buildings and at the FACP.
  2. Smoke Detectors – Paths of egress, electrical rooms, mechanical rooms, MDF,

- IDF, storage rooms, top of stairs, elevator machine room, top of elevator shaft, above each fire alarm panel and remote power supplies terminal cabinets.
- 3. Duct type smoke detectors – all air handling units over 2000 CFM in duct work or return air paths.
- 4. Heat Detectors – Shops, kitchens, coffee bars, central plants, boiler room and truck bays.
- 5. Flow switches – Sprinkler riser.
- 6. Horn - throughout the building.
- 7. Strobes – throughout the building.
- 8. Remote Power supplies: Locate in mechanical rooms, electrical rooms, MDF or other areas approved by Owner.
- 9. Smoke Detectors with low frequency sounder bases in all sleeping rooms.

### 3.2 INSTALLATION

#### A. Wiring:

- 1. All wiring shall be in accordance with NFPA 72 and the National Electrical Code, Local Codes, and article 760 of NFPA Standard 70. All wiring sizes shall conform to recommendations of the equipment manufacturer, and as indicated on the engineered shop drawings.
- 2. All wire shall be U.L. Listed, limited energy (300 volt) FPLP or MPP wire and shall be run open in return air ceiling plenums. The wire shall be listed to U.L. TEST 910 for such applications and is of the low smoke producing fluorocarbon type and complies with NEC Article 760 if so approved by the local authority having jurisdiction. Provide conduit in all inaccessible locations, inside concealed wall, all mechanical/electrical rooms, or other areas where wiring might be exposed and subject to damage.
- 3. Support wire clear of knock out panels, access panels, and maintenance spaces for equipment. Wire and cable shall be run using wire management techniques supporting cable as close as possible to within one foot of the floor or roof rafters. Wire supports shall be directly fastened to the structure on a maximum of five-foot centers. Wire routing shall be parallel and perpendicular to building lines. The wire and cable shall be secured with tie wraps or carrier wire. Sagging more than three inches will not be allowed nor will bending of the supporting ring structure.
- 4. All wiring for SLC signaling circuits shall be of the twisted low capacitance type to guard against outside RF and EMF interference and induced noise.
- 5. All wiring shall be run in a supervised fashion (i.e. no branch wiring or dog-legged wiring) per NFPA requirements such that any wiring disarrangement will initiate the appropriate trouble signals via the main control panel per NFPA and U.L. requirements.
- 6. Wiring splices shall be kept to a minimum with required splices to be made in designated terminal boxes or at field device junction boxes. Transposing or color code changes of wiring will not be permitted. End-of-line supervisory devices shall be installed with the last device on the respective circuit. Said device shall be appropriately marked designating it as the terminating device on the respective circuit.
- 7. No A.C. wiring or any other wiring shall be run in the same conduit as fire alarm wiring.

#### B. Conduit/Raceway

- 1. All wire shall be installed in an approved conduit/raceway system (except where

permitted by NEC and the local authority having jurisdiction). Maximum conduit "fill" shall not exceed 40% per NEC.

2. Conduit and raceway system shall be installed as specified under the general electrical section of the specifications, and per NEC.
3. Minimum conduit size shall be 3/4" EMT. Install conduit per engineered shop drawings.

C. Minimum Wire Sizes Shall Be as Follows:

1. Signaling Line Circuit: 18 AWG
2. Notification Appliance Circuit: 14 AWG
3. Relay Control Circuits: 18 AWG

D. Sprinkler Valves

1. Contractor shall connect all tamper switches and post indicator valves to the supervisory circuit. Connect all water flow switches to the alarm circuit. Coordinate exact locations of water vaults valves and flow switches with sprinkler contractor.

### 3.3 NOTIFICATION APPLIANCE CIRCUITS SYNCHRONIZATION

- A. All visual and audible devices shall be synchronized per the current state adopted version of NFPA 72. Provide all components required.

### 3.4 TEST AND REPORTS

- A. A state licensed factory trained technical representative of the manufacturer shall perform the final control panel connections and supervise testing of the system and it shall be subject to the approval of the responsible engineer and owner. Upon completion of the acceptance tests, the owner and/or his representatives shall be instructed in the proper operation of the system.
- B. The installing contractor shall functionally test each and every device in the entire system for proper operation and response. In addition, each circuit in the system shall be fully tested for wiring supervision to insure proper wiring installation. Any items found not properly installed or non-functioning shall be replaced or repaired and re-tested. All testing shall be supervised by a licensed fire alarm superintendent.
- C. The installing contractor shall provide a complete written report on the functional test of the entire system. The test and report shall verify the function of each device in the system, operation of all auxiliary control functions, and the proper operation of the main fire alarm control panel. A copy of the test report shall be provided with maintenance manuals. The test report shall be signed and dated by the licensed fire alarm superintendent responsible for supervising the final system test and checkout.
- D. The installing contractor's fire alarm superintendent shall test the entire system in the presence of the local authorities having jurisdiction.

### 3.5 SPARE DEVICES

- A. Provide 5% spare field devices including labor to install them. Devices not used shall be given to the Owner at completion of the job.

### 3.6 WARRANTY

- A. The fire alarm system shall be free from defects in workmanship and materials, under normal use and service, for a period of one year from the date of acceptance or beneficial occupancy, whichever shall occur first. Any equipment shown to be defective shall be repaired, replaced or adjusted during normal working hours at no cost to the owner.

### 3.7 GRAPHIC FLOOR PLANS

- A. Provide 1/16" = 1'-0" floor plan showing all devices and zoning. Zoning shall correspond to the zone on the fire alarm control panel. The floor plans shall be framed with a glass cover and located by the fire alarm control panel. This graphic floor plan shall use the actual room numbers based on the architectural graphics package. Verify specific requirements with Owner. Provide a sample for approval.

**END OF SECTION 28 31 00 13**